

On Feature Learning in Neural Networks

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To my parents.

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Abstract

Modern neural networks achieve remarkable performance by extracting useful structure from data through gradient-based optimization. Yet fundamental questions remain: How does this structure emerge during training, and why does it generalize so well? This thesis adopts a feature-learning perspective that studies what representations emerge from optimization and how the training algorithm, network architecture, and data distribution jointly govern this process. The thesis is organized into three parts. Together, they establish feature learning as a unifying perspective for understanding the optimization and generalization of modern neural networks.

Part I: Self-Supervised Representation Learning. We show that data augmentations in contrastive learning decouple spurious correlations from meaningful signals, and that architectural asymmetry in non-contrastive learning biases optimization toward diverse rather than collapsed features. For vision transformers, we further characterize why masked reconstruction learns both global and local features while contrastive objectives preferentially learn global features.

Part II: Reasoning and Post-Training in Transformers. We prove that transformers trained with suitable chain-of-thought traces achieve length generalization, which enables recursive self-improvement. We also show that reinforcement learning with verifiable rewards induces an implicit easy-to-hard curriculum: smooth difficulty spectra yield steady progress, while abrupt gaps produce grokking-type phase transitions.

Part III: Modern Recurrent Architectures. We show how gradient back-propagation through a recurrent-depth architecture trains the model to acquire progressively stronger reasoning capabilities. We then present an empirical study of the memory capacity of recurrent sequence models, measuring the information recoverable from a fixed-size recurrent state and establishing a near-linear scaling of recoverable information with state size, at approximately two bits per state dimension.

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Chapter 1

Introduction

Modern neural networks owe much of their success to a remarkable yet poorly understood ability: when trained by gradient-based optimization on large-scale raw data, they learn internal representations that capture the structure of the training data and generalize broadly to new settings. A network trained to classify images learns shape detectors and texture filters without being told to; a language model trained on next-token prediction acquires the ability to reason and solve problems not seen during training. Understanding *how* neural networks extract useful structure from data and *why* the resulting models generalize as well as they do are among the central questions in machine learning theory.

Classical approaches based on uniform convergence or black-box optimization analysis have yielded foundational insights, yet often do not explain the behavior of neural networks in practice. Uniform convergence bounds are typically too loose to be informative for overparameterized models, while kernel-based analyses such as the neural tangent kernel reduce training to convex optimization over a fixed representation and thereby abstract away the feature learning that gives neural networks much of their expressive power. In practice, the representations a network acquires depend critically on the interplay among the training algorithm, network architecture, and data distribution. A comprehensive theory must account for all three.

This thesis adopts a *feature-learning* perspective. Features are the activation patterns within a neural network that respond to particular components or structures in an input, such as attention patterns that implement content-based retrieval or weights that encode atomic computational operations. Feature learning is the process by which gradient-based optimization shapes the network’s parameters to produce these patterns. The central premise is that superficially different data points share underlying structures that neural networks can discover during training. Once an internal representation of such a structure emerges, the network can recognize and transform it in unseen data and thereby generalize.

Rigorous feature-learning analyses necessarily study simplified settings: shallow or lightly modified architectures, data distributions that admit closed-form descriptions, and training procedures chosen for tractability. These simplifications make it possible to track which features

emerge, in what order, and under what conditions. The resulting analyses bridge the gap between black-box optimization guarantees and often-vacuous uniform generalization bounds while providing mechanistic predictions that can be compared with large-scale practice.

Organization of the Thesis

The remainder of this thesis is organized into three parts, each examining feature learning through a different lens: the pretraining objective, the reasoning capabilities acquired during training and post-training, and the recurrent structure of the architecture itself. Complete proofs for the theoretical chapters are collected in the appendices.

Part I: Self-Supervised Representation Learning. The first part studies how self-supervised objectives shape the features that neural networks learn, and why design choices such as data augmentations, architectural asymmetry, and the choice of pretraining objective determine which features gradient training discovers, even though they leave the set of global optima essentially unchanged.

Chapter 2 analyzes the feature learning process of contrastive learning. In a sparse coding model where sparse meaningful signals coexist with dense spurious correlations, we track the full trajectory of SGD and show that data augmentations govern feature selection through a *feature decoupling* principle: without augmentations, training prefers the spurious dense correlations, whereas augmentations decouple them from the meaningful signals and drive the network through three stages of training toward sparse, semantically meaningful features.

Chapter 3 explains the mechanism of asymmetry in non-contrastive learning. Despite the existence of collapsed global optima, the trainable prediction head, acting together with the stop-gradient and output normalization, creates an implicit optimization bias toward diverse rather than collapsed features. The analysis traces four phases of the learning process, identifying substitution and acceleration effects through which the asymmetric architecture rescues features that would otherwise be discarded.

Chapter 4 turns to self-supervised vision transformers and compares masked reconstruction with contrastive pretraining. We characterize how the two objectives shape attention differently: masked reconstruction learns both global, area-wide patterns and local, fine-grained features, while contrastive objectives preferentially learn global features, explaining the empirical differences between the representations the two paradigms produce.

Part II: Reasoning and Post-Training in Transformers. The second part studies how transformers acquire compositional reasoning ability, both from supervised chain-of-thought training and from reinforcement learning in post-training.

Chapter 5 proves that transformers trained on chain-of-thought data achieve length generalization: they solve problems requiring more compositional steps than any example seen in training, with the extent of extrapolation governed by the algebraic structure of the underlying

task. We further show how this extrapolation enables recursive self-improvement, in which a model’s own outputs on harder problems supply correct training data for the next round of training.

Chapter 6 analyzes the learning dynamics of reinforcement learning with verifiable rewards. We show that a mixed-difficulty training distribution induces an implicit easy-to-hard curriculum: when the difficulty spectrum is smooth, success on easier instances steadily transfers to harder ones, whereas abrupt difficulty gaps produce grokking-type phase transitions in which progress stalls before harder levels are learned rapidly.

Part III: Modern Recurrent Architectures. The final part examines two complementary consequences of recurrence: how shared computation builds and corrects a hierarchy of representations across recurrent depth, and how much information a fixed-size recurrent state makes available across sequence positions.

Chapter 7 studies hierarchical learning in loop transformers, which reuse one shared block across recurrent depth. Its two-stage account first shows how supervision at the final output alone learns a hidden lower-level composition, then establishes, at each fixed recurrent depth, stable forward reuse, constructive full-network gradients, and backward feature correction: because the same computation circuit occurs at every recurrent level, deeper supervision also tightens error bounds for earlier representations. A finite shallow-to-deep transfer path carries the same parameters to greater depths, so that back-propagation through the recurrent architecture trains the model to acquire progressively stronger reasoning capabilities and learn a prescribed deep hierarchy.

Chapter 8 presents an empirical study of the memory capacity of modern recurrent sequence models. Using random sequential retrieval tasks, it measures how much information a trained DeltaNet can recover from its fixed-size recurrent state and finds a near-linear capacity scaling of approximately two retrieved bits per counted state parameter. The result connects architectural memory resources with the information that learning makes accessible at inference time.

Part I

Self-Supervised Representation Learning

Chapter 2

The Feature Learning Process of Contrastive Learning

2.1 Introduction

Self-supervised learning [74, 140, 196, 264] has demonstrated its immense power in different areas of machine learning (e.g. BERT [74] in natural language processing). Recently, it has been discovered that contrastive learning [45, 49, 53, 106, 113, 271], one of the most typical forms of self-supervised learning, can indeed learn representations of image data that achieve superior performance in many downstream vision tasks. Moreover, as shown by the seminal work [113], the learned feature representations can even outperform those learned by supervised learning in several downstream tasks. The remarkable potential of contrastive learning methods poses challenges for researchers to understand and improve upon such simple but effective algorithms.

Contrastive learning in vision learns the feature representations by minimizing pretext task objectives similar to the cross-entropy loss used in supervised learning, where both the inputs and “labels” are derived from the unlabeled data, especially by using augmentations to create multiple views of the same image. The seminal paper [48] has demonstrated the effects of stronger augmentations (comparing to supervised learning) for the improvement of feature quality. [272] showed that as the augmentations become stronger, the quality of representations displayed a U-shaped curve. Such observations provided insights into the inner-workings of contrastive learning. But it remains unclear *what has happened in the learning process* that renders augmentations necessary for successful contrastive learning.

Some recent works have been done to understand contrastive learning from theoretical perspective [20, 284, 295]. However, these works have not analyzed how *data augmentations* affect the *feature learning process* of *neural networks*, which we deem as crucial to understand how

This chapter is adapted from the following joint work:

Zixin Wen and Yuanzhi Li. *Toward Understanding the Feature Learning Process of Self-Supervised Contrastive Learning*.

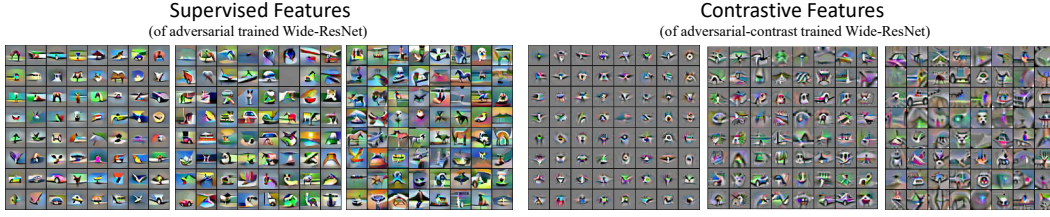


Figure 2.1: Supervised And Contrastive Features. The difference between supervised features and contrastive features (in the higher layers of Wide-ResNet 34x5 over CIFAR10). While both features contain shapes of objects, the supervised features are more colorful than the contrastive features. (here both crop-resize and color distortion were used in contrastive learning, while no color distortion was used in supervised learning. The adversarial-contrast learning follows [148]). And we use the visualization technique in [8].

contrastive learning works in practice. We state the fundamental questions we want to address below, and provide tentative answers to all the questions by building theory on a simplified model that shares similar structures with real scenarios, and we provide some empirical evidence through experiments to verify the validity of our models.

Fundamental Questions

1. How do *neural networks* trained by contrastive learning learn their feature representations *efficiently*, and are the representations similar to those learned in supervised learning?
2. Why does contrastive learning in deep learning collapse in practice when no augmentation is used, and how do standard augmentations on the data help contrastive learning?

2.1.1 Our Contributions

In this paper we directly analyze the *feature learning process* of contrastive learning for neural networks (i.e. learning the hidden layers of the neural network). Our results hold for certain data distributions based on *sparse coding model*. Mathematically, we assume our input data are of the form $x = \mathbf{M}z + \xi$, where $\mathbf{M}z$ is called the sparse signal such that $\|z\|_0 = \widetilde{O}(1)$, and ξ is the *spurious dense noise*, where we simply assume that ξ follows from certain dense distributions (such that $\text{span}(\xi) \equiv \text{span}(x)$) with large norm (e.g., $\|\xi\|_2 = \text{poly}(d) \gg \|\mathbf{M}z\|_2 \approx \widetilde{O}(1)$). Formal definition will be presented in Section 2.2, as we argue that sparse coding model is indeed a proper *provisional* model to study the feature learning process of contrastive learning.

Theoretical results. Over our data distributions based on sparse coding model, when we perform contrastive learning by using stochastic gradient descent (SGD) to train a one-hidden-layer neural networks with ReLU activations:

1. If no augmentation is applied to the data inputs, *the neural networks will learn feature representations that emphasize the spurious dense noise*, which can easily overwhelm the sparse signals.

2. If *natural* augmentation techniques (in particular, the RandomMask defined in [Definition 2.2.3](#)) are applied to the training data, *the neural networks will avoid learning the features associated with dense noise but pick up the features on the sparse signals*. Such a difference of features brought by data augmentation is due to a principle we refer to as “*feature decoupling*”. Moreover, these features can be learned *efficiently* simply by doing a variant of Stochastic Gradient Descent (SGD) over the contrastive training objective (after data augmentations).
3. The features learned by neural networks via contrastive learning (with augmentations) is similar to the features learned via supervised learning (under sparse coding model). This claim holds as long as two requirements are satisfied: (1) The sparse signals in the data have not been corrupted by augmentations in contrastive learning; (2) The labels in supervised learning mostly depends on the sparse signals.

Therefore, our theory indicates that in our model, the success of contrastive learning of neural networks relies essentially on the data augmentations to remove the features associated with the spurious dense noise. We abstract this process into a principle below, which we show to hold in neural networks used in real-world settings as well.

Feature Decoupling

Augmentations in contrastive learning serve to *decouple the correlations of spurious features* between the representations of positive samples. Moreover, after the augmentations, the neural networks will ignore the decoupled features and learn from the similarities of features that are *more resistant to data augmentations*.

We will prove that contrastive learning can successfully learn the desired sparse features using this principle. The intuitions of our proof will be present in [Section 2.4](#).

Empirical evidence of our theory. Empirically, we conduct multiple experiments to justify our theoretical results, and the results indeed matches our theory. We show in contrastive learning:

- *When no proper augmentation is applied to the data, the neural network will learn features with dense patterns*. As shown in [Figure 2.2](#), [Figure 2.3](#) and [Figure 2.4](#): If no augmentations are used, the learned features are completely meaningless and the representations are dense; If only crop-resize augmentations are used, then the mixture of color features (which also generate dense firing patterns) will remain in the neural network and prevent further separation of clusters.
- *Standard augmentations removes features associated with dense patterns, and the remaining features do exhibit sparse firing pattern*. As shown in [Figure 2.3](#) and [Figure 2.4](#), if no (suitable) augmentations are applied, the neural networks will learn dense representations of image data. After the augmentations, neural networks will successfully form separable clusters of representations for image data, and the *learned features indeed emphasizes sparse signals*.
- *The features learned in contrastive learning resemble the features learned in supervised learning*. As shown in [Figure 2.1](#), the shape features (filters that exhibit shape images) of the higher

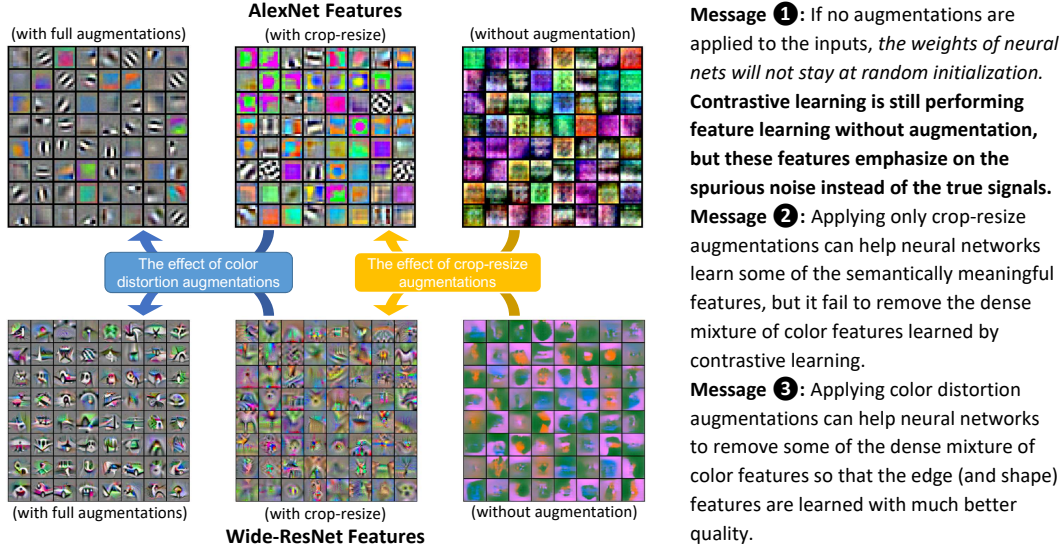


Figure 2.2: Augmentations Decouple Learned Features. Evidence of *feature decoupling*: how do augmentations affect the features learned by neural networks in contrastive learning. The two different augmentations we have conducted here are *color distortions* and *crop-resize*. The color distortions we used consist of color jittering and random grayscale.

layer of Wide-ResNet via supervised learning are similar to those learned in contrastive learning. However, color features learned in supervised learning are much more than those in contrastive learning. This verifies our theoretical results that features preserved under augmentations will be learned by both contrastive and supervised learning.

2.1.2 Related Work

Self-supervised learning. Self/un-supervised representation learning has a long history in the literature. In natural language processing (NLP), self-supervised learning has been the major approach [74, 196]. The initial works [41, 110, 258] of contrastive learning focus on learning the hidden latent variables of the data. Later the attempts to use self-supervised to help pretraining brought the contrastive learning to visual feature learning [45, 49, 53, 106, 113, 206, 271]. On the theoretical side, there has been a lot of papers trying to understand un/self-supervised learning [20, 63, 112, 112, 158, 200, 225, 272, 280, 281, 284, 295]. For contrastive learning, [20] assume that different positive samples are independently drawn from the same latent class, which can be deemed as supervised learning. [295] pointed out the tradeoff between alignment and uniformity. [272, 284] proposed to analyze contrastive learning via information-theoretic techniques. [158] analyzed the optimal solution of a generative self-supervised pretext task, and [281] analyzed contrastive loss from the same perspective. [112] analyzed a spectral version of contrastive loss and analyzed its statistical behaviors. However, the above theoretical works do not study *how features are learned by neural networks* and *how augmentations affect the learned features*, which are

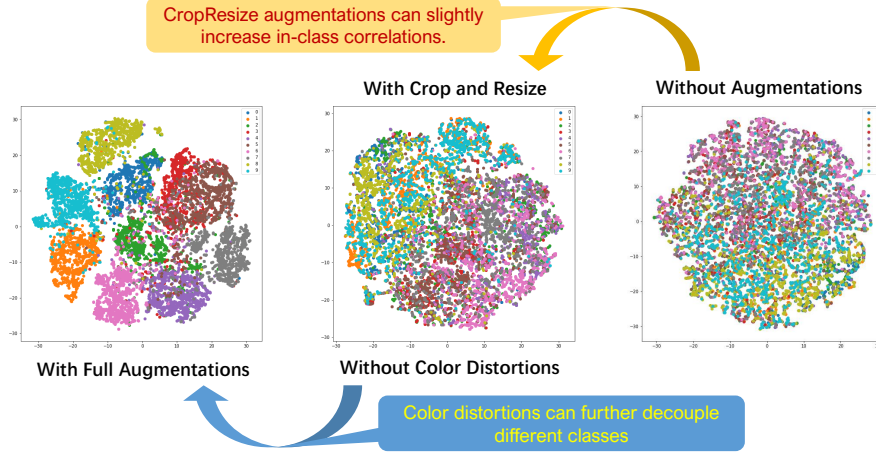


Figure 2.3: Weak Augmentation Produces Dense Representations. Evidence supporting our theoretical framework: the effects of augmentations on the learned representations of Wide-ResNet 34x5 over CIFAR10 visualized via t-SNE. The differences between features learned under different augmentations shows that the neural networks will indeed learn *dense representations* if augmentation is not powerful enough.

essential to understand contrastive learning in practice. [275] tries to analyze the learning process, but their augmentation can fix the class-related node and resample all latent nodes in their generative models, reducing the problem to supervised learning.

Optimization theory of neural networks. There are many prior works on the supervised learning of neural networks. The works [36, 97, 167, 168, 259] focus on the scenarios where data inputs are sampled from Gaussian distributions. *We consider in our paper the Gaussian part of the data to be spurious* and use augmentation to prevent learning from them. Our approach is also fundamentally different from the *neural tangent kernel* (NTK) point of view [11, 12, 13, 56, 78, 132, 166]. The NTK approach relies first order Taylor-expansion with extreme over-parameterization, and cannot explain the *feature learning process* of neural networks, because it is merely linear regression over *prescribed feature map*. Some works consider the regimes beyond NTK [5, 6, 7, 8, 9, 25, 170], which shedded insights to the innerworkings of neural networks in practice.

2.2 Problem Setup

Notations. We use O, Ω, Θ notations to hide universal constants with respect to d and $\tilde{O}, \tilde{\Omega}, \tilde{\Theta}$ notations to hide polylogarithmic factors of d . We use the notations $\text{poly}(d)$, $\text{polylog}(d)$ to represent constant degree polynomials of d or $\log d$. We use $[d]$ as a shorthand for the index set $\{1, \dots, d\}$. For a matrix $\mathbf{M} \in \mathbb{R}^{d' \times d}$, we use $\mathbf{M}_j$, where $j \in [d]$, to denote its j -th column. We say an event happens with high probability (or w.h.p. for short) if the event happens with probability at

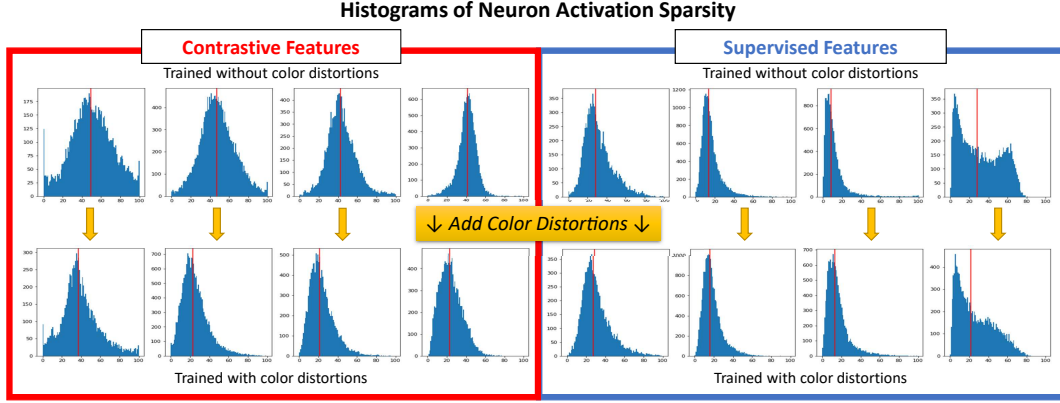


Figure 2.4: Color Distortion Promotes Sparse Features. Another evidence supporting our theoretical framework. After adding the color distortion to augmentation, the neurons of AlexNet (2nd to 5th layer) exhibit sparser firing patterns over input images of CIFAR10. Meanwhile the networks obtained from supervised learning always have sparse activations regardless of augmentations. These observations indicate that (1). In contrastive learning, augmentations can indeed help neural nets focus on the sparse signals. (2). Sparse signals are indeed more important for the downstream tasks (such as supervised classification).

least $1 - e^{-\Omega(\log^2 d)}$. We use $\mathcal{N}(\mu, \Sigma)$ to denote standard normal distribution in with mean μ and covariance matrix Σ .

2.2.1 Data Distribution.

We present our sparse coding model below, which form the basis of our analysis.

Definition 2.2.1 (sparse coding model $(\mathcal{D}_x, \mathcal{D}_z, \mathcal{D}_\xi)$). We assume our raw data samples $x \in \mathbb{R}^{d_1}$ are generated i.i.d. from distribution $\mathcal{D}_x$ in the following form:

$$x = \mathbf{M}z + \xi \sim \mathcal{D}_x, \quad z \sim \mathcal{D}_z, \quad \xi \sim \mathcal{D}_\xi = \mathcal{N}(\mathbf{0}, \sigma_\xi^2 \mathbf{I}_{d_1})$$

Where $z \in \mathbb{R}^d$. We refer to z as the *sparse signal* and ξ as the *spurious dense noise*. We assume $d_1 = \text{poly}(d)$ for simplicity. We have the following assumptions on $\mathbf{M}, z, \xi$ respectively:

- The dictionary matrix $\mathbf{M} = [\mathbf{M}_1, \dots, \mathbf{M}_d] \in \mathbb{R}^{d_1 \times d}$ is a column-orthonormal matrix, and satisfies $\|\mathbf{M}_j\|_\infty \leq \tilde{O}(\frac{1}{\sqrt{d_1}})$ for all $j \in [d]$.
- The sparse latent variable $z = (z_1, \dots, z_d)^\top \in \{-1, 0, 1\}^d$ is sampled from $\mathcal{D}_z$, we assume all z_j 's are symmetric around zero, satisfying $\Pr(|z_j| = 1) = \Theta(\frac{\log \log d}{d})$, and are identically distributed and independent across all $j \in [d]$.¹
- For the spurious dense noise $\xi \sim \mathcal{N}(\mathbf{0}, \sigma_\xi^2 \mathbf{I}_{d_1})$, we assume its variance $\sigma_\xi^2 = \Theta(\frac{\sqrt{\log d}}{d})$.

Why sparse coding model. Sparse coding model was first proposed by neuroscientists to model human visual systems [204, 205], where they provided experimental evidence that sparse codes can produce coding matrices for image patches that resemble known features in certain portion of the visual cortex. It has been further studied by [86, 185, 205, 217, 289, 325] to model images based on the sparse occurrences of objects. For the natural language data, sparse code is also found to be helpful in modelling the polysemy of words [17]. Thus we believe our setting share some similar structures with practical scenarios.

Why sparse signals are more favorable than the dense signal. Theoretically, we argue that sparse signals are more favorable as we can see from the properties of our sparse signals $\mathbf{Mz}$ and dense signals ξ :

1. *The significance of sparse signal.* Since $\sigma_\xi^2 = \Theta(\frac{\sqrt{\log d}}{d})$, the ℓ_2 -norm of ξ becomes $\|\xi\|_2^2 \geq \Omega(\text{poly}(d)) \gg \|\mathbf{Mz}\|_2$ w.h.p. However, whenever there is one $z_j \neq 0$, we have $|\langle \mathbf{Mz}, \mathbf{M}_j \rangle| \geq \Omega(1)$ while $|\langle \xi, \mathbf{M}_j \rangle| \leq \tilde{O}(\frac{1}{\sqrt{d}})$ with high probability. This indicates that even if the dense signal is extremely large in norm, it cannot corrupt the sparse signal.
2. *The individuality of dense signal.* For each $j \in [d]$, the sparse feature $\pm \mathbf{M}_j$ are shared by at least $\tilde{\Omega}(\frac{1}{d})$ of the population. However, for polynomially many independent dense signal ξ_i , with high probability we have $|\langle \frac{\xi_i}{\|\xi_i\|_2}, \frac{\xi_j}{\|\xi_j\|_2} \rangle| \leq \tilde{O}(\frac{1}{\text{poly}(d)})$ for any $i \neq j$, which shows that the dense signal ξ is in some sense “individual to each sample”. This also suggests that any representations of the dense signal can hardly form separable clusters other than isolated points.

2.2.2 Learner Network and Contrastive Learning Algorithm

We use a single-layer neural net $f : \mathbb{R}^{d_1} \rightarrow \mathbb{R}^m$ with ReLU activation as our contrastive learner, where m is the number of neurons. More precisely, it is defined as follows:

$$f(x) = (h_1(x), \dots, h_m(x))^T \in \mathbb{R}^m, \quad h_i(x) = \text{ReLU}(\langle w_i, x \rangle - b_i) - \text{ReLU}(-\langle w_i, x \rangle - b_i)$$

Such activation function h_i is a symmetrized version of ReLU activation. We initialize the parameters by $w_i^{(0)} \sim \mathcal{N}(0, \sigma_0^2 \mathbf{I}_{d_1})$ and $b_i^{(0)} = 0$, where $\sigma_0^2 = \Theta(\frac{1}{d_1 \text{poly}(d)})$ is small (and also theoretically friendly). Corresponding to the two types of signals in Definition 2.2.1, we call the learned weights of neural networks $\{w_i\}_{i \in [m]}$ “features”, and we expand the weight of a neuron as

$$w_i = \sum_{j \in [d]} \langle w_i, \mathbf{M}_j \rangle \mathbf{M}_j + \sum_{j \in [d_1] \setminus [d]} \langle w_i, \mathbf{M}_j^\perp \rangle \mathbf{M}_j^\perp$$

where we name the (unit-norm) directions $\mathbf{M}_j$ and $\mathbf{M}_j^\perp$ as follows:

¹ The choice of $\Pr(|z_j| = 1) = \Theta(\frac{\log \log d}{d})$ instead of $\Theta(\frac{1}{d})$ here is to avoid the scenario where z could be zero with probability $\geq \Omega(1)$. One can also assume the noise vector ξ to be non-spherical Gaussian or has certain directions with larger variance than $\Theta(\sqrt{\log d}/d)$. Although our theory tolerates a wider range of these parameters, we choose to present the simplest setting.

- We call $\mathbf{M} = [\mathbf{M}_j]_{j \in [d]}$ the *sparse features*, which is the features associated with our sparse signals $\mathbf{M}z$. These are the desired features we want our learner network to learn.
- We call $\mathbf{M}^\perp = [\mathbf{M}_j^\perp]_{j \in [d_1] \setminus [d]}$ (the orthogonal complement of $\mathbf{M}$) the *spurious dense features*, which is associated with the dense signal ξ only. These are the undesired features for our learner.

Our contrastive loss function is based on the similarity measure defined as follows: let x and x' be two samples in $\mathbb{R}^{d_1}$, and $f : \mathbb{R}^{d_1} \rightarrow \mathbb{R}^d$ be a feature map, the similarity of the representations of x and x' is defined as

$$\text{Sim}_f(x, x') := \langle f(x), \text{StopGrad}(f(x')) \rangle \quad (2.2.1)$$

The $\text{StopGrad}(\cdot)$ operator here means that we do not compute its gradient in optimization, which is inspired by recent works [53, 106]. Below we present the definition of contrastive loss.

Definition 2.2.2 (Contrastive loss function). Given a pair of positive data samples x_p, x'_p and a batch of negative data samples $\mathfrak{N} = \{x_{n,s}\}_{s \in [\mathfrak{N}]}$, letting τ be the temperature parameter, and denoting $\mathfrak{B} = \{x'_p\} \cup \mathfrak{N}$, the contrastive loss is defined as²

$$\mathcal{L}(f, x_p, x'_p, \mathfrak{N}) := -\tau \log \left(\frac{e^{\text{Sim}_f(x_p, x'_p)/\tau}}{\sum_{x \in \mathfrak{B}} e^{\text{Sim}_f(x_p, x)/\tau}} \right) \quad (2.2.2)$$

Nevertheless, as shown by our experiments (see Figure 2.2 or Figure 2.3), the success of contrastive learning rely on the data augmentations adopted in generating the positive samples. We present our augmentation method RandomMask below, which is an analog of the random cropping data augmentation used in practice.

Definition 2.2.3 (RandomMask and $\mathcal{D}_D$). We first define a distribution $\mathcal{D}_D$ over the space $\mathbb{R}^{d_1 \times d_1}$ of diagonal matrices as follows: let $\mathbf{D} = \text{diag}(\mathbf{D}_{\ell, \ell})_{\ell \in [d_1]} \sim \mathcal{D}_D$ be a diagonal matrix with $\{0, 1\}$ entries, its diagonal entries $\mathbf{D}_{\ell, \ell}$ are sampled from $\text{Bernoulli}(\frac{1}{2})$ independently. Now given a positive sample $x_p \sim \mathcal{D}_x$, we generate $\mathbf{D} \sim \mathcal{D}_D$, and then apply $\mathbf{D}$ to generate x_p^+ and x_p^{++} as follows:

$$x_p^+ := 2\mathbf{D}x_p, \quad x_p^{++} := 2(\mathbf{I} - \mathbf{D})x_p$$

Remark 2.2.1. We do not apply any augmentation to our negative samples for simplicity of theory. And also we point out that adding such augmentations do not reveal any further insights, since we do not expect the augmentation to decouple any correlations other than that between positive samples. Nevertheless our theory can easily adapt to the setting where augmentations are applied to all input data.

² Our contrastive loss (2.2.2) here uses the unnormalized representations instead of the normalized ones, which is simpler to analyze theoretically. As shown in [45], contrastive learning using unnormalized representation can also achieve meaningful (more than 57%) ImageNet top-1 accuracy in linear evaluation of the learned representations.

Intuitions behind the RandomMask augmentation. Intuitively, the RandomMask data augmentation simply masks out roughly a half of the coordinates in the data. The contrastive learning objective asks to learn features that can match *two disjoint set of the coordinates* of given data points. Suppose we can maintain the correlations of desired signals between the disjoint coordinates and remove the undesired correlations, then we can force the algorithm to learn from the desired signals. We will discuss the effects of augmentations with more detail in [Section 2.4](#).

Significance of our analysis on the data augmentations. Our analysis on the data augmentation are fundamentally different from those in previous works [158, 275, 284, 302]. In [275, 284], they argued their data augmentations can change the latent variables unrelated to the downstream tasks, while real-life augmentations can only affect the observables, and cannot identify which latents are the task-specific ones. [302] assumed their augmentations are only picking data points inside a small neighborhood of the original data (in the observable space), which is also untrue in practice. Indeed, common augmentations such as crop-resize and color distortions can considerably change the data, making it very distant to the original data in the observable space. Our analysis of RandomMask makes a step toward understanding realistic data augmentations in deep learning.

Training algorithm using SGD. We consider two cases: training with augmentation and without augmentation:

- *With augmentations.* We perform stochastic gradient descent on the following objectives: let f_t be the contrastive learner at each iterations $t \geq 0$, the objectives is defined as follows:

$$L(f_t) := \mathbb{E}_{x_p^+, x_p^{++}, \mathfrak{R}} \left[\mathcal{L}(f_t, x_p^+, x_p^{++}, \mathfrak{R}) \right],$$

$$\mathbf{Obj}(f_t) := L(f_t) + \frac{\lambda}{2} \sum_{i \in [m]} \|w_i^{(t)}\|_2^2$$

where $\lambda \in [\frac{1}{d^{1.001}}, \frac{1}{d^{1.499}}]$ is the regularization parameter, $L(f_t)$ is the population loss and $x_p, \{x_{n,s}\}_{\mathfrak{R}}$ are sampled from $\mathcal{D}_x$, x_p^+, x_p^{++} are obtained by applying RandomMask to x_p . At each iteration t , let $\eta = \frac{1}{\text{poly}(d)}$ be the learning rate, we update as:

$$w_i^{(t+1)} \leftarrow w_i^{(t)} - \eta \nabla_{w_i} \mathbf{Obj}(f_t)$$

- *Without augmentations.* We perform stochastic gradient descent on the following modified objectives $\mathbf{Obj}_{\text{NA}}(f_t)$:

$$L_{\text{NA}}(f_t) := \mathbb{E}_{x_p, \mathfrak{R}} \left[\mathcal{L}(f_t, x_p, x_p, \mathfrak{R}) \right],$$

$$\mathbf{Obj}_{\text{NA}}(f_t) := L_{\text{NA}}(f_t) + \frac{\lambda}{2} \sum_{i \in [m]} \|w_i^{(t)}\|_2^2$$

where $\lambda \leq O(1/d)$ can be arbitrary. The learning rate $\eta \leq o(1)$ can also be arbitrary. We update as:

$$w_i^{(t+1)} \leftarrow w_i^{(t)} - \eta \nabla_{w_i} \mathbf{Obj}_{\text{NA}}(f_t)$$

We manually tune bias³ $b_i^{(t)}$ during the training process as follows: let $T_1 = \Theta\left(\frac{d \log d_1}{\eta \log \log d}\right)$ be the iteration when all $\|w_i^{(0)}\|_2 \leq \frac{c_0}{1000} \|w_i^{(t)}\|_2$. At $t = T_1$, we reset the bias $b_i^{(t)} = \sqrt{\frac{2 \log d}{d}} \|w_i^{(t)}\|_2$ and update by $b_i^{(t+1)} = b_i^{(t)} (1 + \eta_{b,t})$, where $\eta_{b,t} = \max\{\frac{\eta}{d}, \frac{\|w_i^{(t+1)}\|_2}{\|w_i^{(t)}\|_2} - 1\}$ if $b_i^{(t)} \leq \frac{\text{polylog}(d)}{\sqrt{d}}$.⁴

2.3 Main Results

We now state the main theorems of this paper in our setting. We argue that contrastive learning objective learns completely different features with/without data augmentation. *Moreover, to further illustrate the how these learned features are different with/without data augmentation, we also consider two simple downstream tasks to evaluate the performance of contrastive learning.* We argue that using a linear function taking the learned representation as input to perform these tasks can be more efficient than using raw inputs, it should be considered as successful representation learning.

Definition 2.3.1 (downstream tasks). We consider two simple supervised tasks, regression and classification, based on the label functions defined below:

- Regression: For each $x = \mathbf{M}z + \xi \sim \mathcal{D}_x$, we define its label $y = \langle w^*, z \rangle$, where $w^* \in \mathbb{R}^d$.
 - Classification: For each $x = \mathbf{M}z + \xi \sim \mathcal{D}_x$, we define $y = \text{sign}(\langle w^*, z \rangle)$, where $w^* \in \mathbb{R}^d$.
- where in both cases we assume w^* satisfies $|w_j^*| = \Theta(1)$ for all $j \in [d]$.

Given these downstream tasks, our goal of representations learning is to obtain suitable feature representations and train a linear classifier over them. Specifically, let $f(\cdot)$ be the obtained representation map, we use optimization tool⁵ to find w^* such that

$$w^* = \arg \min_{w \in \mathbb{R}^m} \mathbb{E}[\tilde{\mathcal{L}}(w^\top f(x), y)]$$

where $\tilde{\mathcal{L}}(\cdot, \cdot)$ is the loss function for the downstream tasks considered: For regression, it is the ℓ_2 loss $\tilde{\mathcal{L}}(\hat{y}, y) = (\hat{y} - y)^2$; For classification, it is the logistic loss $\tilde{\mathcal{L}}(\hat{y}, y) = \log(1 + e^{-\hat{y}y})$. It should be noted that these tasks can be done by neural networks via supervised learning as shown in [8], where the sample complexities have not been calculated exactly. However, using linear regression

³ In fact, when trained without augmentations, the biases can be tuned arbitrarily as long as the neurons are not killed. It will not affect our results.

⁴ We manually increase the bias after the weights are updated in order to simplify the proof. It can be verified that the biases will indeed increase over synthetic sparse coding data. More importantly, in synthetic experiments, the bias will decrease if no augmentation is used.

⁵ Since the downstream learning tasks only involve linear learners on convex objectives, for simplicity, we directly argue the properties of the minimizers for these downstream training objectives.

over the input x to find $\mathbf{M}$ requires sample complexity at least $\sqrt{d_1}\sigma_\xi \gg \text{poly}(d)$, which under our setting can be as large as d^{100} , much larger than those of linear regression over contrastive features from our results. Furthermore, even if one can locate the desired features $\mathbf{M}$, the noise level $\sigma_\xi^2 = \Theta(\frac{\sqrt{\log d}}{d})$ is still much larger than the signal size $\mathbb{E}[z_j^2] = \Theta(\frac{\log \log d}{d})$, thus linear models will fail with constant probability.

2.3.1 Contrastive Learning Without Augmentations

We present our theorem for the learned features without using any augmentations.

Theorem 2.3.1 (Contrastive features learned without augmentation). *Let f_t^{NA} be the neural network trained by contrastive learning without any data augmentations, and using $|\mathcal{N}| = \text{poly}(d)$ many negative samples, we have objective guarantees $L_{\text{NA}}(f_t^{\text{NA}}) = o(1)$ for any $t \geq \frac{\text{poly}(d)}{\eta}$. Moreover, given a data sample $x = \mathbf{M}z + \xi \sim \mathcal{D}_x$, with high probability it holds:*

$$\frac{\langle f_t^{\text{NA}}(x), f_t^{\text{NA}}(\xi) \rangle}{\|f_t^{\text{NA}}(x)\|_2 \|f_t^{\text{NA}}(\xi)\|_2} \geq 1 - \tilde{O}\left(\frac{1}{\text{poly}(d)}\right)$$

This results means that in the representations of f_t , the sparse signal $\mathbf{M}z$ are completely overwhelmed by the spurious dense signal ξ . It would be easy to verify the following corollary:

Corollary 2.3.1 (Downstream task performance). *The learned network f_t^{NA} , where $t \geq 0$, fail to achieve meaningful ℓ_2 -loss/accuracy in the downstream tasks in [Definition 2.3.1](#). More specifically, no matter how many labeled data we have for downstream linear evaluation (where f_t^{NA} is frozen):*

- For regression, we have

$$\mathbb{E}_{x \sim \mathcal{D}_x} |y - \langle w^*, f_t^{\text{NA}}(x) \rangle|^2 \geq \Omega(1)$$

- For classification, we have

$$\Pr_{x \sim \mathcal{D}_x} [y = \text{sign}(\langle w^*, f_t^{\text{NA}}(x) \rangle)] = o(1)$$

2.3.2 Contrastive Learning With Augmentation

We present our results of the learned features after successful training with augmentations.

Theorem 2.3.2 (Contrastive features learned with augmentation). *Let $m = d^{1.01}$ be the number of neurons, $\tau = \text{polylog}(d)$, and $|\mathcal{N}| = \text{poly}(d)$ be the number of negative samples. Suppose we train the neural net f_t via contrastive learning with augmentation, then for some small constant $c < \frac{1}{1000}$,*

and some iterations $T \in [T_3, T_4]$, where $T_3 = \frac{d^{1.01}}{\eta}$, $T_4 = \frac{d^{1.99}}{\eta}$, we have objective guarantees

$$\frac{1}{T} \sum_{t \in [T]} \mathbf{Obj}(f_t) \leq o(1), \quad \frac{1}{T} \sum_{t \in [T]} L(f_t) \leq o(1)$$

Moreover, for each neuron $i \in [m]$ and $t \in [T_3, T_4]$, contrastive learning will learn the following set of features:

$$w_i^{(t)} = \sum_{j \in \mathcal{N}_i} \alpha_{i,j} \mathbf{M}_j + \sum_{j \notin \mathcal{N}_i} \alpha'_{i,j} \mathbf{M}_j + \sum_{j \in [d_1] \setminus [d]} \beta_{i,j} \mathbf{M}_j^\perp$$

where $\alpha_{i,j} \in [\frac{\tau}{d^\epsilon}, \tau]$, $|\mathcal{N}_i| = O(1)$, $\alpha'_{i,j} \leq o(\frac{1}{\sqrt{d}}) \|w_i^{(t)}\|_2$ and $|\beta_{i,j}| \leq o(\frac{1}{\sqrt{d_1}}) \|w_i^{(t)}\|_2$. Furthermore, for each dictionary atom $\mathbf{M}_j$, there are at most $o(m/d)$ many $i \in [m]$ such that $j \in \mathcal{N}_i$, and at least $\Omega(1)$ many $i \in [m]$ such that $\mathcal{N}_i = \{j\}$.

This result indicates the following: let $x = \mathbf{M}z + \xi \sim \mathcal{D}_x$ be a data sample and $f_t, t \in [T_3, T_4]$ be the trained network, then $\|f_t(x) - f_t(\mathbf{M}z)\|_2 \leq \tilde{O}(\frac{1}{\sqrt{d}})$ with high probability, while $\|f(\mathbf{M}z)\|_2 \geq \Omega(1)$ with probability at least $1 - \frac{1}{\text{polylog}(d)}$. Thus the learned feature map has successfully removed the spurious dense noise ξ from the model/representation. We have a direct corollary following this theorem.

Corollary 2.3.2 (Downstream task performance). *The learned feature map $f_t, t \in [T_3, T_4]$ obtained by contrastive learning perform well in all the downstream tasks defined in Definition 2.3.1. Specifically, we have*

1. *For the regression task, with sample complexity at most $\tilde{O}(d^{1.001})$, we can obtain $w^* \in \mathbb{R}^m$ such that*

$$\mathbb{E}_{x \sim \mathcal{D}_x} |y - \langle w^*, f_t(x) \rangle|^2 = o(1)$$

2. *For the classification task, again by using logistic regression over feature map f_t , with sample complexity at most $\tilde{O}(d^{1.001})$, we can find $w^* \in \mathbb{R}^d$ such that*

$$\Pr_{x \sim \mathcal{D}_x} [y = \text{sign}(\langle w^*, f_t(x) \rangle)] = 1 - o(1)$$

2.4 Proof Intuition: The Feature Decoupling Principle

Theoretically speaking, contrastive learning objectives can be view as two parts, as is also observed in [295]:

$$\mathcal{L} = -\text{Sim}_f(x_p, x'_p) + \tau \log \left(\sum_{x \in \mathfrak{B}} e^{\text{Sim}_f(x_p, x)/\tau} \right)$$

where the first part $-\text{Sim}_f(x_p, x'_p)$ emphasize similarity between positive samples, and the second part $\tau \log \left\{ \sum_{x \in \mathfrak{B}} e^{\text{Sim}_f(x_p, x)/\tau} \right\}$ emphasize dissimilarities between the positive and negative samples.

To understand what happens in the learning process, we separately discuss the cases of learning with/without augmentations below:

Why does contrastive learning prefer spurious dense noise without augmentation? Without data augmentation, we simply have $x_p = x'_p$. In this case, contrastive learning will learn to emphasize the signals that simultaneously maximize the correlation $\langle f(x_p^{++}), f(x_p^+) \rangle = \|f_t(x_p)\|_2^2$ and minimize $\langle f(x_{n,s}), f(x_p^+) \rangle$ by learning from all the available signals. *However, in our sparse coding model $x = \mathbf{M}z + \xi$, the spurious dense features ξ has much larger ℓ_2 -norm and the least correlations between different samples (see Section 2.2 for discussion).* In contrast, the sparse signals $\mathbf{M}z = \sum_j \mathbf{M}_j z_j$ display larger correlations between different samples because of possible co-occurrences of features $\mathbf{M}_j$ (i.e., at least $\tilde{\Omega}(\frac{1}{d})$ portion of the data contain feature $\mathbf{M}_j$). Thus the our contrastive learner will focus on learning the features associated with the dense noise ξ , and fail to emphasize sparse features.

Feature Decoupling: How does augmentation remove the spurious dense noise: Theoretically, we show how data augmentations help contrastive learning, which demonstrate the principle of *feature decoupling*. The spirit is that the augmentation should be able to making the dense signals completely different between the positive samples while preserve the correlations of sparse signals.

Specifically, under our data model, if no augmentations are applied to the two positive samples x_p^+, x_p^{++} generated from $x_p = \mathbf{M}z_p + \xi_p \sim \mathcal{D}_x$, their correlations will mostly come from the inner product of noise $\langle \xi_p, \xi_p \rangle$, which can easily overwhelm those from the sparse signals $\langle \mathbf{M}z_p, \mathbf{M}z_p \rangle$. Nevertheless, we have a simple observation: different coordinate $\xi_{p,j}$ of our dense noise ξ_p are independent to each other, which enables a simple method to decorrelate the dense noise: by randomly applying two completely opposite masks $\mathbf{D}$ and $\mathbf{I} - \mathbf{D}$ to the data x_p to generate two positive samples $x_p^+ = 2\mathbf{D}x_p$ and $x_p^{++} = 2(\mathbf{I} - \mathbf{D})x_p$. From our observation, such data augmentations can make the dense signals $\mathbf{D}\xi_p$ and $(\mathbf{I} - \mathbf{D})\xi_p$ of x_p^+ and x_p^{++} independent to each other. This independence will decouple the dense features between positive samples, which substantially reduces the gradients of the dense features.

However, the sparse signals are more resistant to data augmentation. As long as the sparse signals $\mathbf{M}z = \sum_{j \in [d]} \mathbf{M}_j z_j$ span across the space, they will show up in both x_p^+ and x_p^{++} , so that their correlations will remain in the representations. More precisely, whenever a sparse signal $\mathbf{M}_j$ is present (meaning its latent variable $z_j \neq 0$), it can be recovered both from $2\mathbf{D}\mathbf{M}z$ and from $2(\mathbf{I} - \mathbf{D})\mathbf{M}z$ with the correct decoding: e.g. we have $T_b(\langle \mathbf{M}_j, 2\mathbf{D}x \rangle) \approx T_b(\langle \mathbf{M}_j, 2(\mathbf{I} - \mathbf{D})x \rangle) \approx z_j$, where $T_b(x) = x \mathbb{1}_{|x| \geq b}$ is a threshold operator with a proper bias $b > 0$. Unless in very rare case $\mathbf{M}_j$ is completely masked by augmentations (that is $\mathbf{D}\mathbf{M}_j = 0$ or $(\mathbf{I} - \mathbf{D})\mathbf{M}_j = 0$), the sparse signals will remain their correlations in the feature representations, which will later be reinforced by neural networks following the SGD trajectory.

2.4.1 The Initial Stage of Training: Initial Feature Decoupling

The initial stage of our training process is defined as the training iterations $t \leq T_1$, where $T_1 = \Theta\left(\frac{d \log d}{\eta \log \log d}\right)$ is the iteration when all $\|w_i^{(t)}\|_2^2 \geq \frac{2(1+c_0)+c_1}{c_1} \|w_i^{(0)}\|_2^2$. Before T_1 , the learning of our neurons focus on emphasizing the entire subspace of sparse features (i.e., focus on learning $\mathbf{M}$), which is enabled by our augmentations RandomMask and **feature decoupling** principle.

More formally, we will investigate for each neuron $i \in [m]$, how the features (the weights) grow at each directions. For the sparse features $\{\mathbf{M}_j\}_{j \in [d]}$, when there is no bias, we shall prove that:

$$\langle \nabla_{w_i} \text{Obj}(f_t), \mathbf{M}_j \rangle \approx \langle w_i^{(t)}, \mathbf{M}_j \rangle \cdot \mathbb{E}[z_j^2]$$

which give exponential rate of growth for our (subspace of) sparse features. Indeed, such an exponential growth will continue to hold until the bias have been pushed up by the negative samples⁶ or until the end of training where the gradient is cancelled by positive-negative contrast. Meanwhile, for the spurious dense features $\{\mathbf{M}_j^\perp\}_{j \in [d_1] \setminus [d]}$, we will prove for each $j \in [d_1] \setminus [d]$, at iterations $t \leq T_1$ (also note that $\lambda = \frac{\text{polylog}(d)}{\sqrt{d_1}}$):

$$\langle \nabla_{w_i} \text{Obj}(f_t), \mathbf{M}_j^\perp \rangle \approx -\lambda \langle w_i^{(t)}, \mathbf{M}_j^\perp \rangle + \tilde{O}\left(\frac{\|w_i^{(t)}\|_2}{d_1}\right)$$

which is only possible because we have used the augmentation RandomMask. Without such augmentation, we shall expect the growth rate of $\langle w_i^{(t)}, \mathbf{M}_j^\perp \rangle$ to be approximately the same with $\langle w_i^{(t)}, \mathbf{M}_j \rangle$, which would collapse the features learned in our neural nets. As the training proceeds, we will prove that:

$$\begin{aligned} \|\mathbf{M}\mathbf{M}^\top w_i^{(t)}\|_2 &\approx \|\mathbf{M}\mathbf{M}^\top w_i^{(0)}\|_2 \cdot \left(1 + O\left(\frac{\eta \log \log d}{d}\right)\right)^t \\ \|\mathbf{M}^\perp (\mathbf{M}^\perp)^\top w_i^{(t)}\|_2 &\approx \|\mathbf{M}^\perp (\mathbf{M}^\perp)^\top w_i^{(0)}\|_2 + o\left(\frac{\|\mathbf{M}^\perp (\mathbf{M}^\perp)^\top w_i^{(0)}\|_2}{\text{poly}(d)}\right) \end{aligned}$$

Therefore, after sufficient iterations (not many comparing to the total training time), the weights $\{w_i^{(t)}\}_{i \in [m]}$ of neurons will mostly consist of the sparse features $\{\mathbf{M}_j\}_{j \in [d]}$ rather than the dense features $\{\mathbf{M}_j^\perp\}_{j \in [d_1] \setminus [d]}$. We can then move into the second stage of training, where we tune the bias to simulate the sparsification process.

2.4.2 The Second Stage of Training: Singletons Emerge

After the initial training stage $t \leq T_1$, we enter the second stage of training, where we will analyze how the growth of bias drive the neurons to become singletons. However, the crucial challenge

⁶ We decide to leave the analysis of how the biases are trained open and based our analysis on manually growing biases. The exact mechanism of bias growth depend on the effects of positive-negative contrast in the second stage, which would significantly complicated our analysis.

here is that as soon as the bias $b_i^{(t)}$ start to grow above zero. The correlations between the sparse features and dense features will emerge to obfuscate our analysis of gradients. Indeed, mathematically we can formulate the problem as follows: Let $b_i^{(t)} \geq 0$, how can we obtain a bound of the following term, which cannot exceed $\langle \nabla_{w_i} \mathbf{Obj}(f_t), \mathbf{M}_j \rangle$ as calculated above:

$$\begin{aligned} \langle \nabla_{w_i} \mathbf{Obj}(f_t), \mathbf{M}_j^\perp \rangle &\approx \mathbb{E} \left[h_{i,t}(x_p^{++}) \langle \nabla_{w_i} h_{i,t}(x_p^+) \mathbf{M}_j^\perp \rangle \right] \\ &\approx \mathbb{E} \left[h_{i,t}(2(\mathbf{I} - \mathbf{D})x_p) \mathbb{1}_{|\langle w_i^{(t)}, 2\mathbf{D}(\mathbf{M}z_p + \xi_p) \rangle| \geq b_i^{(t)}} \langle 2\mathbf{D}\xi_p, \mathbf{M}_j^\perp \rangle \right] \end{aligned}$$

The difficulty, as opposed to what we saw in the initial stage, is that now there is a chain of correlations transmitted through the following line, when $b_i^{(t)} > 0$:

- the term with the masked sparse signals $2(\mathbf{I} - \mathbf{D})\mathbf{M}z_p$ and the term with the masked dense signals $2(\mathbf{I} - \mathbf{D})\xi_p$ in the activation $h_{i,t}(x_p^+)$ are positively correlated;
- the term with the masked sparse signals $2(\mathbf{I} - \mathbf{D})\mathbf{M}z_p$ in the activation $h_{i,t}(x_p^+)$ and the term with the masked sparse signals $2(\mathbf{I} - \mathbf{D})\mathbf{M}z_p$ in the gradient $\nabla_{w_i} h_{i,t}(x_p^{++})$ are positively correlated;
- the term with the masked sparse signals $2\mathbf{D}\mathbf{M}z_p$ and the term with the masked dense signals $2\mathbf{D}\xi_p$ in the gradient $\nabla_{w_i} h_{i,t}(x_p^+)$ are positively correlated.

This chain of correlations will significantly complicated our analysis, we will prove the following lemma that have taken into considerations all the factors affecting the gradients:

Lemma 2.4.1 (sketched). *For neuron $i \in [m]$ and spurious dense feature $\mathbf{M}_j^\perp$, we have*

$$\mathbb{E}[(1 - \ell'_{p,t})h_{i,t}(x_p^{++})\langle \nabla_{w_i} h_{i,t}(x_p^+), \mathbf{M}_j^\perp \rangle] \approx \left(\langle w_i^{(t)}, \mathbf{M}_j^\perp \rangle \pm \tilde{O}\left(\frac{\|w_i^{(t)}\|_2}{\sqrt{d_1}}\right) \right) \mathbb{E}[\langle \xi, \mathbf{M}_j^\perp \rangle^2 \mathbb{1}_{|\langle w_i^{(t)}, x_p \rangle| \geq b_i^{(t)}}]$$

However, we can prove that after the initial stage, both $\langle w_i^{(t)}, 2\mathbf{D}\mathbf{M}_j^\perp \rangle$ and $\tilde{O}\left(\frac{\|w_i^{(t)}\|_2}{\sqrt{d_1}}\right)$ are very small compared to the sparse features $\langle w_i^{(t)}, \mathbf{M}_j \rangle$. Thus $\mathbb{E}[\langle \xi, \mathbf{M}_j^\perp \rangle^2 \mathbb{1}_{|\langle w_i^{(t)}, x_p \rangle| \geq b_i^{(t)}}]$ shall be somehow small since the correlation between $\langle \xi, \mathbf{M}_j^\perp \rangle^2$ and $\mathbb{1}_{|\langle w_i^{(t)}, x_p \rangle| \geq b_i^{(t)}}$ is small.

On the contrary, for some of the features $\langle w_i, \mathbf{M}_j \rangle$ that is “lucky” in the sense that at initialization $\langle w_i^{(0)}, \mathbf{M}_j \rangle^2 \geq (2.02)\sigma_0^2 \log d$, we can maintain such “luckiness” till the stage II and obtain similar gradient approximation as:

Lemma 2.4.2 (sketched). *For neuron $i \in [m]$ and the “lucky” sparse feature $\mathbf{M}_j$, we have*

$$\mathbb{E}[h_{i,t}(x_p^{++})\langle \nabla_{w_i} h_{i,t}(x_p^+), \mathbf{M}_j \rangle] \approx \left(\langle w_i^{(t)}, \mathbf{M}_j \rangle \pm \tilde{O}\left(\frac{\|w_i^{(t)}\|_2}{\sqrt{d_1}}\right) \right) \mathbb{E}[z_j^2 \mathbb{1}_{|\langle w_i^{(t)}, x_p \rangle| \geq b_i^{(t)}}]$$

And we also have two observations: (1) $\langle w_i^{(t)}, \mathbf{M}_j \rangle$ is almost as large as $\Theta(\|w_i^{(t)}\|_2/\sqrt{d})$, which is much larger than $\tilde{O}(\|w_i^{(t)}\|_2/\sqrt{d_1})$; (2) we know $\mathbb{E}[z_j^2 \mathbb{1}_{|\langle w_i^{(t)}, x_p \rangle| \geq b_i^{(t)}}] \gg o(1/d)$ since when the sparse feature $\langle w_i^{(t)}, \mathbf{M}_j \rangle z_j$ is active, it would be much larger than the dense feature $\langle w_i^{(t)}, \mathbf{M}_j^\perp \rangle$.

This paved the way for our feature growth till stage III. Our theory indeed matches what happens in practice, where one can observe the slow emergence of features (in the first layer of AlexNet) during the training process comparing to supervised learning.

2.4.3 The Final Stage of Training: Convergence to Sparse Features

We assume our training proceeds until we reach at least $T_3 = \frac{\text{poly}(d)}{\eta}$, but the stage III start at some $T_2 = \Theta\left(\frac{d \log d}{\eta \log \log d}\right)$ when there exist a neuron $i \in [m]$ such that $\|w_i^{(T_2)}\|_2 \geq d\|w_i^{(t)}\|_2$. At iterations $t \geq T_2$, the negative term will begin to cancel the positive gradient, which drives the learning process to converge. We now sketch the proof here.

When the training process reach $t \geq T_2$, we have the following properties for all the neurons:

- For each $j \in [d]$, there is a set $\mathcal{M}_j = \{i \in [m], |\langle w_i^{(t)}, \mathbf{M}_j \rangle| \geq \Theta(1) \max_{j' \in [d]} |\langle w_i^{(t)}, \mathbf{M}_{j'} \rangle|\}$ such that this set has cardinality $|\mathcal{M}_j| \geq \omega(1)$.
- For all the neurons $i \in [m]$ such that $i \notin \mathcal{M}_j$, we have $|\langle w_i^{(t)}, \mathbf{M}_j \rangle| \leq o\left(\frac{\|w_i^{(t)}\|_2}{\sqrt{d}}\right)$;
- the neuron activations can be written as follows with high probability, which is because now the neurons are truly sparse and can be written as decompositions of $O(1)$ many signals (plus some small mixture): $h_{i,t}(x) \approx \langle w_i^{(t)}, x \rangle \mathbb{1}_{z_j \neq 0 \text{ for } j \in [d]: i \in \mathcal{M}_j} + \tilde{O}(\|w_i^{(t)}\|/d^2)$.

At this stage, for each $j \in [d]$, the gradient $\langle \nabla_{w_i} L_{\text{pos}}(f_t), \mathbf{M}_j \rangle$ does not change much compared to the previous stage, but the negative term changed essentially, which we elaborate as follows:

$$\sum_{x_{n,s} \in \mathfrak{N}} \ell'_{s,t} h_{i,t}(x_{n,s}) \approx \sum_{x_{n,s} \in \mathfrak{N}} \ell'_{s,t} \sum_{j \in [d]: i \in \mathcal{M}_j} \langle w_i^{(t)}, \mathbf{M}_j \rangle z_{n,s,j}$$

For the simplest case where there is only one $j \in [d]$ such that $i \in \mathcal{M}_j$, we can see that

$$\sum_{x_{n,s} \in \mathfrak{N}} \ell'_{s,t} h_{i,t}(x_{n,s}) \approx \sum_{x_{n,s} \in \mathfrak{N}} \ell'_{s,t} \langle w_i^{(t)}, \mathbf{M}_j \rangle z_{n,s,j} \mathbb{1}_{z_{n,s,j} \neq 0}$$

The critical question here is that: The problem at hand is extremely non-convex, how does our algorithm find the minimal of the loss without being trapped in some undesired solutions. We argue that as long as the trajectory of weights following SGD is good in the sense that only “good” features are picked up. The SGD in the final stage will point to the desired solution, then the singletons of our sparse feature $\mathbf{M}_j$ will converge as follows:

$$\langle \nabla_{w_i} L(f_t), \mathbf{M}_j \rangle \approx \mathbb{E} \left[\langle w_i^{(t)}, \mathbf{M}_j \rangle z_{p,j}^2 - \sum_{x_{n,s} \in \mathfrak{N}} \ell'_{s,t} \langle w_i^{(t)}, \mathbf{M}_j \rangle z_{n,s,j} z_{p,j} \mathbb{1}_{z_{p,j} \neq 0, z_{n,s,j} \neq 0} \right] \approx 0$$

While the gradient of other features (including sparse features not favored by the specific neuron $i \in [m]$, and the spurious dense features $\mathbf{M}_j^\perp$) will be smaller to ensure sparse representations. More formal arguments will be presented in the later sections.

2.4.4 Without Augmentations, Dense Features Are Preferred

Now we turn to the case where no augmentations are used. In this scenario, for any ense feature $\mathbf{M}_j^\perp$, we always have:

$$\langle \nabla_{w_i} \mathbf{Obj}(f_t), \mathbf{M}_j^\perp \rangle \approx \langle w_i^{(t)}, \mathbf{M}_j^\perp \rangle \cdot \Theta \left(\frac{\sqrt{\log d}}{d} \cdot \Pr(|h_{i,t}(x_p)| \geq 0) \right)$$

where $\Pr(|h_{i,t}(x_p)| \geq 0) \approx \Pr(|h_{i,t}(\xi_p)| \geq 0)$ which is approximately equal to the growth rate of sparse signals when no augmentations are used. In this case the sparse signals will not be emphasized during any stage of training. And more over, we can easily verify the following condition: at each iteration $t \geq 0$, we have $\|\mathbf{M}^\perp (\mathbf{M}^\perp)^\top w_i^{(t)}\|_2^2 \approx (1 - \frac{1}{\text{poly}(d)}) \|w_i^{(t)}\|_2^2$ based on similar careful characterization of the learning process. Moreover, such learning process can easily converge to low loss: when $\sum_{i \in [m]} \|w_i^{(t)}\|_2^2 \geq \Omega(\tau \log d)$, we simply have

$$\langle f_t(x_p^+), f_t(x_p^{++}) \rangle \approx \Omega(\tau \log d) \quad \text{and} \quad \langle f_t(x_p^+), f_t(x_{n,s}) \rangle \approx \tilde{O}\left(\frac{1}{d}\right) \quad (\text{with high probability})$$

Using this characterization, we immediately obtain the loss (and gradient) convergence.

Appendix

2.A Proof Overview

In this section we present an overview of our full proof. Before going into the proof, we describe some preliminaries.

2.A.1 Preliminaries and Notations

At every iteration $t \geq 0$, we denote the weights of the neurons as $w^{(t)} = \{w_i^{(t)}\}_{i \in [m]}$, and given $x \in \mathbb{R}^{d_1}$ as input, the output of the network is denoted as

$$\begin{aligned} f_t(x) &= (h_{1,t}(x), \dots, h_{m,t}(x))^T \in \mathbb{R}^m, \\ h_{i,t}(x) &= \text{ReLU}(\langle w_i^{(t)}, x \rangle - b_i^{(t)}) - \text{ReLU}(-\langle w_i^{(t)}, x \rangle - b_i^{(t)}) \end{aligned}$$

Data Preparation and Loss Objective. Given a positive sample $x_p \sim \mathcal{D}_x$, the augmented data are defined as follows: we generate random mask $\mathbf{D} \sim \mathcal{D}_{\mathbf{D}}$ (defined in Def. 2.2.3) and apply to x_p as:

$$x_p^+ \leftarrow 2\mathbf{D}x_p, \quad x_p^{++} \leftarrow 2(\mathbf{I} - \mathbf{D})x_p$$

where the 2-factor is to renormalize the data. Now recall our similarity measure is defined as $\text{Sim}_f(x_1, x_2) = \langle f(x_1), \text{StopGrad}(f(x_2)) \rangle$ for inputs x_1, x_2 . Our population contrastive loss objective is defined as follows: suppose in addition to x_p^+, x_p^{++} , we are given a batch of negative samples $\mathfrak{N} = \{x_{n,s}\}_{s \in [\mathfrak{N}]}$, where each $x_{n,s} \sim \mathcal{D}_x$ independently (for short), we write $\mathfrak{B} = \{x_p^{++}\} \cup \mathfrak{N}$ and define

$$L(f_t) := \mathbb{E}_{x_p^+, \mathfrak{B}} \left[-\text{Sim}_{f_t}(x_p^+, x_p^{++}) + \tau \log \left(\sum_{x \in \mathfrak{B}} e^{\text{Sim}_{f_t}(x_p^+, x)/\tau} \right) \right]$$

The Gradient of Weights. We perform stochastic gradient descent on the objective $\text{Obj}(f_t) = L(f_t) + \lambda \|w^{(t)}\|_F^2$ as follows: At iteration $t \geq 0$, we first sample $\{x_{p,\ell}, \mathfrak{N}^\ell = \{x_{n,s,\ell}\}_{s \in [\mathfrak{N}]}\}_{\ell \in [K]}$ independently for $K = \text{poly}(d)$ many batches of data.⁷ We augmented all the positives as $x_{p,\ell}^+, x_{p,\ell}^{++} \leftarrow \text{RandomMask}(x_{p,\ell})$ as defined in Def. 2.2.3. And regroup the data into K batches of data of the form $\{\mathfrak{B}^\ell\}_{\ell \in [K]} = \{\{x_{p,\ell}^{++}\} \cup \mathfrak{N}^\ell\}_{\ell \in [K]}$. Now we evaluate the empirical loss and gradient as

- **empirical objective:** $\widehat{\text{Obj}}(f_t) = \frac{1}{K} \sum_{\ell \in [K]} \mathcal{L}(f_t, x_{p,\ell}^+, \mathfrak{B}^\ell) + \lambda \|w^{(t)}\|_F^2$;
- **empirical gradient of weight w_i :** $\nabla_{w_i} \widehat{\text{Obj}}(f_t) = \frac{1}{K} \sum_{\ell \in [K]} \nabla_{w_i} \mathcal{L}(f_t, x_{p,\ell}^+, \mathfrak{B}^\ell) + \lambda w_i^{(t)}$

⁷ Note that such an assumption of sampling from populations without resorting to a finite dataset is very close to reality, in that the unlabeled data are much cheaper to obtain as opposed to labeled data used in supervised learning. Indeed, [113] have used an unlabeled dataset of one billion images, which is larger than any labeled datasets in vision.

and we update weights $\{w_i^{(t)}\}_{i \in [m]}$ at each iteration $t \geq 0$ as follows:

$$w_i^{(t+1)} \leftarrow w_i^{(t)} - \nabla_{w_i} \widehat{\mathbf{Obj}}(f_t) = (1 - \lambda)w_i^{(t)} - \frac{1}{K} \sum_{\ell \in [K]} \nabla_{w_i} \mathcal{L}(f_t, x_{p,\ell}^+, \mathfrak{B}^\ell)$$

Note that as long as $\|w^{(t)}\|_F^2 = \sum_{i \in [m]} \|w_i^{(t)}\|_2^2 \leq \text{poly}(d)$, the following fact always holds (which can be easily obtained by Bernstein concentration, and note that we do not need to use uniform convergence):

Fact 2.A.1 (approximation of population gradients by empirical gradients). As long as $\|w^{(t)}\|_F^2 \leq \text{poly}(d)$, there exist $K = \text{poly}(d)$ such that the following inequality holds with high probability for all iteration t :

$$\left\| \nabla_{w_i} \widehat{\mathbf{Obj}}(f_t) - \nabla_{w_i} \mathbf{Obj}(f_t) \right\|_2 \leq \frac{\|w_i^{(t)}\|_2}{\text{poly}(d_1)} \quad \text{for all } i \in [m]$$

To compute the gradient of our loss function $\mathcal{L}(f_t, x_{p,\ell}^+, \mathfrak{B}^\ell)$ with respect to the weights $\{w_i^{(t)}\}_{i \in [m]}$, we define the following notations: positive logit $\ell'_{p,t}(x_p^+, \mathfrak{B})$ and negative logits $\ell'_{s,t}(x_p^+, \mathfrak{B})$:

$$\ell'_{p,t}(x_p^+, \mathfrak{B}) := \frac{e^{\text{Sim}_{f_t}(x_p^+, x_p^{++})/\tau}}{\sum_{x \in \mathfrak{B}} e^{\text{Sim}_{f_t}(x_p^+, x)/\tau}} \quad \ell'_{s,t}(x_p^+, \mathfrak{B}) := \frac{e^{\text{Sim}_{f_t}(x_p^+, x_{n,s})/\tau}}{\sum_{x \in \mathfrak{B}} e^{\text{Sim}_{f_t}(x_p^+, x)/\tau}}$$

Then the empirical gradient of $L(f_t)$ with respect to weight $w_i^{(t)}$ at iteration t can be expressed as (recall that we have used StopGrad operation in our similarity measure Sim_{f_t}):

$$\nabla_{w_i} L(f_t) = \mathbb{E} \left[(1 - \ell'_{p,t}) \cdot h_{i,t}(x_p^{++}) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)}} x_p^+ + \sum_{x_{n,s} \in \mathfrak{R}} \ell'_{s,t} \cdot h_{i,t}(x_{n,s}) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)}} x_p^+ \right]$$

These notations will be frequently used in our proof in later sections.

Global Notations. We define some specific notations we will use throughout the proof.

- We let C_z be the constant inside the Θ notation of $\Pr(|z_j| = 1) = \Theta(\frac{\log \log d}{d}) = \frac{C_z \log \log d}{d}$ defined in [Definition 2.2.1](#).
- We assume in our paper all the $\text{polylog}(d)$ explicitly written to be much bigger than the $\text{polylog}(d)$ factors in our $\tilde{O}(\cdot)$ notations.
- For any $j \in [d]$ or set $\mathcal{S} \subset [d]$, let $x = \sum_{j \in [d]} \mathbf{M}_j z_j + \xi \sim \mathcal{D}_x$ be an input, we denote the superscript $\setminus j$ (or $\setminus \mathcal{S}$) as an operation to subtract features $\mathbf{M}_j$ (or $\{\mathbf{M}_j\}_{j \in \mathcal{S}}$) in the data as follows:

$$x^{\setminus j} := x - \mathbf{M}_j z_j \quad \text{or} \quad x^{\setminus \mathcal{S}} := x - \sum_{j \in \mathcal{S}} \mathbf{M}_j z_j$$

Furthermore, for augmented input $x^+ = 2\mathbf{D}x$ (or $x^{++} = 2(\mathbf{I} - \mathbf{D})x$), we also define:

$$x^{\setminus j} := x^+ - \mathbf{M}_j z_j \quad \text{or} \quad x^{\setminus \mathcal{S}} := x - \sum_{j \in \mathcal{S}} \mathbf{M}_j z_j, \quad (\text{and similarly for } x^{++})$$

2.B Some Technical Lemmas

2.B.1 Characterization of Neurons

In this section we give some definitions and lemmas that characterize the neurons at initialization and during the training process. We choose $c_1 = 2 + 2(1 - \gamma)c_0$, $c_2 = c_1 - \gamma c_0$ be two constants. (which we choose $\gamma \in (0, \frac{1}{100})$, similar to the choice in [8]).

Definition 2.B.1. We define several sets of neurons that will be useful for the characterization of the stochastic gradient descent trajectory in later sections.

- For each $j \in [d]$, we define the set $\mathcal{M}_j \subseteq [m]$ of neurons as:

$$\mathcal{M}_j := \left\{ i \in [m] : \langle w_i^{(0)}, \mathbf{M}_j \rangle^2 \geq \frac{c_2 \log d}{d} \|\mathbf{M} \mathbf{M}^\top w_i^{(0)}\|_2^2 \right\}$$

- For each $j \in [d]$, we define the set $\mathcal{M}_j^\star \subseteq [m]$ of neurons as:

$$\mathcal{M}_j^\star := \left\{ i \in [m] : \langle w_i^{(0)}, \mathbf{M}_j \rangle^2 \geq \frac{c_1 \log d}{d} \|\mathbf{M} \mathbf{M}^\top w_i^{(0)}\|_2^2, \right. \\ \left. \langle w_i^{(0)}, \mathbf{M}_{j'} \rangle^2 \leq \frac{c_2 \log d}{d} \|\mathbf{M} \mathbf{M}^\top w_i^{(0)}\|_2^2, \quad \forall j' \in [d], j' \neq j \right\}$$

Properties at initialization: At initialization, where $t = 0$, we need to give several facts concerning our neurons, which will later be useful for the analysis of SGD trajectory.

Lemma 2.B.1. *At iteration $t = 0$, the following properties hold:*

- (a) *With high probability, for every $i \in [m]$, we have*

$$\|w_i^{(0)}\|_2^2 \in \left[\sigma_0^2 d_1 \left(1 - \tilde{O}\left(\frac{1}{\sqrt{d_1}}\right) \right), \sigma_0^2 d_1 \left(1 + \tilde{O}\left(\frac{1}{\sqrt{d_1}}\right) \right) \right]$$

- (b) *With high probability, for every $i \in [m]$, we have*

$$\|\mathbf{M} \mathbf{M}^\top w_i^{(0)}\|_2^2 \in \left[\sigma_0^2 \left(1 - \tilde{O}\left(\frac{1}{\sqrt{d}}\right) \right), \sigma_0^2 d \left(1 + \tilde{O}\left(\frac{1}{\sqrt{d}}\right) \right) \right]$$

- (c) *With probability at least $1 - o(\frac{1}{d^4})$, we have for each $j \in [d]$:*

$$|\mathcal{M}_j^\star| \geq \Omega(d^{\gamma c_0/4}) =: \Xi_1, \quad |\mathcal{M}_j| \leq O(d^{2\gamma c_0}) =: \Xi_2;$$

- (d) For each $i \in [m]$, let $\Lambda_i := \left\{ j \in [d] : |\langle w_i^{(0)}, \mathbf{M}_j \rangle| \leq \sigma_0/d \right\} \subseteq [d]$, then $|\Lambda_i| = O(\frac{d}{\text{polylog}(d)})$;
- (e) For any $j' \neq j$, $|\mathcal{M}_{j'} \cap \mathcal{M}_j| \leq O(\log d)$, with probability at least $1 - o(1/d^4)$.
- (f) For each $i \in [m]$, there are at most $O(1)$ many $j \in [d]$ such that $i \in \mathcal{M}_j$, and at most $O(2^{-\sqrt{\log d}} d)$ many $j \in [d]$ such that $|\langle w_i^{(0)}, \mathbf{M}_j \rangle| \geq \Omega(\sigma_0 \log^{1/4} d)$.

Proof. The proof of (a)–(b) can be derived from simple concentration of chi-squared concentration. The proof of (c) and (f) follow from [8, Lemma B.2]. For (d) it suffices to use basic Gaussian anti-concentration around the mean. For (e) it suffices to use a simple Bernoulli concentration. $\square$

2.B.2 Activation Size and Probability

Lemma 2.B.2 (correlation from augmentation). *Let $\mathbf{M}^\perp \in \mathbb{R}^{d_1 \times (d_1 - d)}$ be an orthonormal complement of $\mathbf{M}$, $\mathbf{D} \sim \mathcal{D}_{\mathbf{D}}$, it holds:*

1. for each $j, j' \in [d]$, with high probability we have

$$\langle \mathbf{M}_j, (\mathbf{I} - 2\mathbf{D})\mathbf{M}_{j'} \rangle \lesssim \tilde{O}\left(\frac{1}{\sqrt{d_1}}\right)$$

2. for each $j \in [d]$, $j' \in [d_1] \setminus [d]$, with high probability we have

$$\langle \mathbf{M}_j, (\mathbf{I} - 2\mathbf{D})\mathbf{M}_{j'}^\perp \rangle \lesssim \tilde{O}\left(\frac{1}{\sqrt{d_1}}\right)$$

Since $\langle \mathbf{M}_j, \mathbf{M}_{j'}^\perp \rangle = 0$, this bound also hold for variables $\langle \mathbf{M}_j, (\mathbf{I} - \mathbf{D})\mathbf{M}_{j'}^\perp \rangle$ and $\langle \mathbf{M}_j, \mathbf{D}\mathbf{M}_{j'}^\perp \rangle$.

Proof. 1. For $j \in [d]$ and $r \in [d_1]$, we denote $(\mathbf{M}_j)_r$ to be the r -th coordinate of $\mathbf{M}_j$. Now we expand

$$\langle \mathbf{M}_j, (\mathbf{I} - 2\mathbf{D})\mathbf{M}_{j'} \rangle = \sum_{r \in [d_1]} (\mathbf{M}_j)_r (\mathbf{M}_{j'})_r (1 - 2\mathbf{D}_{r,r})$$

which can be view as a sub-Gaussian variables with variance parameter $\tilde{O}(1/d_1)$. Applying Chernoff bound concludes the proof.

2. Proof is similar to (1). Since the mean of $\langle \mathbf{M}_j, (\mathbf{I} - 2\mathbf{D})\mathbf{M}_{j'}^\perp \rangle$ is zero, we can compute its variance as

$$\begin{aligned} \mathbb{E} \left[\langle \mathbf{M}_j, (\mathbf{I} - 2\mathbf{D})\mathbf{M}_{j'}^\perp \rangle^2 \right] &= \sum_{r \in [d_1]} (\mathbf{M}_j)_r^2 (\mathbf{M}_{j'}^\perp)_r^2 \mathbb{E}[(1 - 2\mathbf{D}_{r,r})^2] \\ &\leq \tilde{O}\left(\frac{1}{d_1}\right) \sum_{r \in [d_1]} (\mathbf{M}_{j'}^\perp)_r^2 \leq \tilde{O}\left(\frac{1}{d_1}\right) \end{aligned}$$

then again from Chernoff bound we conclude the proof. $\square$

we will give several probability tail bounds for the so defined variables, which will be used in the computations of the training process throughout our analysis.

Lemma 2.B.3 (pre-activation size). *Let $x = \mathbf{M}z + \xi \sim \mathcal{D}_x$, $w_i \in \mathbb{R}^{d_1}$ and $\mathbf{D} \sim \mathcal{D}_{\mathbf{D}}$. Denoting $x^{\setminus j} = \sum_{j' \neq j, j' \in [d]} \mathbf{M}_{j'} z_{j'} + \xi$, we have the following results:*

1. For any $\lambda > 0$, we have

$$\Pr_{\mathbf{D} \sim \mathcal{D}_{\mathbf{D}}} (|\langle w_i, (\mathbf{I} - 2\mathbf{D})\mathbf{M}_j \rangle| > \lambda \|w_i\|_2 \|\mathbf{M}_j\|_{\infty}) \leq 2e^{-\Omega(\lambda^2)}$$

2. (naive Chebychev bound) For any $\lambda > 0$ and $z \in [-1, 1]$, we have

$$\Pr_{z^{\setminus j}, \xi, \mathbf{D}} \left((\langle w_i, (\mathbf{I} - 2\mathbf{D})x^{\setminus j} \rangle + \langle w_i, (\mathbf{I} - 2\mathbf{D})\mathbf{M}_j \rangle |z|)^2 > \frac{\lambda \|w_i\|_2^2 \sqrt{\log d}}{d} \right) \leq O\left(\frac{1}{\lambda}\right)$$

The same tail bound holds for variables $\langle w_i, x \rangle$, $\langle w_i, (\mathbf{I} - 2\mathbf{D})x \rangle$ and $\langle w_i, \xi \rangle$ as well.

3. (high probability bound for sparse signal)

$$\Pr \left(\langle w_i, (\mathbf{I} - 2\mathbf{D})\mathbf{M}z \rangle^2 > \|w_i\|_2^2 \cdot \max_{j \in [d]} \|\mathbf{M}_j\|_{\infty}^2 \log^4 d \right) \lesssim e^{-\Omega(\log^2 d)}$$

4. (high probability bound for dense signal) Let $Z = \langle w_i, (\mathbf{I} - 2\mathbf{D})\xi \rangle$ or $Z = \langle w_i, \xi \rangle$, we have

$$\Pr \left(Z^2 \geq \frac{\|w_i\|_2^2 \log^4 d}{d} \right) \lesssim e^{-\Omega(\log^2 d)}$$

Proof. 1. From the fact that $\mathbf{D} = (\mathbf{D}_{k,k})_{k=1}^{d_1}$, where $\mathbf{D}_{k,k} \sim \text{Bernoulli}(\frac{1}{2})$ are subgaussian variables, we can use the subgaussian tail coupled with Hoeffding's bound to conclude.

2. Since the mean of $\langle w_i, (\mathbf{I} - 2\mathbf{D})x \rangle$ is zero, we can simply compute the variance as

$$\begin{aligned} & \mathbb{E} \left[(\langle w_i, (\mathbf{I} - 2\mathbf{D})x^{\setminus j} \rangle + \langle w_i, (\mathbf{I} - 2\mathbf{D})\mathbf{M}_j \rangle |z|)^2 \right] \\ & \leq \sum_{s=1}^{d_1} (w_i)_s^2 (\mathbf{M}_{j'}^2)_s \mathbb{E} \left[\left(\sum_{j' \neq j} z_{j'}^2 + \xi_s^2 \right) (2\mathbf{D}_{s,s} - 1)^2 \right] + \sum_{s=1}^{d_1} (w_i)_s^2 (\mathbf{M}_j^2)_s \mathbb{E} [z^2 (2\mathbf{D}_{s,s} - 1)^2] \\ & \leq \tilde{O}(\|w_i\|_2^2 / d) \end{aligned}$$

Now we can use Chebychev's inequality to conclude. As to the tail bounds for other variables, it suffices to go through some similar calculations.

3. First we consider for each $j \in [d]$, the variable $Z_j := \langle w_i, (2\mathbf{D} - \mathbf{I})\mathbf{M}_j \rangle$. Note that

$$Z_j = \sum_{s=1}^{d_1} (w_i)_s (\mathbf{M}_j)_s (2\mathbf{D}_{s,s} - 1)$$

is a sum of subgaussian variables, each with variances $O((w_i)_s^2/d_1)$, therefore by using Hoeffding's bound we have with prob $\geq 1 - e^{-\Omega(\log^2 d)}$, we have $|Z_j| \leq \tilde{O}\left(\frac{\|w_i\|_2}{\sqrt{d_1}}\right)$, Now by using a union bound, we conditioned on $\{|Z_j| \leq \tilde{O}\left(\frac{\|w_i\|_2}{\sqrt{d_1}}\right) \text{ for all } j \in [d]\}$ happening, which is still with high probability. We use Bernstein's inequality to show that with high probability over z and $\mathbf{D}$ it holds that $\left|\sum_{j \in [d]} Z_j z_j\right|^2 \lesssim \tilde{O}\left(\frac{\|w_i\|_2^2}{d_1}\right)$.

4. We can first obtain for high probability bounds for each coordinates $\xi_j, j \in [d_1]$ based on the concentration of Gaussian variables, and then use Chernoff bound via the randomness of $\mathbf{D}$ to conclude (when there is no $\mathbf{D}$ involved, the claim is obvious).

□

Lemma 2.B.4 (pre-activation size, II). *Let $i \in [m]$. Suppose the following holds:*

- $\langle w_i^{(t)}, \mathbf{M}_j \rangle^2 \geq \Omega((b_i^{(t)})^2)$ for no more than $O(1)$ many $j \in [d]$;
- $\langle w_i^{(t)}, \mathbf{M}_j \rangle^2 \geq \Omega\left(\frac{(b_i^{(t)})^2}{\sqrt{\log d}}\right)$ for no more than $O(e^{-\Omega(\sqrt{\log d})}d)$ many $j \in [d]$;
- $\|w_i^{(t)}\|_2^2 \leq O\left(\frac{d(b_i^{(t)})^2}{\log d}\right)$.

Then for any $\lambda \geq 0.0001$:

$$\Pr(|\langle w_i^{(t)}, x_p^+ \rangle| \geq \lambda b_i^{(t)}) \lesssim e^{-\Omega(\log^{1/4} d)}, \quad \Pr(|\langle w_i^{(t)}, x_p \rangle| \geq \lambda b_i^{(t)}) \lesssim e^{-\Omega(\log^{1/4} d)}$$

Proof. Our proof follows from similar arguments in [8]. The only difference here is that we have applied RandomMask augmentation to our data x_p^+ and x_p^{++} . We only need to consider two terms:

- The augmented noise $\langle w_i^{(t)}, 2\mathbf{D}\xi_p \rangle$, which follows from Gaussian distribution with variance $O(\|w_i^{(t)}\|_2^2 \sigma_\xi^2)$. We have that for some small constant $c \ll 0.0001$: $|\langle w_i^{(t)}, 2\mathbf{D}\xi_p \rangle| \leq cb_i^{(t)}$;
- The augmented sparse signals $\langle w_i^{(t)}, 2\mathbf{D}\mathbf{M}z_p \rangle = \langle w_i^{(t)}, \mathbf{M}z_p \rangle + \langle w_i^{(t)}, (2\mathbf{D} - \mathbf{I})\mathbf{M}z_p \rangle$. Here the bound for the first term on RHS can be obtained via similar approach in [8], the bound for the second term follows from Lemma 2.B.3.

They conclude the proof. □

Lemma 2.B.5 (pre-activation size, III). *Let $i \in [m]$. Suppose the following holds: there exist a set $\mathcal{N}_i \subseteq [d]$ such that $|\mathcal{N}_i| = O(1)$, and*

- $\langle w_i^{(t)}, \mathbf{M}_j \rangle^2 \leq O\left(\frac{(b_i^{(t)})^2}{\text{polylog}(d)}\right)$ for $j \notin \mathcal{N}_i$;
- $\|w_i^{(t)}\|_2^2 \leq O\left(\frac{d(b_i^{(t)})^2}{\text{polylog}(d)}\right)$.

Then for any $\lambda \in [0.01, 0.99]$,

$$\Pr\left[\left|\sum_{j \notin \mathcal{N}_i} \langle w_i^{(t)}, 2\mathbf{D}\mathbf{M}_j \rangle z_j + \langle w_i^{(t)}, 2\mathbf{D}\xi \rangle\right| \geq \lambda b_i^{(t)}\right] \lesssim e^{-\Omega(\log^2 d)}$$

Proof. The proof is similar to those of Lemma 2.B.4 above and the proof of [8, Lemma C.3]. □

2.C Stage I: Initial Feature Growth

In this section we analyze the training process at the initial stage. Here we define the stage transition time $T_1 = \Theta(\frac{d \log d}{\eta \log \log d})$ to be the iteration when $\|w_i^{(t)}\|_2^2 \geq \frac{2(1+\gamma c_0)+c_1}{c_1} \|w_i^{(0)}\|_2^2$ for all the neurons $i \in [m]$ (where c_0 is a small constant defined in [Lemma 2.B.1](#)). Indeed, we will characterize the trajectory of weights $\{w_i^{(t)}\}_{i \in [m]}$ by calculating the growth of $w_i^{(t)}$ s for all the features $\{\mathbf{M}_j\}_{j \in [d]} \cup \{\mathbf{M}_j^\perp\}_{j \in [d_1] \setminus [d]}$. And also, we keep the bias $b_i^{(t)} = 0$ at this stage to simplify our analysis.

We present our theorem of the initial stage below:

Theorem 2.C.1 (Initial feature decoupling). *At iteration $t = T_1$, we have the following results:*

- (a) $\|\mathbf{M}\mathbf{M}^\top w_i^{(T_1)}\|_2^2 \geq \|w_i^{(T_1)}\|_2^2/2$ for all $i \in [m]$;
- (b) For each $j \in [d]$, and each $i \in \mathcal{M}_j^\star$, we have $|\langle w_i^{(T_1)}, \mathbf{M}_j \rangle| \geq (1 + \gamma c_0) \frac{\sqrt{2 \log d}}{\sqrt{d}} \|w_i^{(T_1)}\|_2$;
- (c) For each $j \in [d]$, and each $i \notin \mathcal{M}_j$, we have $|\langle w_i^{(T_1)}, \mathbf{M}_j \rangle| \leq (1 - \gamma c_0) \frac{\sqrt{2 \log d}}{\sqrt{d}} \|w_i^{(T_1)}\|_2$;
- (d) For each $i \in [m]$, $|\langle w_i^{(T_1)}, \mathbf{M}_j \rangle| \geq \frac{\log^{1/4} d}{\sqrt{d}} \|w_i^{(T_1)}\|_2$, for at most $O(\frac{d}{2\sqrt{\log d}})$ many $j \in [d]$.
- (e) For each $i \in [m]$ and $j \in [d_1] \setminus [d]$, we have $|\langle w_i^{(T_1)}, \mathbf{M}_j^\perp \rangle| \leq O(\sqrt{\frac{\log d}{d_1}}) \|w_i^{(T_1)}\|_2$.

2.C.1 Gradient Computations

Since the each bias $b_i^{(t)}$ remains at zero during this stage, it is easy to compute the positive gradient for each $t \leq T_1$ as the following:

Lemma 2.C.1 (positive gradient, stage I). *Let $h_{i,t}(\cdot)$ be the i -th neuron at iteration $t \leq T_1$ (so that $b_i^{(t)} = 0$), then*

- (a) For each $j \in [d]$, we have

$$\mathbb{E} \left[h_{i,t}(x_p^{++}) \langle \nabla_{w_i} h_{i,t}(x_p^+), \mathbf{M}_j \rangle \right] = \langle w_i^{(t)}, \mathbf{M}_j \rangle \mathbb{E}[z_j^2] \pm \tilde{O} \left(\frac{\|w_i^{(t)}\|_2}{d_1} \right)$$

- (b) For each $j \in [d_1] \setminus [d]$, we have

$$\mathbb{E} \left[h_{i,t}(x_p^{++}) \langle \nabla_{w_i} h_{i,t}(x_p^+), \mathbf{M}_j^\perp \rangle \right] = \pm \tilde{O} \left(\frac{\|w_i^{(t)}\|_2}{d_1} \right)$$

Proof. (a) For each $j \in [d]$ and $t \leq T_1$, since $b_i^{(t)} = 0$ for all $i \in [m]$, we can calculate

$$\begin{aligned} & \mathbb{E} \left[h_i(x_p^{++}) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq 0} \langle x_p^+, \mathbf{M}_j \rangle \right] \\ &= \mathbb{E} \left[(\langle w_i^{(t)}, 2(\mathbf{I} - \mathbf{D})(\mathbf{M}z_p + \xi_p) \rangle) \mathbb{1}_{|\langle w_i^{(t)}, x_p^{++} \rangle| \geq 0} \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq 0} \langle x_p^+, \mathbf{M}_j \rangle \right] \end{aligned}$$

Conditioned on each fixed $z_p \sim \mathcal{D}_z$ and $\mathbf{D} \sim \mathcal{D}_\mathbf{D}$ and use the randomness of ξ_p , we know that events $\{\langle w_i^{(t)}, x_p^+ \rangle = 0\}$ and $\langle w_i^{(t)}, x_p^{++} \rangle$ has probability zero. Thus we can get rid of the indicator functions and compute as follows:

$$\begin{aligned} & \mathbb{E} \left[(\langle w_i^{(t)}, 2(\mathbf{I} - \mathbf{D})(\mathbf{M}z_p + \xi_p) \rangle) \mathbb{1}_{|\langle w_i^{(t)}, x_p^{++} \rangle| \geq 0} \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq 0} \langle x_p^+, \mathbf{M}_j \rangle \right] \\ &= \mathbb{E} \left[\langle w_i^{(t)}, 2(\mathbf{I} - \mathbf{D})(\mathbf{M}z_p + \xi_p) \rangle \langle x_p^+, \mathbf{M}_j \rangle \right] \\ &= \mathbb{E} \left[\langle w_i^{(t)}, 2(\mathbf{I} - \mathbf{D})(\mathbf{M}z_p + \xi_p) \rangle \langle 2\mathbf{D}(\mathbf{M}z_p + \xi_p), \mathbf{M}_j \rangle \right] \end{aligned}$$

From simple observation, conditioned on fixed $\mathbf{D}$, we know that $\mathbf{D}\xi_p$ and $(\mathbf{I} - \mathbf{D})\xi_p$ are independent and both mean zero, and also z_p is independent w.r.t. ξ_p and is mean zero, so we can proceed to compute as

$$\begin{aligned} & \mathbb{E} \left[\langle w_i^{(t)}, 2(\mathbf{I} - \mathbf{D})(\mathbf{M}z_p + \xi_p) \rangle \langle 2\mathbf{D}(\mathbf{M}z_p + \xi_p), \mathbf{M}_j \rangle \right] \\ &= \mathbb{E} \left[\langle w_i^{(t)}, 2(\mathbf{I} - \mathbf{D})\mathbf{M}z_p \rangle \langle \mathbf{D}\mathbf{M}z_p, \mathbf{M}_j \rangle \right] \\ &= \mathbb{E} \left[\langle w_i^{(t)}, 2(\mathbf{I} - \mathbf{D})\mathbf{M}z_p \rangle \left(z_j + \sum_{j' \in [d]} \langle (\mathbf{I} - 2\mathbf{D})\mathbf{M}_{j'}, \mathbf{M}_j \rangle z_{j'} \right) \right] \\ &= \mathbb{E} \left[\sum_{j'' \in [d]} \langle w_i^{(t)}, 2(\mathbf{I} - \mathbf{D})\mathbf{M}_{j''} \rangle z_{p,j''} z_{p,j} \right] \\ & \quad + \mathbb{E} \left[\langle w_i^{(t)}, 2(\mathbf{I} - \mathbf{D})\mathbf{M}z_p \rangle \cdot \sum_{j' \in [d]} \langle (\mathbf{I} - 2\mathbf{D})\mathbf{M}_{j'}, \mathbf{M}_j \rangle z_{p,j'} \right] \end{aligned}$$

Now notice that $z_{p,j''}$ and $z_{p,j}$ are independent to each other if $j'' \neq j$, we have

$$\begin{aligned} \mathbb{E} \left[\sum_{j'' \in [d]} \langle w_i^{(t)}, 2(\mathbf{I} - \mathbf{D})\mathbf{M}_{j''} \rangle z_{p,j''} z_{p,j} \right] &= \mathbb{E} \left[\langle w_i^{(t)}, 2(\mathbf{I} - \mathbf{D})\mathbf{M}_j \rangle z_{p,j}^2 \right] \\ &= \langle w_i^{(t)}, \mathbf{M}_j \rangle \mathbb{E} \left[z_j^2 \right] + \mathbb{E} \left[\langle w_i^{(t)}, (\mathbf{I} - 2\mathbf{D})\mathbf{M}_j \rangle z_{p,j}^2 \right] \\ &= \langle w_i^{(t)}, \mathbf{M}_j \rangle \mathbb{E} \left[z_j^2 \right] \end{aligned}$$

where in the last equality we have used the fact that $\mathbf{I} - 2\mathbf{D}$ is independent to $z_{p,j}$ and has mean zero. Next by using [Lemma 2.B.2](#) and [Lemma 2.B.3](#) (3), we have the bound

$$\left| \mathbb{E} \left[\langle w_i^{(t)}, 2(\mathbf{I} - \mathbf{D})\mathbf{M}z_p \rangle \cdot \sum_{j' \in [d]} \langle (\mathbf{I} - 2\mathbf{D})\mathbf{M}_{j'}, \mathbf{M}_j \rangle z_{p,j'} \right] \right| \leq \tilde{O} \left(\frac{\|w_i^{(t)}\|_2}{d_1} \right)$$

Combining all results above, we obtain the desired approximation.

(b) It is easy to notice that the only difference of this proof with that of (a) is we have

$$\langle x_p^+, \mathbf{M}_j^\perp \rangle = \sum_{j' \in [d]} \langle (\mathbf{I} - 2\mathbf{D})\mathbf{M}_{j', z_{p,j'}}, \mathbf{M}_j^\perp \rangle + \langle (\mathbf{I} - 2\mathbf{D})\xi_p, \mathbf{M}_j^\perp \rangle$$

Now following the same argument as in (1), we can obtain the desired bound. $\square$

In order to analyze the

Lemma 2.C.2 (logits near initialization). *Letting $w_i \in \mathbb{R}^{d_1}$ for each $i \in [m]$, suppose we have $\sum_{i \in [m]} \|w_i^{(t)}\|_2^2 \leq o(\tau/d)$, then with high probability over the randomness of $x_p^+, x_p^{++}, \mathfrak{R}$, it holds:*

$$\left| \ell'_{p,t}(x_p^+, \mathfrak{B}) - \frac{1}{|\mathfrak{B}|} \right|, \left| \ell'_{s,t}(x_p^+, \mathfrak{B}) - \frac{1}{|\mathfrak{B}|} \right| \leq \tilde{O} \left(\frac{\sum_{i \in [m]} \|w_i^{(t)}\|_2^2}{\tau |\mathfrak{B}|} \right)$$

Proof. For the logit $\ell_{s,t}(x_p^+, \mathfrak{B})$ of negative sample $x_{n,s}$, we can simply calculate

$$\begin{aligned} \left| \ell'_{s,t}(x_p^+, \mathfrak{B}) - \frac{1}{|\mathfrak{B}|} \right| &= \left| \frac{e^{\text{Sim}_{f_t}(x_p^+, x_{n,s})/\tau}}{\sum_{x \in \mathfrak{B}} e^{\text{Sim}_{f_t}(x_p^+, x)/\tau}} - \frac{1}{|\mathfrak{B}|} \right| \\ &= \left| \left(\sum_{x \in \mathfrak{B}} e^{\langle f_t(x_p^+), f_t(x) - f_t(x_{n,s}) \rangle / \tau} \right)^{-1} - \frac{1}{|\mathfrak{B}|} \right| \\ &= \left| |\mathfrak{B}| - \sum_{x \in \mathfrak{B}} e^{\langle f_t(x_p^+), f_t(x) - f_t(x_{n,s}) \rangle / \tau} \right| \cdot \left(|\mathfrak{B}| \cdot \sum_{x \in \mathfrak{B}} e^{\langle f_t(x_p^+), f_t(x) - f_t(x_{n,s}) \rangle / \tau} \right)^{-1} \\ &\leq \sum_{x \in \mathfrak{B}} \left| 1 - e^{\langle f_t(x_p^+), f_t(x) - f_t(x_{n,s}) \rangle / \tau} \right| \cdot \left(|\mathfrak{B}| \cdot \sum_{x \in \mathfrak{B}} e^{\langle f_t(x_p^+), f_t(x) - f_t(x_{n,s}) \rangle / \tau} \right)^{-1} \\ &\lesssim \max_{x \in \mathfrak{B}} \left| \langle f_t(x_p^+), f_t(x) - f_t(x_{n,s}) \rangle \right| \cdot \left(\tau \sum_{x \in \mathfrak{B}} e^{\langle f_t(x_p^+), f_t(x) - f_t(x_{n,s}) \rangle / \tau} \right)^{-1} \\ &\stackrel{(1)}{\leq} \tilde{O} \left(\frac{\sum_{i \in [m]} \|w_i^{(t)}\|_2^2}{\tau |\mathfrak{B}|} \cdot \exp \left(\frac{\sum_{i \in [m]} \|w_i^{(t)}\|_2^2}{\tau} \right) \right) \\ &\stackrel{(2)}{\leq} \tilde{O} \left(\frac{\sum_{i \in [m]} \|w_i^{(t)}\|_2^2}{\tau |\mathfrak{B}|} \right) \end{aligned}$$

where (1) is because we have $|1 - e^a| \leq |a|$ for $a \leq 0.1$, and also with high probability

$$|\langle f_t(x_p^+), f_t(x_{n,u}) - f_t(x_{n,s}) \rangle| \leq \tilde{O} \left(\sum_{i \in [m]} \|w_i^{(t)}\|_2^2 \right)$$

and (2) is because $e^x \leq O(1)$ for $x \leq 1/2$. The approximation for $\logit \ell'_{p,t}$ of positive sample can be similarly obtained. $\square$

2.C.2 The Learning Process at Initial Stage

In this subsection we will prove, for every neuron $i \in [m]$, the weights w_i will mostly ignore the spurious features $\mathbf{M}^\perp$ and learn to emphasize the features $\mathbf{M}$. Recall that $T_1 = \Theta(\frac{d \log d}{\eta \log \log d})$ is set to be the time when $\|w_i^{(t)}\|_2^2 \geq \frac{2(1+2^{\gamma c_0})}{c_1} \|w_i^{(0)}\|_2^2$ for all the neurons $i \in [m]$, and that such a T_1 is indeed of order $\Theta(\frac{d \log d}{\eta \log \log d})$.

In order to prove the above theorem, we need the following :

Induction Hypothesis 2.C.1. *The following properties hold for all $t \leq T_1$:*

- (a) $\|\mathbf{M}^\perp(\mathbf{M}^\perp)^\top w_i^{(t)}\|_2^2 \leq (1 + O(1/\text{poly}(d))) \|\mathbf{M}^\perp(\mathbf{M}^\perp)^\top w_i^{(0)}\|_2^2$;
- (b) $\|\mathbf{M}\mathbf{M}^\top w_i^{(t)}\|_2^2 \leq \|\mathbf{M}\mathbf{M}^\top w_i^{(0)}\|_2^2 \left(1 - \eta\lambda + \frac{\eta C_z \log \log d}{d}\right)^{2t} + O(\frac{1}{d}) \|\mathbf{M}\mathbf{M}^\top w_i^{(0)}\|_2^2$, moreover, we have $\|\mathbf{M}\mathbf{M}^\top w_i^{(t)}\|_2^2 \leq O(\|w_i^{(0)}\|_2^2)$;
- (c) $\|\mathbf{M}\mathbf{M}^\top w_i^{(t)}\|_2^2 \geq \|\mathbf{M}\mathbf{M}^\top w_i^{(0)}\|_2^2 \left(1 - \eta\lambda + \frac{\eta C_z \log \log d}{d}\right)^{2t} - O(\frac{1}{d}) \|\mathbf{M}\mathbf{M}^\top w_i^{(0)}\|_2^2$

Proof of Induction Hypothesis 2.C.1. First we need to work out the exact form of gradient for each feature $\mathbf{M}_j$ and $\mathbf{M}_j^\perp$. Fix a neuron $i \in [m]$, for the sparse feature $\mathbf{M}_j$, $j \in [d]$, we can write down the SGD iteration as follows:

$$\begin{aligned} \langle w_i^{(t+1)}, \mathbf{M}_j \rangle &= \langle w_i^{(t)}, \mathbf{M}_j \rangle - \langle \nabla_{w_i} \text{Obj}(f_i), \mathbf{M}_j \rangle \pm \frac{\|w_i^{(t)}\|_2}{\text{poly}(d_1)} \\ &= (1 - \lambda) \langle w_i^{(t)}, \mathbf{M}_j \rangle + \mathbb{E}_{x_p^+, x_p^{++}} \left[(1 - \ell'_{p,t}(x_p^+, \mathfrak{B})) \cdot h_{i,t}(x_p^{++}) \langle \nabla_{w_i} h_{i,t}(x_p^+), \mathbf{M}_j \rangle \right] \\ &\quad - \sum_{x_{n,s} \in \mathfrak{N}} \mathbb{E} \left[\ell'_{s,t}(x_p, \mathfrak{B}) h_{i,t}(x_{n,s}) \langle \nabla_{w_i} h(x_p^+), \mathbf{M}_j \rangle \right] \pm \frac{\|w_i^{(t)}\|_2}{\text{poly}(d_1)} \end{aligned}$$

For the positive term $\mathbb{E} \left[(1 - \ell'_{p,t}(x_p^+, \mathfrak{B})) h_{i,t}(x_p^{++}) \langle \nabla_{w_i} h_{i,t}(x_p^+), \mathbf{M}_j \rangle \right]$, we can use [Lemma 2.C.1](#) and [Lemma 2.C.2](#) to obtain that:

$$\mathbb{E} \left[(1 - \ell'_{p,t}(x_p^+, \mathfrak{B})) \cdot h_{i,t}(x_p^{++}) \langle \nabla_{w_i} h_{i,t}(x_p^+), \mathbf{M}_j \rangle \right] = \langle w_i^{(t)}, \mathbf{M}_j \rangle \Pr(|z_j| = 1) \pm \tilde{O} \left(\frac{\|w_i^{(t)}\|_2}{d_1} \right)$$

And for the negative term $\mathbb{E} \left[\sum_{x_{n,s} \in \mathfrak{N}} \ell'_{s,t}(x_p, \mathfrak{B}) h_{i,t}(x_{n,s}) \langle \nabla_{w_i} h(x_p^+), \mathbf{M}_j \rangle \right]$, we can use [Lemma 2.C.2](#) to bound it as:

$$\mathbb{E} \left[\sum_{x_{n,s} \in \mathfrak{N}} \ell'_{s,t} h_{i,t}(x_{n,s}) \langle \nabla_{w_i} h(x_p^+), \mathbf{M}_j \rangle \right] \stackrel{(1)}{=} \sum_{x_{n,s} \in \mathfrak{N}} \mathbb{E} \left[\left(\ell'_{s,t} - \frac{1}{|\mathfrak{B}|} \right) h_{i,t}(x_{n,s}) \langle \nabla_{w_i} h(x_p^+), \mathbf{M}_j \rangle \right]$$

$$\begin{aligned}
&\leq \sum_{x_{n,s} \in \mathfrak{N}} \mathbb{E} \left[\left| \ell'_{s,t} - \frac{1}{|\mathfrak{B}|} \right| \cdot |h_{i,t}(x_{n,s})| \cdot |\langle \nabla_{w_i} h(x_p^+), \mathbf{M}_j \rangle| \right] \\
&\stackrel{(2)}{\leq} \tilde{O} \left(\frac{\sum_{i \in [m]} \|w_i^{(t)}\|_2^2}{\tau d} \cdot \|w_i^{(t)}\|_2 \right)
\end{aligned}$$

where (2) has applied [Lemma 2.B.3](#) to $h_{i,t}(x_{n,s})$ and $|\langle \nabla_{w_i} h_{i,t}(x_p^+), \mathbf{M}_j \rangle| = |\langle \mathbf{M}_j, x_p^+ \rangle|$. Putting all the above calculations together, we have

$$\langle w_i^{(t+1)}, \mathbf{M}_j \rangle = (1 - \lambda + \Pr(|z_j| = 1)) \langle w_i^{(t)}, \mathbf{M}_j \rangle \pm \tilde{O} \left(\frac{\sum_{i \in [m]} \|w_i^{(t)}\|_2^2}{\tau d} \cdot \|w_i^{(t)}\|_2 \right) \pm \tilde{O} \left(\frac{\|w_i^{(t)}\|_2}{d_1} \right)$$

Before we perform induction, we obtain from similar approach the (stochastic) gradient step of w_i toward the direction of dense feature $\mathbf{M}_j^\perp$ as

$$\begin{aligned}
\langle w_i^{(t+1)}, \mathbf{M}_j^\perp \rangle &= \langle w_i^{(t)}, \mathbf{M}_j^\perp \rangle - \langle \nabla_{w_i} \text{Obj}(f_t), \mathbf{M}_j^\perp \rangle \\
&= (1 - \lambda) \langle w_i^{(t)}, \mathbf{M}_j^\perp \rangle + \mathbb{E} \left[(1 - \ell'_{p,t}) h_{i,t}(x_p^{++}) \langle \nabla_{w_i} h_{i,t}(x_p^+), \mathbf{M}_j^\perp \rangle \right] \\
&\quad - \sum_{x_{n,s} \in \mathfrak{N}} \mathbb{E} \left[\ell'_{s,t} h_{i,t}(x_{n,s}) \langle \nabla_{w_i} h(x_p^+), \mathbf{M}_j^\perp \rangle \right] + \frac{\|w_i^{(t)}\|_2}{\text{poly}(d_1)} \\
&= (1 - \lambda) \langle w_i^{(t)}, \mathbf{M}_j^\perp \rangle \pm \tilde{O} \left(\frac{\sum_{i \in [m]} \|w_i^{(t)}\|_2^2}{\tau d} \cdot \|w_i^{(t)}\|_2 \right) \pm \tilde{O} \left(\frac{\|w_i^{(t)}\|_2}{d_1} \right)
\end{aligned}$$

Then we can begin to perform our induction: at $t = 0$, our properties holds trivially. Now suppose before iteration $t = t_1$, the claimed properties holds, then we can easily obtain that for all $t \leq t_1$:

$$\|w_i^{(t)}\|_2^2 = \|\mathbf{M}\mathbf{M}^\top w_i^{(t)}\|_2^2 + \|\mathbf{M}^\perp (\mathbf{M}^\perp)^\top w_i^{(t)}\|_2^2 \leq O(d) \|w_i^{(0)}\|_2^2 \leq \frac{1}{\text{poly}(d_1)}$$

Thus we have $\sum_{i \in [m]} \|w_i^{(t)}\|_2^2 \leq \frac{1}{\text{poly}(d_1)}$. We now begin to verify all the properties for $t = t_1 + 1$, until t_1 reaches T_1 .

- We first derive an upper bound for $\|\mathbf{M}\mathbf{M}^\top w_i^{(t+1)}\|_2^2$ at iterations $t \leq t_1$. For each $j \in [d]$, as long as $\langle w_i^{(t)}, \mathbf{M}_j \rangle \geq \Omega(\|w_i^{(t)}\|_2 / d \sqrt{d_1})$, then

$$\begin{aligned}
|\langle w_i^{(t)}, \mathbf{M}_j \rangle| &\leq \left(1 - \eta\lambda + \frac{\eta C_z \log \log d}{d} \right) \langle w_i^{(t)}, \mathbf{M}_j \rangle + \tilde{O} \left(\frac{\eta \|w_i^{(t)}\|_2}{d_1} \right) \\
&\leq \left(1 + \frac{\eta C_z \log \log d}{d} + \tilde{O} \left(\frac{\eta}{d^2} \right) \right) |\langle w_i^{(t)}, \mathbf{M}_j \rangle|
\end{aligned}$$

Define set of features: $\mathcal{E}^{(t)} := \{j \in [d] : \langle w_i^{(t)}, \mathbf{M}_j \rangle < O(\|w_i^{(t)}\|_2 / d \sqrt{d_1})\}$, note that $\mathcal{E}^{(t+1)} \subseteq \mathcal{E}^{(t)} \subseteq \Lambda_i$ (where the set Λ_i is defined in [Lemma 2.B.1](#)) in the sense that if $j \notin \mathcal{E}^{(t)}$, then but

$\frac{C_z \log \log d}{d} |\langle w_i^{(t)}, \mathbf{M}_j \rangle| \geq \tilde{O}\left(\frac{\|w_i^{(t)}\|_2}{d_1}\right)$ in the above calculations. Therefore:

$$\begin{aligned} \|\mathbf{M}\mathbf{M}^\top w_i^{(t+1)}\|_2^2 &= \sum_{j \in [d]} \left[\left(1 - \eta\lambda + \frac{\eta C_z \log \log d}{d} \right) \langle w_i^{(t)}, \mathbf{M}_j \rangle \pm \tilde{O}\left(\frac{\eta \|w_i^{(t)}\|_2}{d_1}\right) \right]^2 \\ &\leq \sum_{j \in [d]} \langle w_i^{(0)}, \mathbf{M}_j \rangle^2 \left(1 + \frac{\eta C_z \log \log d}{d} + \tilde{O}\left(\frac{\eta}{d^2}\right) \right)^{2t} \\ &\quad + \sum_{j \in [d]: j \in \mathcal{E}^{(0)}} \tilde{O}\left(\frac{(t+1)^2 \eta^2 \max_{t \leq t_1} \|w_i^{(t)}\|_2^2}{d_1^2}\right) \\ &\leq \|\mathbf{M}\mathbf{M}^\top w_i^{(0)}\|_2^2 \left(1 + \frac{\eta C_z \log \log d}{d} + \tilde{O}\left(\frac{\eta}{d^2}\right) \right)^{2t} + O(1/d) \|\mathbf{M}\mathbf{M}^\top w_i^{(0)}\|_2^2 \end{aligned}$$

which holds for all $t \leq t_1 \leq T_1 = \Theta\left(\frac{d \log d}{\eta \log \log d}\right)$, the last inequality is due to the following calculations:

$$\begin{aligned} \sum_{j \in [d]: j \in \mathcal{E}^{(0)}} \tilde{O}\left(\frac{(t+1)^2 \eta^2 \max_{t \leq t_1} \|w_i^{(t)}\|_2^2}{d_1^2}\right) &\leq \tilde{O}\left(\frac{d^3 \max_{t \leq t_1} \|w_i^{(t)}\|_2^2}{d_1^2}\right) \ll \frac{\max_{t \leq t_1} \|w_i^{(t)}\|_2^2}{d^2 d_1} \\ &\ll \frac{\|w_i^{(0)}\|_2^2}{d d_1} \ll O(1/d) \|\mathbf{M}\mathbf{M}^\top w_i^{(0)}\|_2^2 \end{aligned}$$

- Secondly we give an lower bound for $\|\mathbf{M}\mathbf{M}^\top w_i^{(t+1)}\|_2^2$ for iterations $t \leq t_1$. From the above calculations, we have

$$\begin{aligned} \|\mathbf{M}\mathbf{M}^\top w_i^{(t+1)}\|_2^2 &= \sum_{j \in [d]} \left[\left(1 - \eta\lambda + \frac{\eta C_z \log \log d}{d} \right) \langle w_i^{(t)}, \mathbf{M}_j \rangle \pm \tilde{O}\left(\frac{\eta \|w_i^{(t)}\|_2}{d_1}\right) \right]^2 \\ &\geq \sum_{j \in \mathcal{E}^{(0)}} \langle w_i^{(0)}, \mathbf{M}_j \rangle^2 \left(1 - \eta\lambda + \frac{\eta C_z \log \log d}{d} \right)^{2t} - \tilde{O}\left(\frac{(t+1)^2 \eta^2 d \max_{t \leq t_1} \|w_i^{(t)}\|_2^2}{d_1^2}\right) \\ &\geq \|\mathbf{M}\mathbf{M}^\top w_i^{(0)}\|_2^2 \left(1 - \eta\lambda + \frac{\eta C_z \log \log d}{d} \right)^{2t} - O(1/d) \|\mathbf{M}\mathbf{M}^\top w_i^{(0)}\|_2^2 \end{aligned}$$

where the last inequality follows from our computations of the upper bound.

- Finally we give an upper bound of $\|\mathbf{M}^\perp (\mathbf{M}^\perp)^\top w_i^{(t+1)}\|_2^2$ for iterations $t \leq t_1$. We can calculate similarly, by

$$\begin{aligned} &\|\mathbf{M}^\perp (\mathbf{M}^\perp)^\top w_i^{(t+1)}\|_2^2 \\ &= \sum_{j \in [d_1] \setminus [d]} \left[(1 - \eta\lambda) \langle w_i^{(t)}, \mathbf{M}_j^\perp \rangle \pm \tilde{O}\left(\frac{\eta \|w_i^{(t)}\|_2}{d_1}\right) \right]^2 \end{aligned}$$

$$\begin{aligned}
&\leq \|\mathbf{M}^\perp(\mathbf{M}^\perp)^\top w_i^{(0)}\|_2^2 + \max_{j \in [d_1] \setminus [d]} |\langle w_i^{(0)}, \mathbf{M}_j^\perp \rangle| \tilde{O}(\max_{t \leq t_1} \|w_i^{(t)}\|_2) + \tilde{O}\left(\frac{\eta^2(t+1)^2 \max_{t \leq t_1} \|w_i^{(t)}\|_2^2}{d_1^2}\right) \\
&\stackrel{(1)}{\leq} (1 + \tilde{O}(d/\sqrt{d_1})) \|\mathbf{M}^\perp(\mathbf{M}^\perp)^\top w_i^{(0)}\|_2^2 + O(d^4/d_1^2) \|w_i^{(0)}\|_2^2 \\
&\stackrel{(2)}{\leq} \left(1 + \frac{1}{\text{poly}(d)}\right) \|\mathbf{M}^\perp(\mathbf{M}^\perp)^\top w_i^{(0)}\|_2^2
\end{aligned}$$

where (1) and (2) have used several facts: (1) at initialization, we have $|\langle w_i^{(0)}, \mathbf{M}_j^\perp \rangle| \leq \tilde{O}(\|w_i^{(0)}\|_2/\sqrt{d_1})$ with high probability; (2) from our induction hypothesis, $\tilde{O}(\max_{t \leq t_1} \|w_i^{(t)}\|_2) \leq O(d)\|w_i^{(0)}\|_2$; (3) at initialization we have $\|w_i^{(0)}\|_2^2 \leq O(\|\mathbf{M}^\perp(\mathbf{M}^\perp)^\top w_i^{(0)}\|_2^2)$ with high probability.

Note that for each neuron $i \in [m]$, from [Lemma 2.B.1](#) combined with our upper bound and lower bound, we know when all the weights $\|w_i^{(t)}\|_2$ reach $\Theta(1)\|w_i^{(0)}\|_2$, the maximum $\max_{i \in [m]} \|w_i^{(t+1)}\|_2^2 \leq O(1)\|w_i^{(t+1)}\|_2^2$ for all $t \leq t_1$. Thus we have obtained all the results for $t = t_1 + 1$, and are able to proceed induction. $\square$

Proof of Theorem 2.C.1. The result (a) is easy to verify using [Induction Hypothesis 2.C.1](#). This we only verify (b) and (c). Note that from similar gradient calculations to those in the proof of [Induction Hypothesis 2.C.1](#), we have, for $j \in [d]$ and $i \in \mathcal{M}_j$:

$$\begin{aligned}
|\langle w_i^{(T_1)}, \mathbf{M}_j \rangle| &\geq |\langle w_i^{(0)}, \mathbf{M}_j \rangle| \left(1 - \eta\lambda + \frac{\eta C_z \log \log d}{d}\right)^{T_1} - \tilde{O}\left(\frac{\eta T_1 \|w_i^{(T_1)}\|_2}{d_1}\right) \\
&\stackrel{(1)}{\geq} \frac{\sqrt{c_1 \log d}}{\sqrt{d}} \|\mathbf{M} \mathbf{M}^\top w_i^{(0)}\|_2 \left(1 - \eta\lambda + \frac{\eta C_z \log \log d}{d}\right)^{T_1} - \frac{\|\mathbf{M} \mathbf{M}^\top w_i^{(0)}\|_2}{\text{poly}(d)} \\
&\stackrel{(1)}{\geq} \frac{(1 + c_0) \sqrt{2 \log d}}{\sqrt{d}} \|w_i^{(T_1)}\|_2
\end{aligned}$$

where in (1) we have used [Lemma 2.B.1](#) and the fact that $\frac{\eta T_1}{d_1} \leq \frac{1}{\text{poly}(d)}$. And in the last inequality (2) we argue: when all $\|w_{i'}^{(t)}\|_2, i' \in [m]$ reach $(1 + \frac{2(1+2c_0)}{c_1})\|w_{i'}^{(0)}\|_2$, by using [Induction Hypothesis 2.C.1](#) and our definition of T_1 , combined with the concentrations of initial weight norm $\|\mathbf{M} \mathbf{M}^\top w_i^{(0)}\|_2$ in [Lemma 2.B.1](#) it holds that

$$\begin{aligned}
|\langle w_i^{T_1}, \mathbf{M}_j \rangle|^2 &\geq \frac{c_1 \log d}{d} \cdot \|\mathbf{M} \mathbf{M}^\top w_i^{(T_1)}\|_2^2 \\
&\geq \frac{(2 + 4c_0)^2 \log d}{d} \cdot \|w_i^{(T_1)}\|_2^2 - \|\mathbf{M}^\perp(\mathbf{M}^\perp)^\top w_i^{(T_1)}\|_2^2 \\
&\geq 2(1 + c_0)^2 \frac{\|w_i^{(T_1)}\|_2^2 \log d}{d}
\end{aligned}$$

since all neuron weights $\{w_i^{(t)}\}_{i \in [m]}$ grow in the speed of $\|w_i^{(t)}\|_2 = (1 + \frac{\eta C_z \log \log d}{d} + o(\frac{1}{\text{poly}(d)}))^t \|w_i^{(0)}\|_2$. The property (c) and (d) can be verified via exactly the same approach, combined with [Lemma 2.B.1](#). For (e), noticing that at initialization $|\langle w_i^{(0)}, \mathbf{M}_j^\perp \rangle| \leq O(\sqrt{\frac{\log d}{d_1}}) \|w_i^{(0)}\|_2$, we have

$$|\langle w_i^{(T_1)}, \mathbf{M}_j^\perp \rangle| \leq |\langle w_i^{(0)}, \mathbf{M}_j^\perp \rangle| + O(T_1 \eta) \cdot \max_{t \leq T_1} O\left(\frac{\|w_i^{(t)}\|_2}{d_1}\right) \leq O(\sqrt{\frac{\log d}{d_1}}) \|w_i^{(t)}\|_2$$

□

2.D Stage II: Singleton Emerge

In this section we will present an analysis of how each feature $\mathbf{M}_j$ can be “won” by some subsets of the neurons, which depends on the randomness of random initialization. In this stage, we will prove that the following induction hypothesis holds for all iterations.

Induction Hypothesis 2.D.1. *For all iterations $t \in (T_1, T_2]$, our neurons $i \in [m]$ satisfies:*

- (a) *For $j \in [d]$, if $i \in \mathcal{M}_j^\star$, then $|\langle w_i^{(t)}, \mathbf{M}_j \rangle| \geq (1 + c_0)b_i^{(t)}$;*
- (b) *For $j \in [d]$, if $i \notin \mathcal{M}_j$, then $|\langle w_i^{(t)}, \mathbf{M}_j \rangle| \leq (1 - c_0)b_i^{(t)}$. Furthermore, $|\langle w_i^{(t)}, \mathbf{M}_j \rangle| \leq \tilde{O}(\frac{\|w_i^{(t)}\|_2}{\sqrt{d}})$;*
- (c) *For each $i \in [m]$, there are at most $O(2^{-\sqrt{\log d}}d)$ many $j \in [d]$ such that $\langle w_i^{(t)}, \mathbf{M}_j \rangle^2 \geq \frac{(b_i^{(t)})^2}{\sqrt{\log d}}$;*
- (d) *For each $i \in [m]$, we have $|\langle w_i^{(t)}, \mathbf{M}_j^\perp \rangle| \leq \tilde{O}(\frac{\|w_i^{(t)}\|_2}{\sqrt{d_1}})$ for all $j \in [d_1] \setminus [d]$;*
- (e) $\|w_i^{(t)}\|_2^2 \leq \frac{d(b_i^{(t)})^2}{\log d}$ for all $i \in [m]$.

2.D.1 Gradient Computations

Definition 2.D.1 (notations). For simpler presentation, we define the following notations: given $x = \mathbf{M}z + \xi \sim \mathcal{D}_x$ as in [Definition 2.2.1](#), and $\mathbf{D} \sim \mathcal{D}_\mathbf{D}$ as in [Definition 2.2.3](#), we let (for each $j \in [d]$)

$$x^{\setminus j} := \sum_{j' \neq j, j' \in [d]} \mathbf{M}_{j'} z_{j'} + \xi \quad S_{i,t}^{\setminus j} := \langle w_i^{(t)}, x^{\setminus j} \rangle \quad \tilde{S}_{i,t}^{\setminus j} := \langle w_i^{(t)}, (\mathbf{I} - 2\mathbf{D})x^{\setminus j} \rangle \quad (2.D.1)$$

$$\alpha_{i,j}^{(t)} := \langle w_i^{(t)}, \mathbf{M}_j \rangle \quad \tilde{\alpha}_{i,j}^{(t)} := \langle w_i^{(t)}, (\mathbf{I} - 2\mathbf{D})\mathbf{M}_j \rangle \quad (2.D.2)$$

whenever the neuron index $i \in [m]$ is clear from context, we omit the subscript of neuron index i and time t for simplicity.

First we present our lemma for the gradient of features associated with the sparse signals.

Lemma 2.D.1 (Gradient for sparse features). *Suppose [Induction Hypothesis 2.D.1](#) holds at iteration $t \geq 0$, for $j \in [d]$, we denote events*

$$\begin{aligned} A_1 &:= \{S_{i,t}^{\setminus j} \geq b_i^{(t)} - \alpha_{i,j}^{(t)}\}, & A_2 &:= \{\bar{S}_{i,t}^{\setminus j} \geq b_i^{(t)} - \bar{\alpha}_{i,j}^{(t)}\}; \\ A_3 &:= \{|\bar{S}_{i,t}^{\setminus j} + \bar{\alpha}_{i,j}^{(t)}| \geq (\alpha_{i,j}^{(t)} - b_i^{(t)})/2\}, & A_4 &:= \{S_{i,t}^{\setminus j} \geq (\alpha_{i,j}^{(t)} - b_i^{(t)})/2\}; \end{aligned}$$

and quantities L_1, L_2, L_3, L_4 as

$$L_1 := \sqrt{\frac{\mathbb{E}[|\bar{S}_{i,t}^{\setminus j}|^2(\mathbb{1}_{A_1} + \mathbb{1}_{A_2})]}{\mathbb{E}[\langle w_i^{(t)}, \xi_p \rangle^2]}}, \quad L_2 := \Pr(A_1), \quad L_3 := \sqrt{\frac{\mathbb{E}[|\bar{S}_{i,t}^{\setminus j}|^2(\mathbb{1}_{A_1} + \mathbb{1}_{A_2})]}{\mathbb{E}[\langle w_i^{(t)}, \xi_p \rangle^2]}}, \quad L_4 := \Pr(A_3)$$

then we have the following results:

(a) (all features) For all $i \in [m]$, if $\alpha_{i,j}^{(t)} \geq 0$, we have (when $\alpha_{i,j}^{(t)} \leq 0$ the opposite inequality holds)

$$\begin{aligned} \mathbb{E} \left[h_{i,t}(x_p^{++}) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)}} z_j \right] &\leq \alpha_{i,j}^{(t)} \cdot \mathbb{E}[z_j^2 \mathbb{1}_{|\langle w_i^{(t)}, x_p \rangle| \geq b_i^{(t)} + |\langle w_i^{(t)}, x_p^+ - x_p \rangle|}] \\ &\quad \pm (\alpha_{i,j}^{(t)} + O(\sqrt{\mathbb{E}[\bar{\alpha}_{i,j}^{(t)2}]}) \cdot \mathbb{E}[z_j^2] \cdot O(L_1 + L_2) \end{aligned}$$

(b) (lucky features) If $\alpha_{i,j}^{(t)} > b_i^{(t)}$, we have

$$\begin{aligned} \mathbb{E} \left[h_{i,t}(x_p^{++}) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)}} z_j \right] &= (\alpha_{i,j}^{(t)} - b_i^{(t)}) \cdot \mathbb{E}[z_j^2 \mathbb{1}_{|\langle w_i^{(t)}, x_p \rangle| \geq b_i^{(t)} + |\langle w_i^{(t)}, x_p^+ - x_p \rangle|}] \\ &\quad \pm (\alpha_{i,j}^{(t)} + O(\sqrt{\mathbb{E}[\bar{\alpha}_{i,j}^{(t)2}]}) \cdot \mathbb{E}[z_j^2] \cdot O(L_3 + L_4) \end{aligned}$$

If $\alpha_{i,j}^{(t)} < -b_i^{(t)}$, then the opposite inequality holds with $(\alpha_{i,j}^{(t)} - b_i^{(t)})$ changing to $(\alpha_{i,j}^{(t)} + b_i^{(t)})$.

Proof of [Lemma 2.D.1](#) (a). In the proof we will make the following simplification of notations: we drop the time superscript (t) , and also the subscript for neuron index i in [\(2.D.1\)](#). We start with the case when $0 < \alpha_j < b_i^{(t)}$ and rewrite the expectation as follows:

$$\begin{aligned} &\mathbb{E} \left[h_i(x_p^{++}) \mathbb{1}_{|\langle w_i, x_p^+ \rangle| \geq b_i} z_j \right] \\ &= \mathbb{E} \left[\left(\text{ReLU}(\langle w_i, x_p^{++} \rangle - b_i) - \text{ReLU}(-\langle w_i, x_p^{++} \rangle - b_i) \right) \mathbb{1}_{|\langle w_i, x_p^+ \rangle| \geq b_i} z_j \right] \\ &= \mathbb{E} \left[\left((\langle w_i, x_p^{++} \rangle - b_i) \mathbb{1}_{\langle w_i, x_p^{++} \rangle \geq b_i} - (-\langle w_i, x_p^{++} \rangle - b_i) \mathbb{1}_{-\langle w_i, x_p^{++} \rangle \geq b_i} \right) \mathbb{1}_{|\langle w_i, x_p^+ \rangle| \geq b_i} z_j \right] \\ &= \mathbb{E} \left[\langle w_i, x_p^{++} \rangle \mathbb{1}_{|\langle w_i, x_p^+ \rangle| \geq b_i} \mathbb{1}_{\langle w_i, x_p^+ \rangle \geq b_i} z_j \right] - \mathbb{E} \left[b_i \left(\mathbb{1}_{\langle w_i, x_p^{++} \rangle \geq b_i} - \mathbb{1}_{-\langle w_i, x_p^{++} \rangle \geq b_i} \right) \mathbb{1}_{|\langle w_i, x_p^+ \rangle| \geq b_i} z_j \right] \end{aligned}$$

Notice that the first term on the RHS can be simplified as:

$$\begin{aligned} &\mathbb{E} \left[\langle w_i, x_p^{++} \rangle \mathbb{1}_{|\langle w_i, x_p^+ \rangle| \geq b_i} \mathbb{1}_{\langle w_i, x_p^+ \rangle \geq b_i} z_j \right] \\ &= \mathbb{E} \left[(\langle w_i, x_p \rangle + \langle w_i, (\mathbf{I} - 2\mathbf{D})x_p \rangle) \mathbb{1}_{|\langle w_i, 2(\mathbf{I} - \mathbf{D})x_p \rangle| \geq b_i} \mathbb{1}_{\langle w_i, 2\mathbf{D}x_p \rangle \geq b_i} z_j \right] \end{aligned}$$

$$\stackrel{(1)}{=} \mathbb{E} \left[\langle w_i, x_p \rangle \mathbb{1}_{|\langle w_i, x_p^{++} \rangle| \geq b_i} \mathbb{1}_{|\langle w_i, x_p^+ \rangle| \geq b_i} z_j \right]$$

where (1) is due the fact that $\langle w_i, (\mathbf{I} - 2\mathbf{D})x_p \rangle \mathbb{1}_{|\langle w_i, x_p^{++} \rangle| \geq b_i} \mathbb{1}_{|\langle w_i, x_p^+ \rangle| \geq b_i}$ is symmetric with respect to zero due to the randomness of $\mathbf{D} \sim \mathcal{D}_D$. Thus the expectation can be expanded as:

$$\begin{aligned} & \mathbb{E} \left[h_i(x_p^{++}) \mathbb{1}_{|\langle w_i, x_p^+ \rangle| \geq b_i} z_j \right] \\ &= \mathbb{E} \left[\langle w_i, x_p \rangle \mathbb{1}_{|\langle w_i, x_p^{++} \rangle| \geq b_i} \mathbb{1}_{|\langle w_i, x_p^+ \rangle| \geq b_i} z_j \right] - \mathbb{E} \left[b_i \left(\mathbb{1}_{\langle w_i, x_p^{++} \rangle \geq b_i} - \mathbb{1}_{-\langle w_i, x_p^{++} \rangle \geq b_i} \right) \mathbb{1}_{|\langle w_i, x_p^+ \rangle| \geq b_i} z_j \right] \\ &= \mathbb{E} \left[\alpha_j z_j^2 \mathbb{1}_{|\langle w_i, x_p^{++} \rangle| \geq b_i + |\langle w_i, x_p^+ - x_p \rangle|} \right] \\ &\quad + \mathbb{E} \left[(S^{\setminus j} - b_i) \mathbb{1}_{\langle w_i, x_p^{++} \rangle \geq b_i} \mathbb{1}_{\langle w_i, x_p^+ \rangle \geq b_i} z_j \right] + \mathbb{E} \left[(S^{\setminus j} + b_i) \mathbb{1}_{\langle w_i, x_p^{++} \rangle \leq -b_i} \mathbb{1}_{\langle w_i, x_p^+ \rangle \leq -b_i} z_j \right] \\ &\quad + \mathbb{E} \left[(\alpha_j z_j + S^{\setminus j} - b_i) \mathbb{1}_{\langle w_i, x_p^{++} \rangle \geq b_i} \mathbb{1}_{\langle w_i, x_p^+ \rangle \leq -b_i} z_j \right] + \mathbb{E} \left[(\alpha_j z_j + S^{\setminus j} + b_i) \mathbb{1}_{\langle w_i, x_p^{++} \rangle \leq -b_i} \mathbb{1}_{\langle w_i, x_p^+ \rangle \geq b_i} z_j \right] \\ &= J_1 + J_2 + J_3 \end{aligned}$$

Now we need to obtain absolute bounds for both J_2 and J_3 . We start with J_2 , where

$$\begin{aligned} J_2 &= \mathbb{E} \left[(S^{\setminus j} - b_i) \mathbb{1}_{\langle w_i, x_p^{++} \rangle \geq b_i} \mathbb{1}_{\langle w_i, x_p^+ \rangle \geq b_i} z_j \right] + \mathbb{E} \left[(S^{\setminus j} + b_i) \mathbb{1}_{\langle w_i, x_p^{++} \rangle \leq -b_i} \mathbb{1}_{\langle w_i, x_p^+ \rangle \leq -b_i} z_j \right] \\ &= \mathbb{E} \left[(S^{\setminus j} - b_i) \mathbb{1}_{\langle w_i, x_p \rangle \geq b_i + |\langle w_i, x_p^+ - x_p \rangle|} z_j \right] + \mathbb{E} \left[(S^{\setminus j} + b_i) \mathbb{1}_{\langle w_i, x_p \rangle \leq -b_i - |\langle w_i, x_p^+ - x_p \rangle|} z_j \right] \end{aligned}$$

We proceed with the first term $\mathbb{E} \left[(S^{\setminus j} - b_i) \mathbb{1}_{\langle w_i, x_p \rangle \geq b_i + |\langle w_i, x_p^+ - x_p \rangle|} z_j \right]$. First from a trivial calculation conditioned on the randomness of z_j we have:

$$\begin{aligned} & \mathbb{E} \left[(S^{\setminus j} - b_i) \mathbb{1}_{\langle w_i, x_p \rangle \geq b_i + |\langle w_i, x_p^+ - x_p \rangle|} z_j \right] \\ &= \mathbb{E} \left[(S^{\setminus j} - b_i) |z_j| \left(\mathbb{1}_{S^{\setminus j} \geq b_i - \alpha_j z_j + |\bar{S}^{\setminus j} + \bar{\alpha}_j z_j|} - \mathbb{1}_{S^{\setminus j} \geq b_i + \alpha_j z_j + |\bar{S}^{\setminus j} - \bar{\alpha}_j z_j|} \right) \right] \end{aligned}$$

Now define

$$Z = \frac{1}{2}(|\bar{S}^{\setminus j} + \bar{\alpha}_j z_j| + |\bar{S}^{\setminus j} - \bar{\alpha}_j z_j|) \quad Z' = \frac{1}{2}(|\bar{S}^{\setminus j} + \bar{\alpha}_j z_j| - |\bar{S}^{\setminus j} - \bar{\alpha}_j z_j|)$$

In this case, we always have $|Z'| \leq |\bar{\alpha}_j| |z_j|$, and

$$\left| \mathbb{1}_{S^{\setminus j} \geq b_i - \alpha_j z_j + |\bar{S}^{\setminus j} + \bar{\alpha}_j z_j|} - \mathbb{1}_{S^{\setminus j} \geq b_i + \alpha_j z_j + |\bar{S}^{\setminus j} - \bar{\alpha}_j z_j|} \right| = \mathbb{1}_{S^{\setminus j} - b_i - Z \in [-|\alpha_j z_j - Z'|, |\alpha_j z_j - Z'|]}$$

which allows us to proceed as follows:

$$\begin{aligned} & \left| \mathbb{E} \left[(S^{\setminus j} - b_i) \mathbb{1}_{\langle w_i, x_p \rangle \geq b_i + |\langle w_i, x_p^+ - x_p \rangle|} z_j \right] \right| \\ &= \left| \mathbb{E} \left[(S^{\setminus j} - b_i - Z + Z) |z_j| \left(\mathbb{1}_{S^{\setminus j} \geq b_i - \alpha_j z_j + |\bar{S}^{\setminus j} + \bar{\alpha}_j z_j|} - \mathbb{1}_{S^{\setminus j} \geq b_i + \alpha_j z_j + |\bar{S}^{\setminus j} - \bar{\alpha}_j z_j|} \right) \mathbb{1}_{S^{\setminus j} \geq b_i - \alpha_j z_j} \right] \right| \\ &\leq \mathbb{E} \left[|S^{\setminus j} - b_i - Z| |z_j| \mathbb{1}_{S^{\setminus j} - b_i - Z \in [-|\alpha_j z_j - Z'|, |\alpha_j z_j - Z'|]} \mathbb{1}_{S^{\setminus j} \geq b_i - \alpha_j z_j} \right] \end{aligned}$$

$$\begin{aligned}
& + \mathbb{E} \left[(|\bar{S}^{\setminus j}| + |\bar{\alpha}_j| |z_j|) |z_j| \mathbb{1}_{S^{\setminus j} - b_i - Z \in [-|\alpha_j - Z'|, |\alpha_j - Z'|]} \mathbb{1}_{S^{\setminus j} \geq b_i - \alpha_j} \right] \\
& \leq \mathbb{E} \left[(\alpha_j + 2|\alpha'_j|) |z_j|^2 \mathbb{1}_{S^{\setminus j} \geq b_i - \alpha_j} \right] + \mathbb{E} \left[|\bar{S}^{\setminus j}| |z_j| \mathbb{1}_{S^{\setminus j} - b_i - Z \in [-|\alpha_j - Z'|, |\alpha_j - Z'|]} \mathbb{1}_{S^{\setminus j} \geq b_i - \alpha_j} \right] \\
& = \mathbb{E} \left[(\alpha_j + 2|\alpha'_j|) |z_j|^2 \mathbb{1}_{S^{\setminus j} \geq b_i - \alpha_j} \right] + \sqrt{\mathbb{E} \left[|\bar{S}^{\setminus j}|^2 |z_j|^2 \mathbb{1}_{S^{\setminus j} \geq b_i - \alpha_j} \right]} \cdot \sqrt{\mathbb{E} \left[\mathbb{1}_{S^{\setminus j} - b_i - Z \in [-|\alpha_j - Z'|, |\alpha_j - Z'|]} \right]} \\
& \stackrel{(1)}{\leq} \mathbb{E} \left[(\alpha_j + 2|\alpha'_j|) |z_j|^2 \mathbb{1}_{S^{\setminus j} \geq b_i - \alpha_j} \right] + \sqrt{\mathbb{E} \left[|\bar{S}^{\setminus j}|^2 |z_j|^2 \mathbb{1}_{S^{\setminus j} \geq b_i - \alpha_j} \right]} \cdot \sqrt{\frac{\mathbb{E}[(\alpha_j + |\bar{\alpha}_j|)^2 z_j^2]}{\mathbb{E}[\langle w_i, \xi_p \rangle^2]}} \\
& = 2(\alpha_j + O(\mathbb{E}[|\bar{\alpha}_j|^2]^{1/2})) \mathbb{E}[z_j^2] \left(\sqrt{\frac{\mathbb{E}[|\bar{S}^{\setminus j}|^2 \mathbb{1}_{S^{\setminus j} \geq b_i - \alpha_j}]}{\mathbb{E}[\langle w_i, \xi_p \rangle^2]}} + \Pr(S^{\setminus j} \geq b_i - \alpha_j) \right)
\end{aligned}$$

where in (1) we have used the randomness of ξ_p in the following manner: Fixing the randomness of z and $\mathbf{D}$, we have $S^{\setminus j} - Z$ is a random variable depending solely on the randomness of ξ_p , and thus we have

$$\begin{aligned}
\mathbb{E} \left[\mathbb{1}_{S^{\setminus j} - b_i - Z \in [-|\alpha_j - Z'|, |\alpha_j - Z'|]} \right] & \leq \mathbb{E} \left[\mathbb{1}_{\langle w_i, \xi_p \rangle - |\langle 2(\mathbf{I} - \mathbf{D}) w_i, \xi_p \rangle| \in [|\bar{\alpha}_j| - (\alpha_j + |\bar{\alpha}_j|), \alpha_j + 2|\bar{\alpha}_j|]} \right] \\
& = \mathbb{E} \left[\mathbb{1}_{\langle \mathbf{D} w_i, \xi_p \rangle + \langle (\mathbf{I} - \mathbf{D}) w_i, \xi_p \rangle - |\langle 2(\mathbf{I} - \mathbf{D}) w_i, \xi_p \rangle| \in [-O(\alpha_j + |\bar{\alpha}_j|), O(\alpha_j + |\bar{\alpha}_j|)]} \right] \\
& \leq \mathbb{E} \left[\frac{O(\alpha_j + |\bar{\alpha}_j|)^2 |z_j|}{\langle w_i, \xi_p \rangle^2} \right] = \frac{\mathbb{E}[O(\alpha_j + |\bar{\alpha}_j|)^2 |z_j|]}{\mathbb{E}[\langle w_i, \xi_p \rangle^2]}
\end{aligned}$$

Simultaneously, from similar analysis as above, we have for the second term in J_2 :

$$\begin{aligned}
\left| \mathbb{E} \left[(S^{\setminus j} + b_i) \mathbb{1}_{\langle w_i, x_p \rangle \leq -b_i - \langle w_i, x_p^+ - x_p \rangle |z_j|} \right] \right| & = \left| \mathbb{E} \left[(S^{\setminus j} + b_i) \mathbb{1}_{\langle w_i, x_p \rangle \leq -b_i - |\bar{S}^{\setminus j} + \bar{\alpha}_j z_j| |z_j|} \right] \right| \\
& \leq O(\alpha_j + \mathbb{E}[|\bar{\alpha}_j|^2]^{1/2}) \mathbb{E}[z_j^2] \sqrt{\frac{\mathbb{E}[|\bar{S}^{\setminus j}|^2 \mathbb{1}_{S^{\setminus j} \geq b_i - \alpha_j}]}{\mathbb{E}[\langle w_i, \xi_p \rangle^2]}}
\end{aligned}$$

Now we turn to J_3 , from the symmetry of x_p^+ and x_p^{++} over the randomness of $\mathbf{D} \sim \mathcal{D}_{\mathbf{D}}$, we observe

$$\mathbb{E} \left[b_i \mathbb{1}_{\langle w_i, x_p^+ \rangle \geq b_i} \mathbb{1}_{\langle w_i, x_p^+ \rangle \leq -b_i z_j} \right] = \mathbb{E} \left[b_i \mathbb{1}_{\langle w_i, x_p^+ \rangle \leq -b_i} \mathbb{1}_{\langle w_i, x_p^{++} \rangle \geq b_i z_j} \right]$$

which allows us to drop the b_i terms in J_3 . The analysis of the rest of J_3 is somewhat similar. First we observe that whenever $\mathbb{1}_{\langle w_i, x_p^{++} \rangle \geq b_i} \mathbb{1}_{\langle w_i, x_p^+ \rangle \leq -b_i} \neq 0$, we have

$$\langle w_i, x_p^{++} \rangle \geq b_i \text{ and } \langle w_i, x_p^+ \rangle \leq -b_i \implies \bar{S}^{\setminus j} + \bar{\alpha}_j z_j \geq b_i + |S^{\setminus j} + \alpha_j z_j|$$

When this inequality holds, we always have $\bar{S}^{\setminus j} \geq b_i - \bar{\alpha}_j z_j$. Together with all the above observations, we proceed to compute as:

$$\left| \mathbb{E} \left[(\alpha_j z_j + S^{\setminus j}) \mathbb{1}_{\langle w_i, x_p^{++} \rangle \geq b_i} \mathbb{1}_{\langle w_i, x_p^+ \rangle \leq -b_i z_j} \right] \right|$$

$$\begin{aligned}
&= \left| \mathbb{E} \left[(\alpha_j z_j + S^{\vee j}) \mathbb{1}_{\bar{\alpha}_j z_j + \bar{S}^{\vee j} \geq b_i + |\alpha_j z_j + S^{\vee j}| z_j} \right] \right| \\
&\stackrel{(1)}{\leq} \mathbb{E} \left[|\bar{S}^{\vee j} + \bar{\alpha}_j z_j| |z_j| \left| \mathbb{1}_{\bar{S}^{\vee j} \geq b_i - \bar{\alpha}_j z_j + |\alpha_j z_j + S^{\vee j}|} - \mathbb{1}_{\bar{S}^{\vee j} \geq b_i + \bar{\alpha}_j z_j + |-\alpha_j z_j + S^{\vee j}|} \right| \right] \\
&\leq \mathbb{E} \left[|\bar{S}^{\vee j} + \bar{\alpha}_j z_j| |z_j| \mathbb{1}_{\bar{S}^{\vee j} \geq b_i - \bar{\alpha}_j z_j} \cdot \mathbb{1}_{\bar{S}^{\vee j} \in [b_i - \alpha_j |z_j| + |S^{\vee j} + \bar{\alpha}_j |z_j|, b_i + \alpha_j |z_j| + |S^{\vee j} - \bar{\alpha}_j |z_j|]} \right] \\
&\leq \sqrt{\mathbb{E} \left[|\bar{S}^{\vee j}|^2 |z_j|^2 \mathbb{1}_{\bar{S}^{\vee j} \geq b_i - \bar{\alpha}_j z_j} \right]} \sqrt{\mathbb{E} \left[\mathbb{1}_{\bar{S}^{\vee j} \in [b_i - \alpha_j |z_j| + |S^{\vee j} + \bar{\alpha}_j |z_j|, b_i + \alpha_j |z_j| + |S^{\vee j} - \bar{\alpha}_j |z_j|]} \right]} \\
&\quad + \mathbb{E} \left[\bar{\alpha}_j |z_j|^2 \mathbb{1}_{\bar{S}^{\vee j} \geq b_i - \bar{\alpha}_j z_j} \right] \\
&\leq \sqrt{\mathbb{E} \left[|\bar{S}^{\vee j}|^2 |z_j|^2 \mathbb{1}_{\bar{S}^{\vee j} \geq b_i - \bar{\alpha}_j z_j} \right]} \sqrt{\mathbb{E} \left[\mathbb{1}_{\langle w_i, (2\mathbf{D} - \mathbf{I}) \xi_p \rangle \in [-|\alpha_j z_j + \bar{\alpha}_j |z_j|, |\alpha_j z_j + \bar{\alpha}_j |z_j|]} \right]} + \mathbb{E} \left[\bar{\alpha}_j |z_j|^2 \mathbb{1}_{\bar{S}^{\vee j} \geq b_i - \bar{\alpha}_j z_j} \right] \\
&\leq O(\alpha_j + \mathbb{E}[\bar{\alpha}_j^2]^{1/2}) \mathbb{E}[z_j^2] \cdot \sqrt{\frac{\mathbb{E}[|\bar{S}^{\vee j}|^2 \mathbb{1}_{\bar{S}^{\vee j} \geq b_i - \bar{\alpha}_j z_j}]}{\mathbb{E}[\langle w_i, \xi_p \rangle^2]}} + (\mathbb{E}[\bar{\alpha}_j^2]^{1/2} \mathbb{E}[z_j^2] \Pr(\bar{S}^{\vee j} \geq b_i - \bar{\alpha}_j z_j))
\end{aligned}$$

where in the last inequality, we have use the following reasoning: conditioned on fixed $\mathbf{D} \sim \mathcal{D}_{\mathbf{D}}$, we know that $\langle w_i, (\mathbf{I} - 2\mathbf{D})\xi_p \rangle$ has the same distribution with $\langle w_i, \xi_p \rangle$. We use the randomness to obtain that

$$\begin{aligned}
\mathbb{E}_{\xi_p} \left[\mathbb{1}_{\langle w_i, (2\mathbf{D} - \mathbf{I})\xi_p \rangle \in [-|\alpha_j z_j + \bar{\alpha}_j |z_j|, |\alpha_j z_j + \bar{\alpha}_j |z_j|]} \right] &\leq \frac{|\alpha_j + \bar{\alpha}_j|^2 \cdot |z_j|^2}{\mathbb{E}_{\xi}[\langle w_i, (\mathbf{I} - 2\mathbf{D})\xi_p \rangle^2]} \\
&= \frac{|\alpha_j + \bar{\alpha}_j|^2 \cdot |z_j|^2}{\mathbb{E}_{\xi}[\langle w_i, \xi_p \rangle^2]}
\end{aligned}$$

The second term of J_3 can be similarly bounded by the same quantity. Now by combining all the results of J_1, J_2, J_3 above, we have the desired result for (a). $\square$

Proof of Lemma 2.D.1 (b). This proof is extremely similar to the above proof of Lemma 2.D.1 (a), we describe the differences here and sketch the remaining. First we need to decompose the expectation as follows:

$$\begin{aligned}
\mathbb{E} \left[h_i(x_p^{++}) \mathbb{1}_{\langle w_i, x_p^+ \rangle \geq b_i z_j} \right] &= \mathbb{E} \left[(\alpha_j - b_i) z_j^2 \mathbb{1}_{\langle w_i, x_p \rangle \geq b_i + |\langle w_i, x_p^+ - x_p \rangle|} \right] \\
&\quad + \mathbb{E} \left[S^{\vee j} \mathbb{1}_{\langle w_i, x_p \rangle \geq b_i + |\langle w_i, x_p^+ - x_p \rangle|} |z_j| \right] \\
&\quad + \mathbb{E} \left[(\alpha_j z_j + S^{\vee j}) \mathbb{1}_{\langle w_i, x_p^+ - x_p \rangle \geq b_i + |\langle w_i, x_p \rangle|} |z_j| \right] \\
&= J_1 + J_2 + J_3
\end{aligned}$$

where we have used the following facts:

- $\mathbb{E}[b_i \mathbb{1}_{\langle w_i, x_p^+ \rangle \geq b_i} \mathbb{1}_{\langle w_i, x_p^+ \rangle \geq b_i z_j}] = -\mathbb{E}[b_i \mathbb{1}_{\langle w_i, x_p^+ \rangle \leq -b_i} \mathbb{1}_{\langle w_i, x_p^+ \rangle \leq -b_i z_j}];$
- $\mathbb{E}[b_i \mathbb{1}_{\langle w_i, x_p^+ \rangle \geq b_i} \mathbb{1}_{\langle w_i, x_p^+ \rangle \leq -b_i z_j}] = \mathbb{E}[b_i \mathbb{1}_{\langle w_i, x_p^+ \rangle \geq b_i} \mathbb{1}_{\langle w_i, x_p^+ \rangle \leq -b_i z_j}];$
- $\mathbb{1}_{\langle w_i, x_p^+ \rangle \geq b_i} \mathbb{1}_{\langle w_i, x_p^+ \rangle \geq b_i} = \mathbb{1}_{\langle w_i, x_p \rangle \geq b_i + |\langle w_i, x_p^+ - x_p \rangle|};$
- $\mathbb{1}_{\langle w_i, x_p^+ \rangle \leq -b_i} \mathbb{1}_{\langle w_i, x_p^+ \rangle \leq -b_i} = \mathbb{1}_{\langle w_i, x_p \rangle \leq -b_i - |\langle w_i, x_p^+ - x_p \rangle|};$

$$\bullet \mathbb{1}_{\langle w_i, x_p^+ \rangle \geq b_i} \mathbb{1}_{\langle w_i, x_p^+ \rangle \leq -b_i} = \mathbb{1}_{\langle w_i, x_p^+ \rangle \leq -b_i} \mathbb{1}_{\langle w_i, x_p^+ \rangle \geq b_i} = \mathbb{1}_{|\langle w_i, x_p^+ - x_p \rangle| \geq b_i + |\langle w_i, x_p \rangle|}.$$

Now observe that J_2 can be deal with as follows: define events $A_3 := \{|\bar{S}^j + \bar{\alpha}_j| \geq (\alpha_j - b_i)/2\}$ and $A_4 := \{S^j \geq (\alpha_j - b_i)/2\}$ and notice that $\mathbb{1} \leq \mathbb{1}_{A_3} + \mathbb{1}_{A_4}$, we can compute

$$\begin{aligned} J_2 &= \mathbb{E} \left[S^j \mathbb{1}_{|\langle w_i, x_p \rangle| \geq b_i + |\langle w_i, x_p^+ - x_p \rangle|} |Z_j| \right] \\ &= \mathbb{E} \left[S^j |Z_j| (\mathbb{1}_{S^j \in [b_i - \alpha |Z_j| + |\bar{S}^j + \bar{\alpha}_j|Z_j|, b_i + \alpha |Z_j| + |\bar{S}^j - \bar{\alpha}_j|Z_j|]} \right] \\ &= \mathbb{E} \left[S^j |Z_j| (\mathbb{1}_{A_3} + \mathbb{1}_{A_4} \mathbb{1}_{A_3^c}) \mathbb{1}_{S^j \in [b_i - \alpha |Z_j| + |\bar{S}^j + \bar{\alpha}_j|Z_j|, b_i + \alpha |Z_j| + |\bar{S}^j - \bar{\alpha}_j|Z_j|]} \right] \\ &\stackrel{(1)}{\leq} \mathbb{E} \left[\left(O(\alpha_j + |\bar{\alpha}_j|) |Z_j|^2 \mathbb{1}_{A_3} + |\bar{S}^j| |Z_j| (\mathbb{1}_{A_3} + \mathbb{1}_{A_4}) \right) \mathbb{1}_{S^j \in [b_i - \alpha |Z_j| - b_i + |\bar{S}^j + \bar{\alpha}_j|Z_j|, b_i + \alpha |Z_j| + |\bar{S}^j - \bar{\alpha}_j|Z_j|]} \right] \\ &\leq \mathbb{E}[Z_j^2] \cdot O(\alpha_j + \mathbb{E}[|\bar{\alpha}_j|^2]^{1/2}) \cdot \left(\Pr(A_3) + \sqrt{\mathbb{E}[|\bar{S}^j|^2 (\mathbb{1}_{A_3} + \mathbb{1}_{A_4})] / \mathbb{E}[\langle w_i, \xi_p \rangle^2]} \right) \end{aligned}$$

where (1) relies on the fact that whenever $\mathbb{1}_{|\langle w_i, x_p \rangle| \geq b_i + |\langle w_i, x_p^+ - x_p \rangle|} \neq 0$, we have $|S^j| \leq b_i + (\alpha_j + |\bar{\alpha}_j|) |Z_j| + |\bar{S}^j|$, and also that we have assumed $b_i < \alpha_j$. The term J_3 can be bounded from similar analysis as in the proof of [Lem. 2.D.1](#) (a), but changing the factor from $O(L_1 + L_2)$ to $O(L_3 + L_4)$. Combining these calculations give the desired results. $\square$

Lemma 2.D.2 (Gradient from dense signals). *Let $i \in [m]$ and $j \in [d]$, suppose [Induction Hypothesis 2.D.1](#) holds for the current iteration t , we have*

$$\left| \mathbb{E} \left[h_{i,t}(x_p^{++}) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)}} \langle 2\mathbf{D}\xi_p, \mathbf{M}_j \rangle \right] \right| \leq \tilde{O} \left(\frac{\|w_i^{(t)}\|_2}{d^2} \right) \cdot \max\{\Pr(|\langle w_i^{(t)}, x_p^{++} \rangle| \geq b_i^{(t)}), \tilde{O}(1/\sqrt{d})\}$$

For dense features $\mathbf{M}_j^\perp$, $j \in [d_1] \setminus [d]$, we have similar results:

$$\left| \mathbb{E} \left[h_{i,t}(x_p^{++}) \mathbb{1}_{|\langle w_i, x_p^+ \rangle| \geq b_i} \langle 2\mathbf{D}\xi_p, \mathbf{M}_j^\perp \rangle \right] \right| \leq \tilde{O} \left(\frac{\|w_i^{(t)}\|_2}{d\sqrt{d_1}} \right) \cdot \max\{\Pr(|\langle w_i^{(t)}, x_p^{++} \rangle| \geq b_i^{(t)}), \tilde{O}(1/\sqrt{d})\}$$

Proof. Again in this proof we omit the time superscript (t) . First we deal with the case where the features under consideration is $\mathbf{M}_j$, $j \in [d]$. Since after the RandomMask augmentations, $2\mathbf{D}\xi_p$ and $2(\mathbf{I} - \mathbf{D})\xi_p$ are independent (conditioned on fixed $\mathbf{D}$), we denote $2\mathbf{D}\xi_p = 2\mathbf{D}\xi'$ and $2(\mathbf{I} - \mathbf{D})\xi_p = 2(\mathbf{I} - \mathbf{D})\xi''$, where ξ' and ξ'' are independent. Now we can write as follows:

$$\begin{aligned} &\mathbb{E} \left[h_i(x_p^{++}) \mathbb{1}_{|\langle w_i, x_p^+ \rangle| \geq b_i} \langle 2\mathbf{D}\xi_p, \mathbf{M}_j \rangle \right] \\ &= \mathbb{E} \left[\left(\langle w_i, x_p^{++} \rangle - b_i \right) \mathbb{1}_{\langle w_i, x_p^{++} \rangle \geq b_i} \mathbb{1}_{|\langle w_i, x_p^+ \rangle| \geq b_i} \langle 2\mathbf{D}\xi', \mathbf{M}_j \rangle \right] \\ &\quad + \mathbb{E} \left[\left(\langle w_i, x_p^{++} \rangle + b_i \right) \mathbb{1}_{\langle w_i, x_p^{++} \rangle \leq -b_i} \mathbb{1}_{|\langle w_i, x_p^+ \rangle| \geq b_i} \langle 2\mathbf{D}\xi', \mathbf{M}_j \rangle \right] \end{aligned}$$

For the first term on the RHS, we have

$$\begin{aligned}
& \mathbb{E} \left[\left(\langle w_i, x_p^{++} \rangle - b_i \right) \mathbb{1}_{\langle w_i, x_p^{++} \rangle \geq b_i} \mathbb{1}_{|\langle w_i, x_p^+ \rangle| \geq b_i} \langle 2\mathbf{D}\xi', \mathbf{M}_j \rangle \right] \\
&= \mathbb{E} \left[\left(\langle w_i, x_p^{++} \rangle - b_i \right) \mathbb{1}_{\langle w_i, x_p^{++} \rangle \geq b_i} \mathbb{1}_{|\langle w_i, x_p^+ \rangle| \geq b_i} (\langle \xi', \mathbf{M}_j \rangle + \langle (2\mathbf{D} - \mathbf{I})\xi', \mathbf{M}_j \rangle) \right] \\
&= \mathbb{E} \left[\left(\langle w_i, x_p^{++} \rangle - b_i \right) \mathbb{1}_{\langle w_i, x_p^{++} \rangle \geq b_i} \mathbb{1}_{|\langle w_i, x_p^+ \rangle| \geq b_i} \langle \xi', \mathbf{M}_j \rangle \right] \\
&\quad + \mathbb{E} \left[\left(\langle w_i, x_p^{++} \rangle - b_i \right) \mathbb{1}_{\langle w_i, x_p^{++} \rangle \geq b_i} \mathbb{1}_{|\langle w_i, x_p^+ \rangle| \geq b_i} \sum_{j' \in [d_1]} \langle (2\mathbf{D} - \mathbf{I})\widehat{\mathbf{M}}_{j'}, \mathbf{M}_j \rangle \langle \xi', \widehat{\mathbf{M}}_{j'} \rangle \right] \\
&= I_1 + I_2
\end{aligned}$$

where $\{\widehat{\mathbf{M}}_j\}_{j \in [d_1]}$ is a basis for $\mathbb{R}^{d_1}$ satisfying $\|\widehat{\mathbf{M}}_j\|_\infty \leq O(1/\sqrt{d_1})$. For I_1 , notice that we can use approach similar to the proof of [Lemma 2.C.1](#) as (denoting $[x]_+ := x\mathbb{1}_{x \geq 0}$)

$$\begin{aligned}
& \mathbb{E} \left[\left(\langle w_i, x_p^{++} \rangle - b_i \right) \mathbb{1}_{\langle w_i, x_p^{++} \rangle \geq b_i} \mathbb{1}_{|\langle w_i, x_p^+ \rangle| \geq b_i} \langle \xi', \mathbf{M}_j \rangle \right] \\
&\leq \mathbb{E} \left[\left[\langle w_i, x_p^{++} \rangle - b_i \right]_+ \cdot \mathbb{1}_{h_i(x_p^{++}) \neq 0} \left| \mathbb{1}_{|\langle w_i, x_p^+ \rangle| + 2(\alpha - \alpha')|\langle \xi', \mathbf{M}_j \rangle| \geq b_i} - \mathbb{1}_{|\langle w_i, x_p^+ \rangle| - 2(\alpha - \alpha')|\langle \xi', \mathbf{M}_j \rangle| \geq b_i} \right| |\langle \xi', \mathbf{M}_j \rangle| \right] \\
&\leq \mathbb{E}_{|\mathbf{I} - \mathbf{M}_j \mathbf{M}_j^\top| \xi'} \left[\left[\langle w_i, x_p^{++} \rangle - b_i \right]_+ |\langle \xi', \mathbf{M}_j \rangle| \mathbb{1}_{\langle w_i, x_p^{++} \rangle \geq b_i} \times \right. \\
&\quad \left. \times \mathbb{E}_{\mathbf{I} - \mathbf{M}_j \mathbf{M}_j^\top} \left[\left| \mathbb{1}_{|\langle w_i, x_p^+ \rangle| + 2(\alpha - \alpha')|\langle \xi', \mathbf{M}_j \rangle| \geq b_i} - \mathbb{1}_{|\langle w_i, x_p^+ \rangle| - 2(\alpha - \alpha')|\langle \xi', \mathbf{M}_j \rangle| \geq b_i} \right| \right] \right] \\
&\leq O(1) \cdot \mathbb{E} \left[(\alpha_j + |\tilde{\alpha}_j|)^2 \langle \xi', \mathbf{M}_j \rangle^3 \cdot \frac{[\langle w_i, x_p^{++} \rangle - b_i]_+}{\mathbb{E}[\langle w_i, \xi' \rangle^2]} \mathbb{1}_{\langle w_i, x_p^{++} \rangle \geq b_i} \right]
\end{aligned}$$

where in the last inequality we have used the randomness of $(\mathbf{I} - \mathbf{M}_j \mathbf{M}_j^\top) \xi'$, which allow us to obtain the denominator $\Omega(1) \mathbb{E}[\langle w_i, \xi' \rangle^2]$. For I_2 , notice that w.h.p., we have

$$|\langle (2\mathbf{D} - \mathbf{I})\widehat{\mathbf{M}}_{j'}, \mathbf{M}_j \rangle| \leq \tilde{O} \left(\frac{1}{\sqrt{d_1}} \right)$$

Denote $\alpha'_{j'} = \langle w_i, \widehat{\mathbf{M}}_{j'} \rangle$ $j' \in [d_1]$ and $\tilde{\alpha}'_{j'} = \langle w_i, (\mathbf{I} - 2\mathbf{D})\widehat{\mathbf{M}}_{j'} \rangle$ $j' \in [d_1]$. Noticing that $\sum_{j' \in [d_1]} (|\alpha'_{j'}|^2 + |\tilde{\alpha}'_{j'}|^2) = O(\|w_i\|_2^2)$ coupled with Cauchy-Schwarz inequality, we can similarly obtain:

$$\begin{aligned}
I_2 &\leq \tilde{O} \left(\frac{1}{\sqrt{d_1}} \right) \mathbb{E} \left[\sum_{j' \in [d_1]} (\alpha'_{j'} + |\tilde{\alpha}'_{j'}|)^2 |\langle \xi', \mathbf{M}_j \rangle| |\langle \xi', \widehat{\mathbf{M}}_{j'} \rangle|^2 \cdot \frac{[\langle w_i, x_p^{++} \rangle - b_i]_+}{\mathbb{E}[\langle w_i, \xi' \rangle^2]} \mathbb{1}_{\langle w_i, x_p^{++} \rangle \geq b_i} \right] \\
&\leq \tilde{O} \left(\frac{1}{\sqrt{d_1}} \right) \mathbb{E} \left[O(\|w_i\|_2^2) \cdot |\langle \xi', \mathbf{M}_j \rangle| |\langle \xi', \widehat{\mathbf{M}}_{j'} \rangle|^2 \cdot \frac{1}{\|w_i\|_2^2/d} \times \right. \\
&\quad \left. \times \max \left\{ \tilde{O} \left(\frac{1}{\sqrt{d}} \mathbf{Pr}(|\langle w_i, x_p^{++} \rangle| \geq b_i) \right), \frac{1}{d} \right\} \right]
\end{aligned}$$

$$\leq \tilde{O}\left(\frac{\|w_i\|_2}{d\sqrt{d_1}}\right) \cdot \max\left\{\Pr(|\langle w_i, x_p^{++} \rangle| \geq b_i), \frac{1}{\sqrt{d}}\right\}$$

where in the second inequality we have used the following arguments: first we can compute

$$\begin{aligned} & [\langle w_i, x_p^{++} \rangle - b_i]_+ \\ & \leq \sum_{j \in \mathcal{N}_i} |\langle w_i, \mathbf{M}_j \rangle z_{p,j}| + \left| \sum_{j \notin \mathcal{N}_i} \langle w_i, \mathbf{M}_j \rangle z_{p,j} \right| + |\langle w_i, (2\mathbf{D} - \mathbf{I})\mathbf{M}z_p \rangle| + |\langle w_i, 2(\mathbf{D} - \mathbf{I})\xi_p \rangle| \\ & \leq \clubsuit + \spadesuit + \heartsuit + \diamond \end{aligned}$$

And from [Induction Hypothesis 2.D.1](#), [Lemma 2.B.3](#) and [Lemma 2.B.4](#), we have

$$|\clubsuit| \geq \Omega(\|w_i^{(t)}\|_2) \quad \text{with prob} \leq \tilde{O}\left(\frac{1}{d}\right) \quad |\spadesuit|, |\heartsuit|, |\diamond| \leq \tilde{O}\left(\frac{\|w_i\|_2}{\sqrt{d}}\right) \quad \text{w.h.p.}$$

Summing up over I_1, I_2 , we have the desired bound. For the dense feature $\mathbf{M}_j^\perp$, the analysis is similar and we omit for brevity. $\square$

2.D.2 The Learning Process at the Second Stage

The second stage is defined as the iterations $t \geq T_1$ but $t \leq T_2$, where $T_2 = \Theta\left(\frac{d \log d}{\eta \log \log d}\right)$ is defined as the iteration when one of the neuron $i \in [m]$ satisfies $\|w_i^{(T_2)}\|_2^2 \geq d\|w_i^{(T_1)}\|_2^2$. Our theorem for the training process in this stage is presented below:

Theorem 2.D.1 (Emergence of singletons). *For each neuron $i \in [m]$, not only [Induction Hypothesis 2.D.1](#) but also the following conditions holds at iteration $t = T_2$:*

- (a) *For each $j \in [d]$, if $i \in \mathcal{M}_j^\star$, then $|\langle w_i^{(T_2)}, \mathbf{M}_j \rangle| \geq \Omega(1)\|w_i^{(T_2)}\|_2$;*
- (b) *$b_i^{(T_2)} \geq \frac{\text{polylog}(d)}{\sqrt{d}}\|w_i^{(T_2)}\|_2$;*
- (c) *Let $\alpha_j^\star = \max_{i \in \mathcal{M}_j^\star} |\langle w_i^{(T_2)}, \mathbf{M}_j \rangle|$, then there is a constant $C_j = \Theta(1)$ such that $|\langle w_i^{(t)}, \mathbf{M}_j \rangle| \leq C_j \alpha_j^\star$ for all $i \in \mathcal{M}_j$.*

Before proving this theorem, we prove [Induction Hypothesis 2.D.1](#) as a preliminary step.

Proof of [Induction Hypothesis 2.D.1](#). At iteration $t = T_1$, we have verified all the above properties in [Theorem 2.C.1](#). Now suppose all the properties hold for $t < T_2$, we will verify that it still hold for $t + 1$. In order to calculate the gradient $\nabla_{w_i} \mathbf{Obj}$ along each feature $\mathbf{M}_j$ or $\mathbf{M}_j^\perp$, we have to apply [Lemma 2.D.1](#), [Lemma 2.D.2](#) and [Lemma 2.C.1](#). First we calculate parameters in [Lemma 2.D.1](#) (a) and (b). In order to using [Lemma 2.B.3](#), we have the followings

- $|\bar{S}_{i,t}^{\setminus j}|^2 = |\langle w_i^{(t)}, (\mathbf{I} - 2\mathbf{D})x^{\setminus j} \rangle|^2 \leq \tilde{O}\left(\frac{\|w_i^{(t)}\|_2^2}{d}\right);$

$$\geq \left(\langle w_i^{(t)}, \mathbf{M}_j \rangle - \text{sign}(\langle w_i^{(t)}, \mathbf{M}_j \rangle) \cdot b_i^{(t)} \right) \left(1 + \frac{\eta C_z \log \log d}{d} \left(1 - \frac{\eta}{\text{polylog}(d)} \right) \right)$$

where in the last inequality we have taken into consideration $\sum_{i \in [m]} \|w_i^{(t)}\|_2^2 \leq \frac{1}{\text{poly}(d_1)}$, which follows from our definition of iteration T_2 and the properties at iteration T_1 in [Theorem 2.C.1](#), and also that $\langle w_i^{(t)}, \mathbf{M}_j \rangle / d \geq b_i^{(t)} / d \gg \frac{\|w_i^{(t)}\|_2}{\sqrt{d_1}}$. Next we compare this growth to the growth of bias $b_i^{(t+1)}$. Since we raise our bias by $b_i^{(t+1)} = \max\{b_i^{(t)}(1 + \frac{\eta}{d}), b_i^{(t)} \frac{\|w_i^{(t+1)}\|_2}{\|w_i^{(t)}\|_2}\}$, as long as $\frac{\|w_i^{(t+1)}\|_2}{\|w_i^{(t)}\|_2} \leq \frac{|\langle w_i^{(t+1)}, \mathbf{M}_j \rangle|}{|\langle w_i^{(t)}, \mathbf{M}_j \rangle|}$, we can obtain the desired result ($\frac{\|w_i^{(t+1)}\|_2}{\|w_i^{(t)}\|_2} \leq \frac{|\langle w_i^{(t+1)}, \mathbf{M}_j \rangle|}{|\langle w_i^{(t)}, \mathbf{M}_j \rangle|}$ will be proved later when we prove (d)).

(b) When $i \notin \mathcal{M}_j$, we can similarly obtain that

$$\mathbb{E} \left[z_j^2 \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)} + |\langle w_i^{(t)}, x_p^+ - x_p \rangle|} \right] \leq O \left(\frac{1}{d \text{polylog}(d)} \right)$$

And similarly we can compute the gradient descent dynamics as follows: For $j \in [d]$ such that $|\langle w_i^{(t)}, \mathbf{M}_j \rangle| \geq \frac{\|w_i^{(t)}\|_2 d}{\sqrt{d_1}}$, we have (assume here $\langle w_i^{(t)}, \mathbf{M}_j \rangle > 0$, the opposite is similar)

$$\begin{aligned} & \langle w_i^{(t+1)}, \mathbf{M}_j \rangle \\ &= \langle w_i^{(t)}, \mathbf{M}_j \rangle - \eta \langle \nabla_{w_i} \text{Obj}(f_t), \mathbf{M}_j \rangle + \frac{\eta \|w_i^{(t)}\|_2}{\text{poly}(d_1)} \\ &\leq \langle w_i^{(t)}, \mathbf{M}_j \rangle \left(1 - \eta \lambda + \frac{O(\eta)}{d \text{polylog}(d)} \right) \pm O \left(\frac{\eta \sum_{i' \in [m]} \|w_{i'}^{(t)}\|_2^2 \|w_i^{(t)}\|_2}{d \tau} \right) \pm \tilde{O} \left(\eta \frac{\|w_i^{(t)}\|_2}{d^2} \right) \\ &\leq \langle w_i^{(t)}, \mathbf{M}_j \rangle \left(1 + \frac{O(\eta)}{d \text{polylog}(d)} \right) + \tilde{O} \left(\eta \frac{\|w_i^{(t)}\|_2}{d^2} \right) \end{aligned}$$

Since from our update rule $b_i^{(t+1)} \geq b_i^{(t)}(1 + \frac{\eta}{d})$, we know that $\frac{|\langle w_i^{(t+1)}, \mathbf{M}_j \rangle|}{|\langle w_i^{(t)}, \mathbf{M}_j \rangle|} \leq \frac{b_i^{(t+1)}}{b_i^{(t)}}$. Thus, if $|\langle w_i^{(t)}, \mathbf{M}_j \rangle| \leq (1 - c_0)b_i^{(t)}$ at iteration t , we have

- $|\langle w_i^{(t+1)}, \mathbf{M}_j \rangle| \leq (1 - c_0)b_i^{(t+1)}$ if $|\langle w_i^{(t)}, \mathbf{M}_j \rangle| \geq \frac{\|w_i^{(t)}\|_2 d}{\sqrt{d_1}}$ at iteration t ;
- $|\langle w_i^{(t+1)}, \mathbf{M}_j \rangle| \leq \frac{\|w_i^{(t+1)}\|_2}{\sqrt{d}} \leq (1 - c_0)b_i^{(t+1)}$ if $|\langle w_i^{(t)}, \mathbf{M}_j \rangle| \leq \frac{\|w_i^{(t)}\|_2 d}{\sqrt{d_1}}$ at iteration t .

It is also worth noting that similar calculations also leads to a lower bound

$$|\langle w_i^{(t+1)}, \mathbf{M}_j \rangle| \geq |\langle w_i^{(t)}, \mathbf{M}_j \rangle| (1 - \eta \lambda) - \tilde{O} \left(\eta \frac{\|w_i^{(t)}\|_2}{d^2} \right) \quad (2.D.3)$$

We leave the part of proving $|\langle w_i^{(t+1)}, \mathbf{M}_j \rangle| \leq \tilde{O}(\frac{\|w_i^{(t)}\|_2}{\sqrt{d}})$ to later.

- (c) The result (c) that there exist at most $O(2^{-\sqrt{\log d}} d)$ many $j \in [d]$ such that $|\langle w_i^{(t)}, \mathbf{M}_j \rangle|^2 \geq \frac{(b_i^{(t)})^2}{\log^{1/2} d}$ can be similarly proved.
- (d) Next we consider the learning dynamics for the dense features. We can use [Lemma 2.D.2](#) to calculate its dynamics by

$$\begin{aligned}
& \langle w_i^{(t+1)}, \mathbf{M}_j^\perp \rangle \\
&= \langle w_i^{(t)}, \mathbf{M}_j^\perp \rangle (1 - \eta\lambda) \pm \frac{\eta \|w_i^{(t)}\|_2}{\text{poly}(d_1)} \\
&\quad + \eta \mathbb{E} \left[(1 - \ell'_{p,t}) h_{i,t}(x_p^{++}) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)}} \left(\langle (2\mathbf{D} - \mathbf{I})\mathbf{M}z_p, \mathbf{M}_j^\perp \rangle + \langle 2\mathbf{D}\xi_p, \mathbf{M}_j^\perp \rangle \right) \right] \\
&\quad - \eta \sum_{x_{n,s} \in \mathfrak{N}} \mathbb{E} \left[\ell'_{s,t} h_{i,t}(x_{n,s}) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)}} \left(\langle (2\mathbf{D} - \mathbf{I})\mathbf{M}z_p, \mathbf{M}_j^\perp \rangle + \langle 2\mathbf{D}\xi_p, \mathbf{M}_j^\perp \rangle \right) \right] \\
&= \langle w_i^{(t)}, \mathbf{M}_j^\perp \rangle (1 - \eta\lambda) + \tilde{O} \left(\frac{\eta \|w_i^{(t)}\|_2}{d\sqrt{d_1}} \right) \cdot \Pr(h_{i,t}(x_p^{++}) \neq 0) \\
&\leq \langle w_i^{(t)}, \mathbf{M}_j^\perp \rangle + O \left(\frac{\eta \|w_i^{(t)}\|_2}{d\sqrt{d_1}} e^{-\Omega(\log^{1/4} d)} \right)
\end{aligned}$$

After establishing the bounds of growth speed for each features, we now calculate the proportions they contribute to each neuron weight $i \in [m]$. Namely, we need to prove that when [Induction Hypothesis 2.D.1](#) holds at iteration $t \in [T_1, T_2]$, we have

- To prove $\frac{|\langle w_i^{(t+1)}, \mathbf{M}_j \rangle|}{|\langle w_i^{(t)}, \mathbf{M}_j \rangle|} \geq \frac{\|w_i^{(t+1)}\|_2}{\|w_i^{(t)}\|_2}$ for $i \in \mathcal{M}_j^\star$, we argue as follows: from previous calculations we have

$$\begin{aligned}
& \sum_{j' \in [d], j' \neq j} \langle w_i^{(t+1)}, \mathbf{M}_{j'} \rangle^2 + \sum_{j' \in [d_1] \setminus [d]} \langle w_i^{(t+1)}, \mathbf{M}_{j'}^\perp \rangle^2 \\
&\leq \sum_{j' \in [d], j' \neq j} \langle w_i^{(t)}, \mathbf{M}_{j'} \rangle^2 \left(1 + \frac{O(\eta)}{\text{polylog}(d)}\right) + \sum_{j' \in [d_1] \setminus [d]} \langle w_i^{(t)}, \mathbf{M}_{j'}^\perp \rangle^2 + \tilde{O}\left(\frac{\eta}{d}\right) e^{-\Omega(\log^{1/4} d)} \|w_i^{(t)}\|_2^2
\end{aligned}$$

Therefore by adding $\langle w_i^{(t+1)}, \mathbf{M}_j \rangle^2$ to the LHS we have

$$\|w_i^{(t+1)}\|_2^2 \leq \|w_i^{(t)}\|_2^2 \left(1 + \frac{O(\eta)}{d \text{polylog}(d)}\right)^2 + \left(\frac{|\langle w_i^{(t+1)}, \mathbf{M}_j \rangle|}{|\langle w_i^{(t)}, \mathbf{M}_j \rangle|} - \frac{O(\eta)}{d \text{polylog}(d)} \right) |\langle w_i^{(t)}, \mathbf{M}_j \rangle|^2$$

which implies $\frac{|\langle w_i^{(t+1)}, \mathbf{M}_j \rangle|}{|\langle w_i^{(t)}, \mathbf{M}_j \rangle|} \geq \frac{\|w_i^{(t+1)}\|_2}{\|w_i^{(t)}\|_2}$ as desired.

- To prove $|\langle w_i^{(t+1)}, \mathbf{M}_j \rangle| \leq \tilde{O}(\frac{\|w_i^{(t+1)}\|_2}{\sqrt{d}})$ if $i \notin \mathcal{M}_j$, we first use inequality (2.D.3) to compute

$$\begin{aligned}
\|\mathbf{M}\mathbf{M}^\top w_i^{(t+1)}\|_2 &\geq \|\mathbf{M}\mathbf{M}^\top w_i^{(T_1)}\|_2(1 - \eta\lambda)^{t-T_1} - O\left(\frac{\eta(t - T_1 + 1) \max_{t' \in [T_1, t+1]} \|w_i^{(t')}\|_2}{\sqrt{d}d_1}\right) \\
&\geq \|\mathbf{M}\mathbf{M}^\top w_i^{(T_1)}\|_2(1 - \eta\lambda)^{t-T_1+1} - O\left(\frac{\eta(t - T_1)\|w_i^{(T_1)}\|_2\sqrt{d}}{\sqrt{d_1}}\right) \\
&\geq \|\mathbf{M}\mathbf{M}^\top w_i^{(T_1)}\|_2(1 - o(1)) \quad \text{for } t \leq \Theta\left(\frac{d \log d}{\eta \log \log d}\right).
\end{aligned}$$

Notice that $|\langle w_i^{(T_1)}, \mathbf{M}_j \rangle| \leq O(\sqrt{\frac{\log d}{d}})\|\mathbf{M}\mathbf{M}^\top w_i^{(T_1)}\|_2$ from Theorem 2.C.1. Suppose it also holds for iteration t , we have

$$\begin{aligned}
|\langle w_i^{(t+1)}, \mathbf{M}_j \rangle| &\leq |\langle w_i^{(t)}, \mathbf{M}_j \rangle|(1 + \frac{O(\eta)}{d \text{polylog}(d)}) + \tilde{O}\left(\frac{\eta\|w_i^{(t)}\|_2}{d^2}\right) \\
&\leq |\langle w_i^{(T_1)}, \mathbf{M}_j \rangle|(1 + \frac{O(\eta)}{d \text{polylog}(d)})^{t-T_1} + \tilde{O}\left(\frac{\eta(t - T_1)\|w_i^{(T_1)}\|_2}{d^{3/2}}\right) \\
&\quad \text{(because } \|w_i^{(t)}\|_2 \leq \sqrt{d}\|w_i^{(T_1)}\|_2 \text{ by the definition of } T_2) \\
&\leq |\langle w_i^{(T_1)}, \mathbf{M}_j \rangle|(1 + o(1)) + \tilde{O}\left(\frac{\|w_i^{(T_1)}\|_2}{d}\right) \\
&\leq O\left(\sqrt{\frac{\log d}{d}}\right)\|\mathbf{M}\mathbf{M}^\top w_i^{(T_1)}\|_2 \leq O\left(\sqrt{\frac{\log d}{d}}\right)\|\mathbf{M}\mathbf{M}^\top w_i^{(t+1)}\|_2 \\
&\leq O\left(\sqrt{\frac{\log d}{d}}\right)\|w_i^{(t+1)}\|_2
\end{aligned}$$

- For the dense features, we can compute as follows:

$$\begin{aligned}
&|\langle w_i^{(t+1)}, \mathbf{M}_j^\perp \rangle| \\
&\leq |\langle w_i^{(t)}, \mathbf{M}_j^\perp \rangle| + O\left(\frac{\eta\|w_i^{(t)}\|_2}{d\sqrt{d_1}}e^{-\Omega(\log^{1/4} d)}\right) \\
&\leq |\langle w_i^{(T_1)}, \mathbf{M}_j^\perp \rangle| + \sum_{t'=T_1}^t O\left(\frac{\eta\|w_i^{(t')}\|_2}{d\sqrt{d_1}}e^{-\Omega(\log^{1/4} d)}\right) \\
&\leq O\left(\sqrt{\frac{\log d}{d_1}}\right)\|w_i^{(T_1+1)}\|_2 \left(1 + \frac{O(\eta)}{d \text{polylog}(d)}\right)^2 + \sum_{t'=T_1+1}^t O\left(\frac{\eta\|w_i^{(t')}\|_2}{d\sqrt{d_1}}e^{-\Omega(\log^{1/4} d)}\right) \\
&\leq O\left(\sqrt{\frac{\log d}{d_1}}\right)\|w_i^{(t+1)}\|_2 \left(1 + \frac{O(\eta)}{d \text{polylog}(d)}\right)^{2(t-T_1+1)}
\end{aligned}$$

$$\leq O\left(\sqrt{\frac{\log d}{d_1}}\right) \|w_i^{(t+1)}\|_2$$

where we have used the assumption that $\|w_i^{(t)}\|_2 \leq \|w_i^{(t+1)}\|_2(1 + \frac{O(\eta)}{d \text{polylog}(d)})$ for all $i \in [m]$ and all $t \leq T_2 = \Theta(\frac{d \log d}{\eta \log \log d})$, which we prove here: First of all, from previous calculations we have

$$\|\mathbf{M}\mathbf{M}^\top w_i^{(t+1)}\|_2 \geq \|\mathbf{M}\mathbf{M}^\top w_i^{(t)}\|_2(1 - \eta\lambda) - O\left(\frac{\eta \|w_i^{(t)}\|_2}{d^{3/2}} e^{-\Omega(\log^{1/4} d)}\right)$$

also the trajectory of $\|\mathbf{M}^\perp(\mathbf{M}^\perp)^\top w_i^{(t+1)}\|_2$ can be lower bounded as

$$\|\mathbf{M}^\perp(\mathbf{M}^\perp)^\top w_i^{(t+1)}\|_2 \geq \|\mathbf{M}^\perp(\mathbf{M}^\perp)^\top w_i^{(t)}\|_2(1 - \eta\lambda) - O(\eta/d) e^{-\Omega(\log^{1/4} d)} \|w_i^{(t)}\|_2$$

thus by combining the change of $w_i^{(t)}$ over two subspaces, we have

$$\|w_i^{(t+1)}\|_2^2 \geq \|w_i^{(t)}\|_2^2(1 - \eta\lambda)^2 - O(\eta/d) e^{-\Omega(\log^{1/4} d)} \|w_i^{(t)}\|_2^2 \geq \|w_i^{(t)}\|_2^2(1 - \frac{O(\eta)}{d \text{polylog}(d)})$$

which gives the desired bound.

In the proof above, we have depend on the crucial assumption that $T_2 := \min\{t \in \mathbb{N} : \exists i \in [m] \text{ s.t. } \|w_i^{(t)}\|_2^2 \geq d \|w_i^{(T_1)}\|_2^2\}$ is of order $\Theta(\frac{d \log d}{\eta \log \log d})$. Now we verify it as follows. If $i \in \mathcal{M}_j^\star$ for some $j \in [d]$ (which also means $j' \notin \mathcal{N}_i$ for $j' \neq j$), we have

$$|\langle w_i^{(t)}, \mathbf{M}_j \rangle| \geq |\langle w_i^{(T_1)}, \mathbf{M}_j \rangle| \left(1 + \Omega\left(\frac{\eta \log \log d}{d}\right)\right)^{t-T_1}$$

Thus for some $t = O(\frac{d \log d}{\eta \log \log d})$, we have $|\langle w_i^{(t)}, \mathbf{M}_j \rangle|^2 \geq d \|w_i^{(T_1)}\|_2^2$, which proves that $T_2 \leq O(\frac{d \log d}{\eta \log \log d})$. Conversely, we also have for all $t \leq O(\frac{d \log d}{\eta \log \log d})$

$$\begin{aligned} & \sum_{j' \in [d]: j' \neq j} \langle w_i^{(t)}, \mathbf{M}_{j'} \rangle^2 + \sum_{j' \in [d_1] \setminus [d]} \langle w_i^{(t)}, \mathbf{M}_{j'}^\perp \rangle^2 \\ & \leq \|w_i^{(T_1)}\|_2^2 \left(1 + \frac{O(\eta)}{d \text{polylog}(d)}\right)^{t-T_1} + \max_{t' \leq t} O(\eta(t - T_1)/d) e^{-\Omega(\log^{1/4} d)} \|w_i^{(t')}\|_2^2 \\ & \leq o(d \|w_i^{(T_1)}\|_2^2) \end{aligned}$$

And also

$$|\langle w_i^{(t)}, \mathbf{M}_j \rangle| \leq |\langle w_i^{(T_1)}, \mathbf{M}_j \rangle| \left(1 + \frac{C_z \eta \log \log d}{d} \left(1 - \frac{1}{\text{polylog}(d)}\right)\right)^{t-T_1}$$

$$\leq O\left(\sqrt{\frac{\log d}{d}}\|w_i^{(T_1)}\|_2\left(1 + \frac{C_z\eta \log \log d}{d}\left(1 - \frac{1}{\text{polylog}(d)}\right)\right)^{t-T_1}\right)$$

Therefore we at least need $\frac{d \log(\Omega(\sqrt{d/\log d}))}{\eta C_z \log \log d}(1 - o(1))$ iteration to let any neuron $i \in [m]$ reach $\|w_i^{(t)}\|_2^2 \geq d\|w_i^{(T_1)}\|_2$, which proves that $T_2 = \Theta(\frac{d \log d}{\eta \log \log d})$. $\square$

Proof of Theorem 2.D.1. We follow similar analysis as in the proof of [Induction Hypothesis 2.D.1](#). In order to prove (a) – (d), we have to discuss the two substages of the learning process below.

- **When all $\|w_i^{(t)}\|_2 \leq 2\|w_i^{(T_1)}\|_2$:** From similar analysis in the proof of [Induction Hypothesis 2.D.1](#), the iteration complexity for a neuron $i \in [m]$ to reach $\|w_i^{(t)}\|_2 \geq 2\|w_i^{(T_1)}\|_2$ is no smaller than $T'_{i,1} := \max\{\Omega(\frac{d \log d}{\eta \log \log d}), T_2\}$. At this substage, we have

- the bias growth is large, i.e.,

$$b_i^{(T'_{i,1})} \geq b_i^{(T_1)}(1 + \eta/d)^{T'_{i,1}-T_1} \geq b_i^{(T_1)} \text{polylog}(d) \geq \frac{\text{polylog}(d)}{\sqrt{d}}\|w_i^{(T_1)}\|_2 \geq \frac{\text{polylog}(d)}{\sqrt{d}}\|w_i^{(T'_{i,1})}\|_2$$

- For $j \notin \mathcal{N}_i$ we have

$$\begin{aligned} \sum_{j \in [d], j \notin \mathcal{N}_i} \langle w_i^{(T'_{i,1})}, \mathbf{M}_j \rangle^2 &\leq \sum_{j \in [d], j \notin \mathcal{N}_i} \langle w_i^{(T_1)}, \mathbf{M}_j \rangle^2 \left(1 + \frac{O(\eta)}{d \text{polylog}(d)}\right)^{T_2} + \tilde{O}\left(\frac{\eta \|w_i^{(T_1)}\|_2^2}{d^{3/2}}\right) \\ &\leq (1 + o(1))\|\mathbf{M}\mathbf{M}^\top w_i^{(T_1)}\|_2^2 \quad (\text{since } \|w_i^{(T_1)}\|_2 \lesssim \|\mathbf{M}\mathbf{M}^\top w_i^{(T_1)}\|_2) \end{aligned}$$

- For $j \in [d_1] \setminus [d]$ we have

$$\begin{aligned} \sum_{j \in [d_1] \setminus [d]} \langle w_i^{(T'_{i,1})}, \mathbf{M}_j^\perp \rangle^2 &\leq \sum_{j \in [d_1] \setminus [d]} \langle w_i^{(T_1)}, \mathbf{M}_j^\perp \rangle^2 + O(\eta(T'_{i,1} - T_1)/d)e^{-\Omega(\log^{1/4} d)} \max_{t' \in [T_1, T'_{i,1}]} \|w_i^{(t')}\|_2^2 \\ &\leq (1 + o(1))\|\mathbf{M}^\perp(\mathbf{M}^\perp)^\top w_i^{(T_1)}\|_2^2 \end{aligned}$$

- If $i \in \mathcal{M}_j^*$, there exist $t \leq T_2$ such that $\|w_i^{(t)}\|_2 \geq 2\|w_i^{(T_2)}\|_2$, as we have argued in the proof of [Induction Hypothesis 2.D.1](#). Thus we have

$$\begin{aligned} |\langle w_i^{(T'_{i,1})}, \mathbf{M}_j \rangle|^2 &\geq \|w_i^{(T'_{i,1})}\|_2^2 - \sum_{j \in [d], j \notin \mathcal{N}_i} \langle w_i^{(T'_{i,1})}, \mathbf{M}_j \rangle^2 - \sum_{j \in [d_1] \setminus [d]} \langle w_i^{(T'_{i,1})}, \mathbf{M}_j^\perp \rangle^2 \\ &\geq 2\|w_i^{(T_1)}\|_2^2 - (1 + o(1))\|w_i^{(T_1)}\|_2^2 \geq (1 - o(1))\|w_i^{(T_1)}\|_2^2 \end{aligned}$$

which proves the claim.

- **When some $\|w_i^{(t)}\|_2 \geq 2\|w_i^{(T_1)}\|_2$:** At this substage, we have

- The bias is large consistently, i.e.,

$$b_i^{(t+1)} \geq b_i^{(t)} \cdot \frac{\|w_i^{(t+1)}\|_2}{\|w_i^{(t)}\|_2} \geq \frac{\text{polylog}(d)}{\sqrt{d}} \|w_i^{(t+1)}\|_2 \geq \frac{1}{4} \|w_i^{(T'_{i,1})}\|_2$$

- If $i \in \mathcal{M}_j^\star$, then from similar calculations as above, we can prove by induction that starting from $t = T'_{i,1}$, it holds:

$$\begin{aligned} |\langle w_i^{(t+1)}, \mathbf{M}_j \rangle| &\geq |\langle w_i^{(t)}, \mathbf{M}_j \rangle| \left(1 + \Omega\left(\frac{\eta \log \log d}{d}\right)\right) \geq \|w_i^{(t)}\|_2 \left(1 + \Omega\left(\frac{\eta \log \log d}{d}\right)\right) \\ \sum_{j' \in [d], j' \neq j} \langle w_i^{(t+1)}, \mathbf{M}_{j'} \rangle^2 &\leq \sum_{j' \in [d], j' \neq j} \langle w_i^{(t)}, \mathbf{M}_{j'} \rangle^2 \left(1 + \frac{O(\eta)}{d \text{polylog}(d)}\right)^2 \\ \sum_{j \in [d_1] \setminus [d]} \langle w_i^{(t+1)}, \mathbf{M}_j^\perp \rangle^2 &\leq \sum_{j \in [d_1] \setminus [d]} \langle w_i^{(t)}, \mathbf{M}_j^\perp \rangle^2 \left(1 + \frac{O(\eta)}{d \text{polylog}(d)}\right)^2 \end{aligned}$$

which implies

$$|\langle w_i^{(t+1)}, \mathbf{M}_j \rangle| \geq |\langle w_i^{(t)}, \mathbf{M}_j \rangle| \frac{\|w_i^{(t+1)}\|_2}{\|w_i^{(t)}\|_2} \geq (1 - o(1)) \|w_i^{(t+1)}\|_2$$

Now we only need to prove (c). Assuming $\langle w_i^{(t)}, \mathbf{M}_j \rangle > 0$ (the opposite case is similar), from $t = T_1$, for $i \in \mathcal{M}_j^\star$, we have

$$\begin{aligned} \langle w_i^{(t+1)}, \mathbf{M}_j \rangle &= (\langle w_i^{(t)}, \mathbf{M}_j \rangle - b_i^{(t)}) \left(1 + \frac{\eta C_z \log \log d}{d}\right) \pm O\left(\frac{\eta |\langle w_i^{(t)}, \mathbf{M}_j \rangle|}{d \text{polylog}(d)}\right) \\ &\geq \Omega(1) \langle w_i^{(t)}, \mathbf{M}_j \rangle \left(1 + \frac{\eta C_z \log \log d}{d} \left(1 - \frac{1}{\text{polylog}(d)}\right)\right) \\ &\geq \Omega(1) \langle w_i^{(T_1)}, \mathbf{M}_j \rangle \left(1 + \frac{\eta C_z \log \log d}{d} \left(1 - \frac{1}{\text{polylog}(d)}\right)\right)^{t-T_1} \end{aligned}$$

which implies that after certain iteration $t = T_1 + T'$, where $T' = \Theta(\frac{d}{\eta})$, we shall have

$$|\langle w_i^{(T_1+T')}, \mathbf{M}_j \rangle| \geq |\langle w_i^{(T_1)}, \mathbf{M}_j \rangle| \geq \text{polylog}(d) |\langle w_i^{(T_1)}, \mathbf{M}_j \rangle| \geq b_i^{(T_1)} \text{polylog}(d)$$

However, at iteration $t = T_1 + \Theta(\frac{d}{\eta})$, we can see from previous analysis that $\|w_i^{(t)}\|_2 \leq (1 + o(1)) \|w_i^{(T_1)}\|_2$, so the bias growth can be bounded as

$$b_i^{(t)} \leq b_i^{(T_1)} \left(1 + \frac{\eta}{d}\right)^{\Theta(\frac{d}{\eta})} \cdot \max \left\{ \frac{\|w_i^{(t)}\|_2}{\|w_i^{(T_1)}\|_2}, 1 \right\} \leq O(b_i^{(T_1)})$$

Now from our initialization properties in [Lemma 2.B.1](#), we have that $\langle w_{i'}^{(0)}, \mathbf{M}_j \rangle^2 \leq O(\sigma_0^2 \log d)$ for all $i \in [m]$. Thus via similar arguments, we also have

$$|\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle| \leq \langle w_{i'}^{(0)}, \mathbf{M}_j \rangle \left(1 + \frac{\eta C_z \log \log d}{d} \left(1 \pm \frac{1}{\text{polylog}(d)} \right) \right)^t$$

holds for all $i' \in [m]$. Now it is easy to see that for $t \leq T_2 = \Theta(\frac{d \log d}{\eta \log \log d})$, we have

$$\frac{|\langle w_i^{(t)}, \mathbf{M}_j \rangle|}{|\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle|} \geq \Omega(1) \frac{|\langle w_i^{(0)}, \mathbf{M}_j \rangle| \cdot \left(1 + \frac{\eta C_z \log \log d}{d} \left(1 - \frac{\eta}{\text{polylog}(d)} \right) \right)^t}{|\langle w_{i'}^{(0)}, \mathbf{M}_j \rangle| \left(1 + \frac{\eta C_z \log \log d}{d} + \frac{\eta}{\text{polylog}(d)} \right)^t} \geq \left(1 - O\left(\frac{\eta \log \log d}{d \text{polylog}(d)}\right) \right)^t \geq \Omega(1)$$

Thus the last claim is proved. $\square$

2.E Stage III: Convergence to Sparse Features

At the final stage, we are going to prove that as long as the neurons are sparsely activated, they will indeed converge to sparse solutions, which ensures sparse representations. We present the statement of our convergence theorem below.

Theorem 2.E.1 (Convergence). *At iteration for $t \in [\Omega(\frac{d^{1.01}}{\eta}), O(\frac{d^{1.99}}{\eta})]$, we have the following results:*

- (a) *If $i \in \mathcal{M}_j^*$, then $|\langle w_i^{(t)}, \mathbf{M}_j \rangle| \in [\frac{\tau}{\Xi_2}, O(1)]$;*
- (b) *If $i \notin \mathcal{M}_j$, then $|\langle w_i^{(t)}, \mathbf{M}_j \rangle| \leq O\left(\frac{1}{d^{2\lambda}}\right)$;*
- (c) *For all dense feature $\mathbf{M}_j^\perp$, $j \in [d_1] \setminus [d]$, we have $|\langle w_i^{(t)}, \mathbf{M}_j^\perp \rangle| \leq O(\frac{1}{\sqrt{d_1} d^{1.5\lambda}})$.*
- (d) *We have the loss convergence guarantees: let $T_3 = \Omega(\frac{d^{1.01}}{\eta})$, for any $T \leq O(\frac{d^{1.99}}{\eta})$, we have*

$$\frac{1}{T} \sum_{t=T_3}^{T_3+T-1} \mathbb{E}[\mathcal{L}(f_t, x_p^+, x_p^{++}, \mathfrak{R})] \leq O\left(\frac{1}{\log d}\right)$$

To prove this theorem, we need the following induction hypothesis, which we shall show to hold throughout the final stage.

Induction Hypothesis 2.E.1 (Induction hypothesis at final stage). *For all $t \geq T_2$:*

- 1. *If $i \in \mathcal{M}_j^*$, then $|\langle w_i^{(t)}, \mathbf{M}_j \rangle| \geq \Omega(1) \|w_i^{(t)}\|_2$;*
- 2. *For $i \in [m]$, we have $\|w_i^{(t)}\|_2 \leq O(1)$*
- 3. *For each $j \in [d]$, $\mathfrak{F}_j^{(t)} := \sum_{i \in \mathcal{M}_j} \langle w_i^{(t)}, \mathbf{M}_j \rangle^2 \leq O(\tau \log^3 d)$;*
- 4. *Let $j \in [d]$ and $i \in \mathcal{M}_j^*$, there exist $C = \Theta(1)$ such that $|\langle w_i^{(t)}, \mathbf{M}_j \rangle| \geq C \max_{i' \in \mathcal{M}_j} |\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle|$;*

5. For $i \notin \mathcal{M}_j$, it holds $|\langle w_i^{(t)}, \mathbf{M}_j \rangle| \leq O(\frac{1}{\sqrt{d}\Xi_2^5}) \|w_i^{(t)}\|_2$;
6. For any $i \in [m]$ and any $j \in [d_1] \setminus [d]$, it holds $|\langle w_i^{(t)}, \mathbf{M}_j^+ \rangle| \leq O(\frac{1}{\sqrt{d_1}\Xi_2^5}) \|w_i^{(t)}\|_2$;
7. The bias $b_i^{(t)} \geq \frac{\text{polylog}(d)}{\sqrt{d}} \|w_i^{(t)}\|_2$.

When all the conditions in [Induction Hypothesis 2.E.1](#) hold for some iteration $t \geq T_2$, we have the following fact, which is a simple corollary of [Lemma 2.B.5](#).

Fact 2.E.1. For any $i \in [m]$, we denote $\mathcal{N}_i = \{j \in [d] : i \in \mathcal{M}_j\}$. Suppose [Induction Hypothesis 2.E.1](#) hold at iteration $t \geq T_2$, then with high probability over $x \in \mathcal{D}_x$ and $\mathbf{D} \in \mathcal{D}_{\mathbf{D}}$:

$$\max_{x \in \{x_p, x_p^+, x_p^{++}\}} \mathbb{1}_{h_{i,t}(x) \neq 0} \leq \sum_{j \in \mathcal{N}_i} \mathbb{1}_{z_{p,j} \neq 0}$$

which implies that $\max_{x \in \{x_p, x_p^+, x_p^{++}\} \cup \mathcal{N}} \Pr(h_{i,t}(x) \neq 0) \leq O(\frac{\log \log d}{d})$.

Now for the simplicity of calculations, we define the following notations which are used throughout this section:

Definition 2.E.1 (expansion of gradient). For each $i \in [m]$, $j \in [d]$, we expand $\langle \nabla_{w_i} L(f_t), \mathbf{M}_j \rangle$ as:

$$\begin{aligned} \langle \nabla_{w_i} L(f_t), \mathbf{M}_j \rangle &= \mathbb{E} \left[\left((1 - \ell'_{p,t}) h_{i,t}(x_p^{++}) + \sum_{x_{n,s} \in \mathcal{N}} \ell'_{s,t} h_{i,t}(x_{n,s}) \right) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)}} \langle x_p^+, \mathbf{M}_j \rangle \right] \\ &= \Psi_{i,j}^{(t)} + \Phi_{i,j}^{(t)} + \mathcal{E}_{1,i,j}^{(t)} + \mathcal{E}_{2,i,j}^{(t)} \end{aligned}$$

where the $\Psi^{(t)}, \Phi^{(t)}, \mathcal{E}_1^{(t)}, \mathcal{E}_2^{(t)}$ are defined as follows: for each $x = \sum_j \mathbf{M}_j z_j + \xi \sim \mathcal{D}_x$ (or augmented instance x^+, x^{++}), we write

$$\psi_{i,j}^{(t)}(x) = \left(\langle w_i^{(t)}, \mathbf{M}_j \rangle z_j - b_i^{(t)} \right) \mathbb{1}_{\langle w_i^{(t)}, x \rangle > b_i^{(t)}} - \left(\langle w_i^{(t)}, \mathbf{M}_j \rangle z_j + b_i^{(t)} \right) \mathbb{1}_{\langle w_i^{(t)}, x \rangle < -b_i^{(t)}}$$

and

$$\phi_{i,j}^{(t)}(x) = \langle w_i^{(t)}, x^{\setminus j} \rangle \mathbb{1}_{\langle w_i^{(t)}, x \rangle > b_i^{(t)}} - \langle w_i^{(t)}, x^{\setminus j} \rangle \mathbb{1}_{\langle w_i^{(t)}, x \rangle < -b_i^{(t)}}$$

Now we define

$$\begin{aligned}
\Psi_{i,j}^{(t)} &:= \mathbb{E} \left[\left((1 - \ell'_{p,t}) \cdot \psi_{i,j}^{(t)}(x_p^{++}) + \sum_{x_{n,s} \in \mathfrak{N}} \ell'_{s,t} \cdot \psi_{i,j}^{(t)}(x_{n,s}) \right) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)} Z_{p,j}} \right] \\
\Phi_{i,j}^{(t)} &:= \mathbb{E} \left[\left((1 - \ell'_{p,t}) \cdot \phi_{i,j}^{(t)}(x_p^{++}) + \sum_{x_{n,s} \in \mathfrak{N}} \ell'_{s,t} \cdot \phi_{i,j}^{(t)}(x_{n,s}) \right) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)} Z_{p,j}} \right] \\
\mathcal{E}_{1,i,j}^{(t)} &:= \mathbb{E} \left[\left((1 - \ell'_{p,t}) \cdot h_{i,t}(x_p^{++}) + \sum_{x_{n,s} \in \mathfrak{N}} \ell'_{s,t} \cdot h_{i,t}(x_{n,s}) \right) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)}} \langle \mathbf{M}_j, (2\mathbf{D} - \mathbf{I})\mathbf{M}z_p \rangle \right] \\
\mathcal{E}_{2,i,j}^{(t)} &:= \mathbb{E} \left[\left((1 - \ell'_{p,t}) \cdot h_{i,t}(x_p^{++}) + \sum_{x_{n,s} \in \mathfrak{N}} \ell'_{s,t} \cdot h_{i,t}(x_{n,s}) \right) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)}} \langle \mathbf{M}_j, 2\mathbf{D}\xi_p \rangle \right]
\end{aligned} \tag{2.E.1}$$

Moreover, for $j \in [d_1] \setminus [d]$, we can similarly define the following notations:

$$\begin{aligned}
\Psi_{i,j}^{(t)}, \Phi_{i,j}^{(t)} &\equiv 0, \\
\mathcal{E}_{1,i,j}^{(t)} &:= \mathbb{E} \left[\left((1 - \ell'_{p,t}) \cdot h_{i,t}(x_p^{++}) + \sum_{x_{n,s} \in \mathfrak{N}} \ell'_{s,t} \cdot h_{i,t}(x_{n,s}) \right) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)}} \langle \mathbf{M}_j, (2\mathbf{D} - \mathbf{I})\mathbf{M}z_p \rangle \right] \\
\mathcal{E}_{2,i,j}^{(t)} &:= \mathbb{E} \left[\left((1 - \ell'_{p,t}) \cdot h_{i,t}(x_p^{++}) + \sum_{x_{n,s} \in \mathfrak{N}} \ell'_{s,t} \cdot h_{i,t}(x_{n,s}) \right) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)}} \langle \mathbf{M}_j^\perp, 2\mathbf{D}\xi_p \rangle \right]
\end{aligned} \tag{2.E.2}$$

Equipped with the above definition, we are ready to characterize the training process at the final stage.

2.E.1 Gradient Upper and Lower Bounds for $\Psi^{(t)}$

Lemma 2.E.1 (lower bound for $\Psi_1^{(t)}$). *Suppose [Induction Hypothesis 2.E.1](#) holds at iteration t . For $j \in [d]$ and $i \in \mathcal{M}_j^*$, there exist $G_1 = \Theta(1)$ such that if $\mathfrak{F}_j^{(t)} := \sum_{i' \in \mathcal{M}_j} \langle w_{i'}^{(t)}, \mathbf{M}_j \rangle^2 \leq G_1 \tau \log d$, then we have*

$$\Psi_{i,j}^{(t)} \cdot \text{sign}(\langle w_i^{(t)}, \mathbf{M}_j \rangle) \geq \frac{\mathbb{E}[|z_j|]}{\text{polylog}(d)} \left(1 - O\left(\frac{1}{\Xi_2^3}\right) \right) |\langle w_i^{(t)}, \mathbf{M}_j \rangle|$$

Proof. We begin with the proof of (a). We first decompose $\Psi_{i,j}^{(t)} = \Psi_{i,j,1}^{(t)} + \Psi_{i,j,2}^{(t)}$, where

$$\begin{aligned}\Psi_{i,j,1}^{(t)} &= \mathbb{E} \left[(1 - \ell'_{p,t}) \cdot \psi_{i,j}^{(t)}(x_p^{++}) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)} Z_{p,j}} \right] \\ \Psi_{i,j,2}^{(t)} &= \sum_{x_{n,s} \in \mathfrak{N}} \mathbb{E} \left[\ell'_{s,t} \cdot \psi_{i,j}^{(t)}(x_{n,s}) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)} Z_{p,j}} \right]\end{aligned}$$

We first deal with $\Psi_{i,j,2}^{(t)}$. Using the notation $x_p^{\setminus j,+} := 2\mathbf{D}(\sum_{j' \neq j} \mathbf{M}_{j' Z_{p,j'}} + \xi_p)$, we can rewrite as

$$\begin{aligned}\Psi_{i,j,2}^{(t)} &= \sum_{x_{n,s} \in \mathfrak{N}} \mathbb{E} \left[\ell'_{s,t}(x_p^+, \mathfrak{B}) \cdot \psi_{i,j}^{(t)}(x_{n,s}) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)} Z_{p,j}} \right] \\ &= \sum_{x_{n,s} \in \mathfrak{N}} \mathbb{E} \left[\left(\ell'_{s,t}(x_p^+, \mathfrak{B}) - \ell'_{s,t}(x_p^{\setminus j,+}, \mathfrak{B}) \right) \cdot \psi_{i,j}^{(t)}(x_{n,s}) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)} Z_{p,j}} \right] \\ &\quad + \sum_{x_{n,s} \in \mathfrak{N}} \mathbb{E} \left[\ell'_{s,t}(x_p^{\setminus j,+}, \mathfrak{B}) \cdot \psi_{i,j}^{(t)}(x_{n,s}) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)} Z_{p,j}} \right] \\ &= R_1 + R_2\end{aligned}\tag{2.E.3}$$

We now deal with the term R_1 . Denoting $\widehat{x}_p^+(v) = \widehat{x}_p^+ + 2v\mathbf{D}^{(s)}\mathbf{M}_{j Z_{p,j}}$ for $v \in [0, 1]$, by Newton-Leibniz formula and the basic fact that $\frac{d}{dr} \frac{e^r}{e^r + \sum_{s \neq r} e^s} = \frac{e^r}{e^r + \sum_{s \neq r} e^s} (1 - \frac{e^r}{e^r + \sum_{s \neq r} e^s})$, we can rewrite $\ell'_{s,t}(\widehat{x}_p^+(v), \mathfrak{B})$ and $\ell'_{p,t}(\widehat{x}_p^+(v), \mathfrak{B})$ as

$$\begin{aligned}\widehat{\ell'_{s,t}}(v) &:= \frac{e^{\langle f_t(x_p^{\setminus j,+}) + v(f_t(x_p^+) - f_t(x_p^{\setminus j,+})), f_t(x_{n,s}) \rangle / \tau}}{\sum_{x \in \mathfrak{B}} e^{\langle f_t(x_p^{\setminus j,+}) + v(f_t(x_p^+) - f_t(x_p^{\setminus j,+})), f_t(x) \rangle / \tau}} \equiv \ell'_{s,t}(\widehat{x}_p^+(v), \mathfrak{B}) \\ \widehat{\ell'_{p,t}}(v) &:= \frac{e^{\langle f_t(x_p^{\setminus j,+}) + v(f_t(x_p^+) - f_t(x_p^{\setminus j,+})), f_t(x_p^+) \rangle / \tau}}{\sum_{x \in \mathfrak{B}} e^{\langle f_t(x_p^{\setminus j,+}) + v(f_t(x_p^+) - f_t(x_p^{\setminus j,+})), f_t(x) \rangle / \tau}} \equiv \ell'_{p,t}(\widehat{x}_p^+(v), \mathfrak{B})\end{aligned}$$

and we can then proceed to calculate as follows:

$$\begin{aligned}R_1 &= \sum_{x_{n,s} \in \mathfrak{N}} \mathbb{E} \left[\left(\ell'_{s,t}(x_p^+, \mathfrak{B}) - \ell'_{s,t}(x_p^{\setminus j,+}, \mathfrak{B}) \right) \cdot \psi_{i,j}^{(t)}(x_{n,s}) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)} Z_{p,j}} \right] \\ &= \sum_{x_{n,s} \in \mathfrak{N}} \mathbb{E} \left[\frac{1}{\tau} \left(\int_0^1 \widehat{\ell'_{s,t}}(v) (1 - \widehat{\ell'_{s,t}}(v)) \langle f_t(x_p^+) - f_t(x_p^{\setminus j,+}), f_t(x_{n,s}) \rangle dv \right. \right. \\ &\quad - \sum_{x \in \mathfrak{N} \setminus \{x_{n,s}\}} \int_0^1 \widehat{\ell'_{s,t}}(v) \widehat{\ell'_{u,t}}(v) \langle f_t(x_p^+) - f_t(x_p^{\setminus j,+}), f_t(x_{n,u}) \rangle dv \\ &\quad \left. \left. - \int_0^1 \widehat{\ell'_{s,t}}(v) \widehat{\ell'_{p,t}}(v) \langle f_t(x_p^+) - f_t(x_p^{\setminus j,+}), f_t(x_p^+) \rangle dv \right) \psi_{i,j}^{(t)}(x_{n,s}) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)} Z_{p,j}} \right] \\ &\stackrel{(1)}{\leq} \mathbb{E} \left[\frac{|\mathfrak{N}|}{\tau} \left(\int_0^1 \widehat{\ell'_{s,t}}(v) (1 - \widehat{\ell'_{s,t}}(v)) dv + \sum_{x_{n,u} \in \mathfrak{N} \setminus \{x_{n,s}\}} \int_0^1 \widehat{\ell'_{s,t}}(v) \widehat{\ell'_{u,t}}(v) dv \right) \times \right.\end{aligned}$$

$$\begin{aligned}
& \times \max_{x_{n,u} \in \mathfrak{N} \setminus \{x_{n,s}\}} |\langle f_t(x_p^+) - f_t(x_p^{\setminus j,+}), f_t(x_{n,u}) \rangle| |\psi_{i,j}^{(t)}(x_{n,s})| \mathbb{1}_{|\langle w_{i'}^{(t)}, x_p^+ \rangle| \geq b_i^{(t)} |z_{p,j}|} \Big] \\
& + \mathbb{E} \left[\frac{|\mathfrak{N}|}{\tau} \int_0^1 \widehat{\ell'_{s,t}}(v) \widehat{\ell'_{p,t}}(v) dv |\langle f_t(x_p^+) - f_t(x_p^{\setminus j,+}), f_t(x_p^{++}) \rangle| |\psi_{i,j}^{(t)}(x_{n,s})| \mathbb{1}_{|\langle w_{i'}^{(t)}, x_p^+ \rangle| \geq b_i^{(t)} |z_{p,j}|} \right] \\
& \stackrel{(2)}{\leq} \frac{|\mathfrak{N}|}{\tau} \mathbb{E} \left[\int_0^1 \widehat{\ell'_{s,t}}(v) dv \max_{x \in \mathfrak{B}} \left(\sum_{i \in \mathcal{M}_j} \langle w_{i'}^{(t)}, \mathbf{M}_j \rangle |h_{i,t}(x)| \right) |\psi_{i,j}^{(t)}(x_{n,s})| z_{p,j}^2 \right] \\
& + \widetilde{O}(\Xi_2) \max_{i' \notin \mathcal{M}_j} |\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle| \mathbb{E} \left[\sum_{x_{n,s} \in \mathfrak{N}} \frac{1}{\tau} \int_0^1 \widehat{\ell'_{s,t}}(v) dv |\psi_{i,j}^{(t)}(x_{n,s})| z_{p,j}^2 \right] \\
& + \widetilde{O} \left(\frac{\Xi_2}{\sqrt{d_1} \tau} \right) \mathbb{E} \left[\sum_{x_{n,s} \in \mathfrak{N}} \frac{1}{\tau} \int_0^1 \widehat{\ell'_{s,t}}(v) dv |\psi_{i,j}^{(t)}(x_{n,s})| z_{p,j}^2 \right] + \frac{1}{d^{\Omega(\log d)}} \\
& = R_{1,1} + R_{1,2} + R_{1,3} + \frac{1}{d^{\Omega(\log d)}} \tag{2.E.4}
\end{aligned}$$

where for (1) and (2), we argue as follows:

- for (1), we used the fact that the expectations over $s \in \mathfrak{N}$ in the summation can be view as independently and uniformly selecting from $s \in \mathfrak{N}$. which allow us to equate $\sum_{x_{n,s} \in \mathfrak{N}} = |\mathfrak{N}|$.
- for (2), we use [Fact 2.E.1](#) to ensure that $\sum_{i \in [m]} \mathbb{1}_{h_{i,t}(x_p^+) \neq 0} \leq \widetilde{O}(\Xi_2)$ with high prob. Further noticing that $\max_i \|w_{i'}^{(t)}\|_2 \leq O(1)$ and $|\langle w_{i'}^{(t)}, (\mathbf{I} - 2\mathbf{D})\mathbf{M}_j \rangle| \leq \widetilde{O}(\frac{1}{\sqrt{d_1}} \|w_{i'}^{(t)}\|_2)$ w.h.p, we have for any $x \in \mathfrak{B}$:

$$|\langle f_t(x_p^+) - f_t(x_p^{\setminus j,+}), f_t(x) \rangle| \leq \sum_{i' \in \mathcal{M}_j} |\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle| \cdot |h_{i',t}(x)| + \widetilde{O}(\Xi_2) \max_{i' \in \mathcal{M}_j} |\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle| + \widetilde{O} \left(\frac{1}{\sqrt{d_1}} \right)$$

which gives the desired inequality.

Now we proceed to deal with $R_{1,1}$, since $i \in \mathcal{M}_j^*$, we have automatically $\mathbb{1}_{h_{i,t}(x_{n,s}) \neq 0} = \mathbb{1}_{z_{n,s,j} \neq 0}$ w.h.p., so we can transform $R_{1,1}$ as

$$\begin{aligned}
R_{1,1} &= \mathbb{E} \left[\frac{|\mathfrak{N}|}{\tau} \int_0^1 \widehat{\ell'_{s,t}}(v) dv \max_{x \in \mathfrak{B}} \left(\sum_{i' \in \mathcal{M}_j} \langle w_{i'}^{(t)}, \mathbf{M}_j \rangle |h_{i',t}(x)| \right) |\psi_{i,j}^{(t)}(x_{n,s})| z_{p,j}^2 \right] + \frac{1}{d^{\Omega(\log d)}} \\
&= \mathbb{E} \left[\frac{|\mathfrak{N}|}{\tau} \int_0^1 \widehat{\ell'_{s,t}}(v) dv \left(\sum_{i' \in \mathcal{M}_j} \langle w_{i'}^{(t)}, \mathbf{M}_j \rangle^2 + \Upsilon_j^{(t)} \right) |\psi_{i,j}^{(t)}(x_{n,s})| z_{p,j}^2 \right] + \frac{\Xi_2}{d^{\Omega(\log d)}}
\end{aligned}$$

where $\Upsilon_j^{(t)}$ is defined as the

$$\Upsilon_j^{(t)} = \max_{x \in \mathfrak{B}} \sum_{i' \in \mathcal{M}_j} |h_{i',t}(x)| |\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle| - \sum_{i' \in \mathcal{M}_j} |\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle|^2$$

We proceed to give a high probability bound for $\sum_{i \in \mathcal{M}_j} |h_{i,t}(x_{n,s})| |\langle w_i^{(t)}, \mathbf{M}_j \rangle|$, which lies in the core of our proof. In order to apply [Lemma 2.B.3](#) to the pre-activation in $h_{i,t}(x_{n,s})$, one can first expand as

$$\begin{aligned}
& \sum_{i \in [m]} |h_{i,t}(x_{n,s})| |\langle w_i^{(t)}, \mathbf{M}_j \rangle| \\
& \lesssim \sum_{j' \in [d]} \sum_{i \in \mathcal{M}_{j'}} |\langle w_i^{(t)}, \mathbf{M}_{j'} \rangle| |z_{n,s,j'}| \cdot |\langle w_i^{(t)}, \mathbf{M}_j \rangle| + \sum_{i \in [m]} \tilde{O}\left(\frac{\|w_i^{(t)}\|_2}{\sqrt{d}}\right) |\langle w_i^{(t)}, \mathbf{M}_j \rangle| \\
& = \sum_{j' \in [d]} \sum_{i \in \mathcal{M}_j \cap \mathcal{M}_{j'}} \langle w_i^{(t)}, \mathbf{M}_{j'} \rangle |z_{n,s,j'}| \cdot |\langle w_i^{(t)}, \mathbf{M}_j \rangle| + \sum_{j' \in [d]} \sum_{i \in \mathcal{M}_{j'} \setminus \mathcal{M}_j} |\langle w_i^{(t)}, \mathbf{M}_{j'} \rangle| |z_{n,s,j'}| \cdot |\langle w_i^{(t)}, \mathbf{M}_j \rangle| \\
& \quad + \sum_{i \in [m]} \tilde{O}\left(\frac{\|w_i^{(t)}\|_2}{\sqrt{d}}\right) |\langle w_i^{(t)}, \mathbf{M}_j \rangle| \\
& \stackrel{(1)}{\leq} \sum_{i \in \mathcal{M}_j} \langle w_i^{(t)}, \mathbf{M}_j \rangle^2 |z_{n,s,j}| + \sum_{j' \neq j} \sum_{i \in \mathcal{M}_j \cap \mathcal{M}_{j'}} |\langle w_i^{(t)}, \mathbf{M}_{j'} \rangle| |z_{n,s,j'}| \cdot |\langle w_i^{(t)}, \mathbf{M}_j \rangle| \\
& \quad + \sum_{j' \in [d]} \sum_{i \in \mathcal{M}_{j'} \setminus \mathcal{M}_j} |\langle w_i^{(t)}, \mathbf{M}_{j'} \rangle| \cdot |z_{n,s,j'}| \cdot |\langle w_i^{(t)}, \mathbf{M}_j \rangle| + \sum_{i \in [m]} \tilde{O}\left(\frac{\|w_i^{(t)}\|_2}{\sqrt{d}}\right) |\langle w_i^{(t)}, \mathbf{M}_j \rangle|
\end{aligned}$$

And we proceed to calculate the last two terms on the RHS as follows: firstly, from [Lemma 2.B.1](#) we know for the set of neurons $\Gamma_j := \{j' \neq j, j' \in [d] : \mathcal{M}_j \cap \mathcal{M}_{j'} \neq \emptyset\}$, we have $|\Gamma_j| \leq O(\log d)$, and

$$\begin{aligned}
\left| \sum_{j' \neq j} \sum_{i' \in \mathcal{M}_j \cap \mathcal{M}_{j'}} |\langle w_{i'}^{(t)}, \mathbf{M}_{j'} \rangle| |z_{n,s,j'}| \cdot |\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle| \right| & \leq O\left(\frac{\log d}{\sqrt{\Xi_2}}\right) \cdot \sum_{j' \in \Gamma_j} O(\tau \log^2 d) |z_{n,s,j'}| \\
& \leq O\left(\frac{\tau}{\log d}\right) \quad \text{w.h.p.}
\end{aligned}$$

where in the last inequality we have taken into account the fact that $\mathbb{E}|z_j| = \tilde{O}(\frac{1}{d})$ and have used [Lemma 2.B.4](#). The same techniques also provide the following bound:

$$\left| \sum_{j' \in [d]} \sum_{i' \in \mathcal{M}_{j'} \setminus \mathcal{M}_j} |\langle w_{i'}^{(t)}, \mathbf{M}_{j'} \rangle| |z_{n,s,j'}| \cdot |\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle| \right| \leq O\left(\frac{1}{\sqrt{d}}\right) \sum_{j' \neq j} O(\Xi_2) |z_{n,s,j'}| \leq O\left(\frac{\Xi_2^2}{\sqrt{d}}\right) \quad \text{w.h.p.}$$

Therefore via a union bound, we have

$$\sum_{i' \in [m]} |h_{i',t}(x_{n,s})| |\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle| \leq \sum_{i' \in \mathcal{M}_j} \langle w_{i'}^{(t)}, \mathbf{M}_j \rangle^2 |z_{n,s,j}| + O\left(\frac{\tau}{\log d}\right) \quad \text{w.h.p.}$$

The same arguments also gives (+ further applying [Lemma 2.B.3](#))

$$\max_{x \in \{x_p^+, x_p^{++}\}} \left\{ \sum_{i' \in [m]} |h_{i',t}(x)| |\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle| \right\} \leq \sum_{i' \in \mathcal{M}_j} \langle w_{i'}^{(t)}, \mathbf{M}_j \rangle^2 |z_{p,j}| + O\left(\frac{\tau}{\log d}\right) \quad \text{w.h.p.}$$

which also implies that

$$\Upsilon_j^{(t)} \leq \max_{x \in \mathfrak{B}} \sum_{i' \in [m]} |h_{i',t}(x)| |\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle| - \sum_{i' \in \mathcal{M}_j} |\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle|^2 \leq O\left(\frac{\tau}{\log d}\right)$$

Now we are ready to control the quantity $R_{1,1}$. the idea here is to “decorrelate” the factor $\widehat{\ell'_{s,t}}(v)$ from the others. Definining $\mathfrak{B}^{\setminus s} := \{x_p^{++}\} \cup \mathfrak{N} \setminus \{x_{n,s}\}$ and $\mathfrak{B}'_s := \mathfrak{B}^{\setminus s} \cup \{x_{n,s}^{\setminus j}\}$, there exist a constant $G'_1 > 0$ such that, if $\sum_{i \in \mathcal{M}_j} \langle w_i^{(t)}, \mathbf{M}_j \rangle^2 \leq \tau G'_1 \log d$, we have w.h.p.

$$\begin{aligned} \frac{\ell'_{s,t}(x_p^+, \mathfrak{B})}{\ell'_{s,t}(x_p^+, \mathfrak{B}'_s)} &= \frac{e^{\langle f_t(x_p^+), f_t(x_{n,s}) \rangle / \tau}}{e^{\langle f_t(x_p^+), f_t(x_{n,s}^{\setminus j}) \rangle / \tau}} \cdot \frac{\sum_{x \in \mathfrak{B}'_s} e^{\langle f_t(x_p^+), f_t(x) \rangle / \tau}}{\sum_{x \in \mathfrak{B}} e^{\langle f_t(x_p^+), f_t(x) \rangle / \tau}} \leq \frac{e^{2\langle f_t(x_p^+), f_t(x_{n,s}) \rangle / \tau}}{e^{2\langle f_t(x_p^+), f_t(x_{n,s}^{\setminus j}) \rangle / \tau}} \\ &\leq e^{2G'_1 \log d + O(1/\log d)} \leq o(d/\Xi_2^5) \end{aligned} \quad (2.E.5)$$

Now we define $\mathfrak{N}'_s = \{x_{n,s'} \in \mathfrak{N} \setminus \{x_{n,s}\} : z_{n,s',j} = 0\} \cup \{x_{n,s}^{\setminus j}\}$. Note that from concentration inequality of Bernoulli variables we know $|\mathfrak{N}'_s| = \Omega(|\mathfrak{N}|)$ w.h.p. Thus we have (notice that the outer factor $|\mathfrak{N}|$ can be insert into the expectation by sacrificing some constant factors):

$$\begin{aligned} R_{1,1} &\leq O\left(\frac{|\mathfrak{N}|}{\tau}\right) \cdot \mathbb{E} \left[\widehat{\ell'_{s,t}}(1) \left(\sum_{i \in \mathcal{M}_j} \langle w_i^{(t)}, \mathbf{M}_j \rangle^2 + \Upsilon_j^{(t)} \right) |\psi_{i,j}^{(t)}(x_{n,s})| \cdot z_{p,j}^2 \right] \\ &= O\left(\frac{|\mathfrak{N}| \log \log d}{d\tau}\right) \mathbb{E} \left[\ell'_{s,t}(x_p^+, \mathfrak{B}) \left(\sum_{i \in \mathcal{M}_j} \langle w_i^{(t)}, \mathbf{M}_j \rangle^2 + \Upsilon_j^{(t)} \right) \cdot |\psi_{i,j}^{(t)}(x_{n,s})| \mid |z_{p,j}| \neq 0 \right] \\ &= \widetilde{O}\left(\frac{|\mathfrak{N}|}{d\tau}\right) \mathbb{E} \left[\frac{\ell'_{s,t}(x_p^+, \mathfrak{B})}{\ell'_{s,t}(x_p^+, \mathfrak{B}'_s)} \times \ell'_{s,t}(x_p^+, \mathfrak{B}'_s) \left(\sum_{i \in \mathcal{M}_j} \langle w_i^{(t)}, \mathbf{M}_j \rangle^2 + \Upsilon_j^{(t)} \right) |\psi_{i,j}^{(t)}(x_{n,s})| \mid |z_{p,j}| \neq 0 \right] \\ &\leq O\left(\frac{1}{\tau \Xi_2^4}\right) \mathbb{E} \left[\sum_{x_{n,s} \in \mathfrak{N}'} \ell'_{s,t}(x_p^+, \mathfrak{B}'_s) \left(\sum_{i \in \mathcal{M}_j} \langle w_i^{(t)}, \mathbf{M}_j \rangle^2 + \Upsilon_j^{(t)} \right) |\psi_{i,j}^{(t)}(x_{n,s})| \mid |z_{p,j}| \neq 0 \right] \\ &\stackrel{(1)}{\leq} |\langle w_i^{(t)}, \mathbf{M}_j \rangle - b_i^{(t)}| \cdot O\left(\frac{1}{\Xi_2^3}\right) \cdot \Pr(z_{n,s,j} \neq 0) \\ &\leq |\langle w_i^{(t)}, \mathbf{M}_j \rangle| \cdot O\left(\frac{1}{\Xi_2^3}\right) \Pr(z_j \neq 0) \end{aligned}$$

where in inequality (1) we have used the independence of $z_{n,s,j}$ with respect to $\ell'_{s,t}(x_p^+, \mathfrak{B}'_s)$, and the fact that $\sum_{x_{n,s} \in \mathfrak{N}'} \ell'_{s,t}(x_p^+, \mathfrak{B}'_s) \leq 1$. Now turn back to deal with $R_{1,2}$ and $R_{1,3}$ in ([By Newton-Lebniz](#)).

Indeed, noticing that $\max_{i' \notin \mathcal{M}_j} |\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle| \leq O(\frac{1}{\sqrt{d}})$ from [Induction Hypothesis 2.E.1](#), and that

$$\sum_{x \in \mathfrak{X}} \mathbb{E} \left[\frac{1}{\tau} \int_{0,1} \widehat{\ell'_{s,t}}(v) dv |\psi_{i,j}^{(t)}(x_{n,s}) z_{p,j}^2| \right] \leq \tilde{O}(\frac{1}{d}) |\langle w_i^{(t)}, \mathbf{M}_j \rangle - b_i^{(t)}|$$

we have $R_{1,2}, R_{1,3} \leq o(R_{1,1})$. For R_2 in [\(By Newton-Lebniz\)](#), we can see from the definition of $\ell'_{s,t}(x_p^{j,+}, \mathfrak{B})$ that it is independent to $z_{p,j}$. Notice further that $\mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)}} = \mathbb{1}_{z_{p,j} \neq 0}$ with high probability due to our assumption, and also the fact that $\mathbb{1}_{z_{p,j} \neq 0} z_{p,j}$ has mean zero and is independent to $\ell'_{s,t}(x_p^{j,+}, x_{n,s})$ we have

$$R_2 \leq \text{poly}(d) \cdot e^{-\Omega(\log^2 d)} \lesssim \frac{1}{\text{poly}(d)^{\Omega(\log d)}}$$

Combining the pieces above together, we can have

$$\Psi_{i,j,2}^{(t)} \leq O(\frac{1}{\Xi_2^3} \mathbb{E}[z_j^2]) |\langle w_i^{(t)}, \mathbf{M}_j \rangle - b_i^{(t)}|$$

Now we turn to $\Phi_{1,1}^{(t)}(j)$, whose calculation is similar. Defining $\mathcal{B}_j := \{z_{p,j'} = 0, \forall j' \neq j\}$, we separately discuss the cases when events $\mathcal{B}_j$ or $\mathcal{B}_j^c$ holds:

- When $\mathcal{B}_j^c$ happens, $\text{sign}(\psi_{i,j}^{(t)}(x_p^{++})) = \text{sign}(\langle w_i^{(t)}, \mathbf{M}_j \rangle z_j)$ with high prob by [Fact 2.E.1](#) since we assumed $i \in \mathcal{M}_j^*$. Thus, if $\langle w_i^{(t)}, \mathbf{M}_j \rangle > 0$, we have

$$\mathbb{E} \left[(1 - \ell'_{p,t}) \psi_{i,j}^{(t)}(x_p^{++}) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)}} z_{p,j} \middle| \mathcal{B}_j^c \right] \geq 0$$

if $\langle w_i^{(t)}, \mathbf{M}_j \rangle < 0$, the opposite inequality holds as well.

- When $\mathcal{B}_j$ happens, it is easy to derive that

$$|\langle f_t(x_p^+), f_t(x_p^{++}) \rangle| \leq \sum_{i' \in \mathcal{M}_j} \langle w_{i'}^{(t)}, \mathbf{M}_j \rangle^2 z_{p,j}^2 + O(\Xi_2) \cdot O(\frac{\|w_{i'}^{(t)}\|_2}{\sqrt{d}}) \leq \langle w_i^{(t)}, \mathbf{M}_j \rangle^2 + O(\frac{\Xi_2}{\sqrt{d}})$$

and therefore

$$|\langle f_t(x_p^+), f_t(x_p^{++}) \rangle - \langle f_t(x_p^{j,+}), f_t(x_p^{j,++}) \rangle| \leq \sum_{i' \in \mathcal{M}_j} \langle w_{i'}^{(t)}, \mathbf{M}_j \rangle^2 z_{p,j}^2 + O(\frac{\Xi_2}{\sqrt{d}})$$

from previous analysis, we also have

$$|\langle f_t(x_p^+), f_t(x_{n,s}) \rangle - \langle f_t(x_p^{j,+}), f_t(x_{n,s}) \rangle| \leq \sum_{i' \in \mathcal{M}_j} \langle w_{i'}^{(t)}, \mathbf{M}_j \rangle^2 z_{p,j}^2 + O(\frac{\tau}{\log d}) + O(\frac{\Xi_2}{\sqrt{d}})$$

These inequalities allow us to apply the same techniques in bounding $\Phi_{1,2}^{(t)}$ as follows. We define $\mathfrak{B}'_p := \mathfrak{N} \cup \{x_p^{\setminus j,++}\}$. Then, similar to (2.E.5), for some $G'_2 = \Theta(1)$, we can have

$$\frac{\ell'_{p,t}(x_p^{\setminus j,+}, \mathfrak{B})}{\ell'_{p,t}(x_p^{\setminus j,+}, \mathfrak{B}'_p)} = \frac{e^{\langle f_i(x_p^+), f_i(x_p^{++}) \rangle / \tau}}{e^{\langle f_i(x_p^{\setminus j,+}), f_i(x_p^{\setminus j,++}) \rangle / \tau}} \cdot \frac{\sum_{x \in \mathfrak{B}'_p} e^{\langle f_i(x_p^{\setminus j,+}), f_i(x) \rangle / \tau}}{\sum_{x \in \mathfrak{B}} e^{\langle f_i(x_p^+), f_i(x) \rangle / \tau}} \leq e^{2G'_2 \log d + O(\frac{1}{\log d})} \leq O\left(\frac{d}{\Xi_2^5}\right)$$

Now we can proceed to compute as follows:

$$\begin{aligned} & \mathbb{E} \left[\ell'_{p,t}(x_p^+, \mathfrak{B}) \cdot \psi_{i,j}^{(t)}(x_p^{++}) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)} z_{p,j}} \middle| \mathcal{B}_j \right] \\ &= \mathbb{E} \left[\frac{\ell'_{p,t}(x_p^{\setminus j,+}, \mathfrak{B})}{\ell'_{p,t}(x_p^{\setminus j,+}, \mathfrak{B}'_p)} \times \ell'_{p,t}(x_p^{\setminus j,+}, \mathfrak{B}'_p) \cdot \psi_{i,j}^{(t)}(x_p^{++}) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)} z_{p,j}} \middle| \mathcal{B}_j \right] \\ &\leq |\langle w_i^{(t)}, \mathbf{M}_j \rangle - b_i^{(t)}| O\left(\frac{1}{\Xi_2^5}\right) \Pr(z_j \neq 0) \end{aligned}$$

But from Lemma 2.D.1, and that Induction Hypothesis 2.D.1 still holds for Stage III, we have

$$\mathbb{E} \left[\psi_{i,j}^{(t)}(x_p^{++}) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)} z_{p,j}} \mathbb{1}_{\mathcal{B}_j} \right] = \text{sign}(\langle w_i^{(t)}, \mathbf{M}_j \rangle) (|\langle w_i^{(t)}, \mathbf{M}_j \rangle| - b_i^{(t)}) \frac{1}{\text{polylog}(d)} \mathbb{E}[|z_j|]$$

Combining both cases above gives the bound of $\Psi_{i,j,2}^{(t)}$. Combining results for $\Psi_{i,j,1}^{(t)}$ and $\Psi_{i,j,2}^{(t)}$ concludes the proof. The constant G_1 in the statement can be defined as $G_1 := \min\{G'_1, G'_2\}$. $\square$

Lemma 2.E.2 (upper bound of $\Psi_{i,j}^{(t)}$). *Let $j \in [d]$ and $i \in \mathcal{M}_j^*$. Suppose Induction Hypothesis 2.E.1 hold at iteration t , then there exist a constant $G_2 = \Theta(1)$, if $\mathfrak{F}_j^{(t)} = \sum_{i \in \mathcal{M}_j} \langle w_i^{(t)}, \mathbf{M}_j \rangle^2 \geq G_2 \tau \log d$, we have*

$$\Psi_{i,j}^{(t)} \leq \frac{1}{\text{poly}(d)} |\langle w_i^{(t)}, \mathbf{M}_j \rangle|$$

Similarly, for $i \in \mathcal{M}_j$, we have

$$\Psi_{i,j}^{(t)} \leq \frac{1}{\text{poly}(d)} |\langle w_i^{(t)}, \mathbf{M}_j \rangle| + O\left(\frac{1}{d^2}\right) (b_i^{(t)})$$

Proof. First we deal with the case of $i \in \mathcal{M}_j^*$, we have $\mathbb{1}_{|\langle w_i^{(t)}, x_p \rangle| \geq b_i^{(t)} z_{p,j}} = \mathbb{1}_{z_{p,j} \neq 0} z_{p,j}$ w.h.p. when conditions in Induction Hypothesis 2.E.1 hold. Now by denoting

$$\tilde{\psi}_{i,j}(z_j) := (\langle w_i^{(t)}, \mathbf{M}_j \rangle z_j - b_i^{(t)}) \mathbb{1}_{\langle w_i^{(t)}, \mathbf{M}_j \rangle z_j > 0} - (\langle w_i^{(t)}, \mathbf{M}_j \rangle z_j + b_i^{(t)}) \mathbb{1}_{\langle w_i^{(t)}, \mathbf{M}_j \rangle z_j < 0}$$

we can then easily rewrite $\Psi_{i,j}^{(t)}$ as (by using [Fact 2.E.1](#))

$$\begin{aligned}\Psi_{i,j}^{(t)} &= \mathbb{E} \left[\left((1 - \ell'_{p,t}) \tilde{\psi}_{i,j}^{(t)}(z_{p,j}) - \sum_{x_{n,s} \in \mathfrak{N}} \ell'_{s,t} \cdot \tilde{\psi}_{i,j}^{(t)}(z_{n,s,j}) \right) \mathbb{1}_{z_{p,j} \neq 0} z_{p,j} \right] + \frac{1}{\text{poly}(d)^{\Omega(\log d)}} \\ &= \mathbb{E}_{x_p^+, x_p^{++}} [(\tilde{\psi}_{i,j}^{(t)}(z_{p,j}) - I_1 - I_2) \mathbb{1}_{z_{p,j} \neq 0} z_{p,j}] + \frac{1}{\text{poly}(d)^{\Omega(\log d)}}\end{aligned}$$

where I_1 and I_2 are defined as follows:

$$\begin{aligned}I_1 &:= \mathbb{E}_{\mathfrak{N}} \left[\left((1 - \ell'_{p,t}) \tilde{\psi}_{i,j}^{(t)}(z_{p,j}) - \sum_{x_{n,s} \in \mathfrak{N}} \ell'_{s,t} \cdot \tilde{\psi}_{i,j}^{(t)}(z_{n,s,j}) \right) \mathbb{1}_{z_{n,s,j} = z_{p,j}} \right] \\ &\stackrel{(1)}{=} \tilde{\psi}_{i,j}^{(t)}(z_{p,j}) \mathbb{E}_{\mathfrak{N}} \left[\frac{|\mathfrak{N}| e^{\langle f_i(x_p^+), f_i(x_{n,s}) \rangle / \tau} \mathbb{1}_{z_{n,s,j} = z_{p,j}} + e^{\langle f_i(x_p^+), f_i(x_p^{++}) \rangle / \tau}}{e^{\langle f_i(x_p^+), f_i(x_{n,s}) \rangle / \tau} + \sum_{x \in \mathfrak{N} \setminus \{x_{n,s}\}} e^{\langle f_i(x_p^+), f_i(x) \rangle / \tau}} \right] \\ I_2 &:= \mathbb{E}_{\mathfrak{N}} \left[\sum_{x_{n,s} \in \mathfrak{N}} \ell'_{s,t} \cdot \tilde{\psi}_{i,j}^{(t)}(z_{n,s,j}) \mathbb{1}_{z_{n,s,j} \neq z_{p,j}} \right]\end{aligned}$$

where in (1) we used the identification $z_{p,j} = z_{n,s,j}$. The tricky part here is since all the variables inside the expectation is non-negative we can use Jensen's inequality to move the expectation of $e^{\langle f_i(x_p^+), f_i(x_{n,s}) \rangle / \tau}$ to the denominator. We let $V := e^{\langle f_i(x_p^+), f_i(x_p^{++}) \rangle / \tau}$ and consider it fix when computing I_1 as follows: conditioned on $z_{p,j} \neq 0$, we have

$$\begin{aligned}\frac{I_1}{\tilde{\psi}_{i,j}^{(t)}(z_{p,j})} &= \mathbb{E}_{\mathfrak{N}} \left[\frac{|\mathfrak{N}| e^{\langle f_i(x_p^+), f_i(x_{n,s}) \rangle / \tau} \mathbb{1}_{z_{n,s,j} = z_{p,j}} + V}{e^{\langle f_i(x_p^+), f_i(x_{n,s}) \rangle / \tau} + V + \sum_{x \in \mathfrak{N} \setminus \{x_{n,s}\}} e^{\langle f_i(x_p^+), f_i(x_{n,u}) \rangle / \tau}} \right] \\ &\geq \mathbb{E}_{x_{n,s}} \left[\frac{e^{\langle f_i(x_p^+), f_i(x_{n,s}) \rangle / \tau} \mathbb{1}_{z_{n,s,j} = 1} + \frac{1}{|\mathfrak{N}|} V}{\frac{1}{|\mathfrak{N}|} e^{\langle f_i(x_p^+), f_i(x_{n,s}) \rangle / \tau} + \frac{1}{|\mathfrak{N}|} V + \frac{|\mathfrak{N}|-1}{|\mathfrak{N}|} \mathbb{E}_{x_n} [e^{\langle f_i(x_p^+), f_i(x_n) \rangle / \tau}]} \right] \quad (\text{by Jensen inequality}) \\ &= \mathbb{E}_{x_{n,s}} \left[\frac{e^{\langle f_i(x_p^+), f_i(x_{n,s}) \rangle / \tau} \mathbb{1}_{z_{n,s,j} = z_{p,j}} + \frac{1}{|\mathfrak{N}|} V}{\frac{1}{|\mathfrak{N}|} (e^{\langle f_i(x_p^+), f_i(x_{n,s}) \rangle / \tau} + V) + \frac{|\mathfrak{N}|-1}{|\mathfrak{N}|} \mathbb{E}_{x_n} [e^{\langle f_i(x_p^+), f_i(x_n) \rangle / \tau} (\mathbb{1}_{z_{n,j} = z_{p,j}} + \mathbb{1}_{z_{n,j} \neq z_{p,j}})]} \right] \\ &\stackrel{(1)}{\geq} \mathbb{E}_{x_{n,s}} \left[\frac{X + \frac{1}{|\mathfrak{N}|} V}{\frac{1}{|\mathfrak{N}|} (X + V) + \frac{|\mathfrak{N}|-1}{|\mathfrak{N}|} (1 + \frac{1}{\text{poly}(d)}) \mathbb{E}_{x_n} [X]} \right] \\ &\quad \quad \quad (\text{where } X := e^{\langle f_i(x_p^+), f_i(x_n) \rangle / \tau} \mathbb{1}_{z_{n,s,j} = z_{p,j}} \geq 0) \\ &\stackrel{(2)}{\geq} 1 - O\left(\frac{1}{\text{poly}(d)}\right)\end{aligned}$$

where for the above inequalities, we argue:

- in (1), we need to go through similar analysis as in the proof of [Lemma 2.E.1](#) to obtain that, with high probability over x_p^+ and $x_n^{\setminus j}$:

$$\begin{aligned} \langle f_t(x_p^+), f_t(x_n) \rangle - \langle f_t(x_p^+), f_t(x_n^{\setminus j}) \rangle / \tau &\geq \frac{1}{\tau} \sum_{i \in \mathcal{M}_j} \langle w_i^{(t)}, \mathbf{M}_j \rangle^2 - O\left(\frac{1}{\log d}\right) \\ &\geq G_2 \log d - O\left(\frac{1}{\log d}\right) \end{aligned}$$

for some very large constant $G_2 = \Theta(1)$, which gives (the $\frac{1}{\text{poly}(d)}$ here depends on how large G_2 is)

$$\mathbb{E}_{x_n} [e^{\langle f_t(x_p^+), f_t(x_n) \rangle / \tau} \mathbb{1}_{z_{n,s,j} \neq z_{p,j}}] \leq \frac{1}{\text{poly}(d)} \mathbb{E}_{x_n} [e^{\langle f_t(x_p^+), f_t(x_n) \rangle / \tau} \mathbb{1}_{z_{n,s,j} = z_{p,j}}].$$

- in inequality (2), we need to argue as follows, where $\widetilde{\mathbb{E}}[X] \stackrel{\text{abbr.}}{=} \mathbb{E}_{x_n}[X]$ is only integrated over the randomness of x_n :

$$\begin{aligned} &\mathbb{E}_{x_{n,s}} \left[\frac{(X + \frac{1}{|\mathfrak{N}|} V)}{\frac{1}{|\mathfrak{N}|} (X + V) + \frac{|\mathfrak{N}|-1}{|\mathfrak{N}|} (1 + \frac{1}{\text{poly}(d)}) \widetilde{\mathbb{E}}[X]} \right] \\ &= 1 - \frac{1}{|\mathfrak{N}|} \mathbb{E}_{x_{n,s}} \left[\frac{\widetilde{\mathbb{E}}[X]}{\frac{1}{|\mathfrak{N}|} (X + V) + \frac{|\mathfrak{N}|-1}{|\mathfrak{N}|} (1 + \frac{1}{\text{poly}(d)}) \widetilde{\mathbb{E}}[X]} \right] \\ &\geq 1 - \frac{1}{|\mathfrak{N}|} \cdot \frac{\widetilde{\mathbb{E}}[X]}{\frac{|\mathfrak{N}|-1}{|\mathfrak{N}|} (1 + \frac{1}{\text{poly}(d)}) \widetilde{\mathbb{E}}[X]} \quad (\text{since } X + V \geq 0) \\ &\geq 1 - \frac{1}{\text{poly}(d)} \end{aligned}$$

The same analysis applies to I_2 , which we can bound as

$$\left| \frac{I_1}{\widetilde{\psi}_{i,j}^{(t)}(-z_{p,j})} \right| \leq \frac{1}{\text{poly}(d)}$$

Combining both I_1 and I_2 , we have

$$\Psi_{i,j}^{(t)} \leq \frac{1}{\text{poly}(d)} |\langle w_i^{(t)}, \mathbf{M}_j \rangle|$$

In the case of $i \in \mathcal{M}_j$, we have with prob $\leq \widetilde{O}(\frac{1}{d})$ that $\mathbb{1}_{\langle w_i^{(t)}, x_p^+ \rangle \geq b_i^{(t)}} \neq \mathbb{1}_{\langle w_i^{(t)}, \mathbf{M}_j \rangle_{z_{p,j}} > 0}$ or $\mathbb{1}_{\langle w_i^{(t)}, x_p^+ \rangle \geq b_i^{(t)}} \neq \mathbb{1}_{\langle w_i^{(t)}, \mathbf{M}_j \rangle_{z_{p,j}} > 0}$. When such events happen, we can obtain a bound of $O(\frac{1}{d}) b_i^{(t)}$ over $\Psi_{i,j}^{(t)}$, which times the prob $\widetilde{O}(\frac{1}{d})$ leads to our bound. Combining the above observations and the analyses, we can complete the proof. $\square$

2.E.2 Gradient Computations II

In this section, we give finer characterization of $\Psi_3^{(t)}$ and $\Psi_4^{(t)}$, which is the contributions of the dense features/noisy correlations to the gradient.

Lemma 2.E.3 (bounds for $\mathcal{E}_1^{(t)}$). *At iteration $t \geq T_2$, let $j \in [d]$ and $i \in [m]$, [Induction Hypothesis 2.E.1](#) holds at t , for each $j \in [d_1]$, we have*

$$|\mathcal{E}_{1,i,j}^{(t)}| \leq O\left(\frac{\Xi_2^2 \|w_i^{(t)}\|_2}{d^{3/2} \sqrt{d_1}}\right)$$

Proof. Let $j \in [d]$, since the case of $j \in [d_1] \setminus [d]$ can be similarly dealt with. We first look at the following $\mathcal{E}_{1,1,i,j}^{(t)}$ term in $\mathcal{E}_{1,i,j}^{(t)}$:

$$\mathcal{E}_{1,1,i,j}^{(t)} = \mathbb{E} \left[h_{i,t}(x_p^{++}) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)}} \langle \mathbf{M}_j, (2\mathbf{D} - \mathbf{I}) \mathbf{M}_{z_p} \rangle \right]$$

It is easy to observe that using the randomness and symmetry of $2\mathbf{D} - \mathbf{I}$ w.r.t. zero, we have

$$|\mathcal{E}_{1,1,i,j}^{(t)}| \leq \mathbb{E} \left[|\langle w_i^{(t)}, (2\mathbf{D} - \mathbf{I}) x_p \rangle| \mathbb{1}_{|\langle w_i^{(t)}, x_p \rangle| \geq b_i^{(t)} + |\langle w_i^{(t)}, x_p - x_p^+ \rangle|} |\langle \mathbf{M}_j, (2\mathbf{D} - \mathbf{I}) \mathbf{M}_{z_p} \rangle| \right]$$

When $\{|\langle w_i^{(t)}, x_p \rangle| \geq b_i^{(t)} + |\langle w_i^{(t)}, x_p - x_p^+ \rangle|\}$ happens (which we know from [Fact 2.E.1](#) has prob $\leq \tilde{O}(\frac{1}{d})$), using [Lemma 2.B.3](#), we have

$$|\mathcal{E}_{1,1,i,j}^{(t)}| \leq \tilde{O}\left(\frac{1}{\sqrt{d_1} d^{1.5}}\right) \|w_i^{(t)}\|_2$$

Now we similarly decompose the sum of expectations as follows: let $\mathcal{N}_i := \{j \in [d] : i \in \mathcal{M}_j\}$, which from [Lemma 2.B.1](#) we know are of cardinality at most $O(1)$, then

$$\begin{aligned} \mathcal{E}_{1,2,i,j}^{(t)} &= \mathbb{E} \left[\sum_{x_{n,s} \in \mathfrak{N}} \ell'_{s,t}(x_p^+, \mathfrak{B}) \cdot h_{i,t}(x_{n,s}) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)}} \sum_{j' \in [d]} \langle \mathbf{M}_j, (2\mathbf{D} - \mathbf{I}) \mathbf{M}_{j'} \rangle z_{p,j'} \right] \\ &= \sum_{j' \in \mathcal{N}_i} \mathbb{E} \left[\sum_{x_{n,s} \in \mathfrak{N}} \ell'_{s,t}(x_p^+, \mathfrak{B}) \cdot h_{i,t}(x_{n,s}) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)}} \langle \mathbf{M}_j, (2\mathbf{D} - \mathbf{I}) \mathbf{M}_{j'} \rangle z_{p,j'} \right] \\ &\quad + \sum_{j' \notin \mathcal{N}_i} \mathbb{E} \left[\sum_{x_{n,s} \in \mathfrak{N}} \ell'_{s,t}(x_p^+, \mathfrak{B}) \cdot h_{i,t}(x_{n,s}) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)}} \langle \mathbf{M}_j, (2\mathbf{D} - \mathbf{I}) \mathbf{M}_{j'} \rangle z_{p,j'} \right] \end{aligned}$$

Notice that the major difference between the first and second terms are that the occurrence of features $j' \in \mathcal{N}_i$ has nontrivial probability $\geq \tilde{\Omega}(\frac{1}{d})$ to affect the indicator $\mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)}}$. However, since $|\langle \mathbf{M}_j, (2\mathbf{D} - \mathbf{I}) \mathbf{M}_{j'} \rangle|$ is w.h.p., small due to [Lemma 2.B.2](#), so for the first term, we can use the

symmetry of $2\mathbf{D} - \mathbf{I}$ and $\mathbf{I} - 2\mathbf{D}$ to compute as follows: denote $\mathfrak{B}' = \{x_p^{++}\} \cup \{x_{n,s}\}_{\mathfrak{N}}$, we have

$$\begin{aligned}
& \sum_{j' \in \mathcal{N}_i} \mathbb{E} \left[\sum_{x_{n,s} \in \mathfrak{N}} \ell'_{s,t}(x_p^+, \mathfrak{B}) \cdot h_{i,t}(x_{n,s}) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)}} \langle \mathbf{M}_j, (2\mathbf{D} - \mathbf{I})\mathbf{M}_{j'} \rangle z_{p,j'} \right] \\
&= \sum_{j' \in \mathcal{N}_i} \mathbb{E} \left[\sum_{x_{n,s} \in \mathfrak{N}} \left(\ell'_{s,t}(x_p^+, \mathfrak{B}) - \ell'_{s,t}(x_p^{++}, \mathfrak{B}') \right) \cdot h_{i,t}(x_{n,s}) |\langle \mathbf{M}_j, (2\mathbf{D} - \mathbf{I})\mathbf{M}_{j'} \rangle| z_{p,j'} \right] \\
&\stackrel{(1)}{=} \sum_{j' \in \mathcal{N}_i} \mathbb{E} \left[\sum_{x_{n,s} \in \mathfrak{N}} \frac{1}{\tau} \left(\int_0^1 \widehat{\ell'_{s,t}}(v) (1 - \widehat{\ell'_{s,t}}(v)) \langle f_t(x_p^+) - f_t(x_p^{++}), f_t(x_{n,s}) \rangle dv \right. \right. \\
&\quad \left. \left. - \sum_{u \neq s, x_{n,u} \in \mathfrak{N}} \int_0^1 \widehat{\ell'_{s,t}}(v) \widehat{\ell'_{u,t}}(v) \langle f_t(x_p^+) - f_t(x_p^{++}), f_t(x_{n,u}^+) \rangle dv \right) \times \right. \\
&\quad \left. \times h_{i,t}(x_{n,s}) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)}} |\langle \mathbf{M}_j, (2\mathbf{D} - \mathbf{I})\mathbf{M}_{j'} \rangle| z_{p,j'} \right] \\
&\leq \sum_{j' \in \mathcal{N}_i} \sum_{x_{n,s} \in \mathfrak{N}} \mathbb{E} \left[\left(\int_0^1 \widehat{\ell'_{s,t}}(v) (1 - \widehat{\ell'_{s,t}}(v)) dv + \sum_{u \neq s, x_{n,u} \in \mathfrak{N}} \int_0^1 \widehat{\ell'_{s,t}}(v) \widehat{\ell'_{u,t}}(v) dv \right) \times \right. \\
&\quad \left. \times \max_{x_{n,u} \in \mathfrak{N}} |\langle f_t(x_p^+) - f_t(x_p^{++}), f_t(x_{n,u}^+) \rangle| |h_{i,t}(x_{n,s})| \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)}} z_{p,j'} \right] \\
&\stackrel{(2)}{\leq} \widetilde{O}(\Xi_2) \sum_{j' \in \mathcal{N}_i} \mathbb{E} \left[\sum_{x_{n,s} \in \mathfrak{N}} \frac{1}{\tau} \int_0^1 \widehat{\ell'_{s,t}}(v) dv \cdot \max_{i' \in [m]} |\langle w_{i'}^{(t)}, (\mathbf{I} - 2\mathbf{D})x_p \rangle| |\langle \mathbf{M}_j, (2\mathbf{D} - \mathbf{I})\mathbf{M}_{j'} \rangle| |h_{i,t}(x_{n,s})| |z_{p,j'}| \right] \\
&\stackrel{(3)}{\leq} \widetilde{O} \left(\frac{\Xi_2 \|w_i^{(t)}\|_2 \max_{i' \in [m]} \|w_{i'}^{(t)}\|_2}{d^{3/2} \sqrt{d_1} \tau} \right) + \text{poly}(d) e^{-\Omega(\log^2 d)} \\
&\leq O \left(\frac{\Xi_2 \|w_i^{(t)}\|_2}{d^{1.5} \sqrt{d_1} \tau} \right)
\end{aligned}$$

where in the above calculations:

- In (1) we have defined $\widehat{\ell'_{s,t}}(v)$ as (where $x_p^{\setminus j,+} := 2\mathbf{D}(\sum_{j' \neq j} \mathbf{M}_{j'} z_{p,j'} + \xi_p)$):

$$\widehat{\ell'_{s,t}}(v) := \frac{e^{\langle f_t(x_p^{\setminus j,+}) + v(f_t(x_p^+) - f_t(x_p^{\setminus j,+})), f_t(x_{n,s}) \rangle}}{\sum_{x_{n,u} \in \mathfrak{N}} e^{\langle f_t(x_p^{\setminus j,+}) + v(f_t(x_p^+) - f_t(x_p^{\setminus j,+})), f_t(x_{n,u}) \rangle}};$$

- In (2) we have used the fact that at $t \geq T_2$, it holds $\sum_{i \in [m]} \mathbb{1}_{h_{i,t}(x_{n,s}) \neq 0} \leq \widetilde{O}(\Xi)$ with high probability over all negative samples $\{x_{n,s}\}_{\mathfrak{B}}$, which is from applying [Lemma 2.B.5](#) using the conditions as we assumed in [Lemma 2.B.1](#);

- In (3) we have used mainly [Lemma 2.B.4](#) to obtain that $|\langle w_{i'}^{(t)}, (\mathbf{I} - 2\mathbf{D})x_p \rangle| \leq \tilde{O}(\|w_{i'}^{(t)}\|_2/\sqrt{d})$ holds with high probability, combining with the fact that $\Pr(z_{p,i'} \neq 0) = \tilde{O}(1/d)$, and $\sum_{x_{n,s} \in \mathcal{R}} \widehat{\ell'_{s,t}}(v) \leq 1$ for all $v \in [0, 1]$.

Combining the results of $\mathcal{E}_{1,1,i,j}^{(t)}$ and $\mathcal{E}_{1,2,i,j'}^{(t)}$ we can conclude the proof. $\square$

We can also obtain the following lemmas bounding the gradient contributed by the spurious noise via the same approach as in the proof of [Lemma 2.E.3](#) below. We sketch the proof below.

Lemma 2.E.4 (bounds for $\mathcal{E}_2^{(t)}$). *Let $j \in [d]$ and $i \in [m]$, suppose [Induction Hypothesis 2.E.1](#) holds at t , for all $j \in [d]$, we have*

$$\mathcal{E}_{2,i,j}^{(t)} \leq O\left(\frac{\|w_i^{(t)}\|_2 \Xi_2^2}{d^2 \tau}\right) \cdot \max_{i' \in [m]} \left(|\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle| + \frac{\|w_{i'}^{(t)}\|_2}{\sqrt{d_1}} \right)$$

The same bound holds for $j \in [d_1] \setminus [d]$, with $|\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle|$ changing to $|\langle w_{i'}^{(t)}, \mathbf{M}_j^+ \rangle|$.

Proof. The proof is extremely similar to those in [Lemma 2.E.3](#), which we will omit here, the only differences are: (1) For the first quantity, we do not have a mask applied to $\langle v, \xi_p \rangle$; (2) the variable $\langle \mathbf{M}_j, \xi_p \rangle$ cannot affect the firing probability (prob of being nonzero) of $\mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)}}$ w.h.p due to [Induction Hypothesis 2.E.1](#); (3) one could use a different basis as in the proof of [Lemma 2.D.2](#) to obtain the desired $\tilde{O}(1/\sqrt{d_1})$ factor in the second bound. $\square$

2.E.3 Learning Process at the Final Stage

Before proving [Theorem 2.E.1](#), we need prove [Induction Hypothesis 2.E.1](#), which characterized the trajectory of gradients at iterations $t \geq T_2$. We first prove a lemma, which allow us to obtain the full characterization of $\Phi^{(t)}$ term (defined in [Definition 2.E.1](#)) in gradient calculations.

Lemma 2.E.5 (reduction of $\Phi^{(t)}$ to the bounds of $\Psi^{(t)}$). *Let $j \in [d]$ and $i \in \mathcal{M}_j$. Suppose [Induction Hypothesis 2.E.1](#) hold for all iteration before $t \in [\frac{d^{1.01}}{\eta}, \frac{d^{1.99}}{\eta}]$ and after T_2 , and also we suppose for all $l \in [d]$, $\mathfrak{F}_l^{(t')} = \Omega(\tau \log d)$ at some $t' = \Theta(T_2)$, then*

- for iteration $t \in [\frac{d^{1.01}}{\eta}, \frac{d^{1.495}}{\eta}]$:

$$\Phi_{i,j}^{(t)} \leq \tilde{O}\left(\frac{\Xi_2^2}{d^{3/2}}\right) \|w_i^{(t)}\|_2$$

- for iteration $t \in [\frac{d^{1.495}}{\eta}, \frac{d^{1.99}}{\eta}]$:

$$\Phi_{i,j}^{(t)} \leq \tilde{O}\left(\frac{1}{d^{1.98}}\right) \|w_i^{(t)}\|_2$$

Proof. The proof essentially relies on the condition that [Induction Hypothesis 2.E.1](#) holds for all $t' \in [T_3, t]$. We first consider the case where $i \in \mathcal{M}_j^*$. Similar to how $\psi_{i,j}^{(t)}(x)$ are defined for each x

in [Definition 2.E.1](#), for each $j' \neq j$, we let

$$\rho_{i,j'}^{(t)}(x) := (\langle w_i^{(t)}, x \rangle - \langle w_i^{(t)}, \mathbf{M}z \rangle) \mathbb{1}_{z_{j'} \neq 0}$$

Now it is straightforward to decompose $\Phi_{i,j}^{(t)}$ as follows:

$$\begin{aligned} \Phi_{i,j}^{(t)} &= \mathbb{E} \left[\left((1 - \ell'_{p,t}) \cdot \phi_{i,j}^{(t)}(x_p^{++}) + \sum_{x_{n,s} \in \mathfrak{R}} \ell'_{s,t} \cdot \phi_{i,j}^{(t)}(x_{n,s}) \right) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)} z_{p,j}} \right] \\ &= \sum_{j' \in [d], j' \neq j} \langle w_i^{(t)}, \mathbf{M}_{j'} \rangle \mathbb{E} \left[\left((1 - \ell'_{p,t}) \cdot z_{p,j'} + \sum_{x_{n,s} \in \mathfrak{R}} \ell'_{s,t} \cdot z_{n,s,j'} \mathbb{1}_{z_{n,s,j'} \neq 0} \right) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)} z_{p,j}} \right] \\ &\quad \text{(By [Fact 2.E.1](#))} \\ &\quad + \mathbb{E} \left[\left((1 - \ell'_{p,t}) \cdot \rho_{i,j}^{(t)}(x_p^{++}) + \sum_{x_{n,s} \in \mathfrak{R}} \ell'_{s,t} \cdot \rho_{i,j}^{(t)}(x_{n,s}) \right) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)} z_{p,j}} \right] + \frac{1}{\text{poly}(d)^{\Omega(\log d)}} \\ &= H_1 + H_2 + \frac{1}{\text{poly}(d)^{\Omega(\log d)}} \end{aligned}$$

Indeed, from similar arguments as in the proof of [Lemma 2.E.3](#) and [Lemma 2.E.4](#), we can trivially obtain $|H_2| \leq O(\frac{\Xi_2^2}{d^2}) \|w_i^{(t)}\|_2$. Now we turn to H_1 . Since $\max_{j' \neq j} |\langle w_i^{(t)}, \mathbf{M}_{j'} \rangle| \leq O(\frac{\|w_i^{(t)}\|_2}{\sqrt{d} \Xi_2^5})$, we can simply get (Since w.h.p., $|\{j' \in [d] : z_{p,j'} \neq 0\}| = \tilde{O}(1)$, and if $z_{p,j'} = 0$, the negative terms are small from similar analysis in [Lemma 2.E.1](#))

$$\begin{aligned} |H_1| &\leq O\left(\frac{\|w_i^{(t)}\|_2}{\sqrt{d} \Xi_2^5}\right) \sum_{j' \neq j, j' \in [d]} \mathbb{E} \left[\left((1 - \ell'_{p,t}) \cdot z_{p,j'} + \sum_{x_{n,s} \in \mathfrak{R}} \ell'_{s,t} \cdot z_{n,s,j'} \mathbb{1}_{z_{n,s,j'} \neq 0} \right) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)} z_{p,j}} \right] \\ &\leq O\left(\frac{\|w_i^{(t)}\|_2}{d^{3/2}}\right) \end{aligned}$$

Then we can obtain a crude bound for all $t \in [\frac{d^{1.01}}{\eta}, \frac{d^{1.99}}{\eta}]$ by

$$\Phi_{i,j}^{(t)} \leq (H_1 + H_2) + \frac{1}{\text{poly}(d)^{\Omega(\log d)}} \leq \tilde{O}\left(\frac{\|w_i^{(t)}\|_2}{d^{3/2}}\right)$$

The harder part is to deal with iterations $t \in [\frac{d^{1.495}}{\eta}, \frac{d^{1.498}}{\eta}]$. We first establish a connection between $\Psi^{(t)}$ and $\Phi^{(t)}$. We first assume that for all $j' \neq j, j' \in [d]$, it holds that $|\Psi_{i',j'}^{(t_1)}| / |\langle w_{i'}^{(t_1)}, \mathbf{M}_{j'} \rangle| \leq \Omega(\frac{\Xi_2^2}{\sqrt{d} t \eta})$, which is true for all iteration $t \leq \frac{d \text{polylog}(d)}{\eta}$ from simple calculations. Now suppose at some

$t_1 \geq \frac{d \text{polylog}(d)}{\eta}$, there exist some $j' \neq j, j' \in [d]$ and $i' \in \mathcal{M}_j^\star$ such that

$$|\Psi_{i',j'}^{(t_1)}|/|\langle w_{i'}^{(t_1)}, \mathbf{M}_{j'} \rangle| \geq \Omega\left(\frac{\Xi_2}{\sqrt{dt}\eta}\right)$$

which means we have the followings:

$$\mathbb{E} \left[\left((1 - \ell'_{p,t_1}) z_{p,j'} + \sum_{x_{n,s} \in \mathfrak{R}} \ell'_{s,t_1} z_{n,s,j'} \right) z_{p,j'} \right] \geq \Omega\left(\frac{\Xi_2}{\sqrt{dt}\eta}\right)$$

Letting $\Delta > 0$ be defined as the number such that if $\mathfrak{F}_j^{(t)} = \Delta$, we can have $|\Psi_{i',j'}^{(t)}|/|\langle w_{i'}^{(t)}, \mathbf{M}_{j'} \rangle| \leq O(\frac{\sqrt{\Xi_2} \tau \log d}{\sqrt{dt}\eta})$. Then from the calculations in the proof of [Lemma 2.E.1](#), there must be a constant $\delta > \Omega(1)$ such that $\mathfrak{F}_j^{(t_1)} \geq \Delta - \delta \tau \log d$. However, such growth cannot continue since for some $t' = \Theta(t/\sqrt{\Xi_2})$, we have for each $i' \in \mathcal{M}_{j'}$:

$$\begin{aligned} |\langle w_{i'}^{(t_1+t')}, \mathbf{M}_{j'} \rangle| &\geq |\langle w_{i'}^{(t_1+t')}, \mathbf{M}_{j'} \rangle| (1 - \eta\lambda) + \Psi_{i',j'}^{(t_1+t'-1)} + \Phi_{i',j'}^{(t)} + O\left(\frac{\Xi_2^2}{d^2}\right) \\ &\geq |\langle w_{i'}^{(t)}, \mathbf{M}_{j'} \rangle| (1 - \eta\lambda)^{t'} + \sum_{s=t}^{t+t'-1} \Psi_{i',j'}^{(s)} - O\left(\frac{t' \Xi_2^2}{d^{3/2}}\right) \end{aligned}$$

where the bounds for $\Phi_{i',j'}^{(s)}$ for each $s \in [t_1, t_1 + t']$ are obtained from induction over iterations $s' \in [\frac{d^{1.01}}{\eta}, s]$. Therefore there must exist $t''' \in [t, t + t']$ such that $|\Psi_{i',j'}^{(t)}| \leq O(\frac{\sqrt{\Xi_2}}{\tau \sqrt{dt}\eta})$ or otherwise $\mathfrak{F}_j^{(t+t')} \geq \mathfrak{F}_j^{(t)} + t' \cdot O(\frac{\sqrt{\Xi_2} \tau \log d}{\sqrt{dt}\eta}) \geq \Delta + \delta \tau \log d$, which results in that $|\Psi_{i',j'}^{(t)}| \leq O(\frac{\tau \log d}{\sqrt{dt}\eta}) \|w_i^{(t)}\|_2$, following the same reasoning in [Lemma 2.E.2](#). Above arguments actually proved that $|\Psi_{i',j'}^{(t)}| \leq \Omega(\frac{\Xi_2}{\sqrt{dt}\eta}) \|w_i^{(t)}\|_2$ at all $t \in [\frac{d \text{polylog}(d)}{\eta}, \frac{d^{1.498}}{\eta}]$. Therefore we can use the results of all $\Psi_{i',j'}^{(t)}$, where $j' \neq j, j' \in [d]$ to get (combined with [Fact 2.E.1](#))

$$|H_1| \leq \tilde{O}\left(\max_{j' \neq j, j' \in [d]} \Psi_{i,j}^{(t)}\right) \leq \tilde{O}\left(\frac{\Xi_2^2}{d^2}\right) \|w_i^{(t)}\|_2$$

For iterations $t \geq \frac{d^{1.498}}{\eta}$, the proof is essentially the same: we only need to notice that the difference $\Psi_{i,j}^{(t)} - \lambda \langle w_i^{(t)}, \mathbf{M}_j \rangle$ here will bounce around zero, while the compensation terms in H_1 are bounded by $\tilde{O}(\frac{\|w_i^{(t)}\|_2}{d^{1.98}})$. These observations indeed prove the case $i \in \mathcal{M}_j^\star$. When $i \in \mathcal{M}_j \setminus \mathcal{M}_j^\star$, notice that with prob $\leq \tilde{O}(\frac{1}{d})$ it holds $\mathbb{1}_{|\langle w_i^{(t)}, x \rangle| \geq b_i^{(t)}} = \mathbb{1}_{z_j \neq 0}$ for any $x \in \mathfrak{B}$. Now we expand

$$H_1 = \sum_{j' \in \mathcal{N}_i, j' \neq j} \langle w_i^{(t)}, \mathbf{M}_{j'} \rangle \mathbb{E} \left[\left((1 - \ell'_{p,t}) \cdot z_{p,j'} + \sum_{x_{n,s} \in \mathfrak{R}} \ell'_{s,t} \cdot z_{n,s,j'} \mathbb{1}_{z_{n,s,j'} \neq 0} \right) \mathbb{1}_{|\langle w_i^{(t)}, x_p^* \rangle| \geq b_i^{(t)}} z_{p,j'} \right]$$

$$+ \sum_{j' \notin \mathcal{N}_i, j' \neq j} \langle w_i^{(t)}, \mathbf{M}_{j'} \rangle \mathbb{E} \left[\left((1 - \ell'_{p,t}) \cdot z_{p,j'} + \sum_{x_{n,s} \in \mathfrak{N}} \ell'_{s,t} \cdot z_{n,s,j'} \mathbb{1}_{z_{n,s,j} \neq 0} \right) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)} z_{p,j}} \right]$$

Indeed, the event that there are some $j' \in \mathcal{N}_i$ (which means $i \in \mathcal{M}_j$) such that $z_{p,j'} \neq 0$ has probability $\leq \tilde{O}(\frac{1}{d})$, Thus the first term on the RHS is trivially bounded by $\tilde{O}(\frac{1}{d^2}) \|w_i^{(t)}\|_2$. For the second term of H_1 , we can again go through similar procedure as above to obtain that

$$\mathbb{E} \left[\left((1 - \ell'_{p,t}) z_{p,j'} + \sum_{x_{n,s} \in \mathfrak{N}} \ell'_{s,t} z_{n,s,j'} \mathbb{1}_{z_{n,s,j} \neq 0} \right) \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)} z_{p,j}} \right] \leq \max \left\{ \tilde{O} \left(\frac{\sqrt{\Xi_2}}{\sqrt{d} t \eta} \right), \frac{1}{d^{1.99}} \right\} \|w_i^{(t)}\|_2.$$

Then again we have

$$|H_1| \leq \tilde{O} \left(\max_{j' \neq j, j' \in [d]} \Psi_{i,j}^{(t)} \right) \leq \tilde{O} \left(\max \left\{ \frac{\Xi_2^2}{d^2}, \frac{\Xi_2}{\sqrt{d} t \eta} \right\} \right) \|w_i^{(t)}\|_2,$$

which can be combine with the bound for H_2 to conclude the proof. $\square$

Proof of Induction Hypothesis 2.E.1. First we need to prove all the induction hypothesis hold for $t = T_2$. Indeed, (1), (4), (5), (6), (7) is valid at T_2 from Lemma 2.B.5 and Theorem 2.D.1; (2) and (3) holds at T_2 obviously. Now suppose it hold for some $t \geq T_2$, we will prove that it still hold for $t + 1$. We first deal with the case where $j \in [d]$ and $i \notin \mathcal{M}_j$, where it holds that

$$\begin{aligned} \langle w_i^{(t+1)}, \mathbf{M}_j \rangle &= \langle w_i^{(t)}, \mathbf{M}_j \rangle (1 - \eta \lambda) + \eta \mathbb{E} [h_{i,t}(x_p^{++}) \mathbb{1}_{\langle w_i^{(t)}, x_p^+ \rangle \geq b_i^{(t)}} \langle x_p^+, \mathbf{M}_j \rangle] \\ &\quad - \mathbb{E} \left[\sum_{x_{n,s} \in \mathfrak{N}} \ell'_{s,t} \cdot h_{i,t}(x_{n,s}) \mathbb{1}_{\langle w_i^{(t)}, x_p^+ \rangle \geq b_i^{(t)}} \langle x_p^+, \mathbf{M}_j \rangle \right] \pm \frac{\eta}{\text{poly}(d_1)} \end{aligned}$$

In this case, to calculate the expectation, we need to use Lemma 2.D.1, Lemma 2.E.3 and Lemma 2.E.4. First we compute the probability of events $A_1 - A_4$ by using Lemma 2.B.3, Lemma 2.B.4, Lemma 2.B.5 and our induction hypothesis to obtain

$$\Pr(A_1), \Pr(A_2) \leq \frac{1}{\text{poly}(d)^{\Omega(\log d)}}$$

which implies

$$L_1, L_2 \leq \frac{1}{\text{poly}(d)^{\Omega(\log d)}}$$

Furthermore, from Fact 2.E.1, we also have

$$\mathbb{E} [z_j^2 \mathbb{1}_{|\langle w_i^{(t)}, x_p^+ \rangle| \geq b_i^{(t)} + |\langle w_i^{(t)}, x_p^+ - x_p \rangle|}] \leq \frac{1}{\text{poly}(d)^{\Omega(\log d)}}$$

Now we further take into considerations [Lemma 2.D.2](#), [Lemma 2.E.3](#) and [Lemma 2.E.4](#). We can obtain

$$|\langle w_i^{(t+1)}, \mathbf{M}_j \rangle| \leq \langle w_i^{(t)}, \mathbf{M}_j \rangle (1 - \eta\lambda) + \tilde{O}\left(\frac{\Xi_2^2 \|w_i^{(t)}\|_2}{d^2}\right) \pm \frac{\eta}{\text{poly}(d_1)}$$

Indeed, since we have chosen learning rate $\eta = \frac{1}{\text{poly}(d)}$ and $\lambda \in [\frac{1}{d^{1.01}}, \frac{1}{d^{1.49}}]$, it is easy to prove (5) as follows:

- For $i \notin \mathcal{M}_j$, $|\langle w_i^{(t)}, \mathbf{M}_j \rangle| \leq O(\frac{\|w_i^{(t)}\|_2}{\sqrt{d}\Xi_2^5})$: This is easy since by using [Lemma 2.D.1](#), [Lemma 2.E.3](#) and [Lemma 2.E.4](#), we can prove the following inequality by contradiction⁸

$$|\langle w_i^{(t)}, \mathbf{M}_j \rangle| \leq |\langle w_i^{(t-1)}, \mathbf{M}_j \rangle| (1 + \frac{\eta}{d^2} - \eta\lambda) + \tilde{O}\left(\frac{\eta\Xi_2^2}{d^2}\right) \|w_i^{(t)}\|_2 \leq \dots \leq O\left(\frac{\|w_i^{(t)}\|_2}{\sqrt{d}\Xi_2^5}\right)$$

Now we begin to prove (6). For all $i \in [m]$, we have $\max_{j \in [d_1] \setminus [d]} |\langle w_i^{(t)}, \mathbf{M}_j^\perp \rangle| \leq O(\frac{\|w_i^{(t)}\|_2}{\sqrt{d_1}\Xi_2^5})$ at iteration $t = T_2$; Now, by expanding the gradient updates of $\langle w_i^{(t)}, \mathbf{M}_j^\perp \rangle$, we can see that

$$\begin{aligned} |\langle w_i^{(t+1)}, \mathbf{M}_j^\perp \rangle| &\leq |\langle w_i^{(t)}, \mathbf{M}_j^\perp \rangle| (1 - \eta\lambda) + |\Psi_{i,j}^{(t)}| + |\Phi_{i,j}^{(t)}| + |\mathcal{E}_{1,i,j}^{(t)}| + |\mathcal{E}_{2,i,j}^{(t)}| \\ &\leq |\langle w_i^{(t)}, \mathbf{M}_j^\perp \rangle| (1 - \eta\lambda) + \tilde{O}\left(\frac{\Xi_2^5}{\sqrt{d_1}d^{1.5}}\right) \|w_i^{(t)}\|_2 + \tilde{O}\left(\frac{\Xi_2^5}{\tau\sqrt{d_1}d^2}\right) \|w_i^{(t)}\|_2 \end{aligned}$$

where the last inequality are obtained as follows: first from [Lemma 2.E.3](#) we simply have $|\mathcal{E}_{1,i,j}^{(t)}| \leq \tilde{O}(\frac{\Xi_2^5}{\sqrt{d_1}d^{1.5}}) \|w_i^{(t)}\|_2$, then from [Lemma 2.E.4](#) we have

$$\begin{aligned} |\mathcal{E}_{2,i,j}^{(t)}| &\leq O\left(\frac{\|w_i^{(t)}\|_2 \Xi_2^2}{d^2 \tau}\right) \cdot \max_{i' \in [m]} \left(|\langle w_{i'}^{(t)}, \mathbf{M}_j^\perp \rangle| + \frac{\|w_{i'}^{(t)}\|_2}{\sqrt{d_1}} \right) \\ &\leq \tilde{O}\left(\frac{\Xi_2^5}{\tau\sqrt{d_1}d^2}\right) \|w_i^{(t)}\|_2 \quad (\text{since } \max_{i' \in [m]} |\langle w_{i'}^{(t)}, \mathbf{M}_j^\perp \rangle| \leq O(\frac{1}{\sqrt{d_1}\Xi_2^5}) \text{ from induction}) \end{aligned}$$

After (5) and (6) are proven, it is easy to observe (1) is true at t . Below we shall prove (2), (3) and (4), after which (7) can be also trivially proven. Indeed, (2) is a corollary of (3) and (4), since if $\mathfrak{F}_j^{(t)} \leq O(\tau \log d)$ and (4) holds, we simply have

$$\|w_i^{(t)}\|_2^2 = \sum_{j \in \mathcal{N}_i} \langle w_i^{(t)}, \mathbf{M}_j \rangle^2 + \sum_{j \notin \mathcal{N}_i, j \in [d]} \langle w_i^{(t)}, \mathbf{M}_j \rangle^2 + \sum_{j \in [d_1] \setminus [d]} \langle w_i^{(t)}, \mathbf{M}_j^\perp \rangle^2$$

⁸ Indeed, if the $|\langle w_i^{(t)}, \mathbf{M}_j \rangle| \geq \Omega(\frac{\|w_i^{(t)}\|_2}{\sqrt{d}\Xi_2^5})$ at some iteration t , then by our choice of λ and the calculation of $\Psi_{i,j}^{(t)}$ using [Lemma 2.D.1](#), the gradient sign of $\langle w_i^{(t)}, \mathbf{M}_j \rangle$ will be opposite to itself.

$$\begin{aligned}
&\leq \sum_{j \in \mathcal{N}_i} \langle w_i^{(t)}, \mathbf{M}_j \rangle^2 + O(d) \cdot O\left(\frac{\|w_i^{(t)}\|_2^2}{d \Xi_2^{10}}\right) + O(d_1) \cdot O\left(\frac{\|w_i^{(t)}\|_2^2}{d_1 \Xi_2^{10}}\right) \\
&\leq \sum_{j \in \mathcal{N}_i} \langle w_i^{(t)}, \mathbf{M}_j \rangle^2 + o\left(\frac{1}{\Xi_2^{10}} \|w_i^{(t)}\|_2^2\right)
\end{aligned}$$

which implies (2). Thus we only need to prove (3) and (4). Indeed, for (3), letting $i \in \mathcal{M}_j$, we proceed as follows: we first write the updates of $\langle w_i^{(t)}, \mathbf{M}_j \rangle$ as

$$\begin{aligned}
\langle w_i^{(t+1)}, \mathbf{M}_j \rangle &= \langle w_i^{(t)}, \mathbf{M}_j \rangle (1 - \eta\lambda) + \Psi_{i,j}^{(t)} + \Phi_{i,j}^{(t)} + \mathcal{E}_{1,i,j}^{(t)} + \mathcal{E}_{2,i,j}^{(t)} \\
&= \langle w_i^{(t)}, \mathbf{M}_j \rangle (1 - \eta\lambda) + \Psi_{i,j}^{(t)} + \tilde{O}\left(\frac{\Xi_2^2}{d^2}\right) \|w_i^{(t)}\|_2
\end{aligned}$$

where the last inequality comes from again from [Lemma 2.E.3](#) and [Lemma 2.E.4](#). Now suppose for some t we have $\mathfrak{F}_j^{(t)} \geq \Omega(\tau \log^2 d)$, by [Lemma 2.E.2](#), we have

$$\langle w_i^{(t+1)}, \mathbf{M}_j \rangle = \langle w_i^{(t)}, \mathbf{M}_j \rangle \left(1 + \frac{1}{\text{poly}(d)} - \eta\lambda\right) + \tilde{O}\left(\frac{\Xi_2^2}{d^2}\right) \|w_i^{(t)}\|_2 \leq \langle w_i^{(t)}, \mathbf{M}_j \rangle \left(1 + \frac{1}{\text{poly}(d)} - \eta\lambda/2\right)$$

which means that $\langle w_i^{(t+1)}, \mathbf{M}_j \rangle \leq \langle w_i^{(t)}, \mathbf{M}_j \rangle$. This in fact gives $\mathfrak{F}_j^{(t+1)} \leq \mathfrak{F}_j^{(t)}$, so that (3) is proven.

Now for (4), we need to induct as follows: for $t \leq T'_j := \frac{d \log d}{\eta \log \log d}$ which is the specific iteration when $\mathfrak{F}_j^{(t)} \geq G_1 \tau \log d$, where G_1 is defined in [Lemma 2.E.1](#). The induction of (4) follows from similar proof in [Theorem 2.D.1](#). After T'_j , we discuss as follows

- When $t \in [T'_j, \frac{d^{1.49}}{\eta}]$, from above calculations, for each $i' \in \mathcal{M}_j$, we have

$$\frac{|\langle w_i^{(t+1)}, \mathbf{M}_j \rangle|}{|\langle w_{i'}^{(t+1)}, \mathbf{M}_j \rangle|} = \frac{|\langle w_i^{(t)}, \mathbf{M}_j \rangle| (1 - \eta\lambda) + \eta \Psi_{i,j}^{(t)} \pm O\left(\frac{\sqrt{\Xi_2}}{t\sqrt{d}}\right) \|w_i^{(t)}\|_2}{|\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle| (1 - \eta\lambda) + \eta \Psi_{i',j}^{(t)} \pm O\left(\frac{\sqrt{\Xi_2}}{t\sqrt{d}}\right) \|w_{i'}^{(t)}\|_2}$$

On one hand, for those $i' \in \mathcal{M}_j$ such that $|\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle| \leq b_i^{(t)} \Xi_2^2 \leq O\left(\frac{\Xi_2^2}{t\sqrt{d}}\right) \|w_i^{(t)}\|_2$, we can safely get $|\langle w_i^{(t+1)}, \mathbf{M}_j \rangle| \gg |\langle w_{i'}^{(t+1)}, \mathbf{M}_j \rangle|$. On the other hand, if $|\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle| \geq b_i^{(t)} \Xi_2^2$, then we have

$$\left| \frac{\Psi_{i,j}^{(t)}}{|\langle w_i^{(t)}, \mathbf{M}_j \rangle|} - \frac{\Psi_{i',j}^{(t)}}{|\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle|} \right| \leq O\left(\frac{\Xi_2}{t\sqrt{d}\eta} b_i^{(t)}\right)$$

Thus by letting $\tilde{\Psi}_j := \frac{\Psi_{i,j}^{(t)}}{|\langle w_i^{(t)}, \mathbf{M}_j \rangle|}$, then

$$\frac{|\langle w_i^{(t+1)}, \mathbf{M}_j \rangle|}{|\langle w_{i'}^{(t+1)}, \mathbf{M}_j \rangle|} = \frac{|\langle w_i^{(t)}, \mathbf{M}_j \rangle| (1 + \eta \tilde{\Psi}_j^{(t)} - \eta\lambda) \pm O\left(\frac{\sqrt{\Xi_2}}{t\sqrt{d}}\right) \|w_i^{(t)}\|_2}{|\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle| (1 + \eta \tilde{\Psi}_j^{(t)} - \eta\lambda) \pm O\left(\frac{\sqrt{\Xi_2}}{t\sqrt{d}}\right) \|w_{i'}^{(t)}\|_2}$$

Since at iteration $t \in [T'_j, \frac{d^{1.49}}{\eta}]$, it is easy to obtain that $|\widetilde{\Psi}_j^{(t)} - \lambda| \leq O(\frac{\Xi_2}{\eta t})$.⁹ Thus we have

$$\begin{aligned} \frac{|\langle w_i^{(t+1)}, \mathbf{M}_j \rangle|}{|\langle w_{i'}^{(t+1)}, \mathbf{M}_j \rangle|} &\geq \frac{|\langle w_i^{(t)}, \mathbf{M}_j \rangle|(1 + \eta(\widetilde{\Psi}_j^{(t)} - \lambda)(1 - \frac{\Xi_2^2}{\sqrt{d}}))}{|\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle|(1 + \eta(\widetilde{\Psi}_j^{(t)} - \lambda)(1 + \frac{\Xi_2^2}{\sqrt{d}}))} \geq (1 - \frac{\Xi_2}{t\sqrt{d}}) \frac{|\langle w_i^{(t)}, \mathbf{M}_j \rangle|}{|\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle|} \\ &\geq \prod_{t'=T'_j}^{t-1} (1 - O(\frac{\Xi_2^2}{t'\sqrt{d}})) \frac{|\langle w_i^{(T'_j)}, \mathbf{M}_j \rangle|}{|\langle w_{i'}^{(T'_j)}, \mathbf{M}_j \rangle|} \geq \Omega(1) \end{aligned}$$

where in the last inequality we have used our induction hypothesis at T'_j .

- The proof for iterations $t \in [\frac{d^{1.49}}{\eta}, \frac{d^{1.99}}{\eta}]$ is largely similar to the above. The only difference here is that we rely on a slightly different comparison here: Indeed, we have

$$\frac{|\langle w_i^{(t+1)}, \mathbf{M}_j \rangle|}{|\langle w_{i'}^{(t+1)}, \mathbf{M}_j \rangle|} = \frac{|\langle w_i^{(t)}, \mathbf{M}_j \rangle|(1 + \eta\widetilde{\Psi}_j^{(t)} - \eta\lambda) \pm O(\frac{\Xi_2^2}{d^2})\|w_i^{(t)}\|_2}{|\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle|(1 + \eta\widetilde{\Psi}_j^{(t)} - \eta\lambda) \pm O(\frac{\Xi_2^2}{d^2})\|w_{i'}^{(t)}\|_2}$$

Here we can use similar techniques as above to require $|\widetilde{\Psi}_j^{(t)} - \lambda| \leq \frac{\Xi_2}{t\eta}$. Now we also have

$$\begin{aligned} \frac{|\langle w_i^{(t+1)}, \mathbf{M}_j \rangle|}{|\langle w_{i'}^{(t+1)}, \mathbf{M}_j \rangle|} &\geq \frac{|\langle w_i^{(t)}, \mathbf{M}_j \rangle|(1 + \eta(\widetilde{\Psi}_j^{(t)} - \lambda)(1 - \frac{\Xi_2^2}{\sqrt{d}}))}{|\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle|(1 + \eta(\widetilde{\Psi}_j^{(t)} - \lambda)(1 + \frac{\Xi_2^2}{\sqrt{d}}))} \geq (1 - \frac{\Xi_2}{t\sqrt{d}}) \frac{|\langle w_i^{(t)}, \mathbf{M}_j \rangle|}{|\langle w_{i'}^{(t)}, \mathbf{M}_j \rangle|} \\ &\geq \prod_{t'=d^{1.49}/\eta}^{t-1} (1 - \frac{\Xi_2^2}{t'd^{0.01}}) \frac{|\langle w_i^{(d^{1.49}/\eta)}, \mathbf{M}_j \rangle|}{|\langle w_{i'}^{(d^{1.49}/\eta)}, \mathbf{M}_j \rangle|} \geq \Omega(1) \end{aligned}$$

Now (4) are proven. (7) is an immediate result of our update scheme. $\square$

Definition 2.E.2 (optimal learner). We define a learner network that we deem as the “optimal” feature map for this task. Let $\kappa > 0$, we define $\theta^\star := \{\theta_i^\star\}_{i \in [m]}$ as follows: $\theta_i^\star = \frac{\sqrt{\tau}\kappa}{|\mathcal{M}_j^\star|} \cdot \mathbf{M}_j \cdot \text{sign}(\langle w_i^{(T_2)}, \mathbf{M}_j \rangle)$ if $i \in \mathcal{M}_j^\star$, and $\theta_i^\star = 0$ if $i \notin \cup_{j \in [d]} \mathcal{M}_j^\star$. Furthermore, we define the optimal feature map $f_t^\star$ as follows: for $i \in [m]$, the i -th neuron of $f_{t,\theta}$ given weight $\theta_i \in \mathbb{R}^{d_1}$ is

$$f_{t,\theta,i}(x) = (\langle \theta_i, x \rangle - b_i) \mathbb{1}_{\langle w_i^{(t)}, x \rangle \geq b_i} - (-\langle \theta_i, x \rangle - b_i) \mathbb{1}_{-\langle w_i^{(t)}, x \rangle \geq b_i}$$

and $f_{t,\theta}$ is just $f_{t,\theta}(\cdot) = (f_{t,\theta,1}(\cdot), \dots, f_{t,\theta,m}(\cdot))^\top$.

⁹ The techniques for proving this is extremely similar to the upper bound $\widetilde{\Psi}_j^{(t)} \leq O(\frac{\Xi_2}{t\eta\sqrt{d}})$ in the proof of Lemma 2.E.5.

Indeed, one can assume at some iteration $|\widetilde{\Psi}_j^{(t)} - \lambda| \geq \Omega(\frac{\Xi_2}{\eta t})$, and then proceed to find out that after some iterations

$\tilde{t} = \Theta(\frac{t}{\sqrt{\Xi_2}})$, $|\widetilde{\Psi}_j^{(t)} - \lambda|$ will decrease to $|\widetilde{\Psi}_j^{(\tilde{t})} - \lambda| \leq \frac{\sqrt{\Xi_2}}{\eta \tilde{t}}$, or otherwise the presumption collapse.

Now in order to obtain the loss convergence result in [Theorem 2.E.1](#), we need the following lemma, which characterizes how well the optimal learner performs evaluated by a pseudo objective.

Lemma 2.E.6 (optimality). *Let $\{\theta_i^*\}_{i \in [m]}$ and $f_{t,\theta}$ be defined as in [Definition 2.E.2](#), when [Induction Hypothesis 2.E.1](#) holds, defining a pseudo loss function by*

$$\tilde{L}(f_{t,\theta^*}, f_t) := \mathbb{E} \left[-\tau \log \left(\frac{e^{\langle f_{t,\theta^*}(x_p^+), f_t(x_p^{++}) \rangle / \tau}}{\sum_{x \in \mathfrak{B}} e^{\langle f_{t,\theta^*}(x_p^+), f_t(x) \rangle / \tau}} \right) \right]$$

then by choosing $\kappa = \Theta(\Xi_2)$, and suppose $\sum_{i \in \mathcal{M}_j^*} |\langle w_i^{(t)}, \mathbf{M}_j \rangle| \geq \Omega(\frac{\sqrt{\kappa}}{\Xi_2})$, we have loss guarantee:

$$\tilde{L}(f_{t,\theta^*}, f_t) \leq O\left(\frac{1}{\log d}\right)$$

Proof. The proof of (a) follows from the fact that [Induction Hypothesis 2.E.1](#) holds at iteration $t \geq T_3$, and also from [Lemma 2.B.3](#) and [Lemma 2.B.5](#). (b) can be proven via the following calculations:

$$\tilde{L}(f_{t,\theta^*}, f_t) = \mathbb{E} \left[-\tau \log \left(\frac{e^{\langle f_{t,\theta^*}(x_p^+), f_t(x_p^{++}) \rangle / \tau}}{\sum_{x \in \mathfrak{B}} e^{\langle f_{t,\theta^*}(x_p^+), f_t(x) \rangle / \tau}} \right) \right]$$

where the second inequality follows from the high probability bound (a). Now by using Bernoulli concentration, we know that whenever $\sum_{j \in [d]} \mathbb{1}_{z_{p,j} \neq 0} = \Omega(\log \log d)$ (which happens with constant probability), we have

$$\sum_{j \in [d]} \mathbb{1}_{z_{n,s,j} = z_{p,j}} \leq C \sum_{j \in [d]} \mathbb{1}_{z_{p,j} \neq 0} \quad \left(\text{with prob} \geq 1 - \frac{1}{d^{\Omega(\log \log d)}} \text{ for all } x_{n,s} \in \mathfrak{R}\right)$$

And also from [Definition 2.E.2](#) we know that if for some $j \in [d]$, $z_{p,j} = z_{n,s,j}$, then

$$\sum_{i \in \mathcal{M}_j^*} \left(f_{t,\theta^*,i}(x_p^+) h_{i,t}(x_{n,s}) - f_{t,\theta^*,i}(x_p^+) h_{i,t}(x_p^{++}) \right) \geq \kappa \tau \sum_{j \in \mathcal{M}_j^*} |\langle w_i^{(t)}, \mathbf{M}_j \rangle| + O\left(\frac{1}{\log d}\right)$$

which can be obtained by similar calculations in [Lemma 2.E.1](#). Noticing that the event $z_p \neq 0$ happens with $\text{prob} \geq 1 - \frac{1}{\text{poly}(\log(d))}$, we have

$$\begin{aligned} & \tilde{L}(f_{t,\theta^*}, f_t) \\ & \leq \left(1 - \frac{1}{\text{poly}(\log(d))}\right) \mathbb{E} \left[\log \left(\sum_{x \in \mathfrak{B}} e^{\langle f_{t,\theta^*}(x_p^+), f_t(x) \rangle / \tau - \langle f_{t,\theta^*}(x_p^+), f_t(x_p^{++}) \rangle / \tau} \right) \middle| z_p \neq 0 \right] \\ & \quad + \Pr(z_p = 0) \cdot O(\log |\mathfrak{B}|) \end{aligned}$$

$$\begin{aligned}
&\leq (1 - \frac{1}{\text{polylog}(d)}) \mathbb{E} \left[\log \left(\sum_{x \in \mathfrak{B}} e^{\sum_{j \in [d]} \sum_{i \in \mathcal{M}_j^*} (f_{i,\theta^*,i}(x_p^+) h_{i,t}(x) - f_{i,\theta^*,i}(x_p^+) h_{i,t}(x_p^{++})) / \tau} \right) \middle| z_p \neq 0 \right] \\
&\quad + \Pr(z_p = 0) \cdot O(\log |\mathfrak{B}|) \\
&= (1 - \frac{1}{\text{polylog}(d)}) \mathbb{E} \left[\log \left(1 + \sum_{x_{n,s} \in \mathfrak{N}} e^{-\sum_{j \in \mathcal{M}_j^*} |\langle w_i^{(t)}, \mathbf{M}_j \rangle| (z_{p,j}^2 - z_{p,j} z_{n,s,j}) + O(\frac{1}{\log d})} \right) \middle| z_p \neq 0 \right] + O(\frac{1}{\log d}) \\
&\leq O(\frac{1}{\log d})
\end{aligned}$$

where the last inequality combines the Bernoulli concentration results of $\sum_{j \in [d]} z_{p,j} z_{n,s,j}$ and a union bound for all $s \in [\mathfrak{N}]$, and that $|\langle w_i^{(t)}, \mathbf{M}_j \rangle| \geq \Omega(\frac{\sqrt{\tau}}{\Xi_2})$. $\square$

Proof of Theorem 2.E.1. Due to [Induction Hypothesis 2.E.1](#), we know that as long as training goes on, the neural network will learn the desired features with sparse representations. As a complement of the conditions in [Induction Hypothesis 2.E.1](#), we notice that at some $t \geq \Theta(\frac{d \log d}{\eta \log \log d})$, we have for all $j \in [d]$, $\mathfrak{F}_j^{(t)} \geq \Omega(\tau \log d)$, using [Lemma 2.E.1](#). This can be combined with [Induction Hypothesis 2.E.1](#) (1) and (4) to show (a) [Theorem 2.E.1](#). Now we prove that it actually converge to the desired solutions, rather than bouncing around. Denote $w^{(t)} = (w_1^{(t)}, \dots, w_m^{(t)})$, since our update is $w^{(t+1)} = w^{(t)} - \nabla_w \mathbf{Obj}(f_t) + \frac{1}{\text{poly}(d_1)}$, we have

$$\begin{aligned}
\eta \langle \nabla_w \mathbf{Obj}(f_t), w^{(t)} - \theta^* \rangle &= \frac{\eta^2}{2} \|\nabla \mathbf{Obj}(f_t)\|_F^2 + \frac{1}{2} \|w^{(t)} - \theta^*\|_F^2 - \frac{1}{2} \|w^{(t+1)} - \theta^*\|_F^2 + \frac{\eta^2}{\text{poly}(d_1)} \\
&\leq \eta^2 \text{poly}(d) + \frac{1}{2} \|w^{(t)} - \theta^*\|_F^2 - \frac{1}{2} \|w^{(t+1)} - \theta^*\|_F^2 + \frac{\eta^2}{\text{poly}(d_1)}
\end{aligned}$$

Now we will use the tools from online learning to obtain a loss guarantee: define a pseudo objective for parameter θ

$$\widetilde{\mathbf{Obj}}_t(\theta) := \widetilde{L}(f_t, \theta) + \frac{\lambda}{2} \sum_{i \in [m]} \|\theta_i\|_2^2 = \mathbb{E} \left[-\tau \log \left(\frac{e^{\langle f_t, \theta(x_p^+), f_t(x_p^{++}) \rangle / \tau}}{\sum_{x \in \mathfrak{B}} e^{\langle f_t, \theta(x_p^+), f_t(x) \rangle / \tau}} \right) \right] + \frac{\lambda}{2} \sum_{i \in [m]} \|\theta_i\|_2^2$$

Which is a convex function over θ since it is linear in θ . Moreover, we have $\widetilde{\mathbf{Obj}}_t(w^{(t)}) = \mathbf{Obj}(f_t)$ and $\nabla_{\theta_i} \widetilde{\mathbf{Obj}}_t(w_i^{(t)}) = \nabla_{w_i} \mathbf{Obj}(f_t)$, thus we have

$$\begin{aligned}
\eta \langle \nabla_w \mathbf{Obj}(f_t), w^{(t)} - \theta^* \rangle &= \eta \langle \nabla_{\theta} \widetilde{\mathbf{Obj}}_t(w^{(t)}), w^{(t)} - \theta^* \rangle \\
&\geq \widetilde{\mathbf{Obj}}_t(w^{(t)}) - \widetilde{\mathbf{Obj}}_t(\theta^*) \\
&\geq \widetilde{\mathbf{Obj}}_t(w^{(t)}) - \mathbb{E} \left[-\tau \log \left(\frac{e^{\langle f_t, \theta^*(x_p^+), f_t(x_p^{++}) \rangle / \tau}}{\sum_{x \in \mathfrak{B}} e^{\langle f_t, \theta^*(x_p^+), f_t(x) \rangle / \tau}} \right) \right] + \frac{\lambda}{2} \sum_{i \in [m]} \|\theta_i^*\|_2^2
\end{aligned}$$

$$\begin{aligned}
&\geq \widetilde{\mathbf{Obj}}_t(w^{(t)}) - O\left(\frac{1}{\log d}\right) - O\left(\sum_{i \in [m]} O(\lambda \|\theta_i^*\|_2^2) + O\left(\frac{1}{d^{0.49}}\right)\right) - \frac{O(\log d)}{\text{polylog}(d)} \\
&\geq \mathbf{Obj}(f_t) - O\left(\frac{1}{\log d}\right)
\end{aligned}$$

Now choosing $\kappa = \Theta(\Xi_2) \leq \lambda/d$, and by a telescoping summation, we have

$$\frac{1}{T} \sum_{t=T_3}^{T_3+T-1} \left(\mathbf{Obj}(f_t) - O\left(\frac{1}{\log d}\right) \right) \leq \frac{O(\|w^{(T_3+T)} - \theta^*\|_F^2)}{T\eta} \leq O\left(\frac{m\Xi_2}{T\eta}\right)$$

Since $T\eta \geq m\Xi_2^{10}$, this proves the claim. $\square$

The corollary in the main text can be proven via simple application of linear regression analysis. Because with high probability over polynomially many data $\mathcal{Z}_{\text{sup}} = \{x_i, y_i\}_{i \in [n_{\text{sup}}]}$ independently generated according to our definition [Definition 2.2.1](#) and [Definition 2.3.1](#), $\{x_i\}_{i \in [n_{\text{sup}}]}$ form separable clusters w.r.t. their differences in the latent variables z_i , which dictate their labels.

2.F Results for Learning Without Augmentations

In this section we will sketch the proof when no augmentation is applied to the input data. Indeed, the analysis is similar but much easier compared to the case when augmentations are used. We present the first lemma below.

Lemma 2.F.1 (gradient for features, positive). *Let $i \in [m]$ and $v \in \{\mathbf{M}_j\}_{j \in [d]} \cup \{\mathbf{M}_j^\perp\}_{j \in [d_1] \setminus [d]}$, when bias $b_i = 0$ we have*

$$\mathbb{E} \left[h_i(x_p) \mathbb{1}_{|\langle w_i, x_p \rangle| \geq b_i} \langle x_p, \mathbf{M}_j \rangle \right] = \langle w_i, \mathbf{M}_j \rangle \mathbb{E} \left[\langle x_p, v \rangle^2 \right]$$

when $b_i > 0$, we also have

$$\mathbb{E} \left[h_i(x_p) \mathbb{1}_{|\langle w_i, x_p \rangle| \geq b_i} \langle x_p, \mathbf{M}_j \rangle \right] = \langle w_i, \mathbf{M}_j \rangle \mathbb{E} \left[\langle x_p, v \rangle^2 \mathbb{1}_{|\langle w_i, x_p \rangle| \geq b_i} \right]$$

Proof. The proof is essentially trivial by following the approach in the proof of [Lemma 2.C.1](#) and notice that no compensation term for augmentation is needed here. $\square$

Indeed, since $\mathbb{E}[\langle x_p, \mathbf{M}_j \rangle^2] = (1 + \frac{\log \log d}{\sqrt{\log d}}) \mathbb{E}[\langle x_p, \mathbf{M}_j^\perp \rangle^2]$, for any training phase before close to convergence, the difference does not matter since the growth rate of each feature is exponential, i.e. $\langle w_i^{(t+1)}, v \rangle = \langle w_i^{(t)}, v \rangle (1 + \mathbb{E}[\langle x_p, v \rangle^2])$. Indeed, denoting the neural network trained without augmentation by f_t^{NA} , setting bias $b_i^{(t)} = 0$, we can have a simple lemma:

Lemma 2.F.2 (The superiority of dense feature without augmentations). *For each $i \in [m]$, let w_i has norm $\|w_i\|_2^2$ and orthogonal to each other, and such that $w_i \perp \mathbf{M}_j$ for all $j \in [d]$, then*

$$L(f_t^{\text{NA}}) \leq \tilde{O}\left(\frac{d}{\sum_{i \in [m]} \|w_i^{(t)}\|_2^2}\right)$$

Proof. By Johnson-Lindenstrauss lemma, with high probability we have

$$\begin{aligned}\langle f_t^{\text{NA}}(x_p), f_t^{\text{NA}}(x_p) \rangle &= \|f_t^{\text{NA}}(x_p)\|_2^2 = \Omega(\sigma_\xi^2 \sum_{i \in [m]} \|w_i^{(t)}\|_2^2) \\ \langle f_t^{\text{NA}}(x_p), f_t^{\text{NA}}(x_{n,s}) \rangle &= \|f_t^{\text{NA}}(x_p)\|_2^2 = \tilde{O}\left(\frac{\sigma_\xi^2}{\sqrt{m}} \sum_{i \in [m]} \|w_i^{(t)}\|_2^2\right)\end{aligned}$$

This leads to:

$$\begin{aligned}L(f_t^{\text{NA}}) &= \mathbb{E} \left[\tau \log \left(1 + \sum_{x_{n,s} \in \mathfrak{R}} e^{\langle f_t^{\text{NA}}(x_p), f_t^{\text{NA}}(x_{n,s}) \rangle / \tau - \langle f_t^{\text{NA}}(x_p), f_t^{\text{NA}}(x_p) \rangle / \tau} \right) \right] \\ &\leq \tilde{O}(1) \mathbb{E} \left[\left(\|f_t^{\text{NA}}(x_p)\|_2^2 - \max_{x_{n,s} \in \mathfrak{R}} \langle f_t^{\text{NA}}(x_p), f_t^{\text{NA}}(x_{n,s}) \rangle \right)^{-1} \right] \\ &\leq \tilde{O}\left(\frac{d}{\sum_{i \in [m]} \|w_i^{(t)}\|_2^2}\right) \quad (\text{using above calculations})\end{aligned}$$

which is the claimed result. $\square$

Specifically, if one use the full $\mathbf{M}$ as the feature and raise bias above $\Omega(\|w_i\|_2)$ in this setting, one would not have this superior loss property.

Now we only need to prove a norm result for all the neurons $i \in [m]$, this can be done by similar analysis as in the proof of [Induction Hypothesis 2.C.1](#), which we skip here:

Lemma 2.F.3. *For each $i \in [m]$, for some $t \geq \tilde{\Omega}(\frac{d}{\eta})$, we have $\sum_{i \in [m]} \|w_i^{(t)}\|_2^2 = \Omega(\tau d \text{polylog}(d))$.*

Combining the results above, we can obtain the learning process of contrastive learning without data augmentations, in the presence of a large dense signal in the data. It is easy to see that the representations trained by this method has the following properties:

Fact 2.F.1. At each iteration t , the learned network without augmentations f_t^{NA} satisfies for each $i \in [m]$, we have $\|\mathbf{M}^\perp(\mathbf{M}^\perp)w_i^{(t)}\|_2 = (1 - \frac{1}{\text{poly}(d)})\|w_i^{(t)}\|_2$

Proof. Similar to the proof of [Induction Hypothesis 2.C.1](#), which we skip. $\square$

This fact directly leads to the final result that with high probability over $x \sim \mathcal{D}_x$

$$\left\langle \frac{f_t^{\text{NA}}(x)}{\|f_t^{\text{NA}}(x)\|_2}, \frac{f_t^{\text{NA}}(\xi)}{\|f_t^{\text{NA}}(\xi)\|_2} \right\rangle \geq 1 - \tilde{O}\left(\frac{1}{\text{poly}(d)}\right)$$

And trivially, one cannot perform any linear regression or classification over such feature map f_t^{NA} , to obtain meaningful accuracy in our downstream tasks over the sparse coding data $\mathcal{D}_x$.

Chapter 3

The Mechanism of Asymmetry in Non-contrastive Learning

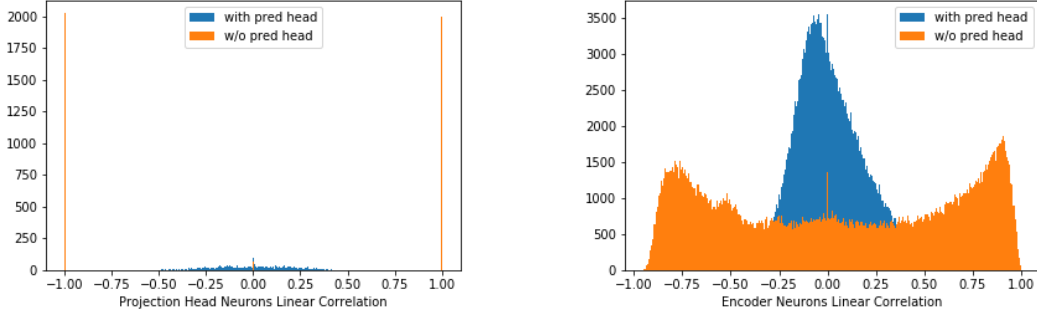
3.1 Introduction

Self-supervised learning is about learning representations of real-world vision or language data without human supervision, and contrastive learning [39, 48, 93, 113, 116, 206] is one of the most successful self-supervised learning approaches. It has been known that the behavior of contrastive learning depends critically on the minimization of the *negative term*, which corresponds to contrasting the representations of *negative pairs*, i.e., pairs of different data points. However, the surprising finding of the *Bootstrap Your Own Latent* (BYOL) method by Grill et al. [106] initiated the research of *non-contrastive self-supervised learning*, which refers to contrastive learning methods without using the negative pairs. BYOL achieved state-of-the-art results in various computer vision benchmarks and there are plenty of follow-up works [30, 40, 53, 82, 109, 122, 203, 336] making improvements in this direction.

On a high level, in non-contrastive self-supervised learning, one wishes to learn a network ϕ such that $\phi(x)$ aligns in direction with $\phi(x')$, where x and x' are called the *positive pair*, generated by random augmentations from the same sample. Without contrasting the negative pairs, it is extremely easy for neural networks to cheat the learning task by learning certain inferior representations. One trivial solution known as the *complete collapse* is when $\phi(\cdot)$ is a constant vector whose variance is zero. Another trivial *global optimal* solution, typically learned by the neural network after training, is when all the coordinates $\phi_i(\cdot)$ are exactly aligned, which is named as *dimensional collapse* by Hua et al. [122]. Nevertheless, adding a trainable prediction head on top of (one branch of) $\phi(x)$ magically avoids learning such solutions, *even though the prediction head can possibly learn the identity mapping and render itself useless*. It is mysterious why even if the network

This chapter is adapted from the following joint work:

Zixin Wen and Yuanzhi Li. *The Mechanism of Prediction Head in Non-Contrastive Self-Supervised Learning*.



(a) Histograms of the correlations of projection head neurons.

(b) Histograms of the correlations of encoder network neurons (before projection head).

Figure 3.1: Dimensional Collapse Without a Prediction Head. *Dimensional collapse.* A network trained without a prediction head learns extremely correlated neurons.

can minimize the training objective by learning an identity prediction head and a collapsed encoder network $\phi(\cdot)$, it still optimizes for a non-collapsed state-of-the-art representation instead when trained by (variants of) stochastic gradient descent (SGD).

Since the proposition of BYOL, there have been lots of empirical studies trying to understand non-contrastive learning. The SimSiam method by Chen and He [53] shows the exponential moving average (EMA) is not necessary for avoiding collapsed solutions while *stop-gradient* is necessary. Richemond et al. [236] empirically disproved the conjecture that information leakage from batch normalization (BN) is the reason why BYOL can avoid collapse. DINO [40] further explored replacing the normalized ℓ_2 -loss by a cross-entropy loss. Zhang et al. [338] gives empirical evidence that using a single bias layer as a prediction head is capable of avoiding collapsed solutions. All the methods above use loss functions that are asymmetric with respect to the positive pair. If one wishes to work without both asymmetry and the negative pairs, one must add extra diversity-enforcing structures say neuron-wise regularization in *Barlow Twins* [336] or a more complicated output normalization scheme than BN [82, 122]. The seminal works [122, 336] provide empirical evidence that the prediction head encourages the network to learn more diversified features. But in theory, the question of how the prediction head helps in learning those diverse features is still unanswered.

Despite the great empirical effort put to investigate these non-contrastive learning methods, there is very little theoretical progress towards explaining them. Most of existing theories focus on contrastive learning, especially from the statistical learning perspective [21, 27, 28, 112, 125, 137, 179, 280, 281, 285, 291]. The theoretical tools used in these paper rely heavily on the properties of the minima of loss function. However, due to the existence of trivial dimensional collapsed *global optimal* solutions (even with the prediction head) of the non-contrastive methods, to the best of our knowledge, *there is no well-established statistical framework for those methods yet*. To explain the non-contrastive learning, it is inevitable to study how the solutions are chosen during the

optimization. Therefore, we consider understanding the optimization process to be crucial for understanding these methods. Our research questions are:

Our theoretical questions: the role of prediction head

Why do most non-contrastive self-supervised methods learn collapsed solutions when the so-called prediction head is absent in the network architecture? How does the *trainable* prediction head help *optimizing* the neural network to learn more diversified representations in non-contrastive self-supervised learning?

Theoretical challenges of our questions. Due to the existence of trivial collapsed optimal solutions of the non-contrastive learning objective, we need to understand the *implicit bias in optimization* posed by the prediction head. However, to the best of our knowledge, all of the previous implicit biases theories focus only on the supervised learning tasks, and thus cannot be applied to our question. Even though [311] has characterized the training trajectory of contrastive learning, its analysis cannot incorporate the training of the prediction head. In theory, the optimization of nonlinear neural networks with at least two trainable layers in self-supervised learning is still intractable. A detailed explanation of our challenges will be given in [Section 3.2](#).

There are already some theoretical papers [215, 276, 296] that try to address similar questions. While none of these papers studied the training process of the prediction head, our results provide a completely different perspective: *We explain why training the prediction head can encourage the network to learn diversified features and avoid dimensional collapses, even when the trivial collapsed optima still exist in the training objective*, which is not covered by the prior works. We defer the detailed comparison of similar works to [Section 3.1.1](#). On a high level, the results in this paper are summarized as follows:

Our empirical contributions. In non-contrastive self-supervised learning, we obtain the following experimental results:

- We discover empirically that even when the prediction head is *linear* and initialized as an identity matrix with only off-diagonal entries being trainable, the performance of learned representation is comparable to using the usual non-linear two-layer MLP or randomly initialized (trainable) linear prediction head. This disproves the belief that non-symmetric initialization of the online and target network is needed. See [Figure 3.2](#).
- We empirically verified that even when the prediction head is an identity-initialized matrix, it does not always converge to a symmetric matrix during training. This proves the trainable prediction head does not need to behave like a symmetric matrix during most of the training process. Therefore the theories based on symmetric prediction head [276, 296] cannot fully explain the behaviors of the trainable prediction head. See [Figure 3.3](#) and [Figure 3.4](#).

Our theoretical contributions. We based our theory on a very simple setting, where the data consist of two features: the strong feature and the weak feature. Intuitively, we can think of the

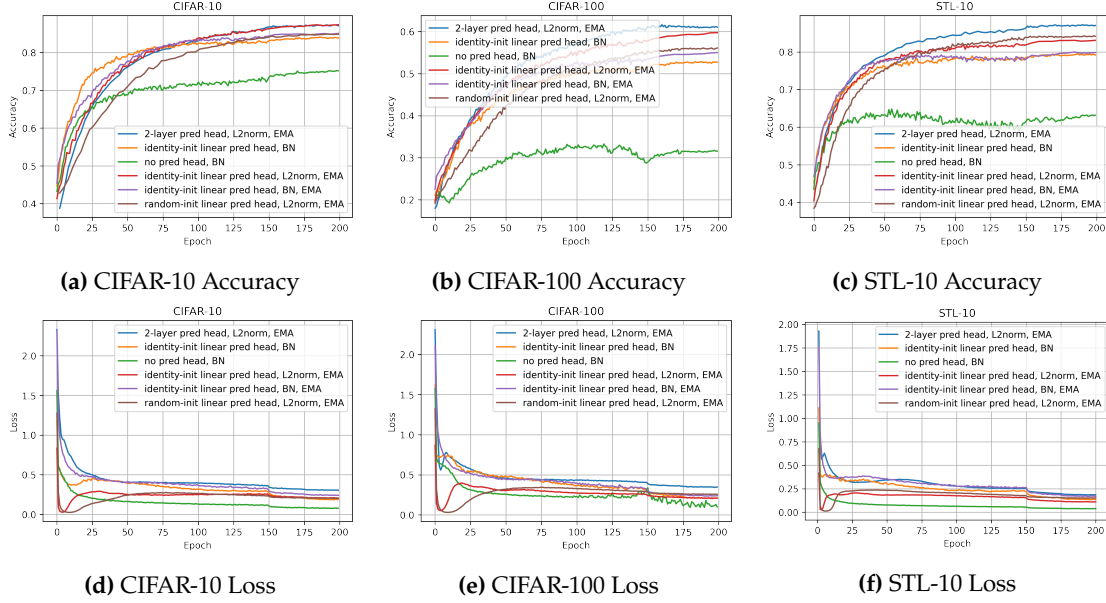


Figure 3.2: Prediction-Head Performance. Performances of using different prediction heads. Here in CIFAR-10, CIFAR-100 and STL-10, identity-initialized linear prediction head can achieve good accuracies comparable to commonly used two-layer non-linear MLP or randomly-initialized linear head. All the prediction heads are trainable, while for identity-initialized prediction head only the off-diagonal entries are trainable. Here BN or L2norm represents the output normalization, and EMA represents using exponential moving average to update the target network as in BYOL [109]. More details of these experiments can be seen in Section 3.7.

strong features in a dataset are the ones that show up more frequently or with large magnitude, and weak features as those that show up rarely or with small magnitude. We consider learning with a *two-layer non-linear neural network with output normalization* using (stochastic) gradient descent. Under this setting, we obtain the following results.

- We prove that without a prediction head, even with BN on the output to avoid complete collapse, the networks will still converge to dimensional collapsed solutions, which provides a theoretical explanation to the dimensional collapse phenomenon observed in [122].
- We prove that the trainable prediction head, combined with suitable output normalization and stop-gradient operation, can learn diversified features to avoid the dimensional collapse problem. We characterize two effects of prediction head: the *substitution effect* and the *acceleration effect*. The intuitions of these two effects are summarized below:

The mechanism of the trainable prediction head

In our setting, we prove that (1) without the prediction head, all the neurons will only learn the strongest feature in the data set thus causing dimensional collapses; (2) the trainable prediction head can help to learn weak features by leveraging two effects: the *substitution effect* and the *acceleration effect*. The substitution effect happens when by learning the prediction head, the learned stronger features in some neurons can substitute for learning the same features in other neurons, which decreases the learning speed of strong features in those neurons. And the acceleration effect happens when the strong features substituted via the prediction head can further accelerate the learning of weaker features in those substituted neurons.

Besides the above effects, we also explain, in our setting, how the two common components in non-contrastive learning: *stop-gradient* operation and *output normalization*, can assist the prediction head in creating those effects during the training process. We point out it is the interactions between these components, rather than their individual effects, that ensure the success of the training. We shall discuss this in more detail in [Section 3.5.3](#).

3.1.1 Comparison to Similar Studies

In this section, we will clarify the differences between our results and some similar studies. Especially the theoretical papers by Tian et al. [276] and Wang et al. [296]. Pokle et al. [215] compared the landscapes between contrastive and non-contrastive learning and points out the existence of non-collapsed bad minima for non-contrastive learning without a prediction head.

We point out that all the claims below are derived *only in our theoretical setting* and are partially verified in experiments over datasets such as CIFAR-10, CIFAR-100, and STL-10.

Can eigenspace alignment explain the effects of training the prediction head? The paper [276] presented a theoretical statement that (symmetric) linear prediction head will *converge* to a matrix that commutes with the covariance matrix of linear representations *at the end of training*, and they provided experiments to support their theory. However, our theory suggests that *the intermediate stage of training the prediction head matters more to the feature learning of the base network than the convergence stage*. Indeed, as shown in Figure 3.3, in many cases, the trainable projection head will *converge back to identity* after training, which commutes with any covariance matrix. However, simply setting the prediction head to identity without training leads to significantly worse results. Therefore, we believe that it is critical to study the entire learning process to understand the role of the prediction head. We prove that in our setting, the substitution effect and the acceleration effect happen during the stage when the networks are trying to learn the weaker features, and after that, the prediction head will converge back to the identity matrix at the end of training (see [Proposition 3.5.1](#)). Again, we emphasize that our characterization of the prediction head trajectory is partially verified by the experiments in [Figure 3.3a](#): *the training trajectory of the prediction head displays a clear two-stage separation*, which demonstrates that the

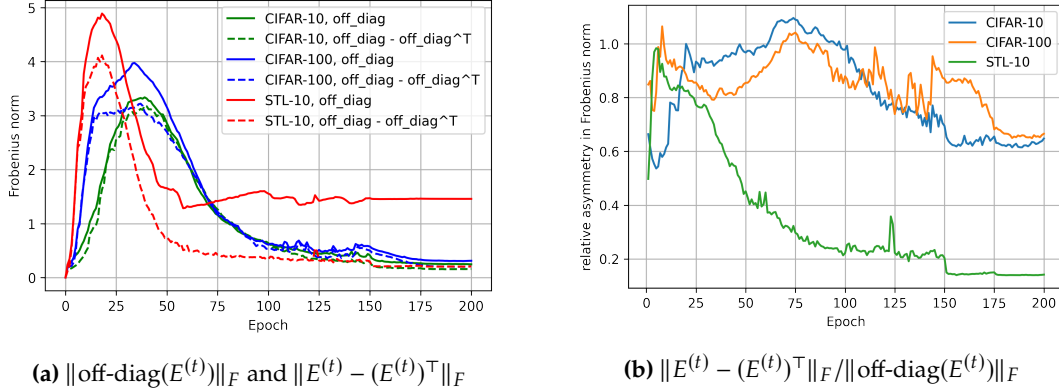


Figure 3.3: Identity-Initialized Head Dynamics. Trajectories of the identity-initialized prediction head. $\text{off-diag}(E)$ is obtained by setting the diagonal of E to be zero. In (a), we discover that over all three datasets considered here, the Frobenius norm of our identity-initialized prediction head’s off-diagonal matrix clearly display a two stage separation, more precisely, a rise and fall pattern; In (b), The off-diagonal matrix of the prediction head is not symmetric in CIFAR-10 and CIFAR-100. Since the diagonal entries are fixed to one, our measure is more accurate in measuring the symmetricity of the prediction head matrix.

convergence result (e.g., the eigenspace alignment result in [276]) is not sufficient to characterize the training process of prediction head. We conjecture the result in [276] on the prediction head is due to a similar convergence result we obtain at the end of training.

Can the symmetric prediction head explain the trainable prediction head? In the paper [276], experiments over the STL-10 dataset showed that the linear prediction head tends to converge to a symmetric matrix during training. And the follow-up paper [296] established a theory under the symmetric prediction head (which is not trained but manually set at each iteration). However, similar to the reason why eigenspace alignment cannot fully explain the effects of the prediction head, the symmetric prediction head given in [296] might not explain the trainable prediction head as well. Under their linear network setting, where W is the weight matrix of the base encoder, they manually set the prediction head W_p at iteration t to be

$$W_p^{(t)} \leftarrow W^{(t)} \mathbb{E}_{x_1} x_1 x_1^\top (W^{(t)})^\top \quad (3.1.1)$$

and the outputs of both online and target network are not normalized. Under this manual update rule of the prediction head, they proved a subspace learning result under gaussian data setting.

Nevertheless, our experiments in Figure 3.2 and Figure 3.3b show that even if we initialize the prediction head using a symmetric matrix (identity), *the trainable prediction head can be very asymmetric at the early training stage when the encoder network learn most of its features*. Moreover, Figure 3.3b demonstrates that the prediction heads in CIFAR-10 and CIFAR-100 experiments do not converge to a symmetric matrix. In accord with these experiments, our theory suggests that the prediction head cannot converge to a symmetric matrix before the encoder network has successfully learned all the features. Moreover, the theory in [296] cannot distinguish between

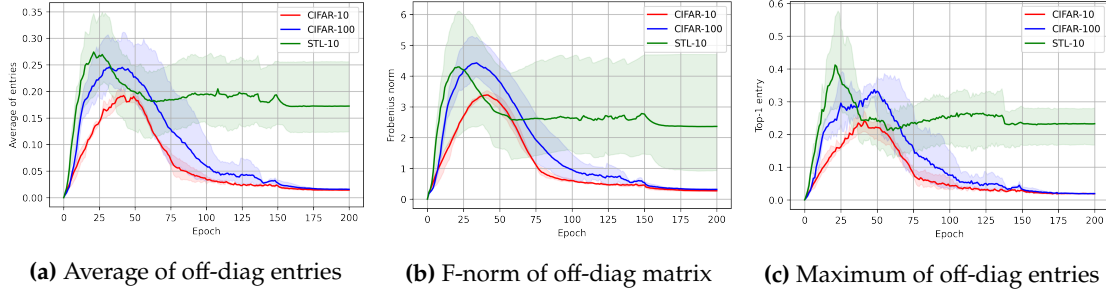


Figure 3.4: Rise-And-Fall Trajectories Across Datasets. Trajectories of the identity-initialized prediction head with a (min, max) confidence band, average over 3 runs. In all three datasets, we observe a consistent rise and fall trajectory pattern.

learning complete collapsed (zero) solutions and learning dimensional collapsed ones, therefore cannot explain why the prediction can help avoid the dimensional collapse. Actually, in the presence of feature imbalance (e.g., $\mathbb{E}_{x_1} x_1 x_1^\top$ has huge eigen-gap), the symmetric prediction head in (3.1.1) is also likely to collapse into a rank-one matrix where W focus on learning the largest eigenvector of the covariance $\mathbb{E}_{x_1} x_1 x_1^\top$.

The differences between our results and [296]’s are in that we are based on nonlinear network architecture and a trainable prediction head. Indeed, our theory and experiments in Figure 3.7 show that when feature imbalance happens (which is very common in vision datasets, see [50]), training a nonlinear network would cause discrepancies in the learning pace between different neurons. We proved that by learning to become asymmetric, the trainable prediction head can leverage such discrepancies and creates the substitution effect (see Lemma 3.5.2) and the acceleration effect (see Theorem 3.5.3). We believe this proves that asymmetry is the key to explaining the implicit bias of the trainable prediction head and our results establish the symmetry-breaking mechanism of the prediction head in non-contrastive learning.

The role of stop-gradient and output-normalization. The seminal work [53] gave empirical results showing that stop-gradient operation is essential for avoiding the collapsed solutions. It is discussed in the theory of Tian et al. [276] that without the stop-gradient, the linear network will learn the zero (constant) solution. [296] also incorporated the stop-gradient into their theory, but they did not explain why stop-gradient is necessary for their setting. We provide a different perspective about why stop-gradient and output normalization (together) are necessary for non-contrastive learning. We proved in our setting, that the stop-gradient and output-normalization together can turn the features substituted via the prediction head into a factor in the gradient of the slower learning neurons, thereby creating the acceleration effect. If either one of these components is missing, the acceleration effect of the prediction head will not happen and all neurons in the network will focus on learning the strongest feature. Formal arguments will be given in Section 3.5.3.

In contrast, [276, 296] did not incorporate the output normalization into their theory, even though their experiments have used certain forms of normalizations. We believe their method is closely related to the whitening method in [82]. To the best of our knowledge, our paper is the first to explain the effects of output-normalization in optimizing nonlinear neural networks in self-supervised learning.

Dimensional collapse Currently the only theoretical investigation on the dimensional collapse is by Jing et al. [139], where they focus on the contrastive learning setting. We believe their result on the role of the projection head is meaningful to understanding non-contrastive learning. But we emphasize that the objective (3.2.2) suffer from much more extreme dimensional collapse, as shown in Figure 3.1. Thus the causes described in Jing et al. [139] such as strong data augmentations cannot fully explain the dimensional collapse in the non-contrastive setting.

3.2 Preliminaries on Non-contrastive Learning

In this section, we formally define what is non-contrastive self-supervised learning. To do this, we first introduce contrastive learning following [48, 311] as background. We use $[N]$ as a shorthand for the index set $\{1, \dots, N\}$.

Background on contrastive learning. Letting $\phi_W(\cdot)$ be the neural networks, contrastive learning aims to learn good representations ϕ_W via contrasting representations of similar data samples to those of dissimilar ones. Usually we are given a batch of data points $\{X_i\}_{i \in [N]}$, and we construct for each $i \in [N]$ a positive pair $(X_i^{(1)}, X_i^{(2)})$ (which are assumed to be similar) by applying random data augmentations to X_i , and collect negative pairs $(X_i^{(1)}, X_j^{(2)})$ for $i \neq j \in [N]$ (which are assumed to be dissimilar). Now given the representations $z_i = \phi_W(X_i^{(1)})$, $z'_i = \phi_W(X_i^{(2)})$, $i \in [N]$, we train the network ϕ_W to minimize the following contrastive loss:

$$L_{\text{contrastive}}(\phi_W) := \frac{1}{N} \sum_{i \in [N]} \underbrace{-\text{sim}(z_i, z'_i)/\tau}_{\text{positive term}} + \log \underbrace{\left[\sum_{j \in [N]} \exp(\text{sim}(z_i, z'_j)/\tau) \right]}_{\text{negative term}} \quad (3.2.1)$$

where $\text{sim}(\cdot, \cdot)$ is the similarity metric, often defined as the cosine similarity, and τ is the so-called temperature hyper-parameter. Intuitively, minimizing the contrastive loss can be roughly viewed as trying to classify the representation z_i as z'_i instead of $z'_j, j \neq i$. It is a common belief that in order for the network ϕ_W to be able to “distinguish” data points X_i from $X_j, j \neq i$, merely minimizing the positive term of contrastive loss is not sufficient.

As shown by the papers [50, 311], the performance of contrastive learning depends critically on the negative term. But the BYOL method [109] managed to remove the negative term without

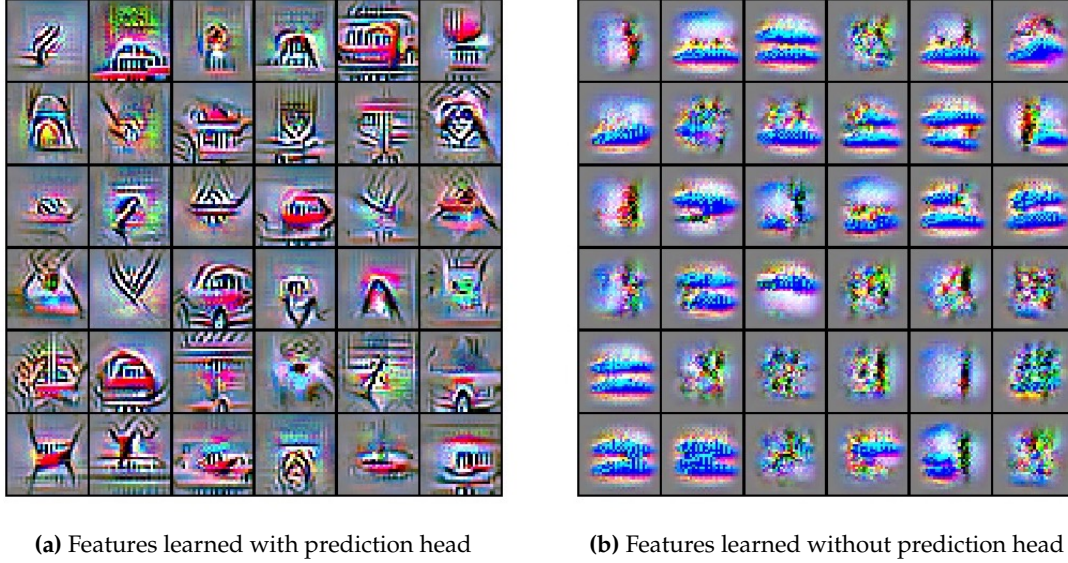


Figure 3.5: Prediction Heads Preserve Feature Diversity. Feature visualization of deep neural network. We visualized the features of an Wide-ResNet-16x5 following the BYOL method by Goyal et al. [103], a adversarial robust version of BYOL. Features learned with prediction head obviously have more variety than features learned without the prediction head. Our feature visualization technique follows from [8].

harm, by adding a trainable prediction head to the network architecture, which opened the new direction of non-contrastive self-supervised learning.

Non-contrastive self-supervised learning. We choose the SimSiam method [53] as our primary framework, whose difference with BYOL is a EMA component that is proven inessential in [53]. Following the same notations as above, except that $z'_i = \text{StopGrad}[\phi_W(X_i^{(2)})]$ is detached from gradient computation, the loss objective become: (the symmetric network version)

$$L'_{\text{SimSiam}} = \frac{1}{N} \sum_{i \in [N]} -\text{sim}(z_i, z'_i) \quad (3.2.2)$$

which is just the positive term in contrastive loss (3.2.1) (not divided by τ). Removing the negative term results in the existence of plenty trivial *global optimal* solutions. For example, the *complete collapse* refers to when $\phi_W(\cdot)$ is some constant vector function with zero variance. Another trivial solution called *dimensional collapse* [122], which is when all the coordinates $[\phi_W(\cdot)]_i$ has correlation ± 1 , meaning $\phi_W(\cdot)$ lies in a one-dimensional subspace of the representation space. The dimensional collapsed solution can minimize the objective (3.2.2) even when the network output $\phi_W(\cdot)$ is normalized by BN to avoid converging to a constant vector [122, 338].

However, by adding a *trainable prediction head* on top of z_i , the training miraculously succeeds and outputs a state-of-the-art feature extractor. Let $g(\cdot)$ be a shallow feed-forward network (often one or two-layer, or even simply linear), we train g and ϕ_W simultaneously on the following

objective:

$$L_{\text{SimSiam}} = \frac{1}{N} \sum_{i \in [N]} -\text{sim}(g(z_i), z'_i) \quad (3.2.3)$$

where z'_i is still detached from gradient computation. The $g(z_i) = g \circ \phi_W(X_i^{(1)})$ and the detached part $z'_i = \text{StopGrad}[\phi_W(X_i^{(2)})]$ are often called the *online* network and the *target* network respectively following [109], known as two branches of non-contrastive learning. Even when such a trainable prediction head is able to represent identity function, the network can still avoid the common collapsed solutions, which presents challenges in understanding their training process and the underlying mechanism of trainable prediction head.

Challenges of understanding non-contrastive learning. Although the non-contrastive losses (3.2.2) and (3.2.3) seems just a term of the contrastive loss (3.2.1), their behaviors are vastly different. As established in [311], the negative pairs are needed for learning all the discriminative features. Without the negative term, the learner has no explicit incentive to learn all the discriminative features from the objective (3.2.3), especially when the trainable prediction head can possibly be an identity map. Indeed, by setting $g(\cdot)$ to the identity map, problem (3.2.3) immediately turn back into (3.2.2) and has the same trivial collapsed global optima. It is one of the most typical examples of *implicit bias* of optimization in deep learning.

Empirically, the seminal paper [53] discovered that *even with trainable linear prediction head which can possibly learn identity mapping*, neural networks trained by SGD still avoid such collapsed solutions. Moreover, as we show in this paper, even with an identity-initialized linear prediction head, as long as we train the prediction head via SGD, it still produces results comparable to when using other types of prediction head. Our empirical evidence in Figure 3.2 suggests that understanding the asymmetry provided by the off-diagonal entries in the identity-initialized linear prediction head suffices to explain (most of) the mechanisms of the prediction head. This observation significantly simplifies the theoretical problem and makes the complete characterization of the training dynamics of the prediction head possible.

Nevertheless, understanding the trainable prediction head urges us to go beyond the traditional statistical framework and optimization landscape analysis. The recent development of the *feature learning theory* of neural networks [6, 8, 133, 134, 311] showed it is possible to directly analyze the training dynamics of neural networks in various supervised or self-supervised tasks. Inspired by this line of research and our observations, we consider understanding the optimization of identity-initialized prediction head the key to understanding the underlying mechanism of these methods, and the characterization of the training dynamics of the full network the major technical challenges.

3.3 Problem Setup

In this section, we present the setting of our theoretical results. We first define the data distribution.

Notations. We use O, Ω, Θ notations to hide universal constants with respect to d and $\tilde{O}, \tilde{\Omega}, \tilde{\Theta}$ notations to hide polynomial factors of $\log d$. We denote $a = o(1)$ if $a \rightarrow 0$ when $d \rightarrow \infty$. We use the notations $\text{poly}(d)$, $\text{polylog}(d)$ to represent large constant degree polynomials of d or $\log d$. We use $\mathcal{N}(\mu, \Sigma)$ to denote standard normal distribution in with mean μ and covariance matrix Σ . We use the bracket $\langle \cdot, \cdot \rangle$ to denote the inner product and $\|\cdot\|_2$ the ℓ_2 -norm in Euclidean space. And for a subspace $V \subset \mathbb{R}^d$, we denote $V^\perp$ as its orthogonal complement. We use $\mathbb{1}_B$ to denote the indicator function of event B .

Following the standard structure of image datasets, we consider data divided into patches, where each patch can contain either features or noises.

Definition 3.3.1 (data distribution and features). Let $X \sim \mathcal{D}$ be $X = (X_1, \dots, X_P) \in \mathbb{R}^{d \times P}$ where each $X_i \in \mathbb{R}^d$ is a patch. We assume that there are two feature vectors v_1, v_2 such that $\|v_\ell\|_2 = 1, \ell = 1, 2$ and are orthogonal to each other. To generate a sample X , we uniformly sampled $\ell \in [2]$ and generate for each $p \in [P]$:

$$X_p = z_p(X)v_\ell + \xi_p \mathbb{1}_{z_p=0}, \quad \mathbb{E}_{X \sim \mathcal{D}}[z_p(X)] = 0, \quad \forall p \in [P]$$

We denote $\mathcal{S}(X) = \{p : z_p(X) \neq 0\} \subseteq [P]$ as the set of feature patches and assume $z_p(X) = z_{p'}(X) \in \{0, \pm\alpha_\ell\}, \forall p, p' \in \mathcal{S}(X)$, i.e., all feature patches have the same direction of v_ℓ within the same X . We assume $P = \text{polylog}(d)$, $S(X) \equiv P_0 = \Theta(\log d)$ for every X . The assumption of ξ_p will be given in [Assumption 3.3.1](#). An intuitive illustration is given in [Figure 3.6](#).

Strong and weak features. We pick $\alpha_1 = 2^{\text{polyloglog}(d)}$ and $\alpha_2 = \alpha_1 / \text{polylog}(d)$. Hence v_1 is the *strong feature* and v_2 is the *weak feature*, and we want the learner network to learn both v_1, v_2 (but by different neurons) as their learning goal. This is a simplification of the real scenario where features show up more consistently across multiple patches of the images, while noises are local and roughly independent across different patches. Intuitively, we can think of the strong features in a dataset are the ones that show up more frequently or with larger magnitude, and weak features as those that show up rarely or with smaller magnitude, which is the common case in any practical dataset.

Remark 3.3.1. Our analysis can be easily generalized to settings of either (1) when $\alpha_1 = \alpha_2$ but the sampling of $\ell \in [2]$ is of non-equal probability (i.e., dataset imbalance setting); or (2) when the two features always co-occur in the same sample but not of the same strength. But we still require $\alpha_1, \alpha_2 \gg \text{polylog}(d)$ to simplify the analysis.

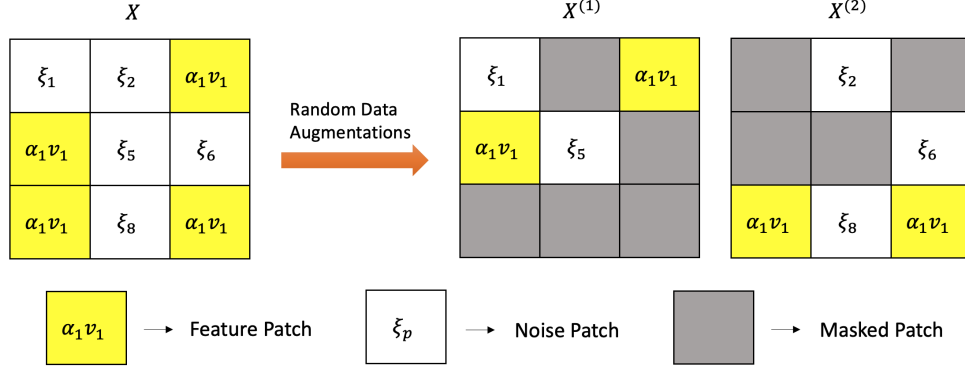


Figure 3.6: Synthetic Data And Augmentations. Illustration of the data distribution and data augmentations. Each data is equipped with a feature, either v_1 or v_2 , and contains a lot of noise patches. After the data augmentations, the positive pair $(X^{(1)}, X^{(2)})$ is constructed by randomly masking out half of non-overlapping patches for each positive sample. The reason for constructing positive pair with non-overlapping patches is because of the strong noise assumption we made in [Assumption 3.3.1](#) and the *feature decoupling* principle in [\[311\]](#).

Assumption 3.3.1 (noise). Denoting $V = \text{span}(v_1, v_2)$, we assume $\xi_p \in V^\perp$ is independent for each $p \in [P] \setminus S(X)$, where $X = (X_p)_{p \in [P]} \sim \mathcal{D}$, and:

- (a) For any unit vector $u \in V^\perp$, $\mathbb{E}[\langle \xi_p, u \rangle] = 0$, and $\mathbb{E}[\langle \xi_p, u \rangle^6] = \sigma^6$ for some $\sigma = \Theta(1)$;
- (b) It holds for some $\varrho \in [0, \frac{1}{d^{\Omega(1)}}]$ it holds $|\mathbb{E}[\langle u_1, \xi_p \rangle^3 \langle u_2, \xi_p \rangle^3]| \leq \varrho$ and $|\mathbb{E}[\langle u_1, \xi_p \rangle^5 \langle u_2, \xi_p \rangle]| \leq \varrho$ for any two vectors $u_1, u_2 \in \mathbb{R}^d$ that are orthogonal to each other.

Remark 3.3.2. A simple example of our noise ξ_p is the spherical Gaussian noise in $V^\perp$. Our [Assumption 3.3.1b](#) ensures that the prediction head cannot be used to cancel the noise correlation between different neurons. We point out that the features in our data can be learned via clustering, but we emphasize that we do not intend to compare our algorithm with any clustering method in this setting since our goal is to study how the prediction head helps in learning the features.

3.3.1 Learner Network

Following the SimSiam framework, the online and target network share the same encoder network in our setting, as explained in [Section 3.2](#). We consider the base encoder network f as a simple convolutional neural network: Let $W = (w_1, \dots, w_m) \in \mathbb{R}^{d \times m}$ be the weight matrix, where $w_i \in \mathbb{R}^d$, the encoder network f is defined by

$$f_j(X) := \sum_{p \in [P]} \sigma(\langle w_j, X_p \rangle), \quad \forall j \in [m]$$

Here we use the cubic activation function $\sigma(z) = z^3$, as polynomial activations are standard in literatures of deep learning theory [9, 14, 47, 96, 145, 165] and also has comparable performance in practice [9]. The (identity initialized) prediction head is defined as a matrix $E = [E_{i,j}]_{(i,j) \in [m]^2}$ with $E_{i,i} \equiv 1, i \in [m]$, where only the off-diagonals $E_{i,j}, i \neq j$ are trainable parameters. The *online network* $\tilde{F}$ is defined by: given $j \in [m]$, we let $F_j(X) := f_j(X) + \sum_{r \neq j} E_{j,r} f_r(X)$, and

$$\tilde{F}_j(X) := \text{BN}(F_j(X)) = \text{BN} \left[\sum_{p \in [P]} \left(\sigma(\langle w_j, X_p \rangle) + \sum_{r \neq j} E_{j,r} \sigma(\langle w_r, X_p \rangle) \right) \right]$$

where the batch normalization BN here is defined as follows: Given a batch of inputs $\{z_i\}_{i \in [N]}$,

$$\text{BN}(z_i) := \frac{z_i - \frac{1}{N} \sum_{i \in [N]} z_i}{\sqrt{\frac{1}{N} \sum_{i \in [N]} z_i^2 - \left(\frac{1}{N} \sum_{i \in [N]} z_i \right)^2}} \quad (3.3.1)$$

And the *target network* G is defined as follows: Given $j \in [m]$

$$\tilde{G}_j(X) := \text{BN}(G_j(X)) = \text{BN} \left[\sum_{p \in [P]} \sigma(\langle w_j, X_p \rangle) \right]$$

3.3.2 Training Algorithm

Data augmentation. We use a very simple data augmentation: for each data $X = (X_p)_{p \in [P]}$, we randomly and uniformly sample half of the patches $\mathcal{P} \subseteq [P]$ to generate two samples (which is the so-called *positive pair* in contrastive learning):

$$X^{(1)} = (X_p \mathbb{1}_{p \in \mathcal{P}})_{p \in [P]}, \quad X^{(2)} = (X_p \mathbb{1}_{p \notin \mathcal{P}})_{p \in [P]} \quad (3.3.2)$$

An intuitive illustration is given in Figure 3.6. Our data augmentation approach is similar to the common cropping augmentation used in contrastive learning [45, 272] and the patch masking strategy in generative pretraining [29, 114] and NLP pretraining [74]. It is also analogous to the data augmentations being studied in theoretical literatures [137, 175, 311] of self-supervised learning, especially the RandomMask augmentation in [311].

Non-contrastive loss function. Now we define the loss function as follows: we sample N data points $\{X_i\}_{i \in [N]}, X_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{D}$ and apply our data augmentation (3.3.2) to obtain $\mathcal{S} = \{X^{(i,1)}, X^{(i,2)}\}_{i \in [N]}$. Now we define

$$L_{\mathcal{S}}(W, E) := \frac{1}{N} \sum_{i \in [N]} \left\| \tilde{F}(X^{(i,1)}) - \text{StopGrad}[\tilde{G}(X^{(i,2)})] \right\|_2^2 \quad (3.3.3)$$

Algorithm 1: Training Algorithm

Input: data distribution $\mathcal{D}$, objective L_S (3.3.3), networks $\tilde{F}, \tilde{G}$, hyper-parameters T, N, η, η_E, m , and a bool variable $\text{TrainPredHead} = \text{True}$.

1. Initialize $w_j^{(0)} \sim \mathcal{N}(0, I_d/d) \forall j \in [m]$ i.i.d., and $E^{(0)} = I_m$;

For $t \in \{0, 1, 2, \dots, T-1\}$:

1. Sample $X^{(t,i)} \leftarrow (X_p^{(t,i)})_{p \in [P]} \sim \mathcal{D}, \forall i \in [N]$ i.i.d.;
2. Sample $\{\mathcal{P}^{(t,i)}\}_{i \in [N]}$ i.i.d., and obtain $\mathcal{S}_t \leftarrow \{X^{(t,i,1)}, X^{(t,i,2)}\}_{i \in [N]}$ via data augmentations

$$X^{(t,i,1)} \leftarrow (X_p^{(t,i)} \mathbb{1}_{p \in \mathcal{P}^{(t,i)}})_{p \in [P]}, \quad X^{(t,i,2)} \leftarrow (X_p^{(t,i)} \mathbb{1}_{p \notin \mathcal{P}^{(t,i)}})_{p \in [P]};$$

3. Perform stochastic gradient descent step to $W^{(t)} = (w_j^{(t)})_{j \in [m]}$ by

$$w_j^{(t+1)} \leftarrow w_j^{(t)} - \eta \nabla_{w_j} L_{\mathcal{S}_t}(W^{(t)}, E^{(t)});$$

If $\text{TrainPredHead} = \text{True}$:

1. update the off diagonal of prediction head $E^{(t)}$ by

$$\begin{aligned} E_{i,i}^{(t+1)} &\leftarrow 1, \\ E_{i,j}^{(t+1)} &\leftarrow E_{i,j}^{(t)} - \eta_E \nabla_{E_{i,j}} L_{\mathcal{S}_t}(W^{(t)}, E^{(t)}), \\ &\quad j \neq i, \quad i, j \in [m]. \end{aligned}$$

Else:

1. keep $E^{(t+1)} = I_m$.

$$= 2 - \frac{1}{N} \sum_{i \in [N]} \langle \tilde{F}(X^{(i,1)}), \text{StopGrad}[\tilde{G}(X^{(i,2)})] \rangle$$

where the StopGrad operator detach gradient computation of the target network $\tilde{G}(\cdot)$. This form of objective (3.3.3) is first defined in Grill et al. [106] and is equivalent to (3.2.3) in Chen and He [53] when $\tilde{F}$ and $\tilde{G}$ share the same encoder network $f(\cdot)$ and their outputs are normalized.

Intuition of the data augmentation. Our data augmentation is an analog of the the standard cropping data augmentation. In Definition 3.3.1, the features v_1, v_2 appear in multiple patches, but the noises are independent across different patches (see Figure 3.6). As our data augmentation produces positive pairs with non-overlapping patches, learning to emphasize noises cannot align the representations of the positive pair, but learning *either one of* the features $\phi(X) = \sum_p \sigma(\langle v_1, X_p \rangle)$ or $\phi(X) = \sum_p \sigma(\langle v_2, X_p \rangle)$ is sufficient. We consider learning the same feature v_i in all the neurons f_j in the encoder network f as the dimensional collapsed solution.

Initialization and hyper-parameters. At $t = 0$, we initialize W and E as $W_{i,j}^{(0)} \sim \mathcal{N}(0, \frac{1}{d})$ and $E^{(0)} = I_m$ and we only train the off-diagonal entries of $E^{(t)}$. For the simplicity of analysis, we let $m = 2$, which suffices to illustrate our main message. For the learning rates, we let $\eta \in (0, \frac{1}{\text{poly}(d)})$ be sufficiently small and $\eta_E \in [\frac{\eta}{\alpha_1^{O(1)}}, \frac{\eta}{\text{polylog}(d)}]$, which is smaller than η^1 .

Optimization algorithm Given the data augmentation and the loss function, we perform (stochastic) gradient descent on the training objective (3.3.3) as follows: at each iteration $t = 0, \dots, T-1$, we sample a new batch of augmented data $\mathcal{S}_t = \{X^{(t,i,1)}, X^{(t,i,2)}\}_{i \in [N]}$ and update

$$W^{(t+1)} = W^{(t)} - \eta \nabla_W L_{\mathcal{S}_t}(W^{(t)}, E^{(t)}), \quad E_{i,j}^{(t+1)} = E_{i,j}^{(t)} - \eta_E \nabla_{E_{i,j}} L_{\mathcal{S}_t}(W^{(t)}, E^{(t)}), \quad \forall i \neq j, i, j \in [m]$$

If we do not train the prediction head, we just simply keep $E^{(t)} \equiv I_m$. We summarize our algorithm in Algorithm 1.

3.4 Statements of Main Results

In this section, we shall present our main theoretical results on the mechanism of learning the prediction head in non-contrastive learning. To measure the correlation between neurons, we introduce the following notion: letting

$$\text{Var}(\psi(X)) := \mathbb{E}_{X \sim \mathcal{D}}[(\psi(X) - \mathbb{E}[\psi(X)])^2]$$

be the variance of any function ψ of $X \sim \mathcal{D}$, we denote the correlation $\text{Corr}(\psi(X), \psi'(X))$ of any two function ψ, ψ' over $\mathcal{D}$ as

$$\text{Corr}(\psi(X), \psi'(X)) := \frac{\mathbb{E}[(\psi(X) - \mathbb{E}[\psi(X)])(\psi'(X) - \mathbb{E}[\psi'(X)])]}{\sqrt{\text{Var}(\psi(X))} \sqrt{\text{Var}(\psi'(X))}}$$

Now we present the main theorem of training with a prediction head, and set $m = 2$.

Theorem 3.4.1 (learning with prediction head and BN, see Theorem 3.F.1). *For every $d > 2$, let $N \geq \text{poly}(d)$, $\eta \in (0, \frac{1}{\text{poly}(d)})$ be sufficiently small, and $\eta_E \in [\frac{\eta}{\alpha_1^{O(1)}}, \frac{\eta}{\text{polylog}(d)}]$. Then with probability $1 - o(1)$, after running Algorithm 1 for $T = \text{poly}(d)/\eta$ many iterations, we shall have for some $\ell \in [2]$:*

$$w_1^{(T)} = \beta_1 v_\ell + \varepsilon_1, \quad w_2^{(T)} = \beta_2 v_{3-\ell} + \varepsilon_2 \quad \text{with} \quad |\beta_1|, |\beta_2| = \Theta(1), \quad \|\varepsilon_1\|_2, \|\varepsilon_2\|_2 \leq \tilde{O}\left(\frac{1}{\sqrt{d}}\right)$$

Furthermore, the objective converges: $\mathbb{E}_{\mathcal{S} \sim \mathcal{D}^N}[L_{\mathcal{S}}(W^{(T)}, E^{(T)})] \leq \text{OPT} + \frac{1}{\text{poly}(d)} \leq O(\frac{1}{\log d})^2$.

¹ We conjecture that by modifying certain assumptions for the noise (especially by allowing the noise to span the feature subspace V), one can prove a similar result for the case $\eta_E = \eta$.

[Theorem 3.4.1](#) clearly shows the network learn all the desired features, even under huge imbalance between v_1 and v_2 . This leads to the following corollary.

Corollary 3.4.1. *Under the same hyper-parameter in [Theorem 3.4.1](#), with probability $1 - o(1)$, after running [Algorithm 1](#) for $T = \text{poly}(d)/\eta$ many iterations, we shall have that the learning avoids dimensional collapse:*

$$|\text{Corr}(f_1(X), f_2(X))| \leq O\left(\frac{1}{\sqrt{d}}\right).$$

In contrast, learning without the prediction head will result in learning only the strong feature v_1 in both neurons, which creates strong correlations between any two neurons. To emphasize that this problem cannot be alleviated by having more neurons, we let the number of neurons m be any positive integer in the following theorem.

Theorem 3.4.2 (learning without prediction head but with BN, see [Theorem 3.G.1](#)). *Let $N \geq \text{poly}(d)$, $\eta = o(1)$ and the number of neurons $m > 0$ be any positive integer. Suppose we freeze $E^{(t)} = I_m$ for all t , then with probability $1 - o(1)$, after running [Algorithm 1](#) with `TrainPredHead = False` for $T = \text{poly}(d)/\eta$ many iterations, we shall have:*

$$w_j^{(T)} = \beta_j v_1 + \varepsilon_j \quad \text{with} \quad |\beta_j| = \Theta(1), \quad \|\varepsilon_j\|_2 \leq \tilde{O}\left(\frac{1}{\sqrt{d}}\right) \quad \text{for all } j \in [m]$$

Furthermore, the objective converges: $\mathbb{E}_{S \sim \mathcal{D}^N}[L_S(W^{(T)}, E^{(T)})] \leq \text{OPT} + \frac{1}{\text{poly}(d)} \leq O\left(\frac{1}{\log d}\right)$. This means the collapsed solution also reaches the global minimum of the objective.

Note that since we have used BN as our output normalization instead of ℓ_2 -norm, the learner is immune to complete collapse and must have a certain variance in the outputs. Immediately, we have the following corollary.

Corollary 3.4.2. *Under the same hyper-parameter in [Theorem 3.4.2](#), with probability $1 - o(1)$, after running [Algorithm 1](#) with `TrainPredHead = False` for $T = \text{poly}(d)/\eta$ many iterations, we shall have dimensional collapse:*

$$|\text{Corr}(f_i(X), f_j(X))| \geq 1 - O\left(\frac{1}{\sqrt{d}}\right), \quad \text{for all } i, j \in [m].$$

² Under our data model [Definition 3.3.1](#), non-overlapping data augmentation [\(3.3.2\)](#) and learner network definition, the global minimum of our objective [\(3.3.3\)](#) in population is the following quantity:

$$\text{OPT} := \min_{W, E} \mathbb{E}_{S \sim \mathcal{D}^N}[L_S(W, E)] = 2 - 2 \frac{\mathbb{E}[|S(X) \cap \mathcal{P}| \cdot |S(X) \setminus \mathcal{P}|]}{\mathbb{E}[|S(X) \cap \mathcal{P}|^2]} = \Theta\left(\frac{1}{\log d}\right)$$

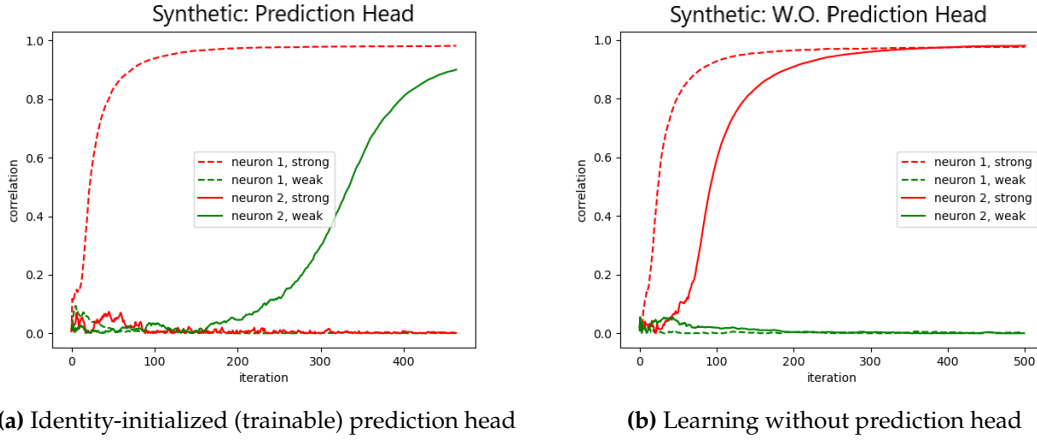


Figure 3.7: Prediction Heads Enable Weak-Feature Learning. The feature learning process over synthetic data. When trained with the prediction head, after the strong feature is learned in the faster learning neuron, the weak feature can be learned in the slower learning neuron. When trained without the prediction head, both neurons will learn the strong feature and ignore the weak feature.

Remark 3.4.1. Note that since we have used BN as our output normalization instead of ℓ_2 -norm, the learner is regularized to avoid complete collapse and must have a certain variance in its neurons. It is easier to obtain a complete collapse result when the network has ℓ_2 -normalized outputs and there is a low-variance feature (but not of smaller magnitude) in the data set, which we refrain from proving here.

How does using the prediction head or not create such a difference in features learned by the non-contrastive methods? We shall give some intuitions by digging through the training process and separately discuss the four phases of the training process.

3.5 The Four Phases of the Learning Process

We divide the complete training process into four phases: phase I for learning the stronger feature, phase II for the substitution effect, phase III for the acceleration effect, and the end phase for convergence. The first three phases explain how the prediction head can help learn the base encoder network, and the last phase of the training explains why the off-diagonal entries often shrink in the later stage of training.

3.5.1 Phase I: Learning the Stronger Feature

At the beginning of training, the stronger feature v_1 enjoys a much larger gradient as opposed to the weaker feature v_2 , so naturally, v_1 will be learned first. However, if for both neurons f_1, f_2 the speed of learning v_1 is the same, then we cannot argue the difference between them and will not be able to show the substitution from either one to another. Indeed, let us assume at

initialization, the neuron $w_1^{(0)}$ won the jackpot of having larger signal-to-noise ratio of $\langle w_j^{(0)}, v_1 \rangle$ between $f_j, j \in [2]$, then we can show the following result under our setting.

Lemma 3.5.1 (learning the stronger feature, see Lemma 3.C.8). *After some $t \geq T_1 = d^{2+o(1)}/\eta$, the feature v_1 in neuron f_1 will be learned to $\langle w_1^{(t)}, v_1 \rangle = \Omega(1)$, while all other features $\langle w_j^{(t)}, v_\ell \rangle = o(1)$ for $(j, \ell) \neq (1, 1)$ are small. And the prediction head $\|E^{(t)} - I_2\|_2 \leq d^{-\Omega(1)}$ is still close to the initialization.*

In this phase, the prediction head has not come into play. The substitution effect can only happen after the feature v_1 in neuron f_1 is learned to a certain degree, and neuron f_2 remains largely unlearned.

3.5.2 Phase II: The Substitution Effect

To illustrate the substitution effect, let us keep assuming that neuron $w_1^{(t)}$ has already learned some significant amount of the strong feature v_1 , say $w_1^{(t)} = \beta_1 v_1 + \text{residual}$ with $|\beta_1| = \Omega(\|\text{residual}\|)$. When this happens, we have the following result: (recall $f_j(\cdot), j \in [2]$ are the neurons of the base encoder network)

Lemma 3.5.2 (substitution effect, formal statement see Lemma 3.D.6). *After $|\langle w_1^{(t)}, v_1 \rangle| = \Omega(1)$ in $O(d^{2+o(1)}/\eta)$ iterations (as shown by Lemma 3.C.8), for much shorter time than learning $\langle w_1^{(t)}, v_1 \rangle$, we shall have $|E_{2,1}^{(t)}|$ increasing until $|E_{2,1}^{(t)} f_1(X^{(1)})| \gg |f_2(X^{(1)})|$ when X is equipped with feature v_1 . In other words, $E_{2,1}^{(t)} f_1(X^{(1)})$ is a substitute for the feature v_1 that should be learned by f_2 .*

Intuition of the substitution effect. After the stronger feature is learned in neuron f_1 , the optimal way to align two positive representations $F_2(X^{(1)}), G_2(X^{(2)})$ is no longer learning features in weight w_2 , but use the prediction head to “borrow” the features in f_1 and incorporate them into F_2 . This is how the substitution effect happens when trained with a prediction head.

Proof sketch for Lemma 3.5.2. Indeed, let us look at the learning of $E_{2,1}^{(t)}$. In this phase, $w_2^{(t)}$ and $E_{2,1}^{(t)}$ are roughly learned to maximize the following quantity:

$$\begin{aligned} \tilde{F}_2(X^{(1)}) \cdot \tilde{G}_2(X^{(2)}) &\propto (f_2(X^{(1)}) + E_{2,1}^{(t)} f_1(X^{(1)})) \times f_2(X^{(2)}) \\ &\approx \sum_{\ell \in [2]} \alpha_\ell \left(\langle w_2^{(t)}, v_\ell \rangle^6 + E_{2,1}^{(t)} \cdot \langle w_1^{(t)}, v_\ell \rangle^3 \cdot \langle w_2^{(t)}, v_\ell \rangle^3 \right) \end{aligned}$$

As the neuron $f_1(\cdot)$ is already learned with feature v_1 , in order to maximize the RHS, we can either try to maximize $\sum_{\ell \in [2]} \langle w_2^{(t)}, v_\ell \rangle^6$, or to maximize $E_{2,1}^{(t)} \langle w_1^{(t)}, v_1 \rangle^3 \cdot \langle w_2^{(t)}, v_1 \rangle^3 \approx E_{2,1}^{(t)} \langle w_2^{(t)}, v_1 \rangle^3$. In this case, the more efficient choice is to learn $|E_{2,1}^{(t)}|$ to substitute for maximizing $\langle w_2^{(t)}, v_\ell \rangle^3$. Actually, because of the high signal-to-noise ratio of learning $w_2^{(t)}$ than $E_{2,1}^{(t)}$, feature $\langle w_2^{(t)}, v_\ell \rangle$ is learned with slower pace than $E_{2,1}^{(t)}$, so that Lemma 3.5.2 can be shown.

3.5.3 Phase III: The Acceleration Effect

After the substitution of v_1 in F_2 , our concern is, whether or not $w_2^{(t)}$ will learn v_2 and only v_2 eventually, so that we can obtain a diverse representation? The answer is yes, as we summarize in the following lemma.

Lemma 3.5.3 (acceleration effect, formal statement see [Lemma 3.E.6](#)). *After $E_{2,1}^{(t)}$ is learned in [Lemma 3.5.2](#), learning v_2 in $w_2^{(t)}$ will be much faster than v_1 , until $\|w_2^{(t)} - \beta_2 v_2\| \leq o(1)$ for some $\beta_2 = \Theta(1)$.*

The acceleration effect is caused by the interactions between the prediction head, the stop gradient operation, and the normalization method (which in this case is the batch normalization). We shall explain these interactions with insights from our theoretical analyses below.

What is the role of the stop-gradient? Thanks to the StopGrad operation, when we compute the gradient $-\nabla_{w_2} F_2(X^{(1)}) \cdot \text{StopGrad}[G_2(X^{(2)})]$ to learn f_2 , this negative gradient will only try to maximize $f_2(X^{(1)}) \cdot f_2(X^{(2)})$, rather than to maximize $f_2(X^{(2)}) \cdot F_2(X^{(1)})$. This is because the stop-gradient is on G not on F : while F_2 has a large component of v_1 borrowed from f_1 using E , G_2 does not have this component. So the gradient of F_2 is to align with the features in G_2 that does not contain many v_1 , while the gradient of G_2 is to align with the features in F_2 that contains a lot of v_1 . Thus the stop gradient on G help ignore the feature borrowed from f_1 using prediction head E and ensures the slower learning neuron f_2 will focus on learning feature v_2 .

What is the role of the output normalization? Again due to the StopGrad operation, the gradient of $\tilde{F}_2$ is taken with respect to the ratio $f_2(X^{(1)})/\sqrt{\text{Var}[F_2(X^{(1)})]}$. As gradient descent tries to maximize this ratio, a direct computation gives

$$\nabla_{w_2} \frac{f_2(X^{(1)})}{\sqrt{\text{Var}(F_2(X^{(1)}))}} = \frac{\nabla_{w_2} f_2(X^{(1)}) \cdot \text{Var}(F_2(X^{(1)})) - f_2(X^{(1)}) \cdot \nabla_{w_2} \text{Var}(F_2(X^{(1)}))}{\text{Var}(F_2(X^{(1)}))^{3/2}}$$

From some calculation, we can obtain the above gradient is proportional to

$$\sum_{\ell \in [2]} \left([E_{2,1}^{(t)} \langle w_1^{(t)}, v_{3-\ell} \rangle]^2 + \text{Var}[f_2(X^{(1)})] \right) \langle \nabla_{w_2} f_2(X^{(1)}), v_\ell \rangle v_\ell$$

which borrow the *substituted feature* $v_{3-\ell}$ from $f_1(\cdot)$ to adjust the gradient of v_ℓ in $f_2(\cdot)$, via the prediction head $E_{2,1}^{(t)}$. Without the output normalization, the learning of v_1 will dominate that of v_2 even when we train the prediction head.

Proof sketch for [Lemma 3.5.3](#). At this stage, when we are updating the weights of $w_2^{(t)}$, we are simultaneously maximizing $f_2(X^{(1)}) \cdot f_2(X^{(2)})$ and also minimizing the normalizing constants $\sqrt{\text{Var}[F_2(X^{(1)})]}$. This two goals are in slight conflict because of the normalization, and by careful

calculation the gradients are roughly given by (interpreting the expectation as empirical)

$$\langle -\nabla_{w_2} L_S, v_\ell \rangle \propto \mathbb{E} \left[\left([E_{2,1}^{(t)} \langle w_1^{(t)}, v_{3-\ell} \rangle^3]^2 + \mathbf{Var}[f_2(X^{(1)})] \right) \cdot f_2(X^{(2)}) \langle -\nabla_{w_2} f_2(X^{(1)}), v_\ell \rangle \right]$$

Because of the learning of f_1 and the substitution effect, we now know $[E_{2,1}^{(t)} \langle w_1^{(t)}, v_{3-\ell} \rangle^3]^2$ is much larger when $\ell = 2$, which accelerates the learning of v_2 in $w_2^{(t)}$ to surpass that of v_1 and leads to [Lemma 3.5.3](#).

3.5.4 The End Phase: Convergence

As the weak features are learned, we have already obtained a good encoder network $f(\cdot)$ as shown in [Theorem 3.4.1](#). The rest of our analysis is to understand what the prediction head converges to in polynomial time. Actually, our [Theorem 3.F.1](#) also contains the following result: **Proposition 3.5.1** (convergence of the prediction head, see [Theorem 3.F.1c](#)). After some $t \geq T = \text{poly}(d)/\eta$ iterations, we shall have $\|E^{(t)} - I_2\|_F \leq \frac{1}{\text{poly}(d)}$.

This result also implies that after learning the weak feature v_2 is complete, the off-diagonal entries of the prediction head will reverse their trajectory and converge to zero at the end of training. While we admit that only some of our real-world experiments show the convergence to zero for the off-diagonal entries of the prediction head, most of the experiments do display a rise and fall trajectory pattern of off-diagonal entries consistently.

3.6 Additional Related Work

Self-supervised learning The area of self-supervised learning has evolved at a tremendous speed in recent years. It has created huge success in natural language processing [[35](#), [74](#), [331](#)] and established a paradigm where the networks are first trained on an unsupervised pretext task and then be finetuned in downstream applications. In vision, supervised pretraining had been the go-to choice until representations learned by contrastive learning [[39](#), [48](#), [54](#), [55](#), [82](#), [93](#), [113](#), [224](#), [271](#)] became dominant in many downstream tasks. Another type of self-supervised learning is the generative learning [[29](#), [114](#), [228](#)], which also gives promising results in downstream adaptations. Interesting applications such as [[224](#), [229](#)] also illustrate the power of contrastive learning in multiple domains.

Theory of self-supervised learning The theoretical side of self-supervised learning developed quickly due to the success of contrastive learning, which is closely related to the methods we are studying. Since Arora et al. [[20](#)], lots of papers have studied the properties of contrastive learning, as mentioned in the introduction. [[50](#), [238](#)] discussed many interesting phenomena associated with the negative term in contrastive learning. Saunshi et al. [[245](#)] provided pieces of evidence that contrastive loss is function class-specific rather than agnostic. Wen and Li [[311](#)] took a feature learning view to understand contrastive learning with neural networks, which inspired

our analysis in the non-contrastive setting. For generative self-supervised learning, [158, 269] provides downstream performance guarantees for generative pretrained models. [244, 303] studied the natural language tasks, where the data are sequentially structured. Liu et al. [175] gave a recovery guarantee for tensors in generative learning under hidden Markov models. [7] analyzed multi-layer generative adversarial networks and provided an optimization guarantee for their stochastic gradient descent ascent algorithm.

Feature learning theory of deep learning Our theoretical results are also inspired by the recent progress of the feature learning theory of neural networks [6, 8, 133, 143, 169, 170, 352]. Li et al. [169] initiate the study of the speed difference in learning different types of features. [170] developed theory for learning two-layer neural networks over Gaussian distribution beyond the *neural tangent kernel* (NTK) [11, 12, 13, 19, 78]. Allen-Zhu and Li [8] studied the origin of adversarial examples and how adversarial training help in robustify the networks. [6] tried to explain ensemble and knowledge distillation under multi-view assumptions. Techniques in this paper are built on this line of research, as the non-convex nature of these analyses allows us to describe the interaction between neural networks, optimization algorithms, and the structures of data. [5, 9] also obtained results separating deep neural networks and shallow models such as kernel methods. Before this recent progress, [36, 77, 167, 168, 259, 273, 345] also studied how shallow neural networks can learn on certain simple data distributions, but all of them focus on the supervised learning. There are also plenty of studies [18, 60, 108, 138, 181, 232, 261] on the implicit bias of optimization in deep learning, but none of their techniques can be applied to the setting of self-supervised learning.

3.7 Experiment Details

The framework we use in our experiments is shown in Figure 3.8. We use a modified version of the codebase shared by the authors of [82], and we use the same data augmentation in their implementation. All our experiments (except for Figure 3.5 and Figure 3.7) use the following architecture and hyper-parameters: we choose standard ResNet-18 as base encoder architecture, 0.003 as the learning rate for Adam optimizer, a two-layer MLP with ReLU activation and 512 hidden neurons as the projection head, an identity-initialized but diagonally froze linear matrix (with shape (64x64)) as the prediction head and a non-tracking-stats, non-affine, non-momentum BN layer as the output normalization. Our experiments in Figure 3.3 use the same architecture and hyper-parameters, but some runs are trained with EMA with momentum 0.99, with output BN replaced by ℓ_2 -norm or using different prediction heads (such as a two-layer MLP or a linear head, with Pytorch default initialization). Evaluation in Figure 3.2 is by training a linear classifier on top of frozen encoder with no data augmentation.

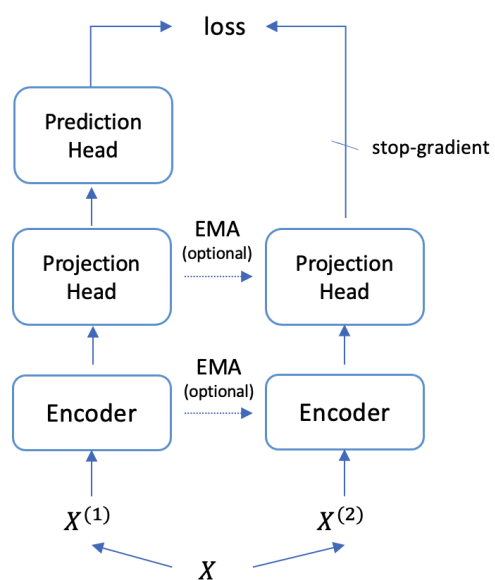


Figure 3.8: **Non-Contrastive Learning Framework.** The framework used in our experiments.

Appendix

3.A Proofs for the Shared Tensor-Power Bound

Several chapters analyze feature growth through closely related tensor-power recursions. The following parameterized statement consolidates the version used in both the non-contrastive learning and length-generalization analyses.

Lemma 3.A.1 (Tensor-power growth, adapted from [6]). *Let $r \geq 2$ be an integer, and consider an increasing sequence $x_t \geq 0$ satisfying*

$$x_{t+1} = x_t + \eta C_t x_t^r, \quad C_t > 0.$$

Suppose that the sequence reaches a threshold $A > x_0$. Then, for every $\delta > 0$ and $\eta \in (0, 1)$,

$$\begin{aligned} \sum_{t \geq 0: x_t \leq A} \eta C_t &\geq \left[\frac{\delta(1+\delta)^{-1}}{(1+\delta)^{r-1} - 1} \left(1 - \left(\frac{(1+\delta)x_0}{A} \right)^{r-1} \right) - \frac{O(\eta A^r) \log(A/x_0)}{x_0 \log(1+\delta)} \right] \frac{1}{x_0^{r-1}}, \\ \sum_{t \geq 0: x_t \leq A} \eta C_t &\leq \left[\frac{(1+\delta)^{r-1}}{r-1} + \frac{O(\eta A^r) \log(A/x_0)}{x_0 \log(1+\delta)} \right] \frac{1}{x_0^{r-1}}. \end{aligned}$$

Chapter 3 applies Lemma 3.A.1 with $r = q$, while Chapter 5 applies it with $r = q - 1$.

We will be working with population gradients throughout the entire appendix. Indeed, since our algorithms use fresh random samples at each iteration, one can easily obtain from standard concentration inequalities an empirical estimate of population gradients up to $\frac{1}{\text{poly}(d)}$ error with $N = \text{poly}(d)$ samples. So we can obtain the same proofs in finite sample case as long as the training ends before some $T = \text{poly}(d)/\eta$. Now we give some notations and warm-up calculations.

3.B Notations and Gradients

In this section, we will give some useful notations and warm-up computations for the technical proofs in subsequent sections. We summarize here the notations that will also be defined in later sections:

Notations. We denote $\mathcal{E}_j = \mathbb{E}[\langle w_j, \xi_p \rangle^6]$, $\mathcal{E}_{j,3-j} = \mathbb{E}[(\langle w_j, \xi_p \rangle^3 + E_{j,3-j} \langle w_{3-j}, \xi_p \rangle^3)^2]$, and

$$\begin{aligned} C_0 &= \frac{\mathbb{E}[|S(X) \cap \mathcal{P}| \cdot |S(X) \setminus \mathcal{P}|]}{2}, & C_1 &= \frac{\mathbb{E}[|S(X) \cap \mathcal{P}|^2]}{2}, & C_2 &= P - |S(X)|, \\ \bar{B}_{j,\ell}^3 &= \text{StopGrad}[\langle w_j, v_\ell \rangle^3], & B_{j,\ell} &= \langle w_j, v_\ell \rangle, & Q_j &= (\mathbb{E}[\text{StopGrad}[G_j^2(X^{(2)})]])^{-1/2}. \end{aligned}$$

and

$$\begin{aligned} U_j &:= \mathbb{E}[F_j^2(X^{(1)})] = \sum_{\ell \in [2]} C_1 \alpha_\ell^6 (B_{j,\ell}^3 + E_{j,3-j} B_{3-j,\ell}^3)^2 + C_2 \mathcal{E}_{j,3-j} \\ H_{j,\ell} &:= C_1 \alpha_\ell^6 (B_{j,\ell}^3 + E_{j,3-j} B_{3-j,\ell}^3)^2 + C_2 \mathcal{E}_{j,3-j}, \\ K_{j,\ell} &:= C_1 \alpha_\ell^6 (B_{j,\ell}^3 + E_{j,3-j} B_{3-j,\ell}^3) (B_{j,3-\ell}^3 + E_{j,3-j} B_{3-j,3-\ell}^3) \end{aligned}$$

Moreover, we denote $\Phi_j := Q_j / U_j^{3/2}$, and (recall $V := \text{span}(v_1, v_2)$)

$$R_j := \langle \Pi_{V^\perp} w_j, w_j \rangle \quad R_{1,2} := \langle \Pi_{V^\perp} w_1, w_2 \rangle \quad \bar{R}_{1,2} := \frac{\langle \Pi_{V^\perp} w_1, w_2 \rangle}{\|\Pi_{V^\perp} w_1\|_2 \|\Pi_{V^\perp} w_2\|_2}$$

For any $j \in [2]$, the gradient $-\nabla_{w_j} L(W, E)$ can be decomposed as

$$\begin{aligned} -\nabla_{w_j} L(W, E) &= \sum_{\ell \in [2]} (\Lambda_{j,\ell} + \Gamma_{j,\ell} - \Upsilon_{j,\ell}) v_\ell - \sum_{(j', \ell) \in [2] \times [2]} \Sigma_{j', \ell} \nabla_{w_j} \mathcal{E}_{j', 3-j'} \\ \Lambda_{j,\ell} &:= C_0 \Phi_j \alpha_\ell^6 B_{j,\ell}^5 H_{j,3-\ell} \\ \Gamma_{j,\ell} &:= C_0 \Phi_{3-j} E_{3-j,j} \alpha_\ell^6 B_{3-j,\ell}^3 B_{j,\ell}^2 H_{3-j,3-\ell} \\ \Upsilon_{j,\ell} &:= C_0 \alpha_{3-\ell}^6 \left(\Phi_j B_{j,3-\ell}^3 B_{j,\ell}^2 K_{j,\ell} + \Phi_{3-j} E_{3-j,j} B_{3-j,3-\ell}^3 B_{j,\ell}^2 K_{3-j,\ell} \right) \\ \Sigma_{j,\ell} &:= C_0 C_2 \Phi_j \alpha_\ell^6 B_{j,\ell}^3 (B_{j,\ell}^3 + E_{j,3-j} B_{3-j,\ell}^3) \end{aligned}$$

Sometimes we need to decompose $\Upsilon_{j,\ell} = \Upsilon_{j,\ell,1} + \Upsilon_{j,\ell,2}$ which is straightforward from its expression.

In [Section 3.E](#), we further define

$$\begin{aligned} \Xi_j^{(t)} &= C_0 C_1 \alpha_1^6 \alpha_2^6 \Phi_j^{(t)} \left((B_{1,1}^{(t)})^6 (B_{2,2}^{(t)})^6 + (B_{2,1}^{(t)})^6 (B_{1,2}^{(t)})^6 \right) \\ \Delta_{j,\ell}^{(t)} &= C_0 \Phi_j^{(t)} \alpha_\ell^6 (B_{j,\ell}^{(t)})^3 (B_{3-j,\ell}^{(t)})^3 C_2 \mathcal{E}_{j,3-j}^{(t)} \end{aligned}$$

for the gradients of the prediction head.

3.B.1 Gradient Computation

Let us $L(W, E)$ to be the population version of the objective. Because $\mathbb{E}[F_j(X^{(1)})]$ and $\mathbb{E}[G_j(X^{(2)})]$ are both zero (which can be verified easily from the zero-mean assumptions of $z_p(X)$ and ξ_p), a direct computation gives:

$$L(W, E) = 2 - \sum_{j \in [2]} \frac{\mathbb{E}[F_j(X^{(1)}) \cdot \text{StopGrad}[G_j(X^{(2)})]]}{\sqrt{\mathbb{E}[F_j^2(X^{(1)})]} \sqrt{\mathbb{E}[\text{StopGrad}[G_j^2(X^{(2)})]]}}$$

We first calculate the normalizing quantity $\mathbb{E}[F_j^2(X^{(1)})]$:

$$\begin{aligned}
\mathbb{E}[F_j^2(X^{(1)})] &= \mathbb{E} \left[\left(\sum_{p \in [P]} \sigma(\langle w_j, X_p^{(1)} \rangle) + E_{j,3-j} \sigma(\langle w_{3-j}, X_p^{(1)} \rangle) \right)^2 \right] \\
&= \frac{1}{2} \sum_{\ell \in [2]} \mathbb{E} [|S(X) \cap \mathcal{P}|^2 \alpha_\ell^6 \langle w_j, v_\ell \rangle^3 + E_{j,3-j} \langle w_{3-j}, v_\ell \rangle^3] \\
&\quad \text{(Because all signal patches has the same sign within the same data)} \\
&\quad + \mathbb{E} [|\mathcal{P} \setminus S(X)| \langle w_j, \xi_p \rangle^3 + E_{j,3-j} \langle w_{3-j}, \xi_p \rangle^3] \\
&\quad \text{(Because noise patches are independent and have mean zero)} \\
&= \sum_{\ell \in [2]} \alpha_\ell^6 \langle w_j, v_\ell \rangle^3 + E_{j,3-j} \langle w_{3-j}, v_\ell \rangle^3 \frac{\mathbb{E} [|S(X) \cap \mathcal{P}|^2]}{2} + (P - |S(X)|) \mathcal{E}_{j,3-j}
\end{aligned}$$

where we let

$$\begin{aligned}
\mathcal{E}_{j,3-j} &\stackrel{\text{def}}{=} \mathbb{E} [(\langle w_j, \xi_p \rangle^3 + E_{j,3-j} \langle w_{3-j}, \xi_p \rangle^3)^2] \\
&= \mathbb{E} [\langle w_j, \xi_p \rangle^6 + 2E_{j,3-j} \langle w_j, \xi_p \rangle^3 \langle w_{3-j}, \xi_p \rangle^3 + E_{j,3-j}^2 \langle w_{3-j}, \xi_p \rangle^6]
\end{aligned}$$

On the other hand, we have

$$\begin{aligned}
&\mathbb{E}[F_j(X^{(1)}) \cdot \text{StopGrad}[G_j(X^{(2)})]] \\
&= \mathbb{E} \left[\left(\sum_{p \in [P]} \sigma(\langle w_j, X_p^{(1)} \rangle) + E_{j,3-j} \sigma(\langle w_{3-j}, X_p^{(1)} \rangle) \right) \times \left(\sum_{p \in [P]} \sigma(\langle w_j, X_p^{(2)} \rangle) \right) \right] \\
&= \frac{1}{2} \sum_{\ell \in [2]} \mathbb{E} \left[\sum_{p \in S(X) \cap \mathcal{P}} \alpha_\ell^3 \langle w_j, v_\ell \rangle^3 + E_{j,3-j} \langle w_{3-j}, v_\ell \rangle^3 \times \sum_{p \in S(X) \setminus \mathcal{P}} \alpha_\ell^3 \text{StopGrad}[\langle w_j, v_\ell \rangle^3] \right] \\
&= \sum_{\ell \in [2]} \alpha_\ell^6 \langle w_j, v_\ell \rangle^3 + E_{j,3-j} \langle w_{3-j}, v_\ell \rangle^3 \cdot \text{StopGrad}[\langle w_j, v_\ell \rangle^3] \cdot \frac{\mathbb{E}[|S(X) \cap \mathcal{P}| \cdot |S(X) \setminus \mathcal{P}|]}{2}
\end{aligned}$$

Now, by denoting

$$\begin{aligned}
C_0 &= \frac{\mathbb{E}[|S(X) \cap \mathcal{P}| \cdot |S(X) \setminus \mathcal{P}|]}{2}, \quad C_1 = \frac{\mathbb{E}[|S(X) \cap \mathcal{P}|^2]}{2}, \quad C_2 = P - |S(X)|, \\
\bar{B}_{j,\ell}^3 &= \text{StopGrad}[\langle w_j, v_\ell \rangle^3], \quad B_{j,\ell} = \langle w_j, v_\ell \rangle, \quad Q_j = (\mathbb{E}[\text{StopGrad}[G_j^2(X^{(2)})]])^{-1/2}.
\end{aligned}$$

we denote $U_j := \mathbb{E}[F_j^2(X^{(1)})]$, where the expanded expression is

$$U_j = \mathbb{E}[F_j^2(X^{(1)})] = \sum_{\ell \in [2]} C_1 \alpha_\ell^6 (B_{j,\ell}^3 + E_{j,3-j} B_{3-j,\ell}^3)^2 + C_2 \mathcal{E}_{j,3-j}$$

and we can rewrite the objective as follows

$$L(W, E) = 2 - \sum_{j \in [2]} \sum_{\ell \in [2]} \frac{Q_j C_0 \alpha_\ell^6 \bar{B}_{j,\ell}^3 (B_{j,\ell}^3 + E_{j,3-j} B_{3-j,\ell}^3)}{U_j^{1/2}} \quad (3.B.1)$$

Now denote

$$\begin{aligned} H_{j,\ell} &= C_1 \alpha_\ell^6 (B_{j,\ell}^3 + E_{j,3-j} B_{3-j,\ell}^3)^2 + C_2 \mathcal{E}_{j,3-j}, \\ K_{j,\ell} &= C_1 \alpha_\ell^6 (B_{j,\ell}^3 + E_{j,3-j} B_{3-j,\ell}^3) (B_{j,3-\ell}^3 + E_{j,3-j} B_{3-j,3-\ell}^3) \end{aligned}$$

It is easy to calculate

$$\begin{aligned} Q_j^{-2} &= \mathbb{E}[\text{StopGrad}[G_j^2(X^{(2)})]] \\ &= \mathbb{E} \left[\left(\sum_{p \in [P]} \sigma(\langle w_j, X_p^{(2)} \rangle) \right)^2 \right] \\ &= \frac{1}{2} \sum_{\ell \in [2]} \alpha_\ell^6 \langle w_j, v_\ell \rangle^6 \mathbb{E}[|S(X) \cap \mathcal{P}|^2] + \mathbb{E}[|\mathcal{P} \setminus S(X)| \langle w_j, \xi_p \rangle^6] \\ &= \sum_{\ell \in [2]} C_1 \alpha_\ell^6 B_{j,\ell}^6 + C_2 \mathcal{E}_j \end{aligned}$$

where $\mathcal{E}_j = \mathbb{E}[\langle w_j, \xi_p \rangle^6]$. And thus the gradient can be computed as (notice $\bar{B}_{j,\ell}^3 = B_{j,\ell}^3$)

$$\begin{aligned} -\nabla_{w_j} L(W, E) &= \sum_{\ell \in [2]} \left(\frac{C_0 Q_j \alpha_\ell^6 H_{j,3-\ell} B_{j,\ell}^5}{U_j^{3/2}} \right) v_\ell + \sum_{\ell \in [2]} \left(\frac{C_0 Q_{3-j} E_{3-j,j} \alpha_\ell^6 B_{3-j,\ell}^3 B_{j,\ell}^2 H_{3-j,3-\ell}}{U_{3-j}^{3/2}} \right) v_\ell \\ &\quad - \sum_{\ell \in [2]} \left(\frac{C_0 Q_j \alpha_{3-\ell}^6 B_{j,3-\ell}^3 B_{j,\ell}^2 K_{j,\ell}}{U_j^{3/2}} + \frac{C_0 Q_{3-j} E_{3-j,j} \alpha_{3-\ell}^6 B_{3-j,3-\ell}^3 B_{j,\ell}^2 K_{3-j,\ell}}{U_{3-j}^{3/2}} \right) v_\ell \\ &\quad - \sum_{j' \in [2]} \sum_{\ell \in [2]} \frac{C_0 C_2 Q_{j'} \alpha_\ell^6 B_{j',\ell}^3 (B_{j',\ell}^3 + E_{j',3-j'} B_{3-j',\ell}^3)}{U_{j'}^{3/2}} \nabla_{w_j} \mathcal{E}_{j',3-j'} \\ &= \sum_{\ell \in [2]} (\Lambda_{j,\ell} + \Gamma_{j,\ell} - \Upsilon_{j,\ell}) v_\ell - \sum_{(j',\ell) \in [2] \times [2]} \Sigma_{j',\ell} \nabla_{w_j} \mathcal{E}_{j',3-j'} \quad (3.B.2) \end{aligned}$$

where

$$\begin{aligned} \nabla_{w_j} \mathcal{E}_{j,3-j} &= 6 \mathbb{E}[\langle w_j, \xi_p \rangle^5 \xi_p + E_{j,3-j} \langle w_j, \xi_p \rangle^2 \langle w_{3-j}, \xi_p \rangle^3 \xi_p] \\ \nabla_{w_j} \mathcal{E}_{3-j,j} &= 6 \mathbb{E}[E_{3-j,j}^2 \langle w_j, \xi_p \rangle^5 \xi_p + E_{3-j,j} \langle w_{3-j}, \xi_p \rangle^3 \langle w_j, \xi_p \rangle^2 \xi_p] \end{aligned}$$

As for the gradient of the prediction head, we can calculate

$$\begin{aligned}
-\nabla_{E_{j,3-j}} L(W, E) &= \sum_{\ell \in [2]} \frac{C_0 Q_j \alpha_\ell^6 B_{j,\ell}^3 B_{3-j,\ell}^3 U_j}{U_j^{3/2}} \\
&\quad - \sum_{\ell \in [2]} \frac{C_0 Q_j \alpha_\ell^6 B_{j,\ell}^3 (B_{j,\ell}^3 + E_{j,3-j} B_{3-j,\ell}^3) \sum_{\ell' \in [2]} C_1 \alpha_{\ell'}^6 (B_{j,\ell'}^3 + E_{j,3-j} B_{3-j,\ell'}^3) B_{3-j,\ell'}^3}{U_j^{3/2}} \\
&\quad - \sum_{\ell \in [2]} \frac{C_0 C_2 Q_j \alpha_\ell^6 B_{j,\ell}^3 (B_{j,\ell}^3 + E_{j,3-j} B_{3-j,\ell}^3)}{U_j^{3/2}} \nabla_{E_{j,3-j}} \mathcal{E}_{j,3-j} \\
&= \sum_{\ell \in [2]} \frac{C_0 Q_j \alpha_\ell^6 B_{j,\ell}^3 (B_{3-j,\ell}^3 H_{j,3-\ell} - B_{3-j,3-\ell}^3 K_{j,3-\ell})}{U_j^{3/2}} \\
&\quad - \sum_{\ell \in [2]} \Sigma_{j,\ell} \mathbb{E} [2 \langle w_j, \xi_p \rangle^3 \langle w_{3-j}, \xi_p \rangle^3 + 2 E_{j,3-j} \langle w_{3-j}, \xi_p \rangle^6]
\end{aligned}$$

where $\Sigma_{j,\ell}$ is defined in (3.B.2). In fact, all the above gradient expressions can be simplified by letting $\Phi_j := Q_j / U_j^{3/2}$ for $j \in [2]$, which is what we shall do in later sections.

Summarizing the notations. We shall define some useful notations to simplify the proof. We define $V = \text{span}(v_1, v_2)$. Let Π_A be the projection operator to subspace $A \subset \mathbb{R}^d$, then

$$R_j := \langle \Pi_{V^\perp} w_j, w_j \rangle \quad R_{1,2} := \langle \Pi_{V^\perp} w_1, w_2 \rangle \quad \bar{R}_{1,2} := \frac{\langle \Pi_{V^\perp} w_1, w_2 \rangle}{\|\Pi_{V^\perp} w_1\|_2 \|\Pi_{V^\perp} w_2\|_2}$$

3.B.2 Some Useful Bounds for Gradients

In this section we use the superscript (t) to denote the iteration t during training. Below we present a claim which comes from direct calculations of $\Sigma_{j,\ell}^{(t)}$ and $\nabla_{w_j} \mathcal{E}_{j,3-j}^{(t)}$, which is very useful in the following sections.

Claim 3.B.1 (on $\Sigma_{j,\ell}^{(t)}$ and $\nabla_{w_j} \mathcal{E}_{j,3-j}^{(t)}$). Let $R_j, R_{1,2}$ be defined as above, then we have

- (a) $\Sigma_{j,\ell}^{(t)} = O(\Sigma_{1,1}^{(t)}) \frac{(B_{j,\ell}^{(t)})^6 + E_{j,3-j}^{(t)} (B_{3-j,\ell}^{(t)})^3 (B_{j,\ell}^{(t)})^3 \Phi_j^{(t)}}{(B_{1,1}^{(t)})^6 \Phi_1^{(t)}}$;
- (b) $\langle \nabla_{w_j} \mathcal{E}_{j,3-j}^{(t)}, \Pi_{V^\perp} w_j^{(t)} \rangle = \Theta([R_j^{(t)}]^3) \pm \Theta(E_{j,3-j}^{(t)}) (\bar{R}_{1,2}^{(t)} + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2}$;
- (c) $\langle \nabla_{w_j} \mathcal{E}_{j,3-j}^{(t)}, w_j^{(t)} \rangle = \Theta((E_{3-j,j}^{(t)})^2) [R_j^{(t)}]^3 \pm O(E_{3-j,j}^{(t)}) (\bar{R}_{1,2}^{(t)} + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2}$;
- (d) $\langle \nabla_{w_j} \mathcal{E}_{j,3-j}^{(t)}, w_{3-j}^{(t)} \rangle = (\Theta(\bar{R}_{1,2}^{(t)}) \pm \varrho) [R_j^{(t)}]^{5/2} [R_{3-j}^{(t)}]^{1/2} + O(E_{j,3-j}^{(t)}) R_j^{(t)} [R_{3-j}^{(t)}]^2$;
- (e) $\langle \nabla_{w_j} \mathcal{E}_{j,3-j}^{(t)}, w_{3-j}^{(t)} \rangle = ((E_{3-j,j}^{(t)})^2 (\Theta(\bar{R}_{1,2}^{(t)}) \pm \varrho) [R_j^{(t)}]^{5/2} [R_{3-j}^{(t)}]^{1/2} + O(E_{3-j,j}^{(t)}) R_j^{(t)} [R_{3-j}^{(t)}]^2)$

Proof. The part on $\Sigma_{j,\ell}^{(t)}$ is trivial from its expression, we shall focus on proving (b) – (d).

On $\langle \nabla_{w_j} \mathcal{E}_{j',3-j'}^{(t)}, w_j^{(t)} \rangle$: If $j = j'$, then

$$\begin{aligned} \langle \nabla_{w_j} \mathcal{E}_{j,3-j}^{(t)}, w_j^{(t)} \rangle &= \Theta(1) \mathbb{E}[\langle w_j^{(t)}, \xi_p \rangle^6 + E_{j,3-j}^{(t)} \langle w_j^{(t)}, \xi_p \rangle^3 \langle w_{3-j}^{(t)}, \xi_p \rangle^3] \\ &= \Theta(1) \mathbb{E}[\langle w_j^{(t)}, \xi_p \rangle^6] + O(E_{j,3-j}^{(t)}) \mathbb{E}[\langle w_j^{(t)}, \xi_p \rangle^3 \langle w_{3-j}^{(t)}, \xi_p \rangle^3] \\ &\quad - \langle (I - \bar{w}_{j,t} \bar{w}_{j,t}^\top) w_{3-j}^{(t)}, \xi_p \rangle^3] \\ &\quad + O(E_{j,3-j}^{(t)}) \mathbb{E}[\langle w_j^{(t)}, \xi_p \rangle^3 \langle (I - \bar{w}_{j,t} \bar{w}_{j,t}^\top) w_{3-j}^{(t)}, \xi_p \rangle^3] \end{aligned}$$

Write $\bar{w}_{j,t} = \frac{\Pi_{V^\perp} w_j^{(t)}}{\|\Pi_{V^\perp} w_j^{(t)}\|_2}$, we can derive

$$\begin{aligned} &\mathbb{E}[\langle w_j^{(t)}, \xi_p \rangle^3 \langle w_{3-j}^{(t)}, \xi_p \rangle^3 - \langle (I - \bar{w}_{j,t} \bar{w}_{j,t}^\top) w_{3-j}^{(t)}, \xi_p \rangle^3] \\ &= \mathbb{E}[\langle w_j^{(t)}, \xi_p \rangle^3 \langle \bar{w}_{j,t} \bar{w}_{j,t}^\top w_{3-j}^{(t)}, \xi_p \rangle O(\langle w_{3-j}^{(t)}, \xi_p \rangle^2)] \\ &= O\left(\frac{R_{1,2}^{(t)}}{\|\Pi_{V^\perp} w_j^{(t)}\|_2^2}\right) \mathbb{E}[\langle w_j^{(t)}, \xi_p \rangle^4 \langle w_{3-j}^{(t)}, \xi_p \rangle^2] \\ &\leq O\left(\frac{R_{1,2}^{(t)}}{\|\Pi_{V^\perp} w_j^{(t)}\|_2^2}\right) \mathbb{E}[\langle w_j^{(t)}, \xi_p \rangle^6]^{\frac{2}{3}} \mathbb{E}[\langle w_{3-j}^{(t)}, \xi_p \rangle^6]^{\frac{1}{3}} \quad (\text{by Hölder's inequality}) \\ &\leq O(\bar{R}_{1,2}^{(t)}) \|\Pi_{V^\perp} w_j^{(t)}\|_2^3 \|\Pi_{V^\perp} w_{3-j}^{(t)}\|_2^3 \end{aligned}$$

and by our assumption on noise ξ_p , we also have

$$\mathbb{E}[\langle w_j^{(t)}, \xi_p \rangle^3 \langle (I - \bar{w}_{j,t} \bar{w}_{j,t}^\top) w_{3-j}^{(t)}, \xi_p \rangle^3] \leq O(\varrho) \|\Pi_{V^\perp} w_j^{(t)}\|_2^3 \|\Pi_{V^\perp} w_{3-j}^{(t)}\|_2^3$$

Combined with the fact that $\mathbb{E}[\langle w_j^{(t)}, \xi_p \rangle^6] = O(\|\Pi_{V^\perp} w_j^{(t)}\|_2^3)$, we can get

$$\langle \nabla_{w_j} \mathcal{E}_{j,3-j}^{(t)}, w_j^{(t)} \rangle = O(\|\Pi_{V^\perp} w_j^{(t)}\|_2^6) \pm O(E_{j,3-j}^{(t)})(R_{1,2}^{(t)} + \varrho) \|\Pi_{V^\perp} w_j^{(t)}\|_2^3 \|\Pi_{V^\perp} w_{3-j}^{(t)}\|_2^3$$

when $j' = 3 - j$, we also have

$$\begin{aligned} \langle \nabla_{w_j} \mathcal{E}_{3-j,j}^{(t)}, w_j^{(t)} \rangle &= \Theta(1) \mathbb{E}[(E_{3-j,j}^{(t)})^2 \langle w_j^{(t)}, \xi_p \rangle^6 + E_{3-j,j}^{(t)} \langle w_j^{(t)}, \xi_p \rangle^3 \langle w_{3-j}^{(t)}, \xi_p \rangle^3] \\ &= O((E_{3-j,j}^{(t)})^2) \|\Pi_{V^\perp} w_j^{(t)}\|_2^6 \pm O(E_{3-j,j}^{(t)})(R_{1,2}^{(t)} + \varrho) \|\Pi_{V^\perp} w_j^{(t)}\|_2^3 \|\Pi_{V^\perp} w_{3-j}^{(t)}\|_2^3 \end{aligned}$$

On $\langle \nabla_{w_j} \mathcal{E}_{j',3-j'}^{(t)}, w_{3-j}^{(t)} \rangle$: when $j' = j$, we have

$$\begin{aligned} \langle \nabla_{w_j} \mathcal{E}_{j,3-j}^{(t)}, w_{3-j}^{(t)} \rangle &= O(1) \mathbb{E}[\langle w_j^{(t)}, \xi_p \rangle^5 \langle w_{3-j}^{(t)}, \xi_p \rangle + E_{j,3-j}^{(t)} \langle w_j^{(t)}, \xi_p \rangle^2 \langle w_{3-j}^{(t)}, \xi_p \rangle^4] \\ &= O(1) \mathbb{E}[\langle w_j^{(t)}, \xi_p \rangle^5 \langle (I - \bar{w}_{j,t} \bar{w}_{j,t}^\top + \bar{w}_{j,t} \bar{w}_{j,t}^\top) w_{3-j}^{(t)}, \xi_p \rangle] \\ &\quad + O(1) \mathbb{E}[E_{j,3-j}^{(t)} \langle w_j^{(t)}, \xi_p \rangle^2 \langle w_{3-j}^{(t)}, \xi_p \rangle^4] \end{aligned} \quad (3.B.3)$$

Using Hölder's inequality and our assumption on ξ_p , we have

$$\mathbb{E}[\langle w_j^{(t)}, \xi_p \rangle^5 \langle (I - \bar{w}_{j,t} \bar{w}_{j,t}^\top + \bar{w}_{j,t} \bar{w}_{j,t}^\top) w_{3-j}^{(t)}, \xi_p \rangle] \lesssim \varrho \|\Pi_{V^\perp} w_j^{(t)}\|_2^5 \|\Pi_{V^\perp} w_{3-j}^{(t)}\|_2$$

In the meantime, we also have

$$\mathbb{E}[\langle w_j^{(t)}, \xi_p \rangle^5 \langle \bar{w}_{j,t} \bar{w}_{j,t}^\top w_{3-j}^{(t)}, \xi_p \rangle] = \Theta(\bar{R}_{1,2}^{(t)}) \mathbb{E}[\langle w_j^{(t)}, \xi_p \rangle^6] [R_j^{(t)}]^{-1/2} [R_{3-j}^{(t)}]^{1/2} = \Theta(\bar{R}_{1,2}^{(t)}) [R_j^{(t)}]^{5/2} [R_{3-j}^{(t)}]^{1/2}$$

for the last term in (3.B.3), we can also use Hölder's inequality to get

$$E_{j,3-j}^{(t)} \mathbb{E}[\langle w_j^{(t)}, \xi_p \rangle^2 \langle w_{3-j}^{(t)}, \xi_p \rangle^4] \lesssim E_{j,3-j}^{(t)} \mathbb{E}[\langle w_j^{(t)}, \xi_p \rangle^6]^{1/3} \mathbb{E}[\langle w_{3-j}^{(t)}, \xi_p \rangle^6]^{2/3} \lesssim E_{j,3-j}^{(t)} R_j^{(t)} [R_{3-j}^{(t)}]^2$$

Therefore, we can combine above analysis to get

$$\langle \nabla_{w_j} \mathcal{E}_{j,3-j}^{(t)}, w_{3-j}^{(t)} \rangle = (\Theta(\bar{R}_{1,2}^{(t)}) \pm \varrho) [R_j^{(t)}]^{5/2} [R_{3-j}^{(t)}]^{1/2} + O(E_{j,3-j}^{(t)}) R_j^{(t)} [R_{3-j}^{(t)}]^2$$

When $j' = 3 - j$, we also have

$$\begin{aligned} \langle \nabla_{w_j} \mathcal{E}_{3-j,j}^{(t)}, w_{3-j}^{(t)} \rangle &= 6 \mathbb{E}[(E_{3-j,j}^{(t)})^2 \langle w_j^{(t)}, \xi_p \rangle^5 \langle w_{3-j}^{(t)}, \xi_p \rangle + E_{3-j,j}^{(t)} \langle w_j^{(t)}, \xi_p \rangle^2 \langle w_{3-j}^{(t)}, \xi_p \rangle^4] \\ &= 6(E_{3-j,j}^{(t)})^2 (\Theta(\bar{R}_{1,2}^{(t)}) \pm \varrho) [R_j^{(t)}]^{5/2} [R_{3-j}^{(t)}]^{1/2} + E_{3-j,j}^{(t)} R_j^{(t)} [R_{3-j}^{(t)}]^2 \end{aligned}$$

which proves the claim. $\square$

3.C Phase I: Learning the Stronger Feature

In this section, we shall discuss the initial phase of learning the stronger feature. Firstly, we establish some properties at the initialization for our induction afterwards.

Initialization properties. We prove the following properties for our network at initialization. Recall our initialization is $w_j^{(0)} \sim \mathcal{N}(0, I_d/d)$, $\forall j \in [2]$ and $E^{(0)} = I_2$.

Lemma 3.C.1 (properties at initialization). *Recall that without loss of generality we let $|B_{1,1}^{(0)}| = \max_{j \in [2]} |B_{j,1}^{(0)}|$. With probability $1 - o(1)$, the following holds:*

- (a) $\|w_j^{(0)}\|_2^2 = 1 \pm \tilde{O}(\frac{1}{\sqrt{d}})$ for all $j \in [2]$, and $|\langle w_1^{(0)}, w_2^{(0)} \rangle| \leq \tilde{O}(\frac{1}{\sqrt{d}})$;

- (b) $\max_{j,\ell} |B_{j,\ell}^{(0)}| \leq O(\sqrt{\log d/d})$ and $\min_{j,\ell} |B_{j,\ell}^{(0)}| \geq \Omega(\frac{1}{\log d}) \max_{j,\ell} |B_{j,\ell}^{(0)}|$;
- (c) $|B_{1,1}^{(0)}| \geq |B_{2,1}^{(0)}|(1 + \frac{1}{\log d})$;
- (d) $\mathcal{E}_j^{(0)} = (1 - O(\frac{1}{d^3}))\sigma^6 \|w_j^{(0)}\|_2^6 = \Theta(1)$ for all $j \in [2]$;
- (e) $H_{j,\ell}^{(0)} = C_2 \mathcal{E}_j^{(0)}(1 + \tilde{O}(\frac{1}{\sqrt{d}}))$ for all $(j, \ell) \in [2] \times [2]$;
- (f) $U_j^{(0)} = C_2 \mathcal{E}_j^{(0)}(1 + \tilde{O}(\frac{\alpha^6}{\sqrt{d}}))$ for all $j \in [2]$;
- (g) $(Q_j^{(0)})^{-2} = C_2 \mathcal{E}_j^{(0)}(1 + \tilde{O}(\frac{\alpha^6}{\sqrt{d}}))$ for all $j \in [2]$;
- (h) $K_{j,\ell}^{(0)} \leq \tilde{O}(\alpha^6/d^3)$ for all $(j, \ell) \in [2] \times [2]$.

Let us first introduce a fact about Gaussian ratio distribution without proof.

Fact 3.C.1 (Gaussian ratio distribution). If X and Y are two independent standard Gaussian variables, then the probability density of $Z = X/Y$ is $p(z) = \frac{1}{\pi(1+z^2)}$, $z \in (-\infty, \infty)$.

- Proof of Lemma 3.C.1.* a. Norm bound comes from simple χ^2 concentration inequality and our initialization $w_j^{(0)} \sim \mathcal{N}(0, \frac{I_d}{d})$. The inner product bound comes from Gaussian concentration.
- b. It is from a direct calculation under our initialization, and some application of Gaussian c.d.f. and a union bound.
- c. It is from a probability distribution of Gaussian ratio distribution from Fact 3.C.1 to bound the probability of $|B_{1,1}^{(0)}|/|B_{2,1}^{(0)}| \leq (1 + \frac{1}{\log d})$ (WLOG we let $|B_{1,1}^{(0)}| = \max_{j \in [2]} |B_{j,1}^{(0)}|$).
- d. It can be directly proven from our assumption on noise ξ_p in the subspace $V^\perp$ and (a).
- e. Since at the initialization we have $B_{j,\ell}^{(0)} = \tilde{O}(\frac{1}{\sqrt{d}})$, $j, \ell \in [2]$ and $E_{j,3-j}^{(0)} = 0$, it is easy to directly upper bound the errors.
- f. Again from $B_{j,\ell}^{(0)} = \tilde{O}(\frac{1}{\sqrt{d}})$, $\forall j, \ell \in [2]$ at initialization and a direct upper bound.
- g. Proof is similar to (e).
- h. Directly from a naive upper bound using (b).

□

3.C.1 Induction in Phase I

We define phase I as all iterations $t \leq T_1$, where $T_1 := \min\{t : B_{1,1}^{(t)} \geq 0.01\}$, we will prove the existence of T_1 at the end of this section. We state the following induction hypotheses, which will hold throughout the phase I:

Induction Hypothesis 3.C.1. For each $t \leq T_1$, all of the followings hold:

- (a). $\|w_j^{(t)}\|_2 = \|w_j^{(0)}\|_2 \pm \tilde{O}(\rho + \frac{1}{\sqrt{d}})$ for each $j \in [2]$;
- (b). $|B_{1,2}^{(t)}|, |B_{2,1}^{(t)}|, |B_{2,2}^{(t)}| = \tilde{\Theta}(\frac{1}{\sqrt{d}})$;
- (c). $|B_{1,1}^{(t)}| \geq \Omega(\frac{1}{\log d}) \max(|B_{1,2}^{(t)}|, |B_{2,2}^{(t)}|, |B_{2,1}^{(t)}|)$;
- (d). $|E_{1,2}^{(t)}| \leq \tilde{O}(\rho + \frac{1}{\sqrt{d}})^{\frac{\eta_E}{\eta}} |B_{1,1}^{(t)}|$ and $|E_{2,1}^{(t)}| \leq \tilde{O}(\frac{1}{d})$;
- (e). $R_1^{(t)}, R_2^{(t)} = \Theta(1)$, $|R_{1,2}^{(t)}| \leq \tilde{O}(\rho + \frac{1}{\sqrt{d}})$

Remark 3.C.1. Since we have chosen $\eta_E \leq \eta$ and $\varrho \leq \frac{1}{d^{\Omega(1)}}$, *Induction 3.C.1d* implies $|E_{j,3-j}^{(t)}| = o(1)$ throughout $t \leq T_1$.

We shall prove the above induction holds in later sections, but first we need some useful claims assuming our induction holds in this phase.

3.C.2 Computing Variables at Phase I

Firstly we establish a claim controlling the noise terms $\mathcal{E}_j, \mathcal{E}_{j,3-j}$ during this phase.

Claim 3.C.1. At each iteration $t \leq T_1$, if *Induction 3.C.1* holds, then

- (a) $\mathcal{E}_1^{(t)} = \mathcal{E}_2^{(t)} \pm O(\sum_{\ell \in [2]} |B_{j,\ell}^{(t)}| + \tilde{O}(\varrho + \frac{1}{\sqrt{d}}))$
- (b) $\mathcal{E}_j^{(t)} = \mathcal{E}_j^{(0)} \pm O(\sum_{\ell \in [2]} |B_{j,\ell}^{(t)}| + \tilde{O}(\varrho + \frac{1}{\sqrt{d}}))$
- (c) $\mathcal{E}_{j,3-j}^{(t)} = \mathcal{E}_j^{(t)} \pm \tilde{O}(E_{j,3-j}^{(t)}(\varrho + \frac{1}{\sqrt{d}}) + (E_{j,3-j}^{(t)})^2);$

Proof. For (a), we can simply write down

$$\mathcal{E}_j^{(t)} = \mathbb{E}[\langle w_j, \xi_p \rangle^6] = \sigma^6 \|\Pi_{V^\perp} w_j^{(t)}\|_2^6$$

Note that by *Induction 3.C.1a* we always have $\|w_j^{(t)}\|_2 = \|w_j^{(0)}\|_2 \pm \tilde{O}(\varrho + \frac{1}{\sqrt{d}})$, and by *Lemma 3.C.1a* we also have $\|w_j^{(0)}\|_2 = (1 \pm \tilde{O}(\frac{1}{\sqrt{d}}))\|w_j^{(0)}\|_2$, which implies

$$\begin{aligned} \|\Pi_{V^\perp} w_j^{(t)}\|_2 - \|\Pi_{V^\perp} w_{3-j}^{(t)}\|_2 &= \|w_j^{(t)}\|_2 - \|w_{3-j}^{(t)}\|_2 \pm O\left(\sum_{j,\ell \in [2]^2} B_{j,\ell}^{(t)}\right) \\ &= \|w_j^{(0)}\|_2 - \|w_{3-j}^{(0)}\|_2 \pm O\left(\sum_{j,\ell \in [2]^2} B_{j,\ell}^{(t)}\right) \pm \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) \\ &= \tilde{O}(\frac{1}{\sqrt{d}}) \pm O\left(\sum_{j,\ell \in [2]^2} B_{j,\ell}^{(t)}\right) \pm \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) \end{aligned}$$

By the elementary equality $x^n - y^n = (x - y) \sum_{0 \leq i \leq n-1} x^i y^{n-1-i}$, we can obtain (a). The proof of (b) is almost the same as (a), and the proof of (c) is just direct calculation. $\square$

Equipped with *Claim 3.C.1*, we can establish the following lemma, which will be frequently applied to bound the gradient in our induction argument.

Lemma 3.C.2 (variables control in phase I). Suppose *Induction 3.C.1* holds at some iteration $t \leq T_1$, then we have:

- (a) if $\forall \ell \in [2], \alpha_\ell |B_{j,\ell}^{(t)}| \leq O(1)$, then $\Phi_j^{(t)} = (C_2 \mathcal{E}_j^{(t)})^{-2} (1 \pm \frac{1}{\text{polylog}(d)})$;
- (b) if $\exists \ell \in [2], |B_{j,\ell}^{(t)}| \geq \Omega(\frac{1}{\alpha_\ell})$, then $\Phi_j^{(t)} = O((C_2 \mathcal{E}_j^{(t)} + \sum_{\ell \in [2]} C_1 \alpha_\ell^6 (B_{j,\ell}^{(t)})^6)^{-2})$;
- (c) if $\alpha_\ell |B_{j,\ell}^{(t)}| \leq O(1)$, $H_{j,\ell}^{(t)} = C_2 \mathcal{E}_j^{(t)} (1 + \frac{1}{\text{polylog}(d)}) = \Theta(C_2)$, otherwise $H_{j,\ell}^{(t)} \in [\Omega(C_2), \tilde{O}(\alpha_\ell^6)]$
- (d) $|K_{j,\ell}^{(t)}| \leq \tilde{O}(\alpha_\ell^6 / d^{3/2})$

Proof. (a) From our assumptions that $|B_{1,2}^{(t)}|, |B_{2,1}^{(t)}|, |B_{2,2}^{(t)}| \leq \tilde{O}(\frac{1}{\sqrt{d}})$ and $\alpha_1 B_{1,1}^{(t)} \leq O(1)$, and also the fact that $\mathcal{E}_j^{(t)} = \Omega(\sigma^6) = \Omega(1)$, $C_2 = \Theta(\text{polylog}(d)) \gg C_1$, we can calculate

$$\begin{aligned} U_j^{(t)} &= \sum_{\ell \in [2]} C_1 \alpha_\ell^6 ((B_{j,\ell}^{(t)})^3 + E_{j,3-j}^{(t)} (B_{3-j,\ell}^{(t)})^3)^2 + C_2 \mathcal{E}_{j,3-j}^{(t)} \\ &= O(C_1) + C_2 \mathcal{E}_j^{(t)} + \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) \\ &= C_2 \mathcal{E}_j^{(t)} (1 \pm \frac{1}{\text{polylog}(d)}) \end{aligned}$$

Meanwhile, we can also compute similarly

$$Q_j^{(t)} = \sum_{\ell \in [2]} C_1 \alpha_\ell^6 (B_{j,\ell}^{(t)})^6 + C_2 \mathcal{E}_j = C_2 \mathcal{E}_j^{(t)} (1 \pm \frac{1}{\text{polylog}(d)})$$

Therefore $\Phi_j^{(t)} = Q_j^{(t)} / (U_j^{(t)})^{3/2} = (C_2 \mathcal{E}_j^{(t)} (1 \pm \frac{1}{\text{polylog}(d)}))^{-2}$ as desired.

(b) The proof is similar to that of (a).

(c) when $\alpha_1 B_{1,1}^{(t)} \leq O(1)$, the proof is similar to (a). When $\alpha_1 B_{1,1}^{(t)} \geq O(1)$, we have from [Induction 3.C.1a](#) and $H_{j,\ell}^{(t)}$'s expression that

$$H_{j,\ell}^{(t)} = C_1 \alpha_\ell^6 ((B_{j,\ell}^{(t)})^3 + E_{j,3-j}^{(t)} (B_{3-j,\ell}^{(t)})^3)^2 + C_2 \mathcal{E}_{j,3-j}^{(t)} \leq \tilde{O}(\alpha_\ell^6)$$

And since $T_1 := \min\{t : B_{1,1}^{(t)} \geq 0.01\}$, so for $t \leq T_1$, we have

$$H_{j,\ell}^{(t)} \geq C_2 \mathcal{E}_{j,3-j}^{(t)} \stackrel{(1)}{\geq} C_2 \mathcal{E}_j^{(t)} - |E_{j,3-j}^{(t)}| \stackrel{(2)}{\geq} \Omega(C_2)$$

where (1) is from [Claim 3.C.1b](#) and (2) is from [Induction 3.C.1d](#).

(d) Since we have assumed $|B_{1,2}^{(t)}|, |B_{2,1}^{(t)}|, |B_{2,2}^{(t)}| \leq \tilde{O}(\frac{1}{\sqrt{d}})$, it is direct to bound $|K_{j,\ell}^{(t)}| \leq \tilde{O}(\alpha_\ell^6 / d^{1.5})$. $\square$

Claim 3.C.2 (about $\Sigma_{j,\ell}^{(t)}$ and $\nabla_{w_j} \mathcal{E}_{j',3-j'}^{(t)}$). If [Induction 3.C.1](#) holds at iteration $t \leq T_1$, then

- (a) $\Sigma_{j,\ell}^{(t)} = O(\Lambda_{1,1}^{(t)} B_{1,1}^{(t)}) \frac{(B_{j,\ell}^{(t)})^6 + E_{j,3-j}^{(t)} (B_{3-j,\ell}^{(t)})^3 (B_{j,\ell}^{(t)})^3 \Phi_j^{(t)}}{(B_{1,1}^{(t)})^6} \frac{\Phi_j^{(t)}}{\Phi_1^{(t)}}$;
- (b) $\langle \nabla_{w_j} \mathcal{E}_{j,3-j}^{(t)}, w_j^{(t)} \rangle = O(1) \pm O(E_{j,3-j}^{(t)}) (R_{1,2}^{(t)} + \varrho)$;
- (c) $\langle \nabla_{w_j} \mathcal{E}_{3-j,j}^{(t)}, w_j^{(t)} \rangle = O((E_{3-j,j}^{(t)})^2) \pm O(E_{3-j,j}^{(t)}) (R_{1,2}^{(t)} + \varrho)$;
- (d) $|\langle \nabla_{w_j} \mathcal{E}_{j,3-j}^{(t)}, w_{3-j}^{(t)} \rangle| = O(R_{1,2}^{(t)} + \varrho) + O(E_{j,3-j}^{(t)})$;
- (e) $|\langle \nabla_{w_j} \mathcal{E}_{3-j,j}^{(t)}, w_{3-j}^{(t)} \rangle| = O(R_{1,2}^{(t)} + \varrho) (E_{3-j,j}^{(t)})^2 + O(E_{3-j,j}^{(t)})$

Proof. Notice that $\|\Pi_{V^\perp} w_j^{(t)}\|_2 = \Theta(1), \forall j \in [2]$ for $t \leq T_1$, which is because of $\|w_j^{(t)}\|_2 = \sqrt{2} \pm o(1)$ from [Induction 3.C.1a](#) and $\max_{j,t} |B_{j,t}^{(t)}| < 0.02^3$. Now we can apply [Claim 3.B.1](#) to obtain the bounds. $\square$

3.C.3 Gradient Lemmas for Phase I

We first present an interesting lemma regarding the effects of Batch-Normalization on the gradients of weights. The following lemma allow us maintain the norm of weights to above a constant throughout phase I.

Lemma 3.C.3 (effects of BN on gradients). *For any $W = (w_1, w_2)$ and E , it holds*

$$(a) \sum_{j \in [2]} \langle \nabla_{w_j} L(W, E), w_j \rangle = 0;$$

Further, if [Induction 3.C.1](#) holds for each $t \leq T_1$, we have

$$(b) |\langle \nabla_{w_j} L(W^{(t)}, E^{(t)}), w_j^{(t)} \rangle| \leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) |\Lambda_{1,1}| \sum_{j \in [2]} |E_{j,3-j}^{(t)}| \text{ for each } j \in [2].$$

Proof. Proof of (a): We first calculate the gradient term as follows:

$$\begin{aligned} \nabla_W L(W, E) &= \nabla_W \sum_{j \in [2]} \frac{\mathbb{E}[F_j(X^{(1)}) \cdot \text{StopGrad}[G_j(X^{(2)})]]}{\sqrt{\mathbb{E}[F_j^2(X^{(1)})]} \sqrt{\mathbb{E}[\text{StopGrad}[G_j^2(X^{(2)})]]}} \\ &= \sum_{j \in [2]} \frac{\mathbb{E}[(\nabla_W F_j(X^{(1)})) \cdot [G(X^{(2)})]_j] \cdot \mathbb{E}[F_j^2(X^{(1)})]}{(\mathbb{E}[F_j^2(X^{(1)})])^{3/2} \sqrt{\mathbb{E}[G_j^2(X^{(2)})]}} \\ &\quad - \sum_{j \in [2]} \frac{\mathbb{E}[(\nabla_W F_j(X^{(1)})) \cdot F_j(X^{(1)})] \cdot \mathbb{E}[[F_j(X^{(1)}) \cdot [G(X^{(2)})]_j]}{(\mathbb{E}[F_j^2(X^{(1)})])^{3/2} \sqrt{\mathbb{E}[G_j^2(X^{(2)})]}} \end{aligned}$$

Since by our definition $\langle \nabla_W F_j(X^{(1)}), W \rangle = \sum_{i \in [2]} \langle \nabla_{w_i} [F_j(X^{(1)})], w_i \rangle = 3[F_j(X^{(1)})]$, we immediately have $\sum_{j \in [2]} \langle \nabla_{w_j} L(W, E), w_j \rangle = 0$.

Proof of (b): Firstly we define a new notion

$$\nabla_{i,j} = \nabla_{w_i} \frac{\mathbb{E}[F_j(X^{(1)}) \cdot \text{StopGrad}[G_j(X^{(2)})]]}{\sqrt{\mathbb{E}[F_j^2(X^{(1)})]} \sqrt{\mathbb{E}[\text{StopGrad}[G_j^2(X^{(2)})]]}}$$

Then it is straightforward to verify that $\sum_{i \in [2]} \langle \nabla_{i,j}, w_i \rangle = 0$ for any $j \in [2]$, which implies that $|\langle \nabla_{j',j}, w_{j'} \rangle| = |\langle \nabla_{3-j',j}, w_{3-j'} \rangle|$. So in order to obtain an upper bound for $|\langle \nabla_{w_j} L(W, E), w_j \rangle| = |\sum_{j' \in [2]} \langle \nabla_{j,j'}, w_{j'} \rangle|$, we only need to upper bound $|\langle \nabla_{j,j'}, w_{3-j'} \rangle|$, each of which can be calculated

³ due to our choice of $\eta = \frac{1}{\text{poly}(d)}$ is small, we can make sure when $T_1 = \min\{t : B_{1,1}^{(t)} \geq 0.01\}, B_{1,1}^{(T_1)} < 0.02$.

as (ignoring all time superscript (t))

$$|\langle \nabla_{3-j,j}, w_{3-j} \rangle| = \frac{\mathbb{E} \left[\sum_{p \in [P] \cap \mathcal{P}} E_{j,3-j} \sigma(\langle w_{3-j}, X_p \rangle) \cdot [G(X^{(2)})]_j \cdot \mathbb{E}[F_j^2(X^{(1)})] \right]}{(\mathbb{E}[F_j^2(X^{(1)})])^{3/2} \sqrt{\mathbb{E}[G_j^2(X^{(2)})]}} \\ - \frac{\mathbb{E} \left[\sum_{p \in [P] \cap \mathcal{P}} E_{j,3-j} \sigma(\langle w_{3-j}, X_p \rangle) \cdot F_j(X^{(1)}) \right] \cdot \mathbb{E}[[F_j(X^{(1)}) \cdot [G(X^{(2)})]_j]}]}{(\mathbb{E}[F_j^2(X^{(1)})])^{3/2} \sqrt{\mathbb{E}[G_j^2(X^{(2)})]}}$$

Now we compute

$$\mathbb{E} \left[\sum_{p \in [P] \cap \mathcal{P}} E_{j,3-j} \sigma(\langle w_{3-j}, X_p \rangle) [G(X^{(2)})]_j \right] = \mathbb{E} \left[\sum_{p \in [P] \cap \mathcal{P}} E_{j,3-j} \sigma(\langle w_{3-j}, X_p \rangle) \sum_{p \in [P] \setminus \mathcal{P}} \sigma(\langle w_j, X_p \rangle) \right] \\ = \sum_{\ell \in [2]} E_{j,3-j} C_0 \alpha_\ell^6 B_{3-j,\ell}^3 B_{j,\ell}^3$$

and

$$\mathbb{E} \left[\sum_{p \in [P] \cap \mathcal{P}} E_{j,3-j} \sigma(\langle w_{3-j}, X_p \rangle) \cdot F_j(X^{(1)}) \right] \\ = \mathbb{E} \left[\sum_{p \in [P] \cap \mathcal{P}} E_{j,3-j} \sigma(\langle w_{3-j}, X_p \rangle) \cdot \sum_{p \in [P] \cap \mathcal{P}} (\sigma(\langle w_j, X_p \rangle) + E_{j,3-j} \sigma(\langle w_{3-j}, X_p \rangle)) \right] \\ = \sum_{\ell \in [2]} E_{j,3-j} C_1 \alpha_\ell^6 B_{3-j,\ell}^3 (B_{j,\ell}^3 + E_{j,3-j} B_{3-j,\ell}^3) + C_2 E_{j,3-j} \mathbb{E}[\langle w_j, \xi_p \rangle^3 \langle w_{3-j}, \xi_p \rangle^3 + E_{j,3-j} \langle w_{3-j}, \xi_p \rangle^6]$$

So we can further obtain the nominator in the expression of $|\langle \nabla_{3-j,j}, w_{3-j} \rangle|$ as

$$\mathbb{E} \left[\sum_{p \in [P] \cap \mathcal{P}} E_{j,3-j} \sigma(\langle w_{3-j}, X_p \rangle) \cdot [G(X^{(2)})]_j \right] \cdot \mathbb{E}[F_j^2(X^{(1)})] \\ - \mathbb{E} \left[\sum_{p \in [P] \cap \mathcal{P}} E_{j,3-j} \sigma(\langle w_{3-j}, X_p \rangle) \cdot F_j(X^{(1)}) \right] \cdot \mathbb{E}[[F_j(X^{(1)}) \cdot [G(X^{(2)})]_j]] \\ = \left(\sum_{\ell \in [2]} E_{j,3-j} C_0 \alpha_\ell^6 B_{3-j,\ell}^3 B_{j,\ell}^3 \right) \cdot \left(\sum_{\ell \in [2]} C_1 \alpha_\ell^6 (B_{j,\ell}^3 + E_{j,3-j} B_{3-j,\ell}^3)^2 + C_2 \mathcal{E}_{j,3-j} \right) \\ - \left(\sum_{\ell \in [2]} E_{j,3-j} C_1 \alpha_\ell^6 B_{3-j,\ell}^3 (B_{j,\ell}^3 + E_{j,3-j} B_{3-j,\ell}^3) \right) \cdot \left(\sum_{\ell \in [2]} C_0 \alpha_\ell^6 B_{j,\ell}^3 (B_{j,\ell}^3 + E_{j,3-j} B_{3-j,\ell}^3) \right)$$

$$\begin{aligned}
& -C_2 E_{j,3-j} \mathbb{E}[\langle w_j, \xi_p \rangle^3 \langle w_{3-j}, \xi_p \rangle^3 + E_{j,3-j} \langle w_{3-j}, \xi_p \rangle^6] \cdot \left(\sum_{\ell \in [2]} C_0 \alpha_\ell^6 B_{j,\ell}^3 (B_{j,\ell}^3 + E_{j,3-j} B_{3-j,\ell}^3) \right) \\
& = E_{j,3-j} \sum_{\ell \in [2]} C_0 \alpha_\ell^6 B_{3-j,\ell}^3 (B_{j,\ell}^3 H_{j,3-\ell} - B_{j,3-\ell}^3 K_{j,3-\ell}) \\
& \quad - C_2 E_{j,3-j} \mathbb{E}[\langle w_j, \xi_p \rangle^3 \langle w_{3-j}, \xi_p \rangle^3 + E_{j,3-j} \langle w_{3-j}, \xi_p \rangle^6] \cdot \left(\sum_{\ell \in [2]} C_0 \alpha_\ell^6 B_{j,\ell}^3 (B_{j,\ell}^3 + E_{j,3-j} B_{3-j,\ell}^3) \right)
\end{aligned}$$

Now can sum over $j' \in [2]$ to get

$$\begin{aligned}
& |\langle \nabla_{w_j} L(W, E), w_j \rangle| \\
& \leq \sum_{j \in [2]} \sum_{\ell \in [2]} C_0 E_{j,3-j} \left| \Phi_j \alpha_\ell^6 B_{3-j,\ell}^3 B_{j,\ell}^3 H_{j,3-\ell} \right| + \sum_{j \in [2]} \sum_{\ell \in [2]} \left| C_0 E_{j,3-j} \Phi_j \alpha_\ell^3 B_{3-j,\ell}^3 B_{j,3-\ell}^3 K_{j,3-\ell} \right| \\
& \quad + \sum_{j \in [2]} \sum_{\ell \in [2]} \left| C_2 E_{j,3-j} \Phi_j \mathbb{E}[\langle w_j, \xi_p \rangle^3 \langle w_{3-j}, \xi_p \rangle^3 + E_{j,3-j} \langle w_{3-j}, \xi_p \rangle^6] C_0 \alpha_\ell^6 B_{j,\ell}^3 (B_{j,\ell}^3 + E_{j,3-j} B_{3-j,\ell}^3) \right|
\end{aligned}$$

Next we are going to bound each term, for the first term of LHS we have

$$\begin{aligned}
\sum_{j \in [2]} \sum_{\ell \in [2]} \left| C_0 E_{j,3-j} \Phi_j \alpha_\ell^6 B_{3-j,\ell}^3 B_{j,\ell}^3 H_{j,3-\ell} \right| & \leq \sum_{j \in [2]} \sum_{\ell \in [2]} |E_{j,3-j}| |\Lambda_{j,\ell}| \left| \frac{B_{3-j,\ell}^3}{B_{j,\ell}^2} \right| \\
& \leq |\Lambda_{1,1}| \sum_{j \in [2]} |E_{j,3-j}| \left| \frac{B_{3-j,\ell}^3 B_{j,\ell}^3 \Phi_j}{B_{1,1}^5 \Phi_1} \right| \\
& \leq \tilde{O}\left(\frac{d^{o(1)}}{\sqrt{d}}\right) |\Lambda_{1,1}| \sum_{j \in [2]} |E_{j,3-j}|
\end{aligned}$$

where the last inequality is because

- By [Lemma 3.C.2a,b](#), we have $\Phi_j^{(t)}/\Phi_1^{(t)} \leq O(\alpha_1^O(1)) \leq d^{o(1)}$ during $t \leq T_1$.
- $(B_{3-j,\ell}^{(t)})^3 (B_{j,\ell}^{(t)})^3 \leq \tilde{O}(\frac{1}{\sqrt{d}}) (B_{1,1}^{(t)})^5$ from [Induction 3.C.1b,c](#).

Similarly, we can also compute

$$\begin{aligned}
\sum_{j \in [2]} \sum_{\ell \in [2]} \left| C_0 E_{j,3-j} \Phi_j \alpha_\ell^3 B_{3-j,\ell}^3 B_{j,3-\ell}^3 K_{j,3-\ell} \right| & \leq \sum_{j \in [2]} \sum_{\ell \in [2]} |E_{j,3-j}| |\Lambda_{1,1}| \left| \frac{B_{3-j,\ell}^3 B_{j,3-\ell}^3 K_{j,3-\ell}}{B_{1,1}^5 H_{j,3-\ell}} \right| \\
& \leq \tilde{O}\left(\frac{d^{o(1)}}{d^2}\right) |\Lambda_{1,1}| \sum_{j \in [2]} |E_{j,3-j}|
\end{aligned}$$

and

$$\begin{aligned}
& \sum_{j \in [2]} \sum_{\ell \in [2]} \left| C_2 E_{j,3-j} \Phi_j \mathbb{E}[\langle w_j, \xi_p \rangle^3 \langle w_{3-j}, \xi_p \rangle^3 + E_{j,3-j} \langle w_{3-j}, \xi_p \rangle^6] C_0 \alpha_\ell^6 B_{j,\ell}^3 (B_{j,\ell}^3 + E_{j,3-j} B_{3-j,\ell}^3) \right| \\
& \stackrel{(1)}{\leq} \sum_{j \in [2]} \sum_{\ell \in [2]} |E_{j,3-j} \Lambda_{j,\ell}| \left| \frac{B_{j,\ell}^3 + E_{j,3-j} B_{3-j,\ell}^3}{B_{j,\ell}^2} \right| |\mathbb{E}[\langle w_j, \xi_p \rangle^3 \langle w_{3-j}, \xi_p \rangle^3 + E_{j,3-j} \langle w_{3-j}, \xi_p \rangle^6]| \\
& \stackrel{(2)}{\leq} \sum_{j \in [2]} \sum_{\ell \in [2]} |E_{j,3-j} \Lambda_{j,\ell}| \left| \frac{B_{j,\ell}^3 + E_{j,3-j} B_{3-j,\ell}^3}{B_{j,\ell}^2} \right| (O(R_{1,2} + \varrho) + O(E_{j,3-j})) \\
& \leq \tilde{O}(R_{1,2} + \varrho) |\Lambda_{1,1}| \sum_{j \in [2]} |E_{j,3-j}|
\end{aligned}$$

Here (1) follows from [Lemma 3.C.2c](#). Step (2) follows from the same calculation in [Claim 3.C.2](#), applied to

$$\mathbb{E}[\langle w_j, \xi_p \rangle^3 \langle w_{3-j}, \xi_p \rangle^3],$$

together with [Induction 3.C.1a](#). Combining these bounds with [Induction 3.C.1e](#) gives

$$|\langle \nabla_{w_j} L(W, E), w_j \rangle| \leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) |\Lambda_{1,1}| \sum_{j \in [2]} |E_{j,3-j}|$$

which gives the desired bound. $\square$

Next we give a lemma characterizing the gradient of feature v_1 in this phase.

Lemma 3.C.4 (learning feature v_1 in phase I). *For each $t \leq T_1$, if Induction 3.C.1 holds at iteration t , then using notations of (3.B.2), we have:*

$$\begin{aligned}
(a) \quad & \langle -\nabla_{w_1} L(W^{(t)}, E^{(t)}), v_1 \rangle = (1 \pm \tilde{O}(\frac{1}{d})) \Lambda_{1,1}^{(t)} \\
(b) \quad & \langle -\nabla_{w_2} L(W^{(t)}, E^{(t)}), v_1 \rangle = (1 \pm O(\frac{1}{\sqrt{d}})) \Lambda_{2,1}^{(t)} + \Gamma_{2,1}^{(t)} \leq (1 \pm O(\frac{1}{\sqrt{d}})) \Lambda_{2,1}^{(t)} \pm \frac{(B_{2,1}^{(t)})^2}{(B_{1,1}^{(t)})^2} E_{1,2}^{(t)} \Lambda_{1,1}^{(t)}
\end{aligned}$$

Proof. From (3.B.2), we write down the gradient formula for $B_{j,1}^{(t)}$ as follows:

$$\langle -\nabla_{w_j} L_{\mathcal{D}}(W^{(t)}, E^{(t)}), v_1 \rangle = \Lambda_{j,1}^{(t)} + \Gamma_{j,1}^{(t)} - \Upsilon_{j,1}^{(t)}$$

where (ignoring the superscript $^{(t)}$ for the RHS)

$$\begin{aligned}
\Lambda_{j,1}^{(t)} &= C_0 \Phi_j \alpha_1^6 B_{j,1}^5 H_{j,2} \\
\Gamma_{j,1}^{(t)} &= C_0 \Phi_{3-j} E_{3-j,j} \alpha_1^6 B_{3-j,1}^3 B_{j,1}^2 H_{3-j,2} \\
\Upsilon_{j,1}^{(t)} &= C_0 \alpha_2^6 \left(\Phi_j B_{j,2}^3 B_{j,1}^2 K_{j,1} + \Phi_{3-j} E_{3-j,j} B_{3-j,2}^3 B_{j,1}^2 K_{3-j,1} \right)
\end{aligned}$$

We first prove (a), and we deal with each term individually:

Comparing $\Lambda_{1,1}^{(t)}$ and $\Gamma_{1,1}^{(t)}$: When $t \leq T_{1,1}$, we have from [Lemma 3.C.2a](#) that

$$\Phi_1^{(t)} H_{1,2}^{(t)} = \frac{1}{C_2 \mathcal{E}_1^{(t)}} (1 \pm \frac{1}{\text{polylog}(d)}) = \frac{1}{C_2 \mathcal{E}_2^{(t)}} (1 \pm \frac{1}{\text{polylog}(d)}) = \Phi_2^{(t)} H_{2,2}^{(t)} (1 \pm \frac{1}{\text{polylog}(d)})$$

Further, by [Induction 3.C.1b,c,d](#) and our definition of stage 1, we know $E_{1,2}^{(t)} \leq \tilde{O}(\frac{1}{d})$. Now from [Induction 3.C.1b](#) that $B_{2,1}^{(t)} \leq \tilde{O}(\frac{1}{\sqrt{d}})$, together we have

$$\Gamma_{1,1}^{(t)} = C_0 \alpha_1^6 E_{2,1}^{(t)} \Phi_2^{(t)} H_{2,2}^{(t)} (B_{2,1}^{(t)})^3 (B_{1,1}^{(t)})^2 \leq \tilde{O}(\frac{1}{d}) C_0 \alpha_1^6 \Phi_1^{(t)} H_{1,2}^{(t)} (B_{1,1}^{(t)})^5 = \tilde{O}(\frac{\Lambda_{1,1}^{(t)}}{d})$$

When $t \in [T_{1,1}, T_1]$, by [Lemma 3.C.2b](#) we have

$$\Phi_1^{(t)} H_{1,2}^{(t)} \geq \Omega(\frac{C_2}{(C_1 \alpha_1^6 (B_{1,1}^{(t)})^6 + O(C_2))^2}) \geq \omega(\frac{1}{d^{0.1}}), \quad \text{and} \quad E_{2,1}^{(t)} \Phi_2^{(t)} H_{2,2}^{(t)} \leq \tilde{O}(\frac{1}{d})$$

Now from our definition of stage 2, it holds that $B_{1,1}^{(t)} \geq \Omega(\frac{1}{\alpha_1})$ while $B_{2,1}^{(t)} \leq \tilde{O}(\frac{1}{\sqrt{d}})$ by [Induction 3.C.1b](#), which gives

$$\Gamma_{1,1}^{(t)} = C_0 \alpha_1^6 E_{2,1}^{(t)} \Phi_2^{(t)} H_{2,2}^{(t)} (B_{2,1}^{(t)})^3 (B_{1,1}^{(t)})^2 \leq \tilde{O}(\frac{1}{d}) C_0 \alpha_1^6 \Phi_1^{(t)} H_{1,2}^{(t)} (B_{1,1}^{(t)})^5 = \tilde{O}(\frac{\Lambda_{1,1}^{(t)}}{d})$$

Comparing $\Lambda_{1,1}^{(t)}$ and $\Upsilon_{1,1}^{(t)}$: Now consider $\Upsilon_{1,1}^{(t)}$, by [Lemma 3.C.2](#), we can follow the same analysis as above to get

$$\Phi_j^{(t)} K_{j,\ell}^{(t)} \leq \tilde{O}(\frac{\alpha_1^{O(1)}}{d^{3/2}}) \Phi_1^{(t)} H_{1,2}^{(t)} \quad \text{for any } (j, \ell) \in [2] \times [2]$$

Combined with $E_{2,1}^{(t)} \leq o(1)$, we can derive

$$\begin{aligned} \Upsilon_{1,1}^{(t)} &= C_0 \alpha_2^6 \left(\Phi_1^{(t)} K_{1,1}^{(t)} (B_{1,2}^{(t)})^3 (B_{1,1}^{(t)})^2 + E_{1,2}^{(t)} \Phi_2^{(t)} K_{2,1}^{(t)} (B_{2,2}^{(t)})^3 (B_{1,1}^{(t)})^2 \right) \\ &\leq \tilde{O}(\frac{\alpha_1^{O(1)} \alpha_2^6}{d^{3/2}}) C_0 \alpha_1^6 \Phi_1^{(t)} H_{1,2}^{(t)} (B_{1,1}^{(t)})^5 \\ &= \tilde{O}(\frac{\Lambda_{1,1}^{(t)}}{d^{3/2-o(1)}}) \quad (\text{since } C_1 = \tilde{O}(1) \text{ and } \alpha_1, \alpha_2 = d^{o(1)}) \end{aligned}$$

Comparing $\Lambda_{2,1}^{(t)}$ and $\Upsilon_{2,1}^{(t)}$: Till now (a) is proved, we can deal with (b) by only comparing $\Lambda_{2,1}^{(t)}$ with $\Upsilon_{2,1}^{(t)}$. Similar to the above arguments, we have by [Induction 3.C.1b](#) we know $K_{j,1}^{(t)} = \tilde{O}(\frac{C_1 \alpha_1^6}{d^{3/2}})$, $\forall j \in$

[2], and thus

$$\Phi_j^{(t)} K_{j,\ell}^{(t)} \leq \tilde{O}\left(\frac{\alpha_1^6}{d^{3/2}}\right) \Phi_2^{(t)} H_{2,2}^{(t)} \quad \text{for any } (j, \ell) \in [2] \times [2]$$

By [Induction 3.C.1e](#) we know $E_{1,2}^{(t)} \leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}})$. Also, note that from [Induction 3.C.1b](#) we have $\tilde{O}((B_{1,2}^{(t)})^3/d) \leq \tilde{O}((B_{2,1}^{(t)})^5)$, and thus

$$E_{1,2}^{(t)} \Phi_1^{(t)} K_{1,1}^{(t)} (B_{1,2}^{(t)})^3 (B_{2,1}^{(t)})^2 \leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) \tilde{O}\left(\frac{\alpha_1^6}{d^{5/2}}\right) \Phi_2^{(t)} H_{2,2}^{(t)} \tilde{O}(B_{1,2}^{(t)})^3 \leq O\left(\frac{1}{d^{3/2}}\right) \Phi_2^{(t)} H_{2,2}^{(t)} (B_{2,1}^{(t)})^5$$

So together we have

$$\begin{aligned} |\Upsilon_{2,1}^{(t)}| &= |C_0 \alpha_2^6 \left(\Phi_2^{(t)} K_{2,1}^{(t)} (B_{2,2}^{(t)})^3 (B_{2,1}^{(t)})^2 + E_{2,1}^{(t)} \Phi_1^{(t)} K_{1,1}^{(t)} (B_{1,2}^{(t)})^3 (B_{2,1}^{(t)})^2 \right)| \\ &\leq O\left(\frac{1}{d^{3/2}}\right) C_0 \alpha_1^6 \Phi_2^{(t)} H_{2,2}^{(t)} (B_{2,1}^{(t)})^5 \\ &= O\left(\frac{1}{d^{3/2}}\right) |\Lambda_{2,1}^{(t)}| \end{aligned}$$

Comparing $\Gamma_{2,1}^{(t)}$ with $\Lambda_{1,1}^{(t)}$: It suffices to notice that

$$|\Gamma_{2,1}^{(t)}| \leq |E_{1,2}^{(t)}| C_0 \alpha_1^6 \Phi_1^{(t)} H_{1,2}^{(t)} |B_{1,1}^{(t)}|^3 (B_{2,1}^{(t)})^2 = \frac{(B_{2,1}^{(t)})^2}{(B_{1,1}^{(t)})^2} |E_{1,2}^{(t)}| |\Lambda_{1,1}^{(t)}|$$

Combining the bounds for $\Lambda_{2,1}^{(t)}$ and $\Gamma_{2,1}^{(t)}$, we obtain the proof of (b). $\square$

Then we can also calculate the gradients of feature v_2 in this phase.

Lemma 3.C.5 (learning feature v_2 in phase I). *For each $t \leq T_1$, if [Induction 3.C.1](#) holds at iteration t , then using notations of [\(3.B.2\)](#), we have for each $j \in [2]$:*

$$\langle -\nabla_{w_j} L(W^{(t)}, E^{(t)}), v_2 \rangle = \left(1 \pm \tilde{O}(\alpha_1^6 (E_{3-j,j}^{(t)} + (B_{j,1}^{(t)})^3)) \right) \Lambda_{j,2}^{(t)} \quad (3.C.1)$$

Proof. Again as in the proof of [Lemma 3.C.4](#), we expand the notations: (ignoring the superscript $^{(t)}$ for the RHS)

$$\begin{aligned} \Lambda_{j,2}^{(t)} &= C_0 \alpha_2^6 \Phi_j H_{j,1} B_{j,2}^5 \\ \Gamma_{j,2}^{(t)} &= C_0 \alpha_2^6 \Phi_j E_{3-j,j} B_{3-j,2}^3 B_{j,2}^2 H_{3-j,1} \\ \Upsilon_{j,2}^{(t)} &= C_0 \alpha_1^6 \left(\Phi_j B_{j,1}^3 B_{j,2}^2 K_{j,2} + \Phi_{3-j} E_{3-j,j} B_{3-j,1}^3 B_{j,2}^2 K_{3-j,2} \right) \end{aligned}$$

We first compare $\Lambda_{j,2}^{(t)}$ and $\Gamma_{j,2}^{(t)}$ as follows: [Lemma 3.C.2](#) we have

- $B_{3-j,2}^{(t)} \leq \tilde{O}(B_{j,2}^{(t)})$ by [Induction 3.C.1b](#);
- From [Lemma 3.C.2a,b](#) we can have $\Phi_{3-j}^{(t)} \leq \tilde{O}(\alpha_1^{O(1)})\Phi_j^{(t)}$, $\forall j \in [2]$.

Together they imply:

$$\begin{aligned} C_0 \alpha_2^6 E_{3-j,j}^{(t)} (B_{3-j,2}^{(t)})^3 (B_{j,2}^{(t)})^2 \Phi_{3-j}^{(t)} H_{3-j,1}^{(t)} &\leq \tilde{O}(\alpha_1^{O(1)} E_{3-j,j}^{(t)}) C_0 \alpha_2^6 \Phi_j^{(t)} H_{j,2}^{(t)} (B_{j,2}^{(t)})^5 \\ &= \tilde{O}(\alpha_1^{O(1)} E_{j,3-j}^{(t)}) \Lambda_{j,2}^{(t)} \end{aligned} \quad (3.C.2)$$

Now we turn to compare $\Lambda_{j,2}^{(t)}$ with $\Upsilon_{j,2}^{(t)}$. We split $\Upsilon_{j,2}^{(t)}$ into two terms $\Upsilon_{j,2,1}^{(t)}, \Upsilon_{j,2,2}^{(t)}$

$$\Upsilon_{j,2,1}^{(t)} = C_0 \alpha_1^6 \Phi_j^{(t)} (B_{j,1}^{(t)})^3 (B_{j,2}^{(t)})^2 K_{j,2}^{(t)}, \quad \Upsilon_{j,2,2}^{(t)} = C_0 \alpha_1^6 \Phi_{3-j}^{(t)} E_{3-j,j}^{(t)} (B_{3-j,1}^{(t)})^3 (B_{j,2}^{(t)})^2 K_{3-j,2}^{(t)}$$

For $\Upsilon_{j,2,1}^{(t)}$, we can calculate

$$\begin{aligned} \Upsilon_{j,2,1}^{(t)} &= C_0 \alpha_1^6 \Phi_j^{(t)} (B_{j,1}^{(t)})^3 (B_{j,2}^{(t)})^2 K_{j,2}^{(t)} \\ &\leq \tilde{O}\left(\frac{C_1 \alpha_2^6}{d^{3/2}}\right) (B_{j,1}^{(t)})^3 \cdot C_0 \alpha_1^6 \Phi_j^{(t)} H_{j,1}^{(t)} (B_{j,2}^{(t)})^2 \quad (K_{j,\ell}^{(t)} \leq \tilde{O}\left(\frac{C_1 \alpha_\ell^6}{d^{3/2}}\right) \text{ from } \text{Lemma 3.C.2d}) \\ &\leq \tilde{O}(\alpha_1^6 (B_{j,1}^{(t)})^3) C_0 \alpha_2^6 \Phi_j^{(t)} H_{j,1}^{(t)} (B_{j,2}^{(t)})^5 \quad (\tilde{O}\left(\frac{C_1}{d^{3/2}}\right) \leq \tilde{O}((B_{j,2}^{(t)})^3) \text{ from } \text{Induction 3.C.1b}) \\ &= \tilde{O}(\alpha_1^6 (B_{j,1}^{(t)})^3) \Lambda_{j,2}^{(t)} \end{aligned} \quad (3.C.3)$$

And for $\Upsilon_{j,2,2}^{(t)}$, we use [Induction 3.C.1b](#) and [Lemma 3.C.2d](#) again to get

$$(B_{3-j,1}^{(t)})^3 (B_{3-j,2}^{(t)})^2 K_{3-j,2}^{(t)} \leq \tilde{O}(C_1 \alpha_2^6 (B_{j,2}^{(t)})^5)$$

and thus combined with $\Phi_{3-j}^{(t)} \leq \tilde{O}(\alpha_1^6) \Phi_j^{(t)}$, $\forall j \in [2]$ from [Lemma 3.C.2a,b](#), we can derive

$$\begin{aligned} \Upsilon_{j,2,2}^{(t)} &= C_0 \alpha_1^6 \Phi_{3-j}^{(t)} E_{j,3-j}^{(t)} (B_{3-j,1}^{(t)})^3 (B_{j,2}^{(t)})^2 K_{3-j,2}^{(t)} \\ &\leq \tilde{O}(\alpha_1^6 E_{3-j,j}^{(t)}) C_0 \alpha_2^6 \Phi_j^{(t)} H_{j,1}^{(t)} (B_{j,2}^{(t)})^5 \\ &= \tilde{O}(\alpha_1^6 E_{3-j,j}^{(t)}) \Lambda_{j,2}^{(t)} \end{aligned} \quad (3.C.4)$$

Now combine the results of (3.C.2), (3.C.3) and (3.C.4) finishes the proof of (3.C.1). $\square$

Lemma 3.C.6 (learning prediction head $E_{1,2}, E_{2,1}$ in phase I). *If [Induction 3.C.1](#) holds at iteration $t \leq T_1$, then we have*

$$\begin{aligned} (a) \quad -\nabla_{E_{1,2}} L(W^{(t)}, E^{(t)}) &= O(\Lambda_{1,1}^{(t)} B_{1,1}^{(t)}) \left(-O(E_{1,2}^{(t)}) + \tilde{O}\left(\frac{(B_{1,2}^{(t)})^3}{(B_{1,1}^{(t)})^3}\right) + O(R_{1,2}^{(t)}) \right); \\ (b) \quad -\nabla_{E_{2,1}} L(W^{(t)}, E^{(t)}) &= \tilde{O}\left(\frac{(B_{1,2}^{(t)})^3}{(B_{1,1}^{(t)})^2}\right) \Lambda_{1,1}^{(t)} + \sum_{\ell \in [2]} C_2 \Lambda_{2,\ell}^{(t)} B_{2,\ell}^{(t)} \left(-O(E_{2,1}^{(t)}) + O(R_{1,2}^{(t)}) \right) \end{aligned}$$

Proof. We first write down the gradient for $E_{j,3-j}^{(t)}$: (ignoring the time superscript (t))

$$-\nabla_{E_{j,3-j}} L(W, E) = \sum_{\ell \in [2]} C_0 \Phi_j \alpha_\ell^6 B_{j,\ell}^3 (B_{3-j,\ell}^3 H_{j,3-\ell} - B_{3-j,3-\ell}^3 K_{j,3-\ell}) - \sum_{\ell \in [2]} \Sigma_{j,\ell} \nabla_{E_{j,3-j}} \mathcal{E}_{j,3-j}$$

where $\nabla_{E_{j,3-j}} \mathcal{E}_{j,3-j} = \mathbb{E} [2\langle w_j, \xi_p \rangle^3 \langle w_{3-j}, \xi_p \rangle^3 + 2E_{j,3-j} \langle w_{3-j}, \xi_p \rangle^6]$. Thus we have

$$\nabla_{E_{j,3-j}} \mathcal{E}_{j,3-j}^{(t)} = O(1) E_{j,3-j}^{(t)} + O(R_{1,2}^{(t)})$$

and by [Claim 3.C.1](#) and [Lemma 3.C.2a,b](#)

$$\Sigma_{j,\ell}^{(t)} = O(\Lambda_{1,1}^{(t)} B_{1,1}^{(t)}) \frac{(B_{j,\ell}^{(t)})^6 + E_{j,3-j}^{(t)} (B_{3-j,\ell}^{(t)})^3 (B_{j,\ell}^{(t)})^3 \Phi_j^{(t)}}{(B_{1,1}^{(t)})^6} \frac{\Phi_j^{(t)}}{\Phi_1^{(t)}} \leq O(\Lambda_{1,1}^{(t)} B_{1,1}^{(t)})$$

Now let us look at $\nabla_{E_{1,2}} L(W^{(t)}, E^{(t)})$, first we consider the term

$$\sum_{\ell \in [2]} C_0 \Phi_1^{(t)} \alpha_\ell^6 (B_{1,\ell}^{(t)})^3 ((B_{2,\ell}^{(t)})^3 H_{1,3-\ell}^{(t)} - (B_{2,3-\ell}^{(t)})^3 K_{1,3-\ell}^{(t)})$$

Using [Lemma 3.C.2](#) and [Induction 3.C.1b,c](#), we know

- $H_{1,1}^{(t)} \leq \tilde{O}(H_{1,2}^{(t)})$ at $t \leq T_{1,1}$ and $H_{1,1}^{(t)} \leq \tilde{O}(\alpha_1^6 H_{1,2}^{(t)})$ for $t \in [T_{1,1}, T_1]$;
- $B_{2,1}^{(t)}, B_{1,2}^{(t)}, B_{2,2}^{(t)} \leq \tilde{O}(B_{2,1}^{(t)}) \leq \tilde{O}(B_{1,1}^{(t)})$;
- $K_{1,3-\ell}^{(t)} \leq \tilde{O}(\alpha_1^6 / d^{3/2})$.

It can be computed that

$$\begin{aligned} C_0 \Phi_1^{(t)} \alpha_2^6 (B_{1,2}^{(t)})^3 (B_{2,2}^{(t)})^3 H_{1,1}^{(t)} &\leq \tilde{O}(1) \left(\frac{B_{2,1}^{(t)}}{B_{1,1}^{(t)}} \right)^3 C_0 \Phi_1^{(t)} \alpha_1^3 (B_{1,1}^{(t)})^6 H_{1,2}^{(t)} \\ \sum_{\ell \in [2]} \left| C_0 \Phi_1^{(t)} \alpha_\ell^6 (B_{1,\ell}^{(t)})^3 (B_{2,\ell}^{(t)})^3 K_{1,3-\ell}^{(t)} \right| &\leq \tilde{O}\left(\frac{\alpha_1^6}{d^{3/2}}\right) \frac{(B_{2,1}^{(t)})^3}{(B_{1,1}^{(t)})^3} C_0 \Phi_1^{(t)} \alpha_1^6 (B_{1,1}^{(t)})^6 H_{1,2}^{(t)} \end{aligned}$$

Now we turn to $\nabla_{E_{2,1}} L(W^{(t)}, E^{(t)})$, similarly we have

$$C_0 \Phi_2^{(t)} \alpha_1^6 (B_{2,1}^{(t)})^3 (B_{1,1}^{(t)})^3 H_{2,2}^{(t)} \leq \tilde{O}(1) \left(\frac{B_{2,1}^{(t)}}{B_{1,1}^{(t)}} \right)^3 C_0 \Phi_1^{(t)} \alpha_1^6 (B_{1,1}^{(t)})^6 H_{1,2}^{(t)}$$

and since $H_{2,1}^{(t)} \leq O(C_2) = O(H_{1,2}^{(t)})$ by Lemma 3.C.2c, we can go through the same arguments again to obtain

$$\begin{aligned} \left| C_0 \Phi_2^{(t)} \alpha_2^6 (B_{1,2}^{(t)})^3 (B_{2,2}^{(t)})^3 H_{2,1}^{(t)} \right| &\leq \tilde{O}(1) \left(\frac{B_{1,2}^{(t)}}{B_{1,1}^{(t)}} \right)^3 C_0 \Phi_1^{(t)} \alpha_1^6 (B_{1,1}^{(t)})^6 H_{1,2}^{(t)} \\ \left| C_0 \Phi_2^{(t)} \alpha_2^6 (B_{1,2}^{(t)})^3 (B_{2,1}^{(t)})^3 K_{2,1}^{(t)} \right| &\leq \tilde{O}\left(\frac{\alpha_1^6}{d^{3/2}}\right) \left(\frac{B_{1,2}^{(t)}}{B_{1,1}^{(t)}} \right)^3 C_0 \Phi_1^{(t)} \alpha_1^6 (B_{1,1}^{(t)})^6 H_{1,2}^{(t)} \end{aligned}$$

Now the proof is complete. $\square$

Also, we will need the following lemma controlling gradient bounds for the noise term.

Lemma 3.C.7 (update of $R_{1,2}^{(t)}$ in phase I). *Suppose Induction 3.C.1 holds at iteration $t \leq T_1$, then we have*

$$\begin{aligned} (a) \quad & |\langle -\nabla_{w_1} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_2^{(t)} \rangle| \leq \tilde{O}\left(\frac{1}{\sqrt{d}} + \varrho\right) \Lambda_{1,1}^{(t)} B_{1,1}^{(t)} \\ (b) \quad & |\langle -\nabla_{w_2} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_1^{(t)} \rangle| \leq \tilde{O}\left(\frac{1}{\sqrt{d}} + \varrho\right) \Lambda_{1,1}^{(t)} B_{1,1}^{(t)} \end{aligned}$$

Proof. **Proof of (a):** Firstly, by Claim 3.C.2a, we can directly write

$$\begin{aligned} \langle \nabla_{w_1} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_2^{(t)} \rangle &= - \sum_{j,\ell} \Sigma_{j,\ell}^{(t)} \langle \nabla_{w_1} \mathcal{E}_{j,3-j}^{(t)}, w_2^{(t)} \rangle \\ &= -\Lambda_{1,1}^{(t)} B_{1,1}^{(t)} \sum_{(j,\ell) \in [2]^2} \frac{(B_{j,\ell}^{(t)})^6 + E_{j,3-j}^{(t)} (B_{3-j,\ell}^{(t)})^3 (B_{j,\ell}^{(t)})^3}{(B_{1,1}^{(t)})^6} \frac{\Phi_j^{(t)}}{\Phi_1^{(t)}} \langle \nabla_{w_1} \mathcal{E}_{j,3-j}^{(t)}, w_1^{(t)} \rangle \end{aligned} \quad (3.C.5)$$

Now we discuss each summand respectively: for $(j, \ell) = (1, 1)$, we have

$$\frac{(B_{j,\ell}^{(t)})^6 + E_{j,3-j}^{(t)} (B_{3-j,\ell}^{(t)})^3 (B_{j,\ell}^{(t)})^3}{(B_{1,1}^{(t)})^6} = 1 + E_{1,2}^{(t)} \frac{(B_{2,1}^{(t)})^3}{(B_{1,1}^{(t)})^3} = 1 + o\left(\frac{1}{d^{3/2} (B_{1,1}^{(t)})^3}\right) \quad (3.C.6)$$

where the last one is due to Induction 3.C.1d. And for $\ell = 2$, we can see from Induction 3.C.1b and d, that $\max_{(j,\ell) \neq (1,1)} |B_{j,\ell}^{(t)}| = \tilde{O}\left(\frac{1}{\sqrt{d}}\right)$ and $E_{j,3-j}^{(t)} \leq o(1)$ to give

$$\frac{(B_{j,2}^{(t)})^6 + E_{j,3-j}^{(t)} (B_{3-j,2}^{(t)})^3 (B_{j,2}^{(t)})^3}{(B_{1,1}^{(t)})^6} \frac{\Phi_j^{(t)}}{\Phi_1^{(t)}} \leq \tilde{O}\left(\frac{1}{d^3}\right) \frac{1}{(B_{1,1}^{(t)})^6} \frac{\Phi_j^{(t)}}{\Phi_1^{(t)}}$$

On one hand, when $t \leq T_{1,1}$, we have $\alpha_\ell B_{j,\ell}^{(t)} \leq O(1)$ for all $(j, \ell) \in [2]^2$, so Lemma 3.C.2a applies for both $\Phi_j^{(t)}$ and results in $\Phi_2^{(t)}/\Phi_1^{(t)} \leq O(1)$. We can also apply Induction 3.C.1c to have $B_{j,2}^{(t)}/B_{1,1}^{(t)} \leq \tilde{O}(1)$. On the other hand, when $t \in [T_{1,1}, T_1]$, we have by Induction 3.C.1b and

Lemma 3.C.2a,b that $\Phi_2^{(t)}/\Phi_1^{(t)} \leq \tilde{O}(\alpha_1^{O(1)}) = d^{o(1)}$, but now $B_{1,1}^{(t)} = d^{-o(1)} \gg \tilde{O}(d^{-1/2})$, therefore

$$\tilde{O}\left(\frac{1}{d^3}\right) \frac{1}{(B_{1,1}^{(t)})^6} \frac{\Phi_2^{(t)}}{\Phi_1^{(t)}} \leq \tilde{O}\left(\frac{1}{d^{3/2}}\right) \frac{1}{(B_{1,1}^{(t)})^3}$$

So together, they imply

$$\frac{(B_{j,2}^{(t)})^6 + E_{j,3-j}^{(t)}(B_{3-j,2}^{(t)})^3(B_{j,2}^{(t)})^3}{(B_{1,1}^{(t)})^6} \frac{\Phi_j^{(t)}}{\Phi_1^{(t)}} \leq \tilde{O}\left(\frac{1}{d^{3/2}(B_{1,1}^{(t)})^3}\right) \quad (3.C.7)$$

and similarly, we have

$$\frac{(B_{2,1}^{(t)})^6 + E_{2,1}^{(t)}(B_{1,1}^{(t)})^3(B_{2,1}^{(t)})^3}{(B_{1,1}^{(t)})^6} \frac{\Phi_2^{(t)}}{\Phi_1^{(t)}} \leq \tilde{O}\left(\frac{1}{d^{3/2}(B_{1,1}^{(t)})^3}\right) \quad (3.C.8)$$

Next we turn to $\langle \nabla_{w_1} \mathcal{E}_{j,3-j}^{(t)}, w_2^{(t)} \rangle$. When $j = 1$, we can apply **Claim 3.C.2d** to get

$$\langle \nabla_{w_1} \mathcal{E}_{1,2}^{(t)}, w_2^{(t)} \rangle = O(R_{1,2}^{(t)} + \varrho) + O(E_{1,2}^{(t)}) = O(\varrho + \frac{1}{\sqrt{d}}) + O(E_{1,2}^{(t)}) \leq O(\varrho + \frac{1}{\sqrt{d}}) \quad (3.C.9)$$

and when $j = 2$, we can apply **Claim 3.C.2e** to get

$$\langle \nabla_{w_1} \mathcal{E}_{2,1}^{(t)}, w_2^{(t)} \rangle = -(E_{2,1}^{(t)})^2 O(R_{1,2}^{(t)} + \varrho) + O(E_{2,1}^{(t)}) = \tilde{O}\left(\frac{1}{d^2}\right)(\varrho + \frac{1}{\sqrt{d}}) + O\left(\frac{1}{d}\right) \quad (3.C.10)$$

Combining (3.C.5), (3.C.6), (3.C.7), (3.C.8), (3.C.9), and (3.C.10) completes the proof of (a).

Proof of (b): The $\Sigma_{j,\ell}^{(t)}$ part is the same as in the proof of (a), so we only deal with $\langle \nabla_{w_2} \mathcal{E}_{1,2}^{(t)}, w_1^{(t)} \rangle$ and $\langle \nabla_{w_2} \mathcal{E}_{2,1}^{(t)}, w_1^{(t)} \rangle$ here. For $\langle \nabla_{w_2} \mathcal{E}_{2,1}^{(t)}, w_1^{(t)} \rangle$, we apply **Claim 3.C.2d** to get

$$\langle \nabla_{w_2} \mathcal{E}_{2,1}^{(t)}, w_1^{(t)} \rangle = O(R_{1,2}^{(t)} + \varrho) + O(1)E_{1,2}^{(t)} \quad (3.C.11)$$

and for $\langle \nabla_{w_2} \mathcal{E}_{1,2}^{(t)}, w_1^{(t)} \rangle$, we have

$$\langle \nabla_{w_2} \mathcal{E}_{1,2}^{(t)}, w_1^{(t)} \rangle = O(R_{1,2}^{(t)} + \varrho)(E_{2,1}^{(t)})^2 + O(1)E_{2,1}^{(t)} \quad (3.C.12)$$

Inserting (3.C.6), (3.C.7), (3.C.8), (3.C.11), and (3.C.12) into

$$\langle -\nabla_{w_2} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_1^{(t)} \rangle$$

finishes the proof of (b). $\square$

3.C.4 At the End of Phase I

Lemma 3.C.8 (Phase I). *Suppose $\eta \leq \frac{1}{\text{poly}(d)}$ is sufficiently small, then [Induction 3.C.1](#) holds for at least all $t \leq T_1 = O(\frac{d^2}{\eta})$, and at iteration $t = T_1$, we have*

- (a) $B_{1,1}^{(T_1)} = \Omega(1)$;
- (b) $\|w_j^{(T_1)}\|_2 = 1 \pm \tilde{O}(\rho + \frac{1}{\sqrt{d}})$;
- (c) $B_{2,1}^{(T_1)} = \tilde{\Theta}(\frac{1}{\sqrt{d}})$ and $B_{j,2}^{(T_1)} = B_{j,2}^{(0)}(1 \pm o(1))$ for $j \in [2]$;
- (d) $E_{2,1}^{(T_1)} = \tilde{O}(\frac{\eta E/\eta}{d})$ and $E_{1,2}^{(T_1)} \leq \tilde{O}(\rho + \frac{1}{\sqrt{d}})$;
- (e) $R_{1,1}^{(T_1)}, R_2^{(T_1)} = \Theta(1)$ and $R_{1,2}^{(T_1)} = \tilde{O}(\rho + \frac{1}{\sqrt{d}})$.

Proof. We begin by first prove the existence of $T_1 := \min\{t : B_{1,1}^{(t)} \geq 0.01\} = O(\frac{d^2}{\eta})$ if [Induction 3.C.1](#) holds whenever $B_{1,1}^{(t)} \leq 0.01$, then we will turn back to prove [Induction 3.C.1](#) holds throughout $t \leq T_1$. We split the analysis into two stages:

Proof of $T_1 \leq O(\frac{d^2}{\eta})$: By [Lemma 3.C.4a](#) we can write down the update of $B_{1,1}^{(t)}$ as

$$B_{1,1}^{(t+1)} = B_{1,1}^{(t)} + \eta(1 \pm \tilde{O}(\frac{1}{d}))\Lambda_{1,1}^{(t)} = B_{1,1}^{(t)} + \eta(1 \pm \tilde{O}(\frac{1}{d}))\Phi_1^{(t)}C_0\alpha_1^6H_{1,2}^{(t)}(B_{1,1}^{(t)})^5 \quad (3.C.13)$$

When $\alpha_1 B_{1,1}^{(t)} \leq O(1)$, by [Lemma 3.C.2a,c](#) we have $\Phi_1^{(t)} = \Theta(\frac{1}{C_2^2})$ and $H_{1,2}^{(t)} = \Omega(C_2)$, this means we can lower bound the update as

$$B_{1,1}^{(t+1)} \geq B_{1,1}^{(t)} + \Omega(\frac{\eta C_0 \alpha_1^6}{C_2})(B_{1,1}^{(t)})^5$$

since $\frac{C_0 \alpha_1^6}{C_2}$ is a constant, we know there exist some $t' \geq 0$ such that $B_{1,1}^{(t')} \geq \Omega(\frac{1}{\alpha_1})$. Also recall that $T_{1,1} := \min\{t : B_{1,1}^{(t)} \geq \Omega(\frac{1}{\alpha_1})\}$. So by [Lemma 3.A.1](#), where $\eta = \frac{1}{\text{poly}(d)}$, $C_t = \Omega(\frac{C_0 \alpha_1^6}{C_2})$, $\delta = \frac{1}{\text{polylog}(d)}$ and $A = \Omega(\frac{1}{\alpha_1})$, $\log(A/B_{1,1}^{(0)}) = \tilde{O}(1)$, we have

$$T_{1,1} = O(\frac{C_2}{\eta C_0 \alpha_1^6}) \sum_{x_t \leq O(\frac{1}{\alpha_1})} \eta C_t \leq O(\frac{C_2}{\eta C_0 \alpha_1^6}) \left(O(1) + \frac{\tilde{O}(\eta)}{B_{1,1}^{(0)}} \right) \frac{1}{(B_{1,1}^{(0)})^4} \leq \tilde{O}\left(\frac{1}{\eta \alpha_1^6 (B_{1,1}^{(0)})^4}\right)$$

Since $(B_{1,1}^{(0)})^4 \geq \tilde{\Omega}(\frac{1}{d^2})$ from our initialization, we have $T_{1,1} \leq O(\frac{d^2}{\eta})$ and thus $T_{1,1}$ exists. Now we consider when $B_{1,1}^{(t)} \geq \Omega(\frac{1}{\alpha_1})$. Now by [Lemma 3.C.2b,c](#), we have $\Phi_1^{(t)} \geq \Omega((C_2 + \alpha_1^6)^{-2})$, which gives an update:

$$B_{1,1}^{(t+1)} \geq B_{1,1}^{(t)} + \Omega(\frac{\eta C_0 \alpha_1^6}{(C_2 + \alpha_1^6)^2})(B_{1,1}^{(t)})^5$$

so again by [Lemma 3.A.1](#), choosing $C_t = \Omega(\frac{C_0 \alpha_1^6}{(C_2 + \alpha_1^6)^2})$,

$$T_1 = \frac{O((C_2 + \alpha_1^6)^2)}{\eta C_0 \alpha_1^6} \sum_{x_t \in [\Omega(\frac{1}{\alpha_1}), 0.01]} \eta C_t \leq \left(O(1) + \frac{\tilde{O}(\eta)}{B_{1,1}^{(T_1,1)}} \right) \frac{\tilde{O}(\alpha_1^{12})}{(B_{1,1}^{(T_1,1)})^4} \leq \tilde{O}\left(\frac{\alpha_1^6}{\eta (B_{1,1}^{(T_1,1)})^4}\right) \leq O\left(\frac{\alpha_1^6}{\eta}\right)$$

where $O(\frac{\alpha_1^6}{\eta}) \ll O(\frac{d^2}{\eta})$, so we have proved that T_1 exist. Now we begin to prove that [Induction 3.C.1](#) holds for all $t \leq T_1$.

Proof of [Induction 3.C.1](#): We first prove (b)–(d), and then come back to prove (a) and (d). At $t = 0$, we know all induction holds from [Properties 3.C.1](#). Now we suppose [Induction 3.C.1](#) holds for all iterations $\leq t - 1$ and prove it holds at t .

The growth of $B_{2,1}^{(t)}$: Applying [Lemma 3.C.4](#), we have for $t \leq T_{1,1}$

$$\begin{aligned} B_{1,1}^{(t+1)} &\geq B_{1,1}^{(t)} + \eta(1 - \tilde{O}(\frac{1}{d}))\Lambda_{1,1}^{(t)} \\ B_{2,1}^{(t+1)} &\leq B_{2,1}^{(t)} + \eta(1 + O(\frac{1}{\sqrt{d}}))\Lambda_{2,1}^{(t)} + \eta \frac{(B_{2,1}^{(t)})^2}{(B_{1,1}^{(t)})^2} E_{1,2}^{(t)} \Lambda_{1,1}^{(t)} \end{aligned}$$

For some $t'_1 := \min\{t : B_{1,1}^{(t)} \geq \frac{\Omega(1)}{d^{0.49}}\}$, we have $E_{1,2}^{(t)} \leq \tilde{O}(B_{1,1}^{(t)} \varrho) \lesssim \frac{1}{d^{0.49}}$ during $t \leq t'_1$, and

$$\frac{(B_{2,1}^{(t)})^2}{(B_{1,1}^{(t)})^2} E_{1,2}^{(t)} \Lambda_{1,1}^{(t)} \lesssim \frac{(B_{2,1}^{(t)})^2}{d^{0.49} (B_{1,1}^{(t)})^2} \Lambda_{1,1}^{(t)} \leq \tilde{O}(\frac{1}{d^{0.49}}) \Lambda_{2,1}^{(t)}$$

which allow us to give an upper bound to $B_{2,1}^{(t+1)}$ as

$$\begin{aligned} B_{2,1}^{(t+1)} &\leq (1 + O(\frac{1}{\sqrt{d}}))\Lambda_{2,1}^{(t)} + \tilde{O}(\frac{1}{d^{0.49}})\Lambda_{2,1}^{(t)} \\ &\leq (1 + \tilde{O}(\frac{1}{d^{0.49}}))\Phi_2^{(t)} C_0 \alpha_1^6 C_2 \mathcal{E}_2^{(t)} (1 + \frac{1}{\text{polylog}(d)})(B_{2,1}^{(t)})^5 \quad (\text{when } t \leq t'_1) \end{aligned}$$

Since we also have

$$B_{1,1}^{(t+1)} \geq (1 - \tilde{O}(\frac{1}{d}))\Lambda_{1,1}^{(t)} \geq (1 - \tilde{O}(\frac{1}{d}))\Phi_1^{(t)} C_0 \alpha_1^6 \mathcal{E}_1^{(t)} (1 - \frac{1}{\text{polylog}(d)})(B_{1,1}^{(t)})^5$$

Since $B_{1,1}^{(0)} \geq B_{2,1}^{(0)}(1 + \Omega(\frac{1}{\log d}))$, we can now apply [Corollary 3.H.1](#) to the two sequence $B_{1,1}^{(t+1)}$ and $B_{2,1}^{(t+1)}$, where $S_t = \frac{\Phi_1^{(t)} \mathcal{E}_1^{(t)}}{\Phi_2^{(t)} \mathcal{E}_2^{(t)}}(1 + \frac{1}{\text{polylog}(d)})$ to get

$$B_{1,1}^{(t'_1)} \geq \frac{1}{d^{0.499}} \quad \text{while} \quad B_{2,1}^{(t'_1)} \leq \tilde{O}(\frac{1}{\sqrt{d}})$$

Note that here the update of $B_{2,1}^{(t)}$ at every step satisfies $\text{sign}(B_{2,1}^{(t+1)} - B_{2,1}^{(t)}) = \text{sign}(B_{2,1}^{(t)})$ which implies $B_{2,1}^{(t'_1)} = \tilde{\Theta}(\frac{1}{\sqrt{d}})$. Now for every $T \in [t'_1, T_1]$, we can apply [Lemma 3.H.1](#) to get that

$$\sum_{t \in [t'_1, T]} \eta \frac{(B_{2,1}^{(t)})^2}{(B_{1,1}^{(t)})^2} E_{1,2}^{(t)} \Lambda_{1,1}^{(t)} \leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) O(\frac{1}{B_{1,1}^{(t'_1)}}) \max_{t \leq T} \{(B_{2,1}^{(t)})^2\} \leq O(\frac{1}{d^{0.5+\Omega(1)}})$$

Suppose we have proved that $B_{2,1}^{(t)} \leq \tilde{O}(\frac{1}{\sqrt{d}})$ for each $t \leq T$, we define a new sequence

$$\begin{aligned} \tilde{B}_{2,1}^{(t+1)} &= \tilde{B}_{2,1}^{(t)} + \eta(1 + \tilde{O}(\frac{1}{d^{0.49}})) \Phi_2^{(t)} C_0 \alpha_1^6 C_2 \mathcal{E}_2^{(t)} (1 + \frac{1}{\text{polylog}(d)}) (\tilde{B}_{2,1}^{(t)})^5, \\ \text{where } \tilde{B}_{2,1}^{(t'_1)} &= B_{2,1}^{(t'_1)} + \sum_{t \in [t'_1, T]} \eta \frac{(B_{2,1}^{(t)})^2}{(B_{1,1}^{(t)})^2} E_{1,2}^{(t)} \Lambda_{1,1}^{(t)} = (1 \pm o(1)) \tilde{B}_{2,1}^{(t'_1)} \end{aligned}$$

It can be directly seen that $|\tilde{B}_{2,1}^{(t)} - \tilde{B}_{2,1}^{(0)}| \geq |B_{2,1}^{(t)} - B_{2,1}^{(0)}|$ for all $t \in [t'_1, T]$. Notice that now $\tilde{B}_{2,1}^{(t'_1)} \leq d^{\Omega(1)} B_{1,1}^{(t'_1)}$, we can now apply [Corollary 3.H.1](#) again to get

$$|B_{2,1}^{(T)} - B_{2,1}^{(0)}| \leq |\tilde{B}_{2,1}^{(T)} - \tilde{B}_{2,1}^{(0)}| \leq \frac{1}{\sqrt{d} \text{polylog}(d)} \quad (\text{for every } T \leq T_{1,1})$$

Now we deal with $t \in [T_{1,1}, T_1]$. During this stage, we can directly apply [Corollary 3.H.1](#) to $\tilde{B}_{2,1}^{(t)}$ and $B_{1,1}^{(t)}$, where $S_t = \frac{\Phi_1^{(t)} H_{1,2}^{(t)}}{\Phi_2^{(t)} H_{2,2}^{(t)}} \leq O(\alpha_1^{O(1)})$, to get that

$$|B_{2,1}^{(T)} - B_{2,1}^{(0)}| \leq |\tilde{B}_{2,1}^{(T)} - \tilde{B}_{2,1}^{(0)}| \leq \frac{1}{\sqrt{d} \text{polylog}(d)} \quad (\text{for every } T \leq T_1)$$

And thus by [Lemma 3.C.1](#), we have $B_{2,1}^{(T)} = B_{2,1}^{(0)}(1 \pm o(1))$.

The growth of $B_{1,2}^{(t)}$ and $B_{2,2}^{(t)}$: By [Lemma 3.C.5](#), we can write down the update as

$$B_{j,2}^{(t+1)} = B_{j,2}^{(t)} + \eta \left(1 \pm \tilde{O}(\alpha_1^6) (E_{3-j,j}^{(t)} + (B_{j,1}^{(t)})^3) \right) \Lambda_{j,2}^{(t)}$$

Since $B_{2,1}^{(t)} \leq \tilde{O}(\frac{1}{\sqrt{d}})$ and $E_{1,2}^{(t)} \leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}})B_{1,1}^{(t)}$, $E_{2,1}^{(t)} \leq \tilde{O}(\frac{1}{d})$ because we chose $\eta_E \leq \eta$, we only need to care about $(B_{1,1}^{(t)})^3$ in the update expression. Now define $t'_2 := \min\{t : B_{1,1}^{(t)} \geq \Omega(\frac{1}{\alpha_1^2})\}$, we have

- For $t \leq t'_2$, by [Corollary 3.H.1](#) and setting $x_t = B_{1,1}^{(t)}$, $C_t = (1 - \tilde{O}(\frac{1}{d}))\Phi_1^{(t)}C_0\alpha_1^6H_{1,2}^{(t)}$, $S_t = O(\frac{\alpha_2^6\Phi_j^{(t)}H_{j,1}^{(t)}}{\alpha_1^6\Phi_1^{(t)}H_{1,2}^{(t)}}) \leq \tilde{O}(\frac{\alpha_2^6}{\alpha_1^6}) \ll \frac{1}{\text{polylog}(d)}$ (by [Lemma 3.C.2a,c](#)), we have $|B_{j,2}^{(t)} - B_{j,2}^{(0)}| \leq O(\frac{\alpha_2^6}{\alpha_1^6} \frac{1}{\sqrt{d}}) \lesssim \frac{1}{\sqrt{d}\text{polylog}(d)}$ for all $t \leq t'_2$, which implies $B_{j,2}^{(t'_2)} = B_{j,2}^{(0)} \pm \frac{1}{\sqrt{d}\text{polylog}(d)} \in [\Omega(\frac{1}{\sqrt{d}\log d}), O(\frac{\sqrt{\log d}}{\sqrt{d}})]$ by [Lemma 3.C.1](#).
- For $t \in [t'_2, T_1]$, we can use [Corollary 3.H.1](#) again and let $x_t = B_{1,1}^{(t)}$, we know $B_{1,1}^{(t'_2)} \geq d^{\Omega(1)}B_{2,1}^{(t'_2)}$. Setting $C_t = (1 - \tilde{O}(\frac{1}{d}))\Phi_1^{(t)}C_0\alpha_1^6H_{1,2}^{(t)}$, $S_t = O((1 + \alpha_1^6)\frac{\alpha_2^6\Phi_j^{(t)}H_{j,1}^{(t)}}{\alpha_1^6\Phi_1^{(t)}H_{1,2}^{(t)}}) \leq O(\alpha^{O(1)})$, we can have $|B_{j,2}^{(t)} - B_{j,2}^{(t'_2)}| \lesssim \frac{1}{\sqrt{d}\text{polylog}(d)}$, which implies $B_{j,2}^{(t)} \in [\Omega(\frac{1}{\sqrt{d}\log d}), O(\frac{\sqrt{\log d}}{\sqrt{d}})]$ for all $t \in [t'_2, T_1]$.

This proves [Induction 3.C.1b](#). Indeed, simple calculations also proves [Induction 3.C.1c](#), since the update of $B_{1,1}^{(t)}$ is always larger than others' during $t \leq T_1$.

For [Induction 3.C.1d](#): From [Lemma 3.C.6](#), we can write down the update

$$-\nabla_{E_{1,2}}L(W^{(t)}, E^{(t)}) = O(\Lambda_{1,1}^{(t)}B_{1,1}^{(t)}) \left(-C_1E_{1,2}^{(t)} + \tilde{O}\left(\frac{(B_{1,2}^{(t)})^3}{(B_{1,1}^{(t)})^3}\right) + C_2(R_{1,2}^{(t)} + \varrho) \right)$$

for some constants $C_1, C_2 = \Theta(1)$. Applying [Lemma 3.H.1](#) to $O(\Lambda_{1,1}^{(t)}B_{1,1}^{(t)})\frac{(B_{1,2}^{(t)})^3}{(B_{1,1}^{(t)})^3}$, we can obtain

$$\sum_{t \leq T} O(\eta_E \Lambda_{1,1}^{(t)} B_{1,1}^{(t)}) \frac{(B_{1,2}^{(t)})^3}{(B_{1,1}^{(t)})^3} = \frac{\eta_E}{\eta} \sum_{t \leq T} O(\eta \Lambda_{1,1}^{(t)}) \frac{(B_{1,2}^{(t)})^3}{(B_{1,1}^{(t)})^2} \leq \tilde{O}\left(\frac{\eta_E/\eta}{d^{3/2}}\right) \frac{1}{B_{1,1}^{(0)}} \leq \tilde{O}\left(\frac{\eta_E/\eta}{d}\right)$$

So here it suffices to notice that whenever $|E_{1,2}^{(t)}| < 2\frac{C_2}{C_1}(R_{1,2}^{(t)} + \varrho)$ (which is obviously satisfied at $t = 0$), we would have

$$O(\Lambda_{1,1}^{(t)}B_{1,1}^{(t)}) \left(-O(E_{1,2}^{(t)}) + C_2(R_{1,2}^{(t)} + \varrho) \right) = -O(\Lambda_{1,1}^{(t)}B_{1,1}^{(t)})\tilde{O}(R_{1,2}^{(t)} + \varrho) \leq O(\Lambda_{1,1}^{(t)}B_{1,1}^{(t)})\tilde{O}\left(\varrho + \frac{1}{\sqrt{d}}\right)$$

In that case, we will always have (since $E_{1,2}^{(0)} = 0$)

$$E_{1,2}^{(t+1)} \leq \left| \sum_{t \leq T} \tilde{O}(\eta_E \Lambda_{1,1}^{(t)} B_{1,1}^{(t)}) \frac{(B_{1,2}^{(t)})^3}{(B_{1,1}^{(t)})^3} \right| + \sum_{s \leq t} O(\eta_E \Lambda_{1,1}^{(s)} B_{1,1}^{(s)}) (R_{1,2}^{(s)} + \varrho) \leq \tilde{O}\left(\varrho + \frac{1}{\sqrt{d}}\right) \frac{\eta_E}{\eta} B_{1,1}^{(t+1)}$$

Similarly for $\nabla_{E_{2,1}} L(W^{(t)}, E^{(t)})$, we can write down

$$-\nabla_{E_{2,1}} L(W^{(t)}, E^{(t)}) = \tilde{O}\left(\frac{(B_{1,2}^{(t)})^3}{(B_{1,1}^{(t)})^2}\right) \Lambda_{1,1}^{(t)} + \sum_{\ell \in [2]} C_2 \Lambda_{2,\ell}^{(t)} B_{2,\ell}^{(t)} \left(-O(E_{2,1}^{(t)} R_2^{(t)}) + O(R_{1,2}^{(t)})\right)$$

by [Lemma 3.H.1](#), we have

$$\sum_{t \leq T_1} \eta_E \tilde{O}\left(\frac{(B_{1,2}^{(t)})^3}{(B_{1,1}^{(t)})^2}\right) \Lambda_{1,1}^{(t)} \leq \tilde{O}\left(\frac{\eta_E/\eta}{d}\right)$$

and since from previous comparison results we know that

$$\sum_{t \leq T_1} \sum_{\ell \in [2]} \eta_E C_2 \Lambda_{2,\ell}^{(t)} B_{2,\ell}^{(t)} = \frac{\eta_E}{\eta} \sum_{t \leq T_1} \sum_{\ell \in [2]} \eta C_2 \Lambda_{2,\ell}^{(t)} B_{2,\ell}^{(t)} \leq \tilde{O}\left(\frac{\eta_E/\eta}{d}\right)$$

we can then prove the claim.

For Induction 3.C.1a: We can write down the update of $\|w_j^{(t)}\|_2^2$ as follows:

$$\begin{aligned} \|w_j^{(t+1)}\|_2^2 &= \|w_j^{(t)} - \eta \nabla_{w_j} L(W^{(t)}, E^{(t)})\|_2^2 \\ &= \|w_j^{(t)}\|_2^2 - \eta \langle \nabla_{w_j} L(W^{(t)}, E^{(t)}), w_j^{(t)} \rangle + \eta^2 \|\nabla_{w_j} L(W^{(t)}, E^{(t)})\|_2^2 \end{aligned}$$

from (3.B.2) and [Induction 3.C.1a,b,c](#) at iteration t and our assumption on ξ_p , we know

$$\|\nabla_{w_j} L(W^{(t)}, E^{(t)})\|_2^2 \leq \tilde{O}(d)$$

which allow us to choose $\eta \leq \frac{1}{\text{poly}(d)}$ to be small enough so that $\eta d T_1 \leq \frac{1}{\eta \text{poly}(d)}$. Then by [Lemma 3.C.3b](#), we have

$$\begin{aligned} \|w_j^{(t+1)}\|_2^2 &= \|w_j^{(0)}\|_2^2 \pm \eta \sum_{s \leq t} |\langle \nabla_{w_j} L(W^{(s)}, E^{(s)}), w_j^{(s)} \rangle| \pm \frac{1}{\text{poly}(d)} \\ &\leq \|w_j^{(0)}\|_2^2 \pm \eta \sum_{s \leq t} \tilde{O}\left(\varrho + \frac{1}{\sqrt{d}}\right) |\Lambda_{1,1}^{(s)}| \sum_{j \in [2]} |E_{j,3-j}^{(s)}| \pm \frac{1}{\text{poly}(d)} \end{aligned}$$

Since from the above analysis of the update of $B_{1,1}^{(t)}$, we know $\sum_{t \leq T_1} \Lambda_{1,1}^{(t)} \leq O(1)$. Moreover, we also know that $|B_{1,1}^{(t)}|$ is increasing and $\text{sign}(\Lambda_{1,1}^{(t)}) = \text{sign}(\Lambda_{1,1}^{(s)})$ for any $s, t \leq T_1$. Thus they imply $\sum_{s \leq t} |\Lambda_{1,1}^{(s)}| = |\sum_{s \leq t} \Lambda_{1,1}^{(s)}| = O(1)$, which can be combine with [Induction 3.C.1d](#) to prove the claim.

Proof of Induction 3.C.1e: We can write down the update of $R_{1,2}^{(t)} = \langle \Pi_{V^\perp} w_1^{(t)}, w_2^{(t)} \rangle$ as fol-

lows

$$\begin{aligned} \langle \Pi_{V^\perp} w_1^{(t+1)}, w_2^{(t+1)} \rangle &= \langle \Pi_{V^\perp} w_1^{(t)} - \Pi_{V^\perp} \eta \nabla_{w_1} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_2^{(t)} - \Pi_{V^\perp} \eta \nabla_{w_2} L(W^{(t)}, E^{(t)}) \rangle \\ &= R_{1,2}^{(t)} - \eta \langle \nabla_{w_1} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_2^{(t)} \rangle - \eta \langle \nabla_{w_2} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_1^{(t)} \rangle \\ &\quad + \eta^2 \langle \Pi_{V^\perp} \nabla_{w_1} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} \nabla_{w_2} L(W^{(t)}, E^{(t)}) \rangle \end{aligned}$$

By Cauchy-Schwarz inequality and the same analysis above we have

$$\begin{aligned} |\langle \Pi_{V^\perp} \nabla_{w_1} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} \nabla_{w_2} L(W^{(t)}, E^{(t)}) \rangle| &\leq \|\nabla_{w_1} L(W^{(t)}, E^{(t)})\|_2 \|\nabla_{w_2} L(W^{(t)}, E^{(t)})\|_2 \\ &\leq \tilde{O}(d) \end{aligned}$$

so by our choice of η

$$\sum_{t \leq T_1} \eta^2 |\langle \Pi_{V^\perp} \nabla_{w_1} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} \nabla_{w_2} L(W^{(t)}, E^{(t)}) \rangle| \leq \frac{1}{\text{poly}(d)}$$

and by [Lemma 3.C.7](#) we have

$$\left| -\eta \langle \nabla_{w_1} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_2^{(t)} \rangle - \eta \langle \nabla_{w_2} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_1^{(t)} \rangle \right| \leq \eta \tilde{O}(\Lambda_{1,1}^{(t)} B_{1,1}^{(t)}) (\varrho + \frac{1}{\sqrt{d}})$$

which implies

$$\begin{aligned} |\langle \Pi_{V^\perp} w_1^{(t+1)}, w_2^{(t+1)} \rangle| &\leq |\langle \Pi_{V^\perp} w_1^{(0)}, w_2^{(0)} \rangle| + \sum_{s \leq t} \sum_{j \in [2]} \eta |\langle \nabla_{w_j} L(W^{(s)}, E^{(s)}), \Pi_{V^\perp} w_{3-j}^{(s)} \rangle| + \frac{1}{\text{poly}(d)} \\ &\leq \tilde{O}(\frac{1}{\sqrt{d}}) + \sum_{s \leq t} \eta \tilde{O}(\Lambda_{1,1}^{(s)} B_{1,1}^{(s)}) + \frac{1}{\text{poly}(d)} \\ &\leq \tilde{O}(\frac{1}{\sqrt{d}}) + \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) B_{1,1}^{(t+1)} \\ &\leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) \end{aligned}$$

which completes the proof of [Induction 3.C.1](#). As for (a) – (e) of [Lemma 3.C.8](#), they are just direct corollary of our induction at $t = T_1$. $\square$

3.D Phase II: The Substitution Effect of Prediction Head

In this phase, As $B_{1,1}^{(t)}$ is learned to become very large ($B_{1,1}^{(t)} \gtrsim \|w_1^{(t)}\|_2$). The focus now shift to grow $E_{2,1}^{(t)}$, because we want $C_1 \alpha_1^6 ((B_{2,1}^{(t)})^3 + E_{2,1}^{(t)} (B_{1,1}^{(t)})^3)^2$ in $H_{2,1}^{(t)}$ to dominate $\mathcal{E}_{2,1}^{(t)}$. We can write

down the gradient of $E_{2,1}^{(t)}$ as

$$-\nabla_{E_{2,1}} L(W^{(t)}, E^{(t)}) = \sum_{\ell \in [2]} C_0 \Phi_2^{(t)} \alpha_\ell^6 (B_{2,\ell}^{(t)})^3 ((B_{1,\ell}^{(t)})^3 H_{2,3-\ell}^{(t)} - (B_{2,3-\ell}^{(t)})^3 K_{2,3-\ell}^{(t)}) - \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} \nabla_{E_{2,1}} \mathcal{E}_{2,1}^{(t)}$$

Now let us define

$$T_2 := \min\{t : R_2^{(t)} < \frac{1}{\log d} |E_{1,2}^{(t)}|\} \quad (3.D.1)$$

We will prove that $E_{2,1}^{(T_2)}$ reaches at most $O(\sqrt{\eta_E/\eta})$ and the following induction hypothesis holds throughout $t \in [T_1, T_2]$. In this phase, the learning of $E_{2,1}^{(t)}$ is much faster than the growth of the first feature v_1 such that $T_2 - T_1 = o(T_1/\sqrt{d})$, which is due to the acceleration effects brought by $B_{1,1}^{(t)} = \Omega(1)$ during this phase.

3.D.1 Induction in Phase II

We will be based on the following induction hypothesis during phase II.

Induction Hypothesis 3.D.1 (Phase II). *When $t \in [T_1, T_2]$, we hypothesize the followings would hold*

- (a) $B_{1,1}^{(t)} = \Theta(1)$, $B_{j,\ell}^{(t)} = B_{j,\ell}^{(T_1)}(1 \pm o(1)) = \tilde{\Theta}(\frac{1}{\sqrt{d}})$ for $(j, \ell) \neq (1, 1)$ and $\text{sign}(B_{j,\ell}^{(t)}) = \text{sign}(B_{j,\ell}^{(T_1)})$;
- (b) $|R_{1,2}^{(t)}| = \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) \alpha_1^{O(1)} [R_1^{(t)}]^{1/2} [R_2^{(t)}]^{1/2}$;
- (c) $R_1^{(t)} \in [\Omega(\frac{1}{d^{3/4}\alpha_1^2}), O(1)]$, $R_2^{(t)} \in [\Omega(\frac{1}{\log d} \sqrt{\eta_E/\eta}), O(1)]$;
- (d) $E_{1,2}^{(t)} \leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) [R_1^{(t)}]^{3/2}$ and $E_{2,1}^{(t)} \leq O(\sqrt{\eta_E/\eta})$.

Under **Induction 3.D.1**, we have some results as direct corollary.

Claim 3.D.1. At each iteration $t \in [T_1, T_2]$, if **Induction 3.C.1** holds, then

- (a) $\mathcal{E}_j^{(t)} = \Theta(C_2 [R_j^{(t)}]^3)$;
- (b) $\mathcal{E}_{j,3-j}^{(t)} = \mathcal{E}_j^{(t)} \pm \tilde{O}(E_{j,3-j}^{(t)} (\varrho + \frac{1}{\sqrt{d}}) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2}) + O((E_{j,3-j}^{(t)})^2 [R_{3-j}^{(t)}]^3)$ for each $j \in [2]$;

Proof. It is trivial to derive (a) from the expression of $\mathcal{E}_j^{(t)}$ and our assumption of ξ_p . For (b) it suffices to directly calculate the expression of $\mathcal{E}_{j,3-j}^{(t)}$ along with **Induction 3.D.1b**. $\square$

Lemma 3.D.1 (variables control in phase II). *In Phase II ($t \in [T_1, T_2]$), if **Induction 3.D.1** holds, then*

- (a) $\Phi_1^{(t)} = \tilde{\Theta}(\frac{1}{\alpha_1^{12}})$, $\Phi_2^{(t)} = \Theta((C_2 [R_2^{(t)}]^3 + C_1 \alpha_1^6 (E_{2,1}^{(t)})^2)^{-2})$;
- (b) $K_{1,\ell}^{(t)} = \tilde{O}(\alpha_\ell^6/d^{3/2})$, $K_{2,\ell}^{(t)} = \tilde{O}(E_{2,1}^{(t)} \alpha_\ell^6/d^{3/2} + \alpha_\ell^6/d^3)$;
- (c) $H_{1,1}^{(t)} = \Theta(C_1 \alpha_1^6)$, $H_{1,2}^{(t)} = \tilde{O}([R_1^{(t)}]^3)$, $H_{2,2}^{(t)} = \Theta(C_2 [R_2^{(t)}]^3)$, $H_{2,1}^{(t)} = \Theta(C_2 [R_2^{(t)}]^3 + C_1 \alpha_1^6 (E_{2,1}^{(t)})^2)$.

Proof. The proof of (a) directly follows from **Induction 3.D.1a,c** and **Claim 3.D.1**. The proof of (b) follows directly from the expression of $K_{j,\ell}$ and **Induction 3.D.1a,d**. The proof of (c) is also similar. $\square$

3.D.2 Gradient Lemmas for Phase II

Lemma 3.D.2 (learning prediction head $E_{1,2}, E_{2,1}$ in phase II). *If [Induction 3.D.1](#) holds at iteration $t \in [T_1, T_2]$, then we have*

$$\begin{aligned}
 (a) \quad -\nabla_{E_{1,2}} L(W^{(t)}, E^{(t)}) &= (1 \pm \tilde{O}(\frac{\alpha_1^{O(1)}}{d^{3/2}})) \Sigma_{1,1}^{(t)} (-2E_{1,2}^{(t)} [R_2^{(t)}]^3 \pm O(\bar{R}_{1,2}^{(t)} + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2}) \\
 &\quad \pm \Sigma_{1,1}^{(t)} \tilde{O}(\frac{\eta_E/\eta}{\sqrt{d}}) \max\{[R_1^{(t)}]^3, \frac{\alpha_1^{O(1)}}{d^{5/2}}\}, \\
 (b) \quad -\nabla_{E_{2,1}} L(W^{(t)}, E^{(t)}) &= (1 \pm \tilde{O}(\frac{\alpha_1^{O(1)}}{d^{3/2}})) C_0 \Phi_2^{(t)} \alpha_1^6 (B_{2,1}^{(t)})^3 (B_{1,1}^{(t)})^3 H_{2,2}^{(t)} \\
 &\quad \pm O(\Sigma_{2,1}^{(t)}) (|E_{2,1}^{(t)}| [R_1^{(t)}]^3 \pm O(\bar{R}_{1,2}^{(t)} + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2})
 \end{aligned}$$

Proof. We first write down the gradient for $E_{j,3-j}^{(t)}$: (ignoring the time superscript (t))

$$-\nabla_{E_{j,3-j}} L(W, E) = \sum_{\ell \in [2]} C_0 \Phi_j \alpha_\ell^6 B_{j,\ell}^3 (B_{3-j,\ell}^3 H_{j,3-\ell} - B_{3-j,3-\ell}^3 K_{j,3-\ell}) - \sum_{\ell \in [2]} \Sigma_{j,\ell} \nabla_{E_{j,3-j}} \mathcal{E}_{j,3-j}$$

where $\nabla_{E_{j,3-j}} \mathcal{E}_{j,3-j} = \mathbb{E} [2\langle w_j, \xi_p \rangle^3 \langle w_{3-j}, \xi_p \rangle^3 + 2E_{j,3-j} \langle w_{3-j}, \xi_p \rangle^6]$. Thus we have

$$\nabla_{E_{j,3-j}} \mathcal{E}_{j,3-j}^{(t)} = 2E_{j,3-j}^{(t)} [R_{3-j}^{(t)}]^3 \pm O(\bar{R}_{1,2}^{(t)} + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2}$$

and by [Claim 3.B.1](#) and [Induction 3.D.1a](#), if $(j, \ell) \neq (1, 1)$

$$\Sigma_{j,\ell}^{(t)} = O(\Sigma_{1,1}^{(t)}) \frac{(B_{j,\ell}^{(t)})^6 + E_{j,3-j}^{(t)} (B_{3-j,\ell}^{(t)})^3 (B_{j,\ell}^{(t)})^3 \Phi_j^{(t)}}{(B_{1,1}^{(t)})^6} \frac{\Phi_j^{(t)}}{\Phi_1^{(t)}} \leq o(\frac{1}{d^{3/2}}) \Sigma_{1,1}^{(t)} \frac{\Phi_j^{(t)}}{\Phi_1^{(t)}}$$

Therefore for $j = 1$:

$$\sum_{\ell \in [2]} \Sigma_{1,\ell}^{(t)} \nabla_{E_{1,2}} \mathcal{E}_{1,2}^{(t)} = (1 \pm \tilde{O}(\frac{\alpha_1^{O(1)}}{d^{3/2}})) \Sigma_{1,1}^{(t)} \nabla_{E_{1,2}} \mathcal{E}_{1,2}^{(t)}$$

Now by [Induction 3.D.1a,c](#) and [Lemma 3.D.1b,c](#) we have $(B_{1,\ell}^{(t)})^3 H_{1,3-\ell}^{(t)} \leq \max\{\Theta(C_2 [R_1^{(t)}]^3), \tilde{O}(\frac{\alpha_1^6}{d^{3/2}})\}$, which leads to the bounds

$$|(B_{1,\ell}^{(t)})^3 (B_{2,\ell}^{(t)})^3 H_{1,3-\ell}^{(t)}| \leq \tilde{O}(\frac{1}{d^{3/2}}) \max\{[R_1^{(t)}]^3, \frac{\alpha_1^6}{d^3}\}, \quad |(B_{1,\ell}^{(t)})^3 (B_{2,3-\ell}^{(t)})^3 K_{1,3-\ell}^{(t)}| \leq \tilde{O}(\frac{1}{d^3})$$

which implies

$$\left| \sum_{\ell \in [2]} C_0 \Phi_1^{(t)} \alpha_\ell^6 (B_{1,\ell}^{(t)})^3 ((B_{2,\ell}^{(t)})^3 H_{1,3-\ell}^{(t)} - (B_{2,3-\ell}^{(t)})^3 K_{1,3-\ell}^{(t)}) \right| \lesssim \tilde{O}\left(\frac{\eta_E/\eta}{\sqrt{d}}\right) \Sigma_{1,1}^{(t)} \max\{[R_1^{(t)}]^3, \frac{\alpha_1^{O(1)}}{d^{5/2}}\}$$

Combining above together, we have

$$\begin{aligned} & -\nabla_{E_{1,2}} L(W^{(t)}, E^{(t)}) \\ &= (1 + o(\frac{1}{d^{3/2}})) \Sigma_{1,1}^{(t)} (-2E_{1,2}^{(t)} [R_2^{(t)}]^3 \pm O(\bar{R}_{1,2}^{(t)} + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2} \pm \tilde{O}(\frac{\eta_E/\eta}{\sqrt{d}}) \max\{[R_1^{(t)}]^3, \frac{\alpha_1^{O(1)}}{d^{5/2}}\}) \end{aligned}$$

For $-\nabla_{E_{2,1}} L(W^{(t)}, E^{(t)})$, the expression is slightly different, we first observe that by [Induction 3.D.1a](#)

$$\Delta_{2,2}^{(t)} \leq \tilde{O}(\frac{1}{d^{3/2}}) \Delta_{2,1}^{(t)}$$

Meanwhile, by [Induction 3.D.1a](#) and [Lemma 3.D.1b,c](#), we have

$$\Xi_2^{(t)} \leq \tilde{O}(\frac{\alpha_1^{O(1)}}{d^3}) C_0 C_2 \Phi_2^{(t)} [R_2^{(t)}]^3,$$

Moreover, we can also calculate $\Sigma_{2,1}^{(t)} = C_0 C_2 \alpha_1^6 E_{2,1}^{(t)} \Phi_2^{(t)} (B_{2,1}^{(t)})^3 = \tilde{O}(\frac{\alpha_1^6}{d^{3/2}}) \Phi_2^{(t)}$, $\Sigma_{2,2}^{(t)} = \tilde{O}(\frac{\alpha_2^6}{d^3}) \Phi_2^{(t)}$, which gives

$$\sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} \nabla_{E_{2,1}} \mathcal{E}_{2,1}^{(t)} = \Sigma_{2,1}^{(t)} (-\Theta(E_{2,1}^{(t)}) [R_1^{(t)}]^3 \pm O(\bar{R}_{1,2}^{(t)} + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2})$$

Now we combine the above results and get

$$\begin{aligned} -\nabla_{E_{2,1}} L(W^{(t)}, E^{(t)}) &= (1 \pm \tilde{O}(\frac{\alpha_1^{O(1)}}{d^{3/2}})) C_0 \Phi_2^{(t)} \alpha_1^6 (B_{2,1}^{(t)})^3 (B_{1,1}^{(t)})^3 \mathcal{E}_{2,1}^{(t)} \\ &\quad \pm O(\Sigma_{2,1}^{(t)}) (|E_{2,1}^{(t)}| [R_1^{(t)}]^3 \pm O(\bar{R}_{1,2}^{(t)} + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2}) \end{aligned}$$

□

Lemma 3.D.3 (reducing noise in phase II). Suppose [Induction 3.D.1](#) holds at $t \in [T_1, T_2]$, then
(a)

$$\begin{aligned} & \langle -\nabla_{w_1} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_1^{(t)} \rangle \\ &= \Sigma_{1,1}^{(t)} \Theta\left(-[R_1^{(t)}]^3 \pm \tilde{O}\left(|E_{1,2}^{(t)}| + \frac{|E_{2,1}^{(t)}|^2}{d^{3/2}}\right) (\bar{R}_{1,2}^{(t)} + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2}\right); \end{aligned}$$

(b)

$$\begin{aligned}
& \langle -\nabla_{w_1} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_2^{(t)} \rangle \\
&= \Sigma_{1,1}^{(t)} \left((-\Theta(\bar{R}_{1,2}^{(t)}) + O(\varrho)) [R_1^{(t)}]^{5/2} [R_2^{(t)}]^{1/2} \right. \\
&\quad \left. + \tilde{O}\left(|E_{1,2}^{(t)}| + \frac{|E_{2,1}^{(t)}|^2}{d^{3/2}}\right) R_1^{(t)} [R_2^{(t)}]^2 \right).
\end{aligned}$$

And furthermore

$$\begin{aligned}
(c) \quad & \langle -\nabla_{w_2} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_2^{(t)} \rangle = -\Theta([R_2^{(t)}]^3) \left(\Sigma_{1,1}^{(t)} \Theta((E_{1,2}^{(t)})^2) + \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} \right) \\
& \quad \pm O\left(\sum_{j,\ell} \Sigma_{j,\ell}^{(t)} E_{j,3-j}^{(t)} (\bar{R}_{1,2}^{(t)} + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2} \right); \\
(d) \quad & \langle -\nabla_{w_2} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_1^{(t)} \rangle = \left(\Sigma_{1,1}^{(t)} \Theta((E_{1,2}^{(t)})^2) + \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} \right) (-\Theta(\bar{R}_{1,2}^{(t)}) \pm O(\varrho)) [R_2^{(t)}]^{5/2} [R_1^{(t)}]^{1/2} \\
& \quad + O\left(\sum_{j,\ell} \Sigma_{j,\ell}^{(t)} E_{j,3-j}^{(t)} R_2^{(t)} [R_1^{(t)}]^2 \right)
\end{aligned}$$

Proof. The proof can be obtained directly from some calculation using [Claim 3.B.1](#) as follows:

Proof of (a): From [\(3.B.2\)](#), we can obtain that

$$\langle -\nabla_{w_1} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_1^{(t)} \rangle = - \sum_{j,\ell} \Sigma_{j,\ell}^{(t)} \langle \nabla_{w_1} \mathcal{E}_{j,3-j}^{(t)}, w_1^{(t)} \rangle$$

Now from [Claim 3.B.1a](#) and [Induction 3.D.1a](#), we know $(B_{j,\ell}^{(t)})^3 \leq \tilde{O}(\frac{1}{d^{3/2}})$ and the following

$$\Sigma_{j,\ell}^{(t)} = O(\Sigma_{1,1}^{(t)}) \frac{(B_{j,\ell}^{(t)})^6 + E_{j,3-j}^{(t)} (B_{3-j,\ell}^{(t)})^3 (B_{j,\ell}^{(t)})^3 \Phi_j^{(t)}}{(B_{1,1}^{(t)})^6 \Phi_1^{(t)}} \leq \tilde{O}\left(\frac{E_{j,3-j}^{(t)}}{d^{3/2}}\right) \Sigma_{1,1}^{(t)} \frac{\Phi_j^{(t)}}{\Phi_1^{(t)}} \quad \text{for any } (j, \ell) \neq (1, 1)$$

From [Induction 3.D.1a,c](#), we know $((B_{2,\ell}^{(t)})^3 + E_{2,1}^{(t)} (B_{1,\ell}^{(t)})^3)^2 \leq \tilde{O}(\frac{1}{d^{3/2}}) E_{2,1}^{(t)}$ and $R_2^{(t)} = \Theta(1)$, which by [Claim 3.D.1a,b](#) and [Lemma 3.D.1a](#) gives $\Phi_2^{(t)} / \Phi_1^{(t)} \leq \tilde{O}(\alpha_1^{O(1)})$. Combine the bounds above, we can obtain $\Sigma_{j,\ell}^{(t)} = \tilde{O}(E_{j,3-j}^{(t)} / d^{3/2}) \Sigma_{1,1}^{(t)}$. We can then directly apply [Claim 3.B.1](#) to prove [Lemma 3.D.3a](#) as follows

$$\begin{aligned}
& \langle -\nabla_{w_1} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_1^{(t)} \rangle \\
&= (1 \pm \tilde{O}(E_{1,2}^{(t)})) \Sigma_{1,1}^{(t)} \left(-\Theta([R_1^{(t)}]^3) \pm O(E_{1,2}^{(t)}) (\bar{R}_{1,2}^{(t)} + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2} \right) \\
&\quad + \tilde{O}(E_{2,1}^{(t)} / d^{3/2}) \Sigma_{1,1}^{(t)} \left(-\Theta((E_{2,1}^{(t)})^2) [R_1^{(t)}]^3 \pm O(E_{2,1}^{(t)}) (\bar{R}_{1,2}^{(t)} + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2} \right)
\end{aligned}$$

$$\begin{aligned}
&= \Theta(\Sigma_{1,1}^{(t)}) \left(-[R_1^{(t)}]^3 \pm \tilde{O}(|E_{1,2}^{(t)}| + \frac{|E_{2,1}^{(t)}|^2}{d^{3/2}})(\bar{R}_{1,2}^{(t)} + \varrho)[R_1^{(t)}]^{3/2}[R_2^{(t)}]^{3/2} \right) \\
&\quad \text{(Since } |E_{1,2}^{(t)}| \leq d^{-\Omega(1)} \text{ by Induction 3.D.1c,d)}
\end{aligned}$$

Proof of (b): For Lemma 3.D.3b, we can use the same analysis for $\Sigma_{1,1}^{(t)}$ above and Claim 3.B.1(d,e) to get (again we have used $\Sigma_{j,\ell}^{(t)} = \tilde{O}(E_{j,3-j}^{(t)})\Sigma_{1,1}^{(t)} = o(\Sigma_{1,1}^{(t)})$)

$$\begin{aligned}
&\langle -\nabla_{w_1} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_2^{(t)} \rangle \\
&= (1 \pm \tilde{O}(E_{1,2}^{(t)}))\Sigma_{1,1}^{(t)} \left((-\Theta(\bar{R}_{1,2}^{(t)}) \pm O(\varrho))[R_1^{(t)}]^{5/2}[R_2^{(t)}]^{1/2} + E_{1,2}^{(t)}R_1^{(t)}[R_2^{(t)}]^2 \right) \\
&\quad + \tilde{O}(E_{2,1}^{(t)}/d^{3/2})\Sigma_{1,1}^{(t)} \left((-\Theta(\bar{R}_{1,2}^{(t)}) + O(\varrho))(E_{2,1}^{(t)})^2[R_1^{(t)}]^{5/2}[R_2^{(t)}]^{1/2} + E_{2,1}^{(t)}R_1^{(t)}[R_2^{(t)}]^2 \right) \\
&= \Sigma_{1,1}^{(t)} \left((-\Theta(\bar{R}_{1,2}^{(t)}) + O(\varrho))[R_1^{(t)}]^{5/2}[R_2^{(t)}]^{1/2} + \tilde{O}(|E_{1,2}^{(t)}| + \frac{|E_{2,1}^{(t)}|^2}{d^{3/2}})R_1^{(t)}[R_2^{(t)}]^2 \right)
\end{aligned}$$

Proof of (c): Similarly to the proof of (a), we can also expand as follows

$$\begin{aligned}
&\langle -\nabla_{w_2} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_2^{(t)} \rangle \\
&= (1 \pm O(E_{1,2}^{(t)}))\Sigma_{1,1}^{(t)} \left(-[R_2^{(t)}]^3 \Theta((E_{1,2}^{(t)})^2) \pm O(E_{1,2}^{(t)})(\bar{R}_{1,2}^{(t)} + \varrho)[R_1^{(t)}]^{3/2}[R_2^{(t)}]^{3/2} \right) \\
&\quad - \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} \left([R_2^{(t)}]^3 \pm O(E_{2,1}^{(t)})(\bar{R}_{1,2}^{(t)} + \varrho)[R_1^{(t)}]^{3/2}[R_2^{(t)}]^{3/2} \right) \\
&= -[R_2^{(t)}]^3 \left(\Sigma_{1,1}^{(t)} \Theta((E_{1,2}^{(t)})^2) + \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} \right) \pm O \left(\sum_{j,\ell} \Sigma_{j,\ell}^{(t)} E_{j,3-j}^{(t)} (\bar{R}_{1,2}^{(t)} + \varrho)[R_1^{(t)}]^{3/2}[R_2^{(t)}]^{3/2} \right)
\end{aligned}$$

Proof of (d): Similarly, we can calculate (again by $\Sigma_{j,\ell}^{(t)} = \tilde{O}(E_{j,3-j}^{(t)})\Sigma_{1,1}^{(t)} = o(\Sigma_{1,1}^{(t)})$)

$$\begin{aligned}
&\langle -\nabla_{w_2} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_1^{(t)} \rangle \\
&= \sum_{\ell \in [2]} \Sigma_{1,\ell}^{(t)} \left((-\Theta(\bar{R}_{1,2}^{(t)}) \pm O(\varrho))(E_{1,2}^{(t)})^2[R_2^{(t)}]^{5/2}[R_1^{(t)}]^{1/2} + E_{1,2}^{(t)}R_2^{(t)}[R_1^{(t)}]^2 \right) \\
&\quad + \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} \left((-\Theta(\bar{R}_{1,2}^{(t)}) \pm O(\varrho))[R_2^{(t)}]^{5/2}[R_1^{(t)}]^{1/2} + E_{2,1}^{(t)}R_2^{(t)}[R_1^{(t)}]^2 \right) \\
&= (1 \pm \tilde{O}(E_{1,2}^{(t)}))\Sigma_{1,1}^{(t)} \left((-\Theta(\bar{R}_{1,2}^{(t)}) \pm O(\varrho))(E_{1,2}^{(t)})^2[R_2^{(t)}]^{5/2}[R_1^{(t)}]^{1/2} + E_{1,2}^{(t)}R_2^{(t)}[R_1^{(t)}]^2 \right) \\
&\quad + \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} \left((-\Theta(\bar{R}_{1,2}^{(t)}) \pm O(\varrho))[R_2^{(t)}]^{5/2}[R_1^{(t)}]^{1/2} + E_{1,2}^{(t)}R_2^{(t)}[R_1^{(t)}]^2 \right) \\
&= \left(\Sigma_{1,1}^{(t)} \Theta((E_{1,2}^{(t)})^2) + \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} \right) (-\Theta(\bar{R}_{1,2}^{(t)}) \pm O(\varrho))[R_2^{(t)}]^{5/2}[R_1^{(t)}]^{1/2} + O \left(\sum_{j,\ell} \Sigma_{j,\ell}^{(t)} E_{j,3-j}^{(t)} R_2^{(t)}[R_1^{(t)}]^2 \right)
\end{aligned}$$

which completes the proof. $\square$

Lemma 3.D.4 (learning feature v_2 in phase II). *For each $t \in [T_1, T_2]$, if Induction 3.D.1 holds at iteration t , then we have for each $j \in [2]$:*

$$|\langle -\nabla_{w_j} L(W^{(t)}, E^{(t)}), v_2 \rangle| \leq \tilde{O}\left(\frac{\alpha_2^6 \alpha_1^6}{d^{5/2}}\right) \left(\Phi_j^{(t)} (|E_{j,3-j}^{(t)}| + [R_j^{(t)}]^3) + \Phi_{3-j}^{(t)} (|E_{3-j,j}^{(t)}| [R_{3-j}^{(t)}]^3 + \frac{|E_{3-j,j}^{(t)}|^2}{d^{3/2}}) \right)$$

Proof. Again as in the proof of Lemma 3.C.4, we expand the notations: (ignoring the superscript $^{(t)}$ for the RHS)

$$\langle -\nabla_{w_j} L(W^{(t)}, E^{(t)}), v_2 \rangle = \Lambda_{j,2}^{(t)} + \Gamma_{j,2}^{(t)} - \Upsilon_{j,2}^{(t)} \quad (3.D.2)$$

where

$$\begin{aligned} \Lambda_{j,2}^{(t)} &= C_0 \alpha_2^6 \Phi_j^{(t)} H_{j,1}^{(t)} (B_{j,2}^{(t)})^5 \\ \Gamma_{j,2}^{(t)} &= C_0 \alpha_2^6 \Phi_{3-j}^{(t)} E_{3-j,j}^{(t)} (B_{3-j,2}^{(t)})^3 (B_{j,2}^{(t)})^2 H_{3-j,1}^{(t)} \\ \Upsilon_{j,2}^{(t)} &= C_0 \alpha_1^6 \left(\Phi_j^{(t)} (B_{j,1}^{(t)})^3 (B_{j,2}^{(t)})^2 K_{j,2}^{(t)} + \Phi_{3-j}^{(t)} E_{3-j,j}^{(t)} (B_{3-j,1}^{(t)})^3 (B_{j,2}^{(t)})^2 K_{3-j,2}^{(t)} \right) \end{aligned}$$

Now we further write $\Upsilon_{j,2}^{(t)} = \Upsilon_{j,2,1}^{(t)} + \Upsilon_{j,2,2}^{(t)}$, where

$$\Upsilon_{j,2,1}^{(t)} = C_0 \alpha_1^6 \Phi_j^{(t)} (B_{j,1}^{(t)})^3 (B_{j,2}^{(t)})^2 K_{j,2}^{(t)}, \quad \Upsilon_{j,2,2}^{(t)} = \Phi_{3-j}^{(t)} E_{3-j,j}^{(t)} (B_{3-j,1}^{(t)})^3 (B_{3-j,2}^{(t)})^2 K_{3-j,2}^{(t)}$$

According to (3.D.2), we can first compute

$$\begin{aligned} \Lambda_{j,2}^{(t)} - \Upsilon_{j,2,1}^{(t)} &= C_0 \alpha_2^6 \Phi_j^{(t)} (B_{j,2}^{(t)})^5 H_{j,1}^{(t)} - C_0 \alpha_1^6 \Phi_j^{(t)} (B_{j,1}^{(t)})^3 (B_{j,2}^{(t)})^2 K_{j,2}^{(t)} \\ &= C_0 \alpha_2^6 \Phi_j^{(t)} (B_{j,2}^{(t)})^5 \left(C_1 \alpha_1^6 ((B_{j,1}^{(t)})^3 + E_{j,3-j}^{(t)} (B_{3-j,1}^{(t)})^3)^2 + C_2 \mathcal{E}_{j,3-j}^{(t)} \right) \\ &\quad - C_0 \alpha_1^6 \Phi_j^{(t)} (B_{j,1}^{(t)})^3 (B_{j,2}^{(t)})^2 C_1 \alpha_2^6 ((B_{j,2}^{(t)})^3 + E_{j,3-j}^{(t)} (B_{3-j,2}^{(t)})^3) ((B_{j,1}^{(t)})^3 + E_{j,3-j}^{(t)} (B_{3-j,1}^{(t)})^3) \\ &= C_0 \alpha_2^6 C_1 \alpha_1^6 \Phi_j^{(t)} (B_{j,2}^{(t)})^5 \left(E_{j,3-j}^{(t)} (B_{3-j,1}^{(t)})^3 (B_{j,1}^{(t)})^3 + (E_{j,3-j}^{(t)})^2 (B_{3-j,1}^{(t)})^6 \right) \\ &\quad - C_0 \alpha_2^6 C_1 \alpha_1^6 \Phi_j^{(t)} (B_{j,2}^{(t)})^2 (B_{3-j,2}^{(t)})^3 E_{j,3-j}^{(t)} \left((B_{j,1}^{(t)})^6 + E_{j,3-j}^{(t)} (B_{3-j,1}^{(t)})^3 (B_{j,1}^{(t)})^3 \right) \\ &\quad + C_0 \alpha_2^6 \Phi_j^{(t)} (B_{j,2}^{(t)})^5 C_2 \mathcal{E}_{j,3-j}^{(t)} \end{aligned}$$

Then we can apply Induction 3.D.1a,c,d, Claim 3.D.1a,b and Lemma 3.D.1a,c to get

$$|\Lambda_{j,2}^{(t)} - \Upsilon_{j,2,1}^{(t)}| \leq \tilde{O}\left(\frac{\alpha_2^6}{\alpha_1^6 d^{5/2}}\right) \Phi_j^{(t)} (|E_{j,3-j}^{(t)}| + [R_j^{(t)}]^3)$$

where the last inequality is due to Lemma 3.D.1a,c. Similarly, we can also compute for $\Gamma_{j,2}^{(t)} - \Upsilon_{j,2,2}^{(t)}$:

$$|\Gamma_{j,2}^{(t)} - \Upsilon_{j,2,2}^{(t)}| \leq \left| C_0 \alpha_2^6 \Phi_{3-j}^{(t)} E_{3-j,j}^{(t)} (B_{3-j,2}^{(t)})^3 (B_{j,2}^{(t)})^2 H_{3-j,1}^{(t)} \right|$$

$$\begin{aligned}
& + \left| C_0 \alpha_1^6 \Phi_{3-j}^{(t)} E_{3-j,j}^{(t)} (B_{3-j,1}^{(t)})^3 (B_{j,2}^{(t)})^2 K_{3-j,2}^{(t)} \right| \\
& \leq \tilde{O}\left(\frac{\alpha_1^6 \alpha_2^6}{d^{5/2}}\right) \Phi_{3-j}^{(t)} |E_{3-j,j}^{(t)}| ([R_{3-j}^{(t)}]^3 + \frac{|E_{3-j,j}^{(t)}|}{d^{3/2}})
\end{aligned}$$

This completes the proof $\square$

Lemma 3.D.5 (learning feature v_1 in Phase II). *For each $t \in [T_1, T_2]$, if Induction 3.D.1 holds at iteration t , then we have:*

$$\begin{aligned}
(a) \quad & \langle -\nabla_{w_1} L(W^{(t)}, E^{(t)}), v_1 \rangle = \Theta(\Sigma_{1,1}^{(t)}) [R_1^{(t)}]^3 + \Gamma_{1,1}^{(t)} \pm \tilde{O}(\alpha_1^{O(1)} / d^{5/2}); \\
(b) \quad & \langle -\nabla_{w_2} L(W^{(t)}, E^{(t)}), v_1 \rangle = \tilde{O}(\alpha_1^{O(1)} / d^{5/2}) + \tilde{O}\left(\frac{\alpha_1^6}{d}\right) E_{1,2}^{(t)} \Phi_1^{(t)} [R_1^{(t)}]^3
\end{aligned}$$

Proof. As in the proof of Lemma 3.D.4, we expand the gradient terms:

$$\langle -\nabla_{w_j} L(W^{(t)}, E^{(t)}), v_1 \rangle = \Lambda_{j,2}^{(t)} + \Gamma_{j,2}^{(t)} - \Upsilon_{j,2}^{(t)} \quad (3.D.3)$$

where

$$\begin{aligned}
\Lambda_{j,1}^{(t)} &= C_0 \alpha_1^6 \Phi_j^{(t)} H_{j,2}^{(t)} (B_{j,1}^{(t)})^5 \\
\Gamma_{j,1}^{(t)} &= C_0 \alpha_1^6 \Phi_{3-j}^{(t)} E_{3-j,j}^{(t)} (B_{3-j,1}^{(t)})^3 (B_{j,1}^{(t)})^2 H_{3-j,2}^{(t)} \\
\Upsilon_{j,1}^{(t)} &= C_0 \alpha_1^6 \left(\Phi_j^{(t)} (B_{j,2}^{(t)})^3 (B_{j,1}^{(t)})^2 K_{j,1}^{(t)} + \Phi_{3-j}^{(t)} E_{3-j,j}^{(t)} (B_{3-j,2}^{(t)})^3 (B_{j,1}^{(t)})^2 K_{3-j,1}^{(t)} \right)
\end{aligned}$$

Indeed, when $j = 1$, by Induction 3.D.1a and Lemma 3.D.1a,c, we can compute

$$\Lambda_{1,1}^{(t)} = C_0 \alpha_1^6 \Phi_1^{(t)} (B_{1,1}^{(t)})^5 H_{1,2}^{(t)} = \Theta(\Sigma_{1,1}^{(t)}) [R_1^{(t)}]^3$$

and with additionally Lemma 3.D.1b, we also have

$$|\Upsilon_{1,1}^{(t)}| = \left| C_0 \alpha_1^6 \left(\Phi_1^{(t)} (B_{1,2}^{(t)})^3 (B_{1,1}^{(t)})^2 K_{1,1}^{(t)} + \Phi_2^{(t)} E_{2,j}^{(t)} (B_{2,2}^{(t)})^3 (B_{1,1}^{(t)})^2 K_{2,1}^{(t)} \right) \right| \leq \tilde{O}\left(\frac{\alpha_1^{O(1)}}{d^{5/2}}\right)$$

which gives the proof of (a). For (b), we can also apply Induction 3.D.1a and Lemma 3.D.1a,c to get

$$\begin{aligned}
\Lambda_{2,1}^{(t)} &= C_0 \alpha_1^6 \Phi_2^{(t)} H_{2,2}^{(t)} (B_{2,1}^{(t)})^5 \leq \tilde{O}(\alpha_1^{O(1)} / d^{5/2}) \\
\Gamma_{2,1}^{(t)} &= C_0 \alpha_1^6 \Phi_1^{(t)} E_{1,2}^{(t)} (B_{1,1}^{(t)})^3 (B_{2,1}^{(t)})^2 H_{1,2}^{(t)} \leq \tilde{O}\left(\frac{1}{d}\right) E_{1,2}^{(t)} \Phi_1^{(t)} \frac{[R_1^{(t)}]^3}{\alpha_1^6} \\
\Upsilon_{2,1}^{(t)} &= C_0 \alpha_1^6 \left(\Phi_2^{(t)} (B_{2,2}^{(t)})^3 (B_{2,1}^{(t)})^2 K_{2,1}^{(t)} + \Phi_1^{(t)} E_{1,2}^{(t)} (B_{1,2}^{(t)})^3 (B_{2,1}^{(t)})^2 K_{1,1}^{(t)} \right) \leq \tilde{O}\left(\frac{\alpha_1^6}{d^4}\right)
\end{aligned}$$

this finishes the proof. $\square$

3.D.3 At the End of Phase II

Now we shall present the main theorem of this section, which gives the result of prediction head $E_{2,1}^{(t)}$ growth after the feature v_1 is learned in the first stage.

Lemma 3.D.6 (Phase II). *Suppose $\eta = \frac{1}{\text{poly}(d)}$ is sufficiently small, then [Induction 3.D.1](#) holds for all iteration $t \in [T_1, T_2]$, and at iteration $t = T_2$, the followings holds:*

- (a) $B_{1,1}^{(T_2)} = \Theta(1)$, $B_{j,\ell}^{(T_2)} = B_{j,\ell}^{(T_1)}(1 \pm o(1)) = \tilde{\Theta}(\frac{1}{\sqrt{d}})$ for $(j, \ell) \neq (1, 1)$
- (b) $R_1^{(T_2)} \leq \tilde{O}(\frac{1}{d^{3/4}})$, $R_2^{(T_2)} = \Theta(\sqrt{\eta_E/\eta})$, and $\bar{R}_{1,2}^{(T_2)} \leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}})$;
- (c) $|E_{1,2}^{(T_2)}| = \tilde{O}(\varrho + \frac{1}{\sqrt{d}})[R_1^{(T_2)}]^{3/2}[R_2^{(T_2)}]^{3/2}$ and $|E_{2,1}^{(T_2)}| = \Theta(\sqrt{\eta_E/\eta})$

Where the part of learning $E_{2,1}^{(t)}$ is what we called substitution effect. One can easily verify that $|E_{2,1}^{(t)} f_1(X^{(1)})| \gg |f_2(X^{(1)})|$ when X is equipped with feature v_1 , as stated in [Lemma 3.5.2](#).

Proof. We first will prove [Induction 3.D.1](#) holds for all iteration $t \in [T_1, T_2]$. We shall first prove that if [Induction 3.D.1](#) continues to hold when $R_2^{(t)} \geq |E_{2,1}^{(t)}|$, we shall have $[R_1^{(t)}]$ decreasing at an exponential rate.

Proof of the decrease of $R_1^{(t)}$: Firstly, we write down the update of $R_1^{(t)}$ using [Lemma 3.D.3a](#):

$$R_1^{(t+1)} = R_1^{(t)} + \eta \Sigma_{1,1}^{(t)} \Theta(-[R_1^{(t)}]^3 \pm O(|E_{1,2}^{(t)}| + \frac{|E_{2,1}^{(t)}|^2}{d^{3/2}})(\bar{R}_{1,2}^{(t)} + \varrho)[R_1^{(t)}]^{3/2}[R_2^{(t)}]^{3/2})$$

from the expression of $\Sigma_{1,1}^{(t)}$ in (3.B.2), and by [Induction 3.D.1a](#) and [Lemma 3.D.1a,c](#), we can compute

$$\Sigma_{1,1}^{(t)} = \Theta(C_0 C_2 \Phi_1^{(t)}) = \Theta(\frac{C_0 C_2}{\alpha_1^{12}})$$

Moreover, from [Induction 3.D.1c](#) we know that

$$\begin{aligned} (|E_{1,2}^{(t)}| + \frac{|E_{2,1}^{(t)}|^2}{d^{3/2}})[R_1^{(t)}]^{3/2}[R_2^{(t)}]^{3/2} &\leq (\tilde{\Theta}(\frac{1}{d^{3/2}}) + \tilde{O}(\varrho + \frac{1}{\sqrt{d}})[R_1^{(t)}]^{3/2})[R_1^{(t)}]^{3/2}[R_2^{(t)}]^{3/2} \\ &\leq (\tilde{\Theta}(\frac{1}{d^{3/2}}) + \tilde{O}(\varrho + \frac{1}{\sqrt{d}})[R_1^{(t)}]^{3/2})[R_1^{(t)}]^{3/2} \end{aligned}$$

Therefore whenever $R_1^{(t)} \geq \frac{\alpha_1^{18}}{d^{3/4}}$ (which $t \leq T_2$ suffices), we shall have always have

$$(\bar{R}_{1,2}^{(t)} + \varrho)(\tilde{\Theta}(\frac{1}{d^{3/2}}) + \tilde{O}(\varrho + \frac{1}{\sqrt{d}})[R_1^{(t)}]^{3/2})[R_1^{(t)}]^{3/2} \leq o([R_1^{(t)}]^3)$$

which implies, if we set $T'_2 := \min\{t : R_1^{(t)} \geq \frac{1}{d^{3/4}\alpha_1^2}\}$, then for all $t \in [T_1, T'_2]$, we will have

$$R_1^{(t+1)} = R_1^{(t)} + \eta \Sigma_{1,1}^{(t)} \Theta(-[R_1^{(t)}]^3 \pm O(|E_{1,2}^{(t)}| + \frac{|E_{2,1}^{(t)}|^2}{d^{3/2}})(\bar{R}_{1,2}^{(t)} + \varrho)[R_1^{(t)}]^{3/2}[R_2^{(t)}]^{3/2})$$

$$\begin{aligned}
&= R_1^{(t)} - \Theta(\eta \Sigma_{1,1}^{(t)}) [R_1^{(t)}]^3 \\
&\leq R_1^{(t)} (1 - \Theta(\frac{\eta C_0 C_2}{\alpha_1^{12}}) \frac{1}{d^{3/2} \alpha_1^2}) \quad (\text{since } R_1^{(t)} \geq \frac{1}{d^{3/4}})
\end{aligned} \tag{3.D.4}$$

From the last inequality we know that after $T_2 = T_1 + \tilde{\Theta}(\frac{d^{1.5}}{\eta \alpha_1^{O(1)}})$, we shall have $R_1^{(t)} \leq O(\frac{\alpha_1^{O(1)}}{d^{3/4}})$. Moreover, suppose $T_2' < T_2$, (which just mean $R_1^{(s)} \leq O(\frac{1}{d^{3/4} \alpha_1^2})$ for some iteration $s \in [T_1, T_2]$) we also have

$$\begin{aligned}
R_1^{(t+1)} &= R_1^{(t)} - \Theta(\eta \Sigma_{1,1}^{(t)}) [R_1^{(t)}]^3 \\
&\geq R_1^{(t)} (1 - \Theta(\frac{\eta C_0 C_2}{\alpha_1^{14}}) \frac{1}{d^{3/2}})
\end{aligned}$$

So when $T_2 \leq T_1 + \tilde{O}(\frac{d^{1.5} \alpha_1^{12}}{\eta})$ iterations, we will have $R_1^{(t)} \geq R_1^{(s)} (1 - \Theta(\frac{\eta C_0 C_2}{d^{3/2} \alpha_1^{14}}))^{T_2 - T_1} \geq \Omega(R_1^{(t)})$ for all $t \in [s, T_2]$, which means we have a lower bound $R_1^{(t)} \geq \frac{1}{d^{3/4} \alpha_1^2}$ throughout $t \in [T_1, T_2]$. This proves [Lemma 3.D.6a](#) and also our induction on $R_1^{(t)}$.

Proof of induction for $E_{1,2}^{(t)}$: By [Lemma 3.D.2a](#), we can write

$$\begin{aligned}
-\nabla_{E_{1,2}} L(W^{(t)}, E^{(t)}) &= (1 + \tilde{O}(\frac{\alpha_1^{O(1)}}{d^{3/2}})) \Sigma_{1,1}^{(t)} (-2E_{1,2}^{(t)} [R_2^{(t)}]^3 \pm O(\bar{R}_{1,2}^{(t)} + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2}) \\
&\quad \pm \Sigma_{1,1}^{(t)} \tilde{O}(\frac{\eta_E / \eta}{\sqrt{d}}) \max\{[R_1^{(t)}]^3, \frac{\alpha_1^{O(1)}}{d^{5/2}}\} \\
&= -\Theta(\Sigma_{1,1}^{(t)} [R_2^{(t)}]^3) E_{1,2}^{(t)} \pm O(\Sigma_{1,1}^{(t)}) \left((\bar{R}_{1,2}^{(t)} + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2} + \tilde{O}(\frac{\eta_E / \eta}{\sqrt{d}}) [R_1^{(t)}]^3 \right)
\end{aligned}$$

Since again from [Induction 3.D.1b,c](#) that $\bar{R}_{1,2}^{(t)} \leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}})$, $R_1^{(t)} = O(1)$, $R_2^{(t)} \in [\sqrt{\eta_E / \eta}, O(1)]$, we can obtain the update of $E_{1,2}^{(t)}$ as

$$\begin{aligned}
E_{1,2}^{(t+1)} &= E_{1,2}^{(t)} (1 - \Theta(\eta_E \Sigma_{1,1}^{(t)} [R_2^{(t)}]^3)) \pm \tilde{O}(\eta_E \Sigma_{1,1}^{(t)}) \left((\varrho + \frac{1}{\sqrt{d}}) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2} + \tilde{O}(\frac{\eta_E / \eta}{\sqrt{d}}) [R_1^{(t)}]^3 \right) \\
&= E_{1,2}^{(t)} (1 - \Theta(\eta_E \Sigma_{1,1}^{(t)} [R_2^{(t)}]^3)) \pm \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) \eta_E \Sigma_{1,1}^{(t)} [R_1^{(t)}]^{3/2} \\
&= E_{1,2}^{(t)} (1 - \Theta(\eta_E \Sigma_{1,1}^{(t)} [R_2^{(t)}]^3)) \pm \eta_E \Sigma_{1,1}^{(t)} J_{1,2}^{(t)}
\end{aligned}$$

where $J_{1,2}^{(t)} = \tilde{C}(\varrho + \frac{1}{\sqrt{d}})[R_1^{(t)}]^{3/2} > 0$ and $\tilde{C} = \tilde{\Theta}(1)$ is larger than the hidden constant (including the $\text{polylog}(d)$ factors) of $E_{2,1}^{(T_1)} \leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}})$ in [Lemma 3.C.8d](#). And then we can compute

$$\begin{aligned} J_{1,2}^{(t+1)} &= \tilde{C}(\varrho + \frac{1}{\sqrt{d}})[R_1^{(t+1)}]^{3/2} \\ &= \tilde{C}(\varrho + \frac{1}{\sqrt{d}})[R_1^{(t)}]^{3/2}(1 - \Theta(\eta \Sigma_{1,1}^{(t)}[R_1^{(t)}]^2))^{3/2} \quad (\text{due to calculations in (3.D.4)}) \\ &= J_{1,2}^{(t)}(1 - \Theta(\eta^{3/2}(\Sigma_{1,1}^{(t)})^{3/2}[R_1^{(t)}]^3)) \quad (\text{because } \eta \Sigma_{1,1}^{(t)} = \frac{\alpha_1^{O(1)}}{\text{poly}(d)} \text{ is very small}) \end{aligned}$$

Now by [Lemma 3.C.8d](#), we know $|E_{1,2}^{(T_1)}| \leq J_{1,2}^{(T_1)}$; then we begin our induction that $|E_{1,2}^{(t)}| < (\log \log d)J_{1,2}^{(t)}$ at for all iterations $t \in [T_1, T_2]$. Now assume we have $|E_{1,2}^{(t)}| = \frac{1}{2}(\log \log d)J_{1,2}^{(t)}$ ⁴, from above calculations it holds that $|E_{1,2}^{(t+1)}| = |E_{1,2}^{(t)}|(1 - \Theta(\eta \Sigma_{1,1}^{(t)}[R_1^{(t)}]^3))$. Then we would have

$$\begin{aligned} \frac{J_{1,2}^{(t+1)}}{J_{1,2}^{(t)}} &\geq (1 - \Theta(\eta^{3/2}(\Sigma_{1,1}^{(t)})^{3/2}[R_1^{(t)}]^3)) \geq (1 - \Theta(\eta \Sigma_{1,1}^{(t)}[R_2^{(t)}]^3)) \geq \frac{|E_{1,2}^{(t+1)}|}{|E_{1,2}^{(t)}|} \\ &\quad (\text{because of the range of } R_1^{(t)} \text{ and } R_2^{(t)}) \end{aligned}$$

This proved that $|E_{1,2}^{(t+1)}| \lesssim \log \log d \cdot J_{1,2}^{(t+1)} \leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}})[R_1^{(t+1)}]^{3/2}$ and also the induction can go on until $t = T_2$.

Proof of the growth of $E_{2,1}^{(t)}$ and $T_2 \leq T_1 + O(\frac{d^{1.5}}{\eta \alpha_1^4})$: According to [Lemma 3.D.2b](#), we can write down the update of $E_{2,1}^{(t)}$ as

$$\begin{aligned} -\nabla_{E_{2,1}} L(W^{(t)}, E^{(t)}) &= (1 \pm O(\frac{\alpha_1^{O(1)}}{d^{3/2}}))\Delta_{2,1}^{(t)} \\ &\quad \pm O(\Sigma_{2,1}^{(t)})(|E_{2,1}^{(t)}|[R_1^{(t)}]^3 \pm O(\bar{R}_{1,2}^{(t)} + \varrho)[R_1^{(t)}]^{3/2}[R_2^{(t)}]^{3/2}) \end{aligned}$$

Then, from [Lemma 3.D.1a,c](#) and [Induction 3.D.1](#), we have

$$O(\Sigma_{2,1}^{(t)})(|E_{2,1}^{(t)}|[R_1^{(t)}]^3 \pm O(\bar{R}_{1,2}^{(t)} + \varrho)[R_1^{(t)}]^{3/2}[R_2^{(t)}]^{3/2}) \leq O(\frac{\text{polylog}(d)}{d^{3/2}\alpha_1^2})\Phi_2^{(t)} \leq O(\frac{1}{d^{3/2}\alpha_1})\Phi_2^{(t)}$$

and also

$$\left| (1 \pm \tilde{O}(\frac{\alpha_1^6}{d^{0.3}}))C_0\Phi_2^{(t)}\alpha_1^6(B_{2,1}^{(t)})^3(B_{1,1}^{(t)})^3H_{2,2}^{(t)} \right| \geq \tilde{\Theta}(\frac{\alpha_1^6}{d^{3/2}})\Phi_2^{(t)}$$

⁴ If we want $|E_{1,2}^{(t)}| > (\log \log d)J_{1,2}^{(t)}$, then as long as $\eta = \frac{1}{\text{poly}(d)}$ is small enough, we can always assume to have found some iteration $t' \in (T_1, t]$ such that $|E_{1,2}^{(t')}| = \frac{1}{2}(\log \log d)J_{1,2}^{(t')}$, and we set $t = t'$ and start our argument from that iteration.

Now by [Lemma 3.D.1a](#) and [Induction 3.D.1a](#), it allow us to simplify the update to

$$\begin{aligned}
E_{2,1}^{(t+1)} &= E_{2,1}^{(t)} - \eta_E \nabla_{E_{2,1}} L(W^{(t)}, E^{(t)}) \\
&= E_{2,1}^{(t)} + (1 \pm \frac{1}{\alpha_1^{\Omega(1)}}) \eta_E C_0 C_2 \alpha_1^6 \Phi_2^{(t)} (B_{2,1}^{(t)})^3 (B_{1,1}^{(t)})^3 \mathcal{E}_{2,1}^{(t)} \\
&\geq E_{2,1}^{(t)} + \eta_E \tilde{\Theta}(\frac{1}{d^{3/2} \alpha_1^6}) \text{sign}(B_{1,1}^{(t)}) \text{sign}(B_{2,1}^{(t)}) \quad (\text{by } \text{Induction 3.D.1} \text{ and } \text{Claim 3.D.1})
\end{aligned}$$

Now since $\text{sign}(B_{j,1}^{(t)}) = \text{sign}(B_{j,1}^{(T_1)})$, we know there is an iteration $T'_{2,1} \leq T_1 + O(\frac{d^{1/2} \alpha_1^{O(1)}}{\eta})$ such that for all $t \in [T'_{2,1}, T_2]$, it holds

$$\begin{aligned}
|E_{2,1}^{(t)}| &= \left| E_{2,1}^{(T_1)} + \sum_{t \in [T_1, T'_{2,1}]} \Theta(\eta_E C_0 C_2 \alpha_1^6) \Phi_2^{(t)} (B_{2,1}^{(t)})^3 (B_{1,1}^{(t)})^3 [R_2^{(t)}]^3 \right| \\
&= \left| |E_{2,1}^{(T_1)}| \pm \sum_{s \in [T_1, T'_{2,1}]} \eta_E \tilde{\Theta}(\frac{1}{d^{3/2} \alpha_1^{O(1)}}) \right| \\
&\in \left[2|E_{2,1}^{(T_1)}|, \tilde{O}(\frac{\alpha_1^{O(1)}}{d}) \right]
\end{aligned}$$

and thus $\text{sign}(E_{2,1}^{(t)}) = \prod_{j \in [2]} \text{sign}(B_{j,1}^{(t)})$ and $|E_{2,1}^{(t)}|$ will be increasing during $t \in [T'_{2,1}, T_2]$. Thus as long as $R_2^{(t)} \geq |E_{2,1}^{(t)}|$ continues to hold, after at most $\tilde{\Theta}(\frac{d^{1.5}}{\eta \alpha_1^6})$ iterations starting from T_1 , we shall have $|E_{2,1}^{(t)}| \geq \Omega(\sqrt{\eta_E/\eta})$.

However, in order to actually prove $|E_{2,1}^{(T_2)}| = \Theta(\sqrt{\eta_E/\eta})$, we will need to ensure that (1) there exist some constant $C = \Omega(\sqrt{\eta_E/\eta})$ such that $|E_{2,1}^{(t)}| > C$ while $R_2^{(s)} \geq \frac{1}{\log d} |E_{2,1}^{(t)}|$ for all $s \in [T_1, t]$; (2) we shall have a upper bound $|E_{2,1}^{(t)}| < O(\sqrt{\eta_E/\eta})$. They will be done below.

Proof of $E_{2,1}^{(T_2)} = \Theta(\sqrt{\eta_E/\eta})$ and $T_2 = T_1 + \tilde{O}(\frac{d^{3/2} \alpha_1^{O(1)}}{\eta})$: In fact, [Induction 3.D.1c](#) are already proved since we have already calculated the dynamics of $R_1^{(t)}$ and its upper bound and lower bound. In this part we are going to prove $T_2 = T_1 + \tilde{\Theta}(\frac{d^{1.5} \alpha_1^{12}}{\eta})$ (which means that $R_2^{(t)} \leq |E_{2,1}^{(t)}|$ can be achieved in $\tilde{O}(\frac{d^{3/2} \alpha_1^{12}}{\eta})$ many iterations). From [Lemma 3.D.3c](#), we can write down the update for $R_2^{(t)}$ as

$$\begin{aligned}
R_2^{(t+1)} &= R_2^{(t)} - 2\eta \langle \nabla_{w_2} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_2^{(t)} \rangle + \eta^2 \|\Pi_{V^\perp} \nabla_{w_2} L(W^{(t)}, E^{(t)})\|_2^2 \\
&= R_2^{(t)} - \eta \Theta([R_2^{(t)}]^3) \left(\Sigma_{1,1}^{(t)} \Theta((E_{1,2}^{(t)})^2) + \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} \right) \\
&\quad \pm \eta O\left(\sum_{j,\ell} \Sigma_{j,\ell}^{(t)} E_{j,3-j}^{(t)} (\bar{R}_{1,2}^{(t)} + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2} \right) + \frac{\eta}{\text{poly}(d)}
\end{aligned}$$

where we have used the fact that $\|\Pi_{V^\perp} \nabla_{w_2} L(W^{(t)}, E^{(t)})\|_2^2 \leq \tilde{O}(d^2)$ from our assumption on the noise ξ_p and a simple bound for $\Sigma_{j,\ell}^{(t)}$ as we have done before. Next we can resort to [Induction 3.D.1d](#) that $|E_{1,2}^{(t)}| \leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}})[R_1^{(t)}]^{3/2}$ to derive

$$\begin{aligned} \sum_{s \in [T_1, t]} \eta \Sigma_{1,1}^{(s)} \Theta((E_{1,2}^{(s)})^2) &\leq \sum_{s \in [T_1, t]} \tilde{O}(\varrho^2 + \frac{1}{d}) \eta \Sigma_{1,1}^{(s)} [R_1^{(s)}]^3 \\ &\leq \tilde{O}(\varrho^2 + \frac{1}{d}) = o(1) \end{aligned}$$

which is because $\sum_{t \in [T_1, T_2]} \Theta(\eta \Sigma_{1,1}^{(t)}) [R_1^{(t)}]^3 \leq O(1)$ and $\Sigma_{1,1}^{(t)} > 0$ as we have calculated in the proof of [Induction 3.D.1a](#) above. Similarly, we can also bound

$$\sum_{s \in [T_1, t]} \Sigma_{1,\ell}^{(s)} |E_{1,2}^{(s)}| (|\bar{R}_{1,2}^{(s)}| + \varrho) [R_1^{(s)}]^{3/2} [R_2^{(s)}]^{3/2} \leq \sum_{s \in [T_1, t]} \tilde{O}(\varrho^2 + \frac{1}{d}) \eta \Sigma_{1,\ell}^{(s)} [R_1^{(s)}]^3 \leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) = o(1)$$

Moreover, because $T_2 \leq T_1 + \tilde{O}(\frac{d^{3/2} \alpha_1^{12}}{\eta})$ and $|E_{2,1}^{(t)}| \leq O(1)$, $\Phi_2^{(t)} \leq \alpha_1^{O(1)}$ from [Induction 3.D.1](#), we have for each $t \leq T_2$:

$$\begin{aligned} \sum_{s \in [T_1, t]} \eta \Sigma_{2,\ell}^{(s)} |E_{2,1}^{(s)}| (|\bar{R}_{1,2}^{(s)}| + \varrho) [R_1^{(s)}]^{3/2} [R_2^{(s)}]^{3/2} &\leq \tilde{O}(\frac{|E_{2,1}^{(s)}|^2}{d^{3/2}}) \sum_{s \in [T_1, t]} \eta \Phi_2^{(s)} \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) \\ &\leq \tilde{O}(\frac{\eta}{d^{3/2}}) \cdot \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) \cdot \tilde{O}(\frac{d^{3/2} \alpha_1^{12}}{\eta}) \\ &\leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) \alpha_1^{O(1)} = o(1) \end{aligned}$$

Thus combining all the bounds above, we have proved that for each $t \in [T_1, T_2]$, it holds

$$\begin{aligned} R_2^{(t)} &= R_2^{(T_1)} - \sum_{s \in [T_1, t]} \Theta(\eta \Sigma_{2,1}^{(s)}) [R_2^{(s)}]^3 \pm o(1) \\ &= R_2^{(T_1)} - \sum_{s \in [T_1, t]} \Theta(\eta C_0 C_2) E_{2,1}^{(s)} \alpha_1^6 \Phi_2^{(s)} (B_{2,1}^{(s)})^3 (B_{1,1}^{(s)})^3 [R_2^{(s)}]^3 \pm o(1) \end{aligned} \quad (3.D.5)$$

$$= R_2^{(T_1)} - \sum_{s \in [T_1, t]} \eta E_{2,1}^{(s)} \tilde{\Theta}(\frac{1}{d^{3/2}}) \Phi_2^{(s)} [R_2^{(s)}]^3 \cdot \text{sign}(E_{2,1}^{(s)}) \cdot \text{sign}(B_{2,1}^{(T_1)}) \cdot \text{sign}(B_{1,1}^{(T_1)}) \pm o(1) \quad (3.D.6)$$

where the last equality is because $\text{sign}(B_{j,\ell}^{(t)}) \equiv \text{sign}(B_{j,\ell}^{(T_1)})$ by [Induction 3.D.1a](#). Now from what we have proved above on the growth of $E_{2,1}^{(t)}$ that $\text{sign}(E_{2,1}^{(t)}) = \text{sign}(B_{1,1}^{(t)} B_{2,1}^{(t)}) \equiv \text{sign}(B_{1,1}^{(T_1)} B_{2,1}^{(T_1)})$ throughout the rest of phase II (which is just $t \in [T'_{2,1}, T_2]$). Recall that

$$R_2^{(T'_{2,1})} = R_2^{(T_1)} \pm o(1), \quad \text{and} \quad E_{2,1}^{(t)} - E_{2,1}^{(T'_{2,1})} = \sum_{s \in [T'_{2,1}, t]} \Theta(\eta E C_0 C_2) \Phi_2^{(s)} (B_{2,1}^{(s)})^3 (B_{1,1}^{(s)})^3$$

The above arguments imply for $t \in [T'_{2,1}, T_2]$:

$$\begin{aligned} R_2^{(t+1)} &= R_2^{(T_1)} - \sum_{s \in [T'_{2,1}, t]} \Theta(\eta C_0 C_2) E_{2,1}^{(s)} \Phi_2^{(s)} (B_{2,1}^{(s)})^3 (B_{1,1}^{(s)})^3 [R_2^{(t)}]^3 \pm o(1) \\ &= R_2^{(T_1)} - \Theta\left(\frac{\eta}{\eta_E} |E_{2,1}^{(t)}|^2\right) - o(1) \end{aligned}$$

Now we can confirm

- (1) there exist a constant $C = \Theta(\sqrt{\eta_E/\eta})$ such that $E_{2,1}^{(t)} = C$ if $R_2^{(t)}$ falls below $\frac{1}{\log d} |E_{2,1}^{(t)}|$;
- (2) $T_2 = T_1 + \tilde{\Theta}\left(\frac{d^{3/2}\alpha_1^{12}}{\eta}\right)$ due to the growth $|E_{2,1}^{(t+1)}| = |E_{2,1}^{(t)}| + \eta_E \tilde{\Theta}\left(\frac{1}{d^{3/2}\alpha_1^{12}\sqrt{\eta_E/\eta}}\right)$ for $t \in [T'_{2,1}, T_2]$.

which are the desired results.

Proof of Induction 3.D.1a: We first obtain from Lemma 3.D.5a that the update of $B_{1,1}^{(t)}$ can be written as

$$B_{1,1}^{(t+1)} = B_{1,1}^{(t)} + \eta \left(\Theta(\Sigma_{1,1}^{(t)}) \text{sign}(B_{1,1}^{(t)}) [R_1^{(t)}]^3 + \Gamma_{1,1}^{(t)} \pm \tilde{O}(\alpha_1^{O(1)}/d^{5/2}) \right)$$

Now by what we have calculated above in (3.D.4), the total decrease of $R_1^{(t)}$ is (since $R_1^{(t)}$ is monotone in this phase)

$$\sum_{t \in [T_1, T_2]} \Theta(\eta \Sigma_{1,1}^{(t)}) [R_1^{(t)}]^3 \leq O(R_1^{(T_1)} - R_1^{(T_2)}) \leq O(1)$$

And also since $T_2 \leq T_1 + \tilde{\Theta}\left(\frac{d^{3/2}\alpha_1^{12}}{\eta}\right)$, we can bound

$$\sum_{t \in [T_1, T_2]} \tilde{O}(\alpha_1^6/d^{5/2}) \leq \tilde{O}(\alpha_1^{O(1)}/d^{5/2}) \cdot \tilde{O}\left(\frac{d^{3/2}}{\eta \alpha_1^6}\right) \leq \tilde{O}(\alpha_1^{O(1)}/d)$$

Now we consider how the $\Gamma_{1,1}^{(t)}$ term accumulates

$$\begin{aligned} \sum_{t \in [T_1, T_2]} \eta \Gamma_{1,1}^{(t)} &= \left(\sum_{t \in [T_1, T'_{2,1}]} + \sum_{t \in [T'_{2,1}, T_2]} \right) \eta C_0 \alpha_1^6 E_{2,1}^{(t)} \Phi_2^{(t)} (B_{2,1}^{(t)})^3 (B_{1,1}^{(t)})^2 H_{2,2}^{(t)} \\ &\stackrel{(1)}{=} \tilde{O}\left(\frac{\alpha_1^{12}}{d}\right) + \sum_{t \in [T'_{2,1}, T_2]} O\left(\eta C_0 \alpha_1^6 \Phi_2^{(t)} |B_{2,1}^{(t)}|^3 |B_{1,1}^{(t)}|^3 H_{2,2}^{(t)}\right) \text{sign}(B_{1,1}^{(t)}) \\ &= \pm o(1) + O(1) \text{sign}(B_{1,1}^{(t)}) \end{aligned}$$

where in (1) we have used $|E_{2,1}^{(t)}| \leq O(1) \leq O(B_{1,1}^{(t)})$ and $\text{sign}(E_{2,1}^{(t)}) = \prod_{j \in [2]} \text{sign}(B_{j,1}^{(t)})$ when $t \in [T'_{2,1}, T_2]$. These calculations tell us $B_{1,1}^{(t)} = B_{1,1}^{(T_1)} + O(1) \text{sign}(B_{1,1}^{(T_1)}) \pm O(\frac{1}{\alpha_1}) = \Theta(1)$ for all

iterations $t \in [T_1, T_2]$. Similarly from Lemma 3.D.5b, for $B_{2,1}^{(t)}$ we can also write

$$B_{2,1}^{(T+1)} = B_{2,1}^{(t)} + \eta \tilde{O}(\alpha_1^{O(1)} / d^{5/2}) + \tilde{O}(\frac{\alpha_1^6}{d}) E_{2,1}^{(t)} \Phi_1^{(t)} [R_1^{(t)}]^3$$

From similar calculations, it holds $B_{2,1}^{(t)} = B_{2,1}^{(T_1)} \pm \tilde{O}(\alpha_1^{O(1)} / d)$, which proves that $B_{2,1}^{(t)} = B_{2,1}^{(T_1)} (1 \pm o(1))$ when $t \in [T_1, T_2]$. Now we turn to feature v_2 . By Lemma 3.D.4 we have for $j \in [2]$:

$$\begin{aligned} |\langle -\nabla_{w_j} L(W^{(t)}, E^{(t)}), v_2 \rangle| &\leq \tilde{O}(\frac{\alpha_2^6 \alpha_1^6}{d^{5/2}}) \left(\Phi_j^{(t)} (|E_{j,3-j}^{(t)}| + [R_j^{(t)}]^3) + \Phi_{3-j}^{(t)} (|E_{3-j,j}^{(t)}| [R_{3-j}^{(t)}]^3 + \frac{|E_{3-j,j}^{(t)}|^2}{d^{3/2}}) \right) \\ &\leq \tilde{O}(\frac{\alpha_2^6 \alpha_1^6}{d^{5/2}}) \end{aligned}$$

where the last inequality is from Lemma 3.D.1a and Induction 3.D.1c,d. Thus when $t \leq T_2 = T_1 + \tilde{O}(\frac{d^{3/2} \alpha_1^{12}}{\eta})$ we would have

$$B_{j,2}^{(t)} = B_{j,2}^{(T_1)} \pm \tilde{O}(\frac{\alpha_1^{O(1)}}{d}) = B_{j,2}^{(T_1)} (1 \pm o(1)) \quad \text{since } B_{j,2}^{(T_1)} = \tilde{\Theta}(\frac{1}{\sqrt{d}}) \text{ by Lemma 3.C.8c}$$

Together they proved Induction 3.D.1a and Lemma 3.D.6a. Moreover, we have also

Proof of Induction 3.D.1b: Firstly, we write down the update of $R_{1,2}^{(t)}$ using Lemma 3.D.3b,d as follows:

$$\begin{aligned} R_{1,2}^{(t+1)} &= R_{1,2}^{(t)} - \eta \langle \nabla_{w_1} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_2^{(t)} \rangle - \eta \langle \nabla_{w_2} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_1^{(t)} \rangle \\ &\quad + \eta^2 \langle \Pi_{V^\perp} \nabla_{w_1} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} \nabla_{w_2} L(W^{(t)}, E^{(t)}) \rangle \\ &= R_{1,2}^{(t)} + \eta \Sigma_{1,1}^{(t)} ((-\Theta(\bar{R}_{1,2}^{(t)}) \pm O(\varrho)) [R_1^{(t)}]^{5/2} [R_2^{(t)}]^{1/2} + \tilde{O}(|E_{1,2}^{(t)}| + \frac{|E_{2,1}^{(t)}|^2}{d^{3/2}}) R_1^{(t)} [R_2^{(t)}]^2) \\ &\quad + \eta \left(\Sigma_{1,1}^{(t)} \Theta((E_{1,2}^{(t)})^2) + \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} \right) (-\Theta(\bar{R}_{1,2}^{(t)}) \pm O(\varrho)) [R_2^{(t)}]^{5/2} [R_1^{(t)}]^{1/2} \\ &\quad + O\left(\sum_{j,\ell} \eta \Sigma_{j,\ell}^{(t)} E_{j,3-j}^{(t)} R_2^{(t)} [R_1^{(t)}]^2 \right) + \frac{\eta}{\text{poly}(d)} \end{aligned}$$

where in the last inequality we have used

$$\begin{aligned} &|\langle \Pi_{V^\perp} \nabla_{w_1} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} \nabla_{w_2} L(W^{(t)}, E^{(t)}) \rangle| \\ &\leq \|\Pi_{V^\perp} \nabla_{w_1} L(W^{(t)}, E^{(t)})\|_2 \|\Pi_{V^\perp} \nabla_{w_2} L(W^{(t)}, E^{(t)})\|_2 \leq \tilde{O}(d) \end{aligned}$$

Now from [Induction 3.D.1c,d](#) that $R_2^{(t)} = \Theta(1)$ and $|E_{1,2}^{(t)}| \leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}})[R_1^{(t)}]^{3/2}$, $|E_{2,1}^{(t)}| \leq O(\sqrt{\eta_E/\eta})$, we can further obtain $|\Sigma_{2,2}^{(t)}| = \tilde{O}(\frac{\alpha_1^{O(1)}}{d^{3/2}})|\Sigma_{2,1}^{(t)}|$, and the bound

$$\begin{aligned} R_{1,2}^{(t+1)} &= R_{1,2}^{(t)} \left(1 - \Theta(\eta \Sigma_{1,1}^{(t)})[R_1^{(t)}]^2 - \Theta(\eta(\Sigma_{1,1}^{(t)}(E_{1,2}^{(t)})^2 + \Sigma_{2,1}^{(t)}))[R_2^{(t)}]^2 \right) \\ &\quad \pm \eta O(\varrho)[R_2^{(t)}]^{1/2}[R_1^{(t)}]^{1/2} \left(O(\Sigma_{1,1}^{(t)})[R_1^{(t)}]^2 + \left(\Sigma_{1,1}^{(t)}\Theta((E_{1,2}^{(t)})^2) + \Sigma_{2,1}^{(t)} \right)[R_2^{(t)}]^2 \right) \end{aligned}$$

Notice here that there exist a constant $C = \Theta(1)$, whenever $|R_{1,2}^{(t)}| \geq C(\varrho + \frac{1}{\sqrt{d}})[R_2^{(t)}]^{1/2}[R_1^{(t)}]^{1/2}$, it will holds

$$\begin{aligned} R_{1,2}^{(t+1)} &= R_{1,2}^{(t)} \left(1 - \Theta(\eta \Sigma_{1,1}^{(t)})[R_1^{(t)}]^2 - \Theta(\eta(\Sigma_{1,1}^{(t)}(E_{1,2}^{(t)})^2 + \Sigma_{2,1}^{(t)}))[R_2^{(t)}]^2 \right) \\ &= R_{1,2}^{(t)} \left(1 - \Theta(\eta \Sigma_{1,1}^{(t)})[R_1^{(t)}]^2 - \Theta(\eta(\Sigma_{1,1}^{(t)}(E_{1,2}^{(t)})^2 + \frac{\alpha_1^6}{d^{3/2}}\Sigma_{2,1}^{(t)}))[R_2^{(t)}]^2 \right) \end{aligned}$$

Thus we can go through the same analysis as in the proof of induction for $E_{1,2}^{(t)}$ to derive that

$$|R_{1,2}^{(t)}| \leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}})[R_2^{(t)}]^{1/2}[R_1^{(t)}]^{1/2}$$

which is the desired result. Note that at the end of phase II

$$\begin{aligned} \text{Induction 3.D.1a} &\implies \text{Lemma 3.D.6a} \\ \text{Induction 3.D.1b,c} &\implies \text{Lemma 3.D.6b} \\ \text{Induction 3.D.1d} &\implies \text{Lemma 3.D.6c} \end{aligned}$$

We now complete the proof of [Lemma 3.D.6](#). □

3.E Phase III: The Acceleration Effect of Prediction Head

We shall prove in this section that the growth of $E_{2,1}^{(t)}$ in the previous phase creates an acceleration effect to the growth of $B_{2,2}^{(t)}$, which will finally outrun the growth of $B_{2,1}^{(t)}$ to win the lottery. We define

$$T_3 := \min \left\{ t : |B_{2,2}^{(t)}| \geq \frac{1}{2} \min\{|B_{1,1}^{(t)}|, \sqrt{\frac{\eta}{\eta_E}}|E_{2,1}^{(t)}|\} \right\} \quad (3.E.1)$$

and we call iterations $t \in [T_2, T_3]$ as the phase III of training and $t \geq T_3$ as the end phase of training.

3.E.1 Induction in Phase III

Induction Hypothesis 3.E.1 (Phase III). *During $t \in [T_2, T_3]$, we hypothesize the following conditions holds.*

- (a) $|B_{1,1}^{(t)}| = \Theta(1)$, $B_{2,1}^{(t)} = B_{2,1}^{(T_2)}(1 \pm o(1))$, $B_{1,2}^{(t)} = B_{1,2}^{(T_2)}(1 \pm o(1))$, $|B_{2,2}^{(t)}| \in [|B_{2,2}^{(T_2)}|, O(1)]$;
- (b) $|E_{2,1}^{(t)}| = \Theta(\sqrt{\eta_E/\eta})$, $\text{sign}(E_{2,1}^{(t)}) = \text{sign}(E_{2,1}^{(T_2)})$ and $|E_{1,2}^{(t)}| \leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}})[R_1^{(t)}]^{3/2}[R_2^{(t)}]^{3/2}$;
- (c) $R_1^{(t)} \in [\Omega(\frac{1}{d}), O(\frac{d^{o(1)}}{d^{3/4}})]$, $[R_2^{(t)}] \in [\frac{1}{\sqrt{d}}, O(\frac{1}{\log d} \sqrt{\eta_E/\eta})]$.

As usual, before we prove the induction, we need to derive some useful claims. But firstly we shall give a much cleaner form of $\nabla_{E_{j,3-j}} L(W^{(t)}, E^{(t)})$ to help us understand the learning process of phase III and the end phase.

Fact 3.E.1. Let us write

$$\begin{aligned}\Xi_j^{(t)} &= C_0 C_1 \alpha_1^6 \alpha_2^6 \Phi_j^{(t)} \left((B_{1,1}^{(t)})^6 (B_{2,2}^{(t)})^6 + (B_{2,1}^{(t)})^6 (B_{1,2}^{(t)})^6 \right) \\ \Delta_{j,\ell}^{(t)} &= C_0 \Phi_j^{(t)} \alpha_\ell^6 (B_{j,\ell}^{(t)})^3 (B_{3-j,\ell}^{(t)})^3 C_2 \mathcal{E}_{j,3-j}^{(t)}\end{aligned}$$

Then the gradient of $E_{j,3-j}^{(t)}$ can be written as

$$-\nabla_{E_{j,3-j}} L(W^{(t)}, E^{(t)}) = -\Xi_j^{(t)} E_{j,3-j}^{(t)} + \sum_{\ell \in [2]} \Delta_{j,\ell}^{(t)} - \sum_{\ell \in [2]} \Sigma_{j,\ell}^{(t)} \nabla_{E_{j,3-j}} \mathcal{E}_{j,3-j}^{(t)}$$

Proof. By expanding the gradients of $E_{j,3-j}^{(t)}$, we can verify by checking each monomial of polynomials of $B_{j,\ell}$ to obtain the first term, and leave the $\mathcal{E}_{j,3-j}^{(t)}$ part for the second term. $\square$

Lemma 3.E.1 (variables control at phase III). *For $t \in [T_2, T_3]$, if [Induction 3.E.1](#) holds at iteration t , then we have*

- (a) $\Phi_1^{(t)} = \tilde{\Theta}(\frac{1}{\alpha_1^{12}})$, $[Q_2^{(t)}]^{-2} = \Theta(C_2[R_2^{(t)}]^3 + C_1\alpha_2^6(B_{2,2}^{(t)})^6)$, $U_2^{(t)} = \Theta(C_1(\alpha_1^6(E_{2,1}^{(t)})^2 + \alpha_2^6(B_{2,2}^{(t)})^6))$;
- (b) $H_{1,1}^{(t)} = \Theta(C_1\alpha_1^6)$, $H_{1,2}^{(t)} \leq O(C_2[R_1^{(t)}]^3) + \tilde{O}(\frac{\alpha_2^6}{d^3})$;
- (c) $H_{2,1}^{(t)} = \Theta(C_1\alpha_1^6(E_{2,1}^{(t)})^2)$, $H_{2,2}^{(t)} = \Theta(C_2[R_2^{(t)}]^3)$;
- (d) $\Sigma_{1,2}^{(t)} \leq \tilde{O}(\frac{|E_{1,2}^{(t)}|}{d^{3/2}})\Sigma_{1,1}^{(t)}$;
- (e) $\mathcal{E}_{j,3-j}^{(t)} = (1 \pm o(1))\mathcal{E}_j^{(t)} = O(C_2[R_j^{(t)}]^3)$

Proof. Assuming [Induction 3.E.1](#) holds at $t \in [T_2, T_3]$, we can recall the expression of these variables and prove their bounds directly. The bounds for Φ_1 and $H_{1,1}$ comes from $|B_{1,1}^{(t)}| = \Theta(1)$ and $|B_{1,2}^{(t)}|, |E_{1,2}^{(t)}| = o(1)$. The bounds for Q_2, U_2 comes from our definition of T_3 in [\(3.E.1\)](#). The rest of the claims can be derived by similar arguments using [Induction 3.E.1](#). $\square$

3.E.2 Gradient Lemmas for Phase III

In this subsection, we would give some gradient lemmas concerning the dynamics of our network in Phase III.

Lemma 3.E.2 (learning feature v_2 in phase III). *For each $t \in [T_2, T_3]$, if Induction 3.E.1 holds at iteration t , then we have:*

$$(a) \quad \langle -\nabla_{w_1} L(W^{(t)}, E^{(t)}), v_2 \rangle = \Theta\left(\frac{(B_{1,2}^{(t)})^2}{(B_{2,2}^{(t)})^2}\right) E_{2,1}^{(t)} \Lambda_{2,2}^{(t)} \pm \tilde{O}\left(\frac{\alpha_1^{O(1)}}{d^4}\right) |E_{2,1}^{(t)}|^2 \Phi_2^{(t)} \pm \tilde{O}\left(\frac{\alpha_1^{O(1)}}{d^{5/2}}\right);$$

$$(b) \quad \langle -\nabla_{w_2} L(W^{(t)}, E^{(t)}), v_2 \rangle = (1 \pm \tilde{O}\left(\frac{1}{d}\right)) \Lambda_{2,2}^{(t)}$$

Proof. Since $\langle -\nabla_{w_j} L(W^{(t)}, E^{(t)}), v_2 \rangle = \Lambda_{j,2}^{(t)} + \Gamma_{j,2}^{(t)} - \Upsilon_{j,2}^{(t)}$, let us write down the definition of $\Lambda_{j,2}^{(t)}, \Gamma_{j,2}^{(t)}, \Upsilon_{j,2}^{(t)}$ respectively:

$$\begin{aligned} \Lambda_{j,2}^{(t)} &= C_0 \alpha_2^6 \Phi_j^{(t)} H_{j,1}^{(t)} (B_{j,2}^{(t)})^5 \\ \Gamma_{j,2}^{(t)} &= C_0 \alpha_2^6 \Phi_{3-j}^{(t)} E_{3-j,j}^{(t)} (B_{3-j,2}^{(t)})^3 (B_{j,2}^{(t)})^2 H_{3-j,1}^{(t)} \\ \Upsilon_{j,2}^{(t)} &= C_0 \alpha_1^6 \left(\Phi_j^{(t)} (B_{j,1}^{(t)})^3 (B_{j,2}^{(t)})^2 K_{j,2}^{(t)} + \Phi_{3-j}^{(t)} E_{3-j,j}^{(t)} (B_{3-j,1}^{(t)})^3 (B_{j,2}^{(t)})^2 K_{3-j,2}^{(t)} \right) \end{aligned}$$

Again we decompose $\Upsilon_{j,2}^{(t)} = \Upsilon_{j,2,1}^{(t)} + \Upsilon_{j,2,2}^{(t)}$ as in the proof of Lemma 3.D.4, where

$$\Upsilon_{j,2,1}^{(t)} = C_0 \alpha_1^6 \Phi_j^{(t)} (B_{j,1}^{(t)})^3 (B_{j,2}^{(t)})^2 K_{j,2}^{(t)}, \quad \Upsilon_{j,2,2}^{(t)} = \Phi_{3-j}^{(t)} E_{3-j,j}^{(t)} (B_{3-j,1}^{(t)})^3 (B_{3-j,2}^{(t)})^2 K_{3-j,2}^{(t)}$$

This gives

$$\begin{aligned} \Lambda_{j,2}^{(t)} - \Upsilon_{j,2,1}^{(t)} &= C_0 \alpha_2^6 \Phi_j^{(t)} (B_{j,2}^{(t)})^5 H_{j,1}^{(t)} - C_0 \alpha_1^6 \Phi_j^{(t)} (B_{j,1}^{(t)})^3 (B_{j,2}^{(t)})^2 K_{j,2}^{(t)} \\ &= C_0 \alpha_2^6 C_1 \alpha_1^6 \Phi_j^{(t)} (B_{j,2}^{(t)})^5 \left(E_{j,3-j}^{(t)} (B_{3-j,1}^{(t)})^3 (B_{j,1}^{(t)})^3 + (E_{j,3-j}^{(t)})^2 (B_{3-j,1}^{(t)})^6 \right) \\ &\quad - C_0 \alpha_2^6 C_1 \alpha_1^6 \Phi_j^{(t)} (B_{j,2}^{(t)})^2 (B_{3-j,2}^{(t)})^3 E_{j,3-j}^{(t)} \left((B_{j,1}^{(t)})^6 + E_{j,3-j}^{(t)} (B_{3-j,1}^{(t)})^3 (B_{j,1}^{(t)})^3 \right) \\ &\quad + C_0 \alpha_2^6 \Phi_j^{(t)} (B_{j,2}^{(t)})^5 C_2 \mathcal{E}_{j,3-j}^{(t)} \end{aligned}$$

When $j = 1$, from Induction 3.E.1 and Lemma 3.E.1a (which gives $\Phi_1^{(t)} \leq \alpha_1^{O(1)} \Phi_2^{(t)}$), we can crudely obtain

$$\begin{aligned} \left| C_0 \alpha_2^6 C_1 \alpha_1^6 \Phi_1^{(t)} (B_{1,2}^{(t)})^5 \left(E_{1,2}^{(t)} (B_{2,1}^{(t)})^3 (B_{1,1}^{(t)})^3 + (E_{1,2}^{(t)})^2 (B_{2,1}^{(t)})^6 \right) \right| &\leq \tilde{O}\left(\frac{\alpha_1^{O(1)}}{d^4}\right) \Phi_1^{(t)} |E_{1,2}^{(t)}| \\ \left| C_0 \alpha_2^6 C_1 \alpha_1^6 \Phi_1^{(t)} (B_{1,2}^{(t)})^2 (B_{2,2}^{(t)})^3 E_{1,2}^{(t)} \left((B_{1,1}^{(t)})^6 + E_{1,2}^{(t)} (B_{2,1}^{(t)})^3 (B_{1,1}^{(t)})^3 \right) \right| &\leq \tilde{O}\left(\frac{\alpha_1^{O(1)}}{d}\right) \Lambda_{2,2}^{(t)} |E_{1,2}^{(t)}| \\ \left| C_0 \alpha_2^6 \Phi_1^{(t)} (B_{1,2}^{(t)})^5 C_2 \mathcal{E}_{1,2}^{(t)} \right| &= \tilde{O}\left(\frac{\alpha_1^6}{d^{5/2}}\right) \Sigma_{1,1}^{(t)} [R_1^{(t)}]^3 \end{aligned}$$

So we have

$$\Lambda_{1,2}^{(t)} - \Upsilon_{1,2,1}^{(t)} = \tilde{O}\left(\frac{\alpha_1^6}{d^{5/2}}\right) \Sigma_{1,1}^{(t)} [R_1^{(t)}]^3 \pm \tilde{O}\left(\frac{\alpha_1^{O(1)}}{d}\right) \Lambda_{2,2}^{(t)} |E_{1,2}^{(t)}|$$

When $j = 2$, we can also derive using [Lemma 3.E.1](#) about $H_{2,1}^{(t)}$ and [Induction 3.E.1](#) about $B_{2,1}^{(t)}$ and some rearrangement to obtain

$$\begin{aligned} C_0 \alpha_2^6 \Phi_2^{(t)} (B_{2,2}^{(t)})^5 \left[C_1 \alpha_1^6 \left(E_{2,1}^{(t)} (B_{1,1}^{(t)})^3 (B_{2,1}^{(t)})^3 + (E_{2,1}^{(t)})^2 (B_{1,1}^{(t)})^6 \right) + C_2 \mathcal{E}_{2,1}^{(t)} \right] &= (1 \pm \tilde{O}\left(\frac{1}{d}\right)) \Lambda_{2,2}^{(t)} \\ \left| C_0 \alpha_2^6 C_1 \alpha_1^6 \Phi_2^{(t)} (B_{2,2}^{(t)})^2 (B_{1,2}^{(t)})^3 E_{2,1}^{(t)} \left((B_{2,1}^{(t)})^6 + E_{2,1}^{(t)} (B_{1,1}^{(t)})^3 (B_{2,1}^{(t)})^3 \right) \right| &\leq \tilde{O}\left(\frac{\alpha_1^{O(1)}}{d^3}\right) |E_{2,1}^{(t)}| \Phi_2^{(t)} \end{aligned}$$

which leads to the approximation

$$\Lambda_{2,2}^{(t)} - \Upsilon_{1,2,2}^{(t)} = (1 \pm \tilde{O}\left(\frac{1}{d}\right)) \Lambda_{2,2}^{(t)} \pm \tilde{O}\left(\frac{\alpha_1^{O(1)}}{d^3}\right) |E_{2,1}^{(t)}| \Phi_2^{(t)}$$

Similarly, we can also calculate

$$\begin{aligned} \Gamma_{j,2}^{(t)} - \Upsilon_{j,2,2}^{(t)} &= C_0 \alpha_2^6 \Phi_{3-j}^{(t)} E_{3-j,j}^{(t)} (B_{3-j,2}^{(t)})^3 (B_{j,2}^{(t)})^2 H_{3-j,1}^{(t)} - C_0 \alpha_1^6 \Phi_{3-j}^{(t)} E_{3-j,j}^{(t)} (B_{3-j,1}^{(t)})^3 (B_{j,2}^{(t)})^2 K_{3-j,2}^{(t)} \\ &= C_0 \alpha_2^6 C_1 \alpha_1^6 \Phi_{3-j}^{(t)} (B_{3-j,2}^{(t)})^3 (B_{j,2}^{(t)})^2 E_{3-j,j}^{(t)} \left(E_{3-j,j}^{(t)} (B_{j,1}^{(t)})^3 (B_{3-j,1}^{(t)})^3 + (E_{3-j,j}^{(t)})^2 (B_{j,1}^{(t)})^6 \right) \\ &\quad - C_0 \alpha_2^6 C_1 \alpha_1^6 \Phi_{3-j}^{(t)} (B_{j,2}^{(t)})^5 (E_{3-j,j}^{(t)})^2 \left((B_{3-j,1}^{(t)})^6 + E_{3-j,j}^{(t)} (B_{j,1}^{(t)})^3 (B_{3-j,1}^{(t)})^3 \right) \\ &\quad + C_0 \alpha_2^6 \Phi_{3-j}^{(t)} E_{3-j,j}^{(t)} (B_{3-j,2}^{(t)})^3 (B_{j,2}^{(t)})^2 C_2 \mathcal{E}_{3-j,j}^{(t)} \end{aligned}$$

When $j = 1$, following similar procedure as above, we can apply [Induction 3.E.1](#) and [Lemma 3.E.1](#) to give

$$\Gamma_{1,2}^{(t)} - \Upsilon_{1,2,2}^{(t)} = \Theta\left(\frac{(B_{1,2}^{(t)})^2}{(B_{2,2}^{(t)})^2}\right) E_{2,1}^{(t)} \Lambda_{2,2}^{(t)} \pm \tilde{O}\left(\frac{\alpha_1^{O(1)}}{d^4}\right) |E_{2,1}^{(t)}|^2 \Phi_2^{(t)}$$

Note that the first term on the RHS dominates the term $\pm \tilde{O}\left(\frac{\alpha_1^{O(1)}}{d}\right) \Lambda_{2,2}^{(t)} |E_{1,2}^{(t)}|$ in the approximation for $\Lambda_{1,2}^{(t)} - \Upsilon_{1,2,1}^{(t)}$ due to [Induction 3.E.1a,b](#). When $j = 2$, since $\Phi_1^{(t)} \leq \tilde{\Theta}\left(\frac{1}{\alpha_1^{12}}\right) \leq \alpha_1^{O(1)} \Phi_2^{(t)} H_{2,1}^{(t)}$ in this phase and $|B_{1,1}^{(t)}| = O(1)$, we can derive

$$|\Gamma_{2,2}^{(t)} - \Upsilon_{2,2,2}^{(t)}| \leq \tilde{\Theta}\left(\frac{\alpha_1^{O(1)}}{d^3}\right) (E_{1,2}^{(t)})^2 \Phi_1^{(t)} + \alpha_1^{O(1)} (E_{1,2}^{(t)})^2 \Lambda_{2,2}^{(t)}$$

It can be seen that $(E_{1,2}^{(t)})^2 \Phi_1^{(t)} \leq (E_{2,1}^{(t)})^2 \Phi_2^{(t)}$ by [Induction 3.E.1](#) and [Lemma 3.E.1](#). And by similar arguments we can have $(1 \pm \tilde{O}(\frac{1}{d})) \Lambda_{2,2}^{(t)} \geq \frac{1}{d^{\Omega(1)}} \tilde{O}(\frac{\alpha_1^{O(1)}}{d^3}) |E_{2,1}^{(t)}| \Phi_2^{(t)}$. Combining all the results above, we can finish the proof. $\square$

Lemma 3.E.3 (learning feature v_1 in Phase III). *For each $t \in [T_2, T_3]$, if [Induction 3.E.1](#) holds at iteration t , then we have: (recall that Δ -notation is from [Fact 3.E.1](#))*

$$\begin{aligned} (a) \quad \langle -\nabla_{w_1} L(W^{(t)}, E^{(t)}), v_1 \rangle &= \Theta(\Sigma_{1,1}^{(t)} [R_1^{(t)}]^3) \pm O\left(\frac{(B_{1,2}^{(t)})^3}{(B_{2,2}^{(t)})^3} + \frac{1}{\sqrt{d}}\right) \alpha_1^{O(1)} \Lambda_{2,2}^{(t)} + \frac{E_{2,1}^{(t)}}{B_{1,1}^{(t)}} \Delta_{2,1}^{(t)} - \frac{B_{2,2}^{(t)}}{B_{1,1}^{(t)}} \Lambda_{2,2}^{(t)}; \\ (b) \quad \langle -\nabla_{w_2} L(W^{(t)}, E^{(t)}), v_1 \rangle &= \tilde{O}(\frac{\alpha_1^{O(1)}}{d^{5/2}}) \Phi_2^{(t)} [R_2^{(t)}]^3 \pm \tilde{O}(\frac{\alpha_1^{O(1)}}{d}) \Lambda_{2,2}^{(t)} \pm \tilde{O}(\frac{\alpha_1^{O(1)}}{d^3}) \end{aligned}$$

Proof. Recall that $\langle -\nabla_{w_j} L(W^{(t)}, E^{(t)}), v_1 \rangle = \Lambda_{j,1}^{(t)} + \Gamma_{j,1}^{(t)} - \Upsilon_{j,1}^{(t)}$. Similar to the proof of [Lemma 3.E.2](#), we can decompose $\Upsilon_{j,1}^{(t)} = \Upsilon_{j,1,1}^{(t)} + \Upsilon_{j,1,2}^{(t)}$ and do similar calculations:

$$\begin{aligned} \Lambda_{j,1}^{(t)} - \Upsilon_{j,1,1}^{(t)} &= C_0 C_1 \alpha_1^6 \alpha_2^6 \Phi_j^{(t)} (B_{j,1}^{(t)})^5 \left(E_{j,3-j}^{(t)} (B_{3-j,2}^{(t)})^3 (B_{j,2}^{(t)})^3 + (E_{j,3-j}^{(t)})^2 (B_{3-j,2}^{(t)})^6 \right) \\ &\quad - C_0 C_1 \alpha_1^6 \alpha_2^6 \Phi_j^{(t)} (B_{j,1}^{(t)})^2 (B_{3-j,1}^{(t)})^3 E_{j,3-j}^{(t)} \left((B_{j,2}^{(t)})^6 + E_{j,3-j}^{(t)} (B_{3-j,2}^{(t)})^3 (B_{j,2}^{(t)})^3 \right) \\ &\quad + C_0 \alpha_1^6 \Phi_j^{(t)} (B_{j,1}^{(t)})^5 C_2 \mathcal{E}_{j,3-j}^{(t)} \end{aligned}$$

When $j = 1$, from [Induction 3.E.1](#) and [Lemma 3.E.1a](#) we know $\Phi_1^{(t)} \leq \alpha_1^{O(1)}$ during $t \in [T_2, T_3]$, which allow us to derive

$$\begin{aligned} &C_0 C_1 \alpha_1^6 \alpha_2^6 \Phi_1^{(t)} (B_{1,1}^{(t)})^5 \left(E_{1,2}^{(t)} (B_{2,2}^{(t)})^3 (B_{1,2}^{(t)})^3 + (E_{1,2}^{(t)})^2 (B_{2,2}^{(t)})^6 \right) \\ &\leq \tilde{O}(\Sigma_{1,1}^{(t)} (E_{1,2}^{(t)})^2) + C_0 C_1 \alpha_1^6 \alpha_2^6 (B_{1,1}^{(t)})^5 E_{1,2}^{(t)} (B_{2,2}^{(t)})^3 (B_{1,2}^{(t)})^3 \\ &\leq O\left(\frac{(B_{1,2}^{(t)})^3}{(B_{2,2}^{(t)})^3}\right) \alpha_1^{O(1)} \Lambda_{2,2}^{(t)} |E_{1,2}^{(t)}| + \Theta(\Sigma_{1,1}^{(t)} [R_1^{(t)}]^3) \end{aligned}$$

And

$$\left| C_0 C_1 \alpha_1^6 \alpha_2^6 \Phi_1^{(t)} (B_{1,1}^{(t)})^2 (B_{2,1}^{(t)})^3 E_{1,2}^{(t)} \left((B_{1,2}^{(t)})^6 + E_{1,2}^{(t)} (B_{2,2}^{(t)})^3 (B_{1,2}^{(t)})^3 \right) \right| \leq \tilde{O}\left(\frac{1}{d^{3/2}}\right) |E_{1,2}^{(t)}| \Lambda_{2,2}^{(t)}$$

which can be summarized as

$$\Lambda_{1,1}^{(t)} - \Upsilon_{1,1,1}^{(t)} = \Theta(\Sigma_{1,1}^{(t)} [R_1^{(t)}]^3) \pm O\left(\frac{(B_{1,2}^{(t)})^3}{(B_{2,2}^{(t)})^3} + (B_{2,1}^{(t)})^3 + \frac{1}{\sqrt{d}}\right) |E_{1,2}^{(t)}| \alpha_1^{O(1)} \Lambda_{2,2}^{(t)}$$

A similar calculation also gives

$$\Lambda_{2,1}^{(t)} - \Upsilon_{2,1,1}^{(t)} = \tilde{O}\left(\frac{\alpha_1^{O(1)}}{d^{5/2}}\right) \Phi_2^{(t)} [R_2^{(t)}]^3 \pm \tilde{O}\left(\frac{\alpha_1^{O(1)}}{d^4}\right) \Phi_2^{(t)} |E_{2,1}^{(t)}| \pm \tilde{O}\left(\frac{\alpha_1^{O(1)}}{d}\right) \Lambda_{2,2}^{(t)} B_{2,2}^{(t)}$$

Now we turn to the other terms in the gradient, from similar calculations in the proof of [Lemma 3.D.4](#), we have

$$\begin{aligned}\Gamma_{j,1}^{(t)} - \Upsilon_{j,1,2}^{(t)} &= C_0 \alpha_2^6 C_1 \alpha_1^6 \Phi_{3-j}^{(t)} (B_{3-j,1}^{(t)})^3 (B_{j,1}^{(t)})^2 E_{3-j,j}^{(t)} \left(E_{3-j,j}^{(t)} (B_{j,2}^{(t)})^3 (B_{3-j,2}^{(t)})^3 + (E_{3-j,j}^{(t)})^2 (B_{j,2}^{(t)})^6 \right) \\ &\quad - C_0 \alpha_2^6 C_1 \alpha_1^6 \Phi_{3-j}^{(t)} (B_{j,1}^{(t)})^5 (E_{3-j,j}^{(t)})^2 \left((B_{3-j,2}^{(t)})^6 + E_{3-j,j}^{(t)} (B_{j,2}^{(t)})^3 (B_{3-j,2}^{(t)})^3 \right) \\ &\quad + C_0 \alpha_2^6 \Phi_{3-j}^{(t)} E_{3-j,j}^{(t)} (B_{3-j,1}^{(t)})^3 (B_{j,1}^{(t)})^2 C_2 \mathcal{E}_{3-j,j}^{(t)}\end{aligned}$$

which also similarly gives

$$\Gamma_{1,1}^{(t)} - \Upsilon_{1,1,2}^{(t)} = \frac{E_{2,1}^{(t)}}{B_{1,1}^{(t)}} \Lambda_{2,1}^{(t)} - \frac{B_{2,2}^{(t)}}{B_{1,1}^{(t)}} \Lambda_{2,2}^{(t)} \pm \tilde{O}\left(\frac{\alpha_1^{O(1)}}{d^{3/2}}\right) \Lambda_{2,2}^{(t)}$$

and

$$|\Gamma_{2,1}^{(t)} - \Upsilon_{2,1,2}^{(t)}| \leq \tilde{O}\left(\frac{\alpha_1^{O(1)}}{d}\right) \Phi_1^{(t)} ((E_{1,2}^{(t)})^2 + |E_{1,2}^{(t)}| [R_1^{(t)}]^3) \leq \tilde{O}\left(\frac{\alpha_1^{O(1)}}{d^3}\right)$$

which finishes the proof. $\square$

Lemma 3.E.4 (reducing noise in phase III). *Suppose [Induction 3.E.1](#) holds at $t \in [T_2, T_3]$, then we have*

$$\begin{aligned}(a) \quad &\langle -\nabla_{w_1} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_1^{(t)} \rangle = -\Theta([R_1^{(t)}]^3) \left(\Sigma_{1,1}^{(t)} + \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} (E_{2,1}^{(t)})^2 \right) \\ &\quad \pm O\left(\sum_{j,\ell} \Sigma_{j,\ell}^{(t)} E_{j,3-j}^{(t)} (\bar{R}_{1,2}^{(t)} + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2} \right); \\ (b) \quad &\langle -\nabla_{w_1} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_2^{(t)} \rangle = \left(\Sigma_{1,1}^{(t)} + \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} (E_{2,1}^{(t)})^2 \right) (-\Theta(\bar{R}_{1,2}^{(t)}) \pm O(\varrho)) [R_1^{(t)}]^{5/2} [R_2^{(t)}]^{1/2} \\ &\quad + O\left(\sum_{(j,\ell) \neq (1,2)} \Sigma_{j,\ell}^{(t)} E_{j,3-j}^{(t)} R_1^{(t)} [R_2^{(t)}]^2 \right) \\ (c) \quad &\langle -\nabla_{w_2} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_2^{(t)} \rangle = -\Theta([R_2^{(t)}]^3) \left(\sum_{\ell \in [2]} \Sigma_{1,1}^{(t)} \Theta((E_{1,2}^{(t)})^2) + \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} \right) \\ &\quad \pm O\left(\sum_{j,\ell} \Sigma_{j,\ell}^{(t)} E_{j,3-j}^{(t)} (\bar{R}_{1,2}^{(t)} + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2} \right); \\ (d) \quad &\langle -\nabla_{w_2} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_1^{(t)} \rangle = \left(\Sigma_{1,1}^{(t)} \Theta((E_{1,2}^{(t)})^2) + \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} \right) (-\Theta(\bar{R}_{1,2}^{(t)}) \pm O(\varrho)) [R_2^{(t)}]^{5/2} [R_1^{(t)}]^{1/2} \\ &\quad + O\left(\sum_{(j,\ell) \neq (1,2)} \Sigma_{j,\ell}^{(t)} E_{j,3-j}^{(t)} R_2^{(t)} [R_1^{(t)}]^2 \right)\end{aligned}$$

Proof. The proof of Lemma 3.E.4 is very similar to Lemma 3.D.3, but we write it down to stress some minor differences. As in (3.B.2), we first write down

$$\langle -\nabla_{w_1} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_1^{(t)} \rangle = - \sum_{j,\ell} \Sigma_{j,\ell}^{(t)} \langle \nabla_{w_1} \mathcal{E}_{j,3-j}^{(t)}, w_1^{(t)} \rangle$$

Proof of (a): Combine the bounds above, we can obtain for each $j \in [2]$: $\Sigma_{1,2}^{(t)} = \tilde{O}(E_{1,2}^{(t)}/d^{3/2})\Sigma_{1,1}^{(t)}$. We can then directly apply Claim 3.B.1 to prove Lemma 3.E.4a as follows

$$\begin{aligned} & \langle -\nabla_{w_1} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_1^{(t)} \rangle \\ &= (1 \pm \tilde{O}(E_{1,2}^{(t)}/d^{3/2}))\Sigma_{1,1}^{(t)} \left(-\Theta([R_1^{(t)}]^3) \pm O(E_{1,2}^{(t)})(\bar{R}_{1,2}^{(t)} + \varrho)[R_1^{(t)}]^{3/2}[R_2^{(t)}]^{3/2} \right) \\ & \quad + (\Sigma_{2,1}^{(t)} + \Sigma_{2,2}^{(t)}) \left(-\Theta((E_{2,1}^{(t)})^2)[R_1^{(t)}]^3 \pm O(E_{2,1}^{(t)})(\bar{R}_{1,2}^{(t)} + \varrho)[R_1^{(t)}]^{3/2}[R_2^{(t)}]^{3/2} \right) \\ &= -\Theta(\Sigma_{1,1}^{(t)} + \Sigma_{2,1}^{(t)} + \Sigma_{2,2}^{(t)})[R_1^{(t)}]^3 \pm O\left(\sum_{j,\ell} \Sigma_{j,\ell}^{(t)} E_{j,3-j}^{(t)} (\bar{R}_{1,2}^{(t)} + \varrho)[R_1^{(t)}]^{3/2}[R_2^{(t)}]^{3/2}\right) \\ & \quad \text{(Since } |E_{1,2}^{(t)}| \leq d^{-\Omega(1)} \text{ by Induction 3.E.1)} \end{aligned}$$

Proof of (b): For Lemma 3.D.3b, we can use the same analysis for $\Sigma_{1,1}^{(t)}$ above and Claim 3.B.1d,e to get (again we have used $\Sigma_{1,2}^{(t)} = \tilde{O}(E_{1,2}^{(t)}/d^{3/2})\Sigma_{1,1}^{(t)}$)

$$\begin{aligned} & \langle -\nabla_{w_1} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_2^{(t)} \rangle \\ &= (1 \pm \tilde{O}(E_{1,2}^{(t)}/d^{3/2}))\Sigma_{1,1}^{(t)} \left((-\Theta(\bar{R}_{1,2}^{(t)}) \pm O(\varrho))[R_1^{(t)}]^{5/2}[R_2^{(t)}]^{1/2} + E_{1,2}^{(t)} R_1^{(t)} [R_2^{(t)}]^2 \right) \\ & \quad + \Theta(\Sigma_{2,1}^{(t)} + \Sigma_{2,2}^{(t)}) \left((-\Theta(\bar{R}_{1,2}^{(t)}) + O(\varrho))(E_{2,1}^{(t)})^2 [R_1^{(t)}]^{5/2}[R_2^{(t)}]^{1/2} + E_{2,1}^{(t)} R_1^{(t)} [R_2^{(t)}]^2 \right) \\ &= \left(\Sigma_{1,1}^{(t)} + \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} (E_{2,1}^{(t)})^2 \right) ((-\Theta(\bar{R}_{1,2}^{(t)}) + O(\varrho))[R_1^{(t)}]^{5/2}[R_2^{(t)}]^{1/2}) \\ & \quad + O\left(\sum_{(j,\ell) \neq (2,1)} \Sigma_{j,\ell}^{(t)} E_{j,3-j}^{(t)} R_1^{(t)} [R_2^{(t)}]^2\right) \end{aligned}$$

Proof of (c): Similarly to the proof of (a), we can also expand as follows

$$\begin{aligned} & \langle -\nabla_{w_2} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_2^{(t)} \rangle \\ &= (1 \pm \tilde{O}(E_{1,2}^{(t)}/d^{3/2}))\Sigma_{1,1}^{(t)} \left(-[R_2^{(t)}]^3 \Theta((E_{1,2}^{(t)})^2) \pm O(E_{1,2}^{(t)})(\bar{R}_{1,2}^{(t)} + \varrho)[R_1^{(t)}]^{3/2}[R_2^{(t)}]^{3/2} \right) \\ & \quad - \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} \left([R_2^{(t)}]^3 \pm O(E_{2,1}^{(t)})(\bar{R}_{1,2}^{(t)} + \varrho)[R_1^{(t)}]^{3/2}[R_2^{(t)}]^{3/2} \right) \\ &= -\Theta([R_2^{(t)}]^3) \left(\Sigma_{1,1}^{(t)} \Theta((E_{1,2}^{(t)})^2) + \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} \right) \pm O\left(\sum_{j,\ell} \Sigma_{j,\ell}^{(t)} E_{j,3-j}^{(t)} (\bar{R}_{1,2}^{(t)} + \varrho)[R_1^{(t)}]^{3/2}[R_2^{(t)}]^{3/2}\right) \end{aligned}$$

Proof of (d): Similarly, we can calculate

$$\begin{aligned}
& \langle -\nabla_{w_2} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_1^{(t)} \rangle \\
&= (1 \pm \widetilde{O}(E_{1,2}^{(t)}/d^{3/2})) \Sigma_{1,1}^{(t)} \left((-\Theta(\bar{R}_{1,2}^{(t)}) \pm O(\varrho))(E_{1,2}^{(t)})^2 [R_2^{(t)}]^{5/2} [R_1^{(t)}]^{1/2} + E_{1,2}^{(t)} R_2^{(t)} [R_1^{(t)}]^2 \right) \\
&\quad + \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} \left((-\Theta(\bar{R}_{1,2}^{(t)}) \pm O(\varrho)) [R_2^{(t)}]^{5/2} [R_1^{(t)}]^{1/2} + E_{1,2}^{(t)} R_2^{(t)} [R_1^{(t)}]^2 \right) \\
&= \left(\Sigma_{1,1}^{(t)} \Theta((E_{1,2}^{(t)})^2) + \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} \right) (-\Theta(\bar{R}_{1,2}^{(t)}) \pm O(\varrho)) [R_2^{(t)}]^{5/2} [R_1^{(t)}]^{1/2} \\
&\quad + O\left(\sum_{(j,\ell) \neq (2,1)} \Sigma_{j,\ell}^{(t)} E_{j,3-j}^{(t)} R_2^{(t)} [R_1^{(t)}]^2 \right)
\end{aligned}$$

which completes the proof. $\square$

Lemma 3.E.5 (learning the prediction head in phase III). *If Induction 3.E.1 holds at iteration $t \in [T_2, T_3]$, then using the notations from Fact 3.E.1, we have*

$$\begin{aligned}
-\nabla_{E_{j,3-j}} L(W^{(t)}, E^{(t)}) &= \Theta\left(\sum_{\ell \in [2]} \Sigma_{j,\ell}^{(t)} (-E_{j,3-j}^{(t)} [R_{3-j}^{(t)}]^3 \pm O(\bar{R}_{1,2}^{(t)} + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2} \right. \\
&\quad \left. - \Xi_j^{(t)} E_{j,3-j}^{(t)} + \sum_{\ell \in [2]} \Delta_{j,\ell}^{(t)} \right)
\end{aligned}$$

Proof. By Fact 3.E.1, we only need to bound the last term $\sum_{\ell \in [2]} \Sigma_{j,\ell}^{(t)} \nabla_{E_{1,2}} \mathcal{E}_{j,3-j}^{(t)}$, which can be directly obtained from applying Claim 3.B.1. $\square$

3.E.3 At the End of Phase III

In order to argue that $B_{2,2}^{(T_2)} = \Omega(1)$ at the end of phase III, we need to define some auxiliary notions. Recall that T_3 is defined in (3.E.1), and now we further define

$$T_{3,1} := \min\{t : C_1 \alpha_2^6 (B_{2,2}^{(t)})^6 \geq C_2 [R_2^{(t)}]^3\}, \quad T_{3,2}^{(t)} = \min\{t : |B_{2,2}^{(t)}| \geq \frac{1}{3} \min\{|E_{2,1}^{(t)}|, |B_{1,1}^{(t)}|\}\} \quad (3.E.2)$$

It can be observed that if Induction 3.E.1 holds for $t \in [T_2, T_3]$ and our learning rate η is small enough, we shall have $T_2 < T_{3,1} \leq T_{3,2} < T_3$. Now we are ready to present the main lemma we want to prove in this phase.

Lemma 3.E.6 (Phase III). *Let T_3 be defined as in (3.E.1). Suppose $\eta = \frac{1}{\text{poly}(d)}$ is sufficiently small, then Induction 3.E.1 holds for all iteration $t \in [T_2, T_3]$, and at iteration $t = T_3$, the followings holds:*

- (a) $|B_{1,1}^{(T_3)}| = \Theta(1)$, $|B_{2,2}^{(T_3)}| = \Theta(1)$, $B_{j,\ell}^{(T_3)} = B_{j,\ell}^{(T_2)} (1 \pm o(1))$ for $j \neq \ell$;
- (b) $R_1^{(T_3)} = \widetilde{O}(\frac{1}{d^{3/4}})$, $R_2^{(T_3)} \in [\widetilde{O}(\frac{1}{d^{1/2}}), \widetilde{O}(\frac{1}{d^{1/4}})]$, and $\bar{R}_{1,2}^{(T_3)} \leq \widetilde{O}(\varrho + \frac{1}{\sqrt{d}})$;
- (c) $|E_{2,1}^{(T_2)}| = \Theta(\sqrt{\eta_E/\eta})$ and $|E_{1,2}^{(T_2)}| = \widetilde{O}(\varrho + \frac{1}{\sqrt{d}}) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2} = \widetilde{O}(\frac{1}{d})$.

Moreover, $|B_{2,2}^{(t)}|$ is increasing and $R_2^{(t)}$ is decreasing. The part of learning $|B_{2,2}^{(t)}|$ till $\Omega(1)$ and keeping $B_{2,1}^{(t)}$ close to its initialization is what's been accelerated by the prediction head $E_{2,1}^{(t)}$.

The proof of [Lemma 3.E.6](#) will be proven after we have proven [Induction 3.E.1](#), which will again be proven after some intermediate results are proven.

Lemma 3.E.7 (The growth of $B_{2,2}^{(t)}$ before $T_{3,1}$). *Let $T_{3,1}$ be defined as in (3.E.2). If [Induction 3.E.1](#) holds for $t \in [T_2, T_{3,1}]$, then we have $R_2^{(T_{3,1})} \leq \frac{\alpha_1^{12}}{d^{1/4}}$ and $B_{2,2}^{(T_{3,1})} \in [\frac{1}{d^{1/4}}, O(\frac{\alpha_1^{O(1)}}{d^{1/4}})]$ and $T_{3,1} \leq T_2 + \tilde{O}(\frac{d^{1.625}\alpha_1^{O(1)}}{\eta})$.*

Proof. Firstly by [Lemma 3.E.4b](#), we can write down the update of $R_2^{(t)}$: (as in [Lemma 3.D.6](#))

$$\begin{aligned} R_2^{(t+1)} &= R_2^{(t)} - \eta \Theta([R_2^{(t)}]^3) \left(\Sigma_{1,1}^{(t)} \Theta((E_{1,2}^{(t)})^2) + \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} \right) \\ &\quad \pm O\left(\sum_{j,\ell} \eta \Sigma_{j,\ell}^{(t)} E_{j,3-j}^{(t)} (\bar{R}_{1,2}^{(t)} + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2} \right) \pm \frac{\eta}{\text{poly}(d)} \end{aligned}$$

Next, by [Claim 3.B.1](#) and [Lemma 3.E.1a](#) combined with [Induction 3.E.1a,b](#), we have $\tilde{O}(\frac{|E_{2,1}^{(t)}|}{d^{3/2}}) \Sigma_{1,1}^{(t)} \frac{\Phi_1^{(t)}}{\Phi_2^{(t)}} \leq \tilde{O}(\Sigma_{2,1}^{(t)})$, which leads to the bound

$$\eta \Sigma_{1,1}^{(t)} \Theta((E_{1,2}^{(t)})^2) \leq \tilde{O}(\varrho^2 + \frac{1}{d}) \alpha_1^{O(1)} \eta \Sigma_{1,1}^{(t)} [R_1^{(t)}]^3 [R_2^{(t)}]^3 \leq O(\frac{1}{d^{9/4}}) \eta \Sigma_{1,1}^{(t)} [R_2^{(t)}]^3 \leq O(\frac{\alpha_1^{O(1)}}{d^{3/4}}) \eta \Sigma_{2,1}^{(t)} [R_2^{(t)}]^3$$

Similarly, we can bound the following term

$$\begin{aligned} \sum_{\ell \in [2]} \eta \Sigma_{1,\ell}^{(t)} |E_{1,2}^{(t)}| (|\bar{R}_{1,2}^{(t)}| + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2} &\leq \tilde{O}(\varrho^2 + \frac{1}{d}) \alpha_1^{O(1)} \sum_{\ell \in [2]} \eta \Sigma_{1,\ell}^{(t)} [R_1^{(t)}]^3 [R_2^{(t)}]^3 \\ &\leq \tilde{O}(\varrho^2 + \frac{1}{d}) \alpha_1^{O(1)} \frac{1}{d^{9/4}} \sum_{\ell \in [2]} \eta \Sigma_{1,\ell}^{(t)} [R_2^{(t)}]^3 \\ &\leq \tilde{O}(\frac{\alpha_1^{O(1)}}{d^{3/4}}) \eta \Sigma_{2,1}^{(t)} [R_2^{(t)}]^3 \end{aligned}$$

Moreover, from [Induction 3.E.1c](#) that $R_2^{(t)} \geq R_1^{(t)}$, we can also calculate for each $t \in [T_2, T_{3,1}]$:

$$\eta \Sigma_{2,\ell}^{(t)} |E_{2,1}^{(t)}| (|\bar{R}_{1,2}^{(t)}| + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2} \leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) \alpha_1^{O(1)} \eta \Sigma_{2,\ell}^{(t)} [R_2^{(t)}]^3$$

Thus by combining the results above, we have the update of $R_2^{(t)}$ at $t \in [T_2, T_3]$ as follows:

$$\begin{aligned} R_2^{(t+1)} &= R_2^{(t)} - \eta \Theta([R_2^{(t)}]^3) \left(\Sigma_{1,1}^{(t)} \Theta((E_{1,2}^{(t)})^2) + \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} \right) \\ &= R_2^{(t)} - \eta (\Sigma_{2,1}^{(t)} + \Sigma_{2,2}^{(t)}) [R_2^{(t)}]^3 \end{aligned} \tag{3.E.3}$$

which implies that $R_2^{(t)}$ is decreasing throughout phase III. From [Lemma 3.E.1a](#) and [Induction 3.E.1b](#), we know that for $t \in [T_2, T_{3,1}]$:

$$\Phi_2^{(t)} = Q_2^{(t)} / [U_2^{(t)}]^{3/2} = \Theta\left(\frac{1}{\sqrt{C_2[R_2^{(t)}]^3 (C_1 \alpha_1^6 (E_{2,1}^{(t)})^2)^{3/2}}}\right)$$

which implies (also using a bit of [Claim 3.B.1](#) and [Induction 3.E.1a](#))

$$\begin{aligned} \Sigma_{2,1}^{(t)} [R_2^{(t)}]^3 &= (1 \pm \tilde{O}\left(\frac{1}{d^{3/2}}\right)) E_{2,1}^{(t)} \Delta_{2,1}^{(t)} \\ &= (1 \pm \tilde{O}\left(\frac{1}{d^{3/2}}\right)) (1 \pm \tilde{O}\left(\frac{1}{d^{3/2}}\right)) C_0 C_2 \alpha_1^6 \Phi_2^{(t)} E_{2,1}^{(t)} (B_{1,1}^{(t)})^3 (B_{2,1}^{(t)})^3 [R_2^{(t)}]^3 \\ &= \Theta\left(\frac{C_2^{1/2} [R_2^{(t)}]^{3/2}}{(U_2^{(t)})^{3/2}}\right) C_0 \alpha_1^6 E_{2,1}^{(t)} (B_{1,1}^{(t)})^3 (B_{2,1}^{(t)})^3 \\ &= \Theta\left(\frac{C_0 C_2^{1/2} |B_{2,1}^{(T_2)}|^3}{C_1^{3/2} \alpha_1^3 |E_{2,1}^{(T_2)}|}\right) [R_2^{(t)}]^{3/2} \\ &\quad (\text{because } B_{2,1}^{(t)} = B_{2,1}^{(T_2)} (1 \pm o(1)), B_{1,1}^{(t)} = \Theta(B_{1,1}^{(T_2)}) \text{ and } E_{2,1}^{(t)} = \Theta(E_{2,1}^{(T_2)}) \text{sign}(B_{1,1}^{(T_2)} B_{2,1}^{(T_2)})) \end{aligned}$$

And for $\Sigma_{2,2}^{(t)}$, from some simple calculations (using [Claim 3.B.1](#)), we have

- when $|B_{2,2}^{(t)}| \leq \frac{\alpha_1}{\alpha_2} \sqrt{|B_{2,1}^{(T_2)}|}$, we would have $\Sigma_{2,2}^{(t)} \leq O(\Sigma_{2,1}^{(t)})$;
- otherwise, we have $\Sigma_{2,1}^{(t)} + \Sigma_{2,2}^{(t)} = \Theta(\Sigma_{2,2}^{(t)})$.

So by [\(3.E.3\)](#), we know R_2 is decreasing for $t \in [T_2, T_{3,1}]$ by at least

$$R_2^{(t+1)} \leq R_2^{(t)} - \eta \Theta\left(\frac{C_0 C_2^{1/2} |B_{2,1}^{(T_2)}|^3}{C_1^{3/2} \alpha_1^3 |E_{2,1}^{(T_2)}|}\right) [R_2^{(t)}]^{3/2} \leq R_2^{(t)} (1 - \eta \zeta [R_2^{(t)}]^{1/2}) \quad (3.E.4)$$

where $\zeta := \Theta\left(\frac{C_0 C_2^{1/2} |B_{2,1}^{(T_2)}|^3}{C_1^{3/2} \alpha_1^3 |E_{2,1}^{(T_2)}|}\right) = \tilde{\Theta}\left(\frac{\sqrt{\eta/\eta_E}}{d^{3/2} \alpha_1^3}\right)$. By this update, we can prove $T_{3,1} \leq T_2 + O\left(\frac{d^{3/2+1/8} \alpha_1^{O(1)}}{\eta}\right)$. In order to do that, we can first see that for some $t'_{3,1} \in [T_2 + \tilde{\Theta}\left(\frac{d^{3/2} \alpha_1^2 \sqrt{\eta_E/\eta}}{\eta}\right), T_2 + \tilde{\Theta}\left(\frac{d^{3/2} \alpha_1^4 \sqrt{\eta_E/\eta}}{\eta}\right)]$, we shall have $R_2^{(t'_{3,1})} \leq d^{-1/4}$. Indeed, suppose otherwise $R_2^{(t'_{3,1}-1)} \geq d^{-1/4}$, then [\(3.E.4\)](#) implies

$$\begin{aligned} R_2^{(t'_{3,1})} &\leq R_2^{(t'_{3,1}-1)} (1 - \eta \zeta [R_2^{(t'_{3,1}-1)}]^{1/2}) \leq R_2^{(t'_{3,1}-1)} (1 - \eta \zeta \frac{1}{d^{1/8}}) \\ &\leq R_2^{(T_2)} \left(1 - \Theta\left(\frac{C_0 C_2^{1/2} \sqrt{\eta/\eta_E}}{C_1^{3/2} d^{3/2} \alpha_1^3}\right) \frac{\eta}{d^{1/8}}\right)^{t'_{3,1}-T_2-1} \\ &\leq O(\sqrt{\eta_E/\eta}) \left(1 - \Theta\left(\frac{C_0 C_2^{1/2} \sqrt{\eta/\eta_E}}{C_1^{3/2} d^{3/2} \alpha_1^3}\right) \frac{\eta}{d^{1/8}}\right)^{t'_{3,1}-T_2-1} \end{aligned}$$

which means there must exist an iteration $t'_{3,1} \in [T_2 + \tilde{\Theta}(\frac{d^{3/2}\alpha_1^2\sqrt{\eta_E/\eta}}{\eta}), T_2 + \tilde{\Theta}(\frac{d^{3/2}\alpha_1^4\sqrt{\eta_E/\eta}}{\eta})]$ such that $R_2^{(t'_{3,1}-1)} \geq d^{-1/4}$ (so the above update bound is still valid when the RHS is for $t \leq t'_{3,1} - 1$) and $R_2^{(t'_{3,1})} < d^{-1/4}$. Next we need to prove that at $t = t'_{3,1}$, it holds $C_1\alpha_2^6(B_{2,2}^{(t)})^6 \geq C_2[R_2^{(t)}]^3$. Let us discuss several possible cases:

1. Suppose $|B_{2,2}^{(t'_{3,1})}| \geq \frac{\alpha_1}{\alpha_2}|B_{2,1}^{(T_1)}|^{1/2} \geq \Theta(\frac{1}{d^{1/4}})$ (by [Induction 3.E.1a](#) and [Lemma 3.E.6](#)), then we already have $C_1\alpha_2^6(B_{2,2}^{(t'_{3,1})})^6 \geq C_2[R_2^{(t'_{3,1})}]^3$ and $T_{3,1} \leq t'_{3,1}$;
2. Suppose otherwise $|B_{2,2}^{(t'_{3,1})}| \leq \frac{\alpha_1}{\alpha_2}|B_{2,1}^{(T_1)}|^{1/2}$, then we shall have $\Sigma_{2,2}^{(t)} \leq O(\Sigma_{2,1}^{(t)})$. So the update of $R_2^{(t)}$ during $t \in [T_2, T_{3,1}]$ can be written as

$$R_2^{(t+1)} = R_2^{(t)} - \Theta(\eta\Sigma_{2,1}^{(t)})[R_2^{(t)}]^3 = R_2^{(t)}(1 - \Theta(\eta\zeta)[R_2^{(t)}]^{1/2})$$

Let $t'_{3,2} = \min\{t : R_2^{(t)} \leq 2d^{-1/4}\}$ be an iteration between T_2 and $t'_{3,1}$, we shall have

$$\sum_{t \in [t'_{3,2}, t'_{3,1}]} \eta\zeta[R_2^{(t)}]^{3/2} = \Theta(R_2^{(t'_{3,2})} - R_2^{(t'_{3,1})}) = \Theta(\frac{1}{d^{1/4}}) \quad \text{and} \quad R_2^{(t)} \in [0.99\frac{1}{d^{1/4}}, 2.01\frac{1}{d^{1/4}}]$$

which also implies $t'_{3,1} - t'_{3,2} = \Theta(\frac{d^{1/8}}{\eta\zeta}) = \tilde{\Theta}(\frac{d^{3/2+1/8}\alpha_1^3\sqrt{\eta_E/\eta}}{\eta})$. In this case, let us look at the update of $B_{2,2}^{(t)}$ at $t \in [T_2, T_3]$. By [Lemma 3.E.22](#), we have

$$B_{2,2}^{(t+1)} = B_{2,2}^{(t)} + \eta(1 \pm \tilde{O}(\frac{1}{d}))\Lambda_{2,2}^{(t)}$$

It is not hard to see $|B_{2,2}^{(t)}|$ is monotonically increasing. Also by [Induction 3.E.1a](#) and [Lemma 3.E.1a](#), if we sum together the update between $t'_{3,2}$ and $t'_{3,1}$ as follows: (suppose the sign of $B_{2,2}^{(t'_{3,2})}$ is positive for now, the negative case can be similarly dealt with)

$$\begin{aligned} B_{2,2}^{(t'_{3,2})} + \sum_{t \in [t'_{3,2}, t'_{3,1}]} \eta(1 \pm \tilde{O}(\frac{1}{d}))\Lambda_{2,2}^{(t)} &= \sum_{t \in [t'_{3,2}, t'_{3,1}]} \Theta(\frac{\eta C_0 C_1 \alpha_1^6 \alpha_2^6 (E_{2,1}^{(T_2)})^2}{\sqrt{C_2 [R_2^{(t)}]^3 (C_1 \alpha_1^6 (E_{2,1}^{(T_2)})^2)^{3/2}}}) (B_{2,2}^{(t)})^5 \\ &\geq B_{2,2}^{(t'_{3,2})} + (B_{2,2}^{(T_2)})^4 \sum_{t \in [t'_{3,2}, t'_{3,1}]} \Theta(\frac{\eta C_0 \alpha_1^3 \alpha_2^6 B_{2,2}^{(t)}}{C_1^{1/2} C_2^{1/2} [R_2^{(t)}]^{3/2} |E_{2,1}^{(T_2)}|}) \\ &\geq B_{2,2}^{(t'_{3,2})} \prod_{t=t'_{3,2}}^{t'_{3,1}} \left(1 + \eta \tilde{\Theta}(\frac{\alpha_1^3 \alpha_2^6}{d^{3/2+1/8} \sqrt{\eta_E/\eta}})\right) \\ &\geq \tilde{\Theta}(\frac{1}{\sqrt{d}}) \left(1 + \eta \tilde{\Theta}(\frac{\alpha_1^3 \alpha_2^6}{d^{3/2+1/8} \sqrt{\eta_E/\eta}})\right)^{\tilde{\Theta}(d^{3/2} \alpha_1^3 \sqrt{\eta_E/\eta}/\eta)} \\ &\geq \Omega(e^{\alpha_1}) \end{aligned}$$

which is a contradiction to our assumption $|B_{2,2}^{(t'_{3,1})}| \leq \frac{\alpha_1}{\alpha_2} |B_{2,1}^{(T_1)}|^{1/2}$. Since $|B_{2,2}^{(t)}|$ is monotonically increasing, we know there must exist some iteration $t \leq t'_{3,1}$ such that $|B_{2,2}^{(t)}| \geq \frac{\alpha_1}{\alpha_2} |B_{2,1}^{(T_1)}|^{1/2}$, which means $T_{3,1} \leq t'_{3,1}$.

Thus we proved the bound of $T_{3,1} \leq T_2 + \tilde{\Theta}(\frac{d^{3/2}\alpha_1^{O(1)}}{\eta})$.

Using similar arguments, we can prove that $R_2^{(T_{3,1})} \leq \frac{\alpha_1^{O(1)}}{d^{1/4}}$. Indeed, we can set $T_{3,3} := \min\{t : |B_{2,2}^{(t'_{3,1})}| \geq \frac{\alpha_1}{\alpha_2} |B_{2,1}^{(T_1)}|^{1/2}\}$. From our arguments in this proof, we know $\Sigma_{2,2}^{(t)} \leq O(\Sigma_{2,2}^{(t)})$ for $t \leq T_{3,3}$. Now we can further choose $t'_{3,3} = \min\{t : R_2^{(t)} \leq a\}$ for some $a = \frac{\alpha_1^{12}}{d^{1/4}}$ to be some iteration with $R_2^{(t)} \geq a$ for $t \in [T_2, t'_{3,3}]$ and $t'_{3,3} - T_2 = \Theta(\frac{\sqrt{a} \log d}{\eta \zeta})$. Now we can work out the update of $B_{2,2}^{(t)}$ during $t \in [T_2, t'_{3,3}]$ again to see that $|B_{2,2}^{(t'_{3,3})}| \leq B_{2,2}^{(T_2)} \left(1 + \eta \tilde{\Theta}(\frac{\alpha_1^3 \alpha_2^6}{d^2 a^{3/2} \sqrt{\eta_E/\eta}})\right)^{\frac{\sqrt{a}}{\eta \zeta}} \leq \tilde{O}(\frac{1}{\sqrt{d}})$. This would prove that $t'_{3,3} \leq T_{3,3}$ and $R_2^{(T_{3,3})} \leq \frac{\alpha_1^{O(1)}}{d^{1/4}}$. So we also have $|B_{2,2}^{(T_{3,3})}| \leq \frac{\alpha_1^{O(1)}}{d^{1/4}}$ because of the definition of $T_{3,1}$. But since $T_{3,3} \geq T_{3,1}$ by our arguments above and the fact that $|B_{2,2}^{(t)}|$ is increasing, we shall have $|B_{2,2}^{(T_{3,1})}| \in [\frac{1}{d^{1/4}}, \frac{\alpha_1^{O(1)}}{d^{1/4}}]$. $\square$

Now we proceed to characterize the learning of $B_{2,2}^{(t)}$ during $t \in [T_{3,1}, T_{3,2}]$.

Lemma 3.E.8 (The growth of $B_{2,2}^{(t)}$ until T_3). *Let $T_{3,1}, T_{3,2}$ be defined as in (3.E.2). If [Induction 3.E.1](#) holds true for all $t \in [T_2, T_3]$, then we have $T_{3,2} = T_{3,1} + \tilde{O}(\frac{d^{1/4}\alpha_1^{O(1)}}{\eta})$ and $T_3 \leq T_{3,2} + \tilde{O}(\frac{\alpha_1^{O(1)}}{\eta})$.*

Proof. We first calculate the bound for $T_{3,2}$. After $T_{3,1}$, since $|B_{2,2}^{(t)}|$ is increasing while $R_2^{(t)}$ is decreasing by [Induction 3.E.1](#). So by [Lemma 3.E.1a](#), we have

$$[Q_2^{(t)}]^{-2} = \Theta(C_1 \alpha_2^6 (B_{2,2}^{(t)})^6), \quad \Phi_2^{(t)} = Q_2^{(t)} / [U_2^{(t)}]^{3/2} = \Theta((C_1^{3/2} \alpha_2^3 \alpha_1^9 |B_{2,2}^{(t)}|)^3 |E_{2,1}^{(t)}|^3)^{-1})$$

So according to [Lemma 3.E.2](#), we would have for all $t \in [T_{3,1}, T_{3,2}]$:

$$\langle -\nabla_{w_2} L(W^{(t)}, E^{(t)}), v_2 \rangle = (1 \pm o(1)) \Lambda_{2,2}^{(t)} = \Theta\left(\frac{1}{C_1^{3/2} \alpha_1^9 |E_{2,1}^{(T_2)}|^3}\right) (B_{2,2}^{(t)})^2 \text{sign}(B_{2,2}^{(t)})$$

where we have used $(E_{2,1}^{(t)})^3 = \Theta((E_{2,1}^{(T_2)})^3)$ from [Induction 3.E.1a](#). So when $t \in [T_{3,1}, T_{3,2}]$, we can write down the explicit form of $\Lambda_{2,2}^{(t)}$ and use [Lemma 3.E.1d](#) to derive

$$\begin{aligned} |B_{2,2}^{(t+1)}| &= |B_{2,2}^{(t)}| + \eta \Theta\left(\frac{C_1 \alpha_1^6 |E_{2,1}^{(T_2)}|^2}{C_1^{3/2} \alpha_1^9 |E_{2,1}^{(T_2)}|^3}\right) (B_{2,2}^{(t)})^2 \\ &\geq |B_{2,2}^{(t)}| \left(1 + \Theta\left(\frac{1}{C_1 \alpha_1^{O(1)}}\right) |B_{2,2}^{(T_{3,1})}|\right) \end{aligned}$$

$$\geq |B_{2,2}^{(t)}| \left(1 + \Theta\left(\frac{1}{C_1 \alpha_1^{O(1)}}\right) \frac{1}{d^{1/4}} \right)$$

Thus after $\tilde{O}(\frac{d^{1/4} \alpha_1^{O(1)}}{\eta})$ many iterations, we would have $|B_{2,2}^{(t)}| \geq \frac{1}{3} \min\{|E_{2,1}^{(t)}|, |B_{1,1}^{(t)}|\}$. Now let us deal with the growth of $|B_{2,2}^{(t)}|$ at $t \in [T_{3,2}, T_{3,3}]$. During this stage, since $B_{2,2}^{(t)}$ is still increasing and $|E_{2,1}^{(t)}| = |E_{2,1}^{(T_2)}|$ by [Induction 3.E.1](#), we have from [Lemma 3.E.1a](#) that

$$\Phi_2^{(t)} = Q_2^{(t)} / [U_2^{(t)}]^{3/2} = \Theta\left(\frac{1}{C_1^2 \alpha_2^{12} (B_{2,2}^{(t)})^{12}}\right) \geq \Theta\left(\frac{1}{C_1^2 \alpha_1^{O(1)}}\right)$$

And we can redo the calculations as above to get $T_3 \leq T_{3,2} + \tilde{O}(\frac{\alpha_1^{O(1)}}{\eta})$ since $\sqrt{\eta/\eta_E} |E_{2,1}^{(t)}|$ and $|B_{1,1}^{(t)}|$ are both $\Theta(1)$ according to [Induction 3.E.1a,b](#) $\square$

Proving The Main Lemma. Now we finally begin to prove [Lemma 3.E.6](#).

Proof of Lemma 3.E.6. We start with proving [Induction 3.E.1](#).

Proof of Induction 3.E.1a: From [Lemma 3.E.3](#), we know the update of $B_{1,1}^{(t)}$ can be written as

$$B_{1,1}^{(t+1)} = B_{1,1}^{(t)} + \Theta(\eta \Sigma_{1,1}^{(t)} [R_1^{(t)}]^3) \pm \eta O\left(\frac{(B_{1,2}^{(t)})^3}{(B_{2,2}^{(t)})^3} + \frac{1}{\sqrt{d}}\right) \alpha_1^{O(1)} \Lambda_{2,2}^{(t)} + \frac{E_{2,1}^{(t)}}{B_{1,1}^{(t)}} \eta \Lambda_{2,1}^{(t)} - \frac{B_{2,2}^{(t)}}{B_{1,1}^{(t)}} \eta \Lambda_{2,2}^{(t)}$$

Since from [Lemma 3.E.7](#) and [Lemma 3.E.8](#), we know $T_3 \leq \tilde{O}(\frac{d^{1.625} \alpha_1^{O(1)}}{\eta})$ and from [Claim 3.B.1](#) and [Induction 3.E.1a,c](#) we have $\Sigma_{1,1}^{(t)} [R_1^{(t)}]^3 \leq \tilde{O}(\frac{\alpha_1^{O(1)}}{d^{2.25}})$, we shall have

$$\sum_{s \in [T_2, t]} \Theta(\eta \Sigma_{1,1}^{(s)} [R_1^{(s)}]^3) \leq \tilde{O}\left(\frac{d^{1.625} \alpha_1^{O(1)}}{\eta}\right) \tilde{O}\left(\frac{\eta \alpha_1^{O(1)}}{d^{2.25}}\right) \leq \frac{1}{\sqrt{d}} = o(1)$$

Further more, by applying [Lemma 3.H.1](#) to $x_t = B_{2,2}^{(t)}$ with $q' = q - 2$, and notice that $\text{sign}(B_{j,2}^{(t)}) = \text{sign}(B_{j,2}^{(T_2)})$ for all $t \in [T_2, T_3]$, we also have

$$\left| \sum_{s \in [T_2, t]} O\left(\frac{(B_{1,2}^{(s)})^3}{(B_{2,2}^{(s)})^3}\right) \alpha_1^{O(1)} \eta \Lambda_{2,2}^{(s)} \right| \leq \tilde{O}\left(\frac{\alpha_1^{O(1)}}{\sqrt{d}}\right)$$

Now we turn to the last two terms. We first see that from the expression (3.E.3) of $R_2^{(t)}$'s update, we have that (note that $\text{sign}(E_{2,1}^{(t)}\Delta_{2,1}^{(t)}) = 1$)

$$\sum_{s \in [T_2, t]} \frac{E_{2,1}^{(s)}}{|B_{1,1}^{(s)}|} \eta \Delta_{2,1}^{(s)} = \sum_{s \in [T_2, t]} \frac{1}{|B_{1,1}^{(s)}|} \Theta(\eta \Sigma_{2,1}^{(s)} [R_2^{(s)}]^3) = \Theta\left(\frac{\sqrt{\eta_E/\eta}}{|B_{1,1}^{(T_2)}|}\right) = \Theta(\sqrt{\eta_E/\eta})$$

where we have used the fact that $\Sigma_{2,1}^{(t)} [R_2^{(t)}]^3 = (1 \pm O(\frac{1}{d})) E_{2,1}^{(t)} \Delta_{2,1}^{(t)}$ and $\sum_{s \in [T_2, t]} \eta \Sigma_{2,1}^{(s)} [R_2^{(s)}]^3 \lesssim R_2^{(T_2)}$ from (3.E.3) (which holds for all $t \in [T_2, T_3]$). And also, the analysis above shows that

$$|B_{1,1}^{(t)}| = |B_{1,1}^{(T_2)}| + O(\sqrt{\eta_E/\eta}) - \sum_{s \in [T_2, t]} \frac{B_{2,2}^{(s)}}{|B_{1,1}^{(s)}|} \eta \Lambda_{2,2}^{(s)}$$

for all $t \in [T_2, T_3]$, which means that either $\sum_{s \in [T_2, t]} \frac{B_{2,2}^{(s)}}{|B_{1,1}^{(s)}|} \eta \Lambda_{2,2}^{(s)} \leq \sum_{s \in [T_2, t]} \frac{E_{2,1}^{(s)}}{|B_{1,1}^{(s)}|} \eta \Delta_{2,1}^{(s)}$ and we have $|B_{1,1}^{(t)}| \geq |B_{1,1}^{(T_2)}|$ holds throughout $t \in [T_2, T_3]$, or that $\sum_{s \in [T_2, t]} \frac{B_{2,2}^{(s)}}{|B_{1,1}^{(s)}|} \eta \Lambda_{2,2}^{(s)} \geq \Omega(\sqrt{\eta_E/\eta})$, in which case we would have $|B_{1,1}^{(t)}|$ to be actually decreasing (as $B_{2,2}^{(t)}$ is increasing). Now that since $B_{1,1}^{(T_2)} = \Theta(1)$, we can easily see by our definition of T_3 and the monotonicity of $B_{1,1}^{(t)}$ after going below $B_{1,1}^{(T_2)} - \Omega(\sqrt{\eta_E/\eta})$ that $B_{1,1}^{(t)} \geq 0.49 B_{1,1}^{(T_2)} = \Omega(1)$ for all $t \in [T_2, T_3]$.

Next let us look at the change of $B_{2,1}^{(t)}$. From Lemma 3.E.3, we can write down the update of $B_{2,1}^{(t)}$:

$$B_{2,1}^{(t+1)} = B_{2,1}^{(t)} + \tilde{O}\left(\frac{\alpha_1^{O(1)}}{d^{5/2}}\right) \eta \Phi_2^{(t)} [R_2^{(t)}]^3 \pm \tilde{O}\left(\frac{\alpha_1^{O(1)}}{d}\right) \eta \Lambda_{2,2}^{(t)} \pm \tilde{O}\left(\frac{\eta \alpha_1^{O(1)}}{d^3}\right)$$

For the first term, according to Lemma 3.E.7 and Lemma 3.E.8 and $R_2^{(t)} \leq O(\sqrt{\eta_E/\eta}) = o(1)$ for all $t \in [T_2, T_3]$ by Induction 3.E.1c, we have $\Phi_2^{(t)} [R_2^{(t)}]^3 \leq \alpha_1^{O(1)}$ for all $t \in [T_2, T_3]$ and

$$\sum_{s \in [T_2, t]} \tilde{O}\left(\frac{\alpha_1^{O(1)}}{d^{5/2}}\right) \eta \Phi_2^{(s)} [R_2^{(s)}]^3 \leq \tilde{O}\left(\frac{d^{1.625} \alpha_1^{O(1)}}{\eta}\right) \eta \tilde{O}\left(\frac{\alpha_1^{O(1)}}{d^{5/2}}\right) \leq \tilde{O}\left(\frac{\alpha_1^{O(1)}}{d^{7/8}}\right)$$

And similarly as in the proof of induction for $B_{1,1}^{(t)}$, we have

$$\sum_{s \in [T_2, t]} \tilde{O}\left(\frac{\alpha_1^{O(1)}}{d}\right) \eta \Lambda_{2,2}^{(s)} \leq \tilde{O}\left(\frac{\alpha_1^{O(1)}}{d}\right), \quad \sum_{s \in [T_2, t]} \tilde{O}\left(\frac{\eta \alpha_1^{O(1)}}{d^3}\right) \leq \tilde{O}\left(\frac{\alpha_1^{O(1)}}{d}\right)$$

which proved the induction for $B_{2,1}^{(t)}$ since $|B_{2,1}^{(T_2)}| = \tilde{\Theta}(\frac{1}{\sqrt{d}})$.

Next we go on for the induction of $B_{1,2}^{(t)}$, we write down its update:

$$B_{1,2}^{(t+1)} = B_{1,2}^{(t)} + \Theta\left(\frac{(B_{1,2}^{(t)})^2}{(B_{2,2}^{(t)})^2}\right)E_{2,1}^{(t)}\eta\Lambda_{2,2}^{(t)} \pm \eta\tilde{O}\left(\frac{\alpha_1^{O(1)}}{d^4}\right)|E_{2,1}^{(t)}|^2\Phi_2^{(t)} \pm \eta\tilde{O}\left(\frac{\alpha_1^{O(1)}}{d^{5/2}}\right)$$

By [Lemma 3.E.7](#) and [Lemma 3.E.8](#), we have for any $t \in [T_2, T_3]$

$$\sum_{s \in [T_2, t]} \eta\tilde{O}\left(\frac{\alpha_1^{O(1)}}{d^{5/2}}\right) \leq \frac{1}{\sqrt{d}\text{polylog}(d)}$$

and also

$$\begin{aligned} \sum_{s \in [T_2, t]} \eta\tilde{O}\left(\frac{\alpha_1^{O(1)}}{d^4}\right)|E_{2,1}^{(s)}|^2\Phi_2^{(s)} &\leq \left(\sum_{s \in [T_2, T_{3,1}]} + \sum_{s \in [T_{3,1}, T_3]} \right) \eta\tilde{O}\left(\frac{\alpha_1^{O(1)}}{d^4}\right)|E_{2,1}^{(s)}|^2\Phi_2^{(s)} \\ &\leq \eta\tilde{O}\left(\frac{\alpha_1^{O(1)}}{d^4}\right) \cdot (T_{3,1} - T_2) \cdot O(\alpha_1^{O(1)}d^{3/8}) + \eta\tilde{O}\left(\frac{\alpha_1^{O(1)}}{d^4}\right)(T_3 - T_{3,1}) \\ &\leq \tilde{O}\left(\frac{\alpha_1^{O(1)}}{d^2}\right) \end{aligned}$$

Now we consider the term $\Theta\left(\frac{(B_{1,2}^{(t)})^2}{(B_{2,2}^{(t)})^2}\right)E_{2,1}^{(t)}\eta\Lambda_{2,2}^{(t)}$, we have by [Induction 3.E.1a](#) that

$$\left| \sum_{s \in [T_2, t]} \Theta\left(\frac{(B_{1,2}^{(s)})^2}{(B_{2,2}^{(s)})^2}\right)E_{2,1}^{(s)}\eta\Lambda_{2,2}^{(s)} \right| \leq O(\sqrt{\eta_E/\eta}(B_{1,2}^{(T_2)})^2) \sum_{s \in [T_2, t]} \eta \frac{|\Lambda_{2,2}^{(s)}|}{(B_{2,2}^{(s)})^2}$$

where we have used our induction hypothesis that $B_{1,2}^{(t)} = B_{1,2}^{(T_2)}(1 \pm o(1))$. Using [Lemma 3.H.1](#) by setting $x_t = B_{2,2}^{(t)}$, $q' = 3$, and $A = \Theta(1) \geq d^{\Omega(1)}B_{2,2}^{(T_2)}$, it holds that

$$\left| \sum_{s \in [T_2, t]} \Theta\left(\frac{(B_{1,2}^{(s)})^2}{(B_{2,2}^{(s)})^2}\right)E_{2,1}^{(s)}\eta\Lambda_{2,2}^{(s)} \right| \leq O(\sqrt{\eta_E/\eta}) \frac{(B_{1,2}^{(T_2)})^2}{|B_{2,2}^{(T_2)}|} \leq O(\sqrt{\eta_E/\eta}) \frac{(B_{1,2}^{(0)})^2}{|B_{2,2}^{(0)}|} \leq \frac{1}{\sqrt{d}\text{polylog}(d)}$$

where in the second inequality we have used [Lemma 3.C.8c](#), [Lemma 3.D.6a](#) and [Lemma 3.C.1](#), and in the last our choice of $\eta_E/\eta \leq \frac{1}{\text{polylog}(d)}$. This ensures the induction can go on until $t = T_3$. And we finished our proof of [Induction 3.E.1a](#).

Proof of Induction 3.E.1b: Let us write down the update of $E_{1,2}^{(t)}$ using Lemma 3.E.5:

$$\begin{aligned}
E_{1,2}^{(t+1)} &= E_{1,2}^{(t)}(1 - \eta_E \Xi_1^{(t)}) + \sum_{\ell \in [2]} \Theta(\eta_E \Sigma_{1,\ell}^{(t)}) (-E_{1,2}^{(t)} [R_2^{(t)}]^3 \pm O(\bar{R}_{1,2}^{(t)} + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2}) + \sum_{\ell \in [2]} \eta_E \Delta_{1,\ell}^{(t)} \\
&= E_{1,2}^{(t)}(1 - \eta_E \Xi_1^{(t)} - \sum_{\ell \in [2]} \Theta(\eta_E \Sigma_{1,\ell}^{(t)}) [R_2^{(t)}]^3) + \tilde{O}\left(\frac{\eta_E}{d^{3/2}}\right) \Phi_1^{(t)} [R_1^{(t)}]^3 \\
&\quad \pm \tilde{O}\left(\varrho + \frac{1}{\sqrt{d}}\right) \sum_{\ell \in [2]} \eta_E \Sigma_{1,\ell}^{(t)} [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2} \\
&= E_{1,2}^{(t)}(1 - \eta_E \Xi_1^{(t)} - \Theta(\eta_E \Sigma_{1,1}^{(t)}) [R_2^{(t)}]^3) \pm \tilde{O}\left(\varrho + \frac{1}{\sqrt{d}}\right) \eta_E \Sigma_{1,1}^{(t)} [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2}
\end{aligned}$$

where in the last inequality we have used $R_2^{(t)} \geq R_1^{(t)}$ from Induction 3.E.1c and $\Sigma_{1,1}^{(t)} \geq \Omega(\Phi_1^{(t)})$, $\Sigma_{2,1}^{(t)} \leq \tilde{O}\left(\frac{1}{d^{3/2}}\right) \Sigma_{1,1}^{(t)}$ from Claim 3.B.1 and Induction 3.E.1a. Now we can use the same analysis in the proof of Lemma 3.D.6 on $E_{1,2}^{(t)}$ to prove the desired claim, which we do not repeat here.

As for $E_{2,1}^{(t)}$, we can obtain similar expressions:

$$\begin{aligned}
E_{2,1}^{(t+1)} &= E_{2,1}^{(t)}(1 - \eta_E \Xi_2^{(t)} - \sum_{\ell \in [2]} \Theta(\eta_E \Sigma_{2,\ell}^{(t)}) [R_1^{(t)}]^3) \\
&\quad \pm \tilde{O}\left(\varrho + \frac{1}{\sqrt{d}}\right) \sum_{\ell \in [2]} \Theta(\eta_E \Sigma_{2,\ell}^{(t)}) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2} + \sum_{\ell \in [2]} \eta_E \Delta_{2,\ell}^{(t)}
\end{aligned}$$

Now we can obtain bounds for each terms as

$$\sum_{s \in [T_2, t]} \sum_{\ell \in [2]} \Theta(\eta_E \Sigma_{2,\ell}^{(s)}) [R_1^{(s)}]^3 \leq \tilde{O}\left(\frac{\eta_E \alpha_1^{O(1)}}{d^2}\right) \cdot \tilde{O}\left(\frac{d^{1.625} \alpha_1^{O(1)}}{\eta}\right) \leq \frac{1}{d^{3/4}}$$

and by (3.E.3) in Lemma 3.E.7, we also have for any $t \in [T_2, T_3]$

$$\begin{aligned}
\sum_{s \in [T_2, t]} \tilde{O}\left(\varrho + \frac{1}{\sqrt{d}}\right) \sum_{\ell \in [2]} \Theta(\eta_E \Sigma_{2,\ell}^{(s)}) [R_1^{(s)}]^{3/2} [R_2^{(s)}]^{3/2} &\leq \tilde{O}\left(\varrho + \frac{1}{\sqrt{d}}\right) \sum_{s \in [T_2, t]} \sum_{\ell \in [2]} \Theta(\eta_E \Sigma_{2,\ell}^{(s)}) [R_2^{(s)}]^3 \\
&\leq \tilde{O}\left(\varrho + \frac{1}{\sqrt{d}}\right) R_2^{(T_2)} \\
&\leq \tilde{O}\left(\varrho + \frac{1}{\sqrt{d}}\right)
\end{aligned}$$

And also by using our induction and by (3.E.3) in Lemma 3.E.7:

$$\sum_{s \in [T_2, t]} \sum_{\ell \in [2]} \eta_E \Delta_{2,\ell}^{(s)} \leq \sum_{s \in [T_2, t]} \frac{\eta_E / \eta}{|E_{2,1}^{(t)}|} \Theta(\eta \Sigma_{2,1}^{(s)} + \eta \Sigma_{2,2}^{(s)}) [R_2^{(s)}]^3 \leq \frac{\eta_E / \eta}{|E_{2,1}^{(T_2)}|} R_2^{(T_2)} \leq O\left(\frac{\eta_E / \eta}{\log d}\right) = o(\sqrt{\eta_E / \eta})$$

Finally, we can calculate

$$\sum_{s \in [T_2, t]} \eta_E \Xi_2^{(t)} E_{2,1}^{(t)} = \sum_{s \in [T_2, t]} \frac{\eta_E}{\eta} \frac{B_{2,2}^{(t)}}{E_{2,1}^{(t)}} \eta \Lambda_{2,2}^{(t)}$$

By resorting to the definition of T_3 and go through similar analysis as for the induction of $B_{1,1}^{(t)}$, we can obtain that $|E_{2,1}^{(t)}|$ is either above $|E_{2,1}^{(T_2)}|(1 + o(1))$ or is decreasing and always above $\frac{1}{2}|E_{2,1}^{(T_2)}|$. This proves [Induction 3.E.1b](#).

Proof of [Induction 3.E.1c](#): The proof of induction of $R_2^{(t)}$ is half done in [Lemma 3.E.7](#), we only need to complete the part when $t \in [T_{3,1}, T_3]$, since by [\(3.E.3\)](#), we always have $R_2^{(t)}$ to be decreasing by

$$R_2^{(t+1)} = R_2^{(t)} (1 - \sum_{\ell \in [2]} \Theta(\eta \Sigma_{2,\ell}^{(s)}) [R_2^{(t)}]^2)$$

And when $t \in [T_{3,1}, T_3]$, we have

$$\sum_{\ell \in [2]} \Theta(\eta \Sigma_{2,\ell}^{(s)}) \leq \tilde{O}(\eta d^{3/8+o(1)})$$

So if we suppose $R_2^{(T_3)} \leq \frac{1}{\sqrt{d}}$, we shall have for $T_3 - T_{3,1} = O(d^{1/4+o(1)}/\eta)$ many iterations that

$$R_2^{(t+1)} \geq R_2^{(T_{3,1})} (1 - \frac{\eta}{d^{5/8}})^{T_3 - T_{3,1}} \geq \Omega(R_2^{(T_{3,1})}) \geq \frac{1}{d^{1/4}} \quad (\text{by [Lemma 3.E.7](#)})$$

So it negates our supposition, which completes the proof of the induction for $R_2^{(t)}$ in $t \in [T_2, T_3]$.

Now we turn to the proof of induction for $R_1^{(t)}$, we write down its update: (as in [Lemma 3.D.6](#))

$$\begin{aligned} R_1^{(t+1)} &= R_1^{(t)} - \Theta(\eta [R_1^{(t)}]^3) \left(\Sigma_{1,1}^{(t)} + \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} (E_{2,1}^{(t)})^2 \right) \\ &\quad \pm O \left(\sum_{j,\ell} \eta \Sigma_{j,\ell}^{(t)} E_{j,3-j}^{(t)} (\bar{R}_{1,2}^{(t)} + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2} \right) \pm \frac{\eta}{\text{poly}(d)} \end{aligned}$$

It is straightforward to derive

$$\sum_{\ell \in [2]} \Sigma_{1,\ell}^{(t)} |E_{1,2}^{(t)}| (\bar{R}_{1,2}^{(t)} + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2} \leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}})^2 \sum_{\ell \in [2]} \Sigma_{1,\ell}^{(t)} [R_1^{(t)}]^3 [R_2^{(t)}]^3$$

and when $t \in [T_2, T_{3,1}]$:

$$\begin{aligned} \sum_{s \in [T_2, t]} \sum_{\ell \in [2]} \eta \Sigma_{2,\ell}^{(s)} |E_{2,1}^{(s)}| (\bar{R}_{1,2}^{(s)} + \varrho) [R_1^{(s)}]^{3/2} [R_2^{(s)}]^{3/2} &\leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) \frac{d^{o(1)} d^{3/8}}{d^{9/8}} \sum_{s \in [T_2, t]} \sum_{\ell \in [2]} \eta \Sigma_{2,\ell}^{(s)} [R_2^{(s)}]^3 \\ &\leq o(\frac{d^{o(1)}}{d^{3/4}}) \end{aligned}$$

and when $t \in [T_{3,1}, T_3]$:

$$\sum_{s \in [T_2, t]} \sum_{\ell \in [2]} \eta \Sigma_{2,\ell}^{(s)} |E_{2,1}^{(s)}| (\bar{R}_{1,2}^{(s)} + \varrho) [R_1^{(s)}]^{3/2} [R_2^{(s)}]^{3/2} \leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) \frac{d^{o(1)} d^{3/8}}{d^{9/8}} \eta \tilde{O}(\frac{d^{1/4+o(1)}}{\eta}) \leq O(\frac{1}{d})$$

So these combined with [Lemma 3.D.6](#) proved that $R_1^{(t)} \leq O(\frac{d^{o(1)}}{d^{3/4}})$ for all $t \in [T_2, T_3]$. We can go through some similar analysis about $R_2^{(t)}$ to get that $R_1^{(t)} \geq \frac{1}{d}$ for all $t \in [T_2, T_3]$.

Finally we begin to prove the induction of $\bar{R}_{1,2}^{(t)}$. Similarly as in the proof of [Lemma 3.D.6](#), we first write down

$$\begin{aligned} R_{1,2}^{(t+1)} &= R_{1,2}^{(t)} - \eta \langle \nabla_{w_1} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_2^{(t)} \rangle - \eta \langle \nabla_{w_2} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} w_1^{(t)} \rangle \\ &\quad + \eta^2 \langle \Pi_{V^\perp} \nabla_{w_1} L(W^{(t)}, E^{(t)}), \Pi_{V^\perp} \nabla_{w_2} L(W^{(t)}, E^{(t)}) \rangle \\ &= R_{1,2}^{(t)} + \eta \left(\Sigma_{1,1}^{(t)} + \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} (E_{2,1}^{(t)})^2 \right) (-\Theta(\bar{R}_{1,2}^{(t)}) \pm O(\varrho)) [R_1^{(t)}]^{5/2} [R_2^{(t)}]^{1/2} \\ &\quad + \eta \left(\Sigma_{1,1}^{(t)} \Theta(E_{1,2}^{(t)})^2 + \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} \right) (-\Theta(\bar{R}_{1,2}^{(t)}) \pm O(\varrho)) [R_2^{(t)}]^{5/2} [R_1^{(t)}]^{1/2} \\ &\quad + O\left(\sum_{(j,\ell) \neq (1,2)} \eta \Sigma_{j,\ell}^{(t)} E_{j,3-j}^{(t)} (R_1^{(t)} [R_2^{(t)}]^2 + R_2^{(t)} [R_1^{(t)}]^2) \right) \pm \frac{\eta}{\text{poly}(d)} \end{aligned}$$

Note that since $|E_{1,2}^{(t)}| \leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) [R_2^{(t)}]^{3/2} [R_1^{(t)}]^{3/2}$ and $R_1^{(t)} \leq O(\frac{1}{d^{3/4}})$, it holds

$$\begin{aligned} \sum_{(j,\ell) \neq (1,2)} \eta \Sigma_{j,\ell}^{(t)} |E_{j,3-j}^{(t)}| R_2^{(t)} [R_1^{(t)}]^2 &\leq \sum_{(j,\ell) \neq (1,2)} \eta \Sigma_{j,\ell}^{(t)} |E_{j,3-j}^{(t)}| R_1^{(t)} [R_2^{(t)}]^2 \\ &\leq o\left(\Sigma_{1,1}^{(t)} [R_1^{(t)}]^2 + \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} [R_2^{(t)}]^2 \right) \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) [R_2^{(t)}]^{1/2} [R_1^{(t)}]^{1/2} \end{aligned}$$

so the update becomes

$$\begin{aligned} R_{1,2}^{(t+1)} &= R_{1,2}^{(t)} \left(1 - \eta \Theta\left(\Sigma_{1,1}^{(t)} + \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} (E_{2,1}^{(t)})^2 \right) [R_1^{(t)}]^2 - \eta \Theta\left(\Sigma_{1,1}^{(t)} (E_{1,2}^{(t)})^2 + \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} \right) [R_2^{(t)}]^2 \right) \\ &\quad \pm \eta \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) [R_1^{(t)}]^{1/2} [R_2^{(t)}]^{1/2} \Theta\left(\Sigma_{1,1}^{(t)} + \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)} (E_{2,1}^{(t)})^2 \right) [R_1^{(t)}]^2 \end{aligned}$$

$$\pm \eta \tilde{O}(\varrho + \frac{1}{\sqrt{d}})[R_1^{(t)}]^{1/2}[R_2^{(t)}]^{1/2} \Theta\left(\Sigma_{1,1}^{(t)}(E_{1,2}^{(t)})^2 + \sum_{\ell \in [2]} \Sigma_{2,\ell}^{(t)}\right)[R_2^{(t)}]^2$$

Now we can use the same arguments as in the proof of $\bar{R}_{1,2}^{(t)}$ in Lemma 3.D.6 to conclude.

Proof of Lemma 3.E.6a,b,c: Indeed, at the end of phase III:

Induction 3.E.1a $\implies$ Lemma 3.E.6a

Induction 3.E.1b $\implies$ Lemma 3.E.6c

Induction 3.E.1c $\implies$ Lemma 3.E.6b

Now we have completed the whole proof. $\square$

3.F The End Phase: Convergence

When we arrive at $t = T_3$, we have already obtained the representation we want for the encoder network $f(X)$, where v_1 and v_2 are satisfactorily learned by different neurons. In the last phase, we prove that such features are the solutions that the algorithm are converging to, which gives a stronger guarantee than just accidentally finding the solution at some intermediate steps.

To prove the convergence, we need to ensure all the good properties that we got through the training still holds. Fortunately, mosts of Induction 3.E.1 still hold, as we summarized below:

Induction Hypothesis 3.F.1. *At the end phase, i.e. when $t \in [T_3, T]$, Induction 3.E.1a continues to hold except that $|B_{2,2}^{(t)}| = \Theta(1)$, Induction 3.E.1b will hold except that for $|E_{2,1}^{(t)}|$ only the upper bound still holds, and the upper bounds in Induction 3.E.1c still hold while the lower bounds for $R_1^{(t)}, R_2^{(t)}$ is $1/\text{poly}(d)$. Moreover, there is a constant $C = O(1)$ such that when $t \geq T_3 + \frac{\alpha_1^c}{\eta}$, we would have $|E_{2,1}^{(t)}| \leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}})[R_1^{(t)}]^{3/2}[R_2^{(t)}]^{3/2}$.*

Now we present the main theorem of the paper, which we shall prove in this section.

Theorem 3.F.1 (End phase: convergence). *For some $T_4 = T_3 + \frac{d^{2+o(1)}}{\eta}$ and $T = \text{poly}(d)/\eta$, we have for all $t \in [T_4, T]$ that Induction 3.F.1 holds true and:*

(a) *Successful learning of both v_1, v_2 : $|B_{1,1}^{(t)}|, |B_{2,2}^{(t)}| = \Theta(1)$ while $|B_{2,1}^{(t)}|, |B_{1,2}^{(t)}| = \tilde{O}(\frac{1}{\sqrt{d}})$.*

(b) *Successful denoising at the end: $R_j^{(t)} \leq R_j^{(T_3)}(1 - \tilde{\Theta}(\frac{1}{\alpha_6^c})[R_j^{(t)}]^2)$ for all $j \in [2]$.*

(c) *Prediction head is close to identity: $|E_{j,3-j}^{(t)}| \leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}})[R_1^{(t)}]^{3/2}[R_1^{(t)}]^{3/2}$ for all $j \in [2]$.*

In fact, (b) and (c) also imply for some sufficiently large $t = \text{poly}(d)/\eta$, it holds $R_j^{(t)} \leq \frac{1}{\text{poly}(d)}$ and $|E_{j,3-j}^{(t)}| \leq \frac{1}{\text{poly}(d)}$ for all $j \in [2]$.

And we have a simple corollary for the objective convergence.

Corollary 3.F.1 (objective convergence, with prediction head). *Let OPT denote the global minimum of the population objective (3.B.1). It is easy to derive that $\text{OPT} = 2 - 2\frac{C_0}{C_1} = \Theta(\frac{1}{\log d})$. We have for some sufficiently large $t \geq \text{poly}(d)/\eta$:*

$$L(W^{(t)}, E^{(t)}) \leq \text{OPT} + \frac{1}{\text{poly}(d)}$$

Now we need to establish some auxiliary lemmas:

Lemma 3.F.1. *For some $t \in [T_3, \text{poly}(d)/\eta]$, if [Induction 3.F.1](#) holds from T_3 to t , we have [Lemma 3.E.4](#) holds at t .*

Proof. Simple from similar calculations in the proof of [Lemma 3.E.4](#). $\square$

Lemma 3.F.2. *For some $t \in [T_3, \text{poly}(d)/\eta]$, if [Induction 3.F.1](#) holds from T_3 to t , we have for each $j \in [2]$ that*

$$\sum_{s \in [T_3, t]} \sum_{\ell \in [2]} \eta \Sigma_{j, \ell}^{(s)} [R_j^{(s)}]^3 \leq O(R_j^{(T_3)}), \quad \forall j \in [2]$$

Proof. Notice that when [Induction 3.F.1](#) holds, we always have

$$\sum_{\ell \in [2]} (\Sigma_{j, \ell}^{(t)} + \Sigma_{3-j, \ell}^{(t)} (E_{3-j, j}^{(t)})^2) = (1 \pm o(1)) \sum_{\ell \in [2]} \Sigma_{j, \ell}^{(t)}$$

we can use [Lemma 3.F.1](#) to obtain the update of $R_2^{(t)}$ as in the calculations when we obtained [\(3.E.3\)](#):

$$R_2^{(t)} = R_2^{(T_3)} - \sum_{s \in [T_3, t]} \sum_{\ell \in [2]} \Theta(\eta \Sigma_{2, \ell}^{(s)}) [R_2^{(s)}]^3$$

which means that $R_2^{(t)}$ is decreasing from T_3 to t . Summing up the update, the part of $R_2^{(t)}$ is solved. For the part of $R_1^{(t)}$, we separately discuss when $|E_{2,1}^{(t)}|$ is larger than or smaller than $\tilde{O}(\varrho + \frac{1}{\sqrt{d}})[R_1^{(t)}]^{3/2}[R_2^{(t)}]^{3/2}$. When the former happens, which we know from [Induction 3.F.1](#) that it cannot last until some $t'_4 = T_3 + \frac{\alpha_1^{O(1)}}{\eta}$ many iterations, we have for $t \in [T_3, t'_4]$

$$\sum_{s \in [T_3, t]} \sum_{(j, \ell) \in [2]^2} \eta \Sigma_{j, \ell}^{(s)} |E_{j, 3-j}^{(s)}| (\bar{R}_{1,2}^{(s)} + \varrho) [R_1^{(s)}]^{3/2} [R_2^{(s)}]^{3/2} \leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) \frac{\alpha_1^{O(1)}}{d} R_1^{(T_3)} \leq \frac{1}{d} R_1^{(T_3)}$$

Now for $t \geq t'_4$ we can simply go through similar calculations as in the proof of [Induction 3.E.1c](#) to obtain

$$\sum_{s \in [t'_4, t]} \sum_{(j, \ell) \in [2]^2} \eta \Sigma_{j, \ell}^{(s)} |E_{j, 3-j}^{(s)}| (\bar{R}_{1,2}^{(s)} + \varrho) [R_1^{(s)}]^{3/2} [R_2^{(s)}]^{3/2} \leq \sum_{s \in [t'_4, t]} \tilde{O}(\varrho + \frac{1}{\sqrt{d}})^2 \sum_{(j, \ell) \in [2]^2} \eta \Sigma_{j, \ell}^{(s)} [R_1^{(s)}]^3 [R_2^{(s)}]^3$$

$$\begin{aligned}
&\leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}})^2 R_2^{(T_3)} \max_{s \in [t'_4, t)} [R_1^{(s)}]^3 \\
&\leq \frac{1}{d} R_1^{(T_3)}
\end{aligned}$$

So by applying [Lemma 3.F.1a](#) and [Lemma 3.E.4](#), we have

$$R_1^{(t)} = (1 \pm o(1)) R_1^{(T_3)} - \sum_{s \in [T_3, t)} \sum_{\ell \in [2]} \Theta(\eta \Sigma_{j, \ell}^{(s)}) [R_1^{(s)}]^3$$

which proves the claim. $\square$

Lemma 3.F.3. *For some $t \in [T_3, \text{poly}(d)/\eta]$, if [Induction 3.F.1](#) holds from T_3 to t . Then we have $|E_{j, 3-j}^{(t)}|$ is decreasing until $|E_{j, 3-j}^{(t)}| \leq O(\bar{R}_{1,2}^{(t)} + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2} + \tilde{O}(\frac{1}{d^{3/2}}) [R_j^{(t)}]^3$. Moreover, we have for each $t \in [T_3, T]$ that*

$$\left| \sum_{s \in [T_3, t]} \eta_E \Xi_j^{(t)} E_{j, 3-j}^{(s)} \right| \leq |E_{j, 3-j}^{(T_3)}| + \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) \leq O(\sqrt{\eta_E/\eta})$$

Proof. We can go through the same calculations in the proof of [Induction 3.E.1b](#) (using [Fact 3.E.1](#)) to obtain

$$\begin{aligned}
E_{j, 3-j}^{(t+1)} &= E_{j, 3-j}^{(t)} (1 - \eta_E \Xi_j^{(t)}) + \sum_{\ell \in [2]} \eta_E \Delta_{j, \ell}^{(t)} \\
&\quad + \sum_{\ell \in [2]} \Theta(\eta_E \Sigma_{j, \ell}^{(t)}) (-E_{j, 3-j}^{(t)} [R_{3-j}^{(t)}]^3 \pm O(\bar{R}_{1,2}^{(t)} + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2}) \\
&= E_{j, 3-j}^{(t)} (1 - \eta_E \Xi_j^{(t)} - \eta_E \Theta(\Sigma_{j, j}^{(t)} [R_{3-j}^{(t)}]^3)) + \tilde{O}(\frac{1}{d^{3/2}}) \sum_{\ell \in [2]} \eta_E \Sigma_{j, \ell}^{(t)} [R_j^{(t)}]^3 \\
&\quad \pm O(\eta_E \Sigma_{j, j}^{(t)}) (\bar{R}_{1,2}^{(t)} + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2}
\end{aligned}$$

where we have used in the second equality that $\sum_{\ell \in [2]} \Delta_{j, \ell}^{(t)} \leq \tilde{O}(\frac{1}{d^{3/2}}) \sum_{\ell \in [2]} \Sigma_{j, \ell}^{(t)} [R_j^{(t)}]^3$ and also $\Sigma_{j, 3-j}^{(t)} \leq O(\frac{1}{d^{3/2}}) \Sigma_{j, j}^{(t)}$ for both $j \in [2]$ when [Induction 3.F.1](#) holds. Note that from above calculations, there exist a constant C such that if $|E_{j, 3-j}^{(t)}| \geq C(\bar{R}_{1,2}^{(t)} + \varrho) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2} + \sum_{\ell \in [2]} \eta_E \Delta_{j, \ell}^{(t)}$, we have $|E_{2,1}^{(t)}|$ to be decreasing. Now it suffices to observe that:

$$\begin{aligned}
\sum_{s \in [T_3, t]} O(\eta_E \Sigma_{j, j}^{(s)}) (\bar{R}_{1,2}^{(s)} + \varrho) [R_1^{(s)}]^{3/2} [R_2^{(s)}]^{3/2} &\leq \sum_{s \in [T_3, t]} O(\eta_E \Sigma_{1,1}^{(s)} + \eta_E \Sigma_{2,2}^{(s)}) (\bar{R}_{1,2}^{(s)} + \varrho) ([R_1^{(s)}]^3 + [R_2^{(s)}]^3) \\
&\leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}})
\end{aligned}$$

which is from [Induction 3.F.1](#), [Induction 3.E.1c](#) and [Lemma 3.F.1](#). Also note that $\Sigma_{j,j}^{(t)}[R_{3-j}^{(t)}]^3 \leq O(\frac{d^{o(1)}}{d^{3/4}})\Xi_j^{(t)}$ at this stage, we have

$$E_{3-j,j}^{(t)} = E_{j,3-j}^{(T_3)} - \sum_{s \in [T_3, t)} \Xi_j^{(s)} E_{j,3-j}^{(s)} + \tilde{O}(\varrho + \frac{1}{\sqrt{d}})$$

Recalling the expression of $\Xi_j^{(t)}$ finishes the proof. $\square$

Lemma 3.F.4. Recall T_2 defined in [\(3.D.1\)](#) and T_3 defined in [\(3.E.1\)](#), we have

$$\sqrt{\eta/\eta_E} \max_{t \leq T_3} |E_{2,1}^{(t)}| \leq \sum_{t \leq T_2} \frac{\eta \Sigma_{1,1}^{(t)}}{|B_{1,1}^{(t)}|} \mathcal{E}_{1,2}^{(t)} + \frac{1}{\alpha_1^{\Omega(1)}}$$

To prove this lemma, we need a simple claim.

Claim 3.F.1. If $\{x_t\}_{t < T}$, $x_t \geq 0$ is an increasing sequence and $C = \Theta(1)$ is a constant such that $x_{t+1} - x_t \leq O(\eta)$ and $\sum_{t < T} x_t(x_{t+1} - x_t) = C$, then for each $\delta \in (\frac{1}{d}, 1)$ it holds $|x_T - \sqrt{C}| \leq O(\delta^2 + x_0^2 + O(\frac{\log d}{d}))$.

Proof. Indeed, for every $g \in 0, 1, \dots$, we define $\mathcal{T}_g := \min\{t : x_t \geq (1 + \delta)^g x_0\}$. and define $b := \min\{g : ((1 + \delta)^g x_0)^2 \geq C - \delta^2\}$. Now for any $g < b$, we have

$$\sum_{t \in [\mathcal{T}_g, \mathcal{T}_{g+1}]} x_t(x_{t+1} - x_t) \geq x_{\mathcal{T}_g}(x_{\mathcal{T}_{g+1}} - x_{\mathcal{T}_g}) \geq (1 + \delta)^g \delta (1 + \delta)^{g-1} x_0^2 - \frac{1}{d} = \delta(1 + \delta)^{2g-1} x_0^2 - \frac{1}{d}$$

By our definition of $\mathcal{T}_g$, we can further get

$$C = \sum_{t < T} x_t(x_{t+1} - x_t) = \sum_{g=1}^b \sum_{t \in [\mathcal{T}_g, \mathcal{T}_{g+1}]} x_t(x_{t+1} - x_t) \geq (1 + \delta)^{2b} x_0^2 - x_0^2 - \frac{b}{d} \geq C - \delta^2 - x_0^2 - \frac{b}{d}$$

And also we have $C \leq (\max_{t \leq T} x_t) \sum_{t < T} (x_{t+1} - x_t) = x_T^2$, so we have $|x_T^2 - C| \leq \delta^2 + x_0^2 + \frac{b}{d}$, where $b = O(\log(C)/\log(1 + \delta)) \leq O(\log d)$, which proves the claim. $\square$

Proof of Lemma 3.F.4. From the proof of [Lemma 3.D.6](#) and [Lemma 3.E.6](#) we know that

$$\max_{t \leq T_3} |E_{2,1}^{(t)}| \leq \sum_{t \leq T_3} (1 \pm \frac{1}{\alpha_1^{\Omega(1)}}) \eta_E |\Delta_{2,1}^{(t)}| + \tilde{O}(\varrho + \frac{1}{\sqrt{d}})$$

And since from the proof of [Lemma 3.D.6](#) we know that

$$\begin{aligned} R_2^{(T_3)} &= R_2^{(0)} - \sum_{t \leq T_3} (1 \pm \tilde{O}(\frac{1}{d^{3/2}})) \eta \Sigma_{2,1}^{(t)} \mathcal{E}_{2,1}^{(t)} \pm \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) \\ &= (1 \pm \tilde{O}(\frac{1}{d^{3/2}})) \sum_{t \leq T_3} E_{2,1}^{(t)} \Delta_{2,1}^{(t)} \pm \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) \end{aligned}$$

We can define some alternative variables $\tilde{E}_{2,1}^{(t)}$ updated as $\tilde{E}_{2,1}^{(t+1)} = \tilde{E}_{2,1}^{(t)} + \eta_E \Delta_{2,1}^{(t)}$ and $\tilde{R}_2^{(t+1)} = \tilde{R}_2^{(t)} - \tilde{E}_{2,1}^{(t)} \Delta_{2,1}^{(t)}$. It is easy to see that $|E_{2,1}^{(t)} - \tilde{E}_{2,1}^{(t)}| \leq \frac{1}{\alpha_1^{O(1)}} \max_{t \leq T_3} |E_{2,1}^{(t)}|$. From above calculations, we know $\frac{\eta}{\eta_E} \sum_{t \in [T_1, T_3]} \tilde{E}_{2,1}^{(t)} (\tilde{E}_{2,1}^{(t+1)} - \tilde{E}_{2,1}^{(t)}) = \tilde{R}_2^{(T_1)} \pm \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) + O(\frac{1}{d^{1/4}})$, which by [Claim 3.F.1](#) implies that

$$\sqrt{\eta/\eta_E} |\tilde{E}_{2,1}^{(T_3)}| = \sqrt{\tilde{R}_2^{(T_1)}} \pm O(\frac{1}{d^{1/4}}) = \sqrt{2} \pm \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) \pm O(\frac{1}{d^{1/4}})$$

And when we turn back, we shall have $\sqrt{\eta/\eta_E} \max_{t \leq T_3} |E_{2,1}^{(t)}| \leq \sqrt{2} + \frac{1}{\alpha_1^{O(1)}}$. Now we can use similar techniques on $B_{1,1}^{(t)}$ and $R_1^{(t)}$. Indeed, from [\(3.D.4\)](#) and similar arguments in phase I, we know for all $t \in [T_1, T_2]$

$$\begin{aligned} R_1^{(t+1)} &= R_1^{(0)} - \sum_{s \leq t} (1 \pm \tilde{O}(\frac{1}{d^{3/2}})) \eta \Sigma_{1,1}^{(s)} \mathcal{E}_{1,2}^{(s)} \pm \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) \\ R_1^{(t+1)} &\leq R_1^{(t)} (1 - \tilde{O}(\frac{\eta}{\alpha_1^6}) [R_1^{(t)}]^2) \end{aligned} \tag{3.F.1}$$

So one can obtain that at some iteration $t' = T_1 + O(\frac{d \alpha_1^{O(1)}}{\eta})$, we shall have $R_1^{(t)} \leq O(\frac{1}{\sqrt{d}})$ for all $t \geq t'$. Now let us consider the growth of $B_{1,1}^{(t)}$ before t' , which clearly constitutes of

$$\begin{aligned} B_{1,1}^{(t')} &= B_{1,1}^{(T_1)} + \sum_{t \in [T_1, t']} (\Lambda_{1,1}^{(t)} + \Gamma_{1,1}^{(t)} - \Upsilon_{1,1}^{(t)}) \\ &= B_{1,1}^{(T_1)} + \sum_{t \in [T_1, t']} \left(\frac{\eta \Sigma_{1,1}^{(t)}}{|B_{1,1}^{(t)}|} \mathcal{E}_{1,2}^{(t)} \text{sign}(B_{1,1}^{(t)}) + \eta \Gamma_{1,1}^{(t)} - \eta \Upsilon_{1,1}^{(t)} \right) \\ &= B_{1,1}^{(0)} + \sum_{t < t'} \frac{\eta \Sigma_{1,1}^{(t)}}{|B_{1,1}^{(t)}|} \mathcal{E}_{1,2}^{(t)} \text{sign}(B_{1,1}^{(t)}) + \sum_{t \in [T_1, t']} \eta (\Gamma_{1,1}^{(t)} - \Upsilon_{1,1}^{(t)}) + \tilde{O}(\frac{1}{\sqrt{d}}) \end{aligned}$$

where the last one comes from the proof of [Lemma 3.C.8](#). Moreover by using the same arguments in the proof of [Lemma 3.D.6](#) we can easily prove that

$$\left| \sum_{t \in [T_1, t']} (\Gamma_{1,1}^{(t)} - \Upsilon_{1,1}^{(t)}) \right| \leq \tilde{O}(\frac{1}{\sqrt{d}}) \implies \sum_{t < t'} \frac{\eta \Sigma_{1,1}^{(t)}}{|B_{1,1}^{(t)}|} \mathcal{E}_{1,2}^{(t)} \geq |B_{1,1}^{(t')}| - |B_{1,1}^{(0)}| - \tilde{O}(\frac{1}{\sqrt{d}})$$

And for $t \in [t', T_2]$, we also have by [\(3.F.1\)](#) that

$$\sum_{t \in [t', T_2]} \frac{\eta \Sigma_{1,1}^{(t)}}{|B_{1,1}^{(t)}|} \mathcal{E}_{1,2}^{(t)} \leq \sum_{t \in [t', T_2]} \eta \Sigma_{1,1}^{(t)} \mathcal{E}_{1,2}^{(t)} \leq O(\frac{1}{\sqrt{d}})$$

Recall $R_1^{(0)} = \sum_{t \in [0, t']} (1 \pm \tilde{O}(\frac{1}{d^{3/2}})) \eta \Sigma_{1,1}^{(t)} \mathcal{E}_{1,2}^{(t)} \pm \tilde{O}(\varrho + \frac{1}{\sqrt{d}})$ by (3.F.1) and $R_1^{(t)} \leq O(\frac{1}{\sqrt{d}})$ for $t \geq t'$. Now we can finally go through the same analysis using Claim 3.F.1 on $B_{1,1}^{(t)}$ and $R_1^{(t)}$ during $t \in [0, t']$ as above to obtain that

$$\sum_{t \leq T_2} \frac{\eta \Sigma_{1,1}^{(t)}}{|B_{1,1}^{(t)}|} \mathcal{E}_{1,2}^{(t)} \geq (1 - \tilde{O}(\frac{1}{d^{3/2}})) \sqrt{R_1^{(0)}} - \tilde{O}(\frac{1}{\sqrt{d}}) = 1 - \tilde{O}(\varrho + \frac{1}{\sqrt{d}})$$

Combining the results, we finishes the proof. $\square$

Now we are prepared to prove Theorem 3.F.1.

3.F.1 Proof of Convergence

Proof of Theorem 3.F.1. First we start with the $B_{j,\ell}^{(t)}$ s. Indeed, we can go through similar calculations to see that all gradients $\langle -\nabla_{w_j} L(W^{(t)}, E^{(t)}), v_\ell \rangle$ can be decomposed into

$$\langle -\nabla_{w_j} L(W^{(t)}, E^{(t)}), v_\ell \rangle = (\Lambda_{j,\ell}^{(t)} - \Upsilon_{j,\ell,1}^{(t)}) + (\Gamma_{j,\ell}^{(t)} - \Upsilon_{j,\ell,2}^{(t)})$$

where $\Lambda_{j,\ell}^{(t)} - \Upsilon_{j,\ell,1}^{(t)}$ and $\Gamma_{j,\ell}^{(t)} - \Upsilon_{j,\ell,2}^{(t)}$ can be expressed as

$$\begin{aligned} \Lambda_{j,\ell}^{(t)} - \Upsilon_{j,\ell,1}^{(t)} &= C_0 \alpha_2^6 C_1 \alpha_1^6 \Phi_j^{(t)} (B_{j,\ell}^{(t)})^5 \left(E_{j,3-j}^{(t)} (B_{3-j,3-\ell}^{(t)})^3 (B_{j,3-\ell}^{(t)})^3 + (E_{j,3-j}^{(t)})^2 (B_{3-j,3-\ell}^{(t)})^6 \right) \\ &\quad - C_0 \alpha_2^6 C_1 \alpha_1^6 \Phi_j^{(t)} (B_{j,\ell}^{(t)})^2 (B_{3-j,\ell}^{(t)})^3 E_{j,3-j}^{(t)} \left((B_{j,3-\ell}^{(t)})^6 + E_{j,3-j}^{(t)} (B_{3-j,3-\ell}^{(t)})^3 (B_{j,3-\ell}^{(t)})^3 \right) \\ &\quad + C_0 \alpha_2^6 \Phi_j^{(t)} (B_{j,\ell}^{(t)})^5 C_2 \mathcal{E}_{j,3-j}^{(t)} \\ \Gamma_{j,\ell}^{(t)} - \Upsilon_{j,\ell,2}^{(t)} &= C_0 \alpha_2^6 C_1 \alpha_1^6 \Phi_{3-j}^{(t)} (B_{3-j,\ell}^{(t)})^3 (B_{j,\ell}^{(t)})^2 E_{3-j,j}^{(t)} \left(E_{3-j,j}^{(t)} (B_{j,3-\ell}^{(t)})^3 (B_{3-j,3-\ell}^{(t)})^3 + (E_{3-j,j}^{(t)})^2 (B_{j,3-\ell}^{(t)})^6 \right) \\ &\quad - C_0 \alpha_2^6 C_1 \alpha_1^6 \Phi_{3-j}^{(t)} (B_{j,\ell}^{(t)})^5 (E_{3-j,j}^{(t)})^2 \left((B_{3-j,3-\ell}^{(t)})^6 + E_{3-j,j}^{(t)} (B_{j,3-\ell}^{(t)})^3 (B_{3-j,3-\ell}^{(t)})^3 \right) \\ &\quad + C_0 \alpha_2^6 \Phi_{3-j}^{(t)} E_{3-j,j}^{(t)} (B_{3-j,\ell}^{(t)})^3 (B_{j,\ell}^{(t)})^2 C_2 \mathcal{E}_{3-j,j}^{(t)} \end{aligned}$$

Firstly, for all the terms that contain factors of $(B_{j,\ell}^{(t)})^2 (B_{3-j,\ell}^{(t)})^2$ (or $(B_{j,\ell}^{(t)})^2 (B_{j,3-\ell}^{(t)})^2$), we can apply Lemma 3.F.3, our Induction 3.F.1 assumption and $|E_{j,3-j}^{(t)}| \leq O(1)$, $\forall t \in [T_3, T]$ to obtain that their (multiplied by η) summation over $t \in [T_3, T]$ is absolutely bounded by $\tilde{O}(\frac{1}{d})$. So we can move on to deal with all other terms. When $j = \ell$, Using Lemma 3.F.3, we have

$$\begin{aligned} \sum_{t \in [T_3, T]} \eta C_0 \alpha_2^6 C_1 \alpha_1^6 \Phi_j^{(t)} |B_{j,\ell}^{(t)}|^5 (E_{j,3-j}^{(t)})^2 (B_{3-j,3-\ell}^{(t)})^6 &= \sum_{t \in [T_3, T]} \frac{\eta \Xi_j^{(t)}}{|B_{j,\ell}^{(t)}|} (E_{j,3-j}^{(t)})^2 \\ &\leq \sqrt{\frac{\eta}{\eta_E}} |E_{j,3-j}^{(T_3)}| + \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) = O(1) \end{aligned}$$

And the sign of LHS is $\text{sign}(B_{j,\ell}^{(t)})$. Moreover, for $j = \ell = 1$, from [Lemma 3.F.4](#) and [Lemma 3.F.3](#) we also have

$$\begin{aligned} \sum_{t \in [T_3, T]} \eta C_0 \alpha_2^6 C_1 \alpha_1^6 \Phi_2^{(t)} |B_{1,1}^{(t)}|^5 (E_{2,1}^{(t)})^2 (B_{2,2}^{(t)})^6 &\leq \sqrt{\frac{\eta}{\eta_E}} \left| \sum_{t \in [T_3, T]} \eta_E \Xi_j^{(t)} E_{j,3-j}^{(t)} \right| \\ &\leq \sqrt{\frac{\eta}{\eta_E}} |E_{2,1}^{(T_3)}| + \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) \\ &\leq \sum_{t \leq T_2} \frac{\eta \Sigma_{1,1}^{(t)}}{|B_{1,1}^{(t)}|} \mathcal{E}_{1,2}^{(t)} + \frac{1}{\alpha_1^{\Omega(1)}} \end{aligned}$$

Since we have

$$B_{1,1}^{(T_2)} = \sum_{s \leq T_2} \frac{\eta \Sigma_{1,1}^{(s)}}{|B_{1,1}^{(s)}|} \mathcal{E}_{1,2}^{(s)} + \sum_{s \leq T_2} \frac{\eta \Sigma_{2,1}^{(s)}}{|B_{1,1}^{(s)}|} \mathcal{E}_{2,1}^{(s)} - \sum_{t \in [T_3, T]} \frac{\eta \Xi_j^{(t)}}{|B_{j,\ell}^{(t)}|} (E_{j,3-j}^{(t)})^2$$

And since by [Induction 3.D.1](#) we have $|B_{1,1}^{(t)}| = \Theta(1)$ during $t \in [T_1, T_2]$ and $\sum_{t \in [T_1, T_2]} \eta \Sigma_{2,1}^{(t)} \geq R^{(T_1)} - o(1) = \sqrt{2} - o(1)$. For all the other terms in the gradient, we can apply [Lemma 3.F.3](#), our [Induction 3.F.1](#) assumption and $|E_{j,3-j}^{(t)}| \leq O(1)$ so we have for $t \in [T_3, T]$

$$\begin{aligned} |B_{1,1}^{(t)}| &= \sum_{s \leq T_2} \frac{\eta \Sigma_{1,1}^{(s)}}{|B_{1,1}^{(s)}|} \mathcal{E}_{1,2}^{(s)} + \sum_{s \leq T_2} \frac{\eta \Sigma_{2,1}^{(s)}}{|B_{1,1}^{(s)}|} \mathcal{E}_{2,1}^{(s)} - \sum_{t \in [T_3, T]} \frac{\eta \Xi_j^{(t)}}{|B_{j,\ell}^{(t)}|} (E_{j,3-j}^{(t)})^2 - o(1) \\ &\geq \sqrt{\eta/\eta_E} \max_{t \leq T_3} |E_{2,1}^{(t)}| + \sum_{s \leq T_2} \frac{\eta \Sigma_{2,1}^{(s)}}{|B_{1,1}^{(s)}|} \mathcal{E}_{2,1}^{(s)} - \sqrt{\frac{\eta}{\eta_E}} |E_{j,3-j}^{(T_3)}| + \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) - o(1) \\ &\geq \sum_{s \leq T_2} \frac{\eta \Sigma_{2,1}^{(s)}}{|B_{1,1}^{(s)}|} \mathcal{E}_{2,1}^{(s)} - o(1) \geq \Omega(1) \end{aligned}$$

which also proved $|B_{1,1}^{(t)}| = O(1)$ since all the terms on the RHS are absolutely $O(1)$ bounded. Since one can see from [Lemma 3.F.3](#) that $|E_{2,1}^{(t)}|$ is decreasing before it reaches $\frac{1}{d}$. Moreover this proves $\sqrt{\eta/\eta_E} |E_{2,1}^{(t)}| \leq |B_{1,1}^{(t)}|$ for all $t \in [T_3, T]$, and also the fact that

$$B_{1,1}^{(t)} \geq \Omega(1), \quad \forall t \in [T_3, T]$$

The case of $B_{2,2}^{(t)}$ is much more simple as $E_{1,2}^{(t)} \leq \tilde{O}(\frac{1}{d})$ throughout $t \in [T_3, T]$ by [Lemma 3.F.3](#) and [Lemma 3.E.6c](#), Now we can go through the similar calculations again to obtain that $B_{2,2}^{(t)} = \Theta(1)$ for all $t \in [T_3, T]$. When $j \neq \ell$, all the terms calculated in the expansion of $\Lambda_{j,\ell}^{(t)} - \Upsilon_{j,\ell,1}^{(t)}$ and $\Gamma_{j,\ell}^{(t)} - \Upsilon_{j,\ell,2}^{(t)}$

contain factors of $(B_{2,1}^{(t)})^2 = \tilde{O}(\frac{1}{d})$ or $(B_{1,2}^{(t)})^2 = \tilde{O}(\frac{1}{d})$. So we can similarly use [Lemma 3.F.3](#) as before to derive that $B_{j,3-j}^{(t)} = B_{j,3-j}^{(T_3)}(1 \pm \tilde{O}(\frac{\alpha_1^{O(1)}}{\sqrt{d}}))$ for all $t \in [T_3, T]$ and $j \in [2]$.

As for the prediction head, the induction of $E_{1,2}^{(t)}$ follows from exactly the same proof in [Lemma 3.E.6](#). The part of $E_{2,1}^{(t)}$ is half done in [Lemma 3.F.3](#). It suffices to notice that $\Xi_2^{(t)} = \tilde{\Theta}(\frac{\alpha_1^6}{\alpha_2^6})$ and if $|E_{2,1}^{(t)}| \geq C(\bar{R}_{1,2}^{(t)} + \varrho)[R_1^{(t)}]^{3/2}[R_2^{(t)}]^{3/2}$ for some $C = O(1)$, then

$$\begin{aligned} E_{2,1}^{(t+1)} &= E_{2,1}^{(t)}(1 - \eta_E \Xi_2^{(t)} - \eta_E \Theta(\Sigma_{2,2}^{(t)}[R_1^{(t)}]^3)) + \tilde{O}(\frac{1}{d^{3/2}}) \sum_{t \in [2]} \eta_E \Sigma_{2,t}^{(t)} [R_2^{(t)}]^3 \\ &\quad \pm O(\eta_E \Sigma_{2,2}^{(t)})(\bar{R}_{1,2}^{(t)} + \varrho)[R_1^{(t)}]^{3/2}[R_2^{(t)}]^{3/2} \\ &\leq E_{2,1}^{(t)}(1 - \tilde{\Theta}(\frac{\eta \alpha_1^6}{\alpha_2^6})) \end{aligned}$$

So after $\frac{\alpha_1^{O(1)}}{\eta}$ many epochs will we have

$$|E_{2,1}^{(t)}| \leq (\log d) \bar{R}_{1,2}^{(t)} + \varrho [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2} \leq \tilde{O}(\varrho + \frac{1}{\sqrt{d}}) [R_1^{(t)}]^{3/2} [R_2^{(t)}]^{3/2}$$

as desired. And the rest of the induction of $E_{2,1}^{(t)}$ is the same as in the induction arguments of $E_{1,2}^{(t)}$ in [Lemma 3.E.6](#).

The induction of $R_1^{(t)}, R_2^{(t)}$ and $R_{1,2}^{(t)}$ is exactly the same as those in the proof of [Lemma 3.E.6](#) except here we only need $R_1^{(t)}/R_2^{(t)} \in [\frac{1}{\alpha_1^{O(1)}}, \alpha_1^{O(1)}]$ after T_4 . Indeed, from the update of $R_j^{(t)}$ (which can be easily worked out), we have

$$R_j^{(t+1)} = R_j^{(t)}(1 - \Theta(\eta \Sigma_{j,j}^{(t)})[R_j^{(t)}]^2) = R_j^{(t)}(1 - \tilde{\Theta}(\frac{\eta}{\alpha_j^6})[R_j^{(t)}]^2)$$

Now after $\frac{d^2 \alpha_1^{O(1)}}{\eta}$ many epochs, we can obtain from similar arguments in [Lemma 3.E.6](#) that $R_1^{(t)}/R_2^{(t)} \in [\frac{1}{\alpha_1^{O(1)}}, \alpha_1^{O(1)}]$ and $R_j^{(t)} \leq \frac{1}{d}$. The induction can go on until $t = \text{poly}(d)/\eta$.

For the convergence of $B_{1,1}^{(t)}$ and $B_{2,2}^{(t)}$ after $t = T_4$, notice that their change depends on $\sum_{t \geq T_4} \frac{E_{j,3-j}^{(t)}}{B_{j,j}^{(t)}} \Xi_j^{(t)}$, which stays very small after T_4 , we have that $|B_{j,j}^{(t)} - B_{j,j}^{(T_4)}| \leq o(1)$ for all $j \in [2]$. This finishes the whole proof. $\square$

3.G Learning Without Prediction Head

When we do not use prediction head in the network architecture, the analysis is much simpler. We can reuse most of the gradient calculations in previous sections as long as we set $E^{(t)}$ to the identity. Note that here we allow $m \geq 1$ to be any positive integer.

Theorem 3.G.1 (learning without the prediction head). *Let m be any positive integer. If we keep $E^{(t)} \equiv I_m$ during the whole training process, then for all $t \in [\tilde{\Omega}(\frac{d^2}{\eta}), \text{poly}(d)/\eta]$, we shall have $|B_{j,1}^{(t)}| = \Theta(1)$, $|B_{j,2}^{(t)}| = \tilde{O}(\frac{1}{\sqrt{d}})$ and $R_j^{(t)} = O(\frac{1}{d^{1-o(1)}})$ for all $j \in [m]$ with probability $1 - o(1)$. Moreover, for a longer training time $t = \text{poly}(d)/\eta$, we would have $R_j^{(t)} \leq \frac{1}{\text{poly}(d)}$ for all $j \in [m]$.*

Moreover, it is direct to obtain a objective convergence result similar to [Corollary 3.F.1](#).

Corollary 3.G.1 (objective convergence, without prediction head). *Let OPT denote the global minimum of the population objective [\(3.B.1\)](#). When trained with $E^{(t)} \equiv I_m$, we have for some sufficiently large $t \geq \text{poly}(d)/\eta$:*

$$L(W^{(t)}, I_m) \leq \text{OPT} + \frac{1}{\text{poly}(d)}$$

Proof of Theorem 3.G.1. The proof is easy to obtain since it is very similar to some proofs in previous sections, and we only sketch it here. Indeed, using the calculations in [Lemma 3.E.3](#) and [Lemma 3.E.2](#) and set $E_{i,j}^{(t)}, i \neq j \in [m]$ to zero. We shall have (note that here $\mathcal{E}_{j,r}^{(t)} \equiv \mathcal{E}_j^{(t)}$ for any $r \neq j$)

$$\langle -\nabla_{w_j} L(W^{(t)}, E^{(t)}), v_\ell \rangle = C_0 C_2 \alpha_\ell^6 (B_{j,\ell}^{(t)})^5 \Phi_j^{(t)} \mathcal{E}_j^{(t)} = \Theta(C_0 C_2 \alpha_\ell^6 \Phi_j^{(t)} (B_{j,\ell}^{(t)})^5 [R_j^{(t)}]^3)$$

Now we can go through the similar induction arguments as in the proof of [Lemma 3.C.8](#) (with TPM lemma to distinguish the learning speed) to obtain that for each $j \in [m]$:

$$|B_{j,1}^{(t)}| = \Theta(1), \quad |B_{j,2}^{(t)}| = |B_{j,2}^{(0)}|(1 \pm o(1)), \quad \forall j \in [m] \quad (\text{when } t \geq \frac{d^2}{\eta})$$

When this is proven, we can also reuse the calculations as in the proof of [Lemma 3.D.3](#) to obtain that

$$R_j^{(t+1)} = R_j^{(t)}(1 - \Theta(\eta \Sigma_{j,1}^{(t)})[R_j^{(t)}]^2) = R_j^{(t)}(1 - \Theta(\eta C_0 C_2 \alpha_1^6 \Phi_j^{(t)} (B_{j,1}^{(t)})^6 [R_j^{(t)}]^2), \quad \forall j \in [m]$$

So again after some $t = \tilde{O}(\frac{d^2}{\eta})$, we shall have $R_j^{(t)} \leq O(\frac{d^{o(1)}}{d})$. While the decrease of $R_j^{(t)}$ is happening, we can make induction that $|B_{j,2}^{(t)}| = |B_{j,2}^{(0)}|(1 \pm o(1))$, since if it holds for all previous

iterations before t , then

$$\begin{aligned} \sum_{s \leq t-1} \eta |\langle -\nabla_{w_j} L(W^{(s)}, E^{(s)}), v_2 \rangle| &= \sum_{s \leq t-1} \eta C_0 \alpha_2^6 \Phi_j^{(s)} |B_{j,2}^{(s)}|^5 C_2 \mathcal{E}_j^{(s)} \\ &\stackrel{(1)}{\leq} \frac{1}{\text{polylog}(d)} |B_{j,2}^{(0)}| \end{aligned}$$

where (1) is due to [Corollary 3.H.1](#), where $x_t = |B_{j,1}^{(t)}|$ and $y_t = |B_{j,2}^{(t)}|$ and $S_t \leq \frac{1}{\text{polylog}(d)}$, $y_0 \leq O(\log d)x_0$. which finishes the proof. $\square$

3.H Tensor Power Method Bounds

In this section, we give two lemmas related to the tensor power method that can help us in previous sections' proofs.

The tensor-power estimate needed here is the shared [Lemma 3.A.1](#), specialized with $r = q$. This lemma has a corollary:

Corollary 3.H.1 (TPM, from [6]). *Let $q \geq 3$ be a constant and $x_0, y_0 = o(1)$ and $A = O(1)$. Let $\{x_t, y_t\}_{t \geq 0}$ be two positive sequences updated as*

- $x_{t+1} = x_t + \eta C_t x_t^q$ for some $C_t > 0$;
- $y_{t+1} = y_t + \eta S_t C_t y_t^q$ for some $S_t > 0$.

Suppose $x_0 \geq y_0 (\max_{t: x_t \leq A} S_t)^{\frac{1}{q-1}} (1 + \frac{1}{\text{polylog}(d)})$, then $y_t \leq \tilde{O}(y_0)$ for all t such that $x_t \leq A$. Moreover, if $x_0 \geq y_0 (\max_{t: x_t \leq A} S_t)^{\frac{1}{q-1}} \log(d)$, we would have $|y_t - y_0| \lesssim \frac{|y_0|}{\text{polylog}(d)}$.

Moreover, we prove the following lemma for comparing the updates of different variables.

Lemma 3.H.1 (TPM of different degrees). *Consider an increasing sequences $x_t \geq 0$ defined by $x_{t+1} = x_t + \eta C_t x_t^q$, for some integer $q > q' \geq 3$ and $q' \leq q - 2$, and $C_t > 0$, and further suppose given $A = O(1)$, there exists $t' \geq 0$, $x_{t'} \geq A$. Then for every $\delta > 0$ and every $\eta \in (0, 1)$:*

$$\begin{aligned} \sum_{t \geq 0, x_t \leq A} \eta C_t x_t^{q'} &\leq (1 + \delta)^{q'} (O(1) + \eta b A^q) \frac{1}{x_0^{q-q'-1}} \\ \sum_{t \geq 0, x_t \leq A} \eta C_t x_t^{q'} &\geq (1 + \delta)^{-q'} \left(\delta (1 + \delta)^{-1} \frac{1 - (1 + \delta)^{-b(q-q'-1)}}{1 - (1 + \delta)^{-(q-q'-1)}} - \eta b A^q \right) \frac{1}{x_0^{q-q'-1}} \end{aligned}$$

where $b = \Theta(\log(A/x_0)/\log(1 + \delta))$. When $A = x_0 d^{\Theta(1)}$, $\eta = o(\frac{1}{A^q \delta})$ and $q = O(1)$, then

$$\sum_{t \geq 0, x_t \leq A} \eta C_t x_t^{q'} = \Theta\left(\frac{1}{x_0^{q-q'-1}}\right)$$

Proof. For every $g \in 0, 1, \dots$, we define $\mathcal{T}_g := \min\{t : x_t \geq (1 + \delta)^g x_0\}$. and define $b := \min\{g : (1 + \delta)^g \geq A\}$, we can write down the following two inequalities according to the update of x_t :

$$\begin{aligned} \sum_{t \in [\mathcal{T}_g, \mathcal{T}_{g+1}]} \eta C_t [(1 + \delta)^g x_0]^q &\leq (1 + \delta)x_{\mathcal{T}_g} - x_{\mathcal{T}_g} + \eta A^q \leq \delta(1 + \delta)^g x_0 + \eta A^q \\ \sum_{t \in [\mathcal{T}_g, \mathcal{T}_{g+1}]} \eta C_t [(1 + \delta)^{g+1} x_0]^q &\geq (1 + \delta)x_{\mathcal{T}_g} - x_{\mathcal{T}_g} - \eta A^q \geq \delta(1 + \delta)^g x_0 - \eta A^q \end{aligned}$$

where $g + 1 \leq b$. Dividing both sides by $[(1 + \delta)^g x_0]^{q-q'}$ in the first inequality and $[(1 + \delta)^{g+1} x_0]^{q-q'}$ in the second, we have

$$\begin{aligned} \sum_{t \in [\mathcal{T}_g, \mathcal{T}_{g+1}]} \eta C_t [(1 + \delta)^g x_0]^{q'} &\leq \frac{\delta}{(1 + \delta)^{g(q-q'-1)}} \frac{1}{x_0^{q-q'-1}} + \frac{\eta A^q}{x_0^{q-q'-1}} \\ \sum_{t \in [\mathcal{T}_g, \mathcal{T}_{g+1}]} \eta C_t [(1 + \delta)^{g+1} x_0]^{q'} &\geq \frac{\delta(1 + \delta)^{-1}}{(1 + \delta)^{(g+1)(q-q'-1)}} \frac{1}{x_0^{q-q'-1}} - \frac{\eta A^q}{x_0^{q-q'-1}} \end{aligned}$$

Therefore if we sum over $g = 0, \dots, b$, then

$$\begin{aligned} \sum_{t \geq 0, x_t \leq A} \eta C_t x_t^{q'} &\leq \sum_{t \geq 0, x_t \leq A} \eta C_t [(1 + \delta)^{g+1} x_0]^{q'} \\ &= (1 + \delta)^{q'} \sum_{t \geq 0, x_t \leq A} \eta C_t [(1 + \delta)^g x_0]^{q'} \\ &\leq (1 + \delta)^{q'} \sum_{0 \leq g \leq b} \left(\frac{\delta}{(1 + \delta)^{g(q-q'-1)}} \frac{1}{x_0^{q-q'-1}} + \frac{\eta A^q}{x_0^{q-q'-1}} \right) \\ &= (1 + \delta)^{q'} O \left(\frac{\delta}{(1 + \delta)^{q-q'-1} - 1} + \eta b A^q \right) \frac{1}{x_0^{q-q'-1}} \\ &\leq (1 + \delta)^{q'} O \left(\frac{1}{q - q' - 1} + \eta b A^q \right) \frac{1}{x_0^{q-q'-1}} \end{aligned}$$

For the lower bound, we also have

$$\begin{aligned} \sum_{t \geq 0, x_t \leq A} \eta C_t x_t^{q'} &\geq (1 + \delta)^{-q'} \sum_{t \geq 0, x_t \leq A} \eta C_t [(1 + \delta)^{g+1} x_0]^{q'} \\ &\geq (1 + \delta)^{-q'} \sum_{0 \leq g \leq b} \left(\frac{\delta(1 + \delta)^{-1}}{(1 + \delta)^{(g+1)(q-q'-1)}} - \eta A^q \right) \frac{1}{x_0^{q-q'-1}} \\ &= (1 + \delta)^{-q'} \left(\delta(1 + \delta)^{-1} \frac{1 - (1 + \delta)^{-b(q-q'-1)}}{1 - (1 + \delta)^{-(q-q'-1)}} - \eta b A^q \right) \frac{1}{x_0^{q-q'-1}} \\ &= (1 + \delta)^{-q'} \left(\delta(1 + \delta)^{-1} \frac{1 - (1 + \delta)^{-b(q-q'-1)}}{1 - (1 + \delta)^{-(q-q'-1)}} - \eta b A^q \right) \frac{1}{x_0^{q-q'-1}} \end{aligned}$$

Inserting $b = \Theta(\log(A/x_0)/\log(1 + \delta))$ proves the lower bound. For the last one we can choose $\delta = \frac{1}{\sqrt{\log d}}$ to get:

$$b = \Theta(\text{polylog}(d)), \quad \frac{\delta(1 - (1 + \delta)^{-b(q-q'-1)})}{1 - (1 + \delta)^{-(q-q'-1)}} = \Omega(1), \quad (1 + \delta)^{-q'} = \Omega(1),$$

which proves the claim. □

Chapter 4

Self-Supervised Vision Transformers

4.1 Introduction

Self-supervised learning (SSL) has been a leading approach to pretrain neural networks for downstream applications since the introduction of BERT [74] and GPT [223] in natural language processing (NLP). On the other hand, in vision, self-supervised learning focused more on *discriminative* methods, which include contrastive learning (CL) [48, 113] and non-contrastive learning methods [40, 48, 106, 336]. Inspired by masked language models in NLP and the seminal work of vision transformers (ViTs) [75], *generative* approaches, such as masked reconstruction-based methods, have gained prominence in self-supervised vision pretraining. The masked autoencoders (MAE) [114] and SimMIM [319] have demonstrated the effectiveness of visual representation learning via reconstruction-based objectives.

Contrastive learning-like objectives promote instance discrimination among samples in the same batch of training. With suitable data augmentation, CL returns well-trained vision encoders like CLIP [224] and DINO [40] that can serve as backbones for state-of-the-art multimodal large language models (MLLMs) [279]. Masked reconstruction objectives (e.g., MAE), on the other hand, enforce neural networks to reconstruct some or all patches of an image given masked inputs. In practice, the MAE-like approach proves to have intriguing generalization properties that differ significantly from the behaviors of CL. The seminal work [114] showed that MAE can visibly conduct visual reasoning to fill missing patches even under very high masking rates. Some critical observations from recent research [212, 306, 320] provide comparative studies of these SSL approaches. They concluded that the ViTs trained via generative objectives display *diverse attention patterns*: different query patches pay attention to distinct local areas. This is in sharp contrast to the discriminative approaches, whose attention heads focus primarily on the most significant global pattern regardless of where the query patches are, as shown in Figure 4.1.

This chapter is adapted from the following joint work:

Yu Huang, Zixin Wen, Yuejie Chi, and Yingbin Liang. *A Theoretical Analysis of Self-Supervised Learning for Vision Transformers*.

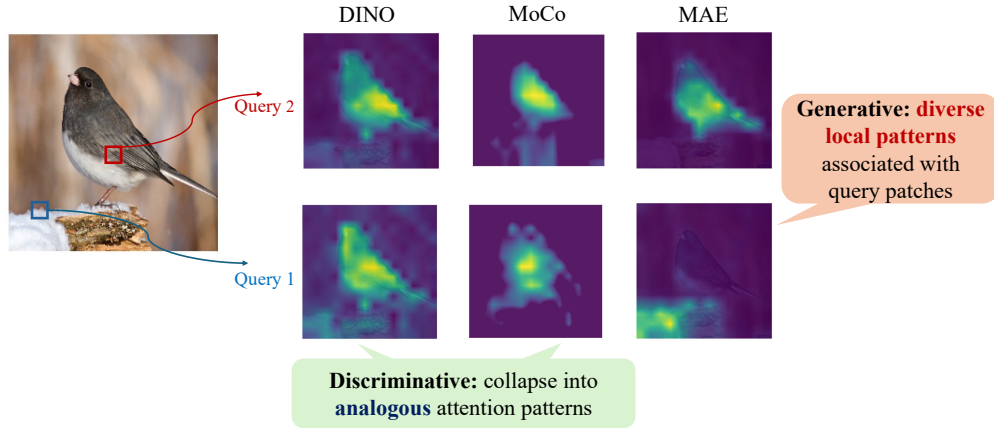


Figure 4.1: Attention Maps Across Learning Objectives. Visualization of attention maps in the last layer of the ViTs for query patches from two different spatial locations, similar to those presented in [212]. The ViTs were trained by the generative self-supervised learning approach of masked reconstruction (MAE) and discriminative methods: DINO [40] and MoCo [55].

These empirical observations motivate the question: from a *theoretical* standpoint, how do ViTs pick up these observed attention patterns during the training process, respectively for different SSL methods?

Despite extensive empirical efforts of studying SSL in vision pretraining, its theoretical understanding is still nascent. Most existing theories of SSL focused on the discriminative approach [20, 50, 112, 238, 276, 296, 311, 312], especially (non-)contrastive learning. There are also a few attempts towards understanding methods using the generative approach like masked reconstructions [38, 112, 208, 340], which mainly adapt the theories developed for CL to their context. In fact, there are two major limitations of these prior works: *i) Transformers*, as the dominant architecture in practice, were not studied in the aforementioned works of self-supervised learning and *ii) there still lacks a suitable theoretical framework that can provide convincing explanations for the empirical findings in [212, 320]*, especially on the difference of the attention patterns learned by different approaches of SSL. The above limitations highlight a significant gap in the literature on SSL for vision pretraining.¹

Motivated by the limited theoretical characterization of SSL for vision with transformers, especially in comparing CL and masked reconstruction objectives, we aim to address the following research questions:

Our Research Questions

Can we *theoretically* characterize the solutions that ViTs converge to in these two mainstream self-supervised learning approaches? How do differences in attention patterns emerge during their respective training processes?

¹ More detailed discussions for related work can be found in Section 4.6.

Contributions. In this paper, we take a step toward answering the above questions. We study the training process of one-layer softmax-based ViTs via gradient descent (GD) for both masked reconstruction and contrastive learning, focusing on *spatially structured data distributions* generalized from supervised learning settings [135]. In our setting, each image is sampled from distinct clusters characterized by unique patch-wise feature associations. Each cluster contains two types of features: a large portion of patches reside in a global area with shared global features, while relatively few patches occupy the remaining local areas with their own local features. We measure the imbalance of feature distribution by a condition called the *information gap* Δ , which is formally defined in Eq. (4.4.1). Under such a setting:

1. We provide global convergence guarantees for training ViTs on both the MAE and the CL loss functions. To the best of our knowledge, this is the first end-to-end guarantee for learning ViTs with self-supervised learning objectives;
2. We provide a comprehensive characterization of the training dynamics of *attention correlations* (see Definition 4.3.1) to illustrate the attention patterns to which ViTs converge:
 - MAE provably learns *diverse* attention patterns, with each patch concentrating its attention on its designated area based on its position, even under a substantial information gap Δ ;
 - CL primarily learns a *global* attention pattern, causing all patches to focus on the global area regardless of their locations, even with a minor information gap Δ .

These qualitative differences in the solutions learned by the two SSL methods provide strong theoretical support for the empirical behavior gaps observed in [212, 320], and highlight the theoretical advantage of MAE in handling highly imbalanced data structures.

Notation. We introduce a few notation to be used throughout the paper. For any two functions $h(x)$ and $g(x)$, we use $h(x) = \Omega(g(x))$ (resp. $h(x) = O(g(x))$) to denote that there exist some universal constants $C_1 > 0$ and a_1 , s.t. $|h(x)| \geq C_1|g(x)|$ (resp. $|h(x)| \leq C_1|g(x)|$) for all $x \geq a_1$; Furthermore, $h(x) = \Theta(g(x))$ indicates $h(x) = \Omega(g(x))$ and $h(x) = O(g(x))$ hold simultaneously. We use $\mathbb{1}\{\cdot\}$ to denote the indicator function, and let $[N] := \{1, 2, \dots, N\}$. We use $\tilde{O}$, $\tilde{\Omega}$, and $\tilde{\Theta}$ to further hide logarithmic factors in the respective notation. We use $\text{poly}(P)$ and $\text{polylog}(P)$ to represent large constant-degree polynomials of P and $\log(P)$, respectively.

4.2 Problem Setup

In this section, we present our problem formulations for studying the training process of ViTs in self-supervised pretraining. We begin with some background information, followed by a description of our data distribution. We then detail the pretraining strategies using MAE and CL, respectively, with the specific transformer architecture considered in this paper.

4.2.1 Background on self-supervised learning

Masked reconstruction-based learning. We follow the masked reconstruction frameworks in [114, 319]. Each data sample $X \in \mathbb{R}^{d \times P}$ has the form $X = (X_p)_{p \in \mathcal{P}}$, which has $|\mathcal{P}| = P$ patches, and each patch $X_p \in \mathbb{R}^d$. Given a collection of images $\{X_i\}_{i \in [n]}$, we select a masking set $\mathcal{M}_i \subset \mathcal{P}$ for each image X_i , and mask these patches to a uniform value $M \in \mathbb{R}^d$. The resulting masked images $\{M(X_i)\}_{i \in [n]}$ are given by

$$M(X_i)_p = \begin{cases} [X_i]_p & p \in \mathcal{U}_i \\ M & p \in \mathcal{M}_i \end{cases}, \quad i \in [n], \quad (4.2.1)$$

where $\mathcal{U}_i = \mathcal{P} \setminus \mathcal{M}_i$ is the index set of unmasked patches. Let $F : X \mapsto \widehat{X}$ be an architecture that outputs a reconstructed image $\widehat{X} \in \mathbb{R}^{d \times P}$ for any given input $X \in \mathbb{R}^{d \times P}$. The pretraining objective is then defined as the mean-squared reconstruction loss over a series of subsets $\mathcal{P}'_i \subset \mathcal{P}$ of the image as follows:

$$\mathcal{L}_{\text{masked}}(F) = \frac{1}{n} \sum_{i=1}^n \sum_{p \in \mathcal{P}'_i} \left\| [X_i]_p - [F(M(X_i))]_p \right\|_2^2. \quad (4.2.2)$$

MAE [114] chose the subset $\mathcal{P}'_i$ as the set of masked patches $\mathcal{M}_i$, whereas SimMIM [319] aimed to reconstruct the full image $\mathcal{P}'_i = \mathcal{P}$. We do not explore the trade-offs between these two approaches in our study.

Contrastive learning. Contrastive learning [48] aims to learn meaningful representations F by distinguishing between similar and dissimilar data points. For a given batch $\{X_i\}_{i \in [n]}$, we generate a positive pair $(X_i^{(1)}, X_i^{(2)})$ for each i by applying random augmentations to X_i . Negative pairs $(X_i^{(1)}, X_j^{(2)})$ for $j \neq i$ are formed from different data points. The model F is trained to minimize the following contrastive loss:

$$\mathcal{L}_{\text{contrastive}}(F) = \frac{1}{n} \sum_{i=1}^n \left[-\tau \log \left(\frac{e^{\text{Sim}_F(X_i^{(1)}, X_i^{(2)})/\tau}}{\sum_{j \in [n]} e^{\text{Sim}_F(X_i^{(1)}, X_j^{(2)})/\tau}} \right) \right], \quad (4.2.3)$$

where Sim_F measures the similarity between two representations, and τ is a temperature parameter controlling the sharpness of the distribution.

4.2.2 Data distribution

We assume the data samples $X \in \mathbb{R}^{d \times P}$ are drawn independently based on some data distribution $\mathcal{D}$. To capture the *feature-position (FP) correlation* in the learning problem, we consider the following setup for vision data. We assume that the data distribution consists of many different clusters, where each cluster captures a distinct spatial pattern, and hence is defined by a different partition of patches with a different set of visual features. We define the data distribution $\mathcal{D}$ formally as follows. An intuitive illustration of data generation is given in Figure 4.2.

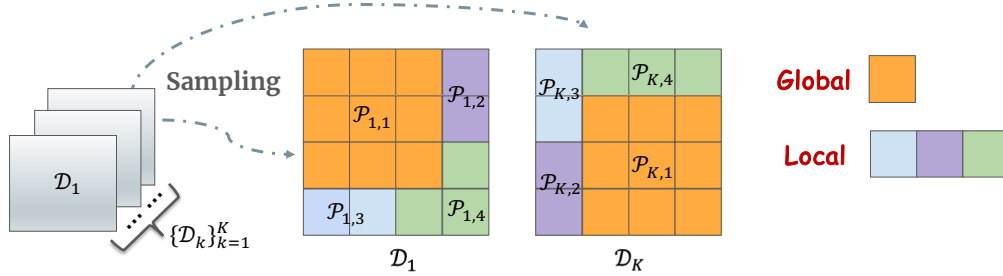


Figure 4.2: Patch-Cluster Data Model. Illustration of our data distribution (see Definition 4.2.1). Each cluster $\mathcal{D}_k$ is segmented into distinct areas $\mathcal{P}_{k,j}$, with squares in the same color representing the same area $\mathcal{P}_{k,j}$. The global area $\mathcal{P}_{k,1}$ (depicted in orange) contains a larger count of patches compared to any other local areas. It is important to note that while we use spatially contiguous partitions for clarity in this illustration, our data model is also applicable to non-contiguous cases.

Definition 4.2.1 (Data distribution $\mathcal{D}$). The data distribution $\mathcal{D}$ has $K = O(\text{polylog}(P))$ different clusters $\{\mathcal{D}_k\}_{k=1}^K$. For every cluster $\mathcal{D}_k$, $k \in [K]$, there is a corresponding partition of $\mathcal{P}$ into N_k disjoint subsets $\mathcal{P} = \bigcup_{j=1}^{N_k} \mathcal{P}_{k,j}$ which we call *areas*. For each sample $X = (X_{\mathbf{p}})_{\mathbf{p} \in \mathcal{P}}$, its sampling process is as follows:

- We draw $\mathcal{D}_k$ uniformly at random from all clusters and draw a sample X from $\mathcal{D}_k$.
- Given $k \in [K]$, for any $j \in [N_k]$, all patches $X_{\mathbf{p}}$ in the area $\mathcal{P}_{k,j}$ are given the same content $X_{\mathbf{p}} = v_{k,j} z_j(X)$, where $v_{k,j} \in \mathbb{R}^d$ is the *visual* feature and $z_j(X)$ is the latent variable. We assume $\bigcup_{k=1}^K \bigcup_{j=1}^{N_k} \{v_{k,j}\}$ are orthogonal to each other with unit norm.
- Given $k \in [K]$, for any $j \in [N_k]$, $z_j(X) \in [L, U]$, where $0 \leq L < U$ are on the order of $\Theta(1)$.²

Area-wide patterns and features. Image data naturally contains two types of features: the global features and the local features. For instance, in an image of an object, global features can capture the shape and texture of the object, such as the fur color of an animal, whereas local features describe specific details of local areas, such as the texture of leaves in the background. Recent empirical studies on self-supervised pretraining with ViTs [212, 306] and observations in Figure 4.1 collectively show that masked pretraining exhibits the capacity to avoid attention collapse concentrating towards those global shapes by identifying diverse local attention patterns. Consequently, unraveling their mechanisms necessitates a thorough examination of data characteristics that embody both global and local features. In this paper, we characterize these two types of features by the following assumption on the data.

² The distribution of $z_j(X)$ can be arbitrary within the above support set.

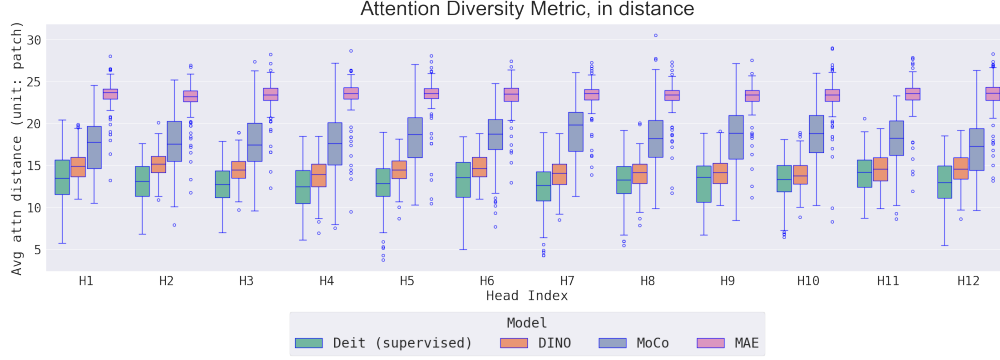


Figure 4.3: Attention Diversity Across Training Methods. *Attention Diversity Metric:* We design a novel empirical metric, the *attention diversity metric*, to probe the last layer of ViTs trained by masked reconstructions (MAE), CL (MoCo), another discriminative SSL approach (DINO), and supervised learning (DeiT). Lower values of this metric signify focused attention on a similar area across different patches, reflecting a global pattern of focus. Conversely, higher values suggest that attention is dispersed, focusing on different, localized areas. The results show that the MAE model excels in capturing *diverse local patterns* compared to discriminative methods like CL. (see Section 4.7 for details).

Assumption 4.2.1 (Global feature vs local feature). Let $\mathcal{D}_k$ with $k \in [K]$ be a cluster from $\mathcal{D}$. We let $\mathcal{P}_{k,1}$ be the *global area* of cluster $\mathcal{D}_k$, and all the other areas $\mathcal{P}_{k,j}, j \in [N_k] \setminus \{1\}$ be the *local areas*. Since each area corresponds to an assigned feature, we also call them the *global* and *local* features, respectively. Moreover, we assume:

- Global area: given $k \in [K]$, we set $C_{k,1} = |\mathcal{P}_{k,1}| = \Theta(P^{\kappa_c})$ with $\kappa_c \in [0.5005, 1]$, where $C_{k,1}$ is the number of patches in the global area $\mathcal{P}_{k,1}$.
- Local area: given $k \in [K]$, we choose $C_{k,j} = |\mathcal{P}_{k,j}| = \Theta(P^{\kappa_s})$ with $\kappa_s \in [0.001, 0.5]$ for $j > 1$, where $C_{k,j}$ denotes the number of patches in the local area $\mathcal{P}_{k,j}$.

The rationale for defining the global feature in this manner stems from observing that patches representing global features ($C_{k,1}$) typically occur more frequently than those representing local features ($C_{k,j}$, for $j > 1$), since global features capture the primary visual information of an image, offering a dominant view, while local features focus on subtler details within the image. Our empirical observations (see Figure 4.3) further substantiate the significance of distinguishing between global and local patterns in data distributions, which is essential for elucidating the distinct behaviors exhibited by MAE and CL.

4.2.3 Masked reconstruction with transformers

Transformer architecture. A transformer block [75, 288] consists of a self-attention layer followed by an MLP layer. The self-attention layer has multiple heads, each of which consists of the following components: a query matrix W^Q , a key matrix W^K , and a value matrix W^V . Given an input X , the output of one head in the self-attention layer can be described by the following

mapping:

$$G(X; W^Q, W^K, W^V) = \text{softmax}((W^Q X)^\top W^K X) \cdot (W^V X)^\top, \quad (4.2.4)$$

where the $\text{softmax}(\cdot)$ function is applied row-wisely and for a vector input $z \in \mathbb{R}^P$, the i -th entry of $\text{softmax}(z)$ is given by $\frac{\exp(z_i)}{\sum_{s=1}^P \exp(z_s)}$.

To simplify the theoretical analysis, we consolidate the product of query and key matrices $(W^Q)^\top W^K$ into one weight matrix denoted as Q . Furthermore, we set W^V to be the identity matrix and fixed during the training. These simplifications are often taken in recent theoretical works [128, 135, 341] in order to allow tractable analysis. With these simplifications in place, Eq. (4.2.4) can be rewritten as

$$G(X; Q) = \text{softmax}(X^\top Q X) \cdot X^\top. \quad (4.2.5)$$

Note that the input tokens in transformers are indistinguishable without explicit spatial information. Therefore, positional encodings should be added to the input embeddings to retain this crucial positional context as in practices [75, 114]. Our assumptions regarding the positional encodings are as follows:

Assumption 4.2.2 (Positional encoding). We assume fixed positional encodings, which is consistent with the implementation in MAE [114]: $E = (e_p)_{p \in \mathcal{P}} \in \mathbb{R}^{d \times P}$ where positional embedding vectors e_p are orthogonal to each other and to all the features $v_{k,j}$, and are of unit-norm.

We now include positional embeddings in Eq. (4.2.5) and introduce the network architecture for masked reconstruction used in this study.

Definition 4.2.2 (ViT architecture for MAE). We assume that our vision transformer $F^{\text{mae}}(X; Q)$ consists of a single-head self-attention layer with an attention weight matrix $Q \in \mathbb{R}^{d \times d}$. For an input image $X \sim \mathcal{D}$, we add positional encoding by letting $\tilde{X} = X + E$. The attention score from patch X_p to patch X_q is denoted by

$$\text{attn}_{p \rightarrow q}^m(X; Q) := \frac{e_p^\top Q \tilde{x}_q}{\sum_{r \in \mathcal{P}} e_p^\top Q \tilde{x}_r}, \quad \text{for } p, q \in \mathcal{P}. \quad (4.2.6)$$

The output of the transformer is given by

$$[F^{\text{mae}}(X; Q)]_p = \sum_{q \in \mathcal{P}} \text{attn}_{p \rightarrow q}^m(X; Q) \cdot X_q, \quad \text{for } p \in \mathcal{P}. \quad (4.2.7)$$

Then we formally define the masking operation and the objective for our masked pretraining task.

Definition 4.2.3 (Random masking). Let $M(X) \rightarrow \mathbb{R}^{d \times P}$ denote the random masking operation, which randomly selects (without replacement) a subset of patches $\mathcal{M}$ in X with a masking ratio $\gamma = \Theta(1) \in (0, 1)$ and masks them to be $M := \mathbf{0} \in \mathbb{R}^d$. The masked samples obey Eq. (4.2.1).

MAE objective. To train the model $F^{\text{mae}}(M(X); Q)$, following the methodology described in MAE practice [114], we minimize the squared reconstruction error in Eq. (4.2.2) only on masked patches, where $M(X)$ follows Definition 4.2.3. The training objective thus can be written as

$$\mathcal{L}_{\text{mae}}(Q) := \frac{1}{2} \mathbb{E} \left[\sum_{\mathbf{p} \in \mathcal{P}} \mathbf{1}\{\mathbf{p} \in \mathcal{M}\} \left\| [F^{\text{mae}}(M(X); Q)]_{\mathbf{p}} - X_{\mathbf{p}} \right\|^2 \right], \quad (4.2.8)$$

where the expectation is with respect to both the data distribution and the masking. Note that our objective remains highly nonconvex with the model defined in Definition 4.2.2.

Training algorithm. The learning objective in Eq. (4.2.8) is minimized via GD with learning rate $\eta > 0$. At $t = 0$, we initialize $Q^{(0)} := \mathbf{0}_{d \times d}$ as the zero matrix. The parameter is updated as follows:

$$Q^{(t+1)} = Q^{(t)} - \eta \nabla_Q \mathcal{L}_{\text{mae}}(Q^{(t)}).$$

Note that the initialization of $Q^{(0)}$ results in any query patch uniformly attending to all patches.

4.2.4 Contrastive learning with transformers

The transformer architecture used for CL is similar to that of MAE, but with a minor modification to accommodate contrastive loss, as outlined below.

Definition 4.2.4 (ViT architecture for CL). We consider a vision transformer $F^{\text{cl}}(X; Q)$ consisting of a single-head self-attention layer with an attention weight matrix $Q \in \mathbb{R}^{d \times d}$. For an input image X , the attention score from patch $X_{\mathbf{p}}$ to patch $X_{\mathbf{q}}$ is denoted by

$$\text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{\text{c}}(X; Q) := \frac{e^{\mathbf{e}_{\mathbf{p}}^{\top} Q X_{\mathbf{q}}}}{\sum_{\mathbf{r} \in \mathcal{P}} e^{\mathbf{e}_{\mathbf{p}}^{\top} Q X_{\mathbf{r}}}}, \quad \text{for } \mathbf{p}, \mathbf{q} \in \mathcal{P}. \quad (4.2.9)$$

The output of the transformer is then computed as

$$F^{\text{cl}}(X; Q) = \frac{1}{P} \sum_{\mathbf{p}, \mathbf{q} \in \mathcal{P}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{\text{c}}(X; Q) \cdot X_{\mathbf{q}} \in \mathbb{R}^d, \quad (4.2.10)$$

which represents the average pooling of all the patches.

The key distinction is that we separate the positional and patch embeddings within the attention mechanism for technical simplicity. However, it is important to emphasize that these two types of embeddings remain coupled for attention calculations.

Definition 4.2.5 (Data augmentation). For a sample $X \in \mathbb{R}^d$, we generate two new samples X^+ and X^{++} by independently applying random masking as in Definition 4.2.3 with a ratio $\gamma_0 = \Theta(1)$, similar to the crop-resize operations used in practice. The unmasked sets for them are denoted as $\mathcal{U}^+$ and $\mathcal{U}^{++}$.

CL objective. Given a sample X , we first generate a pair of positive samples $\{X^+, X^{++}\}$ via Definition 4.2.5. Then we generate a batch of i.i.d. negative samples $\mathfrak{N} = \{X^{-,s}\}_{s \in [N_c]}$. Denoting $\mathfrak{B} = \mathfrak{N} \cup \{X^{++}\}$, we minimize the expected contrastive loss in Eq. (4.2.3) with ℓ_2 -regularization:

$$\mathcal{L}_{\text{cl}}(Q) := \mathbb{E}_{X^+, X^{++}, \mathfrak{N}} \left[-\tau \log \left(\frac{e^{\text{Sim}_{\text{Fcl}}(X^+, X^{++})/\tau}}{\sum_{X' \in \mathfrak{B}} e^{\text{Sim}_{\text{Fcl}}(X^+, X')/\tau}} \right) \right] + \frac{\lambda}{2} \|Q\|_F^2, \quad (4.2.11)$$

where $\|\cdot\|_F$ denotes the Frobenius norm, $\lambda > 0$ is the regularization parameter, and the similarity of the representations of X and X' obtained by $F^{\text{cl}}(\cdot; Q)$ is defined as

$$\text{Sim}_{\text{Fcl}}(X, X') := \langle F^{\text{cl}}(X; Q), \text{StopGrad}(F^{\text{cl}}(X'; Q)) \rangle.$$

The $\text{StopGrad}(\cdot)$ operator ensures that no gradient is computed for this term. Additionally, no augmentation is applied to the negative samples. Both practices are standard in the literature on the theory of contrastive learning [311, 312]. Similar to MAE, we update Q by GD with zero-initialization:

$$Q^{(t+1)} = Q^{(t)} - \eta \nabla_Q \mathcal{L}_{\text{cl}}(Q^{(t)}). \quad (4.2.12)$$

In the following, any variable with a superscript $^{(t)}$ represents that variable at the t -th step of training.

4.3 Warmup of Attention Patterns

To show the significance of the data distribution design and understand the nature of our self-supervised learning tasks, in this section, we will provide some preliminary implications of the spatial structures in Definition 4.2.1. Intuitively, for MAE, for a given cluster $\mathcal{D}_k$, to reconstruct a missing patch $\mathbf{p} \in \mathcal{P}_{k,j} \cap \mathcal{M}$, the attention head should exploit all *unmasked* patches in the *target* area $\mathcal{P}_{k,j}$ to find the same visual feature $v_{k,j}$ to fill in the blank, which emphasizes the *locality* for $\mathbf{p}$ in different areas. However, CL focuses on any discriminative patterns regardless of the location of $\mathbf{p}$, which can align positive pairs but may lead to collapsed attention patterns. We will

elaborate on these points by describing the *area attentions* and illustrating the intuition about how they can be learned via *attention correlations* (Definition 4.3.1).

Area attention. We first define a new notation for a cleaner presentation. For $X \sim \mathcal{D}$ and $\mathbf{p} \in \mathcal{P}$, we write the attention of patch $X_{\mathbf{p}}$ to a subset $\mathcal{A} \subset \mathcal{P}$ of patches by

$$\widetilde{\text{Attn}}_{\mathbf{p} \rightarrow \mathcal{A}}^{\dagger}(X; Q) := \sum_{\mathbf{q} \in \mathcal{A}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{\dagger}(X; Q), \quad \text{for } \dagger \in \{\mathbf{m}, \mathbf{c}\}.$$

MAE’s ability to learn locality with ViTs. Let us first explain why the above notion of area attention matters in understanding how attention works in masked reconstruction. Suppose we have a sample X picked from $\mathcal{D}_k$, and the patch $X_{\mathbf{p}}$ with $\mathbf{p} \in \mathcal{P}_{k,j}$ is masked. Then the prediction of $X_{\mathbf{p}}$ given masked input $M(X)$ can be written as

$$\begin{aligned} [F^{\text{mae}}(M(X); Q)]_{\mathbf{p}} &= \sum_{\mathbf{q} \in \mathcal{P}} M(X)_{\mathbf{q}} \cdot \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{\mathbf{m}}(M(X); Q) \\ &= \sum_{i \in [N_k]} z_i(X) v_{k,i} \cdot \widetilde{\text{Attn}}_{\mathbf{p} \rightarrow \mathcal{U} \cap \mathcal{P}_{k,i}}^{\mathbf{m}}(M(X); Q) \quad (\text{since } M(X)_{\mathbf{q}} = \mathbf{0} \text{ if } \mathbf{q} \in \mathcal{M}). \end{aligned}$$

To reconstruct the original patch $X_{\mathbf{p}}$, the transformer should not only focus on the correct area $\mathcal{P}_{k,j}$, but must also prioritize attention to the *unmasked* patches within this area. This specificity is denoted by the area attention $\widetilde{\text{Attn}}_{\mathbf{p} \rightarrow \mathcal{U} \cap \mathcal{P}_{k,j}}^{\mathbf{m}}$ over $\mathcal{U} \cap \mathcal{P}_{k,j}$, a requirement imposed by masking operations. We refer to these location-dependent attention patterns as *locality*.

To further explain how ViTs perform such prioritization, we introduce the following quantities, which capture the major insights of our analysis to distinguish between MAE and contrastive learning.

Definition 4.3.1. (Attention correlations) Let $\mathbf{p} \in \mathcal{P}$, and we define attention correlations as:

1. Feature-Position (FP) Correlation: $\Phi_{\mathbf{p} \rightarrow v_{k,m}} := e_{\mathbf{p}}^{\top} Q v_{k,m}$, for $k \in [K]$ and $m \in [N_k]$;
2. Position-Position (PP) Correlation: $\Upsilon_{\mathbf{p} \rightarrow \mathbf{q}} := e_{\mathbf{p}}^{\top} Q e_{\mathbf{q}}$, $\forall \mathbf{q} \in \mathcal{P}$.

Due to our (zero) initialization of $Q^{(0)}$, we have $\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(0)} = \Upsilon_{\mathbf{p} \rightarrow \mathbf{q}}^{(0)} = 0$.

These two types of attention correlations, FP correlation $\Phi_{\mathbf{p} \rightarrow v_{k,m}}$ and PP correlation $\Upsilon_{\mathbf{p} \rightarrow \mathbf{q}}$, act as the exponent terms within the softmax calculations for attention scores. Given $\mathbf{p} \in \mathcal{P}_{k,j}$ is masked, the (unnormalized) attention $\text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{\mathbf{m}}$ directed towards an *unmasked* patch $\mathbf{q}$ is influenced jointly by these correlations. Hence, the described attention pattern for MAE can emerge from either a substantial FP correlation $\Phi_{\mathbf{p} \rightarrow v_{k,j}}$ or a significant PP correlation $\Upsilon_{\mathbf{p} \rightarrow \mathbf{q}}$ for $\mathbf{q}$ in the same area as $\mathbf{p}$. However, in our setting, the latter mechanism—learning via PP correlation—fails to produce desired attention patterns: *i).* such a mechanism inadvertently directs attention to the *masked* patches, which is not desirable; *ii).* such position association could be vulnerable to the variation across different clusters, i.e., $\mathbf{p}, \mathbf{q} \in \mathcal{P}_{k,j}$ does not necessarily hold for all $k \in [K]$. This also highlights that prior work [135] that relied solely on pure positional attention cannot fully explain the ViTs’ ability to learn locality when the patch-wise associations are not fixed.

Why CL may fail to learn the locality. Now turning to CL, for $X \in \mathcal{D}_k$, we have the following form of similarity between the positive pair:

$$\begin{aligned} & \langle F^{\text{cl}}(X^+; Q), F^{\text{cl}}(X^{++}; Q) \rangle \\ &= \frac{1}{p^2} \sum_{\mathbf{p}, \mathbf{p}' \in \mathcal{P}} \sum_{i=1}^{N_k} \widetilde{\text{Attn}}_{\mathbf{p} \rightarrow \mathcal{U}^+ \cap \mathcal{P}_{k,i}}^{\text{c}}(X^+; Q) \widetilde{\text{Attn}}_{\mathbf{p}' \rightarrow \mathcal{U}^{++} \cap \mathcal{P}_{k,i}}^{\text{c}}(X^{++}; Q). \end{aligned}$$

Thus, to align the positive representations effectively, the optimal strategy is also to direct attention toward a specific area for each patch $\mathbf{p}$, i.e., greedily ensuring that only one area attention $\widetilde{\text{Attn}}_{\mathbf{p} \rightarrow \mathcal{U}^+ \cap \mathcal{P}_{k,i}}^{\text{c}}$ is activated for some $i \in [N_k]$. However, the above expression suggests that the selected area by the optimal strategy may not necessarily depend on the location $\mathbf{p}$, which could lead to a collapsed attention scenario where all patches focus on the same area. Regarding attention correlations, the attention mechanism defined in Eq. (4.2.9) requires us to handle only the FP correlations among different features for CL. Theorem 4.4.3 in the next section confirms that a collapsed solution indeed occurs: ViTs trained with CL concentrate attention on the global area across all patches by exclusively capturing global FP correlations across all patches, i.e., $\Phi_{\mathbf{p} \rightarrow v_{k,1}}$ becomes large for all $\mathbf{p} \in \mathcal{P}$.

4.4 Statements of Main Results

In this section, we present our main theorems on the learning processes of ViTs in MAE and CL. We begin by introducing notations that will be used in theorem presentations.

Information gap and a technical condition. Based on our data model in Section 4.2.2, we introduce a notion of *information gap* to quantify the degree of imbalance between global and local areas (cf. Assumption 4.2.1). Denoted as Δ , the information gap is defined as follows:

$$\Delta := (1 - \kappa_s) - 2(1 - \kappa_c). \quad (4.4.1)$$

Broadly speaking, a larger Δ means that the number of global features is much greater than local ones, indicating a significant imbalance. In contrast, a smaller value reflects only a slight imbalance.³

Unmasked area attention. Based on the crucial role of those unmasked patches for both reconstruction task and positive contrastive pairs, we further define the *unmasked area attention* as follows:

$$\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{\dagger}(X; Q) := \widetilde{\text{Attn}}_{\mathbf{p} \rightarrow \mathcal{U} \cap \mathcal{P}_{k,m}}^{\dagger}(X; Q), \text{ for } \dagger \in \{\mathfrak{m}, \text{c}\}.$$

³ Our study focuses on the regime where Δ is not too close to zero, i.e., $|\Delta| = \Omega(1)$, which allows for cleaner induction arguments. This condition could be potentially relaxed via more involved analysis.

4.4.1 MAE learns diverse attention patterns

Our results are structured into two parts: *i*). analysis of convergence (Theorem 4.4.1), which includes the global convergence guarantee of the masked reconstruction loss and characterization of the attention pattern at the end of training to demonstrate the diverse locality; *ii*). learning dynamics of attention correlations (Theorem 4.4.2), which shows how transformers capture target FP correlations while downplaying PP correlations as discussed in Section 4.3.

To properly evaluate the reconstruction performance, we further introduce the following notion of the reconstruction loss with respect to a specific patch $\mathbf{p} \in \mathcal{P}$:

$$\mathcal{L}_{\text{mae},\mathbf{p}}(Q) = \frac{1}{2} \mathbb{E} \left[\mathbb{1}\{\mathbf{p} \in \mathcal{M}\} \left\| [F^{\text{mae}}(\mathbf{M}(X); Q)]_{\mathbf{p}} - X_{\mathbf{p}} \right\|^2 \right]. \quad (4.4.2)$$

Now we present our first main result regarding the convergence of MAE.

Theorem 4.4.1 (Training convergence). *Suppose the information gap $\Delta \in [-0.5, -\Omega(1)] \cup [\Omega(1), 1]$. For any $0 < \epsilon < 1$, suppose $\text{polylog}(P) \gg \log(\frac{1}{\epsilon})$. We train the ViTs in Definition 4.2.2 by GD to minimize reconstruction loss in Eq. (4.2.8) with $\eta \ll \text{poly}(P)$. Then for each patch $\mathbf{p} \in \mathcal{P}$, we have*

1. *Loss converges: $\mathcal{L}_{\text{mae},\mathbf{p}}(Q^{(T^*)}) - \mathcal{L}_{\text{mae},\mathbf{p}}^* \leq \epsilon$ in $T^* = O\left(\frac{1}{\eta} \log(P) P^{\max\{2(\frac{U}{L}-1), 1\}(1-\kappa_s)} + \frac{1}{\eta\epsilon} \log\left(\frac{P}{\epsilon}\right)\right)$ iterations, where $\mathcal{L}_{\text{mae},\mathbf{p}}^*$ is the global minimum of the patch-level reconstruction loss in (4.4.2).*
2. *Area-wide pattern of attention: given cluster $k \in [K]$, and $\mathbf{p} \in \mathcal{P}_{k,j}$ for some $j \in [N_k]$, if $X_{\mathbf{p}}$ is masked, then the one-layer transformer nearly “pays all attention” to all unmasked patches in the same area $\mathcal{P}_{k,j}$ as $\mathbf{p}$, i.e.,*

$$\left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,j}}^m(X; Q^{(T^*)})\right)^2 \leq O(\epsilon).$$

Theorem 4.4.1 indicates that, at the time of convergence, for any masked query patch $X_{\mathbf{p}}$ in the k -th cluster, the transformer exhibits an *area-wide* pattern of attention, concentrating on those unmasked patches within the area that $\mathbf{p}$ lies in, as demonstrated in Section 4.3. The location of the patch determines such area-wide attention and can be achieved no matter if $\mathbf{p}$ belongs to the global or local areas, which jointly highlight the *diverse local patterns* for masked vision pretraining no matter degree of the imbalance.

Next, we detail the training phases of attention correlations in the following theorem, which explicitly confirms that the model *learns* target FP correlations while *ignoring* PP correlations to achieve the desirable area-wide attention patterns as suggested in Section 4.3 (illustrated in Figure 4.4).

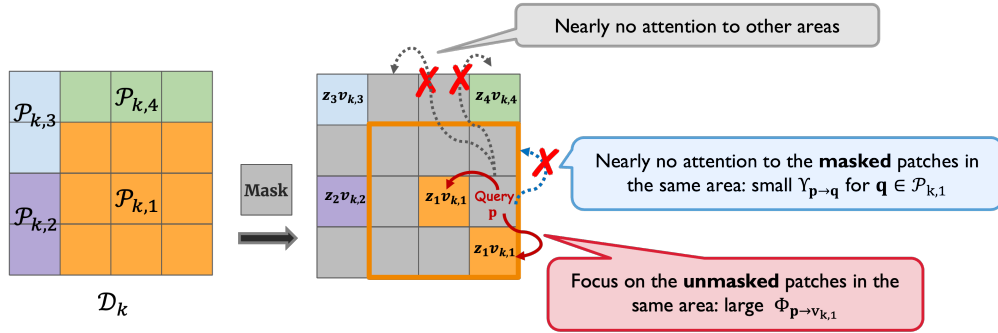


Figure 4.4: Attention-Correlation Mechanism In MAE. The mechanism of how the masked patch attends to other patches via attention correlations in MAE.

Theorem 4.4.2 (Learning Feature-Position correlations). *Following the same assumptions in Theorem 4.4.1, for $\mathbf{p} \in \mathcal{P}$, given $k \in [K]$, if $\mathbf{p} \in \mathcal{P}_{k,j}$ for some $j \in [N_k]$, we have*

For positive information gap $\Delta \in [\Omega(1), 1]$:

- Global areas ($j = 1$) learn FP correlation in one-phase: $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}$ monotonically increases to $O(\log(P/\epsilon))$ throughout the training, with all other attention correlations remain close to 0.
- Local areas ($j > 1$) learn FP correlation in two-phase: In phase one, FP correlation $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}$ between local area and the global area feature quickly decreases to $-\Theta(\log(P))$ whereas all other attention correlation stay close to zero; In phase two, FP correlation $\Phi_{\mathbf{p} \rightarrow v_{k,j}}^{(t)}$ for the target local area starts to grow until convergence with all other attention correlations nearly unchanged.

For negative information gap $\Delta \in [-0.5, -\Omega(1)]$:

- All areas learn FP correlation through one-phase: $\Phi_{\mathbf{p} \rightarrow v_{k,j}}^{(t)}$ monotonically increases to $O(\log(P/\epsilon))$ throughout the training, with all other attention correlations remain close to 0.

The training dynamics are different depending on whether Δ is positive or negative, and further vary for positive Δ depending on whether X_p is situated in global or local areas. Typically, the target FP correlations are learned directly in a single phase. However, for a positive information gap Δ , when patch $\mathbf{p}$ is located in a local area, the learning process contains an additional decoupling phase, to reduce the FP correlation with the non-target global features.

4.4.2 Contrastive learning collapses to global attention patterns

In contrast to MAE's ability to learn diverse local features regardless of the information gap, our results in this section demonstrate that CL inevitably collapses to global attention patterns by solely learning global FP correlations, even under a slight structural imbalance. To prevent trivial solutions in CL, we adopt a noisy variant of the data distribution.

Assumption 4.4.1 (Noisy data). We assume that the data used for contrastive learning is sampled from $\mathcal{D}^{\text{cl}}$. Specifically, to generate a sample $X \sim \mathcal{D}^{\text{cl}}$, we first draw $Z \sim \mathcal{D}$, then add independent and identically distributed (i.i.d.) noise $\zeta_{\mathbf{p}} \sim \mathcal{N}(0, \sigma_0^2 I_d)$ to each patch $Z_{\mathbf{p}}$. The resulting sample is defined as $X_{\mathbf{p}} = Z_{\mathbf{p}} + \zeta_{\mathbf{p}}$. We denote $X \in \mathcal{D}_k^{\text{cl}}$ if $Z \in \mathcal{D}_k$.

Theorem 4.4.3 (Learning with contrastive objective). Suppose the information gap $\Delta \in [-0.5, -\Omega(1)] \cup [\Omega(1), 1]$. We train the ViTs in Definition 4.2.4 by GD to minimize Eq. (4.2.11) with $\eta \ll \text{poly}(P)$, $\sigma_0^2 = \frac{1}{d}$, $\tau = O(\frac{1}{\log d})$. Then after $T^* = O(\frac{\text{poly}(P) \log P}{\eta})$ iterations, we have

1. Loss converges: $\mathcal{L}_{\text{cl}}(Q^{(T^*)}) \leq \mathcal{L}_{\text{cl}}^* + \frac{1}{\text{poly}(P)}$, where $\mathcal{L}_{\text{cl}}^*$ is the global minimum of the contrastive loss in Eq. (4.2.11).
2. Attention concentration on the global area: given $X \in \mathcal{D}_k^{\text{cl}}$ with $k \in [K]$, for any $\mathbf{p} \in \mathcal{P}$, with high probability, we have $1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^c(X'; Q^{(T^*)}) = \frac{1}{\text{poly}(P)}$ for $X' \in \{X^+, X^{++}\}$.⁴
3. All patches learn global FP correlation: given $k \in [K]$, for any $\mathbf{p} \in \mathcal{P}$, $t \in [0, T^*]$, $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} \gg \Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}$ with $m > 1$, and at the convergence, $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(T^*)} = \Theta(\log P)$, $\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(T^*)} = o(1)$.

Intuition behind learning global correlations. As discussed in Section 4.3, the optimal alignment of two positive representations $F^{\text{cl}}(Q; X^+)$ and $F^{\text{cl}}(Q; X^{++})$ involves directing attention towards the same feature for each patch $\mathbf{p}$, possibly irrespective of its location. As long as the imbalanced structure, where global features dominate the data distribution, exists—even to a small degree—it leads to an order-wise stronger concentration of attention on global areas at initialization. Consequently, global FP correlations receive larger gradients compared to local ones. Therefore, global FP correlations are learned first, and focusing on these global correlations is sufficient for the CL objective to converge.

Significance of the results. Theorem 4.4.1 and Theorem 4.4.3 address a critical gap in understanding self-supervised pretraining by offering the first theoretical framework for learning with ViTs, one of the most advanced architectures in vision practice, whereas prior studies have primarily focused on linear models, CNNs, or MLPs [137, 208, 311]. Moreover, by identifying the collapsed solution in CL and emphasizing the effectiveness of MAE in capturing diverse attention patterns, we provide a qualitative comparison between MAE and contrastive learning, validating a non-trivial empirical observation [212]. This offers a comprehensive theoretical analysis of self-supervised learning with ViTs.

⁴ This also holds when no data augmentation is applied to X .

4.5 Overview of the Proof Techniques

In this section, we explain our key proof techniques in analyzing the self-supervised pretraining of transformers, using MAE as an example. We focus on the reconstruction of a specific patch $X_{\mathbf{p}}$ for $\mathbf{p} \in \mathcal{P}$. We aim to elucidate the training phases through which the model learns FP correlations related to the area associated with $\mathbf{p}$ across different clusters $k \in [K]$.

Our characterization of training phases differentiates between whether $X_{\mathbf{p}}$ is located in the global or local areas and further varies based on whether Δ is positive or negative. Specifically, for $\Delta \in [\Omega(1), 1]$, we observe distinct learning dynamics for FP correlations between local and global areas:

- Local area attends to FP correlation in two-phase: given $k \in [K]$, if $a_{k,\mathbf{p}} \neq 1$, then
 1. $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}$ first quickly decreases whereas all other $\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}$ with $m \neq 1$ and $\Upsilon_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)}$ do not change much;
 2. after some point, the increase of $\Phi_{\mathbf{p} \rightarrow v_{k,a_{k,\mathbf{p}}}}^{(t)}$ takes dominance. Such $\Phi_{\mathbf{p} \rightarrow v_{k,a_{k,\mathbf{p}}}}^{(t)}$ will keep growing until convergence with all other FP and PP attention correlations nearly unchanged.
- Global areas learn FP correlation in one-phase: given $k \in [K]$, if $a_{k,\mathbf{p}} = 1$, the update of $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}$ will dominate throughout the training, whereas all other $\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}$ with $m \neq 1$ and learned PP correlations remain close to 0.

For $\Delta \in [-0.5, -\Omega(1)]$, the behaviors of learning FP correlations are uniform for all areas. Namely, all areas learn FP correlation through one-phase: given $k \in [K]$, throughout the training, the increase of $\Phi_{\mathbf{p} \rightarrow v_{k,a_{k,\mathbf{p}}}}^{(t)}$ dominates, whereas all other $\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}$ with $m \neq a_{k,\mathbf{p}}$ and PP correlations $\Upsilon_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)}$ remain close to 0.

For clarity, this section will mainly focus on the learning of *local* feature correlations with a positive information gap $\Delta \geq \Omega(1)$ in Sections 4.5.2 and 4.5.3, which exhibits a two-phase process. The other scenarios will be discussed briefly in Section 4.5.4.

4.5.1 Gradient dynamics of attention correlations

Based on the crucial roles that attention correlations play in determining the reconstruction loss, the main idea of our analysis is to track the dynamics of those attention correlations. We first provide the following GD updates of $\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}$ and $\Upsilon_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)}$ (see Section 4.A.1 for formal statements).

Lemma 4.5.1 (FP correlations, informal). *Given $k \in [K]$, for $\mathbf{p} \in \mathcal{P}$, denote $n = a_{k,\mathbf{p}}$, let $\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)} = \frac{1}{\eta} (\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t+1)} - \Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)})$ for $m \in [N_k]$, and suppose $X_{\mathbf{p}}$ is masked. Then*

1. *for the same area, $\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \approx \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}\right)^2$;*
2. *if $k \in \mathcal{B}_{\mathbf{p}}$, for the global area,*

$$\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} \approx -\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} \cdot \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}\right) + \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}\right) \right);$$

3. for other area $m \notin \{n\} \cup \{1\}$,

$$\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)} \approx \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \left(\mathbb{1}_{\{n \neq 1\}} \left(\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} \right)^2 - \left(1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right) \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right).$$

From Lemma 4.5.1, it is observed that for $\mathbf{p} \in \mathcal{P}_{k,n}$, the feature correlation $\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}$ exhibits a monotonically increasing trend over time because $\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \geq 0$. Furthermore, if $n > 1$, i.e., $\mathcal{P}_{k,n}$ is the local area, $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}$ will monotonically decrease.

Lemma 4.5.2 (PP attention correlations, informal). *Given $\mathbf{p}, \mathbf{q} \in \mathcal{P}$, let $\beta_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = \frac{1}{\eta} (\Upsilon_{\mathbf{p} \rightarrow \mathbf{q}}^{(t+1)} - \Upsilon_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)})$, and suppose $X_{\mathbf{p}}$ is masked. Then $\beta_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = \sum_{k \in [N]} \beta_{k, \mathbf{p} \rightarrow \mathbf{q}}^{(t)}$, where $\beta_{k, \mathbf{p} \rightarrow \mathbf{q}}^{(t)}$ satisfies*

1. if $a_{k,\mathbf{p}} = a_{k,\mathbf{q}} = n$, $\beta_{k, \mathbf{p} \rightarrow \mathbf{q}}^{(t)} \approx \mathbf{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} \left(1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right)^2$;
2. if $k \in \mathcal{B}_{\mathbf{p}} \cap C_{\mathbf{q}}$, where $a_{k,\mathbf{p}} = n > 1$ and $a_{k,\mathbf{q}} = 1$:

$$\beta_{k, \mathbf{p} \rightarrow \mathbf{q}}^{(t)} \approx -\mathbf{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} \cdot \left(\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} \left(1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} \right) + \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \left(1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right) \right);$$

3. if $a_{k,\mathbf{q}} = m \notin \{n\} \cup \{1\}$, where $a_{k,\mathbf{p}} = n$,

$$\beta_{k, \mathbf{p} \rightarrow \mathbf{q}}^{(t)} \approx \mathbf{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} \cdot \left(\mathbb{1}_{\{n \neq 1\}} \left(\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} \right)^2 - \left(1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right) \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right).$$

Based on the above gradient update for $\Upsilon_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)}$, we further introduce the following auxiliary quantity $\Upsilon_{k, \mathbf{p} \rightarrow \mathbf{q}}^{(t)}$, which can be interpreted as the PP attention correlation “projected” on the k -th cluster $\mathcal{D}_k$, and will be useful in the later proof:

$$\Upsilon_{k, \mathbf{p} \rightarrow \mathbf{q}}^{(t+1)} := \Upsilon_{k, \mathbf{p} \rightarrow \mathbf{q}}^{(t)} + \eta \beta_{k, \mathbf{p} \rightarrow \mathbf{q}}^{(t)} \quad \text{with } \Upsilon_{k, \mathbf{p} \rightarrow \mathbf{q}}^{(0)} = 0. \quad (4.5.1)$$

We can directly verify that $\Upsilon_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = \sum_{k \in [K]} \Upsilon_{k, \mathbf{p} \rightarrow \mathbf{q}}^{(t)}$.

The key observation by comparing Lemma 4.5.1 and 4.5.2 is that the gradient of projected PP attention $\beta_{k, \mathbf{p} \rightarrow \mathbf{q}}^{(t)}$ is smaller than the corresponding FP gradient $\alpha_{\mathbf{p} \rightarrow v_{k,a_{k,\mathbf{q}}}}^{(t)}$ in magnitude since

$\mathbf{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} \approx \frac{\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a_{k,\mathbf{q}}}}^{(t)}}{(1-\gamma)C_{k,a_{k,\mathbf{q}}}}$. We will show that the interplay between the increase of $\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}$ and the decrease of $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}$ determines the learning behaviors for the local patch $\mathbf{p} \in \mathcal{P}_{k,n}$ with $n > 1$, and which effect will happen first depends on the initial attention, which is also determined by the value of information gap Δ .

4.5.2 Phase I: decoupling the global FP correlations

We now explain how the attention correlations evolve at the initial phase of the training to decouple the correlations of the non-target global features when $\mathbf{p}$ is located in the local area for the k -th cluster. This phase can be further divided into the following two stages.

Stage 1. At the beginning of training, $\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(0)} = \Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(0)} = 0$, and hence $\text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(0)} = \frac{1}{P}$ for any $\mathbf{q} \in \mathcal{P}$, which implies that the transformer equally attends to each patch. However, with high probability, the number of unmasked global features in the global area $\mathcal{P}_{k,1}$ is much larger than others. Hence, $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(0)} = \frac{|\mathcal{U} \cap \mathcal{P}_{k,1}|}{P} \geq \Omega\left(\frac{1}{P^{1-\kappa_c}}\right) \gg \Theta\left(\frac{1}{P^{1-\kappa_s}}\right) = \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(0)}$ for $m > 1$. Therefore, by Lemma 4.5.1 and 4.5.2, we immediately obtain

- $\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(0)} = -\Theta\left(\frac{1}{P^{2(1-\kappa_c)}}\right)$, whereas $\alpha_{\mathbf{p} \rightarrow v_{k,a_{k,\mathbf{p}}}}^{(0)} = \Theta\left(\frac{1}{P^{(1-\kappa_s)}}\right)$;
- all other FP correlation gradients $\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(0)}$ with $m \neq 1, a_{k,\mathbf{p}}$ are small;
- all projected PP correlation gradients $\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(0)}$ are small.

Since $\Delta = (1 - \kappa_s) - 2(1 - \kappa_c) \geq \Omega(1)$, it can be seen that $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}$ enjoys a much larger decreasing rate initially. This captures the decoupling process of the feature correlations with the global feature $v_{k,1}$ in the global area for $\mathbf{p}$. It can be shown that such an effect will dominate over a certain period that defines stage 1 of phase I. At the end of this stage, we will have $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} \leq -\Omega(\log(P))$, whereas all FP attention correlation $\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}$ with $m > 1$ and all projected PP correlations $\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}$ stay close to 0 (see Section 4.A.3).

During stage 1, the significant decrease of the global FP correlation $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}$ leads to a reduction in the attention score $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}$. Meanwhile, attention scores $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)}$ (where $m > 1$) for other patches remain consistent, reflecting a uniform distribution over unmasked patches within each area. By the end of stage 1, $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}$ drops to a certain level, resulting in a decrease in $|\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}|$ as it approaches $\alpha_{\mathbf{p} \rightarrow v_{k,a_{k,\mathbf{p}}}}^{(t)}$, which indicates that stage 2 begins.

Stage 2. Soon as stage 2 begins, the dominant effect switches as $|\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}|$ reaches the same order of magnitude as $\alpha_{\mathbf{p} \rightarrow v_{k,a_{k,\mathbf{p}}}}^{(t)}$. The following result shows that $\Phi_{\mathbf{p} \rightarrow v_{k,a_{k,\mathbf{p}}}}^{(t)}$ must update during stage 2.

Lemma 4.5.3 (Switching of dominant effects (See Section 4.A.3)). *Under the same conditions as Theorem 4.4.1, for $\mathbf{p} \in \mathcal{P}$, there exists $\tilde{T}_1$, such that at iteration $t = \tilde{T}_1 + 1$, we have*

1. $\Phi_{\mathbf{p} \rightarrow v_{k,a_{k,\mathbf{p}}}}^{(\tilde{T}_1+1)} \geq \Omega(\log(P))$, and $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(\tilde{T}_1+1)} = -\Theta(\log(P))$;
2. all other FP correlations $\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}$ with $m \neq 1, a_{k,\mathbf{p}}$ are small;
3. all projected PP correlations $\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}$ are small.

Intuition of the transition. Once $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}$ decreases to $-\frac{\Delta}{2L} \log(P)$, we observe that $|\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}|$ is approximately equal to $\alpha_{\mathbf{p} \rightarrow v_{k,a_{k,\mathbf{p}}}}^{(t)}$. After this point, reducing $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}$ further is more challenging compared to the increase in $\Phi_{\mathbf{p} \rightarrow v_{k,a_{k,\mathbf{p}}}}^{(t)}$. To illustrate, a minimal decrease of $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}$ by an amount

of $\frac{0.001}{L} \log(P)$ will yield $|\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}| \leq O(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{p^{0.002}})$. Such a discrepancy triggers the switch of the dominant effect.

4.5.3 Phase II: growth of target local FP correlation

Moving beyond phase I, FP correlation $\Phi_{\mathbf{p} \rightarrow v_{k,a_{k,p}}}^{(t)}$ within the target local area $\mathbf{p}$ already enjoys a larger gradient $\alpha_{\mathbf{p} \rightarrow v_{k,a_{k,p}}}^{(t)}$ than other $\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}$ with $m \neq a_{k,p}$ and all projected PP correlations $\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}$. We can show that the growth of $\Phi_{\mathbf{p} \rightarrow v_{k,a_{k,p}}}^{(t)}$ will continue to dominate until the end of training by recognizing the following two stages.

Rapid growth stage. At the beginning of phase II, $\alpha_{\mathbf{p} \rightarrow v_{k,a_{k,p}}}^{(t)}$ is mainly driven by $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a_{k,p}}}^{(t)}$ since $1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a_{k,p}}}^{(t)}$ remains at the constant order. Therefore, the growth of $\Phi_{\mathbf{p} \rightarrow v_{k,a_{k,p}}}^{(t)}$ naturally results in a boost in $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a_{k,p}}}^{(t)}$, thereby promoting an increase in its own gradient $\alpha_{\mathbf{p} \rightarrow v_{k,a_{k,p}}}^{(t)}$, which defines the rapid growth stage. On the other hand, we can prove that the following gap holds for FP and projected PP correlation gradients (see Section 4.A.3):

- all other FP correlation gradients $\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}$ with $m \neq a_{k,p}$ are small;
- all projected PP correlation gradients $\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}$ are small.

Convergence stage. After the rapid growth stage, the desired local pattern with a high target feature-position correlation $\Phi_{\mathbf{p} \rightarrow v_{k,a_{k,p}}}^{(t)}$ is learned. In this last stage, it is demonstrated that the above conditions for non-target FP and projected PP correlations remain valid, while the growth of $\Phi_{\mathbf{p} \rightarrow v_{k,a_{k,p}}}^{(t)}$ starts to decelerate as $\Phi_{\mathbf{p} \rightarrow v_{k,a_{k,p}}}^{(t)}$ reaches $\Theta(\log(P))$, resulting in $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \approx \Omega(1)$, which leads to convergence (see Section 4.A.3).

4.5.4 Learning processes in other scenarios

In this section, we talk about the learning process in other settings, including learning FP correlations for the local area when the information gap is negative, learning FP correlations for the global area, and failure to learn PP correlations.

What is the role of positive information gap? As described in stage 1 of phase 1 in Section 4.5.2, the decoupling effect happens at the beginning of the training because $\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(0)} \gg \alpha_{\mathbf{p} \rightarrow v_{k,a_{k,p}}}^{(0)}$ attributed to $\Delta \geq \Omega(1)$. However, in cases where $\Delta \leq -\Omega(1)$, this relationship reverses, with $\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(0)}$ becoming significantly smaller than $\alpha_{\mathbf{p} \rightarrow v_{k,a_{k,p}}}^{(0)}$. Similarly, other FP gradients $\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(0)}$ with $m \neq 1, a_{k,p}$ and all the projected gradients of PP correlation $\beta_{\mathbf{p} \rightarrow \mathbf{q}}^{(0)}$ are small in magnitude. Consequently, $\Phi_{\mathbf{p} \rightarrow v_{k,a_{k,p}}}^{(t)}$ starts with a larger gradient, eliminating the need to decouple FP correlations for the global area. As a result, training skips the initial phase, and moves directly into Phase II, during which $\Phi_{\mathbf{p} \rightarrow v_{k,a_{k,p}}}^{(t)}$ continues to increase until it converges (see Section 4.A.4).

Learning FP correlations for the global area. When the patch X_p is located in the global area of cluster k , i.e., $a_{k,p} = 1$, the attention score $\text{Attn}_{p \rightarrow \mathcal{P}_{k,1}}^{(0)}$ directed towards the target area $\mathcal{P}_{k,1}$ is initially higher compared to other attention scores due to the presence of a significant number of unmasked patches in the global area. This leads to an initially larger gradient $\alpha_{p \rightarrow v_{k,a_{k,p}}}^{(0)}$. Such an effect is independent of the value of Δ . As a result, the training process skips the initial phase, which is typically necessary for the cases where $a_{k,p} > 1$ with a positive information gap, and moves directly into Phase II (see Section 4.A.5).

All PP correlations are small. Integrating the analysis from all previous discussions, we establish that for every cluster $k \in [K]$, regardless of its association with C_p (global area) or $\mathcal{B}_p$ (local area), and for any patch X_q with $q \in \mathcal{P}$, the projected PP correlation $\Upsilon_{k,p \rightarrow q}^{(t)}$ remains nearly zero in comparison to the significant changes observed in the FP correlation, because the gradient $\beta_{k,p \rightarrow q}^{(t)}$ is relatively negligible. Therefore, the overall PP correlation $\Upsilon_{p \rightarrow q}^{(t)} = \sum_{k=1}^K \Upsilon_{k,p \rightarrow q}^{(t)}$ also stays close to zero, given that the number of clusters $K = \Theta(1)$.

4.6 Related Work

Empirical studies of transformers in vision. A number of works have aimed to understand the transformers in vision from different perspectives: comparison with CNNs [99, 211, 227], robustness [33, 214], and role of positional embeddings [189, 283]. Recent studies [212, 306, 320] have delved into ViTs with self-supervision to uncover the mechanisms at play, particularly through visualization and analysis of metrics related to self-attention. [320] compared the masked image modeling (MIM) method with supervised models, revealing MIM’s capacity to enhance diversity and locality across all ViT layers, which significantly boosts performance on tasks with weak semantics following fine-tuning. Building on MIM’s advantages, [306] further proposed a simple feature distillation method that incorporates locality into various self-supervised methods, leading to an overall improvement in the finetuning performance. [212] conducted a detailed comparison between masked image modeling (MIM) and contrastive learning. They demonstrated that contrastive learning will make the self-attentions collapse into homogeneity for all query patches due to the nature of discriminative learning, while MIM leads to a diverse self-attention map since it focuses on local patterns.

Theory of self-supervised learning. A major line of theoretical studies falls into one of the most successful self-supervised learning approaches, contrastive learning [20, 50, 238, 311], and its variant non-contrastive self-supervised learning [215, 296, 312]. Some other works study the mask prediction approach [158, 175, 303], which is the focus of this paper. [158] provided statistical downstream guarantees for reconstructing missing patches. [303] studied the benefits of head and prompt tuning with masked pretraining under a Hidden Markov Model framework. [175] provided a parameter identifiability view to understand the benefit of masked prediction

tasks, which linked the masked reconstruction tasks to the informativeness of the representation via identifiability techniques from tensor decomposition.

Theory of transformers and attention models. Prior work has studied the theoretical properties of transformers from various aspects: representational power [80, 242, 293, 304, 335], internal mechanism [266, 308], limitations [111, 243], and PAC learning [46]. Recently, there has been a growing body of research studying in-context learning with transformers due to the remarkable emergent in-context ability of large language models [3, 100, 128, 163, 202, 292, 341, 343]. Regarding the training dynamics of attention-based models, [161] studied the training process of shallow ViTs in a classification task. Subsequent research expanded on this by exploring the graph transformer with positional encoding [162] and in-context learning performance of transformers with nonlinear self-attention and nonlinear MLP [163]. However, all of these analyses rely crucially on stringent assumptions on the initialization of transformers and hardly generalize to our setting. [277] mathematically described how the attention map evolves trained by SGD for one-layer transformer but did not provide any convergence guarantee, and the follow-up work [278] considered a generalized case with multiple layers. [267, 287] investigated the implicit bias for self-attention models trained with GD. Furthermore, [128] proved the in-context convergence of a one-layer softmax transformer trained via GD and illustrated the attention dynamics throughout the training process. [328] generalized such an in-context learning problem to a multi-head setting with non-linear task functions. [202] studied GD dynamics on a simplified two-layer attention-only transformer and proved that it can encode the causal structure in the first attention layer. However, none of the previous studies analyzed the training of transformers under self-supervised learning, which is the focus of this paper.

4.7 Experiments

Previous studies on the attention mechanisms of ViT-based pre-training approaches have mainly utilized a metric known as the attention distance [75]. Such a metric quantifies the average spatial distance between the query and key tokens, weighted by their self-attention coefficients. The general interpretation is that larger attention distances indicate global understanding, and smaller values suggest a focus on local features. However, such a metric does not adequately determine if the self-attention mechanism is identifying a unique global pattern. A high attention distance could result from different patches focusing on varied distant areas, which does not necessarily imply that global information is being effectively synthesized. To address this limitation, we introduce a novel and revised version of average attention distance, called the attention diversity metric, which is designed to assess whether various patches are concentrating on a similar region, thereby directly capturing global information.

Attention diversity metric, in distance. This metric is computed for self-attention with a single head of the specific layer. For a given image divided into $N \times N$ patches, the process unfolds

as follows: for each patch, it is employed as the query patch to calculate the attention weights towards all N^2 patches, and those with the top- n attention weights are selected. Subsequently, the coordinates (e.g. (i, j) with $i, j \in [N]$) of these top- n patches are concatenated in sequence to form a $2 \times n$ -dimensional vector. The final step computes the average distance between all these $2n$ -dimensional vectors, i.e., $N^2 \times N^2$ vector pairs.

Setup. In this work, we compare the performance of ViT-B/16 encoder pre-trained on ImageNet-1K [239] among the following four models: masked reconstruction model (MAE), contrastive learning model (MoCo v3 [55]), other self-supervised model (DINO [40]), and supervised model (DeiT [282]). We focus on 12 different attention heads in the last layer of ViT-B on different pre-trained models. The box plot visualizes the distribution of the top-10 averaged attention focus across 152 example images, as similarly done in [75].

Implications. The experiment results based on our new metric are provided in Figure 4.3. Lower values of the attention diversity metric signify a focused attention on a coherent area across different patches, reflecting a global pattern of focus. On the other hand, higher values suggest that attention is dispersed, focusing on different, localized areas. It can be seen that the masked pretraining model is particularly effective in learning more diverse attention patterns, setting it apart from other models that prioritize a uniform global information with less attention diversity. This aligns with and provides further evidence for the findings in [212].

Appendix

4.A Proof of Main Theorems for MAE

4.A.1 Preliminaries

In this section, we will introduce warm-up gradient computations and probabilistic lemmas that establish essential properties of the data and the loss function, which are pivotal for the technical proofs in the upcoming sections for masked pretraining. Throughout the appendix, we assume $N_k = N$ and $C_{k,n} = C_n$ for all $k \in [K]$ for simplicity. We will also omit the explicit dependence on X for $z_n(X)$. We use $k_X \in [K]$ to denote the cluster index that a given image X is drawn from. Furthermore, we will abbreviate $\mathcal{L}_{\text{mae}}(\mathcal{L}_{\text{mae},\mathbf{p}})$ as $\mathcal{L}(\mathcal{L}_{\mathbf{p}})$, and F^{mae} as F for simplicity, when the context makes it clear. We abbreviate $\text{Attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{\text{m}}(X; Q^{(t)})$ ($\text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{\text{m}}(X; Q^{(t)})$) as $\text{Attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)}$ ($\text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)}$), when the context makes it clear.

Gradient computations We first calculate the gradient with respect to Q . We omit the superscript ‘(t)’ and write $\mathcal{L}(Q)$ as $\mathcal{L}$ here for simplicity.

Lemma 4.A.1. *The gradient of the loss function with respect to Q is given by*

$$\frac{\partial \mathcal{L}}{\partial Q} = -\mathbb{E} \left[\sum_{\mathbf{p} \in \mathcal{M}} \sum_{\mathbf{q}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}} \mathbf{M}(X)_{\mathbf{q}}^{\top} (X_{\mathbf{p}} - [F(\mathbf{M}(X); Q)]_{\mathbf{p}}) \cdot \tilde{\mathbf{M}}(X)_{\mathbf{p}} \left(\tilde{\mathbf{M}}(X)_{\mathbf{q}} - \sum_{\mathbf{r}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{r}} \tilde{\mathbf{M}}(X)_{\mathbf{r}} \right)^{\top} \right].$$

Proof. We begin with the chain rule and obtain

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial Q} &= \mathbb{E} \left[\sum_{\mathbf{p} \in \mathcal{M}} \frac{\partial [F(\mathbf{M}(X); Q)]_{\mathbf{p}}}{\partial Q} ([F(\mathbf{M}(X); Q)]_{\mathbf{p}} - X_{\mathbf{p}}) \right] \\ &= \mathbb{E} \left[\sum_{\mathbf{p} \in \mathcal{M}} \sum_{\mathbf{q}} \frac{\partial \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}}{\partial Q} \mathbf{M}(X)_{\mathbf{q}}^{\top} ([F(\mathbf{M}(X); Q)]_{\mathbf{p}} - X_{\mathbf{p}}) \right]. \end{aligned} \quad (4.A.1)$$

We focus on the gradient for each attention score:

$$\begin{aligned} \frac{\partial \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}}{\partial Q} &= \sum_{\mathbf{r}} \frac{\exp \left(\tilde{\mathbf{M}}(X)_{\mathbf{p}}^{\top} Q (\tilde{\mathbf{M}}(X)_{\mathbf{r}} + \tilde{\mathbf{M}}(X)_{\mathbf{q}}) \right)}{\left(\sum_{\mathbf{r}} \exp(\tilde{\mathbf{M}}(X)_{\mathbf{p}}^{\top} Q \tilde{\mathbf{M}}(X)_{\mathbf{r}}) \right)^2} \tilde{\mathbf{M}}(X)_{\mathbf{p}} (\tilde{\mathbf{M}}(X)_{\mathbf{q}} - \tilde{\mathbf{M}}(X)_{\mathbf{r}})^{\top} \\ &= \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}} \sum_{\mathbf{r}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{r}} \tilde{\mathbf{M}}(X)_{\mathbf{p}} (\tilde{\mathbf{M}}(X)_{\mathbf{q}} - \tilde{\mathbf{M}}(X)_{\mathbf{r}})^{\top} \end{aligned}$$

$$= \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}} \tilde{\mathbf{M}}(X)_{\mathbf{p}} \cdot \left[\tilde{\mathbf{M}}(X)_{\mathbf{q}} - \sum_{\mathbf{r}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{r}} \tilde{\mathbf{M}}(X)_{\mathbf{r}} \right]^{\top}.$$

Substituting the above equation into (4.A.1), we complete the proof. $\square$

Recall that the quantities $\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}$ and $\Upsilon_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)}$ are defined in Definition 4.3.1. These quantities are associated with the attention weights for each token, and they play a crucial role in our analysis of learning dynamics. We will restate their definitions here for clarity.

Definition 4.A.1. (Attention correlations) Given $\mathbf{p}, \mathbf{q} \in \mathcal{P}$, for $t \geq 0$, we define two types of attention correlations as follows:

1. Feature Attention Correlation: $\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)} := e_{\mathbf{p}}^{\top} Q^{(t)} v_{k,m}$ for $k \in [K]$ and $m \in [N]$;
2. Positional Attention Correlation: $\Upsilon_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} := e_{\mathbf{p}}^{\top} Q^{(t)} e_{\mathbf{q}}$.

By our initialization, we have $\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(0)} = \Upsilon_{\mathbf{p} \rightarrow \mathbf{q}}^{(0)} = 0$.

Next, we will apply the expression in Lemma 4.A.1 to compute the gradient dynamics of these attention correlations.

Formal statements and proofs of Lemma 4.5.1 and 4.5.2 We first introduce some notation. Given $\mathbf{r} \in \mathcal{U}$, for $\mathbf{p} \in \mathcal{P}$, $k \in [K]$ and $n \in [N]$ define the following quantities:

$$\begin{aligned} J_{\mathbf{r}}^{\mathbf{p}} &:= \mathbf{M}(X)_{\mathbf{r}}^{\top} (X_{\mathbf{p}} - [F(\mathbf{M}(X); Q)]_{\mathbf{p}}), \\ I_{\mathbf{r}}^{\mathbf{p},k,n} &:= \left(\tilde{\mathbf{M}}(X)_{\mathbf{r}} - \sum_{\mathbf{w} \in \mathcal{P}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{w}} \tilde{\mathbf{M}}(X)_{\mathbf{w}} \right)^{\top} v_{k,n}, \\ K_{\mathbf{r}}^{\mathbf{p},\mathbf{q}} &:= \left(\tilde{\mathbf{M}}(X)_{\mathbf{r}} - \sum_{\mathbf{w} \in \mathcal{P}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{w}} \tilde{\mathbf{M}}(X)_{\mathbf{w}} \right)^{\top} e_{\mathbf{q}}. \end{aligned}$$

Lemma 4.A.2 (Formal statement of Lemma 4.5.1). *Given $k \in [K]$, for $\mathbf{p} \in \mathcal{P}$, denote $n = a_{k,\mathbf{p}}$. Letting $\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)} = \frac{1}{\eta} (\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t+1)} - \Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)})$ for $m \in [N_k]$, then*

1. for $m = n$,

$$\begin{aligned} \alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} &= \mathbb{E} \left[\mathbf{1}\{\mathbf{p} \in \mathcal{M}, k_X = k\} \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \cdot \right. \\ &\quad \left. \left(z_n^3 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right)^2 + \sum_{a \neq n} z_a^2 z_n \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)} \right)^2 \right) \right]; \end{aligned}$$

2. for $m \neq n$,

$$\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)} = \mathbb{E} \left[\mathbb{1}\{\mathbf{p} \in \mathcal{M}, k_X = k\} \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \cdot \left(\sum_{a \neq m,n} z_a^2 z_m \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)} \right)^2 - \left(z_m z_n^2 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right) \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} + z_m^3 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \right) \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \right) \right) \right].$$

Proof. From Lemma 4.A.1, we have

$$\begin{aligned} \alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)} &= e_{\mathbf{p}}^\top \left(-\frac{\partial \mathcal{L}}{\partial Q} \right) v_{k,m} \\ &= \mathbb{E}[\mathbb{1}\{\mathbf{p} \in \mathcal{M}\} \sum_{\mathbf{r} \in \mathcal{U}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{r}} J_{\mathbf{r}}^{\mathbf{p}} \cdot I_{\mathbf{r}}^{\mathbf{p},k,m}] \\ &= \mathbb{E}[\mathbb{1}\{\mathbf{p} \in \mathcal{M}, k_X = k\} \sum_{\mathbf{r} \in \mathcal{U}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{r}} J_{\mathbf{r}}^{\mathbf{p}} \cdot I_{\mathbf{r}}^{\mathbf{p},k,m}], \end{aligned}$$

where the last equality holds since when $k_X \neq k$, $I_{\mathbf{r}}^{\mathbf{p},k,m} = 0$ due to orthogonality. Thus, in the following, we only need to consider the case $k_X = k$.

Case 1: $m = n$.

- For $\mathbf{r} \in \mathcal{U} \cap \mathcal{P}_{k,n}$, since $v_{k,n'} \perp v_{k,n}$ for $n' \neq n$, and $v_{k,n} \perp \{e_{\mathbf{q}}\}_{\mathbf{q} \in \mathcal{P}}$ we have

$$\begin{aligned} J_{\mathbf{r}}^{\mathbf{p}} &= z_n v_{k,n}^\top \left(z_n v_{k,n} - \sum_{\mathbf{q} \in \mathcal{U} \cap \mathcal{P}_{k,n}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}} z_n v_{k,n} \right) = z_n^2 (1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}), \\ I_{\mathbf{r}}^{\mathbf{p},k,n} &= (z_n v_{k,n} - \sum_{\mathbf{q} \in \mathcal{U} \cap \mathcal{P}_{k,n}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}} z_n v_{k,n})^\top v_{k,n} = J_{\mathbf{r}}^{\mathbf{p}} / z_n. \end{aligned}$$

- For $\mathbf{r} \in \mathcal{U} \cap \mathcal{P}_{k,n'}$ with $n' \neq n$

$$\begin{aligned} J_{\mathbf{r}}^{\mathbf{p}} &= z_{n'} v_{k,n'}^\top \left(z_n v_{k,n} - \sum_{\mathbf{q} \in \mathcal{U} \cap \mathcal{P}_{k,n'}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}} z_{n'} v_{k,n'} \right) = -z_{n'}^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n'}}, \\ I_{\mathbf{r}}^{\mathbf{p},k,n} &= \left(z_{n'} v_{k,n'} - \sum_{\mathbf{q} \in \mathcal{U} \cap \mathcal{P}_{k,n}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}} z_n v_{k,n} \right)^\top v_{k,n} = -z_n \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}. \end{aligned}$$

Putting it together, we obtain:

$$\begin{aligned} e_{\mathbf{p}}^\top \left(-\frac{\partial \mathcal{L}}{\partial Q} \right) v_{k,n} &= \mathbb{E} \left[\mathbb{1}\{\mathbf{p} \in \mathcal{M}, k_X = k\} \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \cdot \left(z_n^3 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right)^2 + \sum_{a \neq n} z_a^2 z_n \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)} \right)^2 \right) \right]. \end{aligned}$$

Case 2: $m \neq n$. Similarly,

- For $\mathbf{r} \in \mathcal{U} \cap \mathcal{P}_{k,n}$

$$J_{\mathbf{r}}^{\mathbf{p}} = z_n v_{k,n}^{\top} \left(z_n v_{k,n} - \sum_{\mathbf{q} \in \mathcal{U} \cap \mathcal{P}_{k,n}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}} z_n v_{k,n} \right) = z_n^2 (1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}),$$

$$I_{\mathbf{r}}^{\mathbf{p},k,m} = \left(z_n v_{k,n} - \sum_{\mathbf{q} \in \mathcal{U} \cap \mathcal{P}_{k,m}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}} z_m v_{k,m} \right)^{\top} v_{k,m} = -z_m \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}.$$

- For $\mathbf{r} \in \mathcal{U} \cap \mathcal{P}_{k,m}$

$$J_{\mathbf{r}}^{\mathbf{p}} = z_m v_{k,m}^{\top} \left(z_n v_{k,n} - \sum_{\mathbf{q} \in \mathcal{U} \cap \mathcal{P}_{k,m}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} z_m v_{k,m} \right) = -z_m^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}},$$

$$I_{\mathbf{r}}^{\mathbf{p},k,n} = \left(z_m v_{k,m} - \sum_{\mathbf{q} \in \mathcal{U} \cap \mathcal{P}_{k,m}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} z_m v_{k,m} \right)^{\top} v_{k,m} = z_n (1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}).$$

- For $\mathbf{r} \in \mathcal{U} \cap \mathcal{P}_{k,a}$, $a \neq n, m$

$$J_{\mathbf{r}}^{\mathbf{p}} = z_a v_{k,a}^{\top} \left(z_n v_{k,n} - \sum_{\mathbf{q} \in \mathcal{U} \cap \mathcal{P}_{k,a}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} z_a v_{k,a} \right) = -z_a^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}},$$

$$I_{\mathbf{r}}^{\mathbf{p},k,n} = \left(z_a v_{k,a} - \sum_{\mathbf{q} \in \mathcal{U} \cap \mathcal{P}_{k,m}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} z_m v_{k,m} \right)^{\top} v_{k,m} = -z_m \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}.$$

Putting them together, then we complete the proof. $\square$

Lemma 4.A.3 (Formal statement of Lemma 4.5.2). *Given $\mathbf{p}, \mathbf{q} \in \mathcal{P}$, let $\beta_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = \frac{1}{\eta} (\Upsilon_{\mathbf{p} \rightarrow \mathbf{q}}^{(t+1)} - \Upsilon_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)})$, then*

$$\beta_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = \sum_{k \in [K]} \beta_{k, \mathbf{p} \rightarrow \mathbf{q}}^{(t)}$$

where $\beta_{k, \mathbf{p} \rightarrow \mathbf{q}}^{(t)}$ satisfies

1. if $a_{k, \mathbf{p}} = a_{k, \mathbf{q}} = n$,

$$\beta_{k, \mathbf{p} \rightarrow \mathbf{q}}^{(t)} = \mathbb{E} \left[\mathbb{1}\{\mathbf{p} \in \mathcal{M}, k_X = k\} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} \cdot \left(\sum_{a \neq n} z_a^2 \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)} \right)^2 + z_n^2 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right) \left(\mathbb{1}\{\mathbf{q} \in \mathcal{U}\} - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right) \right) \right];$$

2. for $a_{k, \mathbf{p}} = n \neq m = a_{k, \mathbf{q}}$,

$$\beta_{k, \mathbf{p} \rightarrow \mathbf{q}}^{(t)} = \mathbb{E} \left[\mathbb{1}\{\mathbf{p} \in \mathcal{M}, k_X = k\} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} \cdot \right]$$

$$\left(\sum_{a \neq n} z_a^2 \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)} \right)^2 - z_n^2 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right) \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} - \mathbb{1}\{\mathbf{q} \in \mathcal{U}\} z_m^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \right).$$

Proof. First,

$$\beta_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = e_{\mathbf{p}}^\top \left(-\frac{\partial \mathcal{L}}{\partial Q} \right) e_{\mathbf{q}} = \mathbb{E} \left[\mathbb{1}\{\mathbf{p} \in \mathcal{M}\} \sum_{\mathbf{r} \in \mathcal{U}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{r}} J_{\mathbf{r}}^{\mathbf{p}} K_{\mathbf{r}}^{\mathbf{p}, \mathbf{q}} \right].$$

Then we let

$$\beta_{k, \mathbf{p} \rightarrow \mathbf{q}}^{(t)} := \mathbb{E} \left[\mathbb{1}\{\mathbf{p} \in \mathcal{M}, k_X = k\} \sum_{\mathbf{r} \in \mathcal{U}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{r}} J_{\mathbf{r}}^{\mathbf{p}} K_{\mathbf{r}}^{\mathbf{p}, \mathbf{q}} \right].$$

In the following, we denote $a_{k, \mathbf{p}} = n$ and $a_{k, \mathbf{q}} = m$ for simplicity.

Case 1: $m = n$. If $\mathbf{q} \in \mathcal{U} \cap \mathcal{P}_{k,n}$:

- For $\mathbf{r} = \mathbf{q}$

$$J_{\mathbf{r}}^{\mathbf{p}} = z_n v_{k,n}^\top \left(z_n v_{k,n} - \sum_{\mathbf{w} \in \mathcal{U} \cap \mathcal{P}_{k,n}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{w}} z_n v_{k,n} \right) = z_n^2 (1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}),$$

$$K_{\mathbf{r}}^{\mathbf{p}, \mathbf{q}} = (e_{\mathbf{q}} - (\text{attn}_{\mathbf{p} \rightarrow \mathbf{q}} e_{\mathbf{q}} + \sum_{\mathbf{w} \neq \mathbf{q}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{w}} e_{\mathbf{w}}))^\top e_{\mathbf{q}} = 1 - \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}.$$

- For $\mathbf{r} \in \mathcal{U} \cap \mathcal{P}_{k,n}$, and $\mathbf{r} \neq \mathbf{q}$

$$J_{\mathbf{r}}^{\mathbf{p}} = z_n v_{k,n}^\top \left(z_n v_{k,n} - \sum_{\mathbf{w} \in \mathcal{U} \cap \mathcal{P}_{k,n}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{w}} z_n v_{k,n} \right) = z_n^2 (1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}),$$

$$K_{\mathbf{r}}^{\mathbf{p}, \mathbf{q}} = \left(e_{\mathbf{r}} - (\text{attn}_{\mathbf{p} \rightarrow \mathbf{q}} e_{\mathbf{q}} + \sum_{\mathbf{w} \neq \mathbf{q}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{w}} e_{\mathbf{w}}) \right)^\top e_{\mathbf{q}} = -\text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}.$$

Thus

$$\begin{aligned} \sum_{\mathbf{r} \in \mathcal{U} \cap \mathcal{P}_{k,n}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{r}} J_{\mathbf{r}}^{\mathbf{p}} \cdot K_{\mathbf{r}}^{\mathbf{p}, \mathbf{q}} &= z_n^2 \left(1 - \sum_{\mathbf{w} \in \mathcal{U} \cap \mathcal{P}_{k,n}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{w}} \right) \cdot \left(- \sum_{\mathbf{r} \in \mathcal{U} \cap \mathcal{P}_{k,n}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{r}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}} + \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}} \right) \\ &= z_n^2 (1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}})^2 \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)}. \end{aligned}$$

- For $\mathbf{r} \in \mathcal{U} \cap \mathcal{P}_{k,a}$, $a \neq n$

$$J_{\mathbf{r}}^{\mathbf{p}} = z_a v_{k,a}^\top \left(z_n v_{k,n} - \sum_{\mathbf{w} \in \mathcal{U} \cap \mathcal{P}_{k,a}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{w}} z_a v_{k,a} \right) = -z_a^2 \sum_{\mathbf{w} \in \mathcal{U} \cap \mathcal{P}_{k,a}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{w}},$$

$$K_r^{p,q} = \left(e_r - (\text{attn}_{p \rightarrow q} e_q + \sum_{w \neq q} \text{attn}_{p \rightarrow w} e_w) \right)^\top e_q = -\text{attn}_{p \rightarrow q}.$$

Thus

$$\sum_{r \in \mathcal{U}} \text{attn}_{p \rightarrow r} J_r^p K_r^{p,q} = \text{attn}_{p \rightarrow q} \cdot \left(z_n^2 (1 - \text{Attn}_{p \rightarrow \mathcal{P}_{k,n}})^2 + \sum_{a \neq n} z_a^2 (\text{Attn}_{p \rightarrow \mathcal{P}_{k,a}})^2 \right).$$

If $q \in \mathcal{M} \cap \mathcal{P}_{k,n}$:

- For $r \in \mathcal{U} \cap \mathcal{P}_{k,n}$,

$$J_r^p = z_n v_{k,n}^\top \left(z_n v_{k,n} - \sum_{w \in \mathcal{U} \cap \mathcal{P}_{k,n}} \text{attn}_{p \rightarrow w} z_n v_{k,n} \right) = z_n^2 (1 - \text{Attn}_{p \rightarrow \mathcal{P}_{k,n}}),$$

$$K_r^{p,q} = \left(e_r - (\text{attn}_{p \rightarrow q} e_q + \sum_{w \neq q} \text{attn}_{p \rightarrow w} e_w) \right)^\top e_q = -\text{attn}_{p \rightarrow q}.$$

- For $r \in \mathcal{U} \cap \mathcal{P}_{k,a}$, $a \neq n$

$$J_r^p = z_a v_{k,a}^\top \left(z_n v_{k,n} - \sum_{w \in \mathcal{U} \cap \mathcal{P}_{k,a}} \text{attn}_{p \rightarrow w} z_a v_{k,a} \right) = -z_a^2 \sum_{w \in \mathcal{U} \cap \mathcal{P}_{k,a}} \text{attn}_{p \rightarrow w},$$

$$K_r^{p,q} = \left(e_r - (\text{attn}_{p \rightarrow q} e_q + \sum_{w \neq q} \text{attn}_{p \rightarrow w} e_w) \right)^\top e_q = -\text{attn}_{p \rightarrow q}.$$

Thus

$$\begin{aligned} & \sum_{r \in \mathcal{U}} \text{attn}_{p \rightarrow r} J_r^p K_r^{p,q} \\ &= \text{attn}_{p \rightarrow q} \cdot \left(z_n^2 (1 - \text{Attn}_{p \rightarrow \mathcal{P}_{k,n}})^2 - z_n^2 (1 - \text{Attn}_{p \rightarrow \mathcal{P}_{k,n}}) + \sum_{a \neq n} z_a^2 (\text{Attn}_{p \rightarrow \mathcal{P}_{k,a}})^2 \right). \end{aligned}$$

Putting it together,

$$\begin{aligned} \beta_{k,p \rightarrow q}^{(t)} &= \mathbb{E} \left[\mathbb{1}\{p \in \mathcal{M}, k_X = k\} \text{attn}_{p \rightarrow q} \cdot \right. \\ & \quad \left. \left(-z_n^2 (1 - \text{Attn}_{p \rightarrow \mathcal{P}_{k,n}}) \mathbb{1}\{q \in \mathcal{M}\} + z_n^2 (1 - \text{Attn}_{p \rightarrow \mathcal{P}_{k,n}})^2 + \sum_{m \neq n} z_m^2 (\text{Attn}_{p \rightarrow \mathcal{P}_{k,m}})^2 \right) \right]. \end{aligned}$$

Case 2: $m \neq n$. Similarly, if $q \in \mathcal{U} \cap \mathcal{P}_{k,m}$:

- For $r \in \mathcal{U} \cap \mathcal{P}_{k,n}$,

$$J_r^p = z_n v_{k,n}^\top \left(z_n v_{k,n} - \sum_{w \in \mathcal{U} \cap \mathcal{P}_{k,n}} \text{attn}_{p \rightarrow w} z_n v_{k,n} \right) = z_n^2 (1 - \text{Attn}_{p \rightarrow \mathcal{P}_{k,n}}),$$

$$K_r^{p,q} = \left(e_r - \text{attn}_{p \rightarrow q} e_q - \sum_{w \neq q} \text{attn}_{p \rightarrow w} e_w \right)^\top e_q = -\text{attn}_{p \rightarrow q}.$$

- For $r = q$

$$J_r^p = z_m v_{k,m}^\top \left(z_n v_{k,n} - \sum_{w \in \mathcal{U} \cap \mathcal{P}_{k,m}} \text{attn}_{p \rightarrow w} z_m v_{k,m} \right) = -z_m^2 \text{Attn}_{p \rightarrow \mathcal{P}_{k,m}},$$

$$K_r^{p,q} = \left(e_q - \text{attn}_{p \rightarrow q} e_q - \sum_{w \neq q} \text{attn}_{p \rightarrow w} e_w \right)^\top e_q = 1 - \text{attn}_{p \rightarrow q}.$$

- For $r \in \mathcal{U} \cap \mathcal{P}_{k,a}$, $a \neq n$, and $r \neq q$

$$J_r^p = z_a v_{k,a}^\top \left(z_n v_{k,n} - \sum_{w \in \mathcal{U} \cap \mathcal{P}_{k,a}} \text{attn}_{p \rightarrow w} z_a v_{k,a} \right) = -z_a^2 \text{Attn}_{p \rightarrow \mathcal{P}_{k,a}},$$

$$K_r^{p,q} = \left(e_r - \text{attn}_{p \rightarrow q} e_q - \sum_{w \neq q} \text{attn}_{p \rightarrow w} e_w \right)^\top e_q = -\text{attn}_{p \rightarrow q}.$$

Thus

$$\begin{aligned} & \sum_{r \in \mathcal{U}} \text{attn}_{p \rightarrow r} J_r^p K_r^{p,q} \\ &= \text{attn}_{p \rightarrow q} \cdot \left(-z_n^2 (1 - \text{Attn}_{p \rightarrow \mathcal{P}_{k,n}}) \text{Attn}_{p \rightarrow \mathcal{P}_{k,n}} - z_m^2 \text{Attn}_{p \rightarrow \mathcal{P}_{k,m}} + \sum_{a \neq n} z_a^2 (\text{Attn}_{p \rightarrow \mathcal{P}_{k,a}})^2 \right). \end{aligned}$$

If $q \in \mathcal{M} \cap \mathcal{P}_{k,m}$:

- For $r \in \mathcal{U} \cap \mathcal{P}_{k,n}$,

$$J_r^p = z_n v_{k,n}^\top \left(z_n v_{k,n} - \sum_{w \in \mathcal{U} \cap \mathcal{P}_{k,n}} \text{attn}_{p \rightarrow w} z_n v_{k,n} \right) = z_n^2 (1 - \text{Attn}_{p \rightarrow \mathcal{P}_{k,n}}),$$

$$K_r^{p,q} = \left(e_r - \text{attn}_{p \rightarrow q} e_q - \sum_{w \neq q} \text{attn}_{p \rightarrow w} e_w \right)^\top e_q = -\text{attn}_{p \rightarrow q}.$$

- For $r \in \mathcal{U} \cap \mathcal{P}_{k,a}$, $a \neq n$

$$J_r^p = z_a v_{k,a}^\top \left(z_n v_{k,n} - \sum_{w \in \mathcal{U} \cap \mathcal{P}_{k,a}} \text{attn}_{p \rightarrow w} z_a v_{k,a} \right) = -z_a^2 \text{Attn}_{p \rightarrow \mathcal{P}_{k,a}},$$

$$K_r^{p,q} = \left(e_r - \text{attn}_{p \rightarrow q} e_q - \sum_{w \neq q} \text{attn}_{p \rightarrow w} e_w \right)^\top e_q = -\text{attn}_{p \rightarrow q}.$$

Thus

$$\sum_{r \in \mathcal{U}} \text{attn}_{p \rightarrow r} J_r^p K_r^{p,q}$$

$$= \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}} \cdot \left(-z_n^2 (1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}) \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}} + \sum_{a \neq n} z_a^2 (\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}})^2 \right).$$

Therefore

$$\begin{aligned} \beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)} &= \mathbb{E} \left[\mathbb{1}\{\mathbf{p} \in \mathcal{M}, k_X = k\} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}} \cdot \right. \\ &\quad \left. \left(-z_n^2 (1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}) \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}} - \mathbb{1}\{\mathbf{q} \in \mathcal{U}\} z_m^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}} + \sum_{a \neq n} z_a^2 (\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}})^2 \right) \right]. \end{aligned}$$

□

Based on the above gradient update for $\Upsilon_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)}$, we further introduce the following auxiliary quantity, which will be useful in the later proof:

$$\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t+1)} := \Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)} + \eta \beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}, \quad \text{with } \Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(0)} = 0. \quad (4.A.2)$$

It is easy to verify that $\Upsilon_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = \sum_{k \in [K]} \Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}$.

High-probability events We first introduce the following exponential bounds for the hypergeometric distribution $\text{Hyper}(m, D, M)$. $\text{Hyper}(m, D, M)$ describes the probability of certain successes (random draws for which the object drawn has a specified feature) in m draws, without replacement, from a finite population of size M that contains exactly D objects with that feature, wherein each draw is either a success or a failure.

Proposition 4.A.1 ([105]). Suppose $S \sim \text{Hyper}(m, D, M)$ with $1 \leq m, D \leq M$. Define $\mu_M := D/M$. Then for all $t > 0$

$$P(|S - m\mu_M| > t) \leq 2 \exp\left(-\frac{t^2}{4m\mu_M + 2t}\right).$$

We then utilize this property to prove the high-probability set introduced in Section 4.5.1.

Lemma 4.A.4. For $k \in [K]$ $n \in [N]$, define

$$\mathcal{E}_{k,n}(\gamma, P) := \{\mathbf{M} : |\mathcal{P}_{k,n} \cap \mathcal{U}| = \Theta(C_n)\}, \quad (4.A.3)$$

we have

$$\mathbb{P}(\mathbf{M} \in \mathcal{E}_{k,n}) \geq 1 - 2 \exp(-c_{n,1} C_n), \quad (4.A.4)$$

where $c_{n,0} > 0$ is some constant.

Proof. Under the random masking strategy, given $k \in [K]$ and $n \in [N]$, $Y_{k,n} = |\mathcal{U} \cap \mathcal{P}_{k,n}|$ follows the hypergeometric distribution, i.e. $Y_{k,n} \sim \text{Hyper}((1-\gamma)P, C_n, P)$. Then by tail bounds, for $t > 0$,

we have:

$$\mathbb{P}[|Y_{k,n} - (1 - \gamma)C_n| > t] \leq 2 \exp\left(-\frac{t^2}{4(1 - \gamma)C_n + 2t}\right)$$

Letting $t = \Theta(C_n)$, we have

$$\mathbb{P}[Y_{k,n} = \Theta(C_n)] \geq 1 - 2e^{-c_{n,1}C_n}.$$

□

We further have the following fact, which will be useful for proving the property of loss objective in the next subsection.

Lemma 4.A.5. *For $k \in [K]$ and $n \in [N]$, we have*

$$\mathbb{P}(|\mathcal{U} \cap \mathcal{P}_{k,n}| = 0) \leq \exp(-c_{n,0}C_n), \quad (4.A.5)$$

where $c_{n,0} > 0$ is some constant.

Proof. By the form of probability density for $\text{Hyper}((1 - \gamma)P, C_n, P)$, we have

$$\mathbb{P}(|\mathcal{U} \cap \mathcal{P}_{k,n}| = 0) = \frac{\binom{C_n}{0} \binom{(P - C_n)}{(1 - \gamma)P}}{\binom{P}{(1 - \gamma)P}} \leq \gamma^{C_n} = \exp(-c_{n,0}C_n).$$

□

Properties of loss functions Recall the training and regional reconstruction loss we consider are given by:

$$\mathcal{L}(Q) := \frac{1}{2} \mathbb{E} \left[\sum_{\mathbf{p} \in \mathcal{P}} \mathbb{1}\{\mathbf{p} \in \mathcal{M}\} \|[F(\mathbf{M}(X); Q, E)]_{\mathbf{p}} - X_{\mathbf{p}}\|^2 \right], \quad (4.A.6)$$

$$\mathcal{L}_{\mathbf{p}}(Q) = \frac{1}{2} \mathbb{E} \left[\mathbb{1}\{\mathbf{p} \in \mathcal{M}\} \|[F(\mathbf{M}(X), E)]_{\mathbf{p}} - X_{\mathbf{p}}\|^2 \right]. \quad (4.A.7)$$

In this part, we will present several important lemmas for such a training objective. We first single out the following lemma, which connects the loss form with the attention score.

Lemma 4.A.6 (Loss Calculation). *The population loss $L(Q)$ can be decomposed into the following form:*

$$\begin{aligned} \mathcal{L}(Q) &= \sum_{\mathbf{p} \in \mathcal{P}} \mathcal{L}_{\mathbf{p}}(Q), \text{ where} \\ \mathcal{L}_{\mathbf{p}}(Q) &= \frac{1}{2} \sum_{k=1}^K \mathbb{E} \left[\mathbb{1}\{\mathbf{p} \in \mathcal{M}, k_X = k\} \cdot \left(z_{a_{k,\mathbf{p}}}^2 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a_{k,\mathbf{p}}}}^{(t)} \right)^2 + \sum_{a \neq a_{k,\mathbf{p}}} z_a^2 \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)} \right)^2 \right) \right]. \end{aligned}$$

Proof.

$$\begin{aligned}
\mathcal{L}_{\mathbf{p}}(Q) &= \frac{1}{2} \sum_{k=1}^K \mathbb{E} \left[\mathbb{1}\{\mathbf{p} \in \mathcal{M}, k_X = k\} \left\| [F(\mathcal{M}(X), E)]_{\mathbf{p}} - X_{\mathbf{p}} \right\|^2 \right] \\
&= \frac{1}{2} \sum_{k=1}^K \mathbb{E} \left[\mathbb{1}\{\mathbf{p} \in \mathcal{M}, k_X = k\} \left\| \sum_{m \in [N]} \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}} z_m v_{k,m} - z_{a_{k,\mathbf{p}}} v_{k,a_{k,\mathbf{p}}} \right\|^2 \right] \\
&\stackrel{(i)}{=} \frac{1}{2} \sum_{k=1}^K \mathbb{E} \left[\mathbb{1}\{\mathbf{p} \in \mathcal{M}, k_X = k\} \left(z_{a_{k,\mathbf{p}}}^2 \left(1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a_{k,\mathbf{p}}}} \right)^2 + \sum_{m \neq a_{k,\mathbf{p}}} z_m^2 \left(\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}} \right)^2 \right) \right],
\end{aligned}$$

where (i) follows since the features are orthogonal. $\square$

We then introduce some additional crucial notations for the loss objectives.

$$\mathcal{L}_{\mathbf{p}}^* = \min_{Q \in \mathbb{R}^{d \times d}} \mathcal{L}_{\mathbf{p}}(Q), \quad (4.A.8a)$$

$$\mathcal{L}_{\mathbf{p}}^{\text{low}} = \frac{1}{2} (\sigma_z^2 + \frac{L^2}{N-1}) \sum_{k \in [K]} \mathbb{P} \left(|\mathcal{U} \cap \mathcal{P}_{k,a_{k,\mathbf{p}}}| = 0 \right), \quad (4.A.8b)$$

$$\tilde{\mathcal{L}}_{\mathbf{p}}(Q) = \sum_{k=1}^K \tilde{\mathcal{L}}_{k,\mathbf{p}}(Q), \quad \text{where}$$

$$\tilde{\mathcal{L}}_{k,\mathbf{p}}(Q) = \frac{1}{2} \mathbb{E} \left[\mathbb{1}\{\mathbf{p} \in \mathcal{M}, k_X = k, \mathbf{M} \in \mathcal{E}_{k,z_{a_{k,\mathbf{p}}}}\} \left(z_{a_{k,\mathbf{p}}}^2 \left(1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a_{k,\mathbf{p}}}}^{(t)} \right)^2 + \sum_{a \neq a_{k,\mathbf{p}}} z_a^2 \left(\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)} \right)^2 \right) \right]. \quad (4.A.8c)$$

Here $\sigma_z^2 = \mathbb{E}[Z_n(X)^2]$. $\mathcal{L}_{\mathbf{p}}^*$ denotes the minimum value of the population loss in (4.A.7), and $\mathcal{L}_{\mathbf{p}}^{\text{low}}$ represents the unavoidable errors for $\mathbf{p} \in \mathcal{P}$, given that all the patches in $\mathcal{P}_{k,a_{k,\mathbf{p}}}$ are masked. We will show that $\mathcal{L}_{\mathbf{p}}^{\text{low}}$ serves as a lower bound for $\mathcal{L}_{\mathbf{p}}^*$, and demonstrate that the network trained with GD will attain nearly zero error compared to $\mathcal{L}_{\mathbf{p}}^{\text{low}}$. Our convergence will be established by the sub-optimality gap with respect to $\mathcal{L}_{\mathbf{p}}^{\text{low}}$, which necessarily implies the convergence to $\mathcal{L}_{\mathbf{p}}^*$. (It also implies $\mathcal{L}_{\mathbf{p}}^* - \mathcal{L}_{\mathbf{p}}^{\text{low}}$ is small.)

Lemma 4.A.7. For $\mathcal{L}_{\mathbf{p}}^*$ and $\mathcal{L}_{\mathbf{p}}^{\text{low}}$ defined in (4.A.8a) and (4.A.8b), respectively, we have $\mathcal{L}_{\mathbf{p}}^{\text{low}} \leq \mathcal{L}_{\mathbf{p}}^*$ and they are both on the order of $\Theta \left(\exp \left(- (c_1 P^{\kappa_c} + \mathbb{1}\{1 \notin \cup_{k \in [K]} \{a_{k,\mathbf{p}}\}\} c_2 P^{\kappa_s}) \right) \right)$ where $c_1, c_2 > 0$ are some constants.

Proof. We first prove $\mathcal{L}_{\mathbf{p}}^{\text{low}} \leq \mathcal{L}_{\mathbf{p}}^*$:

$$\mathcal{L}_{\mathbf{p}}^* = \min_{Q \in \mathbb{R}^{d \times d}} \frac{1}{2} \sum_{k=1}^K \mathbb{E} \left[\mathbb{1}\{\mathbf{p} \in \mathcal{M}, k_X = k\} \cdot \left(z_{a_{k,\mathbf{p}}}^3 \left(1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a_{k,\mathbf{p}}}}^{(t)} \right)^2 + \sum_{a \neq a_{k,\mathbf{p}}} z_a^2 z_{a_{k,\mathbf{p}}} \left(\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)} \right)^2 \right) \right]$$

$$\geq \min_{Q \in \mathbb{R}^{d \times d}} \frac{1}{2} \sum_{k=1}^K \mathbb{E} \left[\mathbb{1}\{\mathbf{p} \in \mathcal{M}, k_X = k\} \mathbb{1}\{|\mathcal{U} \cap \mathcal{P}_{k,a_{k,p}}| = 0\} \cdot \left(z_{a_{k,p}}^3 \left(1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a_{k,p}}}^{(t)} \right)^2 + \sum_{a \neq a_{k,p}} z_a^2 z_{a_{k,p}} \left(\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)} \right)^2 \right) \right].$$

Notice that when all patches in $\mathcal{P}_{k,a_{k,p}}$ are masked, $\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a_{k,p}}}^{(t)} = 0$. Moreover,

$$\sum_{m \neq a_{k,p}} z_m^2 \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \geq \frac{L^2}{N-1}$$

by Cauchy–Schwarz inequality. Thus

$$\mathcal{L}_{\mathbf{p}}^{\star} \geq \frac{1}{2} \sum_{k=1}^K (\sigma_z^2 + \frac{L^2}{N-1}) \mathbb{P}(|\mathcal{U} \cap \mathcal{P}_{k,a_{k,p}}| = 0) = \mathcal{L}_{\mathbf{p}}^{\text{low}}.$$

Moreover, Lemma 4.A.5 immediately gives

$$\mathcal{L}_{\mathbf{p}}^{\text{low}} = \Theta \left(\exp \left(- (c_1 P^{\kappa_c} + \mathbb{1}\{1 \notin \cup_{k \in [K]} \{a_{k,p}\}\} c_2 P^{\kappa_s}) \right) \right).$$

It remains only to show

$$\mathcal{L}_{\mathbf{p}}^{\star} = O \left(\exp \left(- (c_1 P^{\kappa_c} + \mathbb{1}\{1 \notin \cup_{k \in [K]} \{a_{k,p}\}\} c_2 P^{\kappa_s}) \right) \right).$$

This follows directly by choosing $Q = \sigma I_d$ for some sufficiently large σ , so the details are omitted. $\square$

Lemma 4.A.8. *Given $\mathbf{p} \in \mathcal{P}$, for any Q , we have*

$$\tilde{\mathcal{L}}_{\mathbf{p}}(Q) \leq L_{\mathbf{p}}(Q) - \mathcal{L}_{\mathbf{p}}^{\text{low}} \leq \tilde{\mathcal{L}}_{\mathbf{p}}(Q) + O \left(\exp \left(- (c_3 P^{\kappa_c} + \mathbb{1}\{1 \notin \cup_{k \in [K]} \{a_{k,p}\}\} c_4 P^{\kappa_s}) \right) \right),$$

where $c_3, c_4 > 0$ are some constants.

Proof. The lower bound is directly obtained by the definition and thus we only prove the upper bound.

$$\begin{aligned} & L_{\mathbf{p}}(Q) - \tilde{\mathcal{L}}_{\mathbf{p}}(Q) \\ &= \frac{1}{2} \sum_{k=1}^K \mathbb{E} \left[\mathbb{1}\{\mathbf{p} \in \mathcal{M}, k_X = k, \mathbf{M} \in \mathcal{E}_{k, z_{a_{k,p}}}^c\} \cdot \left(z_{a_{k,p}}^2 \left(1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a_{k,p}}}^{(t)} \right)^2 + \sum_{a \neq a_{k,p}} z_a^2 \left(\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)} \right)^2 \right) \right] \\ &\leq \sum_{k=1}^K U^2 \mathbb{P}(\mathbf{M} \in \mathcal{E}_{k, z_{a_{k,p}}}^c) \end{aligned}$$

$$\leq O\left(\exp\left(-\left(c_3 P^{\kappa_c} + \mathbb{1}\{1 \notin \cup_{k \in [K]} \{a_{k,p}\}\} c_4 P^{\kappa_s}\right)\right)\right),$$

where the last inequality follows from Lemma 4.A.4. $\square$

4.A.2 Overall induction hypotheses and proof outline for MAE

Our main proof utilizes the induction hypotheses. In this section, we introduce the main induction hypotheses for the positive and negative information gaps, which will later be proven to be valid throughout the entire learning process.

Positive information gap We first state our induction hypothesis for the case that the information gap Δ is positive.

Induction Hypothesis 4.A.1. For $t \leq T$, given $\mathbf{p}, \mathbf{q} \in \mathcal{P}$, for $k \in [K]$, the following holds

- a. $\Phi_{\mathbf{p} \rightarrow v_{k,a_{k,p}}}^{(t)}$ is monotonically increasing, and $\Phi_{\mathbf{p} \rightarrow v_{k,a_{k,p}}}^{(t)} \in [0, \tilde{O}(1)]$;
- b. if $a_{k,p} \neq 1$, then $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}$ is monotonically decreasing and $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} \in [-\tilde{O}(1), 0]$;
- c. $|\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}| = \tilde{O}(\frac{1}{p^{1-\kappa_s}})$ for $m \notin \{1\} \cup \{a_{k,p}\}$;
- d. for $\mathbf{q} \neq \mathbf{p}$, $\Upsilon_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = \tilde{O}(\frac{1}{p^{\kappa_s}})$;
- e. $\Upsilon_{\mathbf{p} \rightarrow \mathbf{p}}^{(t)} = \tilde{O}(\frac{1}{p})$.

Negative information gap Now we turn to the case that $\Delta \leq -\Omega(1)$.

Induction Hypothesis 4.A.2. For $t \leq T$, given $\mathbf{p}, \mathbf{q} \in \mathcal{P}$, for $k \in [K]$, the following holds

- a. $\Phi_{\mathbf{p} \rightarrow v_{k,a_{k,p}}}^{(t)}$ is monotonically increasing, and $\Phi_{\mathbf{p} \rightarrow v_{k,a_{k,p}}}^{(t)} \in [0, \tilde{O}(1)]$;
- b. if $a_{k,p} \neq 1$, then $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}$ is monotonically decreasing and $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} \in [-\tilde{O}(\frac{1}{p-\Delta}), 0]$;
- c. $|\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}| = \tilde{O}(\frac{1}{p^{1-\kappa_s}})$ for $m \notin \{1\} \cup \{a_{k,p}\}$;
- d. for $\mathbf{q} \neq \mathbf{p}$, $\Upsilon_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = \tilde{O}(\frac{1}{p^{\kappa_s}})$;
- e. $\Upsilon_{\mathbf{p} \rightarrow \mathbf{p}}^{(t)} = \tilde{O}(\frac{1}{p})$.

Proof outline In both settings, we can classify the process through which transformers learn the feature attention correlation $\Phi_{\mathbf{p} \rightarrow v_{k,a_{k,p}}}^{(t)}$ into two distinct scenarios. These scenarios hinge on the spatial relation of the area $\mathbf{p}$ within the context of the k -th partition $\mathcal{D}_k$, specifically, whether $\mathbf{p}$ is located in the global area of the k -th cluster, i.e. whether $a_{k,p} = 1$. The learning dynamics exhibit different behaviors of learning the local FP correlation in the local area with different Δ , while the behaviors for features located in the global area are very similar, unaffected by the value of Δ . Therefore, through Sections 4.A.3 to 4.A.5, we delve into the learning phases and provide technical proofs for the local area with $\Delta \geq \Omega(1)$, local area with $\Delta \leq -\Omega(1)$ and the global area respectively. Finally, we will put this analysis together to prove that the Induction Hypothesis 4.A.1 (resp. Induction Hypothesis 4.A.2) holds during the entire training process, thereby validating the main theorems in Section 4.A.

4.A.3 Analysis for the local area with positive information gap

In this section, we focus on a specific patch $\mathbf{p} \in \mathcal{P}$ with the k -th cluster for $k \in [K]$, and present the analysis for the case that $X_{\mathbf{p}}$ is located in the local area for the k -th cluster, i.e. $a_{k,p} > 1$. We will analyze the case that $\Delta \geq \Omega(1)$. Throughout this section, we denote $a_{k,p} = n$ for simplicity. We will analyze the convergence of the training process via two phases of dynamics. At the beginning of each phase, we will establish an induction hypothesis, which we expect to remain valid throughout that phase. Subsequently, we will analyze the dynamics under such a hypothesis within the phase, aiming to provide proof of the hypothesis by the end of the phase.

Phase I, stage 1 In this section, we shall discuss the initial stage of phase I. Firstly, we present the induction hypothesis in this stage.

We define the stage 1 of phase I as all iterations $t \leq T_1$, where

$$T_1 \triangleq \max \left\{ t : \Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \geq -\frac{1}{U} \left(\frac{\Delta}{2} - 0.01 \right) \log(P) \right\}.$$

We state the following induction hypotheses, which will hold throughout this period:

Induction Hypothesis 4.A.3. For each $0 \leq t \leq T_1$, $\mathbf{q} \in \mathcal{P} \setminus \{\mathbf{p}\}$, the following holds:

- $\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}$ is monotonically increasing, and $\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \in [0, O\left(\frac{(\frac{\Delta}{2} - 0.01) \log(P)}{P^{0.02}}\right)]$;
- $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}$ is monotonically decreasing and $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} \in [-\frac{1}{U} (\frac{\Delta}{2} - 0.01) \log(P), 0]$;
- $|\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}| = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P^{1-\kappa_S}}\right)$ for $m \neq 1, n$;
- $\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{C_n}\right)$ for $a_{k,\mathbf{q}} = n$, $|\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{p}}^{(t)}| = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right)$;
- $|\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}| = O\left(\frac{|\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}|}{C_1}\right) + O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right)$ for $a_{k,\mathbf{q}} = 1$;
- $|\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}| = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right)$ for $a_{k,\mathbf{q}} \neq 1, n$.

Properties of attention scores. We first introduce several properties of the attention scores if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.3 hold.

Lemma 4.A.9. For $n > 1$, if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.3 hold at iteration $t \leq T_1$, then the following holds

- $1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} \geq \Omega(1)$;
- If $\mathbf{M} \in \mathcal{E}_{k,n}$, $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} = \Theta\left(\frac{1}{P^{1-\kappa_S}}\right)$;
- Moreover, if $\mathbf{M} \in \mathcal{E}_{k,1}$, we have $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} = \Omega\left(\frac{1}{P^{\frac{1-\kappa_S}{2} - 0.01}}\right)$;
- For $\mathbf{q} \in \mathcal{M} \cap (\mathcal{P}_{k,n} \cup \mathcal{P}_{k,1})$, $\text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = O\left(\frac{1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}}{P}\right)$.

Lemma 4.A.10. For $n > 1$, if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.3 hold at iteration $t \leq T_1$, then for $m \neq n, 1$, the following holds:

1. For any $\mathbf{q} \in \mathcal{P}_{k,m}$, $\text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} \leq O\left(\frac{1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}}{P}\right)$.
2. Moreover, $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \leq O\left(\frac{1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}}{N}\right)$.

The above properties can be easily verified through direct calculations by using the definition in (4.2.6) and conditions in Induction Hypothesis 4.A.3, which are omitted here for brevity.

Bounding the gradient updates for FP correlations. We have the following set of lemmas.

Lemma 4.A.11. For $n > 1$, if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.3 hold at iteration $0 \leq t \leq T_1$, then $\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \geq 0$ and satisfies:

$$\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} = \Theta\left(\frac{C_n}{P}\right) = \Theta\left(\frac{1}{P^{1-\kappa_s}}\right).$$

Proof. By Lemma 4.5.2, we have

$$\begin{aligned} & \alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \\ &= \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathbf{p} \in \mathcal{M}\} \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \cdot \left(z_n^3 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right)^2 + \sum_{m \neq n} z_m^2 z_n \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \right)^2 \right) \right] \\ &= \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathcal{E}_{k,n} \cap \mathbf{p} \in \mathcal{M}\} \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \cdot \left(z_n^3 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right)^2 + \sum_{m \neq n} z_m^2 z_n \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \right)^2 \right) \right] \\ &\quad + \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathcal{E}_{k,n}^c \cap \mathbf{p} \in \mathcal{M}\} \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \cdot \left(z_n^3 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right)^2 + \sum_{m \neq n} z_m^2 z_n \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \right)^2 \right) \right] \\ &\leq \mathbb{P}(\mathbf{M} \in \mathcal{E}_{k,n}) \\ &\quad \cdot \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathbf{p} \in \mathcal{M}\} \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \cdot \left(z_n^3 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right)^2 + \sum_{m \neq n} z_m^2 z_n \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \right)^2 \right) \middle| \mathcal{E}_{k,n} \right] \\ &\quad + O(1) \cdot \mathbb{P}(\mathbf{M} \in \mathcal{E}_{k,n}^c) \\ &\leq O\left(\frac{C_n}{P}\right) + O(\exp(-c_{n,1}C_n)) \leq O\left(\frac{C_n}{P}\right), \end{aligned}$$

where the second inequality invokes Lemma 4.A.9 and Lemma 4.A.4, and the last inequality is due to $\exp(-c_{n,1}C_n) \ll \frac{C_n}{P}$. Similarly, we can show that $\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \geq \Omega\left(\frac{C_n}{P}\right)$. $\square$

Lemma 4.A.12. For $n > 1$, if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.3 hold at iteration $0 \leq t \leq T_1$, then $\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} < 0$ and satisfies

$$|\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}| \geq \Omega\left(\frac{1}{P^{2(\frac{1-\kappa_s}{2} - 0.01)}}\right) = \Omega\left(\frac{1}{P^{0.98-\kappa_s}}\right).$$

Proof. We first single out the following fact:

$$\begin{aligned}
& -z_1 z_n^2 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}\right) \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} - z_1^3 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}\right) \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} + \sum_{a \neq 1,n} z_a^2 z_1 \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)}\right)^2 \\
& \leq z_1 \left(\max_{a \neq 1,n} z_a^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)} - z_n^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} - z_1^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} \right) (1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}) \\
& = -z_1 (1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}) \left(z_n^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} + z_1^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} - \max_{a \neq 1,n} z_a^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)} \right). \tag{4.A.9}
\end{aligned}$$

Therefore, by Lemma 4.5.1, we have

$$\begin{aligned}
\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} & \leq \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathcal{E}_{k,1} \cap \mathbf{p} \in \mathcal{M}\} \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} \cdot \right. \\
& \quad \left. \left(-z_1 (1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}) \left(z_n^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} + z_1^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} - \max_{a \neq 1,n} z_a^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)} \right) \right) \right] \\
& \quad + \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathcal{E}_{k,1}^c \cap \mathbf{p} \in \mathcal{M}\} \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} \cdot \sum_{a \neq 1,n} z_a^2 z_1 \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)}\right)^2 \right] \\
& \leq \mathbb{P}(\mathbf{M} \in \mathcal{E}_{k,1}) \cdot \left(-(\Omega(1) \cdot \Omega(\frac{1}{p^{2 \times (\frac{1-\kappa_s}{2} - 0.01)}})) \right) + O(1) \cdot \mathbb{P}(\mathbf{M} \in \mathcal{E}_{k,1}^c) \\
& \leq -\Omega\left(\frac{1}{p^{2 \times (\frac{1-\kappa_s}{2} - 0.01)}}\right) = -\Omega\left(\frac{1}{p^{0.98-\kappa_s}}\right),
\end{aligned}$$

where the second inequality invokes Lemma 4.A.9 and the last inequality comes from Lemma 4.A.4. $\square$

Lemma 4.A.13. *At each iteration $t \leq T_1$, if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.3 hold, then for any $m > 1$ with $m \neq n$, the following holds*

$$|\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}| \leq O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{N}\right) = O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{p^{1-\kappa_s}}\right).$$

Proof. By Lemma 4.5.1, for $m \neq n$, we have

$$\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)} \leq \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathbf{p} \in \mathcal{M}\} \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \cdot \left(\sum_{a \neq m,n} z_a^2 z_m \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)}\right)^2 \right) \right], \tag{4.A.10}$$

$$\begin{aligned}
-\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)} & \leq \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathbf{p} \in \mathcal{M}\} \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \cdot \left(z_m z_n^2 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}\right) \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right. \right. \\
& \quad \left. \left. + z_m^3 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)}\right) \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \right) \right]. \tag{4.A.11}
\end{aligned}$$

For (4.A.10), we have

$$\begin{aligned}
\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)} &\leq \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathcal{E}_{k,1} \cap \mathcal{E}_{k,n} \cap \mathbf{p} \in \mathcal{M}\} \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \cdot \left(\sum_{a \neq m,n} z_a^2 z_m \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)} \right)^2 \right) \right] \\
&\quad + \mathbb{E} \left[\mathbb{1}\{k_X = k, (\mathcal{E}_{k,1} \cap \mathcal{E}_{k,n})^c \cap \mathbf{p} \in \mathcal{M}\} \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \cdot \left(\sum_{a \neq m,n} z_a^2 z_m \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)} \right)^2 \right) \right] \\
&\leq \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathcal{E}_{k,1} \cap \mathcal{E}_{k,n} \cap \mathbf{p} \in \mathcal{M}\} O \left(\frac{1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}}{N} \right) \right. \\
&\quad \left. \cdot \left(z_1^2 z_m \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} \right)^2 + O \left(\frac{1}{N} \right) \right) \right] + O(1) \cdot \mathbb{P}(\mathbf{M} \in (\mathcal{E}_{k,1} \cap \mathcal{E}_{k,n})^c) \\
&\leq O \left(\frac{|\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}|}{N} \right) + O(1) \cdot \mathbb{P}(\mathbf{M} \in (\mathcal{E}_{k,1} \cap \mathcal{E}_{k,n})^c) \\
&\leq O \left(\frac{|\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}|}{p^{1-\kappa_s}} \right),
\end{aligned}$$

where the second inequality is due to Lemma 4.A.10, the last inequality follows from Lemma 4.A.12 and Lemma 4.A.4.

On the other hand, for (4.A.11), we can use the similar argument by invoking Lemma 4.A.10 and Lemma 4.A.11, and thus obtain

$$-\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)} \leq O \left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{p^{1-\kappa_s}} \right).$$

Putting them together, we have

$$|\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}| \leq O \left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{p^{1-\kappa_s}} \right).$$

□

Bounding the gradient updates for positional correlations. We have the following set of lemmas.

Lemma 4.A.14. For $n > 1$, if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.3 hold at iteration $0 \leq t \leq T_1$, then for $\mathbf{q} \in \mathcal{P} \setminus \{\mathbf{p}\}$ and $a_{k,\mathbf{q}} = n$, we have $\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)} \geq 0$ and satisfies:

$$\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = \Theta \left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{C_n} \right).$$

Furthermore, we have $|\beta_{k,\mathbf{p} \rightarrow \mathbf{p}}^{(t)}| = O \left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{p} \right).$

Proof. By Lemma 4.5.2, for $\mathbf{q} \in \mathcal{P}_{k,n}$ with $\mathbf{q} \neq \mathbf{p}$, we have

$$\begin{aligned} \beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = & \underbrace{\mathbb{E} \left[\mathbb{1}\{k_X = k, \mathbf{p} \in \mathcal{M}, \mathbf{q} \in \mathcal{U}\} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} \cdot \left(z_n^2 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right)^2 + \sum_{m \neq n} z_m^2 \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \right)^2 \right) \right]}_{H_1} \\ & + \underbrace{\mathbb{E} \left[\mathbb{1}\{k_X = k, \mathbf{p} \in \mathcal{M}, \mathbf{q} \in \mathcal{M}\} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} \cdot \left(-z_n^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right) \right) \right]}_{H_2} \\ & + \underbrace{\mathbb{E} \left[\mathbb{1}\{k_X = k, \mathbf{p} \in \mathcal{M}, \mathbf{q} \in \mathcal{M}\} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} \cdot \left(\sum_{m \neq n} z_m^2 \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \right)^2 \right) \right]}_{H_3}. \end{aligned}$$

Firstly, for H_1 , notice that

$$\begin{aligned} (C_n - 1)H_1 &= \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathbf{p} \in \mathcal{M}\} \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \cdot \left(z_n^2 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right)^2 + \sum_{m \neq n} z_m^2 \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \right)^2 \right) \right] \\ &= \Theta(\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}). \end{aligned}$$

For H_2 , since $\mathbf{p}, \mathbf{q} \in \mathcal{M}$, by Lemma 4.A.9, we can upper bound $\text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)}$ by $O\left(\frac{1}{P}\right)$, thus

$$-H_2 \leq \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathbf{p} \in \mathcal{M}\} O\left(\frac{1}{P}\right) \cdot \left(z_n^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right) \right) \right] \leq O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{P}\right).$$

Further notice that H_3 can be upper bounded by $O(H_1)$, putting it together, we have

$$\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = \Theta\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{C_n}\right).$$

Turning to $\beta_{k,\mathbf{p} \rightarrow \mathbf{p}}^{(t)}$, when $\mathbf{q} = \mathbf{p}$, it follows

$$\begin{aligned} \beta_n^{(t)} &= \underbrace{\mathbb{E} \left[\mathbb{1}\{k_X = k, \mathbf{p} \in \mathcal{M}\} \text{attn}_{\mathbf{p} \rightarrow \mathbf{p}}^{(t)} \cdot \left(-z_n^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right) \right) \right]}_{J_2} \\ &+ \underbrace{\mathbb{E} \left[\mathbb{1}\{k_X = k, \mathbf{p} \in \mathcal{M}\} \text{attn}_{\mathbf{p} \rightarrow \mathbf{p}}^{(t)} \cdot \left(\sum_{m \neq n} z_m^2 \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \right)^2 \right) \right]}_{J_3}. \end{aligned}$$

We can bound J_2 in a similar way as H_2 . Thus, we only focus on further bounding J_3 :

$$\begin{aligned} J_3 &\leq \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathbf{p} \in \mathcal{M}\} O\left(\frac{1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}}{P}\right) \cdot \left(\sum_{m \neq n} z_m^2 \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)}\right)^2\right) \right] \\ &\leq O\left(\frac{|\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}|}{P}\right). \end{aligned}$$

where the first inequality holds by invoking Lemma 4.A.9 and the last inequality follows similar arguments as analysis for (4.A.10). $\square$

Lemma 4.A.15. *For $n > 1$, if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.3 hold at iteration $0 \leq t \leq T_1$, then for $\mathbf{q} \in \mathcal{P} \setminus \{\mathbf{p}\}$ and $a_{k,\mathbf{q}} = 1$, we have $\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}$ satisfies:*

$$|\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}| = O\left(\frac{|\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}|}{P}\right) + O\left(\frac{|\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}|}{C_1}\right).$$

Proof. By Lemma 4.5.2, for $\mathbf{q} \in \mathcal{P}_{k,1}$, we have

$$\begin{aligned} \beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)} &= \\ &- \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathbf{p} \in \mathcal{M}, \mathbf{q} \in \mathcal{U}\} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} \cdot \right. \\ &\quad \left. \left(z_1^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} (1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}) + z_n^2 (1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}) \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} - \sum_{a \neq 1, n} z_a^2 \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)}\right)^2 \right) \right] \\ &\quad \underbrace{- \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathbf{p} \in \mathcal{M}, \mathbf{q} \in \mathcal{M}\} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} \cdot \left(z_n^2 (1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}) \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right) \right]}_{G_2} \\ &\quad \underbrace{+ \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathbf{p} \in \mathcal{M}, \mathbf{q} \in \mathcal{M}\} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} \cdot \left(\sum_{a \neq n} z_a^2 \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)}\right)^2 \right) \right]}_{G_3}. \end{aligned} \tag{4.A.12}$$

For (4.A.12) denoted as G_1 , following the direct calculations, we have

$$-(C_1 - 1)G_1 = \Theta(\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)})$$

We can further bound G_2 and G_3 in a similar way as H_2 and H_3 in Lemma 4.A.14 and thus obtain

$$-G_2 \leq O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{P}\right),$$

$$G_3 \leq O\left(\frac{|\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}|}{P}\right),$$

which completes the proof. $\square$

Lemma 4.A.16. *For $n > 1$, if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.3 hold at iteration $0 \leq t \leq T_1$, then for $\mathbf{q} \in \mathcal{P} \setminus \{\mathbf{p}\}$ and $n \neq a_{k,\mathbf{q}}$, $\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}$ satisfies:*

$$|\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}| = O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right).$$

Proof. By Lemma 4.5.2, for $\mathbf{q} \in \mathcal{P}_{k,m}$, we have

$$\begin{aligned} \beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = & -\mathbb{E} \left[\mathbb{1}\{k_X = k, \mathbf{p} \in \mathcal{M}, \mathbf{q} \in \mathcal{U}\} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} \right. \\ & \left. \left(z_m^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} (1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)}) + z_n^2 (1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}) \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} - \sum_{a \neq n,m} z_a^2 \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)} \right)^2 \right) \right] \\ & \underbrace{-\mathbb{E} \left[\mathbb{1}\{k_X = k, \mathbf{p} \in \mathcal{M}, \mathbf{q} \in \mathcal{M}\} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} \cdot \left(z_n^2 (1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}) \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right) \right]}_{I_2} \\ & \underbrace{+\mathbb{E} \left[\mathbb{1}\{k_X = k, \mathbf{p} \in \mathcal{M}, \mathbf{q} \in \mathcal{M}\} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} \cdot \left(\sum_{a \neq n} z_a^2 \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)} \right)^2 \right) \right]}_{I_3}. \end{aligned} \quad (4.A.13)$$

It follows that (4.A.13) can be upper bounded by

$$\begin{aligned} O\left(\frac{|\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}|}{C_m}\right) &= O\left(\frac{|\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} - \alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}|}{NC_m}\right) \\ &= O\left(\frac{|\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} - \alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}|}{P}\right), \end{aligned}$$

where the first equality follows from Lemma 4.A.13. The terms I_2 and I_3 can be bounded similarly to G_2 and G_3 ; the details are omitted. $\square$

At the end of Phase I, stage 1. We have the following lemma.

Lemma 4.A.17. *For $n > 1$, if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.3 hold for all $0 \leq t \leq T_1 = O\left(\frac{\log(P)P^{0.98-\kappa_S}}{\eta}\right)$, At iteration $t = T_1 + 1$, we have*

$$a. \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(T_1+1)} \leq -\frac{1}{U} \left(\frac{\Delta}{2} - 0.01 \right) \log(P);$$

$$b. \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(T_1+1)} = O\left(\frac{1}{p^{(1-\kappa_c) + \frac{1}{U}(\frac{\Delta}{2} - 0.01)}}\right).$$

Proof. By comparing Lemma 4.A.11 and Lemma 4.A.12, we have $|\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}| \gg \alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}$. Then the existence of $T_{1,k} = O\left(\frac{\log(P)p^{0.98-\kappa_s}}{\eta}\right)$ directly follows from Lemma 4.A.12. $\square$

Phase I, stage 2 During stage 1, $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}$ significantly decreases to decouple the FP correlations with the global feature, resulting in a decrease in $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}$, while other $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}$ with $m > 1$ remain approximately at the order of $O\left(\frac{1}{p^{1-\kappa_s}}\right) (\Theta\left(\frac{1}{p^{1-\kappa_s}}\right))$. By the end of phase I, $(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)})^2$ decreases to $O(\frac{1}{p^{1.96-2\kappa_s}})$, leading to a decrease in $|\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}|$ as it approaches towards $\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}$. At this point, stage 2 begins. Shortly after entering this phase, the prior dominant role of the decrease of $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}$ in learning dynamics diminishes as $|\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}|$ reaches the same order of magnitude as $\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}$.

We define stage 2 of phase I as all iterations $T_1 < t \leq \tilde{T}_1$, where

$$\tilde{T}_1 \triangleq \max \left\{ t > T_1 : \Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} \leq \left(\frac{\Delta}{2L} + \frac{0.01}{L} + \frac{c_1^*(1-\kappa_s)}{U} \right) \log(P) \right\}$$

for some small constant $c_1^* > 0$.

For computational convenience, we make the following assumptions for κ_c and κ_s , which can be easily relaxed with the cost of additional calculations:

$$\frac{\Delta}{2} \left(\frac{1}{L} - \frac{1}{U} \right) + \frac{0.01}{L} + \frac{0.01}{U} \leq \frac{c_0^*(1-\kappa_s)}{U}, \quad (4.A.14a)$$

$$(1 - \frac{c_1^*L}{U})(1 - \kappa_s) \leq (1 - \kappa_c) + \frac{U}{L} \left(\frac{\Delta}{2} + 0.01 \right). \quad (4.A.14b)$$

Here c_0^* is some small constant. We state the following induction hypotheses, which will hold throughout this period.

Induction Hypothesis 4.A.4. For each $T_1 < t \leq \tilde{T}_1$, $\mathbf{q} \in \mathcal{P} \setminus \{\mathbf{p}\}$, the following holds:

- a. $\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}$ is monotonically increasing, and $\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \in [0, \frac{c_0^* + c_1^*}{U} \log(P)]$;
- b. $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}$ is monotonically decreasing, with

$$\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} \in \left[-\frac{1}{L} \left(\frac{\Delta}{2} + 0.01 \right) \log(P), -\frac{1}{U} \left(\frac{\Delta}{2} - 0.01 \right) \log(P) \right];$$

- c. $|\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}| = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{p^{1-\kappa_s}}\right)$ for $m \neq 1, n$;
- d. $\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{C_n}\right)$ for $a_{k,\mathbf{q}} = n$, $|\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{p}}^{(t)}| = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right)$;
- e. $|\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}| = O\left(\frac{|\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}|}{C_1}\right) + O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right)$ for $a_{k,\mathbf{q}} = 1$.;
- f. $|\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}| = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right)$ for $a_{k,\mathbf{q}} \neq 1, n$.

Property of attention scores. We first single out several properties of attention scores that will be used for the proof of Induction Hypothesis 4.A.4.

Lemma 4.A.18. *if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.4 hold at iteration $T_1 + 1 \leq t \leq \tilde{T}_1$, then the following holds*

1. $1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} \geq \Omega(1)$;
2. if $\mathbf{M} \in \mathcal{E}_{k,n}$, $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \in \left[\Omega\left(\frac{1}{p^{1-\kappa_s}}\right), O\left(\frac{1}{p^{(1-\kappa_s^* - \kappa_0^*)(1-\kappa_s)}}\right) \right]$;
3. Moreover, $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} = O\left(\frac{1}{p^{(1-\kappa_c) + \frac{1}{L}(\frac{1}{2} - 0.01)}}\right)$; if $\mathbf{M} \in \mathcal{E}_{k,1}$, we have $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} = \Omega\left(\frac{1}{p^{(1-\kappa_c) + \frac{1}{L}(\frac{1}{2} + 0.01)}}\right)$;
4. for $\mathbf{q} \in \mathcal{M} \cap (\mathcal{P}_{k,n} \cup \mathcal{P}_{k,1})$, $\text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = O\left(\frac{1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}}{p}\right)$.

Lemma 4.A.19. *if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.4 hold at iteration $T_1 + 1 \leq t \leq \tilde{T}_1$, then for $m \neq n$, the following holds:*

1. for any $\mathbf{q} \in \mathcal{P}_{k,m}$, $\text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} \leq O\left(\frac{1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}}{p}\right)$;
2. Moreover, $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \leq O\left(\frac{1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}}{N}\right)$.

Bounding the gradient updates of FP correlations. We have the following set of lemma.

Lemma 4.A.20. *For $n > 1$, if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.4 hold at iteration $T_1 + 1 \leq t \leq \tilde{T}_1$, then $\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \geq 0$ and satisfies:*

$$\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} = \Omega\left(\frac{1}{p^{1-\kappa_s}}\right).$$

Proof. By Lemma 4.5.2, we have

$$\begin{aligned} & \alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \\ &= \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathbf{p} \in \mathcal{M}\} \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \cdot \left(z_n^3 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right)^2 + \sum_{m \neq n} z_m^2 z_n \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \right)^2 \right) \right] \\ &= \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathcal{E}_{k,n} \cap \mathbf{p} \in \mathcal{M}\} \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \cdot \left(z_n^3 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right)^2 + \sum_{m \neq n} z_m^2 z_n \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \right)^2 \right) \right] \\ &\quad + \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathcal{E}_{k,n}^c \cap \mathbf{p} \in \mathcal{M}\} \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \cdot \left(z_n^3 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right)^2 + \sum_{m \neq n} z_m^2 z_n \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \right)^2 \right) \right] \\ &\geq \mathbb{P}(\mathbf{M} \in \mathcal{E}_{k,n}) \\ &\quad \cdot \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathbf{p} \in \mathcal{M}\} \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \cdot \left(z_n^3 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right)^2 + \sum_{m \neq n} z_m^2 z_n \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \right)^2 \right) \middle| \mathcal{E}_{k,n} \right] \\ &\geq \Omega\left(\frac{C_n}{p}\right), \end{aligned}$$

where the last inequality invokes Lemma 4.A.18. □

Lemma 4.A.21. *For $n > 1$, if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.4 hold at iteration $T_1 + 1 \leq t \leq \tilde{T}_1$, then $\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} < 0$ and satisfies*

$$|\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}| \geq \Omega\left(\frac{1}{p^{2(1-\kappa_c)} + \frac{U}{L}(\Delta+0.02)}\right).$$

Proof. Following (4.A.9), we have

$$\begin{aligned} & -z_1 z_n^2 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}\right) \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} - z_1^3 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}\right) \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} + \sum_{a \neq 1,n} z_a^2 z_1 \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)}\right)^2 \\ & \leq -z_1 (1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}) \left(z_n^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} + z_1^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} - \max_{a \neq 1,n} z_a^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)} \right). \end{aligned}$$

Therefore, by Lemma 4.5.1, we obtain

$$\begin{aligned} \alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} & \leq \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathcal{E}_{k,1} \cap \mathbf{p} \in \mathcal{M}\} \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} \cdot \left(-z_1 (1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}) \right. \right. \\ & \quad \left. \left. \cdot (z_n^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} + z_1^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} - \max_{a \neq 1,n} z_a^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)}) \right) \right] \\ & \quad + \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathcal{E}_{k,1}^c \cap \mathbf{p} \in \mathcal{M}\} \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} \cdot \sum_{a \neq 1,n} z_a^2 z_1 \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)}\right)^2 \right] \\ & \leq \mathbb{P}(\mathbf{M} \in \mathcal{E}_{k,1}) \cdot \left(-\Omega(1) \cdot \Omega\left(\frac{1}{p^{2(1-\kappa_c)} + \frac{2U}{L}(\frac{\Delta}{2} + 0.01)}\right) \right) + O(1) \cdot \mathbb{P}(\mathbf{M} \in \mathcal{E}_{k,1}^c) \\ & \leq -\Omega\left(\frac{1}{p^{2(1-\kappa_c)} + \frac{U}{L}(\Delta+0.02)}\right), \end{aligned}$$

where the second inequality invokes Lemma 4.A.18 and the last inequality comes from Lemma 4.A.4. The upper bound can be obtained by using similar arguments and invoking the upper bound for $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}$ in Lemma 4.A.18. $\square$

Lemma 4.A.22. *For $n > 1$, if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.4 hold at iteration $T_1 + 1 \leq t \leq \tilde{T}_1$, then for any $m > 1$ with $m \neq n$, the following holds*

$$|\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}| \leq O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{p^{1-\kappa_s}}\right).$$

The proof is similar to Lemma 4.A.13, and thus omitted here.

Bounding the gradient updates of positional correlations. We then summarize the properties for gradient updates of positional correlations, which utilize the identical calculations as in Section 4.A.3.

Lemma 4.A.23. For $n > 1$, if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.4 hold at iteration $T_1 + 1 \leq t \leq \tilde{T}_1$, then

- a. if $a_{k,q} = n$ and $q \neq p$, $\beta_{k,p \rightarrow q}^{(t)} \geq 0$; $\beta_{k,p \rightarrow q}^{(t)} = \Theta\left(\frac{\alpha_{p \rightarrow v_{k,n}}^{(t)}}{C_n}\right)$ and $|\beta_n^{(t)}| = O\left(\frac{\alpha_{p \rightarrow v_{k,n}}^{(t)} - \alpha_{p \rightarrow v_{k,1}}^{(t)}}{P}\right)$.
- b. if $a_{k,q} = 1$, $|\beta_{k,p \rightarrow q}^{(t)}| = O\left(\frac{\alpha_{p \rightarrow v_{k,n}}^{(t)} - \alpha_{p \rightarrow v_{k,1}}^{(t)}}{P}\right) + O\left(\frac{|\alpha_{p \rightarrow v_{k,1}}^{(t)}|}{C_1}\right)$.
- c. if $a_{k,q} = m$ and $m \neq 1, n$, $|\beta_{k,p \rightarrow q}^{(t)}| = O\left(\frac{\alpha_{p \rightarrow v_{k,n}}^{(t)} - \alpha_{p \rightarrow v_{k,1}}^{(t)}}{P}\right)$.

End of Phase I, stage 2. We have the following lemma.

Lemma 4.A.24. Induction Hypothesis 4.A.4 holds for all iteration $T_1 + 1 \leq t \leq \tilde{T}_1 = T_1 + O\left(\frac{\log(P)P^{1-\kappa_s}}{\eta}\right)$, and at iteration $t = \tilde{T}_1 + 1$, we have

- a. $\Phi_{p \rightarrow v_{k,n}}^{(\tilde{T}_1+1)} \geq \frac{c_1^*(1-\kappa_s)\log(P)}{U}$;
- b. $\Phi_{p \rightarrow v_{k,1}}^{(\tilde{T}_1+1)} \geq -\left(\frac{\Delta}{2L} + \frac{0.01}{L}\right)\log(P)$.

Proof. The existence of $\tilde{T}_1 = T_1 + O\left(\frac{\log(P)P^{1-\kappa_s}}{\eta}\right)$ directly follows from Lemma 4.A.20 and Lemma 4.A.21. Moreover, since $\alpha_{p \rightarrow v_{k,1}}^{(t)} < 0$, then

$$\Phi_{p \rightarrow v_{k,n}}^{(\tilde{T}_1+1)} \leq \left(\frac{\Delta}{2L} + \frac{0.01}{L} + \frac{c_1^*(1-\kappa_s)}{U}\right)\log(P) - \frac{1}{U}\left(\frac{\Delta}{2} - 0.01\right) \leq \frac{(c_0^* + c_1^*)(1-\kappa_s)}{U}\log(P),$$

where the last inequality invokes (4.A.14a). Now suppose $\Phi_{p \rightarrow v_{k,n}}^{(\tilde{T}_1+1)} < \frac{c_1^*(1-\kappa_s)\log(P)}{U}$, then $\Phi_{p \rightarrow v_{k,1}}^{(\tilde{T}_1+1)} < -\left(\frac{\Delta}{2L} + \frac{0.01}{L}\right)\log(P)$. Denote the first time that $\Phi_{p \rightarrow v_{k,1}}^{(t)}$ reaches $-\left(\frac{\Delta}{2L} + \frac{0.001}{L}\right)\log(P)$ as $\tilde{T}$. Note that $\tilde{T} < \tilde{T}_1$ since $\alpha_{p \rightarrow v_{k,1}}^{(t)}$, the change of $\Phi_{p \rightarrow v_{k,1}}^{(t)}$, satisfies $|\alpha_{p \rightarrow v_{k,1}}^{(t)}| \ll \log(P)$. Then for $t \geq \tilde{T}$, the following holds:

- 1. $\text{Attn}_{p \rightarrow \mathcal{P}_{k,n}}^{(t)} \geq \Omega\left(\frac{1}{P^{1-\kappa_s}}\right)$;
- 2. $\text{Attn}_{p \rightarrow \mathcal{P}_{k,1}}^{(t)} \leq O\left(\frac{1}{P^{\frac{1-\kappa_s}{2} + 0.001}}\right)$.

Therefore,

$$\begin{aligned} |\alpha_{p \rightarrow v_{k,1}}^{(t)}| &\leq \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathcal{E}_{k,1} \cap \mathbf{p} \in \mathcal{M}\} \text{Attn}_{p \rightarrow \mathcal{P}_{k,1}}^{(t)} \right. \\ &\quad \left. z_1 \left(z_n^2 \text{Attn}_{p \rightarrow \mathcal{P}_{k,n}}^{(t)} (1 - \text{Attn}_{p \rightarrow \mathcal{P}_{k,n}}^{(t)}) + z_1^2 \text{Attn}_{p \rightarrow \mathcal{P}_{k,1}}^{(t)} (1 - \text{Attn}_{p \rightarrow \mathcal{P}_{k,1}}^{(t)}) \right) \right] \\ &\quad + \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathcal{E}_{k,1}^c \cap \mathbf{p} \in \mathcal{M}\} \text{Attn}_{p \rightarrow \mathcal{P}_{k,1}}^{(t)} \cdot \sum_{a \neq 1,n} z_1^2 z_a \left(\text{Attn}_{p \rightarrow \mathcal{P}_{k,a}}^{(t)} \right)^2 \right] \\ &\leq O\left(\frac{\alpha_{p \rightarrow v_{k,1}}^{(t)}}{P^{\frac{1-\kappa_s}{2} + 0.001}}\right) + \mathbb{P}(\mathcal{M} \in \mathcal{E}_{k,1}) \cdot \left(O(1) \cdot O\left(\frac{1}{P^{\frac{1-\kappa_s}{2} + 0.001}}\right) \right) + O(1) \cdot \mathbb{P}(\mathcal{M} \in \mathcal{E}_{k,1}^c) \end{aligned}$$

$$\leq O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P^{\frac{1-\kappa_s}{2}+0.001}}\right) + O\left(\frac{1}{P^{(1-\kappa_s)+0.002}}\right).$$

Lemma 4.A.20 still holds, and thus

$$|\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}| \leq O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{P^{0.002}}\right).$$

Since $|\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(\tilde{T}_1+1)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(\tilde{T})}| \geq \Omega(\log(P))$, we have

$$\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(\tilde{T}_1+1)} \geq |\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(\tilde{T}_1+1)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(\tilde{T})}| \cdot \Omega(P^{0.002}) + \Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(\tilde{T})} \gg \Omega(P^{0.002} \log(P)),$$

which contradicts the assumption that $\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(\tilde{T}_1+1)} < \frac{c_1^*(1-\kappa_s) \log(P)}{U}$. $\square$

Phase II, stage 1 For $n > 1$, we define stage 1 of phase II as all iterations $\tilde{T}_1 + 1 \leq t \leq T_2$, where

$$T_2 \triangleq \max \left\{ t : \Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \leq \frac{(1-\kappa_s)}{L} \log(P) \right\}.$$

We state the following induction hypotheses, which will hold throughout this stage:

Induction Hypothesis 4.A.5. For each $\tilde{T}_1 + 1 \leq t \leq T_2$, $\mathbf{q} \in \mathcal{P} \setminus \{\mathbf{p}\}$, the following holds:

- a. $\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}$ is monotonically increasing, and $\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \in \left[\frac{c_1^*(1-\kappa_s)}{U} \log(P), \frac{(1-\kappa_s)}{L} \log(P) \right]$;
- b. $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}$ is monotonically decreasing and

$$\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} \in \left[-\frac{1}{L} \left(\frac{\Delta}{2} + 0.01 \right) \log(P) - o(1), -\frac{1}{U} \left(\frac{\Delta}{2} - 0.01 \right) \log(P) \right];$$

- c. $|\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}| = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P^{1-\kappa_s}}\right)$ for $m \neq 1, n$;
- d. $\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{C_n}\right)$ for $a_{k,\mathbf{q}} = n$, $|\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{p}}^{(t)}| = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right)$;
- e. $|\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}| = O\left(\frac{|\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}|}{C_1}\right) + O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right)$ for $a_{k,\mathbf{q}} = 1$;
- f. $|\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}| = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right)$ for $a_{k,\mathbf{q}} \neq 1, n$.

Property of attention scores. We first single out several properties of attention scores that will be used for the proof of Induction Hypothesis 4.A.5.

Lemma 4.A.25. *if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.5 hold at iteration $\tilde{T}_1 + 1 \leq t \leq T_2$, then the following holds*

1. if $M \in \mathcal{E}_{k,n}$, $\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \geq \Omega\left(\frac{1}{p^{(1-\frac{c_1^* L}{U})(1-\kappa_s)}}\right)$. Moreover, if $\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}$ does not reach the constant level, $1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} = \Omega(1)$; otherwise, $1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} = \Omega\left(\frac{1}{p^{(\frac{U}{L}-1)(1-\kappa_s)}}\right)$.
2. $\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} = O\left(\frac{1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}}{p^{(1-\kappa_c) + \frac{U}{L}(\frac{A}{2} - 0.01)}}\right)$; if $M \in \mathcal{E}_{k,1}$, we have $\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} = \Omega\left(\frac{1}{p^{(1-\kappa_c) + \frac{U}{L}(\frac{A}{2} + 0.01)}}\right)$;
3. for $\mathbf{q} \in \mathcal{M} \cap (\mathcal{P}_{k,n} \cup \mathcal{P}_{k,1})$, $\mathbf{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = O\left(\frac{1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}}{p}\right)$

Lemma 4.A.26. if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.5 hold at iteration $\tilde{T}_1 + 1 \leq t \leq T_2$, then for $m \neq n$, the following holds:

1. for any $\mathbf{q} \in \mathcal{P}_{k,m}$, $\mathbf{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} \leq O\left(\frac{1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}}{p}\right)$.
2. Moreover, $\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \leq O\left(\frac{1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}}{N}\right)$.

Bounding the gradient updates of FP correlations. We have the following set of lemmas.

Lemma 4.A.27. if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.5 hold at iteration $\tilde{T}_1 + 1 \leq t \leq T_2$, then $\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \geq 0$ and satisfies:

$$\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \geq \min \left\{ \Omega\left(\frac{1}{p^{(1-\frac{c_1^* L}{U})(1-\kappa_s)}}\right), \Omega\left(\frac{1}{p^{2(\frac{U}{L}-1)(1-\kappa_s)}}\right) \right\}.$$

Proof. By Lemma 4.5.2, we have

$$\begin{aligned} & \alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \\ &= \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathbf{p} \in \mathcal{M}\} \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \cdot \left(z_n^3 \left(1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right)^2 + \sum_{m \neq n} z_m^2 z_n \left(\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \right)^2 \right) \right] \\ &= \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathcal{E}_{k,n} \cap \mathbf{p} \in \mathcal{M}\} \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \cdot \left(z_n^3 \left(1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right)^2 + \sum_{m \neq n} z_m^2 z_n \left(\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \right)^2 \right) \right] \\ &\quad + \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathcal{E}_{k,n}^c \cap \mathbf{p} \in \mathcal{M}\} \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \cdot \left(z_n^3 \left(1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right)^2 + \sum_{m \neq n} z_m^2 z_n \left(\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \right)^2 \right) \right] \\ &\geq \mathbb{P}(M \in \mathcal{E}_{k,n}) \cdot \\ &\quad \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathbf{p} \in \mathcal{M}\} \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \cdot \left(z_n^3 \left(1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right)^2 + \sum_{m \neq n} z_m^2 z_n \left(\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \right)^2 \right) \middle| \mathcal{E}_{k,n} \right] \\ &\quad + O(1) \cdot \mathbb{P}(M \in \mathcal{E}_{k,n}^c) \\ &\geq \min \left\{ \Omega\left(\frac{1}{p^{(1-\frac{c_1^* L}{U})(1-\kappa_s)}}\right), \Omega\left(\frac{1}{p^{2(\frac{U}{L}-1)(1-\kappa_s)}}\right) \right\}, \end{aligned}$$

where the last inequality invokes Lemma 4.A.25 by observing that for $\mathbf{M} \in \mathcal{E}_{k,n}$,

$$\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} (1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)})^2 \geq \min \left\{ \Omega \left(\frac{1}{P(1 - \frac{c_1^* L}{U})(1 - \kappa_s)} \right) \cdot \Omega(1), \Omega(1) \cdot \Omega \left(\frac{1}{P^{2 \times (\frac{U}{L} - 1)(1 - \kappa_s)}} \right) \right\}.$$

□

Lemma 4.A.28. *For $n > 1$, if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.5 hold at iteration $\tilde{T}_1 + 1 \leq t \leq T_2$, then $\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} < 0$ and satisfies*

$$\begin{aligned} |\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}| &\geq \min \left\{ \Omega \left(\frac{1}{P(1 - \frac{c_1^* L}{U})(1 - \kappa_s)} \right), \Omega \left(\frac{1}{P(\frac{U}{L} - 1)(1 - \kappa_s)} \right) \right\} \cdot \Omega \left(\frac{1}{P(1 - \kappa_c) + \frac{L}{U}(\frac{\Delta}{2} - 0.01)} \right), \\ |\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}| &\leq \max \left\{ O \left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{P(1 - \kappa_c) + \frac{L}{U}(\Delta/2 - 0.01)} \right), O \left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{P^{2(1 - \kappa_c) + \frac{L}{U}(\Delta - 0.02) - (1 - \frac{c_1^* L}{U})(1 - \kappa_s)}} \right) \right\}. \end{aligned}$$

Proof. Following (4.A.9), we have

$$\begin{aligned} &-z_1 z_n^2 \left(1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right) \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} - z_1^3 \left(1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} \right) \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} + \sum_{a \neq 1, n} z_a^2 z_1 \left(\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)} \right)^2 \\ &\leq -z_1 (1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}) \left(z_n^2 \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} + z_1^2 \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} - \max_{a \neq 1, n} z_a^2 \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)} \right). \end{aligned}$$

Therefore, by Lemma 4.5.1, we obtain

$$\begin{aligned} \alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} &\leq \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathcal{E}_{k,1} \cap \mathbf{p} \in \mathcal{M}\} \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} \cdot \right. \\ &\quad \left. \left(-z_1 (1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}) \left(z_n^2 \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} + z_1^2 \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} - \max_{a \neq 1, n} z_a^2 \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)} \right) \right) \right] \\ &\quad + \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathcal{E}_{k,1}^c \cap \mathbf{p} \in \mathcal{M}\} \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} \cdot \sum_{a \neq 1, n} z_a^2 z_1 \left(\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)} \right)^2 \right] \\ &\leq -\min \left\{ \Omega \left(\frac{1}{P(1 - \frac{c_1^* L}{U})(1 - \kappa_s)} \right), \Omega \left(\frac{1}{P(\frac{U}{L} - 1)(1 - \kappa_s)} \right) \right\} \cdot \Omega \left(\frac{1}{P(1 - \kappa_c) + \frac{L}{U}(\frac{\Delta}{2} - 0.01)} \right), \end{aligned}$$

where the second inequality invokes Lemma 4.A.25 and (4.A.14b). Moreover,

$$\begin{aligned} |\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}| &\lesssim \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathcal{E}_{k,1} \cap \mathcal{E}_{k,n} \cap \mathbf{p} \in \mathcal{M}\} \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} \cdot \right. \\ &\quad \left. \left(z_1 z_n^2 \left(1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right) \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} + z_1^3 \left(1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} \right) \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} \right) \right] \end{aligned}$$

$$\begin{aligned}
&= \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathcal{E}_{k,1} \cap \mathcal{E}_{k,n} \cap \mathbf{p} \in \mathcal{M}\} z_1 z_n^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} \cdot \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}\right) \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right] \\
&\quad + \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathcal{E}_{k,1} \cap \mathcal{E}_{k,n} \cap \mathbf{p} \in \mathcal{M}\} z_1^3 (\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)})^2 \cdot \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}\right) \right] \\
&\leq \max \left\{ O \left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{P^{(1-\kappa_c) + \frac{L}{U}(\frac{\Lambda}{2} - 0.01)}} \right), O \left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{P^{2(1-\kappa_c) + \frac{2L}{U}(\frac{\Lambda}{2} - 0.01) - (1 - \frac{c_1^* L}{U})(1-\kappa_s)}} \right) \right\},
\end{aligned}$$

where the second inequality invokes Lemma 4.A.25. $\square$

Lemma 4.A.29. For $n > 1$, if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.5 hold at iteration $\tilde{T}_1 + 1 \leq t \leq T_2$ for any $m > 1$ with $m \neq n$, the following holds

$$|\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}| \leq O \left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P^{1-\kappa_s}} \right).$$

The proof is similar to Lemma 4.A.13, and thus omitted here.

Bounding the gradient updates of positional correlations We then summarize the properties for gradient updates of positional correlations, which utilizes the identical calculations as in Section 4.A.3.

Lemma 4.A.30. For $n > 1$, if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.5 hold at iteration $\tilde{T}_1 + 1 \leq t \leq T_2$, then

- a. if $a_{k,\mathbf{q}} = n$ and $\mathbf{q} \neq \mathbf{p}$, $\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)} \geq 0$; $\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = \Theta \left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{C_n} \right)$ and $|\beta_n^{(t)}| = O \left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P} \right)$.
- b. if $a_{k,\mathbf{q}} = 1$, $|\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}| = O \left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P} \right) + O \left(\frac{|\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}|}{C_1} \right)$.
- c. if $a_{k,\mathbf{q}} = m$ and $m \neq 1, n$, $|\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}| = O \left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P} \right)$.

End of Phase II, stage 1. We have the following lemma.

Lemma 4.A.31. Induction Hypothesis 4.A.5 holds for all $\tilde{T}_1 + 1 \leq t \leq T_2$, and at iteration $t = T_2 + 1$, we have

- a. $\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} > \frac{(1-\kappa_s)}{L} \log(P)$;
- b. $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} = \Omega(1)$ if $\mathbf{M} \in \mathcal{E}_{k,n}$.

Proof. By comparing Lemma 4.A.27 with Lemmas 4.A.28–4.A.31, we have

$$\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \gg |\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}|, |\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}|.$$

The existence of $T_2 = \tilde{T}_1 + O \left(\frac{\log(P)P^\Lambda}{\eta} \right)$ then follows directly from Lemma 4.A.27, where

$$\Lambda = \max \left\{ \left(1 - \frac{c_1^* L}{U}\right), 2 \left(\frac{U}{L} - 1\right) \right\} \cdot (1 - \kappa_s).$$

The second statement can be directly verified by noticing that $\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} > \frac{(1-\kappa_s)}{L} \log(P)$ while all other attention correlations are sufficiently small. $\square$

Phase II, stage 2 In this final stage, we establish that these structures indeed represent the solutions toward which the algorithm converges. Given any $0 < \epsilon < 1$, for $n > 1$, define

$$T_2^\epsilon \triangleq \max \left\{ t > T_2 : \Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \leq \log \left(c_5 \left(\left(\frac{3}{\epsilon} \right)^{\frac{1}{2}} - 1 \right) N \right) \right\}, \quad (4.A.15)$$

where c_5 is some largely enough constant.

We state the following induction hypotheses, which will hold throughout this stage:

Induction Hypothesis 4.A.6. For $n > 1$, suppose $\text{polylog}(P) \gg \log(\frac{1}{\epsilon})$, for each $T_2 + 1 \leq t \leq T_2^\epsilon$, $\mathbf{q} \in \mathcal{P} \setminus \{\mathbf{p}\}$, the following holds:

- $\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}$ is monotonically increasing, and $\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \in [\frac{(1-\kappa_s)}{L} \log(P), O(\log(P/\epsilon))]$;
- $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}$ is mono decreasing and $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} \in \left[-\frac{1}{L} \left(\frac{\Delta}{2} + 0.01 \right) \log(P) - o(1), -\frac{1}{L} \left(\frac{\Delta}{2} - 0.01 \right) \log(P) \right]$;
- $|\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}| = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P^{1-\kappa_s}}\right)$ for $m \neq 1, n$;
- $\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{C_n}\right)$ for $a_{k,\mathbf{q}} = n$, $|\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{p}}^{(t)}| = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right)$;
- $|\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}| = O\left(\frac{|\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}|}{C_1}\right) + O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right)$ for $a_{k,\mathbf{q}} = 1$;
- $|\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}| = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right)$ for $a_{k,\mathbf{q}} \neq 1, n$.

Property of attention scores. We first single out several properties of attention scores that will be used for the proof of Induction Hypothesis 4.A.6.

Lemma 4.A.32. if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.6 hold at iteration $T_{n,2} < t \leq T_{n,2}^\epsilon$, then the following holds

- if $\mathbf{M} \in \mathcal{E}_{k,n}$, $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} = \Omega(1)$ and $(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)})^2 \geq O(\epsilon)$.
- Moreover, $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} = O\left(\frac{1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}}{P^{(1-\kappa_c) + \frac{1}{L}(\frac{\Delta}{2} - 0.01)}}\right)$; if $\mathbf{M} \in \mathcal{E}_{k,1}$, we have $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} = \Omega\left(\frac{1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}}{P^{(1-\kappa_c) + \frac{1}{L}(\frac{\Delta}{2} + 0.01)}}\right)$;
- for $\mathbf{q} \in \mathcal{M} \cap (\mathcal{P}_{k,n} \cup \mathcal{P}_{k,1})$, $\text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = O\left(\frac{1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}}{P}\right)$.

Lemma 4.A.33. if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.6 hold at iteration $T_{n,2} < t \leq T_{n,2}^\epsilon$, then for $m \neq n$, the following holds:

- for any $\mathbf{q} \in \mathcal{P}_{k,m}$, $\text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} \leq O\left(\frac{1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}}{P}\right)$.
- $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \leq O\left(\frac{1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}}{N}\right)$, and if $\mathbf{M} \in \mathcal{E}_{k,m}$, $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} = \Theta\left(\frac{1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}}{N}\right)$.

Bounding the gradient updates of FP correlations. We have the following lemmas.

Lemma 4.A.34. For $n > 1$, if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.6 hold at iteration $T_2 + 1 \leq t \leq T_2^\epsilon$, then $\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \geq 0$ and satisfies:

$$\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \geq \Omega(\epsilon).$$

Proof. By Lemma 4.5.2, we have

$$\begin{aligned} & \alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \\ &= \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathbf{p} \in \mathcal{M}\} \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \cdot \left(z_n^3 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right)^2 + \sum_{m \neq n} z_m^2 z_n \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \right)^2 \right) \right] \\ &= \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathcal{E}_{k,n} \cap \mathbf{p} \in \mathcal{M}\} \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \cdot \left(z_n^3 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right)^2 + \sum_{m \neq n} z_m^2 z_n \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \right)^2 \right) \right] \\ &\quad + \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathcal{E}_{k,n}^c \cap \mathbf{p} \in \mathcal{M}\} \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \cdot \left(z_n^3 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right)^2 + \sum_{m \neq n} z_m^2 z_n \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \right)^2 \right) \right] \\ &\geq \mathbb{P}(\mathcal{M} \in \mathcal{E}_{k,n}) \cdot \\ &\quad \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathbf{p} \in \mathcal{M}\} \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \cdot \left(z_n^3 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} \right)^2 + \sum_{m \neq n} z_m^2 z_n \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \right)^2 \right) \middle| \mathcal{E}_{k,n} \right] \\ &\quad + O(1) \cdot \mathbb{P}(\mathcal{M} \in \mathcal{E}_{k,n}^c) \\ &\geq \Omega(\epsilon), \end{aligned}$$

where the last inequality invokes Lemma 4.A.32, Lemma 4.A.4 and the fact that

$$\epsilon \geq \exp(-\text{polylog}(K)) \gg \exp(-c_{n,1} C_n).$$

□

Lemma 4.A.35. For $n > 1$, if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.6 hold at iteration $T_{n,3} < t \leq T_{n,4}^\epsilon$, then $\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} < 0$ and satisfies

$$|\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}| \leq \max \left\{ O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{P^{(1-\kappa_c) + \frac{L}{U}} (\Delta/2 - 0.01)} \right), O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{P^{2(1-\kappa_c) + \frac{L}{U}} (\Delta - 0.02) - (1 - \frac{c_1^* L}{U})(1 - \kappa_s)} \right) \right\}$$

The proof follows the similar arguments Lemma 4.A.28 by noticing that $\epsilon \gg \mathbb{P}(\mathcal{M} \in \mathcal{E}_{k,m}^c)$ for any $m \neq n$.

Lemma 4.A.36. For $n > 1$, if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.6 hold at iteration $T_2 < t \leq T_2^\epsilon$, then for any $m > 1$ with $m \neq n$, the following holds

$$-O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{P^{1-\kappa_s}} \right) \leq \alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)} \leq 0$$

Proof. We first note that

$$\begin{aligned}
& -z_1 z_n^2 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}\right) \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} - z_m^3 \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)}\right) \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)} + \sum_{a \neq 1,n} z_a^2 z_m \left(\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)}\right)^2 \\
& \leq z_m \left(\max_{a \neq m,n} z_a^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)} - z_n^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} - z_m^2 \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \right) \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)}\right) \\
& \lesssim -\Omega(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)})
\end{aligned}$$

since when $\mathbf{M} \in \mathcal{E}_{k,n}$, we have $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)} = \Omega(1) \gg \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a}}^{(t)}$. Thus, we have

$$\begin{aligned}
0 & \geq \alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)} \gtrsim -\mathbb{E} \left[\mathbb{1}\{k_X = k, \mathcal{E}_{k,n} \cap \mathbf{p} \in \mathcal{M}\} \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)} \cdot \Omega(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(t)}) \right] \\
& \geq -O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{P^{1-\kappa_s}}\right).
\end{aligned}$$

□

Bounding the gradient updates of positional correlations. We then summarize the properties for gradient updates of positional correlations, which utilizes the identical calculations as in Section 4.A.3.

Lemma 4.A.37. *For $n > 1$, if Induction Hypothesis 4.A.1 and Induction Hypothesis 4.A.6 hold at iteration $T_2 + 1 \leq t \leq T_2^\epsilon$, then*

- if $a_{k,\mathbf{q}} = n$ and $\mathbf{q} \neq \mathbf{p}$, $\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)} \geq 0$; $\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = \Theta\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{C_n}\right)$ and $|\beta_n^{(t)}| = O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right)$.*
- if $a_{k,\mathbf{q}} = 1$, $|\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}| = O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right) + O\left(\frac{|\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}|}{C_1}\right)$.*
- if $a_{k,\mathbf{q}} = m$ and $m \neq 1, n$, $|\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}| = O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right)$.*

End of Phase II, stage 2. We have the following lemma.

Lemma 4.A.38. *For $n > 1$, and $0 < \epsilon < 1$, suppose $\text{polylog}(P) \gg \log(\frac{1}{\epsilon})$. Then Induction Hypothesis 4.A.6 holds for all $T_2 < t \leq T_2^\epsilon = T_2 + O\left(\frac{\log(P\epsilon^{-1})}{\eta\epsilon}\right)$, and at iteration $t = T_2^\epsilon + 1$, we have*

- $\tilde{\mathcal{L}}_{k,\mathbf{p}}(Q^{T_2^\epsilon+1}) < \frac{\epsilon}{2K}$;
- If $\mathbf{M} \in \mathcal{E}_{k,n}$, we have $(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(T_2^\epsilon+1)})^2 \leq O(\epsilon)$.

Proof. The existence of $T_{2,k}^\epsilon = T_{2,k} + O\left(\frac{\log(P\epsilon^{-1})}{\eta\epsilon}\right)$ directly follows from Lemma 4.A.34. We further derive

$$\begin{aligned}
\tilde{\mathcal{L}}_{k,\mathbf{p}}(Q^{T_2^\epsilon+1}) &= \frac{1}{2} \mathbb{E} \left[\mathbb{1}\{k_X = k, \mathbf{p} \in \mathcal{M} \cap \mathbf{M} \in \mathcal{E}_{k,n}\} \left(z_n^2 (1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}})^2 + \sum_{m \neq n} z_m^2 (\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}})^2 \right) \right] \\
&\leq \frac{1}{2K} \cdot \gamma \cdot U^2 \cdot (1 + o(1)) \cdot O(\epsilon)
\end{aligned}$$

$$\leq \frac{\epsilon}{2K},$$

where the first inequality is due to direct calculations by the definition of T_2^ϵ , and the second inequality can be obtained by setting $c_{n,2}$ in (4.A.15) sufficiently large. $\square$

4.A.4 Analysis for local areas with negative information gap

In this section, we focus on a specific patch $\mathbf{p} \in \mathcal{P}$ with the k -th cluster for $k \in [K]$, and present the analysis for the case that $X_{\mathbf{p}}$ is located in the local area for the k -th cluster, i.e. $a_{k,p} > 1$. Throughout this section, we denote $a_{k,p} = n$ for simplicity. When $\Delta \leq -\Omega(1)$, we can show that the gap of attention correlation changing rate for the positive case does not exist anymore, and conversely $\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \gg \alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}$ from the beginning. We can reuse most of the gradient calculations in the previous section and only sketch them in this section.

Stage 1: we define stage 1 as all iterations $0 \leq t \leq T_{\text{neg},1}$, where

$$T_{\text{neg},1} \triangleq \max \left\{ t : \Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \leq \frac{(1 - \kappa_s)}{L} \log(P) \right\}.$$

We state the following induction hypothesis, which will hold throughout this stage:

Induction Hypothesis 4.A.7. For each $0 \leq t \leq T_{\text{neg},1}$, $\mathbf{q} \in \mathcal{P} \setminus \{\mathbf{p}\}$, the following holds:

- a. $\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}$ is monotonically increasing, and $\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \in \left[0, \frac{(1 - \kappa_s)}{L} \log(P) \right]$;
- b. $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}$ is monotonically decreasing and $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} \in \left[-O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{P^{-\Delta}}\right), 0 \right]$;
- c. $|\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}| = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P^{1 - \kappa_s}}\right)$ for $m \neq 1, n$;
- d. $\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{C_n}\right)$ for $a_{k,\mathbf{q}} = n$, $|\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{p}}^{(t)}| = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right)$;
- e. $|\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}| = O\left(\frac{|\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}|}{C_1}\right) + O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right)$ for $a_{k,\mathbf{q}} = 1$;
- f. $|\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}| = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right)$ for $a_{k,\mathbf{q}} \neq 1, n$.

Through similar calculations for phase II, stage 1 in Section 4.A.3, we obtain the following lemmas to control the gradient updates for attention correlations.

Lemma 4.A.39. If Induction Hypothesis 4.A.2 and Induction Hypothesis 4.A.7 hold for $0 \leq t \leq T_{\text{neg},1}$, then we have

$$\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \geq \min \left\{ \Omega\left(\frac{1}{P(1 - \kappa_s)}\right), \Omega\left(\frac{1}{P^{2(\frac{L}{L} - 1)(1 - \kappa_s)}}\right) \right\}, \quad (4.A.16a)$$

$$0 \geq \alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} \geq -O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{P^{-\Delta}}\right), \quad (4.A.16b)$$

$$|\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}| \leq O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P^{1-\kappa_s}}\right) \text{ for all } m \neq n, 1 \quad (4.A.16c)$$

$$\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = \Theta\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{C_n}\right) \text{ for } a_{k,\mathbf{q}} = n, \mathbf{q} \neq \mathbf{p} \quad (4.A.16d)$$

$$|\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}| = O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{P}\right) + O\left(\frac{|\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}|}{C_1}\right) \text{ for } a_{k,\mathbf{q}} = 1, \quad (4.A.16e)$$

$$|\beta_{k,\mathbf{p} \rightarrow \mathbf{p}}^{(t)}|, |\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}| = O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right) \text{ for all } a_{k,\mathbf{p}} \neq n, 1. \quad (4.A.16f)$$

Here $\Delta < 0$ implies $|\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}| \ll \alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}$. Induction Hypothesis 4.A.7 can be directly proved by Lemma 4.A.39 and we have

$$T_{\text{neg},1} = O\left(\frac{P^{\max\{1, 2(\frac{L}{L}-1)\} \cdot (1-\kappa_s)} \log(P)}{\eta}\right). \quad (4.A.17)$$

Stage 2: Given any $0 < \epsilon < 1$, define

$$T_{\text{neg},1}^\epsilon \triangleq \max \left\{ t > T_1 : \Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \leq \log \left(c_6 \left(\left(\frac{3}{\epsilon} \right)^{\frac{1}{2}} - 1 \right) P^{1-\kappa_s} \right) \right\}. \quad (4.A.18)$$

where c_6 is some large enough constant. We then state the following induction hypotheses, which will hold throughout this stage:

Induction Hypothesis 4.A.8. For $n > 1$, suppose $\text{polylog}(P) \gg \log(\frac{1}{\epsilon})$, for $\mathbf{q} \in \mathcal{P} \setminus \{\mathbf{p}\}$, and each $T_{\text{neg},1} < t \leq T_{\text{neg},1}^\epsilon$, the following holds:

- a. $\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}$ is monotonically increasing, and $\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \in \left[\frac{(1-\kappa_s)}{L} \log(P), O(\log(P/\epsilon)) \right]$;
- b. $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}$ is monotonically decreasing and $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} \in \left[-O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{P^{-\Delta}}\right), 0 \right]$;
- c. $|\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}| = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P^{1-\kappa_s}}\right)$ for $m \neq 1, n$;
- d. $\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{C_n}\right)$ for $a_{k,\mathbf{q}} = n$, $|\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{p}}^{(t)}| = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right)$;
- e. $|\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}| = O\left(\frac{|\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}|}{C_1}\right) + O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right)$ for $a_{k,\mathbf{q}} = 1$;
- f. $|\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}| = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right)$ for $a_{k,\mathbf{q}} \neq 1, n$.

Lemma 4.A.40. If Induction Hypothesis 4.A.2 and Induction Hypothesis 4.A.8 hold for $T_{\text{neg},1} < t \leq T_{\text{neg},1}^\epsilon$, then we have

$$\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} \geq \Omega(\epsilon), \quad (4.A.19a)$$

$$0 \geq \alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} \geq -O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{P^{-\Delta}}\right), \quad (4.A.19b)$$

$$|\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}| \leq O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P^{1-\kappa_s}}\right) \text{ for all } m \neq n, 1, \quad (4.A.19c)$$

$$\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = \Theta\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{C_n}\right) \text{ for } a_{k,\mathbf{q}} = n, \mathbf{q} \neq \mathbf{p}, \quad (4.A.19d)$$

$$|\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}| = O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}}{P}\right) + O\left(\frac{|\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}|}{C_1}\right) \text{ for } a_{k,\mathbf{q}} = 1, \quad (4.A.19e)$$

$$|\beta_{k,\mathbf{p} \rightarrow \mathbf{p}}^{(t)}|, |\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}| = O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,n}}^{(t)} - \alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right) \text{ for all } a_{k,\mathbf{p}} \neq n, 1. \quad (4.A.19f)$$

Induction Hypothesis 4.A.8 can be directly proved by Lemma 4.A.40. Furthermore, at the end of this stage, we will have:

Lemma 4.A.41. Suppose $\text{polylog}(P) \gg \log(\frac{1}{\epsilon})$, then Induction Hypothesis 4.A.8 holds for all $T_{\text{neg},1} < t \leq T_{\text{neg},1}^\epsilon = T_{\text{neg},1} + O\left(\frac{\log(P\epsilon^{-1})}{\eta\epsilon}\right)$, and at iteration $t = T_{\text{neg},1}^\epsilon + 1$, we have

1. $\tilde{\mathcal{L}}_{k,\mathbf{p}}(Q^{T_{\text{neg},1}^\epsilon+1}) < \frac{\epsilon}{2K}$;
2. If $\mathbf{M} \in \mathcal{E}_{k,n}$, we have $\left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,n}}^{(T_{\text{neg},1}^\epsilon+1)}\right)^2 \leq O(\epsilon)$.

4.A.5 Analysis for the global area

When $a_{\mathbf{p},k} = 1$, i.e. the patch lies in the global area, the analysis is much simpler and does not depend on the value of Δ . We can reuse most of the gradient calculations in Section 4.A.3 and only sketch them in this section.

For $X_{\mathbf{p}}$ in the global region $\mathcal{P}_{k,1}$, since the overall attention $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(0)}$ to the target feature already reaches $\Omega\left(\frac{C_1}{P}\right) = \Omega\left(\frac{1}{P^{1-\kappa_c}}\right)$ due to the large number of unmasked patches featuring $v_{k,1}$ when $\mathbf{M} \in \mathcal{E}_{k,1}$, which is significantly larger than $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(0)} = \Theta\left(\frac{1}{P^{1-\kappa_s}}\right)$ for all other $m > 1$. This results in large $\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}$ initially, and thus the training directly enters phase II.

Stage 1: we define stage 1 as all iterations $0 \leq t \leq T_{c,1}$, where

$$T_{c,1} \triangleq \max \left\{ t : \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} \leq \frac{(1-\kappa_c)}{L} \log(P) \right\}.$$

We state the following induction hypotheses, which will hold throughout this stage:

Induction Hypothesis 4.A.9. For each $0 \leq t \leq T_{c,1}$, $\mathbf{q} \in \mathcal{P} \setminus \{\mathbf{p}\}$, the following holds:

- a. $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}$ is monotonically increasing, and $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} \in \left[0, \frac{(1-\kappa_c)}{L} \log(P)\right]$;
- b. $\Phi_{\mathbf{p} \rightarrow v_{k,m}}$ is monotonically decreasing for $m > 1$ and $\Phi_{\mathbf{p} \rightarrow v_{k,m}} \in \left[-O\left(\frac{\log(P)}{N}\right), 0\right]$;
- c. $\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{C_1}\right)$ for $a_{k,\mathbf{q}} = 1$, $|\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{p}}^{(t)}| = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right)$;
- d. $|\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}| = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right)$ for $a_{k,\mathbf{q}} \neq 1$.

Through similar calculations for phase II, stage 1 in Section 4.A.3, we obtain the following lemmas to control the gradient updates for attention correlations.

Lemma 4.A.42. *If Induction Hypothesis 4.A.1 (or Induction Hypothesis 4.A.2) and Induction Hypothesis 4.A.9 hold for $0 \leq t \leq T_{c,1}$, then we have*

$$\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} \geq \min \left\{ \Omega\left(\frac{1}{P^{(1-\kappa_c)}}\right), \Omega\left(\frac{1}{P^{2(\frac{U}{L}-1)(1-\kappa_c)}}\right) \right\}, \quad (4.A.20a)$$

$$|\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}| \leq O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P^{1-\kappa_s}}\right) \quad \text{for all } m \neq 1, \quad (4.A.20b)$$

$$\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = \Theta\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{C_1}\right), \text{ for } a_{k,\mathbf{q}} = 1, \mathbf{q} \neq \mathbf{p}, \quad (4.A.20c)$$

$$|\beta_{k,\mathbf{p} \rightarrow \mathbf{p}}^{(t)}|, |\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}| = O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right) \quad \text{for all } a_{k,\mathbf{q}} > 1. \quad (4.A.20d)$$

Induction Hypothesis 4.A.9 can be directly proved by Lemma 4.A.42 and we have

$$T_{c,1} = O\left(\frac{P^{\max\{1, 2(\frac{U}{L}-1)\} \cdot (1-\kappa_c)} \log(P)}{\eta}\right). \quad (4.A.21)$$

Stage 2: Given any $0 < \epsilon < 1$, define

$$T_{c,1}^\epsilon \triangleq \max \left\{ t > T_{c,1} : \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} \leq \log \left(c_7 \left(\left(\frac{3}{\epsilon} \right)^{\frac{1}{2}} - 1 \right) P^{1-\kappa_c} \right) \right\}. \quad (4.A.22)$$

where c_7 is some largely enough constant. We then state the following induction hypotheses, which will hold throughout this stage:

Induction Hypothesis 4.A.10. For $n > 1$, suppose $\text{polylog}(P) \gg \log(\frac{1}{\epsilon})$, $\mathbf{q} \in \mathcal{P} \setminus \{\mathbf{p}\}$, for each $T_{c,1} + 1 \leq t \leq T_{c,1}^\epsilon$, the following holds:

- a. $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}$ is monotonically increasing, and $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} \in \left[\frac{(1-\kappa_c)}{L} \log(P), O(\log(P/\epsilon)) \right]$;
- b. $\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}$ is monotonically decreasing for $n > 1$ and $\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)} \in \left[-O\left(\frac{\log(P)}{N}\right), 0 \right]$;
- c. $\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{C_1}\right)$ for $a_{k,\mathbf{q}} = 1$, $|\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{p}}^{(t)}| = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right)$;
- d. $|\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}| = O\left(\frac{\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right)$ for $a_{k,\mathbf{q}} \neq 1$.

We also have the following lemmas to control the gradient updates for attention correlations.

Lemma 4.A.43. *If Induction Hypothesis 4.A.1 (or Induction Hypothesis 4.A.2) and Induction Hypothesis 4.A.9 hold for $T_{c,1} + 1 \leq t \leq T_{c,1}^\epsilon$, then we have*

$$\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} \geq \Omega(\epsilon), |\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}| \leq O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P^{1-\kappa_s}}\right) \quad \text{for all } m \neq 1 \quad (4.A.23a)$$

$$\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = \Theta\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{C_1}\right), \text{ for } a_{k,\mathbf{q}} = 1, \mathbf{q} \neq \mathbf{p} \quad (4.A.23b)$$

$$|\beta_{k,\mathbf{p} \rightarrow \mathbf{p}}^{(t)}|, |\beta_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}| = O\left(\frac{\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}}{P}\right) \text{ for all } a_{k,\mathbf{q}} > 1. \quad (4.A.23c)$$

Induction Hypothesis 4.A.10 can be directly proved by Lemma 4.A.43. Furthermore, at the end of this stage, we will have:

Lemma 4.A.44. Suppose $\text{polylog}(P) \gg \log(\frac{1}{\epsilon})$, then Induction Hypothesis 4.A.10 holds for all $T_{c,1} < t \leq T_{c,1}^\epsilon = T_{c,1} + O\left(\frac{\log(P\epsilon^{-1})}{\eta\epsilon}\right)$, and at iteration $t = T_{c,1}^\epsilon + 1$, we have

1. $\tilde{\mathcal{L}}_{k,\mathbf{p}}(Q^{T_{c,1}^\epsilon+1}) < \frac{\epsilon}{2K}$;
2. If $M \in \mathcal{E}_{k,1}$, we have $\left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(T_{c,1}^\epsilon+1)}\right)^2 \leq O(\epsilon)$.

Proof of Induction Hypotheses We are now ready to show Induction Hypothesis 4.A.1 (resp. Induction Hypothesis 4.A.2) holds through the learning process.

Theorem 4.A.1 (Positive Information Gap). For sufficiently large $P > 0$, $\eta \ll \log(P)$, $\Omega(1) \leq \Delta < 1$, Induction Hypothesis 4.A.1 holds for all iterations $t = 0, 1, \dots, T = O\left(\frac{e^{\text{polylog}(P)}}{\eta}\right)$.

Theorem 4.A.2 (Negative Information Gap). For sufficiently large $P > 0$, $\eta \ll \log(P)$, $-0.5 < \Delta \leq -\Omega(1)$, Induction Hypothesis 4.A.2 holds for all iterations $t = 0, 1, \dots, T = O\left(\frac{e^{\text{polylog}(P)}}{\eta}\right)$.

Proof of Theorem 4.A.1. It is easy to verify Induction Hypothesis 4.A.1 holds at iteration $t = 0$ due to the initialization $Q^{(0)} = \mathbf{0}_{d \times d}$. At iteration $t > 0$:

- Induction Hypothesis 4.A.1a. can be proven by Induction Hypothesis 4.A.3-4.A.6 a and Induction Hypothesis 4.A.9-4.A.10 a, combining with the fact that $\log(1/\epsilon) \ll \text{polylog}(P)$.
- Induction Hypothesis 4.A.1b. can be obtained by invoking Induction Hypothesis 4.A.3-4.A.6 b.
- Induction Hypothesis 4.A.1c. can be obtained by invoking Induction Hypothesis 4.A.3-4.A.6 c and Induction Hypothesis 4.A.9-4.A.10 b.
- To prove Induction Hypothesis 4.A.1d., for $\mathbf{q} \neq \mathbf{p}$, $\Upsilon_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = \sum_{k=1}^K \Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)}$. By item d-f in Induction Hypothesis 4.A.3-4.A.6 and item c-d in Induction Hypothesis 4.A.9-4.A.10, we can conclude that no matter the relative areas $\mathbf{q}$ and $\mathbf{p}$ belong to for a specific cluster, for all $k \in [K]$, throughout the entire learning process, the following upper bound always holds:

$$\Upsilon_{k,\mathbf{p} \rightarrow \mathbf{q}}^{(t)} \leq \max_{t \in [T]} (|\Phi_{\mathbf{p} \rightarrow v_{k,n}}^{(t)}| + |\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}|) \max \left\{ O\left(\frac{1}{C_1}\right), O\left(\frac{1}{C_n}\right), O\left(\frac{1}{P}\right) \right\} \leq \tilde{O}\left(\frac{1}{C_n}\right).$$

Moreover, since $K = O(\text{polylog}(P))$, we then have $\Upsilon_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)} = \widetilde{O}(\frac{1}{C_n})$, which completes the proof.

- The proof for Induction Hypothesis 4.A.1d. is similar as before, by noticing that $\Upsilon_{k, \mathbf{p} \rightarrow \mathbf{p}}^{(t)} = \widetilde{O}(\frac{1}{P})$ for each $k \in [K]$, which is due to Induction Hypothesis 4.A.3-4.A.6 d and Induction Hypothesis 4.A.9-4.A.10 c.

The proof of Theorem 4.A.2 mirrors that of Theorem 4.A.1, with the only difference being the substitution of relevant sections with Induction Hypothesis 4.A.2. For the sake of brevity, this part of the proof is not reiterated here.

Proof of Theorem 4.4.1 and Theorem 4.4.2 with positive information gap

Theorem 4.A.3. Suppose $\Omega(1) \leq \Delta \leq 1$. For any $0 < \epsilon < 1$, suppose $\text{polylog}(P) \gg \log(\frac{1}{\epsilon})$. We apply GD to train the loss function given in (4.2.8) with $\eta \ll \text{poly}(P)$. Then for each $\mathbf{p} \in \mathcal{P}$, we have

1. The loss converges: after $T^* = O\left(\frac{\log(P)P^{\max(2(\frac{L}{L}-1), 1)(1-\kappa_s)}}{\eta} + \frac{\log(P\epsilon^{-1})}{\eta\epsilon}\right)$ iterations, $\mathcal{L}_{\mathbf{p}}(Q^{(T^*)}) - \mathcal{L}_{\mathbf{p}}^* \leq \epsilon$, where $\mathcal{L}_{\mathbf{p}}^*$ is the global minimum of patch-level construction loss in (4.4.2).
2. Attention score concentrates: given cluster $k \in [K]$, if $X_{\mathbf{p}}$ is masked, then the one-layer transformer nearly “pays all attention” to all unmasked patches in the same area $\mathcal{P}_{k, a_{k, \mathbf{p}}}$, i.e., $\left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k, a_{k, \mathbf{p}}}}^{(T^*)}\right)^2 \leq O(\epsilon)$.
3. **Local area learning feature attention correlation through two-phase:** given $k \in [K]$, if $a_{k, \mathbf{p}} > 1$, then we have
 - (a) $\Phi_{\mathbf{p} \rightarrow v_{k, 1}}^{(t)}$ first quickly decrease with all other $\Phi_{\mathbf{p} \rightarrow v_{k, m}}^{(t)}$, $\Upsilon_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)}$ not changing much;
 - (b) after some point, the increase of $\Phi_{\mathbf{p} \rightarrow v_{k, a_{k, \mathbf{p}}}}^{(t)}$ takes dominance. Such $\Phi_{\mathbf{p} \rightarrow v_{k, a_{k, \mathbf{p}}}}^{(t)}$ will keep growing until convergence with all other feature and positional attention correlations nearly unchanged.
4. **Global area learning feature attention correlation through one-phase:** given $k \in [K]$, if $a_{k, \mathbf{p}} = 1$, throughout the training, the increase of $\Phi_{\mathbf{p} \rightarrow v_{k, 1}}^{(t)}$ dominates, whereas all $A_{1, m}^{(t)}$ with $m \neq 1$ and position attention correlations remain close to 0.

Proof. The first statement is obtained by letting $T^* = \max\{T_2^\epsilon, T_{c, 1}^\epsilon\} + 1$ in Lemma 4.A.38 and Lemma 4.A.44, combining with Lemma 4.A.7 and Lemma 4.A.8, which lead to

$$\begin{aligned}
 \mathcal{L}_{\mathbf{p}}(Q^{(T^*)}) - \mathcal{L}_{\mathbf{p}}^* &\leq \mathcal{L}_{\mathbf{p}}(Q^{(T^*)}) - \mathcal{L}_{\mathbf{p}}^{\text{low}} \\
 &\leq \widetilde{\mathcal{L}}_{\mathbf{p}}(Q^{T^*}) + O\left(\exp\left(-\left(c_3 P^{\kappa_c} + \mathbb{1}\{1 \notin \cup_{k \in [K]} \{a_{k, \mathbf{p}}\}\} c_4 P^{\kappa_s}\right)\right)\right) \\
 &\leq K \cdot \frac{\epsilon}{2K} + O\left(\exp\left(-\left(c_3 P^{\kappa_c} + \mathbb{1}\{1 \notin \cup_{k \in [K]} \{a_{k, \mathbf{p}}\}\} c_4 P^{\kappa_s}\right)\right)\right) \\
 &< \epsilon.
 \end{aligned}$$

The second statement follows from Lemma 4.A.38 and Lemma 4.A.44. The third and fourth statements directly follow from the learning process described in Section 4.A.3 and Section 4.A.5 when Induction Hypothesis 4.A.1 holds. $\square$

Proof of Theorem 4.4.1 and Theorem 4.4.2 with negative information gap

Theorem 4.A.4. Suppose $-0.5 \leq \Delta \leq \Omega(1)$. For any $0 < \epsilon < 1$, suppose $\text{polylog}(P) \gg \log(\frac{1}{\epsilon})$. We apply GD to train the loss function given in (4.2.8) with $\eta \ll \text{poly}(P)$. Then for each $\mathbf{p} \in \mathcal{P}$, we have

1. The loss converges: after $T^* = O\left(\frac{\log(P)P^{\max\{2(\frac{U}{L}-1), 1\}(1-\kappa_s)}}{\eta} + \frac{\log(P\epsilon^{-1})}{\eta\epsilon}\right)$ iterations, $\mathcal{L}_{\mathbf{p}}(Q^{(T^*)}) - \mathcal{L}_{\mathbf{p}}^* \leq \epsilon$, where $\mathcal{L}_{\mathbf{p}}^*$ is the global minimum of patch-level construction loss in (4.4.2).
2. Attention score concentrates: given cluster $k \in [K]$, if $X_{\mathbf{p}}$ is masked, then the one-layer transformer nearly “pays all attention” to all unmasked patches in the same area $\mathcal{P}_{k,a_{k,\mathbf{p}}}$, i.e., $\left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,a_{k,\mathbf{p}}}}^{(T^*)}\right)^2 \leq O(\epsilon)$.
3. **All areas learning feature attention correlation through one-phase:** given $k \in [K]$, throughout the training, the increase of $\Phi_{\mathbf{p} \rightarrow v_{k,a_{k,\mathbf{p}}}}^{(t)}$ dominates, whereas all $\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}$ with $m \neq 1$ and position attention correlations $\Upsilon_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)}$ remain close to 0.

Proof. The first statement is obtained by letting $T^* = \max\{T_{\text{neg},1}^{\epsilon}, T_{c,1}^{\epsilon}\} + 1$ in Lemma 4.A.41 and Lemma 4.A.44, combining with Lemma 4.A.7 and Lemma 4.A.8, which lead to

$$\begin{aligned} \mathcal{L}_{\mathbf{p}}(Q^{(T^*)}) - \mathcal{L}_{\mathbf{p}}^* &\leq \mathcal{L}_{\mathbf{p}}(Q^{(T^*)}) - \mathcal{L}_{\mathbf{p}}^{\text{low}} \\ &\leq \tilde{\mathcal{L}}_{\mathbf{p}}(Q^{(T^*)}) + O\left(\exp\left(-\left(c_3 P^{\kappa_c} + \mathbb{1}\{1 \notin \cup_{k \in [K]} \{a_{k,\mathbf{p}}\}\} c_4 P^{\kappa_s}\right)\right)\right) \\ &\leq K \cdot \frac{\epsilon}{2K} + O\left(\exp\left(-\left(c_3 P^{\kappa_c} + \mathbb{1}\{1 \notin \cup_{k \in [K]} \{a_{k,\mathbf{p}}\}\} c_4 P^{\kappa_s}\right)\right)\right) \\ &< \epsilon. \end{aligned}$$

The second statement follows from Lemma 4.A.41 and Lemma 4.A.44. The third and fourth statements directly follow from the learning process described in Section 4.A.4 and Section 4.A.5 when Induction Hypothesis 4.A.2 holds. $\square$

4.B Proof of Main Theorems in Contrastive Learning

Notation. Throughout this section, we abbreviate $\text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}(X; Q^{(t)})$ as $\text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)}(X)$. We also write F^{cl} as F and $\mathcal{L}_{\text{cl}}$ as $\mathcal{L}$ for simplicity. We abbreviate $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^c(X; Q^{(t)})$ (resp. $\text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^c(X; Q^{(t)})$) as $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)}$ (resp. $\text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)}$), when the context makes it clear. Furthermore, we denote

$$\ell_p(X, \mathfrak{B}) := \frac{e^{\text{Sim}_F(X^+, X^{++})/\tau}}{\sum_{X \in \mathfrak{B}} e^{\text{Sim}_F(X^+, X)/\tau}}, \quad \ell_s(X, \mathfrak{B}) := \frac{e^{\text{Sim}_F(X^+, X^{-s})/\tau}}{\sum_{X \in \mathfrak{B}} e^{\text{Sim}_F(X^+, X)/\tau}}.$$

Theorem 4.B.1 (Learning with contrastive objective). *Suppose the information gap $\Delta \in [-0.5, -\Omega(1)] \cup [\Omega(1), 1]$. We train the ViTs in Definition 4.2.4 by GD to minimize (4.2.11) with $\eta \ll \text{poly}(P)$, $\sigma_0^2 = \frac{1}{d}$, $\tau = O(\frac{1}{\log d})$, after $T^* = O(\frac{\text{poly}(P) \log P}{\eta})$ iterations, we have*

1. *Objective converges: $\mathcal{L}_{\text{cl}}(Q^{(T^*)}) \leq \mathcal{L}_{\text{cl}}^* + \frac{1}{\text{poly}(P)}$, where $\mathcal{L}_{\text{cl}}^*$ is the global minimum of the contrastive objective in (4.2.11).*
2. *Attention concentration on global area: given $X \in \mathcal{D}_k^{\text{cl}}$ with $k \in [K]$, for any $\mathbf{p} \in \mathcal{P}$, with high probability, we have $1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}(X'; Q^{(T^*)}) = o(1)$ for $X' \in \{X^+, X^{++}\}$.*
3. *All patches learn global FP correlation: given $k \in [K]$, for any $\mathbf{p} \in \mathcal{P}$, $t \in [0, T^*]$, $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} \gg \Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}$ with $m > 1$, and at the convergence, $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(T^*)} = \Theta(\log P)$, $\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(T^*)} = o(1)$.*

In the following, we will sketch the proof of the above theorem. Indeed, the roadmap of the analysis is similar to the masked reconstruction loss by using the induction argument, where the key difference is the properties for the gradient of the contrastive objective.

4.B.1 Preliminaries

In the following, we denote the contrastive loss without regularization as

$$\overline{\mathcal{L}}(Q) \triangleq \mathbb{E}_{X^+, X^{++}, \mathfrak{N}} \left[-\tau \log \left(\frac{e^{\text{Sim}_{\text{Fcl}}(X^+, X^{++})/\tau}}{\sum_{X' \in \mathfrak{B}} e^{\text{Sim}_{\text{Fcl}}(X^+, X')/\tau}} \right) \right].$$

Lemma 4.B.1 (feature gradient of contrastive loss). *Given $k \in [K]$, for $\mathbf{p} \in \mathcal{P}$, let $\tilde{\alpha}_{\mathbf{p} \rightarrow v_{k,m}}^{(t)} := \frac{1}{\eta} (\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t+1)} - \Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)})$ for $m \in [N_k]$, then*

$$\tilde{\alpha}_{\mathbf{p} \rightarrow v_{k,m}}^{(t)} = e_{\mathbf{p}}^\top \left(-\frac{\partial \overline{\mathcal{L}}}{\partial Q}(Q^{(t)}) \right) v_{k,m} = \alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)} - \lambda \Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)},$$

where

$$\begin{aligned} \alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)} &= e_{\mathbf{p}}^\top \left(-\frac{\partial \overline{\mathcal{L}}}{\partial Q}(Q^{(t)}) \right) v_{k,m} \\ &= \frac{1}{P} \mathbb{E} \left[\sum_{\mathbf{q} \in \mathcal{P}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)}(X^+) X_{\mathbf{q}}^{+\top} \left(F(X^{++}; Q^{(t)}) - \sum_{X' \in \mathfrak{B}} \frac{e^{\text{Sim}_F(X^+, X')/\tau}}{\sum_{X' \in \mathfrak{B}} e^{\text{Sim}_F(X^+, X')/\tau}} F(X'; Q^{(t)}) \right) \right. \\ &\quad \left. \cdot \left[X_{\mathbf{q}}^+ - \sum_{\mathbf{r}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{r}}^{(t)}(X^+) X_{\mathbf{r}}^{+\top} \right]^\top v_{k,m} \right]. \end{aligned}$$

Proof. Notice that

$$-\frac{\partial \mathcal{L}}{\partial Q} = -\frac{\partial \overline{\mathcal{L}}}{\partial Q} + \lambda Q.$$

Then for $-\frac{\partial \bar{\mathcal{L}}}{\partial Q}$, we begin with the chain rule and obtain

$$\begin{aligned}
& -\frac{\partial \bar{\mathcal{L}}}{\partial Q} \\
&= \mathbb{E} \left[\frac{\partial}{\partial Q} \left(\text{Sim}_F(X^+, X^{++}) - \tau \log \left(\sum_{X' \in \mathfrak{B}} e^{\text{Sim}_F(X^+, X')/\tau} \right) \right) \right] \\
&= \mathbb{E} \left[\frac{\partial F(X^+; Q)}{\partial Q} \left(F(X^{++}; Q) - \sum_{X' \in \mathfrak{B}} \frac{e^{\text{Sim}_F(X^+, X')/\tau}}{\sum_{X' \in \mathfrak{B}} e^{\text{Sim}_F(X^+, X')/\tau}} F(X'; Q) \right) \right] \\
&= \frac{1}{P} \mathbb{E} \left[\sum_{\mathbf{p}, \mathbf{q} \in \mathcal{P}} \frac{\partial \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+)}{\partial Q} X_{\mathbf{q}}^{+\top} \left(F(X^{++}; Q) - \sum_{X' \in \mathfrak{B}} \frac{e^{\text{Sim}_F(X^+, X')/\tau}}{\sum_{X' \in \mathfrak{B}} e^{\text{Sim}_F(X^+, X')/\tau}} F(X'; Q) \right) \right]. \quad (4.B.1)
\end{aligned}$$

We focus on the gradient for each attention score:

$$\begin{aligned}
\frac{\partial \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+)}{\partial Q} &= \sum_{\mathbf{r}} \frac{\exp \left(e_{\mathbf{p}}^\top Q (X_{\mathbf{b}}^+ + X_{\mathbf{q}}^+) \right)}{\left(\sum_{\mathbf{r}} \exp(e_{\mathbf{p}}^\top Q X_{\mathbf{r}}) \right)^2} e_{\mathbf{p}} (X_{\mathbf{q}}^+ - X_{\mathbf{r}}^+)^\top \\
&= \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}} \sum_{\mathbf{r}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{r}} e_{\mathbf{p}} (X_{\mathbf{q}}^+ - X_{\mathbf{r}}^+)^\top \\
&= \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) e_{\mathbf{p}} \cdot \left[X_{\mathbf{q}}^+ - \sum_{\mathbf{r}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{r}}(X^+) X_{\mathbf{r}}^+ \right]^\top.
\end{aligned}$$

Substituting the above equation into (4.B.1), we have

$$\begin{aligned}
-\frac{\partial \bar{\mathcal{L}}}{\partial Q} &= \frac{1}{P} \mathbb{E} \left[\sum_{\mathbf{p}, \mathbf{q} \in \mathcal{P}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) X_{\mathbf{q}}^{+\top} \left(F(X^{++}; Q) - \sum_{X' \in \mathfrak{B}} \frac{e^{\text{Sim}_F(X^+, X')/\tau}}{\sum_{X' \in \mathfrak{B}} e^{\text{Sim}_F(X^+, X')/\tau}} F(X'; Q) \right) \right. \\
&\quad \left. \cdot e_{\mathbf{p}} \left[X_{\mathbf{q}}^+ - \sum_{\mathbf{r}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{r}}(X^+) X_{\mathbf{r}}^+ \right]^\top \right].
\end{aligned}$$

Therefore,

$$\begin{aligned}
\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)} &= e_{\mathbf{p}}^\top \left(-\frac{\partial \bar{\mathcal{L}}}{\partial Q} \right) v_{k,m} \\
&= \frac{1}{P} \mathbb{E} \left[\sum_{\mathbf{q} \in \mathcal{P}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) X_{\mathbf{q}}^{+\top} \left(F(X^{++}; Q) - \sum_{X' \in \mathfrak{B}} \frac{e^{\text{Sim}_F(X^+, X')/\tau}}{\sum_{X' \in \mathfrak{B}} e^{\text{Sim}_F(X^+, X')/\tau}} F(X'; Q) \right) \right. \\
&\quad \left. \cdot \left[X_{\mathbf{q}}^+ - \sum_{\mathbf{r}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{r}}(X^+) X_{\mathbf{r}}^+ \right]^\top v_{k,m} \right].
\end{aligned}$$

□

We then present a high-probability event ensuring that the number of common unmasked patches in each area between positive augmented data pairs is proportional to the total number of patches in that area.

Lemma 4.B.2 (masking overlap). *Given a sample $X \sim \mathcal{D}^{c1}$, with probability $1 - e^{-\Theta(P^{\kappa_s})}$ over the randomness of masking augmentation to obtain X^+ , X^{++} , supposing X belongs to the k -th cluster, it holds that*

$$\sum_{\mathbf{p} \in \mathcal{P}_{k,m}} \mathbb{1}\{X_{\mathbf{p}}^+ \neq \mathbf{0}\} \mathbb{1}\{X_{\mathbf{p}}^{++} \neq \mathbf{0}\} = \Theta(C_{k,m}), \quad \forall m \in [N_k].$$

We denote the event that the above relationships hold as $\mathcal{A}_{1,com}$. Similarly, we have the following event for X^+ and X^{++} holds with high probability:

$$\begin{aligned} \mathcal{A}_{1,+} &:= \left\{ \sum_{\mathbf{p} \in \mathcal{P}_{k,m}} \mathbb{1}\{X_{\mathbf{p}}^+ \neq \mathbf{0}\} = \Theta(C_{k,m}), \forall m \in [N_k] \right\}, \\ \mathcal{A}_{1,++} &:= \left\{ \sum_{\mathbf{p} \in \mathcal{P}_{k,m}} \mathbb{1}\{X_{\mathbf{p}}^{++} \neq \mathbf{0}\} = \Theta(C_{k,m}), \forall m \in [N_k] \right\}. \end{aligned}$$

Proof. The proof is similar to the analysis of Lemma 4.A.4 by using the concentration property of hypergeometric distribution. $\square$

4.B.2 Initial stage: emergence of global correlations

For the training process at the initial stage, we define the stage transition time T_1 to be the iteration when $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} \geq (1 - \kappa_c) \log(P)$ for all $\mathbf{p} \in \mathcal{P}$ and $k \in [K]$.

We state the following induction hypothesis, which will hold throughout this stage:

Induction Hypothesis 4.B.1. For each $0 \leq t \leq T_1 = O(\frac{\log(P)p^{3-2\kappa_c}}{\eta})$, $k \in [K]$, letting $\lambda = \frac{2}{p^{3-\kappa_c} \log(P)}$, the following holds:

- $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}$ is monotonically increasing, and $\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} \in [0, (1 - \kappa_c) \log(P)]$;
- $|\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}| \leq O(\max\{P^{\kappa_s-1}, P^{2(\kappa_s-\kappa_c)} \cdot \Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)}\})$ for $m > 1$.

Lemma 4.B.3 (bounding the noise correlation). *Let us define a noiseless version of the attention score and the network output as*

$$\widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{q}}(X) := \frac{e^{e_{\mathbf{p}}^T Q(X_{\mathbf{q}} - \xi_{\mathbf{q}})}}{\sum_{\mathbf{r} \in \mathcal{P}} e^{e_{\mathbf{p}}^T Q(X_{\mathbf{r}} - \xi_{\mathbf{r}})}}, \quad \text{for } \mathbf{p}, \mathbf{q} \in \mathcal{P}. \quad (4.B.2)$$

$$\widehat{F}(X; Q) := \frac{1}{P} \sum_{\mathbf{p}, \mathbf{q} \in \mathcal{P}} \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{q}}(X) \cdot X_{\mathbf{q}} \in \mathbb{R}^d. \quad (4.B.3)$$

Denote

$$\widehat{\ell}_p(X, \mathfrak{B}) := \frac{e^{\text{Sim}_{\widehat{F}}(X^+, X^{++})/\tau}}{\sum_{X \in \mathfrak{B}} e^{\text{Sim}_{\widehat{F}}(X^+, X)/\tau}}, \quad \widehat{\ell}_s(X, \mathfrak{B}) := \frac{e^{\text{Sim}_{\widehat{F}}(X^+, X^{-s})/\tau}}{\sum_{X \in \mathfrak{B}} e^{\text{Sim}_{\widehat{F}}(X^+, X)/\tau}}.$$

Then supposing Induction Hypothesis 4.B.1 holds for $t \leq T_1$, with high probability over the randomness of $X^+, X^{++}, \mathfrak{N}$, then for $X \in \mathfrak{B}$, any $\mathbf{p}, \mathbf{q} \in \mathcal{P}$, $s \in [N_c]$, it holds that

$$\begin{aligned} \left| \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)}(X) - \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^{(t)}(X) \right| &\leq \frac{1}{\text{poly}(d)}; \\ \left\| \widehat{F}^{(t)}(X; Q) - F^{(t)}(X; Q) \right\| &\leq \frac{1}{\text{poly}(d)}; \\ \left| \ell_p^{(t)}(X, \mathfrak{B}) - \widehat{\ell}_p^{(t)}(X, \mathfrak{B}) \right|, \left| \ell_s^{(t)}(X, \mathfrak{B}) - \widehat{\ell}_s^{(t)}(X, \mathfrak{B}) \right| &\leq \frac{1}{\text{poly}(d)}. \end{aligned}$$

We denote the event that the above inequalities hold as $\mathcal{A}_2$.

Proof. The result follows directly from the concentration of Gaussian random variables, the boundedness of the feature vectors and the boundedness of $\|e_{\mathbf{p}}Q\|_2 \leq \Phi_{k \rightarrow v_{k,m}}$ due to the Induction Hypothesis 4.B.1.

□

Lemma 4.B.4 (attention score). *Suppose the Induction Hypothesis 4.B.1 holds for $t \leq T_1$, given $\{X^+, X^{++}, \mathfrak{N}\}$, assuming $X \in \mathcal{D}_k^{c1}$ with $k \in [K]$, then for $m \in [N_k]$, $\mathbf{p} \in \mathcal{P}$, we have*

1. for $a \in \{+, ++\}$, if $X^a \in \mathcal{A}_{2,a}$, then
 - (a) $1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}(X^a) \geq \Omega(1)$ and $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}(X^a) \geq \Omega(\frac{1}{p^{1-\kappa_c}})$;
 - (b) for $m > 1$, $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)}(X^a) = \Theta(\frac{1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}(X^a)}{p^{1-\kappa_s}})$;
2. for $X' \in \mathfrak{N}$, we have
 - (a) $1 - \widehat{\text{Attn}}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}(X') \geq \Omega(1)$ and $\widehat{\text{Attn}}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}(X') \geq \Omega(\frac{1}{p^{1-\kappa_c}})$;
 - (b) for $m > 1$, $\widehat{\text{Attn}}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}^{(t)}(X') = \Theta(\frac{1 - \widehat{\text{Attn}}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}(X')}{p^{1-\kappa_s}})$.

The intuition behind this lemma is that, due to the zero initialization of Q , the attention scores are nearly uniform. As a result, the area attention score $\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}(X^+)$ is proportional to the number of unmasked patches in this area. If Induction Hypothesis 4.B.1 holds, we can easily conclude that only the area attention score for the global area will increase, while the relative relationships among the local area attention scores will be preserved.

Lemma 4.B.5 (logit score). *Suppose the Induction Hypothesis 4.B.1 holds for $t \leq T_1$, given $\{X^+, X^{++}, \mathfrak{N}\}$, suppose $X \in \mathcal{D}_k^{c1}$, we have*

$$\begin{aligned} 1 - \ell_q^{(t)}(X, \mathfrak{B}) &\geq \Omega(1), \quad \ell_q^{(t)}(X, \mathfrak{B}) \geq \Omega(\frac{1}{N_s}), \quad q \in \mathfrak{B} \cap \mathcal{D}_k^{c1} \\ \ell_q^{(t)}(X, \mathfrak{B}) &\leq O(\frac{1}{N_s}), \quad \text{else.} \end{aligned}$$

Lemma 4.B.6 (feature gradient near initialization). *Suppose the Induction Hypothesis 4.B.1 holds for $t \leq T_0$, then for $t \leq T_1$, given $k \in [K]$, $m \in [N_k]$, for $\mathbf{p} \in \mathcal{P}$,*

- *For the global feature $m = 1$,*

$$\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} = \Theta \left(\frac{1}{P} \mathbb{E} \left[z_1 (1 - \ell_p) \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}(X^+) \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}(X^{++}) \right] \right).$$

- *For the local feature $m > 1$*

$$\begin{aligned} \alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)} &= \Theta \left(\frac{1}{P} \mathbb{E} \left[z_m (1 - \ell_p) \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}(X^+) \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}(X^{++}) \right] \right) \\ &\quad + O \left(\frac{1}{P} \mathbb{E} \left[z_m (1 - \ell_p) \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}(X^+) \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}(X^+) \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}(X^{++}) \right] \right). \end{aligned}$$

Proof.

$$\begin{aligned} \alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)} &= \frac{1}{P} \mathbb{E} \left[\sum_{\mathbf{q} \in \mathcal{P}} \mathbf{attn}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) X_{\mathbf{q}}^+{}^\top \left((1 - \ell_p) F(X^{++}; Q) - \sum_{s=1}^{N_c} \ell_s F(X^{-,s}; Q) \right) \right. \\ &\quad \cdot \left[X_{\mathbf{q}}^+ - \sum_{\mathbf{r}} \mathbf{attn}_{\mathbf{p} \rightarrow \mathbf{r}}(X^+) X_{\mathbf{r}}^+ \right]^\top v_{k,m} (\mathbb{1}_{\mathcal{A}_1} + \mathbb{1}_{\mathcal{A}_1^c}) \Big] \\ &\stackrel{(a)}{=} \frac{1}{P} \mathbb{E} \left[\sum_{\mathbf{q} \in \mathcal{P}} \widehat{\mathbf{attn}}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) X_{\mathbf{q}}^+{}^\top \left((1 - \widehat{\ell}_p) \widehat{F}(X^{++}; Q) - \sum_{s=1}^{N_c} \widehat{\ell}_s \widehat{F}(X^{-,s}; Q) \right) \right. \\ &\quad \cdot \left[X_{\mathbf{q}}^+ - \sum_{\mathbf{r}} \widehat{\mathbf{attn}}_{\mathbf{p} \rightarrow \mathbf{r}}(X^+) X_{\mathbf{r}}^+ \right]^\top v_{k,m} (\mathbb{1}_{\mathcal{A}_1} + \mathbb{1}_{\mathcal{A}_1^c}) \Big] + \Xi_{\mathbf{p},k,m,1} \\ &= \frac{1}{P} \mathbb{E} \left[\sum_{i=1}^{N_k} \sum_{\mathbf{q} \in \mathcal{P}_{k,i} \cap \mathcal{U}^+} \widehat{\mathbf{attn}}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) (z_i v_{k,i})^\top \left((1 - \widehat{\ell}_p) \widehat{F}(X^{++}; Q) - \sum_{s=1}^{N_c} \widehat{\ell}_s \widehat{F}(X^{-,s}; Q) \right) \right. \\ &\quad \cdot \left[z_i v_{k,i} - \sum_{j=1}^{N_k} \sum_{\mathbf{r} \in \mathcal{P}_{k,j} \cap \mathcal{U}^+} \widehat{\mathbf{attn}}_{\mathbf{p} \rightarrow \mathbf{r}}(X^+) z_j v_{k,j} \right]^\top v_{k,m} \Big] \tag{J1} \end{aligned}$$

$$\begin{aligned} &+ \frac{1}{P} \mathbb{E} \left[\sum_{i=1}^{N_k} \sum_{\mathbf{q} \in \mathcal{P}_{k,i} \cap \mathcal{U}^+} \widehat{\mathbf{attn}}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) (z_i v_{k,i})^\top \left((1 - \widehat{\ell}_p) \widehat{F}(X^{++}; Q) - \sum_{s=1}^{N_c} \widehat{\ell}_s \widehat{F}(X^{-,s}; Q) \right) \right. \\ &\quad \cdot \left[\xi_{\mathbf{q}} - \sum_{\mathbf{r} \in \mathcal{P} \cap \mathcal{U}^+} \widehat{\mathbf{attn}}_{\mathbf{p} \rightarrow \mathbf{r}}(X^+) \xi_{\mathbf{r}} \right]^\top v_{k,m} \Big] \tag{J2} \end{aligned}$$

$$+ \frac{1}{P} \mathbb{E} \left[\sum_{\mathbf{q} \in \mathcal{P} \cap \mathcal{U}^+} \widehat{\mathbf{attn}}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) \xi_{\mathbf{q}}^\top \left((1 - \widehat{\ell}_p) \widehat{F}(X^{++}; Q) - \sum_{s=1}^{N_c} \widehat{\ell}_s \widehat{F}(X^{-,s}; Q) \right) \right]$$

$$\cdot \left[z_i v_{k,i} - \sum_{j=1}^{N_k} \sum_{\mathbf{r} \in \mathcal{P}_{k,j} \cap \mathcal{U}^+} \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{r}}(X^+) z_j v_{k,j} \right]^\top v_{k,m} \Big] \quad (J_3)$$

$$+ \frac{1}{P} \mathbb{E} \left[\sum_{\mathbf{q} \in \mathcal{P} \cap \mathcal{U}^+} \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) \xi_{\mathbf{q}}^\top \left((1 - \widehat{\ell}_p) \widehat{F}(X^{++}; Q) - \sum_{s=1}^{N_c} \widehat{\ell}_s \widehat{F}(X^{-,s}; Q) \right) \right. \\ \left. \cdot \left[\xi_{\mathbf{q}} - \sum_{\mathbf{r} \in \mathcal{P} \cap \mathcal{U}^+} \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{r}}(X^+) \xi_{\mathbf{r}} \right]^\top v_{k,m} \right] \quad (J_4)$$

$$+ \Xi_{\mathbf{p},k,m,1},$$

where (a) is bounded by Lemma 4.B.3 with error up to $\Xi_{\mathbf{p},k,m,1} \leq \frac{1}{\text{poly}(d)}$, $\mathcal{U}^+$ is the set of masked patches for X^+ . We first look at the term J_1 , notice that $\xi_{\mathbf{q}}$ is the random Gaussian noise with zero mean, and is independent of $\widehat{\text{attn}}$ and $\widehat{\ell}$, we then have

$$\begin{aligned} J_4 &= \frac{1}{P^2} \mathbb{E} \left[\sum_{\mathbf{q} \in \mathcal{P}_{k,m} \cap \mathcal{U}^+} \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) \xi_{\mathbf{q}}^\top \left((1 - \widehat{\ell}_p) \sum_{\mathbf{p}' \in \mathcal{P}} \sum_{\mathbf{r} \in \mathcal{P}_{k,m} \cap \mathcal{U}^{++}} \widehat{\text{attn}}_{\mathbf{p}' \rightarrow \mathbf{r}}(X^{++}) z_m v_{k,m} \right) \right. \\ &\quad \left. \cdot \left[\xi_{\mathbf{q}} - \sum_{\mathbf{r} \in \mathcal{P} \cap \mathcal{U}^+} \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{r}}(X^+) \xi_{\mathbf{r}} \right]^\top v_{k,m} \right] \\ &+ \frac{1}{P^2} \mathbb{E} \left[\sum_{\mathbf{q} \in \mathcal{P}_{k,m} \cap \mathcal{U}^+} \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) \xi_{\mathbf{q}}^\top \left((1 - \widehat{\ell}_p) \sum_{\mathbf{p}' \in \mathcal{P}} \sum_{\mathbf{r} \in \mathcal{U}^{++}} \widehat{\text{attn}}_{\mathbf{p}' \rightarrow \mathbf{r}}(X^{++}) \xi_{\mathbf{r}} \right) \right. \\ &\quad \left. \cdot \left[\xi_{\mathbf{q}} - \sum_{\mathbf{r} \in \mathcal{P} \cap \mathcal{U}^+} \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{r}}(X^+) \xi_{\mathbf{r}} \right]^\top v_{k,m} \right] \\ &= \frac{1}{P^2} \mathbb{E} \left[z_m \sum_{\mathbf{q} \in \mathcal{P}_{k,m} \cap \mathcal{U}^+} \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) \right. \\ &\quad \left. \left((1 - \widehat{\ell}_p) \sum_{\mathbf{p}' \in \mathcal{P}} \sum_{\mathbf{r} \in \mathcal{P}_{k,m} \cap \mathcal{U}^{++}} \widehat{\text{attn}}_{\mathbf{p}' \rightarrow \mathbf{r}}(X^{++}) \right) \left[1 - \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) \right] \right] \\ &= \frac{1}{P^2} \mathbb{E} \left[z_m \left[\text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}(X^+) - \sum_{\mathbf{q} \in \mathcal{P}_{k,m} \cap \mathcal{U}^+} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}^2(X^+) \right] \right. \\ &\quad \left. \left((1 - \ell_p) \sum_{\mathbf{p}' \in \mathcal{P}} \text{Attn}_{\mathbf{p}' \rightarrow \mathcal{P}_{k,m}}(X^{++}) \right) \right] + \Xi_{\mathbf{p},k,m,2} \\ &= \Theta \left(\frac{1}{P} \mathbb{E} \left[z_m (1 - \ell_p) \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}(X^+) \right. \right. \\ &\quad \left. \left. \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}(X^{++}) \right] \right) \end{aligned}$$

where (a) is bounded by invoking Lemma 4.B.3 with error up to $\Xi_{\mathbf{p},k,m,2} \leq \frac{1}{\text{poly}(d)}$, and the last equality is due to Lemma 4.B.4. Turning to J_2 ,

$$\begin{aligned}
J_2 &= \frac{1}{P} \mathbb{E} \left[\mathbb{E} \left[\sum_{i=1}^{N_k} \sum_{\mathbf{q} \in \mathcal{P}_{k,i} \cap \mathcal{U}^+} \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) (z_i v_{k,i})^\top (1 - \widehat{\ell}_p) \widehat{F}(X^{++}; Q) \right. \right. \\
&\quad \left. \left. \cdot \left[\xi_{\mathbf{q}} - \sum_{\mathbf{r} \in \mathcal{P} \cap \mathcal{U}^+} \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{r}}(X^+) \xi_{\mathbf{r}} \right]^\top v_{k,m} \middle| \xi \right] \right] \\
&= \frac{1}{P^2} \mathbb{E} \left[\mathbb{E} \left[\sum_{i=1}^{N_k} \sum_{\mathbf{q} \in \mathcal{P}_{k,i} \cap \mathcal{U}^+} \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) (z_i v_{k,i})^\top (1 - \widehat{\ell}_p) \sum_{\mathbf{p}' \in \mathcal{P}, \mathbf{r} \in \mathcal{P} \cap \mathcal{U}^{++}} \widehat{\text{attn}}_{\mathbf{p}' \rightarrow \mathbf{r}}(X^{++}) \xi_{\mathbf{r}} \right. \right. \\
&\quad \left. \left. \cdot \left[\xi_{\mathbf{q}} - \sum_{\mathbf{r} \in \mathcal{P} \cap \mathcal{U}^+} \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{r}}(X^+) \xi_{\mathbf{r}} \right]^\top v_{k,m} \middle| \xi \right] \right] \\
&= \frac{1}{P^2} \mathbb{E} \left[\mathbb{E} \left[\sum_{\mathbf{q} \in \mathcal{P}_{k,m} \cap \mathcal{U}^+} \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) (z_m v_{k,m})^\top (1 - \widehat{\ell}_p) \sum_{\mathbf{p}' \in \mathcal{P}, \mathbf{r} \in \mathcal{P} \cap \mathcal{U}^{++}} \widehat{\text{attn}}_{\mathbf{p}' \rightarrow \mathbf{r}}(X^{++}) \xi_{\mathbf{r}} \right. \right. \\
&\quad \left. \left. \cdot \left[\xi_{\mathbf{q}} - \sum_{\mathbf{r} \in \mathcal{P} \cap \mathcal{U}^+} \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{r}}(X^+) \xi_{\mathbf{r}} \right]^\top v_{k,m} \middle| \xi \right] \right] \\
&= \frac{1}{P^2} \mathbb{E} \left[z_m \sum_{\mathbf{q} \in \mathcal{P}_{k,m} \cap \mathcal{U}^+} \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) (1 - \widehat{\ell}_p) \right. \\
&\quad \left. \cdot \left(\widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) \mathbb{1}_{\mathbf{q} \in \mathcal{U}^{++}} - \sum_{\mathbf{p}' \in \mathcal{P}, \mathbf{r} \in \mathcal{P} \cap \mathcal{U}^{++} \cap \mathcal{U}^+} \widehat{\text{attn}}_{\mathbf{p}' \rightarrow \mathbf{r}}(X^{++}) \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{r}}(X^+) \right) \right] \\
&= \frac{1}{P^2} \mathbb{E} \left[z_m \sum_{\mathbf{q} \in \mathcal{P}_{k,m} \cap \mathcal{U}^+} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) (1 - \ell_p) \right. \\
&\quad \left. \cdot \left(\text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) \mathbb{1}_{\mathbf{q} \in \mathcal{U}^{++}} - \sum_{\mathbf{p}' \in \mathcal{P}, \mathbf{r} \in \mathcal{P} \cap \mathcal{U}^{++} \cap \mathcal{U}^+} \text{attn}_{\mathbf{p}' \rightarrow \mathbf{r}}(X^{++}) \text{attn}_{\mathbf{p} \rightarrow \mathbf{r}}(X^+) \right) \right] + \Xi_{\mathbf{p},k,m,3}.
\end{aligned}$$

Thus, by invoking Lemma 4.B.4, we have

$$\begin{aligned}
|J_2| &\leq O \left(\frac{1}{P} \mathbb{E} \left[z_m \sum_{\mathbf{q} \in \mathcal{P}_{k,m} \cap \mathcal{U}^+} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) (1 - \ell_p) \cdot \left(\max_{\mathbf{r} \in \mathcal{P} \cap \mathcal{U}^{++} \cap \mathcal{U}^+} \text{attn}_{\mathbf{p} \rightarrow \mathbf{r}}(X^+) \right) \right] \right) + \Xi_{\mathbf{p},k,m,3} \\
&\leq O \left(\frac{1}{P \cdot C_{k,1}} \mathbb{E} \left[z_m (1 - \ell_p) \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}(X^+) \cdot \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}(X^{++}) \right] \right).
\end{aligned}$$

For J_3 ,

$$J_3 = \frac{1}{P} \mathbb{E} \left[\sum_{\mathbf{q} \in \mathcal{P} \cap \mathcal{U}^+} \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) \xi_{\mathbf{q}}^\top \left((1 - \widehat{\ell}_p) \widehat{F}(X^{++}; Q) - \sum_{s=1}^n \widehat{\ell}_s \widehat{F}(X^{-,s}; Q) \right) \right]$$

$$\begin{aligned}
& \cdot \left[z_i v_{k,i} - \sum_{j=1}^{N_k} \sum_{\mathbf{r} \in \mathcal{P}_{k,j} \cap \mathcal{U}^+} \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{r}}(X^+) z_j v_{k,j} \right]^\top v_{k,m} \Big] \\
&= \frac{1}{P} \mathbb{E} \left[\mathbb{E} \left[\sum_{\mathbf{q} \in \mathcal{P}_{k,m} \cap \mathcal{U}^+} \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) \xi_{\mathbf{q}}^\top \left((1 - \widehat{\ell}_p) \widehat{F}(X^{++}; Q) \right) \right. \right. \\
&\quad \left. \left. \cdot z_m \left[1 - \sum_{\mathbf{r} \in \mathcal{P}_{k,m} \cap \mathcal{U}^+} \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{r}}(X^+) \right] \middle| \xi \right] \right] \\
&= \frac{1}{P^2} \mathbb{E} \left[z_m \sum_{\mathbf{q} \in \mathcal{P}_{k,m} \cap \mathcal{U}^+ \cap \mathcal{U}^{++}} \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) (1 - \widehat{\ell}_p) \right. \\
&\quad \left. \cdot \sum_{\mathbf{p}' \in \mathcal{P}} \widehat{\text{attn}}_{\mathbf{p}' \rightarrow \mathbf{q}}(X^{++}) \left[1 - \sum_{\mathbf{r} \in \mathcal{P}_{k,m} \cap \mathcal{U}^+} \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{r}}(X^+) \right] \right] \\
&= \frac{1}{P^2} \mathbb{E} \left[z_m \sum_{\mathbf{q} \in \mathcal{P}_{k,m} \cap \mathcal{U}^+ \cap \mathcal{U}^{++}} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) (1 - \ell_p) \right. \\
&\quad \left. \cdot \sum_{\mathbf{p}' \in \mathcal{P}} \text{attn}_{\mathbf{p}' \rightarrow \mathbf{q}}(X^{++}) \left[1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}(X^+) \right] \right] + \Xi_{\mathbf{p},k,m,4} \\
&\leq O \left(\frac{1}{P^2} \mathbb{E} \left[z_m \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}(X^+) \left(1 - \text{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}(X^+) \right) (1 - \ell_p) \cdot \sum_{\mathbf{p}' \in \mathcal{P}} O \left(\frac{1}{C_{k,m}} \right) \cdot \text{Attn}_{\mathbf{p}' \rightarrow \mathcal{P}_{k,m}}(X^{++}) \right] \right) \\
&\leq O \left(\frac{J_4}{C_{k,m}} \right),
\end{aligned}$$

where the last inequality is due to Lemma 4.B.4. Moreover, for J_1 ,

$$\begin{aligned}
J_1 &= \frac{1}{P^2} \mathbb{E} \left[\sum_{i=1}^{N_k} \sum_{\mathbf{q} \in \mathcal{P}_{k,i} \cap \mathcal{U}^+} \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) (z_i v_{k,i})^\top \right. \\
&\quad \cdot \left((1 - \widehat{\ell}_p) \sum_{\mathbf{p}' \in \mathcal{P}} \sum_{j=1}^{N_k} \sum_{\mathbf{q}' \in \mathcal{P}_{k,j} \cap \mathcal{U}^{++}} \widehat{\text{attn}}_{\mathbf{p}' \rightarrow \mathbf{q}'}(X^{+,+}) z_j v_{k,j} \right. \\
&\quad \left. \left. - \sum_{X^{-,s} \in \mathfrak{N} \cap \mathcal{D}_k^{cl}} \widehat{\ell}_s \sum_{\mathbf{p}' \in \mathcal{P}} \sum_{j=1}^{N_k} \sum_{\mathbf{q}' \in \mathcal{P}_{k,j}} \widehat{\text{attn}}_{\mathbf{p}' \rightarrow \mathbf{q}'}(X^{-,s}) z_{s,j} v_{k,j} \right) \right. \\
&\quad \left. \cdot \left[z_i v_{k,i} - \sum_{j=1}^{N_k} \sum_{\mathbf{r} \in \mathcal{P}_{k,j} \cap \mathcal{U}^+} \widehat{\text{attn}}_{\mathbf{p} \rightarrow \mathbf{r}} z_j v_{k,j} \right]^\top v_{k,m} \right] \\
&= \frac{1}{P^2} \mathbb{E} \left[\sum_{i=1}^{N_k} \sum_{\mathbf{q} \in \mathcal{P}_{k,i} \cap \mathcal{U}^+} \text{attn}_{\mathbf{p} \rightarrow \mathbf{q}}(X^+) (z_i v_{k,i})^\top \right. \\
&\quad \cdot \left((1 - \ell_p) \sum_{\mathbf{p}' \in \mathcal{P}} \sum_{j=1}^{N_k} \sum_{\mathbf{q}' \in \mathcal{P}_{k,j} \cap \mathcal{U}^{++}} \text{attn}_{\mathbf{p}' \rightarrow \mathbf{q}'}(X^{+,+}) z_j v_{k,j} \right.
\end{aligned}$$

$$\begin{aligned}
& - \sum_{X^{-,s} \in \mathfrak{N} \cap \mathcal{D}_k^{cl}} \ell_s \sum_{\mathbf{p}' \in \mathcal{P}} \sum_{j=1}^{N_k} \sum_{\mathbf{q}' \in \mathcal{P}_{k,j}} \mathbf{attn}_{\mathbf{p}' \rightarrow \mathbf{q}'}(X^{-,s}) z_{s,j} v_{k,j} \Big) \\
& \cdot \left[z_i v_{k,i} - \sum_{j=1}^{N_k} \sum_{\mathbf{r} \in \mathcal{P}_{k,j} \cap \mathcal{U}^+} \mathbf{attn}_{\mathbf{p} \rightarrow \mathbf{r}} z_j v_{k,j} \right]^\top v_{k,m} \Big] + \Xi_{\mathbf{p},k,m,5} \\
& = \frac{1}{p^2} \mathbb{E} \left[\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}(X^+) \left(z_m^2 (1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}(X^+)) v_{k,m} - \sum_{i \neq m} z_m z_i \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,i}}(X^+) v_{k,i} \right)^\top \right. \\
& \quad \cdot \left((1 - \ell_p) \sum_{\mathbf{p}' \in \mathcal{P}} \sum_{j=1}^{N_k} \sum_{\mathbf{q}' \in \mathcal{P}_{k,j} \cap \mathcal{U}^{++}} \mathbf{attn}_{\mathbf{p}' \rightarrow \mathbf{q}'}(X^{+,+}) z_j v_{k,j} \right. \\
& \quad \left. \left. - \sum_{X^{-,s} \in \mathfrak{N} \cap \mathcal{D}_k^{cl}} \ell_s \sum_{j=1}^{N_k} \sum_{\mathbf{q}' \in \mathcal{P}_{k,j}} \mathbf{attn}_{\mathbf{p}' \rightarrow \mathbf{q}'}(X^{-,s}) z_{s,j} v_{k,j} \right) \right] + \Xi_{\mathbf{p},k,m,5} \\
& = \frac{1}{p^2} \mathbb{E} \left[\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}(X^+) \left(z_m^2 (1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}(X^+)) v_{k,m} - \sum_{i \neq m} z_m z_i \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,i}}(X^+) v_{k,i} \right)^\top \right. \\
& \quad \cdot \left((1 - \ell_p) \sum_{\mathbf{p}' \in \mathcal{P}} \sum_{j=1}^{N_k} \mathbf{Attn}_{\mathbf{p}' \rightarrow \mathcal{P}_{k,j}}(X^{++}) z_j v_{k,j} - \sum_{X^{-,s} \in \mathfrak{N} \cap \mathcal{D}_k^{cl}} \ell_s \sum_{\mathbf{p}' \in \mathcal{P}} \sum_{j=1}^{N_k} \widetilde{\mathbf{Attn}}_{\mathbf{p}' \rightarrow \mathcal{P}_{k,j}}(X^{-,s}) z_{s,j} v_{k,j} \right) \Big] \\
& \quad + \Xi_{\mathbf{p},k,m,5} \\
& = \frac{1}{p^2} \mathbb{E} \left[\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}(X^+) \left(z_m^2 (1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}(X^+)) \right. \right. \\
& \quad \cdot \left(\sum_{\mathbf{p}' \in \mathcal{P}} ((1 - \ell_p) z_m \mathbf{Attn}_{\mathbf{p}' \rightarrow \mathcal{P}_{k,m}}(X^{++}) - \sum_{X^{-,s} \in \mathfrak{N} \cap \mathcal{D}_k^{cl}} z_{s,m} \ell_s \widetilde{\mathbf{Attn}}_{\mathbf{p}' \rightarrow \mathcal{P}_{k,m}}(X^{-,s})) \right) \Big] \quad (J_{1,1}) \\
& - \frac{1}{p^2} \mathbb{E} \left[\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}(X^+) \left(\sum_{i \neq m} z_m z_i \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,i}}(X^+) \right. \right. \\
& \quad \cdot \left(\sum_{\mathbf{p}' \in \mathcal{P}} ((1 - \ell_p) z_i \mathbf{Attn}_{\mathbf{p}' \rightarrow \mathcal{P}_{k,i}}(X^{++}) - \sum_{X^{-,s} \in \mathfrak{N} \cap \mathcal{D}_k^{cl}} z_{s,i} \ell_s \widetilde{\mathbf{Attn}}_{\mathbf{p}' \rightarrow \mathcal{P}_{k,i}}(X^{-,s})) \right) \Big] \quad (J_{1,2}) \\
& \quad + \Xi_{\mathbf{p},k,m,5}.
\end{aligned}$$

Notice that $J_{1,1} = \Theta(J_4)$. Furthermore, when $m = 1$, $J_{1,2}$ is negligible compared to J_1 , else

$$\begin{aligned}
|J_{1,2}| & \leq O \left(\frac{1}{p^2} \mathbb{E} \left[\mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}(X^+) \left(z_m z_1 \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}(X^+) \right. \right. \right. \\
& \quad \cdot \left(\sum_{\mathbf{p}' \in \mathcal{P}} ((1 - \ell_p) z_1 \mathbf{Attn}_{\mathbf{p}' \rightarrow \mathcal{P}_{k,1}}(X^{++}) - \sum_{X^{-,s} \in \mathfrak{N} \cap \mathcal{D}_k^{cl}} z_{s,1} \ell_s \widetilde{\mathbf{Attn}}_{\mathbf{p}' \rightarrow \mathcal{P}_{k,1}}(X^{-,s})) \right) \Big] \Big) \\
& \leq O \left(\frac{1}{p} \mathbb{E} \left[z_m (1 - \ell_p) \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,m}}(X^+) \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}(X^+) \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}(X^{++}) \right] \right).
\end{aligned}$$

Putting all the terms together, and noticed that

$$\begin{aligned} & O\left(\frac{1}{P}\mathbb{E}\left[z_m(1-\ell_p)\mathbf{Attn}_{\mathbf{p}\rightarrow\mathcal{P}_{k,m}}(X^+)\mathbf{Attn}_{\mathbf{p}\rightarrow\mathcal{P}_{k,1}}(X^+)\mathbf{Attn}_{\mathbf{p}\rightarrow\mathcal{P}_{k,1}}(X^{++})\right]\right) \\ & \geq O\left(\frac{1}{P\cdot C_{k,1}}\mathbb{E}\left[z_m(1-\ell_p)\mathbf{Attn}_{\mathbf{p}\rightarrow\mathcal{P}_{k,m}}(X^+)\cdot\mathbf{Attn}_{\mathbf{p}\rightarrow\mathcal{P}_{k,1}}(X^{++})\right]\right), \end{aligned}$$

then we complete the proof. $\square$

Proof of Induction Hypothesis 4.B.1. By Lemma 4.B.6 and Lemma 4.B.5, at the initial stage of the learning process, we have

$$\begin{aligned} \alpha_{\mathbf{p}\rightarrow v_{k,1}}^{(0)} & \propto \frac{1}{P}\mathbb{E}\left[\mathbf{Attn}_{\mathbf{p}\rightarrow\mathcal{P}_{k,1}}(X^+)\mathbf{Attn}_{\mathbf{p}\rightarrow\mathcal{P}_{k,1}}(X^{++})\right], \\ \alpha_{\mathbf{p}\rightarrow v_{k,1}}^{(0)} & \lesssim \frac{1}{P}\max\left\{\mathbb{E}\left[\mathbf{Attn}_{\mathbf{p}\rightarrow\mathcal{P}_{k,m}}(X^+)\mathbf{Attn}_{\mathbf{p}\rightarrow\mathcal{P}_{k,m}}(X^{++})\right], \right. \\ & \quad \left.\mathbb{E}\left[\mathbf{Attn}_{\mathbf{p}\rightarrow\mathcal{P}_{k,m}}(X^+)\mathbf{Attn}_{\mathbf{p}\rightarrow\mathcal{P}_{k,1}}(X^+)\mathbf{Attn}_{\mathbf{p}\rightarrow\mathcal{P}_{k,1}}(X^{++})\right]\right\}. \end{aligned}$$

Then by the relations of attention score in Lemma 4.B.4, focusing on the high-probability event $A_{1,+}$ and $A_{1,++}$ we have

$$|\alpha_{\mathbf{p}\rightarrow v_{k,m}}^{(t)}| \leq O\left(\max\{P^{\kappa_s-1}, P^{2(\kappa_s-\kappa_c)}\} \cdot \Phi_{\mathbf{p}\rightarrow v_{k,1}}^{(t)}\right) |\alpha_{\mathbf{p}\rightarrow v_{k,1}}^{(t)}| \quad \text{for } m > 1. \quad (4.B.4)$$

By Lemma 4.B.6 and Lemma 4.B.4, we have $\alpha_{\mathbf{p}\rightarrow\mathcal{P}_{k,1}}^{(t)} \geq \Omega(\frac{1}{P^{3-2\kappa_c}}) \geq \lambda\Phi_{\mathbf{p}\rightarrow\mathcal{P}_{k,1}}^{(t)}$, which implies the regularization in this stage is not violated for the dominated FP correlation $\Phi_{\mathbf{p}\rightarrow\mathcal{P}_{k,1}}^{(t)}$. Hence, we could focus on the relation between $\alpha_{\mathbf{p}\rightarrow\mathcal{P}_{k,m}}^{(t)}$ and $\alpha_{\mathbf{p}\rightarrow\mathcal{P}_{k,1}}^{(t)}$ for $m > 1$.

Therefore, the existence of T_1 can be directly obtained by the gradient estimation in Lemma 4.B.6 and the lower bound for the area attention of the global area in Lemma 4.B.4. The induction argument follows directly from 4.B.4. $\square$

The key takeaway from the first stage is that the growth of feature-position attention correlation for the global area is dominant, specifically, $\alpha_{\mathbf{p}\rightarrow v_{k,1}}^{(t)} \gg |\alpha_{\mathbf{p}\rightarrow v_{k,m}}^{(t)}|$. After this initial stage, $\Phi_{\mathbf{p}\rightarrow v_{k,1}}$ reaches $\Omega(\log(P))$, $\mathbf{Attn}_{\mathbf{p}\rightarrow\mathcal{P}_{k,1}}$ has reached $\Omega(1)$ and $1-\ell_p$ still keeps at a constant level. The dominance of global FP correlation will be preserved in the following and the learning process will enter the convergence stage.

4.B.3 Convergence

At this stage, we are going to prove that as long as the ViTs have already learned the global FP correlations, they will indeed converge to these global solutions, which leads to the collapsed global representation. We present the statement of our convergence theorem below.

Theorem 4.B.2 (Convergence guarantees). *Letting $T_2 = \Omega(\frac{P^4 \log P}{\eta})$, for any $T \in [T_2, O(\frac{\text{poly}(P) \log P}{\eta})]$, letting $\lambda = \Theta(\frac{1}{P \log P})$ we have*

$$\frac{1}{T} \sum_{t=T_2}^T \mathcal{L}(Q^{(t)}) \leq \mathcal{L}_{c1}^* + \frac{1}{\text{poly } P},$$

where $\mathcal{L}_{c1}^*$ is the global minimum of the regularized contrastive objective.

We have the following hypothesis for the end of the learning process.

Induction Hypothesis 4.B.2. For $t \in [\Omega(\frac{P^4 \log P}{\eta}), O(\frac{\text{poly}(P) \log P}{\eta})]$, we have the following results:

- For any $k \in [K]$, $\mathbf{p} \in \mathcal{P}$, and $m \in [N_k]$

$$\Phi_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} \in [C_1^*, C_2^*] \log P \quad |\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}| \leq \tilde{O}(\frac{1}{P^{\delta_*}}).$$

where $C_1^*, C_2^* > 0$ are some constants and $\delta_* \in (0, 1)$ is some small constant.

- Attention score from the global area: given $X \in \mathcal{D}_k$, $1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}(X^a) \leq \frac{1}{\text{poly}(P)}$ for $a \in \{+, ++\}$ and $1 - \widetilde{\mathbf{Attn}}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}(X^{n,s}) \leq \frac{1}{\text{poly}(P)}$ for $s \in [N_c]$ with high probability.
- Bounded gradient for the loss:

$$\|\nabla_Q \mathcal{L}(Q^{(t)})\|_F^2 \leq \tilde{O}(\frac{1}{\text{poly } P}).$$

We can reuse most of the calculations in the proof of Induction Hypothesis 4.B.1 to prove the hypothesis and here we only discuss how to bound the gradient of the objective. If the regularization is not violated, i.e., $\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)} \geq \frac{\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}}{\lambda}$, we have $\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)} \leq O(\log(P))$. For $t \geq T_1$, denote the first time when $\alpha_{\mathbf{p} \rightarrow v_{k,m}}^{(t)} - \frac{\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}}{\lambda} \leq O(\frac{1}{P^4})$ as $\tilde{T}_1$, by Lemma 4.B.6, we have $\tilde{\alpha}_{\mathbf{p} \rightarrow v_{k,m}}^{(t)} \geq \Omega(\frac{1}{P^4})$ for $t \in [T_1, \tilde{T}_1]$, and $\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(\tilde{T}_1)} = \tilde{C} \log P$ for some constant $\tilde{C} > 0$. Then we have $\tilde{T}_1 \leq O(\frac{P^4 \log P}{\eta})$. Thus, for $t \geq \tilde{T}_1$,

$$\|\nabla_Q \mathcal{L}(Q^{(t)})\|_F^2 \leq O\left(\sum_{k=1}^K \sum_{\mathbf{p} \in \mathcal{P}} (\alpha_{\mathbf{p} \rightarrow v_{k,1}}^{(t)} - \frac{\Phi_{\mathbf{p} \rightarrow v_{k,m}}^{(t)}}{\lambda})^2\right) \leq O(\frac{1}{\text{poly}(P)}).$$

Proof of convergence. We first define a learning network that we deem as the “optimal” network with the global feature-position attention pattern. Specifically, we define $Q^\star$ as a matrix satisfied $e_{\mathbf{p}}^\top Q^\star v_{k,1} = \sigma_\star$ with $\sigma_\star^2 = \frac{\|\bar{Q}\|_F}{P(\sum_{k=1}^K N_k)}$ and $e_{\mathbf{p}}^\top Q^\star v_{k,m} = 0$ for $\mathbf{p} \in \mathcal{P}$ and $k \in [K]$, $m \in [N_k]$. Furthermore, $w_1^\top Q^\star w_2 = 0$, where $w_1, w_2 \in \text{Span}(\{e_{\mathbf{p}}\}_{\mathbf{p} \in \mathcal{P}} \cap \{v_{k,m}\}_{k \in [K], m \in [N_k]})^\perp$. Here we suppose $\mathcal{L}_{\text{cl}}^\star$ is achieved at the matrix $Q = \bar{Q}$.

Moreover, we consider the following **pseudo** losses and objective: define the linearized learner $\tilde{F}^{(t)}(Q, X) = F(Q^{(t)}, X) + \nabla_Q F(Q^{(t)}, X)(Q - Q^{(t)})$,

$$\begin{aligned}\tilde{\mathcal{L}}_t(Q) &:= \mathbb{E} \left[-\tau \log \left(\frac{e^{\langle \tilde{F}^{(t)}(Q, X^+), F(Q^{(t)}; X^{++}) \rangle / \tau}}{\sum_{X' \in \mathfrak{B}} e^{\langle \tilde{F}^{(t)}(Q, X^+), F(Q^{(t)}; X') \rangle / \tau}} \right) \right], \\ \widetilde{\text{Obj}}_t(Q) &:= \tilde{\mathcal{L}}_t(Q) + \frac{\lambda}{2} \|Q\|_2^2,\end{aligned}$$

and

$$\hat{\mathcal{L}}_t(Q) := \mathbb{E} \left[-\tau \log \left(\frac{e^{\langle F(Q, X^+), F(Q^{(t)}; X^{++}) \rangle / \tau}}{\sum_{X' \in \mathfrak{B}} e^{\langle F(Q, X^+), F(Q^{(t)}; X') \rangle / \tau}} \right) \right].$$

Then we discuss the values of different losses at $Q = Q^\star$. We have the following properties:

$$\mathcal{L}(Q^\star) \leq \mathcal{L}_{\text{cl}}^\star + O\left(\frac{1}{\text{poly}(d)}\right), \quad (4.B.5)$$

$$|\hat{\mathcal{L}}(Q^\star) - \bar{\mathcal{L}}_t(Q^\star)| \leq O\left(\frac{1}{\text{poly}(d)}\right), \quad (4.B.6)$$

$$|\tilde{\mathcal{L}}_t(Q^\star) - \hat{\mathcal{L}}_t(Q^\star)| \leq \frac{1}{\text{poly } d}. \quad (4.B.7)$$

For the first property, we only need to consider the contrastive loss at the global minimum. Notice that for our data distribution, the global minimum of the contrastive loss is achieved when the network can perfectly distinguish the samples from different clusters. Thus, we have $\mathcal{L}_{\text{cl}}^\star = \Theta(\log \frac{N_c}{K})$. Notice that on the event $\mathcal{A}_{1, \text{com}}$, supposing $X \in \mathcal{D}_k^{\text{cl}}$, which happens with prob $\geq 1 - e^{-P^{\kappa_s}}$ we have

$$\begin{aligned}\langle F(Q^\star, X^+), F(Q^\star; X^{++}) \rangle &= \langle v_{k,1}, v_{k,1} \rangle + \frac{1}{|\Theta(C_{k,1})|^2} \sum_{\mathbf{p} \in \mathcal{P}_{k,1} \cap \mathcal{U}^+ \cap \mathcal{U}^{++}} \|\xi_{\mathbf{p}}\|_2^2 \pm o(1), \\ \langle F(Q^\star, X^+), F(Q^\star; X') \rangle &= \langle v_{k,1}, v_{k,1} \rangle \pm o(1) \text{ for } X' \in \mathfrak{N} \cap \mathcal{D}_k^{\text{cl}}.\end{aligned}$$

Furthermore, by Bernstein’s inequality, we have with probability $\geq 1 - \frac{1}{\text{poly}(d)}$, we have $\|\xi_{\mathbf{p}}\|_2^2 = \sigma_0^2 d \pm \tilde{O}(\frac{1}{\text{poly}(d)}) = 1 \pm \tilde{O}(\frac{1}{\text{poly}(d)})$, we denote such an event as $\mathcal{A}_3$. Suppose we consider the temperature $\tau = O(\frac{1}{\log d})$, then conditioned on $\mathcal{A}_{1, \text{com}} \cap \mathcal{A}_3$, we have $\langle F(Q^\star, X^+), F(Q^{(t)}; X') \rangle = \omega(\log d) \pm o(1)$ for $X' \in \mathfrak{B} \cap \mathcal{D}_k^{\text{cl}}$, which could minimize the loss to the level of $\Theta(\log \frac{N_c}{K})$ up to

the error of $O(\frac{1}{\text{poly}(d)})$. Then we have

$$\mathcal{L}(Q^\star) \leq (1 - \frac{1}{\text{poly}(d)})\Theta(\log \frac{N_c}{K}) + \tilde{O}(\frac{1}{\text{poly}(d)}) \leq \mathcal{L}(\bar{Q}) + O(\frac{1}{\text{poly}(d)}).$$

The second property follows from the observation that

$$\begin{aligned} |\widehat{\mathcal{L}}_t(Q^\star) - \bar{\mathcal{L}}(Q^\star)| &\leq O(\|\nabla_Q \bar{\mathcal{L}}(Q^\star)\|_2) \|F(Q^\star, X^{++}) - F(Q^{(t)}, X^{++})\|_2 \\ &\leq O\left(\|\nabla_Q \bar{\mathcal{L}}(Q^\star)\|_2 \cdot \left(1 - \mathbf{Attn}_{\mathbf{p} \rightarrow \mathcal{P}_{k,1}}^{(t)}(X^{++})\right)\right) \leq \tilde{O}(\frac{1}{\text{poly}(P)}). \end{aligned}$$

Similarly, the third property follows from the fact that

$$|\tilde{\mathcal{L}}_t(Q^\star) - \widehat{\mathcal{L}}_t^{cl}(Q^\star)| \leq O(\|\nabla_Q \tilde{\mathcal{L}}_t(Q^\star)\|_2) \|\tilde{F}^{(t)}(Q^\star, X^+) - F(Q^\star, X^+)\|_2 \leq \tilde{O}(\frac{1}{\text{poly}(P)}).$$

Now we will use the tools from online learning to obtain a loss guarantee:

$$\begin{aligned} &\eta \langle \nabla_Q \mathcal{L}(Q^{(t)}), Q^{(t)} - Q^\star \rangle \\ &= \frac{1}{2} \eta^2 \|\nabla_Q \mathcal{L}(Q^{(t)})\|_F^2 - \frac{1}{2} \|Q^{(t)} - Q^\star\|_F^2 + \frac{1}{2} \|Q^{(t+1)} - Q^\star\|_F^2 \\ &= \frac{\eta^2}{2} \cdot \frac{1}{\text{poly}(P)} - \frac{1}{2} \|Q^{(t)} - Q^\star\|_F^2 + \frac{1}{2} \|Q^{(t+1)} - Q^\star\|_F^2. \end{aligned}$$

Notice that $\widetilde{\mathbf{Obj}}_t(Q)$ is a convex function over Q and $\widetilde{\mathbf{Obj}}_t(Q^{(t)}) = \mathcal{L}(Q^{(t)})$, thus

$$\begin{aligned} \langle \nabla_Q \mathcal{L}(Q^{(t)}), Q^{(t)} - Q^\star \rangle &= \langle \nabla_Q \widetilde{\mathbf{Obj}}_t(Q^{(t)}), Q^{(t)} - Q^\star \rangle \\ &\geq \widetilde{\mathbf{Obj}}_t(Q^{(t)}) - \widetilde{\mathbf{Obj}}_t(Q^\star) && \text{(by convexity)} \\ &\geq \widetilde{\mathbf{Obj}}_t(Q^{(t)}) - \mathcal{L}(Q^\star) - \tilde{O}(\frac{1}{\text{poly}(P)}) && \text{(by 4.B.6 and 4.B.7)} \\ &\geq \widetilde{\mathbf{Obj}}_t(Q^{(t)}) - \mathcal{L}_{c1}^\star - \tilde{O}(\frac{1}{\text{poly}(P)}) && \text{(by 4.B.5)} \\ &= \mathcal{L}(Q^{(t)}) - \mathcal{L}_{c1}^\star - \tilde{O}(\frac{1}{\text{poly}(P)}). && \text{(by definition of } \widetilde{\mathbf{Obj}} \text{)} \end{aligned}$$

Thus by a telescoping summation, we have

$$\begin{aligned} \frac{1}{T - T_2} \sum_{t=T_2}^T \mathcal{L}(Q^{(t)}) - \mathcal{L}_{c1}^\star &\leq \sum_{t=T_2}^T \langle \nabla_Q \mathcal{L}(Q^{(t)}), Q^{(t)} - Q^\star \rangle + O(\frac{1}{\text{poly}(P)}) \\ &\leq O(\frac{\|Q^{(T)} - Q^{(\star)}\|_2^2}{T\eta}) \leq O(\frac{1}{\text{poly}(P)}), \end{aligned}$$

which completes the proof. $\square$

Part II

Reasoning and Post-Training in Transformers

Chapter 5

Length Generalization in Transformers Reasoning

5.1 Introduction

Reasoning is a central theme of artificial intelligence [61, 70, 207]. Transformer-based [288] large language models (LLMs) achieve state-of-the-art results on complex reasoning tasks via chain-of-thought (CoT) reasoning [64, 151, 183, 184, 235, 297, 305, 347], where the model generates intermediate steps before delivering a final answer. Recent frontier models such as OpenAI-o1 [207] and DeepSeek-R1 [70] typically produce long CoT traces at inference time, often elicited via reinforcement learning and/or supervised fine-tuning (SFT) that distills from longer chains [197, 219]. These advances have enabled improved performance on more challenging problems [92, 141, 154, 318, 346], yet the mechanisms and limitations underlying CoT reasoning remain poorly understood, posing fundamental theoretical challenges.

Theoretical studies on transformers with CoT have recently advanced along two fronts: expressiveness [44, 85, 171, 190, 191] and statistical learnability [2, 121, 147, 164, 218, 241, 313]. On the expressiveness front, seminal work [85, 171, 191] showed that constant-depth transformers without CoT behave as shallow circuits and are restricted to express the circuit complexity class TC^0 of constant computation depth, whereas with $O(L)$ CoT steps on inputs of length L , they can express log-depth circuits in NC^1 , which is a problem class conjectured to require inherently serial computation.¹ These results reveal that CoT reasoning equips transformers with the expressive power to solve *inherently sequential* problems. In contrast, there remains limited understanding of *how* transformers acquire such reasoning abilities during training. Prior optimization analyses [124, 130, 146, 309] were limited to simple, fully parallelizable tasks in TC^0

This chapter is adapted from the following joint work:

Yu Huang, Zixin Wen, Aarti Singh, Yuejie Chi, and Yuxin Chen. *Transformers Provably Learn Chain-of-Thought Reasoning with Length Generalization and Recursive Self-Improvement*.

¹ For background on circuit complexity and a detailed review of expressiveness results, see Section 5.2 and [16, 290].

that do not require sequential reasoning. This leaves a substantial gap between what transformers can *express* in principle and what they can *learn* through training.

Another key question, motivated by the success of reasoning models, is whether models can extrapolate their reasoning beyond the sequence lengths of the training data: a property known as *length generalization*. In practice, reasoning effectively over long chains of thought relies strongly on the model’s *long-context ability* [156, 172, 321, 330], which can be severely affected by *context rot*, a phenomenon in which model performance degrades as the number of tokens in the context increases [118, 120, 268]. Even for synthetic reasoning tasks, empirical evidence on length generalization of transformers is mixed [119, 136, 174, 317, 342, 348, 349], and several prior studies reported limited extrapolation despite strong in-distribution performance. Architectural choices, including positional encoding and attention variants, can influence generalization considerably [15, 79, 144, 159, 226, 240, 253]. On the theoretical front, previous work established existence or statistical guarantees, showing that transformers can, in principle, represent length-generalizing algorithms or achieve favorable sample complexity independent of the length of the CoT [4, 102, 126, 142, 188]. Nonetheless, it remains unclear whether transformers can provably learn to length-generalize reliably when trained with gradient-based optimization algorithms.

In light of the aforementioned gaps in prior literature, the current paper seeks to make progress towards addressing the following two fundamental questions:

Research Questions

1. *Can transformers, trained via gradient descent (GD), learn CoT reasoning to solve problems requiring inherently sequential reasoning beyond TC^0 ?*
2. *Can the learned reasoning ability generalize to problems that require longer CoTs beyond the lengths of training data?*
3. *Finally, if the learned reasoning ability generalize out-of-distribution, can model utilize this property to self-improve, even recursively?*

To address these questions in a theoretically tractable manner, we analyze a minimally viable transformer: a one-layer transformer block with softmax attention and a feed-forward network (FFN), trained by GD with *no positional encoding* (NoPE). We study this model on synthetic *state-tracking* tasks, namely LEGO [342], which distill core LLM skills such as entity tracking, game-state updates, and code evaluation [149, 195]. This setup is tractable for analysis, while capturing the mechanisms needed for step-by-step computation via CoT. We analyze the training dynamics with CoT on two LEGO task families with distinct *algebraic* action structures: simply transitive group actions, and symmetry group actions. By tracking attention patterns throughout training, we demonstrate how reasoning capabilities emerge and how length generalization is governed by the structural properties of these tasks. More concretely, our main contributions are summarized as follows.

1. *Provable guarantee of learning CoT with length generalization in state-tracking.* We prove that for the LEGO state-tracking task, a one-layer NoPE transformer trained with GD provably solves

constant-length problems using CoT reasoning. Moreover, the learned transformer directly generalizes to problems of substantially longer length, when the group actions in LEGO is *simply transitive*. Conversely, for the canonical action of the symmetry group S_n on $\mathbb{Z}_n$, the learned transformer generalizes only up to constant-factor length. We identify an *attention concentration* mechanism that dictates step-wise retrieval depending on the task structure, leading to distinct length generalization behaviors.

2. *New perspective on recursive self-improvement through out-of-distribution generalization.* When length generalization exists but is limited (for example, under symmetry group actions), we prove that a *self-training* curriculum [159], which trains the model on its own outputs, can bootstrap the CoT ability to solve problems of maximum difficulty. More importantly, our result reveals a new mechanism of self-improvement: out-of-distribution generalization can serve as the engine of recursive self-improvement, where each step of improvement produces new support for the model to sharpen on.
3. *Constant-depth transformers provably learn to solve problems beyond TC^0 via CoT.* The first two results further establish that the model can learn the solution of state-tracking problems for non-solvable groups, which is NC^1 -complete and lies outside TC^0 unless the widely held conjecture $TC^0 \neq NC^1$ in circuit complexity theory fails. Therefore, we provide *the first optimization guarantee* showing that a one-layer transformer learns to solve reasoning tasks beyond TC^0 with CoT, matching the expressivity result of [85, 171, 191] for linear-CoT transformers.
4. *Empirical evidence on synthetic LEGO tasks that supports our theory.* We present a wide range of experiments based on our theoretical setup. Our results corroborate our predictions on length generalization for different group actions, showing a clear separation between the two algebraic structures in the theoretical setting. Moreover, we empirically demonstrate how recursive self-training effectively improves reasoning length. The attention concentration mechanism identified in our theory is also supported by our experimental findings.

5.1.1 Overview of Theoretical Results

To facilitate discussion, let us first briefly introduce the *state-tracking problem*. Given a group $\mathcal{G}$ acting on a state space $\mathcal{Y}$, the goal is to compute the final state y_L by applying a sequence of group actions $g_i \in \mathcal{G}$ starting from an initial state $y_0 \in \mathcal{Y}$:

$$y_0 \xrightarrow{g_1} y_1 \xrightarrow{g_2} \cdots \xrightarrow{g_\ell} y_\ell \cdots \xrightarrow{g_L} y_L.$$

This task naturally lends itself to CoT reasoning, where intermediate steps explicitly track how the state evolves under successive actions.

We consider two types of group action: simply transitive actions and symmetry group actions [88]. A simply transitive action is free and transitive; that is, there is a *unique* group element $g \in \mathcal{G}$ mapping any state to another. An example is a group acting on itself by the group composition.

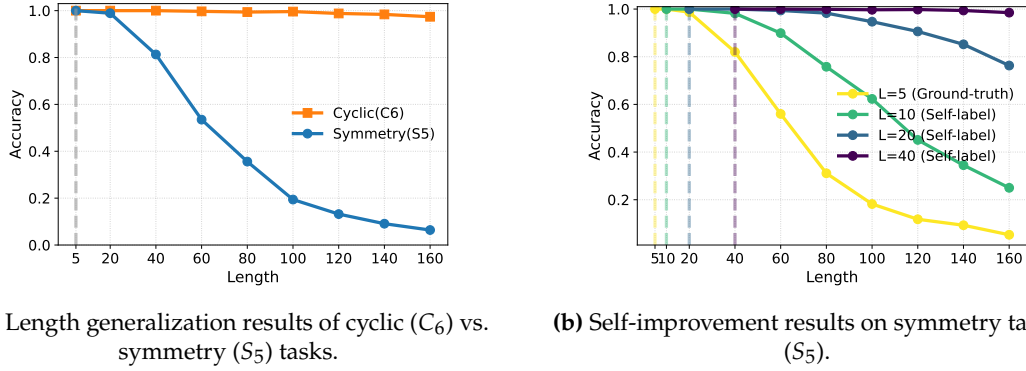


Figure 5.1: Length Generalization Across Group Actions. Empirical results of length generalization on LEGO tasks with different group actions. (a). Transformers length-generalize to solve significantly longer CoT tasks for simply transitive (cyclic) group (Theorem 5.4.1), while generalizing poorly for symmetry group tasks. (b). When direct length generalization falls short for symmetry actions, a recursive self-training scheme that train on the model’s own longer CoT bootstraps the solvable problem length (Theorem 5.5.1). The dashed lines indicate the training length.

In contrast, in the symmetry case, the action is *transitive but not free*: for example, consider the canonical action of S_n on $\mathbb{Z}_n$ by permutations, where *many* group elements send a given number i to j . In group theory, these two types of actions have different sizes of stabilizers, even when the group $\mathcal{G}$ is the same.

Result 1: Provably learning CoT reasoning for state tracking. Our first result shows that transformers can learn to solve these two state-tracking problems via CoT.

Theorem 5.1.1 (Learning CoT, informal). *One-layer transformers, trained via GD, can provably learn to solve state-tracking problems for simply transitive and symmetry group actions via CoT reasoning.*

Since state-tracking problems for symmetry group are NC^1 -complete [31], this result provides the first optimization-based training guarantee for solving problems beyond TC^0 , matching the linear-CoT expressiveness result of [171] with provable learnability under gradient-based training. In comparison, recent theoretical progress only showed that transformers can provably learn tasks in TC^0 such as parity [124, 146, 309] and linear regression [130] with CoT, which, however, are solvable by log-precision, constant-depth transformers without CoT [171, 191].

Result 2: Algebraic structure dictates length generalization. Going beyond training, our next result unveils the qualitative difference in length generalization induced by the difference in the group action structure. More importantly, we provide a mechanistic understanding of the length generalization property learned by gradient-based optimization.

Theorem 5.1.2 (Length generalization, informal). *The algebraic structure of the group actions in the state-tracking task dictates how far the reasoning length generalizes:*

- For simply transitive actions: *standard CoT training on constant-length problems already yields generalization to problems of length d^{c^*} , where $0 < c^* < 1$ is a constant.*
- For symmetry actions: *standard CoT training only generalizes to a constant factor of the training length.*

This provides a rigorous theory of length extrapolation for state-tracking: it characterizes when and how length generalization emerges for different reasoning problems. Formal statements are deferred to Section 5.4 (simply transitive actions) and Section 5.5 (symmetry group actions).

Mechanism of Attention Concentration

Our theory uncovers the *mechanism of attention concentration* that explains the varying degrees of length generalization. At a high level, the performance of long-context reasoning depends on the level at which the attention layer focuses on task-relevant contexts. The algebraic structure of the group action dictates the level of attention concentration at convergence: simply transitive actions enable sharp concentration and strong length generalization, whereas symmetry actions introduce distractors that dilute attention focus, limiting length generalization to constant factors.

Result 3: Recursive self-improvement extends solvable problem length. Although length generalization is limited for symmetry actions, recent empirical studies [107, 159, 257] have shown that neural networks can bootstrap their capability via *self-improvement*. In this work, we prove that a one-layer transformer obtained via a self-training scheme similar to [159], can extend its solvable problem length. To the best of our knowledge, this offers the first optimization guarantee of self-improvement for transformer networks.

Theorem 5.1.3 (Recursive self-improvement, informal). *A one-layer transformer, trained with a recursive self-training scheme, can self-improve: at each stage $k > 1$, the model learning on the self-labeled reasoning traces of length 2^k , length generalizes to solve the next doubled length 2^{k+1} . Consequently, after $\Theta(\log d)$ stages, the model can solve problems of length d , the maximal length in our setting.*

Our work complements recent advances on length generalization in transformers. Prior studies [4, 102, 126, 142, 188] primarily established statistical guarantees and *non-gradient-based* learnability. Notably, [102] identifies conditions under which specialized positional encodings and sparse contextual dependencies enable length extrapolation, provided the model fits the source length. [142] analyzed time-invariant autoregressive models (a fixed next-token generator) and showed that the sample complexity can be independent of the CoT length but on the complexity

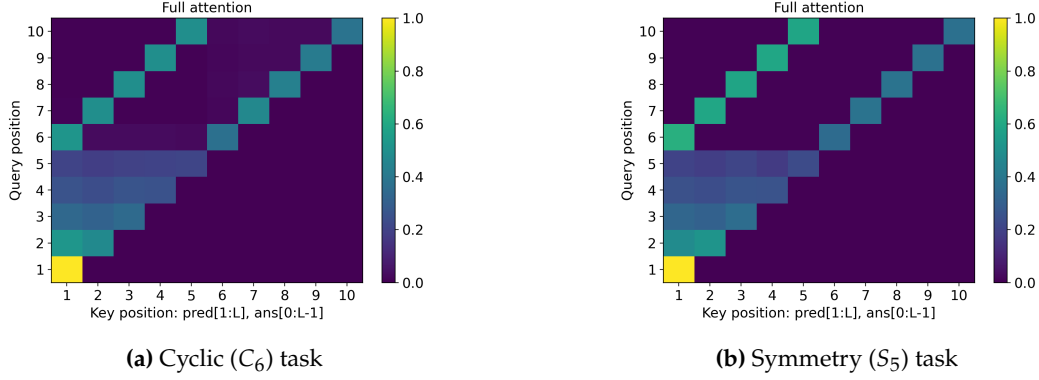


Figure 5.2: Attention Concentration In LEGO Tasks. Attention concentration at convergence for LEGO task with length $L = 5$. The heatmap places the query clause index on the y -axis (keys on the x -axis). For a task of length L , the LEGO sequence prior to the final answer clause has length $2L$; we focus on query positions $L + 1$ to $2L$, corresponding to answer clauses $Z_{\text{ans},0}$ to $Z_{\text{ans},L-1}$. Two diagonal bands in the upper region indicate attention concentrating on the answer clause $Z_{\text{ans},\ell}$ and the predicate clause $Z_{\text{pred},\ell+1}$ when the query is $Z_{\text{ans},\ell}$.

of individual CoT steps. We provide a concrete setting that mirrors these works and, crucially, show that gradient-based optimization *actually finds* such solutions for transformer networks.

5.1.2 Our Empirical Results at a Glance

Empirically, we conduct a series of experiments aligned with our problem setup to test the theory’s predictions, and observe strong agreements with our predicted behavior. In particular:

- *Length generalization comparison.* Figure 5.1a shows that training at a short length yields nearly perfect accuracy at much longer lengths for tasks with simply transitive actions, whereas tasks with symmetry actions exhibit only constant-factor length generalization. These findings are consistent with Theorem 5.1.2.
- *Effectiveness of recursive self-training.* In Figure 5.1b, a double-and-self-labeled curriculum for the task with symmetry actions consistently shifts the length–accuracy curve rightward; after several stages, performance matches the simply transitive case, validating Theorem 5.1.3.
- *Attention concentration patterns.* At convergence, the attention heatmaps in Figure 5.2 exhibit two clear diagonals with concentrated mass, confirming the attention-concentration mechanism predicted by our theory.

Detailed descriptions and discussions, along with additional experiments, are postponed to Section 5.7.

5.2 Background

Computational complexity [16] has been employed to characterize the power of neural networks [101, 255]. Historically, *circuit complexity* has been used extensively to study the power of neural networks [182, 190, 192, 210, 254, 256, 301], due to the structural resemblance of Boolean circuits and neural networks with threshold gates. In a nutshell, circuit complexity evaluates computation models by the size, depth, and gate types of Boolean circuits that implement them. Below we provide a brief introduction of the basics of circuit complexity.

5.2.1 Circuits and Expressiveness

A Boolean circuit is a finite acyclic network of logic gates that computes a Boolean function on $\{0, 1\}^n \rightarrow \{0, 1\}$ for some fixed n . The gates with fan-in 0 are the inputs, which are assigned one of the n Boolean variables. A *circuit family* $\{C_n\}_{n \geq 1}$ computes a language by using C_n on length- n inputs. We measure the *size* of a circuit by the number of gates, and *depth* by the length of the longest input-output path, respectively. For instance, “constant depth” means $\text{depth}(C_n) = O(1)$, while “polynomial size” means $\text{size}(C_n) = O(n^c)$ for some integer c . Standard circuit complexity classes are defined by restricting circuit depth, size, gate set and fan-in:

- TC^0 consists of *constant-depth*, polynomial-size circuits with unbounded fan-in $\{\text{AND}, \text{OR}, \text{NOT}\}$ gates augmented with threshold (e.g., MAJORITY) gates. The circuit class TC^0 captures exactly the complexity of integer multiplication and division, and sorting [42].
- NC^1 consists of *log-depth*, polynomial-size circuits over $\{\text{AND}, \text{OR}, \text{NOT}\}$ with bounded fan-in. This class captures exactly the complexity of recognizing all regular languages.
- P/poly is the class of all languages computable by polynomial-size circuit families.

These classes satisfy the following relations:²

$$\text{TC}^0 \subseteq \text{NC}^1 \subseteq \text{P/poly}.$$

Expressiveness gap of transformers via circuit complexity. One can analyze the expressivity of transformer neural nets through the lens of circuit complexity. Prior work [85, 171, 174, 190, 194] showed that the expressive upper bound of a vanilla constant-depth transformer is limited to TC^0 , the class of problems solvable by extremely shallow, highly parallel circuits. In contrast, when equipped with CoT, recent work [85, 171, 191] proved that $O(n)$ intermediate steps enable transformers to simulate NC^1 -complete language. Further, [171] showed that 2-layer transformers with polynomial CoT steps suffice to express arbitrary polynomial-size circuits, i.e., P/poly . Hence, under the widely believed conjecture that $\text{TC}^0 \neq \text{NC}^1$ [290], CoT strictly extends the expressive power of transformers beyond TC^0 , allowing them to solve problems that inherently require super-constant (here $O(\log n)$) computation depth. A standard example connecting algebraic structure with circuit complexity is the following *word problem*, defined by Dehn [72, 73].

² We intentionally omit uniformity conventions here as this section is only meant to give readers a sense of the classes we refer to and their standard relationships.

Definition 5.2.1 (Word problem for a group G). Let G be a group and let e denote its identity. For a word $w = g_1 \cdots g_k \in G^*$ (each $g_i \in G$), the word problem asks to decide whether $g_1 \circ \cdots \circ g_k = e$.

Barrington [31] proved that the above word problem is NC^1 -complete when the group is non-solvable.

Theorem 5.2.1 (Barrington [31]). *The word problem of every finite non-solvable group is NC^1 -complete. The canonical example is S_n for $n \geq 5$, the symmetry group on n elements that encodes the permutations.*

The word problem captures a fundamental reasoning task: state tracking in a finite world [160, 193, 195]. Given an initial state and a sequence of transformations, the goal of the task is to compute the resulting state. State tracking underlies practical LLM abilities such as narrative entity tracking, chess move analysis, and code execution [149, 195], while exhibiting a rich connection to the circuit complexity theory. Therefore, it has become a standard synthetic testbed for probing the reasoning abilities of language models, both theoretically [174, 193, 195] and empirically [160, 174, 342]. Motivated by this connection, we study the state-tracking task with varying algebraic structures.

5.2.2 LEGO for State Tracking

We focus on a specific formulation of the state-tracking problem, LEGO (*Learning Equality and Group Operations*) [342], which was originally proposed as a synthetic task to study the reasoning behavior of transformers empirically. A typical LEGO instance in [342] takes the following form:

$$b = + a, \quad c = - b, \quad \dots, \quad t = - s, \quad a = -1, \quad \dots$$

Here, $a, b, c, \dots$ are *variables*, each taking a *value* (or state) in $\{-1, +1\}$ in this example. Short expressions such as $b = + a$ are *clauses*, where $= +$ and $= -$ denote *actions*: the action is applied to the *right-hand-side* variable's value to obtain the *left-hand-side* variable's value. For instance, from $b = + a$ and $a = -1$, it follows that $b = -1$. Formally, the LEGO language is defined as follows:

Definition 5.2.2 (LEGO language [342]). Let $\mathcal{X}, \mathcal{G}, \mathcal{Y}$ be finite sets of variables, actions, and values, respectively, where each $g \in \mathcal{G}$ is a map $g : \mathcal{Y} \rightarrow \mathcal{Y}$. The formal language $\text{LEGO}(\mathcal{X}, \mathcal{G}, \mathcal{Y})$ has alphabet $\mathcal{X} \cup \mathcal{G} \cup \mathcal{Y} \cup \{=, (,)\}$ and consists of two types of expressions (called *clauses*):

- (1) *Predicate clause* $x = g(x')$ specifies an action $g \in \mathcal{G}$ linking variables $x, x' \in \mathcal{X}$.
- (2) *Answer clause* $x = y$ assigns a value $y \in \mathcal{Y}$ to a variable $x \in \mathcal{X}$.

A canonical LEGO sentence of length L with answer up to L' concatenates predicate clauses $x_n = g_n(x_{n-1})$ for $n \in [L]$ and answer clauses $x_n = y_n$ for $n \in [L']$ with $L' \leq L$:

$$\underbrace{\boxed{x_1 = g_1(x_0)} \dots \boxed{x_L = g_L(x_{L-1})}}_{\text{predicates}} \underbrace{\boxed{x_0 = y_0} \boxed{x_1 = y_1} \dots \boxed{x_{L'} = y_{L'}}_{\text{answers}}. \quad (5.2.1)$$

which describes the chain of transitions:

$$x_0 \xrightarrow{g_1} x_1 \xrightarrow{g_2} x_2 \xrightarrow{g_3} \dots \xrightarrow{g_{L'}} x_{L'} \dots \xrightarrow{g_L} x_L, \quad \text{starting with } x_0 = y_0.$$

with answers $y_1, \dots, y_{L'}$ up to L'

For semantic validity, any sentence containing a path $x_n = g_n(x_{n-1}), \dots, x_{n-k+1} = g_{n-k+1}(x_{n-k})$ for $k \in [n]$ must satisfy: $y_n = g_n \circ g_{n-1} \circ \dots \circ g_{n-k+1}(y_{n-k})$.

In the LEGO language, predicate clauses encode transformations, while answer clauses encode observed states. Thus, solving a LEGO sentence is exactly state tracking: compose the listed actions along the path and propagate the observed states to predict the next answer consistent with the composition.

5.3 Problem Formulation

As introduced in Section 5.2, the current paper employs the LEGO framework to investigate the reasoning capabilities of transformers. To set the stage, this section presents precise mathematical formulations of the problems to be studied in this paper. Before proceeding, we introduce the following notation:

- *Vocabulary*. Define the vocabulary as $\mathcal{V} := \mathcal{X} \cup \mathcal{G} \cup \mathcal{Y} \cup \{\langle \text{blank} \rangle\}$, where the *blank* token $\langle \text{blank} \rangle$ is a null symbol indicating the absence of other tokens.
- *Vocabulary size*. Let $d := |\mathcal{V}|$ denote the (finite) vocabulary size.

To facilitate theoretical analysis, we concentrate on the asymptotic regime where $d \rightarrow \infty$. Note that $|\mathcal{X}|, |\mathcal{G}|, |\mathcal{Y}|$ may depend on d , and we shall specify any required scaling assumptions as needed.

Assumption 5.3.1 (Asymptotic regime). For a language $\text{LEGO}(\mathcal{X}, \mathcal{G}, \mathcal{Y})$ defined in Definition 5.2.2, we consider the asymptotic regime where both the vocabulary size d and the number of variables $|\mathcal{X}|$ tend to infinity. Assume $|\mathcal{G}| \leq \log^{C_0} d$ for some constant $C_0 \in [1, 100)$, and hence $\mathcal{Y}$ and $\mathcal{G}$ are much smaller than $\mathcal{X}$ in size.

5.3.1 Data Distribution

We begin by specifying how LEGO clauses are tokenized, followed by a definition of the LEGO distribution.

Definition 5.3.1 (LEGO encoding). Each LEGO clause from Definition 5.2.2 is encoded as a fixed-length, 5-token tuple $Z \in \mathcal{V}^5$. More specifically,

- For each predicate clause $x = g(x')$, set $Z_{\text{pred}} := (x, g, x', \langle \text{blank} \rangle, \langle \text{blank} \rangle) \in \mathcal{V}^5$;
- For each answer clause $x = y$, set $Z_{\text{ans}} := (\langle \text{blank} \rangle, \langle \text{blank} \rangle, \langle \text{blank} \rangle, x, y) \in \mathcal{V}^5$.

With Definition 5.3.1 in place, we can encode a LEGO sentence (5.2.1) to a sequence $Z^{L,L'}$:

$$Z^{L,L'} = (Z_{\text{pred},1}, \dots, Z_{\text{pred},L}, Z_{\text{ans},0}, \dots, Z_{\text{ans},L'}), \quad (5.3.1)$$

where $Z_{\text{pred},k}$ and $Z_{\text{ans},k}$ represent the k -th predicate and answer clauses, respectively. For convenience, we denote $\mathcal{I}^{L,L'} = \{(\text{pred}, \ell)\}_{\ell \in [L]} \cup \{(\text{ans}, \ell)\}_{\ell=0}^{L'}$ as the index set associated with the clauses in $Z^{L,L'}$. If $L = L'$, we write the sequence simply as Z^L and the corresponding index set as $\mathcal{I}^L$.

To feed LEGO tokens into a neural network, we first map each symbol to an integer index (*tokenization*), and then map indices to continuous vectors via a learned table (*embedding*). The following definitions formalize these two steps.

Definition 5.3.2 (Tokenization and token embedding). Each $v \in \mathcal{V}$ is assigned a unique index $\tau(v) \in \{1, \dots, d\}$. Denote by $e_i \in \mathbb{R}^d$ the embedding vector associated with index i , and write $e_v \equiv e_{\tau(v)}$ for convenience. The blank token is assigned the zero vector, $e_{\tau(\langle \text{blank} \rangle)} = \mathbf{0}_d \in \mathbb{R}^d$. For technical simplicity, we assume that $\{e_v : v \in \mathcal{V} \setminus \{\langle \text{blank} \rangle\}\}$ forms an orthonormal set in $\mathbb{R}^d$ (note that this assumption can be relaxed to a well-conditioned embedding matrix without affecting our results).

Equipped with Definition 5.3.2, we can transform the LEGO sequence encoding into vector embeddings that can be used as inputs to neural networks.

Definition 5.3.3 (Embedding of LEGO sentences). Let $d_c := 5d$ be the clause embedding dimension. We define an operator $\text{Embed} : \mathcal{V}^5 \rightarrow \mathbb{R}^{d_c}$ that maps a clause to embedding by

$$\mathbf{Z} = \text{Embed}(Z) := (e_{v_1}, e_{v_2}, \dots, e_{v_5}) \in \mathbb{R}^{d_c}, \quad \text{for clause } Z = (v_1, v_2, \dots, v_5) \in \mathcal{V}^5.$$

Specifically, a LEGO sentence $Z^{L,L'}$ defined in (5.3.1) is embedded as

$$\mathbf{Z}^{L,L'} = (\mathbf{Z}_{\text{pred},1}, \dots, \mathbf{Z}_{\text{pred},L}, \mathbf{Z}_{\text{ans},0}, \dots, \mathbf{Z}_{\text{ans},L'}) \in \mathbb{R}^{d_c \times (L+L'+1)}$$

where each column $\mathbf{Z}_{\text{pred},\ell} \in \mathbb{R}^{d_c}$ (resp. $\mathbf{Z}_{\text{ans},\ell}$) is the embedding of clause $Z_{\text{pred},\ell}$ (resp. $Z_{\text{ans},\ell}$). When $L' = L$, we simply write $\mathbf{Z}^L \equiv \mathbf{Z}^{L,L'}$ for simplicity.

Next, we describe the distribution that governs the generation of a LEGO sentence.

Assumption 5.3.2 (LEGO distribution $\mathcal{D}^L, \mathcal{D}^{L,L'}$). Consider $\mathbf{LEGO}(\mathcal{X}, \mathcal{G}, \mathcal{Y})$ as defined in Definition 5.2.2, and let L denote the sequence length. We assume that the distribution $\mathcal{D}^L$ of length- L LEGO sentences satisfies the following properties.

1. All LEGO sentences $Z^L \sim \mathcal{D}^L$ of the form (5.2.1) are encoded by Definition 5.3.1 into representation (5.3.1).
2. The variables $x_0, x_1, \dots, x_L \in \mathcal{X}$ are sampled uniformly at random from $\mathcal{X}$ without replacement.
3. The first value $y_0 \in \mathcal{Y}$ is chosen uniformly at random from $\mathcal{Y}$.
4. The actions $g_1, g_2, \dots, g_L \in \mathcal{G}$ are sampled uniformly at random from $\mathcal{G}$ with replacement.
5. The intermediate values $y_1, y_2, \dots, y_L$ are computed recursively by $y_i = g_i(y_{i-1})$.

For any $L' < L$, we define the truncated distribution $\mathcal{D}^{L,L'}$ of sequences $Z^{L,L'}$ containing all the predicates and the first $L' + 1$ many answer clauses, where $Z^{L,L'}$ is obtained by first sampling $Z^L \sim \mathcal{D}^L$ and then removing the answer clauses $Z_{\text{ans},\ell}$ for all $\ell > L'$.

As can be easily seen, sequences sampled from $\mathcal{D}^L$ or $\mathcal{D}^{L,L'}$ correspond to valid LEGO sentences as defined in Definition 5.2.2. With a slight abuse of notation, we write $\mathbf{Z}^{L,L'} \sim \mathcal{D}^{L,L'}$ to indicate that $\mathbf{Z}^{L,L'}$ is the embedding of a sentence $Z^{L,L'}$ drawn from $\mathcal{D}^{L,L'}$.

5.3.2 Transformer Architecture

In this subsection, we introduce the transformer architecture investigated in this paper. Towards this end, we first introduce a smoothed activation function that will be used in our network.

Definition 5.3.4 (Smooth ReLU). Define a continuously differentiable variant of the ReLU activation function [6, 127] as follows:

$$\mathbf{sReLU}(x) := \begin{cases} \frac{\varrho}{q}, & x \leq -\varrho, \\ \frac{x^q}{\varrho^{q-1}q}, & x \in (-\varrho, \varrho], \\ x - \varrho \left(1 - \frac{1}{q}\right), & x > \varrho, \end{cases}$$

where q and ϱ are some design parameters. Here and throughout, we choose $q = O(1)$ to be a large, even integer, and take $\varrho = \Theta(1/\text{polylog}(d))$.

Transformer architecture. In this work, we focus on an autoregressive transformer [288] whose block consists of a softmax attention layer followed by a position-wise feed-forward network, as described below.

- *Attention layer.* Given LEGO sentence embeddings $\mathbf{Z}^{L,L'}$ and indices $j, k \in \mathcal{I}^{L,L'}$, the attention from clause $\mathbf{Z}_j$ to clause $\mathbf{Z}_k$ is defined through the softmax operator as

$$\text{Attn}_{j \rightarrow k}(\mathbf{Q}, \mathbf{Z}^{L,L'}) := \frac{\exp(\mathbf{Z}_j^\top \mathbf{Q} \mathbf{Z}_k)}{\sum_{r \in \mathcal{I}^{L,L'}} \exp(\mathbf{Z}_j^\top \mathbf{Q} \mathbf{Z}_r)}.$$

Since the model is autoregressive, a standard causal mask is applied to ensure that the latest (answer) token attends only to *preceding* tokens (including itself). The attention output is

$$\text{Attention}(\mathbf{Q}, \mathbf{Z}^{L,L'}) := \sum_{k \in \mathcal{I}^{L,L'}} \text{Attn}_{\text{ans}, L' \rightarrow k}(\mathbf{Q}, \mathbf{Z}^{L,L'}) \cdot \mathbf{Z}_k.$$

Note that in the standard formulation of transformers, the score takes the form $\mathbf{Z}_j^\top \mathbf{W}^Q \mathbf{W}^K \mathbf{Z}_k$ instead for a “query” parameter matrix $\mathbf{W}^Q$ and a “key” parameter matrix $\mathbf{W}^K$. Here, we fold $\mathbf{W}^Q \mathbf{W}^K$ into a single parameter matrix $\mathbf{Q}$, resulting in a score $\mathbf{Z}_j^\top \mathbf{Q} \mathbf{Z}_k$, which can be viewed as an equivalent reparameterization that simplifies analysis without changing expressivity [128, 129, 329].

- *Feed-forward network (FFN).* The FFN with parameter $\mathbf{W} \in \mathbb{R}^{5 \times d \times m \times d_c}$ is defined by

$$\text{FFN}_{i,j}(\mathbf{W}, \mathbf{X}) := \sum_{r \in [m]} \text{sReLU}(\langle \mathbf{W}_{i,j,r}, \mathbf{X} \rangle + b_{i,j,r}), \quad \forall i \in [5], j \in [d],$$

where $\mathbf{X}$ denotes the input, $\mathbf{W}_{i,j,r} \in \mathbb{R}^{d_c}$ are neuron weights, m indicates the number of neurons, and $b_{i,j,r} \in \mathbb{R}$ is some *fixed* bias.

With the above layers defined, we are ready to introduce the transformer model used in this work, which is summarized as follows.

Definition 5.3.5 (Transformer language model). Assume that our learner neural network F is a one-layer decoder transformer block composed of an attention layer with no positional encoding (NoPE) as well as an FFN layer: $F = \text{FFN} \circ \text{Attention}$, with parameters $\mathbf{W} \in \mathbb{R}^{5 \times d \times m \times d_c}$ and $\mathbf{Q} \in \mathbb{R}^{d_c \times d_c}$. Formally, for the i -th token position and the j -th vocabulary index, we have

$$\left[F_i(\mathbf{Z}^{L,L'}) \right]_j := \text{FFN}_{i,j}(\mathbf{W}, \text{Attention}(\mathbf{Q}, \mathbf{Z}^{L,L'})) \in \mathbb{R}, \quad \forall i \in [5], j \in [d]. \quad (5.3.2)$$

We interpret $F(\mathbf{Z}^\ell)$ as five logit vectors $\{F_i(\mathbf{Z}^\ell)\}_{i=1}^5 \subset \mathbb{R}^d$, each parameterizing the distribution of the i -th token of the next clause. Recall that $\mathcal{V}$ is the vocabulary with $|\mathcal{V}| = d$ and $\tau : \mathcal{V} \rightarrow [d]$ denotes the index map. Given an encoded LEGO sequence $\mathbf{Z}^\ell = (Z_1, \dots, Z_\ell)$ with embedding $\mathbf{Z}^\ell$, the model’s predictive distribution for the i -th token of the $(\ell + 1)$ -th

clause is the following softmax:

$$p_{F_i}(Z_{\ell+1,i} = v \mid Z_1, \dots, Z_\ell) = \frac{e^{[F_i(\mathbf{Z}^\ell)]_{\tau(v)}}}{\sum_{j \in [d]} e^{[F_i(\mathbf{Z}^\ell)]_j}}, \quad \forall v \in \mathcal{V}. \quad (5.3.3)$$

Now we can sample the next clause $Z_{\ell+1}$ by sampling from the product distribution

$$Z_{\ell+1} = (Z_{\ell+1,1}, \dots, Z_{\ell+1,5}) \sim \bigotimes_{i=1}^5 p_{F_i} =: p_F,$$

in an autoregressive manner.

5.3.3 LEGO Task as a Reasoning Problem, and Training Objective

With the LEGO distribution and the transformer model in place, we now formalize the task within the LEGO framework. We view solving a length- L LEGO problem as generating a sequence of CoT steps, where each step ℓ produces an intermediate result used to predict $x_\ell = y_\ell$, ultimately leading to the final solution $x_L = y_L$. To be precise, we define the following reasoning task.

Definition 5.3.6 (Reasoning tasks $\mathcal{T}^L$). We define a family $\{\mathcal{T}^L\}_{L \in \mathbb{N}^+}$ that captures the ability to solve sequential reasoning problems. For each $L \geq 1$, task $\mathcal{T}^L$ measures the model's *accuracy along the chain* from step 1 to step L :

$$\text{Acc}_L(F) = \frac{1}{L} \sum_{0 \leq L' < L} \mathbb{E}_{Z^{L'} \sim \mathcal{D}^{L'}} \left[\mathbb{E}_{\widehat{Z}_{\text{ans}, L'+1} \sim p_F(\cdot \mid Z^{L'})} \left[\mathbb{1}\{\widehat{Z}_{\text{ans}, L'+1} = Z_{\text{ans}, L'+1}\} \right] \right], \quad (5.3.4)$$

where p_F is induced by the model F from (5.3.3).

Clearly, $\text{Acc}_L(F) \in [0, 1]$. We say that task $\mathcal{T}^L$ is solved if $\text{Acc}_L(F) \approx 1$. As L grows, $\{\mathcal{T}^L\}_{L \in \mathbb{N}^+}$ poses increasingly difficult state tracking challenges. At step L' , the model conditions on the partial transcript $Z^{L'}$ and predicts the next answer $Z_{\text{ans}, L'+1}$. To enforce step-by-step generation of the CoT trace, we define the following training objective.

Definition 5.3.7 (Next clause loss). The training objective for $\mathcal{T}^L$ with $L \geq 1$ is the *next clause* loss

$$\text{Loss}^L(F) := \sum_{1 \leq L' \leq L} \text{Loss}^{L', L'}(F), \quad (5.3.5a)$$

$$\text{where } \text{Loss}^{L', L'}(F) := \mathbb{E}_{Z^{L'} \sim \mathcal{D}^{L'}} \left[-\log p_F(Z_{\text{ans}, L'} \mid Z^{L', L'-1}) \right]. \quad (5.3.5b)$$

Here, p_F is induced by the model F from (5.3.3). We also define the per-token loss

$$\text{Loss}_i^{L,L'}(F) := \mathbb{E}_{Z^{L,L'} \sim \mathcal{D}^{L,L'}} \left[-\log p_{F_i}(Z_{\text{ans},L',i} \mid Z^{L,L'-1}) \right].$$

This is a teacher forcing style CoT objective: the model is given the ground truth answers for all previous steps and is trained to match the next step's answer [117, 146].

Further, we adopt the following initialization for training.

Assumption 5.3.3 (Initialization). Let F be the transformer network in Definition 5.3.5 with parameters $\mathbf{W}$ and $\mathbf{Q}$. The attention parameter is initialized to zero: $\mathbf{Q}^{(0)} = \mathbf{0}_{d_c \times d_c}$. The FFN weights are initialized randomly and independently as $\mathbf{W}_{i,j,r}^{(0)} \sim \mathcal{N}(\mathbf{0}, \sigma_0^2 \mathbf{I}_d)$ with $\sigma_0 = d^{-1/2}$. The biases are not trained and instead fixed at $b_{i,j,r} = \sigma_0 \log d$ for all i, j, r , chosen to keep most sReLU units active at initialization. All random draws are independent across indices.

Additional Technical Assumptions. Additionally, we introduce a couple of technical assumptions to be used throughout the proof. To control rare large deviations in the logits during training-time analysis, we adopt a bounded-output assumption as stated below.

Assumption 5.3.4 (Logit clipping). There exists $B = C_B \log d$, for some sufficiently large constant $C_B > 0$, such that each coordinate of the raw model output F_i is clipped from above:

$$[F_i]_j \leftarrow \min\{[F_i]_j, B\} \quad \text{for all } i, j.$$

This coordinate-wise clipping is a technical device to control large-deviation tails and simplify the analysis of the dynamics; B can be chosen large enough to avoid interfering with the regimes we study.

Moreover, to simplify the analysis of attention dynamics, we impose a fixed block-sparsity pattern on the attention parameter $\mathbf{Q}$.

Assumption 5.3.5 (Block-sparse attention matrix). Let $\mathbf{Q} = [\mathbf{Q}_{p,q}]_{p,q \in [5]} \in \mathbb{R}^{5d \times 5d}$ be partitioned into 5×5 blocks with $\mathbf{Q}_{p,q} \in \mathbb{R}^{d \times d}$. We assume that

$$\mathbf{Q}_{p,q} \equiv \mathbf{0}_{d \times d} \quad \text{for all } (p, q) \notin \{(4, 3), (4, 4)\},$$

i.e., only the blocks (4, 3) and (4, 4) are trainable.

This block-sparsity pattern, which zeros out most inter-token attention, is standard in recent theoretical analyses of transformer training dynamics [58, 128, 129, 328, 329]. Importantly, although sparse at the 5×5 token level, the two retained blocks (4, 3) and (4, 4) are fully *dense*

Algorithm 2: Curriculum training for simply transitive actions

Input: Model $F^{(0)}$ with parameters $(\mathbf{W}^{(0)}, \mathbf{Q}^{(0)})$; Learning rate η ; Stage snapshots T_1, T_2 .

Stage 1: Learning one-step reasoning ($\mathcal{T}^1$);

for $t = 1$ **to** T_1 **do** // Update the FFN parameter $\mathbf{W}$
 $\mathbf{W}^{(t)} \leftarrow \mathbf{W}^{(t-1)} - \eta \nabla_{\mathbf{W}} \text{Loss}^1(F^{(t-1)});$
 $\mathbf{Q}^{(t)} \equiv \mathbf{Q}^{(t-1)};$

Stage 2: Learning two-step reasoning for length extension ($\mathcal{T}^2$);

for $t = T_1 + 1$ **to** T_2 **do** // Update the attention parameter $\mathbf{Q}$
 $\mathbf{Q}^{(t)} \leftarrow \mathbf{Q}^{(t-1)} - \eta \nabla_{\mathbf{Q}} \sum_{\ell=1}^2 \text{Loss}_5^{2,\ell}(F^{(t-1)});$
 $\mathbf{W}^{(t)} \equiv \mathbf{W}^{(t-1)};$

Output: Model $F^{(T_1+T_2)}$.

$d \times d$ matrices trained without constraints, leaving $2d^2$ free parameters and thus a substantive, non-trivial learning problem.

5.4 Learning CoT on Simply Transitive Actions

In this section, we present our first main results for the case where the group action $\mathcal{G}$ on $\mathcal{Y}$ is simply transitive, which is isomorphic to the action of the cyclic group C_n on $\mathbb{Z}_n$.

Assumption 5.4.1 (Simply transitive group action). Let $\mathcal{Y} = \{0, 1, \dots, n_y - 1\}$ with $n_y \in [\Omega(\log \log d), \log d]$. Assume the group $\mathcal{G}$ acts simply transitively on $\mathcal{Y}$: the action is transitive and for any $y_1, y_2 \in \mathcal{Y}$ there exists a unique $g \in \mathcal{G}$ such that $g \cdot y_1 = y_2$. Equivalently, for any fixed $y \in \mathcal{Y}$ the map $g \mapsto g \cdot y$ is a bijection from $\mathcal{G}$ to $\mathcal{Y}$, so $|\mathcal{G}| = |\mathcal{Y}| = n_y$.

Our first main result demonstrates that, for such a simple task the model obtained via Algorithm 2 for short-chain tasks $\mathcal{T}^1$ and $\mathcal{T}^2$, successfully generalizes to significantly longer tasks.

Theorem 5.4.1. Under Assumptions 5.3.1 to 5.3.5 and 5.4.1, for some constant $0 < c^* < 1$, $n_y < m \ll \log^2 d$, the transformer model $F^{(T_1+T_2)}$ obtained by Algorithm 2 with learning rate $\eta = \frac{1}{\text{poly}(d)}$, and stage 1 and 2 iteration $T_1 = \tilde{O}\left(\frac{d}{\eta(\sigma_0)^{q-2}}\right)$, $T_2 = \tilde{O}\left(\frac{\text{poly}(d)}{\eta\sigma_0}\right)$ satisfies

1. Direct short-to-long length generalization:

$$\text{Acc}_L\left(F^{(T_1+T_2)}\right) \geq 1 - \frac{1}{\text{poly}(d)}, \text{ for every } L \leq O(d^{c^*}), \quad (5.4.1)$$

i.e., $F^{(T_1+T_2)}$, which is trained for task $\mathcal{T}^1$ and $\mathcal{T}^2$, generalizes to solve the tasks $\mathcal{T}^\ell$, $\ell \leq L$.

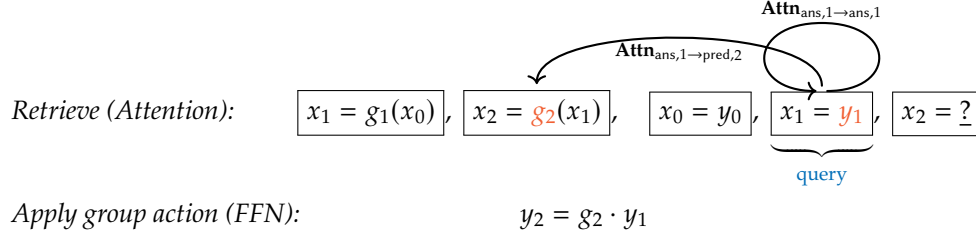


Figure 5.3: LEGO Task Computation. Illustration of how the model solves the LEGO task: given $Z^{2,1}$, the goal is to predict y_2 .

2. *Attention concentration:*³ given $Z^{2,\ell}$ with $\ell \in \{0, 1\}$, we have

$$\text{Attn}_{\text{ans},\ell \rightarrow \text{pred},\ell+1}^{(T_1+T_2)} + \text{Attn}_{\text{ans},\ell \rightarrow \text{ans},\ell}^{(T_1+T_2)} \geq 1 - O\left(\frac{1}{d^{c^*}}\right). \quad (5.4.2)$$

Mechanism of CoT for state-tracking. Given the current sequence $Z^{L,\ell}$ with intermediate steps up ℓ , predicting the next state $y_{\ell+1} = g_{\ell+1}(y_\ell)$ requires two steps:

- (i) *Retrieve* the correct action $g_{\ell+1}$ from the context clause $Z_{\text{pred},\ell+1}$ and the current state y_ℓ from the answer clause $Z_{\text{ans},\ell}$;
- (ii) *Apply the group action*, that is, compute the next state $y_{\ell+1} = g_{\ell+1}(y_\ell)$.

It is well established that attention can implement content-based retrieval [81], and that FFN can represent the group operation [178] (see Figure 5.3 for an example). Algorithm 2 decouples learning the attention (retrieval) and FFN (action) components to simplify the analysis. For task $\mathcal{T}^1$, the transcript $Z^{1,0}$ contains only the two relevant clauses, $Z_{\text{pred},1}$ and $Z_{\text{ans},0}$, without useless contents. Fixed uniform attention ($\mathbf{Q}$ initialized to be zero in Assumption 5.3.3) therefore suffices to expose both clauses to the FFN, and we optimize the FFN to learn the group operation. The high accuracy for $\mathcal{T}^1$ in Theorem 5.4.1 indicates that the FFN has indeed *learned* to apply the operation correctly. For task $\mathcal{T}^2$, with the FFN already trained, the attention layer only needs to learn to route the correct context to the FFN input. The attention concentration result in (5.4.2) confirms that the learned routing pattern is correct. Figures 5.2 and 5.4 also demonstrate the attention concentration patterns empirically.

How does attention concentration induce strong length generalization? As we increase the chain length in $\mathcal{T}^\ell$ for $\ell > 2$, the FFN layer remains largely insensitive to input length since the learned group action is location-invariant. By contrast, the attention layer is affected: more *irrelevant* clauses appear, so retrieval must scan over longer contexts, which risks diluting attention on the relevant clause. Theorem 5.4.1 guarantees that training on short chains already yields attention concentration with error $O(d^{-c^*})$. This “purity” allows the model to tolerate dilution

³ For readability, we abbreviate $\text{Attn}_{\mathbf{k} \rightarrow \mathbf{k}'}(\mathbf{Q}^{(t)}, \mathbf{Z})$ by $\text{Attn}_{\mathbf{k} \rightarrow \mathbf{k}'}^{(t)}$, omitting explicit dependence when immaterial.

and maintain high attention on the relevant clauses for chain lengths up to $O(d^c)$. Technically, this concentration arises because the query matrix $\mathbf{Q}$ learns to locate the same variable x_ℓ that appears simultaneously: the third token of the context clause $Z_{\text{pred},\ell+1}$ and the fourth token of the answer clause $Z_{\text{ans},\ell}$. This co-occurrence furnishes a strong, consistent signal that enables robust retrieval across longer chains, which will be elaborated in the proof overview in Section 5.A.2.

5.5 Learning CoT on Symmetry Group Actions on $\mathbb{Z}_n$

We now turn to the case where the action group $\mathcal{G}$ is isomorphic to the symmetry group, under Assumption 5.5.1. In this case, the problem is NC¹-complete for $n_y \geq 5$.

Assumption 5.5.1 (Symmetry group actions). Let $\mathcal{Y} = \{0, 1, \dots, n_y - 1\}$. We set $\mathcal{G} = \text{Sym}(\mathcal{Y})$, the symmetry group of all permutations of $\mathcal{Y}$, so that $|\mathcal{G}| = n_y!$. We let $\mathcal{G}$ act on $\mathcal{Y}$ in the natural way and write $g \cdot y$ (or $g(y)$) for the image of $y \in \mathcal{Y}$ under $g \in \mathcal{G}$. We assume $n_y = \Theta(\frac{\log \log d}{\log \log \log d})$, and hence $|\mathcal{G}| = n_y! = \text{polylog} d$.

Challenges of learning CoT for the symmetry task. When the model tries to retrieve the group element $g_{\ell+1}$ from the correct context clause $Z_{\text{pred},\ell}$, there can be other context clauses whose group elements also send y_ℓ to $y_{\ell+1}$; we call these *distractor* clauses. In the symmetry case on $\mathcal{Y}$, each pair (i, j) admits $(n_y - 1)!$ elements mapping i to j , so the fraction of distractors is substantial. By contrast, in the simply transitive setting each pair has a unique element, so distractors are unlikely and can be ignored. Attending to distractors still produces the correct next answer, so training may converge with *insufficient attention concentration*. This weaker concentration makes the attention layer less robust to dilution in longer contexts. Hence, for this harder setting, directly proving $d^{\Omega(1)}$ length CoT generalization from constant-length training is difficult.

Self-improvement for reasoning length extension. Recent empirical studies [107, 159, 257] show that *length generalization* can be bootstrapped via model *self-improvement*: models training on their own output can bootstrap their capability to solve longer problems. In particular, the work [159] motivates a *recursive self-training* scheme for the symmetry task. To perform recursive self-training, we adopt the greedy language model as data annotator: the greedy language model $\hat{p}_F$ induced by the network F is defined by

$$\hat{p}_F(Z_{\text{ans},L'+1}|Z^{L,L'}) = \begin{cases} 1, & \text{if } Z_{\text{ans},L'+1} = \arg\max_Z p_F(Z|Z^{L,L'}), \\ 0, & \text{otherwise.} \end{cases} \quad (5.5.1)$$

Now we can define the self-annotated LEGO data distribution:

Algorithm 3: Recursive self-training for symmetry actions

Input: Model $F^{(0)}$ with parameters $(\mathbf{W}^{(0)}, \mathbf{Q}^{(0)})$; Learning rate η ; Error degree $E_1 > 0$ (constant); $\tau_1, \tau_2 = \tilde{O}(\frac{\text{poly}d}{\eta})$; Total Stage K .

Stage 1.1: Train FFN for one-step reasoning ($\mathcal{T}^1$);

```

for  $t = 1$  to  $\tau_1$  do                                     // Update the FFN parameter  $\mathbf{W}$ 
     $\mathbf{W}^{(t)} \leftarrow \mathbf{W}^{(t-1)} - \eta \nabla_{\mathbf{W}} \text{Loss}^1(F^{(t-1)})$ ;
     $\mathbf{Q}^{(t)} \equiv \mathbf{Q}^{(t-1)}$ ;

```

Stage 1.2: Train attention for length extension ($\mathcal{T}^2$);

```

for  $t = \tau_1 + 1$  to  $\tau_1 + \tau_2$  do                             // Update the attention parameter  $\mathbf{Q}$ 
     $\mathbf{Q}^{(t)} \leftarrow \mathbf{Q}^{(t-1)} - \eta \nabla_{\mathbf{Q}} \text{Loss}_{5,2}^{2,2}(F^{(t-1)})$ ;
     $\mathbf{W}^{(t)} \equiv \mathbf{W}^{(t-1)}$ ;

```

$T_1 \leftarrow t$;

Till Stage K : Recursive self-train for length extension;

```

for  $k = 2$  to  $K$  do                                           // Stage  $k$  to solve  $\mathcal{T}^{2^k}$ 
     $L \leftarrow 2^k, \tilde{F}^{(k)} \leftarrow F^{(T_{k-1})}$ ;
    while  $\text{Loss}_{\tilde{F}^{(k)},5}^{L,2}(F^{(t-1)}) > \frac{1}{d^{E_1}}$  do           // Update the attention parameter  $\mathbf{Q}$ 
         $t \leftarrow t + 1$ ;
         $\mathbf{Q}^{(t)} \leftarrow \mathbf{Q}^{(t-1)} - \eta \nabla_{\mathbf{Q}} \text{Loss}_{\tilde{F}^{(k)},5}^{L,2}(F^{(t-1)})$ ;
         $\mathbf{W}^{(t)} \equiv \mathbf{W}^{(t-1)}$ ;
     $T_k \leftarrow t$ ;

```

Output: Models $\{F^{(T_k)}\}_{k=1}^K$.

Definition 5.5.1 (Bootstrapped LEGO distribution). We define $\mathcal{D}_F^{L,L'}$ as the LEGO distribution in Assumption 5.3.2 except that the answers $Z_{\text{ans},\ell}, 1 \leq \ell \leq L'$ is given recursively by sampling the prediction $Z_{\text{ans},\ell} \sim \hat{p}_F(\cdot | Z^{L,\ell-1}), 1 \leq \ell \leq L'$ from the greedy language model $\hat{p}_F$.

Definition 5.5.2 (Self-training loss). Given a (fixed) model $\tilde{F}$ and length L , The self-training next-clause-prediction loss is defined by replacing $\mathcal{D}^{L,L'}$ with $\mathcal{D}_F^{L,L'}$ (Definition 5.5.1) in (5.3.5):

$$\text{Loss}_{\tilde{F}}^{L,L'}(F) \triangleq \mathbb{E}_{Z^{L,L'} \sim \mathcal{D}_{\tilde{F}}^{L,L'}} [-\log p_F(Z_{\text{ans},L',i} | Z^{L,L'-1})], \quad (5.5.2a)$$

$$\text{Loss}_{\tilde{F},i}^{L,L'} \triangleq \mathbb{E}_{Z^{L,L'} \sim \mathcal{D}_{\tilde{F}}^{L,L'}} [-\log p_{F_i}(Z_{\text{ans},L',i} | Z^{L,L'-1})] \quad \text{for } i \in [5]. \quad (5.5.2b)$$

We now present our main results, establishing that a recursive self-training scheme can provably bootstrap the reasoning length for the symmetry LEGO task.

Theorem 5.5.1. Assume the distribution $\mathcal{D}^L$ induced from $\text{LEGO}(\mathcal{X}, \mathcal{G}, \mathcal{Y})$ satisfies Assumption 5.3.1, 5.3.2 and 5.5.1, and assume the transformer network satisfies Assumption 5.3.3, 5.3.4, 5.3.5, and $n_y < m \ll \log^2 d$. Then for any $1 \leq k < \log_2 |\mathcal{X}|$, the transformer $F^{(T_k)}$ trained via Algorithm 3 up to length $L_k = 2^k$ and $T_k = O(\frac{\text{poly}(d)}{\eta})$ satisfies:

1. Constant-factor length generalization: $F^{(T_k)}$ is able to solve $\mathcal{T}^{L_{k+1}}$ with $L_{k+1} = 2^{k+1}$

$$\text{Acc}_{L_{k+1}}(F^{(T_k)}) = 1 - \frac{1}{\text{poly}(d)}. \quad (5.5.3)$$

2. Attention concentration: given $Z^{L_k, \ell}$ with $\ell \in \{0, \dots, L_k - 1\}$, we have

$$\text{Attn}_{\text{ans}, \ell \rightarrow \text{pred}, \ell+1}^{(T_k)} + \text{Attn}_{\text{ans}, \ell \rightarrow \text{ans}, \ell}^{(T_k)} \geq 1 - \tilde{c}, \quad (5.5.4)$$

where $\tilde{c}$ is some sufficiently small constant (smaller than 0.01).

At convergence for the current task $\mathcal{T}^{L_k}$ (at time T_k , the loss has fallen below $1/d^{\text{E}_1}$ in Algorithm 3), (5.5.4) confirms that attention concentration is still insufficient. Nevertheless, Theorem 5.5.1 shows that while this level of concentration cannot withstand the dilution from much longer contexts, it is sufficient for doubling the length. Consequently, a model trained progressively on $\mathcal{T}^L$ for $L = 1, 2, \dots, 2^k$ generalizes to the more challenging task of length 2^{k+1} , yielding the following corollary.

Corollary 5.5.1 (Self-improvement for $|\mathcal{X}|$ -length reasoning). Under the same assumptions as Theorem 5.5.1, letting $K = \Theta(\log d)$, for any length $L \leq |\mathcal{X}|$, the model $F^{(T_K)}$ trained via Algorithm 3 achieves

$$\text{Acc}_L(F^{(T_K)}) \geq 1 - \frac{1}{\text{poly}(d)}.$$

Significance of the result. Note that $\{x_\ell\}_{\ell=0}^L$ is sampled from $\mathcal{X}$ without replacement (Assumption 5.3.2), the longest feasible chain scales with the variable size: $L+1 \leq |\mathcal{X}| = \Theta(d)$. Thus our guarantee attains the best possible length in this setting. Corollary 5.5.1 also demonstrates that the transformer can be trained to solve a task beyond TC^0 with linear-step CoT, matching the expressivity result of [171].⁴ While prior empirical work reports self-improvement in practice, theoretical guarantees, especially for transformers and length generalization, have been scarce [123, 260, 262]. Theorem 5.5.1 provides, to our knowledge, the first rigorous evidence that transformers can *bootstrap* their reasoning via self-training without additional supervision.

⁴ There is a caveat that we do not study when the embedding dimension is logarithmic in the problem length, although it is possible if we allow over-complete basis for the tokens in $\mathcal{X}$ only.

5.6 Proof Roadmap

The proof separates the computation learned by the feed-forward layer from the routing learned by attention. On the one-step task, the feed-forward layer first learns two atomic skills: retrieving the appropriate in-context variable and applying the relevant group action to the current state. The analysis tracks the growth of task-aligned correlations and shows that off-target correlations remain lower order. These learned atomic operations do not depend on the eventual reasoning length.

Attention then learns how to compose the atomic operations. For simply transitive actions, training on two-step chains creates and then balances two diagonal attention blocks: one retrieves the next action and the other retrieves the current state. Once both blocks concentrate, the same routing pattern applies to longer chains. The attention error grows only with the number of irrelevant clauses, which yields the polynomial length-generalization range in Theorem 5.4.1.

For symmetry group actions, stabilizers create ambiguous action–state pairs, so two-step training alone provides only constant-factor extrapolation. Recursive self-training resolves this difficulty. A model that solves length 2^{k-1} produces correct intermediate states for length 2^k ; those states keep the next optimization stage in the basin where the two attention blocks become better concentrated and balanced. Induction over the self-training rounds gives Theorem 5.5.1 and Corollary 5.5.1.

The [appendix to this chapter](#) gives the complete correlation dynamics, gradient estimates, phase inductions, and convergence arguments.

5.7 Experiments

In this section, we conduct synthetic experiments to verify our theoretical claim.

General Setup. We adopt an experimental setting closely aligned with our theoretical setup. Data strictly follow the LEGO distribution in the five-token-per-clause format defined in Definition 5.3.1 and Assumption 5.3.2. The simply transitive task uses the cyclic group of order 6 (denoted C_6), and the symmetry task uses the symmetry group on five elements (denoted S_5). The network is a one-layer, decoder-only transformer with two attention heads and a FFN block. Training optimizes the next-clause loss in (5.3.5a) using Adam [150] with a learning rate of $1e-4$. We train for 300 epochs to ensure the training loss approaches zero and the model converges. For evaluation, to measure the transformer’s ability to solve CoT reasoning tasks, we eschew the teacher-forced accuracy in (5.3.4), which conditions on the ground-truth answer prefix, and instead evaluate final-answer accuracy after autoregressively generating all intermediate answer steps without teacher forcing. This better mimics CoT reasoning and is more challenging because errors can accumulate during self-rollout. For computational efficiency, we report the probability that the model predicts the value of all answer steps correctly, following [171].

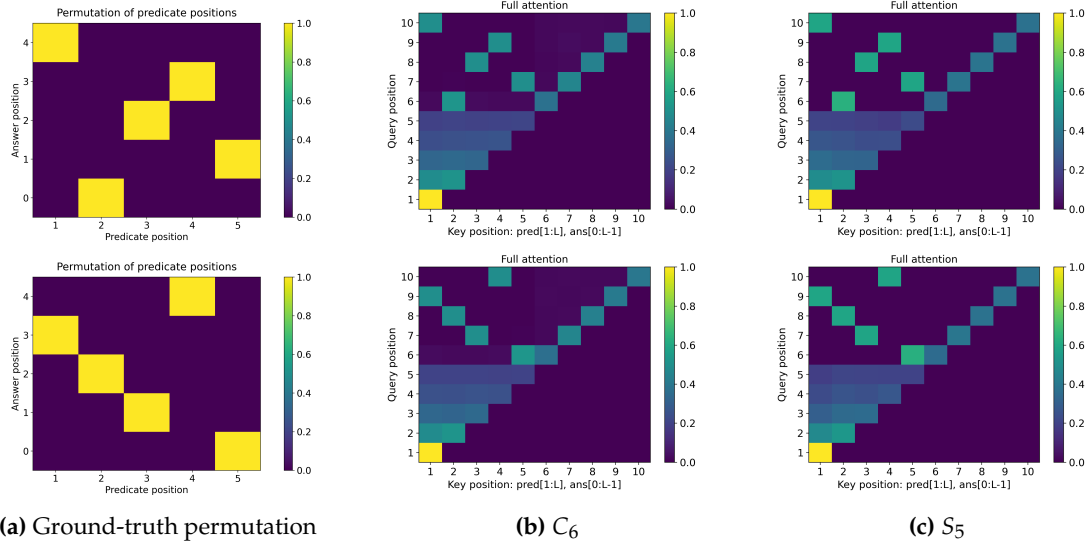


Figure 5.4: Attention Under Permuted Predicate Positions. Attention patterns of the same trained model as Figure 5.2, evaluated with randomly permuted predicate-clause positions. Column (a) gives the ground-truth permutation. Column (b) and Column (c) show the attention heatmaps for the simply transitive and symmetry tasks, respectively.

Length generalization for different group actions. Figure 5.1a shows that when trained at the short length $L = 5$, the model exhibits strong length generalization, achieving near-perfect accuracy at much longer lengths, for example, up to $L = 160$ on the simply transitive task, which corroborates Theorem 5.4.1. By contrast, on the harder symmetry task the generalization is weaker, yielding only constant-factor extensions as expected.

Self-training improves reasoning length. To extend the solvable length on the harder symmetry task, we adopt a recursive self-training curriculum inspired by [159]. We first train at $L = 5$ with ground-truth answer supervision until convergence. At the next stage, we double the length to $L = 10$, use the $L=5$ model to greedily generate answer traces (self-labels) for the $L=10$ data, and retrain on these pseudo-labels. We repeat this doubling process for three stages ($L = 5 \rightarrow 10 \rightarrow 20 \rightarrow 40$), so that the final model is trained on self-labeled data at $L = 40$. Figure 5.1b reports the length-generalization curve for each stage, with the dashed line (matching the curve’s color) marking that stage’s training length. Across stages, we observe at least constant-factor generalization beyond the training length; after multiple rounds of self-training, the model achieves nearly perfect accuracy at lengths far exceeding the initial $L = 5$, ultimately matching the simply transitive task’s performance up to $L = 160$. These results validate Theorem 5.5.1 and demonstrate the effectiveness of recursive self-training for extending reasoning length.

Attention concentration. We visualize attention heatmaps at convergence for models trained at length $L = 5$ on both simply transitive and symmetry group tasks. Each attention matrix is

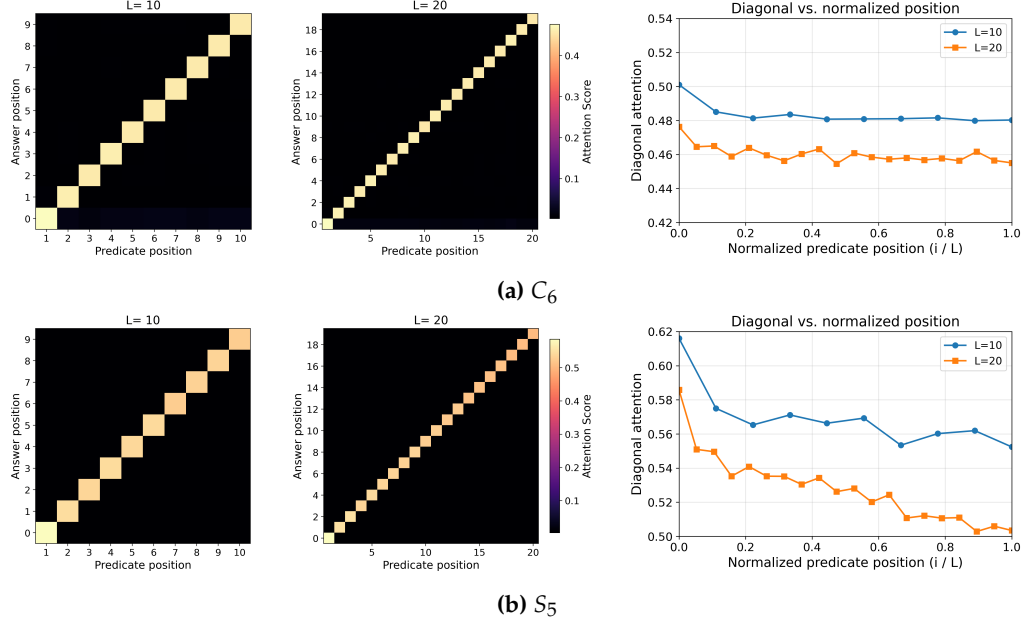


Figure 5.5: Predicate Attention Beyond Training Length. Predicate-clause attention for models trained with LEGO task of length $L = 5$ and tested with $L = 10, 20$. We extract the heatmap of the attention from answer queries to predicate clauses (the upper-left part of the full attention heatmap as in Figure 5.2), and line-plot the diagonal attention against the normalized predicate position. At longer contexts, the symmetry group task displays a visible drop in attention to the target predicate $Z_{\text{pred}, \ell+1}$ at later predicate locations (larger ℓ ; note the absolute query index $L+\ell$ also increases). By contrast, the cyclic group task maintains more consistent attention across lengths and positions.

averaged across the two heads and over 100 independently sampled LEGO sequences. Note that for a task of length L , the LEGO sequence prior to the final answer clause has length $2L$. We focus on the query positions $L+1, \dots, 2L$, which correspond to the answer clauses $Z_{\text{ans}, 0}, \dots, Z_{\text{ans}, L-1}$ for predicting all outputs $y_1, \dots, y_L$.

- In Figure 5.2, the two diagonal structures in the upper region indicate attention concentrating on the answer clause $Z_{\text{ans}, \ell}$ and on the predicate clause $Z_{\text{pred}, \ell+1}$ when the query is $Z_{\text{ans}, \ell}$, validating the attention concentration principle highlighted by Theorem 5.4.1 and Theorem 5.5.1.
- To ablate positional bias, we test on the trained model from Figure 5.2 and only change the input format by randomly permuting the locations of predicate clauses. Results in Figure 5.4 show that retrieval is keyed to the shared variable rather than absolute position.
- Figures 5.5 and 5.6 probe out-of-length (OOL) behavior through attention patterns at test lengths $L \in \{10, 20\}$. As the line plot indicates, the attention scores for both the target predicate and the answer clauses decrease as the task length increases from 10 to 20. This pattern matches the attention dilution predicted by our theory of length generalization. In Figure 5.5, for the symmetry group task, attention to the correct predicate $Z_{\text{pred}, \ell+1}$ decays

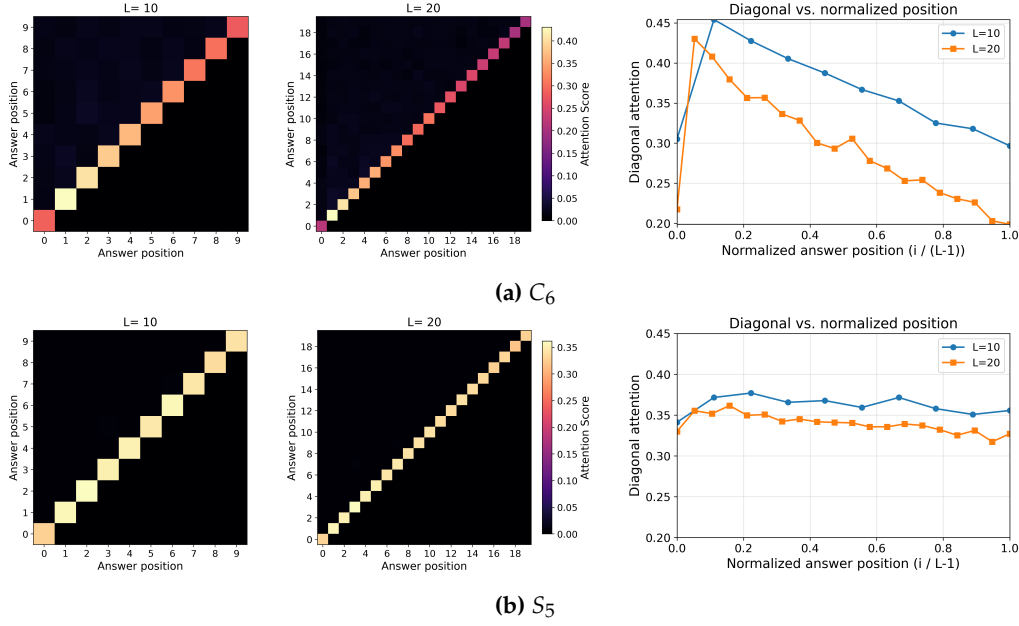


Figure 5.6: Answer Attention Beyond Training Length. Answer-clause attention for models trained with LEGO task of length $L = 5$ and tested with $L = 10, 20$. We extract the heatmap of the attention from answer queries to answer clauses (the upper-right part of the full attention heatmap, as in Figure 5.2), and line-plot the diagonal attention against the normalized answer position. In contrast to the predicate attention in Figure 5.5, at longer contexts, the symmetry group task preserves a stable attention for the answer clause, while the answer attention pattern of the cyclic group task is sensitive to the lengths and positions. Nevertheless, the sharp decay of attention scores for the cyclic group task did not result in a visible performance drop for length generalization, as seen in Figure 5.1.

with position, becoming weaker at later predicates (larger ℓ , hence a larger absolute index $L + \ell$). By contrast, the cyclic group task shows no comparable decay, with attention remaining fairly uniform across lengths and positions. In Figure 5.6, the pattern reverses: attention over answer clauses is more stable for the symmetry group, whereas the cyclic case is more sensitive to length and location. We conjecture that length generalization in the LEGO task hinges on a robust predicate–attention pattern. Accordingly, the observed length-dependent decay of attention on predicate clauses in the symmetry task indicates a non-robust learned attention mechanism, which may underlie its poorer length generalization.

Appendix

5.A Detailed Proof Roadmap

In this section, we outline the proof ideas for the main theorem. Our training schemes in Algorithms 2 and 3 alternate between two phases. First, we train the FFN parameters $\mathbf{W}$ to solve the one-step task $\mathcal{T}^1$. Then, holding $\mathbf{W}$ fixed, we train the attention parameters $\mathbf{Q}$ to solve $\mathcal{T}^2$ and, for symmetry group actions, recursively $\mathcal{T}^{2^k}$. This mirrors the division of labor in our setting (Figure 5.3): the FFN learns the *local update rule*, while the attention layer learns to *route and compose* these updates by locating relevant context over long sequences.

Guided by this picture, our proof overview proceeds in two parts: (1) learning the one-step mechanism for LEGO ($\mathcal{T}^1$): in-context variable retrieval (Section 5.A.1) and group operations (Section 5.A.1). (2) learning the attention layer: direct short-to-long generalization on $\mathcal{T}^2$ under simply transitive actions (Section 5.A.2), and recursive length generalization via self-training on $\mathcal{T}^{2^k}$ under symmetry group actions (Section 5.A.2).

Notations Let us first define a few notations to facilitate the presentation of the proofs. For each $i \in [5]$, $j \in [d]$, $r \in [m]$, define

$$\Lambda_{i,j,r}(\mathbf{Z}^{L,\ell-1}) \triangleq \sum_{\mathbf{k} \in \mathcal{I}^{L,\ell-1}} \text{Attn}_{\text{ans},\ell-1 \rightarrow \mathbf{k}} \cdot \langle \mathbf{W}_{i,j,r}, \mathbf{Z}_{\mathbf{k}} \rangle + b_{i,j,r}. \quad (5.A.1)$$

The quantity $\Lambda_{i,j,r}$ is the FFN pre-activation, i.e., the input to **sReLU**, for token position i , vocabulary index j , and hidden unit r . According to (5.3.2), given $\mathbf{Z}^{L,\ell-1}$, the model’s output at token position i and vocabulary index j is

$$\left[F_i(\mathbf{Z}^{L,\ell-1}) \right]_j = \sum_{r \in [m]} \text{sReLU}(\Lambda_{i,j,r}(\mathbf{Z}^{L,\ell-1})). \quad (5.A.2)$$

We denote $\mathbf{W}_i \triangleq \{\mathbf{W}_{i,j,r}\}_{j \in [d], r \in [m]}$ as the FFN parameters associated with token position i . Each $\mathbf{W}_{i,j,r} \in \mathbb{R}^{5d}$ is written as a vertical concatenation $\mathbf{W}_{i,j,r} = [\mathbf{W}_{i,j,r,1}; \dots; \mathbf{W}_{i,j,r,5}]$, where $\mathbf{W}_{i,j,r,i'} \in \mathbb{R}^d$ for $i' \in [5]$; these five blocks align with the five token-type inputs.

5.A.1 Learning Atomic Skills for Reasoning

For solving task $\mathcal{T}^1$, in stage 1 (stage 1.1 for symmetry group task), we train $\mathbf{W}$ via the full loss Loss^1 , which contains the prediction loss across five tokens in the answer clause and $\mathbf{W}_i$ is responsible for predicting the i -th token in the answer clause. Thus, there are three different types of prediction tasks here: the $\langle \text{blank} \rangle$ tokens (tokens 1, 2, 3), the correct variable x_1 (token 4) and the action update $y_1 = g_1(y_0)$ (token 5). Notice that the $\langle \text{blank} \rangle$ tokens are deterministic, the learning task is straightforward. We therefore focus on the learning dynamics for the 4th and 5th tokens, which involve in-context retrieval ($\mathbf{W}_4$) and one-step group action ($\mathbf{W}_5$). Across these

stage, the attention layer is fixed and keeps as uniform attention due to the zero initialization of the attention matrix, i.e., the input for the FFN layer is $\frac{1}{2}\mathbf{Z}_{\text{pred},1} + \frac{1}{2}\mathbf{Z}_{\text{ans},0}$.

Learning In-Context Retrieval of Variables

Given the input $\frac{1}{2}\mathbf{Z}_{\text{pred},1} + \frac{1}{2}\mathbf{Z}_{\text{ans},0}$ for the FFN layer, the goal is to predict the fourth token in the answer clause $\mathbf{Z}_{\text{ans},1}$, which is the variable x_1 . Intuitively, the network should retrieve the occurrence of x_1 in the first predicate token $\mathbf{Z}_{\text{pred},1}$ and copy it to the target position. Specifically, the FFN pre-activation for predicting the fourth token to be j is

$$\Lambda_{4,j,r}(\mathbf{Z}^{1,0}) = \frac{1}{2} \langle \mathbf{W}_{4,j,r}, \mathbf{Z}_{\text{pred},1} \rangle + \frac{1}{2} \langle \mathbf{W}_{4,j,r}, \mathbf{Z}_{\text{ans},0} \rangle + b_{4,j,r}.$$

Using the 5-vector decomposition of $\mathbf{W}_{4,j,r}$, and the fact that the embedding of (blank) tokens are zero vectors (recall $e_{\tau(\cdot)}$ denote the token embedding vectors), we further obtain:

$$\begin{aligned} \Lambda_{4,j,r}(\mathbf{Z}^{1,0}) &= \frac{1}{2} \left(\langle \mathbf{W}_{4,j,r,1}, e_{\tau(x_1)} \rangle + \langle \mathbf{W}_{4,j,r,2}, e_{\tau(g_1)} \rangle + \langle \mathbf{W}_{4,j,r,3}, e_{\tau(x_0)} \rangle \right) \\ &\quad + \frac{1}{2} \left(\langle \mathbf{W}_{4,j,r,4}, e_{\tau(x_0)} \rangle + \langle \mathbf{W}_{4,j,r,5}, e_{\tau(y_0)} \rangle \right) + b_{4,j,r}. \end{aligned} \quad (5.A.3)$$

Therefore, the main idea of our analysis is to track the training dynamics of $\langle \mathbf{W}_{4,j,r,p}, e_{s'} \rangle$ for $j, s' \in [d]$, $p \in [5]$, and $r \in [m]$. Letting $s \in \tau(X)$ be an embedding index for a variable, our analysis shows the diagonal correlations $\langle \mathbf{W}_{4,s,r,1}, e_s \rangle$ receive strictly larger updates than all other $\langle \mathbf{W}_{4,j,r,p}, e_{s'} \rangle$. This occurs because x_1 co-occurs simultaneously as the first input token at $\mathbf{Z}_{\text{pred},1}$ and as the supervised target at $\mathbf{Z}_{\text{ans},1}$, which amplifies the gradient on $\langle \mathbf{W}_{4,s,r,1}, e_s \rangle$ when $s = \tau(x_1)$. In contrast, non-target coordinates (off-diagonals, wrong variables, value tokens, and group-action tokens) incur negligible gradients and remain $o(1)$ throughout training. Hence the correct variable's diagonal signal becomes order-wise larger, and the *active diagonal mass* $\sum_r \langle \mathbf{W}_{4,s,r,1}, e_s \rangle$ dominantly grows until the end of training.

Given the dominance above, the learned weights align so that $\mathbf{W}_{4,s,r,1}$ points toward e_s for $s \in \tau(X)$. Substituting this alignment into (5.A.3) shows that, when $j = \tau(x_1)$, the red term $\langle \mathbf{W}_{4,\tau(x_1),r,1}, e_{\tau(x_1)} \rangle$ contributes dominantly to $\Lambda_{4,\tau(x_1),r}(\mathbf{Z}^{1,0})$, thereby realizing the intended in-context retrieval: the model copies x_1 from the first position of $\mathbf{Z}_{\text{pred},1}$ to the fourth position of $\mathbf{Z}_{\text{ans},1}$.

Learning the Group Actions

For task $\mathcal{T}^1$, the FFN input $\frac{1}{2}\mathbf{Z}_{\text{pred},1} + \frac{1}{2}\mathbf{Z}_{\text{ans},0}$ already contains the current value y_0 and the action g_1 , with no distracting information. Accordingly, when predicting the next value y_1 , the role of $\mathbf{W}_5$ is to correctly apply the action g_1 to the current value y_0 . In this section, we sketch how the model learns to implement simply transitive group actions, and we briefly discuss the symmetry case.

We first introduce notation to explain what the model should learn.

Definition 5.A.1 (Combinations, simply transitive actions). Assuming the group $\mathcal{G}$ follows Assumption 5.4.1, for each class index $j \in \tau(\mathcal{Y})$, define the *combinations*

$$\Phi := \bigcup_{j \in \tau(\mathcal{Y})} \Phi_j^*, \quad \text{where } \Phi_j^* := \{(g', y') \in \mathcal{G} \times \mathcal{Y} : \tau(g'(y')) = j\}.$$

$|\Phi| = n_y^2$ for the simply transitive case. We call $\phi = (g, y) \in \Phi_j^*$ a *combination* for predicting $j = \tau(g(y))$. Hence, the goal of the model is to correctly identify all of these $\phi = (g, y) \in \Phi$.

Analogously to (5.A.3), the pre-activation at the fifth token for class j and neuron r is

$$\Lambda_{5,j,r}(\mathbf{Z}^{1,0}) = \frac{1}{2} \langle \mathbf{W}_{5,j,r,2}, e_{\tau(g_1)} \rangle + \frac{1}{2} \langle \mathbf{W}_{5,j,r,5}, e_{\tau(y_0)} \rangle + \text{other terms},$$

where slot “2” reads the action token g_1 and slot “5” reads the current value token y_0 . We then define the following feature-magnitude notation:

Definition 5.A.2 (V -Notations). Given $\phi = (g, y) \in \Phi$, for the r -th neuron in the j -th class with $j \in \tau(\mathcal{Y})$ and $r \in [m]$, i.e., $\mathbf{W}_{5,j,r} \in \mathbb{R}^{5d}$, we define

$$V_{j,r}(g) := \langle \mathbf{W}_{5,j,r,2}, e_g \rangle, \quad V_{j,r}(y) := \langle \mathbf{W}_{5,j,r,5}, e_y \rangle,$$

and the composite feature magnitude

$$V_{j,r}(\phi) := \frac{1}{2} (V_{j,r}(g) + V_{j,r}(y)).$$

Then $V_{j,r}(\phi)$ is exactly the contribution of the input pair (g, y) to $\Lambda_{5,j,r}(\mathbf{Z}^{1,0})$ for predicting j . Notice that for each class j , there are m associated neurons $\{\mathbf{W}_{5,j,r}\}_{r=1}^m$, and we index neurons by the pair (j, r) to emphasize that neuron r is specific to class j ; an index r alone has no cross-class meaning in this context.

The main proof idea is to track $V_{j,r}(\phi)$ throughout training and show that it amplifies the correct correlations across the combinations Φ while suppressing the incorrect ones. To make this concrete and to clarify the roles of different neurons, we introduce the following neuron–feature index set:

$$\Psi := \{(j, r, \phi) \mid j \in \tau(\mathcal{Y}), r \in [m], \phi \in \Phi\}.$$

Here, (j, r) again refers to neuron r for class j , as in Definition 5.A.2. With this notation we can write $V_{j,r}(\phi)$ as V_ψ for $\psi = (j, r, \phi) \in \Psi$.

Our proof shows that, given j , for each $\phi \in \Phi_j^*$, there is exactly one neuron r in $\mathbf{W}_{5,j,r}$ that is activated, denoted $r_{g \cdot y}$, to learn ϕ ; that is, $V_{j,r_{g \cdot y}}(\phi)$ will grow to a large value. Therefore, in total n_y^2 distinct neurons will be activated to learn all combinations, i.e., $\{(j, r_{g \cdot y}, (g, y)), \forall j \in \tau(\mathcal{Y}), (g, y) \in \Phi_j^*\}$. The magnitude of remaining V_ψ with non-activated neurons will stay close to the initialization. Our analysis shows that learning follows an *implicit curriculum* induced by

the magnitude of features V_ψ at initialization:

$$\psi < \psi' \iff V_\psi^{(0)} \geq V_{\psi'}^{(0)}, \quad \forall \psi, \psi' \in \Psi.$$

Items on the left under this ordering are learned first, and those on the right are learned later. We denote by $\Sigma^\star$ the *learning curriculum*: the ordered set of neuron–feature indices $\{(j, r_{g \cdot y}, (g, y))\}$ identified above, equipped with this order.

Then, for each $\psi \in \Sigma^\star$, the learning process follows its associated ordering, and when it is the turn of $\psi = (j, r_{g \cdot y}, (g, y))$, the learning process is mainly characterized by the following two phases:

- **Phase I:** Emergence of the feature V_ψ among other features. During this phase, V_ψ grows faster than any $\psi' \neq \psi$ with $\psi < \psi'$, while not affecting the already learned predecessors $\psi' \in \Sigma^\star$ with $\psi' < \psi$, with growth rate

$$V_\psi^{(t+1)} \geq V_\psi^{(t)} + \tilde{\Omega}(\eta) \cdot (b_{j,r} + V_\psi^{(t)} \pm o(\mu))^{q-1}. \quad (5.A.4)$$

This form permits the application of the tensor power method (TPM) [6] and explains the ordering induced by the magnitude of V_ψ at initialization, since TPM implies that a slightly larger initial value leads to dramatically faster growth. By the end of Phase I, for any other feature $\psi' \neq \psi$ with $\psi < \psi'$, the growth of that feature is capped at $\tilde{O}(\sigma_0)$, its initial magnitude.

- **Phase II:** Growth of V_ψ and cancellation of incorrect features. After Phase I, the target feature V_ψ already has a relatively large magnitude; however, features in the set $\{\psi' = (j, r_{g \cdot y}, \phi') \in \Psi \mid \phi' = (g', y) \text{ or } \phi' = (g, y')\}$, namely, the features in the same neuron $(j, r_{g \cdot y})$ that share exactly one component of $\phi = (g, y)$, may also grow as the shared component increases. Note that j is not the correct label for such ϕ' , and we call them **confounding features**. The key characterization in this phase is that, due to a stationarity property of the gradients, although the dynamics are coupled, the wrong half of the confounding features grows in the negative direction while the correct half continues to grow, and ultimately the confounding feature cancels out.

We show that for each $\psi \in \Sigma^\star$ that should be learned, it retains its structure after its own learning process (i.e., at the end of Phase II) and persists through the final convergence while the other features in $\Sigma^\star$ are learned. Specifically, we have the following properties at the end of training:

$$V_{j,r_{g \cdot y}}(g), V_{j,r_{g \cdot y}}(y) \approx B, \quad (5.A.5a)$$

$$V_{j,r_{g \cdot y}}(g'), V_{j,r_{g \cdot y}}(y') \approx -B, \quad (5.A.5b)$$

$$V_{j,r_{g \cdot y}}(g) + V_{j,r_{g \cdot y}}(y') \leq o(1) \text{ and } V_{j,r_{g \cdot y}}(g') + V_{j,r_{g \cdot y}}(y) \leq o(1) \quad \forall g' \neq g, y' \neq y. \quad (5.A.5c)$$

How does this structure perform group actions? For an input pair (g, y) , let the correct answer be $j = \tau(g(y))$. Then the pre-activation for predicting j is around B at neuron $r_{g \cdot y}$, with all other neurons for j near the initial value $\tilde{O}(\sigma_0)$. Thus the model output at j is around B . For all other predictions j' , there exist $\phi_1 = (g', y)$ and $\phi_2 = (g, y')$ such that $j' = \tau(g'(y)) = \tau(g(y'))$. By

cancellation of the incorrect features, we have $V_{j',r_{g',y}}(\phi_1), V_{j',r_{g',y}}(\phi_2) \leq o(1)$. Since $B = \Theta(\log d)$, this implies that the logit on the correct prediction j is very close to 1.

Symmetry Group Actions. The proof strategy for symmetry group actions mirrors the simply transitive case through the emergence, refinement, and convergence phases. However, symmetry actions create richer interactions because multiple group elements can map the same y to the same j , which requires more nuanced control of the training dynamics and leads to different learned feature structures. To illustrate this pattern, we slightly modify the definition of **combinations** used in the simply transitive case by introducing the notion of the **fiber** of a value.

Definition 5.A.3 (Combinations, Symmetry Actions). Assuming the group $\mathcal{G}$ follows Assumption 5.5.1, define $\text{Fiber}_{j,y} := \{g \in \mathcal{G} \mid \tau(g(y)) = j\}$.⁵ This allows us to define the combinations as

$$\Phi = \{\varphi_{j,y} \mid j \in \tau(\mathcal{Y}), y \in \mathcal{Y}\}, \quad \text{where } \varphi_{j,y} = \text{Fiber}_{j,y} \times \{y\}.$$

Moreover, $|\varphi_{j,y}| = (n_y - 1)!$. We continue to call $\phi = (g, y) \in \mathcal{G} \times \mathcal{Y}$ a combination, whereas an element of Φ is now a subset of combinations $\varphi_{j,y}$ (there are n_y^2 such $\varphi_{j,y}$ in total), which includes all pairs (g, y) such that g sends y to j , and which reduces to the single pair (g, y) in the simply transitive case.

Based on these notions, the main difference from the simply transitive case is that previously the basic learning unit is each pair $(g, y) \in \Phi$, whereas now it is the subset $\varphi_{j,y}$ of combinations. All combinations ϕ within $\varphi_{j,y}$ are captured by the same and unique neuron, denoted $r_{j,y}$, in the sense that the feature magnitude $V_{j,r_{j,y}}(\phi)$ is large for any $\phi \in \varphi_{j,y}$. Accordingly, the learning curriculum Σ^* is now based on the initial value of the ensemble of feature magnitudes: $\frac{1}{|\varphi_{j,y}|} \sum_{\phi \in \varphi_{j,y}} V_{j,r_{j,y}}(\phi)$. Finally, at convergence, we have the following imbalance of feature magnitudes:

$$V_{j,r_{j,y}}(g) \approx 2B, \quad V_{j,r_{j,y}}(y) \approx \frac{2B}{n_y} \quad \forall g \in \text{Fiber}_{j,y}; \quad (5.A.6a)$$

$$V_{j,r_{j,y}}(g') \approx \frac{2B}{n_y}, \quad V_{j,r_{j,y}}(y') \approx -2B; \quad (5.A.6b)$$

$$V_{j,r_{j,y}}(g) + V_{j,r_{j,y}}(y') \leq o(1) \text{ and } V_{j,r_{j,y}}(g') + V_{j,r_{j,y}}(y) \leq o(1) \quad \forall g' \notin \text{Fiber}_{j,y}, y' \neq y. \quad (5.A.6c)$$

What Changes for Longer Tasks?

FFN is length invariant. The feed-forward network (with weights $\mathbf{W}$) only acts on the *attended linear combination* at the current output clause, rather than scanning the sequence to retrieve information. Retrieval from the context is delegated to the attention layer. Hence, as sequence

⁵ We use the notion of fiber to denote the left cosets $gG_y \subset \mathcal{G}$, $g \in \mathcal{G}$, where $y \in \mathcal{Y}$ and $G_y = \{g \in \mathcal{G} \mid g(y) = y\}$ is the stabilizer. Since the fiber of the orbit map $f_y(g) = g(y)$ is the preimage of f_y , i.e., $f_y^{-1}(y')$ with $y' = \tau^{-1}(j)$, $j \in \tau(\mathcal{Y})$, this is exactly the set $\text{Fiber}_{j,y}$.

length grows and positions shift, the FFN computation remains the same mapping on its local input, making it length- and position-invariant. On the other hand, uniform attention becomes increasingly diluted as irrelevant clauses accumulate, so selective attention and learning attention layer are required.

Desired attention patterns for predicting each tokens. To predict the fourth token, the model only needs the variable $x_{\ell+1}$ from the predicate clause $\mathbf{Z}_{\text{pred},\ell+1}$. If the attention mass on $\mathbf{Z}_{\text{pred},\ell+1}$ does not vanish with L , the same pattern learned in $\mathcal{T}^1$ with $\mathbf{W}_4$ applies directly. Predicting the fifth token is harder because it requires *two* retrievals: (i) the group action from $\mathbf{Z}_{\text{pred},\ell+1}$, and (ii) the current value from $\mathbf{Z}_{\text{ans},\ell}$, followed by applying the update via $\mathbf{W}_5$. Thus robustness over length hinges on maintaining both attention links. If the attention layer is robust enough to support the fifth-token prediction, then the fourth-token prediction follows as a special case, since it needs only one of the two attention links to persist. Accordingly, in what follows we focus on optimizing the loss for the fifth token; achieving high accuracy there is effectively equivalent to solving the entire task.

5.A.2 Learning the Attention Layer

Successful training on $\mathcal{T}^1$ shows that the model has learned the one-step update $g \circ y$. Turning to task $\mathcal{T}^2$, the remaining difficulty is *routing*: directing attention to the appropriate locations. Thus we train the attention matrix $\mathbf{Q}$ to learn the routing pattern and keep $\mathbf{W}$ fixed. In this part, we will show how training induces the **attention concentration** pattern highlighted in Theorems 5.4.1 and 5.5.1: given an input $\mathbf{Z}^{L,\ell-1}$, the attention mass concentrates on $\text{Attn}_{\text{ans},\ell-1 \rightarrow \text{pred},\ell}$ and $\text{Attn}_{\text{ans},\ell-1 \rightarrow \text{ans},\ell-1}$.

We quantify routing quality via the *attention concentration degree*

$$\epsilon_{\text{attn}}^{L,\ell}(\mathbf{Z}^{L,\ell-1}) = 1 - \text{Attn}_{\text{ans},\ell-1 \rightarrow \text{pred},\ell}(\mathbf{Z}^{L,\ell-1}) - \text{Attn}_{\text{ans},\ell-1 \rightarrow \text{ans},\ell-1}(\mathbf{Z}^{L,\ell-1}), \quad (5.A.7)$$

which measures the fraction of attention mass *not* placed on the two key clauses, and the *attention gap*

$$\Delta^{L,\ell}(\mathbf{Z}^{L,\ell-1}) = \left| \text{Attn}_{\text{ans},\ell-1 \rightarrow \text{pred},\ell}(\mathbf{Z}^{L,\ell-1}) - \text{Attn}_{\text{ans},\ell-1 \rightarrow \text{ans},\ell-1}(\mathbf{Z}^{L,\ell-1}) \right|, \quad (5.A.8)$$

which captures how balanced the two target attentions are. Thus, effective routing corresponds to small $\epsilon_{\text{attn}}^{L,\ell}$ (high concentration) and small $\Delta^{L,\ell}$ (good balance).

Under Assumption 5.3.5, for an input $\mathbf{Z}^{L,\ell-1}$ the (unnormalized) attention score from $\mathbf{Z}_{\text{ans},\ell-1}$ to a clause $\mathbf{Z}_{\mathbf{k}}$ decomposes as

$$\mathbf{Z}_{\text{ans},\ell-1}^\top \mathbf{Q} \mathbf{Z}_{\mathbf{k}} = \mathbf{Z}_{\text{ans},\ell-1,4}^\top \mathbf{Q}_{4,3} \mathbf{Z}_{\mathbf{k},3} + \mathbf{Z}_{\text{ans},\ell-1,4}^\top \mathbf{Q}_{4,4} \mathbf{Z}_{\mathbf{k},4},$$

where $\mathbf{Z}_{\mathbf{k}} = [\mathbf{Z}_{\mathbf{k},1}, \dots, \mathbf{Z}_{\mathbf{k},5}]$ with $\mathbf{Z}_{\mathbf{k},i} \in \mathbb{R}^d$. By design, in the clause embeddings the *fourth* token of a *predicate* clause and the *third* token of an *answer* clause are $\langle \text{blank} \rangle$. Consequently, we further

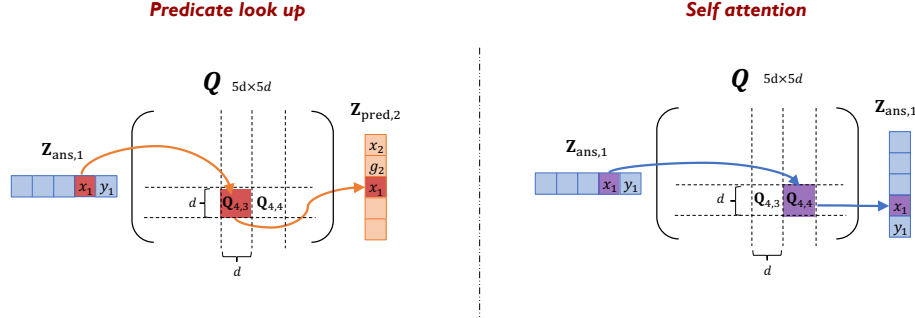


Figure 5.7: Attention Routing In LEGO. The illustration of how different components of the attention matrix $\mathbf{Q}$ are used to route the attention to the appropriate locations. The query clause is $\mathbf{Z}_{\text{ans},1}$ and the goal is to retrieve the correct action g_2 from $\mathbf{Z}_{\text{pred},2}$ and value y_1 from the current answer clause $\mathbf{Z}_{\text{ans},1}$. $[\mathbf{Q}_{4,p}]_{s,s}$ will grow and dominate the learning dynamics for $p \in \{3, 4\}$ and $s \in \tau(\mathcal{X})$. Thus, in this example, large $[\mathbf{Q}_{4,3}]_{\tau(x_1), \tau(x_1)}$ indicates the large attention to the predicate clause $\mathbf{Z}_{\text{pred},2}$ and large $[\mathbf{Q}_{4,4}]_{\tau(x_1), \tau(x_1)}$ indicates the large self-attention to the answer clause $\mathbf{Z}_{\text{ans},1}$.

have

$$\mathbf{Z}_{\text{ans}, \ell-1}^\top \mathbf{Q} \mathbf{Z}_{\text{pred}, \ell'} = \mathbf{Z}_{\text{ans}, \ell-1, 4}^\top \mathbf{Q}_{4,3} \mathbf{Z}_{\text{pred}, \ell', 3}, \quad \mathbf{Z}_{\text{ans}, \ell-1}^\top \mathbf{Q} \mathbf{Z}_{\text{ans}, \ell'} = \mathbf{Z}_{\text{ans}, \ell-1, 4}^\top \mathbf{Q}_{4,4} \mathbf{Z}_{\text{ans}, \ell', 4}, \quad (5.A.9)$$

which means $\mathbf{Q}_{4,3}$ governs attention to predicate clauses, while $\mathbf{Q}_{4,4}$ governs attention to answer clauses. An immediate observation from (5.A.9) is that the desired allocation can be realized by growing the diagonal entries $[\mathbf{Q}_{4,3}]_{s,s}$ and $[\mathbf{Q}_{4,4}]_{s,s}$ ⁶ for $s \in \tau(\mathcal{X})$ since $x_{\ell-1}$ appears as the third token in $\mathbf{Z}_{\text{pred}, \ell}$ and as the fourth token in $\mathbf{Z}_{\text{ans}, \ell-1}$ simultaneously. Our analysis of tracking the attention dynamics will show that these diagonal coordinates indeed receive asymptotically larger gradient magnitudes than all other entries due to strong co-occurrence signal, and the training dynamics are dominated by their growth. See Figure 5.7 for an illustration.

For notational simplicity, we will refer to the relevant diagonal entries $[\mathbf{Q}_{4,3}]_{s,s}$ and $[\mathbf{Q}_{4,4}]_{s,s}$ simply as $\mathbf{Q}_{4,3}$ and $\mathbf{Q}_{4,4}$ below. The remainder of the proof quantifies how the growth of $\mathbf{Q}_{4,3}$ and $\mathbf{Q}_{4,4}$ simultaneously drives the concentration degree $\epsilon_{\text{attn}}^{L, \ell}$ toward zero and controls the gap $\Delta^{L, \ell}$, ensuring that the FFN consistently receives $(g_\ell, y_{\ell-1})$ and thus outputs the correct y_ℓ .

Simply Transitive Actions

For the simply transitive case, we analyze the gradient contribution at position $i = 5$ on task $\mathcal{T}^2$, i.e., the loss $\sum_{\ell=1}^2 \text{Loss}_5^{2, \ell}$. We show that for $\mathcal{T}^2$ the *attention concentration degree* $\epsilon_{\text{attn}}^{2, \ell}$ (for $\ell \in [2]$) can be reduced below $O(1/d^{c^*})$ for some constant $0 < c^* < 1$, indicating highly focused mass on the relevant clauses. When irrelevant entries of $\mathbf{Q}$ are small, we also have $\epsilon_{\text{attn}}^{2, 1} \leq \epsilon_{\text{attn}}^{2, 2}$, since the number of irrelevant clauses doubles from $\ell = 1$ to $\ell = 2$; hence we focus on controlling $\epsilon_{\text{attn}}^{2, 2}$.

⁶ $[\mathbf{A}]_{s,s'}$ denotes the entry of the matrix $\mathbf{A}$ at the s -th row and s' -th column

- **Stage 2.1: Growth of an initial gap.** Early in training, attention is close to uniform, so given $\mathbf{Z}^{2,\ell-1}$ we have the approximations

$$\Lambda_{5,j,r}(\mathbf{Z}^{2,0}) \approx \frac{1}{3} V_{j,r}(g_1) + \frac{1}{3} V_{j,r}(g_2) + \frac{1}{3} V_{j,r}(y_0),$$

$$\Lambda_{5,j,r}(\mathbf{Z}^{2,1}) \approx \frac{1}{4} V_{j,r}(g_1) + \frac{1}{4} V_{j,r}(g_2) + \frac{1}{4} V_{j,r}(y_0) + \frac{1}{4} V_{j,r}(y_1).$$

By the cancellation at the convergence in (5.A.5), for $\ell = 2$ all Λ lie in the small smoothed regime, whereas for $\ell = 1$ we obtain a correct logit for $y_1 = g_1(y_0)$ and a spurious logit for $g_2(y_0)$ of magnitude about $B/3$. Consequently, $-\nabla_{\mathbf{Q}} \text{Loss}_5^{2,2}$ is negligible, while $-\nabla_{\mathbf{Q}} \text{Loss}_5^{2,1}$ is comparatively large and drives $\mathbf{Q}_{4,3}$ to grow faster than $\mathbf{Q}_{4,4}$ (increasing $\mathbf{Q}_{4,4}$ would also amplify the wrong prediction $\tau(g_2(y_0))$ and thus not reduce $\text{Loss}_5^{2,1}$). An $\Omega(1/\log d)$ gap emerges between the diagonals $[\mathbf{Q}_{4,3}]_{s,s}$ and $[\mathbf{Q}_{4,4}]_{s,s}$, yielding an early routing advantage toward right predicate clause.

- **Stage 2.2: Joint growth with a controlled gap.** As $\mathbf{Q}_{4,3}$ increases, the weight $\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}$ becomes large, moving $\Lambda_{5,\tau(g_2(y_1)),r_{g_2:y_1}}$ and $\Lambda_{5,\tau(g_2(y_0)),r_{g_2:y_0}}$ for $\ell = 2$ into the linear regime. Gradients from $\ell = 2$ then dominate, and $\mathbf{Q}_{4,4}$ starts to grow to separate the correct $y_2 = g_2(y_1)$ from the incorrect $\tau(g_2(y_0))$. Throughout, the gap between $\mathbf{Q}_{4,3}$ and $\mathbf{Q}_{4,4}$ stays within $[\Omega(1/\log d), O(1)]$, so the attention gap satisfies $\Delta^{2,2} = \Omega(1/\log d)$.
- **Stage 2.3: Convergence and gap reduction.** Continued joint growth of $\mathbf{Q}_{4,3}$ and $\mathbf{Q}_{4,4}$ concentrates attention near its ideal limit, making $\epsilon_{\text{attn}}^{2,2}$ small. We show $\Delta^{2,2}$ cannot remain above $o(1)$ for long; otherwise an incorrect logit $\text{logit}_{5,\tau(g_2(y_0))}$ would acquire a stronger gradient and force $\mathbf{Q}_{4,4}$ to outpace $\mathbf{Q}_{4,3}$, which contradicts stability. Therefore, at convergence we have (i) $\epsilon_{\text{attn}}^{2,2} = O(d^{-c^*})$ for some $c^* \in (0, 1)$ and (ii) $\Delta^{2,2} = o(1)$; hence both $\mathbf{Q}_{4,3}$ and $\mathbf{Q}_{4,4}$ equal $C \log d \pm o(1)$, while all other entries of the attention matrix remain close to their initial values, where $C > 0$ depends on c^* .

Strong length generalization. The key is that attention concentrates cleanly while remaining stably balanced. For $\mathcal{T}^L$, we obtain

$$\epsilon^{L,\ell} \leq \frac{O(1) \cdot L}{O(1) \cdot L + 1/\epsilon_{\text{attn}}^{2,2}} \leq \frac{1}{1 + \Omega(d^{c^*}/L)}.$$

Hence the model tolerates $O(d^{c^*})$ irrelevant clauses: in particular, if $L = o(d^{c^*})$ then $\epsilon^{L,\ell} = o(1)$; and if $L = \Theta(d^{c^*})$ then $\epsilon^{L,\ell} \leq \frac{1}{1+\Omega(1)}$, which can be made small by choosing the proportionality constant in $L = \Theta(d^{c^*})$ appropriately. Moreover, $\Delta^{L,\ell} \leq \Delta^{2,2} \leq o(1)$, so the two target attention masses remain balanced as L grows, preventing errors from large imbalance (e.g., predicting $\tau(g_2(y_0))$ in Stage 2.3). Consequently, the correct logit satisfies $1 - \text{logit}_{5,\tau(g_{\ell+1}(y_\ell))} \leq \frac{1}{\text{poly}(d)}$, so $\mathcal{T}^L$ is solved with accuracy $1 - 1/\text{poly}(d)$.

Symmetry Group Actions

We now turn to symmetry group tasks $\mathcal{T}^L$ and analyze GD updates with respect to the per-token loss $\text{Loss}_5^{L,2}$ (i.e., predicting the value token in $\mathbf{Z}_{\text{ans},2}$ from $\mathbf{Z}^{L,1}$).

The case $L = 2$. The high-level picture mirrors the simply transitive case, but because multiple group elements can map a given y to the same j , the learned $V_{j,r}(g)$ and $V_{j,r}(y)$ structures are unbalanced. Moreover, as we discussed in the hardness part for the symmetry group in Section 5.5, there will be a non-negligible proportions of distractor clauses. These make it harder to keep a tight balance between $\mathbf{Q}_{4,3}$ and $\mathbf{Q}_{4,4}$ as in the simply transitive case. Nevertheless, we prove that both $\mathbf{Q}_{4,3}$ and $\mathbf{Q}_{4,4}$ grow, and the attention gap $\Delta^{2,2}$ is controlled by a feedback mechanism: if $\Delta^{2,2}$ exceeds a small fixed threshold (in either direction), some incorrect logit receives a stronger gradient, which pushes the system back toward balance. Consequently, after sufficient training: (i) $\epsilon_{\text{attn}}^{2,2} \leq C_1$; (ii) $\Delta^{2,2} \leq C_2$, for sufficiently small constants C_1, C_2 ; and since $B = \Theta(\log d)$, we still obtain $\text{Loss}_5^{2,2} \leq 1/\text{poly}(d)$.

Recursive learning for $\mathcal{T}^{2^k}$, $k \geq 2$. Because $\epsilon_{\text{attn}}^{2^{k-1},2}$ is already a small constant, initialization for $\mathcal{T}^{2^k}$ satisfies $\epsilon_{\text{attn}}^{2^k,2} \leq 2\epsilon_{\text{attn}}^{2^{k-1},2}$ (still small), and $\Delta^{2^k,2} \leq \Delta^{2^{k-1},2}$. Thus the attention pattern remains close to that in $\mathcal{T}^{2^{k-1}}$, and $\mathbf{Z}^{2^k,1}$ follows a bootstrapped LEGO distribution generated by the greedy model $\hat{p}_{F^{(T_{k-1})}}$, which coincides with the original LEGO source. In particular, $y_1 = g_1(y_0)$ and $y_2 = g_2(y_1)$ are correct. We can therefore reuse the convergence analysis from $\mathcal{T}^{2^{k-1}}$ to show that both $\epsilon_{\text{attn}}^{2^k,2}$ and $\Delta^{2^k,2}$ decrease to a small constant, yielding stable, inductive concentration across recursive reasoning depths.

5.A.3 Significance of the Proof

Our proof techniques are inspired by recent advances in understanding the dynamics of feature learning in neural networks [7, 8, 127, 128, 129, 135, 311, 312], which show how gradient-based training induces useful internal patterns and representations. Building on these ideas, we analyze the dynamics of the FFN and attention layers to capture how transformers gradually acquire length-generalizable reasoning through CoT training. We conclude this section by summarizing the technical significance of our analysis for learning in the FFN and attention layers.

Learning the group actions. Most existing optimization-based analyses of CoT [124, 130, 146, 309] hard-code the group action, e.g., parity, as a fixed FFN or an almost-linear map. In contrast, we *train* a nonlinear FFN end-to-end to perform the group action at each step, with parity as a special $n = 2$ instance of our simply transitive framework. This formulation is strictly more general and substantially more challenging than prior setups. Our proof shows that the model learns not only the basic action for simply transitive groups but also more complex actions for richer actions that is transitive but not free. Especially, we precisely characterizing how task-relevant features emerge and spurious features are suppressed during the process. The

technique we used in the proof for learning the FFN on discrete combinations of data is of independent interest beyond analyzing CoT.

Learning the attention patterns. Prior work on transformer training dynamics has established the *attention concentration* principle as a key mechanism for solving various tasks, e.g., in-context learning [128], self-supervised learning [129], and graph learning [202]. In those settings, only a single token needs to be retrieved, so the attention matrix is learned to be diagonal (after a suitable change of basis), and a single pattern suffices. In contrast, our task requires retrieving two different types of clauses, so the model must learn two distinct components (blocks) in the attention matrix, $\mathbf{Q}_{4,3}$ and $\mathbf{Q}_{4,4}$. As we show in Section 5.A.2, beyond growing these blocks, we must maintain a delicate balance between them. This balance is crucial for length generalization across different group actions, introduces substantial technical challenges, and our proof provides a fine-grained control of it.

5.B Detailed Proofs

This section develops the proofs supporting the chapter’s main results. The phase-by-phase arguments are kept in the main chapter so that the optimization mechanisms remain connected to the story; auxiliary estimates are collected in explicitly labeled technical subsections.

5.B.1 Learning In-Context Retrieval of Variables

In this section, we focus on the learning process of $\mathbf{W}_4$ for the task $\mathcal{T}^1$. For this task, $\mathbf{W}_4$ should predict the 4-th token in the answer clause $Z_{\text{ans},1}$, which is the variable x_1 . We will show that, the network learns to retrieve the target variable x_1 from the first token of the first predicate clause $Z_{\text{pred},1}$ to make an accurate prediction. Throughout the rest of the proof, we will omit the subscript for the expectation $\mathbb{E}$, when the context is clear.

Preliminaries

First we define some notations for the presentation of gradients.

Notations for gradient expressions For each $i \in [5]$, $\ell \in [L]$, $j \in [d]$, $r \in [m]$, we denote

$$\mathcal{E}_{i,j}(\mathbf{Z}^{L,\ell-1}) \triangleq \mathbb{1}_{\mathbf{Z}_{\text{ans},\ell,i}=e_j} - \text{logit}_{i,j}(F, \mathbf{Z}^{L,\ell-1}), \quad (5.B.1)$$

$$\Lambda_{i,j,r}(\mathbf{Z}^{L,\ell-1}) \triangleq \sum_{\mathbf{k} \in \mathcal{I}^{L,\ell-1}} \text{Attn}_{\text{ans},\ell-1 \rightarrow \mathbf{k}} \cdot \langle \mathbf{W}_{i,j,r}, \mathbf{Z}_{\mathbf{k}} \rangle + b_{i,j,r}. \quad (5.B.2)$$

where $\mathbf{logit}_{i,j}(F, \mathbf{Z}^{L,\ell-1})$ are defined as

$$\mathbf{logit}_{i,j}(F, \mathbf{Z}^{L,\ell-1}) := \frac{e^{F_{i,j}(\mathbf{Z}^{L,\ell-1})}}{\sum_{j' \in [d]} e^{F_{i,j'}(\mathbf{Z}^{L,\ell-1})}}.$$

Fact 5.B.1. For any $i \in [5]$, $j \in [d]$, $r \in [m]$, we have the following gradient expression:

$$-\nabla_{\mathbf{W}_{i,j,r}} \text{Loss}^L = \mathbb{E} \left[\sum_{\ell=1}^L \mathcal{E}_{i,j}(\mathbf{Z}^{L,\ell-1}) \mathbf{sReLU}'(\Lambda_{i,j,r}(\mathbf{Z}^{L,\ell-1})) \sum_{\mathbf{k} \in \mathcal{I}^{L,\ell-1}} \mathbf{Attn}_{\text{ans},\ell-1 \rightarrow \mathbf{k}} \mathbf{Z}_{\mathbf{k}} \right].$$

For simplicity of notation, we will henceforth denote $\Lambda_{i,j,r}(\mathbf{Z}^{L,\ell-1})$ by $\Lambda_{i,j,r}$ and $\mathcal{E}_{i,j}(\mathbf{Z}^{L,\ell-1})$ by $\mathcal{E}_{i,j}$ when the context is clear.

Given $\mathbf{Z}^L$, we use $\widehat{\mathcal{X}}^L$ to denote the appeared variables in the context clauses, i.e. $\widehat{\mathcal{X}}^L = \{x_0, x_1, \dots, x_L\}$. We write $\widehat{\mathcal{X}}^L$ as $\widehat{\mathcal{X}}$ for simplicity. Throughout this section, we write $[F_i]_j$ as $F_{i,j}$ for simplicity.

Induction Hypothesis

Proof Plan. The main idea is to track the dynamics of different types of weights $\mathbf{W}_{4,j,r,p}$. Specifically, we prove that for each $j \in \tau(\mathcal{X})$, there exists certain neurons $r \in [m]$ such that the corresponding weights $\mathbf{W}_{4,j,r,1}$ grow significantly along the direction e_j , while all others remain small. Specifically, we proceed in four steps:

1. For $j \in \tau(\mathcal{X})$, define

$$\Gamma_{4,j}^{(t)} \triangleq \max_{r \in [m]} \langle \mathbf{W}_{4,j,r,1}^{(t)}, e_j \rangle + \sigma_0 \log d,$$

to track the maximal activation associated with retrieving the correct variable token.

2. **Establish rapid growth (early phase).** Let $\Lambda^- = \Theta(1/m)$, and define the hitting time

$$T_{1,j} \triangleq \min\{t > 0 : \Gamma_{4,j}^{(t)} \geq \Lambda^-\}.$$

We show that for iterations $t \leq T_1 = \Theta(d\sigma_0^{q-2}/\eta)$, the diagonal weights $\langle \mathbf{W}_{4,j,r,1}^{(t)}, e_j \rangle$ grow rapidly, causing $\Gamma_{4,j}^{(t)}$ to enter a linear growth regime. Simultaneously, the model confidently identifies the correct variable, indicated by $1 - \mathbf{logit}_{4,\tau(x_1)}^{(t)} = 1 - o(1)$.

3. **Convergence via dominant neurons (late phase).** For each $j \in \tau(\mathcal{X})$, define active neuron sets and their total activation as:

$$\mathcal{A}_{4,j}^{(t)} \triangleq \{r \in [m] : \langle \mathbf{W}_{4,j,r,1}^{(t)}, e_j \rangle \geq \varrho \log d\}, \quad \Phi_{4,j}^{(t)} \triangleq \sum_{r \in \mathcal{A}_{4,j}^{(t)}} \langle \mathbf{W}_{4,j,r,1}^{(t)}, e_j \rangle.$$

For iterations $t > T_1$, we analyze the refined dynamics, proving that the total diagonal activation $\Phi_{4,j^\star}^{(t)}$, for the weakest activated variable $j^\star$, eventually grows to $\Theta(\log d)$, ensuring successful learning.

4. **Bounding non-target correlations.** We finally show by induction that all other correlations $\langle \mathbf{W}_{4,j,r,p}^{(t)}, e_s \rangle$, including group actions, value tokens, off-diagonal tokens, and non-target variables, remain negligible throughout the training process.

Our proof begins by positing an induction hypothesis expected to hold throughout training. We then analyze the dynamics under this hypothesis and show that it remains valid along the entire training trajectory, establishing the claim at convergence.

Induction 5.B.1. For $t \leq T = \frac{\text{poly}d}{\eta}$, all of the following holds:

- (a). for $j \in \tau(\mathcal{X})$, $\tilde{\Omega}(\sigma_0) \leq \langle \mathbf{W}_{4,j,r,1}^{(t)}, e_j \rangle + \mu \leq \tilde{O}(1)$, where $\langle \mathbf{W}_{4,j,r,1}^{(t)}, e_j \rangle$ is non-decreasing;
(b). for $j \in \tau(\mathcal{X})$, $g \in \mathcal{G}$

$$|\langle \mathbf{W}_{4,j,r,2}^{(t)}, e_{\tau(g)} \rangle| \leq \tilde{O}(\sigma_0) + O\left(\frac{1}{|\mathcal{G}|}\right) \max \left\{ \langle \mathbf{W}_{4,j,r,1}^{(t)}, e_j \rangle, \min_{r' \in \mathcal{A}_{4,j^\star}^{(t)}} \langle \mathbf{W}_{4,j^\star,r',1}^{(t)}, e_{j^\star} \rangle \right\};$$

- (c). for $j \in \tau(\mathcal{X})$, $y \in \mathcal{Y}$

$$|\langle \mathbf{W}_{4,j,r,5}^{(t)}, e_{\tau(y)} \rangle| \leq \tilde{O}(\sigma_0) + O\left(\frac{1}{|\mathcal{Y}|}\right) \max \left\{ \langle \mathbf{W}_{4,j,r,1}^{(t)}, e_j \rangle, \min_{r' \in \mathcal{A}_{4,j^\star}^{(t)}} \langle \mathbf{W}_{4,j^\star,r',1}^{(t)}, e_{j^\star} \rangle \right\};$$

- (d). else, $|\langle \mathbf{W}_{4,j,r,p}^{(t)}, e_{j'} \rangle| \leq \tilde{O}(\sigma_0)$ for any other $j, j' \in [d]$.

Claim 5.B.1. If Induction 5.B.1 holds at iteration t , then for an input $\mathbf{Z}^{1,0}$, we have

1. if $j = \tau(x_1)$,

$$\Lambda_{4,j,r}^{(t)} = \frac{1}{2} \langle \mathbf{W}_{4,j,r,2}^{(t)}, e_j \rangle + \frac{1}{2} \langle \mathbf{W}_{4,j,r,2}^{(t)}, e_{\tau(g_1)} \rangle + \frac{1}{2} \langle \mathbf{W}_{4,j,r,5}^{(t)}, e_{\tau(y_0)} \rangle + \frac{5}{2} \mu + \tilde{O}(\sigma_0);$$

2. else if $j \in \tau(\mathcal{X} \setminus \{x_1\})$,

$$\Lambda_{4,j,r}^{(t)} = \frac{1}{2} \langle \mathbf{W}_{4,j,r,2}^{(t)}, e_{\tau(g_1)} \rangle + \frac{1}{2} \langle \mathbf{W}_{4,j,r,5}^{(t)}, e_{\tau(y_0)} \rangle + \frac{5}{2} \mu + \tilde{O}(\sigma_0);$$

3. otherwise, $0 \leq \Lambda_{4,j,r}^{(t)} \leq \frac{5}{2} \mu + \tilde{O}(\sigma_0)$.

Claim 5.B.2. If Induction 5.B.1 holds at iteration t , then for an input $\mathbf{Z}^{1,0}$,

1. if $j = \tau(x_1)$, $\text{logit}_{4,j}^{(t)} = \frac{e^{\frac{O(\Phi_{4,j}^{(t)})}{e^{O(\Phi_{4,j}^{(t)}) + d}}}}{e^{O(\Phi_{4,j}^{(t)}) + d}};$
2. otherwise, $\text{logit}_{4,j}^{(t)} = O\left(\frac{1}{d}\right) \left(1 - \text{logit}_{\tau(x_1)}^{(t)}\right).$

Proof. If $j = \tau(x_1)$, by Induction 5.B.1 and Claim 5.B.1, we have

$$\begin{aligned} 0 \leq F_{4,j}^{(t)}(\mathbf{Z}^{1,0}) &\leq \sum_{r \in [m]} [\Lambda_{4,j,r}^{(t)}]^+ \leq (\Phi_{4,j}^{(t)} + O(\frac{\max\{\Phi_{4,j}^{(t)}, \Phi_{4,j^*}^{(t)}\}}{|\mathcal{G}|})) + \tilde{O}(\sigma_0) + O(m\varrho \log d) \\ &= (\Phi_{4,j}^{(t)} + O(\frac{\Phi_{4,j}^{(t)}}{|\mathcal{G}|})) + \tilde{O}(\sigma_0) + O(\frac{1}{\text{polylog} d}). \end{aligned}$$

For $j \in \tau(\mathcal{X}) \neq \tau(x_1)$, $F_{4,j}^{(t)}(\mathbf{Z}^{1,0}) \leq \tilde{O}(\sigma_0) + O(\frac{\max\{\Phi_{4,j}^{(t)}, \Phi_{4,j^*}^{(t)}\}}{|\mathcal{G}|})$; else $F_{4,j}^{(t)}(\mathbf{Z}^{1,0}) \leq \tilde{O}(\sigma_0)$. Combining them together, we complete the proof. $\square$

Gradient Lemma

Starting with the gradient computation from Fact 5.B.1:

$$-\nabla_{\mathbf{W}_{4,j,r,p}} \text{Loss}^1 = \frac{1}{2} \mathbb{E} \left[\mathcal{E}_{4,j} \mathbf{sReLU}'(\Lambda_{4,j,r}) \sum_{\mathbf{k} \in \mathcal{I}^{1,0}} \mathbf{Z}_{\mathbf{k},p} \right],$$

we first consider the gradient for $j \in \tau(\mathcal{X})$

Lemma 5.B.1. *For $j \in \tau(\mathcal{X})$, we have*

(a) *for $\mathbf{W}_{4,j,r,1}$, $s \in \tau(\mathcal{X})$*

$$(1) \text{ if } s = j, \langle -\nabla_{\mathbf{W}_{4,j,r,1}}^{(t)} \mathbb{L}^1, e_s \rangle = \frac{1}{2} \mathbb{E} \left[(1 - \text{logit}_{4,j}^{(t)}) \mathbf{sReLU}'(\Lambda_{4,j,r}^{(t)}) \mathbb{1}_{\tau(x_1)=j} \right];$$

$$(2) \text{ } s \neq j, \langle -\nabla_{\mathbf{W}_{4,j,r,1}}^{(t)} \mathbb{L}^1, e_s \rangle = \frac{1}{2} \mathbb{E} \left[-\text{logit}_{4,j}^{(t)} \mathbf{sReLU}'(\Lambda_{4,j,r}^{(t)}) \mathbb{1}_{\tau(x_1)=s} \right].$$

(b) *for $\mathbf{W}_{4,j,r,2}$, $s = \tau(g)$ for $g \in \mathcal{G}$*

$$\begin{aligned} \langle -\nabla_{\mathbf{W}_{4,j,r,2}}^{(t)} \text{Loss}^1, e_s \rangle &= \frac{1}{2} \mathbb{E} \left[(1 - \text{logit}_{4,j}^{(t)}) \mathbf{sReLU}'(\Lambda_{4,j,r}^{(t)}) \mathbb{1}_{\tau(x_1)=j, g_1=g} \right. \\ &\quad \left. - \text{logit}_{4,j}^{(t)} \mathbf{sReLU}'(\Lambda_{4,j,r}^{(t)}) \mathbb{1}_{\tau(x_1) \neq j, g_1=g} \right]. \end{aligned}$$

(c) *for $\mathbf{W}_{4,j,r,p}$ with $p \in \{3, 4\}$, $s \in \tau(\mathcal{X})$*

$$(1) \text{ } s = j, \langle -\nabla_{\mathbf{W}_{4,j,r,3}}^{(t)} \mathbb{L}^1, e_j \rangle = \frac{1}{2} \mathbb{E} \left[-\text{logit}_{4,j}^{(t)} \mathbf{sReLU}'(\Lambda_{4,j,r}^{(t)}) \mathbb{1}_{\tau(x_0)=j} \right];$$

(2) $s \neq j$

$$\begin{aligned} \langle -\nabla_{\mathbf{W}_{4,j,r,3}}^{(t)} \text{Loss}^1, e_s \rangle &= \frac{1}{2} \mathbb{E} \left[(1 - \text{logit}_{4,j}^{(t)}) \mathbf{sReLU}'(\Lambda_{4,j,r}^{(t)}) \mathbb{1}_{\tau(x_0)=s, \tau(x_1)=j} \right. \\ &\quad \left. - \text{logit}_{4,j}^{(t)} \mathbf{sReLU}'(\Lambda_{4,j,r}^{(t)}) \mathbb{1}_{\tau(x_0)=s, j \notin \tau(\widehat{X})} \right]. \end{aligned}$$

(d) *for $\mathbf{W}_{4,j,r,5}$, $s = \tau(y)$ for $g \in \mathcal{Y}$*

$$\langle -\nabla_{\mathbf{W}_{4,j,r,5}}^{(t)} \text{Loss}^1, e_s \rangle = \frac{1}{2} \mathbb{E} \left[(1 - \text{logit}_{4,j}^{(t)}) \mathbf{sReLU}'(\Lambda_{4,j,r}^{(t)}) \mathbb{1}_{\tau(x_1)=j, y_0=y} \right]$$

$$-\mathbf{logit}_{4,j}^{(t)} \mathbf{sReLU}'(\Lambda_{4,j,r}^{(t)}) \mathbb{1}_{\tau(x_1) \neq j, y_0=y} \Big].$$

Move on to $j \notin \tau(X)$, we can obtain

Lemma 5.B.2. *For $j \notin \tau(X)$, we have*

(a) *for $\mathbf{W}_{4,j,r,1}$, $s \in \tau(X)$,*

$$\langle -\nabla_{\mathbf{W}_{4,j,r,1}^{(t)}} \mathbb{L}^1, e_s \rangle = \frac{1}{2} \mathbb{E} \left[-\mathbf{logit}_{4,j}^{(t)} \mathbf{sReLU}'(\Lambda_{4,j,r}^{(t)}) \mathbb{1}_{\tau(x_1)=s} \right].$$

(b) *for $\mathbf{W}_{4,j,r,2}$, $s = \tau(g)$ for $g \in \mathcal{G}$,*

$$\langle -\nabla_{\mathbf{W}_{4,j,r,2}^{(t)}} \mathbb{L}^1, e_s \rangle = \frac{1}{2} \mathbb{E} \left[-\mathbf{logit}_{4,j}^{(t)} \mathbf{sReLU}'(\Lambda_{4,j,r}^{(t)}) \mathbb{1}_{g_1=g} \right].$$

(c) *for $\mathbf{W}_{4,j,r,p}$ with $p \in \{3, 4\}$, $s \in \tau(X)$,*

$$\langle -\nabla_{\mathbf{W}_{4,j,r,p}^{(t)}} \mathbb{L}^1, e_j \rangle = \frac{1}{2} \mathbb{E} \left[-\mathbf{logit}_{4,j}^{(t)} \mathbf{sReLU}'(\Lambda_{4,j,r}^{(t)}) \mathbb{1}_{\tau(x_0)=s} \right].$$

(d) *for $\mathbf{W}_{4,j,r,5}$, $s = \tau(y)$ for $g \in \mathcal{Y}$*

$$\langle -\nabla_{\mathbf{W}_{4,j,r,5}^{(t)}} \mathbb{L}^1, e_s \rangle = \frac{1}{2} \mathbb{E} \left[-\mathbf{logit}_{4,j}^{(t)} \mathbf{sReLU}'(\Lambda_{4,j,r}^{(t)}) \mathbb{1}_{y_0=y} \right].$$

Growth of Gamma

Lemma 5.B.3 (Growth). *Given $j \in \tau(X)$, suppose Induction 5.B.1 holds at iteration t , when $\Phi_{4,j}^{(t)} \leq 0.01 \log d$ or $\Gamma_{4,j}^{(t)} \leq \frac{0.01 \log d}{m}$, then it satisfies*

$$\Gamma_{4,j}^{(t+1)} = \Gamma_{4,j}^{(t)} + \Theta\left(\frac{\eta}{d}\right) \mathbf{sReLU}'(\Gamma_{4,j}^{(t)}).$$

Proof. By Lemma 5.B.1, we have

$$\langle -\nabla_{\mathbf{W}_{4,j,r,1}^{(t)}} \mathbb{L}^1, e_j \rangle = \frac{1}{2} \mathbb{E} \left[(1 - \mathbf{logit}_{4,j}^{(t)}) \mathbf{sReLU}'(\Lambda_{4,j,r}^{(t)}) \mathbb{1}_{\tau(x_1)=j} \right].$$

By Claim 5.B.2, when $\Phi_{4,j}^{(t)} \leq 0.01 \log d$, $\mathbf{logit}_{4,j}^{(t)} = \frac{O(e^{0.01 \log d})}{O(e^{0.01 \log d}) + d} \ll 1$ when $j = \tau(x_1)$; and combining with the fact that the event $\{\tau(x_1) = j\}$ happens with probability $\frac{1}{|X|}$, we complete the proof. $\square$

Lemma 5.B.3, combined with the growth of the tensor power method, immediately gives the following corollary.

Lemma 5.B.4. *Suppose Induction 5.B.1 holds for all iterations. Define threshold $\Lambda^- = \Theta(\frac{1}{m})$. Let $T_{1,j}$ be the first iteration so that $\Gamma_{4,j}^{(t)} \geq \Lambda^-$, and $T_1 \stackrel{\text{def}}{=} \Theta(\frac{d}{\eta \sigma_0^{q-2}})$. Then we have $T_1 \geq T_{1,j}$ for every $j \in \tau(X)$, i.e., for $t \geq T_1$, it satisfies $\Gamma_{4,j}^{(t)} \geq \Lambda^-$.*

Lemma 5.B.5 (Upper bound). *Suppose Induction 5.B.1 holds for all iterations $< t$, we have $\Phi_{4,j}^{(t)} \leq \tilde{O}(1)$, for $j \in \tau(\mathcal{X})$.*

Proof. We only need to consider the time $t \geq T_1$. Notice that the gradient descent update in Lemma 5.B.1 gives

$$\langle -\nabla_{\mathbf{w}_{4,j,r,1}^{(t)}} \mathbb{L}^1, e_j \rangle = \frac{1}{2} \mathbb{E} \left[(1 - \text{logit}_{4,j}^{(t)}) \text{sReLU}'(\Lambda_{4,j,r}^{(t)}) \mathbb{1}_{\tau(x_1)=j} \right]$$

Therefore, for sufficiently small η , we have

$$\begin{aligned} \Phi_{4,j}^{(t+1)} &= \Phi_{4,j}^{(t)} + \sum_{r \in \mathcal{A}_{4,j}^{(t)}} \frac{\eta}{2} \mathbb{E} \left[(1 - \text{logit}_{4,j}^{(t)}) \text{sReLU}'(\Lambda_{4,j,r}^{(t)}) \mathbb{1}_{\tau(x_1)=j} \right] + O(\varrho \log d) \cdot |\mathcal{A}_{4,j}^{(t+1)} \setminus \mathcal{A}_{4,j}^{(t)}| \\ &= \Phi_{4,j}^{(t)} + \sum_{r \in \mathcal{A}_{4,j}^{(t)}} \frac{\eta}{2} \mathbb{E} \left[(1 - \text{logit}_{4,j}^{(t)}) \text{sReLU}'(\Lambda_{4,j,r}^{(t)}) \mathbb{1}_{\tau(x_1)=j} \right] + \frac{1}{\text{polylog} d}. \end{aligned}$$

When there exists $\tilde{T}$, s.t., $\max_{j \in \tau(\mathcal{X})} \Phi_{4,j}^{(\tilde{T})} > \Omega(\log^{1.5} d)$, by Induction 5.B.1 and Claim 5.B.1, given an input sequence $\mathbf{Z}^{1,0}$ with $\tau(x_1) = \tilde{j} = \arg \max_{j \in \tau(\mathcal{X})} \Phi_{4,j}^{(\tilde{T})}$, we have

$$F_{4,j}^{(\tilde{T})}(\mathbf{Z}^{1,0}) \geq \sum_{r \in \mathcal{A}_{4,\tilde{j}}^{(t)}} \Lambda_{4,j,r}^{(\tilde{T})} \geq \left(1 - O\left(\frac{1}{|\mathcal{G}|}\right) \right) \Phi_{4,\tilde{j}}^{(\tilde{T})} - \tilde{O}(\sigma_0) > \Omega(\log^{1.5} d).$$

Following the similar analysis as Claim 5.B.2, $F_{4,j'}^{(\tilde{T})}(\mathbf{Z}^{1,0}) \leq O(\frac{\Phi_{4,j'}^{(\tilde{T})}}{|\mathcal{G}|})$ for other $j' \in \tau(\mathcal{X})$, and $F_{4,j'}^{(\tilde{T})}(\mathbf{Z}) \leq o(1)$ for $j' \notin \tau(\mathcal{X})$, which implies $1 - \text{logit}_{4,j}^{(\tilde{T})} = e^{-\Omega(\log^{1.5} d)}$. Therefore, we derive that for $t \in [\tilde{T} + 1, \frac{\text{poly} d}{\eta})$,

$$\Phi_{4,j}^{(t)} \leq \Phi_{4,j}^{(\tilde{T})} + \tilde{O}(\text{poly} d \cdot e^{-\Omega(\log^{1.5} d)}) + O(\varrho \log d) \cdot m,$$

which implies $\Phi_{4,\tilde{j}}^{(t)} \leq O(\log^{1.5} d)$ since $\varrho \ll \frac{1}{m \log d}$. □

Group and Value Correlations Are Not Large

Lemma 5.B.6. *Suppose Induction 5.B.1 holds for all iterations $< t$, then for any $j \in \tau(\mathcal{X})$ and $s = \tau(g)$, $g \in \mathcal{G}$, we have*

$$|\langle \mathbf{W}_{4,j,r,2}^{(t)}, e_{\tau(g)} \rangle| \leq \tilde{O}(\sigma_0) + O\left(\frac{1}{|\mathcal{G}|}\right) \max \left\{ \langle \mathbf{W}_{4,j,r,1}^{(t)}, e_j \rangle, \min_{r' \in \mathcal{A}_{4,j^*}^{(t)}} \langle \mathbf{W}_{4,j^*,r',1}^{(t)}, e_{j^*} \rangle \right\}$$

Proof. By Lemma 5.B.1, we have

$$\begin{aligned} & \langle -\nabla_{\mathbf{W}_{4,j,r,2}^{(t)}} \text{Loss}^1, e_s \rangle \\ &= \frac{1}{2} \mathbb{E} \left[(1 - \mathbf{logit}_{4,j}^{(t)}) \mathbf{sReLU}'(\Lambda_{4,j,r}^{(t)}) \mathbb{1}_{\tau(x_1)=j, g_1=g} - \mathbf{logit}_{4,j}^{(t)} \mathbf{sReLU}'(\Lambda_{4,j,r}^{(t)}) \mathbb{1}_{\tau(x_1) \neq j, g_1=g} \right]. \end{aligned}$$

Clearly, the positive gradient can be upper-bounded by $O(\frac{1}{|\mathcal{G}|} \langle -\nabla_{\mathbf{W}_{4,j,r,1}^{(t)}} \text{Loss}^1, e_j \rangle)$. Moreover, for the negative gradient, by Claim 5.B.1, we have a naive bound

$$\mathbf{sReLU}'(\Lambda_{4,j,r}^{(t)})|_{\tau(x_1) \neq j, g_1=g} \leq O(1) \mathbf{sReLU}'(\Lambda_{4,j,r}^{(t)})|_{\tau(x_1)=j, g_1=g}.$$

When $t \leq T_1$, by Claim 5.B.2, we have $1 - \mathbf{logit}_{4,j}^{(t)}|_{j=\tau(x_1)} \geq \Omega(1)$ and $\mathbf{logit}_{4,j}^{(t)}|_{j \neq \tau(x_1)} \leq O(\frac{1}{d})$, which implies

$$\mathbb{E} \left[\mathbf{logit}_{4,j}^{(t)} \mathbf{sReLU}'(\Lambda_{4,j,r}^{(t)}) \mathbb{1}_{\tau(x_1) \neq j, g_1=g} \right] \leq O\left(\frac{1}{|\mathcal{G}|}\right) \langle -\nabla_{\mathbf{W}_{4,j,r,1}^{(t)}} \text{Loss}^1, e_j \rangle.$$

Therefore, for $t \leq T_1$, we have

$$|\langle \mathbf{W}_{4,j,r,2}^{(t)}, e_{\tau(g)} \rangle| \leq \tilde{O}(\sigma_0) + O\left(\frac{1}{|\mathcal{G}|}\right) \langle \mathbf{W}_{4,j,r,1}^{(t)}, e_j \rangle.$$

For $t \geq T_1$, notice that by Lemma 5.B.4, $\mathcal{A}_{4,j'}^{(t)} \neq \emptyset$ for $j' \in \tau(\mathcal{X})$, thus for $r' \in \mathcal{A}_{4,j^*}^{(t)}$

$$\mathbf{sReLU}'(\Lambda_{4,j,r}^{(t)})|_{\tau(x_1) \neq j, g_1=g} \leq \mathbf{sReLU}'(\Lambda_{4,j^*,r'}^{(t)})|_{\tau(x_1)=j^*, g_1=g}.$$

Furthermore, $\mathbf{logit}_{4,j}^{(t)}|_{j \neq \tau(x_1)} \leq O(\frac{1}{d})(1 - \mathbf{logit}_{4,j^*}^{(t)}|_{j^*=\tau(x_1)})$, which implies

$$\mathbb{E} \left[\mathbf{logit}_{4,j}^{(t)} \mathbf{sReLU}'(\Lambda_{4,j,r}^{(t)}) \mathbb{1}_{\tau(x_1) \neq j, g_1=g} \right] \leq O\left(\frac{1}{|\mathcal{G}|}\right) \langle -\nabla_{\mathbf{W}_{4,j^*,r',1}^{(t)}} \text{Loss}^1, e_{j^*} \rangle.$$

Due to the arbitrary of r' , we can conclude that

$$|\langle \mathbf{W}_{4,j,r,2}^{(t)}, e_{\tau(g)} \rangle| \leq \tilde{O}(\sigma_0) + O\left(\frac{1}{|\mathcal{G}|}\right) \min_{r' \in \mathcal{A}_{4,j^*}^{(t)}} \langle \mathbf{W}_{4,j^*,r',1}^{(t)}, e_{j^*} \rangle.$$

□

Lemma 5.B.7. Suppose Induction 5.B.1 holds for all iterations $< t$, then for any $j \in \tau(\mathcal{X})$ and $s = \tau(y)$, $y \in \mathcal{Y}$, we have

$$|\langle \mathbf{W}_{4,j,r,5}^{(t)}, e_{\tau(y)} \rangle| \leq \tilde{O}(\sigma_0) + O\left(\frac{1}{|\mathcal{Y}|}\right) \max \left\{ \langle \mathbf{W}_{4,j,r,1}^{(t)}, e_j \rangle, \min_{r' \in \mathcal{A}_{4,j^*}^{(t)}} \langle \mathbf{W}_{4,j^*,r',1}^{(t)}, e_{j^*} \rangle \right\}.$$

Proof. The proof is similar to Lemma 5.B.6, and we omit the details here. □

Off-diagonal Correlations Are Small

Lemma 5.B.8 (off-diagonal bound). *Given $j \in \tau(\mathcal{X})$, suppose Induction 5.B.1 holds at all iterations $< t$, for $s \in \tau(\mathcal{X}) \neq j$*

$$|\langle \mathbf{W}_{4,j,r,1}^{(t)}, e_s \rangle| \leq \tilde{O}(\sigma_0).$$

Proof. By Lemma 5.B.1

$$\langle -\nabla_{\mathbf{W}_{4,j,r,1}^{(t)}} \mathbb{L}^1, e_s \rangle = \frac{1}{2} \mathbb{E} \left[-\text{logit}_{4,j}^{(t)} \text{sReLU}'(\Lambda_{4,j,r}^{(t)}) \mathbb{1}_{\tau(x_1)=s} \right].$$

Notice that by Claim 5.B.1,

$$\text{sReLU}'(\Lambda_{4,j,r}^{(t)})|_{\tau(x_1)=s} \leq O(1) \text{sReLU}'(\Lambda_{4,s,r}^{(t)})|_{\tau(x_1)=s},$$

combined with Claim 5.B.2, $\text{logit}_{4,j} \leq O(\frac{1}{d})(1 - \text{logit}_{4,s}^{(t)})$ when $s = \tau(x_1)$, thus

$$\begin{aligned} \mathbb{E} \left[\text{logit}_{4,j}^{(t)} \text{sReLU}'(\Lambda_{4,j,r}^{(t)}) \mathbb{1}_{\tau(x_1)=s} \right] &\leq \mathbb{E} \left[O(\frac{1}{d})(1 - \text{logit}_{4,s}^{(t)}) \text{sReLU}'(\Lambda_{4,s,r}^{(t)}) \mathbb{1}_{\tau(x_1)=s} \right] \\ &\leq O(\frac{1}{d}) \langle -\nabla_{\mathbf{W}_{4,s,r,1}} \text{Loss}, e_s \rangle. \end{aligned}$$

From Induction 5.B.1, we have

$$|\langle \mathbf{W}_{4,j,r,1}^{(t)}, e_s \rangle| \leq O(\frac{1}{d}) |\langle \mathbf{W}_{4,s,r,1}^{(t)}, e_s \rangle| + \tilde{O}(\sigma_0) \leq \tilde{O}(\frac{1}{d}) + \tilde{O}(\sigma_0) = \tilde{O}(\sigma_0).$$

□

Lemma 5.B.9. *Given $j \in \tau(\mathcal{X})$, suppose Induction 5.B.1 holds at all iterations $< t$, we have*

$$|\langle \mathbf{W}_{4,j,r,p}^{(t)} \text{Loss}^1, e_s \rangle| \leq \tilde{O}(\sigma_0), \quad \text{for } p \in \{3, 4\} \text{ and all } s \in \tau(\mathcal{X})$$

Proof. When $s = j$, we have

$$\begin{aligned} \langle -\nabla_{\mathbf{W}_{4,j,r,p}} \text{Loss}^1, e_j \rangle &= \frac{1}{2} \mathbb{E} \left[-\text{logit}_{4,j}^{(t)} \text{sReLU}'(\Lambda_{4,j,r}^{(t)}) \mathbb{1}_{\tau(x_0)=j} \right] \\ &= \frac{1}{2} \mathbb{E} \left[-\text{logit}_{4,j}^{(t)} \text{sReLU}'(\Lambda_{4,j,r}^{(t)}) \sum_{s \neq j} \mathbb{1}_{\tau(x_0)=j, \tau(x_1)=s} \right]. \end{aligned}$$

Therefore, we can bound the above gradient in the similar way as the off-diagonal case, and obtain

$$|\langle \mathbf{W}_{4,j,r,p}^{(t)}, e_j \rangle| \leq O(\frac{1}{d}) \max_{s \in \tau(\mathcal{X})} |\langle \mathbf{W}_{4,s,r,1}^{(t)}, e_s \rangle| + \tilde{O}(\sigma_0) \leq \tilde{O}(\sigma_0).$$

When $s \neq j$,

$$\begin{aligned} & \langle -\nabla \mathbf{w}_{4,j,r,p} \text{Loss}^1, e_s \rangle \\ &= \frac{1}{2} \mathbb{E} \left[(1 - \text{logit}_{4,j}^{(t)}) \mathbf{sReLU}'(\Lambda_{4,j,r}^{(t)}) \mathbb{1}_{\tau(x_0)=s, \tau(x_1)=j} - \text{logit}_{4,j}^{(t)} \mathbf{sReLU}'(\Lambda_{4,j,r}^{(t)}) \mathbb{1}_{\tau(x_0)=s, j \notin \tau(\bar{X})} \right]. \end{aligned}$$

Noticing that $\{\tau(x_0) = s, \tau(x_1) = j\}$ happens with probability $\frac{1}{|\mathcal{X}|(|\mathcal{X}|-1)}$, thus the positive gradient can be upper bounded by $O(\frac{1}{d}) \cdot |\langle -\nabla \mathbf{w}_{4,j,r,1} \text{Loss}^1, e_j \rangle|$. Furthermore, the negative part can be upper bounded in the similar way as previous off-diagonal negative gradient. Putting it together, we complete the proof. $\square$

Non-target Correlations Are Negligible

Lemma 5.B.10. Suppose Induction 5.B.1 holds at all iterations $< t$, for $j' \notin \tau(\mathcal{X})$, for $p \in [5]$ and $s \in [d]$

$$|\langle \mathbf{W}_{4,j',r,p}^{(t)}, e_s \rangle| \leq \tilde{O}(\sigma_0).$$

Proof. By Lemma 5.B.2, $\langle \mathbf{W}_{4,j',r,p}^{(t)}, e_s \rangle$ for $p \in \{1, 3, 4\}$ and $s \in \tau(\mathcal{X})$ can be bounded in the similar way previous off-diagonal negative gradient.

We can observe that for $j' \notin \tau(\mathcal{X})$, all the non-zero gradient on the different directions are negative gradient, which implies $\langle \mathbf{W}_{4,j',r,p}^{(t)}, e_s \rangle \leq \langle \mathbf{W}_{4,j',r,p}^{(0)}, e_s \rangle = \tilde{O}(\sigma_0)$. Moreover, $\Lambda_{4,j',r}^{(t)} \leq \tilde{O}(\sigma_0)$ is also non-increasing.

For $s = \tau(g)$, $g \in \mathcal{G}$, whenever $\langle \mathbf{W}_{4,j',r,2}^{(t)}, e_s \rangle$ reaches -3μ , we have $\Lambda_{4,j',r}^{(t)}|_{g_1=g} \leq -3\mu + \frac{5}{2}\mu + \tilde{O}(\sigma_0) \leq 0$, and thus $\langle -\nabla_{\mathbf{W}_{4,j',r,2}^{(t)}} \text{Loss}^1, e_s \rangle = \frac{1}{2} \mathbb{E} \left[-\text{logit}_{4,j'}^{(t)} \mathbf{sReLU}'(\Lambda_{4,j',r}^{(t)}) \mathbb{1}_{g_1=g} \right] = 0$, which implies $\langle \mathbf{W}_{4,j',r,2}^{(t)}, e_s \rangle \geq -3\mu$. Hence, $|\langle \mathbf{W}_{4,j',r,2}^{(t)}, e_s \rangle| \leq \tilde{O}(\sigma_0)$. Following the similar argument, we can prove the result for $\langle \mathbf{W}_{4,j',r,5}^{(t)}, e_s \rangle$ for $s \in \tau(\mathcal{Y})$. $\square$

Convergence

Lemma 5.B.11. For $|\mathcal{G}| \geq |\mathcal{Y}| \geq \Omega(\frac{\log \log d}{\log \log \log d})$, $\text{polylog } d \geq m \geq |\mathcal{Y}|$, $\varrho \ll \frac{1}{m \log d}$ and sufficiently small $\eta \leq \frac{1}{\text{poly } d}$, Induction 5.B.1 holds for all iterations $t \leq T = \frac{\text{poly } d}{\eta}$.

Proof. Putting the results in Lemmas 5.B.5 to 5.B.8 and 5.B.10, we can directly establish the results in Induction 5.B.1. $\square$

Lemma 5.B.12 (Convergence). For sufficiently large $T_1 \leq t = \frac{\text{poly } d}{\eta}$, we have

- (a) Objective convergence: $\text{Loss}^1 \leq \frac{1}{\text{poly } d}$;
- (b) Successful learning of diagonal feature: $\Phi_{4,j}^{(t)} \geq \Omega(\log d)$ for any $j \in \tau(\mathcal{X})$.

Proof. Assuming for some sufficiently large constant $n > 0$, $\mathbb{E}[(1 - \mathbf{logit}_{4,j^*}^{(t)}) \mid \tau(x_1) = j^*] \geq \Omega(\frac{1}{d^n})$ for $t \in (T_1, T_1 + \frac{d^{n+1} \log^2 d}{\eta}]$ then by Lemma 5.B.1, we have

$$\Gamma_{4,j^{(*)}}^{(T_1 + \frac{d^2 \log^2 d}{\eta})} \geq \Omega\left(\frac{\eta}{d^{n+1}}\right) \cdot \frac{d^{n+1} \log^2 d}{\eta} + \Gamma_{4,j^{(*)}}^{(t)} \geq \Omega(\log^2 d),$$

which contradicts with $\Gamma_{4,j^{(*)}}^{(t)} \leq \Phi_{4,j^{(*)}}^{(t)} \leq O(\log^{1.5} d) = \tilde{O}(1)$ in the polynomial time. This implies after sufficiently large iteration t , we must have $\mathbb{E}[(1 - \mathbf{logit}_{4,j}^{(t)}) \mid \tau(x_1) = j] \leq O(\frac{1}{d^n})$ for $j \in \tau(X)$. Hence

$$\begin{aligned} \text{Loss}^1 &= \mathbb{E}[-\log \mathbf{logit}_{4,\tau(x_1)}^{(t)}] = \sum_{j \in \tau(X)} \mathbb{E}[-\log \mathbf{logit}_{4,j}^{(t)} \mathbb{1}_{\tau(x_1)=j}] \\ &\leq \sum_{j \in \tau(X)} \mathbb{E}[O(1)(1 - \mathbf{logit}_{4,j}^{(t)}) \mathbb{1}_{\tau(x_1)=j}] \quad (\mathbf{logit}_{4,j}^{(t)} \text{ is very close to } 1) \\ &\leq O\left(\frac{1}{\text{poly}d}\right). \end{aligned}$$

By Claim 5.B.2, at the time of convergence, we must have $\Phi_{4,j}^{(t)} \geq \Omega(\log d)$. $\square$

5.B.2 Learning Simply Transitive Actions

Preliminaries

Definition 5.B.1 (Combinations $\phi, \Phi, \Phi^\dagger$). Let $\mathcal{G}, \mathcal{Y}$ be defined as in Definition 5.2.2. For any pair $(g, y) \in \mathcal{G} \times \mathcal{Y}$, we call $\phi = (g, y)$ a **combination**. We write $\phi_1 = g$ and $\phi_2 = y$ to denote the corresponding components. We further define the set of **correct combinations** $\Phi^\star$ by

$$\Phi_j^\star = \{\phi = (g, y) \mid j = \tau(\alpha(g, y))\}, \forall j \in \tau(\mathcal{Y}); \quad \Phi = \bigcup_{j \in \tau(\mathcal{Y})} \Phi_j^\star \quad (5.B.3)$$

The combinations $\phi \in \Phi_j^\star$ are the ones that can be composed to get $\tilde{y} = \tau^{-1}(j) \in \mathcal{Y}$. We also define the set of **incorrect combinations** $\Phi_j^\dagger$ for each $j \in \tau(\mathcal{Y})$ to be

$$\Phi_j^\dagger = \{\phi \in \Phi \setminus \Phi_j^\star\} \quad (5.B.4)$$

For simply transitive action $\alpha : \mathcal{G} \times \mathcal{Y} \rightarrow \mathcal{Y}$, any combination $\phi \in \Phi_j^\dagger$ is a *incorrect* pair of g and y . That is, it holds that any $y' \neq y \in \mathcal{Y}$ and $g' \neq g \in \mathcal{G}$ satisfy $\alpha(g, y') = \alpha(y, g') \neq \tau^{-1}(j)$.

For each combination, we define the following neural-features:

Definition 5.B.2 (combination features $\psi, \Psi, \Psi^*, \Psi^\dagger$). Given a combination $\phi = (g, y) \in \Phi$, token index $j \in \tau(\mathcal{Y})$ and neuron index $r \in [m]$, we define

$$V_{j,r}(g) = \langle \mathbf{W}_{5,j,r,2}, e_g \rangle, \quad V_{j,r}(y) = \langle \mathbf{W}_{5,j,r,5}, e_y \rangle, \quad V_{j,r}(\phi) := \frac{1}{2}(V_{j,r}(g) + V_{j,r}(y))$$

and we call $V_{j,r}(\phi)$ the *features for combination* ϕ in neuron $(j, r) \in \tau(\mathcal{Y}) \times [m]$. We write $\psi = (j, r, \phi)$ and $V_\psi \equiv V_{j,r}(\phi)$ to make the notation concise. Similar to above, we further define

$$\Psi := \{\psi = (j, r, \phi) \mid \phi \in \Phi\}$$

and Ψ^* that contains the desirable features for each class $j \in \tau(\mathcal{Y})$:

$$\Psi_{j,r}^* = \{\psi = (j, r, \phi) \mid \phi \in \Phi_j^*\}, \quad \Psi^* = \bigcup_{(j,r) \in \tau(\mathcal{Y}) \times [m]} \Psi_{j,r}^* \quad (5.B.5)$$

and $\Psi^\dagger$ that contains the incorrect feature combinations:

$$\Psi_{j,r}^\dagger = \{\psi = (j, r, \phi) \mid \phi \in \Phi \setminus \Phi_j^*\}, \quad \Psi^\dagger = \bigcup_{(j,r) \in \tau(\mathcal{Y}) \times [m]} \widehat{\Psi}_{j,r} \quad (5.B.6)$$

Events of combination appearance. Let us first define some useful notations. For a combination $\phi = (g, y)$, we write $\mathcal{H}_\phi$ to denote the event when ϕ appears in the sequence $\mathbf{Z}^1$:

$$\mathcal{H}_\phi := \{(g_1, y_0) = \phi\}, \quad \mathcal{H}_{\phi_1} = \mathcal{H}_g = \{g_1 = g\}, \quad \mathcal{H}_{\phi_2} = \mathcal{H}_y = \{y_0 = y\}.$$

Further, we write

$$\mathcal{H}_{\phi,1}^\dagger = \{g_1 \neq g, y_0 = y\}, \quad \mathcal{H}_{\phi,2}^\dagger = \{g_1 = g, y_0 \neq y\}, \quad \mathcal{H}_\phi^\dagger = \mathcal{H}_{\phi,1}^\dagger \cup \mathcal{H}_{\phi,2}^\dagger$$

to denote the event where ϕ did not appear but its group element or value is the same.

Finally we define a notion called *learning curriculum*, that sits at the center of our proof.

Definition 5.B.3 (Learning curriculum). We define an order on the set Σ , defined by the following process: Let $\Sigma_0 = \Psi$. At each $i \in [n_y^2]$, we choose

$$\psi_i = \arg \max_{\psi \in \Sigma_i} V_\psi^{(0)}$$

Let's write $\psi_i = (j, r, \phi)$ where $\phi = (g, y)$, then we define Σ_{i+1} by excluding the following features:

1. Exclude the confusing combinations in neuron (j, r) ;

$$\Sigma_{\psi_i}^{\dagger,1} \equiv \Sigma_i^{\dagger,1} = \{\psi' = (j, r, \phi') \in \Sigma_i, \phi' = (g', y') \mid (g' = g) \text{ XOR } (y' = y)\}$$

2. Exclude the unselected combinations in neuron (j, r) ;

$$\Sigma_{\psi_i}^{\dagger,2} \equiv \Sigma_i^{\dagger,2} = \{\psi' = (j, r, \phi') \in \Sigma_i, \phi' = (g', y') \mid g' \neq g, y' \neq y\}$$

3. Exclude the feature indices of the same combination in other neurons $(j, r'), r' \neq r$;

$$\Sigma_{\psi_i}^{\dagger,3} \equiv \Sigma_i^{\dagger,3} = \{\psi' = (j, r', \phi) \in \Sigma_i, \forall r' \neq r \in [m]\}.$$

Which returns $\Sigma_{i+1} = \Sigma_i \setminus \{\psi = (j, r', \phi') \in \Sigma_i \mid \text{either } r' = r \text{ or } \phi' = \phi\}$. Iterate over the whole set Ψ , we will arrive at $\Sigma_{n_y}^2 = \emptyset$. This gives rise to an ordered sequence $\Sigma^\star$ and a set $\Sigma^\dagger$:

$$\Sigma^\star := (\psi_1, \psi_2, \dots, \psi_{n_y^2}); \quad \Sigma^\dagger := \bigcup_{\psi \in \Sigma^\star} \Sigma_\psi^\dagger, \quad \Sigma_\psi^\dagger := \Sigma_\psi^{\dagger,1} \cup \Sigma_\psi^{\dagger,2} \cup \Sigma_\psi^{\dagger,3}$$

By our construction $V_{\psi_i}^{(0)} \geq V_{\psi_{i+1}}^{(0)}$. We write $\psi <_\Sigma \psi'$ to denote that ψ is ahead of ψ' in $\Sigma^\star$.

Intuitively, $\Sigma^\star$ encodes the order at which the features V_ψ grow in magnitude and $\Sigma^\dagger$ contains all the unlearned features during the process.

Throughout the analysis of the FFN layer, we further make the following assumptions.

Assumption 5.B.1. For $(j, r, p) \in [d] \times [m] \times [5]$, we fix $\mathbf{W}_{5,j,r,p}^{(t)} \equiv \mathbf{W}_{5,j,r,p}^{(0)}$ for $p \in \{1, 3, 4\}$ at initialization for simplicity of proof.

Assumption 5.B.2. We use a modified smoothed ReLU as our activation function. This technical assumption is to avoid many pathologies appearing in the learning dynamics.

$$\text{sReLU}(x) := \begin{cases} \omega B & x \leq -B \\ -\omega x & x \in (-B, -\omega] \\ \frac{1}{2}x^2 & x \in (-\omega, 0]; \\ x^q / (\varrho^{q-1}q) & x \in (0, \varrho]; \\ x - \varrho \left(1 - \frac{1}{q}\right) & x \in (\varrho, B] \\ B - \varrho \left(1 - \frac{1}{q}\right) & x > B \end{cases}$$

where $q = O(1)$ is a large even integer and $\varrho = \Theta(1/\text{polylog}(d))$, $\omega \in (d^2\mu^{q-1}, \frac{\lambda}{d^{q/3}})$, $B = C \log d$, $C = \Theta(1) \in (5, \frac{q-1}{3})$ and $\lambda = \frac{d-1}{d-1+e^B}$.

Theorem Statement We will try to prove the following theorem:

Theorem 5.B.1 (Learning Simply Transitive Actions). *Suppose $F^{(t)}$ is returned by Algorithm 2 at $t = T_1$, and let $\delta_1 = d^{c_1}\mu$, $\delta_2 = \omega^{\frac{1}{q-1}}$, it holds that with probability $\geq 1 - o(1)$:*

- A. For all $\psi \in \Sigma^*$, we have $V_\psi^{(T_1)} = B \pm O(\delta_1)$;
- B. For any $\psi \in \Sigma^{+,1}$, $|V_\psi^{(T_1)}| \leq \tilde{O}(\delta_2)$;
- C. For any $\psi \in \Sigma^{+,2}$, $V_\psi^{(T_1)} \leq -B + O(\delta_1)$;
- D. For any $\psi \in \Sigma^{+,3}$, $V_\psi^{(T_1)} \leq O(\delta_1)$;
- E. For $\psi = (j, r, (g, y)) \in \Sigma^*$, $|V_{j,r}^{(t)}(g) - V_{j,r}^{(t)}(y)| \leq \tilde{O}(\delta_2)$.

In fact, we can decompose the statement of Theorem 5.B.1 into the following statement for every feature $\psi \in \Sigma^*$.

Definition 5.B.4 (Feature Shape). Let $\delta = (\delta_1, \delta_2)$ be a tuple of error parameters. Let $\psi \in \Sigma^*$ be a feature in the learning curriculum. We say the feature ψ reached *feature shape* $\mathcal{F}_\psi(\delta)$ with error δ if:

1. $\mathcal{F}_{\psi,1}(\delta_1)$: $V_\psi^{(t)} \geq B - O(\delta_1)$;
2. $\mathcal{F}_{\psi,2}(\delta_2)$: For any $\psi' \in \Sigma_\psi^{+,1}$, it holds that $|V_{\psi'}^{(t)}| \leq \tilde{O}(\delta_2)$;
3. $\mathcal{F}_{\psi,3}(\delta_1)$: For any $\psi' \in \Sigma_\psi^{+,2}$, it holds that $V_{\psi'}^{(t)} \leq -B \pm O(\delta_1)$;
4. $\mathcal{F}_{\psi,4}(\delta_1)$: For any feature $\psi' \in \Sigma_\psi^{+,3}$, it holds that $V_{\psi'}^{(t)} \leq O(\delta_1)$.
5. $\mathcal{F}_{\psi,5}(\delta_2)$: Writing $\psi = (j, r, (g, y))$, then it holds that $|V_{j,r}^{(t)}(g) - V_{j,r}^{(t)}(y)| \leq \tilde{O}(\delta_2)$;

Obviously if we proved the condition $\mathcal{F}_\psi(\delta)$ of all $\psi \in \Sigma^*$ are reached for $\delta = (d^{c_1}\mu, \omega^{\frac{1}{q-1}})$ at T_1 , then Theorem 5.B.1 is proven. We do this by following a induction process sequentially for each $\psi \in \Sigma^*$, and provide a guarantee that all features in Σ^* reach the claimed convergence condition together at $t = T_1$.

Induction Hypotheses and Phase Decomposition To prove Theorem 5.B.1, or equivalently, to prove the convergence condition in Definition 5.B.4 is reached for every feature $\psi \in \Sigma^*$, we shall characterize the dynamics of each feature and prove that they eventually arrive at the desired shape.

Phase Decomposition. Let us define the following timestamps for different phases of learning each $\psi = (j, r, \phi) \in \Psi \cup \widehat{\Psi}$. Let $c_1 = \frac{1}{1000}$ be a small constant:

- (a) Phase I: $t \in [0, T_\psi^1]$, where $T_\psi^1 := \min\{t \geq 0 \mid V_\psi^{(t)} \geq d^{c_1} \mu\}$;
- (b) Phase II.1: $t \in (T_\psi^1, T_\psi^{2,1}]$, where $T_\psi^{2,1} := \min\{t \geq 0 \mid V_\psi^{(t)} \geq \frac{1}{2} \log d\}$;
- (c) Phase II.2: $t \in (T_\psi^{2,1}, T_\psi^{2,2}]$, where $T_\psi^{2,2} := \min\{t \geq 0 \mid \mathbb{E}[\mathbf{logit}_{5,j}^{(t)} \mathbb{1}_{\mathcal{H}_\phi}] \geq 1 - \frac{1}{\sqrt{d}}\}$;
- (d) Phase II.3: $t \in (T_\psi^{2,2}, T_\psi^2]$, where $T_\psi^2 = \max\{T_\psi^{2,3}, T_\psi^{2,2} + \Omega(\frac{1}{\eta(d^{c_1}\mu)^{q-2}})\}$ and $T_\psi^{2,3}$ is defined as:

$$T_\psi^{2,3} := \min \left\{ t \geq T_\psi^{2,2} \mid \mathcal{F}_\psi(\delta_1, \delta_2) \text{ holds, where } \delta_1 = d^{c_1} \sigma_0, \delta_2 = \left(\frac{n_y^2 d \omega}{\lambda} \right)^{\frac{1}{q-1}} \right\}$$

- (e) Phase III.1: $t \in (T_\psi^2, T_{1,1}]$, where $T_{1,1} = T_{\psi_{n_y^2}}^2$ and $\psi_{n_y^2}$ is the last feature in Σ^* . This is the convergence phase where the rest of the features are learned and the feature converge to a perfect shape.
- (f) Phase III.2: $t \in (T_{1,1}, T_1]$, the end phase where all features are in perfect shape and have stablized.

Now we state the following induction hypotheses which naturally results in Theorem 5.B.1.

Induction 5.B.2 (Simply Transitive Actions, All Phases). *For all $t \leq T_1$, the following holds:*

- (a) $V_\psi^{(t)} + b_{i,j,r} \geq \Omega(\mu)$.
- (b) $T_{\psi'}^2 < T_\psi^1$ if $\psi' <_\Sigma \psi$ for $\psi, \psi' \in \Sigma^*$, i.e., the intervals $\{[T_\psi^1, T_\psi^2]\}_{\psi \in \Sigma^*}$ are non-overlapping;

In order to prove Induction 5.B.2 and Theorem 5.B.1, we introduce the following feature based induction hypothesis to prove that $\mathcal{F}_\psi$ defined in Definition 5.B.4 holds for some error parameters at $t = T_\psi^2$.

Induction 5.B.3 (Induction for Individual Feature). *Let $\psi = (j, r, \phi) \in \Sigma^*$, $\phi = (g, y)$, at $t \leq T_1$, the following holds:*

- (a) At $t \leq T_\psi^1$, any $g \neq g', y' \neq y$ has $V_{j,r}(g'), V_{j,r}(y') \leq \widetilde{O}(\mu)$.
- (b) During $t \leq T_\psi^1$, there are at most $\widetilde{O}(\sqrt{d}/\eta)$ iterations where $\mathbf{logit}^{(t)} \geq \frac{1}{\sqrt{d}}$ conditioned on $\mathcal{H}_\phi$.
- (c) Let $\psi' < \psi \in \Sigma^*$, the feature shape $\mathcal{F}_{\psi'}(d^{c_1} \sigma_0, (\frac{n_y^2 d \omega}{\lambda})^{\frac{1}{q-1}})$ holds throughout $t \in [T_\psi^1, T_1]$;
- (d) Let $\psi' > \psi \in \Sigma^*$, then $V_{\psi'}^{(t)} \leq \widetilde{O}(\sigma_0)$ throughout $t \in [0, T_\psi^2]$.

Technical Calculations Let us recall some facts about gradient computation for $\mathbf{W}$.

Fact 5.B.2 (gradient computation). The gradient with respect to $\mathbf{W}_{i,j,r,p}$ for $i = 5, j \in [d], r \in [m], p \in [5]$ and $v \in \mathcal{V}$ is

$$\langle -\nabla_{\mathbf{W}_{5,j,r,p}} \text{Loss}, e_v \rangle = \frac{1}{2} \mathbb{E} \left[\mathcal{E}_{5,j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \sum_{\mathbf{k} \in \mathcal{I}^{1,0}} \langle \mathbf{Z}_{\mathbf{k},p}, e_v \rangle \right] \quad (5.B.7)$$

where $\mathcal{E}_{5,j} = \mathbb{1}_{v=Z_{\text{ans},1,p}} - \mathbf{logit}_{5,j}$ is the loss derivatives. We compute more precise expressions when j, p and v varies:

- (a) when $p = 2$, for $g \in \mathcal{G}$, let $\phi = (g, y) \in \Phi_j^\star$ be the combination with $\phi_1 = g$, we have for all $j \in \tau(\mathcal{Y}), r \in [m]$,

$$\begin{aligned} \langle -\nabla_{\mathbf{w}_{5,j,r,2}} \text{Loss}, e_g \rangle &= \frac{1}{2} \mathbb{E} \left[(1 - \text{logit}_{5,j}) \text{sReLU}'(\Lambda_{5,j,r}) \mathbb{1}_{\mathcal{H}_\phi} \right] \\ &\quad - \frac{1}{2} \mathbb{E} \left[\text{logit}_{5,j} \text{sReLU}'(\Lambda_{5,j,r}) \mathbb{1}_{\mathcal{H}_{\phi,2}^+} \right] \end{aligned} \quad (5.B.8)$$

- (b) when $p = 5$, for $y \in \mathcal{Y}$, let $\phi = (g, y) \in \Phi_j^\star$ be the combination with $\phi_2 = y$, we have for all $j \in \tau(\mathcal{Y}), r \in [m]$,

$$\begin{aligned} \langle -\nabla_{\mathbf{w}_{5,j,r,5}} \text{Loss}, e_y \rangle &= \frac{1}{2} \mathbb{E} \left[(1 - \text{logit}_{5,j}) \text{sReLU}'(\Lambda_{5,j,r}) \mathbb{1}_{\mathcal{H}_\phi} \right] \\ &\quad - \frac{1}{2} \mathbb{E} \left[\text{logit}_{5,j} \text{sReLU}'(\Lambda_{5,j,r}) \mathbb{1}_{\mathcal{H}_{\phi,1}^+} \right] \end{aligned} \quad (5.B.9)$$

- (c) For any combination of p, v not included above, we have due to our assumptions of the distribution and update rule:

$$\langle -\nabla_{\mathbf{w}_{5,j,r,p}} \text{Loss}, e_v \rangle \equiv 0$$

Definition 5.B.5 (Γ -notations). To simplify gradient expression, we define the following notation: let $\psi = (j, r, \phi) \in \Psi$, we write

$$\begin{aligned} \Gamma_{j,r,\phi}^{+, (t)} &= \Gamma_{\psi}^{+, (t)} := \mathbb{E} \left[(1 - \text{logit}_{5,j}) \text{sReLU}'(\Lambda_{5,j,r}) \mathbb{1}_{\mathcal{H}_\phi} \right] \\ \Gamma_{j,r,\phi}^{-, (t)} &= \Gamma_{\psi}^{-, (t)} := \mathbb{E} \left[\text{logit}_{5,j} \text{sReLU}'(\Lambda_{5,j,r}) \mathbb{1}_{\mathcal{H}_\phi^+} \right] \end{aligned} \quad (5.B.10)$$

we can further define Γ for each $g \in \mathcal{G}$ and $y \in \mathcal{Y}$: Let $v \in \mathcal{G} \cup \mathcal{Y}$, we define

$$\begin{aligned} \Gamma_{j,r,v}^{+, (t)} &= \Gamma_{\psi,v}^{+, (t)} := \mathbb{E} \left[(1 - \text{logit}_{5,j}) \text{sReLU}'(\Lambda_{5,j,r}) \mathbb{1}_{\mathcal{H}_v} \right] \\ \Gamma_{j,r,v}^{-, (t)} &= \Gamma_{\psi,v}^{-, (t)} := \mathbb{E} \left[\text{logit}_{5,j} \text{sReLU}'(\Lambda_{5,j,r}) \mathbb{1}_{\mathcal{H}_v^+} \right] \end{aligned} \quad (5.B.11)$$

This allows us to define the following gradient condition:

Definition 5.B.6 (gradient condition). At $t \leq T_1$, letting $\psi = (j, r, \phi) \in \Psi$, we write $C_\psi(\delta)$ for $\delta > 0$ to denote that the gradient update for V_ψ is smaller than δ , formally:

$$\begin{aligned} C_{j,r,v}(\delta) \text{ holds} &\implies |\Gamma_{j,r,v}^{+, (t)} - \Gamma_{j,r,v}^{-, (t)}| \leq \delta \\ C_{j,r,v}^+(\delta) \text{ holds} &\implies |\Gamma_{j,r,v}^{+, (t)}| \leq \delta \\ C_{j,r,v}^-(\delta) \text{ holds} &\implies |\Gamma_{j,r,v}^{-, (t)}| \leq \delta \end{aligned}$$

We further write

$$\begin{aligned} C_\psi(\delta) \text{ holds} &\implies C_{j,r,g}(\delta) \wedge C_{j,r,y}(\delta) \text{ holds} \\ C_\psi^+(\delta) \text{ holds} &\implies C_{j,r,g}^+(\delta) \wedge C_{j,r,y}^+(\delta) \text{ holds} \\ C_\psi^-(\delta) \text{ holds} &\implies C_{j,r,g}^-(\delta) \wedge C_{j,r,y}^-(\delta) \text{ holds} \end{aligned}$$

Now we proceed to characterize the lower bound of logits

Fact 5.B.3 (logit lower bound). We list some basic facts about the architecture Definition 5.3.5 here.

1. For any $Z^1 \in \text{supp}(\mathcal{D}^1)$, we have

$$\lambda := \min_{i,j,F,Z} (1 - \text{logit}_{i,j}(F, Z)) = \frac{d-1}{d-1+e^B}$$

2. Suppose for some $i \in [5]$, $F_{i,j} = o(1)$ for all $j \notin \tau(\mathcal{Y})$, then

$$\min_{i,j,F,Z} \text{logit}_{i,j}(F, Z) = \frac{1}{(1+o(1))d + n_y e^B} = O\left(\frac{\lambda}{n_y d}\right)$$

Phase I: Feature Emergence and Competition

We shall prove the following properties at initialization.

Fact 5.B.4 (initialization range). At $t = 0$, the following holds with probability $\geq 1 - o(1)$:

- (a) $|\langle \mathbf{W}_{5,j,r,p}^{(0)}, e_v \rangle| \leq O(\sigma_0 \sqrt{\log d})$ For all j, r, p and $v \in \mathcal{V}$;
- (b) $V_\psi^{(0)} \geq V_{\psi'}^{(0)} + \gamma$, where $\gamma = \Omega(\frac{\sigma_0}{n_y^4 m^2 \log d})$, for any pair $\psi < \psi' \in \Sigma^*$ in Def. 5.B.3;

Proof. Fact 5.B.4a can easily verified from our initialization $\mathbf{W}_{i,j,r,p}^{(0)} \sim \mathcal{N}(0, \sigma_0 I_d)$. For Fact 5.B.4b, we give a straightforward proof that every pair of $V_\psi^{(0)}, \psi \in \Sigma^*$ has a gap of $\frac{\sigma_0}{n_y^4 m^2 \log d}$. First note that $V_\psi^{(0)}$ of different $\psi \in \Sigma^*$ are independent and identically distributed on the randomness of $\mathbf{W}^{(0)}$, due to the orthogonality of embeddings $e_v, v \in \mathcal{V}$. Then, by the basic property of a Gaussian variable (notice that $V_\psi^{(0)} - V_{\psi'}^{(0)}$ is also Gaussian with variance $2\sigma_0$) (all though different pairs could be dependent), we have with probability $1 - \frac{1}{n_y^4 m^2 \log d}$ that their gap is at least $\gtrsim \frac{\sigma_0}{n_y^4 m^2 \log d}$ for each pair. Then by a union bound over $O(m^2 n_y^4)$ -many all possible pairs we can conclude the proof. $\square$

We give another characterization of the initial activations.

Fact 5.B.5 (activation magnitude). At $t = 0$, with high probability it holds that

$$\Lambda_{5,j,r}^{(0)} = (1 + O(\frac{1}{\sqrt{\log d}}))\mu = \Theta(\sigma \log d) \ll \varrho$$

and thus $\mathbf{sReLU}(\Lambda_{5,j,r}) = \Theta(\frac{1}{q}\mu^q)$, $\mathbf{sReLU}'(\Lambda_{5,j,r}) = \Theta(\mu^{q-1})$ and $F_{5,j} = \Theta(\frac{m}{q}\mu^q)$.

Proof. Combining Definitions 5.3.4 and 5.3.5, Assumption 5.3.3, and Fact 5.B.4 gives the fact. $\square$

We establish some properties in Phase I.

Lemma 5.B.13 (Key properties in Phase I). *Let $\psi = (j, r, (g, y)) \in \Sigma^\star$, and assume Inductions 5.B.2 and 5.B.3 holds at $t \leq T_\psi^1$, then*

- (a) $|V_{j,r}^{(t)}(v)| \leq O(\mu/\sqrt{\log d})$ if $v \neq g, v \neq y$;
- (b) $\Lambda_{5,j,r}^{(t)}(\mathbf{Z}) \geq 0$ for $\mathbf{Z} \in \mathcal{H}_\phi$;
- (c) $\mathbf{logit}_{5,j} = O(1/d)$ whenever $\mathcal{H}_\phi^\dagger$ happens.

Proof. Lemma 5.B.13a is from both the initialization of weights Fact 5.B.4 and Induction 5.B.3a. Lemma 5.B.13b is from Lemma 5.B.13a and Induction 5.B.2a. Lemma 5.B.13c is due to both Induction 5.B.2b and Induction 5.B.2b. $\square$

Competition between feature combinations. In this phase we compare the growth of different features using a proxy.

Lemma 5.B.14 (Approximating gradient with proxy). *Assume Induction 5.B.2. For a fixed $\psi = (j, r, \phi) \in \Sigma^\star$, define the auxiliary sequence*

$$\tilde{V}_\psi^{(0)} := V_\psi^{(0)}, \quad \tilde{V}_\psi^{(t+1)} := \tilde{V}_\psi^{(t)} + \eta \mathbb{E}[\mathbf{sReLU}'(\tilde{\Lambda}_{5,j,r}^{(t)}) \mathbf{1}_{\mathcal{H}_\phi}],$$

where in $\tilde{\Lambda}_{5,j,r}^{(t)}$ the weight V_ψ is replaced by $\tilde{V}_\psi$ (all other weights are unchanged). Then for any feature $\psi \succ \psi_i$ or $\psi \in \Sigma_i \setminus \{\psi_i\}$ and all $t \leq T_\psi^1$, $|V_\psi^{(t)} - \tilde{V}_\psi^{(t)}| \leq O(\tilde{V}_\psi^{(t)} / d^{1/2-3\epsilon_1})$.

Proof. We need to prove the result by induction. Firstly we notice that the newly defined proxy sequence $\tilde{V}_\psi^{(t)}$ is monotonically increasing. Moreover, as $\tilde{V}_\psi^{(t)}$ is initialized the same as $V_\psi^{(t)}$, we can use Lemmas 5.B.23 and 5.B.24 to argue its growth by taking

$$x_t = \tilde{V}_\psi^{(t)} + b_{i,j,r}, \quad \xi = \sum_{p=3,4} \langle \mathbf{W}_{5,j,r,p}, e_{x_0} \rangle + \langle \mathbf{W}_{5,j,r,1}, e_{x_1} \rangle, \quad x_{t+1} \geq x_t + \eta \frac{1}{n_y^2} \mathbb{E}[(x_t + \mu)^{q-1}]$$

And Lemmas 3.A.1 and 5.B.24 guarantee $\tilde{V}_\psi^{(t)} \geq d^{c_1}\mu$ after $\tilde{O}(\frac{1}{\eta\mu^{q-2}})$ steps. Now assume the bound of difference between $V_\psi^{(t)}$ and $\tilde{V}_\psi^{(t)}$ holds at a $t \leq T_\psi^1$. Since $t \leq T_\psi^1$, we have $V_\psi^{(t)} \leq d^{c_1}\sigma_0$ by definition of T_ψ^1 . By Fact 5.B.2, the exact update of V_ψ is

$$V_\psi^{(t+1)} = V_\psi^{(t)} + \eta \left(\mathbb{E}[(1 - \mathbf{logit}_{5,j}^{(t)}) \mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)}) \mathbf{1}_{\mathcal{H}_\phi}] - \frac{1}{2} \mathbb{E}[\mathbf{logit}_{5,j}^{(t)} \mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)}) \mathbf{1}_{\mathcal{H}_\phi^\dagger}] \right).$$

Compare the above with the definition of $\tilde{V}_\psi^{(t)}$, we first notice that $\tilde{V}_\psi^{(t)}$ is dominating $V_\psi^{(t)}$ for all iterations $t \leq T_\psi^1$. Then, we can bound the difference between the two sequences by

$$\begin{aligned} & |\tilde{V}_\psi^{(t+1)} - V_\psi^{(t+1)}| \\ & \leq |\tilde{V}_\psi^{(t)} - V_\psi^{(t)}| + \eta \left| \mathbb{E}[\mathbf{sReLU}'(\tilde{\Lambda}_{5,j,r}^{(t)}) - \mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)}) \mathbb{1}_{\mathcal{H}_\phi}] \right| \\ & \quad + \eta \left| \mathbb{E}[\mathbf{logit}_{5,j}^{(t)} \mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)}) \mathbb{1}_{\mathcal{H}_\phi}] \right| + \frac{\eta}{2} \left| \mathbb{E}[\mathbf{logit}_{5,j}^{(t)} \mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)}) \mathbb{1}_{\mathcal{H}_\phi^c}] \right| \end{aligned}$$

We bound the three error terms for $t \leq T_\psi^1$:

- A. Activation perturbation: by the smoothness of $\mathbf{sReLU}'$ on $[0, \varrho]$ and that both pre-activations are in this range during Phase I, we have

$$\left| \mathbb{E}[(\mathbf{sReLU}'(\tilde{\Lambda}_{5,j,r}^{(t)}) - \mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)})) \mathbb{1}_{\mathcal{H}_\phi}] \right| \leq |\tilde{V}_\psi^{(t)} - V_\psi^{(t)}| \mathbb{E}[(\tilde{\Lambda}_{5,j,r}^{(t)})^{q-2} \mathbb{1}_{\mathcal{H}_\phi}]$$

where we have used the fact that $\tilde{\Lambda}_{5,j,r} \geq \Lambda_{5,j,r}$ because of the dominance of $\tilde{V}_\psi$.

- B. Outside a set of iterations $\mathcal{A}_\phi = \{t \leq T_\psi^1 \mid \mathbf{logit}_{5,j}^{(t)} \geq \frac{1}{\sqrt{d}} \text{ conditioned on } \mathcal{H}_\phi\}$ whose cardinality is no more than $\tilde{O}(d^{\frac{1}{2}+c_1}/\eta)$ by Induction 5.B.3a, we have $\mathbf{logit}_{5,j}^{(t)} \leq d^{-1/2}$. Hence for $t \notin \mathcal{A}_\phi$

$$\left| \mathbb{E}[\mathbf{logit}_{5,j}^{(t)} \mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)}) \mathbb{1}_{\mathcal{H}_\phi}] \right| \leq O(d^{-1/2}) \mathbb{E}[\mathbf{sReLU}'(\tilde{\Lambda}_{5,j,r}^{(t)}) \mathbb{1}_{\mathcal{H}_\phi}]$$

Now we can sum over the iterations to get

$$\begin{aligned} & \left(\sum_{s \leq t, s \in \mathcal{A}_\phi} + \sum_{s \leq t, s \notin \mathcal{A}_\phi} \right) \eta \left| \mathbb{E}[\mathbf{logit}_{5,j}^{(s)} \mathbf{sReLU}'(\Lambda_{5,j,r}^{(s)}) \mathbb{1}_{\mathcal{H}_\phi}] \right| \\ & \leq O\left(\frac{d^{\frac{1}{2}+c_1}}{\eta}\right) \cdot \eta (d^{c_1} \mu)^{q-1} + O(d^{-1/2})(\tilde{V}_\psi^{(t)} - \tilde{V}_\psi^{(0)}) \\ & \leq O(\mu/d^{1/2-c_1}) \end{aligned}$$

- C. Confusing events: before $V_\psi^{(t)} \geq \frac{1}{2} \log d$, we have $\mathbf{logit}_{5,j} = O(1/d)$ in Phase I by Lemma 5.B.13. So we get

$$\sum_{s \leq t} \eta \mathbb{E}[\mathbf{logit}_{5,j}^{(s)} |\mathbf{sReLU}'(\Lambda_{5,j,r}^{(s)})| \mathbb{1}_{\mathcal{H}_\phi^c}] \leq O\left(\frac{1}{\eta \mu^{q-2}}\right) \eta \left(\frac{(d^{c_1} \mu)^{q-1}}{d}\right) \ll O(\mu/\sqrt{d})$$

So we have that the accumulated errors from the last two terms are smaller than $O(\mu/d^{1/2-c_1})$. This allow us to bound the difference by using the more naive approach: we can assume the difference started with $O(\mu/d^{1/2-c_1})$ and now we only need to bound the following sequence for

$$t \leq T_\psi^1$$

$$\delta_{t+1} = \delta_t + O(\eta \delta_t) \mathbb{E} \left[\frac{1}{\tilde{\Lambda}_{5,j,r}^{(t)}} \mathbf{sReLU}'(\tilde{\Lambda}_{5,j,r}^{(t)}) \mathbb{1}_{\mathcal{H}_\phi} \right], \quad \delta_0 \leq O(\mu/d^{1/2-c_1})$$

Now let $\delta' > \eta$ be some small parameter, we can do some slicing of time $t_0, t_1, \dots, t_n$ where $t_i = \min\{t \geq 0 \mid \tilde{V}_\psi^{(t)} \geq (1 + \delta')^i \mu\}$ and $(1 + \delta')^n \mu \leq \tilde{V}_\psi^{(t_{i+1})} \leq (1 + \delta')^{n+1} \mu$. This produce the following bound:

$$\begin{aligned} & \log \delta_{t+1} - \log \delta_0 \\ &= \sum_{1 \leq i \leq n} \sum_{s \in [t_i, t_{i+1}]} \log \left(1 + O(\eta) \mathbb{E} \left[\frac{1}{\tilde{\Lambda}_{5,j,r}^{(s)}} \mathbf{sReLU}'(\tilde{\Lambda}_{5,j,r}^{(s)}) \mathbb{1}_{\mathcal{H}_\phi} \right] \right) \\ &\leq \sum_{1 \leq i \leq n} \sum_{s \in [t_i, t_{i+1}]} O(\eta) \mathbb{E} \left[\frac{1}{\tilde{\Lambda}_{5,j,r}^{(s)}} \mathbf{sReLU}'(\tilde{\Lambda}_{5,j,r}^{(s)}) \mathbb{1}_{\mathcal{H}_\phi} \right] \quad (\log(1+x) \leq x \text{ when } x > 0) \\ &\leq \sum_{1 \leq i \leq n} O \left(\frac{1}{\min_{s \in [t_i, t_{i+1}], \mathbf{Z} \in \mathcal{H}_\phi} \tilde{\Lambda}_{5,j,r}^{(s)}(\mathbf{Z})} \right) \sum_{s \in [t_i, t_{i+1}]} \eta \left| \mathbb{E}[\mathbf{sReLU}'(\tilde{\Lambda}_{5,j,r}^{(s)}) \mathbb{1}_{\mathcal{H}_\phi}] \right| \\ &\leq \sum_{1 \leq i \leq n} O \left(\frac{1}{(1 + \delta')^i \mu} \right) (\tilde{V}_\psi^{(t_{i+1})} - \tilde{V}_\psi^{(t_i)}) \\ &\leq \frac{\delta' \log(\tilde{V}_\psi^{(t)} / \mu)}{\log(1 + \delta')} \\ &\leq 2 \log(\tilde{V}_\psi^{(t)} / \mu) \quad (\log(1 + \delta')^{\frac{1}{\delta'}} < 2 \text{ if } \delta' \text{ is small enough}) \end{aligned}$$

So when $\tilde{V}_\psi^{(t)} \in [d^{c_1} \mu, 2d^{c_1} \mu]$, it holds that $\delta_t \lesssim d^{2c_1} \delta_0 \lesssim \mu/d^{1/2-3c_1}$. This proves the claim. $\square$

Lemma 5.B.15 (competition). *Assuming Induction 5.B.2, and let $\psi \in \Sigma^\star$, we shall have for any feature $\psi' > \psi \in \Sigma^\star$ or $\psi' \in \Sigma_\psi \setminus \{\psi\}$ that $V_{\psi'}^{(T_\psi^1)} \leq \tilde{O}(\sigma_0)$.*

Proof. We can approximate the sequence V_{ψ_i} using $\tilde{V}_{\psi_i}^{(t)}$ and straightforwardly use Lemma 5.B.24 to compare ψ_i with any $\psi' > \psi_i \in \Sigma^\star$. In fact, let's compared the time they reach $V_\psi^{(t)} \geq d^{c_1} \sigma_0$. By Lemma 5.B.14, we can just compare $\tilde{V}_\psi^{(t)}$ before $t = T_\psi^1$, now since by Fact 5.B.4 we know $\tilde{V}_\psi^{(0)} \geq \tilde{V}_{\psi'}^{(0)} + \gamma$, we can use Lemma 5.B.24 to get that the first time when $\tilde{V}_{\psi_i}^{(t)} \geq d^{c_1} \sigma_0$, we have $\tilde{V}_{\psi'}^{(t)} = \tilde{O}(\sigma_0)$. This combined with Lemma 5.B.14 concludes the result. Moreover, we can also obtain bounds on all $\psi \in \Psi$ by discussing different ψ :

- For all $\phi' \in \Phi^\star$ such that there exists $\psi' = (j', r', \phi') < \psi_i$, ϕ' is learned. That means any ψ contains ϕ' has gradient too small because of 5.B.3c and therefore $\mathbb{E}[\mathcal{E}_{5,j} \mathbb{1}_{\mathcal{H}_{\phi'}}] \leq \tilde{O}(\lambda |\Gamma_\psi^{(t)}|)$ and therefore wouldn't be learned.

- All the rest of $\phi' \in \Phi^\star$ has their corresponding $\psi' > \psi \in \Sigma^\star$ and therefore lost in the competition and has small growth.
- All other $\psi \in \Psi \setminus \cup_{i' \leq i} \Sigma_{i'}$ lost the competition as well for the same reason.

This concludes the proof. $\square$

Proof of Inductions 5.B.2 and 5.B.3 in Phase I. Let $\psi_i \in \Sigma^\star$. Assume Induction 5.B.2 holds for some $t \leq T_{\psi_{i-1}}^3$, then we shall prove that it continues to hold for $t \in [T_{\psi_{i-1}}^3, T_{\psi_i}^1]$.

- Induction 5.B.2a: This is simple as all V_ψ for $\psi > \psi_i$ has updates approximated by $\widetilde{V}_\psi^{(t)}$, which is monotonically increasing, and has minimal update by Lemma 5.B.14.
- Induction 5.B.2b: We proved in Lemma 5.B.15 that whenever $\psi < \psi' \in \Sigma^\star$, $T_\psi^1 \leq T_{\psi'}^1$. This does not violate Induction 5.B.2b so the non-overlapping is preserved.
- Induction 5.B.3a: This is proven in Lemma 5.B.15.
- Induction 5.B.3b-c: They will be proven when we analyze the end phase of feature learning. They are the corollaries of the end phase guarantee of the previous features.
- Induction 5.B.3c:
- Induction 5.B.2e: We did not violate Induction 5.B.2e either.
- Induction 5.B.3a-d will be proved in Phase III where we argue the end phase of each feature learned.

$\square$

Phase II: Feature Growth and Cancellations

Below we present the induction hypothesis for phase II.

Induction 5.B.4 (Phase II). *For all $\psi = (j, r, \phi) \in \Sigma^\star$ and $t \in [T_\psi^1, T_\psi^2]$, the following holds*

- (a) *For $t \in [T_\psi^1, T_\psi^{2,1}]$, $\mathcal{E}_{5,j}^{(t)} \geq 1 - \frac{1}{\sqrt{d}}$ conditioned on $\mathcal{H}_\phi$;*
- (b) *For $t \leq T_\psi^{2,1}$, for all $j' \neq j$, we have $\mathbf{logit}_{5,j'}^{(t)} \leq O(\sqrt{d}) \cdot \mathbf{logit}_{5,j'}^{(T_\psi^1)}$.*
- (c) *For any $\phi' = (g', y') \neq \phi$, $V_{j,r}^{(t)}(g'), V_{j,r}^{(t)}(y') \in (-B - O(1), \widetilde{O}(\mu))$;*
- (d) *For $t \leq T_\psi^{2,2}$, the number of iterations where $V_\psi^{(t+1)} - V_\psi^{(t)} < 0$ is bounded by $\widetilde{O}(1/d^{c_1} \eta)$.*

Technical Lemmas

Lemma 5.B.16 (gradient estimation). *Assuming Inductions 5.B.2 and 5.B.4, let $\psi = (j, r, \phi) \in \Sigma^\star$, $\phi = (g, y)$ and $t \in [T_\psi^{2,1}, T_\psi^2]$, then*

- (a) *At any $t \geq T_\psi^{2,1}$, if $V_\psi^{(t)} \in (\frac{1}{3} \log d, \frac{2}{3} \log d)$, then $\Gamma_{j,r,\phi}^{(t)} \geq \Omega(1)$.*
- (b) *Any $g' \neq g \in \mathcal{G}$, $y' \neq y \in \mathcal{Y}$ has the following gradient approximation*

$$\begin{aligned} \Gamma_{j,r,g'}^{(t)} &= -\mathbb{E} \left[\mathbf{logit}_{5,j}^{(t)} \mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)}) \mathbf{1}_{\mathcal{H}_{(g',y)}} \right] \pm O(\omega/n_y) \\ \Gamma_{j,r,y'}^{(t)} &= -\mathbb{E} \left[\mathbf{logit}_{5,j}^{(t)} \mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)}) \mathbf{1}_{\mathcal{H}_{(g,y')}} \right] \pm O(\omega/n_y) \end{aligned}$$

(c) let $\phi' = (g', y') \neq \phi \in \Phi_j^*$ if $V_{j,r}(g, y) \geq B - o(1)$ and $V_{j,r}(g', y') < -B - 2\mu$, we have

$$\Gamma_{j,r,g'}^{(t)} = \sum_{\tilde{y} \in \mathcal{Y}, V_{j,r}(g', \tilde{y}) \in (-B, -\omega - \mu)} \Omega\left(\frac{\lambda \omega}{dn_y^3}\right) - \mathbb{E} \left[\mathbf{logit}_{5,j}^{(t)} \mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)}) \mathbb{1}_{\mathcal{H}_{(g', y')}} \right]$$

and similarly for $\Gamma_{j,r,y'}^{(t)}$.

Proof. Firstly (a) is simple to prove, because when $V_\psi^{(t)} \in (\frac{1}{3} \log d, \frac{2}{3} \log d)$, we have $\mathbf{logit}_{5,j} \mathbb{1}_{\mathcal{H}_\phi^*} \leq O(\frac{1}{d^{c_1}})$, which is from the calculation $F_{5,j} \leq \frac{2}{3} \log d + O(1)$ guaranteed by Inductions 5.B.3 and 5.B.4. For (b), we can estimate the gradient update of $V_{j,r}^{(t)}(g')$ where $g' \neq g$. In fact

$$\begin{aligned} -\nabla_{V_{j,r}(g')} \text{Loss}^{(t)} &= \Gamma_{j,r,g'}^{(t)} = \Gamma_{j,r,g'}^{+, (t)} - \Gamma_{j,r,g'}^{-, (t)} \\ &= \Gamma_{j,r,g'}^{+, (t)} - \sum_{y' \in \mathcal{Y}} \mathbb{E} \left[\mathbf{logit}_{5,j}^{(t)} \mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)}) \mathbb{1}_{\mathcal{H}_{(g', y')}} \right] \\ &= -\mathbb{E} \left[\mathbf{logit}_{5,j}^{(t)} \mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)}) \mathbb{1}_{\mathcal{H}_{(g', y)}} \right] \pm \max\{\tilde{O}(\mu^{q-1}), \omega\} \end{aligned}$$

where the last one is because for $y' \neq y$, the activation gradient $\mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)})$ is either positive and bounded by $\tilde{O}(\mu^{q-1})$ or negative and bounded by ω while $\mathbf{logit}_{5,j} \mathbb{1}_{\mathcal{H}_{\phi'}} \leq 1$. Similarly we can also get the same for $V_{j,r}(y')$ where $y' \neq y$:

$$-\nabla_{V_{j,r}(y')} \text{Loss}^{(t)} = \Gamma_{j,r,y'}^{(t)} = -\mathbb{E} \left[\mathbf{logit}_{5,j}^{(t)} \mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)}) \mathbb{1}_{\mathcal{H}_{(g', y)}} \right] \pm O(\omega/n_y^2)$$

To prove (c), notice that when $V_{j,r}(g', y') < -B - 2\mu$, the part of $\Gamma_{j,r,g'}^{+, (t)} = 0$ due to $\Lambda_{5,j,r}$ exceeding the boundary $-B$. We can group different $\tilde{y} \neq y \in \mathcal{Y}$ by their feature magnitude and obtain the lower bound. The factor of $\frac{\lambda}{dn_y}$ is due to the logit lower bound from Fact 5.B.3. $\square$

Lemma 5.B.17 (Feature Magnitude). *Assume Inductions 5.B.2 and 5.B.4 holds, then for any $\psi \in \Sigma^*$,*

- (a) at $t = T_\psi^{2,1}$, we have $V_{j,r}^{(t)}(g) = \frac{1}{4} \log d \pm o(1)$, $V_{j,r}^{(t)}(y) = \frac{1}{4} \log d \pm o(1)$;
- (b) when $C_\psi^+(\delta)$ hold for some $\delta \leq O(\lambda/d^{c_1})$ at some $t \geq T_\psi^{2,1}$, we have $V_\psi^{(t)} \geq B - O(d^{c_1}\mu)$, that is, $\mathcal{F}_{\psi,1}(d^{c_1}\mu)$ holds.

Proof. Let's proceed the proof one by one.

- Proof of (a) : This can be computed by comparing the gradient of $V_{j,r}^{(t)}(g)$ which is $\Gamma_{j,r,g}^{(t)} = \Gamma_{j,r,g}^{+, (t)} - \Gamma_{j,r,g}^{-, (t)}$ and the gradient of $V_{j,r}^{(t)}(y)$ which is $\Gamma_{j,r,y}^{(t)} = \Gamma_{j,r,y}^{+, (t)} - \Gamma_{j,r,y}^{-, (t)}$. Now since

$$\Gamma_{j,r,g}^{+, (t)} = \Gamma_{j,r,y}^{+, (t)} \geq (1 - O(\frac{1}{\sqrt{d}})) \mathbf{sReLU}'(V_{j,r}^{(t)}(g) + V_{j,r}^{(t)}(y) + O(\mu))$$

and that

$$\Gamma_{j,r,g}^{-(t)} = O\left(\frac{1}{\sqrt{d}} \mathbf{sReLU}'(V_{j,r}^{(t)}(g) + O(\mu))\right), \quad \Gamma_{j,r,y}^{-(t)} = O\left(\frac{1}{\sqrt{d}} \mathbf{sReLU}'(V_{j,r}^{(t)}(y) + O(\mu))\right)$$

one could see that $\sum_{t \leq T_\psi^{2,1}} |\Gamma_{j,r,g}^{(t)} - \Gamma_{j,r,y}^{(t)}| \leq \tilde{O}(\frac{1}{\sqrt{d}})$ and thus (a) holds.

- Proof of (b): from Lemma 5.B.16a we know that when $C_\psi^+(\delta)$ holds for some small $\delta = O(\frac{\lambda}{d^{c_1}})$ it is not because $V_\psi^{(t)}$ dropped below $\frac{1}{2} \log d$, thus when Inductions 5.B.2 and 5.B.4 holds, it can only be because $V_\psi^{(t)} \geq B - O(d^{c_1}\mu)$, otherwise

$$\Gamma_\psi^{+(t)} = \mathbb{E}[(1 - \mathbf{logit}_{5,j}^{(t)}) \mathbb{1}_{\mathcal{H}_\phi}] \geq \lambda \Pr(\mathcal{H}_\phi) \gg \frac{\lambda}{d^{c_1}} \quad (\text{if } V_\psi^{(t)} \leq B - O(d^{c_1}\mu))$$

which concludes the proof. $\square$

Lemma 5.B.18 (Gradient Stationarity). *Let $\psi \in \Sigma^\star$, assume Inductions 5.B.2 and 5.B.3 holds at $t \in [T_\psi^1, T_\psi^2]$, and choose $\delta \gg n_y \omega$, then we have the following guarantees:*

- (a) Define $\mathcal{B}_\delta^+ = \{t \in [T_\psi^{2,1}, T_\psi^2] \mid C_\psi^+(\delta) \text{ doesn't hold}\}$, then $|\mathcal{B}_\delta^+| \leq O(\frac{n_y \log^2 d}{\eta \delta})$;
- (b) Define $\mathcal{B}_\delta^- = \{t \in [T_\psi^{2,1}, T_\psi^2] \mid C_\psi^-(\delta) \text{ doesn't hold}\}$, then $|\mathcal{B}_\delta^-| \leq O(\frac{n_y \log^3 d}{\eta \delta})$;
- (c) Finally, at $t = T_\psi^2$, the feature shape $\mathcal{F}_\psi(\delta_1, \delta_2)$ holds with $\delta_1 = d^{c_1}\mu$ and $\delta_2 = (\frac{n_y^3 d \omega}{\lambda})^{\frac{1}{q-1}}$.

Proof. Let us start with (a). First by writing $\psi = (j, r, \phi)$, we define the following quantity

$$\Upsilon^{(t)} := V_{j,r}^{(t)}(\phi) - \sum_{\phi' \in \Phi_j^\star \setminus \{\phi\}} V_{j,r}^{(t)}(\phi') \quad (5.B.12)$$

which sits at the central of our proof of counter-argument. Now we have for any $T \geq T_\psi^{2,1}$

$$\begin{aligned} \Upsilon^{(t)} &= V_{j,r}^{(T)}(\phi) - \sum_{\phi' \in \Phi_j^\star \setminus \{\phi\}} V_{j,r}^{(T)}(\phi') \\ &= V_{j,r}^{(T_\psi^{2,1})}(\phi) - \sum_{\phi' \in \Phi_j^\star \setminus \{\phi\}} V_{j,r}^{(T_\psi^{2,1})}(\phi') + \sum_{s \in [T_\psi^{2,1}, T)} \eta \Gamma_{j,r,\phi}^{(s)} - \sum_{\phi' \in \Phi_j^\star \setminus \{\phi\}} \sum_{s \in [T_\psi^{2,1}, T)} \eta \Gamma_{j,r,\phi'}^{(s)} \end{aligned}$$

We can further rewrite the Γ s on the RHS to

$$\sum_{t \in [T_\psi^{2,1}, T)} \eta \left(\Gamma_{j,r,\phi}^{(t)} - \sum_{\phi' \in \Phi_j^\star \setminus \{\phi\}} \eta \Gamma_{j,r,\phi'}^{(t)} \right)$$

$$= \sum_{t \in [T_\psi^{2,1}, T)} \eta \left(\Gamma_{j,r,\phi}^{+, (t)} - \Gamma_{j,r,\phi}^{-, (t)} - \sum_{\phi' \neq \phi \in \Phi_j^*} \Gamma_{j,r,\phi'}^{+, (t)} + \sum_{\phi' \neq \phi \in \Phi_j^*} \Gamma_{j,r,\phi'}^{-, (t)} \right) \quad (5.B.13)$$

Now we can start to estimate the RHS. Define $P_1 = \{\phi' \neq \phi \in \Phi^* \mid \phi' \text{ learned}\}^7$ and $P_2 = \{\phi' \neq \phi \in \Phi^* \mid \phi' \text{ not learned}\}$. The sum $\sum_{\phi' \neq \phi \in \Phi_j^*} \Gamma_{j,r,\phi'}^{+, (t)}$ can be decomposed and bounded by

$$- O(\eta n_y \omega) \leq \sum_{\phi' \in P_1} \eta \Gamma_{j,r,\phi'}^{+, (t)} + \sum_{\phi' \in P_2} \eta \Gamma_{j,r,\phi'}^{+, (t)} \leq \tilde{O}(\eta \lambda \mu^{q-2})$$

and then by the same methods of bounding the gradients, we have that

$$\begin{aligned} & \sum_{\phi' \neq \phi \in \Phi_j^*} \Gamma_{j,r,\phi'}^{-, (t)} \\ &= \sum_{\phi' = (g', y') \in \Phi_j^* \setminus \{\phi\}} \mathbb{E} \left[\text{logit}_{5,j}^{(t)} \text{sReLU}'(\Lambda_{5,j,r}^{(t)}) \left(\sum_{g'' \neq g' \in \mathcal{G}} \mathbb{1}_{\mathcal{H}_{(g'', y')}} + \sum_{y'' \neq y' \in \mathcal{Y}} \mathbb{1}_{\mathcal{H}_{(g', y'')}} \right) \right] \\ & \stackrel{(2)}{=} \sum_{g' \neq g} \mathbb{E} [\text{logit}_{5,j}^{(t)} \text{sReLU}'(\Lambda_{5,j,r}^{(t)}) \mathbb{1}_{\mathcal{H}_{(g', y)}}] + \sum_{y' \neq y} \mathbb{E} [\text{logit}_{5,j}^{(t)} \text{sReLU}'(\Lambda_{5,j,r}^{(t)}) \mathbb{1}_{\mathcal{H}_{(g, y')}}] \pm O(\omega) \\ &= \Gamma_{j,r,\phi}^{-, (t)} \pm O(\omega) \end{aligned}$$

where (1) is because for any incorrect feature combination $\phi' = (g', y') \notin \Phi_j^*$ and does not share a component with ϕ , the activation of $\Lambda_{5,j,r}$ conditioned on $\mathcal{H}_{\phi'}$ is within $[-B, \tilde{O}(\mu^{q-1})]$ and thus the negative gradient is upper bounded by ω/η_y^2 and lower bounded by $-\tilde{O}(\mu^{q-1})$. Inserting this back to Eq. (5.B.13), we shall have

$$\sum_{t \in [T_\psi^{2,1}, T)} \eta \left(\Gamma_{j,r,\phi}^{(t)} - \sum_{\phi' \in \Phi_j^* \setminus \{\phi\}} \eta \Gamma_{j,r,\phi'}^{(t)} \right) = \sum_{t \in [T_\psi^{2,1}, T)} \eta \Gamma_{j,r,\phi}^{+, (t)} \pm O((T - T_\psi^{2,1})\eta\omega)$$

Now we have arrived at the desired estimate of the update, finally, we have the following counter-argument: if $C_\psi^+(\delta)$ does not hold for $\Omega(\frac{n_y \log^2 d}{\eta \delta})$ iterations, letting $T - 1$ be the last iteration which $C_\psi^+(\delta)$ does not hold, we have

$$\begin{aligned} \Upsilon^{(T)} &\geq \sum_{t \in [T_\psi^{2,1}, T)} \eta \Gamma_{j,r,\phi}^{+, (t)} \\ &\geq \sum_{t \in [T_\psi^{2,1}, T] \setminus B_\delta^+} \eta \delta + \Omega((T - T_\psi^{2,1})\eta\omega) \quad (\text{For } t \in B_\delta^+, \Gamma_{j,r,\phi}^{+, (t)} \geq 0) \end{aligned}$$

⁷ By ϕ' is learned we mean there is a feature $\psi' < \psi \in \Sigma^*$ such that $\psi' = (j, r, \phi')$.

$$\begin{aligned}
&\geq \sum_{t \in [T_{\psi}^{2,1}, T] \setminus B_{\delta}^+} \eta \delta - O\left(\frac{n_y \log^2 d}{\eta \delta} \eta \omega\right) \\
&\geq \Omega\left(\frac{n_y \log^2 d}{\eta \delta}\right) \eta \delta \quad (\text{by choosing } \delta \gg \omega) \\
&\geq \Omega(n_y \log^2 d)
\end{aligned}$$

which is impossible because the gradient would have vanished when $\Upsilon^{(T)} \geq n_y \log^{1.5} d$ as that would mean $\exists \phi \in \Phi_j^*$ such that $|V_{j,r}^{(t)}(\phi)| \geq \log^2 d$ and it is forbidden in our setting, which proves the claim via contradiction.

Proof of Lemma 5.B.18b. Actually, by applying (a) combined with Lemma 5.B.17b, we know that for some $T \geq T_{\psi}^{2,1} + \tilde{\Omega}(\frac{1}{\eta \lambda})$ it holds $V_{\psi}^{(T)} \geq B - O(d^{c_1} \mu)$. From this iteration forward we shall prove that $V_{\psi}^{(t)}$ stays above $B - O(d^{c_1} \mu)$ and can help. In fact, at $t = T$, $V_{j,r}^{(t)}(V) \in (-O(B), \tilde{O}(\mu))$ for $v \in \mathcal{G} \cup \mathcal{Y} \setminus \{g, y\}$ by Induction 5.B.4. Say $v = g' \neq g$, then due to Lemma 5.B.16, its gradient is approximated by

$$-\mathbb{E} \left[\text{logit}_{5,j}^{(t)} \text{sReLU}'(\Lambda_{5,j,r}^{(t)}) \mathbb{1}_{\mathcal{H}_{(g',y)}} \right] \pm O(\omega/n_y) \quad (5.B.14)$$

Let $\delta \geq \omega$, we will find an iteration $t \leq T_{\psi}^{2,1} + O(\frac{n_y \log^3 d}{\eta \delta})$ that satisfy $C_{\psi}^-(\delta)$. By applying (a), we first get that $C_{\psi}^+(\delta)$ holds for no more than $O(\frac{n_y \log^2 d}{\eta \delta})$ iters after $T_{\psi}^{2,1}$ and before T_{ψ}^2 .

Let's assume this is the case, that $C_{\psi}^-(\delta)$ doesn't hold for more than $\Omega(\frac{n_y \log^3 d}{\eta \delta})$ iterations. In this case, it must be that the gradient $\Gamma_{j,r,\phi}^{-(t)} < -\delta$ is negative, since when the gradient is positive it has an upper bound $O(\omega/n_y) = o(\delta)$ and therefore cannot violate the condition $C_{\psi}^+(\delta)$. Therefore the updates $\sum_{s \leq t} \Gamma_{j,r,\phi}^{-(s)}$ must accumulate to lower than

$$V_{j,r}^{(t)}(\phi) \leq \Upsilon^{(t)} - \Omega\left(\frac{n_y \log^3 d}{\eta \delta}\right) \times \eta \delta = -\Omega(n_y \log^3 d)$$

which is impossible in our setting. This proved the result for any $\delta \gg n_y \omega$.

Proof of Lemma 5.B.18c. Let $\delta = n_y^2 \omega$, which is covered by Lemma 5.B.18a-b, we can find an iteration where $|\Gamma_{j,r,\phi}^{(t)}| \leq \delta$ $\phi' \in \Phi_j^* \setminus \{\phi\}$. This implies we have found a t such that

$$|\mathbb{E}[\text{logit}_{5,j}^{(t)} \text{sReLU}'(\Lambda_{5,j,r}) \mathbb{1}_{\mathcal{H}_{\phi'}}]| \leq n_y^2 \omega \implies |\text{sReLU}'(V_{j,r}^{(t)}(\phi') + O(\mu))| \leq O\left(\frac{n_y^3 d \omega}{\lambda}\right)$$

(We lower bounded the logit by Fact 5.B.3)

Thus we arrive at $|V_{j,r}(\phi')| \leq O((\frac{n_y^3 d \omega}{\lambda})^{\frac{1}{q-1}})$. Remember in the proof of b, $V_{\psi}^{(t)} = \frac{1}{2}(V_{j,r}^{(t)}(g) + V_{j,r}^{(t)}(y)) \geq B - O(d^{c_1} \mu)$ at $t \geq T + \tilde{\Omega}(\frac{d^{c_1}}{\eta \lambda})$. So we have for any $\phi' = (g', y') \in \Phi_j^* \setminus \{\phi\}$,

$$\frac{1}{2}(V_{j,r}^{(t)}(g') + V_{j,r}^{(t)}(y')) \leq -\frac{1}{2}(V_{j,r}^{(t)}(g) + V_{j,r}^{(t)}(y)) + O(\frac{1}{d^{1/3q}}) \leq -B + O((\frac{n_y^3 d \omega}{\lambda})^{\frac{1}{q-1}})$$

Now we have verified that given $\delta = n_y^2 \omega$, there is a iteration $T_{\psi}^{2,3}$ which gives the near-perfect feature shape $\mathcal{F}_{\psi}(\delta_1, \delta_2)$ with $\delta_1 = d^{c_1} \mu$, $\delta_2 = (\frac{n_y^3 d \omega}{\lambda})^{\frac{1}{q-1}}$. We shall prove that this feature shape will hold until $t = T_2$.

For $t \geq T_{\psi}^{2,3}$, we verify $\mathcal{F}_{\psi}(\delta_1, \delta_2)$ in Definition 5.B.4 one by one:

1. $\mathcal{F}_{\psi,1}(d^{c_1} \mu)$: By Lemma 5.B.18a we shall find a step after $t \geq T_{\psi}^{2,1} + O(\frac{d^{c_1}}{\eta \lambda})$ at which $\mathcal{F}_{\psi,1}(d^{c_1} \mu)$ is satisfied. In fact, the majority of the steps after $T_{\psi}^{2,1} + O(\frac{d^{c_1}}{\eta \lambda})$ will satisfy this. We shall prove that combined with the above feature shape guarantee, $\mathcal{F}_{\psi,1}(d^{c_1} \mu)$ will hold true until $t = T_1$.
2. $\mathcal{F}_{\psi,2}((\frac{n_y^3 d \omega}{\lambda})^{\frac{1}{q-1}})$: From the above argument that applies Lemma 5.B.18b, we shall find a time step where $\mathcal{F}_{\psi,1}(d^{c_1} \mu)$ and $\mathcal{F}_{\psi,2}(d^{-1/3q})$ hold. Now we discuss the possible values of the gradient $\Gamma_{\psi'}^{-(t)}$, $\psi' \in \Sigma_{\psi}^{+,1}$. In fact, let $\phi' = (g', y') \in \Phi_j^* \setminus \{\phi\}$, we have
 - Regime A: If $V_{j,r}(\phi') < -B - 2\mu$, then there is no negative gradient for $V_{j,r}(g')$ and $V_{j,r}(y')$ unless $V_{j,r}(g', y)$ or $V_{j,r}(g, y')$ is positive. In fact, only one of the feature can be positive to have negative gradient, and in that case the gradient of $\Gamma_{j,r,v}^{(t)}$, $v \in \{g', y'\}$ is pointing to the convergence direction $|V_{j,r}(g', y)| \rightarrow 0$.
 - Regime B: If $V_{j,r}(\phi') \geq -B - 2\mu$, then there is a possibility of negative gradient for $V_{j,r}(g')$ and $V_{j,r}(y')$ even when $V_{j,r}(g', y)$ or $V_{j,r}(g, y')$ is negative, In this case the feature $V_{j,r}(g', y)$ or $V_{j,r}(g, y')$ might not go the desired direction to converge to zero, but given a $\delta > n_y^2 \omega$, $\Gamma_{j,r,(g',y)}^{-(t)}$ or $\Gamma_{j,r,(g,y')}^{-(t)}$ cannot stay above δ , otherwise the term will dominate and revert the gradient direction. On the other hand, when one side has $V_{j,r}(g, y') < -\Omega((\frac{n_y^3 d \omega}{\lambda})^{\frac{1}{q-1}})$, we have

$$V_{j,r}(\phi') \leq -V_{j,r}(\phi) - \Omega((\frac{n_y^3 d \omega}{\lambda})^{\frac{1}{q-1}}) \leq -B - 2\mu$$

which is back in regime A and therefore will converge back.

Induction over the above two regimes for all $t \in [T_{\psi}^{2,3}, T_{\psi}^2]$ showed that the feature will indeed satisfy $\mathcal{F}_{\psi,2}(\delta)$ for all $\delta > n_y^2 \omega$ at t .

3. $\mathcal{F}_{\psi,3}((\frac{n_y^3 d \omega}{\lambda})^{\frac{1}{q-1}})$: We need to use both $\mathcal{F}_{\psi,1}$ and $\mathcal{F}_{\psi,2}$ to get the desired result. In fact, we have for any $\psi' = (j, r, (g', y')) \in \Sigma_{\psi}^{+,2}$,

$$V_{\psi'}^{(t)} = V_{j,r}^{(t)}(g') + V_{j,r}^{(t)}(y') + V_{j,r}^{(t)}(g) + V_{j,r}^{(t)}(y) - V_{j,r}^{(t)}(\phi) = -B + O((\frac{n_y^3 d \omega}{\lambda})^{\frac{1}{q-1}})$$

because $V_{j,r}^{(t)}(g) + V_{j,r}^{(t)}(y')$ and $V_{j,r}^{(t)}(g') + V_{j,r}^{(t)}(y)$ are bounded due to $\mathcal{F}_{\psi,2}$.

4. $\mathcal{F}_{\psi,4}(d^{c_1}\mu)$: The number of iterations $[T_{\psi}^1, T_{\psi}^2]$ is no more than $\tilde{O}(\frac{1}{\eta(d^{c_1}\mu)^{q-2}})$, so by using Lemma 3.A.1 along with the induction from the result at T_{ψ}^1 shall conclude the proof.
5. $\mathcal{F}_{\psi,5}((\frac{n_y^3 d \omega}{\lambda})^{\frac{1}{q-1}})$: By using Lemma 5.B.19 and $\mathcal{F}_{\psi,2}$ after $t \geq T_{\psi}^{2,3}$, we have the result.

This proved (c). $\square$

Lemma 5.B.19 (Symmetry of Features). *Let $\psi = (j, r, (g, y)) \in \Sigma^*$ and $t \geq T_{\psi}^{2,3}$, we have that if $|V_{\psi'}| \leq \delta$ for all $\psi' \in \Sigma^{+,1}$, then*

$$|V_{j,r}^{(t)}(g) - V_{j,r}^{(t)}(y)| \leq O(\delta)$$

Proof. For a feature $\psi = (j, r, \phi) \in \Sigma^*$, $\phi = (g, y)$, their update can be represented as:

$$\begin{aligned} V_{j,r}^{(t)}(g) &= U_{g,y} - \sum_{y' \neq y} R_{g,y'} + V_{j,r}^{(0)}(g) \\ V_{j,r}^{(t)}(y) &= U_{g,y} - \sum_{g' \neq g} R_{g',y} + V_{j,r}^{(0)}(y) \end{aligned}$$

where $U_{g,y} = \sum_{s \leq t} \eta \Gamma_{j,r,\phi}^{+, (s)}$ and $R_{g,y'}$ is

$$R_{g,y'} = \sum_{s \leq t} \eta \mathbb{E}[\text{logit}_{5,j}^{(s)} \text{sReLU}'(\Lambda_{5,j,r}^{(s)}) \mathbb{1}_{\mathcal{H}_{g',y}}]$$

Therefore:

$$\begin{aligned} \sum_g V_{j,r}^{(t)}(g) &= \sum_{\phi \in \Phi_j^*} U_{\phi} - \sum_g \sum_{y' \neq y} R_{g,y'} = \sum_{\phi \in \Phi_j^*} U_{\phi} - \sum_{\phi^{\dagger} \in \Phi_j^{\dagger}} R_{\phi^{\dagger}} + V_{j,r}^{(0)}(g) \\ \sum_y V_{j,r}^{(t)}(y) &= \sum_{\phi \in \Phi_j^*} U_{\phi} - \sum_y \sum_{g' \neq g} R_{g',y} = \sum_{\phi \in \Phi_j^*} U_{\phi} - \sum_{\phi^{\dagger} \in \Phi_j^{\dagger}} R_{\phi^{\dagger}} + V_{j,r}^{(0)}(y) \end{aligned}$$

Thus

$$\sum_{g'} V_{j,r}^{(t)}(g') = \sum_{y'} V_{j,r}^{(t)}(y') \pm O(\mu)$$

Moreover, we have assumed

$$|V_{j,r}^{(t)}(g) + V_{j,r}^{(t)}(y')| = O(\delta), \forall y' \neq y, \quad |V_{j,r}^{(t)}(y) + V_{j,r}^{(t)}(g')| = O(\delta), \forall g' \neq g$$

Thus

$$\sum_{g'} V_{j,r}^{(t)}(g') = V_{j,r}^{(t)}(g) - (n_y - 1)V_{j,r}^{(t)}(y) \pm O(n_y \delta) = V_{j,r}^{(t)}(y) - (n_y - 1)V_{j,r}^{(t)}(g) \pm O(n_y \delta)$$

Therefore

$$V_{j,r}^{(t)}(g) = V_{j,r}^{(t)}(y) + O(\delta) + O(\mu)$$

inserting $\delta = O((\frac{n_y^2 d \omega}{\lambda})^{\frac{1}{q-1}})$ which is from Lemma 5.B.18 proves the claim. $\square$

Lemma 5.B.20 (Arrival time estimates). *Assuming Inductions 5.B.2 and 5.B.4 for $\psi \in \Sigma^\star$, we have*

- (a) $T_\psi^{2,1} - T_\psi^1 \leq \tilde{O}(1/\eta(d^{c_1} \sigma_0)^{q-2})$;
- (b) $T_\psi^{2,2} - T_\psi^{2,1} \leq O(\frac{d^{\frac{1}{2}+c_1}}{\eta})$;
- (c) $T_\psi^{2,3} - T_\psi^{2,2} \leq \tilde{O}(\frac{1}{\eta \omega})$;
- (d) $T_\psi^2 - T_\psi^1 \leq \tilde{O}(\frac{1}{\eta(d^{2c_1} \mu)^{q-2}})$

Proof. • Proof of (a): Once $V_\psi^{(t)} \geq d^{c_1} \sigma_0$, we can apply Lemma 3.A.1 to compute the number of iteration needed to reach $V_\psi^{(t)} \geq \frac{1}{2} \log d$. In fact, by Induction 5.B.4a-b we know that

$$V_\psi^{(t+1)} \geq V_\psi^{(t)} + \eta(1 - O(\frac{1}{\sqrt{d}})) \text{sReLU}'(V_\psi^{(t)} + O(\mu)) \geq V_\psi^{(t)} + \Omega(\eta(V_\psi^{(t)})^{q-1})$$

Since $V_\psi^{(T_\psi^1)} \geq d^{c_1} \sigma_0$, we have by Lemma 3.A.1 that $T_\psi^{2,1} - T_\psi^1 \leq O(\frac{1}{\eta(d^{c_1} \sigma_0)^{q-2}})$.

- Proof of (b): Simply apply Lemma 5.B.18 with $\delta = \frac{1}{d^{\frac{1}{2}+c_1}}$;
- Proof of (c): Again by applying Lemma 5.B.18 with $\delta = n_y^2 \omega$;
- Proof of (d): Follow the definition of T_ψ^2 .

$\square$

Proof of Induction 5.B.4. The proof of time steps are all in Lemma 5.B.20 and Lemma 5.B.18. We do not repeat here. $\square$

Phase III: Convergence

We present the induction hypothesis in this phase.

Induction 5.B.5 (Phase III, final). *Let $\psi = (j, r, \phi) \in \Sigma^\star$, for $t \in [T_\psi^2, T_1]$, the following holds:*

- (a) At $t \in [T_\psi^2, T_{1,1}]$, we have $\mathcal{F}_\psi(\delta_1, \delta_2)$ holds with $\delta_1 = d^{c_1} \mu$, $\delta_2 = (\frac{n_y^2 d \omega}{\lambda})^{\frac{1}{q-1}}$;
- (b) At $t \in [T_\psi^3, T_1]$, we have $\mathcal{F}_\psi(\delta_1, \delta_2)$ holds with $\delta_1 = d^{c_1} \mu$, $\delta_2 = (n_y^2 \omega)^{\frac{1}{q-1}}$.

We need a lemma to describe the logit distribution when all features are learned.

Lemma 5.B.21 (Logit shape at convergence). *Assuming Inductions 5.B.2 and 5.B.5. Let $\psi \in \Sigma^\star$ be the last of $\Sigma^\star$, then at $t = T_\psi^2$, the followings hold:*

1. For $\phi \in \Phi_j^\star$, $\text{logit}_{5,j}^{(t)} \geq 1 - O(\lambda)$ conditioned on $\mathcal{H}_\phi$.
2. For $\phi \notin \Phi_j^\star$, $\text{logit}_{5,j}^{(t)} \leq O(\lambda/d)$ conditioned on $\mathcal{H}_\phi$.

Proof. Since ψ is the last feature in the learning curriculum $\Sigma^\star$, all feature at $t = T_\psi^2$ has satisfied $\mathcal{F}_\psi(\delta_1, \delta_2)$ for $\delta_1 = d^{c_1}\mu$, $\delta_2 = (\frac{n_y^2 d \omega}{\lambda})^{\frac{1}{q-1}}$. Thus we can bound the logit by

$$\text{logit}_{5,j}^{(t)} = \frac{e^{F_{5,j}^{(t)}(\mathbf{Z})}}{e^{F_{5,j}^{(t)}(\mathbf{Z})} + d} = \frac{e^{B-o(1)}}{e^{B-o(1)} + d} = 1 - O(\lambda) \quad (\text{conditioned on } \mathbf{Z} \in \mathcal{H}_\phi, \phi \in \Phi_j^\star)$$

and

$$\text{logit}_{5,j}^{(t)} = \frac{e^{F_{5,j}^{(t)}(\mathbf{Z})}}{e^{F_{5,j}^{(t)}(\mathbf{Z})} + d} = \frac{e^{o(1)}}{e^{B-o(1)} + d - 1 + e^{o(1)}} = O\left(\frac{\lambda}{d}\right) \quad (\text{conditioned on } \mathbf{Z} \in \mathcal{H}_\phi, \phi \in \Phi_j^\star)$$

which is the desired result. $\square$

Lemma 5.B.22 (Gradient Bounds, Phase III). *Assuming Inductions 5.B.2 and 5.B.5 holds, and let $\psi = (j, r, \phi) \in \Sigma^\star$, then at $t \in [T_{1,1}, T_1]$, the followings hold*

- (a) $\Gamma_\psi^{+, (t)} \geq \Omega(\lambda)$ when $V_\psi^{(t)} \leq B - d^{c_1}\mu$;
- (b) $|\Gamma_{j,r,v}^{-, (t)}| \leq O(\frac{\lambda \omega}{d n_y})$ for any $v \in \mathcal{G} \cup \mathcal{Y}$;
- (c) For $\phi' \in \Phi_j^\star \setminus \{\phi\}$, we have

$$\Gamma_{j,r,\phi'}^{+, (t)} = \begin{cases} -\Theta(\lambda \omega / n_y^2), & \text{if } V_{j,r,\phi'}^{(t)} \in [-B + 2\mu, -\omega] \\ \in [-\Theta(\lambda \omega / n_y^2), -\Theta(\lambda \omega / d n_y^2)] \cup \{0\}, & \text{if } V_{j,r,\phi'}^{(t)} \in [-B - 2\mu, -B + 2\mu] \end{cases}$$

Proof. Lemma 5.B.22a is simple when Lemma 5.B.21a holds. Lemma 5.B.22b is basically redoing the calculations in Lemma 5.B.16. Lemma 5.B.22c is because when $V_{j,r,\phi'}^{(t)} \geq -B + 2\mu$, we have

$$\begin{aligned} \mathbb{E}[\mathcal{E}_{5,j}^{(t)} \mathbf{sReLU}'(\Lambda_{5,j,r}) \mathbb{1}_{\mathcal{H}_{\phi'}}] &= \mathbb{E}[(1 - \text{logit}_{5,j}^{(t)}) \mathbf{sReLU}'(\Lambda_{5,j,r}) \mathbb{1}_{\mathcal{H}_{\phi'}}] \\ &\geq \lambda \mathbf{sReLU}'(V_{j,r,\phi'}^{(t)} + \mu \pm o(\mu)) \Pr(\mathcal{H}_{\phi'}) \\ &\geq -\Theta(\lambda \omega / n_y^2) \end{aligned}$$

And the rest is simply the property of $\mathbf{sReLU}'$ at the negative boundary when some activations of x_0, x_1 crosses the boundary. $\square$

Proof of Induction in Phase III and Theorem 5.B.1

Proof of the Phase III induction. We prove the induction by reusing the proof of Lemma 5.B.18.

- Proof of Induction 5.B.5a: The same as in the proof of Lemma 5.B.18c, with longer time. We do not use any property of T_ψ^3 so no additional handling is needed.
- Proof of Induction 5.B.5b: When all feature is learned, that is, after $t = T_\psi^2$ for the last feature $\psi \in \Sigma^\star$, we have perfect feature shape and therefore have much better logit shape as described by Lemma 5.B.21. We can reuse the proof of Lemma 5.B.18a-b to get the following

result: for any $\delta = \omega(\frac{\lambda\omega}{dn_y})$, we can guarantee that $C_\psi(\delta)$ doesn't hold for more than $\tilde{O}(\frac{1}{\eta\delta})$ iterations. And beyond that, we can just reuse the argument in the proof of Lemma 5.B.18c to obtain the desired result.

- Proof of Induction 5.B.2a-b: They are correct in Phase III because there is no violation.
- Proof of Induction 5.B.3a-b: They are proven in previous phases.
- Proof of Induction 5.B.3c: For predecessor feature $\psi' < \psi \in \Sigma^*$, we proved in this phase that they maintain feature shape until $T_{\psi'}^3$, and improved their feature at $T_{\psi'}^3$.
- Proof of Induction 5.B.3d: This is proven in previous phases.

□

Proof of Theorem 5.B.1. Since all feature in Σ^* have shape $\mathcal{F}_\psi(\delta_1, \delta_2)$ for $\delta_1 = d^{c_1}\mu$, $\delta_2 = \omega^{\frac{1}{q-1}}$ at the end of Induction 5.B.5, we have proven Theorem 5.B.1. □

Auxiliary Technical Tools

First we need a Bernstein inequality for U-statistics

Lemma 5.B.23 (concentration inequality for pseudo-U-statistics). *Let $x_1, \dots, x_n$ be different symbols, and let $m \ll n$ be such that $n \equiv 0 \pmod{m}$. Suppose for some function h with $|h| \leq M$ the random variables $h(x_{i_1}, x_{i_2}, \dots, x_{i_m})$ and $h(x_{i'_1}, x_{i'_2}, \dots, x_{i'_m})$ are independent and identically distributed as long as $\{x_{i_1}, x_{i_2}, \dots, x_{i_m}\} \cap \{x_{i'_1}, x_{i'_2}, \dots, x_{i'_m}\} = \emptyset$, then the pseudo-U-statistic*

$$U_{m,n} = \frac{1}{\binom{n}{m}} \sum_{0 \leq i_1 < i_2 < \dots < i_m \leq n} h(x_{i_1}, x_{i_2}, \dots, x_{i_m})$$

satisfies $\Pr(|U_{n,m} - \mathbb{E}[U_{n,m}]| \geq t) \leq e^{-\frac{mt^2}{mM^2}}$

Proof. The proof is the same as in [237]. □

We present two lemmas related to the tensor power method.

Lemma 5.B.24 (TPM in expectation). *Let $p > 3$ be an integer, and $\mu, \sigma > 0$ satisfying $\mu \gg \sigma\sqrt{\log d}$. Now suppose $\xi \sim \mathcal{N}(0, \sigma)$ is a Gaussian variable and there are two sequences x_t, y_t defined by the following update rules:*

- $x_{t+1} \geq x_t + \eta C_t \mathbb{E}_\xi[(\xi + x_t)^p]$ for some $C_t = \Theta(1)$;
- $y_{t+1} \leq y_t + \eta C_t \mathbb{E}_\xi[(\xi + y_t)^p]$ for some $S = \Theta(1)$.

Then if $x_0, y_0 \geq \mu$, $x_0 - y_0 \geq \varepsilon = \Omega(\frac{1}{\text{polylog}(d)})$, for every $t \geq 0$ such that $x_t \in [d^{0.01}x_0, O(1)]$, we shall have $y_t \leq O(\text{polylog}(d))$.

Proof. Let $\delta > 0$ be such that $\delta \gg \frac{\mu}{\sigma}$ and $\delta = o(1)$. Let $T_i, i \geq 0$ be defined as the first time when $x_t \geq (1 + \delta)^i x_0$, and $b := \min\{i | (1 + \delta)^i x_0 \geq A\}$. Now we can compute a growth lower bound for

$\varepsilon_t := x_t - y_t$ at each step:

$$\begin{aligned}
\varepsilon_{T_{i+1}} &\geq \varepsilon_{T_i} + \sum_{t \in [T_i, T_{i+1})} \eta C_t \mathbb{E}_\xi [(\xi + x_t)^p - (\xi + y_t)^p] \\
&\geq \varepsilon_{T_i} + \varepsilon_{T_i} \sum_{t \in [T_i, T_{i+1})} \eta C_t \mathbb{E}_\xi [(\xi + x_t)^{p-1}] \\
&\geq \varepsilon_{T_i} + \varepsilon_{T_i} \sum_{t \in [T_i, T_{i+1})} \eta C_t \mathbb{E}_\xi [(\xi + x_t)^p] \cdot \frac{1}{(1 + \delta)x_{T_{i+1}}} \\
&\geq \varepsilon_{T_i} + \varepsilon_{T_i} \delta (1 + \delta)^i x_0 \frac{1}{(1 + \delta)^{i+2} x_0} \\
&\geq (1 + \frac{\delta}{(1 + \delta)^2})^i \varepsilon_{T_i}
\end{aligned}$$

Therefore let i' be such that $(1 + \delta)^{i'} \geq \frac{\mu}{\varepsilon}$, we have that

$$\varepsilon_{T_{i'}} \geq (1 + \frac{\delta}{(1 + \delta)^2})^{i'} \varepsilon_0 \geq (1 - o(1))\mu \gg \sigma \sqrt{\log d} \quad (5.B.15)$$

At $t = T_{i'}$ we have $y_{T_{i'}} \leq x_{T_{i'}} \leq (1 + o(1)) \frac{\mu}{\varepsilon} x_0$. After $t = T_{i'}$, we can construct a surrogate sequence $\tilde{y}_t \geq y_t$ for $t \geq T_{i'}$ by initializing $\tilde{y}_{T_{i'}} = y_{T_{i'}} + \frac{\mu}{2}$, and updates by

$$\tilde{y}_{t+1} = \tilde{y}_t + \eta C_t \tilde{y}_t^p$$

This update guarantees that $\tilde{y}_t \geq y_t$ because $\tilde{y}_t^p \geq \mathbb{E}_\xi [(\xi + y_t)^p]$ at every step. Now by applying Corollary 5.B.1 to the sequence of $x_t, \tilde{y}_t$ using the fact that $x_{T_{i'}} \geq \tilde{y}_{T_{i'}} + \frac{\mu}{2} = \tilde{y}_{T_{i'}} (1 + \frac{1}{\text{polylog}(d)})$, we can get the desired result for $\tilde{y}_t$ and thus y_t at $t = T_b$, i.e, when $x_t \geq A$. $\square$

The tensor-power estimate needed here is the shared Lemma 3.A.1, specialized with $r = q - 1$. This lemma has a corollary:

Corollary 5.B.1 (TPM, from [6]). *Let $q \geq 3$ be a constant and $x_0, y_0 = o(1)$ and $A = O(1)$. Let $\{x_t, y_t\}_{t \geq 0}$ be two positive sequences updated as*

- $x_{t+1} = x_t + \eta C_t x_t^{q-1}$ for some $C_t = \Theta(1)$;
- $y_{t+1} = y_t + \eta S C_t y_t^{q-1}$ for some $S = \Theta(1)$.

Suppose $x_0 \geq y_0 S^{\frac{1}{q-2}} (1 + \frac{1}{\text{polylog}(d)})$, letting T_x be the first iteration s.t., $x_t \geq A$, then $y_{T_x} \leq \tilde{O}(y_0)$.

5.B.3 Learning Symmetry Group Actions

Roadmap. We begin by introducing notation and defining fibers, followed by a description of the learning curriculum and the key appearance events. We then break the training process into clearly defined phases and prove the main results for each phase: In Phase I, features emerge and

compete; In Phase II, features grow and mutually cancel, with precise tracking of when different outcomes occur; In Phase III, features converge to a desired shape.

Let us restate the action structure of $\mathcal{G}$ and $\mathcal{Y}$ here.

Assumption 5.B.3 (Assumption 5.5.1, restated). Let $\text{LEGO}(\mathcal{X}, \mathcal{G}, \mathcal{Y})$ be the LEGO language, where $|\mathcal{Y}| = n_Y$ ⁸ and $\mathcal{G}$ is a group acting on $\mathcal{Y}$. We assume the (left) group action $\alpha : \mathcal{G} \times \mathcal{Y} \rightarrow \mathcal{Y}$ is transitive but not free. In particular, we assume $n_Y = |\mathcal{Y}| \in [\omega(1), O(\log d)]$ and the group $\mathcal{G}$ satisfy $n_G = |\mathcal{G}| = \Theta(\text{polylog}(d)) > \frac{1}{\varrho}$ for all $y \in \mathcal{Y}$.

Remark 5.B.1. When the action $\alpha : \mathcal{G} \times \mathcal{Y} \rightarrow \mathcal{Y}$ is transitive, there is only one orbit $\mathcal{Y} = \{g \cdot y \mid g \in \mathcal{G}\} \simeq \mathcal{G}/G_y$ for all $y \in \mathcal{Y}$. By orbit-stabilizer theorem, the stabilizer G_y at any point y has the same cardinality.

Remark 5.B.2. The canonical action of a symmetry group S_n on $\mathbb{Z}_n$ satisfy Assumption 5.B.3 with the choice $n_Y = \Theta(\frac{\log \log d}{\log \log \log d})$ that keeps $n_Y! = O(\text{polylog}(d))$.

Preliminaries

We work with fiber-indexed features. We first define fibers and super-combinations φ , then we define neuron features by its index $\psi = (j, r, \varphi)$, and finally set the curriculum and appearance events in this notation.

Definition 5.B.7 (Fiber of values). Assuming $\mathcal{G}$ follows Assumption 5.B.3. For each $j \in \tau(\mathcal{Y})$ and $y \in \mathcal{Y}$, define the **fiber** of j at y as:

$$\text{Fiber}_{j,y} := \{g \in \mathcal{G} \mid \tau(g \cdot y) = j\}$$

which collects all group elements that send y to $y' = \tau^{-1}(j)$.

Remark 5.B.3. We denote $\text{Fiber}_{j,y}$ to be the set that transport y to $y' = \tau^{-1}(j)$ because it is the pre-image of the predictor map $\phi_y(\cdot) : g \mapsto \tau(g \cdot y)$ over $y \in \mathcal{Y}$. Algebraically the fibers are the left-cosets $\{gG_y\}_{g \in \mathcal{G}}$, where $G_y := \{g \in \mathcal{G} \mid g \cdot y = y\}$ is the stablizer subgroup. Moreover, we have the following fact from the orbit-stablizer theorem.

Fact 5.B.6 (cardinality of the fibers). For every $j \in \tau(\mathcal{Y})$ and $y \in \mathcal{Y}$, $|\text{Fiber}_{j,y}| = |\mathcal{G}|/|\mathcal{Y}| = n_G/n_Y$ from the orbit-stablizer theorem.

Definition 5.B.8 (Super-combinations). Let $(j, y) \in \tau(\mathcal{Y}) \times \mathcal{Y}$, we define the **super-combinations**, which is a set of combinations:

$$\varphi_{j,y} := \text{Fiber}_{j,y} \times \{y\} \subseteq \mathcal{G} \times \mathcal{Y}, \quad \varphi_{j,y}^1 = \text{Fiber}_{j,y}, \quad \varphi_{j,y}^2 = \{y\}$$

⁸ We use n_y or n_Y interchangeably in the proof below.

Let $\Phi := \cup_{j \in \tau(\mathcal{Y}), y \in \mathcal{Y}} \varphi_{j,y}$ be the full set. For a fixed class $j \in \tau(\mathcal{Y})$, we denote the **correct** set to be $\Phi_j^\star := \cup_{y \in \mathcal{Y}} \varphi_{j,y}$, and the **incorrect** set $\Phi_j^\dagger := \cup_{j' \neq j, y \in \mathcal{Y}} \varphi_{j',y}$. A base pair $\phi = (g, y)$ belongs to $\varphi_{j,y}$ if and only if $g \in \text{Fiber}_{j,y}$. To simplify the proof, we also define for each $\varphi_{j,y} \subset \Phi$, the confounding combinations

$$\Phi_\varphi^\dagger = \bigcup_{j' \neq j \text{ XOR } y' \neq y} \varphi_{j',y'}$$

That is, the set of combinations that intersect with exactly one component with $\varphi_{j,y}$.

Definition 5.B.9 (Neuron feature indices). Let us index neuron features by $\psi = (j, r, \varphi)$ with $\varphi \in \Phi$ and $(j, r) \in \tau(\mathcal{Y}) \times [m]$, and define

$$\Psi_{j,r}^\star := \{(j, r, \varphi) : \varphi \in \Phi_j^\star\}, \quad \Psi_{j,r}^\dagger := \{(j, r, \varphi) : \varphi \in \Phi_j^\dagger\}.$$

And we further define

$$\Psi := \tau(\mathcal{Y}) \times [m] \times \Phi, \quad \Psi^\star = \bigcup_{(j,r) \in \tau(\mathcal{Y}) \times [m]} \Psi_{j,r}^\star, \quad \Psi^\dagger = \bigcup_{(j,r) \in \tau(\mathcal{Y}) \times [m]} \Psi_{j,r}^\dagger$$

The set Ψ contains all the neuron indices we care about, and $\Psi^\star$ and $\Psi^\dagger$ contain the correct and incorrect indices for class j .

V-Notations We call a pair $\phi = (g, y)$ with $g \in \mathcal{G}$ and $y \in \mathcal{Y}$ a *base pair*, or following the naming of Section 5.B.2, a *base combination*. For neuron (j, r) , we define

$$V_{j,r}(g) := \langle \mathbf{W}_{5,j,r,2}, e_g \rangle, \quad V_{j,r}(y) := \langle \mathbf{W}_{5,j,r,5}, e_y \rangle, \quad V_{j,r}(\phi) := \frac{1}{2} (V_{j,r}(g) + V_{j,r}(y)).$$

For a fixed value y and class j , the fiber $\text{Fiber}_{j,y}$ collects all permutations sending y to $\tau^{-1}(j)$. For $\psi = (j, r, \varphi_{j,y})$, we define:

$$V_\psi := \frac{1}{n_G/n_Y} \sum_{g \in \text{Fiber}_{j,y}} V_{j,r}(g, y), \quad \bar{V}_\psi := \max_{g \in \text{Fiber}_{j,y}} V_{j,r}(g, y), \quad \underline{V}_\psi := \min_{g \in \text{Fiber}_{j,y}} V_{j,r}(g, y).$$

For compactness of notation, let us also write $V_{j,r}(x) = \sum_{p=3,4} \langle \mathbf{W}_{5,j,r,1}, e_{x_0} \rangle + \langle \mathbf{W}_{5,j,r,1}, e_{x_1} \rangle$ which is a random variable depending on the randomness of x_0, x_1 .

The definition of learning curriculum for the symmetry group actions is essentially the same with the simply transitive case, with a slight difference of using the fiber-combination φ instead of the base combination ϕ . We repeat here for reference.

Definition 5.B.10 (Learning curriculum, symmetry group actions). We generate an feature index set $\Sigma^\star$ over fiber-indexed features. Letting $\Sigma_0 = \Psi$, at each $i \in [n_y^2]$, we choose

$$\psi_i = \arg \max_{\psi=(j,r,\varphi) \in \Sigma_i, \varphi \in \Phi_j^\star} V_\psi^{(0)}.$$

Write $\psi_i = (j, r, \varphi)$, we define the next iteration set Σ_{i+1} by excluding:

- (1) The confusing combinations for φ in the same neuron:

$$\Sigma_{\psi_i}^{+,1} \equiv \Sigma_i^{+,1} = \{(j, r, \varphi') \in \Sigma_i, \varphi' \in \Phi_\varphi^\star\};$$

- (2) All other combinations in the same neuron (j, r) :

$$\Sigma_{\psi_i}^{+,2} \equiv \Sigma_i^{+,2} = \{(j, r, \varphi') \mid \varphi' \in \Phi \setminus (\Phi_\varphi^\star \cup \{\varphi\})\};$$

- (3) The same combination $\varphi_{j,y}$ in other neurons of the same class $j, r', r' \neq r$:

$$\Sigma_{\psi_i}^{+,3} \equiv \Sigma_i^{+,3} = \{(j, r', \varphi_{j,y}) \in \Sigma_i : r' \neq r\}.$$

The next iteration is given by

$$\Sigma_{i+1} = \Sigma_i \setminus (\Sigma_{\psi_i}^{+,1} \cup \Sigma_{\psi_i}^{+,2} \cup \Sigma_{\psi_i}^{+,3} \cup \{\psi_i\})$$

Eventually we obtain $\Sigma^\star = (\psi_1, \dots, \psi_{n_y^2})$ and $\Sigma^\star := \bigcup_{\psi \in \Sigma^\star} \Sigma_\psi^\star$. We write $\psi <_\Sigma \psi'$ (or simply $\psi < \psi'$) if ψ precedes ψ' in $\Sigma^\star$. More generally, we also write $\psi < \psi'$ for $\psi, \psi' \in \Psi$ if $\psi \in \Sigma^\star$ and $\psi' \in \cup_{\psi'' > \psi} \Sigma_{\psi''}^\star$ is from the .

Events of combination appearance. These events specify whether a specific base combination $\phi = (g, y)$ appears, or only one of its component matches.

Definition 5.B.11 (Events of appearance). For $\phi = (g, y) \in \varphi_{j,y}$ for some $j \in \tau(\mathcal{Y})$ and $y \in \mathcal{Y}$, we define

$$\mathcal{H}_\phi := \{g_1 = g, y_0 = y\}, \quad \mathcal{H}_g := \{g_1 = g\}, \quad \mathcal{H}_y := \{y_0 = y\},$$

For the mismatched events $\mathcal{H}^\dagger$, the definition is a little different from the simply transitive actions: since $\varphi_{j,y} = \text{Fiber}_{j,y} \times \{y\}$ and $g \in \text{Fiber}_{j,y}$, we write

$$\mathcal{H}_\phi^\dagger(g) := \{g_1 = g, y_0 \neq y\}, \quad \mathcal{H}_\phi^\dagger(y) := \{g_1 \notin \text{Fiber}_{j,y}, y_0 = y\},$$

When considering the event of all the base pair $\phi \in \varphi_{j,y}$, we also define the union events

$$\mathcal{H}_\varphi := \bigcup_{\phi \in \varphi} \mathcal{H}_\phi, \quad \mathcal{H}_\varphi^\dagger := \mathcal{H}_\varphi^\dagger(y) \cup \left(\bigcup_{\phi=(g,y) \in \varphi} \mathcal{H}_\phi^\dagger(g) \right).$$

Theorem Statement Now we present the theorem for learning symmetric group actions.

Theorem 5.B.2 (Learning Symmetric Group Actions). *Assuming Assumption 5.5.1, and let $F^{(t)}$ be obtained by Algorithm 3 at $t = T_1$, then the loss for the 5th-token is minimized, i.e., $\text{Loss}_5^{(T_1)} \leq \frac{1}{\text{poly}(d)}$, and for each $\psi = (j, r, \varphi) \in \Sigma^\star$, the followings hold:*

- A. For any $\phi \in \varphi$, we have $V_{j,r}^{(T_1)}(\phi) \in [B - O(d^{c_1}\mu), B + O(\mu)]$;
- B. For any $\psi' = (j, r, \varphi') \in \Sigma^{\dagger,1}$ and $\phi' \in \varphi'$, we have $|V_{j,r}^{(T_1)}(\phi')| \leq \tilde{O}(\omega^{\frac{1}{q-1}})$. Or more concretely:

$$\frac{1}{2} \left| V_{j,r}^{(T_1)}(g') + V_{j,r}^{(T_1)}(y') \right| \leq \tilde{O}(\omega^{\frac{1}{q-1}}), \quad \forall (g', y') \in \varphi'$$

- C. For any $\phi = (g, y) \in \varphi$ the following relation holds:

$$\left| V_{j,r}^{(T_1)}(g) - C_\alpha V_{j,r}^{(T_1)}(y) \right| \leq \tilde{O}(\omega^{\frac{1}{q-1}})$$

where $C_\alpha = \frac{1+n_G(n_Y-1)/n_Y}{(n_Y-1)+n_G/n_Y} = \Theta(n_Y)$ due to our choice of n_G, n_Y in Assumption 5.B.3.

- D. For any $\psi' \in \Sigma^{\dagger,3}$, it holds that $\overline{V}_{\psi'}^{(t)} = \tilde{O}(\mu)$.

The assertions of Theorem 5.B.2 described the *feature shapes* of the neurons, which we formally defined in the proof of learning simply transitive actions (Definition 5.B.1). The main difference between the results here and Theorem 5.B.1 is that Theorem 5.B.2C described a different symmetry of features without the factor $\Theta(n_Y)$.

Let us follow the proof in Section 5.B.2 and the feature shape for symmetry group actions.

Definition 5.B.12 (Feature shape). Let $\psi = (j, r, \varphi) \in \Psi$, and let $\delta = (\delta_1, \delta_2)$ be the error parameters. We say the feature ψ has shape $\mathcal{F}_\psi(\delta)$ if the followings are satisfied:

1. $\mathcal{F}_{\psi,1}(\delta_1)$: $\underline{V}_\psi^{(t)} \geq B - O(\delta_1)$ and $\overline{V}_\psi^{(t)} \leq B + O(\delta_1)$;
2. $\mathcal{F}_{\psi,2}(\delta_2)$: For any $\psi' = (j, r, \varphi') \in \Sigma^{\dagger,1}$, it holds that $|V_{j,r}^{(t)}(\phi')| \leq O(\delta_2)$ for all $\phi' \in \varphi'$;
3. $\mathcal{F}_{\psi,3}(\delta_2)$: $|V_{j,r}^{(t)}(g) - C_Y V_{j,r}^{(t)}(y)| \leq O(\delta_2)$ for any $\phi = (g, y) \in \varphi$;
4. $\mathcal{F}_{\psi,4}$: For any feature $\psi' \in \Sigma^{\dagger,3}$, it holds that $\overline{V}_{\psi'}^{(t)} \leq \tilde{O}(\mu)$.

Indeed, we aim to prove that $\mathcal{F}_\psi(\delta)$ holds for suitable parameter δ at the end of training, for every $\psi \in \Sigma^\star$. And this is indeed what the theorem states.

Remark 5.B.4. A major difference between Definition 5.B.12 and Definition 5.B.4 is that now the conditions involve not just one combination of weights, but all combinations $\phi \in \varphi$ given a combination $\varphi \in \Phi$. This creates some significant difficulties for the proof.

Facts and Basic Calculations We now define the gradient notation and conditions.

Definition 5.B.13 (gradient notation). Let $\psi = (j, r, \varphi)$, we write Γ^+ and Γ^- to denote the different terms in the gradient computation. For any base combination $\phi = (g, y) \in \varphi_{j,y}$ where $\varphi_{j,y} \in \Phi$, we write

$$\Gamma_{j,r,\phi}^{+, (t)} := \mathbb{E}[(1 - \text{logit}_{5,j}) \text{sReLU}'(\Lambda_{5,j,r}) \mathbb{1}_{\mathcal{H}_\phi}], \quad \Gamma_{j,r,\phi}^{-, (t)} := \mathbb{E}[\text{logit}_{5,j} \text{sReLU}'(\Lambda_{5,j,r}) \mathbb{1}_{\mathcal{H}_\phi}].$$

When we consider the gradient of $V_{j,r}(g)$ or $V_{j,r}(y)$ for individual tokens $g \in \mathcal{G}$ and $y \in \mathcal{Y}$, we use the following notations: let $\phi = (g, y) \in \varphi_{j,y}$, then we define

$$\begin{aligned} \Gamma_{j,r,g}^{+, (t)} &:= \mathbb{E}[\text{logit}_{5,j} \text{sReLU}'(\Lambda_{5,j,r}) \mathbb{1}_{\mathcal{H}_\phi}] \\ \Gamma_{j,r,g}^{-, (t)} &:= \mathbb{E}[\text{logit}_{5,j} \text{sReLU}'(\Lambda_{5,j,r}) \mathbb{1}_{\mathcal{H}_\phi^+(g)}] \\ \Gamma_{j,r,y}^{-, (t)} &:= \mathbb{E}[\text{logit}_{5,j} \text{sReLU}'(\Lambda_{5,j,r}) \mathbb{1}_{\mathcal{H}_\phi^+(y)}]. \end{aligned}$$

Finally, we sum up over $\phi \in \varphi$ for the gradient of combination $\psi = (j, r, \varphi)$:

$$\begin{aligned} \Gamma_\psi^{+, (t)} &:= \mathbb{E}[(1 - \text{logit}_{5,j}) \text{sReLU}'(\Lambda_{5,j,r}) \mathbb{1}_{\mathcal{H}_\varphi}], \\ \Gamma_\psi^{-, (t)} &:= \mathbb{E}[\text{logit}_{5,j} \text{sReLU}'(\Lambda_{5,j,r}) \mathbb{1}_{\mathcal{H}_\varphi^+}]. \end{aligned}$$

Let $\psi = (j, r, \varphi)$ where $\varphi_1 = \text{Fiber}_{j,y}$. One can check that the defined $\Gamma_\psi^{+, (t)}$ is connected to the base pair Γ by the following relation:

$$\Gamma_\psi^{+, (t)} = \sum_{g \in \text{Fiber}_{j,y}} \Gamma_{j,r,g}^{+, (t)}, \quad \Gamma_\psi^{-, (t)} = \Gamma_{j,r,y}^{-, (t)} + \sum_{g \in \text{Fiber}_{j,y}} \Gamma_{j,r,g}^{-, (t)} \quad (5.B.16)$$

Definition 5.B.14 (gradient condition). At $t \leq T_1$ and $\delta > 0$, write for $\psi = (j, r, \varphi)$

$$C_\psi(\delta) : |\Gamma_\psi^{+, (t)} - \Gamma_\psi^{-, (t)}| \leq \delta, \quad C_\psi^+(\delta) : |\Gamma_\psi^{+, (t)}| \leq \delta, \quad C_\psi^-(\delta) : |\Gamma_\psi^{-, (t)}| \leq \delta.$$

These conditions control the magnitude of the gradient of the feature at iteration t . Next we compute the expressions of the gradient using

Fact 5.B.7 (gradient expressions). Equipped with Definition 5.B.14, the gradient with respect to $\mathbf{W}_{i,j,r,p}$ for $i = 5, j \in [d], r \in [m], p \in [5]$ and $v \in \mathcal{G} \cup \mathcal{Y}$ can be computed by

- (a) when $j \in \tau(\mathcal{Y})$, $r \in [m]$, $p = 5$, for $g \in \mathcal{G}$, let $\phi = (g, y) \in \varphi_{j,y}$ be the combination with $\phi_1 = g$, we have

$$\langle -\nabla_{\mathbf{w}_{5,j,r,2}} \text{Loss}, e_g \rangle = \frac{1}{2} (\Gamma_{j,r,\phi}^{+, (t)} - \Gamma_{j,r,g}^{-, (t)}) \quad (5.B.17)$$

- (b) when $j \in \tau(\mathcal{Y})$, $r \in [m]$, $p = 5$, for $y \in \mathcal{Y}$, let $\varphi_{j,y} \in \Phi_j^*$ be the combination, we have

$$\langle -\nabla_{\mathbf{w}_{5,j,r,5}} \text{Loss}, e_y \rangle = \frac{1}{2} (\Gamma_{\psi}^{+, (t)} - \Gamma_{j,r,y}^{-, (t)}) \quad (5.B.18)$$

- (c) For any combination of p, v not included above, we have due to our assumptions of the distribution and update rule:

$$\langle -\nabla_{\mathbf{w}_{5,j,r,p}} \text{Loss}, e_v \rangle \equiv 0$$

Induction Hypothesis and Training Phases We define the following intermediate time-steps:

Definition 5.B.15 (Phase decomposition). Define timestamps for ψ using the feature:

- (a) Phase I: $t \in [0, T_\psi^1]$, where $T_\psi^1 := \min\{t \geq 0 : \underline{V}_\psi^{(t)} \geq d^{c_1} \mu\}$.
- (b) Phase II.1: $t \in (T_\psi^1, T_\psi^{2,1}]$, where $T_\psi^{2,1} := \min\{t \geq 0 : V_\psi^{(t)} \geq \frac{1}{2} \log d\}$.
- (c) Phase II.2: $t \in (T_\psi^{2,1}, T_\psi^{2,2}]$, where $T_\psi^{2,2} := \min\{t \geq 0 : \mathbb{E}[\mathcal{E}_{5,j}^{(t)} \mathbb{1}_{\mathcal{H}_\varphi}] \leq d^{-1/2}\}$.
- (d) Phase II.3: $t \in (T_\psi^{2,2}, T_\psi^2]$, where $T_\psi^2 = \min\{T_\psi^{2,3}, T_{2,1} + O(\frac{1}{\eta(d^{c_1} \mu)^{q-2}})\}$, and $T_\psi^{2,3}$ is defined by

$$T_\psi^{2,3} := \min\{t \geq 0 : \mathcal{F}_\psi(\delta_1, \delta_2) \text{ holds, where } \delta_1 = d^{c_1} \mu, \delta_2 = \tilde{O}((d\omega/\lambda)^{\frac{1}{q-1}})\}$$

- (e) Phase III.1: $t \in (T_\psi^2, T_{1,1}]$, where $T_{1,1} = T_{\psi_{n_y^2}}^2$ and $\psi_{n_y^2}$ is the last feature in Σ^* .
- (f) Phase III.2: $t \in (T_{1,1}, T_1]$, the end of the training where the feature shape has stabilized

We give several induction hypotheses, each characterizing different aspects of the process.

Induction 5.B.6 (learning symmetric group actions). Let $t \leq T_1$. The following holds:

- (a) $\underline{V}_\psi^{(t)} + b_{i,j,r} \geq \Omega(\mu)$ for all $\psi \in \Sigma^*$.
- (b) $T_\psi^2 < T_{\psi'}^1$, if $\psi < \psi' \in \Sigma^*$. Thus the intervals $\{[T_\psi^1, T_\psi^2]\}_{\psi \in \Sigma^*}$ are non-overlapping.
- (c) Let $\psi' < \psi \in \Sigma^*$, then the feature shape $\mathcal{F}_{\psi'}(d^{c_1} \sigma_0, \omega^{\frac{1}{q-1}})$ holds throughout $t \in [T_{\psi'}^1, T_1]$;
- (d) For any $\psi = (j, r, \varphi) \in \Sigma^*$ There are at most $\tilde{O}(\sqrt{d}/\eta)$ iterations where $\mathbf{logit}^{(t)} \geq \frac{1}{\sqrt{d}}$ conditioned on $\mathcal{H}_\varphi$ in $t \in [0, T_\psi^1]$.

Interpretations of Induction 5.B.6. We explain what the inductions hypothesis mean here:

- (a) This property simply asserts that the *good* features never shrink below a certain level which is crucial for the optimization to work.
- (b) This property describes the *curriculum* nature of Σ^* : each feature V_ψ grows in different time intervals within $0 \leq t \leq T_1$.
- (c) This property asserts that the feature ψ' previous to ψ in the curriculum Σ^* has stable and good feature shapes, which helps with learning the current feature.
- (d) This property means there are very few iteration where the gradient of $V_{j,r}(\phi)$, $\phi \in \varphi$ is disturbed before $T_{j,r,\varphi}^1$ is reached.

Phase I: Emergence of the Feature

We first prove some properties at the beginning of phase I.

Lemma 5.B.25 (Initialization range for features). *Under random initialization and Assumption 5.B.3, with probability at least $1 - o(1)$, the following holds*

- (a) For any $v \in \mathcal{V}$, we have $|\langle \mathbf{W}_{5,j,r,p}^{(0)}, e_x \rangle| \leq O(\mu/\sqrt{\log d})$ for any j, r, p .
- (b) Let $\psi \in \Sigma^*$, $\psi' \in \Psi$ and $\psi < \psi'$, then $V_\psi^{(0)} \geq V_{\psi'}^{(0)} + \Delta_0$ for a gap $\Delta_0 = \Omega(\sigma_0/\text{polylog} d)$.
- (c) We have $\Lambda_{5,j,r}^{(0)}(\mathbf{Z}) = \Omega(\mu)$ for all $j \in [d]$, $r \in [m]$.

Proof. The proof is basically the same as that of Fact 5.B.4. The first one uses basic concentration properties of Gaussian distribution and the second uses a union bound with the anti-concentration property of the Gaussian distribution. (c) is based on the fact that

$$\Lambda_{5,j,r}^{(0)}(\mathbf{Z}) = b_{5,j,r} + \frac{1}{2} \sum_{\mathbf{k} \in \mathcal{I}^{1,0}} \sum_{p \in [5]} \langle \mathbf{W}_{5,j,r,p}^{(0)}, \mathbf{Z}_{\mathbf{k},p} \rangle = \mu \pm O(\mu/\sqrt{\log d})$$

which concludes the proof. $\square$

Corollary 5.B.2. *For any $\psi = (j, r, \varphi) \in \Sigma^*$ and $y \in \varphi^2$, we have with probability $\geq 1 - o(1)$:*

$$\left| \frac{1}{|\varphi|} \sum_{g \in \varphi^1} \mathbb{E}_x[\mathbf{sReLU}'(\Lambda_{5,j,r}^{(0)}) \mid \mathcal{H}_{g,y}] - \mathbb{E}_x[\mathbf{sReLU}'(V_{j,r}^{(0)}(y) + V_{j,r}(x) + b_{5,j,r})] \right| \leq O\left(\frac{1}{\sqrt{n_G/n_Y}}\right) \mathbb{E}_x[\mathbf{sReLU}'(V_{j,r}^{(0)}(y) + V_{j,r}(x) + b_{5,j,r})]$$

Proof. This is true because of Lemma 5.B.25a and Hoeffding's inequality. $\square$

Now we present an induction for phase I.

Induction 5.B.7. *Let $\psi = (j, r, \varphi) \in \Sigma^*$, for $t \leq T_\psi^1$, the gradients satisfy $\Gamma_{j,r,y}^{+, (t)} \geq \Omega(\frac{n_G}{n_Y}) \Gamma_{i,j,g}^{+, (t)}$ for any $(g, y) \in \varphi$.*

Using Fact 5.B.2 with a representative $\phi = (g, y) \in \varphi_{j,y}$ and the definition $V_{j,r}(\phi) = \frac{1}{2}(V_{j,r}(g) + V_{j,r}(y))$, a single gradient step with step size η yields

$$\begin{aligned} V_{j,r}^{(t+1)}(\phi) - V_{j,r}^{(t)}(\phi) &= \frac{\eta}{2} \left(\langle -\nabla_{\mathbf{w}_{5,j,r,5}} \text{Loss}, e_y \rangle + \langle -\nabla_{\mathbf{w}_{5,j,r,2}} \text{Loss}, e_g \rangle \right) \\ &= \frac{\eta}{2} \left(\Gamma_{\psi}^{+, (t)} + \Gamma_{j,r,\phi}^{+, (t)} - \Gamma_{j,r,g}^{-, (t)} - \Gamma_{j,r,y}^{-, (t)} \right) \end{aligned} \quad (5.B.19)$$

Similar to the proof of the simply transitive actions, we define a proxy term and argue that the trajectory of V_{ψ} is similar to the proxy.

Lemma 5.B.26 (Approximating the gradient with proxy). *Assuming Inductions 5.B.6 and 5.B.7 holds at t . Let $\psi = (j, r, \varphi) \in \Psi$, for every $\phi = (g, y) \in \varphi$, we define two proxy sequences $\tilde{V}_{j,r}^{(t)}(v)$, $v \in \{g, y\}$ by the update rule:*

$$\tilde{V}_{j,r}^{(t+1)}(y) = \tilde{V}_{j,r}^{(t)}(y) + \frac{\eta}{2} \tilde{\Gamma}_{\psi}^{+, (t)}, \quad \tilde{V}_{j,r}^{(t+1)}(g) = \tilde{V}_{j,r}^{(t)}(g) + \frac{\eta}{2} \tilde{\Gamma}_{j,r,g}^{+, (t)},$$

where both $\tilde{V}_{j,r}(g)$ and $\tilde{V}_{j,r}(y)$ start from $\tilde{V}_{j,r}^{(0)}(y) := V_{j,r}^{(0)}(y)$, $\tilde{V}_{j,r}^{(0)}(g) := V_{j,r}^{(0)}(g)$. And the $\tilde{\Gamma}$ are defined by replacing the features of $V_{j,r}(v)$, $v \in \varphi^1 \cup \{y\}$ with $\tilde{V}_{j,r}(v)$. Now for all $t \leq T_{\psi}^1$, we have

$$|\tilde{V}_{j,r}^{(t)}(\phi) - V_{j,r}^{(t)}(\phi)| \leq \tilde{O}(\mu/d^{\frac{1}{2}-3c_1}), \quad \text{where } \tilde{V}_{j,r}^{(t)}(\phi) = \tilde{V}_{j,r}^{(t)}(g) + \tilde{V}_{j,r}^{(t)}(y).$$

Proof. Fix $\psi = (j, r, \varphi_{j,y})$. (5.B.19) gives

$$|\tilde{V}_{j,r}^{(t)}(y) - V_{j,r}^{(t)}(y)| = \frac{\eta}{2} \Gamma_{j,r,y'}^{-, (t)}, \quad |\tilde{V}_{j,r}^{(t)}(g) - V_{j,r}^{(t)}(g)| = \frac{\eta}{2} \Gamma_{j,r,g}^{-, (t)}$$

Since Induction 5.B.7 holds at t , we have that $\tilde{V}_{j,r}^{(t+1)}(y) - \tilde{V}_{j,r}^{(t)}(y) \geq \Omega(\frac{n_G}{n_Y})(\tilde{V}_{j,r}^{(t+1)}(g) - \tilde{V}_{j,r}^{(t)}(g))$.

Moreover, since $\sum_{g \in \varphi^1} \Gamma_{j,r,g}^{+, (t)} = \Gamma_{j,r,y}^{+, (t)}$ by Definition 5.B.13. This, combined with Induction 5.B.6, implies that

$$\tilde{V}_{j,r}^{(t+1)}(y) - \tilde{V}_{j,r}^{(t)}(y) \geq \sum_{g \in \varphi^1} \Gamma_{j,r,g}^{+, (t)} \geq \Omega(\frac{n_G}{n_Y})(\tilde{V}_{j,r}^{(t+1)}(g) - \tilde{V}_{j,r}^{(t)}(g)), \quad \forall g \in \varphi^1$$

The above update bound imply, for all $t \leq T_{\psi}^1$, the following property hold: $\tilde{V}_{j,r}^{(t)}(g) - \tilde{V}_{j,r}^{(0)}(g) \leq O(\frac{1}{n_G/n_Y})(\tilde{V}_{j,r}^{(t)}(y) - \tilde{V}_{j,r}^{(0)}(y))$ for all $g \in \varphi^1$. Now, by the same induction procedure in the proof of Lemma 5.B.14, we know that the sum

$$\left| V_{j,r}^{(t)}(y) + \sum_{g \in \varphi^1} V_{j,r}^{(t)}(g) - \tilde{V}_{j,r}^{(t)}(y) + \sum_{g \in \varphi^1} \tilde{V}_{j,r}^{(t)}(g) \right| \leq O(\mu/d^{\frac{1}{2}-3c_1})$$

And then by the above relations we know

$$|\tilde{V}_{j,r}^{(t)}(y) - V_{j,r}^{(t)}(y)| \leq O(\mu/d^{\frac{1}{2}-3c_1})$$

Thus the conclusion holds. $\square$

Lemma 5.B.27 (competition). *Let $\psi \in \Sigma^\star$. Assuming Inductions 5.B.6 and 5.B.7 holds at $t \leq T_\psi^1$. Then at $t = T_\psi^1$, we have for any feature $\psi' \in \Psi$, $\psi' > \psi$, $\bar{V}_{\psi'}^{(t)} \leq \tilde{O}(\mu)$.*

Proof. From Lemma 5.B.26, we know that

$$\tilde{V}_{j,r}^{(t+1)}(y) = \tilde{V}_{j,r}^{(t)}(y) \geq \frac{\eta}{2} \sum_{g \in \varphi^1} \mathbb{E}_x[(\tilde{V}_{j,r}^{(t)}(y) + \tilde{V}_{j,r}^{(t)}(g) + b_{5,j,r} + V_{j,r}(x))^{q-1} \mathbb{1}_{\mathcal{H}_{g,y}}]$$

Suppose there is another $\psi' = (\varphi', y') \in \Sigma^\star$ such that $\psi' > \psi$, then we can also write a comparison based on Lemma 5.B.25:

$$\tilde{V}_{j,r}^{(0)}(y) \geq \tilde{V}_{j,r}^{(0)}(y') + \Delta_0,$$

And therefore one can compute by absolutely bounding the average over $g \in \varphi$ using Corollary 5.B.2:

$$\begin{aligned} \Gamma_\psi^{+, (0)} &= \frac{1}{n_Y |\varphi^1|} \sum_{g \in \varphi^1} \mathbb{E}_x[(\tilde{V}_{j,r}^{(0)}(y) + \tilde{V}_{j,r}^{(0)}(g) + b_{5,j,r} + V_{j,r}(x))^{q-1}] \\ &\geq \frac{1}{n_Y} \left(1 - \frac{1}{\sqrt{n_G/n_Y}}\right) \mathbb{E}_x[(\tilde{V}_{j,r}^{(0)}(y) + b_{5,j,r} + V_{j,r}(x))^{q-1}] \end{aligned}$$

Similarly, for $\psi' = (j', r', \varphi')$ and $\phi' \in \varphi'$, we have

$$\begin{aligned} \Gamma_{\psi'}^{+, (0)} &= \frac{1}{|\varphi'^1|} \sum_{g \in \varphi'^1} \mathbb{E}_x[(\tilde{V}_{j',r'}^{(0)}(y') + \tilde{V}_{j',r'}^{(0)}(g) + b_{5,j',r'} + V_{j',r'}(x))^{q-1} \mid \mathcal{H}_{g,y'}] \\ &\leq \left(1 + \frac{1}{\sqrt{n_G/n_Y}}\right) \mathbb{E}_x[(\tilde{V}_{j',r'}^{(0)}(y') + b_{5,j',r'} + V_{j',r'}(x))^{q-1}] \end{aligned}$$

Now we can compare the proxies $\tilde{V}_{j,r}(y) + b_{5,j,r}$ with $\tilde{V}_{j,r}(y')$ using Lemma 5.B.24 to get that when $V_{j,r}(y) \geq d^{c_1} \mu$ we have $V_{j',r'}(y') \leq \tilde{O}(\mu)$. Now due to Lemma 5.B.26 and Induction 5.B.7, we have the desired result. $\square$

Proof of Inductions 5.B.6 and 5.B.7 in Phase I. Fix $\psi = (j, r, \varphi) \in \Sigma^\star$. We check items (a)-(d) in Induction 5.B.6:

- Induction 5.B.6a: This is simple by combining Lemmas 5.B.25 and 5.B.26, where each $V_{j,r}(\phi)$ is well approximated by a monotonically growing proxy $\tilde{V}_{j,r}(\phi)$.
- Induction 5.B.6b: This is proved later but here Lemma 5.B.27 showed that for any pair $\psi < \psi' \in \Sigma^\star$, that $T_\psi^1 < T_{\psi'}^1$, which is a precondition for Induction 5.B.6b.
- Induction 5.B.6c-d: These two are proved later in future phases and are not violated in phase I.

- **Induction 5.B.7:** We shall show that this is the case. Consider any feature $g \in \varphi^1$, its gradient is clearly bounded by $O(n_Y/n_G)\Gamma_\psi^{+, (t)}$ initially before the feature $V_{j,r}(g) \geq \Omega(\mu)$. In this range the difference of its growth with the update of $V_{j,r}(y)$ averaged over all $g \in \varphi^1$ is bounded by $O(n_Y/n_G)(\tilde{V}_{j,r}^{(t)}(y) + b_{5,j,r})^{q-1}$ because it grows slower than $O(n_Y/n_G)\tilde{V}_{j,r}^{(t)}(y)$ and thus Hoeffding's inequality with variance parameter μ would work to bound the difference between the averaged gradient and the gradient with $V_{j,r}(g)$ in expectation, and thus we obtain a bound $\Gamma_\psi^{+, (t)} \geq \Omega(n_G/n_Y)\Gamma_{j,r,g}^{+, (t)}$. When $V_{j,r}(g) \geq \Omega(\mu)$, the feature $V_{j,r}(y)$ has grown more than $\Omega(n_G/n_Y)\mu$ and the growth ratio still holds. This proved the induction for $t \leq T_\psi^1$. $\square$

Phase II: Feature Growth and Cancellations

Below we present an additional induction hypothesis for phase II.

Induction 5.B.8 (Phase II). *For all features $\psi = (j, r, \varphi) \in \Sigma^\star$ and $t \in [T_\psi^1, T_\psi^2]$, then for any $g' \notin \varphi^1, y' \notin \varphi^2$, we have $V_{j,r}^{(t)}(g'), V_{j,r}^{(t)}(y') \in (-B - \tilde{O}(\mu), \tilde{O}(\mu))$.*

Technical Lemmas

Lemma 5.B.28 (gradient estimation). *Assume Inductions 5.B.6 and 5.B.8. Let $\psi = (j, r, \varphi_{j,y}) \in \Sigma^\star$, let $\phi = (g, y) \in \varphi_{j,y}$, then at $t \in [T_\psi^1, T_\psi^2]$, we have*

- (a) *For $t \in [T_\psi^1, T_\psi^{2,1}]$, we have $\Gamma_\psi^{(t)} \geq \Omega(\Pr(\mathcal{H}_\varphi))\mathbf{sReLU}'(\underline{V}_\psi^{(t)})$;*
- (b) *If $V_{j,r}^{(t)}(\phi) \in (\frac{1}{3} \log d, \frac{2}{3} \log d)$, then $\Gamma_{j,r,\phi}^{(t)} = \Gamma_{j,r,\phi}^{+, (t)} - \Gamma_{j,r,\phi}^{-, (t)} \geq \Omega(\Pr(\mathcal{H}_\phi))$;*
- (c) *For any $g' \notin \varphi^1$ and any $y' \neq y$,*

$$\begin{aligned}\Gamma_{j,r,g'}^{(t)} &= -\mathbb{E}[\mathbf{logit}_{5,j}^{(t)} \mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)}) \mathbb{1}_{\mathcal{H}_{(g',y)}}] \pm O(\omega n_Y/n_G), \\ \Gamma_{j,r,y'}^{(t)} &= -\sum_{g \in \varphi^1} \mathbb{E}[\mathbf{logit}_{5,j}^{(t)} \mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)}) \mathbb{1}_{\mathcal{H}_{(g,y')}}] \pm O(\omega).\end{aligned}$$

- (d) *Any individual component $V_{j,r}^{(t)}(v), v \in \varphi^1 \cup \varphi^2$ cannot fall below $O(\mu)$, and $\Gamma_{j,r,v}^{+, (t)} \geq 0$.*

Proof. Write $\psi = (j, r, \varphi_{j,y})$.

- (a) By the definition of $T_\psi^{2,1}$, we know there exist $\phi \in \varphi$ such that $V_{j,r}(\phi) \leq \frac{1}{2} \log d$. By the same arguments in Lemma 5.B.26, we can get that all $\phi \in \varphi$ satisfy $\frac{1}{2} \log d + o(1)$. Given that Induction 5.B.6 holds at t , conditioned on $\mathcal{H}_\phi$, all other neuron $(j', r') \neq (j, r)$ has activation smaller than ϱ . Therefore $F_{5,j'} \leq o(1)$, and $\mathbf{logit}_{5,j'} = o(1)$ for any $j' \neq j$ conditioned on $\mathcal{H}_\phi$. So conditioned on $\mathbf{Z} \in \mathcal{H}_\phi$, when $V_{j,r}^{(t)}(\phi) \leq \frac{2}{3} \log d$, we have

$$\mathbf{logit}_{5,j} = \frac{e^{F_{5,j}(\mathbf{Z})}}{e^{F_{5,j}(\mathbf{Z})} + (1 + o(1))(d-1)} \leq O(d^{-\Omega(1)})$$

Now we can simply bound all $\Gamma_{j,r,v}^{-(t)}$ for $v \in \varphi^1 \cup \varphi^2$ to be smaller than $\widetilde{O}(\frac{1}{\sqrt{d}})\Gamma_{j,r,v}^{+(t)}$. Thus the proof is obtained by computing $\Gamma_{j,r,v}^{+(t)}$ and sum to get $\Gamma_{\psi}^{+(t)}$.

- (b) Similar to (a), here we also notice that $\mathbf{sReLU}'(\Lambda_{5,j,r}(\mathbf{Z})) = 1$ because $\Lambda_{5,j,r}(\mathbf{Z}) \in [\varrho, B]$ is in the linear regime. Now we can see that the terms $\Gamma_{j,r,\phi}^{+(t)} \geq \Omega(\Pr(\mathcal{H}_{\phi}))$ and $\Gamma_{j,r,\phi}^{-(t)} \leq O(d^{-\Omega(1)})$ and thus we have the result.

- (c) For $g' \notin \text{Fiber}_{j,y'}$, let $y' \in \mathcal{Y}$ such that $g' \in \text{Fiber}_{j,y'}$, then we have

$$\begin{aligned} \Gamma_{j,r,g'}^{(t)} &= \mathbb{E}[(1 - \mathbf{logit}_{5,j})\mathbf{sReLU}'(\Lambda_{5,j})\mathbb{1}_{\mathcal{H}_{(g',y')}}] - \sum_{y' \neq y', y} \mathbb{E}[\mathbf{logit}_{5,j}\mathbf{sReLU}'(\Lambda_{5,j})\mathbb{1}_{\mathcal{H}_{(g',y')}}] \\ &\quad - \mathbb{E}[\mathbf{logit}_{5,j}\mathbf{sReLU}'(\Lambda_{5,j})\mathbb{1}_{\mathcal{H}_{(g',y)}}] \end{aligned}$$

Since $\mathcal{H}_{(g',y)}$ is confusing, the negative term dominates; other appearances contribute at most $O(\omega n_Y/n_G)$ or $O(\mu^{q-1})$. Similar arguments hold for y' .

- (d) This one is proven similarly to (a) but can adapt to any level of $V_{j,r}(\phi)$ for the corresponding $\phi \in \varphi$ for $v \in \varphi^1 \cup \varphi^2$. From Induction 5.B.8c we know that all other feature $V_{j,r}(v') \leq \widetilde{O}(\mu)$ for $v' \notin \varphi^1 \cup \varphi^2$. So for any $v \in \varphi^1 \cup \varphi^2$ such that $V_{j,r}(v) \leq \widetilde{O}(\mu)$, we have that the combination $\phi = (v, v') \in \Phi \setminus \varphi$ has $V_{j,r}(\phi) < O(\mu)$ and thus have gradient

$$\Gamma_{j,r,v}^{(t)} = \Gamma_{j,r,v}^{+(t)} - \Gamma_{j,r,v}^{-(t)} \geq \Gamma_{j,r,v}^{+(t)} - O(\mu^{q-1})$$

Note that $\Gamma_{j,r,v}^{+(t)}$ is either $\geq \lambda/d$ or 0 at this phase. So we can sum up the negative gradient $-O(\mu^{q-1})$ for all iterations $t \in [T_{\psi}^1, T_{\psi}^2]$ (which is fewer than $\widetilde{O}(\frac{1}{\eta(d^{c_1}\mu)^{q-2}})$) to get that the feature $V_{j,r}(v) \gg \mu$. This means $\Lambda_{5,j,r} \geq 0$ when $\mathcal{H}_{\phi}$ happens and thus $\Gamma_{j,r,v}^{+(t)} \geq 0$ for all $t \in [T_{\psi}^1, T_{\psi}^2]$. □

Lemma 5.B.29 (feature magnitude). *Assume Inductions 5.B.6 and 5.B.8 holds. For any feature $\psi = (j, r, \varphi) \in \Sigma^*$, we have*

- (a) *when $C_{\psi}^+(\delta)$ holds for some $\delta \in [n_G\omega, O(\lambda/d^{c_1})]$ at $t \geq T_{\psi}^{2,1}$, we have $V_{j,r}^{(t)}(\phi) \geq B - O(d^{c_1}\mu)$, $\forall \phi \in \varphi$.*
(b) *when $C_{\psi}^-(\delta)$ holds for some $\delta \in [n_G\omega, O(\lambda/d^{1+c_1})]$ at $t \geq T_{\psi}^{2,1}$, we have $V_{j,r}^{(t)}(\phi) \in [-B - O(\mu), \widetilde{O}((\frac{d\delta}{\lambda})^{\frac{1}{q-1}})]$ for all $\phi \in \Phi_{\varphi}^+$.*

Proof. (a) If $C_{\psi}^+(\delta)$ holds with $\delta \leq O(\lambda/d^{c_1})$, then there are two probability: $V_{j,r}^{(t)}(\phi)$ for all $\phi \in \varphi$ is as small as $O(\delta^{\frac{1}{q-1}})$ which is not possible for δ so small because of Lemma 5.B.28a; or that $V_{j,r}^{(t)}(\phi) \geq B - O(d^{c_1}\mu)$ such that $F_{5,j} \geq B$ and thus the gradient has vanished for some events.

- (b) The logic from (a) can be applied here as well. Since the positive gradient is at most $O(\omega)$ by Lemma 5.B.28b, thus whenever $\delta \gg \omega$, the negative gradient conditioned on $\mathcal{H}_{\phi}$, $\phi \in \Phi_{\varphi}^+$ is

bounded, which gives

$$\begin{aligned} \sum_{\phi \in \varphi} \mathbb{E}[\mathbf{logit}_{5,j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \mathbb{1}_{\mathcal{H}_\phi}] &\leq O(\delta) \implies (\Lambda_{5,j,r})^{q-1} \mathbb{1}_{\mathcal{H}_\phi} \leq \delta \lambda / (d \Pr(\mathcal{H}_\phi)) \\ &\quad \text{(because Fact 5.B.3)} \\ \implies V_{j,r}^{(t)}(\phi) &\lesssim \left(\frac{n_G n_Y \lambda \delta}{d} \right)^{\frac{1}{q-1}}, \forall \phi \in \varphi \end{aligned}$$

which proves the desired result. $\square$

Lemma 5.B.30 (gradient stationarity). *Let $\psi = (j, r, \varphi) \in \Sigma^\star$, then for any $\delta > \omega^{q-2}$, the followings hold:*

- (a) Define $\mathcal{B}_\delta^+ := \{t \in [T_\psi^{2,1}, T_\psi^{2,3}] \mid C_\psi^+(\delta) \text{ doesn't hold}\}$ then $|\mathcal{B}_\delta^+| \leq O(\frac{\log^2 d}{\eta \delta})$;
- (b) Define $\mathcal{B}_\delta^- := \{t \in [T_\psi^{2,1}, T_\psi^{2,3}] \mid C_\psi^-(\delta) \text{ doesn't hold}\}$ then $|\mathcal{B}_\delta^-| \leq O(\frac{\log^3 d}{\eta \delta})$;
- (c) For some $t = T_\psi^{2,3} \geq T_\psi^{2,1} + O(\frac{1}{\eta n_G \omega})$, we have $V_{j,r}^{(t)}(\phi) \geq B - O(d^{c_1} \mu)$ for all $\phi \in \varphi$ and $|V_{j,r}^{(t)}(\phi')| \leq \tilde{O}((\frac{d\omega}{\lambda})^{\frac{1}{q-1}})$ for all $\phi' \in \Phi_\varphi^+$, moreover, this will hold until $t = T_\psi^2$

Proof. The proof is very similar in spirit to the proof of Lemma 5.B.18. We first define the following quantity:

$$\Upsilon_\psi^{(t)} = V_{j,r}^{(t)}(y) - \sum_{g \notin \varphi^1} V_{j,r}^{(t)}(g)$$

One can compute that the update of $\Upsilon_\psi^{(t)}$ is simply

$$\eta \left(\Gamma_\psi^{+, (t)} - \sum_{\varphi' \neq \varphi \in \Phi_j^\star} \sum_{\phi \in \varphi'} \Gamma_{j,r,\phi}^{+, (t)} + \sum_{\phi' \notin \varphi \cup \Phi_\varphi^+} \Gamma_{j,r,\phi'}^{-, (t)} \right)$$

The middle term is bounded by $O(\omega)$ and the last term is bounded below by $-O(\omega)$, thus we have that the update satisfy

$$\Upsilon_\psi^{(t+1)} - \Upsilon_\psi^{(t)} \geq \eta \Gamma_\psi^{+, (t)} - O(\eta \omega)$$

Since $\Gamma_\psi^{+, (t)} \geq 0$ For any $\delta \gg \omega$ by Lemma 5.B.28c, we know that the above bound accumulate at $t \geq T_\psi^{2,1}$ at

$$\Upsilon_\psi^{(t)} - \Upsilon_\psi^{(T_\psi^{2,1})} \geq \sum_{s \in [T_\psi^{2,1}, t] \cup \mathcal{B}_\delta^+} \eta \delta - O(\eta \omega)$$

So for no more than $O(\frac{\log^2 d}{\eta\delta})$ iterations we should have the that $\Upsilon^{(t)} \geq \Omega(\log^2 d)$, which is impossible in our setting. This contradiction proves the result for (a). The proof of (b) is similar to (a) and the proof of Lemma 5.B.18b. (c) is also similar to Lemma 5.B.18c by using Lemma 5.B.29a-b with parameter λ/d^{c_1} and parameter $n_G\omega$, along with Lemma 5.B.31. We do not repeat the details here. $\square$

Lemma 5.B.31 (Symmetry of features). *Let $\psi = (j, r, \varphi) \in \Sigma^\star$ and $t \geq T_\psi^{2,3}$, we have that if $\mathcal{F}_{\psi,2}(\delta)$ hold for some $\delta > 0$, then*

$$|V_{j,r}^{(t)}(g) - C_\alpha V_{j,r}^{(t)}(y)| \leq O(\delta)$$

For some $C_\alpha = \frac{1+n_G(n_Y-1)/n_Y}{(n_Y-1)+n_G/n_Y}$.

Proof. For a feature $\psi = (j, r, \varphi) \in \Sigma^\star$, their update can be represented as:

$$\begin{aligned} V_{j,r}^{(t)}(g) - V_{j,r}^{(0)}(g) &= U_{g,y} - \sum_{y' \neq y} R_{g,y'} \\ V_{j,r}^{(t)}(y) - V_{j,r}^{(0)}(y) &= \sum_{g \in \text{Fiber}_{j,y}} U_{g,y} - \sum_{g' \notin \text{Fiber}_{j,y}} R_{g',y} \end{aligned}$$

where $U_{g,y} = \sum_{s \leq t} \eta \Gamma_{j,r,\phi}^{+, (s)}$ and $R_{g,y'}$ is

$$R_{g,y'} = \sum_{s \leq t} \eta \mathbb{E}[\text{logit}_{5,j}^{(s)} \text{sReLU}'(\Lambda_{5,j,r}^{(s)}) \mathbb{1}_{\mathcal{H}_{g',y}}]$$

Therefore:

$$\sum_{g \in \text{Fiber}_{j,y}} V_{j,r}^{(t)}(g) = \sum_{g \in \text{Fiber}_{j,y}} U_{g,y} - \sum_{g \in \text{Fiber}_{j,y}} \sum_{y' \neq y} R_{g,y'}$$

Thus it is true for any $\text{Fiber}_{j,y}$ and $y \in \mathcal{Y}$ that the following holds

$$V_{j,r}^{(t)}(y) - V_{j,r}^{(0)}(y) + \sum_{g' \notin \text{Fiber}_{j,y}} R_{g',y} = \sum_{g \in \text{Fiber}_{j,y}} (V_{j,r}^{(t)}(g) - V_{j,r}^{(0)}(g)) + \sum_{g \in \text{Fiber}_{j,y}} \sum_{y' \neq y} R_{g,y'}$$

By summing both the left and right hand side with all the $(\text{Fiber}_{j,y}, y)$ pairs, we have

$$\sum_{y \in \mathcal{Y}} (V_{j,r}^{(t)}(y) - V_{j,r}^{(0)}(y)) + \sum_{y \in \mathcal{Y}} \sum_{g' \notin \text{Fiber}_{j,y}} R_{g',y} = \sum_{g \in \mathcal{G}} (V_{j,r}^{(t)}(g) - V_{j,r}^{(0)}(g)) + \sum_{g \in \mathcal{G}} \sum_{y': \tau(g \cdot y') \neq j} R_{g,y'}$$

Since $\sum_{y \in \mathcal{Y}} \sum_{g' \notin \text{Fiber}_{j,y}} R_{g',y} = \sum_{g \in \mathcal{G}} \sum_{y': \tau(g \cdot y') \neq j} R_{g,y'}$, we have

$$\sum_{y \in \mathcal{Y}} (V_{j,r}^{(t)}(y) - V_{j,r}^{(0)}(y)) = \sum_{g \in \mathcal{G}} (V_{j,r}^{(t)}(g) - V_{j,r}^{(0)}(g))$$

$$\implies \sum_{y \in \mathcal{Y}} V_{j,r}^{(t)}(y) = \sum_{g \in \mathcal{G}} V_{j,r}^{(t)}(g) \pm O(\mu)$$

Moreover, we have assumed

$$\begin{aligned} |V_{j,r}^{(t)}(g) + V_{j,r}^{(t)}(y')| &= O(\delta), \quad \forall y' \neq y, g \in \text{Fiber}_{j,y}, \\ |V_{j,r}^{(t)}(y) + V_{j,r}^{(t)}(g')| &= O(\delta), \quad \forall g' \notin \text{Fiber}_{j,y} \end{aligned}$$

Thus

$$(1 + \frac{|\mathcal{G}|}{|\mathcal{Y}|}(|\mathcal{Y}| - 1))V_{j,r}^{(t)}(y) = (\frac{|\mathcal{Y}|(|\mathcal{Y}| - 1)}{|\mathcal{G}|} + 1) \sum_{g \in \text{Fiber}_{j,y}} V_{j,r}^{(t)}(g) \pm O(n_G \delta)$$

Therefore choosing whatever $g \in \text{Fiber}_{j,t}$, we have

$$(1 + \frac{|\mathcal{G}|}{|\mathcal{Y}|}(|\mathcal{Y}| - 1))V_{j,r}^{(t)}(y) = (\frac{|\mathcal{Y}|^2}{|\mathcal{G}|} + 1) \frac{|\mathcal{G}|}{|\mathcal{Y}|} V_{j,r}^{(t)}(g) \pm O(n_G \delta_2) = (|\mathcal{Y}| - 1 + \frac{|\mathcal{G}|}{|\mathcal{Y}|})V_{j,r}^{(t)}(g) \pm O(n_G \delta)$$

Now by dividing the right-hand side by the factor on $V_{j,r}^{(t)}(y)$ on the LHS, we have the desired result. $\square$

Proof of Induction

Lemma 5.B.32 (Arrival times). *Assuming Inductions 5.B.6 and 5.B.8 for $\psi = (j, r, \varphi) \in \Sigma^\star$, we have*

- (a) $T_\psi^{2,1} - T_\psi^1 \leq \tilde{O}(\frac{1}{\eta(d^{c_1}\sigma_0)^{q-2}})$;
- (b) $T_\psi^{2,2} - T_\psi^{2,1} \leq O(\frac{d^{\frac{1}{2}+c_1}}{\eta})$;
- (c) $T_\psi^{2,3} \leq \tilde{O}(\frac{1}{\eta\omega})$;
- (d) $T_\psi^2 \leq \tilde{O}(\frac{1}{\eta(d^{2c_1}\mu)^{q-2}})$.

Proof. (a) By Lemma 5.B.28a, after phase I, within $\tilde{O}(1/(\eta(d^{c_1}\sigma_0)^{q-2}))$ we have $V_{j,r}^{(t)}(\phi) \geq \frac{1}{2} \log d$ for all $\phi \in \varphi$.

(b) On $[T^1, T^{2,1}]$, Lemma 5.B.30a gives the rate $O(d^{\frac{1}{2}+c_1}/\eta)$ time for the positive grad to arrive at $1 - \text{logit}_{5,j} \leq \frac{1}{\sqrt{d}}$ conditioned on $\mathcal{H}_\varphi$.

(c) This is again provided by Lemma 5.B.30c by using $\delta = n_G \omega$, combined with Lemma 5.B.29b for the feature shape guarantees.

(d) This is simply by the definition of T_ψ^2 and that $T_\psi^{2,3} \ll \frac{1}{\eta(d^{2c_1}\mu)^{q-2}}$. $\square$

Proof of Inductions 5.B.6 and 5.B.8. Fix $\psi = (j, r, \varphi_{j,y})$ and consider $t \in [T_\psi^1, T_\psi^2]$, then

- Induction 5.B.6a: This is guaranteed by Lemma 5.B.28a-b.
- Induction 5.B.6b: Because the time used is small between T_ψ^1 and T_ψ^2 by Lemma 5.B.32, we have that $T_{\psi'}^1$ is still behind when this happens.

- Induction 5.B.6c: This will be proven in phase III, we do not use any properties that violate this hypothesis.
- Induction 5.B.6d: This is due to Induction 5.B.8a-b and Lemma 5.B.30a with parameter $\delta = 1/\sqrt{d}$.
- Induction 5.B.8: For any $\phi \notin \varphi$, we have that they are either desirable features $\phi \in \varphi_{j,y'}$ for some different $y' \geq y \in \mathcal{Y}$, or that they are the wrong features for the class j , i.e., $\phi \in \varphi_{j',y}$ for all $y \in \mathcal{Y}$. The feature $\psi' = (j, r, \varphi_{j,y'})$ in the former case was out-competed by $\psi = (j, r, \phi)$ so their growth is bounded by $\tilde{O}(\mu)$ in the first two stage and negative in the end due to the feature shape $\mathcal{F}_\psi(d^{c_1}\mu, (\frac{d\omega}{\lambda})^{\frac{1}{q-1}})$ at the end as shown in Lemma 5.B.30c. Combination ϕ in the latter case will be smaller as well due to Lemma 5.B.30c.

The time bounds follow from Lemma 5.B.32. This proves the induction in phase II for any feature $\psi \in \Sigma^*$. $\square$

Phase III: Convergence

We present the induction hypothesis in this phase.

Induction 5.B.9 (Phase III, final). *Let $\psi = (j, r, \varphi) \in \Sigma^*$, for $t \in [T_\psi^2, T_1]$, the following holds:*

- (a) *At $t \in [T_\psi^2, T_{1,1}]$, we have $\mathcal{F}_\psi(\delta_1, \delta_2)$ holds with $\delta_1 = d^{c_1}\mu$, $\delta_2 = \tilde{O}((\frac{d\omega}{\lambda})^{\frac{1}{q-1}})$;*
- (b) *At $t \in [T_{1,1}, T_1]$, we have $\mathcal{F}_\psi(\delta_1, \delta_2)$ holds with $\delta_1 = d^{c_1}\mu$, $\delta_2 = \tilde{O}((\omega)^{\frac{1}{q-1}})$.*

We obtain the shape of logits at convergence.

Lemma 5.B.33 (Logit shape at convergence). *Assuming Induction 5.B.9. Let $\psi_{n_y^2} \in \Sigma^*$ be the last of Σ^* , then at $t \in [T_{\psi_{n_y^2}}^{2,3}, T_1]$, the followings hold:*

- (a) *For $\phi \in \Phi_j^*$, $\text{logit}_{5,j}^{(t)} \geq 1 - O(\lambda)$ conditioned on $\mathcal{H}_\phi$.*
- (b) *For $\phi \notin \Phi_j^*$, $\text{logit}_{5,j}^{(t)} \leq O(\lambda/d)$ conditioned on $\mathcal{H}_\phi$.*

Proof. The proof is very similar to that of Lemma 5.B.21, we do not repeat here. $\square$

Lemma 5.B.34 (Gradient bounds, phase III). *Assuming Inductions 5.B.6 and 5.B.9 holds, and let $\psi = (j, r, \varphi) \in \Sigma^*$, then at $t \in [T_{1,1}, T_1]$, the followings hold*

- (a) $\Gamma_\psi^{(t)} \geq \Omega(\lambda)$ when $V_\psi^{(t)} \leq B - d^{c_1}\mu$;
- (b) $|\Gamma_{j,r,v}^{(t)}| \leq O(\frac{\lambda\omega}{dn_y})$ for any $v \in \mathcal{G} \cup \mathcal{Y}$;
- (c) For $\phi \in \cup_{y \in \mathcal{Y}} \varphi_{j,y}$, we have

$$\Gamma_{j,r,\phi}^{(t)} = \begin{cases} -\Theta(\lambda\omega/n_y^2), & \text{if } V_{j,r,\phi}^{(t)} \in [-B + 2\mu, -\omega] \\ \in [-\Theta(\lambda\omega/n_y^2), -\Theta(\lambda\omega/dn_y^2)] \cup \{0\}, & \text{if } V_{j,r,\phi}^{(t)} \in [-B - 2\mu, -B + 2\mu] \end{cases}$$

Proof. The proof of (a) is simple as $\lambda \gg \omega$ which is the upper bound of all other gradient terms. (b) is because of Lemma 5.B.33b and that Induction 5.B.9b holds which guaranteed suitable feature shape. (c) is simply a combination of (b) and the Lemma 5.B.33a, and Induction 5.B.9b. $\square$

Proof of Induction in Phase III and Theorem 5.B.2

Proof of Induction 5.B.9 and Theorem 5.B.2. The proof is similar to the proof of Induction 5.B.5, thus we leave out some details here.

- Induction 5.B.9a: We know from Induction 5.B.8 and Lemmas 5.B.30 and 5.B.32 that Induction 5.B.9a holds at $t = T_\psi^2$. The feature shape $\mathcal{F}_\psi(d^{c_1}\mu, \tilde{O}((\frac{d\omega}{\lambda})^{\frac{1}{q-1}}))$ guaranteed a starting point for reusing the argument in Lemma 5.B.30c to guarantee that the feature stays at the same feature shape.
- Induction 5.B.9b: After $t = T_{1,1}$, which is $t = T_{\psi_{n_y}^2}$ for the last $\psi \in \Sigma^*$. We shall prove that the feature shape $\mathcal{F}_\psi(d^{c_1}\mu, \tilde{O}((\omega)^{\frac{1}{q-1}}))$ holds for every $\psi \in \Sigma^*$. Since (a) holds for all $\psi \in \Sigma^*$, we have that at $T_{\psi_{n_y}^2}^{2,3}$ that the feature shape in (a) holds for all $\psi = (j, r, \varphi) \in \Sigma^*$. So by Lemma 5.B.34b and similar argument in Lemma 5.B.18c, we can prove that the feature $|V_{j,r}^{(t)}(\phi)| \leq \tilde{O}(\omega^{\frac{1}{q-1}})$ for any $\phi \in \Phi_\varphi^+$, which is $\mathcal{F}_{\psi,2}(\tilde{O}(\omega^{\frac{1}{q-1}}))$. $\mathcal{F}_{\psi,1}(d^{c_1}\mu)$ is guaranteed by Lemma 5.B.34a and $\mathcal{F}_{\psi,3}(\tilde{O}(\omega^{\frac{1}{q-1}}))$ is guaranteed by $\mathcal{F}_{\psi,1}, \mathcal{F}_{\psi,2}$ and Lemma 5.B.31. $\mathcal{F}_{\psi,4}$ is simple as any feature $\psi \in \Sigma_\psi^{+,3}$ lost the competition and are bounded by $\tilde{O}(\mu)$. Moreover, once each $\varphi_{j,y}, j \in \tau(\mathcal{Y}), y \in \mathcal{Y}$ is learned. the feature in $\Sigma_\psi^{+,3}$ has too small gradient bounded by $O(\frac{\lambda}{d}\mu^{q-1})$ which will not have sufficient growth before $t = T_1$.

Since all feature $\psi \in \Sigma^*$ have shape $\mathcal{F}_\psi(\delta_1, \delta_2)$ for $\delta_1 = d^{c_1}\mu$ and $\delta_2 = \tilde{O}(\omega^{\frac{1}{q-1}})$ at $t = T_1$, we have proven Theorem 5.B.2. $\square$

5.B.4 Learning the Attention Layer: Simply Transitive Case

In this section, we consider the case where the group operations form a simply transitive group. According to Assumption 5.4.1, we assume that for any $y_1, y_2 \in \mathcal{Y}$, there exists a unique $g \in \mathcal{G}$ such that $g \cdot y_1 = y_2$. Without loss of generality, we let $\mathcal{Y} = \{0, 1, \dots, n_y - 1\}$, where $n_y \in [\Omega(\log \log d), \log d]$.

We focus on updating only $\mathbf{Q}$, while keeping $\mathbf{W}$ fixed. Combined with the attention structure specified in Assumption 5.3.5, it suffices to consider the updates to the blocks $\mathbf{Q}_{4,3}$ and $\mathbf{Q}_{4,4}$ only. We consider the contribution to the gradient from the position $i = 5$ on task $\mathcal{T}^2$; specifically, the relevant loss function is given by $\sum_{\ell=1}^2 \text{Loss}_5^{2,\ell}$. As $\mathbf{W}$ remains fixed in this section, we omit the superscript (t) in $\mathbf{W}$ and in all related notations that depend solely on $\mathbf{W}$ (e.g., $V_{j,r}$) for notational simplicity.

Gradient Computations

Notations for gradient expressions. We first introduce some notations for the gradients of the attention layer. For $1 \leq \ell \leq L$, given $\mathbf{Z}^{L,\ell-1}$ and $\mathbf{k} \in \mathcal{I}^{L,\ell-1}$, define

$$\Xi_{\ell,i,\mathbf{k}}^L(\mathbf{Z}^{L,\ell-1}) \triangleq \sum_{j \in [d]} \mathcal{E}_{i,j}(\mathbf{Z}^{L,\ell-1}) \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{i,j,r}(\mathbf{Z}^{L,\ell-1})) \langle \mathbf{W}_{i,j,r}, \mathbf{Z}_{\mathbf{k}} \rangle, \quad i \in [5]. \quad (5.B.20)$$

For simplicity of notation, we will henceforth omit the dependence on $\mathbf{Z}^{L,\ell-1}$ in the notation of $\Xi_{\ell,i,\mathbf{k}}^L$ when it is clear from the context.

Fact 5.B.8 (Gradients of $\mathbf{Q}$). For any $p, q \in [5]$, we have

$$\begin{aligned} -\nabla_{\mathbf{Q}_{p,q}} \text{Loss}^L &= \sum_{\ell=1}^L \sum_{i \in [5]} -\nabla_{\mathbf{Q}_{p,q}} \text{Loss}_i^{L,\ell}, \quad \text{where} \\ -\nabla_{\mathbf{Q}_{p,q}} \text{Loss}_i^{L,\ell} &= \\ \mathbb{E} \left[\sum_{\mathbf{k} \in \mathcal{I}^{L,\ell-1}} \mathbf{Attn}_{\text{ans},\ell-1 \rightarrow \mathbf{k}} \cdot \left(\Xi_{\ell,i,\mathbf{k}}^L - \sum_{\mathbf{k}' \in \mathcal{I}^{L,\ell-1}} \mathbf{Attn}_{\text{ans},\ell-1 \rightarrow \mathbf{k}'} \Xi_{\ell,i,\mathbf{k}'}^L \right) \mathbf{Z}_{\text{ans},\ell-1,p} \mathbf{Z}_{\mathbf{k},q}^\top \right]. \end{aligned}$$

Lemma 5.B.35 (Gradients of $\mathbf{Q}_{4,3}$). Given $s \in \tau(X)$, for the diagonal entry $[\mathbf{Q}_{4,3}]_{s,s}$ of the block $\mathbf{Q}_{4,3}$, we have

$$\begin{aligned} \left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_5^{2,1} \right]_{s,s} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1} \cdot \left(\sum_{j \in [d]} \mathcal{E}_{5,j}(\mathbf{Z}^{2,0}) \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \right. \right. \\ &\quad \left. \left(\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_1 \rangle - \Lambda_{5,j,r} + b_{5,j,r} \right) \right) \mathbb{1}_{s=\tau(x_0)} \right], \\ \left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_5^{2,2} \right]_{s,s} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2} \cdot \left(\sum_{j \in [d]} \mathcal{E}_{5,j}(\mathbf{Z}^{2,1}) \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \right. \right. \\ &\quad \left. \left(\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_2 \rangle - \Lambda_{5,j,r} + b_{5,j,r} \right) \right) \mathbb{1}_{s=\tau(x_1)} \right]. \end{aligned}$$

Moreover, for the off-diagonal entries $[\mathbf{Q}_{4,3}]_{s,s'}$ with $s \neq s'$, we have

$$\begin{aligned} \left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_5^{2,1} \right]_{s,s'} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},2} \cdot \left(\sum_{j \in [d]} \mathcal{E}_{5,j}(\mathbf{Z}^{2,0}) \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \right. \right. \\ &\quad \left. \left(\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_2 \rangle - \Lambda_{5,j,r} + b_{5,j,r} \right) \right) \mathbb{1}_{s=\tau(x_0), s'=\tau(x_1)} \right], \\ \left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_5^{2,2} \right]_{s,s'} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1} \cdot \left(\sum_{j \in [d]} \mathcal{E}_{5,j}(\mathbf{Z}^{2,1}) \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \right. \right. \\ &\quad \left. \left(\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_1 \rangle - \Lambda_{5,j,r} + b_{5,j,r} \right) \right) \mathbb{1}_{s=\tau(x_1), s'=\tau(x_0)} \right]. \end{aligned}$$

Proof. For $\ell = 1$, due to Fact 5.B.8, the diagonal entry $[\mathbf{Q}_{4,3}]_{s,s}$ with $s \in \tau(\mathcal{X})$, the expected gradient contribution from $-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_5^{2,1}$ takes the form

$$\mathbb{E} \left[\mathbf{Attn}_{\mathbb{A},0 \rightarrow,1} \cdot \left(\Xi_{\ell,5,,1}^2 - \sum_{\mathbf{k}' \in \mathcal{I}^{2,\ell-1}} \mathbf{Attn}_{\mathbb{A},\ell-1 \rightarrow \mathbf{k}'} \Xi_{\ell,5,\mathbf{k}'}^2 \right) \cdot \mathbb{1}_{s=\tau(x_0)} \right],$$

which is nonzero in expectation only when $s = \tau(x_0)$. Therefore, combined the definition of Ξ in (5.B.20) we have:

$$\begin{aligned} \left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_5^{2,1} \right]_{s,s} &= \mathbb{E} \left[\mathbf{Attn}_{\mathbb{A},0 \rightarrow,1} \cdot \left(\Xi_{\ell,5,,1}^2 - \sum_{\mathbf{k}' \in \mathcal{I}^{2,\ell-1}} \mathbf{Attn}_{\mathbb{A},\ell-1 \rightarrow \mathbf{k}'} \Xi_{\ell,5,\mathbf{k}'}^2 \right) \mathbb{1}_{s=\tau(x_0)} \right] \\ &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1} \cdot \left(\sum_{j \in [d]} \mathcal{E}_{5,j} \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \right. \right. \\ &\quad \left. \left(\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_{,1} \rangle - \sum_{\mathbf{k}' \in \mathcal{I}^{2,\ell-1}} \mathbf{Attn}_{\mathbb{A},\ell-1 \rightarrow \mathbf{k}'} \langle \mathbf{W}_{5,j,r}, \mathbf{Z}_{\mathbf{k}'} \rangle \right) \right) \mathbb{1}_{s=\tau(x_0)} \right] \\ &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1} \cdot \left(\sum_{j \in [d]} \mathcal{E}_{5,j} \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \right. \right. \\ &\quad \left. \left(\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_{,1} \rangle - \Lambda_{5,j,r} + b_{5,j,r} \right) \right) \mathbb{1}_{s=\tau(x_0)} \right]. \end{aligned}$$

Other quantities are computed similarly, and thus is omitted here. $\square$

Lemma 5.B.36 (Gradients of $\mathbf{Q}_{4,4}$). *Given $s \in \tau(\mathcal{X})$, for the diagonal entry $[\mathbf{Q}_{4,4}]_{s,s}$ of the block $\mathbf{Q}_{4,4}$, we have*

$$\begin{aligned} \left[-\nabla_{\mathbf{Q}_{4,4}} \text{Loss}_5^{2,1} \right]_{s,s} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0} \cdot \left(\sum_{j \in [d]} \mathcal{E}_{5,j}(\mathbf{Z}^{2,0}) \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \right. \right. \\ &\quad \left. \left(\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_{\mathbb{A},0} \rangle - \Lambda_{5,j,r} + b_{5,j,r} \right) \right) \mathbb{1}_{s=\tau(x_0)} \right], \\ \left[-\nabla_{\mathbf{Q}_{4,4}} \text{Loss}_5^{2,2} \right]_{s,s} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1} \cdot \left(\sum_{j \in [d]} \mathcal{E}_{5,j}(\mathbf{Z}^{2,1}) \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \right. \right. \\ &\quad \left. \left(\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_{\mathbb{A},1} \rangle - \Lambda_{5,j,r} + b_{5,j,r} \right) \right) \mathbb{1}_{s=\tau(x_1)} \right]. \end{aligned}$$

Moreover, for the off-diagonal entries $[\mathbf{Q}_{4,4}]_{s,s'}$ with $s \neq s'$, we have $[-\nabla_{\mathbf{Q}_{4,4}} \text{Loss}_5^{2,1}]_{s,s'} = 0$, and

$$\left[-\nabla_{\mathbf{Q}_{4,4}} \text{Loss}_5^{2,2} \right]_{s,s'} = \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0} \cdot \left(\sum_{j \in [d]} \mathcal{E}_{5,j}(\mathbf{Z}^{2,1}) \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \right. \right.$$

$$\left(\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_{\mathbb{A},0} \rangle - \Lambda_{5,j,r} + b_{5,j,r} \right) \mathbb{1}_{s=\tau(x_1), s'=\tau(x_0)} \Bigg].$$

Proof. The analysis is similar to that of Lemma 5.B.35, and we omit the details here. $\square$

Some Useful Bounds for Gradients

In this subsection, we establish several useful bounds on the gradients of the attention layer, leveraging the feature structure of the MLP layer learned during stage 1. These bounds will be instrumental for the subsequent analysis.

As established in Lemma 5.B.19 and Theorem 5.B.1, at the end of stage 1, the network exhibits the following activation properties:

- **Sparse activations:** For each $j \in \tau(\mathcal{Y})$, and any feature $(g, y) \in \mathfrak{F}_j$, there exists a unique *activated* neuron $r \in [m]$ such that, under the event $g_1 = g$ and $y_0 = y$, the following holds:

$$\begin{aligned} \Lambda_{5,j,r}^{(T_1)} &\geq B - O(\delta), \quad |V_{j,r}^{(T_1)}(g) - V_{j,r}^{(T_1)}(y)| \leq O(\delta) \\ \Lambda_{5,j,r'}^{(T_1)} &\leq O(\delta^{q-1}) \quad \text{for all } r' \neq r. \end{aligned}$$

- **Cancellation of incorrect features:** let $r \in [m]$ be the activated neuron associated with $(g, y) \in \mathfrak{F}_j$. Then for any $g' \neq g \in \mathcal{G}$ and any $y' \in \mathcal{Y}$, we have:

$$\begin{aligned} |V_{j,r}^{(T_1)}(g) + V_{j,r}^{(T_1)}(y')| &\leq O(\delta), \\ |V_{j,r}^{(T_1)}(g') + V_{j,r}^{(T_1)}(y)| &\leq O(\delta), \end{aligned}$$

In the analysis of FFN layer, we have a pair $\delta = (\delta_1, \delta_2)$; in the expressions above, some terms should use δ_1 and others δ_2 . For brevity, we write $\delta := \max\{\delta_1, \delta_2\}$ and bound all such terms by δ . Since both δ_1 and δ_2 are small, this simplification does not affect our analysis.

Notations for activated neurons. Since in the simply transitive case, the feature sets are disjoint across indices—i.e., $\mathfrak{F}_j \cap \mathfrak{F}_{j'} = \emptyset$ for all $j \neq j'$ —we denote the activated neuron corresponding to $(g, y) \in \mathfrak{F}_j$ by $r_{g,y}$. Moreover, let

$$\mathfrak{A} \triangleq \cup_{j \in \tau(\mathcal{Y})} \mathfrak{A}_j, \text{ where } \mathfrak{A}_j \triangleq \{r \mid \exists (g, y) \in \mathfrak{F}_j, r = r_{g,y}\}.$$

In other words, $\mathfrak{A}$ is the set of all activated neurons across all feature sets $\mathfrak{F}_j$ for $j \in \tau(\mathcal{Y})$. Given $\mathbf{Z}^{L,\ell-1}$, letting $\widehat{\mathcal{G}}(\mathbf{Z}^{L,\ell-1}) = \cup \{g_{\ell'}\}_{\ell'=1}^L$ be the collection of all the chosen group elements in the predicate clauses. Similarly $\widehat{\mathcal{Y}} = \cup \{y_{\ell'}\}_{\ell'=0}^{\ell-1}$. Then define $\widehat{\mathfrak{A}}_j(\mathbf{Z}^{L,\ell-1}) = \{r_{g,y} \mid (g, y) \in \mathfrak{F}_j \wedge (g \in \widehat{\mathcal{G}}(\mathbf{Z}^{L,\ell-1}) \vee y \in \widehat{\mathcal{Y}})\}$. For simplicity, we omit the dependence on $\mathbf{Z}^{L,\ell-1}$ in the notation of $\widehat{\mathfrak{A}}_j$ when it is clear from the context. Equipped with these notations, we can summarize the above properties in the following lemmas.

Lemma 5.B.37 (Properties of target feature magnitude). *Given $(g, y) \in \mathfrak{F}_j$ with $j \in \tau(\mathcal{Y})$, then, the following properties hold.*

$$\frac{1}{2} \left(V_{j,r_{g,y}}(g) + V_{j,r_{g,y}}(y) \right) \geq B - O(\delta), \quad |V_{j,r_{g,y}}(g) - V_{j,r_{g,y}}(y)| \leq O(\delta), \quad (5.B.21)$$

$$|V_{j,r_{g,y}}(g) + V_{j,r_{g,y}}(y')| \leq O(\delta), V_{j,r_{g,y}}(y') < 0 \quad \text{for all } y' \neq y, \quad (5.B.22)$$

$$|V_{j,r_{g,y}}(g') + V_{j,r_{g,y}}(y)| \leq O(\delta), V_{j,r_{g,y}}(g') < 0 \quad \text{for all } g' \neq g. \quad (5.B.23)$$

$$|V_{j,r}(g)|, |V_{j,r}(y)| \leq O(\delta) \quad \text{for all } r \notin \mathfrak{A}_j. \quad (5.B.24)$$

Lemma 5.B.38 (Properties of irrelevant magnitude). *If $(p, v) \notin \{2\} \times \mathcal{G} \cup \{5\} \times \mathcal{Y}$, or $j \notin \tau(\mathcal{Y})$, then for any $r \in [m]$, we have*

$$|\langle \mathbf{W}_{5,j,r,p}, e_v \rangle| \leq \tilde{O}(\sigma_0). \quad (5.B.25)$$

The above lemmas give us some direct computations of the inner products between the weight matrices and input embedding vectors.

Lemma 5.B.39. *Let $j \in \tau(\mathcal{Y})$ and $\ell \in [2]$. Then for any $r \in [m]$, the following holds:*

$$\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_{\text{pred},\ell} \rangle = V_{j,r}(g_\ell) \pm \tilde{O}(\sigma_0), \quad (5.B.26)$$

$$\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_{\text{ans},\ell-1} \rangle = V_{j,r}(y_{\ell-1}) \pm \tilde{O}(\sigma_0). \quad (5.B.27)$$

Moreover, for $j \notin \tau(\mathcal{Y})$ and any $\mathbf{k} \in \mathcal{I}^{2,1}$ and $r \in [m]$, we have

$$|\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_{\mathbf{k}} \rangle| = \tilde{O}(\sigma_0). \quad (5.B.28)$$

Proof. By direct computations and the definition of $\mathbf{Z}_{\text{pred},\ell}$, we have

$$\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_{\text{pred},\ell} \rangle = \langle \mathbf{W}_{5,j,r,1}, e_{x_\ell} \rangle + \langle \mathbf{W}_{5,j,r,2}, e_{g_\ell} \rangle + \langle \mathbf{W}_{5,j,r,3}, e_{x_{\ell-1}} \rangle$$

Plug in Lemma 5.B.38 and the definition of $V_{j,r}(g_\ell)$, we obtain (5.B.26). The proof of (5.B.27) and (5.B.28) is similar, and we omit the details here. $\square$

Furthermore, we can establish some characterizations of the $\Lambda_{5,j,r}(\mathbf{Z}^{2,\ell-1})$ quantities, which are crucial for the following analysis.

Lemma 5.B.40 (Characterizations of Lambda). *Given $\mathbf{Z}^{2,\ell-1}$ with $\ell \in [2]$, with an attention structure $\{\text{Attn}_{\text{ans},\ell-1 \rightarrow \mathbf{k}}\}_{\mathbf{k} \in \mathcal{I}^{2,\ell-1}}$,*

(a) *for $j \in \tau(\mathcal{Y})$, for activated neuron $r \in \mathfrak{A}_j$, we have*

$$\Lambda_{5,j,r}(\mathbf{Z}^{2,\ell-1}) = \sum_{\ell'=1}^2 \text{Attn}_{\text{ans},\ell-1 \rightarrow \text{pred},\ell'} V_{j,r}(g_{\ell'}) + \sum_{\ell'=1}^{\ell} \text{Attn}_{\text{ans},\ell-1 \rightarrow \text{ans},\ell'-1} V_{j,r}(y_{\ell'-1}) \pm \tilde{O}(\sigma_0).$$

(b) for $j \in \tau(\mathcal{Y})$, for any non-activated neuron $r \notin \mathfrak{A}_j$ we have

$$\left| \Lambda_{5,j,r}(\mathbf{Z}^{2,\ell-1}) \right| \leq O(\delta).$$

(c) for $j \notin \tau(\mathcal{Y})$, for any $r \in [m]$, we have

$$\left| \Lambda_{5,j,r}(\mathbf{Z}^{2,\ell-1}) \right| \leq \tilde{O}(\sigma_0).$$

Proof. Recall the definition of $\Lambda_{5,j,r}(\mathbf{Z}^{2,\ell-1})$ in (5.B.2), we have

$$\Lambda_{5,j,r}(\mathbf{Z}^{2,\ell-1}) = \sum_{\mathbf{k} \in I^{2,\ell-1}} \text{Attn}_{\text{ans},\ell-1 \rightarrow \mathbf{k}} \cdot \langle \mathbf{W}_{5,j,r}, \mathbf{Z}_{\mathbf{k}} \rangle + b_{5,j,r}.$$

Thus, the first part follows directly from (5.B.26) and (5.B.27); similarly, the second part holds by plugging (5.B.24) into (5.B.26) and (5.B.27); the last part is a direct consequence of (5.B.28) and the fact $b_{i,j,r} = \sigma_0 \log d$. $\square$

A direct consequence of the above lemma is the following finer characterization of the activated neurons.

Lemma 5.B.41. *Given $j \in \tau(\mathcal{Y})$, for $r \in \mathfrak{A}_j \setminus \widehat{\mathfrak{A}}_j$, we have $\mathbf{sReLU}'(\Lambda_{5,j,r}) = 0$.*

Proof. For $j \in \tau(\mathcal{Y})$, for $r \in \mathfrak{A}_j$, by Lemma 5.B.40, we have

$$\Lambda_{5,j,r}(\mathbf{Z}^{2,\ell-1}) = \sum_{\ell'=1}^2 \text{Attn}_{\text{ans},\ell-1 \rightarrow \text{pred},\ell'} V_{j,r}(g_{\ell'}) + \sum_{\ell'=1}^{\ell} \text{Attn}_{\text{ans},\ell-1 \rightarrow \text{ans},\ell'-1} V_{j,r}(y_{\ell'-1}) \pm \tilde{O}(\sigma_0).$$

By (5.B.24), for $r \in \mathfrak{A}_j \setminus \widehat{\mathfrak{A}}_j$, we have $V_{j,r}(g_{\ell'}), V_{j,r}(y_{\ell'-1}) \leq -B - O(\delta)$. Hence

$$\Lambda_{5,j,r} = -\Omega(B) \pm \tilde{O}(\sigma_0) \ll -\varrho,$$

which implies $\mathbf{sReLU}'(\Lambda_{5,j,r}) = 0$. $\square$

Now we are ready to further derive the gradients of the attention layer starting from Lemmas 5.B.35 and 5.B.36 and the properties established above.

Lemma 5.B.42 (Refined expression for the gradient of $\mathbf{Q}_{4,3}$). *Given $s \in \tau(\mathcal{X})$, for the diagonal entry $[\mathbf{Q}_{4,3}]_{s,s}$ of the block $\mathbf{Q}_{4,3}$, letting $j_1 = \tau(g_1(y_0))$ and $j_2 = \tau(g_2(y_1))$, we have*

$$\begin{aligned} \left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_5^{2,1} \right]_{s,s} &= \mathbb{E} \left[\text{Attn}_{\text{ans},0 \rightarrow \text{pred},1} \cdot \right. \\ &\quad \left. \left((1 - \mathbf{logit}_{5,j_1}) \cdot \left(\sum_{r \in \widehat{\mathfrak{A}}_{j_1}} \mathbf{sReLU}'(\Lambda_{5,j_1,r}) \cdot \left(V_{j_1,r}(g_1) - \Lambda_{5,j_1,r} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \right) \right] \end{aligned}$$

$$\begin{aligned}
& - \sum_{j \neq j_1 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \left(\sum_{r \in \mathfrak{A}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \left(V_{j,r}(g_1) - \Lambda_{5,j,r} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \\
& \pm \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \tilde{O}(\sigma_0^q) \mathbb{1}_{\tau(x_0)=s} \Bigg];
\end{aligned}$$

$$\begin{aligned}
\left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_5^{2,2} \right]_{s,s} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2} \cdot \right. \\
& \left((1 - \mathbf{logit}_{5,j_2}) \cdot \left(\sum_{r \in \mathfrak{A}_{j_2}} \mathbf{sReLU}'(\Lambda_{5,j_2,r}) \cdot \left(V_{j_2,r}(g_2) - \Lambda_{5,j_2,r} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \right. \\
& - \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \left(\sum_{r \in \mathfrak{A}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \left(V_{j,r}(g_2) - \Lambda_{5,j,r} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \\
& \left. \left. \pm \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \tilde{O}(\sigma_0^q) \mathbb{1}_{\tau(x_1)=s} \right) \right].
\end{aligned}$$

Moreover, for the off-diagonal entries $[\mathbf{Q}_{4,3}]_{s,s'}$ with $s \neq s'$, we have

$$\begin{aligned}
\left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_5^{2,1} \right]_{s,s'} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},2} \cdot \right. \\
& \left((1 - \mathbf{logit}_{5,j_1}) \cdot \left(\sum_{r \in \mathfrak{A}_{j_1}} \mathbf{sReLU}'(\Lambda_{5,j_1,r}) \cdot \left(V_{j_1,r}(g_2) - \Lambda_{5,j_1,r} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \right. \\
& - \sum_{j \neq j_1 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \left(\sum_{r \in \mathfrak{A}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \left(V_{j,r}(g_2) - \Lambda_{5,j,r} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \\
& \left. \left. \pm \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \tilde{O}(\sigma_0^q) \mathbb{1}_{\tau(x_0)=s, \tau(x_1)=s'} \right) \right];
\end{aligned}$$

$$\begin{aligned}
\left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_5^{2,2} \right]_{s,s'} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1} \cdot \right. \\
& \left((1 - \mathbf{logit}_{5,j_2}) \cdot \left(\sum_{r \in \mathfrak{A}_{j_2}} \mathbf{sReLU}'(\Lambda_{5,j_2,r}) \cdot \left(V_{j_2,r}(g_1) - \Lambda_{5,j_2,r} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \right. \\
& - \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \left(\sum_{r \in \mathfrak{A}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \left(V_{j,r}(g_1) - \Lambda_{5,j,r} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \\
& \left. \left. \pm \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \tilde{O}(\sigma_0^q) \mathbb{1}_{\tau(x_1)=s, \tau(x_0)=s'} \right) \right].
\end{aligned}$$

Proof. For $\ell = 1$,

- for the diagonal entry $[\mathbf{Q}_{4,3}]_{s,s}$ with $s \in \tau(\mathcal{X})$,

$$\begin{aligned}
\left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_5^{2,1} \right]_{s,s} &= \mathbb{E} \left[\text{Attn}_{\text{ans},0 \rightarrow \text{pred},1} \cdot \left(\sum_{j \in [d]} \mathcal{E}_{5,j}(\mathbf{Z}^{2,0}) \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \right. \right. \\
&\quad \left. \left. \left(\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_1 \rangle - \Lambda_{5,j,r} + b_{5,j,r} \right) \right) \mathbb{1}_{s=\tau(x_0)} \right] \\
&= \mathbb{E} \left[\text{Attn}_{\text{ans},0 \rightarrow \text{pred},1} \cdot \left((1 - \text{logit}_{5,j_1}) \cdot \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j_1,r}) (V_{j_1,r}(g_1) - \Lambda_{5,j_1,r} \pm \tilde{O}(\sigma_0)) \right. \right. \\
&\quad - \sum_{j \neq j_1 \in \tau(\mathcal{Y})} \text{logit}_{5,j} \cdot \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j,r}) (V_{j,r}(g_1) - \Lambda_{5,j,r} \pm \tilde{O}(\sigma_0)) \\
&\quad \left. \left. - \sum_{j \notin \tau(\mathcal{Y})} \text{logit}_{5,j} \cdot \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j,r}) \left(\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_1 \rangle - \Lambda_{5,j,r} \pm \tilde{O}(\sigma_0) \right) \right) \mathbb{1}_{\tau(x_0)=s} \right] \\
&= \mathbb{E} \left[\text{Attn}_{\text{ans},0 \rightarrow \text{pred},1} \cdot \right. \\
&\quad \left((1 - \text{logit}_{5,j_1}) \cdot \left(\sum_{r \in \widehat{\mathfrak{A}}_{j_1}} \mathbf{sReLU}'(\Lambda_{5,j_1,r}) \cdot (V_{j_1,r}(g_1) - \Lambda_{5,j_1,r} \pm \tilde{O}(\sigma_0)) \pm \tilde{O}(\delta^q) \right) \right. \\
&\quad - \sum_{j \neq j_1 \in \tau(\mathcal{Y})} \text{logit}_{5,j} \cdot \left(\sum_{r \in \widehat{\mathfrak{A}}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot (V_{j,r}(g_1) - \Lambda_{5,j,r} \pm \tilde{O}(\sigma_0)) \pm \tilde{O}(\delta^q) \right) \\
&\quad \left. \left. \pm \sum_{j \notin \tau(\mathcal{Y})} \text{logit}_{5,j} \tilde{O}(\sigma_0^q) \right) \mathbb{1}_{\tau(x_0)=s} \right],
\end{aligned}$$

where the last equality follows from Lemma 5.B.40 and Lemma 5.B.41.

- for the non-diagonal entry $[\mathbf{Q}_{4,3}]_{s,s'}$ with $s \neq s' \in \tau(\mathcal{X})$, the analysis is similar unless the condition that the gradient is non-zero only when $s = \tau(x_0)$ and $s' = \tau(x_1)$. Thus, we have

$$\begin{aligned}
\left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_5^{2,1} \right]_{s,s'} &= \mathbb{E} \left[\text{Attn}_{\text{ans},0 \rightarrow \text{pred},2} \cdot \left(\sum_{j \in [d]} \mathcal{E}_{5,j}(\mathbf{Z}^{2,0}) \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \right. \right. \\
&\quad \left. \left. \left(\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_2 \rangle - \Lambda_{5,j,r} + b_{5,j,r} \right) \right) \mathbb{1}_{s=\tau(x_0), s'=\tau(x_1)} \right] \\
&= \mathbb{E} \left[\text{Attn}_{\text{ans},0 \rightarrow \text{pred},2} \cdot \right. \\
&\quad \left((1 - \text{logit}_{5,j_1}) \cdot \left(\sum_{r \in \widehat{\mathfrak{A}}_{j_1}} \mathbf{sReLU}'(\Lambda_{5,j_1,r}) \cdot (V_{j_1,r}(g_2) - \Lambda_{5,j_1,r} \pm \tilde{O}(\sigma_0)) \pm \tilde{O}(\delta^q) \right) \right.
\end{aligned}$$

$$\begin{aligned}
& - \sum_{j \neq j_1 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \left(\sum_{r \in \mathfrak{A}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \left(V_{j,r}(g_2) - \Lambda_{5,j,r} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \\
& \pm \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \tilde{O}(\sigma_0^q) \mathbb{1}_{\tau(x_0)=s, \tau(x_1)=s'} \Bigg].
\end{aligned}$$

For $\ell = 2$,

- for the diagonal entry $[\mathbf{Q}_{4,3}]_{s,s}$ with $s \in \tau(\mathcal{X})$,

$$\begin{aligned}
& \left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_5^{2,2} \right]_{s,s} = \mathbb{E} \left[\mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, 2} \cdot \left(\sum_{j \in [d]} \mathcal{E}_{5,j}(\mathbf{Z}^{2,1}) \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \right. \right. \\
& \quad \left. \left. \left(\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_2 \rangle - \Lambda_{5,j,r} + b_{5,j,r} \right) \right) \mathbb{1}_{s=\tau(x_1)} \right] \\
& = \mathbb{E} \left[\mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, 2} \cdot \left((1 - \mathbf{logit}_{5,j_2}) \cdot \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j_2,r}) \left(V_{j_2,r}(g_2) - \Lambda_{5,j_2,r} \pm \tilde{O}(\sigma_0) \right) \right. \right. \\
& \quad - \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j,r}) \left(V_{j,r}(g_2) - \Lambda_{5,j,r} \pm \tilde{O}(\sigma_0) \right) \\
& \quad \left. \left. - \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j,r}) \left(\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_2 \rangle - \Lambda_{5,j,r} \pm \tilde{O}(\sigma_0) \right) \right) \mathbb{1}_{\tau(x_1)=s} \right] \\
& = \mathbb{E} \left[\mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, 2} \cdot \right. \\
& \quad \left((1 - \mathbf{logit}_{5,j_2}) \cdot \left(\sum_{r \in \mathfrak{A}_{j_2}} \mathbf{sReLU}'(\Lambda_{5,j_2,r}) \cdot \left(V_{j_2,r}(g_2) - \Lambda_{5,j_2,r} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \right. \\
& \quad - \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \left(\sum_{r \in \mathfrak{A}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \left(V_{j,r}(g_2) - \Lambda_{5,j,r} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \\
& \quad \left. \left. \pm \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \tilde{O}(\sigma_0^q) \right) \mathbb{1}_{\tau(x_1)=s} \right],
\end{aligned}$$

where the last equality follows from Lemma 5.B.40 and Lemma 5.B.41.

- for the non-diagonal entry $[\mathbf{Q}_{4,3}]_{s,s'}$ with $s \neq s' \in \tau(\mathcal{X})$, the analysis is similar unless the condition that the gradient is non-zero only when $s = \tau(x_0)$ and $s' = \tau(x_1)$. Thus, we have

$$\begin{aligned}
& \left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_5^{2,2} \right]_{s,s'} = \mathbb{E} \left[\mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, 1} \cdot \left(\sum_{j \in [d]} \mathcal{E}_{5,j}(\mathbf{Z}^{2,1}) \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \right. \right. \\
& \quad \left. \left. \left(\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_1 \rangle - \Lambda_{5,j,r} + b_{5,j,r} \right) \right) \mathbb{1}_{s=\tau(x_1), s'=\tau(x_0)} \right]
\end{aligned}$$

$$\begin{aligned}
&= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1} \cdot \right. \\
&\quad \left((1 - \mathbf{logit}_{5,j_2}) \cdot \left(\sum_{r \in \widehat{\mathfrak{A}}_{j_2}} \mathbf{sReLU}'(\Lambda_{5,j_2,r}) \cdot (V_{j_2,r}(g_1) - \Lambda_{5,j_2,r} \pm \widetilde{O}(\sigma_0)) \pm \widetilde{O}(\delta^q) \right) \right. \\
&\quad - \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \left(\sum_{r \in \widehat{\mathfrak{A}}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot (V_{j,r}(g_1) - \Lambda_{5,j,r} \pm \widetilde{O}(\sigma_0)) \pm \widetilde{O}(\delta^q) \right) \\
&\quad \left. \pm \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \widetilde{O}(\sigma_0^q) \right) \mathbb{1}_{\tau(x_1)=s, \tau(x_0)=s'} \left. \right].
\end{aligned}$$

Therefore, we complete the proof. $\square$

Lemma 5.B.43 (Refined expression for the gradient of $\mathbf{Q}_{4,4}$). *Given $s \in \tau(X)$, for the diagonal entry $[\mathbf{Q}_{4,4}]_{s,s}$ of the block $\mathbf{Q}_{4,4}$, letting $j_1 = \tau(g_1(y_0))$ and $j_2 = \tau(g_2(y_1))$, we have*

$$\begin{aligned}
\left[-\nabla_{\mathbf{Q}_{4,4}} \text{Loss}_5^{2,1} \right]_{s,s} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0} \cdot \right. \\
&\quad \left((1 - \mathbf{logit}_{5,j_1}) \cdot \left(\sum_{r \in \widehat{\mathfrak{A}}_{j_1}} \mathbf{sReLU}'(\Lambda_{5,j_1,r}) \cdot (V_{j_1,r}(y_0) - \Lambda_{5,j_1,r} \pm \widetilde{O}(\sigma_0)) \pm \widetilde{O}(\delta^q) \right) \right. \\
&\quad - \sum_{j \neq j_1 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \left(\sum_{r \in \widehat{\mathfrak{A}}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot (V_{j,r}(y_0) - \Lambda_{5,j,r} \pm \widetilde{O}(\sigma_0)) \pm \widetilde{O}(\delta^q) \right) \\
&\quad \left. \pm \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \widetilde{O}(\sigma_0^q) \right) \mathbb{1}_{\tau(x_0)=s} \left. \right].
\end{aligned}$$

$$\begin{aligned}
\left[-\nabla_{\mathbf{Q}_{4,4}} \text{Loss}_5^{2,2} \right]_{s,s} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1} \cdot \right. \\
&\quad \left((1 - \mathbf{logit}_{5,j_2}) \cdot \left(\sum_{r \in \widehat{\mathfrak{A}}_{j_2}} \mathbf{sReLU}'(\Lambda_{5,j_2,r}) \cdot (V_{j_2,r}(y_1) - \Lambda_{5,j_2,r} \pm \widetilde{O}(\sigma_0)) \pm \widetilde{O}(\delta^q) \right) \right. \\
&\quad - \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \left(\sum_{r \in \widehat{\mathfrak{A}}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot (V_{j,r}(y_1) - \Lambda_{5,j,r} \pm \widetilde{O}(\sigma_0)) \pm \widetilde{O}(\delta^q) \right) \\
&\quad \left. \pm \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \widetilde{O}(\sigma_0^q) \right) \mathbb{1}_{\tau(x_1)=s} \left. \right].
\end{aligned}$$

Moreover, for the off-diagonal entries $[\mathbf{Q}_{4,4}]_{s,s'}$ with $s \neq s'$, we have $[-\nabla_{\mathbf{Q}_{4,4}} \text{Loss}_5^{2,1}]_{s,s'} = 0$, and

$$\begin{aligned} \left[-\nabla_{\mathbf{Q}_{4,4}} \text{Loss}_5^{2,2} \right]_{s,s'} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0} \cdot \right. \\ &\quad \left((1 - \mathbf{logit}_{5,j_2}) \cdot \left(\sum_{r \in \widehat{\mathfrak{V}}_{j_2}} \mathbf{sReLU}'(\Lambda_{5,j_2,r}) \cdot (V_{j_2,r}(y_0) - \Lambda_{5,j_2,r} \pm \widetilde{O}(\sigma_0)) \pm \widetilde{O}(\delta^q) \right) \right. \\ &\quad \left. - \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \left(\sum_{r \in \widehat{\mathfrak{V}}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot (V_{j,r}(y_0) - \Lambda_{5,j,r} \pm \widetilde{O}(\sigma_0)) \pm \widetilde{O}(\delta^q) \right) \right. \\ &\quad \left. \pm \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \widetilde{O}(\sigma_0^q) \right) \mathbb{1}_{\tau(x_1)=s, \tau(x_0)=s'} \left. \right]. \end{aligned}$$

The proof is similar to Lemma 5.B.42 and we omit it here.

Notations for gradient decompositions. We shall define some useful notations to further simplify the expressions of the gradient.

- for $\ell = 1$,
 - for $[\mathbf{Q}_{4,3}]_{s,s}$ with $s \in \tau(\mathcal{X})$, we have $[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_5^{2,1}]_{s,s} = \mathcal{N}_{s,3,1,i} + \mathcal{N}_{s,3,1,ii} + \mathcal{N}_{s,3,1,iii}$, where

$$\begin{aligned} \mathcal{N}_{s,3,1,i} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1} \cdot (1 - \mathbf{logit}_{5,j_1}) \cdot \right. \\ &\quad \left. \left(\sum_{r \in \widehat{\mathfrak{V}}_{j_1}} \mathbf{sReLU}'(\Lambda_{5,j_1,r}) \cdot (V_{j_1,r}(g_1) - \Lambda_{5,j_1,r} \pm \widetilde{O}(\sigma_0)) \pm \widetilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_0)=s} \right], \end{aligned} \quad (5.B.29)$$

$$\begin{aligned} \mathcal{N}_{s,3,1,ii} &= -\mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1} \cdot \sum_{j \neq j_1 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \right. \\ &\quad \left. \left(\sum_{r \in \widehat{\mathfrak{V}}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot (V_{j,r}(g_1) - \Lambda_{5,j,r} \pm \widetilde{O}(\sigma_0)) \pm \widetilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_0)=s} \right], \end{aligned} \quad (5.B.30)$$

$$\mathcal{N}_{s,3,1,iii} = \pm \mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1} \cdot \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \widetilde{O}(\sigma_0^q) \mathbb{1}_{\tau(x_0)=s} \right]. \quad (5.B.31)$$

- for $[\mathbf{Q}_{4,4}]_{s,s}$ with $s \in \tau(\mathcal{X})$, we have $[-\nabla_{\mathbf{Q}_{4,4}} \text{Loss}_5^{2,1}]_{s,s} = \mathcal{N}_{s,4,1,i} + \mathcal{N}_{s,4,1,ii} + \mathcal{N}_{s,4,1,iii}$, where

$$\mathcal{N}_{s,4,1,i} = \mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0} \cdot (1 - \mathbf{logit}_{5,j_1}) \cdot \right. \quad (5.B.32)$$

$$\begin{aligned}
& \left(\sum_{r \in \widehat{\mathfrak{A}}_{j_1}} \mathbf{sReLU}'(\Lambda_{5,j_1,r}) \cdot \left(V_{j_1,r}(y_0) - \Lambda_{5,j_1,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_0)=s} \Bigg], \\
\mathcal{N}_{s,4,1,ii} &= -\mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0} \cdot \sum_{j \neq j_1 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \right. \\
& \quad \left. \left(\sum_{r \in \widehat{\mathfrak{A}}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \left(V_{j,r}(y_0) - \Lambda_{5,j,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_0)=s} \right], \tag{5.B.33}
\end{aligned}$$

$$\mathcal{N}_{s,4,1,iii} = \pm \mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0} \cdot \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \widetilde{O}(\sigma_0^q) \mathbb{1}_{\tau(x_0)=s} \right]. \tag{5.B.34}$$

- for $\ell = 2$,
 - for $[\mathbf{Q}_{4,3}]_{s,s}$ with $s \in \tau(\mathcal{X})$, we have $[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_5^{2,2}]_{s,s} = \mathcal{N}_{s,3,2,i} + \mathcal{N}_{s,3,2,ii} + \mathcal{N}_{s,3,2,iii}$, where

$$\begin{aligned}
\mathcal{N}_{s,3,2,i} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2} \cdot (1 - \mathbf{logit}_{5,j_2}) \cdot \right. \\
& \quad \left. \left(\sum_{r \in \widehat{\mathfrak{A}}_{j_2}} \mathbf{sReLU}'(\Lambda_{5,j_2,r}) \cdot \left(V_{j_2,r}(g_2) - \Lambda_{5,j_2,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s} \right], \tag{5.B.35}
\end{aligned}$$

$$\begin{aligned}
\mathcal{N}_{s,3,2,ii} &= -\mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2} \cdot \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \right. \\
& \quad \left. \left(\sum_{r \in \widehat{\mathfrak{A}}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \left(V_{j,r}(g_2) - \Lambda_{5,j,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s} \right], \tag{5.B.36}
\end{aligned}$$

$$\mathcal{N}_{s,3,2,iii} = \pm \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2} \cdot \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \widetilde{O}(\sigma_0^q) \mathbb{1}_{\tau(x_1)=s} \right]. \tag{5.B.37}$$

- for $[\mathbf{Q}_{4,4}]_{s,s}$ with $s \in \tau(\mathcal{X})$, we have $[-\nabla_{\mathbf{Q}_{4,4}} \text{Loss}_5^{2,2}]_{s,s} = \mathcal{N}_{s,4,2,i} + \mathcal{N}_{s,4,2,ii} + \mathcal{N}_{s,4,2,iii}$, where

$$\mathcal{N}_{s,4,2,i} = \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1} \cdot (1 - \mathbf{logit}_{5,j_2}) \cdot \right. \tag{5.B.38}$$

$$\begin{aligned}
& \left(\sum_{r \in \widehat{\mathfrak{I}}_{j_2}} \mathbf{sReLU}'(\Lambda_{5,j_2,r}) \cdot \left(V_{j_2,r}(y_1) - \Lambda_{5,j_2,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s} \Bigg], \\
\mathcal{N}_{s,4,2,ii} &= -\mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1} \cdot \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \right. \\
& \quad \left. \left(\sum_{r \in \widehat{\mathfrak{I}}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \left(V_{j,r}(y_1) - \Lambda_{5,j,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s} \right], \tag{5.B.39}
\end{aligned}$$

$$\mathcal{N}_{s,4,2,iii} = \pm \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1} \cdot \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \widetilde{O}(\sigma_0^q) \mathbb{1}_{\tau(x_1)=s} \right]. \tag{5.B.40}$$

Probabilistic Events. We conclude this subsection by introducing several probabilistic events that will be used to simplify the characterization of activated neurons in the subsequent analysis.

$$\mathcal{E}_1 \triangleq \{g_1 \neq g_2\}, \tag{5.B.41}$$

$$\mathcal{E}_2 \triangleq \{g_{\ell_1}(y_{\ell'_1}) \neq g_{\ell_2}(y_{\ell'_2}), \text{ for any } (\ell_1, \ell'_1) \neq (\ell_2, \ell'_2), \text{ where } \ell_k \in [2], \ell'_k \in \{0, 1\}\}. \tag{5.B.42}$$

It is easy to see that $\mathcal{E}_1$ and $\mathcal{E}_2$ hold with high probability $1 - O(\frac{1}{\log d})$.

Stage 2.1: Initial Growth of Gap

At the beginning of stage 2, since $\mathbf{Q}$ has not been trained for long, we have the attention score is still close to the uniform structure. Therefore, for $\ell = 1$, we have

$$\Lambda_{5,j,r}(\mathbf{Z}^{2,\ell-1}) \approx \frac{1}{3}V_{j,r}(g_1) + \frac{1}{3}V_{j,r}(g_2) + \frac{1}{3}V_{j,r}(y_0) \pm \widetilde{O}(\sigma_0).$$

If $\mathbf{Z}^{2,\ell-1} \in \mathcal{E}_1$,

- for $j = j_1$, for $r \in \widehat{\mathfrak{I}}_{j_1}$, only $r_{g_1 \cdot y_0}$ is activated since $\Lambda_{5,j_1,r_{g_1 \cdot y_0}} \approx \frac{1}{3}B$ and $\Lambda_{5,j_1,r_{g_2 \cdot g_2^{-1}(j_1)}} \approx -\frac{1}{3}B \ll -\varrho$;
- for $j = j'_1 \triangleq \tau(g_2(y_0))$, only $r_{g_2 \cdot y_0}$ is activated since $\Lambda_{5,j'_1,r_{g_2 \cdot y_0}} \approx \frac{1}{3}B$ and $\Lambda_{5,j'_1,r_{g_1 \cdot g_2^{-1}(j'_1)}} \approx -\frac{1}{3}B \ll -\varrho$;
- for other $j \in \tau(\mathcal{Y})$, we have $\Lambda_{5,j,r} \leq -\frac{1}{3}B$ for all $r \in \widehat{\mathfrak{I}}_j$, thus no activation.

Moreover, for $\ell = 2$, we have

$$\Lambda_{5,j,r}(\mathbf{Z}^{2,\ell-1}) \approx \frac{1}{4}V_{j,r}(g_1) + \frac{1}{4}V_{j,r}(g_2) + \frac{1}{4}V_{j,r}(y_0) + \frac{1}{4}V_{j,r}(y_1) \pm \widetilde{O}(\sigma_0).$$

If $\mathbf{Z}^{2,\ell-1} \in \mathcal{E}_2$,

- for $j \in \{\tau(g_\ell(y_{\ell'}))\}_{\ell \in [2], \ell' \in \{0,1\}}$, only the corresponding $r_{g_\ell \cdot y_{\ell'}}$ is activated in the smoothed regime since $|\Lambda_{5,j,r_{g_\ell \cdot y_{\ell'}}}| = O(\delta)$ and $\Lambda_{5,j,r} \approx -\frac{1}{2}B \ll -\varrho$ for $r \in \widehat{\mathfrak{U}}_j \setminus \{r_{g_\ell \cdot y_{\ell'}}\}$;
- for other $j \in \tau(\mathcal{Y})$, we have $\Lambda_{5,j,r} \leq -\frac{1}{2}B$ for all $r \in \widehat{\mathfrak{U}}_j$, thus no activation.

Here, activation means that the corresponding $\mathbf{sReLU}'(\Lambda_{5,j,r})$ is non-zero, which is crucial for the gradient computation. Based on the above observations, we can see that the gradient from $\ell = 2$ is relatively small since Λ is only activated in the smoothed regime. Thus, initially, the learning process is dominated by $\nabla_{\mathbf{Q}} \text{Loss}_5^{2,1}$. Moreover, if we take a closer look at the gradient from $\ell = 1$, we have

$$\begin{aligned} \mathcal{N}_{s,3,1,ii} &\approx -\mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1} \cdot \mathbf{logit}_{5,j'_1} \cdot \right. \\ &\quad \left. \left(\mathbf{sReLU}'(\Lambda_{5,j'_1,r_{g_2 \cdot y_0}}) \cdot \left(V_{j'_1,r_{g_2 \cdot y_0}}(g_1) - \Lambda_{5,j'_1,r_{g_2 \cdot y_0}} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_0)=s} \right] \geq 0, \\ \mathcal{N}_{s,4,1,ii} &\approx -\mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0} \cdot \mathbf{logit}_{5,j'_1} \cdot \right. \\ &\quad \left. \left(\mathbf{sReLU}'(\Lambda_{5,j'_1,r_{g_2 \cdot y_0}}) \cdot \left(V_{j'_1,r_{g_2 \cdot y_0}}(y_0) - \Lambda_{5,j'_1,r_{g_2 \cdot y_0}} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \right] \leq 0, \end{aligned}$$

since $V_{j'_1,r_{g_2 \cdot y_0}}(y_0) \geq \Omega(B)$ while $V_{j'_1,r_{g_2 \cdot y_0}}(g_1) \leq -\Omega(B)$. Thus, $[\mathbf{Q}_{4,3}]_{s,s}$ will have a significant positive gradient while $[\mathbf{Q}_{4,4}]_{s,s}$ will have a negative counterpart. This will lead to the growth of the gap between $[\mathbf{Q}_{4,3}]_{s,s}$ and $[\mathbf{Q}_{4,4}]_{s,s}$.

We formally characterize this growth behavior within this substage. At the beginning of each sub-stage, we establish an induction hypothesis that we expect to hold throughout. Subsequently, we analyze the dynamics under this hypothesis within the substage, aiming to prove its validity by the end of sub-stage. Due to the symmetry of $[\mathbf{Q}_{4,3}]_{s,s}$ and $[\mathbf{Q}_{4,4}]_{s,s}$ across $s \in \tau(\mathcal{X})$, we may, without loss of generality, focus on a particular $s \in \tau(\mathcal{X})$.

Induction 5.B.10. *Given $s \in \tau(\mathcal{X})$, let $T_{2,1,s}$ denote the first time that $[\mathbf{Q}_{4,3}^{(t)}]_{s,s}$ reaches $\Omega\left(\frac{1}{\log d}\right)$. For all iterations $t < T_{2,1,s}$, we have the following holds*

- $[\mathbf{Q}_{4,3}^{(t)}]_{s,s}$ monotonically increases;
- $|[\mathbf{Q}_{4,4}^{(t)}]_{s,s}| \leq [\mathbf{Q}_{4,3}^{(t)}]_{s,s}$ and $[\mathbf{Q}_{4,3}^{(t)}]_{s,s} - [\mathbf{Q}_{4,4}^{(t)}]_{s,s} = \Theta([\mathbf{Q}_{4,3}^{(t)}]_{s,s})$;
- for $p \in \{3,4\}$, for $s' \in \tau(\mathcal{X}) \neq s$, $|[\mathbf{Q}_{4,p}^{(t)}]_{s,s'}| \leq O\left(\frac{[\mathbf{Q}_{4,p}^{(t)}]_{s,s}}{d}\right)$.

Attention and Logit Preliminaries We first introduce several properties of the attention scores and logits if Induction 5.B.10 holds.

Lemma 5.B.44. *If Induction 5.B.10 holds for all iterations $< t$, given input $\mathbf{Z}^{2,\ell-1}$, then we have*

- for $\ell = 1$,
 - $\mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(t)} \in \left[\frac{1}{3}, \frac{1}{3} + O\left(\frac{1}{\log d}\right) \right]$;

- (b) $\mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},2}^{(t)}, \mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(t)} \leq \mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(t)}$
(c) $|\mathbf{Attn}_{\text{ans},0 \rightarrow \mathbf{k}}^{(t)} - \mathbf{Attn}_{\text{ans},0 \rightarrow \mathbf{k}'}^{(t)}| \leq O(\frac{1}{\log d})$ for $\mathbf{k} \neq \mathbf{k}' \in \mathcal{I}^{(2,0)}$.
2. for $\ell = 2$,
(a) $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \in \left[\frac{1}{4}, \frac{1}{4} + O(\frac{1}{\log d}) \right]$;
(b) $\mathbf{Attn}_{\text{ans},1 \rightarrow \mathbf{k}}^{(t)} \leq \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}$ for $\mathbf{k} \neq (\text{pred}, 2)$;
(c) for $\mathbf{k} \in \{(\text{ans}, 0), (\text{pred}, 1)\}$, $\mathbf{k}' \in \{(\text{ans}, 1), (\text{pred}, 2)\}$,

$$|\mathbf{Attn}_{\text{ans},1 \rightarrow \mathbf{k}}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \mathbf{k}'}^{(t)}| \leq O\left(\frac{1}{\log d}\right);$$

$$(d) |\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)}| \leq O\left(\frac{1}{d}\right).$$

Proof. For $\ell = 1$, given $\mathbf{Z}^{2,\ell-1}$, according to Assumption 5.3.5, we have

$$\begin{aligned} \mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(t)} &= \frac{e^{[\mathbf{Q}_{4,3}^{(t)}]_{\tau(x_0), \tau(x_0)}}}{e^{[\mathbf{Q}_{4,3}^{(t)}]_{\tau(x_0), \tau(x_0)}} + e^{[\mathbf{Q}_{4,3}^{(t)}]_{\tau(x_0), \tau(x_1)}} + e^{[\mathbf{Q}_{4,4}^{(t)}]_{\tau(x_0), \tau(x_0)}}} \\ &= \frac{1}{1 + e^{[\mathbf{Q}_{4,3}^{(t)}]_{\tau(x_0), \tau(x_1)} - [\mathbf{Q}_{4,3}^{(t)}]_{\tau(x_0), \tau(x_0)}} + e^{[\mathbf{Q}_{4,4}^{(t)}]_{\tau(x_0), \tau(x_0)} - [\mathbf{Q}_{4,3}^{(t)}]_{\tau(x_0), \tau(x_0)}}}. \end{aligned}$$

Thus, by Induction 5.B.10,

$$-O\left(\frac{1}{\log d}\right) \leq [\mathbf{Q}_{4,3}^{(t)}]_{\tau(x_0), \tau(x_1)} - [\mathbf{Q}_{4,3}^{(t)}]_{\tau(x_0), \tau(x_0)}, [\mathbf{Q}_{4,4}^{(t)}]_{\tau(x_0), \tau(x_0)} - [\mathbf{Q}_{4,3}^{(t)}]_{\tau(x_0), \tau(x_0)} \leq 0,$$

which implies that $0 \leq \mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(t)} \leq \frac{1}{3} + O\left(\frac{1}{\log d}\right)$. (b) is straightforward since

$$[\mathbf{Q}_{4,3}^{(t)}]_{\tau(x_0), \tau(x_1)}, [\mathbf{Q}_{4,4}^{(t)}]_{\tau(x_0), \tau(x_0)} \leq [\mathbf{Q}_{4,3}^{(t)}]_{\tau(x_0), \tau(x_0)}.$$

(c) is a direct consequence of (a) and (b).

For $\ell = 2$, given $\mathbf{Z}^{2,\ell-1}$, (a)-(c) are very similar to the above analysis, and then for (d), we have

$$\begin{aligned} \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} &= \frac{e^{[\mathbf{Q}_{4,3}^{(t)}]_{\tau(x_1), \tau(x_0)}} - e^{[\mathbf{Q}_{4,4}^{(t)}]_{\tau(x_1), \tau(x_0)}}}{e^{[\mathbf{Q}_{4,3}^{(t)}]_{\tau(x_1), \tau(x_0)}} + e^{[\mathbf{Q}_{4,3}^{(t)}]_{\tau(x_1), \tau(x_1)}} + e^{[\mathbf{Q}_{4,4}^{(t)}]_{\tau(x_1), \tau(x_0)}} + e^{[\mathbf{Q}_{4,4}^{(t)}]_{\tau(x_1), \tau(x_1)}}} \\ &\stackrel{(i)}{\leq} O\left([\mathbf{Q}_{4,3}^{(t)}]_{\tau(x_1), \tau(x_0)} - [\mathbf{Q}_{4,4}^{(t)}]_{\tau(x_1), \tau(x_0)}\right) \stackrel{(ii)}{\leq} O\left(\frac{1}{d}\right), \end{aligned}$$

where (i) is due to the fact that $|e^x - e^y| \leq O(|x - y|)$ when x, y are small, and (ii) is due to Induction 5.B.10 (b). $\square$

Lemma 5.B.45. *If Induction 5.B.10 holds for all iterations $< t$, given input $\mathbf{Z}^{2,\ell-1}$, then we have*

1. for $\ell = 1$, if $\mathbf{Z}^{2,\ell-1} \in \mathcal{E}_1$, then

- (a) for $j = j_1$, $\Lambda_{5,j_1,r}^{(t)} \ll -\varrho$ for $r \in \widehat{\mathfrak{A}}_{j_1} \setminus \{r_{g_1 \cdot y_0}\}$;
 (b) for $j = j'_1 \triangleq \tau(g_2(y_0))$, $\Lambda_{5,j'_1,r}^{(t)} \ll -\varrho$ for $r \in \widehat{\mathfrak{A}}_{j'_1} \setminus \{r_{g_2 \cdot y_0}\}$;
 (c) for other $j \in \tau(\mathcal{Y})$, r is not activated for all $r \in \widehat{\mathfrak{A}}_j$, i.e., $\Lambda_{5,j,r}^{(t)} \ll -\varrho$.
 2. $\ell = 2$, if $\mathbf{Z}^{2,\ell-1} \in \mathcal{E}_2$, then
 (a) for

$$j \in \{\tau(g_\ell(y_{\ell'}))\}_{\ell \in [2], \ell' \in \{0,1\}},$$

only the corresponding $r_{g_\ell \cdot y_{\ell'}}$ may be activated. It obeys

$$|\Lambda_{5,j,r_{g_\ell \cdot y_{\ell'}}}^{(t)}| \leq O(1),$$

while all other $\Lambda_{5,j,r}^{(t)} \ll -\varrho$ for $r \in \widehat{\mathfrak{A}}_j \setminus \{r_{g_\ell \cdot y_{\ell'}}\}$;

- (b) for other $j \in \tau(\mathcal{Y})$, r is not activated for all $r \in \widehat{\mathfrak{A}}_j$, i.e., $\Lambda_{5,j,r}^{(t)} \ll -\varrho$.

Proof. We only prove (a) for $\ell = 2$ since the other cases are straightforward. $j = \tau(g_\ell(y_{\ell'}))$, by Lemma 5.B.40 we have

$$\begin{aligned} \Lambda_{5,j,r_{g_\ell \cdot y_{\ell'}}}^{(t)} &= \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} V_{j,r_{g_\ell \cdot y_{\ell'}}}(g_1) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} V_{j,r_{g_\ell \cdot y_{\ell'}}}(g_2) \\ &\quad + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} V_{j,r_{g_\ell \cdot y_{\ell'}}}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} V_{j,r_{g_\ell \cdot y_{\ell'}}}(y_1) \pm \widetilde{O}(\sigma_0). \end{aligned}$$

Notice that since $\mathbf{Z}^{2,\ell-1} \in \mathcal{E}_2$, we have two V terms are positive and two are negative with magnitude $B \pm O(\delta)$. Therefore, $|\Lambda_{5,j,r_{g_\ell \cdot y_{\ell'}}}^{(t)}| \leq O\left(\frac{1}{\log d}\right) \cdot B = O(1)$. $\square$

Lemma 5.B.46. *If Induction 5.B.10 holds for all iterations $< t$, given input $\mathbf{Z}^{2,0} \in \mathcal{E}_1$, then we have*

- (a) $\Lambda_{5,j_1,r_{g_1 \cdot y_0}}^{(t)} \geq \frac{1}{3}B - O(1)$;
 (b) $-O(\delta) \leq \Lambda_{5,j_1,r_{g_1 \cdot y_0}}^{(t)} - \Lambda_{5,j'_1,r_{g_2 \cdot y_0}}^{(t)} \leq O(1)$;

Proof. By Lemma 5.B.40 we have

$$\begin{aligned} \Lambda_{5,j_1,r_{g_1 \cdot y_0}}^{(t)} &= \\ \mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(t)} V_{j_1,r_{g_1 \cdot y_0}}(g_1) &+ \mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},2}^{(t)} V_{j_1,r_{g_1 \cdot y_0}}(g_2) + \mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(t)} V_{j_1,r_{g_1 \cdot y_0}}(y_0) \pm \widetilde{O}(\sigma_0). \end{aligned}$$

By Lemma 5.B.44 and the cancellation in (5.B.23), we have

$$\begin{aligned} &\mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},2}^{(t)} V_{j_1,r_{g_1 \cdot y_0}}(g_2) + \mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(t)} V_{j_1,r_{g_1 \cdot y_0}}(y_0) \\ &\geq \left(\frac{1}{3} - O\left(\frac{1}{\log d}\right)\right) \cdot O(\delta) - O\left(\frac{1}{\log d}\right) \cdot B. \end{aligned}$$

Putting it back, and using the fact that $\mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(t)} V_{j_1, r_{s_1 \cdot y_0}}(g_1) \geq \frac{1}{3} \cdot (B - O(\delta))$, we have

$$\begin{aligned} \Lambda_{5,j_1, r_{s_1 \cdot y_0}} &\geq \frac{1}{3} \cdot (B - O(\delta)) + \left(\frac{1}{3} - O\left(\frac{1}{\log d}\right) \right) \cdot O(\delta) - O\left(\frac{1}{\log d}\right) \cdot B \pm \tilde{O}(\sigma_0) \\ &\geq \frac{1}{3} B - O(1). \end{aligned}$$

Moving on to (b), we have

$$\begin{aligned} \Lambda_{5,j_1, r_{s_1 \cdot y_0}} - \Lambda_{5,j'_1, r_{s_2 \cdot y_0}} &= \mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(t)}(V_{j_1, r_{s_1 \cdot y_0}}(g_1) - V_{j'_1, r_{s_2 \cdot y_0}}(g_1)) \pm \tilde{O}(\sigma_0) \\ &+ \mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},2}^{(t)}(V_{j_1, r_{s_1 \cdot y_0}}(g_2) - V_{j'_1, r_{s_2 \cdot y_0}}(g_2)) + \mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(t)}(V_{j_1, r_{s_1 \cdot y_0}}(y_0) - V_{j'_1, r_{s_2 \cdot y_0}}(y_0)) \\ &\leq \mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(t)} \cdot (2B + O(\delta)) - \mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},2}^{(t)} \cdot (2B - O(\delta)) + \mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(t)} \cdot O(\delta) \\ &\leq (2B - O(\delta)) \cdot O\left(\frac{1}{\log d}\right) + O(\delta) \leq O(1). \end{aligned}$$

Similarly, we have

$$\begin{aligned} \Lambda_{5,j_1, r_{s_1 \cdot y_0}} - \Lambda_{5,j'_1, r_{s_2 \cdot y_0}} &\geq \mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(t)} \cdot (2B - O(\delta)) - \mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},2}^{(t)} \cdot (2B + O(\delta)) - \mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(t)} \cdot O(\delta) \\ &\geq -\mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},2}^{(t)} \cdot O(\delta) - \mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(t)} \cdot O(\delta) \geq -O(\delta). \end{aligned}$$

□

Lemma 5.B.47. *If Induction 5.B.10 holds for all iterations $< t$, given input $\mathbf{Z}^{2,\ell-1}$, then we have*

1. *for $\ell = 1$, if $\mathbf{Z}^{2,\ell-1} \in \mathcal{E}_1$, $\mathbf{logit}_{5,j}^{(t)} = \Omega(1)$ for $j \in \{j_1, j'_1\}$, $1 - \mathbf{logit}_{5,j_1}^{(t)} - \mathbf{logit}_{5,j'_1}^{(t)} = \frac{1}{\text{poly}d}$.*
2. *for $\ell = 2$, if $\mathbf{Z}^{2,\ell-1} \in \mathcal{E}_2$, $\mathbf{logit}_{5,j}^{(t)} = O(\frac{1}{d})$ for all j .*

Proof. • For $\ell = 1$, by Lemma 5.B.45 and Lemma 5.B.40, we have

$$\begin{aligned} F_{5,j_1}^{(t)}(\mathbf{Z}^{2,\ell-1}) &= \sum_{r \in [m]} \mathbf{sReLU}(\Lambda_{5,j_1,r}^{(t)}) = \Lambda_{5,j_1, r_{s_1 \cdot y_0}}^{(t)} + \varrho(m/q - 1) = \Lambda_{5,j_1, r_{s_1 \cdot y_0}}^{(t)} + O\left(\frac{1}{\text{polylog}d}\right), \\ F_{5,j'_1}^{(t)}(\mathbf{Z}^{2,\ell-1}) &= \sum_{r \in [m]} \mathbf{sReLU}(\Lambda_{5,j'_1,r}^{(t)}) = \Lambda_{5,j'_1, r_{s_2 \cdot y_0}}^{(t)} + \varrho(m/q - 1), \\ F_{5,j}^{(t)}(\mathbf{Z}^{2,\ell-1}) &= \sum_{r \in [m]} \mathbf{sReLU}(\Lambda_{5,j,r}^{(t)}) \leq O\left(\frac{1}{\text{polylog}d}\right) \text{ for } j \neq j_1, j'_1 \in \tau(\mathcal{Y}), \\ F_{5,j}^{(t)}(\mathbf{Z}^{2,\ell-1}) &\leq m \cdot \tilde{O}(\sigma_0^q) \text{ for } j \notin \tau(\mathcal{Y}). \end{aligned}$$

Putting it together, we obtain

$$\mathbf{logit}_{5,j_1}^{(t)} = \frac{1}{1 + e^{F_{5,j'_1}^{(t)} - F_{5,j_1}^{(t)}} + \left(\sum_{j \neq j_1, j'_1 \in \tau(\mathcal{Y})} e^{F_{5,j}^{(t)}} + \sum_{j \notin \tau(\mathcal{Y})} e^{F_{5,j}^{(t)}} \right) \cdot e^{-F_{5,j_1}^{(t)}}}$$

$$= \frac{1}{1 + e^{\Lambda_{5,j'_1}^{(t)}, r_{g_2 \cdot y_0}} - \Lambda_{5,j_1}^{(t)}, r_{g_1 \cdot y_0}} + \left(O(\log d) \cdot e^{O(\frac{1}{\text{polylog} d})} + O(d) \cdot e^{\tilde{O}(m\sigma_0^q)} \right) \cdot e^{-F_{5,j_1}^{(t)}}.$$

Thus, by Lemma 5.B.46, and the fact that $B = C_B \log d$ for some sufficiently large constant $C_B > 0$, we have $\text{logit}_{5,j_1}^{(t)} = \frac{1}{1 + e^{-O(\delta)} + O(1) \cdot e^{-(C_B/3-1)\log d}} = \Omega(1)$. Similarly, we have

$$\begin{aligned} \text{logit}_{5,j'_1}^{(t)} &= \frac{1}{1 + e^{-\Lambda_{5,j'_1}^{(t)}, r_{g_2 \cdot y_0}} + \Lambda_{5,j_1}^{(t)}, r_{g_1 \cdot y_0}} + \left(O(\log d) \cdot e^{O(\frac{1}{\text{polylog} d})} + O(d) \cdot e^{\tilde{O}(m\sigma_0^q)} \right) \cdot e^{-F_{5,j'_1}^{(t)}} \\ &= \frac{1}{1 + e^{O(1)} + O(1) \cdot e^{-(C_B/3-1)\log d}} = \Omega(1). \end{aligned}$$

From the above analysis, it is easy to see that $1 - \text{logit}_{5,j_1}^{(t)} - \text{logit}_{5,j'_1}^{(t)} \leq O\left(\frac{1}{e^{(C_B/3-1)\log d}}\right) = O\left(\frac{1}{\text{poly} d}\right)$.

- For $\ell = 2$, by Lemma 5.B.45 and Lemma 5.B.40, we have

$$F_{5,j}^{(t)}(\mathbf{Z}^{2,\ell-1}) = \sum_{r \in [m]} \text{sReLU}(\Lambda_{5,j,r}^{(t)}) \in [\varrho m/q, O(1) + \varrho(m/q - 1)] \text{ for } j \in \{\tau(g_\ell(y_{\ell'}))\}_{\ell \in [2], \ell' \in \{0,1\}}$$

$$F_{5,j}^{(t)}(\mathbf{Z}^{2,\ell-1}) = \sum_{r \in [m]} \text{sReLU}(\Lambda_{5,j,r}^{(t)}) \leq O\left(\frac{1}{\text{polylog} d}\right) \text{ for } j \in \tau(\mathcal{Y}) \setminus \{\tau(g_\ell(y_{\ell'}))\}_{\ell \in [2], \ell' \in \{0,1\}}$$

$$F_{5,j}^{(t)}(\mathbf{Z}^{2,\ell-1}) \leq m \cdot \tilde{O}(\sigma_0^q) \text{ for } j \notin \tau(\mathcal{Y}).$$

Therefore, for any j , we have $\text{logit}_j^{(t)} = O(\frac{1}{d})$ since $F_{5,j'}^{(t)} \leq O(1)$ for all j' .

□

In the following, we illustrate the activations on the non-high probability event.

Lemma 5.B.48. *If Induction 5.B.10 holds for all iterations $< t$, given input $\mathbf{Z}^{2,\ell-1}$, then we have*

1. for $\ell = 1$, if $\mathbf{Z}^{2,\ell-1} \notin \mathcal{E}_1$, then
 - (a) for $j = j_1$, $\widehat{\mathfrak{A}}_{j_1} = \{r_{g_1 \cdot y_0}\}$, $\Lambda_{5,j_1,r_{g_1 \cdot y_0}}^{(t)} = B \pm O(\delta)$;
 - (b) for $j \neq j_1 \in \tau(\mathcal{Y})$, assuming $j = \tau(g_1(y))$, then $\Lambda_{5,j,r_{g_1 \cdot y}}^{(t)} = \frac{1}{3}B \pm O(1)$ and $\Lambda_{5,j,r}^{(t)} \ll -\varrho$ for $r \in \widehat{\mathfrak{A}}_j \setminus \{r_{g_1 \cdot y}\}$.
2. $\ell = 2$, if $\mathbf{Z}^{2,\ell-1} \notin \mathcal{E}_2$, then
 - (a) if $g_1 = g_2 \wedge y_0 \neq y_1$,
 - i. for $j = j_2$, $\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} = \frac{1}{2}B \pm O(1)$ and $\Lambda_{5,j_2,r}^{(t)} \ll -\varrho$ for $r \in \widehat{\mathfrak{A}}_{j_2} \setminus \{r_{g_2 \cdot y_1}\}$;
 - ii. for $j = \tau(g_2(y_0))$, $\Lambda_{5,j_2,r_{g_2 \cdot y_0}}^{(t)} = \frac{1}{2}B \pm O(1)$ and $\Lambda_{5,j,r}^{(t)} \ll -\varrho$ for $r \in \widehat{\mathfrak{A}}_j \setminus \{r_{g_2 \cdot y_0}\}$;
 - iii. for other $j \in \tau(\mathcal{Y})$, assuming $j = \tau(g_2(y))$, $|\Lambda_{5,j,r_{g_2 \cdot y}}^{(t)}| \leq O(1)$ and $\Lambda_{5,j,r}^{(t)} \ll -\varrho$ for $r \in \widehat{\mathfrak{A}}_j \setminus \{r_{g_2 \cdot y}\}$.
 - (b) if $g_1 \neq g_2 \wedge y_0 = y_1$,
 - i. for $j = j_2$, $\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} = \frac{1}{2}B \pm O(1)$ and $\Lambda_{5,j_2,r}^{(t)} \ll -\varrho$ for $r \in \widehat{\mathfrak{A}}_{j_2} \setminus \{r_{g_2 \cdot y_1}\}$;

- ii. for $j = \tau(g_1(y_1))$, $\Lambda_{5,j,r_{g_1 \cdot y_1}}^{(t)} = \frac{1}{2}B \pm O(1)$ and $\Lambda_{5,j,r}^{(t)} \ll -\varrho$ for $r \in \widehat{\mathfrak{U}}_j \setminus \{r_{g_1 \cdot y_1}\}$;
- iii. for other $j \in \tau(\mathcal{Y})$, assuming $j = \tau(g(y_1))$, $|\Lambda_{5,j,r_{g \cdot y_1}}^{(t)}| \leq O(1)$ and $\Lambda_{5,j,r}^{(t)} \ll -\varrho$ for $r \in \widehat{\mathfrak{U}}_j \setminus \{r_{g \cdot y_1}\}$.
- (c) if $g_1 = g_2 \wedge y_0 = y_1$,
 - i. for $j = j_2$, $\widehat{\mathfrak{U}}_{j_2} = \{r_{g_2 \cdot y_1}\}$, $\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} = B \pm O(\delta)$;
 - ii. for $j \neq j_2 \in \tau(\mathcal{Y})$, $|\Lambda_{5,j,r}^{(t)}| \leq O(1)$ for all $r \in \widehat{\mathfrak{U}}_j$.
- (d) $g_1 \neq g_2 \wedge y_0 \neq y_1 \wedge (g_1(y_0) = g_2(y_1) \vee g_2(y_0) = g_1(y_1))$, $|\Lambda_{5,j,r}^{(t)}| \leq O(1)$ for all $r \in \widehat{\mathfrak{U}}_j$.

With the characterization of activated neurons, we can derive the following logits for the non-high probability event.

Lemma 5.B.49. *If Induction 5.B.10 holds for all iterations $< t$, given input $\mathbf{Z}^{2,\ell-1}$, then we have*

1. for $\ell = 1$, if $\mathbf{Z}^{2,\ell-1} \notin \mathcal{E}_1$, then $1 - \mathbf{logit}_{5,j_1}^{(t)} = O\left(\frac{1}{\text{poly}d}\right)$.
2. $\ell = 2$, if $\mathbf{Z}^{2,\ell-1} \notin \mathcal{E}_2$, then
 - (a) if $g_1 = g_2 \wedge y_0 \neq y_1$, $\mathbf{logit}_{5,j}^{(t)} = \Omega(1)$ for $j \in \{j_2, \tau(g_2(y_0))\}$, $1 - \mathbf{logit}_{5,j_2}^{(t)} - \mathbf{logit}_{5,\tau(g_2(y_0))}^{(t)} = \frac{1}{\text{poly}d}$.
 - (b) if $g_1 \neq g_2 \wedge y_0 = y_1$, $\mathbf{logit}_{5,j}^{(t)} = \Omega(1)$ for $j \in \{j_2, \tau(g_1(y_1))\}$, $1 - \mathbf{logit}_{5,j_2}^{(t)} - \mathbf{logit}_{5,\tau(g_1(y_1))}^{(t)} = \frac{1}{\text{poly}d}$.
 - (c) if $g_1 = g_2 \wedge y_0 = y_1$, $1 - \mathbf{logit}_{5,j_2}^{(t)} = O\left(\frac{1}{\text{poly}d}\right)$.
 - (d) $g_1 \neq g_2 \wedge y_0 \neq y_1 \wedge (g_1(y_0) = g_2(y_1) \vee g_2(y_0) = g_1(y_1))$, $\mathbf{logit}_{5,j_2}^{(t)} = O\left(\frac{1}{d}\right)$ for all j .

Gradient Lemma

Lemma 5.B.50. *If Induction 5.B.10 holds for all iterations $< t$, given $s \in \tau(\mathcal{X})$, for $[\mathbf{Q}_{4,3}^{(t)}]_{s,s}$, we have*

$$\sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,3}^{(t)}} \text{Loss}_5^{2,\ell} \right]_{s,s} = \Theta\left(\frac{\log d}{d}\right).$$

Proof. By gradient decompositions, we have

$$\sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,3}^{(t)}} \text{Loss}_5^{2,\ell} \right]_{s,s} = \sum_{\ell \in [2]} \sum_{\kappa \in \{i, ii, iii\}} \mathcal{N}_{s,3,\ell,\kappa}^{(t)}.$$

By Lemma 5.B.47 and Lemma 5.B.49, it is straightforward to see that $|\mathcal{N}_{s,3,1,iii}^{(t)}|, |\mathcal{N}_{s,3,2,iii}^{(t)}| = O\left(\frac{1}{\text{poly}d}\right)$, and thus we can focus on other terms.

By Lemma 5.B.45 and Lemma 5.B.48, we have

$$\mathcal{N}_{s,3,1,i}^{(t)} = \mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(t)} \cdot (1 - \mathbf{logit}_{5,j_1}^{(t)}) \right].$$

$$\begin{aligned}
& \left(\left(V_{j_1, r_{g_1 \cdot y_0}}(g_1) - \Lambda_{5, j_1, r_{g_1 \cdot y_0}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_0)=s\} \cap \mathcal{E}_1} \Bigg] \\
& + \mathbb{E} \left[\mathbf{Attn}_{\text{ans}, 0 \rightarrow \text{pred}, 1}^{(t)} \cdot (1 - \mathbf{logit}_{5, j_1}^{(t)}) \cdot \right. \\
& \quad \left. \left(\left(V_{j_1, r_{g_1 \cdot y_0}}(g_1) - \Lambda_{5, j_1, r_{g_1 \cdot y_0}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_0)=s\} \cap \mathcal{E}_1^c} \right] \\
& \stackrel{(a)}{=} \Theta\left(\frac{1}{d}\right) \cdot \Omega(1) \cdot \Theta(B) + \Theta\left(\frac{1}{d}\right) \cdot O\left(\frac{1}{\text{poly} d}\right) \cdot \Theta(B) \cdot O\left(\frac{1}{\log d}\right) \\
& = \Theta\left(\frac{\log d}{d}\right),
\end{aligned}$$

where (a) follows from Lemma 5.B.46, Lemma 5.B.47, Lemma 5.B.48 and Lemma 5.B.49, and the fact that $\tau(x_0) = s$ holds with probability $\frac{1}{d}$. $\mathcal{N}_{s, 3, 2, i}^{(t)}$ can be upper bounded similarly.

Moving to $\mathcal{N}_{s, 3, 1, ii}^{(t)}$, noticing that $V_{j'_1, r_{g_2 \cdot y_0}}(g_1) = -B + O(\delta)$ on $\mathcal{E}_1$, we have

$$\begin{aligned}
\mathcal{N}_{s, 3, 1, ii}^{(t)} &= -\mathbb{E} \left[\mathbf{Attn}_{\text{ans}, 0 \rightarrow \text{pred}, 1}^{(t)} \cdot \mathbf{logit}_{5, j'_1}^{(t)} \cdot \right. \\
& \quad \left. \left(\left(V_{j'_1, r_{g_2 \cdot y_0}}(g_1) - \Lambda_{5, j'_1, r_{g_2 \cdot y_0}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_0)=s\} \cap \mathcal{E}_1} \right] \\
&= -\mathbb{E} \left[\mathbf{Attn}_{\text{ans}, 0 \rightarrow \text{pred}, 1}^{(t)} \cdot \sum_{y \neq y_0 \in \mathcal{Y}} \mathbf{logit}_{5, \tau(g_1(y))}^{(t)} \cdot \right. \\
& \quad \left. \left(\left(V_{\tau(g_1(y)), r_{g_1 \cdot y}}(g_1) - \Lambda_{5, \tau(g_1(y)), r_{g_1 \cdot y}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_0)=s\} \cap \mathcal{E}_1^c} \right] \\
&= \Theta\left(\frac{1}{d}\right) \cdot \Omega(1) \cdot \Theta(B) - \Theta\left(\frac{1}{d}\right) \cdot O\left(\frac{1}{\text{poly} d}\right) \cdot \Theta(B) \cdot O\left(\frac{1}{\log d}\right) \\
&= \Theta\left(\frac{\log d}{d}\right).
\end{aligned}$$

For $\mathcal{N}_{s, 3, 2, ii}^{(t)}$, we only need to control the negative gradient, since the positive part can be easily upper bounded by $O\left(\frac{\log d}{d}\right)$.

$$\begin{aligned}
\mathcal{N}_{s, 3, 2, ii}^{(t)} &\geq -\Theta\left(\frac{1}{d}\right) \cdot O\left(\frac{1}{d}\right) \cdot \Theta(B) - \mathbb{E} \left[\mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, 2}^{(t)} \cdot \sum_{y \neq y_1 \in \mathcal{Y}} \mathbf{logit}_{5, \tau(g_2(y))}^{(t)} \cdot \right. \\
& \quad \left. \left(\left(V_{\tau(g_2(y)), r_{g_2 \cdot y}}(g_2) - \Lambda_{5, j, r} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_0)=s, g_1=g_2, y_0 \neq y_1\}} \right] \\
&\stackrel{(a)}{\geq} -\Theta\left(\frac{1}{d}\right) \cdot O\left(\frac{1}{d}\right) \cdot \Theta(B) - \Theta\left(\frac{1}{d}\right) \cdot O(1) \cdot \Theta(B) \cdot O\left(\frac{1}{\log d}\right)
\end{aligned}$$

$$\geq -O\left(\frac{1}{d}\right),$$

where (a) is due to the fact that $\mathbf{logit}_{5,\tau(g_2(y_0))}^{(t)} = \Omega(1)$ and $\mathbf{logit}_{5,\tau(g_2(y))}^{(t)} = O(\frac{1}{d})$ for other $y \neq y_1, y_0$. Putting everything together, we complete the proof. $\square$

Lemma 5.B.51 (Negative gradient). *If Induction 5.B.10 holds for all iterations $< t$, given $s \in \tau(\mathcal{X})$, we have*

$$\sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,4}^{(t)}} \text{Loss}_5^{2,\ell} \right]_{s,s} \geq -O\left(\frac{1}{\log d}\right) \sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,3}^{(t)}} \text{Loss}_5^{2,\ell} \right]_{s,s}.$$

Proof. By gradient decompositions, we have

$$\sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,4}^{(t)}} \text{Loss}_5^{2,\ell} \right]_{s,s} = \sum_{\ell \in [2]} \sum_{\kappa \in \{i, ii, iii\}} \mathcal{N}_{s,4,\ell,\kappa}^{(t)}.$$

Similarly as Lemma 5.B.50, $|\mathcal{N}_{s,4,1,iii}^{(t)}|, |\mathcal{N}_{s,4,2,iii}^{(t)}| = O(\frac{1}{\text{poly}d})$, and thus we can focus on other terms. By Lemma 5.B.45, Lemma 5.B.46, and Lemma 5.B.48, we have

$$\begin{aligned} \mathcal{N}_{s,4,1,i}^{(t)} + \mathcal{N}_{s,4,1,ii}^{(t)} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(t)} \cdot (1 - \mathbf{logit}_{5,j_1}^{(t)}) \cdot \right. \\ &\quad \left. \left((V_{j_1, r_{g_1 \cdot y_0}}(y_0) - \Lambda_{5,j_1, r_{g_1 \cdot y_0}}^{(t)} \pm \tilde{O}(\sigma_0)) \pm \tilde{O}(\delta^q) \right) \mathbf{1}_{\{\tau(x_0)=s\} \cap \mathcal{E}_1} \right] \end{aligned} \quad (5.B.43)$$

$$\begin{aligned} &+ \mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(t)} \cdot (1 - \mathbf{logit}_{5,j_1}^{(t)}) \cdot \right. \\ &\quad \left. \left((V_{j_1, r_{g_1 \cdot y_0}}(y_0) - \Lambda_{5,j_1, r_{g_1 \cdot y_0}}^{(t)} \pm \tilde{O}(\sigma_0)) \pm \tilde{O}(\delta^q) \right) \mathbf{1}_{\{\tau(x_0)=s\} \cap \mathcal{E}_1^c} \right] \end{aligned} \quad (5.B.44)$$

$$\begin{aligned} &- \mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(t)} \cdot \mathbf{logit}_{5,j_1}^{(t)} \cdot \right. \\ &\quad \left. \left((V_{j_1', r_{g_2 \cdot y_0}}(y_0) - \Lambda_{5,j_1', r_{g_2 \cdot y_0}}^{(t)} \pm \tilde{O}(\sigma_0)) \pm \tilde{O}(\delta^q) \right) \mathbf{1}_{\{\tau(x_0)=s\} \cap \mathcal{E}_1} \right] \end{aligned} \quad (5.B.45)$$

$$\begin{aligned} &- \mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(t)} \cdot \sum_{g \neq g_1 \in \mathcal{G}} \mathbf{logit}_{5,\tau(g(y_0))}^{(t)} \cdot \right. \\ &\quad \left. \left((V_{\tau(g(y_0)), r_{g \cdot y_0}}(y_0) - \Lambda_{5,\tau(g(y_0)), r_{g \cdot y_0}}^{(t)} \pm \tilde{O}(\sigma_0)) \pm \tilde{O}(\delta^q) \right) \mathbf{1}_{\{\tau(x_0)=s\} \cap \mathcal{E}_1^c} \right]. \end{aligned} \quad (5.B.46)$$

First, considering the event $\{\tau(x_0) = s\} \cap \mathcal{E}_1$, we have

$$\begin{aligned}
& (5.B.43) + (5.B.45) \\
&= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(t)} \cdot \left((1 - \mathbf{logit}_{5,j_1}^{(t)}) \cdot \left(V_{j_1, r_{s_1 y_1}}(y_0) - \Lambda_{5,j_1, r_{s_1 y_0}}^{(t)} \pm \tilde{O}(\sigma_0) \pm \tilde{O}(\delta^q) \right) \right. \right. \\
&\quad \left. \left. - \mathbf{logit}_{5,j'_1}^{(t)} \cdot \left(V_{j'_1, r_{s_2 y_0}}(y_0) - \Lambda_{5,j'_1, r_{s_2 y_0}}^{(t)} \pm \tilde{O}(\sigma_0) \pm \tilde{O}(\delta^q) \right) \right) \mathbb{1}_{\{\tau(x_0)=s\} \cap \mathcal{E}_1} \right] \\
&\stackrel{(a)}{=} \mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(t)} \cdot \left((1 - \mathbf{logit}_{5,j_1}^{(t)}) \cdot \left(V_{j_1, r_{s_1 y_1}}(y_0) - \Lambda_{5,j_1, r_{s_1 y_0}}^{(t)} \pm \tilde{O}(\sigma_0) \pm \tilde{O}(\delta^q) \right) \right. \right. \\
&\quad \left. \left. - (1 - \mathbf{logit}_{5,j_1}^{(t)} - \frac{1}{d}) \cdot \left(V_{j'_1, r_{s_2 y_0}}(y_0) - \Lambda_{5,j'_1, r_{s_2 y_0}}^{(t)} \pm \tilde{O}(\sigma_0) \pm \tilde{O}(\delta^q) \right) \right) \mathbb{1}_{\{\tau(x_0)=s\} \cap \mathcal{E}_1} \right] \\
&\geq -\mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(t)} \cdot \left((1 - \mathbf{logit}_{5,j_1}^{(t)}) \cdot \right. \right. \\
&\quad \left. \left. \left(V_{j_1, r_{s_1 y_1}}(y_0) - V_{j'_1, r_{s_2 y_0}}(y_0) + \Lambda_{5,j'_1, r_{s_2 y_0}}^{(t)} - \Lambda_{5,j_1, r_{s_1 y_0}}^{(t)} \pm \tilde{O}(\sigma_0) \pm \tilde{O}(\delta^q) \right) \right) \mathbb{1}_{\{\tau(x_0)=s\} \cap \mathcal{E}_1} \right] \\
&\stackrel{(b)}{\geq} -\mathbb{E} \left[\mathbf{Attn}_{\mathbb{A},0 \rightarrow \mathbb{A},0}^{(t)} \cdot \left((1 - \mathbf{logit}_{5,j_1}^{(t)}) \cdot \left(O(\delta) + O(1) \pm \tilde{O}(\sigma_0) \pm \tilde{O}(\delta^q) \right) \right) \mathbb{1}_{\{\tau(x_0)=s\} \cap \mathcal{E}_1} \right] \\
&\geq -O\left(\frac{\mathcal{N}_{s,3,1,i}^{(t)}}{\log d}\right),
\end{aligned}$$

where (a) follows from Lemma 5.B.47; (b) follows from Lemma 5.B.46 and Lemma 5.B.37.

Notice that (5.B.44) ≥ 0 since $V_{j_1, r_{s_1 y_0}}(y_0) - \Lambda_{5,j_1, r_{s_1 y_0}}^{(t)} \geq \Omega(B)$, thus we just need to consider the possible negative gradient from (5.B.46). By Lemma 5.B.49, we have $\sum_{g \neq s_1 \in \mathcal{G}} \mathbf{logit}_{5, \tau(g(y_0))}^{(t)} \leq O(\frac{1}{\text{poly} d})$ on $\{\tau(x_0) = s\} \cap \mathcal{E}_1^c$, and hence (5.B.46) $\ll \mathcal{N}_{s,3,1,i}^{(t)}$.

Moving to the gradient from $\ell = 2$, it is straightforward to see that $\mathcal{N}_{s,4,2,i}^{(t)}$ is non-negative, and thus we can focus on the possible negative gradient from $\mathcal{N}_{s,4,2,ii}^{(t)}$.

$$\begin{aligned}
\mathcal{N}_{s,4,2,ii}^{(t)} &= -\mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \cdot \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j}^{(t)} \cdot \right. \\
&\quad \left. \left(\sum_{r \in \mathfrak{R}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)}) \cdot \left(V_{j,r}(y_1) - \Lambda_{5,j,r}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \mathcal{E}_2} \right] \\
&\quad - \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \cdot \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j}^{(t)} \cdot \right.
\end{aligned}$$

$$\begin{aligned}
& \left(\sum_{r \in \mathfrak{R}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)}) \cdot \left(V_{j,r}(y_1) - \Lambda_{5,j,r}^{(t)} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \mathbf{1}_{\{\tau(x_1)=s\} \cap \mathcal{E}_2^c} \Bigg] \\
& \geq -\Theta\left(\frac{1}{d}\right) \cdot O\left(\frac{1}{d}\right) \cdot \Theta(B) - \Theta\left(\frac{1}{d}\right) \cdot O(1) \cdot \Theta(B) \cdot O\left(\frac{1}{\log d}\right) \geq -O\left(\frac{\mathcal{N}_{s,3,1,i}^{(t)}}{\log d}\right).
\end{aligned}$$

Putting everything together, and combining with the fact that $\sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,3}^{(t)}} \text{Loss}_5^{2,\ell} \right]_{s,s} = \Theta(\mathcal{N}_{s,3,1,i}^{(t)})$ from Lemma 5.B.50, we complete the proof. $\square$

Lemma 5.B.52 (Growth of gap). *If Induction 5.B.10 holds for all iterations $< t$, given $s \in \tau(X)$, we have*

$$\sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,3}^{(t)}} \text{Loss}_5^{2,\ell} \right]_{s,s} + \sum_{\ell=1}^2 \left[\nabla_{\mathbf{Q}_{4,4}^{(t)}} \text{Loss}_5^{2,\ell} \right]_{s,s} \geq \Omega \left(\sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,3}^{(t)}} \text{Loss}_5^{2,\ell} \right]_{s,s} \right).$$

Proof. By gradient decompositions in (5.B.29)-(5.B.40), we have

$$\sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,3}^{(t)}} \text{Loss}_5^{2,\ell} \right]_{s,s} + \sum_{\ell=1}^2 \left[\nabla_{\mathbf{Q}_{4,4}^{(t)}} \text{Loss}_5^{2,\ell} \right]_{s,s} = \sum_{\ell \in [2]} \sum_{\kappa \in \{i,ii,iii\}} \mathcal{N}_{s,3,\ell,\kappa}^{(t)} - \mathcal{N}_{s,4,\ell,\kappa}^{(t)}.$$

Due to Lemma 5.B.47 and Lemma 5.B.49, $|\mathcal{N}_{s,p,\ell,iii}^{(t)}| = O(\frac{1}{\text{poly}d})$ for $p \in \{3,4\}$ and $\ell \in [2]$, we can focus on the gradient difference between $\mathbf{Q}_{4,3}^{(t)}$ and $\mathbf{Q}_{4,4}^{(t)}$ contributed by other terms.

By Lemma 5.B.44, we have

$$\mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(t)} \leq \mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1'}^{(t)} \quad \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \leq \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}.$$

Hence, for $\ell \in [2]$, we have

$$\mathcal{N}_{s,3,\ell,i}^{(t)} - \mathcal{N}_{s,4,\ell,i}^{(t)} \geq -O(\delta) \cdot O\left(\frac{1}{d}\right) \cdot \Theta(1) \geq -O(\delta/d).$$

$$\begin{aligned}
& \mathcal{N}_{s,3,1,ii}^{(t)} - \mathcal{N}_{s,4,1,ii}^{(t)} \\
& = -\mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(t)} \cdot \mathbf{logit}_{5,j_1'}^{(t)} \cdot \right. \\
& \quad \left. \left(\left(V_{j_1',r_{s_2,y_0}}(g_1) - \Lambda_{5,j_1',r_{s_2,y_0}}^{(t)} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \mathbf{1}_{\{\tau(x_0)=s\} \cap \mathcal{E}_1} \right] \\
& + \mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(t)} \cdot \mathbf{logit}_{5,j_1'}^{(t)} \cdot \right. \\
& \quad \left. \left(\left(V_{j_1',r_{s_2,y_0}}(y_0) - \Lambda_{5,j_1',r_{s_2,y_0}}^{(t)} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \mathbf{1}_{\{\tau(x_0)=s\} \cap \mathcal{E}_1} \right]
\end{aligned}$$

$$\begin{aligned}
& -\mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(t)} \cdot \sum_{y \neq y_0 \in \mathcal{Y}} \mathbf{logit}_{5,\tau(g_1(y))}^{(t)} \cdot \right. \\
& \quad \left. \left(\left(V_{\tau(g_1(y)),r_{g_1,y}}(g_1) - \Lambda_{5,\tau(g_1(y)),r_{g_1,y}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbf{1}_{\{\tau(x_0)=s\} \cap \mathcal{E}_1^c} \right] \\
& + \mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(t)} \cdot \sum_{y \neq y_0 \in \mathcal{Y}} \mathbf{logit}_{5,\tau(g_1(y))}^{(t)} \cdot \right. \\
& \quad \left. \left(\left(V_{\tau(g_1(y)),r_{g_1,y}}(y_0) - \Lambda_{5,\tau(g_1(y)),r_{g_1,y}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbf{1}_{\{\tau(x_0)=s\} \cap \mathcal{E}_1^c} \right] \\
& \stackrel{(a)}{\geq} \Theta\left(\frac{1}{d}\right) \cdot \Omega(1) \cdot \Theta(B) - \Theta\left(\frac{1}{d}\right) \cdot O\left(\frac{1}{\text{poly}d}\right) \cdot O(B) \cdot O\left(\frac{1}{\log d}\right) \geq \Omega\left(\frac{\log d}{d}\right),
\end{aligned}$$

where (a) follows from Lemma 5.B.37 that $V_{j'_1, r_{g_2, y_0}}(g_1) - \Lambda_{5, j'_1, r_{g_2, y_0}}^{(t)} \leq -\Omega(B)$, and $V_{j'_1, r_{g_2, y_0}}(y_0) - \Lambda_{5, j'_1, r_{g_2, y_0}}^{(t)} \geq \Omega(B)$.

$$\begin{aligned}
& \mathcal{N}_{s,3,2,ii}^{(t)} - \mathcal{N}_{s,4,2,ii}^{(t)} \\
& = -\mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \cdot \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j}^{(t)} \cdot \right. \\
& \quad \left. \left(\sum_{r \in \widehat{\mathfrak{A}}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)}) \cdot \left(V_{j,r}(g_2) - \Lambda_{5,j,r}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbf{1}_{\{\tau(x_1)=s\} \cap \mathcal{E}_2} \right] \\
& + \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \cdot \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j}^{(t)} \cdot \right. \\
& \quad \left. \left(\sum_{r \in \widehat{\mathfrak{A}}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)}) \cdot \left(V_{j,r}(y_1) - \Lambda_{5,j,r}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbf{1}_{\{\tau(x_1)=s\} \cap \mathcal{E}_2} \right] \\
& - \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \cdot \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j}^{(t)} \cdot \right. \\
& \quad \left. \left(\sum_{r \in \widehat{\mathfrak{A}}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)}) \cdot \left(V_{j,r}(g_2) - \Lambda_{5,j,r}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbf{1}_{\{\tau(x_1)=s\} \cap \mathcal{E}_2^c} \right] \\
& + \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \cdot \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j}^{(t)} \cdot \right. \\
& \quad \left. \left(\sum_{r \in \widehat{\mathfrak{A}}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)}) \cdot \left(V_{j,r}(y_1) - \Lambda_{5,j,r}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbf{1}_{\{\tau(x_1)=s\} \cap \mathcal{E}_2^c} \right]
\end{aligned}$$

$$\stackrel{(a)}{\geq} -\Theta\left(\frac{1}{d}\right) \cdot O\left(\frac{1}{d}\right) \cdot \Theta(B) - \Theta\left(\frac{1}{d}\right) \cdot O(1) \cdot \Theta(B) \cdot O\left(\frac{1}{\log d}\right) \geq -O\left(\frac{1}{d}\right),$$

where (a) follows from Lemma 5.B.45 and Lemma 5.B.47, which together imply that on the event $\mathcal{E}_2$, only a constant number of neurons are activated and $\mathbf{logit}_{5,j}^{(t)} \leq O\left(\frac{1}{d}\right)$ for all $j \neq j_2$; and from Lemma 5.B.49, which implies that on the complement event $\mathcal{E}_2^c$, occurring with probability at most $O\left(\frac{1}{\log d}\right)$, there exists at most one $j \neq j_2$ such that $\mathbf{logit}_{5,j}^{(t)} \geq \Omega(1)$ while $\mathbf{logit}_{5,j}^{(t)} \leq O\left(\frac{1}{\text{poly}d}\right)$ for other $j \in \tau(\mathcal{Y})$.

Putting it all together, we finish the proof. $\square$

Lemma 5.B.53. *If Induction 5.B.10 holds for all iterations $< t$, given $s \in \tau(\mathcal{X})$, for $[\mathbf{Q}_{4,3}^{(t)}]_{s,s} \geq \Omega\left(\frac{\varrho}{\log d}\right)$, we have*

$$\sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,4}^{(t)}} \text{Loss}_5^{2,\ell} \right]_{s,s} \geq \Omega\left(\sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,3}^{(t)}} \text{Loss}_5^{2,\ell} \right]_{s,s} \right).$$

Proof. Notice that by Lemma 5.B.51, when $[\mathbf{Q}_{4,3}^{(t)}]_{s,s} \geq \Omega\left(\frac{\varrho}{\log d}\right)$, we have $[\mathbf{Q}_{4,4}^{(t)}]_{s,s} \geq -O\left(\frac{\varrho}{\log^2 d}\right)$. Hence

$$\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \geq \Omega\left(\frac{\varrho}{\log d}\right),$$

which implies $\Lambda_{5,j_2,r_{\mathcal{S}_2:y_1}}^{(t)}(\mathbf{Z}^{2,1}) \geq \Omega\left(\frac{\varrho}{\log d}\right) \cdot B \pm O(\delta) \geq \varrho$ already lies in the linear regime for $\mathbf{Z}^{2,1} \in \mathcal{E}_2$. Then we have

$$\begin{aligned} \mathcal{N}_{s,4,2,i}^{(t)} &\geq \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \cdot \left((1 - \mathbf{logit}_{5,j_2}^{(t)}) \cdot \right. \right. \\ &\quad \left. \left(V_{j_2,r_{\mathcal{S}_2:y_1}} - \Lambda_{5,j_2,r_{\mathcal{S}_2:y_1}}^{(t)} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \mathcal{E}_2} \right] \\ &\geq \Omega(1) \cdot \Theta(B) \cdot \Theta\left(\frac{1}{d}\right) \geq \Omega\left(\frac{\log d}{d}\right). \end{aligned}$$

Moreover, from Lemma 5.B.51, the magnitude of negative gradient from other $\mathcal{N}$ terms can be upper bounded by $O\left(\frac{\mathcal{N}_{s,3,1,i}^{(t)}}{\log d}\right)$. Therefore, combining with the fact that $\sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,3}^{(t)}} \text{Loss}_5^{2,\ell} \right]_{s,s} = \Theta(\mathcal{N}_{s,3,1,i}^{(t)})$ from Lemma 5.B.50, we complete the proof. $\square$

Lemma 5.B.54. *If Induction 5.B.10 holds for all iterations $< t$, given $s \neq s' \in \tau(\mathcal{X})$, for $p \in \{3, 4\}$, we have*

$$\left| \sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,p}^{(t)}} \text{Loss}_5^{2,\ell} \right]_{s,s'} \right| \leq O\left(\frac{\log d}{d^2}\right) = O\left(\frac{1}{d}\right) \cdot \left| \sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,p}^{(t)}} \text{Loss}_5^{2,\ell} \right]_{s,s} \right|.$$

Proof. The proof follows directly by combining the expressions from Lemma 5.B.42, Lemma 5.B.43, and Lemma 5.B.50, along with the fact that the event $\{\tau(x_0) = s, \tau(x_1) = s'\}$ occurs with probability $O(\frac{1}{d^2})$. $\square$

At the End of Stage 2.1 Putting gradient lemmas together, we can directly prove that Induction 5.B.10 holds for all iterations t until the end of stage 2.1, where we can conclude the following:

Lemma 5.B.55 (End of stage 2.1). *Given $s \in \tau(X)$, Induction 5.B.10 holds for all iterations $t < T_{2,1,s} = O(\frac{d}{\eta \log^2 d})$, then at the end of stage 2.1, we have*

- (a) $[\mathbf{Q}_{4,p}^{(t)}]_{s,s} = \Omega(\frac{1}{\log d})$ for $p \in \{3, 4\}$;
- (b) $[\mathbf{Q}_{4,3}^{(t)}]_{s,s} - [\mathbf{Q}_{4,4}^{(t)}]_{s,s} \geq \Omega(\frac{1}{\log d})$;
- (c) $|\left[\mathbf{Q}_{4,p}^{(t)}\right]_{s,s'}| \leq O(\frac{[\mathbf{Q}_{4,p}^{(t)}]_{s,s}}{d})$ for $s' \in \tau(X) \neq s$ for $p \in \{3, 4\}$; otherwise, $[\mathbf{Q}_{4,p}^{(t)}]_{s,s'} = 0$.

Stage 2.2: Continual Growth of Diagonal Entries

In Stage 2.2, the diagonal entries $[\mathbf{Q}_{4,3}]_{s,s}$ and $[\mathbf{Q}_{4,4}]_{s,s}$ continue to grow until they reach a certain threshold. The analysis in this stage parallels that of Stage 2.1, but our focus now shifts to the gradients contributed by $\ell = 2$, as the logit at $\ell = 1$ is already near-optimal and thus contributes negligibly to the growth of $\mathbf{Q}_{4,3}$ and $\mathbf{Q}_{4,4}$.

Induction 5.B.11. *Given $s \in \tau(X)$, let $T_{2,2,s}$ denote the first time that $[\mathbf{Q}_{4,3}]_{s,s}$ reaches 0.0001. For all iterations $T_{2,1,s} \leq t < T_{2,2,s}$, we have the following holds*

- (a) $[\mathbf{Q}_{4,3}^{(t)}]_{s,s}, [\mathbf{Q}_{4,4}^{(t)}]_{s,s} \leq O(1)$ monotonically increases;
- (b) $[\mathbf{Q}_{4,3}^{(t)}]_{s,s} - [\mathbf{Q}_{4,4}^{(t)}]_{s,s} \in \left[\Omega(\frac{1}{\log d}), O(1)\right]$;
- (c) for $(p, q) \in \{(4, 3), (4, 4)\}$, $|\left[\mathbf{Q}_{p,q}^{(t)}\right]_{s,s'}| \leq O(\frac{[\mathbf{Q}_{p,q}^{(t)}]_{s,s}}{d})$ for $s' \in \tau(X) \neq s$; other $[\mathbf{Q}_{p,q}^{(t)}]_{s,s'} = 0$.

Throughout the following analysis, instead of $\mathcal{E}_2$ defined in (5.B.42), we consider a renewed event $\tilde{\mathcal{E}}_2$ for $\ell = 2$:

$$\tilde{\mathcal{E}}_2 \triangleq \{g_1 \neq g_2 \wedge y_0 \neq y_1\}. \quad (5.B.47)$$

Attention and Logit Preliminaries The proof in this part proceeds analogously to the arguments in Section 5.B.4, with the induction hypothesis from Induction 5.B.11 incorporated. Hence, we omit the details here.

Lemma 5.B.56. *If Induction 5.B.11 holds for all iterations $\in [T_{2,1,s}, t)$, given input $\mathbf{Z}^{2,\ell-1}$, then we have*

1. for $\ell = 1$,

- (a) $\text{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(t)} \in \left[\frac{1}{3} + \Omega(\frac{1}{\log d}), \frac{1}{3} + c_1\right]$, where $c_1 > 0$ is a small constant;
- (b) $\text{Attn}_{\text{ans},0 \rightarrow \text{pred},2}^{(t)} \in \left[\frac{1}{3} - c_2, \frac{1}{3} - \Omega(\frac{1}{\log d})\right]$, where $c_2 > 0$ is a small constant;
- (c) $\text{Attn}_{\text{ans},0 \rightarrow \text{pred},2}^{(t)} + \Omega(\frac{1}{\log d}) \leq \text{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(t)}$.

- (d) $\text{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(t)} - \text{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(t)} \in \left[\Omega\left(\frac{1}{\log d}\right), c_3 \right]$.
2. for $\ell = 2$,
- (a) $\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \in \left[\frac{1}{4} + \Omega\left(\frac{1}{\log d}\right), \frac{1}{4} + c_4 \right]$, where $c_4 > 0$ is a small constant;
- (b) $\text{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)}, \text{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \in \left[\frac{1}{4} - c_5, \frac{1}{4} - \Omega\left(\frac{1}{\log d}\right) \right]$, moreover,
 $|\text{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)}| \leq O\left(\frac{1}{d}\right)$, where $c_5 > 0$ is a small constant;
- (c) $\text{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)}, \text{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \leq \text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \Omega\left(\frac{1}{\log d}\right)$;
- (d) $\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \in \left[\Omega\left(\frac{1}{\log d}\right), c_6 \right]$, where $c_6 > 0$ is a small constant.

Notice that the constant $c_1 - c_6$ depends on the threshold 0.0001 in Induction 5.B.11. We choose the threshold 0.0001 small enough to ensure $2C_B(c_1 + c_2) < 1 - c_6C_B$ and $1 - 4c_5C_B > 0$.

Lemma 5.B.57. If Induction 5.B.11 holds for all iterations $\in [T_{2,1,s}, t)$, given input $\mathbf{Z}^{2,\ell-1}$, then we have

1. for $\ell = 1$, if $\mathbf{Z}^{2,\ell-1} \in \mathcal{E}_1$, then
- (a) for $j = j_1$, $\Lambda_{5,j_1,r}^{(t)} \ll -\rho$ for $r \in \widehat{\mathfrak{A}}_{j_1} \setminus \{r_{g_1 \cdot y_0}\}$;
- (b) for $j = j'_1 \triangleq \tau(g_2(y_0))$, $\Lambda_{5,j'_1,r}^{(t)} \ll -\rho$ for $r \in \widehat{\mathfrak{A}}_{j'_1} \setminus \{r_{g_2 \cdot y_0}\}$;
- (c) for other $j \in \tau(\mathcal{Y})$, r is not activated for all $r \in \widehat{\mathfrak{A}}_j$, i.e., $\Lambda_{5,j,r}^{(t)} \ll -\rho$.
2. $\ell = 2$, if $\mathbf{Z}^{2,\ell-1} \in \widetilde{\mathcal{E}}_2$, then
- (a) for $j = j_2$, $\Lambda_{5,j_2,r}^{(t)} \ll -\rho$ for $r \in \widehat{\mathfrak{A}}_{j_2} \setminus \{r_{g_2 \cdot y_1}\}$;
- (b) for $j = j'_2 \triangleq \tau(g_2(y_0))$, $\Lambda_{5,j'_2,r}^{(t)} \ll -\rho$ for $r \in \widehat{\mathfrak{A}}_{j'_2} \setminus \{r_{g_2 \cdot y_0}\}$;
- (c) for other $j \in \tau(\mathcal{Y})$, r is not activated for all $r \in \widehat{\mathfrak{A}}_j$, i.e., $\Lambda_{5,j,r}^{(t)} \ll -\rho$.

Lemma 5.B.58. If Induction 5.B.11 holds for all iterations $\in [T_{2,1,s}, t)$, given input $\mathbf{Z}^{2,\ell}$, then we have

1. $\ell = 1$, for $\mathbf{Z}^{2,\ell-1} \in \mathcal{E}_1$,
- (a) $\Lambda_{5,j_1,r_{g_1 \cdot y_0}}^{(t)} \geq \left(\frac{1}{3} + \Omega\left(\frac{1}{\log d}\right) \right) B$;
- (b) $\Omega(1) \leq \Lambda_{5,j_1,r_{g_1 \cdot y_0}}^{(t)} - \Lambda_{5,j'_1,r_{g_2 \cdot y_0}}^{(t)} \leq 2(c_1 + c_2)B$.
2. $\ell = 2$, for $\mathbf{Z}^{2,\ell-1} \in \widetilde{\mathcal{E}}_2$,
- (a) $\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} \in \left[\Omega(1), 4c_5B \right]$;
- (b) $\Lambda_{5,j'_2,r_{g_2 \cdot y_0}}^{(t)} \in \left[\Omega(1), c_6B \right]$ and $\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} - \Lambda_{5,j'_2,r_{g_2 \cdot y_0}}^{(t)} \geq \Omega(1)$.

Lemma 5.B.59. If Induction 5.B.11 holds for all iterations $\in [T_{2,1,s}, t)$, given input $\mathbf{Z}^{2,\ell-1}$, then we have

1. for $\ell = 1$, if $\mathbf{Z}^{2,\ell-1} \in \mathcal{E}_1$, $\text{logit}_{5,j'_1}^{(t)} \geq \Omega\left(\frac{1}{d^{2C_B(c_1+c_2)}}\right)$;
2. for $\ell = 2$, if $\mathbf{Z}^{2,\ell-1} \in \widetilde{\mathcal{E}}_2$, $1 - \text{logit}_{5,j_2}^{(t)} = \Omega(1)$, $\text{logit}_{5,j'_2}^{(t)} = O\left(\frac{1}{d^{1-c_6C_B}}\right)$.

Proof. • For $\ell = 1$, we have

$$\text{logit}_{5,j'_1}^{(t)} = \frac{1}{1 + e^{\Lambda_{5,j_1,r_{g_1 \cdot y_0}}^{(t)} - \Lambda_{5,j'_1,r_{g_2 \cdot y_0}}^{(t)} + O(d)} \cdot e^{-\Lambda_{5,j'_1,r_{g_2 \cdot y_0}}^{(t)}}}$$

$$\stackrel{(a)}{\geq} \frac{1}{1 + e^{2(c_1+c_2)B} + O(d) \cdot e^{-\left(\frac{1}{3}-2c_1-2c_2\right)B}} \geq \Omega\left(\frac{1}{d^{2C_B(c_1+c_2)}}\right),$$

where the inequality (a) follows from Lemma 5.B.58 and the last inequality is due to the fact that $(c_1 + c_2)$ is some sufficiently small constant s.t., $e^{-\left(\frac{1}{3}-2c_1-2c_2\right)B} = 1/\text{poly}d$.

- For $\ell = 2$, we have

$$\begin{aligned} \text{logit}_{j_2}^{(t)} &= \frac{1}{1 + e^{-\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} + \Lambda_{5,j_2',r_{g_2 \cdot y_0}}^{(t)} + O(d) \cdot e^{-\Lambda_{5,j_2',r_{g_2 \cdot y_1}}^{(t)}}}} \\ &\stackrel{(a)}{\leq} \frac{1}{1 + O(d) \cdot e^{-4c_5 B}} = O\left(\frac{1}{d^{1-4c_5 C_B}}\right), \end{aligned}$$

where the inequality (a) follows from Lemma 5.B.58. Similarly, we have

$$\begin{aligned} \text{logit}_{j_2'}^{(t)} &= \frac{1}{1 + e^{\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} - \Lambda_{5,j_2',r_{g_2 \cdot y_0}}^{(t)} + O(d) \cdot e^{-\Lambda_{5,j_2',r_{g_2 \cdot y_0}}^{(t)}}}} \\ &\leq \frac{1}{1 + e^{\Omega(1)} + O(d) \cdot e^{-c_6 B}} = O\left(\frac{1}{d^{1-c_6 C_B}}\right). \end{aligned}$$

□

Lemma 5.B.60. *If Induction 5.B.11 holds for all iterations $\in [T_{2,1,s}, t)$, given input $\mathbf{Z}^{2,\ell-1}$, then we have*

1. for $\ell = 1$, if $\mathbf{Z}^{2,\ell-1} \notin \mathcal{E}_1$, then

$$(a) \text{ for } j = j_1, \widehat{\mathfrak{U}}_{j_1} = \{r_{g_1 \cdot y_0}\}, \Lambda_{5,j_1,r_{g_1 \cdot y_0}}^{(t)} = B \pm O(\delta);$$

$$(b) \text{ for } j \neq j_1 \in \tau(\mathcal{Y}), \text{ assuming } j = \tau(g_1(y)), \text{ then } \Lambda_{5,j,r_{g_1 \cdot y}}^{(t)} \in \left[\left(\frac{1}{3} - c_2\right)B, \left(\frac{1}{3} + c_1\right)B\right] \text{ and}$$

$$\Lambda_{5,j,r}^{(t)} \ll -\varrho \text{ for } r \in \widehat{\mathfrak{U}}_j \setminus \{r_{g_1 \cdot y}\}.$$

2. $\ell = 2$, if $\mathbf{Z}^{2,\ell-1} \notin \widetilde{\mathcal{E}}_2$, then

$$(a) \text{ if } g_1 = g_2 \wedge y_0 \neq y_1,$$

$$i. \text{ for } j = j_2, \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} \in \left[\frac{1}{2}B + \Omega(1), \left(\frac{1}{2} + 2c_5\right)B\right] \text{ and } \Lambda_{5,j_2,r}^{(t)} \ll -\varrho \text{ for } r \in \widehat{\mathfrak{U}}_{j_2} \setminus \{r_{g_2 \cdot y_1}\};$$

$$ii. \text{ for } j = \tau(g_2(y_0)), \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} - \Lambda_{5,j_2,r_{g_2 \cdot y_0}}^{(t)} \geq \Omega(1) \text{ and } \Lambda_{5,j,r}^{(t)} \ll -\varrho \text{ for } r \in \widehat{\mathfrak{U}}_j \setminus \{r_{g_2 \cdot y_0}\};$$

$$iii. \text{ for other } j \in \tau(\mathcal{Y}), \text{ assuming } j = \tau(g_2(y)), \Lambda_{5,j,r_{g_2 \cdot y}}^{(t)} \in [\Omega(1), c_6 B] \text{ and } \Lambda_{5,j,r}^{(t)} \ll -\varrho \text{ for } r \in \widehat{\mathfrak{U}}_j \setminus \{r_{g_2 \cdot y}\}.$$

$$(b) \text{ if } g_1 \neq g_2 \wedge y_0 = y_1,$$

$$i. \text{ for } j = j_2, \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} \in \left[\frac{1}{2}B + \Omega(1), \left(\frac{1}{2} + 2c_5\right)B\right] \text{ and } \Lambda_{5,j_2,r}^{(t)} \ll -\varrho \text{ for } r \in \widehat{\mathfrak{U}}_{j_2} \setminus \{r_{g_2 \cdot y_1}\};$$

$$ii. \text{ for } j = \tau(g_1(y_1)), \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} - \Lambda_{5,j_2,r_{g_1 \cdot y_1}}^{(t)} \geq \Omega(1) \text{ and } \Lambda_{5,j,r}^{(t)} \ll -\varrho \text{ for } r \in \widehat{\mathfrak{U}}_j \setminus \{r_{g_1 \cdot y_1}\};$$

$$iii. \text{ for other } j \in \tau(\mathcal{Y}), r \text{ is not activated for all } r \in \widehat{\mathfrak{U}}_j, \text{ i.e., } \Lambda_{5,j,r}^{(t)} \ll -\varrho.$$

$$(c) \text{ if } g_1 = g_2 \wedge y_0 = y_1,$$

$$i. \text{ for } j = j_2, \widehat{\mathfrak{U}}_{j_2} = \{r_{g_2 \cdot y_1}\}, \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} = B \pm O(\delta);$$

ii. for other $j \in \tau(\mathcal{Y})$, assuming $j = \tau(g_2(y))$, $\Lambda_{5,j,r_{g_2 \cdot y}}^{(t)} \in [\Omega(1), c_6 B]$ and $\Lambda_{5,j,r}^{(t)} \ll -\varrho$ for $r \in \widehat{\mathfrak{A}}_j \setminus \{r_{g_2 \cdot y}\}$.

Lemma 5.B.61. *If Induction 5.B.11 holds for all iterations $\in [T_{2,1,s}, t)$, given input $\mathbf{Z}^{2,\ell-1}$, then we have*

1. for $\ell = 1$, if $\mathbf{Z}^{2,\ell-1} \notin \mathcal{E}_1$, then $1 - \mathbf{logit}_{5,j_1}^{(t)} = O\left(\frac{1}{\text{poly}d}\right)$.
2. $\ell = 2$, if $\mathbf{Z}^{2,\ell-1} \notin \widetilde{\mathcal{E}}_2$, then
 - (a) if $g_1 = g_2 \wedge y_0 \neq y_1$, $\mathbf{logit}_{5,j_2}^{(t)} = \Omega(1)$, $1 - \mathbf{logit}_{5,j_2}^{(t)} - \mathbf{logit}_{5,\tau(g_2(y_0))}^{(t)} = \frac{1}{\text{poly}d}$.
 - (b) if $g_1 \neq g_2 \wedge y_0 = y_1$, $\mathbf{logit}_{5,j_2}^{(t)} = \Omega(1)$, $1 - \mathbf{logit}_{5,j_2}^{(t)} - \mathbf{logit}_{5,\tau(g_1(y_1))}^{(t)} = \frac{1}{\text{poly}d}$.
 - (c) if $g_1 = g_2 \wedge y_0 = y_1$, $1 - \mathbf{logit}_{5,j_2}^{(t)} = O\left(\frac{1}{\text{poly}d}\right)$.

Gradient Lemma

Lemma 5.B.62. *If Induction 5.B.11 holds for all iterations $\in [T_{2,1,s}, t)$, given $s \in \tau(\mathcal{X})$, for $[\mathbf{Q}_{4,3}^{(t)}]_{s,s}$, we have*

$$\sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,3}^{(t)}} \text{Loss}_5^{2,\ell} \right]_{s,s} = \Theta\left(\frac{\log d}{d}\right).$$

Proof. The proof is similar to Lemma 5.B.50, but we need to shift our focus to $\left[-\nabla_{\mathbf{Q}_{4,3}^{(t)}} \text{Loss}_5^{2,2} \right]_{s,s}$. By Lemma 5.B.57 and Lemma 5.B.60, we have

$$\begin{aligned} \mathcal{N}_{s,3,2,i}^{(t)} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \cdot (1 - \mathbf{logit}_{5,j_2}^{(t)}) \cdot \right. \\ &\quad \left. \left(\left(V_{j_2, r_{g_2 \cdot y_1}}(g_2) - \Lambda_{5,j_2, r_{g_2 \cdot y_1}}^{(t)} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \widetilde{\mathcal{E}}_2} \right] \\ &+ \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \cdot (1 - \mathbf{logit}_{5,j_2}^{(t)}) \cdot \right. \\ &\quad \left. \left(\left(V_{j_2, r_{g_2 \cdot y_1}}(g_2) - \Lambda_{5,j_2, r_{g_2 \cdot y_1}}^{(t)} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \widetilde{\mathcal{E}}_2^c} \right] \\ &\stackrel{(a)}{=} \Theta\left(\frac{1}{d}\right) \cdot \Omega(1) \cdot \Theta(B) + \Theta\left(\frac{1}{d}\right) \cdot O(1) \cdot \Theta(B) \cdot O\left(\frac{1}{\log d}\right) \\ &= \Theta\left(\frac{\log d}{d}\right), \end{aligned}$$

where the inequality (a) follows from Lemma 5.B.59 and Lemma 5.B.60.

Furthermore, for $\mathcal{N}_{s,3,2,ii}^{(t)}$ we have

$$\left| \mathcal{N}_{s,3,2,ii}^{(t)} \right|$$

$$\begin{aligned}
&\leq \left| \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \cdot \mathbf{logit}_{5,j'_2}^{(t)} \cdot \left(\left(V_{j'_2,r_{g_2,y_0}}(g_2) - \Lambda_{5,j,r}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \tilde{\mathcal{E}}_2} \right] \right| \\
&+ \left| \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \cdot \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j}^{(t)} \cdot \left(\sum_{r \in \mathfrak{A}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)}) \cdot \left(V_{j,r}(g_2) - \Lambda_{5,j,r}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \tilde{\mathcal{E}}_2} \right] \right| \\
&\leq \Theta\left(\frac{1}{d}\right) \cdot O\left(\frac{1}{d^{1-c_6 C_B}}\right) \cdot \Theta(B) + \Theta\left(\frac{1}{d}\right) \cdot O(1) \cdot \Theta(B) \cdot O\left(\frac{1}{\log d}\right) \leq O\left(\frac{1}{d}\right).
\end{aligned}$$

$|\mathcal{N}_{s,3,1,i}^{(t)}|$ and $|\mathcal{N}_{s,3,2,i}^{(t)}|$ can be upper bounded by $O\left(\frac{\log d}{d}\right)$ as Lemma 5.B.50. Thus, we complete the proof. $\square$

Lemma 5.B.63. *If Induction 5.B.11 holds for all iterations $\in [T_{2,1,s}, t)$, given $s \in \tau(\mathcal{X})$, we have*

$$\sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,4}^{(t)}} \text{Loss}_5^{2,\ell} \right]_{s,s} = \Theta\left(\frac{\log d}{d}\right).$$

Proof. The proof follows a similar analysis to Lemma 5.B.53, and we thus omit the details here. $\square$

Lemma 5.B.64. *If Induction 5.B.11 holds for all iterations $\in [T_{2,1,s}, t)$, given $s \in \tau(\mathcal{X})$, we have*

$$\begin{aligned}
&\sum_{t'=T_{2,1,s}}^t \left(\sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,3}^{(t')}} \text{Loss}_5^{2,\ell} \right]_{s,s} - \sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,4}^{(t')}} \text{Loss}_5^{2,\ell} \right]_{s,s} \right) \\
&\geq -O\left(\frac{1}{\log d}\right) \cdot \left(\sum_{t'=T_{2,1,s}}^t \sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,4}^{(t')}} \text{Loss}_5^{2,\ell} \right]_{s,s} \right).
\end{aligned}$$

Proof. Following the analogous analysis as Lemma 5.B.52, we have

$$\sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,3}^{(t)}} \text{Loss}_5^{2,\ell} \right]_{s,s} + \sum_{\ell=1}^2 \left[\nabla_{\mathbf{Q}_{4,4}^{(t)}} \text{Loss}_5^{2,\ell} \right]_{s,s} = \sum_{\ell \in [2]} \sum_{\kappa \in \{i, ii, iii\}} \mathcal{N}_{s,3,\ell,\kappa}^{(t)} - \mathcal{N}_{s,4,\ell,\kappa}^{(t)}.$$

Meanwhile $|\mathcal{N}_{s,p,\ell,iii}^{(t)}| = O\left(\frac{1}{\text{poly} d}\right)$ for $p \in \{3, 4\}$ and $\ell \in [2]$, we can focus on the gradient difference between $\mathbf{Q}_{4,3}^{(t)}$ and $\mathbf{Q}_{4,4}^{(t)}$ contributed by other terms.

For $\ell = 1$, since $\mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(t)} \geq \mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(t)}$ and thus it is straightforward to see that $\mathcal{N}_{s,3,1,i}^{(t)} - \mathcal{N}_{s,4,1,i}^{(t)} \geq 0$. Furthermore, we have

$$\mathcal{N}_{s,3,1,ii}^{(t)} - \mathcal{N}_{s,4,1,ii}^{(t)}$$

$$\begin{aligned}
&= -\mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(t)} \cdot \mathbf{logit}_{5,j'_1}^{(t)} \cdot \left(\left(V_{j'_1, r_{g_2 \cdot y_0}}(g_1) - \Lambda_{5,j'_1, r_{g_2 \cdot y_0}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_0)=s\} \cap \mathcal{E}_1} \right] \\
&+ \mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(t)} \cdot \mathbf{logit}_{5,j'_1}^{(t)} \cdot \left(\left(V_{j'_1, r_{g_2 \cdot y_0}}(y_0) - \Lambda_{5,j'_1, r_{g_2 \cdot y_0}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_0)=s\} \cap \mathcal{E}_1} \right] \\
&- \mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(t)} \cdot \sum_{y \neq y_0 \in \mathcal{Y}} \mathbf{logit}_{5,\tau(g_1(y))}^{(t)} \cdot \left(\left(V_{\tau(g_1(y)), r_{g_1 \cdot y}}(g_1) - \Lambda_{5,\tau(g_1(y)), r_{g_1 \cdot y}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_0)=s\} \cap \mathcal{E}_1^c} \right] \\
&+ \mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(t)} \cdot \sum_{y \neq y_0 \in \mathcal{Y}} \mathbf{logit}_{5,\tau(g_1(y))}^{(t)} \cdot \left(\left(V_{\tau(g_1(y)), r_{g_1 \cdot y}}(y_0) - \Lambda_{5,\tau(g_1(y)), r_{g_1 \cdot y}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_0)=s\} \cap \mathcal{E}_1^c} \right] \\
&\stackrel{(a)}{\geq} \Theta\left(\frac{1}{d}\right) \cdot \Omega\left(\frac{1}{d^{2C_B(c_1+c_2)}}\right) \cdot \Theta(B) - \Theta\left(\frac{1}{d}\right) \cdot O\left(\frac{1}{\text{poly}d}\right) \cdot O(B) \cdot O\left(\frac{1}{\log d}\right) \geq \Omega\left(\frac{\log d}{d^{1+2C_B(c_1+c_2)}}\right).
\end{aligned}$$

where the inequality (a) is due to Lemma 5.B.59 and Lemma 5.B.61.

For $\ell = 2$, since $\mathbf{Attn}_{\text{ans},\text{pred},2 \rightarrow \text{pred},1}^{(t)} \geq \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)}$ and thus it is straightforward to see that $\mathcal{N}_{s,3,2,i}^{(t)} - \mathcal{N}_{s,4,2,i}^{(t)} \geq 0$. Moreover, we have

$$\begin{aligned}
&\mathcal{N}_{s,3,2,ii}^{(t)} - \mathcal{N}_{s,4,2,ii}^{(t)} \\
&= -\mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \cdot \mathbf{logit}_{5,j'_2}^{(t)} \cdot \left(\left(V_{j'_2, r_{g_2 \cdot y_0}}(g_2) - \Lambda_{5,j'_2, r_{g_2 \cdot y_0}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \tilde{\mathcal{E}}_2} \right] \tag{5.B.48}
\end{aligned}$$

$$\begin{aligned}
&+ \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \cdot \mathbf{logit}_{5,j'_2}^{(t)} \cdot \left(\left(V_{j'_2, r_{g_2 \cdot y_0}}(y_1) - \Lambda_{5,j'_2, r_{g_2 \cdot y_0}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \tilde{\mathcal{E}}_2} \right] \tag{5.B.49}
\end{aligned}$$

$$-\mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \cdot \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j}^{(t)} \cdot \left(\left(V_{j, r_{g_2 \cdot y_0}}(g_2) - \Lambda_{j, r_{g_2 \cdot y_0}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \tilde{\mathcal{E}}_2} \right]$$

$$\left(\sum_{r \in \mathfrak{A}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)}) \cdot \left(V_{j,r}(g_2) - \Lambda_{5,j,r}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \tilde{\mathcal{E}}_2^c} \right] \quad (5.B.50)$$

$$+ \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \cdot \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j}^{(t)} \cdot \left(\sum_{r \in \mathfrak{A}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)}) \cdot \left(V_{j,r}(y_1) - \Lambda_{5,j,r}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \tilde{\mathcal{E}}_2^c} \right]. \quad (5.B.51)$$

By Lemma 5.B.59, we obtain that

$$(5.B.48) + (5.B.49) \geq -O\left(\frac{1}{d}\right) \cdot O\left(\frac{1}{d^{1-c_6 C_B}}\right) \cdot \Theta(B) \geq -O\left(\frac{\log d}{d^{2-c_6 C_B}}\right).$$

By Lemma 5.B.60, for $\tilde{\mathcal{E}}_2^c$, we only need to consider the case that $g_1 = g_2 \wedge y_0 \neq y_1$, and we have

$$\begin{aligned} & (5.B.50) + (5.B.51) \\ & \geq -\mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \cdot \sum_{y \neq y_1 \in \mathcal{Y}} \mathbf{logit}_{5,\tau(g_2(y))}^{(t)} \cdot \left(\left(V_{\tau(g_2(y)),r_{g_2,y}}(g_2) - \Lambda_{5,\tau(g_2(y)),r_{g_2,y}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \{g_1=g_2 \wedge y_0 \neq y_1\}} \right] \\ & + \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \cdot \sum_{y \neq y_1 \in \mathcal{Y}} \mathbf{logit}_{5,\tau(g_2(y))}^{(t)} \cdot \left(\left(V_{\tau(g_2(y)),r_{g_2,y}}(y_1) - \Lambda_{5,\tau(g_2(y)),r_{g_2,y}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \{g_1=g_2 \wedge y_0 \neq y_1\}} \right] \\ & \geq -\Theta\left(\frac{1}{d}\right) \cdot O(1) \cdot \Theta(B) \cdot O\left(\frac{1}{\log d}\right) \geq -O\left(\frac{1}{d}\right). \end{aligned}$$

Putting it all together, combining with the fact that c_6 and $(c_1 + c_2)$ are sufficiently small, we finish the proof. $\square$

Lemma 5.B.65 (Lower bound of gap). *If Induction 5.B.11 holds for all iterations $\in [T_{2,1,s}, t)$, given $s \in \tau(\mathcal{X})$, we have $[\mathbf{Q}_{4,3}^{(t)}]_{s,s} - [\mathbf{Q}_{4,4}^{(t)}]_{s,s} \geq \Omega\left(\frac{1}{\log d}\right)$.*

Proof. Letting $\tilde{T}$ denote the first time that $[\mathbf{Q}_{4,3}^{(t)}]_{s,s} - [\mathbf{Q}_{4,4}^{(t)}]_{s,s} \leq \frac{1}{2}([\mathbf{Q}_{4,3}^{(T_{2,1,s})}]_{s,s} - [\mathbf{Q}_{4,4}^{(T_{2,1,s})}]_{s,s})$, which implies that

$$\sum_{t'=T_{2,1,s}}^{\tilde{T}} \left(\sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}^{(t')}_{4,3}} \text{Loss}_5^{2,\ell} \right]_{s,s} - \sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}^{(t')}_{4,4}} \text{Loss}_5^{2,\ell} \right]_{s,s} \right).$$

Hence, by Lemma 5.B.64, we have $[\mathbf{Q}_{4,3}^{(\tilde{T})}]_{s,s}$ and $[\mathbf{Q}_{4,3}^{(\tilde{T})}]_{s,s}$ reaches $\Omega(1)$. Thus, we can have a refined lower bound for (5.B.50) + (5.B.51) in Lemma 5.B.64, and obtain:

$$\begin{aligned}
(5.B.50) + (5.B.51) &\geq -O\left(\frac{1}{\text{poly}d}\right) \\
&\quad - \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \cdot \mathbf{logit}_{5,\tau(g_2(y_0))}^{(t)} \cdot \right. \\
&\quad \left. \left(\left(V_{\tau(g_2(y_0)),r_{g_2 \cdot y_0}}(g_2) - \Lambda_{5,\tau(g_2(y_0)),r_{g_2 \cdot y_0}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \{g_1=g_2 \wedge y_0 \neq y_1\}} \right] \\
&\quad + \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \cdot \mathbf{logit}_{5,\tau(g_2(y_0))}^{(t)} \cdot \right. \\
&\quad \left. \left(\left(V_{\tau(g_2(y_0)),r_{g_2 \cdot y_0}}(y_1) - \Lambda_{5,\tau(g_2(y_0)),r_{g_2 \cdot y_0}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \{g_1=g_2 \wedge y_0 \neq y_1\}} \right] \\
&\stackrel{(a)}{\geq} -O\left(\frac{1}{\text{poly}d}\right) - \Theta\left(\frac{1}{d}\right) \cdot O\left(\frac{1}{d^{\Omega(1)}}\right) \cdot \Theta(B) \cdot O\left(\frac{1}{\log d}\right) \geq -O\left(\frac{1}{d^{1+\Omega(1)}}\right),
\end{aligned}$$

where (a) follows from the fact that on the event $\{g_1 = g_2 \wedge y_0 \neq y_1\}$, we have $\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} - \Lambda_{5,j_2,r_{g_2 \cdot y_0}}^{(t)} \geq \Omega(B)$ once $[\mathbf{Q}_{4,3}^{(t)}]_{s,s}$ and $[\mathbf{Q}_{4,4}^{(t)}]_{s,s}$ reach constant magnitude, and consequently, the logit satisfies $\mathbf{logit}_{5,\tau(g_2(y_0))}^{(t)} \leq O\left(\frac{1}{d^{\Omega(1)}}\right)$.

Therefore,

$$\sum_{t'=\tilde{T}+1}^t \left(\sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}^{(t')}_{4,3}} \text{Loss}_5^{2,\ell} \right]_{s,s} - \sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}^{(t')}_{4,4}} \text{Loss}_5^{2,\ell} \right]_{s,s} \right) \geq -O\left(\frac{1}{\log d \cdot d^{\Omega(1)}}\right) \cdot O(1),$$

which means that $[\mathbf{Q}_{4,3}^{(t)}]_{s,s} - [\mathbf{Q}_{4,4}^{(t)}]_{s,s} \geq \Omega\left(\frac{1}{\log d}\right) - O\left(\frac{1}{\log d \cdot d^{\Omega(1)}}\right) \geq \Omega\left(\frac{1}{\log d}\right)$. $\square$

Lemma 5.B.66. *If Induction 5.B.11 holds for all iterations $\in [T_{2,1,s}, t)$, given $s' \neq s \in \tau(X)$, for $p \in \{3, 4\}$, we have*

$$\left| \sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,p}^{(t)}} \text{Loss}_5^{2,\ell} \right]_{s,s'} \right| \leq O\left(\frac{\log d}{d^2}\right) = O\left(\frac{1}{d}\right) \cdot \left| \sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,p}^{(t)}} \text{Loss}_5^{2,\ell} \right]_{s,s} \right|.$$

At the End of Stage 2.2 Putting gradient lemmas together, we can directly prove that Induction 5.B.11 holds for all iterations t until the end of stage 2.2, where we can conclude the following:

Lemma 5.B.67 (End of Stage 2.2). *Given $s \in \tau(X)$, Induction 5.B.11 holds for all iterations $T_{2,1,s} \leq t < T_{2,2,s} = O\left(\frac{d}{\eta \log d}\right)$, then at the end of stage 2.2, we have*

(a) $[\mathbf{Q}_{4,p}^{(t)}]_{s,s} = \Omega(1)$ for $p \in \{3, 4\}$;

- (b) $[\mathbf{Q}_{4,3}^{(t)}]_{s,s} - [\mathbf{Q}_{4,4}^{(t)}]_{s,s} \in \left[\Omega\left(\frac{1}{\log d}\right), O(1) \right];$
- (c) $||[\mathbf{Q}_{4,p}^{(t)}]_{s,s'}| \leq O\left(\frac{[\mathbf{Q}_{4,p}^{(t)}]_{s,s}}{d}\right)$ for $s' \in \tau(X) \neq s$, and other $[\mathbf{Q}_{4,p}]_{s,s'} = 0$.

Stage 2.3: Decrease of Gap and Convergence

After rapid growth of diagonal entries in stage 2.2, we now focus on the convergence of the attention and logit matrices, and the decrease of the gap between $[\mathbf{Q}_{4,3}^{(t)}]_{s,s}$ and $[\mathbf{Q}_{4,4}^{(t)}]_{s,s}$. Recall that

$$\begin{aligned} \epsilon_{\text{attn}}^{L,\ell}(\mathbf{Z}^{L,\ell-1}) &= 1 - \text{Attn}_{\text{ans},\ell-1 \rightarrow \text{pred},\ell}(\mathbf{Z}^{L,\ell-1}) - \text{Attn}_{\text{ans},\ell-1 \rightarrow \text{ans},\ell-1}(\mathbf{Z}^{L,\ell-1}), \\ \Delta^{L,\ell}(\mathbf{Z}^{L,\ell-1}) &= \text{Attn}_{\text{ans},\ell-1 \rightarrow \text{pred},\ell}(\mathbf{Z}^{L,\ell-1}) - \text{Attn}_{\text{ans},\ell-1 \rightarrow \text{ans},\ell-1}(\mathbf{Z}^{L,\ell-1}). \end{aligned}$$

Through stage 2.3, we will focus on the attention gap $\Delta^{L,\ell}(\mathbf{Z}^{L,\ell-1})$ instead of the gap of the attention matrices. We abbreviate $\epsilon_{\text{attn}}^{L,\ell}(\mathbf{Z}^{L,\ell-1})$ and $\Delta^{L,\ell}(\mathbf{Z}^{L,\ell-1})$ as $\epsilon_{\text{attn}}^{L,\ell}$ and $\Delta^{L,\ell}$ for simplicity.

Induction 5.B.12. Given $\epsilon \geq \tilde{\Omega}(\sigma_0)$, for $s \in \tau(X)$, let $T_{2,3,s}$ denote the first time that $\mathbb{E}[\epsilon_{\text{attn}}^{2,2} \mid \tau(x_1) = s] \leq \epsilon$. For all iterations $T_{2,2,s} \leq t < T_{2,3,s}$, we have the following holds:

- (a) $[\mathbf{Q}_{4,3}^{(t)}]_{s,s}$ and $[\mathbf{Q}_{4,4}^{(t)}]_{s,s}$ monotonically increases $\leq \tilde{O}(1)$;
- (b) $\Delta^{2,\ell} \geq 0$ for any $\mathbf{Z}^{2,\ell}$ with $\ell \in \{1, 2\}$;
- (c) $\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \leq 0.5 + \tilde{c}_1$ for some small constant $\tilde{c}_1 > 0$;
- (d) for $(p, q) \in \{(4, 3), (4, 4)\}$, $|\left[\mathbf{Q}_{p,q}^{(t)}\right]_{s,s'}| \leq O\left(\frac{[\mathbf{Q}_{p,q}^{(t)}]_{s,s}}{d}\right)$ for $s' \in \tau(X) \neq s$; other $[\mathbf{Q}_{p,q}^{(t)}]_{s,s'} = 0$.

Attention and Logit Preliminaries

Lemma 5.B.68. If Induction 5.B.12 holds for all iterations $\in [T_{2,2,s}, t)$, given $\mathbf{Z}^{2,\ell-1}$ then we have

1. for $\ell = 1$,
 - (a) $\text{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(t)} \geq \Omega(1)$, and $\text{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(t)} > \text{Attn}_{\text{ans},0 \rightarrow \text{pred},2}^{(t)}$;
2. for $\ell = 2$,
 - (a) $\text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \geq \Omega(1)$, $\text{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)}, \text{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \leq \text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)}$;
 - (b) $\left| \text{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \right| \leq \tilde{O}\left(\frac{1}{d}\right)$.

Moreover, given $\mathbf{Z}^{2,1}$ and corresponding $\mathbf{Z}^{2,0}$, $\text{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(t)}(\mathbf{Z}^{2,0}) \geq \text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}(\mathbf{Z}^{2,1})$.

Lemma 5.B.69. If Induction 5.B.12 holds for all iterations $\in [T_{2,2,s}, t)$, given input $\mathbf{Z}^{2,\ell-1}$, then we have

1. for $\ell = 1$, if $\mathbf{Z}^{2,\ell-1} \in \mathcal{E}_1$, then
 - (a) for $j = j_1$, $\Lambda_{5,j_1,r}^{(t)} \ll -\varrho$ for $r \in \widehat{\mathfrak{A}}_{j_1} \setminus \{r_{g_1 \cdot y_0}\}$;
 - (b) for $j = j'_1 \triangleq \tau(g_2(y_0)) = \tau(g_1(\bar{y}))$, $\Lambda_{5,j'_1,r}^{(t)} \ll -\varrho$ for $r \in \widehat{\mathfrak{A}}_{j'_1} \setminus \{r_{g_2 \cdot y_0}, r_{g_1 \cdot \bar{y}}\}$;
 - (c) for other $j \in \tau(\mathcal{Y})$, assuming $j = \tau(g_1(y))$ for some $y \neq y_0$, then $\Lambda_{5,j,r}^{(t)} \ll -\varrho$ for $r \in \widehat{\mathfrak{A}}_j \setminus \{r_{g_1 \cdot y}\}$.
2. $\ell = 2$, if $\mathbf{Z}^{2,\ell-1} \in \widetilde{\mathcal{E}}_2$, then
 - (a) for $j = j_2$, $\Lambda_{5,j_2,r}^{(t)} \ll -\varrho$ for $r \in \widehat{\mathfrak{A}}_{j_2} \setminus \{r_{g_2 \cdot y_1}\}$;

(b) for $j = j'_2 \triangleq \tau(g_2(y_0))$, $\Lambda_{5,j'_2,r}^{(t)} \ll -\varrho$ for $r \in \widehat{\mathfrak{A}}_{j'_2} \setminus \{r_{g_2 \cdot y_0}\}$;

(c) for other $j \in \tau(\mathcal{Y})$, if $j = \tau(g_2(y))$ for some $y \in \mathcal{Y}$, then $\Lambda_{5,j,r}^{(t)} \ll -\varrho$ for $r \in \widehat{\mathfrak{A}}_j \setminus \{r_{g_2 \cdot y}\}$.

Lemma 5.B.70. *If Induction 5.B.12 holds for all iterations $\in [T_{2,2,s}, t)$, given input $\mathbf{Z}^{2,\ell-1}$, then we have*

1. $\ell = 1$, for $\mathbf{Z}^{2,\ell-1} \in \mathcal{E}_1$,

$$(a) \Lambda_{5,j_1,r_{g_1 \cdot y_0}}^{(t)} = (1 - 2\epsilon_{\text{attn}}^{2,1})B \pm O(\delta) \geq \left(\frac{1}{3} + c_1\right)B;$$

$$(b) \Lambda_{5,j_1,r_{g_1 \cdot y_0}}^{(t)} - \Lambda_{5,j'_1,r_{g_2 \cdot y_0}}^{(t)} = 2(\text{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(t)} - \text{Attn}_{\text{ans},0 \rightarrow \text{pred},2}^{(t)})B \pm O(\delta) \geq 2(c_1 + c_2)B;$$

$$(c) \text{ for } y \neq y_0, \Lambda_{5,\tau(g_1(y)),r_{g_1 \cdot y}}^{(t)} = \left(2\text{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(t)} - 1\right)B \pm O(\delta), \text{ which is only activated if } \text{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(t)} > \frac{1}{2}$$

2. $\ell = 2$, for $\mathbf{Z}^{2,\ell-1} \in \widetilde{\mathcal{E}}_2$,

$$(a) \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} = (1 - 2\epsilon_{\text{attn}}^{2,2})B \pm O(\delta) \geq 4c_5B;$$

$$(b) \Lambda_{5,j'_2,r_{g_2 \cdot y_0}}^{(t)} = \Delta^{2,2} \cdot B \pm O(\delta), \text{ and}$$

$$\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} - \Lambda_{5,j'_2,r_{g_2 \cdot y_0}}^{(t)} = 2(\text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)})B \pm O(\delta);$$

$$(c) \text{ for } y \neq y_0, y_1, \Lambda_{5,\tau(g_2(y)),r_{g_2 \cdot y}}^{(t)} = \left(2\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - 1\right)B \pm O(\delta), \text{ which is only activated if } \text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} > \frac{1}{2}.$$

Lemma 5.B.71. *If Induction 5.B.12 holds for all iterations $\in [T_{2,2,s}, t)$, given input $\mathbf{Z}^{2,\ell-1}$, then we have*

1. for $\ell = 1$, if $\mathbf{Z}^{2,\ell-1} \in \mathcal{E}_1$,

$$1 - \text{logit}_{5,j_1}^{(t)} = \Theta(1) \cdot \text{logit}_{5,j'_1}^{(t)} = \Theta\left(\frac{1}{d^{2(\text{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(t)} - \text{Attn}_{\text{ans},0 \rightarrow \text{pred},2}^{(t)})C_B}}\right).$$

2. for $\ell = 2$, if $\mathbf{Z}^{2,\ell-1} \in \widetilde{\mathcal{E}}_2$,

$$\text{logit}_{5,j'_2}^{(t)} = \Theta\left(\frac{1}{d^{2(\text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)})C_B + d^{1-\Delta^{2,2}C_B}}}\right),$$

moreover,

$$1 - \text{logit}_{5,j_2}^{(t)} \geq \min\left\{\Omega(1), \Omega\left(\frac{1}{d^{C_B \cdot (1-2\epsilon_{\text{attn}}^{2,2})-1}}\right)\right\}.$$

Lemma 5.B.72. *If Induction 5.B.12 holds for all iterations $\in [T_{2,2,s}, t)$, given input $\mathbf{Z}^{2,\ell-1}$, then we have*

1. for $\ell = 1$, if $\mathbf{Z}^{2,\ell-1} \notin \mathcal{E}_1$, then

$$(a) \text{ for } j = j_1, \widehat{\mathfrak{A}}_{j_1} = \{r_{g_1 \cdot y_0}\}, \Lambda_{5,j_1,r_{g_1 \cdot y_0}}^{(t)} = B \pm O(\delta);$$

$$(b) \text{ for } j \neq j_1 \in \tau(\mathcal{Y}), \text{ assuming } j = \tau(g_1(y)) \text{ for } y \neq y_0, \text{ then } \Lambda_{5,j_1,r_{g_1 \cdot y_0}}^{(t)} - \Lambda_{5,j,r_{g_1 \cdot y}}^{(t)} \geq 2\text{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(t)} \cdot B \geq \Omega(B) \text{ and } \Lambda_{5,j,r}^{(t)} \ll -\varrho \text{ for } r \in \widehat{\mathfrak{A}}_j \setminus \{r_{g_1 \cdot y}\}.$$

2. $\ell = 2$, if $\mathbf{Z}^{2,\ell-1} \notin \widetilde{\mathcal{E}}_2$, then

- (a) if $g_1 = g_2 \wedge y_0 \neq y_1$,
- i. for $j = j_2$, $\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} = (1 - \epsilon_{\text{attn}}^{2,2}) \cdot B \pm O(\delta)$ and $\Lambda_{5,j_2,r}^{(t)} \ll -\varrho$ for $r \in \widehat{\mathfrak{U}}_{j_2} \setminus \{r_{g_2 \cdot y_1}\}$;
 - ii. for $j = \tau(g_2(y_0))$, $\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} - \Lambda_{5,j,r_{g_2 \cdot y_0}}^{(t)} = 2(\text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)})B \pm O(\delta)$
and $\Lambda_{5,j,r}^{(t)} \ll -\varrho$ for $r \in \widehat{\mathfrak{U}}_j \setminus \{r_{g_2 \cdot y_0}\}$;
 - iii. for other $j \in \tau(\mathcal{Y})$, assuming $j = \tau(g_2(y))$ for some $y \neq y_0, y_1$, $\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} - \Lambda_{5,j,r_{g_2 \cdot y}}^{(t)} = 2\text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)}B \pm O(\delta)$ and $\Lambda_{5,j,r}^{(t)} \ll -\varrho$ for $r \in \widehat{\mathfrak{U}}_j \setminus \{r_{g_2 \cdot y}\}$.
- (b) if $g_1 \neq g_2 \wedge y_0 = y_1$,
- i. for $j = j_2$, $\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} = (1 - \epsilon_{\text{attn}}^{2,2}) \cdot B \pm O(\delta)$ and $\Lambda_{5,j_2,r}^{(t)} \ll -\varrho$ for $r \in \widehat{\mathfrak{U}}_{j_2} \setminus \{r_{g_2 \cdot y_1}\}$;
 - ii. for $j = \tau(g_1(y_1)) = \tau(g_2(\tilde{y}))$,

$$\begin{aligned} \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} - \Lambda_{5,j,r_{g_1 \cdot y_1}}^{(t)} &= 2(\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)})B \pm O(\delta) \\ \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} - \Lambda_{5,j,r_{g_2 \cdot \tilde{y}}}^{(t)} &= 2(\text{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} + \text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)})B \pm O(\delta), \end{aligned}$$

where $r_{g_2 \cdot \tilde{y}}$ is only activated if $\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} > \frac{1}{2}$. $\Lambda_{5,j,r}^{(t)} \ll -\varrho$ for $r \in \widehat{\mathfrak{U}}_j \setminus \{r_{g_1 \cdot y_1}, r_{g_2 \cdot \tilde{y}}\}$;

- iii. for other $j \in \tau(\mathcal{Y})$, assuming $j = \tau(g_2(y))$ for $y \neq y_1, \tilde{y}$, then $r_{g_2 \cdot y}$ is only activated if $\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} > \frac{1}{2}$ and

$$\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} - \Lambda_{5,j_2,r_{g_2 \cdot y}}^{(t)} = 2(\text{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} + \text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)})B \pm O(\delta).$$

$$\Lambda_{5,j,r}^{(t)} \ll -\varrho \text{ for } r \in \widehat{\mathfrak{U}}_j \setminus \{r_{g_2 \cdot y}\}.$$

- (c) if $g_1 = g_2 \wedge y_0 = y_1$,
- i. for $j = j_2$, $\widehat{\mathfrak{U}}_{j_2} = \{r_{g_2 \cdot y_1}\}$, $\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} = B \pm O(\delta)$;
 - ii. for other $j \in \tau(\mathcal{Y})$, assuming $j = \tau(g_2(y))$,

$$\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} - \Lambda_{5,j_2,r_{g_2 \cdot y}}^{(t)} = 2(\text{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} + \text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)})B \pm O(\delta).$$

$$\text{and } \Lambda_{5,j,r}^{(t)} \ll -\varrho \text{ for } r \in \widehat{\mathfrak{U}}_j \setminus \{r_{g_2 \cdot y}\}.$$

Lemma 5.B.73. If Induction 5.B.12 holds for all iterations $\in [T_{2,2,s}, t)$, given input $\mathbf{Z}^{2,\ell-1}$, then we have

1. for $\ell = 1$, if $\mathbf{Z}^{2,\ell-1} \notin \mathcal{E}_1$, then $1 - \text{logit}_{5,j_1}^{(t)} = O\left(\frac{1}{d^{2\text{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(t)}} C_B}\right) = O\left(\frac{1}{\text{poly}d}\right)$.
2. $\ell = 2$, if $\mathbf{Z}^{2,\ell-1} \notin \widetilde{\mathcal{E}}_2$, then

$$(a) \text{ if } g_1 = g_2 \wedge y_0 \neq y_1, \text{logit}_{5,\tau(g_2(y_0))}^{(t)} = O\left(\frac{1}{d^{2(\text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)}) C_B}}\right).$$

$$(b) \text{ if } g_1 \neq g_2 \wedge y_0 = y_1, \text{logit}_{5,\tau(g_1(y_1))}^{(t)} = O\left(\frac{1}{d^{2(\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)}) C_B}}\right).$$

$$(c) \text{ if } g_1 = g_2 \wedge y_0 = y_1, 1 - \text{logit}_{5,j_2}^{(t)} = O\left(\frac{1}{d^{2(\text{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} + \text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)}) C_B}}\right) = \frac{1}{\text{poly}d}.$$

Gradient Lemma

Lemma 5.B.74. *If Induction 5.B.12 holds for all iterations $\in [T_{2,2,s}, t)$, given $s \in \tau(\mathcal{X})$, for $[\mathbf{Q}_{4,4}]_{s,s}$, we have*

$$\sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,4}^{(t)}} \text{Loss}_5^{2,\ell} \right]_{s,s} \geq \Omega\left(\frac{\epsilon \log d}{d^{(1-2\epsilon)C_B}}\right).$$

Proof. By gradient decomposition, we have

$$\sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,4}^{(t)}} \text{Loss}_5^{2,\ell} \right]_{s,s} = \sum_{\ell \in [2]} \sum_{\kappa \in \{i, ii, iii\}} \mathcal{N}_{s,4,\ell,\kappa}^{(t)}.$$

Firstly, for $\mathcal{N}_{s,4,2,i}^{(t)}$, by Lemma 5.B.69 and Lemma 5.B.72, we have

$$\begin{aligned} \mathcal{N}_{s,4,2,i}^{(t)} &= \mathbb{E} \left[\text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \cdot (1 - \text{logit}_{5,j_2}^{(t)}) \cdot \right. \\ &\quad \left. \left(\left(V_{j_2, r_{g_2 y_1}}(y_1) - \Lambda_{5,j_2, r_{g_2 y_1}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \tilde{\mathcal{E}}_2} \right] \\ &+ \mathbb{E} \left[\text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \cdot (1 - \text{logit}_{5,j_2}^{(t)}) \cdot \right. \\ &\quad \left. \left(\left(V_{j_2, r_{g_2 y_1}}(y_1) - \Lambda_{5,j_2, r_{g_2 y_1}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \tilde{\mathcal{E}}_2^c} \right] \\ &\stackrel{(a)}{\geq} \Theta\left(\frac{1}{d}\right) \cdot \mathbb{E} \left[\min \left\{ \Omega(1), \Omega\left(\frac{1}{d^{C_B \cdot (1-2\epsilon_{\text{attn}}^{2,2})-1}}\right) \right\} \cdot 2\epsilon_{\text{attn}}^{2,2} \cdot B \mid \tau(x_1) = s \right] \\ &\geq \Omega\left(\frac{\epsilon \log d}{d^{(1-2\epsilon)C_B}}\right), \end{aligned} \tag{5.B.52}$$

where (a) follows from Lemma 5.B.71. Noticing that $\mathcal{N}_{s,4,1,i}^{(t)} > 0$ and $|\mathcal{N}_{s,4,\ell,iii}^{(t)}|$ for $\ell \in [2]$ is sufficiently small, to provide a lower bound for $\sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,4}^{(t)}} \text{Loss}_5^{2,\ell} \right]_{s,s}$, we only need to focus on the negative gradient from $\mathcal{N}_{s,4,1,ii}^{(t)}$ and $\mathcal{N}_{s,4,2,ii}^{(t)}$.

$$\begin{aligned} \mathcal{N}_{s,4,2,ii}^{(t)} &= \\ &- \mathbb{E} \left[\text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \cdot \text{logit}_{5,j'_2}^{(t)} \cdot \right. \\ &\quad \left. \left(\left(V_{j'_2, r_{g_2 y_0}}(y_1) - \Lambda_{5,j'_2, r_{g_2 y_0}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \tilde{\mathcal{E}}_2} \right] \end{aligned}$$

$$\begin{aligned}
& -\mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \cdot \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j}^{(t)} \cdot \left(\sum_{r \in \mathfrak{A}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)}) \cdot \left(V_{j,r}(y_1) - \Lambda_{5,j,r}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \tilde{\mathcal{E}}_2^c} \right] \\
& \geq -\mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \cdot \mathbf{logit}_{5,\tau(g_1(y_1))}^{(t)} \cdot \left(\mathbf{sReLU}'(\Lambda_{5,\tau(g_1(y_1)),r_{g_1 y_1}}^{(t)}) \cdot \left(V_{\tau(g_1(y_1)),r_{g_1 y_1}}(y_1) - \Lambda_{5,\tau(g_1(y_1)),r_{g_1 y_1}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \{g_1 \neq g_2, y_0=y_1\}} \right]. \quad (5.B.53)
\end{aligned}$$

When $\mathbb{E} \left[C_B \cdot (1 - 2\epsilon_{\text{attn}}^{2,2}) \mid \tau(x_1) = s \right] < 1$, by Lemma 5.B.71 and Lemma 5.B.73, we have

$$\begin{aligned}
& \mathbf{logit}_{5,\tau(g_1(y_1))}^{(t)} \mathbb{1}_{\{\tau(x_1)=s\} \cap \{g_1 \neq g_2, y_0=y_1\}} \\
& \leq O \left(\frac{1}{d^2 (\mathbf{Attn}_{\mathbb{A},1 \rightarrow 2}^{(t)} - \mathbf{Attn}_{\mathbb{A},1 \rightarrow 1}^{(t)}) C_B} \right) (1 - \mathbf{logit}_{5,j_2}^{(t)}) \mathbb{1}_{\{\tau(x_1)=s\} \cap \tilde{\mathcal{E}}_2^c}.
\end{aligned}$$

During this time, $\epsilon_{\text{attn}}^{2,2} \geq \Omega(1)$, thus we can lower bound $\mathcal{N}_{s,4,2,ii}^{(t)}$ by $\mathcal{N}_{s,4,2,ii}^{(t)} \geq -O \left(\frac{1}{d^{O(1)} \cdot \log d} \right) \cdot \mathcal{N}_{s,4,2,i'}^{(t)}$ which implies that the negative gradient from $\mathcal{N}_{s,4,2,ii}^{(t)}$ is dominated by the positive gradient from $\mathcal{N}_{s,4,2,i}^{(t)}$.

When $\mathbb{E} \left[C_B \cdot (1 - 2\epsilon_{\text{attn}}^{2,2}) \mid \tau(x_1) = s \right] \geq 1$, since $2(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)}) C_B \geq (1 - 2\eta^{2,2}) C_B$, we obtain

$$\mathbf{logit}_{\tau(g_1(y_1))}^{(t)} \mathbb{1}_{\{\tau(x_1)=s\} \cap \{g_1 \neq g_2, y_0=y_1\}} \leq O \left(\frac{1}{d} \right) \cdot (1 - \mathbf{logit}_{j_2}^{(t)}) \mathbb{1}_{\{\tau(x_1)=s\} \cap \tilde{\mathcal{E}}_2^c}. \quad (5.B.54)$$

For the event $\{g_1 \neq g_2, y_0 = y_1\}$, if $\Lambda_{5,\tau(g_1(y_1)),r_{g_1 y_1}}^{(t)}$ is still in the linear regime, we have

$$\Lambda_{5,\tau(g_1(y_1)),r_{g_1 y_1}}^{(t)} = (1 - 2\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \cdot B \pm O(\delta) \geq \varrho,$$

which implies

$$\epsilon_{\text{attn}}^{2,2} \geq 1 - 2\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \geq \Omega \left(\frac{\varrho}{B} \right). \quad (5.B.55)$$

Hence, putting (5.B.55) back to (5.B.53), and putting (5.B.54) back to (5.B.53), we can lower bound $\mathcal{N}_{s,4,2,ii}^{(t)}$ as follows $\mathcal{N}_{s,4,2,ii}^{(t)} \geq -O \left(\frac{1}{d} \right) \cdot \mathcal{N}_{s,4,2,i}^{(t)}$. If $\Lambda_{5,\tau(g_1(y_1)),r_{g_1 y_1}}^{(t)}$ falls into the smoothed regime, we can upper bound $\mathbf{sReLU}'(\Lambda_{5,\tau(g_1(y_1)),r_{g_1 y_1}}^{(t)})$ by $O \left(\frac{\epsilon_{\text{attn}}^{2,2} B}{\varrho} \right)$, and then similarly, we can obtain $\mathcal{N}_{s,4,2,ii}^{(t)} \geq -O \left(\frac{1}{d} \right) \cdot \mathcal{N}_{s,4,2,i}^{(t)}$.

Following the analogous analysis, the negative gradient from $\mathcal{N}_{s,4,1,ii}^{(\ell)}$ can also be dominated by the positive gradient from $\mathcal{N}_{s,4,2,i}^{(\ell)}$. Hence, we complete the proof. $\square$

Lemma 5.B.75. *If Induction 5.B.12 holds for all iterations $\in [T_{2,2,s}, t)$, given $s \in \tau(\mathcal{X})$, for $[\mathbf{Q}_{4,p}]_{s,s'}$, $p \in \{3, 4\}$, $s' \neq s \in \tau(\mathcal{X})$, $\ell \in [2]$, we have*

$$\left| \left[-\nabla_{\mathbf{Q}_{4,p}^{(\ell)}} \text{Loss}_5^{2,\ell} \right]_{s,s'} \right| \leq O\left(\frac{1}{d}\right) \left| \left[-\nabla_{\mathbf{Q}_{4,p}^{(\ell)}} \text{Loss}_5^{2,\ell} \right]_{s,s} \right|.$$

Non-negative Gap

Lemma 5.B.76. *If Induction 5.B.12 holds for all iterations $\in [T_{2,2,s}, t)$, then at time t , we have $\Delta^{2,\ell} \geq 0$ for any $\mathbf{Z}^{2,\ell}$ with $\ell \in \{1, 2\}$.*

Proof. Let $\tilde{T}$ denote the first time that $\mathbb{E}[\Delta^{2,2} \mid \tau(x_1) = s] < \alpha$, where $\alpha = \frac{1}{\frac{\epsilon^{\alpha-2}}{\text{poly}d}}$.

Following the analogous analysis as Lemma 5.B.64, we have

$$\sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,3}^{(\tilde{T})}} \text{Loss}_5^{2,\ell} \right]_{s,s} + \sum_{\ell=1}^2 \left[\nabla_{\mathbf{Q}_{4,4}^{(\tilde{T})}} \text{Loss}_5^{2,\ell} \right]_{s,s} = \sum_{\ell \in [2]} \sum_{\kappa \in \{i, ii, iii\}} \mathcal{N}_{s,3,\ell,\kappa}^{(\tilde{T})} - \mathcal{N}_{s,4,\ell,\kappa}^{(\tilde{T})}.$$

We can ignore the negligible difference introduced by $\mathcal{N}_{s,p,\ell,iii}^{(\tilde{T})}$ for $p \in \{3, 4\}$ and $\ell \in [2]$.

For $\ell = 1$, since $\mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(\tilde{T})} \geq \mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(\tilde{T})}$ and thus it is straightforward to see that $\mathcal{N}_{s,3,1,i}^{(\tilde{T})} - \mathcal{N}_{s,4,1,i}^{(\tilde{T})} \geq 0$. Furthermore, we have

$$\begin{aligned} & \mathcal{N}_{s,3,1,ii}^{(\tilde{T})} - \mathcal{N}_{s,4,1,ii}^{(\tilde{T})} \\ &= -\mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(\tilde{T})} \cdot \mathbf{logit}_{5,j'_1}^{(\tilde{T})} \cdot \mathbf{sReLU}'(\Lambda_{5,j'_1,r_{s_2,y_0}}^{(t)}) \right. \\ & \quad \left. \left(\left(V_{j'_1,r_{s_2,y_0}}(g_1) - \Lambda_{5,j'_1,r_{s_2,y_0}}^{(\tilde{T})} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_0)=s\} \cap \mathcal{E}_1} \right] \end{aligned} \quad (5.B.56)$$

$$\begin{aligned} & + \mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(\tilde{T})} \cdot \mathbf{logit}_{5,j'_1}^{(\tilde{T})} \cdot \mathbf{sReLU}'(\Lambda_{5,j'_1,r_{s_2,y_0}}^{(\tilde{T})}) \right. \\ & \quad \left. \left(\left(V_{j'_1,r_{s_2,y_0}}(y_0) - \Lambda_{5,j'_1,r_{s_2,y_0}}^{(\tilde{T})} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_0)=s\} \cap \mathcal{E}_1} \right] \end{aligned} \quad (5.B.57)$$

$$\begin{aligned} & - \mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(\tilde{T})} \cdot \sum_{y \neq y_0 \in \mathcal{Y}} \mathbf{logit}_{5,\tau(g_1(y))}^{(\tilde{T})} \cdot \mathbf{sReLU}'(\Lambda_{5,\tau(g_1(y)),r_{s_1,y}}^{(\tilde{T})}) \right. \\ & \quad \left. \left(\left(V_{\tau(g_1(y)),r_{s_1,y}}(g_1) - \Lambda_{5,\tau(g_1(y)),r_{s_1,y}}^{(\tilde{T})} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_0)=s\} \cap \mathcal{E}_1^c} \right] \end{aligned}$$

$$+ \mathbb{E} \left[\mathbf{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(\tilde{T})} \cdot \sum_{y \neq y_0 \in \mathcal{Y}} \mathbf{logit}_{5,\tau(g_1(y))}^{(\tilde{T})} \cdot \mathbf{sReLU}'(\Lambda_{5,\tau(g_1(y)),r_{g_1 \cdot y}}^{(\tilde{T})}) \right. \\ \left. \left(\left(V_{\tau(g_1(y)),r_{g_1 \cdot y}}(y_0) - \Lambda_{5,\tau(g_1(y)),r_{g_1 \cdot y}}^{(\tilde{T})} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbf{1}_{\{\tau(x_0)=s\} \cap \mathcal{E}_1^c} \right].$$

By Lemma 5.B.73, we obtain that $\mathcal{N}_{s,3,1,ii}^{(\tilde{T})} - \mathcal{N}_{s,4,1,ii}^{(\tilde{T})}$ is dominated by (5.B.56) + (5.B.57). Moreover, by Lemma 5.B.71, we have

$$\mathbf{logit}_{5,\tau(g_1(y))}^{(\tilde{T})}(\mathbf{Z}^{2,0}) \geq \Omega \left(\frac{1}{d^{2(\mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(\tilde{T})} - \mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},2}^{(\tilde{T})}) C_B}} \right) \\ \geq \Omega \left(\frac{1}{d^{2(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(\tilde{T})}) C_B}} \right). \quad (5.B.58)$$

For $\ell = 2$, we have

$$\mathcal{N}_{s,3,2,i}^{(\tilde{T})} - \mathcal{N}_{s,4,2,i}^{(\tilde{T})} \\ \geq \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} \cdot (1 - \mathbf{logit}_{5,j_2}^{(\tilde{T})}) \cdot \right. \\ \left. \left(\left(V_{j_2,r_{g_2 \cdot y_1}}(g_2) - \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(\tilde{T})} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbf{1}_{\{\tau(x_1)=s\} \cap \tilde{\mathcal{E}}_2} \right] \\ - \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(\tilde{T})} \cdot (1 - \mathbf{logit}_{5,j_2}^{(\tilde{T})}) \cdot \right. \\ \left. \left(\left(V_{j_2,r_{g_2 \cdot y_1}}(y_1) - \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(\tilde{T})} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbf{1}_{\{\tau(x_1)=s\} \cap \tilde{\mathcal{E}}_2} \right] \\ \geq \Omega(\alpha \epsilon \log d) \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(\tilde{T})}) \cdot \mathbf{1}_{\tau(x_1)=s} \right], \quad (5.B.59)$$

where the last inequality follows from the definition of $\tilde{T}$.

$$\mathcal{N}_{s,3,2,ii}^{(\tilde{T})} - \mathcal{N}_{s,4,2,ii}^{(\tilde{T})} \\ = -\mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} \cdot \mathbf{logit}_{5,j'_2}^{(\tilde{T})} \cdot \mathbf{sReLU}'(\Lambda_{5,j'_2,r_{g_2 \cdot y_0}}^{(\tilde{T})}) \right. \\ \left. \left(\left(V_{j'_2,r_{g_2 \cdot y_0}}(g_2) - \Lambda_{5,j'_2,r_{g_2 \cdot y_0}}^{(\tilde{T})} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbf{1}_{\{\tau(x_1)=s\} \cap \tilde{\mathcal{E}}_2} \right] \\ + \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(\tilde{T})} \cdot \mathbf{logit}_{5,j'_2}^{(\tilde{T})} \cdot \mathbf{sReLU}'(\Lambda_{5,j'_2,r_{g_2 \cdot y_0}}^{(\tilde{T})}) \right. \\ \left. \left(\left(V_{j'_2,r_{g_2 \cdot y_0}}(y_1) - \Lambda_{5,j'_2,r_{g_2 \cdot y_0}}^{(\tilde{T})} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbf{1}_{\{\tau(x_1)=s\} \cap \tilde{\mathcal{E}}_2} \right] \quad (5.B.60)$$

$$\left(\left(V_{j_2', r_{g_2 \cdot y_0}}(y_1) - \Lambda_{5, j_2', r_{g_2 \cdot y_0}}^{(\tilde{T})} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \tilde{\mathcal{E}}_2} \right] \quad (5.B.61)$$

$$- \mathbb{E} \left[\mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, 2}^{(\tilde{T})} \cdot \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5, j}^{(\tilde{T})} \cdot \left(\sum_{r \in \mathfrak{A}_j} \mathbf{sReLU}'(\Lambda_{5, j, r}^{(\tilde{T})}) \cdot \left(V_{j, r}(g_2) - \Lambda_{5, j, r}^{(\tilde{T})} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \tilde{\mathcal{E}}_2^c} \right] \quad (5.B.62)$$

$$+ \mathbb{E} \left[\mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{ans}, 1}^{(\tilde{T})} \cdot \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5, j}^{(\tilde{T})} \cdot \left(\sum_{r \in \mathfrak{A}_j} \mathbf{sReLU}'(\Lambda_{5, j, r}^{(\tilde{T})}) \cdot \left(V_{j, r}(y_1) - \Lambda_{5, j, r}^{(\tilde{T})} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \tilde{\mathcal{E}}_2^c} \right]. \quad (5.B.63)$$

Notice that by Lemma 5.B.70 and the definition of $\tilde{T}$, we have

$$|(\text{5.B.60}) + (\text{5.B.61})| \leq O((\alpha \log d)^{q-1}) \cdot \mathbb{E} \left[\left(1 - \mathbf{logit}_{5, j_2}^{(\tilde{T})} \right) \cdot \mathbb{1}_{\tau(x_1)=s} \right],$$

which is dominated by (5.B.59) due to the choice of α .

Furthermore, by Lemma 5.B.72, for $\tilde{\mathcal{E}}_2^c$, we only need to focus on the output $\mathbf{logit}_{5, \tau(g_2(y_0))}^{(t)}$ from the case $g_1 = g_2 \wedge y_0 \neq y_1$, and obtain

$$\begin{aligned} & (\text{5.B.62}) + (\text{5.B.63}) \\ & \geq - \mathbb{E} \left[\mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, 2}^{(\tilde{T})} \cdot \mathbf{logit}_{5, \tau(g_2(y_0))}^{(\tilde{T})} \cdot \mathbf{sReLU}'(\Lambda_{5, \tau(g_2(y_0)), r_{g_2 \cdot y_0}}^{(\tilde{T})}) \right. \\ & \quad \left. \left(\left(V_{\tau(g_2(y_0)), r_{g_2 \cdot y_0}}(g_2) - \Lambda_{5, \tau(g_2(y_0)), r_{g_2 \cdot y_0}}^{(\tilde{T})} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \{g_1=g_2 \wedge y_0 \neq y_1\}} \right] \\ & + \mathbb{E} \left[\mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{ans}, 1}^{(\tilde{T})} \cdot \mathbf{logit}_{5, \tau(g_2(y_0))}^{(\tilde{T})} \cdot \mathbf{sReLU}'(\Lambda_{5, \tau(g_2(y_0)), r_{g_2 \cdot y_0}}^{(\tilde{T})}) \right. \\ & \quad \left. \left(\left(V_{\tau(g_2(y_0)), r_{g_2 \cdot y_0}}(y_1) - \Lambda_{5, \tau(g_2(y_0)), r_{g_2 \cdot y_0}}^{(\tilde{T})} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \{g_1=g_2 \wedge y_0 \neq y_1\}} \right]. \end{aligned}$$

By Lemma 5.B.73, we have

$$\mathbf{logit}_{5, \tau(g_2(y_0))}^{(\tilde{T})}(\mathbf{Z}^{2,1}) = O\left(\frac{1}{d^2 (\mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{ans}, 1}^{(\tilde{T})} - \mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{ans}, 0}^{(\tilde{T})}) C_B} \right).$$

Since $\mathbb{E}[\Delta^{2,2} \mid \tau(x_1) = s] < \alpha$, where α is sufficiently small, then combining with (5.B.58), we have

$$\mathbf{logit}_{5,\tau(g_2(y_0))}^{(\tilde{T})}(\mathbf{Z}^{2,1}) = O(1) \cdot \mathbf{logit}_{5,\tau(g_2(y_0))}^{(\tilde{T})}(\mathbf{Z}^{2,0}).$$

Therefore, since $\tilde{\mathcal{E}}_2^c$ happens with probability $O(\frac{1}{\log d})$, we obtain that

$$(5.B.62) + (5.B.63) \geq -O\left(\frac{1}{\log d}\right) \left(\mathcal{N}_{s,3,1,ii}^{(t)} - \mathcal{N}_{s,4,1,ii}^{(t)} \right).$$

Putting it all together, we can conclude that when $\mathbb{E}[\Delta^{2,2} \mid \tau(x_1) = s]$ reaches below α , the gap in non-decreasing direction is guaranteed, i.e.,

$$\sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,3}^{(\tilde{T})}} \text{Loss}_5^{2,\ell} \right]_{s,s} + \sum_{\ell=1}^2 \left[\nabla_{\mathbf{Q}_{4,4}^{(\tilde{T})}} \text{Loss}_5^{2,\ell} \right]_{s,s} > 0,$$

which completes the proof. $\square$

Upper Bound for $\mathbf{Q}$

Lemma 5.B.77. *If Induction 5.B.12 holds for all iterations $\in [T_{2,2,s}, t]$, given $s \in \tau(X)$, then at time t , we have $[\mathbf{Q}_{4,p}^{(t)}]_{s,s} \leq \tilde{O}(1)$ for $p \in \{3, 4\}$.*

Proof. Denote the first time that $[\mathbf{Q}_{4,3}^{(t)}]_{s,s}$ reaches $\Omega(\log^{1+c} d)$ for some small constant $c > 0$ as $\tilde{T}$. Then by direct calculations, we have

$$\begin{aligned} \mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},2}^{(t)} &\leq O\left(\frac{1}{e^{\Omega(\log^{1+c} d)}}\right), \\ \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} &\leq O\left(\frac{1}{e^{\Omega(\log^{1+c} d)}}\right). \end{aligned}$$

Moreover, by Lemma 5.B.71, we can simply bound the logits as follows:

$$\begin{aligned} 1 - \mathbf{logit}_{j_1}^{(t)} &\leq O\left(\frac{1}{d^{C_B(1-2\mathbf{Attn}_{\text{ans},0 \rightarrow \text{pred},2}^{(t)})-1}}\right) \leq O\left(\frac{1}{d^{C_B/2-1}}\right) \text{ for } \mathbf{Z}^{2,0} \in \mathcal{E}_1; \\ 1 - \mathbf{logit}_{j_2}^{(t)} &\leq O\left(\frac{1}{d^{C_B(1-2\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - 2\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)})-1}}\right) \\ &\quad + O\left(\frac{1}{d^{2(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)})C_B}}\right) \leq O\left(\frac{1}{d^{C_B/2-1}}\right) \text{ for } \mathbf{Z}^{2,1} \in \tilde{\mathcal{E}}_2. \end{aligned}$$

Thus, by focusing on $\mathcal{N}_{s,3,1,i}$ and $\mathcal{N}_{s,4,1,i}$, we have

$$\left| \sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,3}^{(t)}} \text{Loss}_5^{2,\ell} \right]_{s,s} \right| \leq \frac{1}{d} \cdot O\left(\frac{1}{d^{C_B/2-1}}\right) \cdot O\left(\frac{1}{e^{\Omega(\log^{1+c} d)}}\right) \cdot B \leq O\left(\frac{\log d}{e^{\Omega(\log^{1+c} d)} d^{C_B/2}}\right),$$

which implies

$$\mathbf{Q}_{4,3}^{(T)} \leq \mathbf{Q}_{4,3}^{(\tilde{T})} + O\left(\frac{T \log d}{e^{\Omega(\log^{1+c} d)} d^{C_B/2}}\right) \leq \mathbf{Q}_{4,3}^{(\tilde{T})} + O\left(\frac{\text{poly} d \cdot \log d}{e^{\Omega(\log^{1+c} d)} d^{C_B/2}}\right) \leq O(\log^{1+c} d).$$

□

Attention Upper Bound

Lemma 5.B.78. *If Induction 5.B.12 holds for all iterations $\in [T_{2,2,s}, t)$, given $s \in \tau(X)$, then at time t , for any $\mathbf{Z}^{2,1}$, we have $\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \leq 0.5 + \tilde{c}_1$, where $\tilde{c}_1 > 0$ is some small constant.*

Proof. Let $\tilde{T}$ denote the first time that $\mathbb{E}[\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \mid \tau(x_1) = s]$ reaches $0.5 + \tilde{c}$, where $\tilde{c} > 0$ is some small constant s.t., $2\tilde{c} \cdot C_B > 1$. At $\tilde{T}$, we have $\text{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(\tilde{T})} \mathbb{1}_{\tau(x_0)=s} \geq \text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} \mathbb{1}_{\tau(x_1)=s}$; moreover, $\text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(\tilde{T})}$ and $\text{Attn}_{\text{ans},0 \rightarrow \text{ans},0}^{(\tilde{T})}$ is still at the constant level.

- For $\ell = 1$, by Lemma 5.B.69 and Lemma 5.B.72, for $y \in \mathcal{Y} \setminus \{y_0\}$, we have only $r_{g_1 \cdot y}$ has been activated to the linear regime for the prediction $\tau(g_1(y))$. Furthermore, we obtain

$$\begin{aligned} \sum_{y \in \mathcal{Y} \setminus \{y_0\}} \text{logit}_{5,\tau(g_1(y))}^{(\tilde{T})} &\geq \frac{\log d \cdot e^{2\tilde{c}C_B \log d}}{\log d \cdot e^{2\tilde{c}C_B \log d} + O(d)} \left(1 - \text{logit}_{5,j_1}^{(\tilde{T})}\right) \\ &= (1 - o(1)) \cdot \left(1 - \text{logit}_{5,j_1}^{(\tilde{T})}\right). \end{aligned}$$

Thus,

$$\begin{aligned} &\left[-\nabla_{\mathbf{Q}_{4,3}^{(\tilde{T})}} \text{Loss}_5^{2,1} \right]_{s,s} \\ &= \mathbb{E} \left[\text{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(\tilde{T})} \cdot \left(\left(1 - \text{logit}_{5,j_1}^{(\tilde{T})}\right) \cdot \left(\left(V_{j_1, r_{g_1 \cdot y_0}}(g_1) - \Lambda_{5,j_1, r_{g_1 \cdot y_0}}^{(\tilde{T})} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \right. \right. \\ &\quad - \sum_{y \in \mathcal{Y} \setminus \{y_0\}} \text{logit}_{5,\tau(g_1(y))}^{(\tilde{T})} \cdot \left(\left(V_{\tau(g_1(y)), r_{g_1 \cdot y}}(g_1) - \Lambda_{5,\tau(g_1(y)), r_{g_1 \cdot y}}^{(\tilde{T})} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \\ &\quad \left. \pm \frac{1}{d} \left(1 - \text{logit}_{5,j_1}^{(\tilde{T})}\right) \cdot \tilde{O}(\sigma_0^q) \right) \mathbb{1}_{\tau(x_0)=s} \right] \\ &\leq \mathbb{E} \left[\text{Attn}_{\text{ans},0 \rightarrow \text{pred},1}^{(\tilde{T})} \cdot \left(\left(1 - \text{logit}_{5,j_1}^{(\tilde{T})}\right) \cdot \left(\left(2\text{Attn}_{\text{ans},0 \rightarrow \text{pred},2}^{(\tilde{T})} \cdot B \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \right) \right] \end{aligned}$$

$$\begin{aligned}
& - \sum_{y \in \mathcal{Y} \setminus \{y_0\}} \mathbf{logit}_{5, \tau(g_1(y))}^{(\tilde{T})} \cdot \left(\left(2 \mathbf{Attn}_{\text{ans}, 0 \rightarrow \text{ans}, 0}^{(\tilde{T})} \cdot B \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \\
& \pm \frac{1}{d} \left(1 - \mathbf{logit}_{5, j_1}^{(\tilde{T})} \right) \cdot \tilde{O}(\sigma_0^q) \mathbb{1}_{\tau(x_0)=s} \Bigg] < 0.
\end{aligned}$$

- For $\ell = 2$, for $\mathbf{Z}^{2,1} \in \tilde{\mathcal{E}}_2 \cup \{g_1 = g_2, y_0 \neq y_1 x\}$, we have

$$\begin{aligned}
\sum_{y \in \mathcal{Y} \setminus \{y_0, y_1\}} \mathbf{logit}_{5, \tau(g_2(y))}^{(\tilde{T})} & \geq \frac{\log d \cdot e^{2\tilde{c}C_B \log d}}{\log d \cdot e^{2\tilde{c}C_B \log d} + O(d)} \left(1 - \mathbf{logit}_{5, j_2}^{(\tilde{T})} - \mathbf{logit}_{5, j'_2}^{(\tilde{T})} \right) \\
& = (1 - o(1)) \cdot \left(1 - \mathbf{logit}_{5, j_2}^{(\tilde{T})} - \mathbf{logit}_{5, j'_2}^{(\tilde{T})} \right).
\end{aligned}$$

Otherwise, we have

$$\sum_{y \in \mathcal{Y} \setminus \{y_1\}} \mathbf{logit}_{5, \tau(g_2(y))}^{(\tilde{T})} = (1 - o(1)) \cdot \left(1 - \mathbf{logit}_{5, j_2}^{(\tilde{T})} - \mathbf{logit}_{5, j'_2}^{(\tilde{T})} \right).$$

Therefore, similar to $\ell = 1$, we obtain $\left[-\nabla_{\mathbf{Q}_{4,3}^{(\tilde{T})}} \text{Loss}_5^{2,2} \right]_{s,s} < 0$.

Combing the above two cases, we have $\sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,3}^{(\tilde{T})}} \text{Loss}_5^{2,\ell} \right]_{s,s} < 0$. It is also clear from Lemma 5.B.74 that $\sum_{\ell=1}^2 \left[-\nabla_{\mathbf{Q}_{4,4}^{(\tilde{T})}} \text{Loss}_5^{2,\ell} \right]_{s,s} \geq 0$. Hence $\mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, 2}^{(\tilde{T})}$ cannot further grow once it reaches $0.5 + \tilde{c}$. $\square$

Decreasing Gap at the End of Convergence Let $\tilde{T}$ denote the first time that $\mathbb{E}[\epsilon_{\text{attn}}^{2,2} \mid \tau(x_1) = s] \leq 3\epsilon$, if $\mathbb{E}[\Delta^{2,2} \mid \tau(x_1) = s] \leq O\left(\frac{(d^{1.01}\epsilon)^{\frac{1}{q-1}}}{\log d}\right)$, then we can let $T^* = \tilde{T}$ and stop the training. Otherwise, we have $\mathbb{E}[\Delta^{2,2} \mid \tau(x_1) = s] \geq \Omega\left(\frac{(d^{1.01}\epsilon)^{\frac{1}{q-1}}}{\log d}\right)$. Following the similar argument as in Lemma 5.B.74, we have the gradient contribution from $\ell = 1$ is dominated by the gradient contribution from $\ell = 2$. Thus, we focus on $\ell = 2$, and obtain

$$\begin{aligned}
& \left[-\nabla_{\mathbf{Q}_{4,3}^{(\tilde{T})}} \text{Loss}_5^{2,2} \right]_{s,s} \\
& = \mathbb{E} \left[\mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, 2}^{(\tilde{T})} \cdot \left(\left(1 - \mathbf{logit}_{5, j_2}^{(\tilde{T})} \right) \cdot \left(\left(V_{j_2, r_{g_2 y_1}}(g_2) - \Lambda_{5, j_2, r_{g_2 y_1}}^{(\tilde{T})} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \right. \right. \\
& \quad \left. \left. \pm \frac{1}{d} \left(1 - \mathbf{logit}_{5, j_2}^{(\tilde{T})} \right) \cdot \tilde{O}(\sigma_0^q) \right) \mathbb{1}_{\tau(x_0)=s} \right] \\
& - \mathbb{E} \left[\mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, 2}^{(\tilde{T})} \cdot \left(\sum_{y \in \mathcal{Y} \setminus \{y_1\}} \mathbf{logit}_{5, \tau(g_2(y))}^{(\tilde{T})} \cdot \mathbf{sReLU}'(\Lambda_{5, \tau(g_2(y)), r_{g_2 y}}^{(\tilde{T})}) \right) \right]
\end{aligned}$$

$$\begin{aligned}
& \cdot \left(\left(V_{\tau(g_2(y)), r_{g_2 \cdot y}}(g_2) - \Lambda_{5, \tau(g_2(y)), r_{g_2 \cdot y}}^{(\tilde{T})} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_0)=s\} \cap \tilde{\mathcal{E}}_2} \Big] \\
& - \mathbb{E} \left[\mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, 2}^{(t)} \cdot \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5, j}^{(t)} \cdot \right. \\
& \quad \left. \left(\sum_{r \in \mathfrak{A}_j} \mathbf{sReLU}'(\Lambda_{5, j, r}^{(t)}) \cdot \left(V_{j, r}(g_2) - \Lambda_{5, j, r}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \tilde{\mathcal{E}}_2^c} \right] \\
& \leq \mathbb{E} \left[\left(1 - \mathbf{logit}_{5, j_2}^{(t)} \right) \cdot O(\epsilon \log d) \mathbb{1}_{\tau(x_1)=s} \right].
\end{aligned}$$

Turning to $\mathbf{Q}_{4,4}$, we have

$$\begin{aligned}
& \left[-\nabla_{\mathbf{Q}_{4,4}^{(\tilde{T})}} \text{Loss}_5^{2,2} \right]_{s,s} \\
& = \mathbb{E} \left[\mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{ans}, 1}^{(\tilde{T})} \cdot \left(\left(1 - \mathbf{logit}_{5, j_2}^{(\tilde{T})} \right) \cdot \left(\left(V_{j_2, r_{g_2 \cdot y_1}}(y_1) - \Lambda_{5, j_2, r_{g_2 \cdot y_1}}^{(\tilde{T})} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \right. \right. \\
& \quad \left. \left. \pm \frac{1}{d} \left(1 - \mathbf{logit}_{5, j_2}^{(\tilde{T})} \right) \cdot \tilde{O}(\sigma_0^q) \right) \mathbb{1}_{\{\tau(x_0)=s\}} \right] \\
& - \mathbb{E} \left[\mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{ans}, 1}^{(\tilde{T})} \cdot \left(\sum_{y \in \mathcal{Y} \setminus \{y_1\}} \mathbf{logit}_{5, \tau(g_2(y))}^{(\tilde{T})} \cdot \mathbf{sReLU}'(\Lambda_{5, \tau(g_2(y)), r_{g_2 \cdot y}}^{(\tilde{T})}) \right. \right. \\
& \quad \left. \left. \cdot \left(\left(V_{\tau(g_2(y)), r_{g_2 \cdot y}}(y_1) - \Lambda_{5, \tau(g_2(y)), r_{g_2 \cdot y}}^{(\tilde{T})} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \right) \mathbb{1}_{\{\tau(x_0)=s\} \cap \tilde{\mathcal{E}}_2} \right] \quad (5.B.64)
\end{aligned}$$

$$\begin{aligned}
& - \mathbb{E} \left[\mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{ans}, 1}^{(t)} \cdot \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5, j}^{(t)} \cdot \right. \\
& \quad \left. \left(\sum_{r \in \mathfrak{A}_j} \mathbf{sReLU}'(\Lambda_{5, j, r}^{(t)}) \cdot \left(V_{j, r}(y_1) - \Lambda_{5, j, r}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \tilde{\mathcal{E}}_2^c} \right]. \quad (5.B.65)
\end{aligned}$$

For (5.B.64), we have

$$\begin{aligned}
(5.B.64) & \geq \mathbb{E} \left[\mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{ans}, 1}^{(\tilde{T})} \cdot \left(-\mathbf{logit}_{5, j_2'}^{(\tilde{T})} \cdot \right. \right. \\
& \quad \left. \left. \min \left\{ \Omega(\log d), \Omega \left(\left((\mathbf{Attn}_{\mathbb{A}, 1 \rightarrow 2}^{(\tilde{T})} - \mathbf{Attn}_{\mathbb{A}, 1 \rightarrow \mathbb{A}, 1}^{(\tilde{T})}) \log d \right)^{q-1} \log d \right) \right\} \right) \mathbb{1}_{\{\tau(x_0)=s\} \cap \tilde{\mathcal{E}}_2} \right] \\
& \geq \mathbb{E} \left[\left(1 - \mathbf{logit}_{5, j_2}^{(\tilde{T})} \right) \cdot \Omega \left(\frac{1}{d} \right) \cdot \right]
\end{aligned}$$

$$\min \left\{ \Omega(\log d), \Omega \left(\left((\mathbf{Attn}_{\mathbb{A},1 \rightarrow 2}^{(\tilde{T})} - \mathbf{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1}^{(\tilde{T})}) \log d \right)^{q-1} \log d \right) \right\} \mathbb{1}_{\{\tau(x_0)=s\} \cap \tilde{\mathcal{E}}_2}.$$

Moreover, for (5.B.65), we have

$$\begin{aligned} (5.B.65) &\geq -\mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(\tilde{T})} \cdot \mathbf{logit}_{5,\tau(g_1(y_1))}^{(\tilde{T})} \cdot \mathbf{sReLU}'(\Lambda_{5,\tau(g_1(y_1)),r_{g_1 \cdot y_1}}^{(\tilde{T})}) \right. \\ &\quad \left. \left(\left(V_{\tau(g_1(y_1)),r_{g_1 \cdot y_1}}(y_1) - \Lambda_{5,\tau(g_1(y_1)),r_{g_1 \cdot y_1}}^{(\tilde{T})} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\{\tau(x_1)=s\} \cap \{g_1 \neq g_2 \wedge y_0=y_1\}} \right] \\ &\geq -\mathbb{E} \left[\left(1 - \mathbf{logit}_{5,j_2}^{(\tilde{T})} \right) \cdot O\left(\frac{1}{d}\right) \cdot O\left(\frac{1}{\log d}\right) \cdot O\left((\epsilon \log d)^{q-1}\right) \mathbb{1}_{\{\tau(x_0)=s\} \cap \tilde{\mathcal{E}}_2} \right], \end{aligned}$$

where the last inequality holds since $\Lambda_{5,\tau(g_1(y_1)),r_{g_1 \cdot y_1}}^{(\tilde{T})} \leq \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(\tilde{T})} \right) B \leq O(\epsilon \log d)$.

Since $\mathbb{E} \left[\Delta^{2,2} \mid \tau(x_1) = s \right] \geq \Omega\left(\frac{(d^{1.01}\epsilon)^{\frac{1}{q-1}}}{\log d}\right)$ we have

$$(5.B.64) \geq d^{0.01} \cdot \left| \left[-\nabla_{\mathbf{Q}_{4,3}^{(\tilde{T})}} \text{Loss}_5^{2,2} \right]_{s,s} \right|,$$

and thus (5.B.64) $\gg$ (5.B.65). Thus, for $\epsilon \ll \frac{1}{d}$, if the attention gap does not decrease to the level of $O\left(\frac{(d^{1.01}\epsilon)^{\frac{1}{q-1}}}{\log d}\right)$, $[\mathbf{Q}_{4,4}]_{s,s}$ will start to dominantly increase while $[\mathbf{Q}_{4,3}]_{s,s}$ will not change too much. On the other hand, if the gap of attention holds, then $[\mathbf{Q}_{4,3}]_{s,s} \geq [\mathbf{Q}_{4,4}]_{s,s}$, we have

$$\begin{aligned} \epsilon_{\text{attn}}^{2,2} &= 1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} = \frac{O(1)}{O(1) + e^{[\mathbf{Q}_{4,4}^{(t)}]_{s,s}} + e^{[\mathbf{Q}_{4,3}^{(t)}]_{s,s}}} \\ &\geq \frac{O(1)}{O(1) + 2e^{[\mathbf{Q}_{4,3}^{(\tilde{T})}]_{s,s}}} \geq \frac{1}{2} \cdot 3\epsilon > \epsilon. \end{aligned}$$

This implies that, we can find some time between $\tilde{T}$ and $T_{2,3,s}$, s.t., the gap will decrease to $O\left(\frac{(d^{1.01}\epsilon)^{\frac{1}{q-1}}}{\log d}\right)$. We denote this time as $T^\star$ and stop the training.

At the End of the Training Putting everything together, we have that at the end of the training, we have

Lemma 5.B.79. *Given $s \in \tau(\mathcal{X})$, at $T^\star = \tilde{O}\left(\frac{d^{(1-2\epsilon)C_B}}{\eta\epsilon}\right)$, if $\tilde{\Omega}(\sigma_0) \ll \epsilon \leq O(\frac{1}{d^{1.01}})$, we have*

- (a) *Attention convergence: $\epsilon_{\text{attn}}^{2,\ell} \leq O(\epsilon)$ for $\ell \in [2]$, and $\mathbb{E}[\Delta^{2,2} \mid \tau(x_1) = s] \leq O\left(\frac{(d^{1.01}\epsilon)^{\frac{1}{q-1}}}{\log d}\right)$;*
- (b) *$\left[\mathbf{Q}_{4,p}^{(T_{2,3,s})}\right]_{s,s'} \geq \Omega(\log d)$ for $p \in \{3,4\}$ if $s = s' \in \tau(\mathcal{X})$, otherwise, $\left[\mathbf{Q}_{4,p}^{(T_{2,3,s})}\right]_{s,s'} \leq \tilde{O}\left(\frac{1}{d}\right)$;*

(c) *Loss convergence:* $\sum_{\ell=1}^2 \text{Loss}_5^{2,\ell} \leq \frac{1}{\text{poly}d}$.

Proof of Main Theorem

Theorem 5.B.3 (Restatement of Theorem 5.4.1). *Under Assumptions 5.3.1, 5.3.2, 5.3.3, 5.3.4, 5.3.5 and 5.4.1, for some constant $0 < c^* < 1$, $n_y < m \ll \log^2 d$, the transformer model $F^{(T_1+T_2)}$ obtained by Algorithm 2 with learning rate $\eta = \frac{1}{\text{poly}(d)}$, and stage 1 and 2 iteration $T_1 = \tilde{O}\left(\frac{d}{\eta(\sigma_0)^{q-2}}\right)$, $T_2 = \tilde{O}\left(\frac{\text{poly}(d)}{\eta\sigma_0}\right)$ satisfies*

$$\text{Acc}_L\left(F^{(T_1+T_2)}\right) \geq 1 - \frac{1}{\text{poly}(d)}, \text{ for every } L \leq O(d^{c^*}), \quad (5.B.66)$$

i.e., $F^{(T_1+T_2)}$, which is trained for task $\mathcal{T}^1$ and $\mathcal{T}^2$, generalizes to solve the tasks $\mathcal{T}^\ell$, $\ell \leq L$.

Proof. By Lemma 5.B.79, at the end of Stage 2 training, we have $\left[\mathbf{Q}_{4,p}^{(T_{2,3,s})}\right]_{s,s} \geq \Omega(\log d)$ for all $p \in \{3, 4\}$ and $s \in \tau(X)$.

Therefore, for task $\mathcal{T}^L$ with $L \leq O(d^{c^*})$, where $0 < c^* < 1$ is a constant depending on the value of ϵ in Lemma 5.B.79, we obtain

$$\epsilon_{\text{attn}}^{L,\ell} \leq \frac{O(1) \cdot L}{O(1) \cdot L + e \left[\mathbf{Q}_{4,3}^{(T_{2,3,s})}\right]_{s,s} + e \left[\mathbf{Q}_{4,4}^{(T_{2,3,s})}\right]_{s,s}} = o(1).$$

Moreover, we have

$$\Delta^{L,\ell} \leq \Delta^{2,1} = \text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(T_{2,3,s})} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(T_{2,3,s})} \leq o(1).$$

These together guarantee that

$$1 - \text{logit}_{5,\tau(g_{\ell+1}(y_\ell))}(F^{(T^*)}, \mathbf{Z}^{(L,\ell)}) \leq \frac{O(1) \cdot d + e^{o(1)}}{O(1) \cdot d + e^{o(1)} + e^{\Omega(\log d)}} \leq \frac{1}{\text{poly}(d)},$$

which implies

$$\text{Acc}_L(F^{(T^*)}) \geq 1 - \frac{1}{\text{poly}(d)}.$$

□

5.B.5 Learning the Attention Layer: Symmetry Case for Short-Length

In this part, we consider the scenario where the group operations form a symmetry group. Specifically, following Assumption 5.5.1, we assume that $\mathcal{Y} = \{0, 1, \dots, n_y - 1\}$, and let $\mathcal{G}$ be the symmetry group for $\mathcal{Y}$, with order $|\mathcal{G}| = n_y! = \Theta(\text{polylog}d) \gg \frac{1}{\varrho}$, where $n_y = \Theta\left(\frac{\log \log d}{\log \log \log d}\right)$.

Similar to the simply transitive case, we restrict our analysis to updating only $\mathbf{Q}$, specifically the blocks $\mathbf{Q}_{4,3}$ and $\mathbf{Q}_{4,4}$.

Throughout this section, we focus on the simple task $\mathcal{T}^2$ and analyze gradient descent updates with respect to the per-token loss $\text{Loss}_5^{2,2}$. Given $s \in \tau(\mathcal{X})$, let $\text{Loss}_{5,s}^{2,2} = -\mathbb{E} \left[\log p_F(\mathbf{Z}_{\text{ans},2,5} \mid \mathbf{Z}^{2,1}) \mid \tau(x_1) = s \right]$. Due to the symmetry of $[\mathbf{Q}_{4,3}]_{s,s}$ and $[\mathbf{Q}_{4,4}]_{s,s}$ across $s \in \tau(\mathcal{X})$, we may, without loss of generality, focus on a particular $s \in \tau(\mathcal{X})$ and analyze the corresponding loss $\text{Loss}_{5,s}^{2,2}$ in what follows.

Gradient Computations

We start with the gradient computations for the attention layer.

Notations for gradient expressions. We first introduce some notations for the gradients of the attention layer. For $1 \leq \ell \leq L$, given $\mathbf{Z}^{L,\ell-1}$ and $\mathbf{k} \in \mathcal{I}^{L,\ell-1}$, define

$$\Xi_{\ell,i,\mathbf{k}}^L(\mathbf{Z}^{L,\ell-1}) \triangleq \sum_{j \in [d]} \mathcal{E}_{i,j}(\mathbf{Z}^{L,\ell-1}) \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{i,j,r}(\mathbf{Z}^{L,\ell-1})) \langle \mathbf{W}_{i,j,r}, \mathbf{Z}_{\mathbf{k}} \rangle, \quad i \in [5]. \quad (5.B.67)$$

For simplicity of notation, we will henceforth omit the dependence on $\mathbf{Z}^{L,\ell-1}$ in the notation of $\Xi_{\ell,i,\mathbf{k}}^L$ when it is clear from the context.

Fact 5.B.9 (Gradients of $\mathbf{Q}$). For any $p, q \in [5]$, we have

$$\begin{aligned} -\nabla_{\mathbf{Q}_{p,q}} \text{Loss}^L &= \sum_{\ell=1}^L \sum_{i \in [5]} -\nabla_{\mathbf{Q}_{p,q}} \text{Loss}_i^{L,\ell}, \quad \text{where} \\ -\nabla_{\mathbf{Q}_{p,q}} \text{Loss}_i^{L,\ell} &= \\ &\mathbb{E} \left[\sum_{\mathbf{k} \in \mathcal{I}^{L,\ell-1}} \mathbf{Attn}_{\text{ans},\ell-1 \rightarrow \mathbf{k}} \cdot \left(\Xi_{\ell,i,\mathbf{k}}^L - \sum_{\mathbf{k}' \in \mathcal{I}^{L,\ell-1}} \mathbf{Attn}_{\text{ans},\ell-1 \rightarrow \mathbf{k}'} \Xi_{\ell,i,\mathbf{k}'}^L \right) \mathbf{Z}_{\text{ans},\ell-1,p} \mathbf{Z}_{\mathbf{k},q}^\top \right]. \end{aligned}$$

Lemma 5.B.80 (Gradients of $\mathbf{Q}_{4,3}$). Given $s \in \tau(\mathcal{X})$, for the diagonal entry $[\mathbf{Q}_{4,3}]_{s,s}$ of the block $\mathbf{Q}_{4,3}$, we have

$$\begin{aligned} \left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_5^{2,2} \right]_{s,s} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2} \cdot \left(\sum_{j \in [d]} \mathcal{E}_{5,j}(\mathbf{Z}^{2,1}) \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \right. \right. \\ &\quad \left. \left. \left(\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_{2,2} \rangle - \Lambda_{5,j,r} + b_{5,j,r} \right) \mathbb{1}_{s=\tau(x_1)} \right) \right]. \end{aligned}$$

Moreover, for the off-diagonal entries $[\mathbf{Q}_{4,3}]_{s,s'}$ with $s \neq s'$, we have

$$\left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_5^{2,2} \right]_{s,s'} = \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1} \cdot \left(\sum_{j \in [d]} \mathcal{E}_{5,j}(\mathbf{Z}^{2,1}) \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \right. \right.$$

$$\left(\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_{\mathbb{A},1} \rangle - \Lambda_{5,j,r} + b_{5,j,r} \right) \mathbb{1}_{s=\tau(x_1), s'=\tau(x_0)} \Bigg].$$

Lemma 5.B.81 (Gradients of $\mathbf{Q}_{4,4}$). *Given $s \in \tau(X)$, for the diagonal entry $[\mathbf{Q}_{4,4}]_{s,s}$ of the block $\mathbf{Q}_{4,4}$, we have*

$$\begin{aligned} \left[-\nabla_{\mathbf{Q}_{4,4}} \text{Loss}_5^{2,2} \right]_{s,s} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1} \cdot \left(\sum_{j \in [d]} \mathcal{E}_{5,j}(\mathbf{Z}^{2,1}) \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \right. \right. \\ &\quad \left. \left. \left(\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_{\mathbb{A},1} \rangle - \Lambda_{5,j,r} + b_{5,j,r} \right) \mathbb{1}_{s=\tau(x_1)} \right) \right]. \end{aligned}$$

Moreover, for the off-diagonal entries $[\mathbf{Q}_{4,4}]_{s,s'}$ with $s \neq s'$, we have

$$\begin{aligned} \left[-\nabla_{\mathbf{Q}_{4,4}} \text{Loss}_5^{2,2} \right]_{s,s'} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0} \cdot \left(\sum_{j \in [d]} \mathcal{E}_{5,j}(\mathbf{Z}^{2,1}) \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \right. \right. \\ &\quad \left. \left. \left(\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_{\mathbb{A},0} \rangle - \Lambda_{5,j,r} + b_{5,j,r} \right) \mathbb{1}_{s=\tau(x_1), s'=\tau(x_0)} \right) \right]. \end{aligned}$$

Some Useful Bounds for Gradients

In this subsection, we establish several useful bounds on the gradients of the attention layer, leveraging the feature structure of the MLP layer learned during stage 1.1. These bounds will be instrumental for the subsequent analysis.

Recall that for $j \in \tau(\mathcal{Y})$ and $y \in \mathcal{Y}$, the fiber $\text{Fiber}_{j,y}$, and the set of feature combinations for predicting $y = \tau^{-1}(j)$, are defined as

$$\text{Fiber}_{j,y} = \{g \in \mathcal{G} \mid \tau(g(y)) = j\}, \quad \mathfrak{F}_j = \{(\text{Fiber}_{j,y}, y) \in 2^{\mathcal{G}} \times \mathcal{Y}\}.$$

As established in Theorem 5.B.2, at the end of stage 1.1, we have

- **Sparse activations:** For $j \in \tau(\mathcal{Y})$, let $\phi = (\text{Fiber}_{j,y}, y) \in \mathfrak{F}_j$, then there exists exactly one *activated* neuron $r \in [m]$ such that when $g_1 = g \in \text{Fiber}_{j,y}$, $y_0 = y$ happens:

$$\begin{aligned} \Lambda_{5,j,r}^{(T_{1.1})} &\geq B - O(\delta), \quad \left| C_\alpha V_{j,r}^{(t)}(y) - V_{j,r}^{(t)}(g) \right| \leq O(\delta) \\ \Lambda_{5,j,r'}^{(T_{1.1})} &\leq O(d^{-\Omega(1)}) \quad \forall r' \neq r. \end{aligned}$$

for some $C_\alpha = \Theta(n_y)$;

- **Cancellation of incorrect features:** For $j \in \tau(\mathcal{Y})$, let $\phi = (\text{Fiber}_{j,y}, y) \in \mathfrak{F}_j$, and let the $r \in [m]$ be the activated neuron, then for any $g' \notin \text{Fiber}_{j,y}$, and any $y' \neq y \in \mathcal{Y}$, we have

$$\left| V_{j,r}^{(T_{1.1})}(g) + V_{j,r}^{(T_{1.1})}(y') \right| \leq O(\delta) \quad \text{and} \quad \left| V_{j,r}^{(T_{1.1})}(g') + V_{j,r}^{(T_{1.1})}(y) \right| \leq O(\delta).$$

In the analysis of FFN layer, we have a pair $\delta = (\delta_1, \delta_2)$; in the expressions above, some terms should use δ_1 and others δ_2 .

Notations for activated neurons. We denote by $r_{j,y}$ the unique activated neuron corresponding to $\phi = (\text{Fiber}_{j,y}, y) \in \mathfrak{F}_j$. For any $g \in \text{Fiber}_{j,y}$, we also write $r_{g,y}$ for the same neuron $r_{j,y}$. Note that $r_{g_1,y} = r_{g_2,y}$ for distinct $g_1, g_2 \in \text{Fiber}_{j,y}$. Moreover, define

$$\mathfrak{A} \triangleq \cup_{j \in \tau(\mathcal{Y})} \mathfrak{A}_j, \text{ where } \mathfrak{A}_j \triangleq \{r_{j,y} \mid y \in \mathcal{Y}\}.$$

In other words, $\mathfrak{A}$ is the set of all activated neurons across all feature sets $\mathfrak{F}_j$ for $j \in \tau(\mathcal{Y})$. Given $\mathbf{Z}^{L,\ell-1}$, letting $\widehat{\mathcal{G}}(\mathbf{Z}^{L,\ell-1}) = \cup \{g_{\ell'}\}_{\ell'=1}^L$ be the collection of all the chosen group elements in the predicate clauses. Similarly $\widehat{\mathcal{Y}} = \cup \{y_{\ell'}\}_{\ell'=0}^{\ell-1}$. Then define $\widehat{\mathfrak{A}}_j(\mathbf{Z}^{L,\ell-1}) = \left\{ r_{g,y} \mid g \in \text{Fiber}_{j,y} \wedge \left(g \in \widehat{\mathcal{G}}(\mathbf{Z}^{L,\ell-1}) \vee y \in \widehat{\mathcal{Y}} \right) \right\}$. For simplicity, we omit the dependence on $\mathbf{Z}^{L,\ell-1}$ in the notation of $\widehat{\mathfrak{A}}_j$ when it is clear from the context. Equipped with these notations, we can summarize the above properties in the following lemmas.

Lemma 5.B.82 (Properties of target feature magnitude). *Given $j \in \tau(\mathcal{Y})$ and $y \in \mathcal{Y}$ then for $g \in \text{Fiber}_{j,y}$, the following properties hold.*

$$V_{j,r_{g,y}}(g) + V_{j,r_{g,y}}(y) \geq 2B - O(\delta), \quad \left| C_\alpha V_{j,r}^{(t)}(y) - V_{j,r}^{(t)}(g) \right| \leq O(\delta), \quad (5.B.68)$$

$$\left| V_{j,r_{g,y}}(g) + V_{j,r_{g,y}}(y') \right| \leq O(\delta), \quad V_{j,r_{g,y}}(y') < 0 \quad \text{for all } y' \neq y, \quad (5.B.69)$$

$$\left| V_{j,r_{g,y}}(g') + V_{j,r_{g,y}}(y) \right| \leq O(\delta), \quad V_{j,r_{g,y}}(g') < 0 \quad \text{for all } g' \notin \text{Fiber}_{j,y}. \quad (5.B.70)$$

$$\left| V_{j,r}(g) \right|, \left| V_{j,r}(y) \right| \leq O(\delta) \quad \text{for all } r \notin \mathfrak{A}_j. \quad (5.B.71)$$

Lemma 5.B.83 (Properties of irrelevant magnitude). *If $(p, v) \notin \{2\} \times \mathcal{G} \cup \{5\} \times \mathcal{Y}$, or $j \notin \tau(\mathcal{Y})$, then for any $r \in [m]$, we have*

$$\left| \langle \mathbf{W}_{5,j,r,p}, e_v \rangle \right| \leq \widetilde{O}(\sigma_0). \quad (5.B.72)$$

The above lemmas give us some direct computations of the inner products between the weight matrices and input embedding vectors.

Lemma 5.B.84. *Let $j \in \tau(\mathcal{Y})$ and $\ell \in [2]$. Then for any $r \in [m]$, the following holds:*

$$\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_{\text{pred},\ell} \rangle = V_{j,r}(g_\ell) \pm \widetilde{O}(\sigma_0), \quad (5.B.73)$$

$$\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_{\text{ans},\ell-1} \rangle = V_{j,r}(y_{\ell-1}) \pm \widetilde{O}(\sigma_0). \quad (5.B.74)$$

Moreover, for $j \notin \tau(\mathcal{Y})$ and any $\mathbf{k} \in \mathcal{I}^{2,1}$ and $r \in [m]$, we have

$$\left| \langle \mathbf{W}_{5,j,r}, \mathbf{Z}_{\mathbf{k}} \rangle \right| = \widetilde{O}(\sigma_0). \quad (5.B.75)$$

Furthermore, we can establish some characterizations of the $\Lambda_{5,j,r}(\mathbf{Z}^{2,\ell-1})$ quantities, which are crucial for the following analysis.

Lemma 5.B.85 (Characterizations of Lambda). *Given $\mathbf{Z}^{2,1}$ with $\{\mathbf{Attn}_{\text{ans},1 \rightarrow \mathbf{k}}\}_{\mathbf{k} \in \mathcal{I}^{2,1}}$, then*

(a) *for $j \in \tau(\mathcal{Y})$, for activated neuron $r \in \mathfrak{A}_j$, we have*

$$\Lambda_{5,j,r} = \sum_{\ell'=1}^2 \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},\ell'} V_{j,r}(g_{\ell'}) + \sum_{\ell'=1}^2 \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},\ell'-1} V_{j,r}(y_{\ell'-1}) \pm \tilde{O}(\sigma_0).$$

(b) *for $j \in \tau(\mathcal{Y})$, for any non-activated neuron $r \notin \mathfrak{A}_j$ we have*

$$|\Lambda_{5,j,r}| \leq O(\delta).$$

(c) *for $j \notin \tau(\mathcal{Y})$, for any $r \in [m]$, we have*

$$|\Lambda_{5,j,r}| \leq \tilde{O}(\sigma_0).$$

A direct consequence of the above lemma is the following finer characterization of the activated neurons.

Lemma 5.B.86. *Given $j \in \tau(\mathcal{Y})$, for $r \in \mathfrak{A}_j \setminus \widehat{\mathfrak{A}}_j$, we have $\mathbf{sReLU}'(\Lambda_{5,j,r}) = 0$.*

Now we are ready to further derive the gradients of the attention layer starting from Lemmas 5.B.80 and 5.B.81 and the properties established above.

Lemma 5.B.87 (Refined expression for the gradient of $\mathbf{Q}_{4,3}$). *Given $s \in \tau(\mathcal{X})$, for the diagonal entry $[\mathbf{Q}_{4,3}]_{s,s}$ of the block $\mathbf{Q}_{4,3}$, letting $j_2 = \tau(g_2(y_1))$, we have*

$$\begin{aligned} \left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_5^{2,2} \right]_{s,s} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2} \cdot \right. \\ &\quad \left((1 - \mathbf{logit}_{5,j_2}) \cdot \left(\sum_{r \in \widehat{\mathfrak{A}}_{j_2}} \mathbf{sReLU}'(\Lambda_{5,j_2,r}) \cdot (V_{j_2,r}(g_2) - \Lambda_{5,j_2,r} \pm \tilde{O}(\sigma_0)) \pm \tilde{O}(\delta^q) \right) \right. \\ &\quad \left. - \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \left(\sum_{r \in \mathfrak{A}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot (V_{j,r}(g_2) - \Lambda_{5,j,r} \pm \tilde{O}(\sigma_0)) \pm \tilde{O}(\delta^q) \right) \right. \\ &\quad \left. \left. \pm \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \tilde{O}(\sigma_0^q) \right) \mathbb{1}_{\tau(x_1)=s} \right]. \end{aligned}$$

Moreover, for the off-diagonal entries $[\mathbf{Q}_{4,3}]_{s,s'}$ with $s \neq s'$, we have

$$\begin{aligned} \left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_5^{2,2} \right]_{s,s'} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1} \cdot \right. \\ &\quad \left((1 - \mathbf{logit}_{5,j_2}) \cdot \left(\sum_{r \in \widehat{\mathfrak{A}}_{j_2}} \mathbf{sReLU}'(\Lambda_{5,j_2,r}) \cdot (V_{j_2,r}(g_1) - \Lambda_{5,j_2,r} \pm \tilde{O}(\sigma_0)) \pm \tilde{O}(\delta^q) \right) \right. \end{aligned}$$

$$\begin{aligned}
& - \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \left(\sum_{r \in \mathfrak{A}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \left(V_{j,r}(g_1) - \Lambda_{5,j,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \\
& \pm \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \widetilde{O}(\sigma_0^q) \mathbb{1}_{\tau(x_1)=s, \tau(x_0)=s'} \Bigg].
\end{aligned}$$

Lemma 5.B.88 (Refined expression for the gradient of $\mathbf{Q}_{4,4}$). *Given $s \in \tau(\mathcal{X})$, for the diagonal entry $[\mathbf{Q}_{4,4}]_{s,s}$ of the block $\mathbf{Q}_{4,4}$, letting $j_2 = \tau(g_2(y_1))$, we have*

$$\begin{aligned}
\left[-\nabla_{\mathbf{Q}_{4,4}} \text{Loss}_5^{2,2} \right]_{s,s} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{ans}, 1} \cdot \right. \\
& \left((1 - \mathbf{logit}_{5,j_2}) \cdot \left(\sum_{r \in \mathfrak{A}_{j_2}} \mathbf{sReLU}'(\Lambda_{5,j_2,r}) \cdot \left(V_{j_2,r}(y_1) - \Lambda_{5,j_2,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \right. \\
& - \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \left(\sum_{r \in \mathfrak{A}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \left(V_{j,r}(y_1) - \Lambda_{5,j,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \\
& \left. \left. \pm \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \widetilde{O}(\sigma_0^q) \right) \mathbb{1}_{\tau(x_1)=s} \right].
\end{aligned}$$

Moreover, for the off-diagonal entries $[\mathbf{Q}_{4,3}]_{s,s'}$ with $s \neq s'$, we have

$$\begin{aligned}
\left[-\nabla_{\mathbf{Q}_{4,4}} \text{Loss}_5^{2,2} \right]_{s,s'} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{ans}, 0} \cdot \right. \\
& \left((1 - \mathbf{logit}_{5,j_2}) \cdot \left(\sum_{r \in \mathfrak{A}_{j_2}} \mathbf{sReLU}'(\Lambda_{5,j_2,r}) \cdot \left(V_{j_2,r}(y_0) - \Lambda_{5,j_2,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \right. \\
& - \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \left(\sum_{r \in \mathfrak{A}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \left(V_{j,r}(y_0) - \Lambda_{5,j,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \\
& \left. \left. \pm \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \widetilde{O}(\sigma_0^q) \right) \mathbb{1}_{\tau(x_1)=s, \tau(x_0)=s'} \right].
\end{aligned}$$

Following the above calculations, we can further obtain the gradient summation of $\mathbf{Q}_{4,3}$ and $\mathbf{Q}_{4,4}$ as follows:

Lemma 5.B.89 (Gradient sum of $\mathbf{Q}_{4,3}$ and $\mathbf{Q}_{4,4}$). *Given $s \in \tau(\mathcal{X})$, letting $j_2 = \tau(g_2(y_1))$, we have*

$$\begin{aligned}
& \left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_5^{2,2} \right]_{s,s} + \left[-\nabla_{\mathbf{Q}_{4,4}} \text{Loss}_5^{2,2} \right]_{s,s} \\
&= \mathbb{E} \left[\left((1 - \mathbf{logit}_{5,j_2}) \cdot \left(\sum_{r \in \mathfrak{A}_{j_2}} \mathbf{sReLU}'(\Lambda_{5,j_2,r}) \cdot \right. \right. \right.
\end{aligned}$$

$$\begin{aligned}
& \left(-\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0} \cdot V_{j_2,r}(y_0) - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1} \cdot V_{j_2,r}(g_1) \right. \\
& \quad \left. + (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}) \Lambda_{5,j_2,r} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \\
& + \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \left(\sum_{r \in \hat{\mathbb{A}}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \right. \\
& \quad \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0} \cdot V_{j,r}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1} \cdot V_{j,r}(g_1) \right. \\
& \quad \left. - (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}) \Lambda_{5,j,r} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \\
& \quad \left. \pm \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot (\mathbf{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1} + \mathbf{Attn}_{\mathbb{A},1 \rightarrow 2}) \tilde{O}(\sigma_0^q) \right) \mathbb{1}_{\tau(x_1)=s} \Bigg].
\end{aligned}$$

Notations for gradient decompositions. We shall define some useful notations to further simplify the expressions of gradient.

Lemma 5.B.90. *For any $s \in \tau(\mathcal{X})$, we define the following notations for the gradient decompositions:*

1. *for $[\mathbf{Q}_{4,3}]_{s,s}$ we have $[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_5^{2,2}]_{s,s} = \mathbb{E}[\mathcal{N}_{s,3,2,i} + \mathcal{N}_{s,3,2,ii} + \mathcal{N}_{s,3,2,iii}]$, where*

$$\begin{aligned}
\mathcal{N}_{s,3,2,i} &= \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2} \cdot (1 - \mathbf{logit}_{5,j_2}) \cdot \\
& \left(\sum_{r \in \hat{\mathbb{A}}_{j_2}} \mathbf{sReLU}'(\Lambda_{5,j_2,r}) \cdot (V_{j_2,r}(g_2) - \Lambda_{5,j_2,r} \pm \tilde{O}(\sigma_0)) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s},
\end{aligned} \tag{5.B.76}$$

$$\begin{aligned}
\mathcal{N}_{s,3,2,ii} &= -\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2} \cdot \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \\
& \left(\sum_{r \in \hat{\mathbb{A}}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot (V_{j,r}(g_2) - \Lambda_{5,j,r} \pm \tilde{O}(\sigma_0)) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s},
\end{aligned} \tag{5.B.77}$$

$$\mathcal{N}_{s,3,2,iii} = \pm \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2} \cdot \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \tilde{O}(\sigma_0^q) \mathbb{1}_{\tau(x_1)=s}. \tag{5.B.78}$$

2. *for $[\mathbf{Q}_{4,4}]_{s,s}$, we have $[-\nabla_{\mathbf{Q}_{4,4}} \text{Loss}_5^{2,2}]_{s,s} = \mathbb{E}[\mathcal{N}_{s,4,2,i} + \mathcal{N}_{s,4,2,ii} + \mathcal{N}_{s,4,2,iii}]$, where*

$$\begin{aligned}
\mathcal{N}_{s,4,2,i} &= \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1} \cdot (1 - \mathbf{logit}_{5,j_2}) \cdot \\
& \left(\sum_{r \in \hat{\mathbb{A}}_{j_2}} \mathbf{sReLU}'(\Lambda_{5,j_2,r}) \cdot (V_{j_2,r}(y_1) - \Lambda_{5,j_2,r} \pm \tilde{O}(\sigma_0)) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s},
\end{aligned} \tag{5.B.79}$$

$$\mathcal{N}_{s,4,2,ii} = -\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1} \cdot \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \tag{5.B.80}$$

$$\left(\sum_{r \in \widehat{\mathcal{A}}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \left(V_{j,r}(y_1) - \Lambda_{5,j,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s},$$

$$\mathcal{N}_{s,4,2,iii} = \pm \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1} \cdot \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \widetilde{O}(\sigma_0^q) \mathbb{1}_{\tau(x_1)=s}. \quad (5.B.81)$$

3. for the summation of $[\mathbf{Q}_{4,3}]_{s,s}$ and $[\mathbf{Q}_{4,4}]_{s,s}$, we have $[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_5^{2,2}]_{s,s} + [-\nabla_{\mathbf{Q}_{4,4}} \text{Loss}_5^{2,2}]_{s,s} = \mathbb{E}[\mathcal{N}_{s,2,i} + \mathcal{N}_{s,2,ii} + \mathcal{N}_{s,2,iii}]$, where

$$\begin{aligned} \mathcal{N}_{s,2,i} &= (1 - \mathbf{logit}_{5,j_2}) \cdot \left(\sum_{r \in \widehat{\mathcal{A}}_{j_2}} \mathbf{sReLU}'(\Lambda_{5,j_2,r}) \cdot \right. \\ &\quad \left(-\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0} \cdot V_{j_2,r}(y_0) - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1} \cdot V_{j_2,r}(g_1) \right. \\ &\quad \left. + (1 - \mathbf{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1} - \mathbf{Attn}_{\mathbb{A},1 \rightarrow 2}) \Lambda_{5,j_2,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \Big) \mathbb{1}_{\tau(x_1)=s} \\ \mathcal{N}_{s,2,ii} &= \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \left(\sum_{r \in \widehat{\mathcal{A}}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \right. \\ &\quad \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0} \cdot V_{j,r}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1} \cdot V_{j,r}(g_1) \right. \\ &\quad \left. - (1 - \mathbf{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1} - \mathbf{Attn}_{\mathbb{A},1 \rightarrow 2}) \Lambda_{5,j,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \Big) \mathbb{1}_{\tau(x_1)=s} \\ \mathcal{N}_{s,2,iii} &= \pm \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot (\mathbf{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1} + \mathbf{Attn}_{\mathbb{A},1 \rightarrow 2}) \widetilde{O}(\sigma_0^q) \mathbb{1}_{\tau(x_1)=s}. \end{aligned}$$

Probabilistic events. We conclude this subsection by introducing several probabilistic events that will be used to simplify the characterization of activated neurons in the subsequent analysis. We first define some events that may contribute non-trivially to the gradient of $\mathbf{Q}_{4,3}$ and $\mathbf{Q}_{4,4}$:

$$\mathcal{E}_1 = \left\{ y_0 \neq y_1, g_1(y_{\ell-1}) \neq g_2(y_{\ell-1}), \text{ for all } \ell \in [2] \right\}, \quad (5.B.82)$$

$$\mathcal{E}_2 = \left\{ g_1, g_2 \in \text{Fiber}_{\tau(g_2(y_0)), y_0}, g_1(y_1) \neq g_2(y_1) \right\}, \quad (5.B.83)$$

$$\mathcal{E}_3 = \left\{ y_0 = y_1, g_1(y_1) \neq g_2(y_1) \right\}, \quad (5.B.84)$$

$$\mathcal{E}_4 = \left\{ y_0 \neq y_1, g_1(y_{\ell-1}) = g_2(y_{\ell-1}), \text{ for all } \ell \in [2] \right\}. \quad (5.B.85)$$

We first observe that event $\mathcal{E}_1$ occurs with high probability $1 - O(\frac{1}{n_c})$, and serves as the primary regime of interest. In contrast, events $\mathcal{E}_2$ and $\mathcal{E}_3$ each occur with probability $\Theta(1/n_y)$ and correspond to instances of initial prediction ambiguity, where certain incorrect classes may exhibit disproportionately large logits. Similarly, $\mathcal{E}_4$ corresponds to instances of initial prediction ambiguity, while occurs with probability $\Theta(1/n_y^2)$. Furthermore, we define the following events, which yield negligible gradient contributions to $\mathbf{Q}_{4,3}$ and $\mathbf{Q}_{4,4}$, as they do not lead to significant

confusion among the incorrect predictions:

$$\mathcal{E}_5 = \left\{ y_0 \neq y_1, g_1(y_1) = g_2(y_1), g_1(y_0) \neq g_2(y_0) \right\}, \quad (5.B.86)$$

$$\mathcal{E}_6 = \left\{ y_0 = y_1, g_1(y_1) = g_2(y_1) \right\}. \quad (5.B.87)$$

Here, $\mathcal{E}_5$ occurs with probability $\Theta(1/n_y)$, and $\mathcal{E}_6$ occurs with probability $\Theta(1/n_c^2)$. Together, the events $\cup_{i \in [6]} \mathcal{E}_i$ form a partition of the entire sample space.

Stage 1.2.1: Initial Growth of Q

We define the following notations:

$$\begin{aligned} \epsilon_{\text{attn}}^{L,\ell}(\mathbf{Z}^{L,\ell-1}) &= 1 - \text{Attn}_{\text{ans},\ell-1 \rightarrow \text{pred},\ell}(\mathbf{Z}^{L,\ell-1}) - \text{Attn}_{\text{ans},\ell-1 \rightarrow \text{ans},\ell-1}(\mathbf{Z}^{L,\ell-1}), \\ \Delta^{L,\ell}(\mathbf{Z}^{L,\ell-1}) &= \left| \text{Attn}_{\text{ans},\ell-1 \rightarrow \text{pred},\ell}(\mathbf{Z}^{L,\ell-1}) - \text{Attn}_{\text{ans},\ell-1 \rightarrow \text{ans},\ell-1}(\mathbf{Z}^{L,\ell-1}) \right|. \end{aligned}$$

We abbreviate $\epsilon_{\text{attn}}^{L,\ell}(\mathbf{Z}^{L,\ell-1})$ and $\Delta^{L,\ell}(\mathbf{Z}^{L,\ell-1})$ as $\epsilon_{\text{attn}}^{L,\ell}$ and $\Delta^{L,\ell}$ for simplicity. Since we only focus on the input $\mathbf{Z}^{2,1}$ in the following analysis, we will omit the notation related to the length, i.e., we use ϵ_{attn} and Δ when the context is clear.

Induction 5.B.13. *Given $s \in \tau(X)$, let $T_{1,2,1,s}$ denote the first time that $\mathbb{E}[\epsilon_{\text{attn}} \mid \tau(x_1) = s] \leq 0.4$. For all iterations $t \leq T_{1,2,1,s}$, we have the following holds*

- (a) $[\mathbf{Q}_{4,3}^{(t)}]_{s,s} + [\mathbf{Q}_{4,4}^{(t)}]_{s,s} \leq O(1)$ monotonically increases;
- (b) for $p \in \{3, 4\}$, for $s' \in \tau(X) \neq s$, $[\mathbf{Q}_{4,p}^{(t)}]_{s,s'} \leq O\left(\frac{[\mathbf{Q}_{4,p}^{(t)}]_{s,s'}}{d}\right)$; otherwise $[\mathbf{Q}_{4,p}^{(t)}]_{s,s'} = 0$;
- (c) for any sample $\mathbf{Z}^{2,1}$, we have $\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \geq -O\left(\frac{\log \log d}{\log d}\right)$;
- (d) for any sample $\mathbf{Z}^{2,1}$, we have $\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \leq c_1$ for some small constant $c_1 > 0$.

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Lemma 5.B.91. *If Induction 5.B.13 holds for all iterations $< t$, then we have*

- 1. $\text{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} + \text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \in [0.4 \pm \tilde{O}\left(\frac{1}{d}\right), 0.5]$;
- 2. $|\text{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)}| \leq \tilde{O}\left(\frac{1}{d}\right)$.

Lemma 5.B.92. *If Induction 5.B.13 holds for all iterations $< t$, then given $\mathbf{Z}^{2,1} \in \mathcal{E}_1$,*

- 1. *for the prediction j_2 , we have*

$$\begin{aligned} \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} &= \\ &\left(\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \right) \cdot 2B + \left(\text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \right) \cdot \frac{2B}{n_y} + O(\delta). \end{aligned}$$

2. for the prediction $j'_2 = \tau(g_2(y_0))$, we have

$$\Lambda_{5,j'_2,r_{g_2 \cdot y_0}}^{(t)} = \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \right) \cdot 2B + \tilde{O}\left(\frac{B}{d \cdot n_y}\right) + O(\delta).$$

3. for the prediction $\tau(g_1(y_0))$, we have

$$\Lambda_{5,\tau(g_1(y_0)),r_{g_1 \cdot y_0}}^{(t)} = \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \right) \cdot 2B + \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \right) \cdot \frac{2B}{n_y} + O(\delta).$$

furthermore, we have $\Lambda_{5,\tau(g_1(y_0)),r_{g_1 \cdot y_0}}^{(t)} \leq \Lambda_{5,\tau(g_2(y_1)),r_{g_2 \cdot y_1}}^{(t)}$.

4. for the prediction $\tau(g_1(y_1))$, we have

$$\Lambda_{5,\tau(g_1(y_1)),r_{g_1 \cdot y_1}}^{(t)} = \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \right) \cdot \frac{2B}{n_y} + \tilde{O}\left(\frac{B}{d}\right) + O(\delta).$$

5. for other $j \in \tau(\mathcal{Y})$, if there are j and y , s.t., $g_1, g_2 \in \text{Fiber}_{j,y}$ (notice that $y \neq y_0, y_1$ for $\mathcal{E}_1$), then for such a j , $r \in \widehat{\mathfrak{A}}_j \setminus \{r_{g_2 \cdot y}\}$ cannot be activated, moreover, we have

$$\Lambda_{5,j,r_{g_2 \cdot y}}^{(t)} = \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \right) \cdot 2B + \tilde{O}\left(\frac{B}{d \cdot n_y}\right) + O(\delta);$$

else, none of $r \in \widehat{\mathfrak{A}}_j$ can be activated.

Notice that for $\mathcal{E}_1$, the neurons mentioned in 1-4 should be different neurons, while the predictions in 1 and 3, or 2 and 4 can be the same in some cases, e.g., $g_1(y_1) = g_2(y_0)$. In all cases, except for the neurons mentioned above, i.e., $\cup_{\ell, \ell' \in [2]} \{r_{g_\ell \cdot y_{\ell'-1}}\}$ (which may not be activated), all other neurons $r \in \cup_{\ell, \ell' \in [2]} (\widehat{\mathfrak{A}}_{\tau(g_\ell \cdot y_{\ell'-1})} \setminus \{r_{g_\ell \cdot y_{\ell'-1}}\})$ cannot be activated.

Lemma 5.B.93. If Induction 5.B.13 holds for all iterations $< t$, then given $\mathbf{Z}^{2,1} \in \mathcal{E}_2$, we have

1. for the prediction j_2 , $r \in \widehat{\mathfrak{A}}_{j_2} \setminus \{r_{g_2 \cdot y_1}\}$ cannot be activated, moreover, we have

$$\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} = \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \right) \cdot 2B + \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \right) \cdot \frac{2B}{n_y} + O(\delta).$$

2. for the prediction $j'_2 = \tau(g_2(y_0))$, $r \in \widehat{\mathfrak{A}}_{j'_2} \setminus \{r_{g_2 \cdot y_0}\}$ (notice that in this case $r_{g_2 \cdot y_0} = r_{g_1 \cdot y_0}$) cannot be activated, moreover, we have

$$\Lambda_{5,j'_2,r_{g_2 \cdot y_0}}^{(t)} = \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \right) \cdot 2B + O(\delta).$$

3. for the prediction $\tau(g_1(y_1))$, $r \in \widehat{\mathfrak{A}}_{\tau(g_1(y_1))} \setminus \{r_{g_1 \cdot y_1}\}$ cannot be activated, moreover, we have

$$\Lambda_{5, \tau(g_1(y_1)), r_{g_1 \cdot y_1}}^{(t)} = \left(\text{Attn}_{\text{ans}, 1 \rightarrow \text{ans}, 1}^{(t)} - \text{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, 2}^{(t)} \right) \cdot \frac{2B}{n_y} + O(\delta).$$

4. for other $j \in \tau(\mathcal{Y})$, if there are j and y , s.t., $g_1, g_2 \in \text{Fiber}_{j, y}$ (notice that $y \neq y_0, y_1$ for $\mathcal{E}_2$), then for such a j , $r \in \widehat{\mathfrak{A}}_j \setminus \{r_{g_2 \cdot y}\}$ cannot be activated, moreover, we have

$$\Lambda_{5, j, r_{g_2 \cdot y}}^{(t)} = \left(\text{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, 2}^{(t)} - \text{Attn}_{\text{ans}, 1 \rightarrow \text{ans}, 1}^{(t)} \right) \cdot 2B + \widetilde{O}\left(\frac{B}{d \cdot n_y}\right) + O(\delta);$$

else, none of $r \in \widehat{\mathfrak{A}}_j$ can be activated.

Lemma 5.B.94. If Induction 5.B.13 holds for all iterations $< t$, then given $\mathbf{Z}^{2,1} \in \mathcal{E}_3$, we have

1. for the prediction j_2 , $r \in \widehat{\mathfrak{A}}_{j_2} \setminus \{r_{g_2 \cdot y_1}\}$ cannot be activated, moreover, we have

$$\Lambda_{5, j_2, r_{g_2 \cdot y_1}}^{(t)} = \text{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, 2}^{(t)} \cdot 2B \pm O(\delta).$$

2. for the prediction $\tau(g_1(y_1))$, $r \in \widehat{\mathfrak{A}}_{\tau(g_1(y_1))} \setminus \{r_{g_1 \cdot y_1}\}$ cannot be activated, moreover, we have

$$\Lambda_{5, \tau(g_1(y_1)), r_{g_1 \cdot y_1}}^{(t)} = \text{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, 1}^{(t)} \cdot 2B \pm O(\delta).$$

3. for other $j = g(y_1)$, where $g_1, g_2 \notin \text{Fiber}_{j, y_1}$, we have

$$\Lambda_{5, j, r_{g \cdot y_1}}^{(t)} = \left(\text{Attn}_{\text{ans}, 1 \rightarrow \text{ans}, 1}^{(t)} - \text{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, 2}^{(t)} \right) \cdot \frac{2B}{n_y} + \widetilde{O}\left(\frac{B}{d \cdot n_y}\right) + O(\delta);$$

moreover, if there exists y , s.t., $g_1, g_2 \in \text{Fiber}_{j, y}$ (notice that $y \neq y_1$ for $\mathcal{E}_3$), we have

$$\Lambda_{5, j, r_{g_2 \cdot y}}^{(t)} = \left(\text{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, 2}^{(t)} - \text{Attn}_{\text{ans}, 1 \rightarrow \text{ans}, 1}^{(t)} \right) \cdot 2B + \widetilde{O}\left(\frac{B}{d \cdot n_y}\right) + O(\delta);$$

besides, other $r \in \widehat{\mathfrak{A}}_j$ cannot be activated.

Lemma 5.B.95. If Induction 5.B.13 holds for all iterations $< t$, then given $\mathbf{Z}^{2,1} \in \mathcal{E}_4$, we have

1. for the prediction j_2 , $r \in \widehat{\mathfrak{A}}_{j_2} \setminus \{r_{g_2 \cdot y_1}\}$ cannot be activated, moreover, we have

$$\Lambda_{5, j_2, r_{g_2 \cdot y_1}}^{(t)} = \text{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, 2}^{(t)} \cdot 2B + O(\delta).$$

2. for the prediction $\tau(g_2(y_0))$, $r \in \widehat{\mathfrak{A}}_{\tau(g_2(y_0))} \setminus \{r_{g_2 \cdot y_0}\}$ cannot be activated, moreover, we have

$$\Lambda_{5, j_2', r_{g_2 \cdot y_0}}^{(t)} = \left(\text{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, 2}^{(t)} + \text{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, 1}^{(t)} - \text{Attn}_{\text{ans}, 1 \rightarrow \text{ans}, 1}^{(t)} \right) \cdot 2B + O(\delta).$$

3. for other $j \in \tau(\mathcal{Y})$, if there exist j and y , s.t., $g_1, g_2 \in \text{Fiber}_{j,y}$, then for such a j , $r \in \widehat{\mathfrak{U}}_j \setminus \{r_{g_2 \cdot y}\}$ cannot be activated, moreover, we have

$$\Lambda_{5,j,r_{g_2 \cdot y}}^{(t)} = \left(\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \right) \cdot 2B + \widetilde{O}\left(\frac{B}{d \cdot n_y}\right) + O(\delta);$$

else, none of $r \in \widehat{\mathfrak{U}}_j$ can be activated.

Lemma 5.B.96. If Induction 5.B.13 holds for all iterations $< t$, then given $\mathbf{Z}^{2,1} \in \mathcal{E}_5$, we have

1. for the prediction j_2 , $r \in \widehat{\mathfrak{U}}_{j_2} \setminus \{r_{g_2 \cdot y_1}\}$ cannot be activated, moreover, we have

$$\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} = \text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \cdot 2B \pm O(\delta).$$

2. for the prediction $\tau(g_2(y_0))$, $r \in \widehat{\mathfrak{U}}_{\tau(g_2(y_0))} \setminus \{r_{g_2 \cdot y_0}\}$ cannot be activated, moreover, we have

$$\Lambda_{5,j'_2,r_{g_2 \cdot y_0}}^{(t)} = \left(\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \right) \cdot 2B + \widetilde{O}\left(\frac{B}{d \cdot n_y}\right) + O(\delta).$$

3. for the prediction $\tau(g_1(y_0))$, $r \in \widehat{\mathfrak{U}}_{\tau(g_2(y_0))} \setminus \{r_{g_1 \cdot y_0}\}$ cannot be activated, moreover, we have

$$\begin{aligned} \Lambda_{5,\tau(g_1(y_0)),r_{g_1 \cdot y_0}}^{(t)} &= \left(\text{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \right) \cdot 2B \\ &+ \left(\text{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \right) \cdot \frac{2B}{n_y} + O(\delta). \end{aligned}$$

4. for other $j \in \tau(\mathcal{Y})$, if there are j and y , s.t., $g_1, g_2 \in \text{Fiber}_{j,y}$, then for such a j , $r \in \widehat{\mathfrak{U}}_j \setminus \{r_{g_2 \cdot y}\}$ cannot be activated, moreover, we have

$$\Lambda_{5,j,r_{g_2 \cdot y}}^{(t)} = \left(\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \right) \cdot 2B + \widetilde{O}\left(\frac{B}{d \cdot n_y}\right) + O(\delta);$$

else, none of $r \in \widehat{\mathfrak{U}}_j$ can be activated.

Lemma 5.B.97. If Induction 5.B.13 holds for all iterations $< t$, then given $\mathbf{Z}^{2,1} \in \mathcal{E}_6$, we have

1. for the prediction j_2 , $r \in \widehat{\mathfrak{U}}_{j_2} \setminus \{r_{g_2 \cdot y_1}\}$ cannot be activated, moreover, we have

$$\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} = \left(\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} + \text{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \right) \cdot 2B \pm O(\delta).$$

2. for other $j = g(y_1)$, where $g_2 \notin \text{Fiber}_{j,y_1}$, we have

$$\Lambda_{5,j,r_{g \cdot y_1}}^{(t)} = \left(\text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \right) \cdot \frac{2B}{n_y} + \widetilde{O}\left(\frac{B}{d \cdot n_y}\right) + O(\delta);$$

moreover, if there exists $y \neq y_1$, s.t., $g_1, g_2 \in \text{Fiber}_{j,y}$, we have

$$\Lambda_{5,j,r_{g_2 \cdot y}}^{(t)} = \left(\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \right) \cdot 2B + \widetilde{O}\left(\frac{B}{d \cdot n_y}\right) + O(\delta);$$

besides, other $r \in \widehat{\mathfrak{A}}_j$ cannot be activated.

Gradient Lemma

Lemma 5.B.98. *If Induction 5.B.13 holds for all iterations $< t$, given $s \in \tau(\mathcal{X})$, we have*

$$\begin{aligned} & \left[-\nabla_{\mathbf{Q}_{4,3}^{(t)}} \text{Loss}_5^{2,2} \right]_{s,s} + \left[-\nabla_{\mathbf{Q}_{4,4}^{(t)}} \text{Loss}_5^{2,2} \right]_{s,s} \\ & \geq \min \left\{ \Omega\left(\frac{1}{d \cdot n_y}\right), \Omega\left(\frac{B}{d}\right) \cdot \mathbb{E} \left[(1 - \text{logit}_{5,j_2}^{(t)}) | \tau(x_1) = s, \mathcal{E}_1 \right] \right\} > 0. \end{aligned}$$

Proof. By Lemma 5.B.89 and Lemma 5.B.90, we have $\left[-\nabla_{\mathbf{Q}_{4,3}^{(t)}} \text{Loss}_5^{2,2} \right]_{s,s} + \left[-\nabla_{\mathbf{Q}_{4,4}^{(t)}} \text{Loss}_5^{2,2} \right]_{s,s} = \mathbb{E}[\mathcal{N}_{s,2,i} + \mathcal{N}_{s,2,ii} + \mathcal{N}_{s,2,iii}]$. Based on Lemma 5.B.92-Lemma 5.B.97, we can first directly bound the term $\mathcal{N}_{s,2,iii}^{(t)}$ as follows:

$$\mathbb{E} \left[\mathcal{N}_{s,2,iii}^{(t)} \right] \leq \widetilde{O}(\sigma_0^q) = \frac{1}{\text{poly}d}.$$

In the following discussion, we focus on $\mathcal{N}_{s,2,i}^{(t)}$ and $\mathcal{N}_{s,2,ii}^{(t)}$, and consider two regimes for the gradient lower bound: (i) $\mathbb{E}[\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} | \tau(x_1) = s] \leq \frac{\rho}{B}$, and (ii) $\mathbb{E}[\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} | \tau(x_1) = s] \geq \frac{\rho}{B}$. The proof strategy is to analyze $\mathcal{N}_{s,2,i}^{(t)}$ and $\mathcal{N}_{s,2,ii}^{(t)}$ under different event conditions in two regimes.

1. For regime (i),

(a) For $\mathbf{Z}^{2,1} \in \mathcal{E}_1$, by Induction 5.B.13 and Lemma 5.B.92, we have an immediate logit upper bound for $j \in \tau(\mathcal{Y})$: $\text{logit}_{5,j}^{(t)} \leq O(\frac{1}{d})$. Then by Lemma 5.B.89, we have

$$\begin{aligned} & \mathbb{E} \left[\mathcal{N}_{s,2,i}^{(t)} \mathbb{1}_{\mathcal{E}_1} \right] \\ & = \mathbb{E} \left[(1 - \text{logit}_{5,j_2}^{(t)}) \cdot \left(\sum_{r \in \widehat{\mathfrak{A}}_{j_2}} \text{sReLU}'(\Lambda_{5,j_2,r}^{(t)}) \cdot \right. \right. \\ & \quad \left. \left(-\text{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j_2,r}(y_0) - \text{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j_2,r}(g_1) \right. \right. \\ & \quad \left. \left. + (1 - \text{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1}^{(t)} - \text{Attn}_{\mathbb{A},1 \rightarrow 2}^{(t)}) \Lambda_{5,j_2,r}^{(t)} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_1} \right]. \end{aligned}$$

By Lemma 5.B.92, we can obtain

$$\begin{aligned} & \mathbb{E} \left[(1 - \text{logit}_{5,j_2}^{(t)}) \cdot \left(\text{sReLU}'(\Lambda_{5,j_2,r_{s_2 y_1}}^{(t)}) \cdot \right. \right. \\ & \quad \left. \left(-\text{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j_2,r_{s_2 y_1}}(y_0) - \text{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j_2,r_{s_2 y_1}}(g_1) \right. \right. \\ & \quad \left. \left. + (1 - \text{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1}^{(t)} - \text{Attn}_{\mathbb{A},1 \rightarrow 2}^{(t)}) \Lambda_{5,j_2,r_{s_2 y_1}}^{(t)} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_1} \right] \end{aligned}$$

$$\geq \Omega\left(\frac{B}{d}\right) \cdot \mathbb{E}\left[\mathbf{sReLU}'(\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)}) \middle| \tau(x_1) = s, \mathcal{E}_1\right]. \quad (5.B.88)$$

On the other hand,

$$\begin{aligned} & - \mathbb{E}\left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) \cdot \mathbf{sReLU}'(\Lambda_{5,j_2,r_{g_1 \cdot y_0}}^{(t)}) \cdot \right. \\ & \quad \left. \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j_2,r_{g_1 \cdot y_0}}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j_2,r_{g_1 \cdot y_0}}(g_1)\right) \right. \\ & \quad \left. \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_1} \mathbb{1}_{g_1(y_0)=g_2(y_1)}\right] \\ & \geq -O\left(\frac{B}{d \cdot n_y}\right) \cdot \mathbb{E}\left[\mathbf{sReLU}'(\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)}) \middle| \tau(x_1) = s, \mathcal{E}_1\right], \end{aligned}$$

which implies that $\mathbb{E}\left[\mathcal{N}_{s,2,i}^{(t)} \mathbb{1}_{\mathcal{E}_1}\right] \geq \Omega\left(\frac{B}{d}\right) \cdot \mathbb{E}\left[\mathbf{sReLU}'(\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)}) \middle| \tau(x_1) = s, \mathcal{E}_1\right] \geq 0$.

Moving to $\mathcal{N}_{s,2,ii}^{(t)}$, by Lemma 5.B.89 and Lemma 5.B.92, we have

$$\begin{aligned} & \mathbb{E}\left[\mathcal{N}_{s,2,ii}^{(t)} \mathbb{1}_{\mathcal{E}_1}\right] \\ & = \mathbb{E}\left[\sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j}^{(t)} \cdot \left(\sum_{r \in \mathfrak{A}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}^{(t)}) \cdot \right. \right. \\ & \quad \left. \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j,r}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j,r}(g_1)\right) \right. \\ & \quad \left. - (1 - \mathbf{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1}^{(t)} - \mathbf{Attn}_{\mathbb{A},1 \rightarrow 2}^{(t)}) \Lambda_{5,j,r}^{(t)} \pm \tilde{O}(\sigma_0) \pm \tilde{O}(\delta^q)\right) \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_1}\right] \\ & \geq -O\left(\frac{n_y \cdot B}{d^2}\right). \end{aligned}$$

- (b) For $\mathbf{Z}^{2,1} \in \mathcal{E}_2$, by Induction 5.B.13 and Lemma 5.B.93, we can also obtain a crude bound of logit: $\mathbf{logit}_{5,j}^{(t)} \leq O\left(\frac{1}{d}\right)$ for $j \neq j'_2$. Then $\mathcal{N}_{s,2,i}^{(t)}$ can be bounded in the same way as (5.B.88), and we have

$$\mathbb{E}\left[\mathcal{N}_{s,2,i}^{(t)} \mathbb{1}_{\mathcal{E}_2}\right] \geq \Omega\left(\frac{B}{d \cdot n_y}\right) \cdot \mathbb{E}\left[\mathbf{sReLU}'(\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)}) \middle| \tau(x_1) = s, \mathcal{E}_2\right] \geq 0.$$

Moving to $\mathcal{N}_{s,2,ii'}^{(t)}$

- for $j = \tau(g_1(y_1))$, we have

$$\begin{aligned} & = \mathbb{E}\left[\mathbf{logit}_{5,j}^{(t)} \left(\mathbf{sReLU}'(\Lambda_{5,j,r_{g_1 \cdot y_1}}^{(t)}) \cdot \right. \right. \\ & \quad \left. \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j,r_{g_1 \cdot y_1}}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j,r_{g_1 \cdot y_1}}(g_1)\right) \right. \end{aligned}$$

$$\begin{aligned}
& - \left(1 - \mathbf{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1}^{(t)} - \mathbf{Attn}_{\mathbb{A},1 \rightarrow 2}^{(t)} \right) \Lambda_{5,j,r}^{(t)} \pm \tilde{O}(\sigma_0) \Big) \pm \tilde{O}(\delta^q) \Big) \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_2} \Big] \\
& \geq -O\left(\frac{B}{n_y^2 \cdot d^2}\right).
\end{aligned}$$

where the last inequality is due to the cancellation of the term $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j,r_{\mathcal{S}_1 y_1}}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j,r_{\mathcal{S}_1 y_1}}(g_1)$ and the fact that

$$\Lambda_{5,\tau(g_1(y_1)),r_{\mathcal{S}_1 y_1}}^{(t)} = \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \right) \cdot \frac{2B}{n_y} + O(\delta) \leq O\left(\frac{B}{n_y}\right).$$

- for $j = \tau(g_2(y))$ if there exists $y \neq y_0, y_1$ s.t., $g_1(y) = g_2(y)$
 - when $\mathbb{E}[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \mid \tau(x_1) = s] \leq 0.01$, by Induction 5.B.13 and Lemma 5.B.93, we have $\mathbf{logit}_{5,\tau(g_2(y))}^{(t)} \ll O\left(\frac{1}{d}\right) \cdot \mathbf{logit}_{5,j'_2}^{(t)}$ and $\mathbf{logit}_{5,j'_2}^{(t)} = \Omega(1)$. Then

$$\begin{aligned}
& \left| \mathbb{E} \left[\mathbf{logit}_{5,j}^{(t)} \cdot \left(\mathbf{sReLU}'(\Lambda_{5,j,r_{\mathcal{S}_1 y}}^{(t)}) \cdot \right. \right. \right. \\
& \quad \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j,r_{\mathcal{S}_1 y}}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j,r_{\mathcal{S}_1 y}}(g_1) \right. \\
& \quad \left. \left. \left. - (1 - \mathbf{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1}^{(t)} - \mathbf{Attn}_{\mathbb{A},1 \rightarrow 2}^{(t)}) \Lambda_{5,j,r_{\mathcal{S}_1 y}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_2} \right] \Big| \\
& \leq O\left(\frac{B}{n_y \cdot d^2}\right).
\end{aligned}$$

On the other hand,

$$\begin{aligned}
& \mathbb{E} \left[\mathbf{logit}_{5,j'_2}^{(t)} \cdot \left(\mathbf{sReLU}'(\Lambda_{5,j'_2,r_{\mathcal{S}_1 y_0}}^{(t)}) \cdot \right. \right. \\
& \quad \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j'_2,r_{\mathcal{S}_1 y_0}}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j'_2,r_{\mathcal{S}_1 y_0}}(g_1) \right. \\
& \quad \left. \left. - (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5,j'_2,r_{\mathcal{S}_1 y_0}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \right. \\
& \quad \left. \left. \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_1} \right] \\
& \geq \Omega\left(\frac{B}{n_y \cdot d}\right),
\end{aligned}$$

where the last inequality follows from the fact that

$$\begin{aligned}
& \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j'_2,r_{\mathcal{S}_1 y_0}}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j'_2,r_{\mathcal{S}_1 y_0}}(g_1) \\
& - (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5,j'_2,r_{\mathcal{S}_1 y_0}}^{(t)}
\end{aligned}$$

$$= \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot \left(2B - 2\Lambda_{5,j'_2,r_{g_1 \cdot y_0}}^{(t)} \right) \pm O(\delta) \geq \Omega(B).$$

- when $\mathbb{E}[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \mid \tau(x_1) = s] \geq 0.01$, by Lemma 5.B.93, we have $\Lambda_{5,j,r_{g_1 \cdot y}}^{(t)}$ cannot be activated. Furthermore,

$$\begin{aligned} & \mathbb{E} \left[\mathbf{logit}_{5,j'_2}^{(t)} \cdot \left(\mathbf{sReLU}'(\Lambda_{5,j'_2,r_{g_1 \cdot y_0}}^{(t)}) \cdot \right. \right. \\ & \quad \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j'_2,r_{g_1 \cdot y_0}}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j'_2,r_{g_1 \cdot y_0}}(g_1) \right. \\ & \quad \left. \left. - (1 - \mathbf{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1}^{(t)} - \mathbf{Attn}_{\mathbb{A},1 \rightarrow 2}^{(t)}) \Lambda_{5,j'_2,r_{g_1 \cdot y_0}}^{(t)} \pm \widetilde{O}(\sigma_0) \pm \widetilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_1} \right] \\ & \geq 0. \end{aligned}$$

Putting the above discussion together, we have

$$\mathbb{E} \left[\mathcal{N}_{s,2,ii}^{(t)} \mathbb{1}_{\mathcal{E}_2} \right] \geq -O\left(\frac{B}{n_y^2 \cdot d^2}\right).$$

- (c) For $\mathbf{Z}^{2,1} \in \mathcal{E}_3$, by Induction 5.B.13 and Lemma 5.B.94, we can first have some facts of $\mathbf{logit}$:

$$\begin{aligned} 1 - \mathbf{logit}_{5,j_2}^{(t)} &= \Omega(1), \quad \mathbf{logit}_{5,\tau(g_1(y_1))}^{(t)} = \Omega(1) \\ \mathbf{logit}_{5,j}^{(t)} &\leq \frac{1}{\text{poly}d} \text{ for other } j. \end{aligned}$$

Therefore, we have

$$\begin{aligned} & \mathbb{E} \left[\mathcal{N}_{s,2,i}^{(t)} \mathbb{1}_{\mathcal{E}_3} \right] \\ &= \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) \cdot \left(-\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j_2,r_{g_2 \cdot y_1}}(y_1) - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j_2,r_{g_2 \cdot y_1}}(g_1) \right. \right. \\ & \quad \left. \left. + (1 - \mathbf{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1}^{(t)} - \mathbf{Attn}_{\mathbb{A},1 \rightarrow 2}^{(t)}) \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} \pm \widetilde{O}(\sigma_0) \pm \widetilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_3} \right] \\ &\geq O\left(\frac{B}{n_y \cdot d}\right). \end{aligned}$$

Moving to $\mathcal{N}_{s,2,ii}^{(t)}$, we have

$$\begin{aligned} & \mathbb{E} \left[\mathcal{N}_{s,2,ii}^{(t)} \mathbb{1}_{\mathcal{E}_3} \right] \\ &\geq \mathbb{E} \left[\mathbf{logit}_{5,\tau(g_1(y_1))}^{(t)} \right]. \end{aligned}$$

$$\begin{aligned}
& \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{\tau(g_1(y_1)), r_{g_1 \cdot y_1}}(y_1) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{\tau(g_1(y_1)), r}(g_1) \right. \\
& - (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5, \tau(g_1(y_1)), r_{g_1 \cdot y_1}}^{(t)} \pm \tilde{O}(\sigma_0) \\
& \left. \pm \tilde{O}(\delta^q) \right) 1_{\tau(x_1)=s} 1_{\mathcal{E}_3} \Big] - O\left(\frac{n_y \cdot B}{d \cdot \text{poly}d}\right) \\
& \geq \Omega\left(\frac{B}{n_y \cdot d}\right) - O\left(\frac{B}{d \cdot \text{poly}d}\right) = \Omega\left(\frac{B}{n_y \cdot d}\right).
\end{aligned}$$

- (d) For $\mathbf{Z}^{2,1} \in \mathcal{E}_4$, by comparing Lemma 5.B.95 with Lemma 5.B.93, we can directly bound $\mathcal{N}_{s,2,ii}^{(t)}$ in the same way as $\mathcal{E}_2$, where the only difference is that we do not need to consider $\tau(g_1(y_1))$ for $\mathcal{E}_4$. Thus $\mathbb{E}\left[\mathcal{N}_{s,2,ii}^{(t)} 1_{\mathcal{E}_4}\right] \geq 0$. Moreover, it is clear that

$$\begin{aligned}
& \mathbb{E}\left[\mathcal{N}_{s,2,i}^{(t)} 1_{\mathcal{E}_4}\right] \\
& = \mathbb{E}\left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) \cdot \left(\mathbf{sReLU}'(\Lambda_{5,j_2, r_{g_2 \cdot y_1}}^{(t)}) \cdot \right. \right. \\
& \quad \left. \left(- \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j_2, r_{g_2 \cdot y_1}}(y_0) - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j_2, r_{g_2 \cdot y_1}}(g_1) \right. \right. \\
& \quad \left. \left. + (1 - \mathbf{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1}^{(t)} - \mathbf{Attn}_{\mathbb{A},1 \rightarrow 2}^{(t)}) \Lambda_{5,j_2, r_{g_2 \cdot y_1}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right] 1_{\tau(x_1)=s} 1_{\mathcal{E}_4} \Big] \\
& \geq 0.
\end{aligned}$$

where the last inequality is due to the cancellation of $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j_2, r_{g_2 \cdot y_1}}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j_2, r_{g_2 \cdot y_1}}(g_1)$, and the fact that

$$\Lambda_{5,j_2, r_{g_2 \cdot y_1}}^{(t)} = \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \cdot 2B + O(\delta) \geq \Omega(B).$$

- (e) For $\mathbf{Z}^{2,1} \in \mathcal{E}_5 \cup \mathcal{E}_6$, by Induction 5.B.13, Lemma 5.B.96 and Lemma 5.B.97, we can derive the following logit condition: $1 - \mathbf{logit}_{5,j_2}^{(t)}, \mathbf{logit}_{5,j}^{(t)} \leq \frac{1}{\text{poly}d}$ for $j \neq j_2$. Hence, we can simply bound $\mathcal{N}_{s,2,i}^{(t)}$ and $\mathcal{N}_{s,2,ii}^{(t)}$ as follows:

$$\left| \mathbb{E}\left[\mathcal{N}_{s,2,i}^{(t)} 1_{\mathcal{E}_m}\right] \right|, \left| \mathbb{E}\left[\mathcal{N}_{s,2,ii}^{(t)} 1_{\mathcal{E}_m}\right] \right| \leq O\left(\frac{B}{dn_y} \cdot \frac{1}{\text{poly}d}\right) \text{ for } m \in \{5, 6\}.$$

Putting it all together, we have for the regmie (1),

$$\left[-\nabla_{\mathbf{Q}_{4,3}^{(t)}} \mathbb{L}_5^{2,2} \right]_{s,s} + \left[-\nabla_{\mathbf{Q}_{4,4}^{(t)}} \mathbb{L}_5^{2,2} \right]_{s,s} = \sum_{\kappa \in \{i, ii, iii\}} \sum_{i \in [6]} \mathbb{E}\left[\mathcal{N}_{s,2,\kappa}^{(t)} 1_{\mathcal{E}_i}\right] \geq \Omega\left(\frac{B}{d \cdot n_y}\right).$$

2. In regime (ii), the analysis follows a structure analogous to that of regime (i). The primary distinction lies in the fact that, under the main event $\mathcal{E}_1$, the term $\Lambda_{5,j_2, r_{g_2 \cdot y_1}}^{(t)}$ is guaranteed

to remain within the linear regime, thereby serving as the dominant component driving the overall gradient.

(a) For $\mathbf{Z}^{2,1} \in \mathcal{E}_1$, by Induction 5.B.13 and Lemma 5.B.92, we have

$$\begin{aligned} & \mathbb{E} \left[\mathcal{N}_{s,2,i}^{(t)} \mathbb{1}_{\mathcal{E}_1} \right] \\ &= \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) \cdot \left(\sum_{r \in \widehat{\mathfrak{I}}_{j_2}} \mathbf{sReLU}'(\Lambda_{5,j_2,r}^{(t)}) \cdot \right. \right. \\ & \quad \left. \left(-\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j_2,r}(y_0) - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j_2,r}(g_1) \right. \right. \\ & \quad \left. \left. + (1 - \mathbf{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1}^{(t)} - \mathbf{Attn}_{\mathbb{A},1 \rightarrow 2}^{(t)}) \Lambda_{5,j_2,r}^{(t)} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_1} \right]. \end{aligned}$$

By Lemma 5.B.92, we can obtain

$$\begin{aligned} & \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) \cdot \left(\left(-\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j_2,r_{g_2 \cdot y_1}}(y_0) - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j_2,r_{g_2 \cdot y_1}}(g_1) \right. \right. \right. \\ & \quad \left. \left. + (1 - \mathbf{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1}^{(t)} - \mathbf{Attn}_{\mathbb{A},1 \rightarrow 2}^{(t)}) \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_1} \right] \quad (5.B.89) \\ & \geq \Omega\left(\frac{B}{d}\right) \cdot \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) \middle| \tau(x_1) = s, \mathcal{E}_1 \right]. \end{aligned}$$

On the other hand,

$$\begin{aligned} & -\mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) \cdot \mathbf{sReLU}'(\Lambda_{5,j_2,r_{g_1 \cdot y_0}}^{(t)}) \cdot \right. \\ & \quad \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j_2,r_{g_1 \cdot y_0}}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j_2,r_{g_1 \cdot y_0}}(g_1) \right) \\ & \quad \left. \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_1} \mathbb{1}_{g_1(y_0)=g_2(y_1)} \right] \\ & \geq -O\left(\frac{B}{d \cdot n_y}\right) \cdot \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) \middle| \tau(x_1) = s, \mathcal{E}_1 \right], \end{aligned}$$

which implies that $\mathbb{E} \left[\mathcal{N}_{s,2,i}^{(t)} \mathbb{1}_{\mathcal{E}_1} \right] \geq \Omega\left(\frac{B}{d}\right) \cdot \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) \middle| \tau(x_1) = s, \mathcal{E}_1 \right] \geq 0$. Moving

to $\mathcal{N}_{s,2,iii}^{(t)}$, by Lemma 5.B.92, we have

- for $j = \tau(g_1(y_0))$, $r = r_{g_1 \cdot y_0}$

$$\begin{aligned} & \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j,r}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j,r}(g_1) \right. \\ & \quad \left. - (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5,j,r}^{(t)} \pm \widetilde{O}(\sigma_0) \right) \geq 0. \end{aligned}$$

- For $j = \tau(g_1(y_1))$ and $r = r_{g_1 \cdot y_1}$, the terms

$$\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j,r}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j,r}(g_1)$$

cancel, so

$$\begin{aligned} & \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j,r}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j,r}(g_1) \right. \\ & \quad \left. - (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5,j,r}^{(t)} \pm \tilde{O}(\sigma_0) \right) \\ & \geq -(1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5,j,r}^{(t)}. \end{aligned}$$

- for $j = \tau(g_2(y_0))$, $r = r_{g_2 \cdot y_0}$

$$\begin{aligned} & \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j,r}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j,r}(g_1) \right. \\ & \quad \left. - (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5,j,r}^{(t)} \pm \tilde{O}(\sigma_0) \right) \\ & \geq (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5,j,r}^{(t)}. \end{aligned}$$

- for $j = g_1(y)$, where $\exists y \neq y_0, y_1$, s.t., $g_2(y) = g_1(y)$, we have

$$\begin{aligned} & \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j,r}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j,r}(g_1) \right. \\ & \quad \left. - (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5,j,r}^{(t)} \pm \tilde{O}(\sigma_0) \right) \\ & \geq -(1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5,j,r}^{(t)}. \end{aligned}$$

Putting them together, and upper bound $\sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j}^{(t)}$ by $1 - \mathbf{logit}_{5,j_2}^{(t)}$, we have

$$\begin{aligned} & \mathbb{E} \left[\mathcal{N}_{s,2,ii}^{(t)} \mathbb{1}_{\mathcal{E}_1} \right] \\ & = \mathbb{E} \left[\sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j}^{(t)} \cdot \left(\sum_{r \in \mathfrak{R}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \right. \right. \\ & \quad \left. \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j,r}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j,r}(g_1) \right. \right. \\ & \quad \left. \left. - (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5,j,r}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_1} \right] \\ & \geq -\mathbb{E} \left[\left(1 - \mathbf{logit}_{5,j_2}^{(t)} \right) \cdot (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \right. \\ & \quad \cdot \max_{y \neq y_0, y_1} \left\{ \Lambda_{5,\tau(g_1(y_1)),r_{g_1 \cdot y_1}}^{(t)}, \Lambda_{5,\tau(g_2(y_0)),r_{g_2 \cdot y_0}}^{(t)} \right\} \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_1} \right], \end{aligned}$$

where the last inequality is due to the fact that $\Lambda_{5,\tau(g_2(y_0)),r_{g_2 \cdot y_0}}^{(t)} = \Lambda_{5,\tau(g_2(y)),r_{g_2 \cdot y}}^{(t)} \pm \tilde{O}(\frac{B}{d \cdot n_y})$ for $y \neq y_0, y_1$ s.t., $g_1(y) = g_2(y)$. Notice that

$$\max_{y \neq y_0, y_1} \left\{ \Lambda_{5,\tau(g_1(y_1)),r_{g_1 \cdot y_1}}^{(t)}, \Lambda_{5,\tau(g_2(y_0)),r_{g_2 \cdot y_0}}^{(t)} \right\} \leq \max \left\{ \text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)}, \Theta\left(\frac{1}{n_y}\right) \right\} 2B,$$

while

$$\begin{aligned} (5.B.89) &\geq \mathbb{E} \left[\left(1 - \text{logit}_{5,j_2}^{(t)} \right) \cdot \text{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot 2B \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_1} \right] \\ &= \frac{1}{2} \mathbb{E} \left[\left(1 - \text{logit}_{5,j_2}^{(t)} \right) \cdot \left(1 - \text{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1}^{(t)} - \text{Attn}_{\mathbb{A},1 \rightarrow 2}^{(t)} \right) 2B \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_1} \right]. \end{aligned}$$

Since by Induction 5.B.13, $\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \leq c_1 \ll \frac{1}{2}$, thus we have

$$\mathbb{E} \left[\mathcal{N}_{s,2,i}^{(t)} \mathbb{1}_{\mathcal{E}_1} \right] + \mathbb{E} \left[\mathcal{N}_{s,2,ii}^{(t)} \mathbb{1}_{\mathcal{E}_1} \right] \geq \Omega\left(\frac{B}{d}\right) \cdot \mathbb{E} \left[\left(1 - \text{logit}_{5,j_2}^{(t)} \right) \mathbb{1}_{\tau(x_1)=s, \mathcal{E}_1} \right].$$

(b) For $\mathbf{Z}^{2,1} \in \mathcal{E}_2$, $\mathcal{N}_{s,2,i}^{(t)}$ can be bounded in the same way as (5.B.89), and we have

$$\mathbb{E} \left[\mathcal{N}_{s,2,i}^{(t)} \mathbb{1}_{\mathcal{E}_2} \right] \geq \Omega\left(\frac{B}{d \cdot n_y}\right) \cdot \mathbb{E} \left[\left(1 - \text{logit}_{5,j_2}^{(t)} \right) \mathbb{1}_{\tau(x_1)=s, \mathcal{E}_2} \right] \geq 0.$$

Moving to $\mathcal{N}_{s,2,ii'}^{(t)}$,

- for $j = \tau(g_1(y_1))$, we have

$$\begin{aligned} &= \mathbb{E} \left[\text{logit}_{5,j}^{(t)} \left(\text{sReLU}'(\Lambda_{5,j,r_{g_1 \cdot y_1}}) \cdot \right. \right. \\ &\quad \left(\text{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j,r_{g_1 \cdot y_1}}(y_0) + \text{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j,r_{g_1 \cdot y_1}}(g_1) \right. \\ &\quad \left. \left. - \left(1 - \text{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1}^{(t)} - \text{Attn}_{\mathbb{A},1 \rightarrow 2}^{(t)} \right) \Lambda_{5,j,r}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_2} \right] \\ &\geq -O\left(\frac{B}{n_y^2 \cdot d}\right) \cdot \mathbb{E} \left[\left(1 - \text{logit}_{5,j_2}^{(t)} \right) \mathbb{1}_{\tau(x_1)=s, \mathcal{E}_2} \right]. \end{aligned} \tag{5.B.90}$$

where the last inequality is due to the cancellation of the term $\text{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j,r_{g_1 \cdot y_1}}(y_0) + \text{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j,r_{g_1 \cdot y_1}}(g_1)$ and the fact that

$$\Lambda_{5,\tau(g_1(y_1)),r_{g_1 \cdot y_1}}^{(t)} = \left(\text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \right) \cdot \frac{2B}{n_y} + O(\delta) \leq O\left(\frac{B}{n_y}\right).$$

- for $j'_2 = \tau(g_2(y_0)) = \tau(g_1(y_0))$, clearly, we have

$$\begin{aligned} & \mathbb{E} \left[\mathbf{logit}_{5,j'_2}^{(t)} \cdot \left(\mathbf{sReLU}'(\Lambda_{5,j'_2,r_{g_1 y_0}}^{(t)}) \cdot \right. \right. \\ & \quad \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j'_2,r_{g_1 y_0}}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j'_2,r_{g_1 y_0}}(g_1) \right. \\ & \quad \left. \left. - (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5,j'_2,r_{g_1 y_0}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \\ & \quad \left. \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_2} \right] \geq 0, \end{aligned}$$

where the last inequality is due to the fact that

$$\begin{aligned} & \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j'_2,r_{g_1 y_0}}(g_1) - (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \\ & \quad - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5,j'_2,r_{g_1 y_0}}^{(t)} \\ & \geq \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \\ & \quad \cdot \left(1 - 2 \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \right) \right) 2B \\ & \geq \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \left(1 - 2(c_1 + 0.25) \right) 2B \geq 0. \end{aligned}$$

- for $j = \tau(g_2(y))$ if there exists $y \neq y_0, y_1$ s.t., $g_1(y) = g_2(y)$
 - when $\mathbb{E}[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \mid \tau(x_1) = s] \leq 0.01$, by Induction 5.B.13 and Lemma 5.B.93, we have $\mathbf{logit}_{5,\tau(g_2(y))}^{(t)} \ll O(\frac{1}{d}) \cdot \mathbf{logit}_{5,j'_2}^{(t)}$. Then

$$\begin{aligned} & \left| \mathbb{E} \left[\mathbf{logit}_{5,j}^{(t)} \cdot \left(\mathbf{sReLU}'(\Lambda_{5,j,r_{g_1 y}}^{(t)}) \cdot \right. \right. \right. \\ & \quad \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j,r_{g_1 y}}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j,r_{g_1 y}}(g_1) \right. \\ & \quad \left. \left. - (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5,j,r_{g_1 y}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_2} \right] \Big| \\ & \leq O\left(\frac{1}{d}\right) \cdot (5.B.90). \end{aligned}$$

- when $\mathbb{E}[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \mid \tau(x_1) = s] \geq 0.01$, by Lemma 5.B.93, we have $\Lambda_{5,j,r_{g_1 y}}^{(t)}$ cannot be activated.

Putting the above discussion together, we have

$$\mathbb{E} \left[\mathcal{N}_{s,2,ii}^{(t)} \mathbb{1}_{\mathcal{E}_1} \right] + \mathbb{E} \left[\mathcal{N}_{s,2,ii}^{(t)} \mathbb{1}_{\mathcal{E}_2} \right] \geq \Omega\left(\frac{B}{d \cdot n_y}\right) \cdot \mathbb{E} \left[(1 - \mathbf{logit}_{5,j'_2}^{(t)}) \mid \tau(x_1) = s, \mathcal{E}_2 \right].$$

- (c) For $\mathbf{Z}^{2,1} \in \mathcal{E}_3$, by Induction 5.B.13 and Lemma 5.B.94, we can first have the following logit bound:

$$\mathbf{logit}_{5,j}^{(t)} \leq \frac{1}{\text{poly}d} \cdot \left(1 - \mathbf{logit}_{5,j_2}^{(t)}\right) \text{ for } j \neq j_2, \tau(g_1(y_1)).$$

Therefore, we have

$$\begin{aligned} & \mathbb{E} \left[\mathcal{N}_{s,2,i}^{(t)} \mathbb{1}_{\mathcal{E}_3} \right] \\ &= \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) \cdot \left(-\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j_2, r_{g_2 \cdot y_1}}(y_1) - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j_2, r_{g_2 \cdot y_1}}(g_1) \right. \right. \\ & \quad \left. \left. + (1 - \mathbf{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1}^{(t)} - \mathbf{Attn}_{\mathbb{A},1 \rightarrow 2}^{(t)}) \Lambda_{5,j_2, r_{g_2 \cdot y_1}}^{(t)} \pm \tilde{O}(\sigma_0) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_3} \right] \\ &\geq \Omega\left(\frac{B}{n_y \cdot d}\right) \cdot \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) | \tau(x_1) = s, \mathcal{E}_3 \right]. \end{aligned}$$

Moving to $\mathcal{N}_{s,2,ii}^{(t)}$, we have

$$\begin{aligned} & \mathbb{E} \left[\mathcal{N}_{s,2,ii}^{(t)} \mathbb{1}_{\mathcal{E}_3} \right] \\ &\geq \mathbb{E} \left[\mathbf{logit}_{5,\tau(g_1(y_1))}^{(t)} \cdot \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{\tau(g_1(y_1)), r_{g_1 \cdot y_1}}(y_1) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{\tau(g_1(y_1)), r}(g_1) \right. \right. \\ & \quad \left. \left. - (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5,\tau(g_1(y_1)), r_{g_1 \cdot y_1}}^{(t)} \pm \tilde{O}(\sigma_0) \pm \tilde{O}(\delta^q) \right) \right. \\ & \quad \left. \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_3} \right] \\ &- O\left(\frac{n_y \cdot B}{n_y \cdot d \cdot \text{poly}d}\right) \cdot \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) | \tau(x_1) = s, \mathcal{E}_3 \right]. \end{aligned}$$

Thus,

$$\mathbb{E} \left[\mathcal{N}_{s,2,ii}^{(t)} \mathbb{1}_{\mathcal{E}_3} \right] + \mathbb{E} \left[\mathcal{N}_{s,2,i}^{(t)} \mathbb{1}_{\mathcal{E}_3} \right] \geq \Omega\left(\frac{B}{n_y \cdot d}\right) \cdot \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) | \tau(x_1) = s, \mathcal{E}_3 \right].$$

- (d) For $\mathbf{Z}^{2,1} \in \mathcal{E}_4$, we can bound the gradient in a manner similar to regime (i), and obtain

$$\mathbb{E} \left[\mathcal{N}_{s,2,ii}^{(t)} \mathbb{1}_{\mathcal{E}_4} \right] + \mathbb{E} \left[\mathcal{N}_{s,2,i}^{(t)} \mathbb{1}_{\mathcal{E}_4} \right] \geq 0.$$

- (e) For $\mathbf{Z}^{2,1} \in \mathcal{E}_5 \cup \mathcal{E}_6$, by Induction 5.B.13, Lemma 5.B.96 and Lemma 5.B.97, we can derive the following logit condition: $1 - \mathbf{logit}_{5,j_2}^{(t)}, \mathbf{logit}_{5,j}^{(t)} \leq \frac{1}{\text{poly}d} \cdot \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) | \tau(x_1) = s, \mathcal{E}_1 \right]$

for $j \neq j_2$. Hence, we can simply bound $\mathcal{N}_{s,2,i}^{(t)}$ and $\mathcal{N}_{s,2,ii}^{(t)}$ as follows:

$$\begin{aligned} & \left| \mathbb{E} \left[\mathcal{N}_{s,2,i}^{(t)} \mathbb{1}_{\mathcal{E}_m} \right] \right|, \left| \mathbb{E} \left[\mathcal{N}_{s,2,ii}^{(t)} \mathbb{1}_{\mathcal{E}_m} \right] \right| \\ & \leq O\left(\frac{B}{dn_y} \cdot \frac{1}{\text{poly}d}\right) \cdot \mathbb{E} \left[(1 - \text{logit}_{5,j_2}^{(t)}) | \tau(x_1) = s, \mathcal{E}_1 \right], \text{ for } m \in \{5, 6\}. \end{aligned}$$

Putting everything together, we have for the regmie (ii),

$$\begin{aligned} \left[-\nabla_{\mathbf{Q}_{4,3}^{(t)}} \mathbb{L}_5^{2,2} \right]_{s,s} + \left[-\nabla_{\mathbf{Q}_{4,4}^{(t)}} \mathbb{L}_5^{2,2} \right]_{s,s} &= \sum_{\kappa \in \{i, ii, iii\}} \sum_{i \in [6]} \mathbb{E} \left[\mathcal{N}_{s,2,\kappa}^{(t)} \mathbb{1}_{\mathcal{E}_i} \right] \\ &\geq \Omega\left(\frac{B}{d}\right) \cdot \mathbb{E} \left[(1 - \text{logit}_{5,j_2}^{(t)}) | \tau(x_1) = s, \mathcal{E}_1 \right]. \end{aligned}$$

□

Lemma 5.B.99. *If Induction 5.B.13 holds for all iterations $< t$, given $s \in \tau(\mathcal{X})$, for $[\mathbf{Q}_{4,p}]_{s,s'}$, $p \in \{3, 4\}$, $s' \neq s \in \tau(\mathcal{X})$, we have*

$$\left| \left[-\nabla_{\mathbf{Q}_{4,p}^{(t)}} \text{Loss}_5^{2,2} \right]_{s,s'} \right| \leq O\left(\frac{1}{d}\right) \left| \left[-\nabla_{\mathbf{Q}_{4,p}^{(t)}} \text{Loss}_5^{2,2} \right]_{s,s} \right|.$$

Bounded Decrease of Attention to Related Context Clause

Lemma 5.B.100. *If Induction 5.B.13 holds for all iterations $< t$, then for any sample $\mathbf{Z}^{2,1}$, we have*

$$\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \geq -O\left(\frac{\log d \log d}{\log d}\right).$$

Proof. Denote the first time that $\mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} | \tau(x_1) = s \right] \leq -\Omega\left(\frac{\log d \log d}{\log d}\right)$ as $\tilde{T}$. Notice that $\left| [\mathbf{Q}_{4,p}^{(t)}]_{s,s'} \right| \leq O\left(\frac{[\mathbf{Q}_{4,p}^{(t)}]_{s,s}}{d}\right)$ for $p \in \{3, 4\}$, thus for any sample $\mathbf{Z}^{2,1}$ satisfying $\tau(x_1) = s$, at time $\tilde{T}$, we have

$$\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(\tilde{T})} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} \leq -\Omega\left(\frac{\log d \log d}{\log d}\right).$$

Based on the the gradient compositions from Lemma 5.B.90, we have $\left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_5^{2,2} \right]_{s,s} = \mathbb{E} \left[\mathcal{N}_{s,3,2,i} + \mathcal{N}_{s,3,2,ii} + \mathcal{N}_{s,3,2,iii} \right]$, and we will discuss $\mathcal{N}_{s,3,2,\kappa}$ for $\kappa \in \{i, ii, iii\}$ on different samples $\mathbf{Z}^{2,1}$. Following the similar argument as Lemma 5.B.98, we can first directly bound the term

$\mathcal{N}_{s,3,2,iii}^{(\tilde{T})}$ as follows:

$$\mathbb{E} \left[\mathcal{N}_{s,3,2,iii}^{(\tilde{T})} \right] \leq \tilde{O}(\sigma_0^q) = \frac{1}{\text{poly}d}.$$

- for $\mathbf{Z}^{2,1} \in \mathcal{E}_1$, at time $\tilde{T}$, since the neurons for predicting j_2 cannot be activated, and $\mathbf{logit}_{5,j}^{(\tilde{T})} \leq O(\frac{1}{d})$ for $j \neq j_2$, thus we can naively bound the gradient on the event $\mathcal{E}_1$ as follows:

$$\left| \sum_{\kappa \in \{i,ii\}} \mathbb{E}[\mathcal{N}_{s,3,2,\kappa}^{(\tilde{T})} 1_{\mathcal{E}_1}] \right| \leq O\left(\frac{n_y B}{d^2}\right) + \tilde{O}(\delta^q).$$

- for $\mathbf{Z}^{2,1} \in \mathcal{E}_2$, similarly, $\Lambda_{5,j_2,r}^{(t)}$ is not activated, and thus we can focus on the term $\mathcal{N}_{s,3,2,ii}$, and specifically, the prediction of $j'_2 = \tau(g_2(y_0))$. By (5.B.77) and Lemma 5.B.93, we have

$$\begin{aligned} \mathbb{E} \left[\mathcal{N}_{s,3,2,ii}^{(\tilde{T})} 1_{\mathcal{E}_2} \right] &\geq -\mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} \right. \\ &\quad \cdot \left(1 + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(\tilde{T})} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(\tilde{T})} \right) \mid \tau(x_1) = s \left. \right] \\ &\quad \cdot \mathbb{E} \left[\mathbf{logit}_{5,j'_2}^{(\tilde{T})} \mid \tau(x_1) = s, \mathcal{E}_2 \right] \frac{2B}{n_y \cdot d} \pm \tilde{O}(\sigma_0^q). \end{aligned} \quad (5.B.91)$$

- for $\mathbf{Z}^{2,1} \in \mathcal{E}_3$, by Lemma 5.B.94, we can mainly focus on the term $\mathcal{N}_{s,3,2,i}$, and the prediction of $\tau(g_1(y_1))$ in $\mathcal{N}_{s,3,2,ii}^{(\tilde{T})}$ since $\mathbf{logit}_{5,\tau(g_1(y_1))}^{(\tilde{T})} = 1 - O(\frac{1}{\log d})$. Hence, we have

$$\begin{aligned} \mathbb{E} \left[\left(\mathcal{N}_{s,3,2,ii}^{(\tilde{T})} + \mathcal{N}_{s,3,2,i}^{(\tilde{T})} \right) 1_{\mathcal{E}_3} \right] &\geq \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} \right. \\ &\quad \cdot \left(1 + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(\tilde{T})} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} \right) \mid \tau(x_1) = s \left. \right] \cdot \left(1 - O\left(\frac{1}{\log d}\right) \right) \frac{2B}{n_y \cdot d} \pm \tilde{O}(\sigma_0^q). \end{aligned} \quad (5.B.92)$$

- for $\mathbf{Z}^{2,1} \in \mathcal{E}_4$, the negative gradient can be bounded in the same way as (5.B.91), however, the probability of $\mathcal{E}_4$ is order-wise smaller than $\mathcal{E}_2$ and $\mathcal{E}_3$, which can be neglected. Moreover, for $\mathbf{Z}^{2,1} \in \mathcal{E}_5 \cup \mathcal{E}_6$, the overall gradient is also negligible since $\mathbf{logit}_{5,j_2}^{(t)}$ is very close to 1.

Putting it all together, we have

$$\begin{aligned} \left[-\nabla_{\mathbf{Q}_{4,3}^{(\tilde{T})}} \text{Loss}_5^{2,2} \right]_{s,s} &\geq -\mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} \right. \\ &\quad \cdot \left(1 + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(\tilde{T})} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(\tilde{T})} \right) \mid \tau(x_1) = s \left. \right] \\ &\quad \cdot \mathbb{E} \left[\mathbf{logit}_{5,j'_2}^{(\tilde{T})} \mid \tau(x_1) = s, \mathcal{E}_2 \right] \frac{B}{n_y \cdot d} \end{aligned}$$

$$\begin{aligned}
& + \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} \right. \\
& \cdot \left. \left(1 + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(\tilde{T})} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} \right) \mid \tau(x_1) = s \right] \cdot \left(1 - O\left(\frac{1}{\log d}\right) \right) \frac{2B}{n_y \cdot d} \pm \tilde{O}(\sigma_0^q).
\end{aligned}$$

Notice that

$$\begin{aligned}
& \left(1 + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(\tilde{T})} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} \right) \\
& - \left(1 + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(\tilde{T})} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(\tilde{T})} \right) \\
& = 2\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(\tilde{T})} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(\tilde{T})}.
\end{aligned}$$

If $2\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(\tilde{T})} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(\tilde{T})} < c$ for some small constant $c > 0$, s.t., $\frac{cB}{\log d} < 1$, then $\Lambda_{5,j_2',r_{s_2 \cdot y_0}}^{(\tilde{T})} \leq cB$. Hence $\mathbf{logit}_{5,j_2'}^{(\tilde{T})} \leq \frac{1}{d^{1-\frac{cB}{\log d}}} = o(1)$, and (5.B.91) is dominated by the positive term (5.B.92). Else, clearly, (5.B.91) can be cancelled out by the positive term (5.B.92). Therefore,

$$\left[-\nabla_{\mathbf{Q}_{4,3}^{(\tilde{T})}} \text{Loss}_5^{2,2} \right]_{s,s} \geq \Omega\left(\frac{B}{n_y \cdot d}\right).$$

As a consequence, together with the growth of $\mathbf{Q}_{4,3}^{(\tilde{T})} + \mathbf{Q}_{4,4}^{(\tilde{T})}$, and nearly no change of $[\mathbf{Q}_{4,p}^{(t)}]_{s,s'}$ in Lemma 5.B.99, we have $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}$ must start to decrease and cannot be larger than $O\left(\frac{\log \log d}{\log d}\right)$. $\square$

Attention gap is small

Lemma 5.B.101. *If Induction 5.B.13 holds for all iterations $< t$, then for any sample $\mathbf{Z}^{2,1}$, we have*

$$\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \leq c_1,$$

where $c_1 > 0$ is a small constant.

Proof. Let $\tilde{T}$ denote the first time $\mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \mid \tau(x_1) = s \right] \geq \frac{1.0005 \log d}{2B}$. Notice that $\left| [\mathbf{Q}_{4,p}^{(t)}]_{s,s'} \right| \leq O\left(\frac{[\mathbf{Q}_{4,p}^{(t)}]_{s,s}}{d}\right)$ for $p \in \{3,4\}$, thus for any sample $\mathbf{Z}^{2,1}$ satisfying $\tau(x_1) = s$, at time $\tilde{T}$, we have

$$\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(\tilde{T})} \geq \frac{1.0005 \log d}{2B}.$$

Based on Lemma 5.B.90, we will discuss the gradients of $\mathbf{Q}_{4,3}$ and $\mathbf{Q}_{4,4}$ on different samples $\mathbf{Z}^{2,1}$. Since $\Lambda_{5,j_2',r_{s_2 \cdot y_1}}^{(\tilde{T})}, \Lambda_{5,j_2',r_{s_2 \cdot y_0}}^{(\tilde{T})}$ is guaranteed to be activated to the linear regime for $\mathbf{Z}^{2,1} \in \mathcal{E}_1$, we only need to focus on the main event $\mathcal{E}_1$.

From Lemma 5.B.92, at time $\tilde{T}$, we have

$$\mathbf{logit}_{5,j'_2}^{(\tilde{T})} = \frac{e^{\frac{1.0005 \log d}{2B} \cdot 2B}}{e^{\frac{1.0005 \log d}{2B} \cdot 2B} + O(d)} \left(1 - \mathbf{logit}_{5,j_2}^{(\tilde{T})}\right) = \left(1 - O(1/d^{0.0005})\right) \left(1 - \mathbf{logit}_{5,j_2}^{(\tilde{T})}\right).$$

Then by Lemma 5.B.90, we can obtain the gradient on the event $\mathcal{E}_1$ as follows:

$$\begin{aligned} & \mathbb{E} \left[\left(\mathcal{N}_{s,3,2,i}^{(\tilde{T})} + \mathcal{N}_{s,3,2,ii}^{(\tilde{T})} + \mathcal{N}_{s,3,2,iii}^{(\tilde{T})} \right) \mathbb{1}_{\mathcal{E}_1} \right] \\ &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} \cdot \left((1 - \mathbf{logit}_{5,j_2}^{(\tilde{T})}) \cdot \left(V_{j_2,r_{s_2 \cdot y_1}}(g_2) - \Lambda_{5,j_2,r_{s_2 \cdot y_1}}^{(\tilde{T})} \pm \tilde{O}(\sigma_0) \right) \right. \right. \\ &\quad \left. \left. - \left(1 - O\left(\frac{1}{d^{0.0005}}\right) \right) (1 - \mathbf{logit}_{5,j_2}^{(t)}) \cdot \left(V_{j'_2,r_{s_2 \cdot y_0}}(g_2) - \Lambda_{5,j_2,r_{s_2 \cdot y_0}}^{(\tilde{T})} \pm \tilde{O}(\sigma_0) \right) \right. \right. \\ &\quad \left. \left. \pm \tilde{O}(\sigma_0^q) \right) \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_1} \right] \\ &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} \cdot \left(\left(1 - O\left(\frac{1}{d^{0.0005}}\right) \right) (1 - \mathbf{logit}_{5,j_2}^{(\tilde{T})}) \cdot \left(\Lambda_{5,j'_2,r_{s_2 \cdot y_0}}^{(\tilde{T})} - \Lambda_{5,j_2,r_{s_2 \cdot y_1}}^{(\tilde{T})} \right) \right. \right. \\ &\quad \left. \left. + O\left(\frac{1}{d^{0.0005}}\right) (1 - \mathbf{logit}_{5,j_2}^{(\tilde{T})}) \cdot \left(V_{5,j_2,r_{s_2 \cdot y_1},2}(g_2) - \Lambda_{5,j_2,r_{s_2 \cdot y_1}}^{(\tilde{T})} \right) \right) \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_1} \right], \end{aligned}$$

on the other hand, we have

$$\begin{aligned} & \mathbb{E} \left[\left(\mathcal{N}_{s,4,2,i}^{(\tilde{T})} + \mathcal{N}_{s,4,2,ii}^{(\tilde{T})} + \mathcal{N}_{s,4,2,iii}^{(\tilde{T})} \right) \mathbb{1}_{\mathcal{E}_1} \right] \\ &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} \cdot \left((1 - \mathbf{logit}_{5,j_2}^{(\tilde{T})}) \cdot \left(V_{j_2,r_{s_2 \cdot y_1}}(y_1) - \Lambda_{5,j_2,r_{s_2 \cdot y_1}}^{(\tilde{T})} \pm \tilde{O}(\sigma_0) \right) \right. \right. \\ &\quad \left. \left. - \left(1 - O\left(\frac{1}{d^{0.0005}}\right) \right) (1 - \mathbf{logit}_{5,j_2}^{(t)}) \cdot \left(V_{j'_2,r_{s_2 \cdot y_0}}(y_1) - \Lambda_{5,j_2,r_{s_2 \cdot y_0}}^{(\tilde{T})} \pm \tilde{O}(\sigma_0) \right) \right. \right. \\ &\quad \left. \left. \pm \tilde{O}(\sigma_0^q) \right) \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_1} \right] \\ &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} \cdot \left(\left(1 - O\left(\frac{1}{d^{0.0005}}\right) \right) (1 - \mathbf{logit}_{5,j_2}^{(\tilde{T})}) \right. \right. \\ &\quad \cdot \left(-V_{j'_2,r_{s_2 \cdot y_0}}(y_1) + \Lambda_{5,j'_2,r_{s_2 \cdot y_0}}^{(\tilde{T})} - \Lambda_{5,j_2,r_{s_2 \cdot y_1}}^{(\tilde{T})} \right) \\ &\quad \left. \left. - O\left(\frac{1}{d^{0.0005}}\right) (1 - \mathbf{logit}_{5,j_2}^{(\tilde{T})}) \cdot \Lambda_{5,j_2,r_{s_2 \cdot y_1}}^{(\tilde{T})} \right) \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_1} \right]. \end{aligned}$$

Since at time $\tilde{T}$, we have $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(\tilde{T})} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(\tilde{T})} \leq \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(\tilde{T})} \leq \frac{1.005 \log d}{2B}$, we have $\Lambda_{5,j_2',r_{s_2 \cdot y_0}}^{(\tilde{T})} - \Lambda_{5,j_2,r_{s_2 \cdot y_1}}^{(\tilde{T})} \ll -V_{j_2',r_{s_2 \cdot y_0}}(y_1)$. Therefore,

$$\left[-\nabla_{\mathbf{Q}_{4,4}^{(\tilde{T})}} \text{Loss}_5^{2,2} \right]_{s,s} + \left[\nabla_{\mathbf{Q}_{4,3}^{(\tilde{T})}} \text{Loss}_5^{2,2} \right]_{s,s} \geq \Omega\left(\frac{B}{d}\right) \cdot \mathbb{E} \left[1 - \mathbf{logit}_{5,j_2}^{(\tilde{T})} | \tau(x_1) = s \right],$$

which implies that $[\mathbf{Q}_{4,4}]_{s,s}$ will grow faster than $[\mathbf{Q}_{4,3}]_{s,s}$, and thus the attention gap cannot be further increased. $\square$

At the End of Stage 1.2.1

Lemma 5.B.102. *For all iterations $t \leq T_{1,2,1,s} = O\left(\frac{n_y d}{\eta \log d}\right) + O\left(\frac{(0.1+c_1/2)B}{\eta \log d}\right)$, we have Induction 5.B.13 holds, and at time $T_{1,2,1,s}$, we have*

- (a) $[\mathbf{Q}_{4,3}^{(T_{1,2,1,s})}]_{s,s}, [\mathbf{Q}_{4,4}^{(T_{1,2,1,s})}]_{s,s} \geq \Omega(1)$;
- (b) other $[\mathbf{Q}_{4,p}^{(T_{1,2,1,s})}]_{s,s'}$ for $p \in \{3,4\}$, $s' \in \tau(X) \neq s$ are at most $\tilde{O}(\frac{1}{d})$.

Proof. The existence of $T_{1,2,1,s}$ can be directly obtained by using Lemmas 5.B.98 to 5.B.101. Furthermore, $[\mathbf{Q}_{4,4}^{(T_{1,2,1,s})}]_{s,s} \geq \Omega(1)$ can be guaranteed since Lemma 5.B.101 implies that $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(T_{1,2,1,s})} \geq 0.3 - \frac{c_1}{2} \pm \tilde{O}(1/d) > 0.2$, which means that $[\mathbf{Q}_{4,4}^{(T_{1,2,1,s})}]_{s,s}$ should at least grow to a constant level compared to $|\mathbf{Q}_{4,p}^{(T_{1,2,1,s})}]_{s,s'}| = \tilde{O}(1/d)$ with $p \in \{3,4\}$.

We will handle $[\mathbf{Q}_{4,3}^{(T_{1,2,1,s})}]_{s,s}$ by means of a proof by contradiction. Suppose that $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(T_{1,2,1,s})} \geq 0.4 - \tilde{c}$, where $\tilde{c} = \frac{\log d}{8B}$ is a sufficiently small constant. Then denote $\tilde{T}$ the first time that $\mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} | \tau(x_1) = s \right] \geq 0.4 - 2\tilde{c}$. Notice that $\left| [\mathbf{Q}_{4,p}^{(t)}]_{s,s'} \right| \leq O\left(\frac{[\mathbf{Q}_{4,p}^{(t)}]_{s,s}}{d}\right)$ for $p \in \{3,4\}$, thus for any sample $\mathbf{Z}^{2,1}$ satisfying $\tau(x_1) = s$, at time $\tilde{T}$, we have

$$\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(\tilde{T})} \geq 0.4 - 2\tilde{c} \pm \tilde{O}(1/d).$$

- If $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(\tilde{T})} \geq 2\varrho$, then
 - for $\mathbf{Z}^{2,1} \in \mathcal{E}_1$, $\Lambda_{5,j_2,r_{s_2 \cdot y_1}}^{(\tilde{T})}$ has already been activated to the linear regime. Furthermore, $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(\tilde{T})} \leq 0.2 + 2\tilde{c} - 0.2 \leq 2\tilde{c}$, which implies $1 - \mathbf{logit}_{5,j_2}^{(1)} = 1 - o(1)$. Thus, by Lemma 5.B.90, we have

$$\mathbb{E} \left[\left(\mathcal{N}_{s,3,2,i}^{(\tilde{T})} + \mathcal{N}_{s,3,2,ii}^{(\tilde{T})} + \mathcal{N}_{s,3,2,iii}^{(\tilde{T})} \right) \mathbb{1}_{\mathcal{E}_1} \right] \geq \Omega\left(\frac{B}{d}\right),$$

while

$$\mathbb{E} \left[\left(\mathcal{N}_{s,4,2,i}^{(\tilde{T})} + \mathcal{N}_{s,4,2,ii}^{(\tilde{T})} + \mathcal{N}_{s,4,2,iii}^{(\tilde{T})} \right) \mathbb{1}_{\mathcal{E}_1} \right] \leq -\Omega\left(\frac{B}{d}\right).$$

- for $\mathbf{Z}^{2,1} \in \mathcal{E}_2$, $\Lambda_{5,j_2',r_{g_2,y_0}}^{(\tilde{T})} = (\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(\tilde{T})} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(\tilde{T})})2B \leq (0.4 + \frac{4}{3}\tilde{c} - 0.4 + 2\tilde{c})2B = \frac{5 \log d}{6}$. Thus $\mathbf{logit}_{5,j_2}^{(1)} = O(\frac{1}{d^{1/6}})$. Hence by Lemma 5.B.90, we have

$$\mathbb{E} \left[\left(\mathcal{N}_{s,3,2,i}^{(\tilde{T})} + \mathcal{N}_{s,3,2,ii}^{(\tilde{T})} + \mathcal{N}_{s,3,2,iii}^{(\tilde{T})} \right) \mathbb{1}_{\mathcal{E}_1} \right] \geq -O\left(\frac{B}{d^{7/6}n_y}\right),$$

while

$$\mathbb{E} \left[\left(\mathcal{N}_{s,4,2,i}^{(\tilde{T})} + \mathcal{N}_{s,4,2,ii}^{(\tilde{T})} + \mathcal{N}_{s,4,2,iii}^{(\tilde{T})} \right) \mathbb{1}_{\mathcal{E}_1} \right] \leq -\Omega\left(\frac{B}{n_y d}\right).$$

- $\mathbf{Z}^{2,1} \in \mathcal{E}_3$, we can use the following naive bounds for $p \in \{3, 4\}$:

$$|\mathbb{E} \left[\left(\mathcal{N}_{s,p,2,i}^{(\tilde{T})} + \mathcal{N}_{s,p,2,ii}^{(\tilde{T})} + \mathcal{N}_{s,p,2,iii}^{(\tilde{T})} \right) \mathbb{1}_{\mathcal{E}_1} \right]| \leq O\left(\frac{B}{dn_y}\right).$$

Putting them together, combining with the fact that the gradient contributed by $\mathcal{E}_4 \cup \mathcal{E}_5 \cup \mathcal{E}_6$ is negligible, we can conclude that

$$\left[-\nabla_{\mathbf{Q}_{4,3}^{(\tilde{T})}} \text{Loss}_5^{2,2} \right]_{s,s} \geq \Omega\left(\frac{B}{d}\right), \quad \left[-\nabla_{\mathbf{Q}_{4,4}^{(\tilde{T})}} \text{Loss}_5^{2,2} \right]_{s,s} \leq -\Omega\left(\frac{B}{d}\right),$$

which implies $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(\tilde{T})}$ cannot further increase above $0.4 - 2\tilde{c}$.

- If $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(\tilde{T})} < 2\varrho$, we shift our focus to the comparison between event $\mathcal{E}_2 \mathcal{E}_3$,

- for $\mathbf{Z}^{2,1} \in \mathcal{E}_1$, we have

$$\mathbb{E} \left[\left(\mathcal{N}_{s,3,2,i}^{(\tilde{T})} + \mathcal{N}_{s,3,2,ii}^{(\tilde{T})} + \mathcal{N}_{s,3,2,iii}^{(\tilde{T})} \right) \mathbb{1}_{\mathcal{E}_1} \right] \geq -\tilde{O}(\sigma_0^q),$$

while

$$\mathbb{E} \left[\left(\mathcal{N}_{s,4,2,i}^{(\tilde{T})} + \mathcal{N}_{s,4,2,ii}^{(\tilde{T})} + \mathcal{N}_{s,4,2,iii}^{(\tilde{T})} \right) \mathbb{1}_{\mathcal{E}_1} \right] \leq \tilde{O}(\sigma_0^q).$$

- for $\mathbf{Z}^{2,1} \in \mathcal{E}_2$, similarly as previous case, we have

$$\mathbb{E} \left[\left(\mathcal{N}_{s,3,2,i}^{(\tilde{T})} + \mathcal{N}_{s,3,2,ii}^{(\tilde{T})} + \mathcal{N}_{s,3,2,iii}^{(\tilde{T})} \right) \mathbb{1}_{\mathcal{E}_2} \right] \geq -O\left(\frac{B}{d^{7/6}n_y}\right),$$

while

$$\mathbb{E} \left[\left(\mathcal{N}_{s,4,2,i}^{(\tilde{T})} + \mathcal{N}_{s,4,2,ii}^{(\tilde{T})} + \mathcal{N}_{s,4,2,iii}^{(\tilde{T})} \right) \mathbb{1}_{\mathcal{E}_2} \right] \leq O\left(\frac{B}{d^{7/6}n_y}\right).$$

▪ $\mathbf{Z}^{2,1} \in \mathcal{E}_3$, $\left| \Lambda_{5,j'_2,r_{s_1 \cdot y_1}}^{(\tilde{T})} - \Lambda_{5,j_2,r_{s_2 \cdot y_1}}^{(\tilde{T})} \right| \leq 4\rho B = o(1)$, hence, we have

$$\mathbb{E} \left[\left(\mathcal{N}_{s,3,2,i}^{(\tilde{T})} + \mathcal{N}_{s,3,2,ii}^{(\tilde{T})} + \mathcal{N}_{s,3,2,iii}^{(\tilde{T})} \right) \mathbb{1}_{\mathcal{E}_3} \right] \geq \Omega\left(\frac{B}{dn_y}\right),$$

while

$$\mathbb{E} \left[\left(\mathcal{N}_{s,4,2,i}^{(\tilde{T})} + \mathcal{N}_{s,4,2,ii}^{(\tilde{T})} + \mathcal{N}_{s,4,2,iii}^{(\tilde{T})} \right) \mathbb{1}_{\mathcal{E}_3} \right] \leq O\left(\frac{B \log \log d}{dn_y \cdot \log d}\right).$$

Putting them together, again we can conclude that

$$\left[-\nabla_{\mathbf{Q}_{4,3}^{(\tilde{T})}} \text{Loss}_5^{2,2} \right]_{s,s} \geq \Omega\left(\frac{B}{n_y d}\right), \quad \left[-\nabla_{\mathbf{Q}_{4,4}^{(\tilde{T})}} \text{Loss}_5^{2,2} \right]_{s,s} \leq O\left(\frac{B \log \log d}{dn_y \cdot \log d}\right),$$

which implies $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(\tilde{T})}$ cannot further increase above $0.4 - 2\tilde{c}$.

Consequently, this leads to a contradiction, and $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(T_{1,2,1,s})} < 0.4 - \tilde{c}$, where $\tilde{c} = \frac{\log d}{8B}$. Then it would follow that $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(T_{1,2,1,s})} \geq \tilde{c}$, and thus $[\mathbf{Q}_{4,3}^{(T_{1,2,1,s})}]_{s,s} \geq \Omega(1)$. $\square$

Stage 1.2.2: Convergence with Small Wrong Attention

Recall that $\text{Loss}_{5,s}^{2,2} = -\mathbb{E} \left[\log p_F(\mathbf{Z}_{\text{ans},2,5} | \mathbf{Z}^{(2,1)}) | \tau(x_1) = s \right]$ for $s \in \tau(\mathcal{X})$.

Induction 5.B.14. Given $s \in \tau(\mathcal{X})$, let $T_{1,2,2,s}$ denote the first time that $\text{Loss}_{5,s}^{2,2}$ decreases below $\Theta\left(e^{(-\frac{1}{2} + 3.01c_1) \cdot 2B}\right)$. For all iterations $T_{1,2,1,s} \leq t < T_{1,2,2,s}$, we have the following holds

- (a) $[\mathbf{Q}_{4,3}^{(t)}]_{s,s} + [\mathbf{Q}_{4,4}^{(t)}]_{s,s}$ monotonically increases;
- (b) for $p \in \{3, 4\}$, for $j \in \tau(\mathcal{X}) \neq s$, $|[\mathbf{Q}_{4,p}^{(t)}]_{s,j}| \leq O\left(\frac{[\mathbf{Q}_{4,p}^{(t)}]_{s,j}}{d}\right)$;
- (c) for any sample $\mathbf{Z}^{2,1}$, we have $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \leq c_1$ for some sufficiently small constant $c_1 = \frac{1.005 \log d}{B} > 0$;
- (d) for any $\mathbf{Z}^{2,1}$, we have $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \leq \min \left\{ \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - c_2, 0 \right\}$, where $c_2 = \frac{\log d}{4B} > 0$ is some sufficiently small constant.

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Lemma 5.B.103. If Induction 5.B.14 holds for all iterations $[T_{1,2,1,s}, t)$, then for any sample $\mathbf{Z}^{2,1}$, we have

1. $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \in \left[\frac{4}{3}c_1, 0.4 \pm \tilde{O}\left(\frac{1}{d}\right) \right]$;
2. $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \geq \Omega(1)$;
3. $|\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)}| \leq \tilde{O}\left(\frac{1}{d}\right)$.

Proof. In the following, we focus on the main events $\mathcal{E}_1$, $\mathcal{E}_2$, and $\mathcal{E}_3$, which correspond to cases where some confused wrong predictions occur. We denote

$$\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)}, \quad \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} = c + \tilde{O}\left(\frac{1}{d}\right).$$

- If $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \geq \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)}$, then
 - If $\mathbf{Z}^{2,1} \in \mathcal{E}_1$, we have

$$\begin{aligned} & \log p_F(\mathbf{Z}_{\text{ans},2,5} | \mathbf{Z}^{(2,1)}) \\ & \leq \Theta(1) \cdot \max \left\{ e^{(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)})2B - (\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)})2B}, \right. \\ & \quad \left. e^{\log d - (\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)})2B} \right\} \\ & \leq \Theta \left(e^{2c_1 B - (\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)})2B} \right) \\ & \leq \Theta \left(e^{-2\left(\frac{1}{2} - 2c - c_1\right)B} \right), \end{aligned}$$

where the last inequality follows from the fact that $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \geq \frac{1}{2}(1 - 2c) = \frac{1}{2} - c$.

- If $\mathbf{Z}^{2,1} \in \mathcal{E}_2$, we have

$$\begin{aligned} & \log p_F(\mathbf{Z}_{\text{ans},2,5} | \mathbf{Z}^{(2,1)}) \leq \Theta(1) \\ & \cdot \max \left\{ e^{(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)})B - (\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)})2B}, \right. \\ & \quad \left. e^{\log d - (\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)})2B} \right\} \\ & \leq \Theta \left(e^{2(c+c_1)B - (\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)})2B} \right) \\ & \leq \Theta \left(e^{-2\left(\frac{1}{2} - 3c - c_1\right)B} \right). \end{aligned}$$

- If $\mathbf{Z}^{2,1} \in \mathcal{E}_3$, we have

$$\begin{aligned} & \log p_F(\mathbf{Z}_{\text{ans},2,5} | \mathbf{Z}^{(2,1)}) \\ & \leq \Theta(1) \cdot \max \left\{ e^{(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)})2B}, e^{\log d - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}2B} \right\} \\ & \leq \Theta(1) \cdot \max \left\{ e^{-2\left(\frac{1}{2} - c_1 - c\right)B}, e^{-2\left(\frac{1}{2} - 2c\right)B} \right\}. \end{aligned}$$

- If $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \leq \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)}$, then
 - If $\mathbf{Z}^{2,1} \in \mathcal{E}_1$, we have

$$\begin{aligned} \log p_F(\mathbf{Z}_{\text{ans},2,5} | \mathbf{Z}^{(2,1)}) &\leq \Theta(1) \cdot e^{\log d - (\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)})2B} \\ &\leq \Theta \left(e^{2c_1 B - (\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)})2B} \right) \\ &\leq \Theta \left(e^{-2\left(\frac{1}{2} - \frac{5}{2}c - \frac{c_2}{2} - c_1\right)B} \right), \end{aligned}$$

where the last inequality uses the fact that $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \leq c - c_2$, implying $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \geq \frac{1}{2} - \frac{3}{2}c + \frac{c_2}{2}$.

- If $\mathbf{Z}^{2,1} \in \mathcal{E}_2$, we have

$$\begin{aligned} \log p_F(\mathbf{Z}_{\text{ans},2,5} | \mathbf{Z}^{(2,1)}) &\leq \Theta(1) \cdot \\ \max \left\{ e^{(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)})2B - (\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)})2B}, \right. \\ &\quad \left. e^{\log d - (\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)})2B} \right\} \\ &\leq \Theta \left(e^{\max\{c, c_1\}2B - (\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)})2B} \right) \\ &\leq \Theta \left(e^{-2\left(\frac{1}{2} - 2c - \max\{c, c_1\}\right)B} \right). \end{aligned}$$

- If $\mathbf{Z}^{2,1} \in \mathcal{E}_3$, we have

$$\begin{aligned} \log p_F(\mathbf{Z}_{\text{ans},2,5} | \mathbf{Z}^{(2,1)}) &\leq \Theta(1) \cdot \max \left\{ e^{(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)})2B}, e^{\log d - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}2B} \right\} \\ &\leq \Theta(1) \cdot \max \left\{ e^{-2\left(\frac{1}{2} - c_1 - c\right)B}, e^{-2\left(\frac{1}{2} - 2c\right)B} \right\}. \end{aligned}$$

Putting all cases together, if $c \leq \frac{2}{3}c_1$, we must have

$$\text{Loss}_{5,s}^{2,2} \leq \Theta \left(e^{-2\left(\frac{1}{2} + 3 \cdot \frac{2c_1}{3} + c_1\right)B} \right) = \Theta \left(e^{(-\frac{1}{2} + 3c_1)2B} \right),$$

which contradicts the definition of $T_{1,2,2,s}$. Therefore, it must hold that $c \geq \frac{2}{3}c_1$ for all $t \leq T_{1,2,2,s}$. $\square$

Lemma 5.B.104. *If Induction 5.B.14 holds for all iterations $[T_{1,2,1,s}, t)$, then given $\mathbf{Z}^{2,1} \in \mathcal{E}_1$,*

1. for the prediction j_2 , we have

$$\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} = \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \right) \cdot 2B + O\left(\frac{B}{n_y}\right) + O(\delta).$$

2. for the prediction $j'_2 = \tau(g_2(y_0))$, we have

$$\Lambda_{5,j'_2,r_{g_2 \cdot y_0}}^{(t)} = \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \right) \cdot 2B + \tilde{O}\left(\frac{B}{d \cdot n_y}\right) + O(\delta).$$

3. for the prediction $\tau(g_1(y_0))$, $r_{g_1 \cdot y_0}$ cannot be activated.

4. for the prediction $\tau(g_1(y_1))$, we have

$$\Lambda_{5,\tau(g_1(y_1)),r_{g_1 \cdot y_1}}^{(t)} = \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \right) \cdot \frac{2B}{n_y} + \tilde{O}\left(\frac{B}{d}\right) + O(\delta).$$

5. for other $j \in \tau(\mathcal{Y})$, if there are j and y , s.t., $g_1, g_2 \in \text{Fiber}_{j,y}$ (notice that $y \neq y_0, y_1$ for $\mathcal{E}_1$), then for such a j , $r \in \widehat{\mathfrak{U}}_j \setminus \{r_{g_2 \cdot y}\}$ cannot be activated, moreover, we have

$$\Lambda_{5,j,r_{g_2 \cdot y}}^{(t)} = \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \right) \cdot 2B + \tilde{O}\left(\frac{B}{d \cdot n_y}\right) + O(\delta);$$

else, none of $r \in \widehat{\mathfrak{U}}_j$ can be activated.

Notice that for $\mathcal{E}_1$, the neurons mentioned in 1-4 should be different neurons, while the predictions in 1 and 3, or 2 and 4 can be the same in some cases, e.g., $g_1(y_1) = g_2(y_0)$. In all cases, except for the neurons mentioned above, i.e., $\cup_{\ell, \ell' \in [2]} \{r_{g_\ell \cdot y_{\ell'-1}}\}$ (which may not be activated), all other neurons $r \in \cup_{\ell, \ell' \in [2]} \left(\widehat{\mathfrak{U}}_{\tau(g_\ell \cdot y_{\ell'-1})} \setminus \{r_{g_\ell \cdot y_{\ell'-1}}\} \right)$ cannot be activated.

Lemma 5.B.105. If Induction 5.B.14 holds for all iterations $t \in [T_{1,2,1,s}, T_{1,2,2,s})$, then given $\mathbf{Z}^{2,1} \in \mathcal{E}_2$, we have

1. for the prediction j_2 , $r \in \widehat{\mathfrak{U}}_{j_2} \setminus \{r_{g_2 \cdot y_1}\}$ cannot be activated, moreover, we have

$$\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} = \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \right) \cdot 2B + \tilde{O}\left(\frac{B}{n_y}\right) + O(\delta).$$

2. for the prediction $j'_2 = \tau(g_2(y_0))$, $r \in \widehat{\mathfrak{U}}_{j'_2} \setminus \{r_{g_2 \cdot y_0}\}$ (notice that in this case $r_{g_2 \cdot y_0} = r_{g_1 \cdot y_0}$) cannot be activated, moreover, we have

$$\Lambda_{5,j'_2,r_{g_2 \cdot y_0}}^{(t)} = \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \right) \cdot 2B + O(\delta).$$

3. for the prediction $\tau(g_1(y_1))$, $r \in \widehat{\mathfrak{U}}_{\tau(g_1(y_1))} \setminus \{r_{g_1 \cdot y_1}\}$ cannot be activated, moreover, we have

$$\Lambda_{5,\tau(g_1(y_1)),r_{g_1 \cdot y_1}}^{(t)} = \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \right) \cdot \frac{2B}{n_y} + O(\delta).$$

4. for other $j \in \tau(\mathcal{Y})$, if there are j and y , s.t., $g_1, g_2 \in \text{Fiber}_{j,y}$ (notice that $y \neq y_0, y_1$ for $\mathcal{E}_2$), then for such a j , $r \in \widehat{\mathfrak{A}}_j \setminus \{r_{g_2 \cdot y}\}$ cannot be activated, moreover, we have

$$\Lambda_{5,j,r_{g_2 \cdot y}}^{(t)} = \left(\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \right) \cdot 2B + \widetilde{O}\left(\frac{B}{d \cdot n_y}\right) + O(\delta);$$

else, none of $r \in \widehat{\mathfrak{A}}_j$ can be activated.

Lemma 5.B.106. If Induction 5.B.14 holds for all iterations $t \in [T_{1,2,1,s}, T_{1,2,2,s})$, then given $\mathbf{Z}^{2,1} \in \mathcal{E}_3$, we have

1. for the prediction j_2 , $r \in \widehat{\mathfrak{A}}_{j_2} \setminus \{r_{g_2 \cdot y_1}\}$ cannot be activated, moreover, we have

$$\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} = \text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \cdot 2B \pm O(\delta).$$

2. for the prediction $\tau(g_1(y_1))$, $r \in \widehat{\mathfrak{A}}_{\tau(g_1(y_1))} \setminus \{r_{g_1 \cdot y_1}\}$ cannot be activated, moreover, we have

$$\Lambda_{5,\tau(g_1(y_1)),r_{g_1 \cdot y_1}}^{(t)} = \text{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot 2B \pm O(\delta).$$

3. for other $j = g(y_1)$, where $g_1, g_2 \notin \text{Fiber}_{j,y_1}$, we have

$$\Lambda_{5,j,r_{g \cdot y_1}}^{(t)} = \left(\text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \right) \cdot \frac{2B}{n_y} + \widetilde{O}\left(\frac{B}{d \cdot n_y}\right) + O(\delta);$$

moreover, if there exists y , s.t., $g_1, g_2 \in \text{Fiber}_{j,y}$ (notice that $y \neq y_1$ for $\mathcal{E}_3$), we have

$$\Lambda_{5,j,r_{g_2 \cdot y}}^{(t)} = \left(\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \right) \cdot 2B + \widetilde{O}\left(\frac{B}{d \cdot n_y}\right) + O(\delta);$$

besides, other $r \in \widehat{\mathfrak{A}}_j$ cannot be activated.

Lemma 5.B.107. If Induction 5.B.14 holds for all iterations $t \in [T_{1,2,1,s}, T_{1,2,2,s})$, then given $\mathbf{Z}^{2,1} \in \mathcal{E}_4$, we have

1. for the prediction j_2 , $r \in \widehat{\mathfrak{A}}_{j_2} \setminus \{r_{g_2 \cdot y_1}\}$ cannot be activated, moreover, we have

$$\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} = \text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \cdot 2B + O(\delta).$$

2. for the prediction $\tau(g_2(y_0))$, $r \in \widehat{\mathfrak{A}}_{\tau(g_2(y_0))} \setminus \{r_{g_2 \cdot y_0}\}$ cannot be activated, moreover, we have

$$\Lambda_{5,j'_2,r_{g_2 \cdot y_0}}^{(t)} = \left(\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} + \text{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \right) \cdot 2B + O(\delta).$$

3. for other $j \in \tau(\mathcal{Y})$, if there are j and y , s.t., $g_1, g_2 \in \text{Fiber}_{j,y}$, then for such a j , $r \in \widehat{\mathfrak{A}}_j \setminus \{r_{g_2 \cdot y}\}$ cannot be activated, moreover, we have

$$\Lambda_{5,j,r_{g_2 \cdot y}}^{(t)} = \left(\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \right) \cdot 2B + \widetilde{O}\left(\frac{B}{d \cdot n_y}\right) + O(\delta);$$

else, none of $r \in \widehat{\mathfrak{A}}_j$ can be activated.

Lemma 5.B.108. *If Induction 5.B.14 holds for all iterations $t \in [T_{1,2,1,s}, T_{1,2,2,s})$, then given $\mathbf{Z}^{2,1} \in \mathcal{E}_5$, we have*

1. *for the prediction j_2 , $r \in \widehat{\mathfrak{A}}_{j_2} \setminus \{r_{g_2 \cdot y_1}\}$ cannot be activated, moreover, we have*

$$\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} = \text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \cdot 2B \pm O(\delta).$$

2. *for the prediction $\tau(g_2(y_0))$, $r \in \widehat{\mathfrak{A}}_{\tau(g_2(y_0))} \setminus \{r_{g_2 \cdot y_0}\}$ cannot be activated, moreover, we have*

$$\Lambda_{5,j'_2,r_{g_2 \cdot y_0}}^{(t)} = \left(\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \right) \cdot 2B + \widetilde{O}\left(\frac{B}{d \cdot n_y}\right) + O(\delta).$$

3. *for the prediction $\tau(g_1(y_0))$, none of $r \in \widehat{\mathfrak{A}}_{\tau(g_2(y_0))}$ can be activated.*
4. *for other $j \in \tau(\mathcal{Y})$, if there are j and y , s.t., $g_1, g_2 \in \text{Fiber}_{j,y}$, then for such a j , $r \in \widehat{\mathfrak{A}}_j \setminus \{r_{g_2 \cdot y}\}$ cannot be activated, moreover, we have*

$$\Lambda_{5,j,r_{g_2 \cdot y}}^{(t)} = \left(\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \right) \cdot 2B + \widetilde{O}\left(\frac{B}{d \cdot n_y}\right) + O(\delta);$$

else, none of $r \in \widehat{\mathfrak{A}}_j$ can be activated.

Lemma 5.B.109. *If Induction 5.B.14 holds for all iterations $t \in [T_{1,2,1,s}, T_{1,2,2,s})$, then given $\mathbf{Z}^{2,1} \in \mathcal{E}_6$, we have*

1. *for the prediction j_2 , $r \in \widehat{\mathfrak{A}}_{j_2} \setminus \{r_{g_2 \cdot y_1}\}$ cannot be activated, moreover, we have*

$$\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} = \left(\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} + \text{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \right) \cdot 2B \pm O(\delta).$$

2. *for other $j = g(y_1)$, where $g_2 \notin \text{Fiber}_{j,y_1}$, we have $r_{g \cdot y_1}$ cannot be activated, moreover, if there exists $y \neq y_1$, s.t., $g_1, g_2 \in \text{Fiber}_{j,y}$, we have*

$$\Lambda_{5,j,r_{g_2 \cdot y}}^{(t)} = \left(\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \right) \cdot 2B + \widetilde{O}\left(\frac{B}{d \cdot n_y}\right) + O(\delta);$$

besides, other $r \in \widehat{\mathfrak{A}}_j$ cannot be activated.

Gradient Lemma

Lemma 5.B.110. *If Induction 5.B.14 holds for all iterations $t \in [T_{1,2,1,s}, T_{1,2,2,s})$, given $s \in \tau(X)$, if $\mathbb{E}[\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \mid \tau(x_1) = s] \leq \frac{1}{2}$ we have*

$$\begin{aligned} & \left[-\nabla_{\mathbf{Q}_{4,3}^{(t)}} \text{Loss}_5^{2,2} \right]_{s,s} + \left[-\nabla_{\mathbf{Q}_{4,4}^{(t)}} \text{Loss}_5^{2,2} \right]_{s,s} \\ & \geq \Omega\left(\frac{B}{d}\right) \cdot \mathbb{E}\left[(1 - \text{logit}_{5,j_2}^{(t)}) \mid \tau(x_1) = s, \mathcal{E}_1\right] + \Omega\left(\frac{B}{n_y d}\right) \cdot \sum_{m=2}^3 \mathbb{E}\left[(1 - \text{logit}_{5,j_2}^{(t)}) \mid \tau(x_1) = s, \mathcal{E}_m\right]. \end{aligned}$$

Proof. The analysis follows the similar idea as Lemma 5.B.98, while Induction 5.B.14 ensures that $\mathbb{E}[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \mid \tau(x_1) = s] \geq \Omega(1) > \frac{\rho}{B}$, which falls into the regime (ii). Thus for the main event $\mathcal{E}_1$, the term $\Lambda^{(t)}_{5,j_2,r_{g_2} \cdot y_1}$ is guaranteed to remain within the linear regime. We can first directly upper bound the term $\mathcal{N}_{s,2,iii}^{(t)}$ by $\tilde{O}(\sigma_0^q)$.

(a) For $\mathbf{Z}^{2,1} \in \mathcal{E}_1$, by Induction 5.B.14 and Lemma 5.B.104, we can obtain

$$\begin{aligned} & \mathbb{E} \left[\mathcal{N}_{s,2,i}^{(t)} \mathbb{1}_{\mathcal{E}_1} \right] \\ &= \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) \cdot \left(\left(-\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j_2,r_{g_2} \cdot y_1}(y_0) - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j_2,r_{g_2} \cdot y_1}(g_1) \right. \right. \right. \\ & \quad \left. \left. + (1 - \mathbf{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1}^{(t)} - \mathbf{Attn}_{\mathbb{A},1 \rightarrow 2}^{(t)}) \Lambda_{5,j_2,r_{g_2} \cdot y_1}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_1} \right] \quad (5.B.93) \\ &\geq \Omega\left(\frac{B}{d}\right) \cdot \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) \mid \tau(x_1) = s, \mathcal{E}_1 \right]. \end{aligned}$$

Moving to $\mathcal{N}_{s,2,iii}^{(t)}$ by Lemma 5.B.104, we have

- For $j = \tau(g_1(y_1))$ and $r = r_{g_1 \cdot y_1}$, the terms

$$\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j,r}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j,r}(g_1)$$

cancel, so

$$\begin{aligned} & \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j,r}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j,r}(g_1) \right. \\ & \quad \left. - (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5,j,r}^{(t)} \pm \tilde{O}(\sigma_0) \right) \\ & \geq -(1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5,j,r}^{(t)}. \end{aligned}$$

- for $j = \tau(g_2(y_0))$, $r = r_{g_2 \cdot y_0}$

$$\begin{aligned} & \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j,r}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j,r}(g_1) \right. \\ & \quad \left. - (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5,j,r}^{(t)} \pm \tilde{O}(\sigma_0) \right) \\ & \geq -(1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5,j,r}^{(t)}. \end{aligned}$$

- for $j = g_1(y)$, where $\exists y \neq y_0, y_1$, s.t., $g_2(y) = g_1(y)$, we have

$$\begin{aligned} & \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j,r}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j,r}(g_1) \right. \\ & \quad \left. - (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5,j,r}^{(t)} \pm \tilde{O}(\sigma_0) \right) \\ & \geq -(1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5,j,r}^{(t)}. \end{aligned}$$

Putting them together, and upper bound $\sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j}^{(t)}$ by $1 - \mathbf{logit}_{5,j_2}^{(t)}$, we have

$$\begin{aligned}
& \mathbb{E} \left[\mathcal{N}_{s,2,ii}^{(t)} \mathbb{1}_{\mathcal{E}_1} \right] \\
&= \mathbb{E} \left[\sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j}^{(t)} \cdot \left(\sum_{r \in \widehat{\mathfrak{A}}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \right. \right. \\
&\quad \left. \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j,r}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j,r}(g_1) \right. \right. \\
&\quad \left. \left. - (1 - \mathbf{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1}^{(t)} - \mathbf{Attn}_{\mathbb{A},1 \rightarrow 2}^{(t)}) \Lambda_{5,j,r}^{(t)} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_1} \Big] \\
&\geq -\mathbb{E} \left[\left(1 - \mathbf{logit}_{5,j_2}^{(t)} \right) \cdot (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \right. \\
&\quad \left. \cdot \max_{y \neq y_0, y_1} \left\{ \Lambda_{5,\tau(g_1(y_1)),r_{g_1 \cdot y_1}}^{(t)}, \Lambda_{5,\tau(g_2(y_0)),r_{g_2 \cdot y_0}}^{(t)} \right\} \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_1} \right],
\end{aligned}$$

where the last inequality is due to the fact that $\Lambda_{5,\tau(g_2(y_0)),r_{g_2 \cdot y_0}}^{(t)} = \Lambda_{5,\tau(g_2(y)),r_{g_2 \cdot y}}^{(t)} \pm \widetilde{O}(\frac{B}{d \cdot n_y})$ for $y \neq y_0, y_1$ s.t., $g_1(y) = g_2(y)$. Notice that

$$\begin{aligned}
& \max_{y \neq y_0, y_1} \left\{ \Lambda_{5,\tau(g_1(y_1)),r_{g_1 \cdot y_1}}^{(t)}, \Lambda_{5,\tau(g_2(y_0)),r_{g_2 \cdot y_0}}^{(t)} \right\} \leq \\
& \max \left\{ \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)}, \Theta\left(\frac{1}{n_y}\right) \right\} 2B,
\end{aligned}$$

while

$$\begin{aligned}
(5.B.93) &\geq \mathbb{E} \left[\left(1 - \mathbf{logit}_{5,j_2}^{(t)} \right) \cdot \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot 2B \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_1} \right] \\
&= \mathbb{E} \left[\left(1 - \mathbf{logit}_{5,j_2}^{(t)} \right) \cdot (1 - \mathbf{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1}^{(t)} - \mathbf{Attn}_{\mathbb{A},1 \rightarrow 2}^{(t)}) B \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_1} \right].
\end{aligned}$$

Since by Induction 5.B.14, $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \leq c_1 \ll \frac{1}{2}$, thus we have

$$\mathbb{E} \left[\mathcal{N}_{s,2,i}^{(t)} \mathbb{1}_{\mathcal{E}_1} \right] + \mathbb{E} \left[\mathcal{N}_{s,2,ii}^{(t)} \mathbb{1}_{\mathcal{E}_1} \right] \geq \Omega\left(\frac{B}{d}\right) \cdot \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) \mathbb{1}_{\tau(x_1)=s, \mathcal{E}_1} \right].$$

(b) For $\mathbf{Z}^{2,1} \in \mathcal{E}_2$, $\mathcal{N}_{s,2,i}^{(t)}$ can be bounded analogously to (5.B.93), yielding

$$\mathbb{E} \left[\mathcal{N}_{s,2,i}^{(t)} \mathbb{1}_{\mathcal{E}_2} \right] \geq \Omega\left(\frac{B}{d \cdot n_y}\right) \cdot \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) \mathbb{1}_{\tau(x_1)=s, \mathcal{E}_2} \right] \geq 0.$$

Moving to $\mathcal{N}_{s,2,iii}^{(t)}$

- For $j = \tau(g_1(y_1))$, we obtain

$$\begin{aligned}
&= \mathbb{E} \left[\mathbf{logit}_{5,j}^{(t)} \left(\mathbf{sReLU}'(\Lambda_{5,j,r_{\mathcal{S}_1 \cdot y_1}}) \cdot \right. \right. \\
&\quad \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j,r_{\mathcal{S}_1 \cdot y_1}}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j,r_{\mathcal{S}_1 \cdot y_1}}(g_1) \right. \\
&\quad \left. \left. - (1 - \mathbf{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1}^{(t)} - \mathbf{Attn}_{\mathbb{A},1 \rightarrow 2}^{(t)}) \Lambda_{5,j,r}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_2} \Big] \\
&\geq -O\left(\frac{B}{n_y^2 \cdot d}\right) \cdot \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) \mid \tau(x_1) = s, \mathcal{E}_2 \right],
\end{aligned} \tag{5.B.94}$$

where the inequality follows from the cancellation of the term

$$\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j,r_{\mathcal{S}_1 \cdot y_1}}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j,r_{\mathcal{S}_1 \cdot y_1}}(g_1),$$

together with the fact that

$$\Lambda_{5,\tau(g_1(y_1)),r_{\mathcal{S}_1 \cdot y_1}}^{(t)} = \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \right) \cdot \frac{2B}{n_y} + O(\delta) \leq O\left(\frac{B}{n_y}\right).$$

- For $j'_2 = \tau(g_2(y_0)) = \tau(g_1(y_0))$, it is clear that

$$\begin{aligned}
&\mathbb{E} \left[\mathbf{logit}_{5,j'_2}^{(t)} \left(\mathbf{sReLU}'(\Lambda_{5,j'_2,r_{\mathcal{S}_1 \cdot y_0}}) \cdot \right. \right. \\
&\quad \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j'_2,r_{\mathcal{S}_1 \cdot y_0}}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j'_2,r_{\mathcal{S}_1 \cdot y_0}}(g_1) \right. \\
&\quad \left. \left. - (1 - \mathbf{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1}^{(t)} - \mathbf{Attn}_{\mathbb{A},1 \rightarrow 2}^{(t)}) \Lambda_{5,j'_2,r_{\mathcal{S}_1 \cdot y_0}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_2} \Big] \\
&\geq 0,
\end{aligned}$$

where the last inequality is due to the fact that

$$\begin{aligned}
&\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j'_2,r_{\mathcal{S}_1 \cdot y_0}}(g_1) - (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5,j'_2,r_{\mathcal{S}_1 \cdot y_0}}^{(t)} \\
&\geq \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \left(1 - 2 \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \right) \right) 2B \\
&\geq \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \left(1 - 2(c_1 + 0.2) \right) 2B \geq 0.
\end{aligned}$$

- For $j = \tau(g_2(y))$ with some $y \neq y_0, y_1$ such that $g_1(y) = g_2(y)$, we distinguish two cases:

▪ If

$$\mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \mid \tau(x_1) = s \right] \leq \frac{c_1}{4},$$

then by Induction 5.B.14 and Lemma 5.B.105, we have we have $\Lambda_{5,j_2',r_{g_2 \cdot y_0}}^{(t)} - \Lambda_{5,j,r_{g_2 \cdot y}}^{(t)} \geq (-\frac{c_1}{6} + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)})2B \geq (-\frac{c_1}{4} + \frac{3}{4}c_1)2B = c_1B$, which implies

$$\sum_{y \neq y_0, y_1} \mathbf{logit}_{5,\tau(g_2(y))}^{(t)} \mathbb{1}_{g_1(y)=g_2(y)} \ll \tilde{O}\left(\frac{1}{d^{1/2}}\right) \cdot \mathbf{logit}_{5,j_2'}^{(t)}.$$

Consequently,

$$\begin{aligned} \sum_{y \neq y_0, y_1} \left| \mathbb{E} \left[\mathbf{logit}_{5,j}^{(t)} \left(\mathbf{sReLU}'(\Lambda_{5,j,r_{g_1 \cdot y}}^{(t)}) \cdot \right. \right. \right. \\ \left. \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j,r_{g_1 \cdot y}}(y_0) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j,r_{g_1 \cdot y}}(g_1) \right. \right. \\ \left. \left. - (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5,j,r_{g_1 \cdot y}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \\ \left. \left. \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_2} \mathbb{1}_{g_1(y)=g_2(y)} \right] \right| \leq O\left(\frac{1}{d^{1/2}}\right) \cdot (5.B.94). \end{aligned}$$

▪ If

$$\mathbb{E}[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \mid \tau(x_1) = s] \geq \frac{c_1}{4},$$

then by Lemma 5.B.105, the activation $\Lambda_{5,j,r_{g_1 \cdot y}}^{(t)}$ cannot be triggered.

Putting the above discussion together, we have

$$\mathbb{E} \left[\mathcal{N}_{s,2,ii}^{(t)} \mathbb{1}_{\mathcal{E}_1} \right] + \mathbb{E} \left[\mathcal{N}_{s,2,ii}^{(t)} \mathbb{1}_{\mathcal{E}_2} \right] \geq \Omega\left(\frac{B}{d \cdot n_y}\right) \cdot \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) \mid \tau(x_1) = s, \mathcal{E}_2 \right].$$

(c) For $\mathbf{Z}^{2,1} \in \mathcal{E}_3$, the gradient admits an analysis analogous to that in Lemma 5.B.98. In particular, by Induction 5.B.14 and Lemma 5.B.106, we first obtain the following logit bound:

$$\mathbf{logit}_{5,j}^{(t)} \leq \frac{1}{\text{poly}d} \cdot (1 - \mathbf{logit}_{5,j_2}^{(t)}), \quad j \neq j_2, \tau(g_1(y_1)).$$

Therefore,

$$\begin{aligned} & \mathbb{E} \left[\mathcal{N}_{s,2,i}^{(t)} \mathbb{1}_{\mathcal{E}_3} \right] \\ &= \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) \cdot \left(-\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j_2,r_{g_2 \cdot y_1}}(y_1) - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{j_2,r_{g_2 \cdot y_1}}(g_1) \right. \right. \\ & \quad \left. \left. + (1 - \mathbf{Attn}_{\text{A},1 \rightarrow \text{A},1}^{(t)} - \mathbf{Attn}_{\text{A},1 \rightarrow \text{A},2}^{(t)}) \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} \pm \tilde{O}(\sigma_0) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_3} \right] \\ &\geq \Omega\left(\frac{B}{n_y \cdot d}\right) \cdot \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) \mid \tau(x_1) = s, \mathcal{E}_3 \right]. \end{aligned}$$

Moving to $\mathcal{N}_{s,2,ii}^{(t)}$, we have

$$\begin{aligned} & \mathbb{E} \left[\mathcal{N}_{s,2,ii}^{(t)} \mathbb{1}_{\mathcal{E}_3} \right] \\ & \geq \mathbb{E} \left[\mathbf{logit}_{5,\tau(g_1(y_1))}^{(t)} \cdot \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{\tau(g_1(y_1)),r_{g_1 \cdot y_1}}(y_1) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} \cdot V_{\tau(g_1(y_1)),r}(g_1) \right. \right. \\ & \quad \left. \left. - (1 - \mathbf{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1}^{(t)} - \mathbf{Attn}_{\mathbb{A},1 \rightarrow 2}^{(t)}) \Lambda_{5,\tau(g_1(y_1)),r_{g_1 \cdot y_1}}^{(t)} \pm \tilde{O}(\sigma_0) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_3} \right] \\ & \quad - O \left(\frac{B}{d \cdot \text{poly}d} \right) \cdot \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) \mid \tau(x_1) = s, \mathcal{E}_3 \right]. \end{aligned}$$

Combining the two bounds, we conclude

$$\mathbb{E} \left[\mathcal{N}_{s,2,i}^{(t)} \mathbb{1}_{\mathcal{E}_3} \right] + \mathbb{E} \left[\mathcal{N}_{s,2,ii}^{(t)} \mathbb{1}_{\mathcal{E}_3} \right] \geq \Omega \left(\frac{B}{n_y \cdot d} \right) \cdot \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) \mid \tau(x_1) = s, \mathcal{E}_3 \right].$$

(d) For $\mathbf{Z}^{2,1} \in \mathcal{E}_4$, in analogy with Lemma 5.B.98, we obtain the following crude bound:

$$\mathbb{E} \left[\mathcal{N}_{s,2,i}^{(t)} \mathbb{1}_{\mathcal{E}_4} \right] + \mathbb{E} \left[\mathcal{N}_{s,2,ii}^{(t)} \mathbb{1}_{\mathcal{E}_4} \right] \geq 0.$$

(e) For $\mathbf{Z}^{2,1} \in \mathcal{E}_5 \cup \mathcal{E}_6$, by Induction 5.B.14, Lemma 5.B.108 and Lemma 5.B.109, we obtain the following logit condition:

$$1 - \mathbf{logit}_{5,j_2}^{(t)}, \mathbf{logit}_{5,j}^{(t)} \leq \frac{1}{\text{poly}d} \cdot \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) \mid \tau(x_1) = s, \mathcal{E}_1 \right], \quad j \neq j_2.$$

Hence, $\mathcal{N}_{s,2,i}^{(t)}$ and $\mathcal{N}_{s,2,ii}^{(t)}$ can be bounded as

$$\begin{aligned} & \left| \mathbb{E} \left[\mathcal{N}_{s,2,i}^{(t)} \mathbb{1}_{\mathcal{E}_m} \right] \right|, \quad \left| \mathbb{E} \left[\mathcal{N}_{s,2,ii}^{(t)} \mathbb{1}_{\mathcal{E}_m} \right] \right| \\ & \leq O \left(\frac{B}{dn_y} \cdot \frac{1}{\text{poly}d} \right) \cdot \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) \mid \tau(x_1) = s, \mathcal{E}_1 \right], \quad m \in \{5, 6\}. \end{aligned}$$

Moreover, for $\mathbf{Z}^{2,1} \in \mathcal{E}_6$, if

$$\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} > \frac{1}{2},$$

then $\mathcal{N}_{s,2,i}^{(t)} = 0$. Nevertheless, due to the above order-wise bound, this observation does not affect the overall analysis.

Putting everything together, we have

$$\begin{aligned} \left[-\nabla_{\mathbf{Q}_{4,3}^{(t)}} \mathbb{L}_5^{2,2} \right]_{s,s} + \left[-\nabla_{\mathbf{Q}_{4,4}^{(t)}} \mathbb{L}_5^{2,2} \right]_{s,s} &= \sum_{\kappa \in \{i, ii, iii\}} \sum_{m \in [6]} \mathbb{E} \left[\mathcal{N}_{s,2,\kappa}^{(t)} \mathbb{1}_{\mathcal{E}_m} \right] \\ &\geq \Omega\left(\frac{B}{d}\right) \cdot \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) | \tau(x_1) = s, \mathcal{E}_1 \right] + \Omega\left(\frac{B}{n_y d}\right) \cdot \sum_{m=2}^3 \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) | \tau(x_1) = s, \mathcal{E}_m \right]. \end{aligned}$$

□

Lemma 5.B.111. *If Induction 5.B.14 holds for all iterations $t \in [T_{1,2,1,s}, T_{1,2,2,s})$, given $s \in \tau(\mathcal{X})$, for $[\mathbf{Q}_{4,p}]_{s,s'}, p \in \{3, 4\}$, $s' \neq s \in \tau(\mathcal{X})$, we have*

$$\left| \left[-\nabla_{\mathbf{Q}_{4,p}^{(t)}} \text{Loss}_5^{2,2} \right]_{s,s'} \right| \leq O\left(\frac{1}{d}\right) \left| \left[-\nabla_{\mathbf{Q}_{4,p}^{(t)}} \text{Loss}_5^{2,2} \right]_{s,s} \right|.$$

Attention gap is small

Lemma 5.B.112. *If Induction 5.B.14 holds for all iterations $t \in [T_{1,2,1,s}, T_{1,2,2,s})$, then for any sample $\mathbf{Z}^{2,1}$, we have*

$$\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \leq c_1,$$

where $c_1 > 0$ is a small constant.

The proof follows along the same lines as Lemma 5.B.101, and hence the details are omitted for brevity.

Lemma 5.B.113. *If Induction 5.B.14 holds for all iterations $t \in [T_{1,2,1,s}, T_{1,2,2,s})$, then for any sample $\mathbf{Z}^{2,1}$, we have*

$$\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \leq \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - c_2,$$

where $c_2 = \frac{\log d}{4B} > 0$ is a small constant.

Proof. Let $\tilde{T}$ denote the first time such that

$$\mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \mid \tau(x_1) = s \right] \leq \frac{\log d}{4.02B}.$$

Since $|\left[\mathbf{Q}_{4,p}^{(t)}\right]_{s,s'}| \leq O\left(\frac{|\mathbf{Q}_{4,p}^{(t)}|_{s,s}}{d}\right)$ for $p \in \{3, 4\}$, it follows that for any sample $\mathbf{Z}^{2,1}$ with $\tau(x_1) = s$, at time $\tilde{T}$ we have

$$\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \geq \frac{\log d}{4.02B} \pm \tilde{O}(1/d).$$

By Lemma 5.B.103, we also have $\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(\tilde{T})} \geq \Omega(1) \gg \varrho$. We then consider the following cases:

- If $\mathbf{Z}^{2,1} \in \mathcal{E}_1$, then $\Lambda_{5,j_2,r_{s_2,y_1}}^{(\tilde{T})}$ is already activated into the linear regime. Moreover, since $\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(\tilde{T})} \leq \frac{\log d}{4.02B} - \text{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} < 0$, it follows that $\Lambda_{5,j'_2,r_{s_2,y_0}}^{(\tilde{T})}$ cannot be activated. Thus, by Lemma 5.B.90,

$$\mathbb{E} \left[(\mathcal{N}_{s,3,2,i}^{(\tilde{T})} + \mathcal{N}_{s,3,2,ii}^{(\tilde{T})} + \mathcal{N}_{s,3,2,iii}^{(\tilde{T})}) \mathbb{1}_{\mathcal{E}_1} \right] \geq \Omega\left(\frac{B}{d}\right) \cdot \mathbb{E} \left[1 - \text{logit}_{5,j_2}^{(\tilde{T})} \mid \tau(x_1) = s, \mathcal{E}_1 \right],$$

while

$$\mathbb{E} \left[(\mathcal{N}_{s,4,2,i}^{(\tilde{T})} + \mathcal{N}_{s,4,2,ii}^{(\tilde{T})} + \mathcal{N}_{s,4,2,iii}^{(\tilde{T})}) \mathbb{1}_{\mathcal{E}_1} \right] \leq -\Omega\left(\frac{B}{dn_y}\right) \cdot \mathbb{E} \left[1 - \text{logit}_{5,j_2}^{(\tilde{T})} \mid \tau(x_1) = s, \mathcal{E}_1 \right].$$

- If $\mathbf{Z}^{2,1} \in \mathcal{E}_2$, then

$$\Lambda_{5,j'_2,r_{s_2,y_0}}^{(\tilde{T})} = (\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} + \text{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(\tilde{T})} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(\tilde{T})})2B \leq \frac{\log d}{2.01}.$$

Hence $\text{logit}_{5,j_2}^{(1)} \leq O(d^{-1.01/2.01})(1 - \text{logit}_{5,j_2}^{(t)})$. By Lemma 5.B.90,

$$\mathbb{E} \left[(\mathcal{N}_{s,3,2,i}^{(\tilde{T})} + \mathcal{N}_{s,3,2,ii}^{(\tilde{T})} + \mathcal{N}_{s,3,2,iii}^{(\tilde{T})}) \mathbb{1}_{\mathcal{E}_2} \right] \geq \Omega\left(\frac{B}{dn_y}\right) \cdot \mathbb{E} \left[1 - \text{logit}_{5,j_2}^{(\tilde{T})} \mid \tau(x_1) = s, \mathcal{E}_2 \right],$$

and

$$\mathbb{E} \left[(\mathcal{N}_{s,4,2,i}^{(\tilde{T})} + \mathcal{N}_{s,4,2,ii}^{(\tilde{T})} + \mathcal{N}_{s,4,2,iii}^{(\tilde{T})}) \mathbb{1}_{\mathcal{E}_2} \right] \leq -\Omega\left(\frac{B}{dn_y}\right) \cdot \mathbb{E} \left[1 - \text{logit}_{5,j_2}^{(\tilde{T})} \mid \tau(x_1) = s, \mathcal{E}_2 \right].$$

- If $\mathbf{Z}^{2,1} \in \mathcal{E}_3$, we can apply the crude bounds

$$\mathbb{E} \left[(\mathcal{N}_{s,3,2,i}^{(\tilde{T})} + \mathcal{N}_{s,3,2,ii}^{(\tilde{T})} + \mathcal{N}_{s,3,2,iii}^{(\tilde{T})}) \mathbb{1}_{\mathcal{E}_3} \right] \geq 0,$$

and

$$\mathbb{E} \left[(\mathcal{N}_{s,4,2,i}^{(\tilde{T})} + \mathcal{N}_{s,4,2,ii}^{(\tilde{T})} + \mathcal{N}_{s,4,2,iii}^{(\tilde{T})}) \mathbb{1}_{\mathcal{E}_3} \right] \leq 0.$$

Combining the above, and noting that the gradient contribution from $\mathcal{E}_4 \cup \mathcal{E}_5 \cup \mathcal{E}_6$ is negligible, we conclude

$$\left[-\nabla_{\mathbf{Q}_{4,3}^{(\tilde{T})}} \text{Loss}_5^{2,2} \right]_{s,s} \geq \Omega\left(\frac{B}{d}\right) \cdot \mathbb{E} \left[1 - \text{logit}_{5,j_2}^{(\tilde{T})} \mid \tau(x_1) = s, \mathcal{E}_1 \right] \quad (5.B.95)$$

$$+ \Omega\left(\frac{B}{dn_y}\right) \cdot \mathbb{E} \left[1 - \text{logit}_{5,j_2}^{(\tilde{T})} \mid \tau(x_1) = s, \mathcal{E}_2 \right],$$

$$\left[-\nabla_{\mathbf{Q}_{4,4}^{(\tilde{T})}} \text{Loss}_5^{2,2} \right]_{s,s} \leq -\Omega\left(\frac{B}{d}\right) \cdot \mathbb{E} \left[1 - \text{logit}_{5,j_2}^{(\tilde{T})} \mid \tau(x_1) = s, \mathcal{E}_1 \right] \quad (5.B.96)$$

$$- \Omega\left(\frac{B}{dn_y}\right) \cdot \mathbb{E} \left[1 - \text{logit}_{5,j_2}^{(\tilde{T})} \mid \tau(x_1) = s, \mathcal{E}_2 \right].$$

Finally, observe that

$$\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} = \frac{e^{[\mathbf{Q}_{4,3}^{(t)}]_{s,s}} - e^{[\mathbf{Q}_{4,4}^{(t)}]_{s,s}} - e^{\tilde{O}(1/d)}}{e^{[\mathbf{Q}_{4,3}^{(t)}]_{s,s}} + e^{[\mathbf{Q}_{4,4}^{(t)}]_{s,s}} + e^{\tilde{O}(1/d)}}.$$

Hence, (5.B.95) and (5.B.96) together imply that

$$\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)}$$

must decrease at time $\tilde{T} + 1$. □

At the End of Training

Lemma 5.B.114 (At the end of stage 1.2.2). *Induction 5.B.14 holds for all iterations $T_{1,2,1,s} < t \leq T_{1,2,2,s} = O\left(\frac{\text{poly}(d)}{\eta B}\right)$. At the end of training, we have*

- (a) *Attention concentration: $\mathbb{E}[\epsilon_{\text{attn}} \mid \tau(x_1) = s] \leq 2.52c_1$ for some small constant $0 < c_1 = \frac{1.005 \log d}{2B}$;*
- (b) *Loss convergence: $\text{Loss}_{5,s}^{2,2} \leq e^{(-\frac{1}{2} + 3.01c_1) \cdot 2B} = \frac{1}{\text{poly} d}$.*

Proof. Lemma 5.B.103 and Lemma 5.B.110 guarantee the continual growth of $[\mathbf{Q}_{4,3}^{(t)}]$ and $[\mathbf{Q}_{4,4}^{(t)}]$ until the attention weight ϵ_{attn} reaches $\frac{4c_1}{3}$. Combining Lemma 5.B.110, Lemma 5.B.111, Lemma 5.B.112, and Lemma 5.B.113, we conclude that there exists a stopping time

$$T_{1,2,2,s} = O\left(\frac{\text{poly}(d)}{\eta B}\right),$$

and that Induction 5.B.14 holds for all $t \in [T_{1,2,1,s}, T_{1,2,2,s})$.

Next, we establish an upper bound for ϵ_{attn} at the end of training. In this stage, it suffices to focus on the main event $\mathcal{E}_1$. We denote

$$\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1'}^{(t)}, \quad \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} = c + \tilde{O}\left(\frac{1}{d}\right).$$

- If $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \geq \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)}$, then for $\mathbf{Z}^{2,1} \in \mathcal{E}_1$ we have

$$\begin{aligned} \log p_F(\mathbf{Z}_{\text{ans},2,5} \mid \mathbf{Z}^{(2,1)}) &\geq \Theta(1) \cdot e^{\log d - (\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)}) 2B} \\ &\geq \Theta\left(e^{-\left(\frac{1}{2} + \frac{c_1}{2} - 2c - \frac{\log d}{B}\right) 2B}\right), \end{aligned}$$

where the last inequality follows from

$$\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \leq \frac{1}{2}(1 - 2c + c_1) = \frac{1+c_1}{2} - c.$$

- If $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \leq \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)}$, then for $\mathbf{Z}^{2,1} \in \mathcal{E}_1$ we have

$$\begin{aligned} \log p_F(\mathbf{Z}_{\text{ans},2,5} | \mathbf{Z}^{2,1}) &\geq \Theta(1) \cdot e^{\log d - (\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)})2B} \\ &\geq \Theta \left(e^{-\left(\frac{1}{2} - 2c - \frac{\log d}{B}\right)2B} \right), \end{aligned}$$

where the last inequality follows from the fact that $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \leq \frac{1}{2} - c$.

Putting the above cases together, we obtain

$$\frac{1}{2} + \frac{c_1}{2} - 2c - \frac{\log d}{B} \geq \frac{1}{2} - 3.01c_1,$$

which implies

$$c \leq \frac{1}{2}(3.02c_1 + 0.5c_1 - c_1) = 1.26c_1.$$

□

5.B.6 Recursive Learning the Attention Layer: Symmetry Case

In this section, we analyze the recursive learning dynamics of the attention layer in the symmetry case. To this end, we recall the definitions of the greedy data annotator, the bootstrapped LEGO data distribution, and the associated self-training loss.

Definition 5.B.16 (Greedy data annotator). The greedy language model $\widehat{p}_F$ induced by a network F is defined as

$$\widehat{p}_F(Z_{\text{ans},L'+1} | Z^{L,L'}) = \begin{cases} 1, & \text{if } Z_{\text{ans},L'+1} = \arg\max_Z p_F(Z | Z^{L,L'}), \\ 0, & \text{otherwise.} \end{cases} \quad (5.B.97)$$

Definition 5.B.17 (Bootstrapped LEGO distribution). We define $\mathcal{D}_F^{L,L'}$ as the LEGO distribution in Assumption 5.3.2, except that the answers $Z_{\text{ans},\ell}$, $1 \leq \ell \leq L'$ are generated recursively by sampling $Z_{\text{ans},\ell} \sim \widehat{p}_F(\cdot | Z^{L,\ell-1})$ from the greedy language model $\widehat{p}_F$.

Definition 5.B.18 (Self-training loss). Given a (fixed) model $\widetilde{F}$ and length L , the self-training next-clause-prediction loss is defined by replacing $\mathcal{D}^{L,L'}$ in Definition 5.3.7 with the bootstrapped distribution $\mathcal{D}_F^{L,L'}$ from Definition 5.B.17:

$$\text{Loss}_{\widetilde{F}}^{L,L'}(F) \triangleq \mathbb{E}_{Z^{L,L'} \sim \mathcal{D}_{\widetilde{F}}^{L,L'}} \left[-\log p_F(Z_{\text{ans},L',i} | Z^{L,L'-1}) \right]. \quad (5.B.98)$$

Similarly, the per-token loss is defined by

$$\text{Loss}_{\bar{F},i}^{L,L'} = \mathbb{E}_{Z^{L,L'} \sim \mathcal{D}_{\bar{F}}^{L,L'}} \left[-\log p_{F_i}(Z_{\text{ans},L',i} \mid Z^{L,L'-1}) \right], \quad i \in [5].$$

At stage $k \geq 2$, let $F^{(T_{k-1})}$ denote the model obtained from the previous stage, trained on the task $\mathcal{T}^{L_{k-1}}$ with $L_{k-1} = 2^{k-1}$. Using the greedy annotator $\hat{p}_{F^{(T_{k-1})}}$, we construct the bootstrapped LEGO distribution $\mathcal{D}_{F^{(T_{k-1})}}^{L_k, L_k}$ with total length $L_k = 2^k$. The corresponding self-training loss $\text{Loss}_{F^{(T_{k-1})}}^{L_k, L'}$ is then defined as in Definition 5.B.18. In this stage, we focus on training via gradient descent on $\text{Loss}_{F^{(T_{k-1})},i}^{L_k, L'}$ with sequence length $L' = 2$ and target token $i = 5$, initialized from the previous-stage model $F^{(T_{k-1})}$.

For notational simplicity, throughout the discussion we drop the subscript of the expectation operator $\mathbb{E}$ when $\mathbf{Z}$ is sampled from the ground-truth LEGO distribution; the subscript will be explicitly included only when the expectation is taken over the bootstrapped distribution. Moreover, we abbreviate the wrong attention error $\epsilon_{\text{attn}}^{L_k, 2}$ as $\epsilon_{\text{attn}}^{L_k}$ and the attention gap $\Delta^{L_k, 2}$ as Δ^{L_k} when the context is clear.

Preliminaries

We first present preliminaries on the recursive learning attention layer, including its gradient computations and some useful probability lemmas.

Gradient Computations

Fact 5.B.10 (Gradients of $\mathbf{Q}$). Given $F^{(T_{k-1})}$ from the previous stage $\mathcal{T}^{L_{k-1}}$ with $L_{k-1} = 2^{k-1}$ for $k \geq 2$, for $(p, q) \in \{(4, 3), (4, 4)\}$, we have

$$-\nabla_{\mathbf{Q}_{p,q}} \text{Loss}_{F^{(T_{k-1})},5}^{L_k, 2} = -\nabla_{\mathbf{Q}_{p,q}} \text{Loss}_5^{L_k, 2}.$$

Remark 5.B.5. This fact is straightforward to verify. Since the attention error $\epsilon_{\text{attn}}^{L_{k-1}, 2}$ has already been controlled at a small constant level in the previous stage for the model $F^{(T_{k-1})}$, when transitioning from length 2^{k-1} to 2^k , the incorrect attention produced by $F^{(T_{k-1})}$ cannot increase significantly. In particular, we have

$$\epsilon_{\text{attn}}^{L_k} \leq 2\epsilon_{\text{attn}}^{L_{k-1}},$$

which remains small. Moreover, the attention gap $\Delta^{L_k} = |\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2} - \text{Attn}_{\text{ans},1 \rightarrow \text{ans},1}|$ is non-increasing due to the doubling of irrelevant clauses. As a result, the probability assigned by $F^{(T_{k-1})}$ to the correct prediction of $Z_{\text{ans},2,5}$ given $Z^{L_{k-1}}$ remains significantly larger than that of any incorrect prediction. Consequently, under the greedy data annotator, the self-training loss $\text{Loss}_{F^{(T_{k-1})},5}^{L_k, 2}$ coincides with the original loss $\text{Loss}_5^{L_k, 2}$ on the ground-truth LEGO distribution.

Lemma 5.B.115 (Gradients of $\mathbf{Q}_{4,3}$). *Given $F^{(T_{k-1})}$ from the previous stage and $s \in \tau(X)$, for the diagonal entry $[\mathbf{Q}_{4,3}]_{s,s}$ of the block $\mathbf{Q}_{4,3}$, we have*

$$\begin{aligned} \left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_{F^{(T_{k-1})},5}^{L_k,2} \right]_{s,s} &= \mathbb{E} \left[\text{Attn}_{\text{ans},1 \rightarrow \text{pred},2} \cdot \left(\sum_{j \in [d]} \mathcal{E}_{5,j}(\mathbf{Z}^{2,1}) \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \right. \right. \\ &\quad \left. \left. \left(\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_2 \rangle - \Lambda_{5,j,r} + b_{5,j,r} \right) \right) \mathbb{1}_{s=\tau(x_1)} \right]. \end{aligned}$$

Moreover, for the off-diagonal entries $[\mathbf{Q}_{4,3}]_{s,s'}$ with $s \neq s'$, we have

$$\begin{aligned} \left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_{F^{(T_{k-1})},5}^{L_k,2} \right]_{s,s'} &= \mathbb{E} \left[\text{Attn}_{\text{ans},1 \rightarrow \text{pred},1} \cdot \left(\sum_{j \in [d]} \mathcal{E}_{5,j}(\mathbf{Z}^{2,1}) \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \right. \right. \\ &\quad \left. \left. \left(\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_1 \rangle - \Lambda_{5,j,r} + b_{5,j,r} \right) \right) \mathbb{1}_{s=\tau(x_1), s'=\tau(x_0)} \right]. \end{aligned}$$

Lemma 5.B.116 (Gradients of $\mathbf{Q}_{4,4}$). *Given $F^{(T_{k-1})}$ from the previous stage and $s \in \tau(X)$, for the diagonal entry $[\mathbf{Q}_{4,4}]_{s,s}$ of the block $\mathbf{Q}_{4,4}$, we have*

$$\begin{aligned} \left[-\nabla_{\mathbf{Q}_{4,4}} \text{Loss}_{F^{(T_{k-1})},5}^{L_k,2} \right]_{s,s} &= \mathbb{E} \left[\text{Attn}_{\text{ans},1 \rightarrow \text{ans},1} \cdot \left(\sum_{j \in [d]} \mathcal{E}_{5,j}(\mathbf{Z}^{2,1}) \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \right. \right. \\ &\quad \left. \left. \left(\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_{\mathbb{A},1} \rangle - \Lambda_{5,j,r} + b_{5,j,r} \right) \right) \mathbb{1}_{s=\tau(x_1)} \right]. \end{aligned}$$

Moreover, for the off-diagonal entries $[\mathbf{Q}_{4,4}]_{s,s'}$ with $s \neq s'$, we have

$$\begin{aligned} \left[-\nabla_{\mathbf{Q}_{4,4}} \text{Loss}_{F^{(T_{k-1})},5}^{L_k,2} \right]_{s,s'} &= \mathbb{E} \left[\text{Attn}_{\text{ans},1 \rightarrow \text{ans},0} \cdot \left(\sum_{j \in [d]} \mathcal{E}_{5,j}(\mathbf{Z}^{2,1}) \sum_{r \in [m]} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \right. \right. \\ &\quad \left. \left. \left(\langle \mathbf{W}_{5,j,r}, \mathbf{Z}_{\mathbb{A},0} \rangle - \Lambda_{5,j,r} + b_{5,j,r} \right) \right) \mathbb{1}_{s=\tau(x_1), s'=\tau(x_0)} \right]. \end{aligned}$$

Notations for activated neurons. Recall that

$$\mathfrak{A} = \bigcup_{j \in \tau(\mathcal{Y})} \mathfrak{A}_j, \quad \text{where } \mathfrak{A}_j = \{ r_{j,y} \mid y \in \mathcal{Y} \}.$$

Given $\mathbf{Z}^{L,\ell-1}$, let

$$\widehat{\mathcal{G}}(\mathbf{Z}^{L,\ell-1}) = \bigcup_{\ell'=1}^L \{g_{\ell'}\}$$

be the collection of all group elements chosen in the predicate clauses, and similarly define

$$\widehat{\mathcal{Y}} = \bigcup_{\ell'=0}^{\ell-1} \{y_{\ell'}\}.$$

We then introduce

$$\widehat{\mathfrak{A}}_j(\mathbf{Z}^{L,\ell-1}) = \left\{ r_{g \cdot y} \mid g \in \text{Fiber}_{j,y}, g \in \widehat{\mathcal{G}}(\mathbf{Z}^{L,\ell-1}) \vee y \in \widehat{\mathcal{Y}} \right\}.$$

For simplicity, we omit the dependence on $\mathbf{Z}^{L,\ell-1}$ in the notation of $\widehat{\mathfrak{A}}_j$ when it is clear from context.

The above recalls the relevant notations from Section 5.B.5. With these preliminaries in place, we now state the following lemma, which also relies on the feature-magnitude bounds established therein.

Lemma 5.B.117 (Characterizations of Lambda). *Given $\mathbf{Z}^{L_k,1}$ with $\{\mathbf{Attn}_{\text{ans},1 \rightarrow \mathbf{k}}\}_{\mathbf{k} \in \mathcal{I}^{L_k,1}}$, then*

(a) *for $j \in \tau(\mathcal{Y})$, for activated neuron $r \in \mathfrak{A}_j$, we have*

$$\Lambda_{5,j,r} = \sum_{\ell'=1}^2 \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},\ell'} V_{j,r}(g_{\ell'}) + \sum_{\ell'=1}^{L_k} \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},\ell'-1} V_{j,r}(y_{\ell'-1}) \pm \widetilde{O}(\sigma_0).$$

(b) *for $j \in \tau(\mathcal{Y})$, for any non-activated neuron $r \notin \mathfrak{A}_j$ we have*

$$|\Lambda_{5,j,r}| \leq O(\delta).$$

(c) *for $j \notin \tau(\mathcal{Y})$, for any $r \in [m]$, we have*

$$|\Lambda_{5,j,r}| \leq \widetilde{O}(\sigma_0).$$

Lemma 5.B.118. *Given $j \in \tau(\mathcal{Y})$, for $r \in \mathfrak{A}_j \setminus \widehat{\mathfrak{A}}_j$, we have $\mathbf{sReLU}'(\Lambda_{5,j,r}) = 0$.*

We are now ready to derive the gradients of the attention layer, building on Lemmas 5.B.115 and 5.B.116 and the properties established above.

Lemma 5.B.119 (Refined expression for the gradient of $\mathbf{Q}_{4,3}$). *Given $F^{(T_{k-1})}$ from the previous stage and $s \in \tau(\mathcal{X})$, for the diagonal entry $[\mathbf{Q}_{4,3}]_{s,s}$ of the block $\mathbf{Q}_{4,3}$, letting $j_2 = \tau(g_2(y_1))$, we have*

$$\begin{aligned} \left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_{F^{(T_{k-1})},5}^{L_k,2} \right]_{s,s} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2} \cdot \right. \\ &\quad \left. \left((1 - \mathbf{logit}_{5,j_2}) \cdot \left(\sum_{r \in \widehat{\mathfrak{A}}_{j_2}} \mathbf{sReLU}'(\Lambda_{5,j_2,r}) \cdot \left(V_{j_2,r}(g_2) - \Lambda_{5,j_2,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \right) \right] \end{aligned}$$

$$\begin{aligned}
& - \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \left(\sum_{r \in \mathfrak{A}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \left(V_{j,r}(g_2) - \Lambda_{5,j,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \\
& \pm \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \widetilde{O}(\sigma_0^q) \mathbb{1}_{\tau(x_1)=s} \Bigg].
\end{aligned}$$

Moreover, for the off-diagonal entries $[\mathbf{Q}_{4,3}]_{s,s'}$ with $s \neq s'$, we have

$$\begin{aligned}
\left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_{F^{(T_{k-1})}, 5}^{L_k, 2} \right]_{s,s'} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, 1} \cdot \right. \\
& \left((1 - \mathbf{logit}_{5,j_2}) \cdot \left(\sum_{r \in \mathfrak{A}_{j_2}} \mathbf{sReLU}'(\Lambda_{5,j_2,r}) \cdot \left(V_{j_2,r}(g_1) - \Lambda_{5,j_2,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \right. \\
& - \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \left(\sum_{r \in \mathfrak{A}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \left(V_{j,r}(g_1) - \Lambda_{5,j,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \\
& \left. \left. \pm \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \widetilde{O}(\sigma_0^q) \right) \mathbb{1}_{\tau(x_1)=s, \tau(x_0)=s'} \right].
\end{aligned}$$

Lemma 5.B.120 (Refined expression for the gradient of $\mathbf{Q}_{4,4}$). *Given $F^{(T_{k-1})}$ from the previous stage and $s \in \tau(X)$, for the diagonal entry $[\mathbf{Q}_{4,4}]_{s,s}$ of the block $\mathbf{Q}_{4,4}$, letting $j_2 = \tau(g_2(y_1))$, we have*

$$\begin{aligned}
\left[-\nabla_{\mathbf{Q}_{4,4}} \text{Loss}_{F^{(T_{k-1})}, 5}^{L_k, 2} \right]_{s,s} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{ans}, 1} \cdot \right. \\
& \left((1 - \mathbf{logit}_{5,j_2}) \cdot \left(\sum_{r \in \mathfrak{A}_{j_2}} \mathbf{sReLU}'(\Lambda_{5,j_2,r}) \cdot \left(V_{j_2,r}(y_1) - \Lambda_{5,j_2,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \right. \\
& - \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \left(\sum_{r \in \mathfrak{A}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \left(V_{j,r}(y_1) - \Lambda_{5,j,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \\
& \left. \left. \pm \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \widetilde{O}(\sigma_0^q) \right) \mathbb{1}_{\tau(x_1)=s} \right].
\end{aligned}$$

Moreover, for the off-diagonal entries $[\mathbf{Q}_{4,3}]_{s,s'}$ with $s \neq s'$, we have

$$\begin{aligned}
\left[-\nabla_{\mathbf{Q}_{4,4}} \text{Loss}_{F^{(T_{k-1})}, 5}^{L_k, 2} \right]_{s,s'} &= \mathbb{E} \left[\mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{ans}, 0} \cdot \right. \\
& \left((1 - \mathbf{logit}_{5,j_2}) \cdot \left(\sum_{r \in \mathfrak{A}_{j_2}} \mathbf{sReLU}'(\Lambda_{5,j_2,r}) \cdot \left(V_{j_2,r}(y_0) - \Lambda_{5,j_2,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \right. \\
& - \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \left(\sum_{r \in \mathfrak{A}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \left(V_{j,r}(y_0) - \Lambda_{5,j,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \\
& \left. \left. \pm \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \widetilde{O}(\sigma_0^q) \right) \mathbb{1}_{\tau(x_1)=s} \right].
\end{aligned}$$

$$\pm \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \tilde{O}(\sigma_0^q) \mathbb{1}_{\tau(x_1)=s, \tau(x_0)=s'} \Bigg].$$

Following the above calculations, we can further obtain the gradient summation of $\mathbf{Q}_{4,3}$ and $\mathbf{Q}_{4,4}$ as follows:

Lemma 5.B.121 (Gradient sum of $\mathbf{Q}_{4,3}$ and $\mathbf{Q}_{4,4}$). *Given $F^{(T_{k-1})}$ from the previous stage and $s \in \tau(\mathcal{X})$, letting $j_2 = \tau(g_2(y_1))$, we have*

$$\begin{aligned} & \left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_{F^{(T_{k-1})}, 5}^{L_k, 2} \right]_{s, s} + \left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_{F^{(T_{k-1})}, 5}^{L_k, 2} \right]_{s, s} \\ &= \mathbb{E} \left[\left((1 - \mathbf{logit}_{5, j_2}) \cdot \left(\sum_{r \in \mathfrak{A}_{j_2}} \mathbf{sReLU}'(\Lambda_{5, j_2, r}) \cdot \right. \right. \right. \\ & \quad \left(-\mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{ans}, 0} \cdot V_{j_2, r}(y_0) - \sum_{\ell \neq 2} \mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, \ell} \cdot V_{j_2, r}(g_\ell) \right. \\ & \quad \left. \left. + (1 - \mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{ans}, 1} - \mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, 2}) \Lambda_{5, j_2, r} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \\ & + \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5, j} \cdot \left(\sum_{r \in \mathfrak{A}_j} \mathbf{sReLU}'(\Lambda_{5, j, r}) \cdot \right. \\ & \quad \left(\mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{ans}, 0} \cdot V_{j, r}(y_0) + \sum_{\ell \neq 2} \mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, \ell} \cdot V_{j, r}(g_\ell) \right. \\ & \quad \left. \left. - (1 - \mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{ans}, 1} - \mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, 2}) \Lambda_{5, j, r} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \\ & \left. \pm \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5, j} \cdot (\mathbf{Attn}_{\mathbb{A}, 1 \rightarrow \mathbb{A}, 1} + \mathbf{Attn}_{\mathbb{A}, 1 \rightarrow 2}) \tilde{O}(\sigma_0^q) \right] \mathbb{1}_{\tau(x_1)=s}. \end{aligned}$$

Notations for gradient decompositions. Next we define some useful notations to further simplify the expressions of gradient.

Lemma 5.B.122. *Given $F^{(T_{k-1})}$ from the previous stage and $s \in \tau(\mathcal{X})$, we define the following notations for the gradient decompositions:*

1. for $[\mathbf{Q}_{4,3}]_{s, s}$ we have $\left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_{F^{(T_{k-1})}, 5}^{L_k, 2} \right]_{s, s} = \mathbb{E}[\mathcal{N}_{s, 3, L_k, i} + \mathcal{N}_{s, 3, L_k, ii} + \mathcal{N}_{s, 3, L_k, iii}]$, where

$$\mathcal{N}_{s, 3, L_k, i} = \mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, 2} \cdot (1 - \mathbf{logit}_{5, j_2}). \quad (5.B.99)$$

$$\left(\sum_{r \in \mathfrak{A}_{j_2}} \mathbf{sReLU}'(\Lambda_{5, j_2, r}) \cdot \left(V_{j_2, r}(g_2) - \Lambda_{5, j_2, r} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s},$$

$$\mathcal{N}_{s, 3, L_k, ii} = -\mathbf{Attn}_{\text{ans}, 1 \rightarrow \text{pred}, 2} \cdot \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5, j}. \quad (5.B.100)$$

$$\left(\sum_{r \in \widehat{\mathfrak{V}}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \left(V_{j,r}(g_2) - \Lambda_{5,j,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s},$$

$$\mathcal{N}_{s,3,L_k,iii} = \pm \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2} \cdot \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \widetilde{O}(\sigma_0^q) \mathbb{1}_{\tau(x_1)=s}. \quad (5.B.101)$$

2. for $[\mathbf{Q}_{4,4}]_{s,s}$, we have $\left[-\nabla_{\mathbf{Q}_{4,4}} \text{Loss}_{F^{(T_k-1)},5}^{L_k,2} \right]_{s,s} = \mathbb{E}[\mathcal{N}_{s,4,L_k,i} + \mathcal{N}_{s,4,L_k,ii} + \mathcal{N}_{s,4,L_k,iii}]$, where

$$\begin{aligned} \mathcal{N}_{s,4,L_k,i} &= \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1} \cdot (1 - \mathbf{logit}_{5,j_2}) \cdot \\ &\quad \left(\sum_{r \in \widehat{\mathfrak{V}}_{j_2}} \mathbf{sReLU}'(\Lambda_{5,j_2,r}) \cdot \left(V_{j_2,r}(y_1) - \Lambda_{5,j_2,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s}, \end{aligned} \quad (5.B.102)$$

$$\begin{aligned} \mathcal{N}_{s,4,L_k,ii} &= -\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1} \cdot \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \\ &\quad \left(\sum_{r \in \widehat{\mathfrak{V}}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \left(V_{j,r}(y_1) - \Lambda_{5,j,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s}, \end{aligned} \quad (5.B.103)$$

$$\mathcal{N}_{s,4,L_k,iii} = \pm \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1} \cdot \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \widetilde{O}(\sigma_0^q) \mathbb{1}_{\tau(x_1)=s}. \quad (5.B.104)$$

3. for the summation of $[\mathbf{Q}_{4,3}]_{s,s}$ and $[\mathbf{Q}_{4,4}]_{s,s}$, we have $\left[-\nabla_{\mathbf{Q}_{4,3}} \text{Loss}_5^{2,2} \right]_{s,s} + \left[-\nabla_{\mathbf{Q}_{4,4}} \text{Loss}_5^{2,2} \right]_{s,s} = \mathbb{E}[\mathcal{N}_{s,2,i} + \mathcal{N}_{s,2,ii} + \mathcal{N}_{s,2,iii}]$, where

$$\begin{aligned} \mathcal{N}_{s,L_k,i} &= (1 - \mathbf{logit}_{5,j_2}) \cdot \left(\sum_{r \in \widehat{\mathfrak{V}}_{j_2}} \mathbf{sReLU}'(\Lambda_{5,j_2,r}) \cdot \right. \\ &\quad \left(-\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0} \cdot V_{j_2,r}(y_0) - \sum_{\ell \neq 2} \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},\ell} \cdot V_{j_2,r}(g_\ell) \right. \\ &\quad \left. \left. + (1 - \mathbf{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1} - \mathbf{Attn}_{\mathbb{A},1 \rightarrow 2}) \Lambda_{5,j_2,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s}, \\ \mathcal{N}_{s,L_k,ii} &= \sum_{j \neq j_2 \in \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot \left(\sum_{r \in \widehat{\mathfrak{V}}_j} \mathbf{sReLU}'(\Lambda_{5,j,r}) \cdot \right. \\ &\quad \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0} \cdot V_{j,r}(y_0) + \sum_{\ell \neq 2} \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},\ell} \cdot V_{j,r}(g_\ell) \right. \\ &\quad \left. \left. - (1 - \mathbf{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1} - \mathbf{Attn}_{\mathbb{A},1 \rightarrow 2}) \Lambda_{5,j,r} \pm \widetilde{O}(\sigma_0) \right) \pm \widetilde{O}(\delta^q) \right) \mathbb{1}_{\tau(x_1)=s}, \\ \mathcal{N}_{s,L_k,iii} &= \pm \sum_{j \notin \tau(\mathcal{Y})} \mathbf{logit}_{5,j} \cdot (\mathbf{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1} + \mathbf{Attn}_{\mathbb{A},1 \rightarrow 2}) \widetilde{O}(\sigma_0^q) \mathbb{1}_{\tau(x_1)=s}. \end{aligned}$$

Probabilistic Events We introduce the following events:

$$\begin{aligned}\mathcal{E}_{L_k,1} &= \left\{ \frac{1}{L_k} \sum_{\ell \in [L_k] \setminus \{2\}} \mathbb{1}\{g_\ell(y_1) = g_2(y_1)\} \leq U_k \right\}, \\ \mathcal{E}_{L_k,2} &= \{y_0 = y_1\}, \\ \mathcal{E}_{L_k,3} &= \left\{ \max_{y \in \mathcal{Y}} \sum_{\ell \in [L_k] \setminus \{2\}} \mathbb{1}\{g_\ell(y) = g_2(y)\} \leq W_k \right\}.\end{aligned}$$

We set

$$U_k = \frac{1}{L_k} \left\lceil 1 + \frac{L_k}{8} \right\rceil = \Theta(1), \quad W_k = \begin{cases} \Theta\left(\frac{\log n_y}{\log\left(\frac{4n_y}{L_k} \log n_y\right)}\right), & L_k \leq n_y \log n_y, \\ \Theta\left(\frac{L_k}{n_y}\right), & L_k \geq n_y \log n_y. \end{cases}$$

By standard balls-into-bins tail bounds and maximum-load estimates (e.g., [221]), we obtain

$$\Pr(\mathbf{Z}^{L_k,1} \notin \mathcal{E}_{L_k,1}) = \begin{cases} O(n_y^{-U_k L_k + 1}), & L_k \ll n_y, \\ \exp(-\Theta(n_y \log n_y)), & L_k = \Theta(n_y), \\ \exp(-\Theta(L_k \log n_y)), & L_k \gg n_y, \end{cases}$$

and

$$\Pr(\mathbf{Z}^{L_k,1} \in \mathcal{E}_{L_k,3}) = 1 - n_y^{-\Omega(1)}.$$

Reducing the Wrong Attention

As discussed in Remark 5.B.5, both $\epsilon_{\text{attn}}^{L_k}$ and the attention gap remain bounded by a small constant at the beginning of stage k . Hence, we can directly proceed to stage 1.2.2 of the convergence analysis as $\mathcal{T}^2$. Moreover, by the symmetry between $[\mathbf{Q}_{4,3}]_{s,s}$ and $[\mathbf{Q}_{4,4}]_{s,s}$, we may, without loss of generality, restrict attention to a particular $s \in \tau(\mathcal{X})$ and analyze the corresponding loss $\text{Loss}_{5,s}^{L_k,2}$ in what follows.

Induction 5.B.15. Given $F^{(T_{k-1})}$ from the previous stage $\mathcal{T}^{L_{k-1}}$ with $L_{k-1} = 2^{k-1}$ for $k \geq 2$, let T_k denote the first time that $\text{Loss}_{5,s}^{L_k,2}$ decreases below $e^{(-\frac{1}{2} + 3.01c_1)2B}$. For all iterations $t \leq T_k$, we have the following holds

1. $[\mathbf{Q}_{4,3}]_{s,s}^{(t)} + [\mathbf{Q}_{4,4}]_{s,s}^{(t)}$ monotonically increases;
2. for any sample $\mathbf{Z}^{2,1}$, we have $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \leq c_1$ for some sufficiently small constant $c_1 = \frac{1.005 \log d}{2B} > 0$;
3. for any $\mathbf{Z}^{2,1}$, we have $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \leq \min\left\{\frac{L_k-1}{L_k} \epsilon_{\text{attn}}^{L_k,2} - c_2, 0\right\}$, where $c_2 = \frac{\log d}{4B} > 0$ is some sufficiently small constant.

4. for $p \in \{3, 4\}$, for $s' \in \tau(X) \neq s$, $|\mathbf{Q}_{4,p}^{(t)}]_{s,s'}| \leq O\left(\frac{[\mathbf{Q}_{4,p}^{(t)}]_{s,s}}{d}\right)$; otherwise $[\mathbf{Q}_{4,p}^{(t)}]_{s,s'} = 0$.

Attention and Lambda Preliminaries

Lemma 5.B.123. *If Induction 5.B.15 holds for all iterations $[T_{k-1}, t)$, then for any sample $\mathbf{Z}^{L_k,1}$, we have*

1. $\epsilon_{\text{attn}}^{L_k} \in \left[\frac{4}{3}c_1, 2\epsilon_{\text{attn}}^{L_{k-1}}\right]$;
2. $\mathbf{Attn}_{\text{ans},1 \rightarrow \mathbf{k}}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \mathbf{k}'}^{(t)} \geq \Omega(1)$ for any $\mathbf{k}, \mathbf{k}' \in \mathcal{I}^{L_k,1} \setminus \{(\text{ans}, 1), (\text{pred}, 2)\}$;
3. $|\mathbf{Attn}_{\text{ans},1 \rightarrow \mathbf{k}}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \mathbf{k}'}^{(t)}| \leq \tilde{O}\left(\frac{1}{d}\right)$ for any $\mathbf{k}, \mathbf{k}' \in \mathcal{I}^{L_k,1} \setminus \{(\text{ans}, 1)\}$.

Proof. The highest loss occurs when the sample $\mathbf{Z}^{L_k}$ makes the wrong answer highly confused with the correct answer. Thus, we consider the case that $g_\ell \cdot y_0 = g_2 \cdot y_0$ for all $\ell \in [L_k] \setminus \{2\}$ and $y_0 \neq y_1$.

$$\begin{aligned}
& \log p_F(\mathbf{Z}_{\text{ans},2,5} | \mathbf{Z}^{(L_k,1)}) \\
& \leq \Theta(1) \cdot \max \left\{ e^{\left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} + \frac{L_k-1}{L_k} \epsilon_{\text{attn}}^{L_k,2} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \right) 2B - \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \frac{1}{L_k} \epsilon_{\text{attn}}^{L_k,2} \right) 2B}, \right. \\
& \quad \left. e^{\log d - \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \frac{1}{L_k} \epsilon_{\text{attn}}^{L_k,2} \right) 2B} \right\} \\
& \leq \Theta \left(e^{\left(\frac{L_k-1}{L_k} \epsilon_{\text{attn}}^{L_k,2} + c_1 \right) 2B - \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \frac{1}{L_k} \epsilon_{\text{attn}}^{L_k,2} \right) 2B} \right) \\
& \leq \Theta \left(e^{-\left(\frac{1}{2} - \frac{3}{2} \epsilon_{\text{attn}}^{L_k,2} - c_1 \right) 2B} \right).
\end{aligned}$$

Thus, if $\epsilon_{\text{attn}}^{L_k} \leq \frac{4}{3}c_1$, we must have

$$\text{Loss}_{5,s}^{2,2} \leq \Theta \left(e^{\left(-\frac{1}{2} + 3 \cdot \frac{2c_1}{3} + c_1 \right) 2B} \right) = \Theta \left(e^{-(\frac{1}{2} + 3c_1) 2B} \right),$$

which contradicts the definition of $T_{1,2,2,s}$. Therefore, it must hold that $\epsilon_{\text{attn}}^{L_k} \geq \frac{4}{3}c_1$ for all $t \leq T_k$. $\square$

Lemma 5.B.124. *If Induction 5.B.15 holds for all iterations $[T_{k-1}, t)$, then given $\mathbf{Z}^{L_k,1} \notin \mathcal{E}_{L_k,2}$, we have*

1. for the prediction j_2 , we have

$$\Lambda_{5,j_2,r_{g_2 y_1}}^{(t)} = \left(\sum_{\ell \in [L_k]} \mathbb{1}_{g_\ell(y_1)=g_2(y_1)} \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},\ell}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \right) \cdot 2B + O\left(\frac{B}{n_y}\right) + O(\delta).$$

2. for the prediction $j'_2 = \tau(g_2(y_0))$, we have

$$\Lambda_{5,j'_2,r_{g_2 y_0}}^{(t)} = \left(\sum_{\ell \in [L_k]} \mathbb{1}_{g_\ell(y_0)=g_2(y_0)} \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},\ell}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \right) \cdot 2B + O\left(\frac{B}{n_y}\right) + O(\delta).$$

3. for other $j = g_2(y) \in \tau(\mathcal{Y})$, noticing that for this case $y \neq y_0, y_1$, then we have

$$\begin{aligned} \Lambda_{5,j,r_{g_2 y}}^{(t)} &= \left(\sum_{\ell \in [L_k]} \mathbb{1}_{g_\ell(y)=g_2(y)} \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},\ell}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \right) \cdot 2B \\ &\quad + O\left(\frac{B}{n_y}\right) + O(\delta). \end{aligned}$$

Lemma 5.B.125. If Induction 5.B.15 holds for all iterations $[T_{k-1}, t)$, then given $\mathbf{Z}^{L_k,1} \in \mathcal{E}_{L_k,2}$, we have

1. for the prediction j_2 , we have

$$\Lambda_{5,j_2,r_{g_2 y_1}}^{(t)} = \sum_{\ell \in [L_k]} \mathbb{1}_{g_\ell(y_1)=g_2(y_1)} \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},\ell}^{(t)} \cdot 2B + O\left(\frac{B}{n_y}\right) + O(\delta).$$

2. for other $j = \tau(g_2(y)) \in \tau(\mathcal{Y})$ with $y \neq y_1$, then we have

$$\begin{aligned} \Lambda_{5,j,r_{g_2 y}}^{(t)} &= \left(\sum_{\ell \in [L_k]} \mathbb{1}_{g_\ell(y)=g_2(y)} \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},\ell}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \right) \cdot 2B \\ &\quad + O\left(\frac{B}{n_y}\right) + O(\delta). \end{aligned}$$

Gradient Lemma

Lemma 5.B.126. If Induction 5.B.15 holds for all iterations $t \in [T_{k-1}, T_k)$, given $s \in \tau(X)$, we have

$$\begin{aligned} &\left[-\nabla_{\mathbf{Q}_{4,3}^{(t)}} \text{Loss}_5^{2,2} \right]_{s,s} + \left[-\nabla_{\mathbf{Q}_{4,4}^{(t)}} \text{Loss}_5^{2,2} \right]_{s,s} \\ &\geq \Omega(B/d) \mathbb{E} \left[\epsilon_{\text{attn}}^{L_k} \cdot (1 - \mathbf{logit}_{5,j_2}^{(t)}) | \tau(x_1) = s, \mathcal{E}_{L_k,1} \cap \mathcal{E}_{L_k,2}^c \right]. \end{aligned}$$

Proof. By Induction 5.B.15, Lemma 5.B.124, and Lemma 5.B.125, we have

$$\begin{aligned} &\mathbb{E} \left[\mathcal{N}_{s,L_k,i}^{(t)} \right] \\ &= \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) \cdot \left(- \sum_{\ell \neq 2} \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},\ell}^{(t)} \cdot V_{j_2,r_{g_2 y_1}}(g_\ell) - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j_2,r_{g_2 y_1}}(y_0) \right. \right. \\ &\quad \left. \left. + (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5,j_2,r_{g_2 y_1}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \right] \quad (5.B.105) \\ &\mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\Lambda_{5,j_2,r_{g_2 y_1}}^{(t)} \leq B}. \end{aligned}$$

Similarly,

$$\begin{aligned} &\mathbb{E} \left[\mathcal{N}_{s,L_k,ii}^{(t)} \right] \\ &= \mathbb{E} \left[\sum_{y \neq y_1} \mathbf{logit}_{5,\tau(g_2(y))}^{(t)} \cdot \mathbf{sReLU}'(\Lambda_{5,\tau(g_2(y)),r_{g_2 y}}^{(t)}) \right] \end{aligned}$$

$$\begin{aligned} & \left(\sum_{\ell \neq 2} \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},\ell}^{(t)} \cdot V_{\tau(g_2(y)),r_{g_2,y}}(g_\ell) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{\tau(g_2(y)),r_{g_2,y}}(y_0) \right. \\ & \quad \left. - (1 - \mathbf{Attn}_{\mathbb{A},1 \rightarrow \mathbb{A},1}^{(t)} - \mathbf{Attn}_{\mathbb{A},1 \rightarrow 2}^{(t)}) \Lambda_{5,\tau(g_2(y)),r_{g_2,y}}^{(t)} \pm \tilde{O}(\sigma_0) \right) \pm \tilde{O}(\delta^q) \Big] \mathbb{1}_{\tau(x_1)=s}. \end{aligned}$$

Consider the event $\mathcal{E}_{L_k,1}$. For $\mathbf{Z}^{L_k,1} \in \mathcal{E}_{L_k,1} \cap \mathcal{E}_{L_k,2}^c$, we have $\Lambda_{5,j_2,r_{g_2,y_1}}^{(t)} \leq B$, since

$$\begin{aligned} \Lambda_{5,j_2,r_{g_2,y_1}}^{(t)} &= \sum_{\ell \neq 2} \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},\ell}^{(t)} \cdot V_{j_2,r_{g_2,y_1}}(g_\ell) \mathbb{1}_{g_\ell(y_1)=j_2} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j_2,r_{g_2,y_1}}(y_0) \\ &\leq 2 \left(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} + (U_k - 1/L_k) \cdot \epsilon_{\text{attn}}^{L_k} \right) B \\ &\leq 2 \left(\frac{1 - \epsilon_{\text{attn}}^{L_k} + c_1}{2} + (U_k - 1/L_k) \cdot \epsilon_{\text{attn}}^{L_k} \right) B \\ &= 2 \left(\frac{1+c_1}{2} - \left(\frac{1}{2} + 1/L_k - U_k \right) \cdot \epsilon_{\text{attn}}^{L_k} \right) B. \end{aligned}$$

Thus, provided $\epsilon_{\text{attn}}^{L_k} \geq 4c_1/3$ and choosing $U_k = \left\lfloor 1 + \frac{L_k}{8} \right\rfloor / L_k$, we indeed have $\Lambda_{5,j_2,r_{g_2,y_1}}^{(t)} \leq B$.

- For $j = \tau(g_2(y))$ in $\mathcal{N}_{s,L_k,ii}^{(t)}$:
 - $y = y_0 \neq y_1$,

$$\begin{aligned} & \sum_{\ell \neq 2} \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},\ell}^{(t)} \cdot V_{\tau(g_2(y)),r_{g_2,y}}(g_\ell) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{\tau(g_2(y)),r_{g_2,y}}(y_0) \\ & \geq \sum_{\ell \neq 2} \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},\ell}^{(t)} \cdot V_{\tau(g_2(y)),r_{g_2,y}}(g_\ell) \mathbb{1}_{g_\ell(y) \neq g_2(y)} \\ & \geq -O\left(\frac{B}{n_y}\right) \cdot (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}). \end{aligned} \tag{5.B.106}$$

- $y \neq y_0, y_1$. If there exists at least one $\ell \neq 2$ such that $g_\ell(y) = g_2(y)$, then by cancellation,

$$\begin{aligned} & \sum_{\ell \neq 2} \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},\ell}^{(t)} \cdot V_{\tau(g_2(y)),r_{g_2,y}}(g_\ell) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{\tau(g_2(y)),r_{g_2,y}}(y_0) \\ & \geq -O\left(\frac{B}{n_y}\right) \cdot (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}). \end{aligned} \tag{5.B.107}$$

Else, pick $y' \neq y, y_1$ so that $\Lambda_{5,\tau(g_2(y')),r_{g_2,y'}}^{(t)} \geq \Lambda_{5,\tau(g_2(y)),r_{g_2,y}}^{(t)}$. Then

$$\text{logit}_{5,\tau(g_2(y))} \leq O\left(\frac{1}{n_y}\right) (1 - \text{logit}_{5,j_2}),$$

which implies

$$\begin{aligned} & \text{logit}_{5,j}^{(t)} \left(\sum_{\ell \neq 2} \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},\ell}^{(t)} \cdot V_{\tau(g_2(y)),r_{g_2,y}}(g_\ell) + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{\tau(g_2(y)),r_{g_2,y}}(y_0) \right) \\ & \geq -O\left(\frac{B}{L_k n_y}\right) \cdot (1 - \text{logit}_{5,j_2}) \cdot (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}). \end{aligned} \tag{5.B.108}$$

- For $\mathcal{N}_{s,L_k,i}^{(t)}$:
 - If $\mathbf{Z}^{L_k,1} \in \mathcal{E}_{L_k,1} \cap \mathcal{E}_{L_k,2}^c$, then

$$\begin{aligned}
& - \sum_{\ell \neq 2} \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},\ell}^{(t)} \cdot V_{j_2,r_{g_2 \cdot y_1}}(g_\ell) - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j_2,r_{g_2 \cdot y_1}}(y_0) \\
& \quad + (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} \\
& \geq - \sum_{\ell \neq 2} \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},\ell}^{(t)} \mathbb{1}_{g_\ell = j_2} \cdot V_{j_2,r_{g_2 \cdot y_1}}(g_\ell) - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j_2,r_{g_2 \cdot y_1}}(y_0) \\
& \quad + (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} \\
& \geq \left(- (U_k - \frac{1}{L_k}) \cdot 2B + \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} \right) \cdot \epsilon_{\text{attn}}^{L_k} \geq \left(\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} - \frac{B}{4} \right) \cdot \epsilon_{\text{attn}}^{L_k}. \quad (5.B.109)
\end{aligned}$$

- Else, for $\mathbf{Z}^{L_k,1} \in \mathcal{E}_{L_k,1}^c \cup \mathcal{E}_{L_k,2}$, note that having more identical predictions $g_\ell(y_1) = j_2$ or less distraction (i.e., $y_0 = y_1$) increases $\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)}$. Hence

$$\mathbb{E}[(1 - \mathbf{logit}_{5,j_2}) \mid \mathcal{E}_{L_k,1}^c \cup \mathcal{E}_{L_k,2}] \leq O(1) \cdot \mathbb{E}[(1 - \mathbf{logit}_{5,j_2}) \mid \mathcal{E}_{L_k,1} \cap \mathcal{E}_{L_k,2}^c],$$

and

$$\begin{aligned}
& \left| - \sum_{\ell \neq 2} \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},\ell}^{(t)} \cdot V_{j_2,r_{g_2 \cdot y_1}}(g_\ell) - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)} \cdot V_{j_2,r_{g_2 \cdot y_1}}(y_0) \right. \\
& \quad \left. + (1 - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)}) \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} \right| \leq O(1) \cdot \epsilon_{\text{attn}}^{L_k} \cdot B. \quad (5.B.110)
\end{aligned}$$

Putting the above together, we have

$$\begin{aligned}
& \left[- \nabla_{\mathbf{Q}_{4,3}^{(t)}} \text{Loss}_5^{2,2} \right]_{s,s} + \left[- \nabla_{\mathbf{Q}_{4,4}^{(t)}} \text{Loss}_5^{2,2} \right]_{s,s} \\
& = \mathbb{E} \left[\mathcal{N}_{s,L_k,1}^{(t)} \mathbb{1}_{\mathcal{E}_{L_k,1} \cap \mathcal{E}_{L_k,2}^c} \right] + \mathbb{E} \left[\mathcal{N}_{s,L_k,1}^{(t)} \mathbb{1}_{\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} \leq B} \mathbb{1}_{\mathcal{E}_{L_k,1}^c \cup \mathcal{E}_{L_k,2}} \right] \\
& \quad + \mathbb{E} \left[\mathcal{N}_{s,L_k,2}^{(t)} \right] \pm \widetilde{O}(\sigma_0^q) \\
& = \mathbb{E} \left[(\mathcal{N}_{s,L_k,1}^{(t)} + \mathcal{N}_{s,L_k,2}^{(t)}) \mathbb{1}_{\mathcal{E}_{L_k,1} \cap \mathcal{E}_{L_k,2}^c} \right] \\
& \quad + \mathbb{E} \left[\mathcal{N}_{s,L_k,1}^{(t)} \mathbb{1}_{\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} \leq B} \mathbb{1}_{\mathcal{E}_{L_k,1}^c \cup \mathcal{E}_{L_k,2}} \right] + \mathbb{E} \left[\mathcal{N}_{s,L_k,2}^{(t)} \mathbb{1}_{\mathcal{E}_{L_k,1}^c \cup \mathcal{E}_{L_k,2}} \right] \pm \widetilde{O}(\sigma_0^q).
\end{aligned}$$

First, by (5.B.106), (5.B.107), (5.B.109),

$$\begin{aligned}
& \mathbb{E} \left[(\mathcal{N}_{s,L_k,1}^{(t)} + \mathcal{N}_{s,L_k,2}^{(t)}) \mathbb{1}_{\mathcal{E}_{L_k,1} \cap \mathcal{E}_{L_k,2}^c} \right] \\
& \geq \mathbb{E} \left[(1 - \mathbf{logit}_{5,j_2}^{(t)}) \left(\left(\Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(t)} - \frac{B}{4} \right) \epsilon_{\text{attn}}^{L_k} \right) - (1 - \mathbf{logit}_{5,j_2}^{(t)}) \left(O\left(\frac{B}{n_y}\right) \epsilon_{\text{attn}}^{L_k} + \max_{y \neq y_1} \Lambda_{5,\tau(g_2(y)),r_{g_2 \cdot y}}^{(t)} \right) \right. \right. \\
& \quad \left. \left. \cdot \mathbb{1}_{\tau(x_1)=s} \mathbb{1}_{\mathcal{E}_{L_k,1} \cap \mathcal{E}_{L_k,2}^c} \right] \right]
\end{aligned}$$

$$\geq \Omega(B/d) \cdot \mathbb{E} \left[\epsilon_{\text{attn}}^{L_k} \cdot (1 - \mathbf{logit}_{5,j_2}^{(t)}) \mid \tau(x_1) = s, \mathcal{E}_{L_k,1} \cap \mathcal{E}_{L_k,2}^c \right].$$

Moreover, by (5.B.110),

$$\begin{aligned} & \mathbb{E} \left[\mathcal{N}_{s,L_k,1}^{(t)} \mathbb{1}_{\Lambda_{5,j_2,r_{g_2},y_1}^{(t)} \leq B/2} \mathbb{1}_{\mathcal{E}_{L_k,1}^c \cup \mathcal{E}_{L_k,2}} \right] \\ & \geq -\Pr(\mathbf{Z}^{L_k,1} \notin \mathcal{E}_{L_k,1}^c) \cdot O(B/d) \cdot \mathbb{E} \left[\epsilon_{\text{attn}}^{L_k} \cdot (1 - \mathbf{logit}_{5,j_2}^{(t)}) \mid \tau(x_1) = s, \mathcal{E}_{L_k,1}^c \cup \mathcal{E}_{L_k,2} \right] \\ & \geq -\left(O(1/n^{\Omega(1)}) + 1/n_y \right) \cdot O(B/d) \cdot \mathbb{E} \left[\epsilon_{\text{attn}}^{L_k} \cdot (1 - \mathbf{logit}_{5,j_2}^{(t)}) \mid \tau(x_1) = s, \mathcal{E}_{L_k,1} \cap \mathcal{E}_{L_k,2}^c \right], \end{aligned}$$

and, by (5.B.106), (5.B.107), (5.B.108),

$$\mathbb{E} \left[\mathcal{N}_{s,L_k,2}^{(t)} \mathbb{1}_{\mathcal{E}_{L_k,1}^c \cup \mathcal{E}_{L_k,2}} \right] \geq \frac{1}{n_y} \cdot O(B/d) \cdot \mathbb{E} \left[\epsilon_{\text{attn}}^{L_k} \cdot (1 - \mathbf{logit}_{5,j_2}^{(t)}) \mid \tau(x_1) = s, \mathcal{E}_{L_k,1}^c \cup \mathcal{E}_{L_k,2} \right].$$

Therefore,

$$\begin{aligned} & \left[-\nabla_{\mathbf{Q}_{4,3}^{(t)}} \text{Loss}_5^{2,2} \right]_{s,s} + \left[-\nabla_{\mathbf{Q}_{4,4}^{(t)}} \text{Loss}_5^{2,2} \right]_{s,s} \\ & \geq \Omega(B/d) \cdot \mathbb{E} \left[\epsilon_{\text{attn}}^{L_k} \cdot (1 - \mathbf{logit}_{5,j_2}^{(t)}) \mid \tau(x_1) = s, \mathcal{E}_{L_k,1} \cap \mathcal{E}_{L_k,2}^c \right]. \end{aligned}$$

□

Lemma 5.B.127. *If Induction 5.B.15 holds for all iterations $t \in [T_{k-1}, T_k)$, given $s \in \tau(\mathcal{X})$, for $[\mathbf{Q}_{4,p}]_{s,s'}$, $p \in \{3, 4\}$, $s' \neq s \in \tau(\mathcal{X})$, we have*

$$\left| \left[-\nabla_{\mathbf{Q}_{4,p}^{(t)}} \text{Loss}_5^{2,2} \right]_{s,s'} \right| \leq O\left(\frac{1}{d}\right) \left| \left[-\nabla_{\mathbf{Q}_{4,p}^{(t)}} \text{Loss}_5^{2,2} \right]_{s,s} \right|.$$

Attention gap is small

Lemma 5.B.128. *If Induction 5.B.15 holds for all iterations $t \in [T_{k-1}, T_k)$, then for any sample $\mathbf{Z}^{L_k,1}$, we have*

$$\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \leq c_1,$$

where $c_1 > 0$ is a small constant.

Proof. The proof follows the same high-level idea as Lemma 5.B.101. The key observation is that for each $\mathbf{Z}^{L_k,1}$, there always exists at least one $y \neq y_1$ such that, for some index j appearing in the gradient term $\mathcal{N}_{s,L_k,2}^{(t)}$, we have

$$\Lambda_{5,j,r_{g_2},y}^{(t)} \geq 2(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)})B.$$

Let $\widehat{\mathcal{J}}$ denote the set of indices j achieving $\arg \max_{y \notin \{y_1\}} \Lambda_{5,j,r_{g_2},y}^{(t)}$, and pick one such index $j' = g_2(y')$. Define $\widetilde{T}$ to be the first time when $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \geq c_1$. At this time,

we have

$$\sum_{j \in \widehat{\mathcal{J}}} \mathbf{logit}_{5,j}^{(\widetilde{T})} = \left(1 - O(d^{-\Omega(1)})\right) (1 - \mathbf{logit}_{5,j_2}^{(\widetilde{T})}).$$

Plugging this into the gradient expressions yields

$$\begin{aligned} & \mathcal{N}_{s,3,L_k,i}^{(\widetilde{T})} + \mathcal{N}_{s,3,L_k,ii}^{(\widetilde{T})} \\ &= \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\widetilde{T})} \cdot \left(\left(1 - O(d^{-\Omega(1)})\right) (1 - \mathbf{logit}_{5,j_2}^{(\widetilde{T})}) \cdot \left(\Lambda_{5,j',r_{g_2 \cdot y'}}^{(\widetilde{T})} - \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(\widetilde{T})} \right) \right. \\ & \quad \left. + O(d^{-\Omega(1)}) (1 - \mathbf{logit}_{5,j_2}^{(\widetilde{T})}) \cdot \left(V_{5,j_2,r_{g_2 \cdot y_1},2}(g_2) - \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(\widetilde{T})} \right) \right) \mathbb{1}_{\tau(x_1)=s}, \end{aligned}$$

and similarly

$$\begin{aligned} & \mathcal{N}_{s,4,L_k,i}^{(\widetilde{T})} + \mathcal{N}_{s,4,L_k,ii}^{(\widetilde{T})} \\ &= \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\widetilde{T})} \cdot \left(\left(1 - O(d^{-\Omega(1)})\right) (1 - \mathbf{logit}_{5,j_2}^{(\widetilde{T})}) \cdot \left(-V_{j_2',r_{g_2 \cdot y_0}}(y_1) + \Lambda_{5,j',r_{g_2 \cdot y'}}^{(\widetilde{T})} - \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(\widetilde{T})} \right) \right. \\ & \quad \left. + O(d^{-\Omega(1)}) (1 - \mathbf{logit}_{5,j_2}^{(\widetilde{T})}) \cdot \left(V_{5,j_2,r_{g_2 \cdot y_1},2}(y_1) - \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(\widetilde{T})} \right) \right) \mathbb{1}_{\tau(x_1)=s}. \end{aligned}$$

Since $\Lambda_{5,j',r_{g_2 \cdot y'}}^{(\widetilde{T})} - \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(\widetilde{T})} \leq 0$ and

$$-V_{j_2',r_{g_2 \cdot y_1}}(y_1) - \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(\widetilde{T})} + \Lambda_{5,j_2,r_{g_2 \cdot y'}}^{(\widetilde{T})} \geq \Omega(B),$$

we obtain

$$\left[-\nabla_{\mathbf{Q}_{4,4}^{(\widetilde{T})}} \text{Loss}_5^{2,2} \right]_{s,s} + \left[\nabla_{\mathbf{Q}_{4,3}^{(\widetilde{T})}} \text{Loss}_5^{2,2} \right]_{s,s} \geq \Omega\left(\frac{B}{d}\right) \cdot \mathbb{E}[1 - \mathbf{logit}_{5,j_2}^{(\widetilde{T})} \mid \tau(x_1) = s].$$

Therefore, the quantity $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)}$ cannot further increase, and we conclude that

$$\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} \leq c_1.$$

□

Lemma 5.B.129. *If Induction 5.B.15 holds for all iterations $t \in [T_{k-1}, T_k)$, then for any sample $\mathbf{Z}^{L_k,1}$, we have*

$$\mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \leq \frac{L_k - 1}{L_k} \epsilon_{\text{attn}}^{L_k} - c_2,$$

where $c_2 = \frac{\log d}{2B} > 0$ is a small constant.

Proof. The argument parallels Lemma 5.B.113, but in the recursive setting we first note that for every $\mathbf{Z}^{L_k,1} \in \mathcal{E}_{L_k,2}^c$,

$$\Lambda_{5,\tau(g_2(y_0)),r_{g_2 \cdot y_0}}^{(t)} \leq \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} + \frac{L_k - 1}{L_k} \epsilon_{\text{attn}}^{L_k}.$$

Let $\tilde{T}$ be the first time such that

$$\mathbb{E} \left[\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} + \frac{L_k - 1}{L_k} \epsilon_{\text{attn}}^{L_k} \mid \tau(x_1) = s \right] \leq \frac{\log d}{4.02 B}.$$

Then, for any y ,

$$\begin{aligned} & \Lambda_{5,\tau(g_2(y)),r_{g_2 \cdot y}}^{(\tilde{T})} \\ &= 2(\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} + \sum_{\ell \neq 2} \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},\ell}^{(\tilde{T})} \mathbf{1}_{g_\ell(y)=g_2(y)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(\tilde{T})}) B \leq \frac{\log d}{2.01}. \end{aligned}$$

Hence the corresponding wrong-class logit mass is small:

$$\mathbf{logit}_{5,j_2}^{(1)} \leq O(d^{-1.01/2.01})(1 - \mathbf{logit}_{5,j_2}^{(t)}).$$

Plugging this into the gradient decomposition, we obtain

$$\begin{aligned} & \mathcal{N}_{s,3,L_k,i}^{(\tilde{T})} + \mathcal{N}_{s,3,L_k,ii}^{(\tilde{T})} \\ &= \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} \left((1 - \mathbf{logit}_{5,j_2}^{(\tilde{T})})(V_{5,j_2,r_{g_2 \cdot y_1},2}(g_2) - \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(\tilde{T})}) \right. \\ & \quad \left. - O(d^{-1.01/2.01})(1 - \mathbf{logit}_{5,j_2}^{(\tilde{T})}) \sum_{y \neq y_0} (V_{5,j_2,r_{g_2 \cdot y},2}(g_2) - \Lambda_{5,j_2,r_{g_2 \cdot y}}^{(\tilde{T})}) \right) \mathbf{1}_{\tau(x_1)=s} \\ &\geq \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} \cdot \Omega(1) \cdot (1 - \mathbf{logit}_{5,j_2}^{(\tilde{T})}) \cdot (V_{5,j_2,r_{g_2 \cdot y_1},2}(g_2) - \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(\tilde{T})}) \mathbf{1}_{\tau(x_1)=s}, \end{aligned}$$

and

$$\begin{aligned} & \mathcal{N}_{s,4,L_k,i}^{(\tilde{T})} + \mathcal{N}_{s,4,L_k,ii}^{(\tilde{T})} \\ &= \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} \left((1 - \mathbf{logit}_{5,j_2}^{(\tilde{T})})(V_{5,j_2,r_{g_2 \cdot y_1},2}(y_1) - \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(\tilde{T})}) \right. \\ & \quad \left. - O(d^{-1.01/2.01})(1 - \mathbf{logit}_{5,j_2}^{(\tilde{T})}) \sum_{y \neq y_0} (V_{5,j_2,r_{g_2 \cdot y},2}(y_1) - \Lambda_{5,j_2,r_{g_2 \cdot y}}^{(\tilde{T})}) \right) \mathbf{1}_{\tau(x_1)=s} \\ &\leq -\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(\tilde{T})} \cdot \Omega(1) \cdot (1 - \mathbf{logit}_{5,j_2}^{(\tilde{T})}) \cdot \Lambda_{5,j_2,r_{g_2 \cdot y_1}}^{(\tilde{T})} \mathbf{1}_{\tau(x_1)=s}. \end{aligned}$$

Therefore,

$$\left[-\nabla_{\mathbf{Q}_{4,3}^{(\tilde{T})}} \text{Loss}_5^{2,2} \right]_{s,s} \geq \Omega\left(\frac{B}{d}\right) \cdot \mathbb{E} \left[1 - \mathbf{logit}_{5,j_2}^{(\tilde{T})} \mid \tau(x_1) = s, \mathcal{E}_{L_k,2}^c \right], \quad (5.B.111)$$

$$\left[-\nabla_{\mathbf{Q}_{4,4}^{(\tilde{T})}} \text{Loss}_5^{2,2} \right]_{s,s} \leq -\Omega\left(\frac{B}{d}\right) \cdot \mathbb{E}\left[1 - \mathbf{logit}_{5,j_2}^{(\tilde{T})} \mid \tau(x_1) = s, \mathcal{E}_{L_k,2}^c\right]. \quad (5.B.112)$$

Finally, recall that

$$\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} + \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},1}^{(t)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)} = \frac{e^{[\mathbf{Q}_{4,3}^{(t)}]_{s,s}} - e^{[\mathbf{Q}_{4,4}^{(t)}]_{s,s}} - e^{\tilde{O}(1/d)}}{e^{[\mathbf{Q}_{4,3}^{(t)}]_{s,s}} + e^{[\mathbf{Q}_{4,4}^{(t)}]_{s,s}} + e^{\tilde{O}(1/d)}}.$$

Combining the two gradient inequalities shows that the quantity

$$\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} + \frac{L_k - 1}{L_k} \epsilon_{\text{attn}}^{L_k} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},1}^{(t)}$$

must strictly decrease at time $\tilde{T} + 1$. □

At the end of stage

Lemma 5.B.130 (At the end of stage). *Induction 5.B.15 holds for all iterations $T_{k-1} < t \leq T_k = O(\frac{\text{poly}(d)}{\eta B})$. At the end of stage k , we have*

- (a) *Attention concentration: $\epsilon_{\text{attn}}^{L_k} \leq 5.04c_1$ for some small constant $c_1 = \frac{1.005 \log d}{B} > 0$;*
- (b) *Loss convergence: $\text{Loss}_{F(T_{k-1}),5}^{L_k,2} \leq e^{(-\frac{1}{2} + 3.01c_1)2B} = \frac{1}{\text{poly}d}$.*

Proof. By Lemmas 5.B.123 and 5.B.126, the diagonal entries $[\mathbf{Q}_{4,3}^{(t)}]_{s,s}$ and $[\mathbf{Q}_{4,4}^{(t)}]_{s,s}$ keep increasing until the total wrong-attention mass satisfies $\epsilon_{\text{attn}} \geq \frac{4c_1}{3}$. Combining Lemmas 5.B.126 to 5.B.129, there exists a stopping time

$$T_k = O\left(\frac{\text{poly}(d)}{\eta B}\right),$$

such that Induction 5.B.15 holds for all $t \in [T_1, T_k]$.

To obtain an upper bound on $\epsilon_{\text{attn}}^{L_k}$ at the end of stage k , it suffices to work on the main event $\mathcal{E}_{L_k,3}$.

Case 1: $L_k = O(1)$. For each $y \in \mathcal{Y}$, at most one index $\ell \neq 2$ can satisfy $g_\ell(y) = g_2(y)$, and thus the number of collisions per y is uniformly bounded by W_k . Hence,

$$\begin{aligned} & \log p_F(Z_{\text{ans},2,5} \mid Z^{(2,1)}) \\ & \geq \Theta(1) \cdot \exp\left(\log d - (\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} + \sum_{\ell \neq 2} \mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},\ell}^{(t)} \mathbb{1}_{g_\ell(y_1)=g_2(y_1)} - \mathbf{Attn}_{\text{ans},1 \rightarrow \text{ans},0}^{(t)}) 2B\right) \\ & \geq \Theta(1) \cdot \exp\left(\log d - (\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} + (1/L_k - 1/L_k) \epsilon_{\text{attn}}^{L_k}) 2B\right) \\ & \geq \Theta\left(\exp\left(-\left(\frac{1}{2} + \frac{c_1}{2} - \frac{\epsilon_{\text{attn}}^{L_k}}{2} - \frac{\log d}{B}\right) 2B\right)\right), \end{aligned}$$

where the last line uses $\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} \leq \frac{1}{2}(1 - \epsilon_{\text{attn}}^{L_k} + c_1) = \frac{1+c_1}{2} - \frac{\epsilon_{\text{attn}}^{L_k}}{2}$.

Case 2: $L_k = \omega(1)$. If $L_k \leq n_y \log n_y$ then $W_k/L_k = O(1/L_k)$; otherwise $W_k/L_k = O(1/n_y)$. In either regime $W_k/L_k = o(1)$, and we similarly obtain

$$\begin{aligned} \log p_F(Z_{\text{ans},2,5} \mid Z^{(2,1)}) &\geq \Theta(1) \cdot \exp\left(\log d - (\mathbf{Attn}_{\text{ans},1 \rightarrow \text{pred},2}^{(t)} + (o(1) - 1/L_k) \epsilon_{\text{attn}}^{L_k}) 2B\right) \\ &\geq \Theta\left(\exp\left(-\left(\frac{1}{2} + \frac{c_1}{2} - \frac{\epsilon_{\text{attn}}^{L_k}}{2} - \frac{\log d}{B}\right) 2B\right)\right). \end{aligned}$$

At convergence we therefore have

$$\frac{1}{2} + \frac{c_1}{2} - \frac{\epsilon_{\text{attn}}^{L_k}}{2} - \frac{\log d}{B} \geq \frac{1}{2} - 3.01 c_1,$$

which rearranges to

$$\epsilon_{\text{attn}}^{L_k} \leq 2(3.02c_1 + 0.5c_1 - c_1) = 5.04c_1.$$

In particular, since $B/\log d$ can be taken as a sufficiently large constant, we obtain the clean stage-end bound

$$\epsilon_{\text{attn}}^{L_k} = O(c_1).$$

□

Proof of Main Theorem

Theorem 5.B.4 (Recursive self-training (Restatement)). *Assume the distribution $\mathcal{D}^L$ induced from $\text{LEGO}(X, \mathcal{G}, \mathcal{Y})$ satisfies Assumption 5.3.1, 5.3.2 and 5.5.1, and assume the transformer network satisfies Assumption 5.3.3, 5.3.4, 5.3.5, and $n_y < m \ll \log^2 d$. Then for any $1 \leq k < \log_2 |\mathcal{X}|$, the transformer $F^{(T_k)}$ trained via Algorithm 3 up to length $L_k = 2^k$ and $T_k = O(\frac{\text{poly}(d)}{\eta})$ satisfies: $F^{(T_k)}$ is able to solve $\mathcal{T}^{L_{k+1}}$ with $L_{k+1} = 2^{k+1}$, i.e.,*

$$\text{Acc}_{L_{k+1}}(F^{(T_k)}) = 1 - O(1/\text{poly}(d)).$$

Proof. By Lemma 5.B.130, at the time T_k , we have $\epsilon_{\text{attn}}^{L_k} \leq 5.04c_1$, combining with Induction 5.B.15, non-diagonal entry of $\mathbf{Q}_{p,q}$ remains close to 0, thus moving to the next stage, we have $\epsilon_{\text{attn}}^{L_{k+1},\ell} \leq 4\epsilon_{\text{attn}}^{L_k} \leq 20.16c_1$ (notice that $20.16c_1$ could be sufficiently small e.g., 0.01 since we can set a reasonably large B) for all $\ell < L_{k+1}$. Moreover, $\Delta_{L_{k+1},\ell} \leq \Delta_{L_k,1}$. Hence, we have

$$\begin{aligned} &\mathbb{E}_{Z^{L_{k+1}} \sim \mathcal{D}^{L_{k+1}}} \left[\mathbb{E}_{\widehat{Z}_{\text{ans},\ell+1} \sim p_F^{(T_k)}(\cdot \mid Z^{L_{k+1},\ell})} [\mathbb{1}_{\{\widehat{Z}_{\text{ans},\ell+1} \neq Z_{\text{ans},\ell+1}\}}] \right] \\ &\leq O(1) \cdot \mathbb{E}_{Z^{L_{k+1}} \sim \mathcal{D}^{L_{k+1}}} \left[1 - \text{logit}_{5,\tau(Z_{\text{ans},\ell+1,5})} \right] \\ &\leq O(1) \cdot e^{\left(-\left(\frac{1}{2} - \epsilon_{\text{attn}}^{L_{k+1}} - (\epsilon_{\text{attn}}^{L_{k+1}} - c_2)/2 - \epsilon_{\text{attn}}^{L_{k+1}}\right) 2B\right)} \end{aligned}$$

$$\leq O(1) \cdot e^{(-\frac{1}{2} - \frac{\log d}{4B} + \frac{5}{2} \cdot 0.01)2B} = O\left(\frac{1}{\text{poly}d}\right).$$

Thus $\text{Acc}_{L_{k+1}}(F^{T_k}) \geq 1 - O\left(\frac{1}{\text{poly}d}\right)$.

□

Chapter 6

Learning Dynamics of RL with Verifiable Rewards

6.1 Introduction

Large language models (LLMs) such as OpenAI-o3 [207] and DeepSeek [70] have shown striking performance in complex reasoning tasks. A key enabler of this recent progress is reinforcement learning with verifiable rewards (RLVR) [91, 157, 252], which fine-tunes pre-trained base models via reinforcement learning (RL) using automatically verifiable, outcome-based feedback, such as a binary signal indicating whether the final answer is correct or not.

This raises an immediate question: if RLVR relies on outcome-based feedback only at the end of a reasoning trajectory, how can such a sparse reward mechanism drive effective learning on long-horizon problems? As the horizon grows, RL algorithms encounter an inherent search barrier, because useful signals are buried within an exponentially expanding space of trajectories. While recent studies have sought to understand the mechanism of RLVR [263, 310, 315, 332, 333, 334], existing findings often provide mixed, inconclusive results across different tasks and setups. It remains unclear under what conditions outcome-only rewards are sufficient to enable effective RL.

A recent controlled study [339] offered an important insight: RLVR is most effective near the model’s *edge of competence*, where problems are neither already solved nor impossible. They conducted experiments showing that RLVR improves model’s reasoning performance the most when the data sits right above the model’s capability frontier, while other difficulty levels induce early plateaus. The work [339] suggests picking problems close to the model’s capability frontier, but did not answer why RLVR in practice can work on the general mixture of problems. In practice, RLVR is not done on a single fixed difficulty level but on a spectrum of problems, in

This chapter is adapted from the following joint work:

Yu Huang, Zixin Wen, Yuejie Chi, Yuting Wei, Aarti Singh, Yingbin Liang, and Yuxin Chen. *On the Emergence of Implicit Curriculum in RLVR Learning Dynamics*.

which some are already learnable and others still beyond reach [70, 252]. This motivates the following question:

Research Question

How does the difficulty spectrum of data shape RLVR learning dynamics?

To address this question, we study how policy-gradient dynamics evolve across controlled difficulty spectra. We adopt a minimal transformer model [288], which is the backbone architecture of most LLMs. It consists of a softmax-based attention layer followed by a multilayer perceptron (MLP) layer. We freeze the MLP layer in a way that perfectly implements the atomic operation, modeling the regime where the model already possesses the requisite atomic skills and RLVR only needs to learn how to compose these skills [333]. We study RL training on this task under outcome-based rewards via the standard policy gradient algorithm REINFORCE [314]. Within this setting, we track the learning dynamics of the transformer model and identify the factors that govern progress across increasing horizons. Our main answer is that mixed-difficulty training naturally gives rise to an *implicit curriculum* (illustrated in Figure 6.1): if the difficulty spectrum is smooth enough, the gradient signal is relayed from easier problems to slightly harder ones, keeping optimization near the edge of competence; if the spectrum contains large jumps, the same curriculum develops delayed handoffs and training exhibits grokking-like plateaus before abrupt progress. An overview of our main contributions is provided below.

1. *A comparative study between short-horizon learning and long-horizon barriers.* We show that with outcome-based rewards, REINFORCE-style policy gradient algorithms provably learn short-horizon compositions. Meanwhile, beyond a critical horizon, the gradient field at initialization becomes exponentially flat, indicating an optimization barrier for near-random policies. In contrast, supervised fine-tuning (SFT) can provably learn beyond this critical horizon by providing intermediate feedback for sequential compositional reasoning.
2. *A theoretical characterization of the learning dynamics in RLVR.* On an easy-to-hard mixture over horizons, we establish polynomial-time convergence guarantees for outcome-based RL training. We show that an *implicit curriculum* emerges in RL: in the smooth regime, progress transfers across adjacent horizons through a *relay effect*, which keeps training near the edge of competence and yields steady progress toward harder problems; when the spectrum contains large discontinuities, the same curriculum instead exhibits delayed progress, long plateaus, and abrupt, grokking-like phase transitions [263].
3. *Novel techniques from Fourier analysis on groups.* We introduce a Fourier analysis [270] framework that transforms the problem of trajectory-level success conditioning into tractable calculations based on Fourier analysis for convolutions of measures. Our new framework allows us to compute the magnitudes of policy gradients in long-horizon group composition problems by resorting to the spectral properties of the group representations, which greatly simplifies the analysis of combinatorial event probabilities.

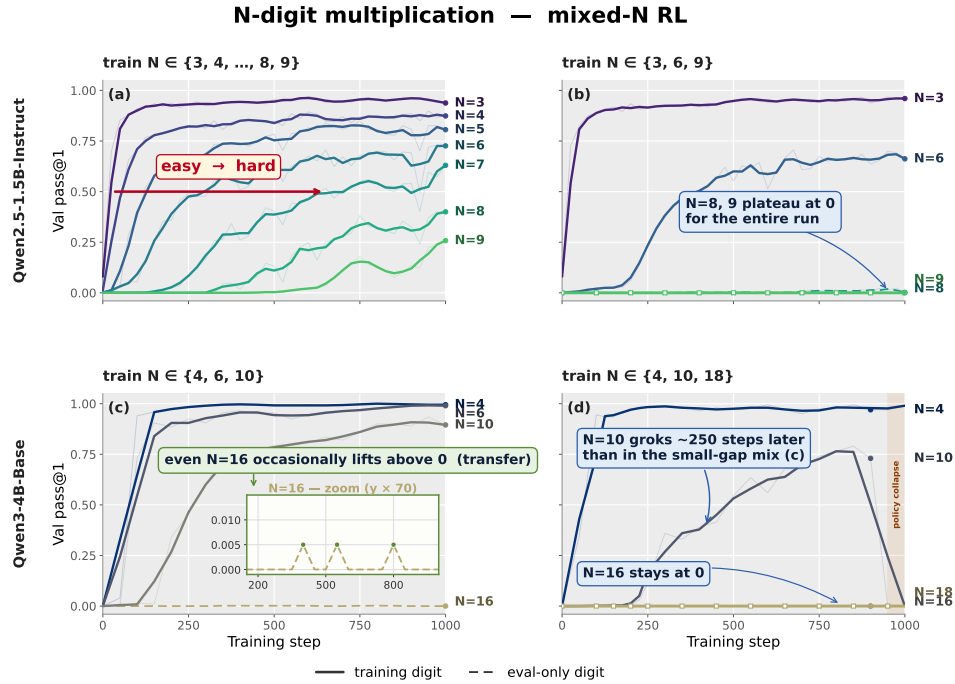


Figure 6.1: Implicit Curricula From Mixed-Difficulty RL. *Mixed-difficulty RL induces an implicit curriculum.* We train on different digit mixtures of N -digit by N -digit multiplication tasks, where larger N indicates greater difficulty. Without any explicit schedule, easier digits become learnable first and then help unlock harder digits. A smooth ladder of difficulties yields a steady easy-to-hard progression, whereas sparse anchors leave harder digits silent for long periods before delayed jumps or persistent plateaus. Detailed setup and discussion are given in Section 6.7.

The implicit curriculum: induced learning dynamics over difficulty spectra. Collectively, our results provide an optimization-based explanation for how RLVR improves model’s capability frontier. The high-level phenomenon is an *implicit curriculum*: when the training distribution displays a spectrum of difficulties, optimization naturally progresses from already-learnable problems to slightly harder ones without any explicit schedule. When the difficulty spectrum is smooth, the underlying mechanism is a *relay effect*: easier horizons continue to provide non-vanishing gradient signal while the next harder horizon is becoming learnable, so gradient dominance is handed off before training stalls. When the gap between adjacent difficulty levels is too large, i.e., when the spectrum is non-smooth, the same curriculum becomes delayed and unpredictable: the handoff arrives only after a prolonged plateau, followed by a grokking-style jump. Thus, we conclude that RLVR’s effectiveness is governed not only by exploration efficiency, but also by whether the difficulty spectrum supports a smooth learning progression. Figure 6.2 depicts the contrast between these two regimes.

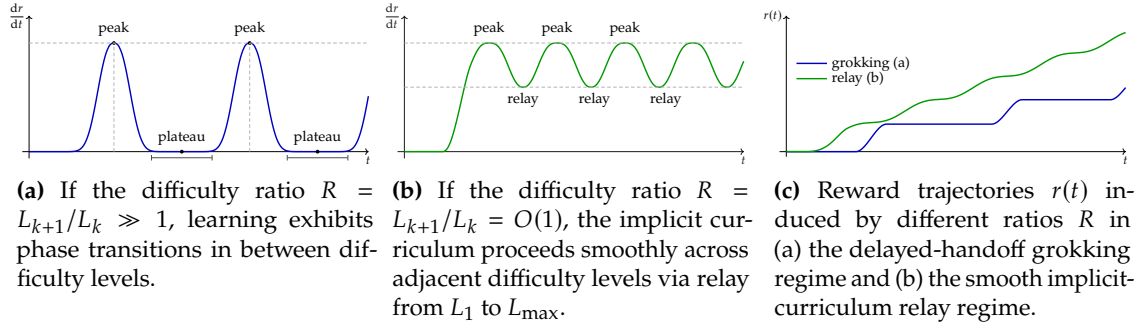


Figure 6.2: Reward Dynamics Across Difficulty Ratios. *Reward dynamics in mixed-difficulty RL.* A schematic illustration of the reward growth rate dr/dt and $r(t)$ for mixed-difficulty RL, showing how the difficulty ratio $R = L_{k+1}/L_k$ determines whether the implicit curriculum at the edge of competence proceeds smoothly or through delayed plateau-and-jump transitions. Small ratios yield smooth relay-based handoffs across horizons, while large ratios produce delayed handoffs and grokking-type phase transitions.

6.2 Related Work

Empirical understanding of RLVR mechanisms. With RLVR’s recent success and apparent scalability, there has been growing interest in understanding its mechanics, sparking an active debate: *what does RLVR actually teach base LLMs beyond pre-training?* The existing literature provides mixed evidence. Some works characterize RL primarily as a capability refiner or reranker [315, 332, 334, 344], whereas others argue that RL can induce substantial reasoning gains beyond the base model [176, 263, 310, 333]. Related findings suggest that RLVR can yield notable improvements even from spurious rewards [251] or extremely limited RL data (e.g., one-shot) [298]. Complementary evidence further highlights an entropy-based mechanism as a potential driver of such gains [66, 294]. Recently, [339] helps reconcile these views by showing that genuine capability gains arise mainly when there is sufficient headroom beyond pre-training and the RL data are calibrated to the model’s edge of competence.

Theory of RL training for LLMs. The empirical success of RLVR has spurred theoretical studies from various perspectives [37, 43, 52, 69, 180, 222, 231, 286, 350]. Of particular relevance are gradient-based analyses with transformer policies [37, 180, 231]. [37] formalize the benefits of curriculum-style RL post-training, while [180] studies learnability in settings with intermediate supervision. In contrast, we focus on outcome-based RL where no dense feedback is available. A closely related work [231] shows that outcome-based RL can induce step-by-step reasoning under gradient flow, with asymptotic guarantees under static data conditions. We go beyond this by analyzing realistic gradient dynamics along the full learning trajectory and characterizing phase transitions in both reward and gradient.

Curriculum and difficulty-aware RL. Curriculum learning in RL studies how sequencing tasks or samples can make hard problems learnable [199]. Recent LLM reasoning work uses easy-to-hard schedules [209], learned curriculum policies [51], and online filtering or frontier-of-

learnability sampling to focus training on intermediate-difficulty problems with useful success signals [22, 87]. Since generalization across difficulty levels is limited [153], broad and smooth difficulty spectra can be preferable to relying only on easy or hard subsets. Related hard-problem exploration methods reduce effective difficulty through privileged or off-policy prefixes, as in POPE [220] and PrefixRL [250]. These works support the real-world relevance of our mechanism: sparse terminal rewards become useful when data or prompting keeps training near the current competence frontier.

Grokking in supervised learning and RL. Grokking characterizes delayed generalization in supervised learning, where performance stays flat for long periods before improving abruptly [216]. Similar plateau-to-jump dynamics also appear in RLVR training [263], and have been described as “aha moments” in RL systems such as DeepSeek-R1-Zero [70]. Prior work investigates mechanisms behind such phase changes [155, 177, 198, 274], while our theory on the learning dynamics provides a mechanistic explanation for both the plateau and the subsequent transition in RLVR.

6.3 Problem Setup

In this section, we present the formal setup for our theoretical investigation. We first define the compositional reasoning task, and then describe a minimal transformer architecture as well as the policy gradient objective used to study RL training dynamics.

Notation. For functions h, g , write $h(x) = \Omega(g(x))$ (resp. $O(g(x))$) if there exist universal constants $C > 0$ and a such that $|h(x)| \geq C|g(x)|$ (resp. $\leq C|g(x)|$) for all $x \geq a$; write $h(x) = \Theta(g(x))$ if both bounds hold. We write $h(x) = o(g(x))$ if $\lim_{x \rightarrow \infty} \frac{h(x)}{g(x)} = 0$, and $h(x) = \omega(g(x))$ if $\lim_{x \rightarrow \infty} \frac{h(x)}{g(x)} = \infty$. Let $\mathbb{1}\{\cdot\}$ be the indicator and $[L] = \{1, \dots, L\}$. We use $\tilde{O}$, $\tilde{\Theta}$ and $\tilde{\Omega}$ to suppress logarithmic factors, and use $\text{poly}(d)$ and $\text{polylog}(d)$ for polynomials in d and $\log d$, respectively.

6.3.1 Compositional Reasoning

To study the mechanistic challenges of multi-step reasoning, we consider the state-tracking task that has received much recent attention [130, 174, 195]. This setting serves as a simple example of compositional reasoning: while each individual step is computationally simple, the task requires the precise sequential composition of L transitions in order to compute the final result.

Definition 6.3.1 (L -step compositional reasoning). Let $\mathcal{Y}$ be a finite set (the *state space*) and $\mathcal{G}$ a finite group, acting on $\mathcal{Y}$ via $(g, y) \mapsto g(y)$. For any initial state $y_0 \in \mathcal{Y}$ and sequence of transitions $g_1, \dots, g_L \in \mathcal{G}$, we can obtain a trajectory $y_\ell = g_\ell(y_{\ell-1})$ where $\ell = 1, \dots, L$. The goal is to predict the final state y_L given the sequence $(y_0, g_1, \dots, g_L)$. We define L as the *length* or *horizon* of the problem.

Our analysis rests on some structural assumptions on the group actions of $\mathcal{G}$ on $\mathcal{Y}$, described below.

Assumption 6.3.1 (Group structure and action). We assume $\mathcal{G}$ is a finite non-abelian simple group that acts simply transitively on the set $\mathcal{Y}$, which implies that there is a bijective correspondence between group elements and states, such that $|\mathcal{G}| = |\mathcal{Y}| = d$. We focus on the asymptotic regime where the state-space size tends to infinity, i.e., $d \rightarrow \infty$.

Assumption 6.3.2 (Bounded-horizon collision sparsity). For every fixed horizon $L = O(1)$, let $g_1, \dots, g_L$ be sampled uniformly without replacement from $\mathcal{G}$. Define

$$\rho_L(d) \triangleq \sum_{\mathbf{i}=(i_1, \dots, i_L) \in [L]^L: \mathbf{i} \neq (1, \dots, L)} \mathbb{P}(g_{i_L} \circ \dots \circ g_{i_1} = g_L \circ \dots \circ g_1). \quad (6.3.1)$$

We assume $\rho_L(d) = o(1)$ as $d = |\mathcal{G}| \rightarrow \infty$ for every fixed L .

Remark 6.3.1. The non-abelian condition in Assumption 6.3.1 rules out the most degenerate source of path collisions, since abelian groups identify all orderings of the same multiset of operations. However, the outcome-based RL task also needs a mild quantitative separation between the intended trajectory and accidental shortcut trajectories. Assumption 6.3.2 captures this separation: for any fixed horizon, using the sampled prompt operators in a noncanonical order, or with repetitions, should almost never reach the same final state. Thus a terminal reward is, with high probability, evidence of following the intended composition rather than an accidental shortcut. Many standard group families satisfy this type of sparsity; see Remark 6.B.1.

Next, we specify the format of reasoning data and the distribution over such instances.

Definition 6.3.2 (Reasoning problems). Fix a set of positional identifiers $\mathcal{X} \subset \mathbb{R}^d$ consisting of mutually orthogonal vectors. We encode each token as a position–symbol pair (x, s) where $x \in \mathcal{X}$ and $s \in \mathcal{G} \cup \mathcal{Y}$. A length- L reasoning instance is $Z^L = (Z_p^L, Z_a^L)$ consisting of:

- *problem description (prompt)*: a sequence of L transition tokens:

$$Z_p^L = ((x_{p,1}, g_1), (x_{p,2}, g_2), \dots, (x_{p,L}, g_L)),$$

where $\{x_{p,\ell}\}_{1 \leq \ell \leq L} \subset \mathcal{X}$ are distinct and $\{g_\ell\}_{1 \leq \ell \leq L} \subset \mathcal{G}$.

- *compositional solution*: a sequence of $(L + 1)$ state tokens:

$$Z_a^L = ((x_{a,0}, y_0), (x_{a,1}, y_1), \dots, (x_{a,L}, y_L)),$$

where $y_\ell = g_\ell(y_{\ell-1})$ for all $\ell \in [L]$.

- *position alignment*: the prompt and solution positions are related by a fixed *unknown* permutation $\varsigma : \mathcal{X} \rightarrow \mathcal{X}$ such that $x_{a,\ell-1} = \varsigma(x_{p,\ell})$, $\forall \ell \in [L]$.

Since the network we will use is a transformer (Section 6.3.2), token positions carry no intrinsic meaning in the absence of absolute positional information. Hence, the prompt/solution format encodes only a relative correspondence and does not provide any positional shortcut. Now we are ready to define the data distribution for the reasoning problems.

Definition 6.3.3 (Data distribution $\mathcal{D}^L$). Given a problem length L , we sample a reasoning instance $Z^L = (Z_p^L, Z_a^L)$ as in Definition 6.3.2 by the following process:

1. $g_1, \dots, g_L$ is sampled from $\mathcal{G}$ uniformly without replacement;
2. the initial state y_0 is sampled uniformly at random from the set $\mathcal{Y}$;
3. we sample distinct prompt identifiers $\{x_{p,1}, \dots, x_{p,L}\}$ uniformly from $\mathcal{X}$;
4. Set $x_{a,\ell-1} = \varsigma(x_{p,\ell})$ for all $\ell \in \{1, \dots, L\}$. Additionally, $x_{a,L}$ is sampled from $\mathcal{X} \setminus \{x_{a,k}\}_{k=0}^{L-1}$.
5. the intermediate states are computed via the group action: $y_\ell = g_\ell(y_{\ell-1})$.

An instance $Z^L \triangleq (Z_p^L, Z_a^L)$ generated by the above procedure is said to be sampled from $\mathcal{D}^L$. Since both $\{g_\ell\}_{\ell=1}^L$ and $\{x_{p,\ell}\}_{\ell=1}^L$ are sampled without replacement, the length is bounded by $L \leq \min\{|\mathcal{X}| - 1, d\}$.

Assumption 6.3.3. We assume $|\mathcal{X}| = \Theta(d^{c_x})$ for some constant $c_x \in (0.1, 1)$. We denote by $L_{\max} = |\mathcal{X}| - 1$ the maximum problem length.

Embeddings and tokenizer. We embed each group symbol $g \in \mathcal{G}$ and each state symbol $y \in \mathcal{Y}$ into $\mathbb{R}^{2d}$ via a shared orthonormal map: $\text{emb} : \mathcal{G} \cup \mathcal{Y} \rightarrow \mathbb{R}^{2d}$. With a slight abuse of notation, whenever a symbol $g \in \mathcal{G}$ or $y \in \mathcal{Y}$ is used as a model input or appears in the network computation, we write the symbol itself to denote its embedding $\text{emb}(g)$ or $\text{emb}(y)$, respectively, whenever this causes no ambiguity. In the sequel, we use $s \in \mathcal{G} \cup \mathcal{Y}$ to refer generically to a symbol from either set. We also fix a bijective tokenizer $\tau : \mathcal{Y} \rightarrow [d]$ to index states for the next-state prediction objective, so that predicting $y \in \mathcal{Y}$ is equivalent to predicting the class label $\tau(y) \in [d]$.

6.3.2 Transformer Architecture

Building on Definitions 6.3.2 and 6.3.3, we now define a simple transformer, aimed at predicting the next state in the solution trace given the prompt and the current state token.

Definition 6.3.4 (One-layer transformer [288]). We consider a simplified transformer consisting of a single attention module followed by a one-hidden-layer MLP. For a prompt

$Z_p^L = ((x_{p,\ell}, g_\ell))_{\ell=1}^L$ and the current solution token $Z_{a,k} = (x_{a,k}, y_k)$, the transformer with parameter $\theta = (W, Q)$ outputs an (unnormalized) score vector over the next-state index in $[d]$:

$$\text{TF}_\theta(Z_{a,k}, Z_p^L) = \text{MLP}_W \left(\text{Attn}_Q(Z_{a,k}, Z_p^L) \right) \in \mathbb{R}^d, \quad (6.3.2)$$

where the operators Attn_Q and MLP_W are defined shortly in Eq. (6.3.3) and Eq. (6.3.4), respectively.

Attention Layer. The attention module uses the current solution position $x_{a,k}$ to form weights over the prompt positions $\{x_{p,\ell}\}_{\ell=1}^L$, and returns a vector that aggregates the transition embeddings. Specifically, given a query weight $Q \in \mathbb{R}^{d \times d}$, we define

$$\text{Attn}_Q(Z_{a,k}, Z_p^L) \triangleq \frac{1}{2} \left(y_k + \sum_{\ell=1}^L \mathbf{Attn}_{a,k \rightarrow p,\ell}(Z_{a,k}, Z_p^L) \cdot g_\ell \right) \in \mathbb{R}^{2d}, \quad (6.3.3a)$$

where the attention weight $\mathbf{Attn}_{a,k \rightarrow p,\ell}(Z_{a,k}, Z_p^L)$ is obtained by softmax-normalizing the inner products $\langle Qx_{a,k}, x_{p,\ell} \rangle$:

$$\text{softmax} \left(\left(\langle Qx_{a,k}, x_{p,1} \rangle, \dots, \langle Qx_{a,k}, x_{p,L} \rangle \right) \right)_\ell. \quad (6.3.3b)$$

Here, for any $u \in \mathbb{R}^n$, we denote the i -th entry of the softmax output as $\text{softmax}(u)_i \triangleq \frac{\exp(u_i)}{\sum_{j=1}^n \exp(u_j)}$. In standard transformer architectures, the score typically takes the form $\langle W_Q x_{a,k}, W_K x_{p,\ell} \rangle$ rather than using a single matrix Q . We adopt the equivalent reparameterization commonly used in theoretical studies to simplify analysis without changing expressivity [128, 328, 341]. The factor $1/2$ normalizes the combined contribution of the residual term y_k and the attention output.

Remark 6.3.2. We compute attention scores using only positional identifiers $x_{a,k}, x_{p,\ell}$, a standard simplification in theoretical analyses of transformers [59, 129, 135, 146, 309]. This decoupling separates token association from reasoning with vector embeddings (states and transitions), which will be carried out by the subsequent MLP computation. Since the attention scores depend only on $(x_{a,k}, x_{p,1:L})$, we often suppress the arguments and write $\mathbf{Attn}_{a,k \rightarrow p,\ell}$ instead of $\mathbf{Attn}_{a,k \rightarrow p,\ell}(Z_{a,k}, Z_p^L)$ when it is clear from the context.

MLP layer. Given the attention output in $\mathbb{R}^{2d}$, the MLP operator with parameter W maps it to logits in $\mathbb{R}^d$ for next-state prediction (indexed by $\tau : \mathcal{Y} \rightarrow [d]$). With m hidden units and ReLU activation $\sigma(z) = \max\{0, z\}$, for each $j \in [d]$,

$$\left[\text{MLP}_W(Z) \right]_j = \sum_{r=1}^m \sigma \left(\langle W_{j,r}, Z \rangle \right), \quad W_{j,r} \in \mathbb{R}^{2d}. \quad (6.3.4)$$

In the present work, we keep W fixed and assume that the MLP has already acquired pre-trained *atomic skills* for one-step transitions. Specifically, given a state y and a transition symbol g

appearing in the form $\frac{1}{2}(g + y)$, the MLP implements the map $y \mapsto g(y)$. The structural details of the MLP and the existence of such an implementation are deferred to Section 6.3.3. Accordingly, our focus is on how the attention module enables long-horizon composition once the model already possesses this one-step atomic skill. This perspective aligns with recent RLVR studies that investigate compositional reasoning beyond single-step transitions [213, 333].

Induced next-state distribution. Given the transformer’s output $\text{TF}_\theta(Z_{a,k}, Z_p^L)$, we define the induced next-state distribution (policy) by softmax normalization:

$$\pi_\theta(j \mid Z_{a,k}, Z_p^L) \triangleq \text{softmax}\left(\text{TF}_\theta(Z_{a,k}, Z_p^L)\right)_j \quad (6.3.5)$$

for all $j \in [d]$. Equivalently, we write $\pi_\theta(y \mid Z_{a,k}, Z_p^L) \triangleq \pi_\theta(\tau(y) \mid Z_{a,k}, Z_p^L)$ for $y \in \mathcal{Y}$. In our reasoning format, the answer positions $(x_{a,1}, \dots, x_{a,L})$ are part of the instance and are not predicted; the model only predicts the next state symbol at each step. Starting from the prefix $Z^{L,0} = [Z_{a,0}, Z_p^L]$, the induced distribution over the generated state sequence $\hat{y}^L = (\hat{y}_1, \dots, \hat{y}_L)$ factorizes autoregressively as

$$\begin{aligned} \pi_\theta(\hat{y}^L \mid Z^{L,0}) &= \prod_{k=0}^{L-1} \pi_\theta(\hat{y}_{k+1} \mid \hat{Z}_{a,k}, Z_p^L), \\ \text{where } \hat{Z}_{a,k} &= (x_{a,k}, \hat{y}_k), \hat{y}_0 = y_0. \end{aligned} \quad (6.3.6)$$

When it is clear from the context, we abbreviate the conditioning as (y_0, G^L) , where $G^L = (g_1, \dots, g_L)$. Formally, $\pi_\theta^L(\cdot \mid y_0, G^L)$ still conditions on the full instance Z^L (including $(x_{p,1:L}, x_{a,0:L})$ and the fixed permutation $\mathfrak{s}$); we simply suppress these positional variables in the notation.

6.3.3 Pretrained Atomic Skills

As alluded to previously, we assume that the MLP module provides a pre-trained atomic skill for single-step transitions, and we keep its parameter W fixed throughout RL training. This allows us to focus on long-horizon composition in the attention dynamics.

For each output index $j \in [d]$ and hidden neuron $r \in [m]$, let us define the feature magnitude

$$V_{j,r}(s) \triangleq \langle W_{j,r}, s \rangle, \quad s \in \mathcal{G} \cup \mathcal{Y}.$$

For each pair (g, y) , let $j = \tau(g(y))$ be the correct next-state index. Within the neuron group $\{W_{j,r}\}_{r \in [m]}$, we designate a unique neuron $r_{g,y} \in [m]$ associated with this pair¹. We define

$$B = C_B \log d \quad (6.3.7)$$

¹ Our theory can also accommodate the setting where, for each feature, there is a non-overlapping group of neurons $r_1, r_2, \dots, r_k \in [m]$ jointly satisfying the same activation pattern (e.g., $\sum_{i=1}^k \sigma(V_{j,r_i}(g) + V_{j,r_i}(y)) = 2B$). We do not adopt this setting to simplify the proof.

for some sufficiently large integer $C_B = O(1)$ and $\sigma_0 = d^{-1/2}$, and assume the features satisfy

$$V_{j,r_{g,y}}(g) = B, \quad V_{j,r_{g,y}}(y) = B + 2\sigma_0; \quad (6.3.8a)$$

$$V_{j,r_{g,y}}(s) = -B, \quad \forall s \in (\mathcal{G} \cup \mathcal{Y}) \setminus \{g, y\}; \quad (6.3.8b)$$

$$V_{j,r}(s) = 0, \quad \forall r \notin \{r_{g,y}\}_{\tau(g(y))=j}, \quad \forall s \in \mathcal{G} \cup \mathcal{Y}. \quad (6.3.8c)$$

Proposition 6.3.1. Under Assumptions 6.3.1–6.3.3, if the MLP weight W satisfies Eq. (6.3.8a)–Eq. (6.3.8c), then given any Q , for any $y_0 \in \mathcal{Y}$ and $G^1 = (g_1)$ with $g_1 \in \mathcal{G}$, we have

$$\pi_\theta(g_1(y_0) \mid y_0, G^1) = 1 - \frac{1}{\text{poly}d}.$$

Note that for $L = 1$, the model necessarily attends to the only prompt $Z_{p,1}$. Combined with the residual connection, the MLP receives an aggregate input $\frac{1}{2}(g_1 + y_0)$. Proposition 6.3.1 thus guarantees that an MLP equipped with the above structural properties can perfectly implement the atomic group action.

Remark 6.3.3. *At a high level, when the MLP input contains the correct pair (g, y) , a unique activated neuron creates a large positive margin for the correct logit, while mismatched symbols produce canceling (negative) contributions. Consequently, the MLP predicts $g(y)$ with near-perfect accuracy. Prior analysis [130] shows that an MLP can learn such a feature-separated structure under supervised training with suitable initialization.*

Therefore, we formalize this one-step capability into the following structural assumption on W .

Assumption 6.3.4 (Pretrained MLP). The MLP weight W is fixed and satisfies Eq. (6.3.8a)–Eq. (6.3.8c).

How does the transformer reason sequentially? Solving the L -step state-tracking task reduces to carrying out the *single-step* operation L times: retrieve the required transition g_ℓ from the prompt and apply it to the current predicted state $\hat{y}_{\ell-1}$. Under the above assumption, the pre-trained MLP contains all the atomic skills for one-step transitions, and the remaining challenge lies in *association*: the attention layer must find the correct g_ℓ for the current reasoning step, which is learned by RL or SFT.

6.3.4 Outcome-based RL Objective

We train the induced policy π_θ using an outcome-based RL objective with a terminal reward. Given an instance $Z^L \sim \mathcal{D}^L$ (equivalently, (y_0, G^L)), we generate a state sequence $\hat{y}^L = (\hat{y}_1, \dots, \hat{y}_L)$ with the policy π_θ via Eq. (6.3.6). We assign reward 1 if the final prediction $\hat{y}_L$ matches the true final state y_L , and 0 otherwise:

$$r(\hat{y}^L \mid y_0, G^L) \triangleq \mathbb{1}\{\hat{y}_L = y_L\},$$

where $y_L = g_L(\cdots g_1(y_0))$. The RL objective is defined as the expected terminal reward:

$$\mathcal{J}_L(\theta) = \mathbb{E}_{Z^L} \left[\mathbb{E}_{\hat{y}^L \sim \pi_\theta^L(\cdot | y_0, G^L)} [r(\hat{y}^L | y_0, G^L)] \right]. \quad (6.3.9)$$

For comparison, we also consider an SFT-type objective [62]. Unlike the outcome-based RL objective, which provides a terminal reward only after the full rollout, this supervised objective employs teacher forcing [130, 146, 309, 329]: at each step k we condition on the ground-truth current state y_{k-1} and apply immediate supervision to the next state y_k . Formally, the SFT objective is written as

$$\text{Loss}_L(\theta) \triangleq \mathbb{E}_{Z^L} \left[\frac{1}{L} \sum_{k=1}^L -\log \pi_\theta(y_k | y_{k-1}, G^L) \right]. \quad (6.3.10)$$

Learning algorithm. We consider the REINFORCE algorithm of policy gradient [314]:

$$\nabla \mathcal{J}_L(\theta) = \mathbb{E}_{Z^L, \hat{y}^L} \left[r(\hat{y}^L | y_0, G^L) \nabla \log \pi_\theta^L(\hat{y}^L | y_0, G^L) \right].$$

Given that W is kept fixed, we study gradient ascent on Q with *length-normalized* [91, 115] policy gradient:

$$Q^{(t+1)} = Q^{(t)} + \eta \nabla_Q \tilde{\mathcal{J}}_L(\theta^{(t)}), \quad (6.3.11)$$

where $\tilde{\mathcal{J}}_L(\theta^{(t)}) = \frac{1}{L} \mathcal{J}_L(\theta^{(t)})$ and $\eta > 0$ represents the step size. In addition, we consider gradient descent for optimizing the SFT loss:

$$Q^{(t+1)} = Q^{(t)} - \eta \nabla_Q \text{Loss}_L(\theta^{(t)}). \quad (6.3.12)$$

Assumption 6.3.5 (Initialization). At $t = 0$, $Q^{(0)}$ is initialized to be the zero matrix $\mathbf{0}_{d \times d}$.

For simplicity, in the ensuing discussions, we let $A^{(t)}$ represent the value of A at iteration t , dropping the explicit dependence on the $\theta^{(t)}$ or $Q^{(t)}$ as long as it is clear from the context.

6.4 Learning Short-horizon Compositional Reasoning

In this section, we examine the RL dynamics of transformers on short-horizon compositional tasks. We prove that RL successfully learns compositional reasoning up to a critical horizon, beyond which a flat-gradient barrier emerges due to the nature of sparse, outcome-based rewards. In contrast, we demonstrate that SFT can overcome this limitation by leveraging immediate supervision.

6.4.1 RL for Short-horizon Compositions

Following the setup in Section 6.3.3, the transformer $\text{TF}_{\theta(0)}$ at initialization can execute the atomic one-step skills. We assume that the attention is approximately uniform at $t = 0$, so the induced policy does not reliably implement multi-step compositions. Our first result shows that, for any short horizon $L \leq C_B = O(1)$ where C_B is determined by the pretrained MLP parameters, policy-gradient RL successfully learns the L -step composition and yields the desired attention concentration pattern.

Theorem 6.4.1 (RL for short-horizon problems). *Suppose Assumptions 6.3.1-6.3.4 hold, and assume that $\text{TF}_{\theta(0)}$ is initialized according to Assumption 6.3.5. Consider any $L \in [2, C_B]$ (where $C_B > 0$ is some sufficiently large constant as mentioned in Section 6.3.3), $\eta = \frac{1}{\text{poly}(d)}$ and $\epsilon \in (\frac{1}{\log^{\Omega(1)}(d)}, \frac{1}{4})$. Then the transformer $\text{TF}_{\theta(t)}$ trained via Eq. (6.3.11) on the objective $\mathcal{J}_L$ after $T_{L,\epsilon} = O(\frac{L_{\max} \log(L/\epsilon)}{\eta \log d} \cdot d^{(1-\epsilon)C_B-1})$ iterations attains:*

(a) Reward optimality: At $t = T_{L,\epsilon}$, the reward is optimal in the sense that

$$\mathcal{J}_L^{(t)} \geq 1 - O\left(\frac{1}{d^{C_B(1-\epsilon)-1}}\right).$$

(b) Optimal short-horizon attention: At $t = T_{L,\epsilon}$, for any $\ell \leq L$, we have

$$\text{Attn}_{a,\ell-1 \rightarrow p,\ell}^{(t)} \geq 1 - \epsilon.$$

Theorem 6.4.1 provides the first provable guarantee that a transformer can learn multi-step *compositional* reasoning via outcome-only policy gradients, even though the initialization implements only the atomic one-step skill. Our theory unveils an explicit short-horizon regime ($L \leq C_B$) in which learning is provably efficient. Beyond reward optimality, this theorem also uncovers an emergent attention concentration pattern, providing a mechanistic characterization of how the learned transformer implements the L -step composition.

6.4.2 Critical Horizon and Exponentially Flat Region

When L exceeds the critical threshold C_B , the near-uniform attention at initialization yields little informative signal to the MLP layer. As a consequence, the model behaves nearly randomly on long compositional instances. Under outcome-only rewards, the resulting policy-gradient signal becomes exponentially small even when a nonzero reward is received, as formalized in the result below.

Proposition 6.4.1 (Exponentially flat region). *Suppose Assumptions 6.3.1-6.3.4 hold, and assume that the $\text{TF}_{\theta(0)}$ is initialized according to Assumption 6.3.5. Then for any horizon $L > 2C_B$,*

whenever the feature magnitude $\max_{x, x' \in \mathcal{X}} \langle Q^{(t)} x, x' \rangle \leq 0.01$, we have $\mathcal{J}_L^{(t)} = \frac{1}{d}(1 \pm o(1))$, and

$$\max_{x, x' \in \mathcal{X}} \left| \left\langle [\nabla_Q \tilde{\mathcal{J}}_L^{(t)}] x, x' \right\rangle \right| \leq \tilde{O} \left(\frac{1}{L_{\max}} \right) \cdot d^{-\Omega(L)}.$$

Why is the landscape flat for RL initially? Conceptually, the initial training horizon controls the concentration of signals the model could learn from each sample, which dilutes due to the $O(d^L)$ possible trajectories if the model uniformly traverses the actions specified by the problem instance. In this case, outcome-based rewards, due to *lack of process-level feedback*, make it extremely difficult to extract out sufficient signals from policy gradients.

6.4.3 SFT Succeeds Beyond the Critical Horizon

For comparison purposes, we also consider an SFT objective as in Eq. (6.3.10). Since SFT provides intermediate supervision rather than only an outcome-based reward, it remains effective even for long-horizon problems.

Theorem 6.4.2 (SFT provably escapes initial flat region). *Suppose Assumptions 6.3.1-6.3.4 hold, and assume that $\text{TF}_{\theta^{(0)}}$ is initialized according to Assumption 6.3.5. Then for any length $2 \leq L \leq \text{polylog}(d)$, $\eta = \frac{1}{\text{poly}(d)}$ and $\epsilon \in (\frac{1}{\log^{\Omega(1)}(d)}, \frac{1}{4})$, the transformer $\text{TF}_{\theta^{(t)}}$ trained via Eq. (6.3.12) on the objective Loss_L (cf. Eq. (6.3.10)) for*

$$T_{L,\epsilon} = O \left(\frac{L_{\max} \log(L/\epsilon) d^{(1-\epsilon)C_B-1}}{\eta \epsilon \log d} + \frac{L_{\max} L}{\eta \log d} \right)$$

iterations, satisfies:

(a) Loss convergence: *At $t = T_{L,\epsilon}$, the loss converges in the sense that*

$$\text{Loss}_L^{(t)} \leq O \left(\frac{1}{d^{C_B(1-\epsilon)-1}} \right).$$

(b) Optimal attention: *For any $\ell \leq L$, we have*

$$\text{Attn}_{a,\ell-1 \rightarrow p,\ell}^{(t)} \geq 1 - \epsilon.$$

In words, Theorem 6.4.2 demonstrates that SFT can successfully train the transformer to solve the composition reasoning beyond the critical horizon.

6.5 Learning Dynamics of Mixed-difficulty RL

In practice, RL datasets typically contain instances of mixed complexity, due to either heterogeneous data collection or explicit curriculum strategies [51, 209, 337], which can fundamentally

affect optimization dynamics. Motivated by this observation, we study policy-gradient training under *mixed-difficulty* distributions, which is close to how large-scale LLM RL is performed in production [70]. Our central message is that large gaps in difficulty level lead to delayed, grokking-like handoffs, whereas smoother spectra enable a relay effect that transfers learning progress from easier to harder horizons.

6.5.1 Easy-to-Hard Mixture

To model mixed difficulty, we consider a mixture over multiple reasoning horizons. This geometric horizon schedule serves as our formalization of a *difficulty spectrum*: adjacent horizons play the role of neighboring difficulty levels through which the implicit curriculum unfolds. Let us choose *difficulty ratio* $R > 1$ and fix a starting horizon $L_1 \geq 2$. Define a set of horizons $\mathcal{L}_R = \{L_1, L_2, \dots, L_K\}$ by the recursion:

$$L_k = \begin{cases} \lceil RL_{k-1} \rceil, & \text{if } L_{\max} > \lceil RL_{k-1} \rceil \\ L_{\max}, & \text{otherwise} \end{cases}$$

where $K = \min\{k : L_k = L_{\max}\}$ so that $L_K = L_{\max}$. We then define the mixed-difficulty objective as the uniform mixture

$$\mathcal{J}_{\text{mix},R}(\theta) = \mathbb{E}_{L \sim \text{Unif}(\mathcal{L}_R)}[\mathcal{J}_L(\theta)]. \quad (6.5.1)$$

To optimize $\mathcal{J}_{\text{mix},R}$, we consider a length-normalized update:

$$Q^{(t+1)} = Q^{(t)} + \eta \nabla_Q \tilde{\mathcal{J}}_{\text{mix},R}(\theta^{(t)}), \quad (6.5.2)$$

where $\nabla_Q \tilde{\mathcal{J}}_{\text{mix},R}(\theta) = \mathbb{E}_{L \sim \text{Unif}(\mathcal{L}_R)}[\nabla_Q \tilde{\mathcal{J}}_L(\theta)]$, and recall $\tilde{\mathcal{J}}_L(\theta) = \frac{1}{L} \mathcal{J}_L(\theta)$.

6.5.2 Grokking Dynamics Under Large Difficulty Ratios

Intuitively, under mixed-difficulty training, the shorter-horizon tasks are simpler and are therefore solved first, after which learning attempts to extend to longer horizons. When the gap between adjacent difficulty levels is large, i.e., when the mixture ratio R is large, this inter-horizon handoff becomes severely delayed: the implicit curriculum turns jagged because the next horizon remains too far beyond the current competence frontier. As a result, the policy can spend an extended period with near-zero reward on the next horizon before making noticeable improvements. This plateau-and-jump pattern resembles an empirical phase-transition phenomenon dubbed *grokking* [216, 263]: after a prolonged phase of receiving near-zero reward, the policy abruptly rises to near-perfect accuracy. Our next theorem formalizes this behavior in the mixed-horizon setting by quantitatively characterizing the length of the plateau and the ensuing transition.

To state our result, we introduce two observable states for each horizon that capture (i) when progress first becomes noticeable and (ii) when the horizon is essentially solved.

Stopping times for mastery and visible states. For any horizon $L_k \in \mathcal{L}_R$, we say that the task at horizon L_k has *visible return* at iteration t if $\mathcal{J}_{L_k}^{(t)} \geq 0.01$:

$$T_{\text{vis},k} \triangleq \min \{t : \mathcal{J}_{L_k}^{(t)} \geq 0.01\}. \quad (6.5.3)$$

We say that the horizon L_k is *mastered* at iteration t if $\mathcal{J}_{L_k}^{(t)} \geq 0.99$:

$$T_{\text{mas},k} \triangleq \min \{t : \mathcal{J}_{L_k}^{(t)} \geq 0.99\}. \quad (6.5.4)$$

Theorem 6.5.1 (Grokking dynamics). *Let $\mathcal{J}_{\text{mix},R}$ be the mixed-difficulty objective with ratio $\omega(1) \leq R \leq \frac{L_{\max}}{2C_B}$ and starting horizon $L_1 = C_B$. Under Assumptions 6.3.1–6.3.5, consider the RL training process under the length-normalized update Eq. (6.5.2) with step size $\eta = 1/\text{poly}(d)$. Then for each $1 \leq k \leq K-2$, the following hold:*

- (a) Long inter-difficulty plateaus. *Before the next horizon L_{k+1} makes noticeable progress (i.e., before it enters the visible-return state), the inter-horizon plateau length satisfies*

$$T_{\text{vis},k+1} - T_{\text{mas},k} = \tilde{\Theta}\left(\frac{L_{\max}}{\eta}\right) \cdot d^{C_B-1}. \quad (6.5.5)$$

- (b) Grokking-like phase transitions. *Once L_{k+1} enters the visible reward state, it reaches mastery quickly: $T_{\text{mas},k+1} - T_{\text{vis},k+1} \leq \tilde{O}\left(\frac{L_{\max}}{\eta}\right) \cdot L_{k+1}$.*

Theorem 6.5.1 shows that each transition $L_k \rightarrow L_{k+1}$ consists of a long near-zero-return plateau followed by a rapid rise to mastery once return becomes visible. Aggregating these transitions yields a time-to-mastery bound for the longest horizon, in which the total runtime is dominated by the plateaus.

Corollary 6.5.1. *Under the assumptions of Theorem 6.5.1, suppose $c_x < \frac{C_B-2}{C_B+2}$, where c_x is defined in Assumption 6.3.3. Then the first time the longest horizon $L_{\max}$ reaches mastery satisfies*

$$T_{\text{mas},K} = \tilde{\Theta}\left(\frac{L_{\max}}{\eta}\right) \cdot d^{C_B-1} \triangleq \mathcal{T}_{\text{plat}}. \quad (6.5.6)$$

Why does grokking happen in RL? For long-horizon tasks, reward can either come from fully correct traces or from rare lucky guesses that reach the correct final answer despite intermediate mistakes. Before the policy can reliably generate correct traces at long horizons, the reward stays near-zero, and the gradient signal mainly consists of those from the lucky guesses, which are random and uninformative. Meanwhile, gradient updates from shorter horizons keep sharpening the internal features long after their rewards have saturated. In other words, shorter horizons continue to supply hidden structural progress, but this progress is handed off to the next horizon only after a long delay because the difficulty gap is too large. Once the longer horizon finally

becomes learnable, rewards improve rapidly within a few iterations. Our analysis in Lemma 6.A.2 makes explicit this delayed-handoff mechanism by tracking how the hidden feature sharpening eventually translates into long-horizon policy improvement.

6.5.3 Relay Dynamics under Moderate Difficulty Ratios

The grokking dynamics above highlight that when the mixture contains well-separated horizons, training can stall at each new horizon: even after L_k is mastered, the next horizon L_{k+1} may remain in the near-zero-reward regime for a long plateau before its reward becomes visible. We now show the smooth counterpart: when the difficulty spectrum is sufficiently smooth (i.e., when R is a moderate constant), the implicit curriculum enters a well-behaved relay regime. The underlying optimization mechanism is a *relay* effect, in which progress on easier horizons continuously supports the next harder horizon, preventing prolonged plateaus.

Our next theorem formalizes the relay regime by providing an upper bound on $T_{\text{vis},k+1} - T_{\text{mas},k}$, which will be significantly smaller than the long plateaus in the regime with large difficulty gaps.

Theorem 6.5.2 (Relay dynamics). *Let $\mathcal{J}_{\text{mix},R}$ be the mixed-difficulty objective with ratio $2 \leq R \leq O(1)$ and starting horizon $L_1 = C_B$. Under Assumptions 6.3.1–6.3.5, consider the RL training process under the length-normalized update Eq. (6.5.2) with step size $\eta = 1/\text{poly}(d)$. Then for each $k \leq K-2$, before L_{k+1} enters the visible-return state, the inter-horizon plateau length satisfies*

$$T_{\text{vis},k+1} - T_{\text{mas},k} \leq \tilde{O}\left(\frac{L_{\max}}{\eta}\right) \cdot d^{(1 - \frac{C_B}{C_B+R})C_B-1}.$$

Moreover, once L_{k+1} enters the visible-return state, it reaches mastery rapidly: $T_{\text{mas},k+1} - T_{\text{vis},k+1} \leq \tilde{O}\left(\frac{L_{\max}}{\eta}\right) \cdot L_{k+1}$.

Compared with the grokking regime, Theorem 6.5.2 shortens each inter-horizon plateau by a factor $d^{\Theta(1)}$. Although a smoother spectrum induces more horizons, we have $K = O(\log_R(L_{\max}/C_B)) \leq O(\log d)$. Therefore, the total time to reach mastery at the longest horizon is still governed by the (much shorter) relay plateaus, leading to a strictly faster overall convergence bound than in the large-ratio regime.

Corollary 6.5.2. *Under the assumptions of Theorem 6.5.2, suppose $c_x < \frac{C_B-2}{C_B+2}$, where c_x is defined in Assumption 6.3.3. Then the first time that the longest horizon $L_{\max}$ reaches mastery satisfies $T_{\text{mas},K} \leq \mathcal{T}_{\text{relay}}$, where*

$$\mathcal{T}_{\text{relay}} \leq \tilde{O}\left(\frac{L_{\max}}{\eta}\right) \cdot d^{(1 - \frac{C_B}{C_B+R})C_B-1} \leq \tilde{O}\left(d^{-\frac{C_B^2}{C_B+R}}\right) \mathcal{T}_{\text{plat}}.$$

Implicit curriculum at the edge of competence. Theorem 6.5.1 and Theorem 6.5.2 show that the mixed-horizon dynamics are governed by how quickly training can move from mastering L_k to making noticeable progress on L_{k+1} . These are two regimes of the same implicit curriculum. When $R = O(1)$, the handoff is overlapping and smooth: progress on L_k starts benefiting L_{k+1} before L_k fully saturates, so the next horizon gains a visible return much sooner and training relays steadily across horizons. This is the relay regime. The policy remains “just competent enough” on L_k that success on L_{k+1} is no longer purely random, while L_k still provides a strong learning signal; the two horizons therefore improve in tandem, keeping optimization near the edge of competence [339]. In the large- R regime, by contrast, the same curriculum has a delayed handoff: the policy must become overwhelmingly competent on L_k before L_{k+1} ceases to be random, so training exhibits a long plateau followed by a grokking-like phase transition. See Section 6.A.2 for a more detailed, gradient-level mechanism explanation.

6.6 Proof Roadmap

The analysis reduces the transformer dynamics to a scalar attention-strength parameter that measures how sharply the policy retrieves the action needed at the current reasoning step. The terminal-reward policy gradient for horizon L is informative only when the current attention strength makes correct or nearly correct trajectories sufficiently likely. This yields the critical horizon in Proposition 6.4.1: short tasks generate a usable gradient, while longer tasks begin in an exponentially flat region.

For a mixture of horizons, shorter tasks therefore increase the shared attention strength first. Their updates continue after their visible reward has saturated and gradually move the next horizon toward its learnable regime. If adjacent horizons are separated by a large ratio, the next task remains reward-flat for a long interval and then improves abruptly, producing the delayed handoff and grokking behavior in Theorem 6.5.1. With a moderate ratio, the next horizon becomes gradient-effective earlier, giving the smooth relay dynamics of Theorem 6.5.2.

The exact gradient characterization is obtained by expressing the induced trajectory distribution as a convolution of measures on the underlying group and controlling its nontrivial Fourier components. The [appendix to this chapter](#) contains this harmonic-analysis argument, the posterior and reward estimates, and the complete fixed- and mixed-horizon inductions.

6.7 Experiments

In this section, we provide empirical support for our theoretical findings. We first use a controlled synthetic setting that mirrors our theory, and then report real-model evidence on N -digit multiplication.

6.7.1 Synthetic Experiments

General setup. The experimental setting is designed to mirror our theoretical framework and assumptions in Section 6.3. We consider a cyclic group action over $\mathbb{Z}_{96}$, and study two training paradigms:

- fixed-length training with reasoning lengths $L \in \{5, 15, 45\}$;
- mixed-length training: $\mathcal{L}_3 = \{5, 15, 45\}$ and $\mathcal{L}_7 = \{5, 35\}$. The reasoning depth L is stochastically assigned within each training batch ($N = 512$), where the problem length of each individual sample is uniformly drawn from $\mathcal{L}_R$ with R the difficulty ratio.

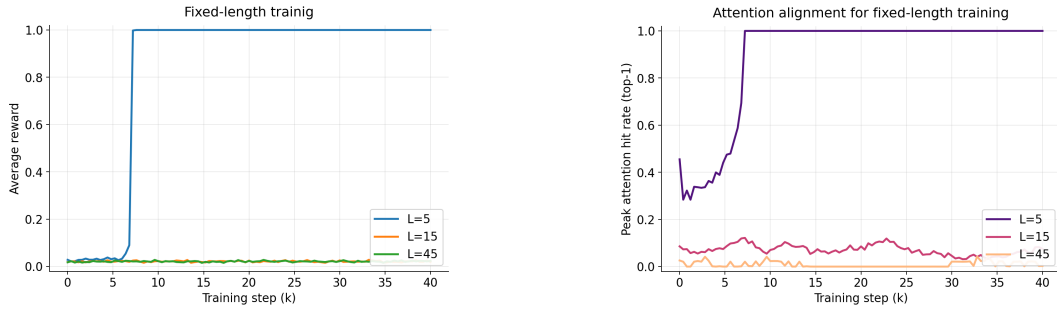
We use an abelian cyclic action for experimental convenience; for the lengths considered here ($L \geq 5$), the commutativity-induced shortcut effects discussed in Remark 6.3.1 are mild.

Model and training settings. We employ a one-layer detached attention layer paired with a fixed MLP transition head. First, the transition MLP is pretrained via supervised learning to master the one-step group operation $(y, g) \mapsto y \cdot g$, after which its parameters are frozen. The attention parameters Q are subsequently trained using the REINFORCE algorithm to maximize the terminal reward. We utilize an exponential moving-average baseline (momentum 0.95) and an entropy penalty (10^{-3}) to facilitate stable policy gradients.

Evaluation. We periodically compute the per-length average success rate by running greedy rollouts and calculating the fraction of episodes where the model correctly predicts the entire trajectory $\{\hat{y}_L = y_L\}$ over 30 batches of size 512. We additionally report an attention-alignment metric, peak attention-hit rate, defined as the fraction of steps where the argmax attention weight selects the unique prompt position corresponding to the current action token, i.e., the prompt index $s^{-1}(x_{a,k})$ within the sampled set $\{x_{p,\ell}\}_{\ell=1}^L$. A higher hit rate indicates that the attention layer recovers the underlying permutation s by locating the correct prompt position at each step, consistent with attention concentration ($\text{Attn}_L \rightarrow 1$).

Results for fixed-length training. The average reward and peak attention hit rate under fixed-length training are reported in Figure 6.3. We overlay three curves corresponding to training runs at different lengths in the same plot. We observe that the short-horizon setting ($L_1 = 5$) achieves nearly optimal reward together with strong attention concentration, whereas longer horizons exhibit near-flat plateaus in both reward and attention. This behavior is consistent with the predictions of Theorem 6.4.1 and Proposition 6.4.1.

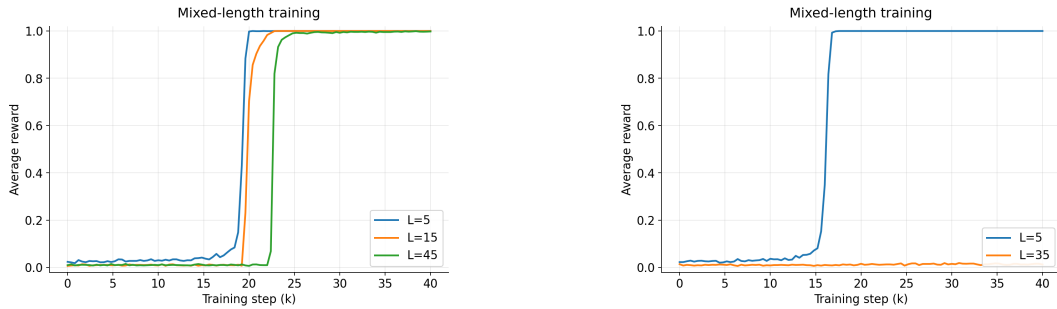
Results for mixed-length training. For mixed-length training, we consider two difficulty-ratio regimes: moderate ($R = 3$) and large ($R = 7$). As shown in Figure 6.4, both settings exhibit the paper’s implicit curriculum, but in different forms. In the moderate-ratio setting, Figure 6.4a shows the smooth relay regime: the plateau between consecutive horizons is shortened, so progress relays efficiently from easier to harder tasks. In contrast, Figure 6.4b shows the delayed-handoff regime: when the difficulty ratio is too large, the longer horizon ($L = 35$) remains at



(a) RL rapidly learns short-horizon compositions, whereas longer horizons exhibit a near-flat reward plateau.

(b) Attention concentrates for short horizons, but saturates at a lower level for longer horizons.

Figure 6.3: Fixed-Length RL Dynamics. Average reward and peak attention hit rate during fixed-length RL training.



(a) $R = 3$: the implicit curriculum stays smooth, with relay across horizons.

(b) $R = 7$: a larger difficulty ratio yields a prolonged plateau at longer horizons.

Figure 6.4: Mixed-Length RL Dynamics. Average reward during mixed-length RL training under different difficulty ratios.

near-zero reward throughout training, reflecting a prolonged plateau before the next stage of progress becomes visible. Taken collectively, these observations validate the predictions from Theorem 6.5.1 and Theorem 6.5.2 in the mixed-difficulty regime.

6.7.2 Real-world LLM RL Behaviors on N-digit Multiplication

We complement the controlled experiments with RLVR runs on large-scale models. These experiments use N -digit by N -digit integer multiplication as a scalable reasoning task with a natural difficulty parameter N .

General setup. Each prompt asks for the product of two N -digit operands sampled uniformly from $[10^{N-1}, 10^N)$, and the reward is binary exact match on the final answer. We run experiments on Qwen2.5-1.5B-Instruct [322] and Qwen3-4B-Base [323]. We study:

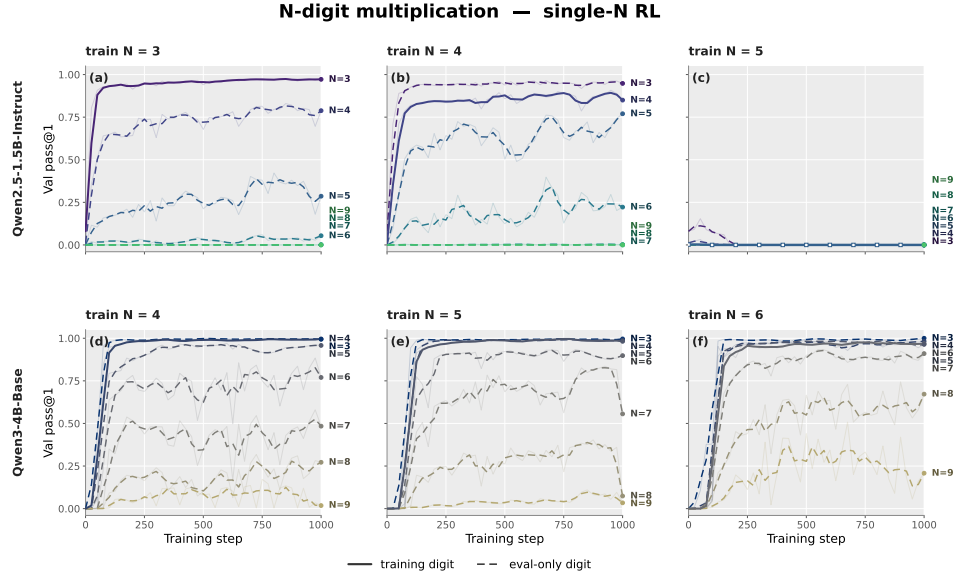


Figure 6.5: Single-Length RL Generalization. RL training on single- N dataset for Qwen2.5-1.5B-Instruct and Qwen3-4B-Base across different digit lengths. Across settings, models rapidly improve on the training length and often transfer to nearby or shorter lengths, but performance drops sharply for longer unseen lengths. Qwen3-4B-Base exhibits stronger in-distribution learning and broader near-length generalization than Qwen2.5-1.5B-Instruct, yet both models show limited extrapolation to substantially larger N , indicating that single-length RL primarily induces generalization to nearby lengths.

- *Single- N RL*: training on a single fixed digit length $N \in \{3, 4, 5\}$ for Qwen2.5-1.5B-Instruct and $N \in \{4, 5, 6\}$ for Qwen3-4B-Base;
- *Mixed- N RL*: training on prompts whose digit length is drawn uniformly per sample from a configured set within each batch, with no explicit curriculum schedule. We consider smooth-gap and large-gap mixtures across the two model scales: Qwen2.5-1.5B-Instruct uses $N \in \{3, 4, \dots, 9\}$ and $N \in \{3, 6, 9\}$, while Qwen3-4B-Base uses $N \in \{4, 6, 10\}$ and $N \in \{4, 10, 18\}$.

RL training and evaluation. All experiments use GRPO [252] with $n = 8$ rollouts per prompt, temperature 1.0, clip ratio 0.2, no KL penalty, token-level loss, and an over-length penalty. We train with Adam [150] at learning rate 10^{-6} for 1000 update steps, using maximum response lengths of 4096 tokens for the 1.5B model and 8192 tokens for the 4B model. Checkpoints and per-digit evaluations are produced every 25 steps, with greedy decoding at evaluation. We report per-digit pass@1 on held-out splits with 200 problems per digit length. Dashed lines indicate digit lengths that are evaluated but not trained.

Results for single- N training. Single- N RL reveals a *capability frontier* that depends sharply on model scale. In Figure 6.5, for Qwen2.5-1.5B-Instruct (the top panel), training succeeds only at less

difficult digits $N \in \{3, 4\}$: in-domain accuracy saturates above 0.9 within 100 steps and transfers broadly to neighboring digits, whereas training on $N = 5$ stays at zero accuracy over the entire 1000-step budget, supporting our theory in Theorem 6.4.1 and Proposition 6.4.1. Qwen3-4B-Base (the bottom panel) pushes the frontier substantially further: all three runs ($N=4, 5, 6$) succeed and transfer down to $N=3$ and up to $N=9$.

Results for mixed-N training. For mixed-N training, both smooth-gap and large-gap regimes exhibit the implicit curriculum, but in distinct forms (Figure 6.1). In the *smooth relay* regime (left panel), successive training digits emerge in an easy-to-hard staircase: each harder N takes off shortly after its easier predecessor significantly grows, and the hardest training digit climbs to non-trivial accuracy. In the *grokking* regime (right panel), a larger gap between consecutive training digits induces a *prolonged plateau* for the next horizon and the hardest in-domain digit remains at zero throughout training. This result validates the predictions of Theorem 6.5.1 and Theorem 6.5.2 in mixed-difficulty settings for large-scale models.

Appendix

6.A Detailed Proof Roadmap

This section explains the main proof ideas behind our learning-dynamic analysis. The central technical ingredient is a characterization of the policy-gradient signal as a function of a *step-wise probability*, which reveals an explicit long-horizon thresholding mechanism.

6.A.1 Technical Preliminaries

We first map our reasoning mechanism to some step-invariant quantities. At step ℓ , attention weights over prompt tokens are given by a softmax of scores $\langle Qx_{a,\ell-1}, x_{p,\ell'} \rangle$. Since $x_{a,\ell-1} = \mathfrak{s}(x_{p,\ell})$, correct retrieval means that the *aligned* prompt token $x_{p,\ell}$ receives a strictly larger score than all misaligned tokens $x_{p,\ell'}$ with $\ell' \neq \ell$. With initialization $Q^{(0)} = \mathbf{0}$ and the symmetry of $\mathcal{D}^L$, the expected policy gradient update preserves a two-level score structure: for all $x \in \mathcal{X}$ and $x' \neq x$,

$$\langle Q^{(t)} \mathfrak{s}(x), x \rangle = q^{(t)}, \quad \langle Q^{(t)} \mathfrak{s}(x), x' \rangle = r^{(t)}.$$

Consequently, the attention weight is step-invariant:

$$\mathbf{Attn}_{a,\ell-1 \rightarrow p,\ell}^{(t)} \equiv \mathbf{Attn}_L^{(t)}.$$

The following lemma translates $\mathbf{Attn}_L^{(t)}$ into the corresponding three classes of step-invariant next-state probabilities.

Lemma 6.A.1 (Step-invariant probability). *Given $\mathbf{Attn}_L^{(t)}$, with a fixed MLP in Assumption 6.3.4, the next-state distribution assigns step-invariant probability masses to:*

- *Target:* $p_{L,1}^{(t)} \triangleq \pi_{\theta^{(t)}}(g_\ell(\widehat{y}_{\ell-1}) \mid \widehat{y}_{\ell-1}, G^L)$;
- *Context distractor:* $p_{L,2}^{(t)} \triangleq \pi_{\theta^{(t)}}(g_{\ell'}(\widehat{y}_{\ell-1}) \mid \widehat{y}_{\ell-1}, G^L)$ for any $\ell' \neq \ell$;
- *Other distractor:* $p_{L,3}^{(t)} \triangleq \pi_{\theta^{(t)}}(y \mid \widehat{y}_{\ell-1}, G^L)$ for any $y \notin \{g_{\ell'}(\widehat{y}_{\ell-1}) \mid \ell' \in [L]\}$.

Moreover, it holds that $p_{L,1}^{(t)} \propto d^{C_B \mathbf{Attn}_L^{(t)}}$. For later use, define the effective probability margin:

$$\Delta_L^{(t)} := p_{L,1}^{(t)} - p_{L,3}^{(t)} \quad \text{and} \quad \delta_L^{(t)} := p_{L,2}^{(t)} - p_{L,3}^{(t)}. \quad (6.A.1)$$

6.A.2 Key Lemma for Gradient Estimation

The preceding discussion reduces the model's one-step behavior to two scalars $(q^{(t)}, r^{(t)})$. We now state the main technical result: for length- L tasks, the policy-gradient signal in the $(q^{(t)}, r^{(t)})$ coordinates admits an explicit characterization in terms of the step-level margin $\Delta_L^{(t)}$. This turns the learning dynamics into an effective one-dimensional evolution of attention concentration.

Lemma 6.A.2 (Gradient characterization). *Throughout the mixed-training process, given $L \in \mathcal{L}_R$, if the step-wise probability satisfies $\frac{\delta_L^{(t)}}{\Delta_L^{(t)}} \ll \frac{1}{L^2}(1 - \Delta_L^{(t)})$ and $\frac{p_{L,2}^{(t)}}{p_{L,1}^{(t)}} \ll 1 - p_{L,1}^{(t)}$, then we have*

$$\nabla_q \tilde{\mathcal{J}}_L^{(t)} \propto (\Delta_L^{(t)})^L (1 - \Delta_L^{(t)}), \quad (6.A.2)$$

$$\mathcal{J}_L^{(t)} = \frac{1}{d} + \left(1 - \frac{1}{d}\right)(1 \pm o(1)) \cdot (\Delta_L^{(t)})^L. \quad (6.A.3)$$

Moreover, it holds that $|\nabla_r \tilde{\mathcal{J}}_L^{(t)}| \leq O(\frac{1}{L_{\max}}) \nabla_q \tilde{\mathcal{J}}_L^{(t)}$.

Remark 6.A.1. Lemma 6.A.2 shows that the policy gradient is essentially driven by the q -direction, with $|\nabla_r \tilde{\mathcal{J}}_L|$ being a lower-order term. Its magnitude exhibits two regimes: when Δ_L is small, the long-horizon factor $(\Delta_L)^L$ exponentially suppresses the gradient; when $\Delta_L \approx 1$, the update is in a convergence regime and is controlled by the shrinking term $1 - \Delta_L$. Since larger q increases the target attention weight $\mathbf{Attn}_L$ and hence the margin Δ_L via Lemma 6.A.1, the takeaway is that length- L learning is negligible until q is large enough to make Δ_L moderate, after which progress slows down again as $\Delta_L \rightarrow 1$.

Two alternating phases in mixed-horizon training. Under mixed-horizon training, learning typically proceeds from shorter to longer horizons: shorter tasks suffer from weaker attention dilution, so they contribute usable gradient earlier and drive up q ; as q grows, longer horizons gradually become gradient-effective and then start to show visible return. Consequently, learning alternates between (i) a gradient transfer phase, where $\nabla_q \tilde{\mathcal{J}}_{L_k}$ dominates while $\nabla_q \tilde{\mathcal{J}}_{L_{k+1}}$ remains negligible, and (ii) a reward-emergence phase, where $\nabla_q \tilde{\mathcal{J}}_{L_{k+1}}$ becomes substantial and L_{k+1} quickly reaches mastery.

Regime comparison: smooth vs. delayed implicit curriculum. The key difference is whether $\nabla_q \tilde{\mathcal{J}}_{L_{k+1}}$ becomes non-negligible before $\nabla_q \tilde{\mathcal{J}}_{L_k}$ has decayed to a saturated signal. With Lemmas 6.A.1 and 6.A.2 in mind, activating L_{k+1} in the first place requires its target attention $\mathbf{Attn}_{L_{k+1}}$ to reach a constant-level regime so that the gate $(\Delta_{L_{k+1}})^{L_{k+1}}$ is not exponentially suppressed; the handoff is then controlled by when

$$(\Delta_{L_{k+1}})^{L_{k+1}} \approx 1 - \Delta_{L_k}.$$

For large $R = \omega(1)$, reaching this regime for L_{k+1} forces q so large that L_k is already driven to $\mathbf{Attn}_{L_k} = 1 - o(1)$, making $1 - \Delta_{L_k}$ small (on the order of $d^{-(C_B-1)}$). Thus $\nabla_q \tilde{\mathcal{J}}_{L_{k+1}}$ stays negligible over a long plateau, and reward emerges via a grokking-style jump: the curriculum handoff is delayed until the next horizon activates only after the previous one has nearly saturated. For moderate $R = O(1)$, the same catch-up happens while L_k is still away from full saturation, so $\nabla_q \tilde{\mathcal{J}}_{L_k}$ and $\nabla_q \tilde{\mathcal{J}}_{L_{k+1}}$ overlap and jointly drive progress. This overlap is the relay mechanism, and it is what makes the implicit curriculum smooth near the edge of competence.

6.A.3 Proof of Lemma 6.A.2: Fourier Analysis on Groups

We begin by discussing the central technical challenge in analyzing long-horizon policy gradients, and then introduce our Fourier-based techniques for tackling the challenges.

Key technical challenges. The starting point is to express the policy gradient in terms of the one-step *action distribution* on the group. By simple transitivity, each transition $\widehat{y}_{\ell-1} \rightarrow \widehat{y}_\ell$ corresponds to a unique group element $u_\ell \in \mathcal{G}$ such that $\widehat{y}_\ell = u_\ell(\widehat{y}_{\ell-1})$. Let μ_ℓ denote the one-step action law of u_ℓ on $\mathcal{G}$ (under the current policy), and write $\mu_\ell(g) = \mathbb{P}(u_\ell = g)$. Armed with this notation, the step- ℓ gradient reduces to a posterior-vs-prior gap for the target action:

$$\nabla_q \mathcal{J}_L \propto \sum_{\ell \in [L]} \left(\mathbb{P}(u_\ell = g_\ell \mid \widehat{y}_L = y_L) - \mu_\ell(g_\ell) \right).$$

Let $G_* = g_L \circ \dots \circ g_1$. Simple transitivity also implies that terminal success is exactly the group-product constraint

$$\widehat{y}_L = y_L \iff u_L \circ \dots \circ u_1 = G_*.$$

Hence, the success probability and the numerator in the posterior can be written as point evaluations of convolution products, for instance:²

$$\mathbb{P}(\widehat{y}_L = y_L) = (\mu_L * \dots * \mu_1)(G_*).$$

The challenge is that these are high-order convolutions evaluated at a specific group element. When L is large, a direct expansion in the group domain involves exponentially many mixed terms and offers no clean control: conditioning on $u_L \circ \dots \circ u_1 = G_*$ couples all steps, so the posterior $\mathbb{P}(u_\ell = \cdot \mid \widehat{y}_L = y_L)$ is inherently trajectory-level and does not factorize into per-step statistics.

Fourier analysis to estimate the dominant signal. To make these convolution powers tractable, we pass to the Fourier domain on $\mathcal{G}$ [152, 270], where convolution becomes multiplication, turning the L -fold convolution into a structured product of Fourier operators: $\prod_{\ell \in [L]} \widehat{\mu}_\ell(\lambda)$. Here $\widehat{\mu}_\ell(\lambda)$ is the Fourier transform of μ_ℓ at an irreducible unitary representation λ ². Notice that the step-invariant three-way partition of next-token outcomes in Lemma 6.A.1 is equivalently a three-way partition of u_ℓ , and thus

$$\mu_\ell(g) = p_{L,1} \mathbb{1}\{g = g_\ell\} + p_{L,2} \mathbb{1}\{g \in G^L \setminus \{g_\ell\}\} + p_{L,3} \mathbb{1}\{g \in \mathcal{G} \setminus G^L\}.$$

² see Section 6.B.2 for formal definitions of convolution $*$ and Fourier transforms.

Exploiting this structure, we obtain

$$\widehat{\mu}_\ell(\lambda) = \Delta_L \lambda(g_\ell) + \delta_L \sum_{g \in G^L \setminus \{g_\ell\}} \lambda(g),$$

which splits into a target-aligned contribution and a residual contribution from context distractors. Taking products across L steps, the leading contribution arises from selecting the aligned component at each step (under mild separation conditions on Δ_L and δ_L), which yields the characteristic $(\Delta_L)^L(1 - \Delta_L)$ structure in the resulting posterior deviation.

Prior use of group representations in machine learning. Group representation theory has been widely explored in machine learning to model and exploit symmetry [83, 186] and to analyze structured distributions [45, 152]. Our use is different in both setting and goal: we bring Fourier analysis into the study of *long-horizon, compositional* RL objectives, where the terminal success event couples all steps and makes policy-gradient estimation inherently trajectory-dependent. The group structure turns this global conditioning into an L -fold convolution object that can be controlled spectrally.

6.B Detailed Proofs

This section develops the proofs supporting the chapter’s main results. The phase-by-phase arguments are kept in the main chapter so that the optimization mechanisms remain connected to the story; auxiliary estimates are collected in explicitly labeled technical subsections.

6.B.1 Preliminaries

In this section, we introduce some useful notation and derive several preliminary policy gradient lemmas, which will be repeatedly used in the subsequent training-dynamics analysis. Throughout the proof, we use d_p to denote $|\mathcal{X}|$.

Gradient Computations

Notation for gradient expressions. Consider a problem instance of length L , we denote a full answer trajectory with initial prefix $Z_{a,0}$ by

$$\widehat{Z}_a^L \triangleq (Z_{a,0}, \widehat{Z}_{a,1}, \dots, \widehat{Z}_{a,L}).$$

For each $1 \leq \ell \leq L$, we denote the partial trajectory up to step ℓ (including the initial prefix) by

$$\widehat{Z}_a^{L,\ell} \triangleq (Z_{a,0}, \widehat{Z}_{a,1}, \dots, \widehat{Z}_{a,\ell}).$$

We introduce the following shorthand notation (for $j \in \tau(\mathcal{Y})$, $r \in [m]$, and $\ell, k \in [L]$):

$$\mathcal{E}_j(\widehat{Z}_a^{L,\ell}, Z_p^L) \triangleq \mathbb{1}_{\tau(\widehat{y}_\ell)=j} - \pi_\theta(j \mid \widehat{y}_{\ell-1}, G^L), \quad (6.B.1a)$$

$$\Lambda_{j,r}(\widehat{Z}_a^{L,\ell-1}, Z_p^L) \triangleq \frac{1}{2} \left(\langle W_{j,r}, \widehat{Z}_{a,\ell-1} \rangle + \sum_{k \in [L]} \mathbf{Attn}_{a,\ell-1 \rightarrow p,k} \cdot \langle W_{j,r}, Z_{p,k} \rangle \right). \quad (6.B.1b)$$

$$\Xi_{\ell,k}(\widehat{Z}_a^{L,\ell}, Z_p^L) \triangleq \frac{1}{2} \sum_{j \in \tau(\mathcal{Y})} \mathcal{E}_j(\widehat{Z}_a^{L,\ell}, Z_p^L) \sum_{r \in [m]} \sigma'(\Lambda_{j,r}(\widehat{Z}_a^{L,\ell-1}, Z_p^L)) \langle W_{j,r}, Z_{p,k} \rangle. \quad (6.B.1c)$$

Here, $\pi_\theta(j \mid \widehat{y}_{\ell-1}, G^L) = \text{softmax} \left(\text{TF}_\theta \left(\widehat{Z}_{a,\ell-1}, Z_p^L \right) \right)_j$.

Fact 6.B.1 (Gradients of Q). Given a problem length L , we have the following expression for the policy gradient w.r.t. the attention matrix Q :

$$\nabla_Q \widetilde{\mathcal{J}}_L = \frac{1}{L} \mathbb{E}_{Z^L, \widehat{y}^L} \left[\mathbb{1}_{\widehat{y}_L = y_L} \sum_{\ell=1}^L \nabla_Q \log \pi_\theta(\widehat{y}_\ell \mid \widehat{y}_{\ell-1}, G^L) \right],$$

where

$$\begin{aligned} & \nabla_Q \log \pi_\theta(\widehat{y}_\ell \mid \widehat{y}_{\ell-1}, G^L) \\ &= \sum_{k \in [L]} \mathbf{Attn}_{a,\ell-1 \rightarrow p,k} \cdot \left(\Xi_{\ell,k}(\widehat{Z}_a^{L,\ell}, Z_p^L) - \sum_{k' \in [L]} \mathbf{Attn}_{a,\ell-1 \rightarrow p,k'} \Xi_{\ell,k'}(\widehat{Z}_a^{L,\ell}, Z_p^L) \right) x_{a,\ell-1} x_{p,k}^\top. \end{aligned}$$

Moreover, the gradient of Q for the supervised loss Loss_L can be written as:

$$\begin{aligned} & -\nabla_Q \text{Loss}_L \\ &= \frac{1}{L} \mathbb{E}_{Z^L} \left[\sum_{\ell=1}^L \sum_{k=1}^L \mathbf{Attn}_{a,\ell-1 \rightarrow p,k} \cdot \left(\Xi_{\ell,k}(Z_a^{L,\ell}, Z_p^L) - \sum_{k' \in [L]} \mathbf{Attn}_{a,\ell-1 \rightarrow p,k'} \Xi_{\ell,k'}(Z_a^{L,\ell}, Z_p^L) \right) x_{a,\ell-1} x_{p,k}^\top \right]. \end{aligned}$$

Then we consider the gradient of $x^\top \nabla_Q \log \pi_\theta(\widehat{y}_\ell \mid \widehat{y}_{\ell-1}, G^L) x'$ for $x, x' \in \mathcal{X}$.

Lemma 6.B.1. Given $x, x' \in \mathcal{X}$, if $x = \mathfrak{s}(x')$, when $x_{a,\ell-1} = x$ for $\ell \in [L]$, then we have

$$\begin{aligned} & x^\top \nabla_Q \log \pi_\theta(\widehat{y}_\ell \mid \widehat{y}_{\ell-1}, G^L) x' \\ &= \mathbf{Attn}_{a,\ell-1 \rightarrow p,\ell} \cdot \left(\Xi_{\ell,\ell}(\widehat{Z}_a^{L,\ell}, Z_p^L) - \sum_{k' \in [L]} \mathbf{Attn}_{a,\ell-1 \rightarrow p,k'} \Xi_{\ell,k'}(\widehat{Z}_a^{L,\ell}, Z_p^L) \right) \\ &= \mathbf{Attn}_{a,\ell-1 \rightarrow p,\ell} \cdot \left(\sum_{j \in \tau(\mathcal{Y})} \mathcal{E}_j(\widehat{Z}_a^{L,\ell}, Z_p^L) \sum_{r \in [m]} \sigma'(\Lambda_{j,r}(\widehat{Z}_a^{L,\ell-1}, Z_p^L)) \cdot \left(\langle W_{j,r}, Z_{p,\ell} \rangle - \Lambda_{j,r}(\widehat{Z}_a^{L,\ell-1}, Z_p^L) \right) \right). \end{aligned}$$

Lemma 6.B.2. *Given $x, x' \in \mathcal{X}$, if $x \neq \mathfrak{s}(x')$, when $x_{p,\ell'} = x'$ and $x_{a,\ell-1} = x$ for $\ell, \ell' \in [L]$, noticing that $\ell' \neq \ell$, then we have*

$$\begin{aligned} & x^\top \nabla_Q \log \pi_\theta(\hat{y}_\ell \mid \hat{y}_{\ell-1}, G^L) x' \\ &= \mathbf{Attn}_{a,\ell-1 \rightarrow p,\ell'} \cdot \left(\Xi_{\ell,\ell'}(\hat{Z}_a^{L,\ell}, Z_p^L) - \sum_{k' \in [L]} \mathbf{Attn}_{a,\ell-1 \rightarrow p,k'} \Xi_{\ell,k'}(\hat{Z}_a^{L,\ell}, Z_p^L) \right) \\ &= \mathbf{Attn}_{a,\ell-1 \rightarrow p,\ell'} \cdot \left(\sum_{j \in \tau(\mathcal{Y})} \mathcal{E}_j(\hat{Z}_a^{L,\ell}, Z_p^L) \sum_{r \in [m]} \sigma'(\Lambda_{j,r}(\hat{Z}_a^{L,\ell-1}, Z_p^L)) \cdot (\langle W_{j,r}, Z_{p,\ell'} \rangle - \Lambda_{j,r}(\hat{Z}_a^{L,\ell-1}, Z_p^L)) \right). \end{aligned}$$

Observe that there are d^{L-1} intermediate-state sequences $(\hat{y}_1, \dots, \hat{y}_{L-1}) \in \mathcal{Y}^{L-1}$ that lead to $\hat{y}_L = y_L$. Only these trajectories yield a nonzero terminal reward. Hence, it suffices to restrict our attention to their contributions. Combining this with the preceding lemmas, a direct calculation gives

Lemma 6.B.3. *Given $x, x' \in \mathcal{X}$, we have*

- if $x = \mathfrak{s}(x')$, then

$$\begin{aligned} x^\top \nabla_Q \tilde{\mathcal{J}}_L x' &= \frac{1}{2Ld_p} \mathbb{E}_{Z^L} \left[\sum_{v_{1:L} \in \mathcal{Y}^{L-1} \times \{y_L\}} \left(\prod_{\ell'=1}^L \pi_\theta(v_{\ell'} \mid v_{\ell'-1}, G^L) \right) \cdot \left(\sum_{\ell=1}^L \mathbf{Attn}_{a,\ell-1 \rightarrow p,\ell} \right. \right. \\ &\quad \left. \left. \left(\sum_{j \in \tau(\mathcal{Y})} \mathcal{E}_j(\hat{Z}_a^{L,\ell}, Z_p^L) \sum_{r \in [m]} \sigma'(\Lambda_{j,r}(\hat{Z}_a^{L,\ell-1}, Z_p^L)) \cdot (\langle W_{j,r}, Z_{p,\ell} \rangle - \Lambda_{j,r}(\hat{Z}_a^{L,\ell-1}, Z_p^L)) \right) \right) \Big|_{\hat{y}_{1:\ell}=v_{1:\ell}} \right]. \end{aligned}$$

- else,

$$\begin{aligned} x^\top \nabla_Q \tilde{\mathcal{J}}_L x' &= \frac{1}{2Ld_p(d_p-1)} \mathbb{E}_{Z^L} \left[\sum_{v_{1:L} \in \mathcal{Y}^{L-1} \times \{y_L\}} \left(\prod_{\ell'=1}^L \pi_\theta(v_{\ell'} \mid v_{\ell'-1}, G^L) \right) \cdot \left(\sum_{\ell=1}^L \sum_{\ell' \neq \ell} \mathbf{Attn}_{a,\ell-1 \rightarrow p,\ell'} \right. \right. \\ &\quad \left. \left. \left(\sum_{j \in \tau(\mathcal{Y})} \mathcal{E}_j(\hat{Z}_a^{L,\ell}, Z_p^L) \sum_{r \in [m]} \sigma'(\Lambda_{j,r}(\hat{Z}_a^{L,\ell-1}, Z_p^L)) \cdot (\langle W_{j,r}, Z_{p,\ell'} \rangle - \Lambda_{j,r}(\hat{Z}_a^{L,\ell-1}, Z_p^L)) \right) \right) \Big|_{\hat{y}_{1:\ell}=v_{1:\ell}} \right]. \end{aligned}$$

Proof. The two results are obtained by invoking Lemma 6.B.1 and Lemma 6.B.2, respectively. Specifically, for a given ℓ , the event $\{x_{a,\ell-1} = x\}$ occurs with probability $1/d_p$, while for $k \neq \ell$, the event $\{x_{p,\ell} = \mathfrak{s}^{-1}(x), x_{p,k} = x', x' \neq \mathfrak{s}^{-1}(x)\}$ occurs with probability $\frac{1}{d_p(d_p-1)}$. The remaining calculation details are omitted here for brevity. $\square$

Similarly, for the gradient of SFT loss Loss_L , we have

Lemma 6.B.4. *Given $x, x' \in \mathcal{X}$, we have*

- if $x = \mathfrak{s}(x')$, then

$$-x^\top \nabla_Q \text{Loss}_L x' = \frac{1}{2Ld_p} \mathbb{E}_{Z^L} \left[\sum_{\ell=1}^L \mathbf{Attn}_{a,\ell-1 \rightarrow p,\ell} \right]$$

$$\left(\sum_{j \in \tau(\mathcal{Y})} \mathcal{E}_j(Z_a^{L,\ell}, Z_p^L) \sum_{r \in [m]} \sigma'(\Lambda_{j,r}(Z_a^{L,\ell-1}, Z_p^L)) \cdot \left(\langle W_{j,r}, Z_{p,\ell} \rangle - \Lambda_{j,r}(Z_a^{L,\ell-1}, Z_p^L) \right) \right).$$

• *else,*

$$-x^\top \nabla_Q \text{Loss}_L x' = \frac{1}{2Ld_p(d_p - 1)} \mathbb{E}_{Z^L} \left[\sum_{\ell=1}^L \sum_{\ell' \neq \ell} \mathbf{Attn}_{a,\ell-1 \rightarrow p,\ell'} \right. \\ \left. \left(\sum_{j \in \tau(\mathcal{Y})} \mathcal{E}_j(Z_a^{L,\ell}, Z_p^L) \sum_{r \in [m]} \sigma'(\Lambda_{j,r}(Z_a^{L,\ell-1}, Z_p^L)) \cdot \left(\langle W_{j,r}, Z_{p,\ell'} \rangle - \Lambda_{j,r}(Z_a^{L,\ell-1}, Z_p^L) \right) \right) \right].$$

We further introduce additional notation to simplify the presentation. Given $G^L = \{g_1, \dots, g_L\}$ and the initial value y_0 with induced $\{y_1, \dots, y_L\}$, define

$$\mathfrak{J}(\theta; y_0, G^L) \triangleq \sum_{v \in \mathcal{Y}^{L-1} \times \{y_L\}} \left(\prod_{\ell'=1}^L \pi_\theta(v_{\ell'} \mid v_{\ell'-1}, G^L) \right) \left(\sum_{\ell=1}^L \mathbf{Attn}_{a,\ell-1 \rightarrow p,\ell} \mathfrak{G}_\ell(\theta; v) \right) \quad (6.B.2)$$

$$\text{where } \mathfrak{G}_\ell(\theta; v) \triangleq \sum_{j \in \tau(\mathcal{Y})} \mathcal{E}_j(\widehat{Z}_a^{L,\ell}, Z_p^L) \sum_{r \in [m]} \sigma'(\Lambda_{j,r}(\widehat{Z}^{L,\ell-1})) \cdot \left(\langle W_{j,r}, Z_{p,\ell} \rangle - \Lambda_{j,r}(\widehat{Z}^{L,\ell-1}) \right) \Big|_{\widehat{y}_{1:\ell} = v_{1:\ell}}. \quad (6.B.3)$$

In what follows, we suppress the dependence on θ and write $\mathfrak{J}(y_0, G^L)$ and $\mathfrak{G}_\ell(v)$ for brevity.

Notations for scalarized attention dynamics. Based on the gradient update, the quantity $\langle x, Qx' \rangle$ takes only two possible values, depending on whether $x = s(x')$ (the matched position) or $x \neq s(x')$ (a mismatched position). Accordingly, we define the (unnormalized) *target* and *non-target* attention scores as

$$q \triangleq \langle Q s(x), x \rangle, \quad x \in \mathcal{X}, \quad (6.B.4a)$$

$$r \triangleq \langle Q s(x), x' \rangle, \quad x' \in \mathcal{X} \setminus \{x\}. \quad (6.B.4b)$$

With this notation, Lemma 6.B.3 can be viewed as a policy-gradient update on (q, r) . Hence, it suffices to track the dynamics of these two scalars in the sequel. Thus, following Eq. (6.B.2), we have

$$\nabla_q \widetilde{\mathcal{J}}_L = \frac{1}{2Ld_p} \mathbb{E}_{Z^L} [\mathfrak{J}(y_0, G^L)]. \quad (6.B.5)$$

Furthermore, under this reduction, for a fixed problem length L , the attention weights $\mathbf{Attn}_{a,\ell-1 \rightarrow p,\ell'}$ (for $\ell, \ell' \in [L]$) take only two distinct values depending on whether the prompt position matches:

$$\mathbf{Attn}_{a,\ell-1 \rightarrow p,\ell} = \frac{e^q}{e^q + (L-1)e^r}, \quad (6.B.6a)$$

$$\mathbf{Attn}_{a,\ell-1 \rightarrow p,\ell'} = \frac{e^r}{e^q + (L-1)e^r}, \quad \ell' \neq \ell. \quad (6.B.6b)$$

When the context is clear, we denote the target attention weight $\mathbf{Attn}_{a,\ell-1 \rightarrow p,\ell}$ by $\mathbf{Attn}_L$ for brevity.

Some Useful Bounds

Notation for activated neurons. Fix an output index $j \in \tau(\mathcal{Y})$, define the fiber $\mathfrak{F}_j \triangleq \{(g, y) \in \mathcal{G} \times \mathcal{Y} : \tau(g(y)) = j\}$, i.e., the set of transition–state pairs whose next state is tokenized as j . For each $(g, y) \in \mathfrak{F}_j$, let $r_{g \cdot y}$ denote the (unique) neuron in the pre-trained MLP that is activated for predicting $j = \tau(g(y))$ as defined in Eq. (6.3.8). We further define the set of all activated neurons

$$\mathfrak{A} \triangleq \cup_{j \in \tau(\mathcal{Y})} \mathfrak{A}_j, \text{ where } \mathfrak{A}_j \triangleq \{r \mid \exists (g, y) \in \mathfrak{F}_j, r = r_{g \cdot y}\}.$$

Equivalently, $\mathfrak{A}$ collects the activated neurons across all fibers $\{\mathfrak{F}_j\}_{j \in [n_Y]}$.

Substituting the conditions from Eq. (6.3.8) yields the following characterizations of $\Lambda_{j,r}$.

Lemma 6.B.5 (Characterizations of Λ). *Given $\ell \in [L]$ and $(\widehat{Z}_a^{L,\ell-1}, Z_p^L)$. Let $\{\mathbf{Attn}_{a,\ell-1 \rightarrow p,k}\}_{k=1}^L$ denote the attention weights from the answer token at step $\ell-1$ to the L prompt tokens. Then we have:*

(a) *For any $j \in \tau(\mathcal{Y})$ and any activated neuron $r \in \mathfrak{A}_j$,*

$$\Lambda_{j,r}(\widehat{Z}_a^{L,\ell-1}, Z_p^L) = \frac{1}{2} \left(V_{j,r}(\widehat{y}_{\ell-1}) + \sum_{k=1}^L \mathbf{Attn}_{a,\ell-1 \rightarrow p,k} V_{j,r}(g_k) \right).$$

(b) *For any $j \in \tau(\mathcal{Y})$ and any non-activated neuron $r \notin \mathfrak{A}_j$,*

$$\Lambda_{j,r}(\widehat{Z}_a^{L,\ell-1}, Z_p^L) = 0.$$

Lemma 6.B.6 (Values of Λ at step ℓ). *Fix $\ell \in [L]$ and an input $(\widehat{Z}_a^{L,\ell-1}, Z_p^L)$. Let $\{\mathbf{Attn}_{a,\ell-1 \rightarrow p,k}\}_{k=1}^L$ denote the attention weights. Then the following properties hold.*

(a) **target transition** g_ℓ . Let $j := \tau(g_\ell \cdot \widehat{y}_{\ell-1})$. Then

$$\Lambda_{j,r_{g_\ell \cdot \widehat{y}_{\ell-1}}}(\widehat{Z}_a^{L,\ell-1}, Z_p^L) = \mathbf{Attn}_{a,\ell-1 \rightarrow p,\ell} B + \sigma_0,$$

and for any $r \in \mathfrak{A}_j \setminus \{r_{g_\ell \cdot \widehat{y}_{\ell-1}}\}$,

$$\Lambda_{j,r}(\widehat{Z}_a^{L,\ell-1}, Z_p^L) < 0.$$

(b) **in-context distractor** $g_k, k \neq \ell$. Fix $k \in [L] \setminus \{\ell\}$ and let $j := \tau(g_k \cdot \widehat{y}_{\ell-1})$. Then

$$\Lambda_{j,r_{g_k \cdot \widehat{y}_{\ell-1}}}(\widehat{Z}_a^{L,\ell-1}, Z_p^L) = \mathbf{Attn}_{a,\ell-1 \rightarrow p,k} B + \sigma_0,$$

and for any $r \in \mathfrak{A}_j \setminus \{r_{g_k \cdot \widehat{y}_{\ell-1}}\}$,

$$\Lambda_{j,r}(\widehat{Z}_a^{L,\ell-1}, Z_p^L) < 0.$$

(c) *vocabulary distractor* g . For any $g \in \mathcal{G} \setminus G^L$ and $j := \tau(g \cdot \widehat{y}_{\ell-1})$,

$$\Lambda_{j, r_{g \cdot \widehat{y}_{\ell-1}}}(\widehat{Z}_a^{L, \ell-1}, Z_p^L) = \sigma_0,$$

and for any $r \in \mathfrak{A}_j \setminus \{r_{g \cdot \widehat{y}_{\ell-1}}\}$,

$$\Lambda_{j, r}(\widehat{Z}_a^{L, \ell-1}, Z_p^L) < 0.$$

Proof. By Lemma 6.B.5, for any $j \in \tau(\mathcal{Y})$ and any neuron $r \in \mathfrak{A}_j$, we have

$$\Lambda_{j, r}(\widehat{Z}_a^{L, \ell-1}, Z_p^L) = \frac{1}{2} \left(V_{j, r}(\widehat{y}_{\ell-1}) + \sum_{k=1}^L \mathbf{Attn}_{a, \ell-1 \rightarrow p, k} V_{j, r}(g_k) \right).$$

By the simple transitivity assumption, there exists a unique $g^* \in \mathcal{G}$ such that $\tau(g^*(\widehat{y}_{\ell-1})) = j$. Invoking Eq. (6.3.8), we have

$$V_{j, r_{g^* \cdot \widehat{y}_{\ell-1}}}(\widehat{y}_{\ell-1}) = B + 2\sigma_0.$$

If $g^* = g_k \in G^L$ for some $k \in [L]$, then $V_{j, r_{g^* \cdot \widehat{y}_{\ell-1}}}(g_k) = B$ and $V_{j, r_{g^* \cdot \widehat{y}_{\ell-1}}}(g_{k'}) = -B$ for all $k' \neq k$. Therefore,

$$\begin{aligned} 2\Lambda_{j, r_{g^* \cdot \widehat{y}_{\ell-1}}}(\widehat{Z}_a^{L, \ell-1}, Z_p^L) &= \mathbf{Attn}_{a, \ell-1 \rightarrow p, k} B + \sum_{k' \neq k} \mathbf{Attn}_{a, \ell-1 \rightarrow p, k'} (-B) + (B + 2\sigma_0) \\ &= \mathbf{Attn}_{a, \ell-1 \rightarrow p, k} B - (1 - \mathbf{Attn}_{a, \ell-1 \rightarrow p, k}) B + B + 2\sigma_0 \\ &= 2\mathbf{Attn}_{a, \ell-1 \rightarrow p, k} B + 2\sigma_0. \end{aligned}$$

Otherwise, if $g^* \notin G^L$, then $V_{j, r_{g^* \cdot \widehat{y}_{\ell-1}}}(g_k) = -B$ for all $k \in [L]$, and hence

$$2\Lambda_{j, r_{g^* \cdot \widehat{y}_{\ell-1}}}(\widehat{Z}_a^{L, \ell-1}, Z_p^L) = \sum_{k=1}^L \mathbf{Attn}_{a, \ell-1 \rightarrow p, k} (-B) + (B + 2\sigma_0) = 2\sigma_0.$$

Finally, consider any other pair (g, y) such that $\tau(g(y)) = j$ but $y \neq \widehat{y}_{\ell-1}$. For the corresponding neuron $r_{g \cdot y} \in \mathfrak{A}_j$, we have $V_{j, r_{g \cdot y}}(\widehat{y}_{\ell-1}) = -B$. Moreover, among $\{g_k\}_{k=1}^L$, at most one index can contribute $+B$ and the remaining contribute $-B$, so

$$\begin{aligned} 2\Lambda_{j, r_{g \cdot y}}(\widehat{Z}_a^{L, \ell-1}, Z_p^L) &\leq \left(2 \max_{k \in [L]} \mathbf{Attn}_{a, \ell-1 \rightarrow p, k} - 1 \right) B - B \\ &< 0. \end{aligned}$$

This concludes the proof. $\square$

Throughout the following analysis, we suppress the dependence on the underlying instance. When the context is clear, we abbreviate $\Lambda_{j, r}(\widehat{Z}_a^{L, \ell-1}, Z_p^L)$ as $\Lambda_{j, r}$ and $\mathcal{E}_j(\widehat{Z}_a^{L, \ell}, Z_p^L)$ as $\mathcal{E}_j$.

Hence, combining the reduced attention pattern in Eq. (6.B.6), namely, one target receiving weight $\mathbf{Attn}_L^{(t)}$ and $L - 1$ symmetric non-targets, the above characterization of the activations $\Lambda_{j, r}$

implies a step-invariant, context-level structure for the next-state distribution $\pi_\theta(\cdot \mid \widehat{y}_{\ell-1}, G^L)$. In particular, the candidates decompose into three groups: (i) the target $g_\ell \cdot \widehat{y}_{\ell-1}$; (ii) the $L - 1$ symmetric non-targets $\{g_{\ell'} \cdot \widehat{y}_{\ell-1} : \ell' \neq \ell\}$; and (iii) the remaining $d - L$ states outside the context induced set (i.e., vocabulary distractors). We formalize this decomposition in the following lemma.

Lemma 6.B.7. *At step ℓ , conditioning on $\widehat{y}_{\ell-1}$ and G^L , the policy $\pi_\theta^{(t)}(\cdot \mid \widehat{y}_{\ell-1}, G^L)$ satisfies:*

(i) For $j = \tau(g_\ell(\widehat{y}_{\ell-1}))$,

$$\pi_\theta^{(t)}(j \mid \widehat{y}_{\ell-1}, G^L) = \frac{d^{\text{Attn}_L^{(t)} C_B}}{d^{\text{Attn}_L^{(t)} C_B} + (L - 1)d^{\frac{1 - \text{Attn}_L^{(t)}}{L - 1} C_B} + (d - L)} \triangleq p_{L,1}^{(t)}. \quad (6.B.7)$$

(ii) For $j \in \tau(\{g \cdot \widehat{y}_{\ell-1} : g \in G^L, g \neq g_\ell\})$,

$$\pi_\theta^{(t)}(j \mid \widehat{y}_{\ell-1}, G^L) = \frac{d^{\frac{1 - \text{Attn}_L^{(t)}}{L - 1} C_B}}{d^{\text{Attn}_L^{(t)} C_B} + (L - 1)d^{\frac{1 - \text{Attn}_L^{(t)}}{L - 1} C_B} + (d - L)} \triangleq p_{L,2}^{(t)}. \quad (6.B.8)$$

(iii) For any other $j \in \tau(\mathcal{Y})$,

$$\pi_\theta^{(t)}(j \mid \widehat{y}_{\ell-1}, G^L) = \frac{1}{d^{\text{Attn}_L^{(t)} C_B} + (L - 1)d^{\frac{1 - \text{Attn}_L^{(t)}}{L - 1} C_B} + (d - L)} \triangleq p_{L,3}^{(t)}. \quad (6.B.9)$$

Moreover, $\pi_\theta^{(t)}(j \mid \widehat{y}_{\ell-1}, G^L)$ does not depend on ℓ . Hence, we suppress the index ℓ and write $p_{L,1}^{(t)}$, $p_{L,2}^{(t)}$, and $p_{L,3}^{(t)}$ for brevity.

Probabilistic Event. We conclude this subsection by introducing a probabilistic event that characterizes the potential for *path collisions*, where an incorrect sequence of operations inadvertently leads to the same outcome as the intended compositional path. Such an event serves as a key tool for bounding the interference from distracting trajectories in our subsequent analysis:

$$\mathfrak{E}_L \triangleq \left\{ \exists \mathbf{i} = (i_1, \dots, i_L) \in [L]^L, \mathbf{i} \neq (1, \dots, L), \text{ s.t. } g_{i_L} \circ \dots \circ g_{i_1}(y_0) = y_L \right\}.$$

Lemma 6.B.8 (Probability of Trajectory Collision). *Under Assumptions 6.3.1 and 6.3.2, for every fixed $L = O(1)$, the probability that any noncanonical sequence of L actions formed by the same set of available operators hits the target state y_L satisfies*

$$\mathbb{P}(\mathfrak{E}_L) \leq \rho_L(d) = o(1). \quad (6.B.10)$$

Proof. By simple transitivity, for any fixed y_0 , the equality $g_{i_L} \circ \dots \circ g_{i_1}(y_0) = y_L$ is equivalent to $g_{i_L} \circ \dots \circ g_{i_1} = g_L \circ \dots \circ g_1$. Applying the union bound over all noncanonical index tuples $\mathbf{i} \neq (1, \dots, L)$ gives $\mathbb{P}(\mathfrak{E}_L) \leq \rho_L(d)$, which is $o(1)$ for fixed L by Assumption 6.3.2. $\square$

Remark 6.B.1 (Two-step collisions). *For $L = 2$, the only nontrivial noncanonical collision is commutation of the two sampled operators. Under sampling without replacement, this probability is*

$$\mathbb{P}(g_1 \circ g_2 = g_2 \circ g_1) = \frac{k(\mathcal{G}) - 1}{d - 1}, \quad (6.B.11)$$

where $k(\mathcal{G})$ is the number of conjugacy classes of $\mathcal{G}$. Thus the two-step part of Assumption 6.3.2 asks only for sublinear conjugacy-class growth, rather than a uniformly bounded number of conjugacy classes. Indeed, the number of ordered commuting pairs in $\mathcal{G}^2$ is $\sum_{g \in \mathcal{G}} |C_{\mathcal{G}}(g)| = d k(\mathcal{G})$; removing the d diagonal pairs and normalizing by the $d(d - 1)$ ordered distinct pairs gives the displayed probability. For example, for $\mathcal{G} = \text{PSL}_2(q)$ one has $k(\mathcal{G}) = O(q)$ while $|\mathcal{G}| = \Theta(q^3)$, so this two-step collision probability is $O(q^{-2})$; more generally, conjugacy-class bounds for finite groups of Lie type are given in Fulman and Guralnick [90].

6.B.2 Gradient Characterization for General Length

In this section, we use spectral analysis to derive explicit gradient formulas for $q^{(t)}$ and $r^{(t)}$ under the step-invariant constraints in Lemma 6.B.7. For each fixed problem length L , these characterizations provide the analytical foundation for our subsequent study of training dynamics across different training settings in later sections.

Action distribution induced by π_{θ} . We introduce an action variable $u_{\ell} \in \mathcal{G}$ as the unique group element applied at step ℓ . By the simply-transitive action of $\mathcal{G}$ on $\mathcal{Y}$, each transition $\hat{y}_{\ell-1} \rightarrow \hat{y}_{\ell}$ corresponds to a unique u_{ℓ} such that $\hat{y}_{\ell} = u_{\ell}(\hat{y}_{\ell-1})$. With this notation, Lemma 6.B.7 can be equivalently stated as a step-invariant action distribution.

Lemma 6.B.9. *Fix a step ℓ and condition on $(\hat{y}_{\ell-1}, G^L)$. Then the policy $\pi_{\theta}(\cdot \mid \hat{y}_{\ell-1}, G^L)$ can be partitioned into the following three classes:*

- *target action $(p_{L,1})$: applying the correct in-context rule g_{ℓ} ,*

$$p_{L,1} := \pi_{\theta}(j \mid \hat{y}_{\ell-1}, G^L), \quad j = \tau(g_{\ell}(\hat{y}_{\ell-1})). \quad (6.B.12a)$$

- *in-context distractor actions $(p_{L,2})$: applying an incorrect rule from the context,*

$$p_{L,2} := \pi_{\theta}(j \mid \hat{y}_{\ell-1}, G^L), \quad j \in \tau\left(\{g(\hat{y}_{\ell-1}) : g \in G^L, g \neq g_{\ell}\}\right). \quad (6.B.12b)$$

- *vocabulary distractor actions $(p_{L,3})$: any other token not corresponding to an in-context transition,*

$$p_{L,3} := \pi_{\theta}(j \mid \hat{y}_{\ell-1}, G^L), \quad j \in \tau\left(\{g(\hat{y}_{\ell-1}) : g \notin G^L\}\right). \quad (6.B.12c)$$

Trajectory measure induced by TF_{θ}

In this part, we formally define the trajectory measure induced by the model TF_{θ} and derive an explicit gradient representation under this probabilistic framework. This formulation allows

us to express the optimization objective in terms of conditional (posterior) probabilities over trajectories.

Given a problem instance (y_0, G^L) , for any trajectory $\mathbf{v} = (v_1, \dots, v_L)$, we define the trajectory measure $\tilde{\mathbb{P}}_{\theta, (y_0, G^L)}$ induced by TF_θ as

$$\tilde{\mathbb{P}}_{\theta, (y_0, G^L)}(\mathbf{v}) := \prod_{\ell'=1}^L \pi_\theta(v_{\ell'} \mid v_{\ell'-1}, G^L). \quad (6.B.13)$$

Under the induced measure Eq. (6.B.13), we can rewrite the gradient expression in Eq. (6.B.5) as

$$\mathfrak{J}(y_0, G^L) = \mathbf{Attn}_L \cdot \tilde{\mathbb{P}}_{\theta, (y_0, G^L)}(v_L = y_L) \cdot \tilde{\mathbb{E}}_{\theta, (y_0, G^L)} \left[\sum_{\ell=1}^L \mathfrak{G}_\ell(\mathbf{v}) \middle| v_L = y_L \right], \quad (6.B.14)$$

where

$$\tilde{\mathbb{P}}_{\theta, (y_0, G^L)}(v_L = y_L) = \sum_{\mathbf{v} \in \mathcal{Y}^{L-1} \times \{y_L\}} \tilde{\mathbb{P}}_{\theta, (y_0, G^L)}(\mathbf{v}), \quad (6.B.15)$$

$$\tilde{\mathbb{E}}_{\theta, (y_0, G^L)} \left[\sum_{\ell=1}^L \mathfrak{G}_\ell(\mathbf{v}) \middle| v_L = y_L \right] = \frac{\sum_{\mathbf{v} \in \mathcal{Y}^{L-1} \times \{y_L\}} \tilde{\mathbb{P}}_{\theta, (y_0, G^L)}(\mathbf{v}) \sum_{\ell=1}^L \mathfrak{G}_\ell(\mathbf{v})}{\tilde{\mathbb{P}}_{\theta, (y_0, G^L)}(v_L = y_L)}. \quad (6.B.16)$$

We now turn to the term $\mathfrak{G}_\ell$. By Lemma 6.B.6, we can rewrite $\mathfrak{G}_\ell(\mathbf{v})$ as

$$\begin{aligned} \mathfrak{G}_\ell(\mathbf{v}) &= \sum_{j \in \tau(\mathcal{Y})} \mathcal{E}_j \sum_{r \in [m]} \sigma'(\Lambda_{j,r}) \left(V_{j,r}(g_\ell) - \Lambda_{j,r} \right) \\ &= \sum_{g \in \mathcal{G}} \mathcal{E}_{\tau(g(\widehat{y}_{\ell-1}))} \left(V_{\tau(g(\widehat{y}_{\ell-1})), r_{g(\widehat{y}_{\ell-1})}}(g_\ell) - \Lambda_{\tau(g(\widehat{y}_{\ell-1})), r_{g(\widehat{y}_{\ell-1})}} \right) \\ &= \mathbb{1}\{g(v_{\ell-1}) = v_\ell\} \left(V_{\tau(g(\widehat{y}_{\ell-1})), r_{g(\widehat{y}_{\ell-1})}}(g_\ell) - \Lambda_{\tau(g(\widehat{y}_{\ell-1})), r_{g(\widehat{y}_{\ell-1})}} \right) \\ &\quad - \sum_{g \in \mathcal{G}} \pi_\theta(\tau(g(v_{\ell-1})) \mid v_{\ell-1}, G^L) \left(V_{\tau(g(\widehat{y}_{\ell-1})), r_{g(\widehat{y}_{\ell-1})}}(g_\ell) - \Lambda_{\tau(g(\widehat{y}_{\ell-1})), r_{g(\widehat{y}_{\ell-1})}} \right). \end{aligned}$$

Using the step-invariant three-way partition in Lemma 6.B.7, the last term further simplifies to

$$\begin{aligned} \mathfrak{G}_\ell(\mathbf{v}) &= \mathbb{1}\{g(v_{\ell-1}) = v_\ell\} \left(V_{\tau(g(\widehat{y}_{\ell-1})), r_{g(\widehat{y}_{\ell-1})}}(g_\ell) - \Lambda_{\tau(g(\widehat{y}_{\ell-1})), r_{g(\widehat{y}_{\ell-1})}} \right) \\ &\quad - p_{L,1} \left((1 - \mathbf{Attn}_L)B - \sigma_0 \right) \\ &\quad + (L-1)p_{L,2} \left(\left(1 + \frac{1 - \mathbf{Attn}_L}{L-1} \right) B + \sigma_0 \right) \\ &\quad + (d-L)p_{L,3} \left(B + \sigma_0 \right), \end{aligned}$$

where we used that the in-context distractor set has size $L-1$ and the remaining vocabulary set has size $d-L$.

Next, define the posterior probabilities (under the trajectory measure conditioned on success)

$$\rho_{\ell,1} \triangleq \widetilde{\mathbb{P}}_{\theta,(y_0,G^L)}(u_\ell = g_\ell | v_L = y_L), \quad (6.B.17)$$

$$\rho_{\ell,2} \triangleq \widetilde{\mathbb{P}}_{\theta,(y_0,G^L)}(u_\ell \in G^L \setminus \{g_\ell\} | v_L = y_L). \quad (6.B.18)$$

Taking the conditional expectation of $\mathfrak{G}_\ell(v)$ given $v_L = y_L$ yields

$$\begin{aligned} \widetilde{\mathbb{E}}_{\theta,(y_0,G^L)}[\mathfrak{G}_\ell(v) | v_L = y_L] &= \rho_{\ell,1} \left((1 - \mathbf{Attn}_L)B - \sigma_0 \right) - \rho_{\ell,2} \left(\left(1 + \frac{1 - \mathbf{Attn}_L}{L-1}\right)B + \sigma_0 \right) \\ &\quad - (1 - \rho_{\ell,1} - \rho_{\ell,2}) \left(B + \sigma_0 \right) \\ &\quad - p_{L,1} \left((1 - \mathbf{Attn}_L)B - \sigma_0 \right) + (L-1)p_{L,2} \left(\left(1 + \frac{1 - \mathbf{Attn}_L}{L-1}\right)B + \sigma_0 \right) \\ &\quad + (1 - p_{L,1} - (L-1)p_{L,2}) \left(B + \sigma_0 \right) \\ &= \left[(\rho_{\ell,1} - p_{L,1})(2 - \mathbf{Attn}_L) + \left(p_{L,2} - \frac{\rho_{\ell,2}}{L-1} \right) (1 - \mathbf{Attn}_L) \right] B. \end{aligned}$$

Putting it back to Eq. (6.B.14), we have

$$\begin{aligned} \nabla_q \widetilde{\mathcal{J}}_L &= \frac{1}{2Ld_p} \mathbb{E}_{Z^L} [\mathfrak{J}(y_0, G^L)] \\ &= \frac{1}{2Ld_p} \mathbb{E}_{Z^L} \left[\mathbf{Attn}_L \widetilde{\mathbb{P}}_{\theta,(y_0,G^L)}(v_L = y_L) B \right. \end{aligned} \quad (6.B.19)$$

$$\left. \cdot \sum_{\ell} \left((\rho_{\ell,1} - p_{L,1})(2 - \mathbf{Attn}_L) + \left(p_{L,2} - \frac{\rho_{\ell,2}}{L-1} \right) (1 - \mathbf{Attn}_L) \right) \right]. \quad (6.B.20)$$

Intuitively, the gradient is driven by the gap between the posterior action probabilities conditioned on success and their unconditional counterparts. Therefore, controlling the gradient reduces to estimating the posterior probabilities $\rho_{\ell,1}$ and $\rho_{\ell,2}$ as functions of the probability tuple $(p_{L,1}, p_{L,2}, p_{L,3})$. In what follows, we suppress the dependence on $(\theta, (y_0, G^L))$ whenever the context is clear.

Preliminaries: Harmonic Analysis on $\mathcal{G}$

The trajectory measure introduced in the previous part involves cumulative products of group actions, which correspond to repeated convolutions of measures on the underlying group. A direct combinatorial analysis of these convolutions is often intractable. To address this, we work in the Fourier domain via the irreducible representations of the group.

In this section, we briefly review the basic facts of harmonic analysis on finite groups and collect the spectral tools we will use to decouple these convolution operations. For background and more detailed treatments, see Kondor [152], Serre [249].

Definition 6.B.1 (Irreducible Representations). Let $\mathcal{G}$ be a finite group of order $|\mathcal{G}| = N$. Let Λ denote the set of equivalence classes of irreducible unitary representations of $\mathcal{G}$. For each $\lambda \in \Lambda$, $\lambda : \mathcal{G} \rightarrow U(d_\lambda)$ is a homomorphism, where $U(d_\lambda)$ is the group of $d_\lambda \times d_\lambda$ unitary matrices.

- The trivial representation is denoted by $\mathbf{1}$, with $d_1 = 1$ and $\mathbf{1}(g) = 1, \forall g$.
- Orthogonality Relations: For $\lambda, \eta \in \Lambda$:

$$\langle \lambda_{ij}, \eta_{kl} \rangle = \sum_{g \in \mathcal{G}} \lambda_{ij}(g) \overline{\eta_{kl}(g)} = \frac{N}{d_\lambda} \delta_{\lambda\eta} \delta_{ik} \delta_{jl}$$

Specifically, for $\lambda \neq \mathbf{1}$, $\sum_{g \in \mathcal{G}} \lambda(g) = \mathbf{0}_{d_\lambda \times d_\lambda}$.

Definition 6.B.2 (Fourier Transform). Let Λ be the set of irreducible unitary representations of $\mathcal{G}$. For any $\lambda \in \Lambda$, let d_λ be its dimension. For a function $f : \mathcal{G} \rightarrow \mathbb{C}$, the Fourier transform is:

$$\widehat{f}(\lambda) \triangleq \sum_{h \in \mathcal{G}} f(h) \lambda(h).$$

Definition 6.B.3 (Convolution). The convolution of two functions $f, v : \mathcal{G} \rightarrow \mathbb{C}$ is defined as:

$$(f * v)(g) := \sum_{h \in \mathcal{G}} f(gh^{-1})v(h).$$

The Fourier transform maps convolution to matrix multiplication:

$$\widehat{f * v}(\lambda) = \widehat{f}(\lambda) \widehat{v}(\lambda).$$

Lemma 6.B.10 (Fourier Inversion Formula & Plancherel Identity). Function $f : \mathcal{G} \rightarrow \mathbb{C}$ can be reconstructed from its Fourier coefficients:

$$f(g) = \frac{1}{N} \sum_{\lambda \in \Lambda} d_\lambda \text{Tr} \left(\widehat{f}(\lambda) \lambda(g)^{-1} \right).$$

Using the inversion formula at $g = e$ (identity), we have the identity:

$$\sum_{\lambda \in \Lambda} d_\lambda^2 = N.$$

We distinguish the trivial representation $\mathbf{1}$ (where $\lambda(h) = 1$) from non-trivial representations $\lambda \neq \mathbf{1}$. Note that $\sum_{\lambda \neq \mathbf{1}} d_\lambda^2 = N - 1$.

Definition 6.B.4 (Character Value & Spectral Decay Factor). Let $\mathcal{G}$ be a finite group and let Λ denote the set of its irreducible unitary representations. The **character** of a representation $\lambda \in \Lambda$, denoted by $\chi_\lambda : \mathcal{G} \rightarrow \mathbb{C}$, is defined as the trace of the linear operator $\lambda(g)$ for each $g \in \mathcal{G}$:

$$\chi_\lambda(g) := \text{Tr}(\lambda(g)).$$

The scalar $\chi_\lambda(g)$ is referred to as the character value of the element g corresponding to λ . Furthermore, we define the **spectral decay factor**, denoted by $\gamma(\mathcal{G})$, as the maximum normalized character value over all non-trivial representations and non-identity elements:

$$\gamma(\mathcal{G}) := \max_{\substack{\lambda \in \Lambda, \lambda \neq \mathbf{1} \\ g \in \mathcal{G}, g \neq e}} \frac{|\chi_\lambda(g)|}{d_\lambda},$$

where d_λ denotes the dimension of the representation λ .

Remark 6.B.2 (Magnitude of $\gamma(\mathcal{G})$). For any finite group $\mathcal{G}$, it holds that $0 \leq \gamma(\mathcal{G}) \leq 1$. Specifically, if $\mathcal{G}$ is abelian or has a non-trivial center $Z(\mathcal{G}) \neq \{e\}$, then $\gamma(\mathcal{G}) = 1$. If $\mathcal{G}$ is a non-abelian simple group, then $\gamma(\mathcal{G}) < 1$. Furthermore, for many sequences of simple groups (e.g., $\text{PSL}_2(q)$), $\gamma(\mathcal{G}) \rightarrow 0$ as $|\mathcal{G}| \rightarrow \infty$, indicating rapid spectral decay.

Spectral Decomposition of the Step Measure

With the harmonic analysis framework in place, we translate conditional expectations under the trajectory measure induced by TF_θ into convolution equations on the group.

Reduction to Group Actions. Since $\mathcal{G}$ acts simply transitively on $\mathcal{Y}$, for any trajectory v with fixed $v_0 = y_0$ there exists a unique sequence of group actions $(u_1, \dots, u_L) \in \mathcal{G}^L$ such that $v_\ell = u_\ell(v_{\ell-1})$ for $\ell \in [L]$. Consequently,

$$v_L = (u_L \cdots u_1)(y_0).$$

Let the target path be given by $(g_1, \dots, g_L)$ so that $y_L = (g_L \cdots g_1)(y_0)$, and define the target composition $G_* \triangleq g_L \cdots g_1$. Then the endpoint constraint is equivalent to the group equation

$$v_L = y_L \iff u_L \cdots u_1 = G_*.$$

Under this representation, the posteriors of interest can be written as

$$\rho_{\ell,1} = \tilde{\mathbb{P}}(u_\ell = g_\ell | u_L \cdots u_1 = G_*), \quad \rho_{\ell,2} = \tilde{\mathbb{P}}(u_\ell \in G^L \setminus \{g_\ell\} | u_L \cdots u_1 = G_*).$$

Definition 6.B.5 (One-step measure). For each step $\ell \in [L]$, define a probability measure μ_ℓ on $\mathcal{G}$ by

$$\mu_\ell(h) \triangleq \tilde{\mathbb{P}}(u_\ell = h) = \begin{cases} p_{L,1}, & h = g_\ell, \\ p_{L,2}, & h \in G^L \setminus \{g_\ell\}, \\ p_{L,3}, & h \in \mathcal{G} \setminus G^L. \end{cases}$$

Equivalently,

$$\mu_\ell = p_{L,1}\delta_{g_\ell} + p_{L,2}\delta_{G^L \setminus \{g_\ell\}} + p_{L,3}\delta_{\mathcal{G} \setminus G^L},$$

where δ_S denotes the (unnormalized) uniform measure on a set $S \subseteq \mathcal{G}$. Moreover, the marginal probability of the endpoint is given by

$$\tilde{\mathbb{P}}(v_L = y_L) = (\mu_L * \cdots * \mu_1)(G_*).$$

We now compute the Fourier transform $\widehat{\mu}_\ell(\lambda)$ for a nontrivial irreducible representation $\lambda \neq \mathbf{1}$.

Definition 6.B.6 (Spectral objects and effective parameters). Given $\ell \in [L]$, for an irreducible representation λ of $\mathcal{G}$, define the sample operator

$$W_\ell(\lambda) \triangleq \sum_{h \in G^L \setminus \{g_\ell\}} \lambda(h).$$

We also define the effective parameters

$$\Delta_L \triangleq p_{L,1} - p_{L,3}, \quad \delta_L \triangleq p_{L,2} - p_{L,3}, \quad \sigma_{G^L} \triangleq \max_{\lambda \neq \mathbf{1}} \max_{\ell \in [L]} \|W_\ell(\lambda)\|_{\text{op}},$$

where $\|\cdot\|_{\text{op}}$ denotes the operator norm induced by $\|\cdot\|_2$, i.e., $\|A\|_{\text{op}} := \sup_{\|x\|_2=1} \|Ax\|_2$.

Lemma 6.B.11 (Fourier transform of the one-step measure). *Let μ_ℓ be the one-step measure in Definition 6.B.5. For any nontrivial irreducible representation $\lambda \neq \mathbf{1}$,*

$$\widehat{\mu}_\ell(\lambda) = \Delta_L \lambda(g_\ell) + \delta_L W_\ell(\lambda).$$

Proof. By definition of μ_ℓ ,

$$\widehat{\mu}_\ell(\lambda) = p_{L,1}\lambda(g_\ell) + p_{L,2} \sum_{h \in G^L \setminus \{g_\ell\}} \lambda(h) + p_{L,3} \sum_{h \in \mathcal{G} \setminus G^L} \lambda(h).$$

For $\lambda \neq \mathbf{1}$, by Lemma 6.B.10, we have $\sum_{h \in \mathcal{G}} \lambda(h) = 0$, hence $\sum_{h \in \mathcal{G} \setminus G^L} \lambda(h) = -(\lambda(g_\ell) + W_\ell(\lambda))$. Substituting and collecting terms yields the claim. $\square$

Posterior Estimation

Building on the operator decomposition, we compute the posterior probabilities by evaluating traces of the resulting spectral products. For notational convenience, we define the events

$$E \triangleq \{v_L = y_L\}, \quad A_\ell \triangleq \{u_\ell = g_\ell\}, \quad B_\ell \triangleq \{u_\ell \in G^L \setminus \{g_\ell\}\}.$$

Thus, the main task is to control $\widetilde{\mathbb{P}}(E)$ as well as the joint probabilities $\widetilde{\mathbb{P}}(A_\ell \cap E)$ and $\widetilde{\mathbb{P}}(B_\ell \cap E)$.

Lemma 6.B.12. *With E, A_ℓ, B_ℓ defined above, we have the expansions*

$$\widetilde{\mathbb{P}}(E) = \frac{1}{d} + \left(1 - \frac{1}{d}\right) \Delta_L^L + \mathcal{R}_E, \quad (6.B.21a)$$

$$\widetilde{\mathbb{P}}(A_\ell \cap E) = \frac{p_{L,1}}{d} + \left(1 - \frac{1}{d}\right) p_{L,1} \Delta_L^{L-1} + \mathcal{R}_A, \quad (6.B.21b)$$

$$\widetilde{\mathbb{P}}(B_\ell \cap E) = \frac{(L-1)p_{L,2}}{d} + \mathcal{R}_B. \quad (6.B.21c)$$

Moreover, the remainders satisfy

$$|\mathcal{R}_E| \leq \left(1 - \frac{1}{d}\right) \left[(\Delta_L + \sigma_{G^L} \delta_L)^L - \Delta_L^L - L \sigma_{G^L} \delta_L \Delta_L^{L-1} + (L-1) L \gamma(\mathcal{G}) \delta_L \Delta_L^{L-1} \right], \quad (6.B.22a)$$

$$|\mathcal{R}_A| \leq p_{L,1} \left(1 - \frac{1}{d}\right) \left[(\Delta_L + \sigma_{G^L} \delta_L)^{L-1} - \Delta_L^{L-1} - (L-1) \sigma_{G^L} \delta_L \Delta_L^{L-2} + (L-1)^2 \gamma(\mathcal{G}) \delta_L \Delta_L^{L-2} \right], \quad (6.B.22b)$$

$$|\mathcal{R}_B| \leq p_{L,2} \left(1 - \frac{1}{d}\right) \left[\sigma_{G^L} ((\Delta_L + \sigma_{G^L} \delta_L)^{L-1} - \Delta_L^{L-1}) + (L-1) \gamma(\mathcal{G}) \Delta_L^{L-1} \right]. \quad (6.B.22c)$$

Proof. By Lemma 6.B.10 and the convolution theorem, each quantity $\widetilde{\mathbb{P}}(\cdot)$ can be written as a sum of traces of products of Fourier operators. We isolate the trivial-representation contribution and bound the remaining terms using the decomposition $\widehat{\mu}_k(\lambda) = \Delta_L \lambda(g_k) + \delta_L W_k(\lambda)$. We spell out the details for $\widetilde{\mathbb{P}}(E)$; the bounds for $\widetilde{\mathbb{P}}(A_\ell \cap E)$ and $\widetilde{\mathbb{P}}(B_\ell \cap E)$ follow analogously.

Estimation of $\widetilde{\mathbb{P}}(E)$. Let us start with $\widetilde{\mathbb{P}}(E)$. By Lemma 6.B.10, we have

$$\begin{aligned} \widetilde{\mathbb{P}}(E) &= (\mu_L * \cdots * \mu_1)(G_*) \\ &= \frac{1}{d} \sum_{\lambda \in \Lambda} d_\lambda \operatorname{Tr} \left(\mu_L * \cdots * \mu_1(\lambda) \lambda(G_*)^{-1} \right) \\ &= \frac{1}{d} \sum_{\lambda \in \Lambda} d_\lambda \operatorname{Tr} \left(\underbrace{\left[\prod_{k=L}^1 \widehat{\mu}_k(\lambda) \right]}_{=: \Pi(\lambda)} \lambda(G_*)^{-1} \right). \end{aligned}$$

- For $\lambda = \mathbf{1}$: $\widehat{\mu}_k(\mathbf{1}) = 1$. Hence $\Pi(\lambda) = 1$, and we can obtain

$$\frac{1}{d} d_\lambda \operatorname{Tr}(\Pi(\lambda) \lambda(G_*)^{-1}) = \frac{1}{d} d_{\mathbf{1}} \cdot d_{\mathbf{1}} = \frac{1}{d}.$$

- For $\lambda \neq \mathbf{1}$: by the decomposition $\widehat{\mu}_k(\lambda) = \Delta_L \lambda(g_k) + \delta_L W_k(\lambda)$ from Lemma 6.B.11, we have

$$\Pi(\lambda) = \prod_{k=L}^1 (\Delta_L \lambda(g_k) + \delta_L W_k(\lambda)) = \Delta_L^L \prod_{k=L}^1 \lambda(g_k) + T_{\text{res}}(\lambda).$$

Then the trace contribution by the first term is:

$$\operatorname{Tr}\left(\Delta_L^L \left(\prod_{k=L}^1 \lambda(g_k)\right) \lambda(G_*)^{-1}\right) = (\Delta_L)^L \operatorname{Tr}(\lambda(\prod_{k=L}^1 g_k) \lambda(G_*)^{-1}) = d_\lambda (\Delta_L)^L.$$

Summing this over all $\lambda \neq \mathbf{1}$, we obtain

$$\begin{aligned} \frac{1}{d} \sum_{\lambda \neq \mathbf{1}} d_\lambda \operatorname{Tr}\left(\Delta_L^L \left(\prod_{k=L}^1 \lambda(g_k)\right) \lambda(G_*)^{-1}\right) &= \frac{1}{d} \sum_{\lambda \neq \mathbf{1}} d_\lambda \cdot d_\lambda (\Delta_L)^L \\ &= \left(1 - \frac{1}{d}\right) (\Delta_L)^L, \end{aligned}$$

where the last equality holds since $\sum_{\lambda \neq \mathbf{1}} d_\lambda^2 = d - 1$ by Lemma 6.B.10. Therefore, it suffices to control the operator norm of the residual term $T_{\text{res}}(\lambda)$. Notice that $T_{\text{res}}(\lambda)$ can be expanded into $2^L - 1$ terms, each of the form $M_L \cdots M_1$, where for each k ,

$$M_k \in \{\Delta_L \lambda(g_k), \delta_L W_k(\lambda)\},$$

and at least one factor M_k equals $\delta_L W_k(\lambda)$. We further decompose $T_{\text{res}}(\lambda)$ into two parts, $T_{\text{res}}(\lambda) = T_{\text{res},1}(\lambda) + T_{\text{res},2}(\lambda)$:

- $T_{\text{res},1}(\lambda)$ consists of the terms for which there exists a unique $k^* \in [L]$ such that $M_{k^*} = \delta_L W_{k^*}(\lambda)$. In this case,

$$\begin{aligned} &\left| \operatorname{Tr}\left(\delta_L \Delta_L^{L-1} \left(\prod_{k=L}^{k^*+1} \lambda(g_k)\right) W_{k^*}(\lambda) \left(\prod_{k=k^*-1}^1 \lambda(g_k)\right) \lambda(G_*)^{-1}\right) \right| \\ &= \left| \delta_L \Delta_L^{L-1} \operatorname{Tr}(W_{k^*}(\lambda) \lambda(g_{k^*})^{-1}) \right| \\ &= \left| \delta_L \Delta_L^{L-1} \sum_{g \in G^L \setminus \{g_{k^*}\}} \operatorname{Tr}(\lambda(g g_{k^*}^{-1})) \right| \\ &= \delta_L \Delta_L^{L-1} \sum_{g \in G^L \setminus \{g_{k^*}\}} \left| \chi_\lambda(g g_{k^*}^{-1}) \right| \\ &\leq \delta_L \Delta_L^{L-1} (L-1) \cdot d_\lambda \gamma(\mathcal{G}), \end{aligned} \tag{6.B.23}$$

where the last inequality uses $g g_{k^*}^{-1} \neq e$ (here e denotes the identity element of $\mathcal{G}$) and the definition of $\gamma(\mathcal{G})$. Since there are L such terms in $T_{\text{res},1}(\lambda)$, we obtain

$$\left| \frac{1}{d} \sum_{\lambda \neq 1} d_\lambda \text{Tr} \left(T_{\text{res},1}(\lambda) \lambda(G_*)^{-1} \right) \right| \leq \left(1 - \frac{1}{d} \right) \delta_L \Delta_L^{L-1} (L-1) L \gamma(\mathcal{G}).$$

- $T_{\text{res},2}(\lambda)$ collects the remaining $2^L - 1 - L$ terms, i.e., those for which at least two factors M_k equal $\delta_L W_k(\lambda)$. Then

$$\begin{aligned} \|T_{\text{res},2}(\lambda)\|_{\text{op}} &= \left\| \sum_{M_k \in \{\Delta_L \lambda(g_k), \delta_L W_k(\lambda)\}, \sum_{k=1}^L \mathbb{1}_{M_k = \delta_L W_k(\lambda)} \geq 2} M_L, \dots, M_1 \right\|_{\text{op}} \\ &\leq \sum_{M_k \in \{\Delta_L \lambda(g_k), \delta_L W_k(\lambda)\}, \sum_{k=1}^L \mathbb{1}_{M_k = \delta_L W_k(\lambda)} \geq 2} \prod_{k=L}^1 \|M_k\|_{\text{op}} \\ &\stackrel{(a)}{\leq} \sum_{i=2}^L \binom{L}{i} \Delta_L^{L-i} (\sigma_{G^L} \delta_L)^i = (\Delta_L + \sigma_{G^L} \delta_L)^L - \Delta_L^L - L \sigma_{G^L} \delta_L \Delta_L^{L-1}, \end{aligned}$$

Here (a) holds since $\|\Delta_L \lambda(g_k)\|_{\text{op}} \leq \Delta_L$, $\|\delta_L W_k(\lambda)\|_{\text{op}} \leq \sigma_{G^L} \delta_L$, and there are exactly $\binom{L}{i}$ choices of indices for which i different M_k 's equal $\delta_L W_k(\lambda)$. Consequently,

$$\begin{aligned} \left| \frac{1}{d} \sum_{\lambda \neq 1} d_\lambda \text{Tr} \left(T_{\text{res},2}(\lambda) \lambda(G_*)^{-1} \right) \right| &\leq \frac{1}{d} \sum_{\lambda \neq 1} d_\lambda \cdot d_\lambda \|T_{\text{res},2}(\lambda)\|_{\text{op}} \|\lambda(G_*)^{-1}\|_{\text{op}} \\ &= \left(1 - \frac{1}{d} \right) \left((\Delta_L + \sigma_{G^L} \delta_L)^L - \Delta_L^L - L \sigma_{G^L} \delta_L \Delta_L^{L-1} \right). \end{aligned}$$

Putting everything together, we obtain

$$\widetilde{\mathbb{P}}(E) = \frac{1}{d} + \left(1 - \frac{1}{d} \right) \Delta_L^L + \mathcal{R}_E,$$

where the remainder term $\mathcal{R}_E$ satisfies

$$|\mathcal{R}_E| \leq \left(1 - \frac{1}{d} \right) \left((\Delta_L + \sigma_{G^L} \delta_L)^L - \Delta_L^L - L \sigma_{G^L} \delta_L \Delta_L^{L-1} + \delta_L \Delta_L^{L-1} (L-1) L \gamma(\mathcal{G}) \right).$$

Estimation of $\widetilde{\mathbb{P}}(A_\ell \cap E)$. The analysis is similar to that of $\widetilde{\mathbb{P}}(E)$. The key difference is that we replace the measure at step ℓ by the Dirac measure $p_{L,1} \delta_{g_\ell}$, since the ℓ -th step takes the action g_ℓ . Correspondingly, its Fourier transform becomes $p_{L,1} \lambda(g_\ell)$. Hence,

$$\widetilde{\mathbb{P}}(A_\ell \cap E) = (\mu_L * \dots * p_{L,1} \delta_{g_\ell} * \dots * \mu_1)(G_*)$$

$$= \frac{1}{d} \sum_{\lambda \in \Lambda} d_\lambda \operatorname{Tr} \left(\underbrace{[\widehat{\mu}_L(\lambda) \cdots p_{L,1} \lambda(g_\ell) \cdots \widehat{\mu}_1(\lambda)]}_{=: \Pi_{A_\ell}(\lambda)} \lambda(G_*)^{-1} \right).$$

- For $\lambda = \mathbf{1}$: since $\widehat{\mu}_k(\mathbf{1}) = 1$, we have $\Pi_{A_\ell}(\mathbf{1}) = p_{L,1}$ and thus

$$\frac{1}{d} d_\lambda \operatorname{Tr} (\Pi_{A_\ell}(\lambda) \lambda(G_*)^{-1}) = \frac{p_{L,1}}{d}.$$

- For $\lambda \neq \mathbf{1}$: we can write

$$\begin{aligned} \Pi_{A_\ell}(\lambda) &= \left(\prod_{k=L}^{\ell+1} (\Delta_L \lambda(g_k) + \delta_L W_k(\lambda)) \right) (p_{L,1} \lambda(g_\ell)) \left(\prod_{k=\ell-1}^1 (\Delta_L \lambda(g_k) + \delta_L W_k(\lambda)) \right) \\ &= p_{L,1} \Delta_L^{L-1} \left(\prod_{k=L}^1 \lambda(g_k) \right) \lambda(g_\ell) \left(\prod_{k=L}^1 \lambda(g_k) \right) + T_{\text{res}, A_\ell}(\lambda) \end{aligned}$$

The trace contribution of the leading term is

$$\begin{aligned} \operatorname{Tr} \left(p_{L,1} \Delta_L^{L-1} \left(\prod_{k=L}^1 \lambda(g_k) \right) \lambda(G_*)^{-1} \right) &= p_{L,1} (\Delta_L)^{L-1} \operatorname{Tr} \left(\lambda \left(\prod_{k=L}^1 g_k \right) \lambda(G_*)^{-1} \right) \\ &= d_\lambda p_{L,1} (\Delta_L)^{L-1}. \end{aligned}$$

Summing over all $\lambda \neq \mathbf{1}$ yields

$$\begin{aligned} \frac{1}{d} \sum_{\lambda \neq \mathbf{1}} d_\lambda \operatorname{Tr} \left(p_{L,1} (\Delta_L)^{L-1} \left(\prod_{k=L}^1 \lambda(g_k) \right) \lambda(G_*)^{-1} \right) &= \frac{1}{d} \sum_{\lambda \neq \mathbf{1}} d_\lambda \cdot d_\lambda p_{L,1} (\Delta_L)^{L-1} \\ &= \left(1 - \frac{1}{d} \right) p_{L,1} (\Delta_L)^{L-1}. \end{aligned}$$

The residual term $T_{\text{res}, A_\ell}(\lambda)$ can be controlled exactly as in the analysis of $T_{\text{res}}(\lambda)$, which gives

$$\begin{aligned} &\left| \frac{1}{d} \sum_{\lambda \neq \mathbf{1}} d_\lambda \operatorname{Tr} \left(T_{\text{res}, A_\ell}(\lambda) \lambda(G_*)^{-1} \right) \right| \\ &\leq p_{L,1} \left(1 - \frac{1}{d} \right) \left((\Delta_L + \sigma_{G^L} \delta_L)^{L-1} - \Delta_L^{L-1} - (L-1) \sigma_{G^L} \delta_L \Delta_L^{L-2} + \delta_L \Delta_L^{L-2} (L-1)^2 \gamma(\mathcal{G}) \right). \end{aligned}$$

Putting the above bounds together, we conclude that

$$\widetilde{\mathbb{P}}(A_\ell \cap E) = \frac{p_{L,1}}{d} + \left(1 - \frac{1}{d} \right) p_{L,1} \Delta_L^{L-1} + \mathcal{R}_{A_\ell},$$

where

$$|\mathcal{R}_A| \leq p_{L,1} \left(1 - \frac{1}{d}\right) \left((\Delta_L + \sigma_{G^L} \delta_L)^{L-1} - \Delta_L^{L-1} - (L-1) \sigma_{G^L} \delta_L \Delta_L^{L-2} + \delta_L \Delta_L^{L-2} (L-1)^2 \gamma(\mathcal{G}) \right).$$

Estimation of $\widetilde{\mathbb{P}}(B_\ell \cap E)$. For $B_\ell \cap E$, at step ℓ we use the measure $p_{L,2} \delta_{G^L \setminus \{g_\ell\}}$, whose Fourier operator is

$$\widehat{\mu}_{B_\ell}(\lambda) = p_{L,2} \sum_{g \in G^L \setminus \{g_\ell\}} \lambda(g) = p_{L,2} W_\ell(\lambda).$$

Hence,

$$\begin{aligned} \widetilde{\mathbb{P}}(B_\ell \cap E) &= (\mu_L * \cdots * p_{L,2} \delta_{G^L \setminus \{g_\ell\}} * \cdots * \mu_1)(G_*) \\ &= \frac{1}{d} \sum_{\lambda \in \Lambda} d_\lambda \operatorname{Tr} \left(\underbrace{[\widehat{\mu}_L(\lambda) \cdots \widehat{\mu}_{B_\ell}(\lambda) \cdots \widehat{\mu}_1(\lambda)]}_{=: \Pi_{B_\ell}(\lambda)} \lambda(G_*)^{-1} \right). \end{aligned}$$

- For $\lambda = \mathbf{1}$: since $\widehat{\mu}_k(\mathbf{1}) = 1$, we have $\Pi_{B_\ell}(\mathbf{1}) = p_{L,2}(L-1)$, and thus

$$\frac{1}{d} d_\lambda \operatorname{Tr} (\Pi_{B_\ell}(\lambda) \lambda(G_*)^{-1}) = \frac{(L-1) \cdot p_{L,2}}{d}.$$

- For $\lambda \neq \mathbf{1}$: analogous to the decomposition of $T_{\text{res}}(\lambda)$, the operator $\Pi_{B_\ell}(\lambda)$ can be expanded into 2^{L-1} terms of the form $M_L \cdots (p_{L,2} W_\ell(\lambda)) \cdots M_1$, where, for each $k \neq \ell$,

$$M_k \in \{\Delta_L \lambda(g_k), \delta_L W_k(\lambda)\}.$$

We further split $\Pi_{B_\ell}(\lambda)$ into two parts,

$$\Pi_{B_\ell}(\lambda) = T_{B_\ell,1}(\lambda) + T_{B_\ell,2}(\lambda).$$

- $T_{B_\ell,1}$ consists of the unique term for which $M_k = \Delta_L \lambda(g_k)$ for all $k \neq \ell$. In this case,

$$\begin{aligned} & \left| \operatorname{Tr} \left(\Delta_L^{L-1} \left(\prod_{k=L}^{k^*+1} \lambda(g_k) \right) p_{L,2} W_\ell(\lambda) \left(\prod_{k=k^*-1}^1 \lambda(g_k) \right) \lambda(G_*)^{-1} \right) \right| \\ & \leq p_{L,2} \Delta_L^{L-1} (L-1) \cdot d_\lambda \gamma(\mathcal{G}), \end{aligned}$$

where the inequality follows by an argument analogous to Eq. (6.B.23). Therefore, we have:

$$\left| \frac{1}{d} \sum_{\lambda \neq \mathbf{1}} d_\lambda \operatorname{Tr} \left(T_{B_\ell,1}(\lambda) \lambda(G_*)^{-1} \right) \right| \leq \left(1 - \frac{1}{d} \right) p_{L,2} \Delta_L^{L-1} (L-1) \gamma(\mathcal{G}).$$

- $T_{B_\ell,2}$ collects the remaining terms, i.e., those for which at least one index $k \neq \ell$ satisfies $M_k = \delta_L W_k(\lambda)$. Then

$$\begin{aligned} \|T_{\text{res},2}(\lambda)\|_{\text{op}} &= \left\| \sum_{M_k \in \{\Delta_L \lambda(g_k), \delta_L W_k(\lambda)\} \text{ for } k \neq \ell, \sum_{k \neq \ell} \mathbb{1}_{M_k = \delta_L W_k(\lambda)} \geq 1} M_L \cdots p_{L,2} W_\ell(\lambda) \cdots M_1 \right\|_{\text{op}} \\ &\leq p_{L,2} \sigma_{G^L} \sum_{i=1}^L \binom{L}{i} \Delta_L^{L-i} (\sigma_{G^L} \delta_L)^i \\ &= p_{L,2} \sigma_{G^L} \left((\Delta_L + \sigma_{G^L} \delta_L)^{L-1} - \Delta_L^{L-1} \right), \end{aligned}$$

which can be shown by the same argument as in the bound for $T_{\text{res},2}$. Consequently,

$$\begin{aligned} \left| \frac{1}{d} \sum_{\lambda \neq 1} d_\lambda \text{Tr} \left(T_{B_\ell,2}(\lambda) \lambda (G_\star)^{-1} \right) \right| &\leq \frac{1}{d} \sum_{\lambda \neq 1} d_\lambda \cdot d_\lambda \|T_{B_\ell,2}\|_{\text{op}} \|\lambda (G_\star)^{-1}\|_{\text{op}} \\ &= \left(1 - \frac{1}{d} \right) p_{L,2} \sigma_{G^L} \left((\Delta_L + \sigma_{G^L} \delta_L)^{L-1} - \Delta_L^{L-1} \right). \end{aligned}$$

Putting everything together, we obtain

$$\widetilde{\mathbb{P}}(B_\ell \cap E) = \frac{(L-1)p_{L,2}}{d} + \mathcal{R}_B,$$

where the remainder term $\mathcal{R}_B$ satisfies

$$|\mathcal{R}_B| \leq \left(1 - \frac{1}{d} \right) p_{L,2} \left(\sigma_{G^L} \left((\Delta_L + \sigma_{G^L} \delta_L)^{L-1} - \Delta_L^{L-1} \right) + \Delta_L^{L-1} (L-1) \gamma(\mathcal{G}) \right).$$

□

The expansions in Lemma 6.B.12 immediately imply the following deviations of the posterior probabilities $\rho_{\ell,1}$ and $\rho_{\ell,2}$ from their corresponding priors.

Proposition 6.B.1 (Posterior deviation and dominant term). The posterior deviations admit the exact identities

$$\rho_{\ell,1} - p_{L,1} = \frac{p_{L,1} \Delta_L^{L-1} (1 - \Delta_L) \left(1 - \frac{1}{d} \right) + \mathcal{R}_A - p_{L,1} \mathcal{R}_E}{\widetilde{\mathbb{P}}(E)}, \quad (6.B.24a)$$

$$p_{L,2} - \frac{\rho_{\ell,2}}{L-1} = \frac{p_{L,2} \Delta_L^L \left(1 - \frac{1}{d} \right) + p_{L,2} \mathcal{R}_E - \frac{\mathcal{R}_B}{L-1}}{\widetilde{\mathbb{P}}(E)}. \quad (6.B.24b)$$

Moreover, if

$$\frac{\sigma_{G^L} \delta_L}{\Delta_L} \ll \frac{1}{L}, \quad (6.B.25)$$

then the remainder terms satisfy

$$|\mathcal{R}_A| \leq p_{L,1} \Delta_L^{L-1} \left(1 - \frac{1}{d}\right) \left(O \left(\frac{\sigma_{G^L}^2 \delta_L^2}{\Delta_L^2} \right) + \frac{(L-1)^2 \gamma(\mathcal{G}) \delta_L}{\Delta_L} \right), \quad (6.B.26a)$$

$$|\mathcal{R}_E| \leq \Delta_L^L \left(1 - \frac{1}{d}\right) \left(O \left(\frac{\sigma_{G^L}^2 \delta_L^2}{\Delta_L^2} \right) + \frac{(L-1)L \gamma(\mathcal{G}) \delta_L}{\Delta_L} \right), \quad (6.B.26b)$$

$$\frac{|\mathcal{R}_B|}{L-1} \leq p_{L,2} \Delta_L^{L-1} \left(1 - \frac{1}{d}\right) \left(O \left(\frac{\sigma_{G^L}^2 \delta_L}{(L-1)\Delta_L} \right) + \gamma(\mathcal{G}) \right). \quad (6.B.26c)$$

Proof. For $\rho_{\ell,1} - p_{L,1}$, by Lemma 6.B.12 we have

$$\begin{aligned} \rho_{\ell,1} - p_{L,1} &= \frac{\widetilde{\mathbb{P}}(A_\ell \cap E) - p_{L,1} \widetilde{\mathbb{P}}(E)}{\widetilde{\mathbb{P}}(E)} \\ &= \frac{\frac{p_{L,1}}{d} + p_{L,1} \Delta_L^{L-1} \left(1 - \frac{1}{d}\right) + \mathcal{R}_A - p_{L,1} \left(\frac{1}{d} + \Delta_L^L \left(1 - \frac{1}{d}\right) + \mathcal{R}_E \right)}{\widetilde{\mathbb{P}}(E)} \\ &= \frac{p_{L,1} \Delta_L^{L-1} (1 - \Delta_L) \left(1 - \frac{1}{d}\right) + \mathcal{R}_A - p_{L,1} \mathcal{R}_E}{\widetilde{\mathbb{P}}(E)}, \end{aligned}$$

which gives Eq. (6.B.24a). For $p_{L,2} - \frac{\rho_{\ell,2}}{L-1}$, we similarly obtain

$$\begin{aligned} p_{L,2} - \frac{\rho_{\ell,2}}{L-1} &= \frac{p_{L,2}(L-1) \widetilde{\mathbb{P}}(E) - \widetilde{\mathbb{P}}(B_\ell \cap E)}{(L-1) \widetilde{\mathbb{P}}(E)} \\ &= \frac{p_{L,2}(L-1) \left(\frac{1}{d} + \Delta_L^L \left(1 - \frac{1}{d}\right) + \mathcal{R}_E \right) - \frac{(L-1)p_{L,2}}{d} - \mathcal{R}_B}{(L-1) \widetilde{\mathbb{P}}(E)} \\ &= \frac{p_{L,2} \Delta_L^L \left(1 - \frac{1}{d}\right) + p_{L,2} \mathcal{R}_E - \mathcal{R}_B / (L-1)}{\widetilde{\mathbb{P}}(E)}, \end{aligned}$$

which gives Eq. (6.B.24b).

It remains to bound $\mathcal{R}_E$, $\mathcal{R}_A$, and $\mathcal{R}_B$ under Eq. (6.B.25). For notational simplicity, let $x \triangleq \sigma_{G^L} \delta_L / \Delta_L$. Then

$$(\Delta_L + \sigma_{G^L} \delta_L)^k = \Delta_L^k (1 + x)^k.$$

Bounds for $\mathcal{R}_E$ and $\mathcal{R}_A$. The expressions in the brackets for $\mathcal{R}_E$ and $\mathcal{R}_A$ contain

$$(1 + x)^k - 1 - kx, \quad k \in \{L, L-1\}.$$

Under $x \ll 1/L$, the second-order Taylor remainder gives $(1+x)^k - 1 - kx = O(k^2 x^2)$, which implies

$$(\Delta_L + \sigma_{G^L} \delta_L)^k - \Delta_L^k - k\sigma_{G^L} \delta_L \Delta_L^{k-1} = \Delta_L^k \cdot O(k^2 x^2) = O(k^2 \sigma_{G^L}^2 \delta_L^2 \Delta_L^{k-2}).$$

Substituting this estimate into the displayed bounds for $\mathcal{R}_E$ and $\mathcal{R}_A$ yields the claimed controls for $|\mathcal{R}_E|$ and $|\mathcal{R}_A|$.

Bound for $\mathcal{R}_B$. Here the bracket contains $(\Delta_L + \sigma_{G^L} \delta_L)^{L-1} - \Delta_L^{L-1} = \Delta_L^{L-1}((1+x)^{L-1} - 1)$. Under $x \ll 1/L$, the first-order estimate gives $(1+x)^{L-1} - 1 = O((L-1)x)$, hence

$$\sigma_{G^L}((\Delta_L + \sigma_{G^L} \delta_L)^{L-1} - \Delta_L^{L-1}) = \sigma_{G^L} \Delta_L^{L-1} \cdot O((L-1)\sigma_{G^L} \delta_L / \Delta_L) = O((L-1)\sigma_{G^L}^2 \delta_L \Delta_L^{L-2}).$$

Plugging this into the displayed bound for $\mathcal{R}_B$ yields the stated control on $|\mathcal{R}_B|$.

□

Gradient Characterization: Proof of Lemma 6.A.2

Combining the posterior deviations in Proposition 6.B.1 with Eq. (6.B.20), we obtain the following characterization of the gradient, which is a formal version of Lemma 6.A.2.

Proposition 6.B.2 (Gradient characterization). Given problem length L , suppose that

$$\frac{L^2 \delta_L}{\Delta_L} = o(1) \cdot (1 - \Delta_L) \quad \text{and} \quad \frac{p_{L,2}}{p_{L,1}} = o(1) \cdot (1 - \Delta_L).$$

Then

$$\begin{aligned} \nabla_q \tilde{\mathcal{J}}_L &= \Theta(\log d/d_p) \cdot p_{L,1} \Delta_L^{L-1} \left(1 - \frac{1}{d}\right) (1 - \Delta_L), \\ |\nabla_r \tilde{\mathcal{J}}_L| &= O(1/d_p) \cdot \nabla_q \tilde{\mathcal{J}}_L. \end{aligned}$$

Proof. Recall that

$$\nabla_q \tilde{\mathcal{J}}_L = \frac{1}{2Ld_p} \mathbb{E}_{Z^L} \left[\underbrace{\mathbf{Attn}_L \cdot B \cdot \tilde{\mathbb{P}}(E) \sum_{\ell=1}^L \left((\rho_{\ell,1} - p_{L,1})(2 - \mathbf{Attn}_L) + \left(p_{L,2} - \frac{\rho_{\ell,2}}{L-1} \right) (1 - \mathbf{Attn}_L) \right)}_{=: J_{\text{gap}}} \right]. \quad (6.B.27)$$

Note that $2 - \mathbf{Attn}_L \geq 1$ and $1 - \mathbf{Attn}_L \in (0, 1)$. Under the stated assumptions, since $\sigma_{G^L} \leq L-1$, the condition $\sigma_{G^L} \delta_L / \Delta_L \ll 1/L$ holds. Hence Proposition 6.B.1 applies. We now verify that for J_{gap} , all remainder contributions are negligible compared to the leading term $p_{L,1} \Delta_L^{L-1} (1 - \frac{1}{d})(1 - \Delta_L)$.

- **Bounding $\mathcal{R}_A$.** By Eq. (6.B.26a) in Proposition 6.B.1,

$$|\mathcal{R}_A| \leq p_{L,1} \Delta_L^{L-1} \left(1 - \frac{1}{d}\right) \left(O \left(\frac{\sigma_{G^L}^2 \delta_L^2}{\Delta_L^2} \right) + \frac{(L-1)^2 \gamma(\mathcal{G}) \delta_L}{\Delta_L} \right).$$

Using $\sigma_{G^L} \leq L-1$ and $\frac{L^2 \delta_L}{\Delta_L} = o(1) \cdot (1 - \Delta_L)$, we have

$$\frac{\sigma_{G^L}^2 \delta_L^2}{\Delta_L^2} \leq \frac{L^2 \delta_L}{\Delta_L} \cdot \frac{\delta_L}{\Delta_L} = o(1) \cdot (1 - \Delta_L) \cdot \frac{\delta_L}{\Delta_L} = o(1) \cdot (1 - \Delta_L),$$

and similarly

$$\frac{(L-1)^2 \gamma(\mathcal{G}) \delta_L}{\Delta_L} \leq \gamma(\mathcal{G}) \cdot o(1) \cdot (1 - \Delta_L).$$

Therefore,

$$|\mathcal{R}_A| \leq p_{L,1} \Delta_L^{L-1} \left(1 - \frac{1}{d}\right) (1 - \Delta_L) (o(1) + o(1) \gamma(\mathcal{G})) \ll p_{L,1} \Delta_L^{L-1} \left(1 - \frac{1}{d}\right) (1 - \Delta_L).$$

- **Bounding $\mathcal{R}_E$.** By Eq. (6.B.26b) in Proposition 6.B.1,

$$|\mathcal{R}_E| \leq \Delta_L^L \left(1 - \frac{1}{d}\right) \left(O \left(\frac{\sigma_{G^L}^2 \delta_L^2}{\Delta_L^2} \right) + \frac{(L-1)L \gamma(\mathcal{G}) \delta_L}{\Delta_L} \right).$$

Using the same estimates as above and $\Delta_L \leq 1$, we obtain

$$|\mathcal{R}_E| \ll \Delta_L^L \left(1 - \frac{1}{d}\right) (1 - \Delta_L).$$

Consequently, $p_{L,1} |\mathcal{R}_E|$ is dominated by $p_{L,1} \Delta_L^L \left(1 - \frac{1}{d}\right) (1 - \Delta_L)$.

- **Bounding $\mathcal{R}_B$.** By Eq. (6.B.26c) in Proposition 6.B.1,

$$\frac{|\mathcal{R}_B|}{L-1} \leq p_{L,2} \Delta_L^{L-1} \left(1 - \frac{1}{d}\right) \left(O \left(\frac{\sigma_{G^L}^2 \delta_L}{(L-1) \Delta_L} \right) + \gamma(\mathcal{G}) \right).$$

Using $\sigma_{G^L} \leq L-1$ and $\frac{L^2 \delta_L}{\Delta_L} = o(1) \cdot (1 - \Delta_L)$, we have

$$\frac{\sigma_{G^L}^2 \delta_L}{(L-1) \Delta_L} \leq \frac{L^2 \delta_L}{\Delta_L} = o(1) \cdot (1 - \Delta_L).$$

Moreover, $\frac{p_{L,2}}{p_{L,1}} = o(1) \cdot (1 - \Delta_L)$ implies

$$p_{L,2} \Delta_L^{L-1} \leq p_{L,1} \Delta_L^{L-1} \cdot o(1) \cdot (1 - \Delta_L).$$

Thus,

$$\frac{|\mathcal{R}_B|}{L-1} \leq p_{L,1} \Delta_L^{L-1} \left(1 - \frac{1}{d}\right) (1 - \Delta_L) \left(o(1) + \gamma'(\mathcal{G}) \cdot o(1)\right) \ll p_{L,1} \Delta_L^{L-1} \left(1 - \frac{1}{d}\right) (1 - \Delta_L).$$

Finally, the same assumption $\frac{p_{L,2}}{p_{L,1}} = o(1) \cdot (1 - \Delta_L)$ also yields

$$p_{L,2} \Delta_L^L \left(1 - \frac{1}{d}\right) \ll p_{L,1} \Delta_L^{L-1} \left(1 - \frac{1}{d}\right) (1 - \Delta_L),$$

so the contribution of the second posterior deviation term is dominated by the first term.

Plugging the above bounds into the expression for $\nabla_q \mathcal{J}_L$, and using that $\mathbf{Attn}_L = \Theta(1)$, which is implied by $\frac{p_{L,2}}{p_{L,1}} = o(1) \cdot (1 - \Delta_L)$, we conclude that

$$\nabla_q \tilde{\mathcal{J}}_L = \Theta(\log d/d_p) \cdot p_{L,1} \Delta_L^{L-1} \left(1 - \frac{1}{d}\right) (1 - \Delta_L).$$

The analysis for $|\nabla_r \tilde{\mathcal{J}}_L|$ is similar. Alternatively, we may invoke the direct comparison bound in Lemma 6.B.3 to obtain $|\nabla_r \tilde{\mathcal{J}}_L| = O(1/d_p) \cdot \nabla_q \tilde{\mathcal{J}}_L$. $\square$

Exponentially Flat Region for Long-Horizon Tasks: Proof of Proposition 6.4.1

Following the same decomposition underlying Lemma 6.B.12, we show that when the step-invariant probability tuple $(p_{L,1}, p_{L,2}, p_{L,3})$ has small effective margins $\Delta_L := p_{L,1} - p_{L,3}$ and $\delta_L := p_{L,2} - p_{L,3}$, the resulting policy gradient is upper bounded by a quantity that decays exponentially in the horizon length L . We then specialize this general exponential barrier to our concrete setting, which immediately yields Proposition 6.4.1.

Proposition 6.B.3. Under Assumptions 6.3.1–6.3.4, for any $2 \leq L \leq L_{\max}$, suppose the step-invariant probability tuple $(p_{L,1}, p_{L,2}, p_{L,3})$ satisfies, with $\Delta_L := p_{L,1} - p_{L,3}$ and $\delta_L := p_{L,2} - p_{L,3}$,

$$\Delta_L + L\delta_L \leq \tilde{O}\left(d^{-\Omega(1)}\right), \quad p_{L,i} \leq d^{-\Omega(1)} \text{ for } i \in [3]. \quad (6.B.28)$$

Then,

$$\left|\nabla_q \tilde{\mathcal{J}}_L\right| \leq \tilde{O}\left(\frac{1}{d_p}\right) \cdot d^{-\Omega(L)}, \quad \left|\nabla_r \tilde{\mathcal{J}}_L\right| \leq \tilde{O}\left(\frac{1}{d_p^2}\right) \cdot d^{-\Omega(L)}. \quad (6.B.29)$$

Proof. A key takeaway from Lemma 6.B.12 is that, when bounding the remainder contributions (e.g., $\mathcal{R}_E$), we decompose the remainder term $T_{\text{res}}(\lambda)$ into several parts. Independent of this finer decomposition, its operator norm admits the crude bound

$$\|T_{\text{res}}(\lambda)\|_{\text{op}} \leq (\Delta_L + \sigma_{G_L} \delta_L)^L - \Delta_L^L.$$

Using this bound directly gives

$$\mathcal{R}_E \leq \left(1 - \frac{1}{d}\right)(\Delta_L + \sigma_{G_L} \delta_L)^L.$$

The same argument applies to $\mathcal{R}_A$ and $\mathcal{R}_B$, yielding

$$\mathcal{R}_A \leq \left(1 - \frac{1}{d}\right)p_{L,1}(\Delta_L + \sigma_{G_L} \delta_L)^{L-1}, \quad \mathcal{R}_B \leq \left(1 - \frac{1}{d}\right)p_{L,2}\sigma_{G_L}(\Delta_L + \sigma_{G_L} \delta_L)^{L-1}.$$

Invoking Eq. (6.B.24) from Proposition 6.B.1 and substituting the above bounds into Eq. (6.B.27), we obtain

$$\begin{aligned} |\nabla_q \tilde{\mathcal{J}}_L| &\leq \tilde{O}\left(\frac{B}{d_p}\right) \left(p_{L,1}\Delta_L^{L-1} + p_{L,1}(\Delta_L + \sigma_{G_L} \delta_L)^{L-1} + p_{L,2}\frac{\sigma_{G_L}}{L-1}(\Delta_L + \sigma_{G_L} \delta_L)^{L-1}\right) \\ &\leq \tilde{O}\left(\frac{1}{d_p}\right) \cdot d^{-\Omega(L)}, \end{aligned}$$

where in the last step we use $\sigma_{G_L} \leq L-1$ together with the assumptions $\Delta_L + L\delta_L \leq \tilde{O}(d^{-\Omega(1)})$ and $p_{L,i} \leq d^{-\Omega(1)}$ for $i \in [3]$. The bound for $|\nabla_r \tilde{\mathcal{J}}_L|$ follows by the same reasoning and is omitted. $\square$

Proposition 6.B.4 (Proposition 6.4.1 restated). Under Assumptions 6.3.1–6.3.4, suppose $\text{TF}_{\theta(0)}$ is initialized according to Assumption 6.3.5. Then for any horizon $L > 2C_B$, whenever the feature magnitudes satisfy $\max\{|r^{(t)}|, |q^{(t)}|\} \leq 0.01$, we have $\mathcal{J}_L^{(t)} = \frac{1}{d}(1 \pm o(1))$, and

$$|\nabla_q \tilde{\mathcal{J}}_L^{(t)}| \leq \tilde{O}\left(\frac{1}{d_p}\right) \cdot d^{-\Omega(L)}, \quad |\nabla_r \tilde{\mathcal{J}}_L^{(t)}| \leq \tilde{O}\left(\frac{1}{d_p^2}\right) \cdot d^{-\Omega(L)}.$$

Proof. Since $\max\{|r^{(t)}|, |q^{(t)}|\} \leq 0.01$, the attention weights satisfy

$$\text{Attn}_L^{(t)} C_B \leq \frac{C_B e^{0.02}}{e^{0.02} + L - 1} < 1, \quad \frac{1 - \text{Attn}_L^{(t)}}{L - 1} C_B \leq \frac{C_B}{e^{-0.02} + L - 1} < 1.$$

In particular, this implies $p_{L,i}^{(t)} \leq d^{-\Omega(1)}$ for all $i \in [3]$. It remains to bound $\Delta_L^{(t)} + L\delta_L^{(t)}$, which we do by considering two regimes.

Case 1: $L < d^{0.01}$. By Lemma 6.B.7,

$$p_{L,1}^{(t)} \leq O\left(\frac{1}{L + d^{1 - \frac{C_B e^{0.02}}{e^{0.02} + L - 1}}}\right) = d^{-\Omega(1)}, \quad p_{L,2}^{(t)} \leq O\left(\frac{1}{L + d^{1 - \frac{C_B}{e^{-0.02} + L - 1}}}\right) \leq d^{-0.5}.$$

Therefore,

$$\Delta_L^{(t)} + L\delta_L^{(t)} \leq p_{L,1}^{(t)} + Lp_{L,2}^{(t)} \leq d^{-\Omega(1)}.$$

Case 2: $L \geq d^{0.01}$. In this regime, we bound $\Delta_L^{(t)}$ and $\delta_L^{(t)}$ directly. In particular,

$$\Delta_L^{(t)} \leq O\left(\frac{e^{\frac{C_B e^{0.02}}{e^{0.02} + L - 1} \log d} - 1}{d}\right) \leq \tilde{O}\left(\frac{1}{Ld}\right),$$

and

$$L\delta_L^{(t)} \leq O\left(L \cdot \frac{e^{\frac{C_B}{e^{-0.02} + L - 1} \log d} - 1}{d}\right) \leq \tilde{O}\left(\frac{1}{d}\right).$$

Thus,

$$\Delta_L^{(t)} + L\delta_L^{(t)} \leq \tilde{O}\left(\frac{1}{d}\right).$$

In both regimes, the conditions of Proposition 6.B.3 are satisfied. Therefore, applying Proposition 6.B.3 yields the desired gradient bound. Moreover, $\mathcal{J}_L^{(t)} = \frac{1}{d}(1 \pm o(1))$ follows directly from Lemma 6.B.14.

□

Reward Characterization

Note that $\tilde{\mathbb{P}}(E)$ is exactly the expected reward for a fixed instance (y_0, G_L) . Consequently,

$$\mathcal{J}_L = \mathbb{E}_{Z^L}[\tilde{\mathbb{P}}(E)].$$

Therefore, the gradient characterization in Proposition 6.B.2 immediately yields a corresponding characterization of the reward.

Lemma 6.B.13. *Given a problem of length L , suppose that*

$$\frac{L^2 \delta_L}{\Delta_L} = o(1) \cdot (1 - \Delta_L) \quad \text{and} \quad \frac{p_{L,2}}{p_{L,1}} = o(1) \cdot (1 - \Delta_L).$$

Then,

$$\mathcal{J}_L = \frac{1}{d} + \left(1 - \frac{1}{d}\right)(1 \pm o(1)) \cdot \Delta_L^L.$$

Moreover, by adapting the argument in Proposition 6.B.3 to control the residual term $\mathcal{R}_E$, we obtain the following coarse upper bound.

Lemma 6.B.14. *Given a problem of length L , we have*

$$\left|\mathcal{J}_L - \frac{1}{d}\right| \leq \left(1 - \frac{1}{d}\right)(\Delta_L + \sigma_{G_L} \delta_L)^L.$$

6.B.3 Learning Dynamics of Short-horizon RL

In this section, we focus on the regime $L \leq C_B$. Our analysis tracks the training dynamics of the two scalar quantities q and r defined in Eq. (6.B.4a) and Eq. (6.B.4b). We proceed in three steps.

First, we state an induction hypothesis that is maintained throughout training. Second, under this hypothesis, we derive one-step update bounds for q and r . Finally, we close the induction by showing that the hypothesis holds for all iterations.

We will focus on the RL training dynamics; the same proof structure and bookkeeping apply to SFT training. Accordingly, at the end of this section, we briefly list the key lemmas and the corresponding induction for SFT, and omit the details.

Induction 6.B.1. *Given $\Omega(\frac{1}{\text{polylog} d}) < \epsilon < \frac{1}{4}$, and let T_1 be the first iteration such that $\text{Attn}_L^{(t)} \geq 1 - \epsilon$. Then for every iteration $t < T_1$, the following statements hold:*

- (a) $O(\log \frac{L}{\epsilon}) \geq q^{(t)} \geq 0$, and $q^{(t)}$ is monotonically nondecreasing in t (starting from 0);
- (b) $|r^{(t)}| \leq O(1/d_p)q^{(t)}$.

Attention and Logit Preliminaries

We first introduce several properties of the attention scores and logits if Induction 6.B.1 holds.

Lemma 6.B.15. *If Induction 6.B.1 holds for all iterations $< t$, then we have*

- (a) $\text{Attn}_L^{(t)} = \frac{e^{q^{(t)}-r^{(t)}}}{e^{q^{(t)}-r^{(t)}} + (L-1)} \geq \frac{1}{L}$;
- (b) $\text{Attn}_{a,\ell-1 \rightarrow p,k}^{(t)} = \frac{1}{(L-1) + e^{q^{(t)}-r^{(t)}}} = \frac{1}{L-1} (1 - \text{Attn}_L^{(t)})$ for $k \neq \ell$.

Therefore, direct calculations by combining Lemma 6.B.15 and Lemma 6.B.7 yield the following lemma.

Lemma 6.B.16. *Assume that Induction 6.B.1 holds for all iterations $< t$. We have*

$$p_{L,1}^{(t)} \geq \Omega(1), \quad 1 - p_{L,1}^{(t)} \geq \Omega\left(\frac{1}{d^{(1-\epsilon)C_B-1}}\right)$$

and the following bounds on the transition probabilities $p_{L,2}^{(t)}$ and $p_{L,3}^{(t)}$.

- (1) **Regime I:** if $\text{Attn}_L^{(t)} < 1 - \frac{L-1}{C_B}$, then

- (a) in-context distractor transition

$$p_{L,2}^{(t)} = \Theta\left(d^{-\left(\text{Attn}_L^{(t)} - \frac{1-\text{Attn}_L^{(t)}}{L-1}\right)C_B}\right) = O\left(\frac{1}{L}\right) \left(1 - p_{L,1}^{(t)}\right).$$

- (b) vocabulary distractor transition

$$p_{L,3}^{(t)} = O\left(d^{-\text{Attn}_L^{(t)}C_B}\right) = O\left(\frac{1}{d^{\frac{C_B}{L-1}(1-\text{Attn}_L^{(t)})}}\right) \left(1 - p_{L,1}^{(t)}\right).$$

- (2) **Regime II:** if $\text{Attn}_L^{(t)} \geq 1 - \frac{L-1}{C_B}$, then

(a) in-context distractor transition

$$p_{L,2}^{(t)} = O\left(d^{-\text{Attn}_L^{(t)} C_B}\right) = O\left(\frac{1}{d}\right) \left(1 - p_{L,1}^{(t)}\right).$$

(b) vocabulary distractor transition

$$p_{L,3}^{(t)} = O\left(d^{-\text{Attn}_L^{(t)} C_B}\right) = O\left(\frac{1}{d}\right) \left(1 - p_{L,1}^{(t)}\right).$$

Gradient Lemma

Since the initialization is uniform, the initial step-wise probabilities satisfy $p_{L,1}^{(0)} = p_{L,2}^{(0)}$, so the gradient characterization in Proposition 6.B.2 is not directly applicable. We therefore need finer control of the gradients at the very beginning of training.

Lemma 6.B.17. *Assume that Induction 6.B.1 holds for all iterations $< t$, when $1 - \text{Attn}_L^{(t)} \geq \Omega(1)$, we have*

$$\nabla_q \tilde{\mathcal{J}}_L^{(t)} \geq \Omega\left(\frac{\log d}{d_p d^{\text{Attn}_L^{(t)} C_B - 1}}\right).$$

Proof. By Eq. (6.B.5), we have

$$\begin{aligned} \nabla_q \tilde{\mathcal{J}}_L^{(t)} &= \frac{1}{2Ld_p} \mathbb{E}_{Z^L} [\mathfrak{J}^{(t)}(y_0, G^L)] \\ &= \frac{1}{2Ld_p} \mathbb{E}_{Z^L} \left[\sum_{v \in \mathcal{Y}^{L-1} \times \{y_L\}} \left(\prod_{\ell'=1}^L \pi_{\theta}^{(t)}(v_{\ell'} \mid v_{\ell'-1}, G^L) \right) \left(\sum_{\ell=1}^L \cdot \text{Attn}_{a,\ell-1 \rightarrow p,\ell} \mathfrak{G}_{\ell}(v) \right) \right]. \end{aligned} \quad (6.B.30)$$

Therefore, in the following, we will divide the trajectory $v \in \mathcal{V}^{L-1} \times \{y_L\}$ into three different categories:

- $v = v^* \triangleq (y_1, \dots, y_L)$, by Lemma 6.B.16, we can obtain

$$\prod_{\ell'=1}^L \pi_{\theta}^{(t)}(v_{\ell'} \mid v_{\ell'-1}, G^L) = p_{L,1}^{(t)} \geq \Omega(1).$$

Along the trajectory v^* , by Lemma 6.B.6, we have

$$\begin{aligned} \mathfrak{G}_{\ell}(v) &= \sum_{j \in \tau(\mathcal{Y})} \mathcal{E}_j^{(t)} \sum_{r \in [m]} \sigma'(\Lambda_{j,r}^{(t)}) \cdot \left(\langle W_{j,r}, Z_{p,\ell} \rangle - \Lambda_{j,r}^{(t)} \right) \\ &= (1 - p_{L,1}^{(t)}) \left(B - \text{Attn}_L^{(t)} (B + \sigma_0) \right) - (L-1) p_{L,2}^{(t)} \left(-B - \text{Attn}_L^{(t)} (B + \sigma_0) \right) \\ &\quad - (d-L) p_{L,3}^{(t)} \left(-B - \sigma_0 \right) \end{aligned}$$

$$\geq \Omega(B) \cdot (1 - \mathbf{Attn}_L^{(t)}) \cdot (1 - p_{L,1}^{(t)}). \quad (6.B.31)$$

Therefore, we have

$$\mathbb{E}_{Z^L} \left[\left(\prod_{\ell'=1}^L \pi_{\theta}^{(t)}(y_{\ell'} \mid y_{\ell'-1}, G^L) \right) \left(\sum_{\ell=1}^L \mathbf{Attn}_{a,\ell-1 \rightarrow p,\ell}(\mathfrak{G}_{\ell}(v^*)) \right) \right] \geq \Omega\left(\frac{\log d}{d_p d^{\mathbf{Attn}_L^{(t)} C_{B-1}}}\right). \quad (6.B.32)$$

- On the event $\mathfrak{E}_L$, there exists other trajectories (may more than one) $v' \neq v^*$, s.t., $v'_{\ell} = g'_{\ell}(v'_{\ell-1})$ with some $g'_{\ell} \in G^L$ for all $\ell \in [L]$ (letting $v_0 = y_0$). We denote the collection of such v' as $\mathbf{V}'$. In the following, we denote $g_{i_{\ell}} = g'_{\ell}$. For $v' \in \mathbf{V}'$, there will be at least one $\tilde{\ell}$, s.t., $i_{\tilde{\ell}} \neq \tilde{\ell}$. Thus, by Lemma 6.B.16, we have

$$\pi_{\theta}^{(t)}(v'_{\tilde{\ell}} \mid v'_{\tilde{\ell}-1}, G^L) \leq O(1/L) \cdot (1 - p_{L,1}^{(t)}).$$

Hence,

$$\prod_{\ell'=1}^L \pi_{\theta}^{(t)}(v'_{\ell'} \mid v'_{\ell'-1}, G^L) \leq O(1) \cdot (1 - p_{L,1}^{(t)}) \cdot \prod_{\ell'=1}^L \pi_{\theta}^{(t)}(y_{\ell'} \mid y_{\ell'-1}, G^L). \quad (6.B.33)$$

Moreover, along the trajectory v' , the analysis is similar as v^* , we have

$$\begin{aligned} & \left| \sum_{j \in \tau(\mathcal{Y})} \mathcal{E}_j^{(t)} \sum_{r \in [m]} \sigma'(\Lambda_{j,r}^{(t)}) \cdot (\langle W_{j,r}, Z_{p,\ell} \rangle - \Lambda_{j,r}^{(t)}) \right| \\ & \leq (1 - \pi_{\theta}^{(t)}(g_{i_{\ell}}(\widehat{y}_{\ell-1}) \mid v'_{\ell-1}, G^L)) (B + \mathbf{Attn}_L^{(t)} B + \sigma_0) \\ & \quad + \sum_{\ell' \neq i_{\ell}} \pi_{\theta}^{(t)}(g_{\ell'}(\widehat{y}_{\ell-1}) \mid v'_{\ell-1}, G^L) (B + \mathbf{Attn}_L^{(t)} B + \sigma_0) \\ & \quad + \sum_{g \notin G^L} \pi_{\theta}^{(t)}(g(\widehat{y}_{\ell-1}) \mid v'_{\ell-1}, G^L) (B + \sigma_0) \end{aligned} \quad (6.B.34)$$

$$\leq O(B). \quad (6.B.35)$$

Hence, combining Eq. (6.B.31) Eq. (6.B.35) and the probability of $\mathfrak{E}_L$ from Lemma 6.B.8 together, it holds that

$$\begin{aligned} & \left| \mathbb{E}_{Z^L} \left[\sum_{v \in \mathbf{V}'} \left(\prod_{\ell'=1}^L \pi_{\theta}^{(t)}(v_{\ell'} \mid v_{\ell'-1}, G^L) \right) \left(\sum_{\ell=1}^L \mathbf{Attn}_{a,\ell-1 \rightarrow p,\ell}(\mathfrak{G}_{\ell}(v)) \right) \right] \right| \\ & \leq o(1) \mathbb{E}_{Z^L} \left[|\mathbf{V}'| \left(\prod_{\ell'=1}^L \pi_{\theta}^{(t)}(y_{\ell'} \mid y_{\ell'-1}, G^L) \right) \left(\sum_{\ell=1}^L \mathbf{Attn}_{a,\ell-1 \rightarrow p,\ell}(\mathfrak{G}_{\ell}(v^*)) \right) \right] \\ & \leq o(1) \mathbb{E}_{Z^L} \left[\left(\prod_{\ell'=1}^L \pi_{\theta}^{(t)}(y_{\ell'} \mid y_{\ell'-1}, G^L) \right) \left(\sum_{\ell=1}^L \mathbf{Attn}_{a,\ell-1 \rightarrow p,\ell}(\mathfrak{G}_{\ell}(v^*)) \right) \right], \end{aligned} \quad (6.B.36)$$

where the last inequality follows the fact that $|\mathcal{V}'| \leq L^L = O(1)$.

- for other $v \in (\mathcal{Y}^{L-1} \times \{y_L\}) \setminus (\mathcal{V}' \cup \{v^*\})$, there will be at least one $\widehat{\ell} \in [L]$, s.t., $v_{\widehat{\ell}} = g(v_{\widehat{\ell}-1})$ with $g \neq \widehat{\mathcal{G}}$.
 - if there exists only one such $\widehat{\ell}$, then since the group is simply transitive, there exists another $\widehat{\ell}' \neq \widehat{\ell}$ s.t., $v_{\widehat{\ell}'} = g(v_{\widehat{\ell}'-1})$ with $g \neq g_{\widehat{\ell}'}$. By Lemma 6.B.16, for such $\widehat{\ell}$ and $\widehat{\ell}'$, we have

$$\begin{aligned} & \pi_{\theta}^{(t)}(v_{\widehat{\ell}} | v_{\widehat{\ell}-1}, G^L) \cdot \pi_{\theta}^{(t)}(v_{\widehat{\ell}'} | v_{\widehat{\ell}'-1}, G^L) \\ & \leq O\left(d^{-\text{Attn}_L^{(t)} C_B}\right) \cdot (1 - p_{L,1}^{(t)}) = O\left(\frac{1}{d^{1+\Omega(1)}}\right) (1 - p_{L,1}^{(t)}). \end{aligned} \quad (6.B.37)$$

- if there exists exact $k > 1$ such $\widehat{\ell}$, denoted as $\widehat{\ell}_1, \dots, \widehat{\ell}_k$. By Lemma 6.B.16, we have

$$\begin{aligned} & \prod_{i=1}^k \pi_{\theta}^{(t)}(v_{\widehat{\ell}_i} | v_{\widehat{\ell}_i-1}, G^L) \\ & \leq O\left(d^{-(k-1)\text{Attn}_L^{(t)} C_B}\right) \cdot O\left(\frac{1}{d^{\frac{C_B}{L-1}(1-\text{Attn}_L^{(t)})}}\right) (1 - p_{L,1}^{(t)}) \\ & \leq O\left(\frac{1}{d^{k+\Omega(1)}}\right) (1 - p_1^{(t)}). \end{aligned} \quad (6.B.38)$$

Here, the last inequality holds since $((k-1)x + \frac{1-x}{L-1})C_B$ is monotonically increase for $x \geq \frac{1}{L}$ and the minimum value is $\frac{kC_B}{L} = k + \Omega(1)$.

Moreover, we have

$$\left| \sum_{j \in \tau(\mathcal{Y})} \mathcal{E}_j^{(t)} \sum_{r \in [m]} \sigma'(\Lambda_{j,r}^{(t)}) \cdot (\langle W_{j,r}, Z_{p,\ell} \rangle - \Lambda_{j,r}^{(t)}) \right| \leq O(B), \quad (6.B.39)$$

which in turn leads to

$$\begin{aligned} & \left| \mathbb{E}_{Z^L} \left[\sum_{v \in (\mathcal{Y}^{L-1} \times \{y_L\}) \setminus (\mathcal{V}' \cup \{v^*\})} \left(\prod_{\ell'=1}^L \pi_{\theta}^{(t)}(v_{\ell'} | v_{\ell'-1}, G^L) \right) \left(\sum_{\ell=1}^L \cdot \text{Attn}_{a,\ell-1 \rightarrow p,\ell}(\mathfrak{G}_{\ell}(v)) \right) \right] \right| \\ & \leq \sum_{k=1}^L \binom{L}{k} O(N^k) O\left(\frac{1}{d^{k+\Omega(1)}}\right) \mathbb{E}_{Z^L} \left[\left(\prod_{\ell'=1}^L \pi_{\theta}^{(t)}(y_{\ell'} | y_{\ell'-1}, G^L) \right) \left(\sum_{\ell=1}^L \cdot \text{Attn}_{a,\ell-1 \rightarrow p,\ell}(\mathfrak{G}_{\ell}(v^*)) \right) \right] \\ & \leq O\left(\frac{1}{d^{\Omega(1)}}\right) \mathbb{E}_{Z^L} \left[\left(\prod_{\ell'=1}^L \pi_{\theta}^{(t)}(y_{\ell'} | y_{\ell'-1}, G^L) \right) \left(\sum_{\ell=1}^L \cdot \text{Attn}_{a,\ell-1 \rightarrow p,\ell}(\mathfrak{G}_{\ell}(v^*)) \right) \right]. \end{aligned} \quad (6.B.40)$$

Therefore, we put Eq. (6.B.32), Eq. (6.B.36), Eq. (6.B.40) together, and thus conclude that

$$\nabla_q \tilde{\mathcal{J}}_L^{(t)} = \frac{1}{L d_p} \mathbb{E}_{Z^L} [\mathfrak{J}^{(t)}(y_0, G^L)] \geq \Omega\left(\frac{\log d}{d_p d^{\text{Attn}_L^{(t)} C_B - 1}}\right).$$

□

Lemma 6.B.18. Assume that Induction 6.B.1 holds for all iterations $< t$, when $\text{Attn}_L^{(t)} \geq 1 - \frac{L-1}{C_B}$, we have

$$\nabla_q \tilde{\mathcal{J}}_L^{(t)} = \Theta\left(\frac{\log d}{d_p d^{\text{Attn}_L^{(t)} C_B - 1}}\right).$$

Proof. By Lemma 6.B.16, when $\text{Attn}_L^{(t)} \geq 1 - \frac{L-1}{C_B}$, we have $\Delta_L^{(t)}, p_{L,1}^{(t)} = \Omega(1)$, and $\delta_L^{(t)} \leq p_{L,2}^{(t)} \leq O(1/d)(1 - \Delta_L^{(t)})$. Hence, the condition in Proposition 6.B.2 holds, and we can directly apply it to complete the proof. □

Lemma 6.B.19. Assume that Induction 6.B.1 holds for all iterations $< t$. Then,

$$|\nabla_r \tilde{\mathcal{J}}_L^{(t)}| \leq O\left(\frac{1}{d_p}\right) |\nabla_q \tilde{\mathcal{J}}_L^{(t)}|.$$

Proof. The claim follows from Lemma 6.B.3, we thus omit the details. □

Proof of Theorem 6.4.1

By combining the gradient bounds in Lemmas 6.B.17 to 6.B.19, we show that the induction hypothesis Induction 6.B.1 is maintained throughout this stage until $q^{(t)}$ reaches the target scale $\Omega(\log(L/\epsilon))$. At that point, we obtain $\text{Attn}_L^{(t)} \geq 1 - \epsilon$, which leads to the following lemma.

Lemma 6.B.20 (End of Constant-Length Training). For any $\Omega(1/\text{polylog} d) < \epsilon < \frac{L-1}{C_B}$, the induction hypothesis Induction 6.B.1 holds for all iterations

$$t < T_1 = O\left(d_p d^{(1-\epsilon)C_B - 1} \cdot \frac{\log(L/\epsilon)}{\eta \log d}\right).$$

Moreover, at $t = T_1$ we have: $q^{(T_1)} \geq \Omega(\log(L/\epsilon)); |r^{(T_1)}| \leq O\left(\frac{1}{d}\right) q^{(T_1)}$.

Proof. Assume Induction 6.B.1 holds up to iteration t . Then Lemmas 6.B.17 and 6.B.18 imply that the policy gradient for q is strictly positive and satisfies a lower bound of the form

$$\nabla_q \tilde{\mathcal{J}}_L^{(t)} \geq \Omega\left(\frac{\log d}{d_p}\right) \cdot d^{-(1-\epsilon)C_B + 1},$$

where we used that along this stage $\mathbf{Attn}_L^{(t)} \leq 1 - \epsilon$. Under policy gradient update with step size η , we therefore have the per-iteration increase

$$q^{(t+1)} - q^{(t)} = \eta \nabla_q \tilde{\mathcal{J}}_L^{(t)} \geq \eta \cdot \Omega\left(\frac{\log d}{d_p}\right) \cdot d^{-(1-\epsilon)C_B+1}.$$

Summing over iterations until $q^{(t)}$ reaches $\Omega(\log(L/\epsilon))$ yields

$$T_1 = O\left(\frac{d_p}{\eta \log d} \cdot \log(L/\epsilon) \cdot d^{(1-\epsilon)C_B-1}\right),$$

as claimed.

Finally, Lemma 6.B.19 gives $|\nabla_r \tilde{\mathcal{J}}_L^{(t)}| \leq O(\frac{1}{d}) \nabla_q \tilde{\mathcal{J}}_L^{(t)}$ throughout this stage, and hence $r^{(t)}$ remains slaved to $q^{(t)}$, i.e., $|r^{(t)}| \leq O(\frac{1}{d})q^{(t)}$ for all $t \leq T_1$. $\square$

Theorem 6.4.1 follows immediately from Lemma 6.B.20.

Proof of Theorem 6.4.2

The proof follows the same template as the RL case: we (i) set up an induction hypothesis controlling the key parameters, (ii) derive lower/upper bounds on the relevant gradients under this hypothesis, and (iii) combine these bounds to upper bound the time needed for $q^{(t)}$ to reach the target scale, at which point the attention satisfies $\mathbf{Attn}_L^{(t)} \geq 1 - \epsilon$. In other words, Theorem 6.4.2 is obtained by assembling the lemmas below.

By comparing Lemma 6.B.3 and Lemma 6.B.4, we observe that while the gradient forms are structurally similar, the supervised analysis is more direct. This simplification arises because we only need to track the ground-truth trajectory defined by $v = (y_1, \dots, y_L)$. Consequently, we can establish an induction hypothesis analogous to Induction 6.B.1.

Induction 6.B.2. For any length $2 \leq L \leq \text{polylog} d$, fix any $\Omega(\frac{1}{\text{polylog} d}) < \epsilon < \min\{\frac{1}{2}(1 - \frac{1}{C_B}), \frac{L-1}{2L}\}$, and let T_1 be the first iteration such that $\mathbf{Attn}_L^{(t)} \geq 1 - \epsilon$. Then for every iteration $t < T_1$, the following statements hold:

- (a) $O(\log \frac{L}{\epsilon}) \geq q^{(t)} \geq 0$, and $q^{(t)}$ is monotonically nondecreasing in t (starting from 0);
- (b) $|r^{(t)}| \leq O(1/d_p)q^{(t)}$.

Building on this induction, we characterize the gradient dynamics through the following lemmas.

Lemma 6.B.21. Assume that Induction 6.B.2 holds for all iterations $< t$, then we have

- if $\mathbf{Attn}_L^{(t)} \leq \frac{1}{C_B}$, then

$$-\nabla_q \text{Loss}_L^{(t)} \geq \Omega\left(\frac{\log d}{d_p L}\right).$$

- else, we have

$$-\nabla_q \text{Loss}_L^{(t)} \geq \Omega\left(\frac{\epsilon \log d}{d_p d^{(1-\epsilon)C_B-1}}\right).$$

Proof. The claim follows from a similar analysis as Eq. (6.B.31) in the proof of Lemma 6.B.17, we thus omit the details. $\square$

Lemma 6.B.22. Assume that Induction 6.B.2 holds for all iterations $< t$. Then,

$$|\nabla_r \text{Loss}_L^{(t)}| \leq O\left(\frac{1}{d_p}\right) |\nabla_q \text{Loss}_L^{(t)}|.$$

By combining the results above, we obtain the total training time required to reach the target attention level:

Lemma 6.B.23 (End of Training). For any constant length $2 \leq L \leq \text{polylog} d$, fix any $\Omega(\frac{1}{\text{polylog} d}) < \epsilon < \min\{\frac{1}{2}(1 - \frac{1}{C_B}), \frac{L-1}{2L}\}$, the induction hypothesis Induction 6.B.2 holds for all iterations

$$t < T_1 = O\left(d_p d^{(1-\epsilon)C_B-1} \cdot \frac{\log(L/\epsilon)}{\eta \epsilon \log d} + \frac{L d_p}{\eta \log d}\right).$$

Moreover, at $t = T_1$ we have: $q^{(T_1)} \geq \Omega(\log(L/\epsilon))$; $|r^{(T_1)}| \leq O(\frac{1}{d}) q^{(T_1)}$.

Theorem 6.4.2 follows immediately from Lemma 6.B.23 by noting that $q^{(T_1)} = \Omega(\log(L/\epsilon))$ implies $\text{Attn}_L^{(T_1)} \geq 1 - \epsilon$ (by the definition of T_1) and the stated bound on T_1 matches the claimed iteration complexity.

6.B.4 Learning Dynamics of Mixed-difficulty RL

In this section, we study the mixed-difficulty setting, where tasks of different lengths are interleaved. By combining the constant-length analysis in Section 6.B.3 with the gradient characterizations from Section 6.B.2, we analyze two regimes of the difficulty ratio R : (i) the large difficulty ratio regime $R = \omega(1)$, which gives rise to grokking-style dynamics, and (ii) the moderate difficulty ratio regime $R = O(1)$, which leads to smoother relay dynamics.

We begin by reviewing the mixed-difficulty setup and introducing some timestamps that will be useful for characterizing the overall learning dynamics.

Mixed-difficulty setup. Let $R > 1$ denote the *difficulty ratio*, and set the starting (effectively short) horizon to be $L_1 := C_B$. Define the horizon set $\mathcal{L}_R = \{L_1, L_2, \dots, L_K\}$ recursively by

$$L_k = \min\{\lceil R L_{k-1} \rceil, L_{\max}\}, \quad 2 \leq k \leq K,$$

where $K = \lceil \log_R(L_{\max}/L_1) \rceil$, so that $L_K = L_{\max}$. For simplicity, we focus on the case $R \geq 2$ throughout.

Mastery and visible return states. For any $L_i \in \mathcal{L}_R$, we say the horizon L_i has *visible return* at time t if

$$\mathcal{J}_{L_i}^{(t)} \geq 0.01. \quad (6.B.41)$$

Denote the first iteration such that L_i has visible return as $T_{\text{vis},i}$. We say the horizon L_i is *mastered* at time t if

$$\mathcal{J}_{L_i}^{(t)} \geq 0.99. \quad (6.B.42)$$

Denote the first iteration such that L_i is mastered as $T_{\text{mas},i}$.

Plateau between consecutive horizons. For $k \in \{1, \dots, K-1\}$, define

$$\mathcal{T}_k \triangleq T_{\text{vis},k+1} - T_{\text{mas},k} = \left| \left\{ t \mid \mathcal{J}_{L_k}^{(t)} \geq 0.99, \mathcal{J}_{L_{k+1}}^{(t)} < 0.01 \right\} \right|. \quad (6.B.43)$$

In words, $\mathcal{T}_k$ counts the number of iterations during which L_k is already mastered while L_{k+1} has not yet achieved a visible return.

Analysis of Large Difficulty Gap Regime

In this subsection, we analyze the large difficulty ratio regime, where $R = \omega(1)$. Following a similar proof strategy as in Section 6.B.3, we start with the induction hypothesis that is expected to hold through the training process.

Induction 6.B.3. Given $\Omega\left(\frac{1}{\text{poly log } d}\right) < \epsilon < \frac{1}{4}$, and let T^* be the first iteration such that $\text{Attn}_{L_{\max}}^{(t)} \geq 1 - \epsilon$. Then, for all iterations $t < T^*$, the following hold:

- (a) $0 \leq q^{(t)} \leq O\left(\log \frac{L_{\max}}{\epsilon}\right)$, and $q^{(t)}$ monotonically increases.
- (b) $|r^{(t)}| \leq O(1/d_p)q^{(t)}$.

Properties of the Attention Scores and Critical Thresholds We record some properties of the attention scores and critical thresholds.

Lemma 6.B.24. If Induction 6.B.3 holds for all iterations $< t$, then we have

- (a) $\text{Attn}_L^{(t)} = \frac{e^{q^{(t)} - r^{(t)}}}{e^{q^{(t)} - r^{(t)}} + (L-1)} \geq \frac{1}{L}$;
- (b) $\text{Attn}_{a,\ell-1 \rightarrow p,k}^{(t)} = \frac{1}{(L-1) + e^{q^{(t)} - r^{(t)}}} = \frac{1}{L-1} (1 - \text{Attn}_L^{(t)})$ for $k \neq \ell$.

Lemma 6.B.25 (Critical threshold of q). If Induction 6.B.3 holds, then given $L \in \mathcal{L}_R$, the critical threshold of q required to satisfy $\mathcal{J}_L \geq 1 - \xi$ for some constant $0 < \xi \leq 1$ is given by:

$$\begin{aligned} q &\geq \log \frac{L-1}{C_B-1} + f\left(\frac{\log L - \log \log \frac{1}{1-\xi}}{\log d}\right) \\ &\geq \log \frac{L-1}{C_B-1} + \frac{C_B}{C_B-1} \cdot \frac{\log L - \log \log \frac{1}{1-\xi}}{\log d} + O\left(\frac{\log^2 L}{\log^2 d}\right). \end{aligned}$$

where $f(x) = \log\left(\frac{1+x}{1-x/(C_B-1)}\right)$. Similarly, the critical threshold of q required to satisfy $\mathbf{Attn}_L \geq 1 - \xi$ for any $0 < \xi \leq 1$ is given by:

$$q \geq \log \frac{(1-\xi)(L-1)}{\xi}.$$

Proof. Given $\mathcal{J}_L \geq 1 - \xi$, by Lemma 6.B.13, we have

$$\Delta_L \geq (1 - \xi)^{1/L} = 1 - \frac{-\log(1 - \xi)}{L}.$$

Then, by Lemma 6.B.7, we can derive that

$$\mathbf{Attn}_L \geq \frac{1}{C_B} + \frac{\log L - \log(-\log(1 - \xi))}{C_B \log d}.$$

Hence, applying Lemma 6.B.24, we have

$$q \geq \log(L-1) + \log\left(\frac{\mathbf{Attn}_L}{1 - \mathbf{Attn}_L}\right) \geq \log \frac{L-1}{C_B-1} + f\left(\frac{\log L - \log \log \frac{1}{1-\xi}}{\log d}\right).$$

Here, we then use the first-order Taylor expansion for $f(x)$ to get the second inequality. $\square$

Notice that in the large difficulty ratio regime, the changes in $\log L$ between two consecutive horizons are $\Omega(\log R) \gg 1$, which is much larger than the $\frac{\log L}{\log d} \leq O(1)$ term. Therefore, the above lemma implies that the change in q between two consecutive horizons is dominated by $\Omega(\log R)$.

Warm-up Stage for L_1 We define the warm-up stage as the period during which the starting horizon L_1 reaches the mastery state, namely $0 \leq t < T_{\text{mas},1}$. At initialization, the attention scores are essentially uniform across horizons. We will show that, during this stage, the only non-negligible gradient contribution comes from the effectively short horizon L_1 .

We first record several basic properties of q , r , and the attention scores throughout the warm-up stage.

Lemma 6.B.26. *If Induction 6.B.3 holds, then for all iterations $0 \leq t < T_{\text{mas},1}$:*

- (a) $0 \leq q^{(t)} \leq O\left(\frac{L_1}{\log d}\right)$, and $q^{(t)}$ is monotonically increasing in t .
- (b) $|r^{(t)}| \leq O\left(\frac{1}{d_p}\right) q^{(t)}$.

Proof. The range of $q^{(t)}$ is a direct consequence of Lemma 6.B.25. The monotonicity of $q^{(t)}$ and the bound on $r^{(t)}$ follow directly from Induction 6.B.3. $\square$

Lemma 6.B.27. *If Induction 6.B.3 holds, then for all iterations $0 \leq t < T_{\text{mas},1}$ and for any horizon L_i with $i \geq 2$, we have*

$$\mathbf{Attn}_{\text{ans}, \ell-1 \rightarrow k}^{(t)} \leq O\left(\frac{1}{L_i}\right) = o(1), \quad \forall \ell \in [L_i], k \in [\ell].$$

Proof. This follows directly from Lemma 6.B.24. Moreover, since $R = \omega(1)$ in this regime, we have $L_i \geq \omega(1)$ for all $i \geq 2$, so the bound is indeed $o(1)$ as d grows. $\square$

Combining the above with the same reasoning as in Proposition 6.B.4, we can verify that the condition Eq. (6.B.28) in Proposition 6.B.3 holds for all longer horizons during warm-up, which yields the following.

Lemma 6.B.28. *If Induction 6.B.3 holds, then for all iterations $0 \leq t < T_{\text{mas},1}$ and for any horizon L_i with $i \geq 2$, we have*

$$|\nabla_q \tilde{\mathcal{J}}_{L_i}^{(t)}| \leq \tilde{O}\left(\frac{1}{d_p}\right) \cdot d^{-\Omega(L_i)}, \quad |\nabla_r \tilde{\mathcal{J}}_{L_i}^{(t)}| \leq \tilde{O}\left(\frac{1}{d_p^2}\right) \cdot d^{-\Omega(L_i)}.$$

Compared with Lemmas 6.B.17 and 6.B.18 in Section 6.B.3, Lemma 6.B.28 shows that during warm-up, the gradients contributed by longer horizons L_i (for $i \geq 2$) are negligible relative to the shortest horizon $L_1 = C_B$. Therefore, we can apply the constant-length analysis from Section 6.B.3 to the warm-up stage for L_1 , which yields the following characterization at the end of warm-up.

Lemma 6.B.29. *Induction 6.B.3 holds through $0 \leq t < T_{\text{mas},1}$ with*

$$T_{\text{mas},1} = O\left(\frac{KL_{\max}L_1}{\eta \log^2 d}\right),$$

and at time $T_{\text{mas},1}$ we have $q^{(T_{\text{mas},1})} \geq \Omega\left(\frac{\log L_1}{\log d}\right)$.

Transition Between Mastery States Since we have established that the initial horizon can reach the mastery state, we next analyze how mastery propagates across *consecutive* horizons. Specifically, we study the transition from horizon i to horizon $i + 1$ over the time interval $[T_{\text{mas},i}, T_{\text{mas},i+1})$.

Recall the definition $K = \lceil \log_R (L_{\max}/L_1) \rceil$. By construction, the horizons grow by a factor R up to index $K - 1$, while the last step may be truncated so that $L_K = L_{\max}$; consequently, L_K/L_{K-1} is not necessarily equal to R . For notational convenience, we therefore restrict attention to $i \in \{1, \dots, K - 2\}$, and fix an arbitrary $i^* \in \{1, \dots, K - 2\}$ for the remainder of the analysis. Moreover, we absorb the gradient term $\nabla_q \mathcal{J}_{L_K}$ into $\nabla_q \mathcal{J}_{L_{K-1}}$, since for all times prior to $T_{\text{mas},K-1}$, $\nabla_q \mathcal{J}_{L_K}$ can be upper bounded by $\nabla_q \mathcal{J}_{L_{K-1}}$.

By the critical threshold of q in Lemma 6.B.25, we have the following characterization of the attention scores:

Lemma 6.B.30. *If Induction 6.B.3 holds, then for all iterations $T_{\text{mas},i^*} \leq t < T_{\text{mas},i^*+1}$:*

(a) *if $i^* > 1$, then for any $i < i^*$, we have*

$$\text{Attn}_{L_i}^{(t)} \geq 1 - O\left(\frac{1}{R^{i^*-i}}\right) \cdot (1 - \text{Attn}_{L_{i^*}}^{(t)}) = 1 - o(1).$$

(b) for $i = i^*$, we have

$$\frac{1}{C_B} + \Omega\left(\frac{\log L_i}{\log d}\right) < \mathbf{Attn}_{L_i}^{(t)} \leq 1 - \Omega\left(\frac{1}{R}\right).$$

(c) if $i^* < K - 2$, then for any $i > i^* + 1$, we have

$$\mathbf{Attn}_{L_i}^{(t)} \leq O\left(\frac{1}{L_i}\right).$$

This immediately implies the following characterization of the logits:

Lemma 6.B.31. *If Induction 6.B.3 holds, then for all iterations $T_{\text{mas}, i^*} \leq t < T_{\text{mas}, i^*+1}$:*

(a) if $i^* > 1$, then for any $i < i^*$, we have $(p_{L_i,1}^{(t)})^{L_i} \geq \Omega(\mathcal{J}_{L_i^*}^{(t)}) = \Omega(1)$, and also

$$\Omega\left(\frac{1}{d^{C_B-1}}\right) \leq 1 - p_{L_i,1}^{(t)} \leq O\left(\frac{1}{d^{(1-e^{-q^{(t)}})R-(i^*-i)L_i^*}C_B-1}}\right).$$

(b) for $i = i^*$, we have

$$1 - p_{L_i,1}^{(t)} \geq \Omega\left(\frac{1}{d^{(1-\Theta(e^{-q^{(t)}})L_i))C_B-1}}\right).$$

(c) if $i^* < K - 2$, then for any $i > i^* + 1$, we have

$$p_{L_i,1}^{(t)} \leq O\left(\frac{1}{d}\right).$$

The logit conditions imply that for any $i < i^*$, we can invoke the gradient characterization in Proposition 6.B.2, and for any $i > i^* + 1$, we can invoke the gradient characterization in Proposition 6.B.3. Therefore, we have the following characterization of the gradient:

Lemma 6.B.32. *If Induction 6.B.3 holds, then for all iterations $T_{\text{mas}, i^*} \leq t < T_{\text{mas}, i^*+1}$,*

(a) if $i^* > 1$, then for any $i < i^*$, we have

$$\Omega\left(\frac{1}{d^{C_B-1}}\right) \cdot \frac{\log d}{d_p} \leq \nabla_q \tilde{\mathcal{J}}_{L_i}^{(t)} \leq O\left(\frac{1}{d^{(1-\frac{L_i^*}{e^{q^{(t)}}R}i^*-i)C_B-1}}\right) \cdot \frac{\log d}{d_p}$$

(b) for $i = i^*$, we have

$$\nabla_q \tilde{\mathcal{J}}_{L_i}^{(t)} = \Omega\left(\frac{1}{d^{(1-\Theta(e^{-q^{(t)}})L_i))C_B-1}}\right) \cdot \frac{\log d}{d_p}$$

(c) if $i^* < K - 2$, then for any $i > i^* + 1$, we have

$$|\nabla_q \tilde{\mathcal{J}}_{L_i}^{(t)}| \leq \tilde{O}\left(\frac{1}{d_p}\right) \cdot d^{-\Omega(L_i)}.$$

Lemma 6.B.32 immediately implies a gradient lower bound for $\mathcal{J}_{\text{mix},R}$ during $[T_{\text{mas},i^*}, T_{\text{mas},i^*+1})$:
Lemma 6.B.33. *If Induction 6.B.3 holds, then for all iterations $T_{\text{mas},i^*} \leq t < T_{\text{mas},i^*+1}$, we have*

$$\nabla_q \tilde{\mathcal{J}}_{\text{mix},R}^{(t)} \geq \frac{\log d}{Kd_p} \Omega\left(\frac{i^*}{d^{C_B-1}}\right).$$

Proof. By Lemma 6.B.32, when $R \leq o(\log d)$, we have

$$\nabla_q \tilde{\mathcal{J}}_{L_{i^*}}^{(t)} / \nabla_q \tilde{\mathcal{J}}_{L_i}^{(t)} \leq O(d^{1/R}) = O(e^{\log d/R}) \gg 1.$$

Thus $\nabla_q \tilde{\mathcal{J}}_{L_{i^*}}^{(t)}$ dominates the gradient of short horizons, which leads to the following lower bound:

$$\nabla_q \tilde{\mathcal{J}}_{\text{mix},R}^{(t)} \geq \frac{\log d}{Kd_p} \Omega\left(\frac{1}{d^{(1-\frac{1}{R})C_B-1}}\right).$$

On the other hand, we have

$$\nabla_q \tilde{\mathcal{J}}_{\text{mix},R}^{(t)} \geq \frac{1}{K} \sum_{i=1}^{i^*} \nabla_q \tilde{\mathcal{J}}_{L_i}^{(t)} \geq \Omega\left(\frac{i^* \log d}{Kd_p d^{C_B-1}}\right).$$

Further noting that $i^* \leq K-2 \leq O(\log d)$, thus when $R \leq o(\log d)$, we have $d^{\frac{C_B}{R}} \geq i^*$, which implies that in both cases, we have

$$\nabla_q \tilde{\mathcal{J}}_{\text{mix},R}^{(t)} \geq \frac{\log d}{Kd_p} \Omega\left(\frac{i^*}{d^{C_B-1}}\right).$$

□

So far, we have already controlled the gradient for the horizons before or after the current consecutive mastery state. In the following, we are going to examine $\nabla_q \tilde{\mathcal{J}}_{L_{i^*}}^{(t)} + \nabla_r \tilde{\mathcal{J}}_{L_{i^*+1}}^{(t)}$.

Lemma 6.B.34. *If Induction 6.B.3 holds, then for all iterations $T_{\text{vis},i^*+1} \leq t < T_{\text{mas},i^*+1}$, we have*

$$\nabla_q \tilde{\mathcal{J}}_{L_{i^*}}^{(t)} + \nabla_q \tilde{\mathcal{J}}_{L_{i^*+1}}^{(t)} \geq \Omega\left(\frac{\log d}{L_{i^*+1} d_p}\right)$$

Proof. By the critical threshold of q in Lemma 6.B.25, when $t \geq T_{\text{vis},i^*+1}$, we have $p_{L_{i^*+1},1}^{(t)} \geq 1 - O(\frac{1}{L_{i^*+1}})$. Hence, the conditions of Proposition 6.B.2 are satisfied, and invoking it, we then obtain

$$\nabla_q \tilde{\mathcal{J}}_{L_{i^*+1}}^{(t)} \geq \Omega\left(\frac{\log d}{d_p}\right)(1 - p_{L_{i^*+1},1}^{(t)}).$$

On the other hand, since $t \leq T_{\text{mas}, i^*+1}$, again by Lemma 6.B.25, we have $p_{L_{i^*+1}, 1}^{(t)} \leq 1 - \Omega(\frac{1}{L_{i^*}})$. Thus, we have

$$\nabla_q \tilde{\mathcal{J}}_{L_{i^*+1}}^{(t)} \geq \Omega\left(\frac{\log d}{L_{i^*+1} d_p}\right),$$

which completes the proof. $\square$

In the following, we are going to show that during the time period $[T_{\text{mas}, i^*}, T_{\text{vis}, i^*+1})$, there exists a major period during which the gradient is dominated by the current mastery state L_{i^*} .

Lemma 6.B.35. *If Induction 6.B.3 holds, then during $[T_{\text{mas}, i^*}, T_{\text{vis}, i^*+1})$, when*

$$q^{(t)} \in [\Omega(\log R^{0.01} L_{i^*}), O(\log R^{0.99} L_{i^*})] \quad (6.B.44)$$

we have

$$\nabla_q \tilde{\mathcal{J}}_{L_{i^*}}^{(t)} + \nabla_q \tilde{\mathcal{J}}_{L_{i^*+1}}^{(t)} = (1 + o(1)) \nabla_q \tilde{\mathcal{J}}_{L_{i^*}}^{(t)}.$$

Moreover,

$$\nabla_q \tilde{\mathcal{J}}_{L_{i^*}}^{(t)} = \Theta\left(\frac{1}{d^{(1-e^{-q^{(t)}} L_{i^*}) C_B - 1}}\right) \cdot \frac{\log d}{d_p}.$$

Proof. By the critical threshold of q in Lemma 6.B.25,

$$\Omega(\log L_{i^*}) \leq q^{(t)} \leq O(\log R L_{i^*}) = O(\log L_{i^*+1}).$$

So the condition Eq. (6.B.44) is well-defined. Furthermore, by Lemma 6.B.25, when Eq. (6.B.44) holds, we have $p_{L_{i^*+1}, 1}^{(t)} \leq O(\frac{1}{d})$. Hence applying Proposition 6.B.3, and we have

$$|\nabla_q \tilde{\mathcal{J}}_{L_{i^*+1}}^{(t)}| \leq \tilde{O}\left(\frac{1}{d_p}\right) \cdot d^{-\Omega(L_{i^*+1})}.$$

Furthermore, Eq. (6.B.44) combined with Lemma 6.B.25 implies that

$$1 - \text{Attn}_{L_{i^*}}^{(t)} = \Theta(e^{-q^{(t)} L_{i^*}}).$$

Hence,

$$1 - p_{L_{i^*+1}, 1}^{(t)} = \Theta\left(\frac{1}{d^{(1-e^{-q^{(t)}} L_{i^*}) C_B - 1}}\right).$$

Therefore, invoking Proposition 6.B.2, we complete the proof. $\square$

Putting everything together, we can then characterize the grokking-style behaviour happening during the transition period $[T_{\text{mas}, i^*}, T_{\text{mas}, i^*+1})$.

Lemma 6.B.36. *Induction 6.B.3 holds through $[T_{\text{mas},i^*}, T_{\text{mas},i^*+1})$, where*

$$T_{\text{mas},i^*+1} = T_{\text{mas},i^*} + O\left(\frac{d^{C_B-1} K d_p \log R}{\eta i^* \log d}\right).$$

(a) *the reward of $J_{L_{i^*+1}}$ saturates below 0.01 for a time period of*

$$\mathcal{T}_k \geq \Omega\left(\frac{d^{C_B-1} K d_p}{i^* \eta \log d}\right) \cdot \frac{\log R}{1 + C_B R^{-0.01} \log d}.$$

(b) $T_{\text{mas},i^*+1} - T_{\text{vis},i^*+1} \leq O\left(\frac{L_{i^*+1} d_p K}{\eta \log d}\right).$

Proof. The existence of $T_{\text{mas},i^*+1} = T_{\text{mas},i^*} + O\left(\frac{d^{C_B-1} K d_p \log R}{\eta i^* \log d}\right)$ is guaranteed by the gradient lower bound in Lemma 6.B.33. Moreover, the second item is guaranteed by the gradient lower bound in Lemma 6.B.34. Then we focus on the first statement. We approximate the total number of iterations $\mathcal{T}_{i^*}$ by the integral

$$\mathcal{T}_{i^*} \gtrsim \int_{\Omega(\log R^{0.01} L_{i^*})}^{O(\log R^{0.99} L_{i^*})} \frac{dq}{\eta \nabla_q \tilde{\mathcal{J}}_{\text{mix},R}}.$$

By Lemma 6.B.35, we can have a naive upper bound on the gradient:

$$\nabla_q \tilde{\mathcal{J}}_{\text{mix},R} \leq \frac{i^*}{K} \cdot \nabla_q \tilde{\mathcal{J}}_{L_{i^*}}^{(t)} \leq O\left(\frac{i^* \log d}{K d_p}\right) \cdot \frac{1}{d^{(1-e^{-q^{(t)}} L_{i^*}) C_B - 1}}.$$

Plugging this into the integral, we have

$$\begin{aligned} \mathcal{T}_{i^*} &\geq \Omega\left(\frac{d^{C_B-1} K d_p}{i^* \eta \log d}\right) \int_{\log R^{0.01}}^{\log R^{0.99}} d^{-C_B e^{-q}} dq = \Omega\left(\frac{d^{C_B-1} K d_p}{i^* \eta \log d}\right) \int_{R^{-0.99}}^{R^{-0.01}} \frac{e^{-(C_B \log d)u}}{u} du \\ &\geq \Omega\left(\frac{d^{C_B-1} K d_p}{i^* \eta \log d}\right) \cdot e^{-(C_B R^{-0.01} \log d)} \int_{R^{-0.99}}^{R^{-0.01}} \frac{1}{u} du \\ &\geq \Omega\left(\frac{d^{C_B-1} K d_p}{i^* \eta \log d}\right) \cdot e^{-(C_B R^{-0.01} \log d)} \cdot \log R \\ &\geq \Omega\left(\frac{d^{C_B-1} K d_p}{i^* \eta \log d}\right) \cdot \frac{\log R}{1 + C_B R^{-0.01} \log d}, \end{aligned}$$

where we use the Taylor expansion for the last inequality. □

Proof of Theorem 6.5.1 and Corollary 6.5.1

Proof. For any $k \in \{1, \dots, K-2\}$, Lemma 6.B.36 implies

$$\mathcal{T}_k \geq \Omega\left(\frac{d^{C_B-1} K d_p}{k \eta \log d}\right) \cdot \frac{\log R}{1 + C_B R^{-0.01} \log d}$$

$$\geq \tilde{\Omega}\left(\frac{d^{C_B-1} d_p}{\eta \log d}\right),$$

where the last inequality uses $\frac{\log R}{1+C_B R^{-0.01} \log d} = \Omega(1/\log d)$ and $K/k = \tilde{\Omega}(1)$.

Moreover,

$$T_{\text{mas},k+1} - T_{\text{vis},k+1} \leq O\left(\frac{L_{k+1} d_p K}{\eta \log d}\right) \leq O\left(\frac{d_p^2 K}{\eta \log d}\right) \ll \mathcal{T}_k,$$

since $d_p = d^{c_x}$ and $c_x < C_B - 1$. Therefore,

$$T_{\text{mas},k+1} - T_{\text{mas},k} \geq \tilde{\Omega}\left(\frac{d^{C_B-1} d_p}{\eta \log d}\right).$$

On the other hand, Lemma 6.B.36 also gives the matching upper bound

$$T_{\text{mas},k+1} - T_{\text{mas},k} \leq \tilde{O}\left(\frac{d^{C_B-1} d_p}{\eta \log d}\right).$$

Summing over $k \in \{1, \dots, K-2\}$, we obtain

$$T_{\text{mas},K-1} - T_{\text{mas},1} = \tilde{\Theta}\left(\frac{d^{C_B-1} d_p}{\eta \log d}\right).$$

Finally, by Lemma 6.B.29, the time spent in the warm-up stage is negligible compared to $T_{\text{mas},K-1} - T_{\text{mas},1}$. Moreover, the final step can be bounded as

$$T_{\text{mas},K} - T_{\text{mas},K-1} \leq O(T_{\text{mas},K-1} - T_{\text{mas},K-2}).$$

This completes the proof. $\square$

Analysis of Moderate Difficulty Ratio Regime

In this subsection, we analyze the moderate difficulty ratio regime, where $2 \leq R = O(1)$. Our overall proof strategy follows that of the large difficulty ratio regime, with several adjustments to account for the smaller gap between two consecutive scales.

We begin by stating the induction hypothesis, which we expect to remain valid throughout training.

Induction 6.B.4. Given $\Omega\left(\frac{1}{\text{poly} \log d}\right) < \epsilon < \frac{1}{4}$, and let T^* be the first iteration such that $\text{Attn}_{L_{\max}}^{(t)} \geq 1 - \epsilon$. Then, for all iterations $t < T^*$, the following hold:

- (a) $0 \leq q^{(t)} \leq O\left(\log \frac{L_{\max}}{\epsilon}\right)$, and $q^{(t)}$ monotonically increases.
- (b) $|r^{(t)}| \leq O(1/d_p)q^{(t)}$.

Properties of the Attention Scores and Critical Thresholds We record several basic properties of the attention scores and the critical thresholds.

Lemma 6.B.37. *If Induction 6.B.4 holds for all iterations $< t$, then:*

(a)

$$\mathbf{Attn}_L^{(t)} = \frac{e^{q^{(t)} - r^{(t)}}}{e^{q^{(t)} - r^{(t)}} + (L - 1)} \geq \frac{1}{L};$$

(b) *for any $k \neq \ell$,*

$$\mathbf{Attn}_{a, \ell-1 \rightarrow p, k}^{(t)} = \frac{1}{(L - 1) + e^{q^{(t)} - r^{(t)}}} = \frac{1}{L - 1} (1 - \mathbf{Attn}_L^{(t)}).$$

Lemma 6.B.38 (Critical threshold of q). *If Induction 6.B.3 holds, then for any $L \in \mathcal{L}_R$, a sufficient threshold on q for $\mathcal{J}_L \geq 1 - \xi$, where $0 < \xi \leq 1$ is a constant, is*

$$q \geq \log \frac{L - 1}{C_B - 1} + f\left(\frac{\log L - \log \log \frac{1}{1 - \xi}}{\log d}\right),$$

where $f(x) = \log\left(\frac{1+x}{1-x/(C_B-1)}\right)$. Similarly, a sufficient threshold on q for $\mathbf{Attn}_L \geq 1 - \xi$ is

$$q \geq \log \frac{(1 - \xi)(L - 1)}{\xi}.$$

The above lemmas mirror their counterparts in the large-difficulty regime. However, to track the variation of the critical threshold when L increases only by a constant factor R , we need a more careful comparison than the coarse Taylor-expansion argument used for widely separated scales.

Lemma 6.B.39. *If Induction 6.B.4 holds and $0 < \xi \leq 1$ is a constant, let $q_\xi(L)$ denote the critical threshold of q required to ensure $\mathcal{J}_{L,1} \geq 1 - \xi$. Then for any $L_k, L_{k+1} \in \mathcal{L}_R$,*

$$q_\xi(L_{k+1}) - q_\xi(L_k) = \log R \cdot \left(1 + O(1/\log d)\right).$$

Proof. By Lemma 6.B.38, we have

$$\frac{dq_\xi(L)}{dL} = \frac{1}{L} \left[1 + \frac{1}{\log d} \cdot \frac{C_B}{(1 + X(L))(C_B - 1 - X(L))} \right],$$

where $X(L) = \frac{\log L - \log \log \frac{1}{1 - \xi}}{\log d}$. Since $0 \leq X(L) \leq 1 + O(1/\log d)$, it follows that $\frac{dq_\xi(L)}{dL} = \frac{1}{L} (1 + O(1/\log d))$. Therefore,

$$\begin{aligned} q_\xi(L_{k+1}) - q_\xi(L_k) &= \int_{L_k}^{L_{k+1}} \frac{1}{L} (1 + O(1/\log d)) dL \\ &= \log R \cdot \left(1 + O(1/\log d)\right). \end{aligned}$$

□

Warm-up Stage for L_1 We define the warm-up stage as the period during which the starting horizon L_1 reaches the mastery state, namely $0 \leq t < T_{\text{mas},1}$. The analysis is similar to the large difficulty ratio regime, but with some modifications since L_i with $i \geq 2$ could be relatively small and still at the constant-length regime.

Lemma 6.B.40. *If Induction 6.B.4 holds, then for all iterations $0 \leq t < T_{\text{mas},1}$:*

$$(a) \ 0 \leq q^{(t)} \leq O\left(\frac{L_1}{\log d}\right), \text{ and } q^{(t)} \text{ is monotonically increasing in } t.$$

$$(b) \ |r^{(t)}| \leq O\left(\frac{1}{d_p}\right) q^{(t)}.$$

Lemma 6.B.41. *If Induction 6.B.4 holds, then for all iterations $0 \leq t < T_{\text{mas},1}$, and for any $L_i \in \mathcal{L}_R$ with $i \geq 2$, we have*

$$(a) \ \text{if } L_i = O(1), \text{ then } 1 - C_B \text{Attn}_{L_i}^{(t)} \geq 1 - \frac{1}{0.99R^{i-1}+1};$$

$$(b) \ \text{else, } \text{Attn}_{L_i}^{(t)} \leq O\left(\frac{1}{L_i}\right) = o(1).$$

Proof. The second item is similar to the large difficulty ratio regime. For the first item, by Lemma 6.B.40, we have

$$\text{Attn}_{L_i}^{(t)} \leq \frac{1}{(L_i - 1) \cdot e^{-O(L_1/\log d)} + 1} = \frac{1}{R^{i-1}C_B \cdot e^{-O(L_1/\log d)} + 1}.$$

Hence,

$$1 - C_B \text{Attn}_{L_i}^{(t)} \geq 1 - \frac{1}{0.99R^{i-1} + 1}.$$

□

Similarly as Lemma 6.B.28, the condition Eq. (6.B.28) in Proposition 6.B.3 holds for $L_i = \omega(1)$, which yields the following.

Lemma 6.B.42. *If Induction 6.B.4 holds, then for all iterations $0 \leq t < T_{\text{mas},1}$ and for any horizon $L_i = \omega(1)$, we have*

$$|\nabla_q \tilde{\mathcal{F}}_{L_i}^{(t)}| \leq \tilde{O}\left(\frac{1}{d_p}\right) \cdot d^{-\Omega(L_i)}, \quad |\nabla_r \tilde{\mathcal{F}}_{L_i}^{(t)}| \leq \tilde{O}\left(\frac{1}{d_p^2}\right) \cdot d^{-\Omega(L_i)}.$$

Lemma 6.B.41 shows that even some longer horizons L_i are still at the constant-length regime, its target attention scores are still below $\frac{1}{C_B}$, which means $p_{L_i,1}^{(t)} \leq d^{-\Omega(1)}$ is still close to 0. However, directly applying Proposition 6.B.3 to $L_i = O(1)$ only gives a bound of $d^{-\Omega(L_i)}$, which may be too loose in the constant-length regime. Thus we use a variant of Proposition 6.B.3 and precise characterization of $1 - C_B \text{Attn}_{L_i}^{(t)}$ to get a more precise bound.

Lemma 6.B.43. *If Induction 6.B.4 holds, then for all iterations $0 \leq t < T_{\text{mas},1}$ and for any horizon $L_i = O(1)$ with $i \geq 2$, we have*

$$|\nabla_q \tilde{\mathcal{F}}_{L_i}^{(t)}| \leq \tilde{O}\left(\frac{1}{d_p}\right) \cdot d^{-L_i(1 - \frac{1}{0.99R^{i-1}+1})}, \quad |\nabla_r \tilde{\mathcal{F}}_{L_i}^{(t)}| \leq \tilde{O}\left(\frac{1}{d_p^2}\right) \cdot d^{-L_i(1 - \frac{1}{0.99R^{i-1}+1})}.$$

Since $L_i \geq R^{i-1}C_B$, we have $L_i(1 - \frac{1}{0.99R^{i-1}+1}) \geq C_B + \Omega(1)$. Compared with Lemma 6.B.17 in Section 6.B.3, Lemma 6.B.42 and Lemma 6.B.43 show that during warm-up the gradients contributed by longer horizons L_i (for $i \geq 2$) are negligible relative to the shortest horizon $L_1 = C_B$. Therefore, we can apply the constant-length analysis from Section 6.B.3 to the warm-up stage for L_1 , which yields the following characterization at the end of warm-up.

Lemma 6.B.44. *Induction 6.B.4 holds through $0 \leq t < T_{\text{mas},1}$ with*

$$T_{\text{mas},1} = O\left(\frac{KL_{\max}L_1}{\eta \log^2 d}\right),$$

and at time $T_{\text{mas},1}$ we have $q^{(T_{\text{mas},1})} \geq \Omega\left(\frac{\log L_1}{\log d}\right)$.

Transition between mastery states In this part, we analyze the transition of the mastery state across consecutive horizons. Concretely, we focus on the time interval $[T_{\text{mas},i}, T_{\text{mas},i+1})$ for $i \in \{1, \dots, K-2\}$. As before, we fix an arbitrary $i^* \in \{1, \dots, K-2\}$ for the remainder of the analysis (The restriction $i \leq K-2$ excludes the final truncated step where L_K/L_{K-1} is not necessarily R .)

By the critical threshold of q in Lemma 6.B.38, we have the following characterization of the attention scores:

Lemma 6.B.45. *If Induction 6.B.4 holds, then for all iterations $T_{\text{mas},i^*} \leq t < T_{\text{mas},i^*+1}$:*

(a) *if $i^* > 1$, then for any $i < i^*$, we have*

$$\mathbf{Attn}_{L_i}^{(t)} - \mathbf{Attn}_{L_{i^*}}^{(t)} \geq \Omega(1).$$

(b) *for $i = i^*$, we have*

$$\frac{1}{C_B} + \Omega\left(\frac{\log L_i}{\log d}\right) < \mathbf{Attn}_{L_i}^{(t)} \leq 1 - \Omega(1).$$

(c) *if $i = i^* + 1$, we have*

$$\frac{1}{RC_B} \leq \mathbf{Attn}_{L_i}^{(t)} \leq \frac{1}{C_B} + O\left(\frac{\log L_i}{\log d}\right).$$

(d) *if $i^* < K-2$, then for any $i > i^* + 1$, we have*

- *if $L_i = O(1)$, then $1 - C_B \mathbf{Attn}_{L_i}^{(t)} \geq 1 - \frac{1}{0.99R^{i-i^*-1}+1}$;*
- *else, $\mathbf{Attn}_{L_i}^{(t)} \leq O\left(\frac{1}{L_i}\right) = o(1)$.*

This immediately implies the following characterization of the logits:

Lemma 6.B.46. *If Induction 6.B.4 holds, then for all iterations $T_{\text{mas},i^*} \leq t < T_{\text{mas},i^*+1}$:*

(a) if $i^\star > 1$, then for any $i < i^\star$, we have $(p_{L_i,1}^{(t)})^{L_i} \geq \Omega(\mathcal{J}_{L_{i^\star}}^{(t)}) = \Omega(1)$, and also

$$1 - p_{L_i,1}^{(t)} \leq d^{-\Omega(1)} (1 - p_{L_{i^\star},1}^{(t)}).$$

(b) for $i = i^\star$, we have

$$1 - p_{L_i,1}^{(t)} = \Theta\left(d^{-\left(1 - \frac{1}{e^{q(t)}/(L_i-1)+1}\right)C_B+1}\right) \geq \Omega(d^{-(1-\Omega(1))C_B+1}).$$

(c) if $i = i^\star + 1$, we have

$$p_{L_i,1}^{(t)}/p_{L_i,2}^{(t)} \geq d^{\Omega(1)}.$$

(d) if $i^\star < K - 2$, then for any $i > i^\star + 1$, we have

- if $L_i = O(1)$, then $p_{L_i,1}^{(t)} \leq O\left(d^{-\left(1 - \frac{1}{0.99R^{i-i^\star-1}+1}\right)}\right)$;
- else, $p_{L_i,1}^{(t)} \leq O(1/d)$.

The logit conditions also guarantee that for any $i \leq i^\star + 1$, we can invoke the gradient characterization in Proposition 6.B.2, and for any $i > i^\star + 1$, we can invoke the gradient characterization in Proposition 6.B.3 and the variant in Lemma 6.B.43. Therefore, we have the following characterization of the gradient:

Lemma 6.B.47. *If Induction 6.B.3 holds, then for all iterations $T_{\text{mas},i^\star} \leq t < T_{\text{mas},i^\star+1}$,*

(a) if $i^\star > 1$, then for any $i < i^\star$, we have

$$\frac{\log d}{d_p} \cdot \frac{1}{d^{C_B-1}} \leq \nabla_q \tilde{\mathcal{J}}_{L_i}^{(t)} \leq d^{-\Omega(1)} \nabla_q \tilde{\mathcal{J}}_{L_{i^\star}}^{(t)}.$$

(b) for $i = i^\star$, we have

$$\nabla_q \tilde{\mathcal{J}}_{L_{i^\star}}^{(t)} = \Theta(1 - p_{L_{i^\star},1}^{(t)}) \cdot \frac{\log d}{d_p}.$$

(c) if $i = i^\star + 1$, we have

$$\nabla_q \tilde{\mathcal{J}}_{L_{i^\star+1}}^{(t)} = \Theta\left(\left(1 - p_{L_{i^\star+1},1}^{(t)}\right)\left(p_{L_{i^\star+1},1}^{(t)}\right)^{L_{i^\star+1}}\right) \cdot \frac{\log d}{d_p}.$$

(d) if $i^\star < K - 2$, then for any $i > i^\star + 1$, we have

$$|\nabla_q \tilde{\mathcal{J}}_{L_i}^{(t)}| \leq d^{-\Omega(1)} \nabla_q \tilde{\mathcal{J}}_{L_{i^\star}}^{(t)}.$$

Thus, to control the gradient of $\mathcal{J}_{\text{mix},R}$ during $[T_{\text{mas},i^*}, T_{\text{mas},i^*+1})$, we only need to focus on the gradient of L_{i^*} and L_{i^*+1} . Similar to Lemma 6.B.34, we have the following lower bound for the period that the reward of L_{i^*+1} becomes visible.

Lemma 6.B.48. *If Induction 6.B.4 holds, then for all iterations $T_{\text{vis},i^*+1} \leq t < T_{\text{mas},i^*+1}$, we have*

$$\nabla_q \tilde{\mathcal{J}}_{L_{i^*}}^{(t)} + \nabla_q \tilde{\mathcal{J}}_{L_{i^*+1}}^{(t)} \geq \Omega\left(\frac{\log d}{L_{i^*+1} d_p}\right).$$

Now we turn to the period $[T_{\text{mas},i^*}, T_{\text{vis},i^*+1})$. The main difference from the large difficulty ratio regime is that in this stage, L_{i^*} and L_{i^*+1} will jointly decide a gradient lower bound for $\mathcal{J}_{\text{mix},R}$, which is significantly larger than the one in the long-plateau stage in the large difficulty ratio regime.

Lemma 6.B.49. *If Induction 6.B.4 holds, then during $[T_{\text{mas},i^*}, T_{\text{vis},i^*+1})$, we have*

$$\nabla_q \tilde{\mathcal{J}}_{L_{i^*}}^{(t)} + \nabla_q \tilde{\mathcal{J}}_{L_{i^*+1}}^{(t)} \geq \Omega\left(d^{-\frac{RC_B}{R+C_B}+1}\right) \cdot \frac{\log d}{d_p}.$$

Proof. Notice that during $[T_{\text{mas},i^*}, T_{\text{vis},i^*+1})$, by Lemma 6.B.47, $\nabla_q \tilde{\mathcal{J}}_{L_{i^*}}^{(t)}$ is dominated by the term $1 - p_{L_{i^*},1}^{(t)}$. On the other hand, for L_{i^*+1} , by Lemma 6.B.45 and Lemma 6.B.47, we have $\nabla_q \tilde{\mathcal{J}}_{L_{i^*+1}}^{(t)}$ firstly dominated by the term $(p_{L_{i^*+1},1}^{(t)})^{L_{i^*+1}}$, which will increase as q increases, and then by the term $(1 - p_{L_{i^*+1},1}^{(t)})$ when $(p_{L_{i^*+1},1}^{(t)})^{L_{i^*+1}}$ reaches the constant level, and $(1 - p_{L_{i^*+1},1}^{(t)})$ is lower bounded by $\Omega(1/L_{i^*+1})$. Therefore, to lower bound the gradient summation, we only need to find the time t when $(p_{L_{i^*+1},1}^{(t)})^{L_{i^*+1}}$ reaches the same level as $1 - p_{L_{i^*},1}^{(t)}$. Thus, consider

$$C_B \cdot \text{Attn}_{L_{i^*}} - 1 = L_{i^*+1}(1 - C_B \cdot \text{Attn}_{L_{i^*+1}}),$$

which can be rewritten as

$$C_B \frac{e^q}{e^q + L_{i^*} - 1} - 1 = L_{i^*+1} \left(1 - C_B \frac{e^q}{e^q + L_{i^*+1} - 1}\right) \quad (6.B.45)$$

which is a quadratic equation in e^q . Denoting $W_L(x) = C_B \frac{x}{x+L-1}$, then solving Eq. (6.B.45) is equivalent to finding the solution x^* of $W_{L_{i^*}}(x^*) - 1 = L_{i^*+1}(1 - W_{L_{i^*+1}}(x^*))$. Consider the point $x_0 = \frac{L_{i^*+1}-1}{C_B-1}$. Note that $1 - W_{L_{i^*+1}}(x_0) = 0$ and $W_L(x)$ is monotonically increasing. Thus $x^* < x_0$. Hence,

$$W_{L_{i^*}}(x^*) - 1 \leq W_{L_{i^*}}(x_0) - 1 = \frac{C_B \frac{L_{i^*+1}-1}{L_{i^*}-1}}{\frac{L_{i^*+1}-1}{L_{i^*}-1} + (C_B - 1)} - 1.$$

Hence,

$$\nabla_q \tilde{\mathcal{J}}_{L_{i^*}}^{(t)} + \nabla_q \tilde{\mathcal{J}}_{L_{i^*+1}}^{(t)} \geq \Omega\left(d^{W_{L_{i^*}}(x^*)-1}\right) \cdot \frac{\log d}{d_p} \geq \left(d^{-C_B(1-\frac{C_B}{R+C_B})-1}\right) \cdot \frac{\log d}{d_p}.$$

□

Putting everything together, we can then characterize the relay behaviour happening during the transition period $[T_{\text{mas},i^*}, T_{\text{mas},i^*+1})$.

Lemma 6.B.50. *Induction 6.B.3 holds through $[T_{\text{mas},i^*}, T_{\text{mas},i^*+1})$, where*

$$T_{\text{mas},i^*+1} = T_{\text{mas},i^*} + O\left(\frac{d^{C_B-1} K d_p \log R}{\eta i^* \log d}\right).$$

(a) *the reward of $J_{L_{i^*+1}}$ only saturates below 0.01 for a time period of at most*

$$\mathcal{T}_k \leq O\left(\frac{d^{\frac{RC_B}{R+C_B}-1} K d_p}{\eta \log d}\right) \cdot \log R.$$

(b) $T_{\text{mas},i^*+1} - T_{\text{vis},i^*+1} \leq O\left(\frac{L_{i^*+1} d_p K}{\eta \log d}\right)$.

Proof. The proof is straightforward by Lemma 6.B.48 and Lemma 6.B.49 and the fact that $q^{(t)}$ changes $\Theta(\log R)$ during $[T_{\text{mas},i^*}, T_{\text{vis},i^*+1})$ and $O(1)$ during $[T_{\text{vis},i^*+1}, T_{\text{mas},i^*+1})$ due to Lemma 6.B.25. □

Proof of Theorem 6.5.2 and Corollary 6.5.2

Proof. Theorem 6.5.2 follows immediately from Lemmas 6.B.48 and 6.B.50 together with the bound $K = O(\log d)$.

For Corollary 6.5.2, we apply Lemma 6.B.50 iteratively for $K - 2$ transitions (from horizon 1 up to horizon $K - 1$):

$$\begin{aligned} T_{\text{mas},K-1} - T_{\text{mas},1} &\leq O\left(\frac{d^{\frac{RC_B}{R+C_B}-1} K d_p}{\eta \log d}\right) \cdot (K-2) \log R + O\left(\frac{d_p K}{\eta \log d}\right) \cdot L_1 \left(\frac{R^{K-1} - 1}{R - 1}\right) \\ &\leq \tilde{O}\left(\frac{d_p}{\eta}\right) \cdot d^{\frac{RC_B}{R+C_B}-1} + \tilde{O}\left(\frac{d_p}{\eta}\right) \cdot L_{\max}, \end{aligned}$$

where the last inequality uses $K = O(\log d)$ and $R^{K-1} = O(L_{\max})$. Combining this with the condition $L_{\max} = O(d^{c_x})$ and $c_x < \frac{2C_B}{2+C_B} \leq \frac{RC_B}{R+C_B}$, we obtain

$$T_{\text{mas},K-1} - T_{\text{mas},1} \leq \tilde{O}\left(\frac{d_p}{\eta}\right) \cdot d^{\frac{RC_B}{R+C_B}-1}.$$

Finally, by Lemma 6.B.44, the time spent in the warm-up stage is negligible compared to $T_{\text{mas},K-1} - T_{\text{mas},1}$. Moreover, we can bound the final step by $T_{\text{mas},K} - T_{\text{mas},K-1} \leq O(T_{\text{mas},K-1} - T_{\text{mas},K-2})$. This completes the proof. □

Part III

Modern Recurrent Architectures

Chapter 7

Hierarchical Learning in Loop Transformers

7.1 Introduction

Many algorithms perform additional computation by repeatedly applying the same update rule. Recurrent depth implements this idea in neural networks: a shared block updates the hidden state over multiple computational steps, so the amount of computation can increase without introducing a separate set of parameters at every depth. Universal Transformers introduced this form of depth-wise recurrence into self-attention architectures [71], providing a natural model for iterative computation with shared parameters.

Since then, looped and recursive transformers have been studied in a range of settings. They can be explicitly programmed to implement general algorithms [100], trained to perform iterative in-context learning [326], and used to improve length extrapolation or latent reasoning by increasing the number of recurrent steps [84, 246]. Recent language models also use recurrent depth for parameter efficiency and test-time computation [23, 98, 351]. These results show that weight-shared transformers can represent and execute useful multistep computations.

A separate question is whether these computations can be learned through gradient-based optimization. Existing constructions primarily establish expressivity by specifying parameters that implement a desired algorithm. Recent optimization analyses have obtained learning guarantees in more restricted settings. For example, looped linear transformers can learn iterative algorithms related to preconditioned gradient descent [95], and gradient descent can select the power method in a layer-normalized looped linear transformer [316]. However, these results do not address how a loop transformer learns a nonlinear hierarchical computation whose intermediate states are not directly supervised.

This chapter is based on unpublished work¹.

¹ Most of the proof in this chapter is written by GPT-5.6.

The main difficulty is credit assignment across recurrent depth. In an end-to-end task, the loss is evaluated only at the final output, while the intermediate states used to produce that output remain hidden. The gradient must therefore determine both which intermediate computations should be learned and how they should be coordinated across recurrent steps. Moreover, because the same parameters are used at every depth, a single gradient update changes all recurrent levels simultaneously. The resulting gradient aggregates contributions from the full unrolled computation, and it is not immediate that these contributions reinforce the routing decisions required by the target hierarchy.

This motivates the following research question:

Research Question

Can gradient-based training learn a hierarchical computation in a loop transformer using only final-output supervision? If so, how do the shared-parameter gradients behave as recurrent depth increases?

To study this question in a theoretically tractable setting, we consider a sequential-composition task over a finite group. The target admits a natural hierarchical computation in which the same pairwise operation is repeatedly applied to neighboring partial results. The model uses one shared module across all recurrent levels, and the training objective observes only the final outputs. We derive the population gradient-flow dynamics of the shared routing matrix and track the resulting trajectory. Overall, our results provide an optimization-based account of how hierarchical computation can be learned under recurrent weight sharing.

7.1.1 Our Contributions

More concretely, our main contributions are summarized as follows.

1. *Provable learning of hierarchical computation from final-output supervision.* Given a frozen module that supplies group composition, we show that population gradient flow on a separated length-three base distribution learns the equality-routing rule using only the final answer. Although the intermediate computations are never labeled, training discovers a local routing rule that can be reused across recurrent levels.
2. *A characterization of computation and gradient propagation across recurrent depth.* We give an explicit depth-dependent margin sufficient for accurate repeated use of the learned routing rule. We further analyze the full end-to-end gradient through the unrolled network and show that the useful learning signal remains constructive despite the additional interactions introduced by weight sharing.
3. *Backward feature correction and progressive learning across depth.* We establish a shallow-to-deep procedure that carries one parameter matrix across tasks and trains each active depth until its attention-error envelope reaches its $\varepsilon_{\text{risk}}$ -calibrated forward budget. While a deeper task is active, an error-refinement form of *backward feature correction* (BFC) [9] holds across depths:

reevaluating completed shallower tasks under the updated shared matrix yields strictly tighter valid positional- and state-error bounds.

Collectively, our results provide an optimization-based explanation for how hierarchical computation can emerge in a loop transformer. They show that recurrent weight sharing does more than make additional computation possible: under end-to-end training, it allows a local rule learned on shallow instances to be reused while each newly active objective decreases its own risk and lowers its attention-error envelope.

7.2 Related Work

Recurrent depth and loop-transformer architectures. Adaptive Computation Time and PonderNet learn how long a recurrent model should compute [26, 104]. Universal Transformers combine depth-wise parameter sharing with self-attention, while Deep Equilibrium Models differentiate through a fixed point of an effectively infinite-depth weight-tied network [24, 71]. A complementary line uses a finite number of repeated transformer blocks. Looped transformers have been programmed as general computers [100], trained to implement learning algorithms [326], adapted to input length by changing the number of loops [84], and interpreted as latent reasoning [246]. Recursive language models further use depth recurrence for parameter efficiency and test-time compute [23, 98, 351]. Each training phase uses a fixed finite recurrent depth, and the end-task loss trains the computation shared across the corresponding unrolled levels. Thus, the relevant comparison is not recurrence alone, but recurrence whose shared computation is learned through the final task loss.

Learnability of iterative computation. The closest optimization analyses recover linear iterative algorithms from end-task supervision. Gatmiry et al. [95] characterize looped linear transformers whose population-risk minimizers implement multi-step preconditioned gradient descent. Wu et al. [316] show that gradient descent selects the power method in a layer-normalized looped linear transformer. Work on attention selection establishes max-margin or sparse-retrieval behavior in single-layer settings [267, 300]. In the recurrent model, each selection changes the state consumed by later uses of the shared block, so the proof must retain the nonlinear recurrent dependencies rather than analyze an isolated attention update. This distinction is what makes the sign of the full shared-parameter gradient the central technical question for the analysis.

Hierarchical feature learning and BFC. Hierarchical targets can guide feature learning through Fourier staircases, intermediate nonlinear features, effective-dimension reduction, and spectral recovery [1, 67, 89, 201, 265, 299]. Recent analyses extend this picture to shallow-to-deep chaining and separations between constant and logarithmic depth [68, 233, 234]. These results complement classical studies of coupled training in deep linear or homogeneous networks [76, 247] and rich feature evolution in multilayer limits [34, 57, 324]. Allen-Zhu and Li [9] identify BFC as a mechanism through which higher-level training repairs lower-level features. Recurrence realizes

this coupling directly: the parameter receiving the deeper signal is literally the parameter that produced the earlier representations. The guarantee is a strict improvement of deterministic error envelopes, not pointwise monotonic improvement of each realized state. The recurrent setting therefore gives BFC a concrete temporal meaning: deeper supervision tightens error bounds when completed shallower tasks are reevaluated under the updated shared computation.

Provable compositional reasoning. Compositional reasoning repeatedly combines simple operations rather than memorizing each input–output map. Sequential group composition makes this distinction precise by separating lookup-table memorization from repeated computation that exploits associativity [187]. Curricula can develop such computations through easy-to-hard ordering, intermediate subtask labels, mixed-difficulty rewards, or depth-increasing schedules [32, 37, 65, 131, 301, 307, 313]; intermediate outputs can likewise aid autoregressive compositional learning [230]. In contrast, only task depth changes, intermediate products remain hidden, and the same shared matrix is trained from final labels. This isolates a curriculum over computational depth without adding direct supervision for the intermediate hierarchy.

Taken together, the literature explains why looped computation is expressive and why hierarchical targets can shape features, but it leaves open whether the exact end-to-end gradient of a nonlinear, weight-shared computation remains constructive. Section 7.3 specifies the model used to address that question.

7.3 Problem Setup

7.3.1 Hierarchical Composition Task

Let $(G, \circ)$ be a finite group of order d . For a sequence $m_{1:n} = (m_1, \dots, m_n) \in G^n$, define

$$\text{eval}(m_{1:n}) = m_n \circ m_{n-1} \circ \dots \circ m_1, \quad y^\star = \text{eval}(m_{1:n}). \quad (7.3.1)$$

Fix a maximum dyadic depth $L_{\max} \geq 2$, set $N = 2^{L_{\max}}$, and choose an orthonormal positional dictionary $\mathcal{X} = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(p)}\} \subset \mathbb{R}^p$ with $p \geq N + 1$. For $n \leq N$, an input is $X = (m_{1:n}, \iota)$, where the elements are independent and uniform and $\iota : \{0, 1, \dots, n\} \rightarrow \{1, \dots, p\}$ is an independent, uniformly sampled injection. Write $\mathbf{x}_i = \mathbf{x}^{(\iota(i))}$; $\mathbf{x}_0$ marks the boundary before the first token. Relabeling changes the positional names but preserves equality between positional features. Throughout, positional vectors and matrices are written in the orthonormal dictionary basis $\mathcal{X}$. In particular, $\mathbf{1}$ denotes the all-ones coefficient vector in that basis.

Associativity supplies the target dyadic computation tree. For $n = 2^L$, level zero contains the singleton blocks $[i, i]$. Level one composes adjacent singleton blocks, level two composes adjacent length-two blocks, and level L produces the root block $[1, n]$. A block ending at position i and formed at level ℓ represents

$$y_{i,\ell}^\star = m_i \circ m_{i-1} \circ \dots \circ m_{i-2^\ell+1}. \quad (7.3.2)$$

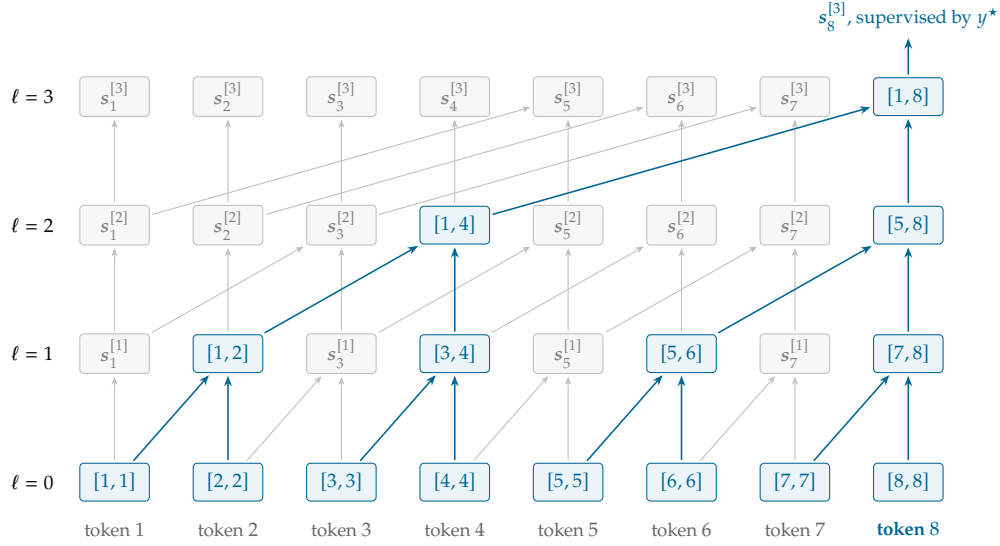


Figure 7.1: The Target Computation Tree Is Embedded in a Larger Looped Computation. Each row is one recurrent level and each column is one token position. The blue target tree feeds the supervised output at token 8; the gray nodes are simultaneous off-tree states generated by the same recurrent update, and their backward contributions must also be controlled. Only blue nodes are labeled by target block intervals. Gray retrieval edges are schematic rather than exhaustive: when a query has reached x_0 and no key matches it, the resulting softmax retrieval from all causal keys is not drawn as a fan of edges. A vertical edge denotes the current-state input, not a carry gate.

Its adjacent left block ends at $i - 2^\ell$. Thus every merge applies the same semantic rule at a different scale.

The length-three base task is intentionally separate from the dyadic family. It is the shortest sequence for which the final token must make two successive hidden identifications. Once that computation is learned, the schedule uses lengths $4, 8, \dots, 2^{L_{\max}}$.

Figure 7.1 previews how the target computation tree is embedded in the full looped computation. The blue subgraph is the target tree needed by the supervised root, while the gray computations are produced by the same shared block and therefore also contribute to its gradient.

7.3.2 Loop-Transformer Architecture

Token i stores three fields at recurrent level ℓ :

$$\mathbf{h}_i^{[\ell]} = (\mathbf{x}_{i,\text{qry}}^{[\ell]}, \mathbf{x}_i, \mathbf{s}_i^{[\ell]}). \quad (7.3.3)$$

The fixed key $\mathbf{x}_i$ marks the block's right endpoint. The dynamic query-side feature marks the position immediately to the left of the represented block. The categorical state $\mathbf{s}_i^{[\ell]} \in \Delta(G)$, where $\Delta(G)$ is the probability simplex over G , represents the block product. Let $\mathbf{e}_g \in \mathbb{R}^d$ be the one-hot vector indexed by $g \in G$. For $\xi \in [0, 1]$, call a state ξ -concentrated on g if its g -coordinate is at

least $1 - \xi$. Initialization is

$$\mathbf{h}_i^{[0]} = (\mathbf{x}_{i-1}, \mathbf{x}_i, \mathbf{e}_{m_i}). \quad (7.3.4)$$

Consequently, each singleton token begins with the correct group element and with a query-side feature pointing to its immediate predecessor.

The only trained parameter is a shared matrix $\mathbf{Q} \in \mathbb{R}^{p \times p}$. At each recurrent level, token $i \geq 2$ assigns strict-causal attention to $j < i$:

$$\alpha_{ij}^{[\ell]}(\mathbf{Q}) = \frac{\exp\left(\left\langle \mathbf{Q}\mathbf{x}_{i,\text{qry}}^{[\ell]}, \mathbf{x}_j \right\rangle\right)}{\sum_{k < i} \exp\left(\left\langle \mathbf{Q}\mathbf{x}_{i,\text{qry}}^{[\ell]}, \mathbf{x}_k \right\rangle\right)}. \quad (7.3.5)$$

Call $u = (i, \ell)$ a *query instance*, with candidate-key set $\mathcal{K}_u = \{j : j < i\}$. It consumes level ℓ and produces level $\ell + 1$; abbreviate $\alpha_{u,j} := \alpha_{ij}^{[\ell]}$ and $\mathbf{x}_{u,\text{qry}} := \mathbf{x}_{i,\text{qry}}^{[\ell]}$. Token 1 remains fixed because it has no predecessor.

Each query instance uses the resulting distribution both to retrieve a state and to move the query-side feature:

$$\mathbf{a}_i^{[\ell]} = \sum_{j < i} \alpha_{ij}^{[\ell]} \mathbf{s}_j^{[\ell]}, \quad \mathbf{x}_{i,\text{qry}}^{[\ell+1]} = \sum_{j < i} \alpha_{ij}^{[\ell]} \mathbf{x}_{j,\text{qry}}^{[\ell]}. \quad (7.3.6)$$

The atomic module composes the current distribution with the retrieved one and returns a new categorical state. On exact one-hot inputs $\mathbf{e}_a, \mathbf{e}_b$, its output is concentrated on $a \circ b$; on the mixtures produced by soft attention, it remains differentiable so that the final loss can train through the hidden computations. The explicit pair-indexed construction and its quantitative concentration bounds are given in Section 7.E.

We write

$$B = C_B \log d, \quad \tau = d^{-1/2}, \quad C_B > 8, \quad (7.3.7)$$

for the module's logit and smoothing scales. Here B controls the sharpness of the atomic composition logits, C_B fixes their logarithmic growth with group size, and τ is the smoothing scale that keeps the module differentiable on soft mixtures.

The atomic skill supplies the composition rule; learning determines which predecessor supplies its second operand. Under the ideal computation, the query-side feature equals the adjacent block's fixed key, so the same equality test solves every merge.

7.3.3 Final-Output Training

For a depth- L unroll, the prediction is the root distribution $F_Q(X) = \mathbf{s}_n^{[L]}(\mathbf{Q})$; write $f_{y^\star} := F_Q(X)_{y^\star}$ for the final probability of the label. The population objective is

$$\mathcal{R}_{n,L}(\mathbf{Q}) = \mathbb{E}[-\log F_Q(X)_{y^\star}], \quad (7.3.8)$$

and the shared weight matrix Q is trained by population gradient flow,

$$\frac{dQ(t)}{dt} = -\nabla_Q \mathcal{R}_{n,L}(Q(t)), \quad Q(0) = \mathbf{0}. \quad (7.3.9)$$

At initialization every available predecessor has equal attention. At a depth transition the active risk and unrolled graph change, but Q does not. For the dyadic depths we write

$$\mathcal{R}_L(Q) := \mathcal{R}_{2^L, L}(Q), \quad 2 \leq L \leq L_{\max}. \quad (7.3.10)$$

Let $\mathcal{A}_L$ denote all query instances $u = (i, \ell)$ in this depth- L unroll, with $2 \leq i \leq 2^L$ and $0 \leq \ell \leq L-1$.

Here t is optimization time, ℓ is a recurrent level within one forward pass, and L is the active task depth. A training phase holds L fixed while t evolves, then increases L while carrying the current parameter forward.

7.4 Main Results

The four formal guarantees are organized around the scalar attention margin defined next. Sections 7.5 and 7.6 explain their mechanisms, and the [appendix to this chapter](#) gives complete proofs.

7.4.1 Target Computation Tree and Learning Coordinate

At level ℓ , a correct query points to the adjacent left block. Equality matching retrieves that block, the atomic skill composes its product with the current one, and the copied query-side feature advances the boundary by 2^ℓ positions. This invariant repeats at every scale.

Random positional relabeling makes equality matching the only population-level direction distinguished by the data. Let I be the $p \times p$ identity and $\mathbf{1} \in \mathbb{R}^p$ the all-ones vector in dictionary coordinates. Define

$$\Pi = I - \frac{1}{p} \mathbf{1} \mathbf{1}^\top, \quad Q_\gamma = \gamma \Pi, \quad \kappa_{\text{sym}} = \frac{1}{p-1}. \quad (7.4.1)$$

Under Q_γ , an equal pair has logit advantage γ over an unequal pair; thus γ is the attention margin. Positional permutation symmetry confines population gradient flow to this family.

7.4.2 Learning the Base Routing Rule

The base task uses (m_1, m_2, m_3) . The final token first retrieves token 2 to form $m_3 \circ m_2$, then retrieves token 1 to produce

$$y^\star = m_3 \circ m_2 \circ m_1. \quad (7.4.2)$$

To state a nondegenerate learning condition, name the two leading states supplied by the hidden retrievals:

$$h_{\text{cur}}^{\star} = m_3 \circ m_2, \quad h_{\text{cur}}^{\text{alt}} = m_3 \circ m_1, \quad h_{\text{ret}}^{\star} = m_1, \quad h_{\text{ret}}^{\text{alt}} = m_2 \circ m_1.$$

Assumption 7.4.1 (Base-task output separation). Let $\mathcal{E}$ be the event that $h_{\text{cur}}^{\star} \neq h_{\text{cur}}^{\text{alt}}$ and $h_{\text{ret}}^{\star} \neq h_{\text{ret}}^{\text{alt}}$, and the four group elements

$$c^{\sigma, \sigma'} = h_{\text{cur}}^{\sigma} \circ h_{\text{ret}}^{\sigma'}, \quad \sigma, \sigma' \in \{\star, \text{alt}\},$$

are pairwise distinct. We assume $\mathbb{P}(\mathcal{E}) > 0$.

Condition the length-three population risk on this event and define

$$\mathcal{R}_{\text{base}}(Q) = \mathbb{E}[-\log F_Q(X)_{y^{\star}} \mid \mathcal{E}], \quad V_{\text{base}}(\gamma) = -\kappa_{\text{sym}} \frac{d\mathcal{R}_{\text{base}}(Q_{\gamma})}{d\gamma}. \quad (7.4.3)$$

Theorem 7.4.1 (Learning equality routing on the base task). *Suppose Assumption 7.4.1 holds and fix the atomic-skill scale. For all sufficiently large d ,*

$$V_{\text{base}}(\gamma) > 0 \quad \text{for every finite } \gamma \geq 0.$$

Consequently, the population gradient flow in (7.3.9) strictly decreases the conditioned length-three population risk and reaches every prescribed finite attention margin $\gamma = \Gamma$ in finite optimization time.

The theorem supplies the first finite checkpoint in the curriculum using only the final label. Operationally, the shared attention rule escapes uniform attention, learns the two hidden retrievals needed by the length-three task, and reaches any finite margin required to initialize the first dyadic stage.

7.4.3 Fixed-Depth Reuse and Constructive Updates

At depth L , soft attention lets positional error accumulate, whereas a successful atomic composition restores state concentration. Fix a target tolerance $\varepsilon_{\text{risk}} > 0$. Call a query-side feature *aligned* if its coefficient on the correct predecessor's key exceeds every other dictionary coefficient by at least $1/2$. Set

$$\xi = \min\{\varepsilon_{\text{risk}}/16, \tau/(16B), 1/128\}.$$

The atomic module's non-target activation leakage is bounded by

$$\theta := \frac{4B^2\xi^2}{\tau}. \quad (7.4.4)$$

Define the attention-error envelope and the forward budget

$$\bar{e}_{\text{att}}(n, \gamma) := (n - 1)e^{-\gamma/2}, \quad (7.4.5)$$

$$b_{\text{state}}(\varepsilon_{\text{risk}}) := \min \left\{ \frac{1}{4} - \xi, \frac{1}{2} \left[1 - \frac{4}{3B} \left(\log \frac{d-1}{\xi} + \theta \right) \right] - \xi \right\}, \quad (7.4.6)$$

$$b_{\text{fwd}}(\varepsilon_{\text{risk}}, L) := \min \left\{ b_{\text{state}}(\varepsilon_{\text{risk}}), \frac{1}{4L} \right\}. \quad (7.4.7)$$

Use the shorthand

$$\Gamma_{\text{fwd}, L} := 2 \log \frac{2^L - 1}{b_{\text{fwd}}(\varepsilon_{\text{risk}}, L)}. \quad (7.4.8)$$

The schedule uses $\bar{e}_{\text{att}}$ as a monotone scalar tracking envelope. It becomes a proved upper bound on realized wrong-key mass once the positional-alignment induction closes; no such bound is assumed at the start of a newly activated depth.

Proposition 7.4.1 (Forward stability and accuracy at fixed depth L). Under the token initialization in (7.3.4), for all sufficiently large d with $b_{\text{state}}(\varepsilon_{\text{risk}}) > 0$, if

$$\bar{e}_{\text{att}}(2^L, \gamma) \leq b_{\text{fwd}}(\varepsilon_{\text{risk}}, L), \quad \text{equivalently} \quad \gamma \geq \Gamma_{\text{fwd}, L}, \quad (7.4.9)$$

then every target-tree query-side feature remains aligned, every target-tree state remains ξ -concentrated on its correct block product, and $\mathcal{R}_L(\mathbf{Q}_\gamma) \leq \varepsilon_{\text{risk}}$.

Corollary 7.4.1 (Forward-margin scaling). *Under the conditions of the preceding proposition, and for fixed $\varepsilon_{\text{risk}}$, d , and module-scale constant C_B , the sufficient margin in (7.4.9) has the asymptotic form*

$$\gamma \geq 2L \log 2 + 2 \log L + O_{\varepsilon_{\text{risk}}, d, C_B}(1) = O(L).$$

Forward stability alone does not determine whether the actual shared-parameter gradient is constructive. For the dyadic objective, define

$$\mathcal{R}_L^{\text{sym}}(\gamma) = \mathcal{R}_L(\mathbf{Q}_\gamma), \quad V_L(\gamma) = -\kappa_{\text{sym}} \frac{d\mathcal{R}_L^{\text{sym}}(\gamma)}{d\gamma}. \quad (7.4.10)$$

Let $\mathcal{T}_L \subset \mathcal{A}_L$ be the target-tree query instances and j_v^* the correct predecessor at $v \in \mathcal{T}_L$. In the hard-attention limit $\gamma \rightarrow \infty$, each target-tree query places unit mass on its correct predecessor. To make the downstream sensitivity concrete, write

$$\beta_{v,j}(\gamma) = \frac{\partial \log f_{y^*}}{\partial \alpha_{v,j}}, \quad \beta_{v,j}^\infty = \lim_{\gamma \rightarrow \infty} \beta_{v,j}(\gamma). \quad (7.4.11)$$

Thus $\beta_{v,j}^\infty$ measures how an infinitesimal increase in the attention paid to key j at query v affects the final correct-label log probability under ideal routing. Set

$$A_L = \kappa_{\text{sym}} \mathbb{E} \left[\sum_{v \in \mathcal{T}_L} \sum_{j \in \mathcal{K}_v \setminus \{j_v^*\}} (\beta_{v,j_v^*}^\infty - \beta_{v,j}^\infty) \right]. \quad (7.4.12)$$

Theorem 7.4.2 (Constructive full-network gradients at fixed depth). *Fix $L_{\max} < \infty$. For all sufficiently large d , there is a finite positive-update threshold Γ_{upd} , common to $2 \leq L \leq L_{\max}$, such that*

$$V_L(\gamma) \geq \frac{A_L}{2} e^{-\gamma} > 0 \quad \text{for every } \gamma \geq \Gamma_{\text{upd}}.$$

Hence the depth- L population gradient-flow dynamics strictly increase the inherited attention margin throughout this region.

This guarantee applies to the full end-to-end gradient, not merely to a gradient restricted to the target computation tree. Once the inherited margin enters the stated region, every active depth up to $L_{\max}$ supplies a strictly constructive update whose leading scale is $A_L e^{-\gamma}$.

7.4.4 Cross-Depth Backward Feature Correction

The same scalar increase changes the result of reevaluating a completed shallower task. For a completed depth j , let $n_j = 2^j$ and use its own wrong-key attention envelope from (7.4.5). Define the deterministic envelopes

$$\bar{e}_{\ell j}^{\text{pos}}(\gamma) = \ell \bar{e}_{\text{att}}(n_j, \gamma), \quad \bar{e}_j^{\text{state}}(\gamma) = (d-1) \exp \left[-\frac{3B}{4} (1 - 2(\bar{e}_{\text{att}}(n_j, \gamma) + \xi)) + \theta \right]. \quad (7.4.13)$$

The first envelope bounds missing mass on the correct positional coordinate at level ℓ ; the second bounds missing mass on the correct state after a stable composition.

Corollary 7.4.2 (Cross-depth backward feature correction (BFC)). *Under the standing model setting and the conditions of Theorem 7.4.2, for every sufficiently large d with $b_{\text{state}}(\varepsilon_{\text{risk}}) > 0$, fix a completed depth j and a later active depth K , with $2 \leq j < K \leq L_{\max}$. Let $t_1 < t_2$ lie on one symmetric population-gradient-flow interval for depth K , with $\gamma(t_1) \geq \Gamma_{\text{fwd},j}$ and $\gamma(t) \geq \Gamma_{\text{upd}}$ throughout. Then, for every $1 \leq \ell \leq j$,*

$$\bar{e}_{\ell j}^{\text{pos}}(\gamma(t_2)) < \bar{e}_{\ell j}^{\text{pos}}(\gamma(t_1)), \quad \bar{e}_j^{\text{state}}(\gamma(t_2)) < \bar{e}_j^{\text{state}}(\gamma(t_1)), \quad \mathcal{R}_j(Q(t_2)) < \mathcal{R}_j(Q(t_1)).$$

The corollary establishes an error-refinement form of *backward feature correction*: deeper training tightens both the positional and state-error envelopes of every already completed shallower depth. The bound conclusion does not claim pointwise improvement of every realized hidden

state. This is a cross-depth statement: the shallower depth has already reached its own forward budget, so its bounds remain valid and its risk and bounds decrease while the later objective raises the shared margin.

7.4.5 Progressive Training Across Recurrent Depths

Base learning, fixed-depth reuse, and constructive deep updates combine into a progressive-training guarantee. First fix $L_{\max} \geq 2$, $\varepsilon_{\text{risk}} > 0$, a dictionary size $p \geq 2^{L_{\max}} + 1$, and a module-scale constant $C_B > 8$. For each group under consideration, suppose Assumption 7.4.1 holds. The lower threshold on d is chosen after these quantities are fixed and, in particular, ensures $b_{\text{state}}(\varepsilon_{\text{risk}}) > 0$.

Assume additionally that the common tail estimate in Theorem 7.4.2 admits a choice of update threshold Γ_{upd} satisfying

$$\Gamma_{\text{upd}} < \Gamma_{\text{fwd},2}. \quad (7.4.14)$$

This compatibility requirement is an additional finite-margin hypothesis; it does not follow merely by taking d sufficiently large.

Choose the finite base handoff in (7.6.7), at which the length-three risk is at most $\varepsilon_{\text{risk}}$ and the routing margin is at least Γ_{upd} . The compatibility inequality places this handoff strictly below the first dyadic forward threshold. Each dyadic stage runs until its own attention-error envelope first reaches its forward budget. We call T_L the corresponding *forward-envelope stopping time*. Only the active objective and unrolled graph change at a handoff; the same learned weights $\mathbf{Q}$ are carried forward. Let $S_2, \dots, S_{L_{\max}}$ and $T_{\text{base}}, T_2, \dots, T_{L_{\max}}$ denote the corresponding global start and stopping times. Their exact definitions and the proof of finiteness are given in Section 7.6.2.

Theorem 7.4.3 (Progressive training across recurrent depths). *For every sufficiently large d for which an admissible common Γ_{upd} satisfying (7.4.14) has been fixed, each of $T_{\text{base}}, T_2, \dots, T_{L_{\max}}$ is finite, and*

$$\mathcal{R}_{\text{base}}(\mathbf{Q}(T_{\text{base}})) \leq \varepsilon_{\text{risk}}, \quad \mathcal{R}_L(\mathbf{Q}(T_L)) \leq \varepsilon_{\text{risk}} \quad (2 \leq L \leq L_{\max}).$$

Every dyadic stage has positive duration and satisfies

$$\gamma(S_L) < \gamma(T_L) = \Gamma_{\text{fwd},L}.$$

Its true risk and deterministic attention envelope decrease strictly on the whole interval $[S_L, T_L]$, at whose endpoint the attention envelope equals $b_{\text{fwd}}(\varepsilon_{\text{risk}}, L)$ and the depth- L forward bounds become valid. While any depth K is active, reevaluating every completed shallower task $j < K$ under the updated shared matrix gives valid positional- and state-error bounds that decrease strictly, and its completed-task risk also decreases strictly.

Remark 7.4.1. *The theorem applies to a fixed finite $L_{\max}$; it does not give guarantees uniform as $L_{\max}$ grows or assert that this schedule is necessary.*

7.5 Stage I: Learning the Base Routing Rule

To prove Theorem 7.4.1, we reduce gradient flow by symmetry, differentiate through the two unsupervised mixed states, and convert strict finite-margin progress into a finite handoff time.

Symmetry reduction. Permutation invariance confines gradient flow to the family $Q(t) = Q_{\gamma(t)}$, while softmax shift invariance removes the all-ones direction. Projecting $\nabla_Q \mathcal{R}_{\text{base}}$ onto Π gives $\dot{\gamma}(t) = V_{\text{base}}(\gamma(t))$. The separation assumption is nonvacuous: for the additive group $\mathbb{Z}_d$ with prime $d \geq 5$, its event has probability at least $1 - 4/d$.

Mixed-state derivative. Write $F_c(\gamma)$ for the final pre-softmax logit of class c and $f_c(\gamma)$ for its softmax probability. At margin γ , the probabilities of the desired first and second retrievals are

$$r_1(\gamma) = \frac{e^\gamma}{1 + e^\gamma}, \quad r_2(\gamma) = \frac{e^{\gamma r_1(\gamma)}}{1 + e^{\gamma r_1(\gamma)}}.$$

The first soft retrieval produces a current-state mixture concentrated on $h_{\text{cur}}^\star$ and $h_{\text{cur}}^{\text{alt}}$; the second produces a retrieved-state mixture concentrated on $h_{\text{ret}}^\star$ and $h_{\text{ret}}^{\text{alt}}$. Direct differentiation gives $r'_1, r'_2 > 0$ and shows that the target-versus-alternative margin in each mixture increases with γ . At $\gamma = 0$, differentiating through both composition modules yields

$$V_{\text{base}}(0) = \kappa_{\text{sym}} \left(\frac{B^2 + B}{16} + o(B^2) \right) > 0;$$

the B^2 term passes through both recurrent uses, while the B term comes from the final retrieval.

Strict progress and finite handoff. For a finite $\gamma > 0$, the same mixed-state calculation shows that every target logit gap grows:

$$\frac{d}{d\gamma} (F_{y^\star}(\gamma) - F_c(\gamma)) > 0, \quad c \neq y^\star.$$

Therefore

$$\frac{d}{d\gamma} \log f_{y^\star} = \sum_{c \neq y^\star} f_c \frac{d}{d\gamma} (F_{y^\star} - F_c) > 0,$$

and averaging over the separated base event proves $V_{\text{base}}(\gamma) > 0$. On any compact interval $[0, \Gamma]$, continuity gives a positive minimum $V_{\min}(\Gamma)$, so the hitting time is at most $\Gamma/V_{\min}(\Gamma)$. The composition-module derivative bounds used in the logit-gap inequality are proved in Section 7.E.

This establishes the base handoff used in Stage II.

7.6 Stage II: Learning Across Recurrent Depth

7.6.1 Fixed-Depth Guarantee

Fix $2 \leq L \leq L_{\max}$ and activate the length- 2^L , depth- L task at its inherited stage start. The argument has three steps: the inherited margin lies in a common positive-gradient region; the active objective raises that margin until its attention-error envelope reaches its own forward budget; and, throughout this deeper stage, the valid error bounds of already completed shallower tasks tighten under the shared update.

A sufficient forward-validity condition for recurrent reuse. Because every query-side feature is a convex combination of dictionary vectors, let e_ℓ^{pos} be the maximum, over level- ℓ target-tree query instances, of one minus the coefficient on the correct predecessor’s key. An inaccurate retrieval increases this error, whereas a successful composition restores state concentration. Once the correct query coordinate exceeds every incorrect coordinate by at least $1/2$, the total wrong-key mass is bounded by (7.4.5). Induct over recurrent levels. If the level- ℓ positional error is at most $\ell \bar{e}_{\text{att}}$, the correct coordinate remains the unique maximum and the next wrong-key mass is again at most $\bar{e}_{\text{att}}(2^L, \gamma)$. Copying the retrieved positional feature then gives

$$e_{\ell+1}^{\text{pos}} \leq e_\ell^{\text{pos}} + \bar{e}_{\text{att}}(2^L, \gamma).$$

For the state field, concentrated current and predecessor states plus small wrong-key mass give a fixed positive activation gap for the unit corresponding to the correct operand pair. The output softmax therefore restores the same state-concentration bound after every merge. Choosing $\bar{e}_{\text{att}}(2^L, \gamma) = O_{\varepsilon_{\text{risk}}}(1/L)$ closes both inductions and makes the root cross-entropy at most $\varepsilon_{\text{risk}}$. Solving this inequality for γ gives the margin condition in (7.4.9): $2L \log 2$ offsets the largest causal window and $2 \log L$ controls accumulation across levels.

Why weight sharing does not destroy the inherited margin. Forward stability alone does not control the actual shared-parameter gradient. Every recurrent use of the shared matrix contributes, so a calculation restricted to the target computation tree would be insufficient. A single local-to-global identity captures all local contributions.

For an arbitrary query instance $u \in \mathcal{A}_L$ and candidate predecessor $j \in \mathcal{K}_u$, set

$$\omega_{u,j} = \langle \mathbf{x}_{u,\text{qry}}, \mathbf{x}_j \rangle, \quad \beta_{u,j} = \frac{\partial \log f_{y^\star}}{\partial \alpha_{u,j}}. \quad (7.6.1)$$

The first quantity is the positional match; the second, already anchored in (7.4.11), contains every downstream dependence of the final log probability on that local attention weight. Differentiating

the local softmax and summing all recurrent uses gives the covariance identity

$$\frac{d \log f_{y^*}}{d\gamma} = \frac{1}{2} \sum_{u \in \mathcal{A}_L} \sum_{j,k \in \mathcal{K}_u} \alpha_{u,j} \alpha_{u,k} (\beta_{u,j} - \beta_{u,k}) (\omega_{u,j} - \omega_{u,k}). \quad (7.6.2)$$

The identity says that the margin grows when keys with stronger positional matches also have more favorable downstream sensitivities. Its exact derivation appears in Section 7.D. In the hard-routing limit, the target-tree terms reduce to A_L . The root query supplies the dominant positive contribution, while the nearly one-hot composition Jacobians control the other target-tree queries; hence $A_L > 0$ for all sufficiently large d .

Write $\rho = e^{-\gamma}$. On the blue target tree in Figure 7.1, each correct-versus-wrong attention pair has probability product $\rho + O((1 + \gamma)\rho^2)$. Replacing the downstream sensitivities by their hard-routing limits therefore yields the leading term

$$A_L \rho = A_L e^{-\gamma}.$$

The nearly one-hot composition Jacobians keep all remaining target-tree terms below this leading order.

Why off-tree paths remain lower order. The gray computations in Figure 7.1 can affect the supervised root only by first crossing into the blue target tree. At that first crossing, the gray state is necessarily selected as a wrong candidate, which contributes an $O(\rho)$ attention factor. Its local covariance also requires an independent routing defect—either leakage away from the limiting maximizers or a residual split among tied maximizers—that vanishes as $\gamma \rightarrow \infty$. Consequently, each gray path contributes approximately

$$O(\rho) \times o_\gamma(1) = o_\gamma(e^{-\gamma}).$$

Crossings through logit derivatives introduce at most an additional factor of γ and remain lower order after the same leakage or split is included. The complete path classification and four-factor Jacobian calculation appear in Section 7.F.3.

Summing the finite graph and averaging over examples yields, as $\gamma \rightarrow \infty$ uniformly over the fixed finite family of depths,

$$V_L(\gamma) = A_L e^{-\gamma} + o_\gamma(e^{-\gamma}), \quad A_L > 0.$$

The symmetry projection gives $\dot{\gamma}(t) = V_L(\gamma(t))$ when the depth- L risk is active. Because the family of depths is finite, we choose one common Γ_{upd} beyond which the remainder is at most $A_L e^{-\gamma}/2$ at every depth. From that point onward, the inherited margin continues to grow under the full gradient.

Cross-depth tightening of the error bounds. The same increase in γ changes the result of reevaluating every completed shallower task with the shared block. Suppose depth K is active and $j < K$ is complete. Then $\gamma \geq \Gamma_{\text{fwd},j}$ keeps the depth- j envelopes in their valid regime, while Theorem 7.4.2 gives $\dot{\gamma} = V_K(\gamma) > 0$. Differentiating (7.4.13) along the depth- K trajectory gives

$$\frac{d}{dt} \bar{e}_{\ell j}^{\text{pos}} = -\frac{1}{2} \bar{e}_{\ell j}^{\text{pos}}(\gamma(t)) V_K(\gamma(t)) < 0, \quad \frac{d}{dt} \bar{e}_j^{\text{state}} = -\frac{3B}{4} \bar{e}_{\text{att}}(2^j, \gamma(t)) \bar{e}_j^{\text{state}}(\gamma(t)) V_K(\gamma(t)) < 0.$$

The completed risk improves as well, because

$$\frac{d}{dt} \mathcal{R}_j(Q_{\gamma(t)}) = -\frac{V_j(\gamma(t)) V_K(\gamma(t))}{\kappa_{\text{sym}}} < 0.$$

These derivatives prove Corollary 7.4.2. They concern reevaluation of the completed depth- j task under the updated shared matrix; they do not assert that the earlier levels inside the larger active depth- K unroll have entered the depth- K forward-validity regime.

7.6.2 Transfer Across Recurrent Depths

The transfer argument has four parts: neighboring forward thresholds are strictly nested, the base handoff initializes the dyadic schedule below its first target, each active stage reaches its own target in finite time, and the same entry condition propagates to every later stage.

Why the next forward target is unmet at handoff. Recall the own-depth budget and shorthand threshold in (7.4.7)–(7.4.8), and write $b_L := b_{\text{fwd}}(\varepsilon_{\text{risk}}, L)$. The sequence (b_L) is nonincreasing. By construction, $\Gamma_{\text{fwd},L}$ is exactly the margin at which the depth- L attention envelope reaches b_L and the fixed-depth forward condition becomes valid.

For $2 \leq K < L_{\text{max}}$, the neighboring-depth envelopes satisfy

$$\bar{e}_{\text{att}}(2^K, \Gamma_{\text{fwd},K}) = b_K, \tag{7.6.3}$$

$$\bar{e}_{\text{att}}(2^{K+1}, \gamma) = r_K \bar{e}_{\text{att}}(2^K, \gamma), \tag{7.6.4}$$

$$r_K := \frac{2^{K+1} - 1}{2^K - 1} = 2 + \frac{1}{2^K - 1} \in (2, 3]. \tag{7.6.5}$$

Consequently,

$$\bar{e}_{\text{att}}(2^{K+1}, \Gamma_{\text{fwd},K}) = r_K b_K > b_K \geq b_{K+1}. \tag{7.6.6}$$

Equivalently, the inherited margin lies strictly below the next depth's forward threshold:

$$\Gamma_{\text{fwd},K} < \Gamma_{\text{fwd},K+1}.$$

Thus the next depth's forward target has not been reached at handoff. This envelope comparison does not determine whether the inherited depth- $(K+1)$ risk is above or below $\varepsilon_{\text{risk}}$.

Base handoff and stage boundaries. Define the explicit base handoff margin

$$\Gamma_{\text{base}} := \max \left\{ \Gamma_{\text{upd}}, 2 \log \frac{2}{b_{\text{fwd}}(\varepsilon_{\text{risk}}, 2)} \right\}. \quad (7.6.7)$$

The base theorem reaches this finite margin, where the base risk is at most $\varepsilon_{\text{risk}}$. Compatibility and $2 \log(2/b_2) < 2 \log(3/b_2) = \Gamma_{\text{fwd},2}$ imply $\Gamma_{\text{base}} < \Gamma_{\text{fwd},2}$.

Define the global stage boundaries by

$$\begin{aligned} T_{\text{base}} &= \inf\{t \geq 0 : \gamma(t) = \Gamma_{\text{base}}\}, & S_2 &= T_{\text{base}}, \\ S_L &= T_{L-1} \quad (3 \leq L \leq L_{\text{max}}), \\ T_L &= \inf\{t \geq S_L : \bar{e}_{\text{att}}(2^L, \gamma(t)) \leq b_{\text{fwd}}(\varepsilon_{\text{risk}}, L)\}. \end{aligned}$$

Thus the depth-two stage starts in the positive-update region and strictly below its own forward target.

Why each stage stops in finite time. Suppose an active depth- L stage begins in the entry range

$$\Gamma_{\text{upd}} \leq \gamma(S_L) < \Gamma_{\text{fwd},L} \quad (2 \leq L \leq L_{\text{max}}).$$

Then Theorem 7.4.2 gives

$$\dot{\gamma}(t) = V_L(\gamma(t)) > 0, \quad \frac{d}{dt} \bar{e}_{\text{att}}(2^L, \gamma(t)) = -\frac{1}{2} \bar{e}_{\text{att}}(2^L, \gamma(t)) V_L(\gamma(t)) < 0,$$

and also

$$\frac{d}{dt} \mathcal{R}_L(\mathbf{Q}_{\gamma(t)}) = -\frac{V_L(\gamma(t))^2}{\kappa_{\text{sym}}} < 0.$$

Continuity and the positive lower bound on each compact margin interval give a finite hitting time for the forward threshold, so T_L is finite. At T_L , the attention envelope equals b_L and the fixed-depth proposition gives $\mathcal{R}_L \leq \varepsilon_{\text{risk}}$. The strict entry inequality makes the stage duration positive, while the displayed derivatives give strict envelope and risk decrease throughout the stage.

Propagation across stages and completed-task tightening. The base handoff establishes the entry range for $L = 2$. If it holds at depth L , the preceding argument gives $\gamma(T_L) = \Gamma_{\text{fwd},L}$. Since $S_{L+1} = T_L$, strict threshold nesting places the next stage in the same entry range. This closes the induction through L_{max} .

Finally, if K is active and $j < K$ is complete, then $\gamma(S_K) = \Gamma_{\text{fwd},K-1} \geq \Gamma_{\text{fwd},j}$. The depth- j forward bounds therefore remain valid, and Corollary 7.4.2 shows that those bounds and $\mathcal{R}_j$ decrease strictly throughout $[S_K, T_K]$. Together with the endpoint guarantee at every active depth, this proves Theorem 7.4.3.

7.7 Roadmap for the Complete Proofs

The proofs follow the order of the main results. Section 7.B establishes the model and scalar reduction; Section 7.C proves base-task progress and forward accuracy; and Section 7.D proves constructive deeper gradients and curriculum transfer. Supporting composition, asymptotic, symmetry, and extension results appear in Sections 7.E to 7.H.

Appendix

7.A Proof overview and notation

The proof has three parts. Section 7.B defines the model and scalar reduction; Section 7.C proves base-task progress and forward accuracy; and Section 7.D proves constructive deeper gradients and curriculum transfer. The composition, finite-graph, symmetry, and extension ingredients are established in Sections 7.E to 7.H.

Notation guide. A bar denotes a deterministic error envelope, and a superscript ∞ denotes the ideal-routing limit as $\gamma \rightarrow \infty$. The main error family is

$$e_v^{\text{att}}, \quad e_v^{\text{pos}}, \quad e_v^{\text{state}}, \quad \bar{e}_{\text{att}}(n, \gamma), \quad \bar{e}_{\ell j}^{\text{pos}}(\gamma), \quad \bar{e}_j^{\text{state}}(\gamma).$$

Here the unbarred quantities describe a realized query or state node, whereas the barred quantities are analytic envelopes; they become valid deterministic bounds under the stated forward hypotheses. The budgets b_{state} and b_{fwd} respectively control the state reset and one-stage forward accuracy.

Two near-identical symbol pairs recur throughout the graph argument:

$$\mathcal{A}_L \text{ (all query instances)} \quad \text{versus} \quad A_L \text{ (the leading gradient coefficient),}$$

$$\mathcal{T}_L \text{ (intended-tree query instances)} \quad \text{versus} \quad T_L \text{ (the stopping time for stage } L\text{).}$$

Finally, $\gamma(t)$ is the learned routing margin along training, $V_L(\gamma)$ is its static depth- L growth function, Γ_{upd} is the common positive-update threshold, Γ_{base} is a base handoff margin, and $\Gamma_{\text{fwd}, L}$ is the forward-envelope threshold for depth L . In the progressive schedule, T_L is the first time the active attention envelope reaches its designated forward budget.

For asymptotic notation, $O_{L_{\text{max}}, C_B}(\cdot)$ in the large-group comparison denotes a constant independent of d , while $o_d(\cdot)$ denotes the limit $d \rightarrow \infty$. In the later large-margin expansion, after d is fixed, $O_d(\cdot)$ may depend on d and the limiting variable is γ .

7.B Task, model, and scalar training dynamics (Stages I and II)

The full looped predictor is defined first, followed by the reduction of population gradient flow to the one-dimensional routing-margin family.

7.B.1 Supervised task and loop-transformer setting

Definition 7.B.1 (Finite-group composition). Let $(G, \circ)$ be a finite group of order $d = |G|$ with identity 1_G . For $m_{1:n} = (m_1, \dots, m_n) \in G^n$, define

$$\text{eval}(m_{1:n}) = m_n \circ m_{n-1} \circ \dots \circ m_1.$$

The supervised target is the group element

$$y^\star = \text{eval}(m_{1:n}).$$

Fix a maximum length N and an orthonormal dictionary of positional features

$$\mathcal{X} = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(p)}\} \subset \mathbb{R}^p, \quad p \geq N + 1.$$

We use $\mathcal{X}$ as the coordinate basis for the positional feature space. Accordingly, every positional vector and matrix below—including $\mathbf{Q}$, $\mathbf{I}$, $\mathbf{1}$, and $\mathbf{\Pi}$ —is expressed in dictionary coordinates. An input is

$$X = (m_{1:n}, \iota),$$

where the symbols $m_1, \dots, m_n$ are independent and uniform on G and $\iota : \{0, 1, \dots, n\} \rightarrow [p]$ is a uniformly sampled injection, independent of the symbols. We write $\mathbf{x}_i = \mathbf{x}^{(\iota(i))}$. Thus the label depends on the group sequence, whereas the relabeling determines how that sequence is presented to the router.

Definition 7.B.2 (Randomly relabeled group-composition example). The length- n data distribution is the law of

$$(X, y^\star) = ((m_{1:n}, \iota), \text{eval}(m_{1:n}))$$

under the sampling rule above.

Definition 7.B.3 (Token initialization). Token i is initialized as

$$\mathbf{h}_i^{[0]} = (\mathbf{x}_{i,\text{qry}}^{[0]}, \mathbf{x}_i, \mathbf{s}_i^{[0]}) = (\mathbf{x}_{i-1}, \mathbf{x}_i, \mathbf{e}_{m_i}), \quad (7.B.1)$$

where $\mathbf{e}_g \in \mathbb{R}^d$ is the one-hot vector indexed by $g \in G$.

The middle field is token i 's attention key. It is never updated, so the key of token j is always its dictionary vector $\mathbf{x}_j$, and we write it without a role tag. Only the query-side feature carries the qry tag, because it is a different, dynamic object: its level-zero value is the *preceding* token's dictionary vector, $\mathbf{x}_{i,\text{qry}}^{[0]} = \mathbf{x}_{i-1}$.

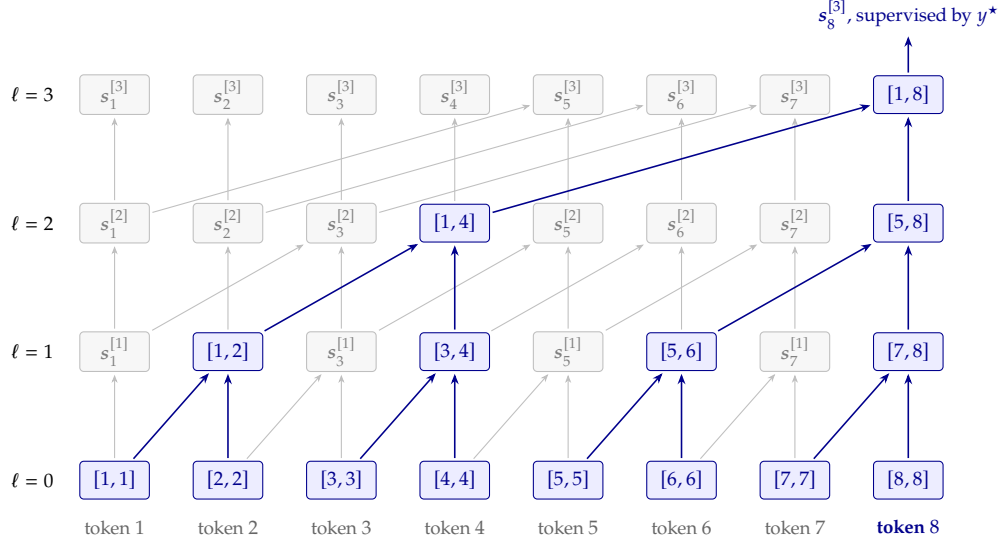


Figure 7.2: Ideal Parallel Forward Pass. The ideal parallel forward pass on a length-eight input. Column i is token position i ; row ℓ shows the recurrent state at that token and level. Blue nodes lie on the ideal dyadic computation tree and are labeled by their target blocks. Gray nodes are simultaneous off-tree states generated by the same recurrent update and are therefore labeled only by their state indices; in particular, the formal model continues to update a token after its query-side feature reaches x_0 . At level ℓ , a vertical edge carries a token’s current-state input and a drawn diagonal edge marks its ideal retrieved predecessor. Gray retrieval edges are schematic: when no key matches x_0 , the actual softmax retrieval from all causal keys is not expanded into a fan of edges. Thus a vertical edge never denotes a carry gate. The highlighted subgraph feeding token 8 is exactly the dyadic block tree of Fig. 7.3.

The random injection removes absolute-position shortcuts: across examples, no dictionary vector is permanently associated with “position 3” or “the previous token.” Equality of positional features is the only positional relation preserved by the data distribution. We study dyadic lengths $n = 2^L$ up to

$$L_{\max} = \lfloor \log_2 N \rfloor.$$

The ideal computation gives every token three semantic fields: a left endpoint, a right endpoint, and the product of the symbols between them. In the neural representation, the left endpoint is encoded by a dynamic query-side positional feature, the right endpoint by a fixed key-side positional feature, and the block product by a categorical state feature. Forming a larger block moves the query-side feature leftward and updates the state, while the key-side feature remains fixed. Figure 7.2 displays the resulting parallel forward pass on a length-eight input.

A length-eight example displays the ground-truth recursion. At token 8, the ideal equality router produces the sequence of blocks shown in Fig. 7.3.

The invariant behind this table is the learning target for attention: the query-side feature marking the current left endpoint equals the fixed key-side feature of the adjacent predecessor block. The ideal router therefore needs to recognize equality, not memorize an absolute position or a task-specific offset.

recurrent level ℓ	block represented at token 8	query-side feature	correct predecessor
0	$[8, 8]$	$\mathbf{x}_7$	token 7 representing $[7, 7]$
1	$[7, 8]$	$\mathbf{x}_6$	token 6 representing $[5, 6]$
2	$[5, 8]$	$\mathbf{x}_4$	token 4 representing $[1, 4]$
3	$[1, 8]$	$\mathbf{x}_0$	computation complete

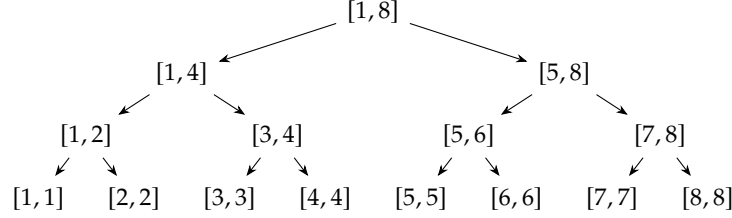


Figure 7.3: Length-Eight Dyadic Computation. The table follows token 8 through the recurrent levels; the block tree shows the same composition recursion and coincides with the highlighted sub-graph of the parallel forward pass in Fig. 7.2.

Model and training objective

Because the dictionary is an orthonormal basis of the positional feature space, it provides the coordinate basis for $\mathbf{Q} \in \mathbb{R}^{p \times p}$; no parameter direction acts outside its span. Lemma 7.G.1 gives the corresponding projection argument for an embedding in a larger feature space.

At recurrent level ℓ , token $i \geq 2$ attends to its strict-causal window $j < i$ according to

$$\alpha_{ij}^{[\ell]}(\mathbf{Q}) = \frac{\exp(\langle \mathbf{Q} \mathbf{x}_{i, \text{qry}}^{[\ell]}, \mathbf{x}_j \rangle)}{\sum_{k < i} \exp(\langle \mathbf{Q} \mathbf{x}_{i, \text{qry}}^{[\ell]}, \mathbf{x}_k \rangle)}. \quad (7.B.2)$$

We call $u = (i, \ell)$ a *query instance* and write $\mathcal{K}_u = \{j : j < i\}$ for its candidate-key set. Under the *consuming-state convention*, (i, ℓ) denotes the attention application that consumes level- ℓ inputs and produces level- $(\ell + 1)$ outputs, where $0 \leq \ell \leq L - 1$ in a depth- L unroll. Its logits are

$$z_{u,j} = \langle \mathbf{Q} \mathbf{x}_{i, \text{qry}}^{[\ell]}, \mathbf{x}_j \rangle, \quad j \in \mathcal{K}_u,$$

and we abbreviate its attention distribution by $\alpha_u = (\alpha_{u,j})_{j \in \mathcal{K}_u}$, with $\alpha_{u,j} := \alpha_{ij}^{[\ell]}$.

The retrieved state feature and the updated query-side positional feature are

$$\mathbf{a}_i^{[\ell]} = \sum_{j < i} \alpha_{ij}^{[\ell]} \mathbf{s}_j^{[\ell]}, \quad \mathbf{x}_{i, \text{qry}}^{[\ell+1]} = \sum_{j < i} \alpha_{ij}^{[\ell]} \mathbf{x}_{j, \text{qry}}^{[\ell]}. \quad (7.B.3)$$

Attention keys are not updated, and token 1 remains fixed because it has no strict-causal predecessor. A frozen MLP, defined next, updates $\mathbf{s}_i^{[\ell+1]}$ from the current state and the retrieved state.

The same matrix Q is used at every token and recurrent level. It is the only trained component. Its optimization-time initialization is

$$Q(0) = \mathbf{0}. \quad (7.B.4)$$

Thus the initialized predictor assigns every available predecessor equal attention. In contrast, an ideal equality router puts unit mass on the key whose positional feature equals the current query-side feature. Training must move the shared matrix from the former behavior toward the latter using only the final label.

The attention layer chooses a predecessor; the frozen MLP composes the current and retrieved states. Because the two inputs have different operand roles, we use mutually orthonormal, role-tagged copies

$$\{\mathbf{y}_a : a \in G\}, \quad \{\mathbf{g}_b : b \in G\}$$

of the group vocabulary. For distributions $\mathbf{p}, \mathbf{q} \in \Delta(G)$, write

$$\mathbf{y}(\mathbf{p}) = \sum_{a \in G} p_a \mathbf{y}_a, \quad \mathbf{g}(\mathbf{q}) = \sum_{b \in G} q_b \mathbf{g}_b.$$

Definition 7.B.4 (Smoothed ReLU). For $\tau > 0$, define

$$\phi_\tau(z) = \begin{cases} 0, & z \leq 0, \\ \frac{z^2}{2\tau}, & 0 < z < \tau, \\ z - \frac{\tau}{2}, & z \geq \tau. \end{cases}$$

Then ϕ_τ is continuously differentiable, nondecreasing, and agrees with ReLU up to a constant shift above τ .

Fix

$$B = C_B \log d, \quad \tau = d^{-1/2}, \quad (7.B.5)$$

where $C_B > 8$ is a constant. The pair-indexed weights are given in Section 7.E.1; the results below require only the formal interface and the activation in Definition 7.B.4.

At each recurrent level, the state update is

$$\mathbf{Z}_i^{[\ell]} = \frac{1}{2} \left(\mathbf{y}(\mathbf{s}_i^{[\ell]}) + \mathbf{g}(\mathbf{a}_i^{[\ell]}) \right), \quad \mathbf{s}_i^{[\ell+1]} = \text{softmax} \left(\text{MLP}_W(\mathbf{Z}_i^{[\ell]}) \right). \quad (7.B.6)$$

This residual-plus-attention interface and frozen atomic-composition module follow Huang et al. [131]. For later use, write

$$F_c(\mathbf{p}, \mathbf{q}) = [\text{MLP}_W(\frac{1}{2}(\mathbf{y}(\mathbf{p}) + \mathbf{g}(\mathbf{q})))]_c$$

for the pre-softmax logit of output class c .

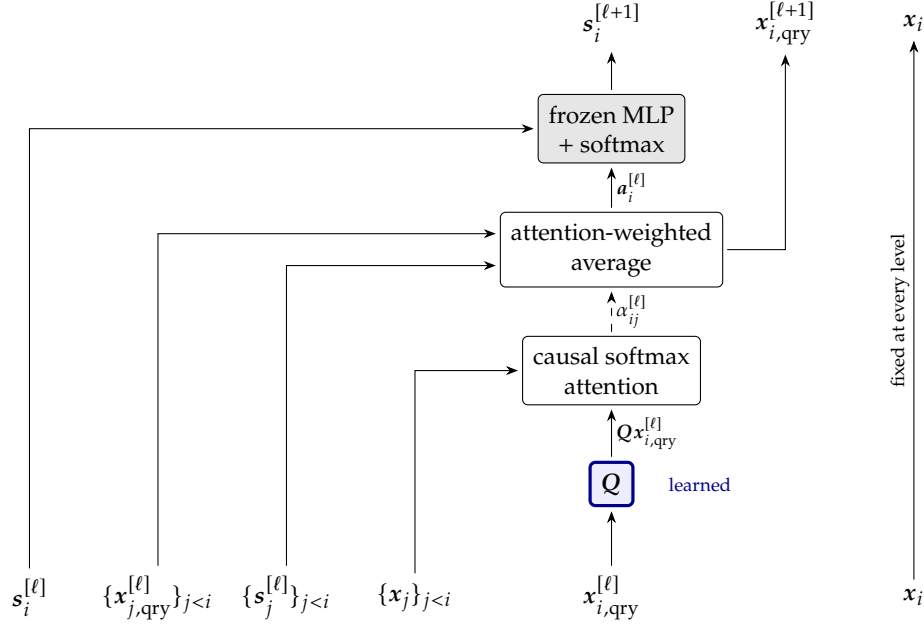


Figure 7.4: A Single Recurrent Level. One recurrent level at token i , read bottom to top. The shared matrix Q is the only learned parameter. Attention retrieves a predecessor state and moves the query-side positional feature; the frozen MLP then composes the current and retrieved states.

The formal properties of this interface are stated and proved separately in Proposition 7.B.1. One recurrent level is shown in Fig. 7.4.

For a depth- L evaluation, abbreviate the full network map $F_{W,Q}$ by F . The predictor is the final state distribution

$$F(X) = s_n^{[L]}(Q) \in \Delta(G).$$

For an output class $c \in G$, write

$$f_c := [F(X)]_c. \quad (7.B.7)$$

We train only Q using final-output cross-entropy:

$$\mathcal{R}_{n,L}(Q) = \mathbb{E} \left[-\log F(X)_{y^*} \right]. \quad (7.B.8)$$

The population gradient flow for the weight matrix Q is

$$\frac{dQ(t)}{dt} = -\nabla_Q \mathcal{R}_{n,L}(Q(t)), \quad Q(0) = \mathbf{0}. \quad (7.B.9)$$

Thus t denotes optimization time. Recurrent depth is always indexed by ℓ and written in square brackets.

For the dyadic curriculum, define

$$\mathcal{R}_L(\mathbf{Q}) := \mathcal{R}_{2^L, L}(\mathbf{Q}) \quad (2 \leq L \leq L_{\max}). \quad (7.B.10)$$

Thus $\mathcal{R}_2$ is the length-four, depth-two dyadic objective. Every $\mathcal{R}_L$ uses the unconditioned data law in Definition 7.B.2. The distinct length-three base objective is introduced in Section 7.C together with its conditioning event; it is never identified with the unconditioned risk $\mathcal{R}_{3,2}$.

At a curriculum switch, the objective and unrolled depth change, but the parameter does not: if a stage stops at time T , the next stage begins from $\mathbf{Q}(T)$.

For later use, cross-entropy controls squared error of any predicted distribution $\mathbf{p}$:

$$\|\mathbf{p} - \mathbf{e}_{y^*}\|_2^2 \leq 2(1 - p_{y^*}) \leq 2[-\log p_{y^*}]. \quad (7.B.11)$$

7.B.2 Structural properties and scalar training dynamics

The frozen MLP implements the required composition interface on state mixtures, while positional relabeling confines population gradient flow to one equality-matching direction.

Composition interface and scalar symmetry reduction

The explicit module constructed in Section 7.E.1 is C^1 on state mixtures. Subsequent arguments use only the following proposition.

Proposition 7.B.1 (Frozen composition interface). For every $\mathbf{p}, \mathbf{q} \in \Delta(G)$, the hidden unit indexed by (a, b) has preactivation

$$\Lambda_{a,b}(\mathbf{p}, \mathbf{q}) = B(p_a + q_b - 1) + \tau p_a. \quad (7.B.12)$$

If $\mathbf{p} = \mathbf{e}_a$, then for every $\mathbf{q} \in \Delta(G)$,

$$[\mathbf{s}^+]_{a \circ b} = \frac{\exp(Bq_b)}{\sum_{h \in G} \exp(Bq_h)}, \quad \mathbf{s}^+ = \text{softmax}(\text{MLP}_{\mathbf{W}}(\mathbf{Z})). \quad (7.B.13)$$

Proof. Section 7.E.1 verifies both formulas from the explicit weights. $\square$

In particular, setting $\mathbf{q} = \mathbf{e}_b$ gives

$$[\mathbf{s}^+]_{a \circ b} = \pi_{\text{comp}} := \frac{e^B}{e^B + d - 1}, \quad 1 - \pi_{\text{comp}} \leq d^{1-C_B}. \quad (7.B.14)$$

Equation (7.B.12) converts simultaneous operand concentration into an output-logit advantage, while (7.B.14) gives nearly exact one-hot composition. Together with differentiability on mixtures, these are the module properties used below. Their quantitative consequence is the state-concentration reset in Lemma 7.C.6 and Corollary 7.C.3.

The data law does not privilege any positional dictionary vector. In the dictionary basis, let

$$\mathbf{\Pi} := \mathbf{I} - \frac{1}{p} \mathbf{1}\mathbf{1}^\top.$$

Its diagonal-minus-off-diagonal gap is one. The next lemma shows that population training moves the full matrix only in this equality-matching direction. The reduction has two steps: permutation invariance confines the matrix to a two-dimensional fixed subspace, and softmax shift invariance removes one behaviorally invisible direction. The group-action and well-posedness arguments in Section 7.G yield the scalar family and flow identity.

For $\gamma \in \mathbb{R}$, set

$$\mathbf{Q}_\gamma := \gamma \mathbf{\Pi}.$$

Because $\mathbf{\Pi}$ has diagonal-minus-off-diagonal gap one, γ is exactly the matched-versus-mismatched logit margin: for any distinct dictionary vectors $\mathbf{x} \neq \mathbf{x}'$,

$$\gamma = \langle \mathbf{Q}_\gamma \mathbf{x}, \mathbf{x} \rangle - \langle \mathbf{Q}_\gamma \mathbf{x}, \mathbf{x}' \rangle. \quad (7.B.15)$$

If a query-side positional feature is $\mathbf{x}_{\text{qry}} = \sum_{\mathbf{x} \in \mathcal{X}} \omega(\mathbf{x}) \mathbf{x}$, then under $\mathbf{Q}_\gamma$ the attention logit of key $\mathbf{x}_j$ and its attention probability are

$$\langle \mathbf{Q}_\gamma \mathbf{x}_{\text{qry}}, \mathbf{x}_j \rangle = \gamma \left(\omega(\mathbf{x}_j) - \frac{1}{p} \sum_{\mathbf{x} \in \mathcal{X}} \omega(\mathbf{x}) \right), \quad \alpha_j = \frac{\exp(\gamma \omega(\mathbf{x}_j))}{\sum_k \exp(\gamma \omega(\mathbf{x}_k))}.$$

Thus $\gamma = 0$ gives uniform attention, while increasing γ strengthens the preference for keys with larger positional matches. In particular, the entire forward pass on this family depends on $\mathbf{Q}_\gamma$ only through γ .

Along the learned trajectory, $\gamma(t)$ is the routing margin measured from $\mathbf{Q}(t)$; it is neither a second parameter nor a surrogate optimization clock.

For a fixed routing margin γ , define the symmetric pullback of the matrix risk by

$$\mathcal{R}_{n,L}^{\text{sym}}(\gamma) := \mathcal{R}_{n,L}(\gamma \mathbf{\Pi}).$$

For a dyadic stage we abbreviate $\mathcal{R}_L^{\text{sym}}(\gamma) := \mathcal{R}_{2^L,L}^{\text{sym}}(\gamma)$. Define the corresponding routing-margin growth rate by

$$V_{n,L}(\gamma) := -\kappa_{\text{sym}} \frac{d\mathcal{R}_{n,L}^{\text{sym}}(\gamma)}{d\gamma}, \quad \kappa_{\text{sym}} := \frac{1}{p-1}. \quad (7.B.16)$$

Thus $V_{n,L}(\gamma) > 0$ means that gradient flow increases the matched logit relative to every mismatched logit at margin γ .

Lemma 7.B.1 (One-margin population trajectory). *Under the random positional relabeling in Definition 7.B.2, population gradient flow from $\mathbf{Q}(0) = \mathbf{0}$ remains in the family above:*

$$\mathbf{Q}(t) = \mathbf{Q}_{\gamma(t)} = \gamma(t)\mathbf{\Pi}, \quad \gamma(0) = 0.$$

In the original optimization time, its routing margin satisfies

$$\frac{d\gamma(t)}{dt} = V_{n,L}(\gamma(t)). \quad (7.B.17)$$

Proof. In dictionary coordinates, a permutation π acts by a permutation matrix P_π and sends $\mathbf{Q}$ to $P_\pi \mathbf{Q} P_\pi^\top$. The population risk is invariant under this conjugation and its gradient is equivariant. The gradient vector field is locally Lipschitz, so population gradient flow has a unique solution; since $\mathbf{Q}(0) = \mathbf{0}$ is fixed by every conjugation, uniqueness keeps $\mathbf{Q}(t)$ in the permutation-fixed subspace. That subspace consists of matrices with one common diagonal entry and one common off-diagonal entry, and hence is

$$\text{span}\{\mathbf{I}, \mathbf{1}\mathbf{1}^\top\}.$$

These invariance, regularity, and fixed-subspace claims are proved in Lemmas 7.G.2 and 7.G.3 and Proposition 7.G.1.

Every query-side positional feature is a convex combination of dictionary vectors, and its coefficient sum remains one under the attention-weighted positional update. Consequently, adding a multiple of $\mathbf{1}\mathbf{1}^\top$ to $\mathbf{Q}$ adds the same offset to every candidate-key logit at a fixed query instance. Softmax cancels that offset, so the risk is constant in this all-ones direction and the population gradient is Frobenius-orthogonal to it. The intersection of the permutation-fixed subspace with that orthogonal complement is

$$\text{span}\left\{\mathbf{I} - \frac{1}{p}\mathbf{1}\mathbf{1}^\top\right\} = \text{span}\{\mathbf{\Pi}\};$$

see Lemma 7.G.4 and Corollary 7.G.1. Starting from zero, the matrix trajectory therefore has the form $\mathbf{Q}(t) = \gamma(t)\mathbf{\Pi}$. Equation (7.B.15) identifies its coefficient with the routing margin.

Finally, write $\nabla_{\mathbf{Q}} \mathcal{R}_{n,L} = \lambda_{\mathbf{Q}}(t)\mathbf{\Pi}$ along the trajectory. Since $\mathbf{\Pi}$ is an orthogonal projector with $\|\mathbf{\Pi}\|_{\text{F}}^2 = p - 1$, the chain rule gives

$$\left. \frac{d\mathcal{R}_{n,L}^{\text{sym}}(\gamma)}{d\gamma} \right|_{\gamma=\gamma(t)} = \langle \lambda_{\mathbf{Q}}(t)\mathbf{\Pi}, \mathbf{\Pi} \rangle_{\text{F}} = (p-1)\lambda_{\mathbf{Q}}(t).$$

On the other hand, substituting $\mathbf{Q}(t) = \gamma(t)\mathbf{\Pi}$ into (7.B.9) gives

$$\frac{d\gamma(t)}{dt} \mathbf{\Pi} = -\lambda_{\mathbf{Q}}(t)\mathbf{\Pi}.$$

Combining the two identities with (7.B.16) proves (7.B.17) without rescaling optimization time. $\square$

For the dyadic objective, write $V_L := V_{2^L, L}$. The fixed argument γ evaluates an update at a fixed symmetric router; during training the learned margin is written $\gamma(t)$. Training progress therefore reduces to positivity of the applicable growth function on the required margin interval.

Global indexing and graph notation

For a fixed dyadic stage, let $\mathcal{A}_L$ be the set of all uses of the shared attention block in the full unrolled computation, equivalently the set of all query instances, and let $\mathcal{T}_L \subseteq \mathcal{A}_L$ be the query instances in the intended dyadic computation tree. We use $u \in \mathcal{A}_L$ for an arbitrary query instance and $v \in \mathcal{T}_L$ for a target-tree query instance. Only the latter has a designated correct predecessor key $j_v^\star$.

A superscript $\star$ denotes a target quantity. Forward errors and margins are introduced in the forward-analysis section, and backpropagated sensitivities are introduced with the covariance identity.

7.C Base-task learning and forward accuracy (Stage I and forward Stage II)

Length two involves no routing decision: token 2 has a single strict-causal predecessor. Length three is therefore the smallest task on which routing is supervised, and it supplies the initial router for the dyadic curriculum. We prove that final-label backpropagation from $Q(0) = \mathbf{0}$ increases the logit of a matching positional key relative to a mismatching one.

7.C.1 Base-task learning argument

Base-task setup and forward pass

Fix an input $X = (m_{1,3}, \iota)$. Its label is

$$y^\star = m_3 \circ m_2 \circ m_1. \quad (7.C.1)$$

The desired computation has two routing decisions at token 3. At the first recurrent level, token 3 has query feature x_2 and should retrieve token 2, whose key is also x_2 . After copying token 2's left endpoint, its new query should match token 1 at the second recurrent level. The frozen MLP then composes m_3 with m_2 and subsequently with m_1 .

This desired route is a rule, not a unique finite ground-truth matrix. Along the symmetric family, letting $\gamma \rightarrow \infty$ approaches its hard equality-matching behavior. The initialized network is the opposite endpoint: at $\gamma(0) = 0$, both causal query instances at token 3 induce uniform attention distributions. The hidden states are therefore mixtures of desired and alternative computations. The analysis differentiates through those mixtures to determine the update to the shared matrix.

Fix a base-task input. To describe its forward pass under the symmetric router $\mathbf{Q} = \gamma \mathbf{\Pi}$, introduce the following one-parameter quantities. Let

$$r_1(\gamma) = \frac{e^\gamma}{1 + e^\gamma}. \quad (7.C.2)$$

Name the desired and alternative current states

$$h_{\text{cur}}^\star = m_3 \circ m_2, \quad h_{\text{cur}}^{\text{alt}} = m_3 \circ m_1,$$

and define the distribution $p(\gamma)$ by

$$\begin{aligned} p_\star(\gamma) &= \frac{e^{Br_1}}{Z_1}, & p_{\text{alt}}(\gamma) &= \frac{e^{B(1-r_1)}}{Z_1}, & p_{\text{other}}(\gamma) &= \frac{1}{Z_1}, \\ Z_1 &= e^{Br_1} + e^{B(1-r_1)} + d - 2. \end{aligned} \quad (7.C.3)$$

Here $p_\star$ indexes the desired length-two block state, p_{alt} indexes the state created by the wrong first retrieval, and each remaining group element receives the coordinate p_{other} .

For token 2's first-level state, set

$$\pi_{\text{comp}} = \frac{e^B}{e^B + d - 1} \quad \text{on } h_{\text{ret}}^{\text{alt}} = m_2 \circ m_1, \quad \pi_{\text{other}} = \frac{1}{e^B + d - 1} \quad \text{on each other element.} \quad (7.C.4)$$

Define

$$r_2(\gamma) = \frac{e^{\gamma r_1(\gamma)}}{1 + e^{\gamma r_1(\gamma)}}, \quad (7.C.5)$$

and define the retrieved-state distribution $q(\gamma)$ by

$$\begin{aligned} q_\star(\gamma) &= r_2 + (1 - r_2)\pi_{\text{other}} & \text{on } h_{\text{ret}}^\star &= m_1, \\ q_{\text{alt}}(\gamma) &= (1 - r_2)\pi_{\text{comp}} & \text{on } h_{\text{ret}}^{\text{alt}} &= m_2 \circ m_1, \\ q_{\text{other}}(\gamma) &= (1 - r_2)\pi_{\text{other}} & \text{on each other retrieved state.} \end{aligned} \quad (7.C.6)$$

Set

$$\mathbf{Z}_3^{[1]} = \frac{1}{2} (\mathbf{y}(\mathbf{p}) + \mathbf{g}(\mathbf{q})). \quad (7.C.7)$$

Define the restriction of the two-input logit map to this base path by

$$\widehat{F}_c(\gamma) := F_c(\mathbf{p}(\gamma), \mathbf{q}(\gamma)).$$

For $\sigma, \sigma' \in \{\star, \text{alt}\}$, also set

$$\Lambda_{\sigma, \sigma'} = B(p_\sigma + q_{\sigma'} - 1) + \tau p_\sigma. \quad (7.C.8)$$

To state the exact forward pass without collisions among its four leading routes, impose the following base-task condition.

Assumption 7.C.1 (Base-task output separation). For $\sigma, \sigma' \in \{\star, \text{alt}\}$, define the four route outputs

$$c^{\sigma, \sigma'} = h_{\text{cur}}^{\sigma} \circ h_{\text{ret}}^{\sigma'}.$$

Let $\mathcal{E}$ be the event that $h_{\text{cur}}^{\star} \neq h_{\text{cur}}^{\text{alt}}, h_{\text{ret}}^{\star} \neq h_{\text{ret}}^{\text{alt}}$, and the four elements $\{c^{\sigma, \sigma'} : \sigma, \sigma' \in \{\star, \text{alt}\}\}$ are pairwise distinct. We assume $\mathbb{P}(\mathcal{E}) > 0$.

For the additive group $G = \mathbb{Z}_d$ with odd prime $d \geq 5$, the excluded equalities are $m_2 = m_1$, $m_2 = 0$, $m_1 = 0$, and $2m_2 = m_1$. Each has probability $1/d$, and hence

$$\mathbb{P}(\mathcal{E}) \geq 1 - \frac{4}{d} > 0. \quad (7.C.9)$$

The condition only separates the four leading base routes into distinct output classes; the arbitrary-depth forward and covariance arguments use the unconditioned law in Definition 7.B.2.

Conditioning on this event, define the base population risk by

$$\mathcal{R}_{\text{base}}(\mathbf{Q}) := \mathbb{E}[-\log F(X)_{y^{\star}} \mid \mathcal{E}].$$

Its symmetric pullback and routing-margin growth rate are

$$\mathcal{R}_{\text{base}}^{\text{sym}}(\gamma) := \mathcal{R}_{\text{base}}(\gamma \mathbf{\Pi}), \quad V_{\text{base}}(\gamma) := -\kappa_{\text{sym}} \frac{d\mathcal{R}_{\text{base}}^{\text{sym}}(\gamma)}{d\gamma}.$$

Proposition 7.C.1 (Exact symmetric base-task forward pass). On the separation event $\mathcal{E}$ of Assumption 7.C.1, for every $\gamma \geq 0$, the two attention probabilities are $r_1(\gamma)$ and $r_2(\gamma)$, the current- and retrieved-state distributions are $\mathbf{p}(\gamma)$ and $\mathbf{q}(\gamma)$, and the final MLP input is (7.C.7). Its four leading pair-unit preactivations are (7.C.8), and the pair $(\star, \star)$ produces the label $y^{\star}$.

Proof. The first query's matching key has logit advantage γ , which gives (7.C.2). The alternative token receives probability $1 - r_1$. Thus $r_1(0) = 1/2$, while $r_1(\gamma) \rightarrow 1$ under ideal equality routing. Applying Proposition 7.B.1 to the two possible retrieved states gives (7.C.3).

Token 2 has only one predecessor, so exact attention and the frozen composition interface give (7.C.4). The first positional update at token 3 is

$$\mathbf{x}_{3, \text{qry}}^{[1]} = r_1 \mathbf{x}_1 + (1 - r_1) \mathbf{x}_0.$$

At the second recurrent level, token 1 is the correct predecessor. Its key $\mathbf{x}_1$ has matching coefficient r_1 , whereas token 2's key $\mathbf{x}_2$ has coefficient zero. This gives (7.C.5), and mixing the two predecessor-state distributions gives (7.C.6).

The concatenation rule gives (7.C.7). Finally, Proposition 7.B.1 applied to each leading current-retrieved pair gives (7.C.8). $\square$

Definition 7.C.1 (Depth-two input margins). Define

$$\Delta_{\text{cur}}(\gamma) = p_{\star}(\gamma) - p_{\text{alt}}(\gamma), \quad \Delta_{\text{ret}}(\gamma) = q_{\star}(\gamma) - q_{\text{alt}}(\gamma). \quad (7.C.10)$$

These are the target-versus-non-target margins in the current-state and retrieved-state halves of the final MLP input.

Corollary 7.C.1 (Exact leading pair-unit gaps). *The input margins in Definition 7.C.1 determine the leading-pair preactivation gaps by*

$$\Lambda_{\star,\star} - \Lambda_{\star,\text{alt}} = B\Delta_{\text{ret}}, \quad (7.C.11)$$

$$\Lambda_{\star,\star} - \Lambda_{\text{alt},\star} = (B + \tau)\Delta_{\text{cur}}, \quad (7.C.12)$$

$$\Lambda_{\star,\star} - \Lambda_{\text{alt},\text{alt}} = B\Delta_{\text{ret}} + (B + \tau)\Delta_{\text{cur}}. \quad (7.C.13)$$

Proof. Subtract the corresponding formulas in (7.C.8) and use (7.C.10). $\square$

These identities explain the scalar quantities used later: increasing either input margin raises the desired pair relative to a corresponding alternative.

Backward signal and positive base-stage dynamics

The separation condition in Assumption 7.C.1 ensures that the four leading routes do not collapse into the same output class. The following activation audit makes their contribution to output cross-entropy explicit.

Lemma 7.C.1 (Active hidden units at uniform attention). *Suppose the four leading pairs in (7.C.8) produce distinct output classes. At $\gamma = 0$ and for sufficiently large d , these four pairs are the only active pair-indexed units. Every other pair-indexed unit has strictly negative preactivation and therefore contributes neither to an output logit nor to its γ -derivative in a neighborhood of zero.*

Proof. The complete activation audit is in Section 7.E.2. Its key comparison is that the four leading preactivations equal $\tau/2 + o(\tau)$, whereas every nonleading pair has preactivation at most $-B/3 + \tau < 0$ for sufficiently large d . $\square$

Theorem 7.C.1 (Symmetry breaking at uniform attention). *Suppose Assumption 7.C.1 holds and $C_B > 8$. For sufficiently large d ,*

$$V_{\text{base}}(0) = \kappa_{\text{sym}} \left(\frac{B^2 + B}{16} + o(B^2) \right) > 0. \quad (7.C.14)$$

Consequently,

$$\left. \frac{d\gamma(t)}{dt} \right|_{t=0} > 0,$$

so the initial matrix update increases the logit of every matching positional pair relative to every mismatching pair.

Proof. The uniform-point derivatives are collected in Lemma 7.E.1. These derivatives identify the leading contribution to the routing update. At uniform attention, both routing probabilities equal $1/2$. Differentiating through the first attention distribution gives a state-mixture velocity of scale B , while differentiating the second gives a retrieved-state velocity of scale one. Substitution into (7.C.8) yields the four leading preactivation velocities in Lemma 7.E.1. The four units share smoothed-ReLU slope $1/2 + o(1)$, so the target-logit derivative is

$$\left. \frac{d}{d\gamma} \widehat{F}_{y^*} \right|_0 = \frac{B^2 + B}{16} + o(B^2).$$

All nonleading derivatives vanish by Lemma 7.C.1, and the output softmax average of the four active derivatives is $O(B^2/d) = o(B^2)$. Projecting the expected log-likelihood derivative onto the symmetric matrix direction proves (7.C.14). The complete derivative and smoothing-branch calculation appears in Section 7.E.2; only its leading sign is used subsequently. $\square$

The leading B^2 term records two successive uses of the composition module: the first converts a routing preference into a state preference, and the final pair unit reads that state preference. The second query instance contributes the lower-order B term. Both contributions update the same shared matrix direction. Equation (7.B.17) converts this static directional calculation into the time derivative of the learned routing margin.

Lemma 7.C.2 (Monotone depth-two input quantities). *For every finite $\gamma \geq 0$,*

$$\frac{d}{d\gamma} r_1(\gamma) > 0, \quad \frac{d}{d\gamma} r_2(\gamma) > 0, \quad \frac{d}{d\gamma} \Delta_{\text{cur}}(\gamma) > 0, \quad \frac{d}{d\gamma} \Delta_{\text{ret}}(\gamma) > 0.$$

Proof. The logistic map is strictly increasing, so (7.C.2) gives $\frac{d}{d\gamma} r_1 > 0$. The exponent in (7.C.5) is $\gamma r_1(\gamma)$, whose derivative is $r_1 + \gamma \frac{d}{d\gamma} r_1 > 0$; hence $\frac{d}{d\gamma} r_2 > 0$.

In (7.C.3), increasing r_1 raises the desired exponent $B r_1$ and lowers the alternative exponent $B(1 - r_1)$. Thus $p_\star$ increases and p_{alt} decreases. Equation (7.C.6) similarly shows that $q_\star$ increases and q_{alt} decreases with r_2 . The two input margins therefore increase with the routing margin. $\square$

The uniform-point calculation explains how training first breaks symmetry. A finite handoff additionally requires the same sign at every intervening routing margin. It is enough to prove the stronger pointwise fact that every target-versus-competitor output-logit gap increases with γ .

Lemma 7.C.3 (Pointwise growth of the base target-logit gaps). *Suppose Assumption 7.C.1 holds and $C_B > 8$. For sufficiently large d , every example in $\mathcal{E}$, every finite $\gamma \geq 0$, and every output class $c \neq y^\star$*

satisfy

$$\frac{d}{d\gamma}(\widehat{F}_{y^\star}(\gamma) - \widehat{F}_c(\gamma)) > 0. \quad (7.C.15)$$

Proof sketch. This sketch isolates the comparison that drives the argument; the detailed competitor-by-competitor and smoothing-branch audit appears in Section 7.E.2. Write $H_{a,b} = \phi_\tau(\Lambda_{a,b})$. By Lemma 7.C.2, the desired state and retrieval coordinates increase, while their leading alternatives decrease. The target pair therefore gains activation relative to each of the other three leading pairs. The derivative and smoothing-branch estimates in Lemma 7.E.2 strengthen these comparisons uniformly with the numerical constant $1/108$:

$$\min \left\{ \frac{dH_{\star,\star}}{d\gamma}, \min_{(\sigma,\sigma') \neq (\star,\star)} \frac{d}{d\gamma}(H_{\star,\star} - H_{\sigma,\sigma'}) \right\} \geq \frac{B}{108} \left(\frac{dr_1}{d\gamma} + \frac{dr_2}{d\gamma} \right) \quad (7.C.16)$$

for every $\gamma \geq 0$ and all sufficiently large d .

Only two nonleading pair types can have positive derivative: $(\star, \text{other})$ and $(\text{other}, \star)$. Their combined positive velocity is

$$o_d(1)B \left(\frac{dr_1}{d\gamma} + \frac{dr_2}{d\gamma} \right),$$

uniformly for $\gamma \geq 0$. Finally, the Latin-square property of the group table places at most one leading competitor and at most one unit of each growing nonleading type in any output class. The target class contains the target pair and none of those opposing nonleading types. Classwise aggregation therefore gives (7.C.15). The detailed competitor and smoothing-branch calculations are deferred to Section 7.E.2; their required conclusion is precisely the uniform inequality just displayed. $\square$

Theorem 7.C.2 (Base-stage progress). *Suppose Assumption 7.C.1 holds, $C_B > 8$, and d is sufficiently large. Then*

$$V_{\text{base}}(\gamma) > 0 \quad \text{for every finite } \gamma \geq 0. \quad (7.C.17)$$

Consequently, along the gradient-flow trajectory from $Q(0) = \mathbf{0}$,

$$\frac{d\gamma(t)}{dt} = V_{\text{base}}(\gamma(t)) > 0 \quad (7.C.18)$$

at every finite iterate. The trajectory reaches every prescribed finite routing-margin threshold Γ in finite optimization time.

Proof. For a fixed example, differentiating the final log softmax gives

$$\frac{d}{d\gamma} \log f_{y^\star} = \frac{d}{d\gamma} \widehat{F}_{y^\star} - \sum_c f_c \frac{d}{d\gamma} \widehat{F}_c = \sum_{c \neq y^\star} f_c \left(\frac{d}{d\gamma} \widehat{F}_{y^\star} - \frac{d}{d\gamma} \widehat{F}_c \right).$$

Every summand is positive by Lemma 7.C.3. Taking the conditional population expectation and multiplying by the positive symmetry factor κ_{sym} proves (7.C.17). On a compact interval $[0, \Gamma]$, continuity supplies a positive minimum $V_{\min}(\Gamma)$ of the routing-margin growth rate. Equation (7.B.17) therefore reaches Γ by a hitting time bounded by

$$T_{\Gamma} \leq \frac{\Gamma}{V_{\min}(\Gamma)}.$$

□

Along the base trajectory, the population risk decreases strictly. Until any finite threshold is reached, r_1 , r_2 , Δ_{cur} , and Δ_{ret} all increase strictly by Lemma 7.C.2.

7.C.2 From routing alignment to forward accuracy

This section determines a routing margin sufficient for accurate depth- L composition by controlling the following chain:

$$\begin{aligned} \text{routing margin} &\longrightarrow \text{attention error} \longrightarrow \text{positional alignment} \\ &\longrightarrow \text{retrieved-state concentration} \longrightarrow \text{final risk}. \end{aligned}$$

Positional alignment determines the next attention-logit gap, and the resulting attention bound controls the state mixture supplied to the frozen composition module.

Forward-error framework

For any length n and fixed routing margin γ , define the common attention-error envelope

$$\bar{e}_{\text{att}}(n, \gamma) := (n - 1)e^{-\gamma/2}. \quad (7.C.19)$$

Once the correct positional coordinate exceeds every other coordinate by at least $1/2$, this envelope bounds the total attention assigned to wrong keys. The proof first propagates that quantitative positional alignment, then uses the resulting attention bound to control the retrieved state, and finally invokes the frozen MLP's concentration reset. The separate arguments reflect the separate roles of these three network components.

Definition 7.C.2 (Target dyadic computation tree). For $n = 2^L$ and $0 \leq \ell \leq L$, define the level- ℓ target state nodes by

$$\mathcal{W}_{L,\ell} := \{(k2^\ell, \ell) : 1 \leq k \leq 2^{L-\ell}\}, \quad \mathcal{W}_L := \bigcup_{\ell=0}^L \mathcal{W}_{L,\ell}. \quad (7.C.20)$$

A level- ℓ state node $w = (i, \ell) \in \mathcal{W}_{L, \ell}$ represents the recurrent state $s_w := s_i^{[\ell]}$ and query-side positional feature $x_w := x_{i, \text{qry}}^{[\ell]}$ associated with the contiguous block

$$[i - 2^\ell + 1, i].$$

Its target block product is

$$y_w^\star := y_{i, \ell}^\star = m_i \circ m_{i-1} \circ \cdots \circ m_{i-2^\ell+1},$$

and its target positional feature is $x_w^\star := x_{i-2^\ell}$. The root state node is (n, L) . For $0 \leq \ell \leq L-1$, a non-leaf state node $(i, \ell+1)$ has current child (i, ℓ) and target predecessor child $(i-2^\ell, \ell)$.

Definition 7.C.3 (Target-tree query instances). For $0 \leq \ell \leq L-1$, define

$$\mathcal{T}_{L, \ell} := \{(k2^{\ell+1}, \ell) : 1 \leq k \leq 2^{L-\ell-1}\}, \quad \mathcal{T}_L := \bigcup_{\ell=0}^{L-1} \mathcal{T}_{L, \ell}. \quad (7.C.21)$$

Each query instance $(i, \ell) \in \mathcal{T}_{L, \ell}$ produces the non-leaf state node $(i, \ell+1)$ by consuming its two level- ℓ children. These are the *target-tree query instances*.

Thus state-node levels run from 0 through L , whereas query-instance levels run from 0 through $L-1$.

Lemma 7.C.4 (Target-computation invariant). *Suppose the attention distribution at every target-tree query instance assigns unit mass to its target predecessor and the state update applies the exact group operation $(a, b) \mapsto a \circ b$. Then every target-tree state node $(i, \ell) \in \mathcal{W}_L$ has state $y_{i, \ell}^\star$ and positional feature x_{i-2^ℓ} . In particular, the root state after L recurrent levels is $m_n \circ \cdots \circ m_1$.*

Proof. The base case follows from the token initialization (7.B.1) and the level-zero case of Definition 7.C.2. Assume the claim holds at level ℓ . The current child of $(i, \ell+1)$ represents the right block $[i-2^\ell+1, i]$ and has positional feature x_{i-2^ℓ} . The predecessor child ending at $i-2^\ell$ has positional key x_{i-2^ℓ} and represents the adjacent left block $[i-2^{\ell+1}+1, i-2^\ell]$. Target attention selects this child. The MLP composes the two block products in the order required by Definition 7.B.1, and the positional-feature update copies the predecessor's positional feature $x_{i-2^{\ell+1}}$. Induction proves the claim. $\square$

Let $v = (i, \ell) \in \mathcal{T}_L$ denote a target-tree query instance. In the consuming-state convention, its target quantities are

$$x_{v, \text{qry}}^\star = x_{i-2^\ell}, \quad y_v^\star = y_{i, \ell}^\star, \quad y_{v, \text{ret}}^\star = y_{i-2^\ell, \ell}^\star, \quad j_v^\star = i - 2^\ell.$$

Thus $y_v^\star$ is the current right-block product consumed by v , while $y_{v,\text{ret}}^\star$ is the adjacent left-block product it should retrieve. Query instances in $\mathcal{A}_L \setminus \mathcal{T}_L$ remain part of the full forward and backward computation graphs but are not assigned these target-tree quantities.

Every positional feature is a convex combination of dictionary vectors. For any query instance $u \in \mathcal{A}_L$, write

$$\mathbf{x}_{u,\text{qry}} = \sum_{\mathbf{x} \in \mathcal{X}} \omega_u(\mathbf{x}) \mathbf{x}, \quad \sum_{\mathbf{x} \in \mathcal{X}} \omega_u(\mathbf{x}) = 1, \quad \omega_u(\mathbf{x}) \geq 0.$$

By orthonormality, the matching coefficient of candidate key j is

$$\omega_{u,j} := \omega_u(\mathbf{x}_j) = \langle \mathbf{x}_{u,\text{qry}}, \mathbf{x}_j \rangle. \quad (7.C.22)$$

Let $\mathbf{s}_v$ be the current-state distribution and $\mathbf{a}_v$ the retrieved-state distribution. Define

$$e_v^{\text{att}} = 1 - \alpha_{v,j_v^\star}, \quad (7.C.23)$$

$$e_v^{\text{pos}} = 1 - \omega_v(\mathbf{x}_{v,\text{qry}}^\star), \quad (7.C.24)$$

$$\Delta_v^{\text{pos}} = \omega_v(\mathbf{x}_{v,\text{qry}}^\star) - \max_{\mathbf{x} \neq \mathbf{x}_{v,\text{qry}}^\star} \omega_v(\mathbf{x}), \quad (7.C.25)$$

$$e_v^{\text{state}} = 1 - [\mathbf{s}_v]_{y_v^\star}, \quad (7.C.26)$$

$$\Delta_v^{\text{act}} = [\mathbf{s}_v]_{y_v^\star} + [\mathbf{a}_v]_{y_{v,\text{ret}}^\star} - 1. \quad (7.C.27)$$

These quantities appear in the order used by the recurrent block. e_v^{pos} and Δ_v^{pos} describe whether the query still identifies the correct key; e_v^{att} is the resulting wrong-key mass; and e_v^{state} measures the current state entering the composition update. The remaining quantity Δ_v^{act} links the retrieved-state error to the nonlinear pair unit used by the frozen MLP. The quantity Δ_v^{act} is the activation margin of the target pair-indexed unit, since

$$\Lambda_{y_v^\star, y_{v,\text{ret}}^\star} = B \Delta_v^{\text{act}} + \tau [\mathbf{s}_v]_{y_v^\star}.$$

Fix a target risk tolerance $\varepsilon_{\text{risk}} > 0$ and define

$$\xi = \min \left\{ \frac{\varepsilon_{\text{risk}}}{16}, \frac{\tau}{16B}, \frac{1}{128} \right\}. \quad (7.C.28)$$

Definition 7.C.4 (Forward-stability conditions). The positional feature at a target query instance v is *aligned* if $\Delta_v^{\text{pos}} \geq 1/2$. A target state node w is ξ -*concentrated* if $[\mathbf{s}_w]_{y_w^\star} \geq 1 - \xi$.

These local errors and margins satisfy two elementary deterministic inequalities. If a positional feature has error at most e , its target coefficient is at least $1 - e$, while every individual non-target coefficient is at most the total non-target mass e . Hence

$$\Delta_v^{\text{pos}} \geq 1 - 2e_v^{\text{pos}}. \quad (7.C.29)$$

If both the current state and target predecessor state are ξ -concentrated and the attention error is e_{att} , then the target retrieved-state coordinate receives probability at least $(1 - e_{\text{att}})(1 - \xi)$. Therefore

$$\Delta_v^{\text{act}} \geq (1 - \xi) + (1 - e_{\text{att}})(1 - \xi) - 1 \geq 1 - e_{\text{att}} - 2\xi. \quad (7.C.30)$$

Thus positional-feature alignment controls the attention error, while a small attention error together with concentrated current and target predecessor states yields a positive target pair-unit activation margin.

The forward proof will produce a positive budget $b_{\text{fwd}}(\varepsilon_{\text{risk}}, L)$ such that

$$\bar{e}_{\text{att}}(2^L, \gamma) \leq b_{\text{fwd}}(\varepsilon_{\text{risk}}, L)$$

implies, simultaneously, that every target-tree query remains aligned, every target-tree attention error is at most $\bar{e}_{\text{att}}(2^L, \gamma)$, every target-tree state is ξ -concentrated, and the final cross-entropy is at most $\varepsilon_{\text{risk}}$.

Error propagation and completion

For $0 \leq \ell \leq L$, expand each node in the level set $\mathcal{W}_{L,\ell}$ from (7.C.20) as

$$\mathbf{x}_w = \sum_{\mathbf{x} \in \mathcal{X}} \omega_w(\mathbf{x}) \mathbf{x}$$

and set

$$e_w^{\text{pos}} := 1 - \omega_w(\mathbf{x}_w^*).$$

The corresponding query-level sets $\mathcal{T}_{L,\ell}$ are defined in (7.C.21). Define

$$E_\ell = \max_{w \in \mathcal{W}_{L,\ell}} e_w^{\text{pos}}. \quad (7.C.31)$$

Lemma 7.C.5 (Positional-feature error recursion). *Suppose $\gamma \geq 0$, $0 \leq \ell \leq L - 1$, $E_\ell \leq 1/4$, and the causal window contains at most $n - 1$ keys. Then every $v \in \mathcal{T}_{L,\ell}$ satisfies*

$$e_v^{\text{att}} \leq \bar{e}_{\text{att}}(n, \gamma). \quad (7.C.32)$$

The next-level positional-feature error satisfies

$$E_{\ell+1} \leq E_\ell + \bar{e}_{\text{att}}(n, \gamma). \quad (7.C.33)$$

Proof. By the elementary bound in (7.C.29), the current state node of each $v \in \mathcal{T}_{L,\ell}$ satisfies

$$\Delta_v^{\text{pos}} \geq 1 - 2E_\ell \geq \frac{1}{2}.$$

Under the two-logit parameterization, the logit difference between the target key and any non-target key is

$$\gamma(\omega_{v,j_v^*} - \omega_{v,j}) \geq \gamma \Delta_v^{\text{pos}} \geq \frac{\gamma}{2},$$

where (7.C.22) identifies the coordinate coefficients with the query–key inner products. Therefore

$$\begin{aligned} e_v^{\text{att}} &= \frac{\sum_{j \neq j_v^*} \exp(z_{v,j} - z_{v,j_v^*})}{1 + \sum_{j \neq j_v^*} \exp(z_{v,j} - z_{v,j_v^*})} \\ &\leq \sum_{j \neq j_v^*} e^{-\gamma/2} \leq (n-1)e^{-\gamma/2}, \end{aligned}$$

which proves (7.C.32).

The target predecessor is also a node in $\mathcal{W}_{L,\ell}$, so its positional feature has at least $1 - E_\ell$ mass on the next target positional feature. The attention distribution assigns that predecessor weight $1 - e_v^{\text{att}}$. Ignoring any helpful contribution from non-target predecessors, the target coefficient at the output node in $\mathcal{W}_{L,\ell+1}$ is at least

$$(1 - e_v^{\text{att}})(1 - E_\ell).$$

Hence

$$\begin{aligned} E_{\ell+1} &\leq 1 - (1 - e_v^{\text{att}})(1 - E_\ell) \\ &= E_\ell + e_v^{\text{att}} - e_v^{\text{att}} E_\ell \\ &\leq E_\ell + \bar{e}_{\text{att}}(n, \gamma), \end{aligned}$$

which is (7.C.33). □

Corollary 7.C.2 (Accumulated positional-feature error). *If*

$$L\bar{e}_{\text{att}}(n, \gamma) \leq \frac{1}{4},$$

then all target-tree positional features remain aligned and

$$E_\ell \leq \ell \bar{e}_{\text{att}}(n, \gamma) \quad (0 \leq \ell \leq L). \quad (7.C.34)$$

Proof. By (7.B.1), the level-zero positional features are exact. Iterating (7.C.33), the assumed bound keeps every intermediate error below 1/4, so the hypothesis of Lemma 7.C.5 remains valid and the induction is self-consistent. □

The positional recursion allows errors to add across levels. State errors have a more favorable behavior: the next technical lemma establishes an output-logit margin, and its corollary shows that the composition MLP returns every new state to the same concentration region.

Put

$$\theta := \frac{4B^2\xi^2}{\tau}. \quad (7.C.35)$$

Lemma 7.C.6 (Composition logit margin for concentrated inputs). *Suppose the current-state distribution $\mathbf{p}$ satisfies $p_{a_\star} \geq 1 - \xi$, and the retrieved-state distribution $\mathbf{q}$ satisfies*

$$q_{b_\star} \geq 1 - e_{\text{ret}}, \quad \max_{b \neq b_\star} q_b \leq e_{\text{ret}}, \quad e_{\text{ret}} \leq \frac{1}{4}. \quad (7.C.36)$$

For sufficiently large d , the target class $c_\star := a_\star \circ b_\star$ is separated from every non-target class by

$$F_{c_\star} - \max_{c \neq c_\star} F_c \geq \frac{3B}{4}(1 - 2e_{\text{ret}}) - \theta. \quad (7.C.37)$$

The resulting softmax confidence satisfies

$$[s^+]_{c_\star} \geq \frac{1}{1 + (d-1) \exp[-\frac{3B}{4}(1 - 2e_{\text{ret}}) + \theta]}. \quad (7.C.38)$$

The pair-unit estimates and their Latin-square aggregation are deferred to Section 7.E.1. The forward argument uses only the resulting one-step reset stated next.

Define the largest attention error that preserves state concentration:

$$b_{\text{state}}(\varepsilon_{\text{risk}}) = \min \left\{ \frac{1}{4} - \xi, \frac{1}{2} \left[1 - \frac{4}{3B} \left(\log \frac{d-1}{\xi} + \theta \right) \right] - \xi \right\}. \quad (7.C.39)$$

For every fixed $\varepsilon_{\text{risk}} > 0$ and $C_B > 8$, this budget is positive for all sufficiently large d . Indeed, eventually $\xi = \tau/(16B)$, $\theta = \tau/64$, and

$$\frac{4}{3B} \left(\log \frac{d-1}{\xi} + \theta \right) = \frac{2}{C_B} + o(1) < \frac{1}{4}.$$

The theorem below records positivity explicitly so that its finite- d hypothesis is unambiguous.

Corollary 7.C.3 (One-step reset of state concentration). *For all sufficiently large d , suppose the current state and target predecessor state are ξ -concentrated. If the target predecessor receives attention at least $1 - e_{\text{att}}$ with $e_{\text{att}} \leq b_{\text{state}}(\varepsilon_{\text{risk}})$, then the retrieved-state distribution satisfies (7.C.36) with $e_{\text{ret}} = e_{\text{att}} + \xi$, and the next state is ξ -concentrated.*

Proof. The target predecessor state assigns probability at least $1 - \xi$ to $b_\star$, and it receives attention probability at least $1 - e_{\text{att}}$. Hence

$$q_{b_\star} \geq (1 - e_{\text{att}})(1 - \xi) \geq 1 - (e_{\text{att}} + \xi).$$

Each non-target retrieved-state coordinate is at most the total error $e_{\text{att}} + \xi$. Thus Lemma 7.C.6 applies with $e_{\text{ret}} = e_{\text{att}} + \xi \leq 1/4$. Moreover, the second term in (7.C.39) gives

$$1 - 2e_{\text{ret}} \geq \frac{4}{3B} \left(\log \frac{d-1}{\xi} + \theta \right).$$

Substitution into (7.C.37) yields

$$F_{c_\star} - \max_{c \neq c_\star} F_c \geq \log \frac{d-1}{\xi}.$$

Equation (7.C.38) therefore gives

$$1 - [s^+]_{c_\star} \leq \frac{\xi}{1 + \xi} \leq \xi,$$

which is the claimed concentration reset. $\square$

The corollary is a one-step stability statement: small errors in the retrieved-state mixture do not accumulate directly in the state representation. Instead, each state update returns to the same ξ -concentrated region.

Define the forward budget

$$b_{\text{fwd}}(\varepsilon_{\text{risk}}, L) := \min \left\{ b_{\text{state}}(\varepsilon_{\text{risk}}), \frac{1}{4L} \right\}. \quad (7.C.40)$$

The state-reset condition uses $b_{\text{fwd}} \leq b_{\text{state}}$, while positional alignment uses $Lb_{\text{fwd}} \leq 1/4$. The progressive schedule below stops exactly when each active depth first reaches this forward budget.

Theorem 7.C.3 (Forward stability and accuracy at depth L). *Under the standing data and token initialization in Definitions 7.B.2 and 7.B.3, for all sufficiently large d , fix $\gamma \geq 0$ and evaluate the symmetric router $\mathcal{Q}_\gamma$. Assume $b_{\text{state}}(\varepsilon_{\text{risk}}) > 0$. If*

$$\bar{e}_{\text{att}}(2^L, \gamma) \leq b_{\text{fwd}}(\varepsilon_{\text{risk}}, L), \quad (7.C.41)$$

then every target-tree positional feature remains aligned, every target-tree state is ξ -concentrated, and

$$\mathcal{R}_L(\mathcal{Q}_\gamma) \leq \varepsilon_{\text{risk}}. \quad (7.C.42)$$

Proof. Let $e_{\text{att}} = \bar{e}_{\text{att}}(2^L, \gamma)$. Since every causal candidate-key set has size at most $2^L - 1$, Lemma 7.C.5 and Corollary 7.C.2 give

$$E_\ell \leq \ell e_{\text{att}} \leq L e_{\text{att}} \leq \frac{1}{4}.$$

Thus the attention-logit-gap induction closes at every level and every target-tree positional feature remains aligned.

It remains to induct over the levels of the target computation tree. By (7.B.1), the level-zero states are ξ -concentrated. Suppose the current and target predecessor states at one update are concentrated. The local attention error is at most

$$\bar{e}_{\text{att}}(2^L, \gamma) = e_{\text{att}} \leq b_{\text{fwd}}(\varepsilon_{\text{risk}}, L) \leq b_{\text{state}}(\varepsilon_{\text{risk}}).$$

By Corollary 7.C.3, the output state is also concentrated. Hence all target-tree states, including the root, satisfy the same bound.

At the root, the probability of $y^\star$ is at least $1 - \xi$. Since $\xi \leq 1/128$,

$$-\log(1 - \xi) \leq 2\xi \leq \frac{\varepsilon_{\text{risk}}}{8} < \varepsilon_{\text{risk}},$$

which proves (7.C.42). □

Corollary 7.C.4 (Explicit sufficient accuracy margin). *Under the assumptions of Theorem 7.C.3, it is sufficient that the symmetric routing margin satisfy*

$$\gamma \geq 2 \log \frac{2^L - 1}{b_{\text{fwd}}(\varepsilon_{\text{risk}}, L)}. \quad (7.C.43)$$

For fixed $\varepsilon_{\text{risk}}$, d , and C_B such that $b_{\text{state}}(\varepsilon_{\text{risk}}) > 0$, this sufficient margin has the asymptotic form

$$\gamma \geq 2L \log 2 + 2 \log L + O_{\varepsilon_{\text{risk}}, d, C_B}(1). \quad (7.C.44)$$

Proof. Substituting (7.C.43) into (7.C.19) gives $\bar{e}_{\text{att}}(2^L, \gamma) \leq b_{\text{fwd}}(\varepsilon_{\text{risk}}, L)$, so Theorem 7.C.3 applies. Finally, $2 \log(2^L - 1) = 2L \log 2 + O(1)$, while the $1/(4L)$ term in (7.C.40) contributes $2 \log L + O(1)$; the other budget term is absorbed in $O_{\varepsilon_{\text{risk}}, d, C_B}(1)$. □

The term $2L \log 2$ offsets the 2^L candidate keys in a causal window, while $2 \log L$ makes the per-step attention error small enough that positional-feature errors remain controlled across L recurrent updates. State-feature errors do not accumulate linearly because each composition step returns the output to the same concentration region.

7.D Deeper-stage positivity and progressive training (backward Stage II and transfer)

This section aggregates every query-wise contribution to the shared-matrix update, including mixed states outside the intended tree, and proves positivity of the deeper-stage routing-margin growth rate.

Theorem 7.D.1 (Positive deeper-stage update). *Fix $L_{\max} < \infty$ and $C_B > 8$. For sufficiently large d , there are positive coefficients A_L and a finite positive-update threshold Γ_{upd} , common to $2 \leq L \leq L_{\max}$, such that*

$$V_L(\gamma) \geq \frac{A_L}{2} e^{-\gamma} > 0 \quad (\gamma \geq \Gamma_{\text{upd}}). \quad (7.D.1)$$

Consequently, whenever the depth- L risk is active and $\gamma(t) \geq \Gamma_{\text{upd}}$,

$$\frac{d\gamma(t)}{dt} \geq \frac{A_L}{2} e^{-\gamma(t)} > 0. \quad (7.D.2)$$

Remark 7.D.1 (Dependence on the inherited checkpoint). *The guarantee applies on the symmetric family $\mathbf{Q}_\gamma = \gamma \mathbf{\Pi}$, which contains every checkpoint inherited by the progressive schedule. It does not transfer to an arbitrary parameter vector with the same scalar risk value or the same nominal routing statistic.*

The proof has four steps. First, an exact covariance identity sums the backward signal over every use of the shared matrix. Second, ideal equality routing isolates the positive root-query coefficient. Third, saturated MLP Jacobians control the remaining target-tree coefficient, while every off-tree influence crosses an additional wrong-key edge. Fourth, a finite-margin expansion makes the positive sign uniform over the fixed collection of depths. The theorem is proved after these ingredients are established.

7.D.1 Covariance and ideal-routing analysis

Deeper-stage covariance framework

Recall from (7.B.10) that $\mathcal{R}_L$ denotes the length- 2^L , depth- L population risk. At every global time t during this stage, the matrix and routing-margin dynamics are

$$\frac{d\mathbf{Q}(t)}{dt} = -\nabla_{\mathbf{Q}} \mathcal{R}_L(\mathbf{Q}(t)), \quad \frac{d\gamma(t)}{dt} = V_L(\gamma(t)). \quad (7.D.3)$$

Each stage begins from the matrix left by the preceding objective. The global matrix $\mathbf{Q}(t)$ is continuous at every boundary; only the data distribution, unrolled depth, and population risk change. The formal stage times are introduced with the curriculum in Section 7.D.3. The entire router—not merely its margin or its preceding risk value—is reused.

An *intermediate iterate* is a matrix $\mathbf{Q}(t_0)$ at a global time inside this stage, together with $\gamma_0 = \gamma(t_0)$ and the recurrent states produced by evaluating the depth- L network at that matrix. All local backward quantities below are evaluated at this iterate.

Unroll the depth- L recurrent computation into a finite acyclic graph. For query instance $u \in \mathcal{A}_L$ and candidate key $j \in \mathcal{K}_u$, use the query-key matching coefficient $\omega_{u,j}$ from (7.C.22). For a fixed symmetric router with routing margin γ , the local attention logit is a common offset across candidate keys plus $\gamma \omega_{u,j}$.

Let $f_{y^\star}$ denote the network's final probability of the label. The backpropagated sensitivity to one local attention weight is

$$\beta_{u,j} = \frac{\partial \log f_{y^\star}}{\partial \alpha_{u,j}}, \quad (7.D.4)$$

where the recurrent graph is fully unrolled before differentiation. Thus $\beta_{u,j}$ includes every downstream state and positional-feature effect of perturbing $\alpha_{u,j}$.

At this query instance, the local attention logit is a common offset plus $\gamma \omega_{u,j}$. Holding the query instance's recurrent inputs fixed while differentiating its attention distribution gives

$$\frac{\partial \alpha_{u,j}}{\partial \gamma} = \alpha_{u,j} \left(\omega_{u,j} - \sum_k \alpha_{u,k} \omega_{u,k} \right). \quad (7.D.5)$$

Multiplying this local attention-weight response by $\beta_{u,j}$ converts it into the effect of that query instance on the final log likelihood. Because every query instance uses the same $\mathbf{Q}$, the network-level update adds these effects over the entire unrolled graph.

Lemma 7.D.1 (Exact covariance identity). *For every example,*

$$\frac{d \log f_{y^\star}}{d\gamma} = \sum_{u \in \mathcal{A}_L} \sum_j \alpha_{u,j} \beta_{u,j} \left(\omega_{u,j} - \sum_k \alpha_{u,k} \omega_{u,k} \right), \quad (7.D.6)$$

$$= \frac{1}{2} \sum_{u \in \mathcal{A}_L} \sum_{j,k} \alpha_{u,j} \alpha_{u,k} (\beta_{u,j} - \beta_{u,k}) (\omega_{u,j} - \omega_{u,k}). \quad (7.D.7)$$

Consequently, the depth- L projected routing-margin growth function is

$$V_L(\gamma) := \kappa_{\text{sym}} \mathbb{E} \left[\frac{d \log f_{y^\star}}{d\gamma} \right] = -\kappa_{\text{sym}} \frac{d \mathcal{R}_L^{\text{sym}}(\gamma)}{d\gamma}. \quad (7.D.8)$$

Proof. Summing $\beta_{u,j} \frac{\partial \alpha_{u,j}}{\partial \gamma}$ from (7.D.5) over all $u \in \mathcal{A}_L$ gives (7.D.6). The indirect dependence of later query features on earlier values is already included in $\beta_{u,j}$ and is not differentiated a second time in this local partial derivative. The pairwise covariance identity in Lemma 7.F.3 gives (7.D.7); taking the population expectation and multiplying by κ_{sym} gives (7.D.8). $\square$

For any $u \in \mathcal{A}_L$, define

$$C_u = \frac{1}{2} \sum_{j,k} \alpha_{u,j} \alpha_{u,k} (\beta_{u,j} - \beta_{u,k}) (\omega_{u,j} - \omega_{u,k}). \quad (7.D.9)$$

Then

$$V_L(\gamma) = \kappa_{\text{sym}} \mathbb{E} \left[\sum_{v \in \mathcal{T}_L} C_v + \sum_{u \in \mathcal{A}_L \setminus \mathcal{T}_L} C_u \right]. \quad (7.D.10)$$

source	scale	role in the sign argument
root target-tree query instance	$e^{-\gamma} \chi_B$	positive dominant part of A_L
earlier target-tree query instances	$e^{-\gamma} o_d(\chi_B)$	same routing order, but suppressed by a saturated MLP Jacobian
finite-routing and off-tree corrections	$o_\gamma(e^{-\gamma})$	strictly higher order in the routing-margin expansion

Table 7.1: Asymptotic hierarchy of the contributions to the routing-margin growth rate.

The first sum contains the target-tree query instances for which correct-versus-wrong notation is defined. The second retains every off-tree query instance, including those with tied limiting matching coefficients, in the full backward graph. Gradient flow strengthens equality routing exactly when the complete projected update in (7.D.10) is positive.

Thus a query instance helps the target when keys with larger matching coefficients also have larger downstream sensitivities. The pairwise form is especially useful because it separates target–non-target comparisons from comparisons among non-target keys without dropping any update in the unrolled graph.

For the large-group comparison, define the one-step composition error and the saturated log-likelihood sensitivity scale by

$$\zeta_B = 1 - \pi_{\text{comp}} = \frac{d-1}{e^B + d - 1}, \quad \chi_B = \frac{Bd}{e^B + d - 1}. \quad (7.D.11)$$

The asymptotic hierarchy in Table 7.1 keeps the large-margin and large-group limits distinct.

Thus the root dominates the coefficient as $d \rightarrow \infty$. Earlier target-tree query instances can still contribute to the coefficient of $e^{-\gamma}$; the strictly faster decay in γ applies to deviations from the ideal-routing coefficient and to off-tree propagation. This distinction prevents the two asymptotic comparisons from being conflated.

The covariance formula suggests a simple mechanism. At a target-tree query instance, the target key has the largest matching coefficient. If it also has the largest backpropagated sensitivity, the target–non-target covariance is positive to first order in the attention error, whereas the signed comparison among non-target keys is only second order.

Remark 7.D.2 (A local bound without a sign). *At an intermediate iterate and target-tree query instance v , write $\star = j_v^\star$, $e_v = e_v^{\text{att}}$,*

$$\Delta_v^\beta := \min_{j \neq \star} (\beta_{v,\star} - \beta_{v,j}), \quad R_v^\beta := \max_{j,k \neq \star} |\beta_{v,j} - \beta_{v,k}|.$$

Splitting (7.D.9) into target–non-target and non-target–non-target pairs gives, whenever $\Delta_v^\beta > 0$,

$$C_v \geq (1 - e_v) e_v \Delta_v^\beta \Delta_v^{\text{pos}} - \frac{1}{2} R_v^\beta e_v^2. \quad (7.D.12)$$

The first term is the desired first-order update and the second is quadratic in the attention error. The hypothesis $\Delta_v^\beta > 0$, however, is a backward condition: forward bounds do not identify every mixed non-target state and cannot imply $\beta_{v,j_v^\star} > \beta_{v,j}$ for every wrong key. This local estimate is diagnostic and is not used later. The ideal-routing comparison below instead proves the sign after population aggregation and controls the remaining target-tree and off-tree effects.

The leading coefficient A_L

The learned network always has a finite routing margin, but its update is most transparent when compared with the limit $\gamma \rightarrow \infty$. This *ideal-routing reference* hardens only attention. It is not a finite ground-truth parameter matrix, and it retains the composition MLP and its output softmax. The coefficient A_L below is designed to be the limit of $e^\gamma V_L(\gamma)$; the subsequent expansion proves that interpretation.

Definition 7.D.1 (Ideal-routing reference). Fix a dyadic stage L and one example. A superscript ∞ denotes the pointwise limit as $\gamma \rightarrow \infty$ in the full recurrent graph. Thus s_u^∞ is the limiting current state consumed at query instance u and $\beta_{u,j}^\infty$ is the limiting backpropagated sensitivity. For every target-tree query instance v , the ideal-routing reference satisfies $\alpha_{v,j_v^\star}^\infty = 1$.

Definition 7.D.2 (Leading update coefficient). Define

$$A_L = \kappa_{\text{sym}} \mathbb{E} \left[\sum_{v \in \mathcal{T}_L} \sum_{j \neq j_v^\star} (\beta_{v,j_v^\star}^\infty - \beta_{v,j}^\infty) \right]. \quad (7.D.13)$$

Query instances outside the target ancestry can have tied maximal matching coefficients in the limit. Their limiting attention distribution is the limit of the finite- γ softmax, not an arbitrarily selected key. The common-rate lemma below proves, by induction over the finite forward and backward graphs, that every such quantity has a limit and approaches it at a positive exponential rate up to a polynomial factor in γ . The rate need not be an integer power of $e^{-\gamma}$: an off-tree query instance can have a unique limiting matching-coefficient gap strictly between zero and one. Target-tree query instances instead have unit limiting matching-coefficient gap and require no tie convention. The full rate induction is in Section 7.F.3; the argument at this point uses only the additional vanishing defect carried by every off-tree contribution.

Lemma 7.D.2 (Leading coefficient). Fix $L_{\max} < \infty$ and suppose $C_B > 8$. Uniformly over $2 \leq L \leq L_{\max}$, the coefficient in (7.D.13) satisfies

$$A_L = \kappa_{\text{sym}} (2^L - 2) \left(1 - \frac{1}{d} \right) \chi_B + o_d(\kappa_{\text{sym}} \chi_B). \quad (7.D.14)$$

Consequently, for sufficiently large d ,

$$A_L > 0 \quad (2 \leq L \leq L_{\max}). \quad (7.D.15)$$

Lemma 7.D.3 (Hard-routing target-state saturation). *Uniformly over $2 \leq L \leq L_{\max}$, every target-tree state node $w \in \mathcal{W}_L$ in the ideal-routing reference satisfies*

$$\|\mathbf{s}_w^\infty - \mathbf{e}_{y_w^\star}\|_1 = O_{L_{\max}, C_B}(\zeta_B). \quad (7.D.16)$$

Its proof is the Latin-square and softmax calculation in Section 7.F.2.

To keep the loss and log-likelihood signs explicit, write the composition output distribution and its single-example cross-entropy as

$$p_c^{\text{out}}(\mathbf{p}, \mathbf{r}) := [\text{softmax}((F_h(\mathbf{p}, \mathbf{r}))_{h \in G})]_c, \quad \ell_c^{\text{comp}}(\mathbf{p}, \mathbf{r}) := -\log p_c^{\text{out}}(\mathbf{p}, \mathbf{r}). \quad (7.D.17)$$

Definition 7.D.3 (Root retrieved-state sensitivity). For $a, b \in G$ and a candidate retrieved-state distribution $\mathbf{q} \in \Delta(G)$, let

$$S_{a,b}(\mathbf{p}, \mathbf{r}; \mathbf{q}) := D_2 \ell_{a \circ b}^{\text{comp}}(\mathbf{p}, \mathbf{r})[\mathbf{q} - \mathbf{r}], \quad (7.D.18)$$

where D_2 denotes the directional derivative with respect to the retrieved state. Thus $S_{a,b}$ measures the increase in final cross-entropy caused by shifting infinitesimal retrieved-state mass from $\mathbf{r}$ toward $\mathbf{q}$.

For $a, b \in G$, we retain the pair subscript $S_{a,b}$, rather than writing $S_{a \circ b}$, because the evaluation point $(\mathbf{e}_a, \mathbf{e}_b)$ uses a and b separately.

Lemma 7.D.4 (Stable root retrieved-state sensitivity). *At the exact composition input,*

$$S_{a,b}(\mathbf{e}_a, \mathbf{e}_b; \mathbf{q}) = \chi_B(1 - q_b). \quad (7.D.19)$$

For every fixed $C_0 > 0$ and all sufficiently large d , uniformly over $a, b \in G$ and $\mathbf{p}, \mathbf{r}, \mathbf{q} \in \Delta(G)$ satisfying

$$\|\mathbf{p} - \mathbf{e}_a\|_1 + \|\mathbf{r} - \mathbf{e}_b\|_1 \leq C_0 \zeta_B, \quad (7.D.20)$$

one has

$$S_{a,b}(\mathbf{p}, \mathbf{r}; \mathbf{q}) = \chi_B(1 - q_b) + O_{C_0}(B^2 d^2 \zeta_B^2). \quad (7.D.21)$$

The perturbation is negligible at the leading sensitivity scale because

$$\frac{B^2 d^2 \zeta_B^2}{\chi_B} = O(B d^3 e^{-B}) = o_d(1). \quad (7.D.22)$$

Proof. At the exact input (e_a, e_b) , Proposition 7.B.1 gives

$$p_{a \circ b}^{\text{out}}(e_a, e_b) = \frac{e^B}{e^B + d - 1}, \quad p_{a \circ h}^{\text{out}}(e_a, e_b) = \frac{1}{e^B + d - 1} \quad (h \neq b).$$

Set $r(s) = e_b + s(q - e_b)$. Differentiating the exact response (7.B.13) gives

$$\begin{aligned} \left. \frac{d}{ds} \log p_{a \circ b}^{\text{out}}(e_a, r(s)) \right|_{s=0} &= B(q_b - 1) - B \sum_{h \in G} p_{a \circ h}^{\text{out}}(e_a, e_b) (q_h - \mathbb{1}\{h = b\}) \\ &= B(q_b - 1) - \frac{B(e^B - 1)(q_b - 1)}{e^B + d - 1} \\ &= -\frac{Bd}{e^B + d - 1} (1 - q_b) = -\chi_B(1 - q_b). \end{aligned} \quad (7.D.23)$$

The middle equality uses $\sum_{h \neq b} q_h = 1 - q_b$, proving (7.D.19). $\square$

The perturbation estimate, including the two Hessian terms and the absolute-continuity argument at the smoothing breakpoints, is proved in Section 7.F.2.

Lemma 7.D.5 (Non-target root-key averaging). *Throughout this lemma, let $v = v_{\text{root}}$ with $v_{\text{root}} := (2^L, L - 1)$, let $a = y_v^*$ and $b = y_{v, \text{ret}}^*$, and, for a root candidate key j , abbreviate the level- $(L - 1)$ node state $s_{(j, L-1)}^\infty$ as s_j^∞ . For every non-target causal key j at the root, the ideal-routing reference satisfies*

$$\mathbb{E}[[s_j^\infty]_b] = \frac{1}{d}. \quad (7.D.24)$$

Proof. The hard state at key j is independent of at least one uniform group factor in b . If $j < 2^{L-1}$, the missing factor can be taken from the end of the target left half. If $j > 2^{L-1}$, hard equality-based attention through $L - 1$ recurrent steps uses only symbols in $[j - 2^{L-1} + 1, j]$, so the factor m_1 is missing. Conditioning on the non-target-key state and all remaining symbols therefore makes b uniform on G , which proves the claim. $\square$

Lemma 7.D.6 (Saturated hard-routing composition Jacobian). *For the composition state map $\mathcal{U}(p, q) := \text{softmax}((F_c(p, q))_{c \in G})$, every pair of target-tree inputs satisfying (7.D.16) has*

$$\|D\mathcal{U}(p, q)\|_{(\oplus, 1) \rightarrow 1} = O_{L_{\max}, C_B}(Bd\zeta_B), \quad (7.D.25)$$

where the simplex-tangent norms are defined in Section 7.F.2. The same bound holds for the transpose Jacobian in the dual norms used by backpropagation.

Its proof is deferred to Section 7.F.2. The leading-coefficient argument below uses only the displayed Jacobian order.

Proof sketch of Lemma 7.D.2. This sketch identifies the root contribution and the extra Jacobian on earlier queries. The detailed pathwise calculation appears in Section 7.F.2. Throughout this proof sketch, $v = v_{\text{root}} = (2^L, L - 1)$, $a = y_v^*$, $b = y_{v, \text{ret}}^*$, and, for a root candidate key j , we abbreviate

the level- $(L-1)$ node state $s_{(j,L-1)}^\infty$ as s_j^∞ . At the root query, Lemma 7.D.4 converts the sensitivity difference for a non-target key j into $\chi_B(1 - [s_j^\infty]_b) + o_d(\chi_B)$. Averaging with Lemma 7.D.5 and summing over the $2^L - 2$ non-target root keys gives the leading term in (7.D.14).

Every earlier target-tree contribution contains at least one additional composition-state Jacobian and is therefore smaller by $O_{L_{\max}, C_B}(Bd\zeta_B) = o_d(1)$. Query instances outside the hard target ancestry have zero limiting sensitivity. The full calculation is given in Section 7.F.2. $\square$

The leading term has a direct interpretation. At the root query instance, each non-target key receives an infinitesimal amount of attention. Its state agrees on average with the target retrieved-state coordinate only with mass $1/d$; the remaining mass reduces the target log likelihood at the composition-softmax sensitivity scale χ_B . Earlier deviations are suppressed by at least one additional saturated MLP Jacobian.

7.D.2 The finite-margin expansion

The ideal-routing coefficient is not itself a routing-margin growth rate: at the exact hard-attention limit, the attention softmax has zero derivative. In a trained network with large but finite γ , each wrong key still receives mass of order $e^{-\gamma}$. That small mass converts the ideal-routing sensitivity coefficient into the leading finite-network update. Throughout this subsection, write

$$\rho := e^{-\gamma}. \quad (7.D.26)$$

Definition 7.D.4 (Elementary unrolled graph and query output). For one example, let $\mathcal{G}_{L,\gamma}$ be the following fixed finite directed acyclic computation graph. Bundle the two recurrent variables at a token and level into the single node

$$h_i^{[\ell]} := (s_i^{[\ell]}, x_{i,\text{qry}}^{[\ell]}).$$

For a query instance $u = (i, \ell)$, the graph has the matching-coefficient vector node $\omega_u = (\omega_{u,j})_{j < i}$, the attention-vector node α_u , the retrieved-state node a_u , the copied-positional node $x_{u,\text{copy}}$, and the next recurrent node $h_i^{[\ell+1]}$. Their elementary maps are

$$\omega_{u,j} = \langle x_{i,\text{qry}}^{[\ell]}, x_j \rangle, \quad \alpha_u = \text{softmax}(\gamma \omega_u), \quad (7.D.27)$$

$$a_u = \sum_{j < i} \alpha_{u,j} s_j^{[\ell]}, \quad x_{u,\text{copy}} = \sum_{j < i} \alpha_{u,j} x_{j,\text{qry}}^{[\ell]}, \quad (7.D.28)$$

$$h_i^{[\ell+1]} = (\mathcal{U}(s_i^{[\ell]}, a_u), x_{u,\text{copy}}). \quad (7.D.29)$$

Here projections from a recurrent bundle onto its state and positional coordinates are included in the displayed maps. Thus, for each candidate j , there are separate direct edges $h_j^{[\ell]} \rightarrow a_u$ and $h_j^{[\ell]} \rightarrow x_{u,\text{copy}}$; there are also the current-input edge $h_i^{[\ell]} \rightarrow h_i^{[\ell+1]}$, the matching and

attention edges in (7.D.27), the two α_u -to-aggregate edges, and the aggregate-to-output edges in (7.D.29). Fixed-key, token-1 identity, and final output/log-likelihood maps are included as deterministic edges. Coordinate derivatives with respect to $\omega_{u,j}$ or $\alpha_{u,j}$ mean the corresponding column of the local Jacobian of this fixed vector-node graph; they do not change its node or edge set.

The graph structure is independent of γ ; only its node values and local Jacobians depend on γ . The phrase *output of query instance u* always means the single product-valued recurrent node

$$\text{out}(u) := h_i^{[\ell+1]},$$

not one unspecified member of its state and positional coordinates. The attention vector and the two aggregates are internal nodes of the query.

Definition 7.D.5 (Hard support graph). The *hard support graph* $\mathcal{G}_L^\infty$ is the spanning subgraph of $\mathcal{G}_{L,\gamma}$ obtained by deleting, for every $u = (i, \ell)$ and candidate j , the direct value edge $h_j^{[\ell]} \rightarrow a_u$ and the direct positional-copy edge $h_j^{[\ell]} \rightarrow x_{u,\text{copy}}$ when $\alpha_{u,j}^\infty = 0$. All other edges are retained, in particular the current-input, matching-coefficient, matching-to-attention, attention-to-aggregate, composition, projection, identity, and output edges. Existence of the hard attention limits, including at tied off-tree queries, is proved in Lemma 7.F.6; the target-tree limits used in the ancestry argument are already fixed by ideal equality routing. This is a forward-routing support graph, not the graph of nonzero limiting Jacobians: for example, a retained target-tree matching-to-attention edge has a Jacobian that tends to zero.

Definition 7.D.6 (Hard target ancestry). The *hard target ancestry* $\mathcal{H}_L$ consists of the nodes having a retained directed path in $\mathcal{G}_L^\infty$ to the final log likelihood, together with the retained edges on such paths. Thus membership of a query instance in $\mathcal{H}_L$ means membership of its unambiguous recurrent-output node $\text{out}(u)$.

Definition 7.D.7 (Off-tree query instance). A query instance u is *off-tree* if $\text{out}(u) \notin \mathcal{H}_L$.

Query instances and their attention branches have the following concrete description in the hard support graph.

Lemma 7.D.7 (Hard-ancestry query instances). *The query instances whose recurrent-output nodes belong to $\mathcal{H}_L$ are exactly the target-tree query instances $\mathcal{T}_L$. Hence the off-tree query instances are exactly $\mathcal{A}_L \setminus \mathcal{T}_L$.*

Proof. Trace the retained elementary edges backward from the final log likelihood. At a target-tree query $v = (i, \ell)$, the current-state coordinate of $h_i^{[\ell]}$ enters $\text{out}(v)$ deterministically, while the only recurrent candidate bundles entering its two aggregates through retained direct edges are those

with positive hard attention weight. Ideal routing at v gives unit weight to $j_v^\star$ and zero weight to every other candidate. The matching and attention nodes add no further recurrent bundle: their only recurrent parent is the current bundle $h_i^{[\ell]}$. Backward traversal from $\text{out}(v)$ therefore reaches exactly the two recurrent child bundles $h_i^{[\ell]}$ and $h_{j_v^\star}^{[\ell]}$.

Starting at the root and iterating this observation produces exactly the current and predecessor children in the dyadic recursion of Lemma 7.C.4. Their producing query instances are precisely $\mathcal{T}_L$. Conversely, each such producer has a retained path through its parent queries to the root. No other recurrent-output node can enter that backward traversal because every other candidate-to-aggregate edge at a target-tree query is deleted. The final claim follows from Definition 7.D.7. $\square$

Lemma 7.D.8 (Hard-ancestry boundary edges). *Let $v = (i, \ell) \in \mathcal{T}_L$. The two direct edges from the designated candidate bundle $h_{j_v^\star}^{[\ell]}$ to the retrieved-state and copied-positional aggregates are retained. For every $j \neq j_v^\star$, the direct value edge $h_j^{[\ell]} \rightarrow a_v$ and direct positional-copy edge $h_j^{[\ell]} \rightarrow x_{v, \text{copy}}$ are deleted. All current-input and internal matching, attention, and output edges of v are retained.*

Proof. Ideal routing assigns unit weight to $j_v^\star$ and zero weight to every $j \neq j_v^\star$. Every assertion therefore follows directly from Definitions 7.D.5 and 7.D.6. $\square$

Although an off-tree query instance remains present at finite routing margin, any path by which it affects the root must cross one of these deleted boundary edges.

Lemma 7.D.9 (Off-tree path cut on the elementary graph). *Every directed path in $\mathcal{G}_{L, \gamma}$ from the recurrent output $\text{out}(u)$ of an off-tree query instance u to the final log likelihood contains, at a later target-tree query instance v , either the direct value edge or the direct positional-copy edge of a candidate $j \neq j_v^\star$. Equivalently, the path contains an edge deleted from $\mathcal{G}_L^\infty$.*

More generally, if a directed path starts and ends in $\mathcal{H}_L$ but uses a node or edge outside that retained subgraph, then it either uses such a deleted edge with both endpoints in $\mathcal{H}_L$, or its first re-entry into $\mathcal{H}_L$ is through such a deleted edge.

Proof. For the first assertion, take the first path edge $e = (a, b)$ whose target lies in $\mathcal{H}_L$. Such an edge exists because the final log likelihood lies in $\mathcal{H}_L$. If e were retained, then concatenating e with the retained path from b to the loss would put a in $\mathcal{H}_L$, contrary to the choice of the first entry. Thus e is deleted. By Definition 7.D.5, the only deleted edges are direct candidate-to-retrieval and candidate-to-positional-copy edges with zero hard weight. Since their aggregate target lies in $\mathcal{H}_L$, its producing query v lies in $\mathcal{T}_L$ by Lemma 7.D.7; at such a query the zero-weight candidates are exactly $j \neq j_v^\star$. This proves the claimed cut.

For the second assertion, if the path uses a deleted edge whose endpoints both lie in $\mathcal{H}_L$, that edge is already the required cut. Otherwise choose a maximal subpath whose interior nodes lie outside $\mathcal{H}_L$ and apply the same first-re-entry argument to its last edge. The explicit edge list in Definitions 7.D.4 and 7.D.5 leaves no unclassified deterministic or attention-mediated re-entry. $\square$

Lemma 7.D.10 (Target-query direct and logit-mediated Jacobians). *Let $v \in \mathcal{T}_L$ and $j \neq j_v^\star$. In the elementary graph:*

1. *the direct edge from the candidate recurrent bundle to either the retrieved state or the copied positional feature has Jacobian $O(\rho)$;*
2. *the matching-to-attention edge satisfies*

$$\|D_{\omega_v} \alpha_v\| = O(\gamma \rho).$$

Consequently, a local route from the current positional coordinate through the matching coefficients and attention logits has Jacobian $O(\gamma \rho)$;

3. *the source edges from α_v to the two aggregates are bounded but do not carry an attention-weight factor. In particular, a derivative with respect to the coordinate $\alpha_{v,j}$ inserts the candidate state or positional feature itself.*

Proof. The target-tree expansion in Lemma 7.F.7 gives $\alpha_{v,j} = O(\rho)$ independently of the present cut argument. Equations (7.D.28) show, with P_s and P_x denoting the two coordinate projections, that

$$D_{h_j^{[l]}} \mathbf{a}_v = \alpha_{v,j} P_s, \quad D_{h_j^{[l]}} \mathbf{x}_{v,\text{copy}} = \alpha_{v,j} P_x.$$

These are $O(\rho)$. For the logit-mediated route, differentiating the query-wise softmax gives

$$\frac{\partial \alpha_{v,k}}{\partial \omega_{v,j}} = \gamma \alpha_{v,k} (\mathbf{1}_{\{j=k\}} - \alpha_{v,j}).$$

At a target-tree query, the total non-target mass is $O(\rho)$, so every entry, and hence the operator norm in the fixed finite dimension, is $O(\gamma \rho)$. The matching map in (7.D.27) has a bounded Jacobian, proving the combined-route assertion. Finally, (7.D.28) gives $D_{\alpha_v} \mathbf{a}_v = (\mathbf{s}_j^{[l]})_j$ and $D_{\alpha_v} \mathbf{x}_{v,\text{copy}} = (\mathbf{x}_{j,\text{qry}}^{[l]})_j$; compactness of the state and positional simplices proves the last item. $\square$

Corollary 7.D.1 (Off-tree cut factor). *Every pathwise forward influence from the recurrent output of an off-tree query instance to the final log likelihood contains a direct candidate-edge Jacobian of order $O(\rho)$, and hence one of order $O((1 + \gamma)\rho)$. Every directed path from the attention-vector node of an off-tree query to the final log likelihood passes through its recurrent-output node and contains the same later cut. A path that additionally traverses the matching-to-attention edge of a target-tree query has the independently specified $O(\gamma \rho)$ logit-mediated factor from Lemma 7.D.10.*

Proof. By Lemma 7.D.9, each path from the off-tree recurrent output crosses a deleted direct value or positional-copy edge of a later target-tree query instance. Its local bound follows from the first item of Lemma 7.D.10. By (7.D.28)–(7.D.29), every path from the query’s attention node first reaches that recurrent output. Reverse-mode differentiation traverses the transpose of these same local Jacobians. The final sentence of the corollary is the second item of Lemma 7.D.10. $\square$

The cut factor is counted once. A forward perturbation and its reverse-mode sensitivity are two descriptions of the same chain-rule transmission, not two independent factors of ρ . The strictly higher-order off-tree bound comes from multiplying this one cut factor by a separate local leakage or splitting defect.

Remark 7.D.3 (Length-eight cut example). *For $L = 3$, the hard target ancestry contains the query instances*

$$(8, 2), \quad (8, 1), (4, 1), \quad (8, 0), (6, 0), (4, 0), (2, 0).$$

The query instance $u = (7, 1)$ is off-tree: under hard equality routing it constructs the length-four block ending at token 7, rather than one of the two length-four blocks used by the root. Its level-2 recurrent-output bundle can affect the root query instance $v = (8, 2)$ only through the direct value or positional-copy edge of the root's non-target candidate key 7; the target predecessor key is $j_v^ = 4$. Hence*

$$\alpha_{v,7} = \rho + O((1 + \gamma)\rho^2).$$

The cut argument shows that $\text{Sens}_u(\gamma)$ contains an additional order- ρ factor, before the remaining finite path Jacobians are applied. If u has a unique limiting maximizer, its local covariance also contains the leakage outside that maximizer. If it has tied maximizers, it instead contains their vanishing matching-coefficient split. In either case the local defect converges exponentially up to polynomial factors in γ . The next lemma therefore shows that this query instance contributes $o_\gamma(\rho)$, strictly below the root query instance's order- ρ contribution.

There are only two ways for an off-tree query instance to have a nonzero local covariance at large routing margin. Its attention distribution can leak outside the limiting maximal-key set, or the matching coefficients of keys inside a tied maximal set can remain slightly unequal. The following algebraic lemma treats both effects without splitting off-tree query instances into separate cases.

Fix $u \in \mathcal{A}_L \setminus \mathcal{T}_L$ and let

$$\mathbf{M}_u := \arg \max_j \omega_{u,j}^\infty \tag{7.D.30}$$

be its limiting maximal-key set. Define

$$\text{Leak}_u(\gamma) := \sum_{j \notin \mathbf{M}_u} \alpha_{u,j}, \tag{7.D.31}$$

$$\text{Split}_u(\gamma) := \max_{j,k \in \mathbf{M}_u} |\omega_{u,j} - \omega_{u,k}|, \tag{7.D.32}$$

$$\text{Sens}_u(\gamma) := \max_{j,k} |\beta_{u,j} - \beta_{u,k}|. \tag{7.D.33}$$

The maximum in (7.D.32) is zero when $|\mathbf{M}_u| = 1$.

Lemma 7.D.11 (Off-tree covariance classification). *Every off-tree query instance satisfies*

$$|C_u| \leq \text{Sens}_u(\gamma) \left(\text{Leak}_u(\gamma) + \frac{1}{2} \text{Split}_u(\gamma) \right). \tag{7.D.34}$$

Proof. Split the ordered pairs in (7.D.9) according to whether both indices lie in M_u . Positional matching coefficients belong to $[0, 1]$. The total absolute contribution from pairs outside M_u^2 is therefore at most

$$\begin{aligned} & \frac{1}{2} \text{Sens}_u(\gamma) \sum_{(j,k) \notin M_u^2} \alpha_{u,j} \alpha_{u,k} \\ &= \frac{1}{2} \text{Sens}_u(\gamma) \left[1 - (1 - \text{Leak}_u(\gamma))^2 \right] \leq \text{Sens}_u(\gamma) \text{Leak}_u(\gamma). \end{aligned}$$

For $j, k \in M_u$, the matching-coefficient difference is at most $\text{Split}_u(\gamma)$, while $\sum_{j,k \in M_u} \alpha_{u,j} \alpha_{u,k} \leq 1$. These pairs contribute at most

$$\frac{1}{2} \text{Sens}_u(\gamma) \text{Split}_u(\gamma).$$

Adding the two estimates proves (7.D.34). $\square$

Thus a query instance with a unique limiting maximizer has only a leakage term, a query instance whose candidate keys are all limiting maximizers has only a splitting term, and a query instance with tied maximal matching coefficients and lower-scoring keys has both.

Fix finite $L_{\max}$ and d . In the next two lemmas, all estimates are uniform over $2 \leq L \leq L_{\max}$, all examples, and all applicable query instances as $\gamma \rightarrow \infty$. In accordance with the standing convention, their hidden constants are written with the sole dependence subscript d and are independent of the stage, example, query instance, and γ .

Lemma 7.D.12 (Target-tree convergence rate). *At every target-tree query instance, each non-target key satisfies*

$$\alpha_{v,j} = \rho + O_d((1 + \gamma)\rho^2), \quad \alpha_{v,j^*} = 1 - (|\mathcal{K}_v| - 1)\rho + O_d((1 + \gamma)\rho^2). \quad (7.D.35)$$

At the same query instance, the backward sensitivities obey

$$\beta_{v,j} = \beta_{v,j}^\infty + o_\gamma(1), \quad (7.D.36)$$

for every candidate key j .

Proof. The attention expansion is Lemma 7.F.7, and the sensitivity expansion is the pathwise conclusion of Lemma 7.F.10. The uniform common choice of constants is proved in Section 7.F.3; only the resulting common first-order expansion is needed here. $\square$

Lemma 7.D.13 (Off-tree convergence rate). *Every off-tree query instance satisfies*

$$\text{Sens}_u(\gamma) (\text{Leak}_u(\gamma) + \text{Split}_u(\gamma)) = o_\gamma(\rho). \quad (7.D.37)$$

Proof. The local-defect estimate from Lemma 7.F.6 converges exponentially up to polynomial factors in γ . The off-tree sensitivity estimate from Lemma 7.F.11 contains the additional cut factor

ρ , also up to polynomial factors. The uniform rate choice is deferred to Section 7.F.3; its product estimate gives (7.D.37). $\square$

Theorem 7.D.2 (Deeper-stage expansion). *Fix $L_{\max} < \infty$ and $C_B > 8$. For sufficiently large d , uniformly over $2 \leq L \leq L_{\max}$ as $\gamma \rightarrow \infty$,*

$$V_L(\gamma) = A_L \rho + o_\gamma(\rho). \quad (7.D.38)$$

Consequently,

$$\lim_{\gamma \rightarrow \infty} e^\gamma V_L(\gamma) = A_L > 0 \quad (2 \leq L \leq L_{\max}), \quad (7.D.39)$$

uniformly over this finite set of stages.

Proof. This proof records the short covariance-level argument. The common-rate induction and complete term-by-term expansion are given in Section 7.F.3. Insert the target-tree attention and sensitivity expansions from Lemma 7.D.12 into the covariance identity (7.D.7). Target–non-target pairs contribute ρA_L ; non-target–non-target pairs have order $O((1 + \gamma)\rho^2)$.

For off-tree instances, Lemmas 7.D.11 and 7.D.13 combine the cut factor with an independent leakage or splitting defect, giving the strictly higher-order remainder in (7.D.38). The example-wise estimates are uniform before taking the population expectation, and stage uniformity follows by taking common constants over the finite collection of depths. The full covariance expansion is given in Section 7.F.3. $\square$

The theorem controls the full backward computation graph, not only the target-tree query instances. The ideal-routing reference supplies the leading coefficient; every difference between the finite trained network and that reference enters a remainder that is uniformly smaller than the leading order ρ term.

Proof of Theorem 7.D.1. The scaled limit and positivity of A_L follow from Lemma 7.D.2 and Theorem 7.D.2. There are finitely many positive coefficients A_L . The uniform little- o remainder in (7.D.38) therefore gives a common finite threshold Γ_{upd} such that

$$|V_L(\gamma) - A_L \rho| \leq \frac{A_L}{2} \rho$$

for every stage and every $\gamma \geq \Gamma_{\text{upd}}$. Subtracting the remainder in (7.D.38) from its positive leading term gives (7.D.1). The warm-started scalar-flow identity in Corollary 7.G.2 then gives (7.D.2) for every symmetrically initialized active stage. $\square$

quantity	notation	guarantee
update threshold	Γ_{upd}	every dyadic risk has positive routing-margin update above this threshold
forward budget	b_K	attention-envelope level used by the forward theorem and stage stopping rule
forward threshold	$\Gamma_{\text{fwd},K}$	margin at which the depth- K envelope reaches b_K
base handoff	Γ_{base}	the base risk is small and the update threshold is reached

Table 7.2: Budgets and thresholds used in the progressive training schedule.

7.D.3 Progressive training with warm starts

Assume throughout this section that $L_{\max} \geq 2$ and $b_{\text{state}}(\varepsilon_{\text{risk}}) > 0$. Write $b_K = b_{\text{fwd}}(\varepsilon_{\text{risk}}, K)$ and use the shorthand

$$\Gamma_{\text{fwd},K} := 2 \log \frac{2^K - 1}{b_K}, \quad (2 \leq K \leq L_{\max}). \quad (7.D.40)$$

This is the routing margin at which the depth- K attention envelope reaches its forward budget.

The positive-update theorem supplies at least one common tail threshold. For the full progressive result, assume that an admissible choice can be made with

$$\Gamma_{\text{upd}} < \Gamma_{\text{fwd},2}. \quad (7.D.41)$$

This compatibility is a separate finite-margin hypothesis and is not implied by the sufficiently-large- d assumption.

The quantities and their roles are summarized in Table 7.2.

Threshold ordering and stage hitting

For sufficiently large d , Theorem 7.C.3 shows that $\gamma \geq \Gamma_{\text{fwd},K}$ implies $\mathcal{R}_K(Q_\gamma) \leq \varepsilon_{\text{risk}}$.

Lemma 7.D.14 (Nested forward thresholds). *For every $2 \leq K < L_{\max}$, the budgets satisfy $b_K \geq b_{K+1}$ and*

$$\bar{e}_{\text{att}}(2^K, \Gamma_{\text{fwd},K}) = b_K, \quad (7.D.42)$$

$$\bar{e}_{\text{att}}(2^{K+1}, \gamma) = r_K \bar{e}_{\text{att}}(2^K, \gamma), \quad (7.D.43)$$

$$r_K := \frac{2^{K+1} - 1}{2^K - 1} = 2 + \frac{1}{2^K - 1} \in (2, 3]. \quad (7.D.44)$$

Consequently,

$$\bar{e}_{\text{att}}(2^{K+1}, \Gamma_{\text{fwd},K}) = r_K b_K > b_{K+1}, \quad (7.D.45)$$

$$\Gamma_{\text{fwd},K+1} - \Gamma_{\text{fwd},K} = 2 \log \left(r_K \frac{b_K}{b_{K+1}} \right) > 0. \quad (7.D.46)$$

Indeed, the threshold gap is greater than $2 \log 2$. Thus the next stage's stopping condition is not met at the handoff. This failure of a sufficient condition does not determine the true risk at entry.

Proof. The definition of b_K is the minimum of a K -independent term and $1/(4K)$, so it is nonincreasing. Direct substitution proves (7.D.42). Multiplication by $b_K \geq b_{K+1}$ gives (7.D.45); taking logarithms gives (7.D.46). Since $r_K > 2$ and $b_K/b_{K+1} \geq 1$, the gap is greater than $2 \log 2$. $\square$

For the base task, choose the explicit handoff

$$\Gamma_{\text{base}} := \max\{\Gamma_{\text{upd}}, 2 \log(2/b_{\text{fwd}}(\varepsilon_{\text{risk}}, 2))\}. \quad (7.D.47)$$

The base stage therefore makes its own risk small and enters the common positive-update region.

Definition 7.D.8 (Progressive training schedule). The progressive training schedule is the following population training procedure.

- (i) Initialize the single global matrix by $\mathbf{Q}(0) = \mathbf{0}$ and train with the base risk according to

$$\frac{d\mathbf{Q}(t)}{dt} = -\nabla_{\mathbf{Q}} \mathcal{R}_{\text{base}}(\mathbf{Q}(t)) \quad (0 \leq t < T_{\text{base}}), \quad T_{\text{base}} = \inf\{t \geq 0 : \gamma(t) = \Gamma_{\text{base}}\}.$$

- (ii) Set $S_2 := T_{\text{base}}$ and $S_L := T_{L-1}$ for $3 \leq L \leq L_{\text{max}}$. On each interval $S_L \leq t < T_L$, use the depth- L risk and define its forward-envelope stopping time by

$$\frac{d\mathbf{Q}(t)}{dt} = -\nabla_{\mathbf{Q}} \mathcal{R}_L(\mathbf{Q}(t)), \quad T_L = \inf\{t \geq S_L : \bar{e}_{\text{att}}(2^L, \gamma(t)) \leq b_{\text{fwd}}(\varepsilon_{\text{risk}}, L)\}.$$

If $L < L_{\text{max}}$, activate the depth- $(L+1)$ risk and unrolled graph at T_L while keeping the same matrix and routing margin.

Theorem 7.D.3 (Progressive training across recurrent depths). Assume $L_{\text{max}} \geq 2$, Assumption 7.C.1, and $C_B > 8$. For every sufficiently large d for which $b_{\text{state}}(\varepsilon_{\text{risk}}) > 0$ and an admissible common Γ_{upd} satisfying (7.D.41) have been fixed, every stopping time in Definition 7.D.8 is finite, and no matrix is reinitialized at a switch. Moreover,

$$\mathcal{R}_{\text{base}}(\mathbf{Q}(T_{\text{base}})) \leq \varepsilon_{\text{risk}}, \quad \mathcal{R}_L(\mathbf{Q}(T_L)) \leq \varepsilon_{\text{risk}} \quad (2 \leq L \leq L_{\text{max}}).$$

Every dyadic stage has positive duration and satisfies

$$\Gamma_{\text{upd}} \leq \gamma(S_L) < \gamma(T_L) = \Gamma_{\text{fwd}, L}, \quad \bar{e}_{\text{att}}(2^L, \gamma(T_L)) = b_L. \quad (7.D.48)$$

Its true risk and deterministic attention envelope decrease strictly throughout the stage. While depth K is active, reevaluating every completed shallower task $j < K$ under the updated shared matrix yields

valid positional- and state-error bounds that decrease strictly, and its completed-task risk also decreases strictly.

Lemma 7.D.15 (Base handoff bounds). *For every sufficiently large d with a fixed admissible common Γ_{upd} satisfying (7.D.41),*

$$\mathcal{R}_{\text{base}}^{\text{sym}}(\Gamma_{\text{base}}) \leq \varepsilon_{\text{risk}}, \quad \Gamma_{\text{upd}} \leq \Gamma_{\text{base}} < \Gamma_{\text{fwd},2}. \quad (7.D.49)$$

Proof. The proof of Theorem 7.C.3 applies to the length-three target chain with two query instances. At Γ_{base} its uniform attention error obeys

$$\bar{e}_{\text{att}}(3, \Gamma_{\text{base}}) = 2e^{-\Gamma_{\text{base}}/2} \leq b_2.$$

The positional induction, state-concentration reset, and final root estimate therefore give the base-risk bound pointwise and hence under the conditional length-three law.

The definition of Γ_{base} gives its lower bound. Finally, $2 \log(2/b_2) < 2 \log(3/b_2) = \Gamma_{\text{fwd},2}$ together with compatibility gives the strict upper bound in (7.D.49). $\square$

Lemma 7.D.16 (One-stage forward-envelope hitting). *Under the standing assumptions of this section, the token initialization in Definition 7.B.3, and the conditions of Theorem 7.D.1, initialize an active depth- K stage at $\mathbf{Q}(S_K) = \gamma_K^- \mathbf{\Pi}$ with*

$$\Gamma_{\text{upd}} \leq \gamma_K^- < \Gamma_{\text{fwd},K}. \quad (7.D.50)$$

Until the forward-envelope threshold is reached,

$$\frac{d\gamma(t)}{dt} = V_K(\gamma(t)) \geq \frac{A_K}{2} e^{-\gamma(t)} > 0, \quad \frac{d}{dt} \bar{e}_{\text{att}}(2^K, \gamma(t)) = -\frac{1}{2} \bar{e}_{\text{att}}(2^K, \gamma(t)) V_K(\gamma(t)) < 0, \quad (7.D.51)$$

and

$$\frac{d}{dt} \mathcal{R}_K(\mathbf{Q}(t)) = -\frac{V_K(\gamma(t))^2}{\kappa_{\text{sym}}} < 0. \quad (7.D.52)$$

The stopping time at $\Gamma_{\text{fwd},K}$ is finite and obeys

$$T_K - S_K \leq \frac{2}{A_K} (e^{\Gamma_{\text{fwd},K}} - e^{\gamma_K^-}), \quad (7.D.53)$$

where the endpoint satisfies (7.D.48) and $\mathcal{R}_K(\mathbf{Q}(T_K)) \leq \varepsilon_{\text{risk}}$.

Proof. The warm-started symmetry reduction gives $\mathbf{Q}(t) = \gamma(t) \mathbf{\Pi}$ and $\dot{\gamma} = V_K(\gamma)$. Since $\gamma_K^- \geq \Gamma_{\text{upd}}$, (7.D.1) proves the first inequality in (7.D.51). Direct differentiation gives the envelope derivative, and $V_K = -\kappa_{\text{sym}}(\mathcal{R}_K^{\text{sym}})'$ gives (7.D.52). Moreover,

$$\frac{d}{dt} e^{\gamma(t)} = e^{\gamma(t)} V_K(\gamma(t)) \geq \frac{A_K}{2}.$$

Integration proves (7.D.53). Continuity and strict monotonicity give the endpoint envelope equality, and Theorem 7.C.3 then gives the endpoint risk bound. $\square$

Remark 7.D.4 (Quantitative compatibility for the full schedule). *The positive-update theorem supplies a tail region, but its threshold is not canonical: any larger value is also admissible. The hypothesis (7.D.41) requires the existence and choice of an admissible common value below the first forward-envelope threshold; it is not a statement about every enlarged choice. Without this additional finite-margin compatibility, the present argument does not prove that the depth-two stage starts in the positive-update region while still lying below its own stopping target.*

Cross-depth backward feature correction

Suppose depth K is active and depth $j < K$ is already complete. The schedule has $\gamma(S_K) \geq \Gamma_{\text{fwd},j}$, so reevaluating the completed depth- j task under the updated shared matrix remains in its valid regime. Use its fixed sequence length $n_j = 2^j$ to define the deterministic positional-feature error bound

$$\bar{e}_{\ell j}^{\text{pos}}(\gamma) = \ell \bar{e}_{\text{att}}(n_j, \gamma). \quad (7.D.54)$$

Define the one-step state-feature error bound

$$\bar{e}_j^{\text{state}}(\gamma) = (d-1) \exp \left[-\frac{3B}{4} (1 - 2(\bar{e}_{\text{att}}(n_j, \gamma) + \xi)) + \theta \right]. \quad (7.D.55)$$

The first follows from Lemma 7.C.5 and Corollary 7.C.2. For the second, the target-coordinate error of the retrieved-state distribution in Lemma 7.C.6 is at most $\bar{e}_{\text{att}}(n_j, \gamma) + \xi$.

This deterministic error-bound improvement is termed *backward feature correction* (BFC). Its precise mechanism is *recursive error refinement*: information obtained at greater recurrent depth sharpens the available control of feature errors on completed shallower tasks. More precisely, this means a strict simultaneous decrease of their levelwise positional-feature and state-feature error bounds. It does not assert pointwise monotonicity of every realized hidden state.

Corollary 7.D.2 (Cross-depth deterministic BFC). *For every sufficiently large d under the standing assumptions and compatibility condition of this section, fix an active depth K and a completed depth $2 \leq j < K \leq L_{\text{max}}$ along the schedule in Definition 7.D.8. If $t_1 < t_2$ lie in $[S_K, T_K]$, then $\gamma(t_1) < \gamma(t_2)$ and*

$$\bar{e}_{\ell j}^{\text{pos}}(\gamma(t_2)) < \bar{e}_{\ell j}^{\text{pos}}(\gamma(t_1)), \quad \bar{e}_j^{\text{state}}(\gamma(t_2)) < \bar{e}_j^{\text{state}}(\gamma(t_1)), \quad \mathcal{R}_j(\mathbf{Q}(t_2)) < \mathcal{R}_j(\mathbf{Q}(t_1)) \quad (7.D.56)$$

for every $1 \leq \ell \leq j$. These are bounds for reevaluating the completed depth- j task with its own length 2^j . The statement does not make the earlier levels inside the active length- 2^K unroll valid before T_K .

Proof. Threshold nesting gives $\gamma(S_K) = \Gamma_{\text{fwd},K-1} \geq \Gamma_{\text{fwd},j} > \Gamma_{\text{upd}}$. Hence the depth- j forward theorem applies at entry and remains valid as the margin increases. The positional recursion and

state-concentration reset make (7.D.54)–(7.D.55) valid upper bounds on the completed target-tree errors throughout the interval. The active depth- K update gives $\dot{\gamma} = V_K(\gamma) > 0$. Differentiating (7.D.54) with respect to optimization time gives

$$\frac{d}{dt} \bar{e}_{\ell|j}^{\text{pos}}(\gamma(t)) = -\frac{1}{2} \bar{e}_{\ell|j}^{\text{pos}}(\gamma(t)) \frac{d\gamma(t)}{dt} < 0.$$

The state-feature bound obeys

$$\frac{d}{dt} \bar{e}_j^{\text{state}}(\gamma(t)) = -\frac{3B}{4} (n_j - 1) e^{-\gamma(t)/2} \bar{e}_j^{\text{state}}(\gamma(t)) \frac{d\gamma(t)}{dt} < 0.$$

Both bounds therefore decrease throughout the training interval. Finally, $V_j(\gamma) > 0$ in the same common tail region, so

$$\frac{d}{dt} \mathcal{R}_j(Q(t)) = -\frac{V_j(\gamma(t))V_K(\gamma(t))}{\kappa_{\text{sym}}} < 0.$$

□

The length-three theorem supplies the base handoff, the one-stage lemma supplies the stage induction, and the BFC corollary supplies strict tightening for completed shallower tasks throughout each deeper stage.

Proof of Theorem 7.D.3. By Theorem 7.C.2, the length-three trajectory reaches the finite routing margin Γ_{base} in finite time, and Lemma 7.D.15 gives base accuracy and the full depth-two entry range $\Gamma_{\text{upd}} \leq \Gamma_{\text{base}} < \Gamma_{\text{fwd},2}$. The procedure carries $Q(T_{\text{base}}) = \Gamma_{\text{base}} \mathbf{\Pi}$ into the first dyadic stage. The warm-start-capable symmetry statement in Corollary 7.G.2 keeps the newly active trajectory in the same one-margin family. The one-stage lemma therefore gives a positive-duration depth-two stage, strict loss and attention-envelope decrease, and a finite hit of $\Gamma_{\text{fwd},2}$.

Inductively, suppose stage K ends at $\gamma(T_K) = \Gamma_{\text{fwd},K}$. Exact weight reuse initializes stage $K+1$ with that same margin. The nested-threshold lemma gives

$$\Gamma_{\text{upd}} < \gamma(S_{K+1}) < \Gamma_{\text{fwd},K+1}.$$

The one-stage lemma again supplies a positive-duration stage, a strictly decreasing risk and attention envelope, and a finite hit of the next forward-envelope threshold. This completes the induction through L_{max} . The cross-depth BFC corollary gives strict positional- and state-error-bound decrease, as well as strict completed-risk decrease, for every completed $j < K$ while stage K is active. The base and endpoint risk statements follow from the base lemma and the fixed-depth forward theorem, respectively. □

Extensions: gradient descent and stage skipping

The main theorem concerns the population-gradient-flow induction. This subsection collects its timing bounds and two variations: a one-stage small-step discrete analogue and a direct length-three-to-length-eight handoff.

Corollary 7.D.3 (Stage-duration bounds). *Under the hypotheses of Theorem 7.D.3, write $\gamma_K^- = \gamma(S_K)$. Every depth- K stage has positive duration and obeys*

$$T_K - S_K \leq \frac{2}{A_K} (e^{\Gamma_{\text{fwd},K}} - e^{\gamma_K^-}). \quad (7.D.57)$$

Proof. This is (7.D.53); positivity follows from the strict entry inequality in (7.D.48). $\square$

Corollary 7.D.4 (Total dyadic training time). *Under the hypotheses of Theorem 7.D.3, the total gradient-flow time over the dyadic stages is bounded by*

$$T_{L_{\max}} - T_{\text{base}} \leq \sum_{K=2}^{L_{\max}} \frac{2e^{\Gamma_{\text{fwd},K}}}{A_K}. \quad (7.D.58)$$

Proof. Sum Corollary 7.D.3 over the dyadic stages and discard the nonnegative terms $e^{\gamma_K^-}$. $\square$

Corollary 7.D.5 (One stage under gradient descent). *Under the conditions of Theorem 7.D.1, fix $2 \leq L \leq L_{\max}$ and any compact interval $[a, b] \subseteq [\Gamma_{\text{upd}}, \infty)$ for one depth- L risk, and let $H_L > 0$ be a Lipschitz constant for the scalar derivative on this interval:*

$$|(\mathcal{R}_L^{\text{sym}})'(\gamma) - (\mathcal{R}_L^{\text{sym}})'(\tilde{\gamma})| \leq H_L |\gamma - \tilde{\gamma}| \quad (\gamma, \tilde{\gamma} \in [a, b]).$$

Such a finite constant exists by Proposition 7.G.1 and compactness. Initialize full-matrix population gradient descent at $\mathbf{Q}_0 = a\mathbf{\Pi}$ and choose $0 < \eta \leq (\kappa_{\text{sym}} H_L)^{-1}$. Under the symmetric reduction in Lemma 7.H.1, its routing margin increases strictly until it reaches or crosses b , and the number of steps is at most

$$1 + \frac{2e^b(b-a)}{\eta A_L}. \quad (7.D.59)$$

Proof. See Section 7.H. $\square$

Before the crossing, the population risk also decreases at every noncrossing step. That descent calculation is deferred to Section 7.H; the corollary uses only its monotonicity conclusion. This is a one-stage result; it does not claim a complete discrete analogue of the base stage and progressive curriculum. In particular, taking $a = \gamma(S_L)$ and $b = \Gamma_{\text{fwd},L}$ for an entry margin in (7.D.50) makes the first crossing the discrete analogue of the stopping rule. The crossing iterate may overshoot,

but its attention envelope is at most $b_{\text{fwd}}(\varepsilon_{\text{risk}}, L)$, so the fixed-depth forward theorem gives risk at most $\varepsilon_{\text{risk}}$.

Corollary 7.D.6 (Direct length-three to length-eight transition). *Under the hypotheses of Theorem 7.D.3, suppose $L_{\max} \geq 3$. Then the stopping matrix at Γ_{base} may initialize the length-eight, depth-three objective directly. At this handoff,*

$$\Gamma_{\text{upd}} \leq \Gamma_{\text{base}} < \Gamma_{\text{fwd},3}.$$

The depth-three risk and attention envelope decrease strictly until the attention envelope first reaches b_3 , which occurs in finite time at routing margin $\Gamma_{\text{fwd},3}$. The forward theorem then gives $\mathcal{R}_3 \leq \varepsilon_{\text{risk}}$.

Proof. See Section 7.H. □

7.E Composition module and base-task derivative calculations

7.E.1 The composition module: construction and activation estimates

This section gives the full realization of the differentiable composition module summarized in Section 7.B and the pair-unit activation estimates used in the forward-stability proof. The two role-tagged vocabularies and the scales B and τ are defined in (7.B.5), and the activation ϕ_τ is defined in Definition 7.B.4.

Definition 7.E.1 (Explicit pair-indexed composition module). For every pair $(a, b) \in G^2$, assign one pair-indexed hidden unit $r_{a,b}$ to output class $c = a \circ b$. Its weights satisfy

$$\langle W_{c,r_{a,b}}, y_a \rangle = B + 2\tau, \quad \langle W_{c,r_{a,b}}, g_b \rangle = B, \quad (7.E.1)$$

$$\langle W_{c,r_{a,b}}, u \rangle = -B, \quad u \in (\{y_h\}_{h \in G} \cup \{g_h\}_{h \in G}) \setminus \{y_a, g_b\}. \quad (7.E.2)$$

All unassigned hidden units have zero weight. The output logit for c is

$$[\text{MLP}_W(Z)]_c = \sum_{a \circ b = c} \phi_\tau(\langle W_{c,r_{a,b}}, Z \rangle). \quad (7.E.3)$$

Proof of Proposition 7.B.1. The activation ϕ_τ is C^1 by Definition 7.B.4. The module in Definition 7.E.1 is a finite sum of compositions of this activation with affine maps, so it is C^1 on state mixtures.

Pair activation. On $Z = \frac{1}{2}(y(p) + g(q))$, the state coordinates contribute

$$\frac{1}{2} [p_a(B + 2\tau) - (1 - p_a)B],$$

and the retrieved-state coordinates contribute

$$\frac{1}{2}[q_b B - (1 - q_b)B].$$

Adding and simplifying gives

$$\frac{1}{2}[2B(p_a + q_b - 1) + 2\tau p_a],$$

which is the pair-unit preactivation in (7.B.12).

Exact-state response. When $p = e_a$, the pair-indexed units in the true first-operand row satisfy

$$\Lambda_{a,b} = Bq_b + \tau \geq \tau,$$

so their activations are $Bq_b + \tau/2$. Every pair-indexed unit with first-operand index $a' \neq a$ has nonpositive preactivation and is inactive. The common shift $\tau/2$ cancels in the output softmax, giving (7.B.13). Substituting $q = e_b$ gives (7.B.14). $\square$

Proof of Lemma 7.C.6. For sufficiently large d , the scales in (7.B.5) satisfy $\tau \leq B$. We compare the target pair-indexed unit with two classes of competitors and then aggregate those comparisons within each output class. Write

$$H_{a,b} := \phi_\tau(\Lambda_{a,b}).$$

Separation within the target first-operand row. Because $q_b \geq 0$ and $p_{a_\star} \geq 1 - \xi$, each pair in row $a_\star$ has preactivation

$$\begin{aligned} \Lambda_{a_\star,b} &= B(p_{a_\star} + q_b - 1) + \tau p_{a_\star} \\ &\geq -B\xi + \tau(1 - \xi) = \tau - (B + \tau)\xi \\ &\geq \tau - 2B\xi \geq \frac{7\tau}{8} \geq \frac{3\tau}{4}, \end{aligned}$$

where the penultimate inequality uses $\xi \leq \tau/(16B)$ from (7.C.28). By Definition 7.B.4,

$$\phi'_\tau(z) = \begin{cases} z/\tau, & 0 < z < \tau, \\ 1, & z \geq \tau. \end{cases}$$

Thus $\phi'_\tau(z) \geq 3/4$ at every point between any two preactivations in row $a_\star$. For $b \neq b_\star$,

$$\Lambda_{a_\star,b_\star} - \Lambda_{a_\star,b} = B(q_{b_\star} - q_b) \geq B(1 - 2e_{\text{ret}}).$$

Integrating the derivative along this interval gives

$$\begin{aligned} H_{a_\star, b_\star} - H_{a_\star, b} &= \int_{\Lambda_{a_\star, b}}^{\Lambda_{a_\star, b_\star}} \phi'_\tau(z) dz \\ &\geq \frac{3}{4}(\Lambda_{a_\star, b_\star} - \Lambda_{a_\star, b}) \\ &\geq \frac{3B}{4}(1 - 2e_{\text{ret}}). \end{aligned} \tag{7.E.4}$$

Leakage from non-target first-operand rows. For $a \neq a_\star$, the total off-target state mass gives $p_a \leq \xi$. If also $b \neq b_\star$, then

$$\begin{aligned} \Lambda_{a, b} &\leq B(\xi + e_{\text{ret}} - 1) + \tau\xi \\ &\leq B(2\xi + e_{\text{ret}} - 1) < 0, \end{aligned}$$

where the last inequality uses $\xi \leq 1/128$ and $e_{\text{ret}} \leq 1/4$. Hence

$$H_{a, b} = 0 \quad (a \neq a_\star, b \neq b_\star). \tag{7.E.5}$$

The only remaining non-target-row unit type is $(a, b_\star)$ with $a \neq a_\star$. Since $q_{b_\star} \leq 1$,

$$\Lambda_{a, b_\star} \leq (B + \tau)p_a \leq 2B\xi \leq \frac{\tau}{8} < \tau.$$

If this preactivation is nonpositive, its activation is zero. If it is positive, it lies in the quadratic branch of Definition 7.B.4, and therefore

$$H_{a, b_\star} = \frac{\Lambda_{a, b_\star}^2}{2\tau} \tag{7.E.6}$$

$$\leq \frac{(B + \tau)^2 \xi^2}{2\tau} \leq \frac{2B^2 \xi^2}{\tau} \leq \theta. \tag{7.E.7}$$

The factor four in θ is a convenient slack constant.

Aggregation within an output class. The group multiplication table is a Latin square: for fixed a and c , the equation $a \circ b = c$ determines a unique b , and for fixed b and c , it determines a unique a . It follows from (7.E.5) that the target class contains only its target pair:

$$F_{c_\star} = H_{a_\star, b_\star}.$$

For a non-target class $c \neq c_\star$, let b_c be the unique element with $a_\star \circ b_c = c$. Then $b_c \neq b_\star$. The class c contains exactly one target-row unit $(a_\star, b_c)$, at most one unit of the form $(a, b_\star)$, and otherwise only inactive units with both indices non-target. Hence

$$F_c \leq H_{a_\star, b_c} + \theta.$$

Using (7.E.4), we conclude that

$$F_{c_\star} - F_c \geq \frac{3B}{4}(1 - 2e_{\text{ret}}) - \theta,$$

uniformly over $c \neq c_\star$. This proves (7.C.37).

Finally, if the right-hand side of (7.C.37) is denoted by Δ_F , then

$$[s^+]_{c_\star} = \frac{1}{1 + \sum_{c \neq c_\star} e^{F_c - F_{c_\star}}} \geq \frac{1}{1 + (d-1)e^{-\Delta_F}},$$

which is (7.C.38). $\square$

7.E.2 Base-task derivative details

This section records the calculations used in Theorem 7.C.1 and Lemma 7.C.3.

The following direct calculations provide the uniform-point quantities needed by the initialization theorem. From (7.C.2),

$$\frac{d}{d\gamma} r_1(\gamma) = r_1(\gamma)(1 - r_1(\gamma)),$$

so

$$r_1(0) = \frac{1}{2}, \quad \left. \frac{d}{d\gamma} r_1 \right|_0 = \frac{1}{4}. \quad (7.E.8)$$

Since $r_2(\gamma) = \sigma(\gamma r_1(\gamma))$ with $\sigma(z) = e^z / (1 + e^z)$,

$$\frac{d}{d\gamma} r_2(\gamma) = r_2(\gamma)(1 - r_2(\gamma))(r_1(\gamma) + \gamma \frac{d}{d\gamma} r_1(\gamma)),$$

and therefore

$$r_2(0) = \frac{1}{2}, \quad \left. \frac{d}{d\gamma} r_2 \right|_0 = \frac{1}{8}. \quad (7.E.9)$$

Let

$$Z_0 = 2e^{B/2} + d - 2, \quad s_0 = \frac{e^{B/2}}{Z_0}.$$

At $r_1 = 1/2$, the derivative of the normalizer Z_1 with respect to r_1 vanishes by symmetry. Hence

$$\left. \frac{\partial p_\star}{\partial r_1} \right|_{r_1=1/2} = Bs_0, \quad \left. \frac{\partial p_{\text{alt}}}{\partial r_1} \right|_{r_1=1/2} = -Bs_0.$$

Because $s_0 = \frac{1}{2} + O(de^{-B/2})$, combining with (7.E.8) gives

$$\left. \frac{d}{d\gamma} p_\star \right|_0 = \frac{B}{8}(1 + o(1)), \quad \left. \frac{d}{d\gamma} p_{\text{alt}} \right|_0 = -\frac{B}{8}(1 + o(1)). \quad (7.E.10)$$

From (7.C.6) and (7.E.9),

$$\left. \frac{d}{d\gamma} q_{\star} \right|_0 = \frac{1}{8}(1 - \pi_{\text{other}}), \quad \left. \frac{d}{d\gamma} q_{\text{alt}} \right|_0 = -\frac{1}{8}\pi_{\text{comp}}. \quad (7.E.11)$$

At the uniform point,

$$p_{\sigma} + q_{\sigma'} - 1 = O(de^{-B/2}), \quad \sigma, \sigma' \in \{\star, \text{alt}\},$$

so the common four-pair preactivation is

$$\Lambda_0 = \frac{\tau}{2} + O(Bde^{-B/2}). \quad (7.E.12)$$

Lemma 7.E.1 (Uniform-point base derivative audit). *Suppose $C_B > 8$. At $\gamma = 0$, the four leading pair units have smoothed-ReLU slope $1/2 + o(1)$, and their preactivation velocities satisfy*

$$\begin{aligned} \left. \frac{d}{d\gamma} \Lambda_{\star, \star} \right|_0 &= \frac{B^2 + B}{8}(1 + o(1)), & \left. \frac{d}{d\gamma} \Lambda_{\star, \text{alt}} \right|_0 &= \frac{B^2 - B}{8}(1 + o(1)), \\ \left. \frac{d}{d\gamma} \Lambda_{\text{alt}, \star} \right|_0 &= \frac{-B^2 + B}{8}(1 + o(1)), & \left. \frac{d}{d\gamma} \Lambda_{\text{alt}, \text{alt}} \right|_0 &= \frac{-B^2 - B}{8}(1 + o(1)). \end{aligned} \quad (7.E.13)$$

Proof. When $C_B > 3$, the error term in (7.E.12) is $o(\tau)$, so

$$\phi'_{\tau}(\Lambda_0) = \frac{\Lambda_0}{\tau} = \frac{1}{2} + o(1).$$

Substituting (7.E.10)–(7.E.11) into (7.C.8) gives (7.E.13). $\square$

Activation and persistent-progress details

The following argument makes the base-task proof sketch uniform over finite margins. The argument first separates the target pair from the leading competitors, then bounds the only nonleading pair types whose activations can increase.

Proof of Lemma 7.C.1. At $\gamma = 0$, the two leading state masses satisfy

$$p_{\star} = p_{\text{alt}} = \frac{e^{B/2}}{2e^{B/2} + d - 2} = \frac{1}{2} + o(1), \quad p_{\text{other}} = e^{-B/2}(1/2 + o(1)).$$

Similarly,

$$q_{\star} = q_{\text{alt}} = \frac{1}{2} + o(1), \quad q_{\text{other}} = o(1).$$

The four leading preactivations are $\tau/2 + o(\tau)$ and hence lie in the positive smoothing branch. Every nonleading pair contains a nonleading state or retrieved-state coordinate, so its two

probability coordinates sum to at most $1/2 + o(1)$. Equation (7.B.12) then gives

$$\Lambda_{a,b} \leq -\frac{B}{3} + \tau < 0$$

for sufficiently large d . Strict negativity persists in a neighborhood of $\gamma = 0$, proving Lemma 7.C.1. $\square$

Write

$$P = p_{\star}, \quad M = p_{\text{alt}}, \quad O = p_{\text{other}}, \quad U = q_{\star}, \quad V = q_{\text{alt}}, \quad W = q_{\text{other}},$$

set $H_{a,b} = \phi_{\tau}(\Lambda_{a,b})$, and use a prime for $d/d\gamma$. Direct differentiation of (7.C.3) and (7.C.6) gives

$$P' = Br_1'P(1 - P + M), \quad M' = -Br_1'M(1 + P - M), \quad O' = -Br_1'O(P - M), \quad (7.E.14)$$

$$U' = (1 - \pi_{\text{other}})r_2', \quad V' = -\pi_{\text{comp}}r_2', \quad W' = -\pi_{\text{other}}r_2'. \quad (7.E.15)$$

In particular, $P', U' > 0$, while $M', O', V', W' \leq 0$. The exact attention velocity ratio is

$$\frac{r_2'}{r_1'} = \frac{r_2(1 - r_2)}{r_1(1 - r_1)}(r_1 + \gamma r_1'). \quad (7.E.16)$$

It is positive and continuous, equals $1/2$ at zero, and tends to one as $\gamma \rightarrow \infty$: indeed, $r_1' = e^{-\gamma}(1 + o(1))$, $r_2(1 - r_2) = e^{-\gamma r_1}(1 + o(1))$, and $\gamma(1 - r_1) \rightarrow 0$. The comparison can in fact be made explicit. For $\gamma \geq 0$,

$$\frac{1}{4}e^{-\gamma} \leq r_1' \leq e^{-\gamma}, \quad \frac{1}{4}e^{-\gamma r_1} \leq r_2(1 - r_2) \leq e^{-\gamma r_1},$$

while

$$0 \leq \gamma(1 - r_1) \leq \gamma e^{-\gamma} \leq \frac{1}{e}, \quad \frac{1}{2} \leq r_1 + \gamma r_1' = r_1(1 + \gamma(1 - r_1)) \leq 1 + \frac{1}{e}.$$

Substitution in (7.E.16), using $e^{1/e} < 2$ and $1 + 1/e < 3/2$, gives the uniform numerical bounds

$$\frac{1}{8}r_1' \leq r_2' \leq 12r_1' \quad (\gamma \geq 0). \quad (7.E.17)$$

Lemma 7.E.2 (Uniform base activation-velocity estimates). *Suppose $C_B > 8$. For all sufficiently large d , uniformly over $\gamma \geq 0$,*

$$\min \left\{ H'_{\star, \star}, \min_{(\sigma, \sigma') \neq (\star, \star)} (H_{\star, \star} - H_{\sigma, \sigma'})' \right\} \geq \frac{1}{108} B(r_1' + r_2'). \quad (7.E.18)$$

The only nonleading pair types that can have positive velocity are $(\star, \text{other})$ and $(\text{other}, \star)$. Their total positive velocity satisfies

$$[H'_{\star, \text{other}}]_+ + [H'_{\text{other}, \star}]_+ \leq \varepsilon_d B(r_1' + r_2'), \quad \varepsilon_d = o_d(1). \quad (7.E.19)$$

One may take

$$\varepsilon_d = 4\tau + 4Be^{-B} + 2eB\tau^{-1}e^{-B}. \quad (7.E.20)$$

Proof. **Step 1: Keep the target unit uniformly active.** Set $T = \Lambda_{\star,\star}$. By (7.E.12), $T(0) = \tau/2 + o(\tau)$, while

$$T' = (B + \tau)P' + BU' > 0. \quad (7.E.21)$$

Increase the lower threshold on d so that

$$\begin{aligned} \pi_{\text{other}} &\leq \frac{1}{2}, & \tau &\leq B, & \left|T(0) - \frac{\tau}{2}\right| &\leq \frac{\tau}{6}, \\ 2Be^{96-B} &\leq \frac{\tau}{6}, & 64Be^{-B} &\leq 1, & 2eBe^{-B} &\leq \tau. \end{aligned} \quad (7.E.22)$$

All six conditions hold eventually because $B = C_B \log d$, $\tau = d^{-1/2}$, and $C_B > 8$. It follows from the third condition and $T' > 0$ that, for every $\gamma \geq 0$,

$$\phi'_\tau(T) \geq \frac{1}{3}. \quad (7.E.23)$$

Since $U' = (1 - \pi_{\text{other}})r'_2 \geq r'_2/2$ and $r'_1 + r'_2 \leq 9r'_2$ by (7.E.17), we obtain the explicit bound

$$H'_{\star,\star} \geq \frac{B}{6}r'_2 \geq \frac{1}{54}B(r'_1 + r'_2). \quad (7.E.24)$$

Step 2: Compare the target with the easy leading competitors. The target has the same-scale advantage over each leading competitor. Whenever a competitor preactivation C has $C' \geq 0$, monotonicity of ϕ'_τ and $T \geq C$ give the identity and lower bound

$$\begin{aligned} H'_{\star,\star} - \phi'_\tau(C)C' &= \phi'_\tau(T)(T' - C') + C'(\phi'_\tau(T) - \phi'_\tau(C)) \\ &\geq \phi'_\tau(T)(T' - C'). \end{aligned} \quad (7.E.25)$$

If $C' < 0$, its activation velocity is nonpositive and (7.E.24) applies directly.

For $(\star, \text{alt})$, the preactivation-gap derivative is

$$(T - \Lambda_{\star,\text{alt}})' = B(U' - V') = B(1 - \pi_{\text{other}} + \pi_{\text{comp}})r'_2.$$

Equations (7.E.23), (7.E.17), and (7.E.25) give the lower bound $B(r'_1 + r'_2)/54$ when the competitor velocity is nonnegative; the negative-velocity case follows from (7.E.24). For (alt, alt) , the competitor preactivation velocity $(B + \tau)M' + BV'$ is negative, so (7.E.24) again applies.

Step 3: Control the remaining leading competitor. It remains to treat $(\text{alt}, \star)$. Put

$$C = \Lambda_{\text{alt},\star} = B(M + U - 1) + \tau M.$$

If $C' \leq 0$, use (7.E.24); if $C' > 0$ but $C \leq 0$, the competitor is inactive. Suppose therefore that $C, C' > 0$. The velocity inequality

$$(B + \tau)(-M') < BU' \quad (7.E.26)$$

and $P \geq M$ give

$$(B + \tau)Br_1' M \leq (B + \tau)(-M') < BU' \leq 12Br_1'.$$

Because $r_1' > 0$ at every finite γ , this yields the explicit estimate

$$M < \frac{12}{B + \tau}. \quad (7.E.27)$$

Activation gives

$$1 - U < (1 + \tau/B)M. \quad (7.E.28)$$

For sufficiently large d ,

$$1 - U = (1 - r_2)(1 - \pi_{\text{other}}) \geq \frac{1}{4}e^{-\gamma r_1}.$$

Combining this with (7.E.27) and (7.E.28) gives

$$\frac{1}{4}e^{-\gamma r_1} < 1 - U < \frac{12}{B}, \quad \gamma \geq \gamma r_1 > \log(B/48).$$

Consequently

$$1 - r_1 < e^{-\gamma} < \frac{48}{B}. \quad (7.E.29)$$

The exact ratio

$$\frac{M}{P} = e^{-B(2r_1-1)}$$

and $P \leq 1$ then give

$$M < e^{96-B}. \quad (7.E.30)$$

A second use of (7.E.28), now with $1 + \tau/B \leq 2$, gives

$$\frac{1}{4}e^{-\gamma r_1} < 2e^{96-B}, \quad \gamma \geq \gamma r_1 > B - 96 - \log 8.$$

Moreover, (7.E.22) yields

$$0 < C < (B + \tau)e^{96-B} \leq 2Be^{96-B} \leq \frac{\tau}{6}. \quad (7.E.31)$$

Thus $\phi'_\tau(C) = C/\tau \leq 1/6$, so $\phi'_\tau(T) - \phi'_\tau(C) \geq 1/6$. Since

$$(T - C)' = (B + \tau)(P' - M') \geq (B + \tau)(-M')$$

and $(B + \tau)(-M') + C' = BU'$, the exact identity preceding (7.E.25) gives

$$H'_{\star,\star} - H'_{\text{alt},\star} \geq \frac{1}{6}BU' \geq \frac{1}{108}B(r'_1 + r'_2) \quad (7.E.32)$$

where the last inequality again uses $U' \geq r'_2/2$ and $r'_1 + r'_2 \leq 9r'_2$.

Combining the three cases proves (7.E.18).

Step 4: Bound the growing nonleading pairs. It remains to control nonleading pairs. The types (alt, other), (other, alt), and (other, other) are inactive at zero by the activation audit, and remain inactive because both of their relevant coordinates are nonincreasing by (7.E.14)–(7.E.15). Only $(\star, \text{other})$ and $(\text{other}, \star)$ can have positive velocity.

If $H'_{\star,\text{other}} > 0$, its pair unit is active, and therefore

$$1 - P < W + \frac{\tau}{B}.$$

Since $M \leq 1 - P$, (7.E.14) gives

$$P' \leq 2Br'_1(1 - P) \leq 2Br'_1\left(W + \frac{\tau}{B}\right).$$

The W' term in the pair preactivation velocity is nonpositive, while $W \leq \pi_{\text{other}} \leq e^{-B}$. Since $0 \leq \phi'_\tau \leq 1$ and $B + \tau \leq 2B$,

$$[H'_{\star,\text{other}}]_+ \leq 4(B\tau + B^2e^{-B})r'_1. \quad (7.E.33)$$

If $H'_{\text{other},\star} > 0$, activation gives

$$1 - U < (1 + \tau/B)O. \quad (7.E.34)$$

Because $O \leq e^{-Br_1}$ and $1 - U \geq \frac{1}{4}e^{-\gamma r_1}$, this implies $e^{-\gamma r_1} < 8e^{-Br_1}$ and therefore

$$\gamma > B - \frac{\log 8}{r_1} \geq B - 2 \log 8. \quad (7.E.35)$$

Hence

$$1 - r_1 < e^{-\gamma} < 64e^{-B}, \quad O \leq e^{-Br_1} < e^{-B+64Be^{-B}} \leq e^{1-B},$$

where the last inequality uses (7.E.22). The active preactivation satisfies

$$0 < \Lambda_{\text{other},\star} \leq (B + \tau)O \leq 2eBe^{-B} \leq \tau,$$

so the unit lies in the quadratic branch and its slope is at most $2eB\tau^{-1}e^{-B}$. Its O' term is nonpositive, and therefore

$$[H'_{\text{other},\star}]_+ \leq 2eB^2\tau^{-1}e^{-B}r'_2. \quad (7.E.36)$$

Combining these bounds and using $\tau = d^{-1/2}$ and $B = C_B \log d$ proves (7.E.19), with

$$\varepsilon_d = 4\tau + 4Be^{-B} + 2eB\tau^{-1}e^{-B} = o_d(1).$$

□

Detailed proof of Lemma 7.C.3. This proof expands the comparison summarized in the earlier proof sketch of Lemma 7.C.3 and supplies its uniform constants. For a fixed output class, the Latin-square property permits at most one pair from a fixed first-operand row or second-operand column. Separation prevents two leading pairs from sharing a class. The target class contains $H_{\star,\star}$ but neither growing nonleading type, while a competitor contains at most one leading pair and one of each growing nonleading type. Consequently, (7.E.18) and (7.E.19) give

$$\widehat{F}'_{y^\star} - \widehat{F}'_c \geq \left(\frac{1}{108} - \varepsilon_d \right) B(r'_1 + r'_2).$$

Choose $d_0(C_B)$ large enough that (7.E.22) holds and $\varepsilon_d < 1/216$. Then $\widehat{F}'_{y^\star} - \widehat{F}'_c > 0$ for every $d \geq d_0(C_B)$, every finite $\gamma \geq 0$, and every $c \neq y^\star$. This proves (7.C.15) and completes the detailed proof of Lemma 7.C.3. □

The base-task calculation is now complete: the target logit gains uniformly against every competitor at every finite routing margin, which is the input needed by the Stage I progress theorem.

7.F Auxiliary estimates for the deeper-stage analysis

This section supplies the supporting estimates consumed by the covariance, ideal-routing, and finite-margin arguments in Section 7.D. We begin with local softmax bounds, then control the ideal-routing sensitivities, and finally establish common forward and reverse rates on the finite unrolled graph.

7.F.1 Elementary softmax and covariance bounds

The proof repeatedly uses three elementary facts.

Lemma 7.F.1 (Softmax margin bound). *Let $z_\star$ be one logit among d classes and suppose*

$$z_\star - \max_{c \neq \star} z_c \geq \Delta.$$

Then

$$\text{softmax}(z)_\star \geq \frac{1}{1 + (d-1)e^{-\Delta}}, \quad (7.F.1)$$

and

$$-\log \text{softmax}(z)_\star \leq \log(1 + (d-1)e^{-\Delta}). \quad (7.F.2)$$

Proof. Divide the softmax numerator and denominator by $e^{z_\star}$. Every competing exponential is at most $e^{-\Delta}$, and there are at most $d - 1$ of them. $\square$

Lemma 7.F.2 (Attention error from a logit margin). *Suppose a query instance has one target key and at most K non-target candidate keys, and the target logit exceeds every non-target logit by at least Δ . Then its non-target attention mass satisfies*

$$1 - \alpha_\star \leq Ke^{-\Delta}. \quad (7.F.3)$$

Proof. As in the preceding proof,

$$1 - \alpha_\star = \frac{\sum_{j \neq \star} e^{z_j - z_\star}}{1 + \sum_{j \neq \star} e^{z_j - z_\star}} \leq \sum_{j \neq \star} e^{-\Delta} \leq Ke^{-\Delta}.$$

$\square$

Lemma 7.F.3 (Pairwise covariance identity). *For weights $a_j \geq 0$ with $\sum_j a_j = 1$ and scalar sequences b_j, c_j ,*

$$\sum_j a_j b_j c_j - \left(\sum_j a_j b_j \right) \left(\sum_k a_k c_k \right) = \frac{1}{2} \sum_{j,k} a_j a_k (b_j - b_k)(c_j - c_k). \quad (7.F.4)$$

Proof. Expand the right-hand side. The two diagonal products $\sum_{j,k} a_j a_k b_j c_j$ and $\sum_{j,k} a_j a_k b_k c_k$ each equal $\sum_j a_j b_j c_j$, while the two cross terms each equal $(\sum_j a_j b_j)(\sum_k a_k c_k)$. The factor $1/2$ completes the identity. $\square$

7.F.2 Saturation and sensitivity estimates for the ideal-routing reference

This section collects the norm estimates used in the ideal-routing comparison of Section 7.D. Fix $L_{\max} < \infty$ throughout. Let

$$\mathbb{T}_\Delta := \left\{ \mathbf{u} \in \mathbb{R}^G : \sum_{c \in G} u_c = 0 \right\}, \quad \|(\mathbf{u}, \mathbf{v})\|_{\oplus, 1} := \|\mathbf{u}\|_1 + \|\mathbf{v}\|_1.$$

Thus $\mathbb{T}_\Delta$ is the tangent space of the probability simplex with the ℓ_1 norm, and the two-input tangent space carries the displayed sum norm. The dual norm of a simplex-tangent covector λ is

$$\|\lambda\|_{\mathbb{T}_\Delta^*} := \sup_{\substack{\mathbf{u} \in \mathbb{T}_\Delta \\ \|\mathbf{u}\|_1 \leq 1}} |\lambda[\mathbf{u}]| = \frac{1}{2} \left(\max_c \lambda_c - \min_c \lambda_c \right), \quad (7.F.5)$$

where covectors differing by a constant vector are identified. The dual of the input sum norm is the maximum of the two component dual norms. For the ambient perturbation estimates, equip $\mathbb{R}^{2d}$ with ℓ_∞ and write $\|H\|_{\infty \rightarrow \infty} = \max_i \sum_j |H_{ij}|$.

Hard-routing saturation and the leading coefficient

At ideal routing, target-tree states remain close to their exact products, the root sensitivity is positive, and every earlier target-tree contribution carries an additional saturated-composition Jacobian.

Proof of Lemma 7.D.3. Under hard attention at target-tree query instances, the positional features are exactly the target positional features in Lemma 7.C.4. An induction using (7.B.12) proves the claim. Indeed, suppose the two input states $\mathbf{p}, \mathbf{q}$ satisfy $\|\mathbf{p} - \mathbf{e}_a\|_1 + \|\mathbf{q} - \mathbf{e}_b\|_1 = O(\zeta_B)$, and write $H_{g,h} = \phi_\tau(\Lambda_{g,h})$. Since $B\zeta_B = o(\tau)$, the pair units fall into four classes:

$$\begin{aligned} H_{a,b} &= B + \frac{\tau}{2} + O(B\zeta_B), \\ H_{a,h} &= \frac{\tau}{2} + O(B\zeta_B) & (h \neq b), \\ H_{g,b} &= O(B\zeta_B) & (g \neq a), \\ H_{g,h} &= 0 & (g \neq a, h \neq b). \end{aligned}$$

For the last line, (7.B.12) gives $\Lambda_{g,h} = -B + O(B\zeta_B) < 0$; the preceding three lines follow from the same formula and the one-Lipschitz property of ϕ_τ .

For an output class c , let $\mathcal{I}_c = \{(g, h) : g \circ h = c\}$. The Latin-square property implies that $\mathcal{I}_c$ contains exactly one pair in row a and one in column b . For $c = a \circ b$ these are the same pair and every other unit is inactive, whereas for $c \neq a \circ b$ they are distinct and all remaining units are inactive. Consequently,

$$F_{a \circ b}(\mathbf{p}, \mathbf{q}) = B + \frac{\tau}{2} + O(B\zeta_B), \quad F_c(\mathbf{p}, \mathbf{q}) = \frac{\tau}{2} + O(B\zeta_B) \quad (c \neq a \circ b).$$

After subtracting the common softmax shift $\tau/2$, the target logit is $B + O(B\zeta_B)$ and every competing logit is $O(B\zeta_B)$. Hence

$$1 - [\mathbf{s}^+]_{a \circ b} \leq (d-1)e^{-B+O(B\zeta_B)} = O(\zeta_B), \quad \|\mathbf{s}^+ - \mathbf{e}_{a \circ b}\|_1 = 2(1 - [\mathbf{s}^+]_{a \circ b}) = O(\zeta_B).$$

Because $L_{\max}$ is fixed and $B\zeta_B = o(1)$, the induction closes uniformly over the target tree. This concerns the finite-softmax MLP state; it does not replace that state by a one-hot vector. $\square$

Proof of Lemma 7.D.2. This is the detailed calculation announced by the earlier proof sketch. We make the root and below-root contributions explicit and verify the additional Jacobian suppression for every earlier query. We separate the root query instance from all earlier updates.

Contribution of the root query instance. Throughout this paragraph, $v = v_{\text{root}} = (2^L, L-1)$, $a = y_v^*$, $b = y_{v,\text{ret}}^*$, and, for a root candidate key j , we abbreviate the level- $(L-1)$ node state $\mathbf{s}_{(j,L-1)}^\infty$ as $\mathbf{s}_j^\infty$. Because the root is the final update, (7.D.4) and the linear attention-value map give

$$\beta_{v,j_v^*}^\infty - \beta_{v,j}^\infty = S_{a,b}(\mathbf{s}_v^\infty, \mathbf{s}_{j_v^*}^\infty; \mathbf{s}_j^\infty). \quad (7.F.6)$$

By Lemma 7.D.3, the two target-tree inputs satisfy (7.D.20) for a constant depending only on $L_{\max}$. The perturbation estimate in Lemma 7.D.4 and the comparison in (7.D.22) therefore yield, uniformly over root keys,

$$\beta_{v,j_v^\star}^\infty - \beta_{v,j}^\infty = \chi_B(1 - [\mathbf{s}_j^\infty]_b) + o_d(\chi_B). \quad (7.F.7)$$

The root has $2^L - 2$ non-target causal keys. By Lemma 7.D.5, each satisfies (7.D.24). Combining (7.F.7) and (7.D.24), the total root contribution to (7.D.13) is

$$\kappa_{\text{sym}}(2^L - 2) \left(1 - \frac{1}{d}\right) \chi_B + o_d(\kappa_{\text{sym}} \chi_B). \quad (7.F.8)$$

Earlier target-tree query instances. The root-loss covector has tangent-dual norm $O_{L_{\max}, C_B}(\chi_B)$ by Lemma 7.F.4. This is the sharper tangent scale $O_{L_{\max}, C_B}(B\zeta_B)$, not the ambient $O(Bd\zeta_B)$ scale.

For an earlier target-tree query instance, decompose the derivative with respect to its attention distribution into the retrieved-state and copied-positional channels in (7.B.3). The retrieved-state channel immediately enters a composition state update. A copied-positional perturbation can affect the final state only by changing a later attention distribution. Along the hard target ancestry, Lemma 7.F.7 and (7.F.34) give

$$D_\omega \alpha = \gamma (\text{diag}(\alpha) - \alpha \alpha^\top) = O_d(\gamma e^{-\gamma}) \longrightarrow 0.$$

If such a path leaves the hard target ancestry, it must re-enter through a deleted direct value or positional-copy edge of a target-tree query, and the one-cut path bound in Corollary 7.D.1 and Lemma 7.F.9 instead supplies $O(\text{poly}(\gamma)e^{-\gamma}) \rightarrow 0$. Thus the copied-positional channel contributes zero to the limiting sensitivity β^∞ . Every surviving below-root contribution therefore contains at least one composition-state Jacobian. Combining (7.F.17) with Lemma 7.D.6 gives one factor $O_{L_{\max}, C_B}(Bd\zeta_B)$ beyond the root scale. Since $Bd\zeta_B = o_d(1)$, all later composition-state Jacobians are at most one for sufficiently large d ; selected-value maps have norm at most one, and the number of remaining paths depends only on $L_{\max}$. There are at most $L_{\max} 2^{2L_{\max}}$ target-key-non-target-key pairs in the finite unrolled graphs, so their total absolute contribution is

$$O_{L_{\max}, C_B}(\kappa_{\text{sym}} \chi_B B d \zeta_B) = o_d(\kappa_{\text{sym}} \chi_B).$$

Query instances outside the target ancestry have zero hard-attention backpropagated sensitivity to the final output and do not enter A_L . Adding this bound to (7.F.8) proves (7.D.14). $\square$

Sensitivity and composition-Jacobian estimates

The remaining ideal-routing claims require quantitative perturbation bounds. We control the root loss in a neighborhood of one-hot inputs and then bound the Jacobian of the saturated composition map in the simplex-tangent norms used by backpropagation.

Proof of the perturbation claims in Lemma 7.D.4. Step 1: Preserve the root logit gap. Let $\mathbf{x} = (\mathbf{p}, \mathbf{r}) \in \mathbb{R}^{2d}$ and, for $c \in G$, let $\mathcal{I}_c = \{(g, h) \in G^2 : g \circ h = c\}$. The group table is a Latin square: $|\mathcal{I}_c| = d$, and both coordinate projections from $\mathcal{I}_c$ to G are bijections. At the exact input, Proposition 7.B.1 gives

$$F_{a \circ b}(\mathbf{e}_a, \mathbf{e}_b) = B + \frac{\tau}{2}, \quad F_c(\mathbf{e}_a, \mathbf{e}_b) = \frac{\tau}{2} \quad (c \neq a \circ b). \quad (7.F.9)$$

Because ϕ_τ is one-Lipschitz, (7.B.12) and the two bijections give, for every $c \in G$,

$$\begin{aligned} |F_c(\mathbf{p}, \mathbf{r}) - F_c(\mathbf{e}_a, \mathbf{e}_b)| &\leq \sum_{(g, h) \in \mathcal{I}_c} |\Lambda_{g, h}(\mathbf{p}, \mathbf{r}) - \Lambda_{g, h}(\mathbf{e}_a, \mathbf{e}_b)| \\ &\leq (B + \tau) \|\mathbf{p} - \mathbf{e}_a\|_1 + B \|\mathbf{r} - \mathbf{e}_b\|_1 \leq (B + \tau) C_0 \zeta_B. \end{aligned} \quad (7.F.10)$$

It follows that, for $c \neq a \circ b$,

$$F_{a \circ b}(\mathbf{p}, \mathbf{r}) - F_c(\mathbf{p}, \mathbf{r}) \geq B - 2(B + \tau) C_0 \zeta_B. \quad (7.F.11)$$

Since $B \zeta_B = o(1)$, the softmax margin bound gives

$$1 - p_{a \circ b}^{\text{out}}(\mathbf{p}, \mathbf{r}) \leq (d - 1) e^{-B + 2(B + \tau) C_0 \zeta_B} = O_{C_0}(\zeta_B), \quad (7.F.12)$$

using $(d - 1) e^{-B} = \zeta_B (1 + (d - 1) e^{-B}) = O(\zeta_B)$.

Step 2: Bound the loss derivatives in logit space. Differentiate the logit-space cross-entropy. If $\mathbf{p}^{\text{out}} = (p_h^{\text{out}})_{h \in G}$ and $c_\star = a \circ b$, then

$$\nabla_F \ell_{c_\star}^{\text{comp}} = \mathbf{p}^{\text{out}} - \mathbf{e}_{c_\star}, \quad \nabla_F^2 \ell_{c_\star}^{\text{comp}} = \text{diag}(\mathbf{p}^{\text{out}}) - \mathbf{p}^{\text{out}} (\mathbf{p}^{\text{out}})^\top.$$

For the softmax Hessian, the absolute row sum in row h is $2p_h^{\text{out}}(1 - p_h^{\text{out}})$. Every non-target probability is at most $1 - p_{c_\star}^{\text{out}}$, so (7.F.12) yields

$$\|\nabla_F \ell_{c_\star}^{\text{comp}}\|_1 = 2(1 - p_{c_\star}^{\text{out}}) = O_{C_0}(\zeta_B), \quad \|\nabla_F^2 \ell_{c_\star}^{\text{comp}}\|_{\infty \rightarrow \infty} = O_{C_0}(\zeta_B). \quad (7.F.13)$$

The same $O_{C_0}(\zeta_B)$ bound holds for the loss gradient in ℓ_∞ .

Step 3: Propagate the bounds to the composition inputs. For $(g, h) \in G^2$, define the constant preactivation gradient

$$\mathbf{g}_{g, h} := \nabla_x \Lambda_{g, h} = (B + \tau) \mathbf{e}_{p_g} + B \mathbf{e}_{r_h}.$$

Then

$$\nabla_x F_c(\mathbf{x}) = \sum_{(g, h) \in \mathcal{I}_c} \phi'_\tau(\Lambda_{g, h}(\mathbf{x})) \mathbf{g}_{g, h}. \quad (7.F.14)$$

Since $0 \leq \phi'_\tau \leq 1$, each output row of J_F has absolute sum at most $d(2B + \tau)$. A fixed input coordinate appears in exactly one pair of $\mathcal{I}_c$ for each output class, so each column has absolute

sum at most $d(B + \tau)$. Hence

$$\|J_F\|_{\infty \rightarrow \infty} \leq d(2B + \tau) = O(Bd), \quad \|J_F^T\|_{\infty \rightarrow \infty} \leq d(B + \tau) = O(Bd). \quad (7.F.15)$$

Thus the chain-rule term containing the output-softmax Hessian satisfies

$$\|J_F^T \nabla_F^2 \ell_{c_\star}^{\text{comp}} J_F\|_{\infty \rightarrow \infty} = O_{C_0}(B^2 d^2 \zeta_B).$$

For the remaining curvature term, use that ϕ'_τ is τ^{-1} -Lipschitz. Equation (7.F.14) gives

$$\begin{aligned} \|\nabla F_c(x) - \nabla F_c(x')\|_1 &\leq \frac{1}{\tau} \sum_{(g,h) \in \mathcal{I}_c} |g_{g,h}^T(x - x')| \|g_{g,h}\|_1 \\ &\leq \frac{d(2B + \tau)^2}{\tau} \|x - x'\|_\infty. \end{aligned}$$

Thus ∇F_c is $O(B^2 d/\tau)$ -Lipschitz from ℓ_∞ to ℓ_∞ , and $\|D^2 F_c\|_{\infty \rightarrow \infty} = O(B^2 d/\tau)$ wherever the Hessian exists. Together with (7.F.13) and $\tau = d^{-1/2}$, this yields

$$\left\| \sum_{h \in G} \frac{\partial \ell_{c_\star}^{\text{comp}}}{\partial F_h} D^2 F_h \right\|_{\infty \rightarrow \infty} = O_{C_0}(B^2 d^{3/2} \zeta_B) = O_{C_0}(B^2 d^2 \zeta_B).$$

The smoothed ReLU is C^1 and its derivative is Lipschitz, although its second derivative is undefined at the two smoothing breakpoints. The neighborhood in (7.D.20) is convex, and the loss gradient is absolutely continuous on each line segment in it. Integrating the preceding almost-everywhere Hessian bounds therefore gives a genuine Lipschitz bound. The gradient chain rule and the two Hessian terms show

$$\|D_2 \ell_{a \circ b}^{\text{comp}}\|_\infty = O_{C_0}(Bd \zeta_B), \quad \text{Lip}_\infty(D \ell_{a \circ b}^{\text{comp}}) = O_{C_0}(B^2 d^2 \zeta_B). \quad (7.F.16)$$

Step 4: Compare the perturbed and exact sensitivities. Let (p, r) satisfy (7.D.20). Since $\|q - r\|_1 \leq 2$, the mean-value estimate and (7.F.16) give

$$\begin{aligned} |S_{a,b}(p, r; q) - S_{a,b}(e_a, e_b; q)| &\leq 2 \text{Lip}_\infty(D_2 \ell_{a \circ b}^{\text{comp}}) (\|p - e_a\|_\infty + \|r - e_b\|_\infty) \\ &\quad + \|D_2 \ell_{a \circ b}^{\text{comp}}(e_a, e_b)\|_\infty \|r - e_b\|_1 \\ &= O_{C_0}(B^2 d^2 \zeta_B^2). \end{aligned}$$

Together with (7.D.19), this proves (7.D.21). Finally,

$$\frac{B^2 d^2 \zeta_B^2}{\chi_B} = \frac{Bd(d-1)^2}{e^B + d - 1} \leq Bd^3 e^{-B} = C_B (\log d) d^{3-C_B} = o_d(1),$$

which proves (7.D.22). Under $C_B > 8$ the upper bound is at most $C_B(\log d)d^{-5}$; for this isolated comparison, $C_B > 3$ would already suffice. $\square$

Lemma 7.F.4 (Limiting root loss sensitivity in tangent dual norm). *Throughout this lemma, $v = v_{\text{root}} = (2^L, L-1)$, $a = y_v^*$, $b = y_{v, \text{ret}}^*$, and, for a root candidate key j , we abbreviate the level- $(L-1)$ node state $s_{(j, L-1)}^\infty$ as s_j^∞ . Uniformly over $2 \leq L \leq L_{\max}$,*

$$\left\| D\ell_{a \circ b}^{\text{comp}}(s_v^\infty, s_{j_v^*}^\infty) \right\|_{(\mathbb{T}_\Delta \oplus \mathbb{T}_\Delta)^*} = O_{L_{\max}, C_B}(\chi_B). \quad (7.F.17)$$

In fact, the current-input component is $o_d(\chi_B)$.

Proof. For the retrieved input, evaluate (7.D.21) at simplex vertices and take differences. The coordinate range of the resulting covector is $O_{L_{\max}, C_B}(\chi_B)$ by (7.D.22); hence (7.F.5) gives the same tangent-dual bound.

For the current input, $D_1\ell_{a \circ b}^{\text{comp}}(e_a, e_b)$ vanishes on $\mathbb{T}_\Delta$: every $h \in \mathbb{T}_\Delta$ satisfies $D_1F_c(e_a, e_b)[h] = (B + \tau)h_a$ for all $c \in G$, a common logit shift annihilated by cross-entropy. By Lemma 7.D.3, the ideal root inputs are within $O_{L_{\max}, C_B}(\zeta_B)$ of (e_a, e_b) . The joint-gradient Lipschitz bound in (7.F.16) therefore gives

$$\left\| D_1\ell_{a \circ b}^{\text{comp}}(s_v^\infty, s_{j_v^*}^\infty) \right\|_{\mathbb{T}_\Delta^*} = O_{L_{\max}, C_B}(B^2 d^2 \zeta_B^2) = o_d(\chi_B).$$

The dual of the input sum norm is the maximum of the component dual norms, which proves (7.F.17). $\square$

Proof of Lemma 7.D.6. Write $H_{g,h}(\mathbf{p}, \mathbf{q}) := \phi_\tau(\Lambda_{g,h}(\mathbf{p}, \mathbf{q}))$. For a tangent direction (h, k) ,

$$DH_{g,h}(\mathbf{p}, \mathbf{q})[h, k] = \phi'_\tau(\Lambda_{g,h})((B + \tau)h_g + Bk_h). \quad (7.F.18)$$

Since $0 \leq \phi'_\tau \leq 1$, the norm of this scalar functional on the two-input tangent space is at most $B + \tau$. The logit for class c is the sum of the units with $g \circ h = c$. Hence, without using cancellation,

$$\begin{aligned} \|DF(\mathbf{p}, \mathbf{q})[h, k]\|_1 &\leq \sum_{g, h \in G} |DH_{g,h}(\mathbf{p}, \mathbf{q})[h, k]| \\ &\leq d(B + \tau) \|h\|_1 + dB \|k\|_1 \leq d(B + \tau) \|(h, k)\|_{\oplus, 1}. \end{aligned} \quad (7.F.19)$$

The factor d appears because each current-state and retrieved-state coordinate occurs in d pair units.

Let $C_{\text{sat}} = C_{\text{sat}}(L_{\max})$ be a constant in (7.D.16). If the target labels of the two inputs are a and b , then $\|\mathbf{p} - e_a\|_1 + \|\mathbf{q} - e_b\|_1 \leq 2C_{\text{sat}}\zeta_B$. The exact logits in (7.F.9) have target-versus-competitor margin B . Equation (7.F.10), with $C_0 = 2C_{\text{sat}}$, therefore gives

$$F_{a \circ b}(\mathbf{p}, \mathbf{q}) - F_c(\mathbf{p}, \mathbf{q}) \geq B - O_{L_{\max}, C_B}(B\zeta_B) \quad (c \neq a \circ b).$$

Consequently, if $s = \mathcal{U}(p, q)$ and $\epsilon = 1 - s_{a \circ b}$, then

$$\epsilon \leq (d-1)e^{-B+O_{L_{\max}, C_B}(B\zeta_B)} = O_{L_{\max}, C_B}(\zeta_B). \quad (7.F.20)$$

This is uniform in the node, the labels, and the particular states satisfying (7.D.16).

The softmax Jacobian at s is $J_{\text{sm}} = \text{diag}(s) - ss^\top$. The absolute sum of column j is $2s_j(1 - s_j)$. For $j = a \circ b$ this is at most 2ϵ , while for $j \neq a \circ b$ it is at most $2s_j \leq 2\epsilon$. Thus

$$\|J_{\text{sm}}\|_{1 \rightarrow 1} \leq 2\epsilon = O_{L_{\max}, C_B}(\zeta_B). \quad (7.F.21)$$

This ambient logit-space bound is stronger than a tangent-space bound, and J_{sm} takes values in $\mathbb{T}_\Delta$. Combining (7.F.19) and (7.F.21), and using $\tau \leq B$ for large d , proves (7.D.25). By duality, the transpose Jacobian used in backpropagation obeys the same operator bound. $\square$

These estimates complete the ideal-routing layer of the argument: the root retains a positive sensitivity scale, whereas every earlier intended query is suppressed by at least one saturated composition Jacobian.

7.F.3 Common-rate induction for the finite unrolled graph

This section proves Lemmas 7.D.12 and 7.D.13. Fix finite d and $L_{\max}$ throughout. Constants in the proof may depend on these fixed quantities, including $B = C_B \log d$, but not on the stage, example, or query instance. For the unit target-tree routing gap, write $\rho := e^{-\gamma}$ throughout.

Rate bookkeeping and forward induction

The forward induction yields rates for local states and attention distributions that are uniform over the fixed family of depths; the reverse induction then propagates sensitivity bounds from the root.

Definition 7.F.1 (Exponential-polynomial rate class). For a finite-dimensional quantity $X(\gamma)$ with limit X^∞ , write

$$X - X^\infty \in \mathfrak{R}(k, \kappa) \quad (7.F.22)$$

when there are $C, \Gamma < \infty$ such that

$$\|X(\gamma) - X^\infty\| \leq C(1 + \gamma)^k e^{-\kappa\gamma} \quad (\gamma \geq \Gamma).$$

Here k is a nonnegative integer and $\kappa > 0$.

Lemma 7.F.5 (Closure of the rate classes). *The rate classes obey the following closure rules.*

1. *Enlarging k or decreasing κ preserves the relation in Definition 7.F.1.*

2. Finite sums satisfy

$$\mathfrak{R}(k_1, \kappa_1) + \mathfrak{R}(k_2, \kappa_2) \subseteq \mathfrak{R}(\max\{k_1, k_2\}, \min\{\kappa_1, \kappa_2\}),$$

after enlarging the hidden constant.

3. Multiplication by γ gives

$$\gamma \mathfrak{R}(k, \kappa) \subseteq \mathfrak{R}(k + 1, \kappa). \quad (7.F.23)$$

4. A C^1 map with bounded derivative on a neighborhood of the relevant compact range preserves a rate class. If its derivative is Lipschitz, its Jacobian also converges in the same rate class.

5. A finite product of bounded matrix-valued quantities with limits preserves a common rate class: if every $X_r - X_r^\infty \in \mathfrak{R}(k_r, \kappa_r)$, then

$$X_m \cdots X_1 - X_m^\infty \cdots X_1^\infty \in \mathfrak{R}\left(\max_r k_r, \min_r \kappa_r\right),$$

after enlarging the hidden constant.

Proof. The sum and multiplication rules follow directly from the defining bound, after increasing the hidden constant. The map and Jacobian claims follow from the mean-value theorem applied to the map and then to its derivative. For products, use the telescoping identity

$$X_m \cdots X_1 - X_m^\infty \cdots X_1^\infty = \sum_{r=1}^m X_m \cdots X_{r+1} (X_r - X_r^\infty) X_{r-1}^\infty \cdots X_1^\infty$$

and boundedness of the finitely many factors. $\square$

Every state, positional, matching-coefficient, attention, and aggregate node of $\mathcal{G}_{L,\gamma}$ lies in a compact probability simplex, a compact interval, or a finite product of these sets. The parameter-scaled raw logits are not separate graph nodes; their effect is part of the ω -to- α edge. A C^1 map on a neighborhood of the remaining compact ranges has bounded derivative. The positional updates are linear. The composition map is C^1 , and its Jacobian is Lipschitz because the explicit smoothed ReLU has a Lipschitz derivative. These facts supply the closure rules needed in both directions of the graph.

Lemma 7.F.6 (Forward convergence and local attention defects). *Every forward node variable in every fixed elementary graph $\mathcal{G}_{L,\gamma}$ has a limit and belongs to a rate class in Definition 7.F.1 with a positive exponential exponent. Moreover, for every query instance u , there are $k_u < \infty$ and $\kappa_u > 0$ such that*

$$\text{Leak}_u(\gamma) + \text{Split}_u(\gamma) = O((1 + \gamma)^{k_u} e^{-\kappa_u \gamma}). \quad (7.F.24)$$

This includes query instances with unique, tied, or fully tied limiting matching coefficients.

Proof. Topologically order the elementary graph $\mathcal{G}_{L,\gamma}$. By (7.B.1), its input states, fixed key features, and initial query-side positional features are independent of γ , so they equal their

limiting values exactly. Suppose inductively that all inputs to a query instance u have limits and converge in rate classes with positive exponents. The matching coefficients $\omega_{u,j}$ then do as well.

Let

$$M_u = \arg \max_j \omega_{u,j}^\infty.$$

If there are keys outside M_u , finiteness of the candidate-key set gives the positive limiting gap

$$\Delta_u := \min_{j \notin M_u} \left(\max_k \omega_{u,k}^\infty - \omega_{u,j}^\infty \right) > 0.$$

For all sufficiently large γ , each such matching-coefficient gap is at least $\Delta_u/2$, and therefore

$$\sum_{j \notin M_u} \alpha_{u,j} \leq K_u e^{-\Delta_u \gamma / 2}, \quad (7.F.25)$$

where $K_u = |\mathcal{K}_u|$ is the number of candidate keys.

For $j, k \in M_u$, the limiting coefficients agree. After taking a common incoming rate for the finite set of coordinates,

$$\max_{j,k \in M_u} |\omega_{u,j} - \omega_{u,k}| = O((1 + \gamma)^k e^{-\kappa \gamma}) \quad (7.F.26)$$

for some $k < \infty$ and $\kappa > 0$. Multiplying by γ still tends to zero by (7.F.23). The softmax restricted to M_u is smooth at the zero vector of attention-logit contrasts, so its probabilities converge to the uniform distribution on M_u at rate $O((1 + \gamma)^{k+1} e^{-\kappa \gamma})$. Together with (7.F.25), this proves that the complete attention distribution has a limit and approaches it at some positive exponential rate, up to a polynomial factor.

The retrieved state and copied positional feature are finite weighted sums of already convergent inputs. They inherit a positive exponential rate. The smooth composition map preserves that rate, closing the recurrent induction. The final coordinate projection and logarithm preserve the rate because the target probability ranges in a compact subinterval of $(0, 1]$ for fixed d . Thus every forward node variable in every fixed elementary graph has a limit and a rate of the form (7.F.22). Equations (7.F.25)–(7.F.26) also give, query instance by query instance,

$$\text{Leak}_u(\gamma) + \text{Split}_u(\gamma) = O((1 + \gamma)^{k_u} e^{-\kappa_u \gamma}) \quad (7.F.27)$$

with $k_u < \infty$ and $\kappa_u > 0$. This argument covers a unique maximizer, a proper tied maximal set, and the fully tied case. $\square$

Target-tree and reverse-induction rates

This subsection combines the forward rates with a finite-DAG path expansion. Target-tree sensitivities converge to their ideal limits, whereas any off-tree path must pay a cut factor when it re-enters the supervised ancestry.

Lemma 7.F.7 (First-order target-tree expansion). *Every target-tree query instance satisfies*

$$\mathbf{x}_{v,\text{qry}} = \mathbf{x}_{v,\text{qry}}^\infty + O_d(\rho), \quad \mathbf{s}_v = \mathbf{s}_v^\infty + O_d(\rho), \quad (7.F.28)$$

and its attention weights satisfy (7.D.35). In particular, each non-target target-tree weight equals $\rho + O((1 + \gamma)\rho^2)$, uniformly over the fixed finite family of stages and examples.

Proof. Target-tree query instances admit the sharper expansion in powers of ρ . Their base case follows from (7.B.1). Suppose inductively that the current and target predecessor variables entering a target-tree query instance differ from their hard limits by $O(\rho)$. Its query-side positional feature then has target matching coefficient $1 + O(\rho)$ and every non-target matching coefficient is $O(\rho)$. Hence every target–non-target logit gap is

$$\gamma(1 + O(\rho)).$$

For a non-target key,

$$\begin{aligned} \exp[-\gamma(1 + O(\rho))] &= \rho \exp(O(\gamma\rho)) \\ &= \rho [1 + O(\gamma\rho)]. \end{aligned}$$

Normalizing the finite attention-logit vector gives

$$\alpha_{v,j} = \rho + O((1 + \gamma)\rho^2) \quad (j \neq j_v^\star), \quad (7.F.29)$$

$$\alpha_{v,j_v^\star} = 1 - (|\mathcal{K}_v| - 1)\rho + O((1 + \gamma)\rho^2). \quad (7.F.30)$$

The positional copy is linear, so its deviation from the selected predecessor feature is $O(\rho)$. The retrieved state differs from the hard selected state by $O(\rho)$: the target predecessor itself has $O(\rho)$ error by induction, and the total non-target attention mass is $O(\rho)$. Lipschitz continuity of the composition map then gives an $O(\rho)$ output-state error. Induction over the target tree proves (7.F.28) and (7.D.35). This part of the argument is independent of the off-tree cut lemma. $\square$

Lemma 7.F.8 (Finite-DAG path expansion on the elementary graph). *Fix a stage and example, and use the node and edge set of $\mathcal{G}_{L,\gamma}$ from Definition 7.D.4. For a node a , write its finite-dimensional variable as z_a . For an edge $e = (a, b)$, write*

$$J_e(\gamma) := D_{z_a} z_b$$

for the corresponding partial Jacobian, evaluated at the forward pass. If $\mathcal{P}(a, b)$ is the set of directed paths from a to b , and if the edges of P occur in forward order as $e_1, \dots, e_m$, then

$$D_{z_a} z_b = \sum_{P \in \mathcal{P}(a,b)} J_{e_m} \cdots J_{e_1}, \quad \|D_{z_a} z_b\| \leq \sum_{P \in \mathcal{P}(a,b)} \prod_{e \in P} \|J_e\|. \quad (7.F.31)$$

Here and below, vector spaces carry Euclidean norms and Jacobians carry the induced operator norms. The path set is taken in this same elementary graph; no coarse-graph projection or subsequent edge refinement is involved.

Proof. Topologically order the descendants of a and set $D_{z_a} z_a = I$. For every later node b , the chain rule gives

$$D_{z_a} z_b = \sum_{c \in \text{pa}(b)} J_{(c,b)} D_{z_a} z_c,$$

with a zero term when c is not a descendant of a . Substituting the inductive path expansion for each parent c appends the edge (c, b) to every path from a to c and produces every path from a to b exactly once. This proves the identity in (7.F.31); submultiplicativity and the triangle inequality prove its norm bound. $\square$

Lemma 7.F.9 (One-cut path bound). *For every off-tree query instance u , there are finite constants C_u and k_u^{cut} such that*

$$\max_j |\beta_{u,j}| \leq \|D_{\alpha_u} \ell\| \leq C_u (1 + \gamma)^{k_u^{\text{cut}}} e^{-\gamma}, \quad \ell := \log f_{y^*}. \quad (7.F.32)$$

Proof. Apply Lemma 7.F.8 directly to $\mathcal{G}_{L,\gamma}$ from Definition 7.D.4. The state, positional, matching, attention, and aggregate variables remain in fixed compact sets. The explicit maps in (7.D.27)–(7.D.29) therefore show that every non-softmax local Jacobian is bounded: this includes the two α -to-aggregate source edges, all candidate value and positional-copy edges, and the composition and output edges. At a softmax attention map,

$$D_{\omega} \alpha = \gamma (\text{diag}(\alpha) - \alpha \alpha^\top),$$

so a matching-to-attention edge contributes at most one power of $1 + \gamma$. This supplies exponents $q_e \in \{0, 1\}$, with $q_e = 1$ only needed for those softmax edges. By Corollary 7.D.1, every path from α_u to ℓ first passes through $\text{out}(u)$ and then contains a deleted direct value or positional-copy edge of a later target-tree query. By the first item of Lemma 7.D.10, its Jacobian is $O(e^{-\gamma})$ (and hence is bounded by $C_{e_p} (1 + \gamma) e^{-\gamma}$). Select the first such edge. Bound every other factor polynomially, including any later cut or logit-mediated transmission. The graph has finitely many paths, each of finite length, so taking the maximum polynomial degree and summing the path constants gives

$$\begin{aligned} \|D_{\alpha_u} \ell\| &\leq \sum_{P \in \mathcal{P}(\alpha_u, \ell)} \prod_{e \in P} \|J_e\| \\ &\leq e^{-\gamma} \sum_{P \in \mathcal{P}(\alpha_u, \ell)} \left(\prod_{e \in P} C_e \right) (1 + \gamma)^{1 + \sum_{e \in P \setminus \{e_p\}} q_e} \\ &\leq C_u (1 + \gamma)^{k_u^{\text{cut}}} e^{-\gamma}, \end{aligned}$$

which proves (7.F.32). The estimate uses one cut factor on each path; any additional small factors are harmlessly bounded by their polynomial upper bounds. $\square$

Lemma 7.F.10 (Pathwise target-sensitivity convergence). *For every target-tree query instance v and candidate key j , there are a finite integer $k_{v,j}$ and a positive exponent $\kappa_{v,j}$ such that*

$$\beta_{v,j} - \beta_{v,j}^\infty = O((1 + \gamma)^{k_{v,j}} e^{-\kappa_{v,j}\gamma}). \quad (7.F.33)$$

The conclusion remains valid even when a path from α_v to the final log likelihood passes through off-tree attention maps with tied limiting maximizers.

Proof. Fix one target-tree query instance v and use $\mathcal{G}_{L,\gamma}$ itself. Its structure and finite path set do not depend on γ . By Lemma 7.D.7, the node α_v , its two aggregate children, and the final log likelihood belong to $\mathcal{H}_L$. Partition the directed paths from α_v to $\ell = \log f_{y^*}$ into two classes. An *ancestry-confined path* uses only retained edges of $\mathcal{H}_L$. An *excursion path* uses at least one node outside $\mathcal{H}_L$ or at least one edge not retained in $\mathcal{H}_L$. This edge-wise definition includes a deleted non-target direct edge whose endpoints both happen to lie in $\mathcal{H}_L$. Every path belongs to exactly one class on this single graph.

Ancestry-confined paths. Every attention map traversed by such a path belongs to a target-tree query with a unique hard maximizer; otherwise its recurrent output would be an off-tree ancestor of the loss, contrary to Lemma 7.D.7. By Lemma 7.F.7,

$$D_\omega \alpha = \gamma (\text{diag}(\alpha) - \alpha \alpha^\top), \quad \|D_\omega \alpha\| = O((1 + \gamma)e^{-\gamma}), \quad (7.F.34)$$

so this retained logit-mediated Jacobian converges to its zero hard-reference Jacobian at a positive exponential-polynomial rate. Retained direct value and positional-copy edges have coefficients $1 + O(\rho)$; a nonselected direct edge is deleted and therefore belongs to the excursion class. The two source-edge Jacobians from α_v to its aggregates are, by Lemma 7.D.10, the matrices of candidate states and candidate positional features, with no attention-probability factor. Those variables have limits and positive rates by Lemma 7.F.6, including when the candidate itself is off-tree. Every remaining linear or composition Jacobian converges at a positive rate by the forward induction, the Lipschitz-Jacobian property, and Lemma 7.F.5. The derivative at the final log likelihood has the same property because the root probability converges to a strictly positive limit.

It follows from the finite-product rule in Lemma 7.F.5 that the Jacobian product along each ancestry-confined path converges to its hard-reference product at a positive exponential-polynomial rate. Denote the finite sum of those limiting products by $\beta_{v,j}^{\mathcal{H}}$. This is the derivative obtained by summing the hard limits of the retained elementary paths, including the local perturbation of the attention coordinate at v . The excursion estimate below proves that the complete finite-margin sensitivity converges to this value, thereby both establishing the pointwise limit and identifying $\beta_{v,j}^{\mathcal{H}} = \beta_{v,j}^\infty$.

Excursion paths. Fix such a path. The second assertion of Lemma 7.D.9 applies directly, with no projection to another graph. If the path has a maximal subpath whose interior nodes lie outside $\mathcal{H}_L$, its first re-entry is a deleted direct value or positional-copy edge of a later target-tree query. If all its nodes lie in $\mathcal{H}_L$ but the path is not confined, it uses a deleted direct edge whose endpoints both lie there. In either case Lemmas 7.D.8 and 7.D.10 identify an edge on the path with Jacobian

$$O((1 + \gamma)e^{-\gamma}). \quad (7.F.35)$$

All non-softmax Jacobians on the rest of the path are bounded. At an arbitrary matching-to-attention map,

$$\|D_\omega \alpha\| \leq C(1 + \gamma).$$

This polynomial bound is the point needed for a tied off-tree attention: when its limiting maximal set has more than one key, the local Jacobian need not converge and can grow linearly in γ , but it contributes only one polynomial factor. Since the path has finite length, multiplying all its local bounds with the re-entry factor in (7.F.35) gives

$$\|J_P(\gamma)\| \leq C_P(1 + \gamma)^{q_P} e^{-\gamma}$$

for finite C_P, q_P . Thus every excursion-path contribution converges to zero in a positive exponential-polynomial rate class.

There are finitely many paths. Taking a common positive exponent and a common polynomial degree over the ancestry-confined differences and the vanishing excursion terms proves convergence of the complete derivative $D_{\alpha_v} \ell$ to the hard-ancestry path sum. In particular, $\beta_{v,j}^{\mathcal{H}} = \beta_{v,j}^\infty$. Taking the j th coordinate yields (7.F.33). $\square$

Lemma 7.F.11 (Reverse sensitivity rates). *For every target-tree query instance and candidate key, the backward sensitivity converges to its hard-routing limit at a positive exponential rate up to a polynomial factor in γ . For every off-tree query instance u , there is a finite integer k_u^{cut} such that*

$$\text{Sens}_u(\gamma) = O((1 + \gamma)^{k_u^{\text{cut}}} e^{-\gamma}).$$

Proof. The target-tree assertion, including paths containing tied off-tree attentions, is exactly (7.F.33) from Lemma 7.F.10. In particular, it is not inferred from convergence of every local attention Jacobian.

For an off-tree query instance, the hard-reference sensitivity is zero because the instance is not an ancestor of the root in $\mathcal{H}_L$. By Lemma 7.F.9, the complete derivative with respect to its attention vector is the sum of the pathwise Jacobian products, with one $O((1 + \gamma)\rho)$ cut factor selected on every path and only polynomial growth in all remaining factors. Since $\text{Sens}_u(\gamma) \leq 2 \max_j |\beta_{u,j}|$, (7.F.32) gives

$$\text{Sens}_u(\gamma) = O((1 + \gamma)^{k_u^{\text{cut}}} \rho), \quad (7.F.36)$$

as claimed.

The factor ρ in (7.F.36) is the one cut factor. It is not counted again for reverse traversal. The additional $e^{-\kappa\gamma}$ used in the covariance expansion is instead the independent local defect in (7.F.27). $\square$

Uniformization for Lemmas 7.D.12 and 7.D.13. For fixed d , positional dictionary size, and $L_{\max}$, there are finitely many stages, examples, query instances, coordinates, and directed paths. There are also finitely many positive limiting gaps Δ_u ; query instances with no lower limiting matching coefficient contribute no such gap. Take the minimum of one and all positive exponents produced by the forward induction and the reverse path classifications; it remains positive. Taking the maximum polynomial degree, enlarged to cover finite products, and common maxima of the hidden constants and thresholds proves all displays in Lemmas 7.D.12 and 7.D.13 uniformly over the stated family.

The quantifier order is therefore: fix $L_{\max}$ and C_B , choose and fix the finite d required by the leading-coefficient argument, and then choose the common rate constants. No uniformity in an additional $d \rightarrow \infty$ limit is asserted. $\square$

The common-rate induction therefore gives uniform asymptotic bounds for every forward quantity and backward path in the fixed finite family.

Proof of the finite-margin growth-rate expansion

The common forward and reverse rates separate the leading term $A_L e^{-\gamma}$ from the strictly smaller off-tree remainder.

Proof of Theorem 7.D.2. Apply Lemmas 7.D.12 and 7.D.13. The target-tree estimates hold uniformly over the fixed collection of stages and examples. In particular, the pathwise ancestry/excursion decomposition in Lemma 7.F.10 makes the sensitivity deviations uniformly $o_\gamma(1)$ even when an excursion contains tied off-tree attention maps. At every off-tree query instance, (7.D.37) supplies the uniform $o_\gamma(\rho)$ product of the cut sensitivity and the independent local defect.

Covariance expansion. Insert (7.D.35) and (7.D.36) into the exact pairwise covariance (7.D.7). At a target-tree query instance, the target-key–non-target-key terms give

$$\rho \sum_{j \neq j_v^*} (\beta_{v,j_v^*}^\infty - \beta_{v,j}^\infty) + o_\gamma(\rho),$$

because the hard matching-coefficient gap is one. Non-target–non-target pairs are $O((1 + \gamma)\rho^2) = o_\gamma(\rho)$. For every off-tree query instance, Lemma 7.D.11 together with (7.D.37) gives

$$|C_u| = o_\gamma(\rho).$$

Thus the complete off-tree sum has the same strictly higher-order bound without a separate case analysis for unique and tied maximizers. Summing over the finite set of query instances, taking

the population expectation, and multiplying by κ_{sym} gives (7.D.38). Multiplying that estimate by e^γ and letting $\gamma \rightarrow \infty$ proves (7.D.39), using $A_L > 0$ from Lemma 7.D.2. $\square$

7.G Ambient-space and ODE details for the symmetry reduction

7.G.1 Symmetry reduction and gradient-flow invariance

This section supplies the linear-algebra and gradient-flow details used in Lemma 7.B.1. The reduction is a statement about the full learned matrix, not an assumption that the matrix is scalar: the random positional relabeling first confines the population trajectory to the matrices that treat all dictionary elements identically, and softmax shift invariance then removes one direction that cannot affect attention probabilities.

Ambient symmetry and the fixed subspace

The dictionary $\mathcal{X} = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(p)}\} \subset \mathbb{R}^p$ contains p orthonormal vectors, so it is an orthonormal basis of $\mathbb{R}^p$. Both the query-side and key-side positional features lie in this space, and the learned router is therefore a full matrix $\mathbf{Q} \in \mathbb{R}^{p \times p}$. In particular, there are no unobserved positional directions outside the dictionary span in the stated model.

Lemma 7.G.1 (Reduction to the positional dictionary span). *Suppose, more generally, that the positional dictionary is embedded in $\mathbb{R}^D$ with $D \geq p$. Let $S = \text{span}(X)$ and let P_S be the orthogonal projector onto S . Then the population risk depends on the router $\mathbf{Q}$ only through $P_S \mathbf{Q} P_S$. Its gradient vanishes on the other three matrix blocks, and gradient flow from zero stays in the compressed block $P_S \mathbb{R}^{D \times D} P_S$.*

Proof. Every query and key remains in S : this is true at initialization and is preserved because the query update in (7.B.3) is a linear combination of earlier queries. Hence every attention logit satisfies

$$\langle \mathbf{Q} \mathbf{x}_{\text{qry}}, \mathbf{x}_j \rangle = \langle P_S \mathbf{Q} P_S \mathbf{x}_{\text{qry}}, \mathbf{x}_j \rangle. \quad (7.G.1)$$

Thus replacing $\mathbf{Q}$ by $P_S \mathbf{Q} P_S$ leaves every attention logit, attention weight, recurrent state, and loss unchanged. The risk is constant in each of the other three block directions, so its gradient has zero projection onto those blocks. Gradient flow initialized with all three blocks equal to zero therefore keeps them zero. $\square$

The argument below consequently applies to the compressed $p \times p$ matrix without loss. Under the standing choice $D = p$, $P_S = \mathbf{I}$ and this compression is automatic.

Use the dictionary as an orthonormal coordinate basis. For a permutation $\pi \in S_p$, let P_π be its permutation matrix, so that $P_\pi \mathbf{e}_a = \mathbf{e}_{\pi(a)}$. Relabeling the dictionary sends every positional feature $\mathbf{x}$ to $P_\pi \mathbf{x}$ and acts on the router by conjugation,

$$\mathbf{Q} \mapsto P_\pi \mathbf{Q} P_\pi^\top. \quad (7.G.2)$$

Lemma 7.G.2 (Risk invariance and gradient equivariance). *For every positional permutation π and every matrix Q ,*

$$\mathcal{R}_{n,L}(P_\pi Q P_\pi^\top) = \mathcal{R}_{n,L}(Q), \quad (7.G.3)$$

$$\nabla_Q \mathcal{R}_{n,L}(P_\pi Q P_\pi^\top) = P_\pi \nabla_Q \mathcal{R}_{n,L}(Q) P_\pi^\top. \quad (7.G.4)$$

The same statements hold for the conditional base risk $\mathcal{R}_{\text{base}}$.

Proof. For relabeled query and key features, conjugation preserves each attention logit:

$$\langle P_\pi Q P_\pi^\top (P_\pi \mathbf{x}_{\text{qry}}), P_\pi \mathbf{x}_j \rangle = \langle Q \mathbf{x}_{\text{qry}}, \mathbf{x}_j \rangle.$$

Induction over recurrent levels therefore shows that simultaneous relabeling of the positional features and conjugation of Q leaves all attention probabilities and state features unchanged, while it relabels later query-side positional features by the same P_π . The injection ι in Definition 7.B.2 is uniform, so its distribution is invariant under postcomposition by π . Averaging the identical per-example losses proves (7.G.3). The base conditioning event in Assumption 7.C.1 depends only on the group symbols and is therefore unchanged by positional relabeling, so the same proof applies to $\mathcal{R}_{\text{base}}$.

Differentiate (7.G.3) in an arbitrary matrix direction and use that P_π is orthogonal. Identifying directional derivatives with the Frobenius inner product gives (7.G.4). $\square$

Write $J = \mathbf{1}\mathbf{1}^\top$.

Lemma 7.G.3 (Permutation-fixed matrices). *The matrices fixed by all conjugations in (7.G.2) form the two-dimensional subspace*

$$\{Q : P_\pi Q P_\pi^\top = Q \text{ for every } \pi \in S_p\} = \text{span}\{I, J\}. \quad (7.G.5)$$

Proof. If a matrix is fixed by every permutation conjugation, transpositions show that all its diagonal entries are equal. Permutations act transitively on ordered pairs of distinct indices, so all its off-diagonal entries are also equal. A matrix with common diagonal entry a and common off-diagonal entry b equals $(a - b)I + bJ$, proving (7.G.5). The converse is immediate because both I and J are fixed by permutation conjugation. $\square$

Lemma 7.G.4 (Softmax gauge invariance). *For every $c \in \mathbb{R}$,*

$$\mathcal{R}_{n,L}(Q + cJ) = \mathcal{R}_{n,L}(Q), \quad \langle \nabla_Q \mathcal{R}_{n,L}(Q), J \rangle_F = 0. \quad (7.G.6)$$

The analogous identities hold for $\mathcal{R}_{\text{base}}$.

Proof. In dictionary coordinates, every query-side positional feature has the form

$$\mathbf{x}_{\text{qry}} = \sum_{a=1}^p \omega_a \mathbf{e}_a, \quad \sum_{a=1}^p \omega_a = 1.$$

The coefficient sum is one at level zero and remains one because (7.B.3) is an attention-weighted convex combination. Therefore $J\mathbf{x}_{\text{qry}} = \mathbf{1}$, and for every candidate key $\mathbf{e}_j$,

$$\langle (Q + cJ)\mathbf{x}_{\text{qry}}, \mathbf{e}_j \rangle = \langle Q\mathbf{x}_{\text{qry}}, \mathbf{e}_j \rangle + c.$$

The added term is the same for every candidate-key logit at the fixed query instance and cancels in softmax. This proves the first identity in (7.G.6); differentiating it in c proves the Frobenius orthogonality of the gradient. $\square$

Corollary 7.G.1 (Centered fixed direction). *The permutation-fixed subspace orthogonal to the gauge direction is*

$$\text{span}\{I, J\} \cap J^\perp = \text{span}\left\{I - \frac{1}{p}J\right\} = \text{span}\{\Pi\}. \quad (7.G.7)$$

Proof. If $aI + bJ$ is orthogonal to J , then

$$0 = \langle aI + bJ, J \rangle_F = ap + bp^2,$$

so $b = -a/p$. $\square$

The direction J is therefore a gauge direction: it changes the matrix but not the attention map. For example, γI and $\gamma \Pi$ produce the same attention probabilities. Euclidean gradient flow from zero selects the centered representative because its velocity is always orthogonal to J .

Well-posedness and invariance of population gradient flow

The final structural step verifies that the population trajectory exists for all time and remains in the centered permutation-fixed direction. This turns the matrix ODE into the scalar routing-margin dynamics used throughout the analysis.

Proposition 7.G.1 (Well-posed population gradient flow). For every fixed finite task defined above, the population gradient vector field $-\nabla_Q \mathcal{R}(Q)$ is locally Lipschitz. For every finite initial time t_0 and every finite matrix Q_0 , the initial-value problem

$$\dot{Q}(t) = -\nabla_Q \mathcal{R}(Q(t)), \quad Q(t_0) = Q_0, \quad (7.G.8)$$

has a unique forward-global solution for $t \geq t_0$.

Proof. For fixed n, p, d , the population risk is a finite average over group-symbol sequences and positional injections. Every unrolled computation graph is finite. Attention softmax is smooth, and the smoothed ReLU in Definition 7.B.4 is C^1 with globally Lipschitz derivative. Finite composition therefore makes each per-example loss, and hence the population risk, continuously differentiable with locally Lipschitz gradient. The Picard–Lindelof theorem gives a unique solution on a maximal time interval.

The solution cannot escape to infinity in finite time. Indeed, cross-entropy is nonnegative and along gradient flow

$$\frac{d}{dt}\mathcal{R}(\mathbf{Q}(t)) = -\|\nabla_{\mathbf{Q}}\mathcal{R}(\mathbf{Q}(t))\|_{\mathbb{F}}^2.$$

For every finite $T > t_0$ in the maximal forward interval,

$$\int_{t_0}^T \|\dot{\mathbf{Q}}(t)\|_{\mathbb{F}}^2 dt \leq \mathcal{R}(\mathbf{Q}_0), \quad \int_{t_0}^T \|\dot{\mathbf{Q}}(t)\|_{\mathbb{F}} dt \leq \sqrt{(T - t_0)\mathcal{R}(\mathbf{Q}_0)}.$$

More particularly, Cauchy–Schwarz gives

$$\|\mathbf{Q}(t) - \mathbf{Q}(s)\|_{\mathbb{F}} \leq \sqrt{(t - s)\mathcal{R}(\mathbf{Q}_0)} \quad (t_0 \leq s < t).$$

Thus $\mathbf{Q}(t)$ is Cauchy and has a finite limit at any hypothetical finite endpoint of its maximal interval. Local existence there would extend the solution, so the solution is global. $\square$

Corollary 7.G.2 (Warm-started symmetric population trajectory). *Let $\mathcal{R}$ be any fixed population risk defined above and suppose that, at some finite time t_0 , the active stage is initialized at $\mathbf{Q}(t_0) = \gamma_0 \mathbf{\Pi}$. Write $\mathcal{R}^{\text{sym}}(\gamma) := \mathcal{R}(\gamma \mathbf{\Pi})$ and let V be its routing-margin growth rate as in (7.B.16). Then the unique solution of (7.G.8) obeys*

$$\mathbf{Q}(t) = \gamma(t) \mathbf{\Pi}, \quad \frac{d\gamma(t)}{dt} = V(\gamma(t)), \quad \gamma(t_0) = \gamma_0, \quad (7.G.9)$$

for $t \geq t_0$. Thus $V = V_{n,L}$ for a depth- (n, L) risk and $V = V_{\text{base}}$ for the conditional base risk.

Proof. For any $\pi \in S_p$, gradient equivariance in Lemma 7.G.2 shows that $P_\pi \mathbf{Q}(t) P_\pi^\top$ solves the same ODE as $\mathbf{Q}(t)$ and has the same initial condition $\gamma_0 \mathbf{\Pi}$, because $\mathbf{\Pi}$ is fixed by every permutation conjugation. Uniqueness implies

$$P_\pi \mathbf{Q}(t) P_\pi^\top = \mathbf{Q}(t) \quad (\pi \in S_p),$$

so the trajectory lies in $\text{span}\{\mathbf{I}, \mathbf{J}\}$ by Lemma 7.G.3. At every such point, gradient equivariance keeps the gradient in the same fixed subspace, while (7.G.6) makes it orthogonal to $\mathbf{J}$. Corollary 7.G.1 therefore makes the velocity a multiple of $\mathbf{\Pi}$. Since the initial matrix already lies in this direction, there is a scalar function $\gamma(t)$ with $\mathbf{Q}(t) = \gamma(t) \mathbf{\Pi}$ and $\gamma(t_0) = \gamma_0$.

Write $\nabla_{\mathbf{Q}} \mathcal{R}(\gamma(t) \mathbf{\Pi}) = \lambda_{\mathbf{Q}}(t) \mathbf{\Pi}$. Since $\|\mathbf{\Pi}\|_{\mathbb{F}}^2 = p - 1$, the chain rule gives

$$\left. \frac{d\mathcal{R}^{\text{sym}}(\gamma)}{d\gamma} \right|_{\gamma=\gamma(t)} = (p - 1) \lambda_{\mathbf{Q}}(t).$$

The matrix ODE gives $\dot{\gamma}(t) = -\lambda_{\mathbf{Q}}(t)$, and the definition $\kappa_{\text{sym}} = 1/(p - 1)$ therefore yields $\dot{\gamma}(t) = V(\gamma(t))$. This proves (7.G.9). $\square$

Because $\mathbf{\Pi}$ has diagonal entries $1 - 1/p$ and off-diagonal entries $-1/p$, its diagonal-minus-off-diagonal gap is exactly one. The scalar coefficient $\gamma(t)$ in (7.G.9) is consequently the matched-key logit advantage in (7.B.15), rather than an arbitrary reparameterization of the gradient-flow trajectory.

At a curriculum switch, apply the corollary anew with the next population risk, the switch time as t_0 , and the inherited checkpoint matrix as the initial condition. Thus changing the active objective does not require reinitializing the matrix or restarting the scalar reduction at zero.

7.H Proofs for curriculum extensions

Two local variants accompany the progressive-training result: a small-step population-gradient-descent analogue for one stage and a direct length-three-to-length-eight warm start.

Lemma 7.H.1 (Symmetric reduction of population gradient descent). *Consider the full-matrix population-gradient update*

$$\mathbf{Q}_{k+1} = \mathbf{Q}_k - \eta \nabla_{\mathbf{Q}} \mathcal{R}_L(\mathbf{Q}_k).$$

If $\mathbf{Q}_0 = \gamma_0 \mathbf{\Pi}$, then every iterate has the form $\mathbf{Q}_k = \gamma_k \mathbf{\Pi}$, and its coefficient satisfies

$$\gamma_{k+1} = \gamma_k + \eta V_L(\gamma_k).$$

Proof. Conjugation equivariance keeps the gradient in the permutation-fixed subspace, while gauge orthogonality makes it orthogonal to $\mathbf{J}$. By Corollary 7.G.1, the gradient is therefore a multiple of $\mathbf{\Pi}$ at every symmetric iterate. The same Frobenius-inner-product calculation as in the proof of Lemma 7.B.1 identifies the coefficient update with $\eta V_L(\gamma_k)$. $\square$

Proof of Corollary 7.D.5. The assumed interval lies above Γ_{upd} . Hence, while $\gamma_k \in [a, b]$, the static bound (7.D.1) gives

$$\gamma_{k+1} - \gamma_k \geq \frac{\eta A_L}{2} e^{-b} > 0.$$

Summing this lower bound proves (7.D.59). On a noncrossing step, the full segment from γ_k to γ_{k+1} lies in $[a, b]$. Since $\frac{d\mathcal{R}_L^{\text{sym}}(\gamma)}{d\gamma} = -V_L(\gamma)/\kappa_{\text{sym}}$, the scalar descent lemma gives

$$\mathcal{R}_L^{\text{sym}}(\gamma_{k+1}) \leq \mathcal{R}_L^{\text{sym}}(\gamma_k) - \eta \left(\frac{1}{\kappa_{\text{sym}}} - \frac{\eta H_L}{2} \right) V_L(\gamma_k)^2 < \mathcal{R}_L^{\text{sym}}(\gamma_k).$$

$\square$

Skipping directly to a deeper stage. The positive-update invariant and nested forward-envelope thresholds allow depth three to start directly from the base handoff. Its forward budget is not yet met.

Proof of Corollary 7.D.6. The base-handoff lemma gives $\Gamma_{\text{base}} \geq \Gamma_{\text{upd}}$, while compatibility and nested thresholds give $\Gamma_{\text{base}} < \Gamma_{\text{fwd},2} < \Gamma_{\text{fwd},3}$. Thus Lemma 7.D.16 applies with $K = 3$, and

supplies strict risk and attention-envelope decrease until the finite hit of $\Gamma_{\text{fwd},3}$. The fixed-depth forward theorem applies at that endpoint. $\square$

Together, the two extensions establish a one-stage small-step discrete analogue and show that the positive-update invariant permits a particular intermediate depth to be skipped. They do not assert a complete gradient-descent version of the progressive curriculum.

Chapter 8

Memory Capacity of Recurrent Neural Networks

8.1 Introduction

Modern recurrent sequence models replace an ever-growing attention cache with a fixed-size state. This gives linear-time decoding and bounded memory, but it also creates an information bottleneck: after an input token has passed, its effect on future predictions must be mediated by the recurrent state. The size of that state is easy to count; the amount of information in it that a trained model can use is not.

Classical storage-capacity results characterize information encoded in learned parameters. For example, a perceptron’s asymptotic capacity is two bits per synapse under Gardner’s analysis [94], and pretrained transformers empirically store about two bits of factual knowledge per weight parameter [10]. A recurrent state is different: it is sequence dependent, updated once per token, and reset between sequences. The numerical units are comparable, but the encoding processes and timescales differ. The question here is how much information about preceding tokens can be recovered through the model’s learned output head, and how this quantity scales with recurrent-state size.

This chapter measures that quantity with a synthetic sequence-recall task. A model observes a uniformly random token sequence and, after a separator, must reproduce it. Because the tokens are independent and uniformly distributed, the information content of the target sequence is known by construction. The reduction in cross-entropy relative to uniform prediction then provides an operational measure of how much sequence information the trained model recovers.

The main experiment studies DeltaNet, whose recurrent matrices implement a delta-rule associative memory [248, 327]. The evidence is organized in three levels. The primary reference set provides 30 selected measurements across a $32\times$ range of state sizes; seven training settings

This chapter is based on unpublished joint work with Tianyuan Zhang, Zixuan Wang, Jiawei Yang, and Hao Tan.

characterize dependence on the training distribution; and five additional experiments examine transformed recall, selective recall, repeated recall, vocabulary-weighted training, and reverse-order recall. These comparisons are descriptive diagnostics rather than matched causal ablations.

8.2 Measuring Recurrent-State Capacity

8.2.1 Random-sequence recall

Let the target sequence contain $N = sn$ tokens drawn independently from a usable vocabulary of size v , arranged as n strings of length s . An example is shown in Figure 8.1. The encoding segment is not included in the training loss. During training, the realized length is sampled from $\{N/8, N/4, N/2, N\}$ with relative frequencies $1 : 1 : 2 : 4$. The initial token announces that length, and all reported measurements are evaluated at length N .

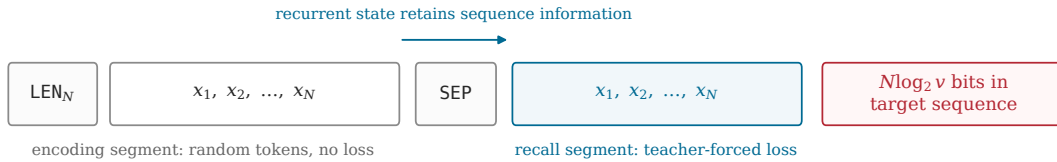


Figure 8.1: Random-Sequence Recall Task. Reference random-sequence recall task. The model encodes a random token sequence of announced length, carries information about it across a separator in its fixed-size recurrent state, and is evaluated on its reproduction of the sequence. Loss is applied only during recall.

The entropy of the target sequence is

$$H_{\text{seq}} = N \log_2 v. \quad (8.2.1)$$

For the reference generator, the configured vocabulary size is $V = 1024$. One separator and four length indicators are reserved, leaving $v = V - 5 = 1019$ available token types. A legacy experiment index uses $sn \log_2(V - 1)$ to group sequences; at this vocabulary size, the difference from the exact sequence entropy is approximately 0.006 bit per token. We retain the legacy index for run matching but use the generator’s usable vocabulary in the information calculation.

8.2.2 Recovered information

Let ℓ_{CE} be the mean teacher-forced cross-entropy in nats over the N_{recall} scored tokens. Uniform prediction incurs $\log_2 v$ bits per token. We therefore measure recovered information as

$$B_{\text{rec}} = N_{\text{recall}} \left(\log_2 v - \frac{\ell_{\text{CE}}}{\ln 2} \right). \quad (8.2.2)$$

Equation (8.2.2) is the total reduction in cross-entropy relative to a uniform predictor, expressed in bits. It measures information recovered through the model’s learned output head; it neither captures all information that may be present in the recurrent state nor implies that an arbitrary external decoder can recover the same information. The same distinction is important when comparing this quantity with measurements of information stored in model weights [10].

To compare models, we normalize recovered information by the recurrent-state size,

$$\rho = \frac{B_{\text{rec}}}{N_{\text{state}}}. \quad (8.2.3)$$

We report ρ as recovered bits per unit of state size.

8.2.3 DeltaNet state size

Each DeltaNet head maintains a matrix S_t . Using simplified notation, its delta-rule update is

$$S_t = S_{t-1} - \beta_t (S_{t-1} k_t - v_t) k_t^\top, \quad (8.2.4)$$

where k_t and v_t are the key and value at position t , and β_t is a learned update coefficient. The update corrects the value stored under the current key rather than merely adding another outer product.

The evaluated models use a sliding-window first layer followed by DeltaNet layers. We define recurrent-state size as the total number of scalar elements in the recurrent matrices. For L total layers, hidden width d , and head dimension h , there are d/h heads per recurrent layer and h^2 elements per head. Following this convention,

$$N_{\text{state}} = (L - 1) \frac{d}{h} h^2 = (L - 1) dh. \quad (8.2.5)$$

This denominator excludes learned weights, the first layer’s local-attention cache, and optional short-convolution buffers. Variants that double the query/key or value dimension use rectangular state matrices and are counted as twice the corresponding state size.

8.3 Experimental Protocol

The experimental archive contains logs from 5,297 training runs across 17 experiment groups and 179 reported configurations. Each run corresponds to a particular architecture, target length, learning rate, seed, and restart. We reconstructed every reported measurement directly from these logs using the following procedure:

1. We averaged each run’s accuracy and recovered information over its final ten valid evaluation records.

2. We grouped runs by experiment type, architecture, and nominal target-sequence information.
3. We retained the run with the largest recovered-information value in each group.

Exact ties in recovered information are resolved using the accuracy tie-breaker from the accompanying report. Because this rule applies only to exact ties in the primary metric, it does not alter the best-run curve. All 179 reconstructed measurements agree with the report at its displayed precision. The primary reference set contains 30 configurations selected from 1,672 training runs. The 30 selected measurements represent the best observed result for each configuration rather than 30 independent replications. Selection across learning rates, seeds, and restarts is useful for characterizing achieved performance, but it introduces upward selection bias and does not provide inferential uncertainty.

8.4 Results

8.4.1 Recovered information scales nearly linearly

Figure 8.2 plots the 30 reference measurements, including base DeltaNet configurations and variants with modified recurrent-state dimensions. The recurrent-state size ranges from 4,096 to 131,072. Recovered information remains close to two bits per unit of state size across this $32\times$ range. The fitted log-log slope is 0.925. The normalized measure has median 2.012, mean 2.045, and range [1.394, 2.409].

Recovered information and recurrent-state size

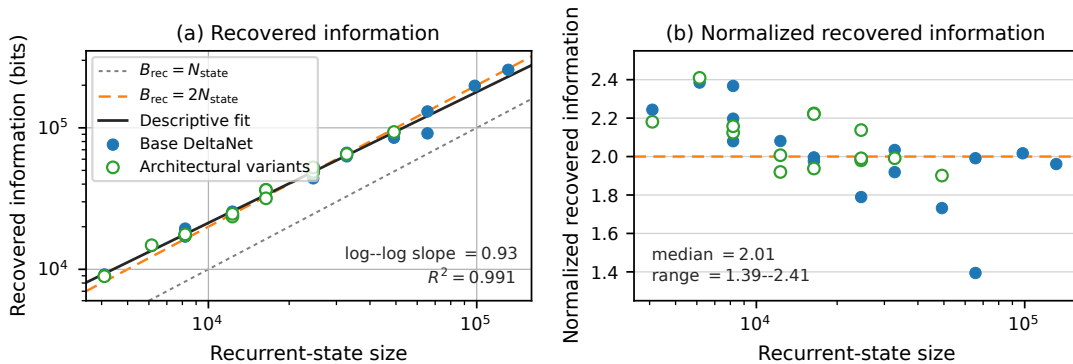


Figure 8.2: Recovered Information Scales With State Size. Scaling of recovered information with recurrent-state size. *Left:* recovered information for the 30 selected reference measurements. The solid line is an unweighted fit in log-log coordinates; dashed lines denote constant values of recovered information divided by state size. *Right:* the same measurements normalized by state size. The approximately constant ratio is an empirical finite-range summary, not an information-theoretic upper bound.

The high R^2 indicates that the log-log fit captures most of the observed variation in recovered information. It does not establish asymptotic linear scaling: the experiment covers one architecture family, one numerical precision regime, and a finite range of state sizes and sequence lengths. Within that scope, however, the absence of visible saturation supports using approximately two recovered bits per unit of state size as a practical capacity scale.

As a complementary result, changing the key-to-value dimension ratio does not appear to affect DeltaNet’s capacity per state element. Across the tested value-widened and key/query-widened configurations, normalized capacity remains between 2.12 and 2.22.

8.4.2 Dependence on the training setting

The primary reference setting is the final stage of a longer experimental progression. Figure 8.3 re-aggregates standard DeltaNet configurations from seven training settings. Fixed-length training has a median normalized value of 1.11. The length-curriculum setting has a median of 1.63, while the linear-curriculum setting has a median of 1.53. Across the 14 exponential-curriculum configurations, the median normalized value is 1.87.

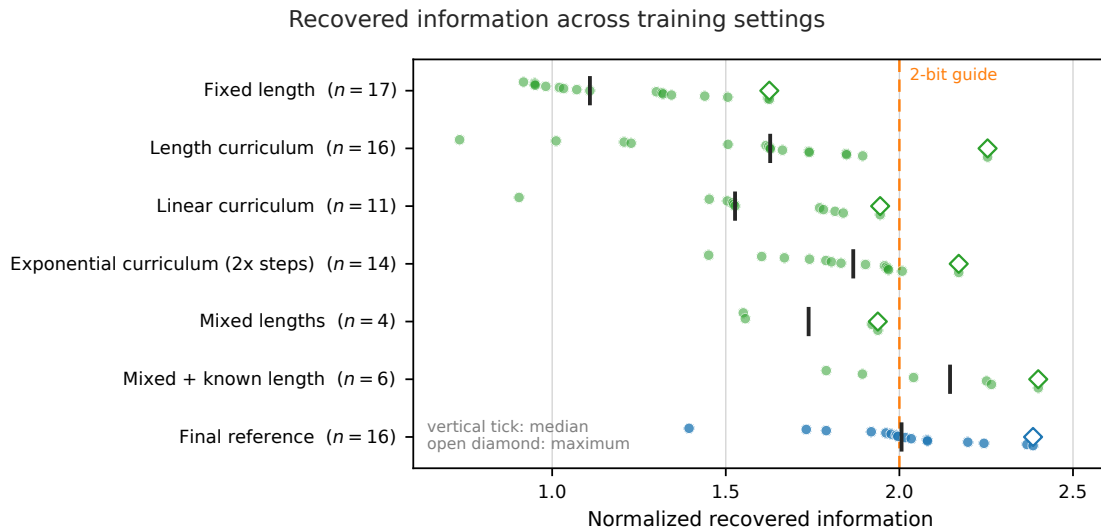


Figure 8.3: Normalized Information Across Training Settings. Recovered information normalized by state size across seven training settings, restricted to standard DeltaNet configurations. Each dot is one selected best-run measurement, the black tick is the group median, and the open diamond is its maximum.

Mixing lengths without announcing the realized length has a median of 1.74. With a length token, the median is 2.15; the final reference set has a median of 2.01 across a substantially broader state-size grid. This sequence is not a controlled monotone ablation, because the data generators, search budgets, and configuration coverage also changed. It nevertheless shows that the best observed results depend jointly on the architecture, training distribution, and evaluation

protocol. The two-bit result should therefore be stated specifically for mixed-length training with an announced length, rather than as an unconditional property of DeltaNet.

The exponential-curriculum experiments also include four configurations with the MLP blocks removed. Their token accuracies are only 0.062–0.149, although their normalized values range from 1.32 to 2.11 for unusually long target sequences. Thus, a substantial reduction in cross-entropy can coexist with poor exact-token accuracy. This observation is consistent with an important role for the surrounding blocks in reliable encoding and recall, but the four unmatched configurations do not isolate a causal MLP effect.

8.4.3 Additional recall tasks

We next compare the reference task with five extensions. Every figure uses the same log–log axes and shows all reference measurements in gray. The plotted points are selected best-run results, and the summaries in Table 8.1 are descriptive rather than inferential.

Table 8.1: Descriptive summaries of the reference and additional recall tasks. For normalized information and token accuracy, each cell reports the median followed by the observed range.

Recall task	Measurements	Normalized information	Token accuracy
Reference	30	2.01 [1.39, 2.41]	0.55 [0.24, 0.82]
Transformed recall	10	1.99 [1.61, 2.29]	0.52 [0.15, 0.65]
Selective recall	9	2.10 [1.86, 2.41]	0.61 [0.45, 0.92]
Repeated recall	5	1.86 [1.47, 2.25]	0.35 [0.32, 0.46]
Weighted vocabulary	9	1.90 [1.44, 2.15]	0.54 [0.34, 0.78]
Reverse-order recall	9	0.96 [0.71, 1.90]	0.11 [0.03, 0.43]

Transformed recall. After the separator, the model receives a shift $a \in \{0, 1, 2, 3\}$ and predicts $y_j = (x_j + a) \bmod V$, where V is the configured vocabulary size. As shown in Figure 8.4, its median normalized value of 1.99 is close to the reference median. This particular global token transformation can therefore be incorporated into recall with no apparent reduction under the present metric; the result does not imply that arbitrary computation on the encoded sequence has no cost.

Selective recall. During encoding, substrings are marked with either a MEMORIZE or FORGET control token, and only the former are queried. The median normalized value in Figure 8.5 is 2.10, with a maximum of 2.41. Because the control tokens are supplied before recall and only queried tokens contribute to the target-sequence information, the experiment cannot distinguish among active forgetting, selective storage, and the simpler requirement of retaining fewer tokens. The results therefore support the conclusion that selective recall achieves a normalized result comparable to the reference task under this state-size definition.

Repeated recall. The repeated-recall experiment primarily evaluates four successive copies of the target: 95 of 101 source configurations use four recall segments, while six early configurations use two. Recovered information is normalized to one copy of the target sequence and is not summed

Transformed recall and the reference task

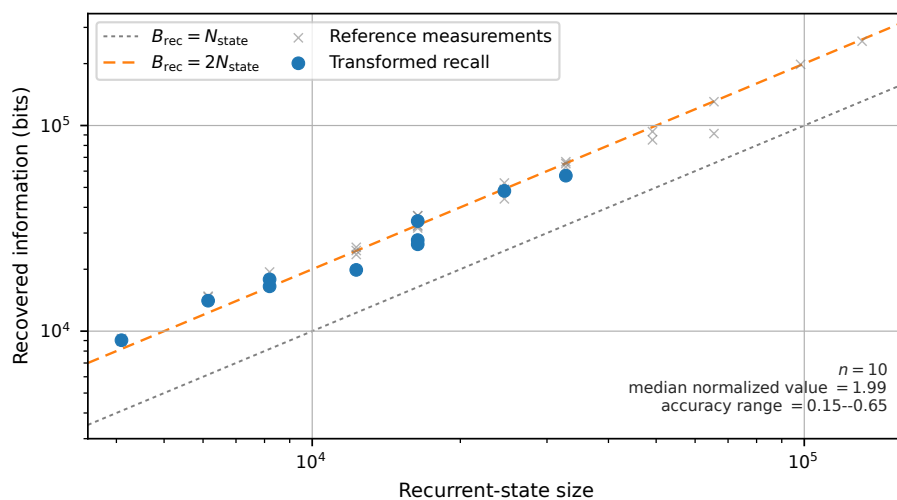


Figure 8.4: Transformed Recall. Transformed recall compared with the reference measurements on common axes. The ten measurements remain near the two-bit guide over the tested state sizes.

Selective recall and the reference task

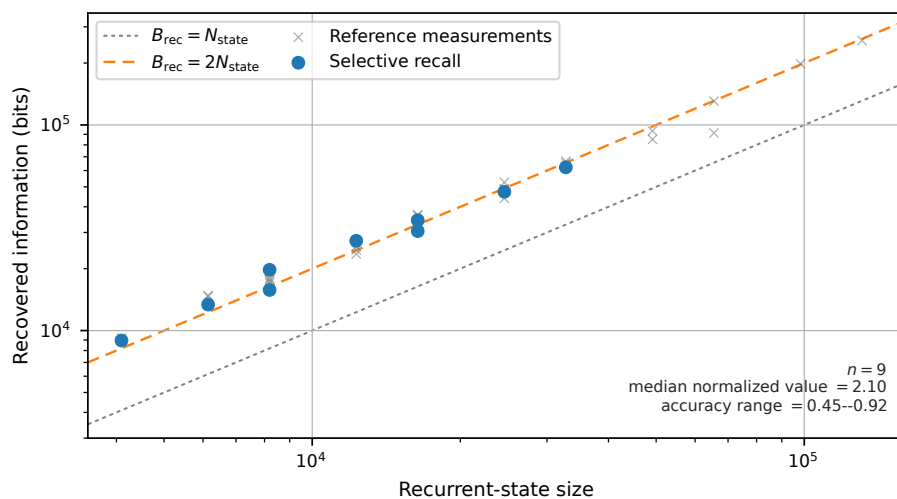


Figure 8.5: Selective Recall. Selective recall compared with the reference measurements. Recovered information normalized by state size remains within the reference range; the experiment does not distinguish active forgetting from selective storage.

across repetitions. In Figure 8.6, the median normalized value is 1.86, while median accuracy falls to 0.35. Because evaluation is teacher-forced, later recall segments occur after ground-truth target tokens have re-entered the model’s input. The task therefore measures stability under repeated presentation and recall, not repeated decoding from a frozen post-encoding state.

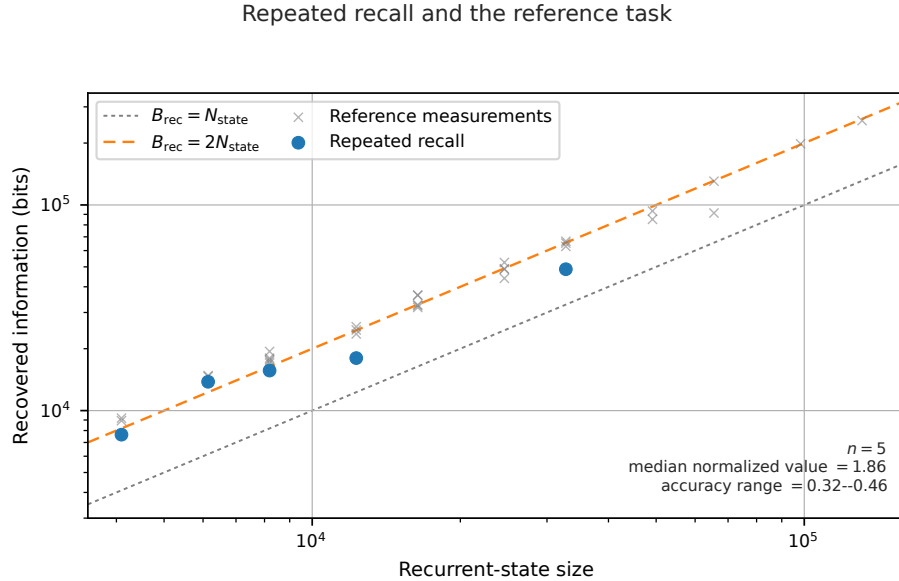


Figure 8.6: Repeated Recall. Repeated recall compared with the reference measurements. The five measurements show a modest downward shift in recovered information relative to state size and a narrow range of low accuracies under teacher forcing.

Weighted vocabulary. Tokens from the lowest quarter of the vocabulary receive ten times the loss weight. The nine selected measurements have a median normalized value of 1.90 and overlap the reference range in Figure 8.7. This sample shows no clearly separated upward shift, but no formal equivalence test or seed-level uncertainty analysis was performed; the available results therefore do not establish equivalence.

Reverse-order recall. The reverse-order task has median normalized information of 0.96 and median accuracy of 0.11, as shown in Figure 8.8. This difference might be attributed to the requested output order; however, the available experiments do not provide a controlled order-only ablation. The reverse-order generator uses fixed-length batches in a staged curriculum, omits the length token, places 32 separators between encoding and recall, and has 1,023 usable token types. The result establishes that this particular reverse-order task is substantially harder than the reference task; it does not isolate output order as the sole cause.

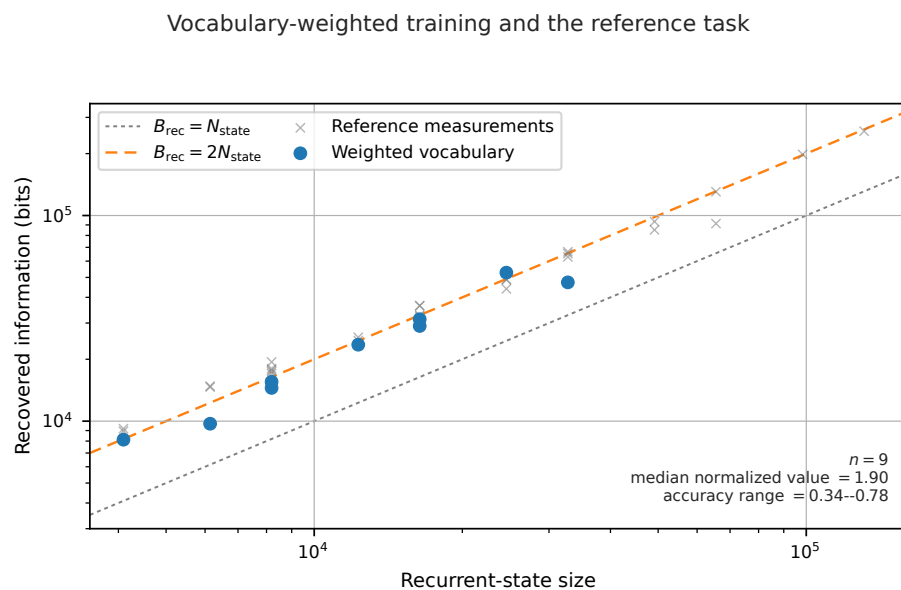


Figure 8.7: Vocabulary-Weighted Recall. Vocabulary-weighted training compared with the reference measurements. The nine results overlap the reference range and do not show a clearly separated upward shift.

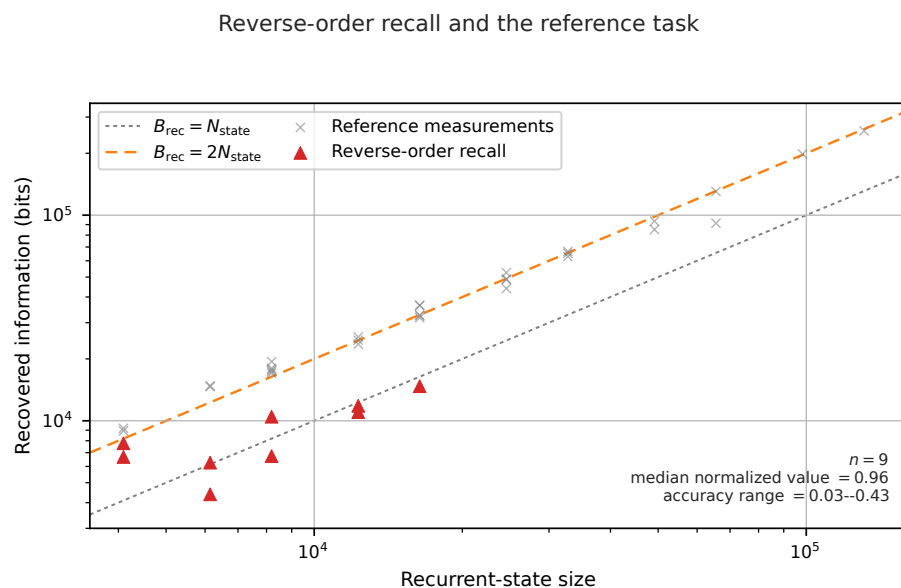


Figure 8.8: Reverse-Order Recall. Reverse-order recall compared with the reference measurements. It performs substantially worse, but also differs in length scheduling, separators, and usable vocabulary size.

8.5 Scope and Limitations

Cross-entropy-based estimate. Equation (8.2.2) measures the reduction in cross-entropy under the model’s learned output head with teacher forcing. Free-running generation can accumulate errors, and a separately trained decoder may expose a different amount of information.

Best-run selection and uncertainty. Each point is a maximum over a nonuniform collection of learning rates, seeds, and restarts. The resulting curve characterizes the best performance achieved in the available runs but is subject to upward selection bias. The available runs do not support confidence intervals, a hierarchical seed analysis, or preregistered hypothesis tests.

State-size definition. The two-bit scale depends on how N_{state} is defined. The state size in Equation (8.2.5) includes DeltaNet matrices but excludes learned weights, local-attention caches, and short-convolution buffers. The numerator is nevertheless obtained from the complete model, whose performance can benefit from those components. Other explicit counting conventions would therefore rescale the normalized measure while leaving the underlying recovered-information measurements unchanged.

Differences across task configurations. The training settings and additional recall tasks do not form a balanced factorial experiment. They differ in search coverage and, in some cases, in length schedules, separators, control tokens, target structure, or loss weighting. The figures summarize performance under these configurations but do not identify a single intervention as the sole cause.

Data coverage. One corrected copy-sequence experiment group has no parseable measurement in the report. An earlier version is present in the 178-point audit but is not used for the main result because its sequence construction differs from the final reference task. The central claim rests on the 30-point reference set, not on treating all experiment groups as exchangeable evidence.

Architecture and domain. The clean scaling statement is established for DeltaNet on uniformly random token sequences. By removing semantic structure and compressibility, the random-sequence construction makes the target-sequence entropy exactly calculable, but it also limits direct transfer to natural language. Denser matched sweeps across other recurrent architectures, state precisions, and free-running evaluation settings are needed before interpreting the observed ratio as an architecture-class constant.

Chapter 9

Conclusion

This thesis has studied how gradient-based training turns structure in data into structure inside a neural network. Across self-supervised representation learning, transformer reasoning, and recurrent architectures, the central finding is that feature learning is governed not by the objective alone, but by the interaction among the objective, the data distribution, and the architecture. Data augmentations determine which correlations survive contrastive training; architectural asymmetry changes which solutions are reached in non-contrastive learning; attention objectives select different spatial patterns; the organization of reasoning examples determines how capability expands; and recurrent weight sharing determines both what can be remembered and how computation is refined across depth.

The analyses in this thesis make these interactions explicit by following the training process. Rather than treating a trained network only as a global optimizer or a fixed feature map, they identify the intermediate mechanisms through which useful representations emerge: feature decoupling, substitution and acceleration, attention concentration, easy-to-hard relays, backward feature correction, and the allocation of information within a finite recurrent state. Together, these mechanisms support a common view of generalization. A model generalizes when training discovers a reusable rule or representation shared by many examples, and the training setup provides a path along which that structure can be learned.

Lessons and Implications

The results of this thesis support a common lesson: gradient-based training generalizes when three elements align:

1. The data must contain a structure that recurs across examples;
2. the learning signal must make that structure accessible to optimization;
3. and the architecture must be able to represent and reuse it.

None of these elements is sufficient by itself. A useful pattern may exist in the data yet remain overshadowed by stronger spurious correlations; an architecture may be expressive enough to implement a computation yet fail to discover it in gradient-based optimization. Generalization is therefore not just a property of the function class or the final objective. It is a consequence of the optimization trajectory by which training constructs a usable internal representation.

Optimization should be understood as the structure discovery process. Several results in this thesis show that low loss or global optimality does not determine what a network learns internally. In self-supervised learning, augmentations, prediction heads, and reconstruction targets change which features are reinforced and how they are organized. In reinforcement learning, the spacing of difficulty levels changes whether optimization progresses smoothly or remains stalled before an abrupt transition. These findings suggest that a theory of neural networks should explain not only whether optimization converges, but which features or computations emerge, in what order, and through which intermediate mechanisms.

Data distribution is part of the learning algorithm. In the practice of deep learning, data engineering is as important as tuning the learning algorithm for neural networks to learn desirable features. In contrastive learning, augmented views determine which correlations remain visible to be extracted; in reasoning, the spectrum of examples determines whether gradient signal can transfer between data points; in random recall tasks, the retrieval task structure affects how much information a recurrent state can recover. Thus, data should not be viewed merely as samples drawn from a fixed environment. It actively shapes the optimization landscape and the path through capability space. A practical implication is that data design should preserve the desired structure while weakening shortcuts and maintaining learnable transitions between levels of difficulty.

Reusable internal structure is the basis of extrapolation. Across the settings studied here, successful generalization relies on learning a unit of computation at the appropriate granularity: a semantic feature that survives augmentation, an attention pattern across tokens, or an atomic action that can be composed repeatedly. Once learned, such a unit can be applied to new examples and, in some cases, repeated beyond the length, difficulty, or depth encountered during training. This suggests a design principle for systematic generalization: training procedures should encourage models to discover modular computations whose domain of reuse is broader than the individual training instances that reveal them.

These lessons also suggest a broader agenda for learning theory. The central object of study should not be only the final predictor, but the relationship between the structure present in the data, the gradient paths created by the training procedure, and the representations made possible by the architecture. The central lesson of this thesis is therefore that optimization does more than finding minima: it selects and constructs reusable internal structure, and that structure determines what the trained model can do beyond its training setting.

Limitations and Future Directions

Self-supervised feature learning. The self-supervised analyses rely on simplified models and data distributions that isolate individual learning mechanisms. Their conclusions therefore describe principles of feature selection and organization more directly than the full behavior of large-scale systems. Future work should determine how these mechanisms interact in deeper networks, richer data, and realistic training pipelines, and whether they continue to explain representation quality at practical scale.

Transformer reasoning and post-training. The reasoning chapters focus on controlled forms of compositional generalization and difficulty. Real-world reasoning also depends on semantic knowledge, planning, search, and the ability to identify relevant information in long contexts. A broader theory should explain how these dimensions interact, when progress on simpler problems transfers to harder ones, and how training can encourage reusable computations rather than solutions tied to a particular task format or context length.

Modern recurrent architectures. The recurrent chapters present two complementary powers of recurrence: reusing computation across depth and maintaining information across sequence positions. The looped-transformer analysis relies on assumptions that separate feature construction, routing, and refinement, while the capacity study examines one architecture using a deliberately controlled recall task. Future work should study how recurrent computation and usable memory interact when these components are learned jointly, and how the resulting behavior varies across architectures, data distributions, decoding tasks, and numerical precision. Extending this analysis to adaptive or unbounded computation could clarify when recurrence produces stable systematic generalization and how that ability is constrained by the information available in the recurrent state.

Despite these limitations, the studies in this thesis show that training-dynamics analysis can connect optimization to the representations and patterns found inside neural networks. A promising future direction is to apply the same approach toward deeper architectures, and less idealized training procedures. Such a theory would not only predict whether a model can perform well, but explain which internal structures and algorithmic components make the generalization possible and how the training process discovers them.

Bibliography

- [1] Emmanuel Abbe, Enric Boix-Adserà, Matthew S. Brennan, Guy Bresler, and Dheeraj Nagaraj. The staircase property: How hierarchical structure can guide deep learning. In *Advances in Neural Information Processing Systems*, 2021. [7.2](#)
- [2] Emmanuel Abbe, Samy Bengio, Aryo Lotfi, Colin Sandon, and Omid Saremi. How far can transformers reason? the globality barrier and inductive scratchpad. *Advances in Neural Information Processing Systems*, 37:27850–27895, 2024. [5.1](#)
- [3] Kwangjun Ahn, Xiang Cheng, Hadi Daneshmand, and Suvrit Sra. Transformers learn to implement preconditioned gradient descent for in-context learning. *arXiv preprint arXiv:2306.00297*, 2023. [4.6](#)
- [4] Kartik Ahuja and Amin Mansouri. On provable length and compositional generalization. *arXiv preprint arXiv:2402.04875*, 2024. [5.1](#), [5.1.1](#)
- [5] Zeyuan Allen-Zhu and Yuanzhi Li. What can resnet learn efficiently, going beyond kernels? In *NeurIPS 2019 : Thirty-third Conference on Neural Information Processing Systems*, pages 9017–9028, 2019. [2.1.2](#), [3.6](#)
- [6] Zeyuan Allen-Zhu and Yuanzhi Li. Towards understanding ensemble, knowledge distillation and self-distillation in deep learning. *arXiv preprint arXiv:2012.09816*, 2020. [2.1.2](#), [3.2](#), [3.6](#), [3.A.1](#), [3.H.1](#), [5.3.4](#), [5.A.1](#), [5.B.1](#)
- [7] Zeyuan Allen-Zhu and Yuanzhi Li. Forward super-resolution: How can GANs learn hierarchical generative models for real-world distributions. *arXiv preprint arXiv:2106.02619*, 2021. [2.1.2](#), [3.6](#), [5.A.3](#)
- [8] Zeyuan Allen-Zhu and Yuanzhi Li. Feature purification: How adversarial training performs robust deep learning. In *2021 IEEE 62nd Annual Symposium on Foundations of Computer Science (FOCS)*, pages 977–988. IEEE, 2022. [\(document\)](#), [2.1](#), [2.1.2](#), [2.3](#), [2.B.1](#), [2.B.1](#), [2.B.2](#), [2.B.2](#), [3.5](#), [3.2](#), [3.6](#), [5.A.3](#)
- [9] Zeyuan Allen-Zhu and Yuanzhi Li. Backward feature correction: How deep learning performs deep (hierarchical) learning. In *Proceedings of the 36th Conference on Learning Theory*, 2023. [2.1.2](#), [3.3.1](#), [3.6](#), [3](#), [7.2](#)

- [10] Zeyuan Allen-Zhu and Yuanzhi Li. Physics of language models: Part 3.3, knowledge capacity scaling laws. *arXiv preprint arXiv:2404.05405*, 2024. 8.1, 8.2.2
- [11] Zeyuan Allen-Zhu, Yuanzhi Li, and Yingyu Liang. Learning and generalization in overparameterized neural networks, going beyond two layers. In *NeurIPS 2019 : Thirty-third Conference on Neural Information Processing Systems*, pages 6158–6169, 2019. 2.1.2, 3.6
- [12] Zeyuan Allen-Zhu, Yuanzhi Li, and Zhao Song. A convergence theory for deep learning via over-parameterization. In *ICML 2019 : Thirty-sixth International Conference on Machine Learning*, pages 242–252, 2019. 2.1.2, 3.6
- [13] Zeyuan Allen-Zhu, Yuanzhi Li, and Zhao Song. On the convergence rate of training recurrent neural networks. In *Advances in Neural Information Processing Systems*, pages 6676–6688, 2019. 2.1.2, 3.6
- [14] Alexandr Andoni, Rina Panigrahy, Gregory Valiant, and Li Zhang. Learning polynomials with neural networks. In *International conference on machine learning*, pages 1908–1916. PMLR, 2014. 3.3.1
- [15] Cem Anil, Yuhuai Wu, Anders Andreassen, Aitor Lewkowycz, Vedant Misra, Vinay Ramasesh, Ambrose Slone, Guy Gur-Ari, Ethan Dyer, and Behnam Neyshabur. Exploring length generalization in large language models. *Advances in Neural Information Processing Systems*, 35:38546–38556, 2022. 5.1
- [16] S. Arora and B. Barak. *Computational Complexity: A Modern Approach*. Cambridge University Press, 2006. ISBN 978-0-521-42426-4. URL <https://theory.cs.princeton.edu/complexity/book.pdf>. 1, 5.2
- [17] Sanjeev Arora, Yuanzhi Li, Yingyu Liang, Tengyu Ma, and Andrej Risteski. Linear algebraic structure of word senses, with applications to polysemy. *Transactions of the Association for Computational Linguistics*, 6:483–495, 2018. 2.2.1
- [18] Sanjeev Arora, Nadav Cohen, Wei Hu, and Yuping Luo. Implicit regularization in deep matrix factorization. *Advances in Neural Information Processing Systems*, 32, 2019. 3.6
- [19] Sanjeev Arora, Simon Du, Wei Hu, Zhiyuan Li, and Ruosong Wang. Fine-grained analysis of optimization and generalization for overparameterized two-layer neural networks. In *International Conference on Machine Learning*, pages 322–332. PMLR, 2019. 3.6
- [20] Sanjeev Arora, Hrishikesh Khandeparkar, Mikhail Khodak, Orestis Plevrakis, and Nikunj Saunshi. A theoretical analysis of contrastive unsupervised representation learning. *arXiv preprint arXiv:1902.09229*, 2019. 2.1, 2.1.2, 3.6, 4.1, 4.6
- [21] Jordan T Ash, Surbhi Goel, Akshay Krishnamurthy, and Dipendra Misra. Investigating the role of negatives in contrastive representation learning. *arXiv preprint arXiv:2106.09943*, 2021. 3.1
- [22] Sanghwan Bae, Jiwoo Hong, Min Young Lee, Hanbyul Kim, JeongYeon Nam, and Donghyun Kwak. Online difficulty filtering for reasoning oriented reinforcement learning. *arXiv*

preprint *arXiv:2504.03380*, 2025. 6.2

- [23] Sangmin Bae, Adam Fisch, Hrayr Harutyunyan, Ziwei Ji, Sungwoong Kim, and Tal Schuster. Relaxed recursive transformers: Effective parameter sharing with layer-wise LoRA. In *International Conference on Learning Representations*, 2025. 7.1, 7.2
- [24] Shaojie Bai, J. Zico Kolter, and Vladlen Koltun. Deep equilibrium models. In *Advances in Neural Information Processing Systems*, 2019. 7.2
- [25] Yu Bai and Jason D. Lee. Beyond Linearization: On Quadratic and Higher-Order Approximation of Wide Neural Networks. *arXiv:1910.01619 [cs, math, stat]*, February 2020. 2.1.2
- [26] Andrea Banino, Jan Balaguer, and Charles Blundell. PonderNet: Learning to ponder. *arXiv preprint arXiv:2107.05407*, 2021. 7.2
- [27] Yamini Bansal, Gal Kaplun, and Boaz Barak. For self-supervised learning, rationality implies generalization, provably. *arXiv preprint arXiv:2010.08508*, 2020. 3.1
- [28] Han Bao, Yoshihiro Nagano, and Kento Nozawa. Sharp learning bounds for contrastive unsupervised representation learning. *arXiv preprint arXiv:2110.02501*, 2021. 3.1
- [29] Hangbo Bao, Li Dong, Songhao Piao, and Furu Wei. Beit: Bert pre-training of image transformers. *arXiv preprint arXiv:2106.08254*, 2021. 3.3.2, 3.6
- [30] Adrien Bardes, Jean Ponce, and Yann LeCun. Vicreg: Variance-invariance-covariance regularization for self-supervised learning. *arXiv preprint arXiv:2105.04906*, 2021. 3.1
- [31] David A Barrington. Bounded-width polynomial-size branching programs recognize exactly those languages in nc. In *Proceedings of the eighteenth annual ACM symposium on Theory of computing*, pages 1–5, 1986. 5.1.1, 5.2.1, 5.2.1
- [32] Yoshua Bengio, Jérôme Louradour, Ronan Collobert, and Jason Weston. Curriculum learning. In *Proceedings of the 26th International Conference on Machine Learning*, 2009. 7.2
- [33] Srinadh Bhojanapalli, Ayan Chakrabarti, Daniel Glasner, Daliang Li, Thomas Unterthiner, and Andreas Veit. Understanding robustness of transformers for image classification. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 10231–10241, 2021. 4.6
- [34] Blake Bordelon and Cengiz Pehlevan. Self-consistent dynamical field theory of kernel evolution in wide neural networks. In *Advances in Neural Information Processing Systems*, 2022. 7.2
- [35] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020. 3.6
- [36] Alon Brutzkus and Amir Globerson. Globally optimal gradient descent for a convnet with gaussian inputs. In *Proceedings of the 34th International Conference on Machine Learning-Volume*

- 70, pages 605–614. JMLR. org, 2017. [2.1.2](#), [3.6](#)
- [37] Dake Bu, Wei Huang, Andi Han, Atsushi Nitanda, Hau-San Wong, Qingfu Zhang, and Taiji Suzuki. Provable benefit of curriculum in transformer tree-reasoning post-training. In *Proceedings of the 43rd International Conference on Machine Learning*, 2026. [6.2](#), [7.2](#)
 - [38] Shuhao Cao, Peng Xu, and David A Clifton. How to understand masked autoencoders. *arXiv preprint arXiv:2202.03670*, 2022. [4.1](#)
 - [39] Mathilde Caron, Ishan Misra, Julien Mairal, Priya Goyal, Piotr Bojanowski, and Armand Joulin. Unsupervised learning of visual features by contrasting cluster assignments. *Advances in Neural Information Processing Systems*, 33:9912–9924, 2020. [3.1](#), [3.6](#)
 - [40] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 9650–9660, 2021. ([document](#)), [3.1](#), [3.1](#), [4.1](#), [4.1](#), [4.7](#)
 - [41] Miguel Á. Carreira-Perpiñán and Geoffrey E. Hinton. On contrastive divergence learning. In *AISTATS*, 2005. [2.1.2](#)
 - [42] Ashok K Chandra, Larry Stockmeyer, and Uzi Vishkin. Constant depth reducibility. *SIAM Journal on Computing*, 13(2):423–439, 1984. [5.2.1](#)
 - [43] Fan Chen, Audrey Huang, Noah Golowich, Sadhika Malladi, Adam Block, Jordan T Ash, Akshay Krishnamurthy, and Dylan J Foster. The coverage principle: How pre-training enables post-training. *arXiv preprint arXiv:2510.15020*, 2025. [6.2](#)
 - [44] Lijie Chen, Binghui Peng, and Hongxun Wu. Theoretical limitations of multi-layer transformer. *arXiv preprint arXiv:2412.02975*, 2024. [5.1](#)
 - [45] Shuxiao Chen, Edgar Dobriban, and Jane H Lee. A group-theoretic framework for data augmentation. *Journal of Machine Learning Research*, 21(245):1–71, 2020. [2.1](#), [2.1.2](#), [2](#), [3.3.2](#), [6.A.3](#)
 - [46] Sitan Chen and Yuanzhi Li. Provably learning a multi-head attention layer. *arXiv preprint arXiv:2402.04084*, 2024. [4.6](#)
 - [47] Sitan Chen, Jerry Li, Yuanzhi Li, and Anru R Zhang. Learning polynomial transformations. *arXiv preprint arXiv:2204.04209*, 2022. [3.3.1](#)
 - [48] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *International conference on machine learning*, pages 1597–1607. PmLR, 2020. [2.1](#), [3.1](#), [3.2](#), [3.6](#), [4.1](#), [4.2.1](#)
 - [49] Ting Chen, Simon Kornblith, Kevin Swersky, Mohammad Norouzi, and Geoffrey Hinton. Big Self-Supervised Models are Strong Semi-Supervised Learners. *Arxiv*, page 13, 2020. [2.1](#), [2.1.2](#)
 - [50] Ting Chen, Calvin Luo, and Lala Li. Intriguing properties of contrastive losses. *Advances in Neural Information Processing Systems*, 34:11834–11845, 2021. [3.1.1](#), [3.2](#), [3.6](#), [4.1](#), [4.6](#)

- [51] Xiaoyin Chen, Jiarui Lu, Minsu Kim, Dinghuai Zhang, Jian Tang, Alexandre Piché, Nicolas Gontier, Yoshua Bengio, and Ehsan Kamalloo. Self-evolving curriculum for llm reasoning. *arXiv preprint arXiv:2505.14970*, 2025. [6.2](#), [6.5](#)
- [52] Xingwu Chen, Tianle Li, and Difan Zou. Reshaping reasoning in LLMs: A theoretical analysis of RL training dynamics through pattern selection. *arXiv preprint arXiv:2506.04695*, 2025. [6.2](#)
- [53] Xinlei Chen and Kaiming He. Exploring simple siamese representation learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 15750–15758, 2021. [2.1](#), [2.1.2](#), [2.2.2](#), [3.1](#), [3.1](#), [3.1.1](#), [3.2](#), [3.2](#), [3.3.2](#)
- [54] Xinlei Chen, Haoqi Fan, Ross Girshick, and Kaiming He. Improved baselines with momentum contrastive learning. *arXiv preprint arXiv:2003.04297*, 2020. [3.6](#)
- [55] Xinlei Chen, Saining Xie, and Kaiming He. An empirical study of training self-supervised vision transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 9640–9649, 2021. [\(document\)](#), [3.6](#), [4.1](#), [4.7](#)
- [56] Zixiang Chen, Yuan Cao, Difan Zou, and Quanquan Gu. How much over-parameterization is sufficient to learn deep relu networks? *arXiv preprint arXiv:1911.12360*, 2019. [2.1.2](#)
- [57] Zixiang Chen, Greg Yang, Qian Zhao, and Quanquan Gu. Global convergence and rich feature learning in l -layer infinite-width neural networks under μ parametrization. In *Proceedings of the 42nd International Conference on Machine Learning*, 2025. [7.2](#)
- [58] Yuan Cheng, Yu Huang, Zhe Xiong, Yingbin Liang, and Vincent YF Tan. Transformers provably learn directed acyclic graphs via kernel-guided mutual information. *arXiv preprint arXiv:2510.25542*, 2025. [5.3.3](#)
- [59] Yuan Cheng, Fengzhuo Zhang, Yunlong Hou, Cunxiao Du, Chao Du, Tianyu Pang, Aixin Sun, and Zhuoran Yang. Demystifying the slash pattern in attention: The role of rope. *arXiv preprint arXiv:2601.08297*, 2026. [6.3.2](#)
- [60] Lenaic Chizat and Francis Bach. Implicit bias of gradient descent for wide two-layer neural networks trained with the logistic loss. In *Conference on Learning Theory*, pages 1305–1338. PMLR, 2020. [3.6](#)
- [61] Francois Chollet, Mike Knoop, Gregory Kamradt, Bryan Landers, and Henry Pinkard. Arc-agi-2: A new challenge for frontier ai reasoning systems. *arXiv preprint arXiv:2505.11831*, 2025. [5.1](#)
- [62] Tianzhe Chu, Yuexiang Zhai, Jihan Yang, Shengbang Tong, Saining Xie, Dale Schuurmans, Quoc V. Le, Sergey Levine, and Yi Ma. SFT memorizes, RL generalizes: A comparative study of foundation model post-training. *arXiv preprint arXiv:2501.17161*, *arXiv:2501.17161*, 2025. [6.3.4](#)
- [63] Adam Coates, Andrew Y. Ng, and Honglak Lee. An analysis of single-layer networks in unsupervised feature learning. In *AISTATS*, 2011. [2.1.2](#)

- [64] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, and Reiichiro Nakano. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021. 5.1
- [65] Elisabetta Cornacchia and Elchanan Mossel. A mathematical model for curriculum learning for parities. In *Proceedings of the 40th International Conference on Machine Learning*, 2023. 7.2
- [66] Ganqu Cui, Yuchen Zhang, Jiacheng Chen, Lifan Yuan, Zhi Wang, Yuxin Zuo, Haozhan Li, Yuchen Fan, Huayu Chen, Weize Chen, et al. The entropy mechanism of reinforcement learning for reasoning language models. *arXiv preprint arXiv:2505.22617*, 2025. 6.2
- [67] Yatin Dandi, Luca Pesce, Lenka Zdeborová, and Florent Krzakala. The computational advantage of depth: Learning high-dimensional hierarchical functions with gradient descent. In *Advances in Neural Information Processing Systems*, 2025. 7.2
- [68] Yatin Dandi, Matteo Vilucchio, Lorenzo Arnaboldi, Hugo Tabanelli, and Florent Krzakala. Deep learning as neural low-degree filtering: A spectral theory of hierarchical feature learning. *arXiv preprint arXiv:2605.13612*, 2026. 7.2
- [69] Damek Davis and Benjamin Recht. What is the objective of reasoning with reinforcement learning? *arXiv preprint arXiv:2510.13651*, 2025. 6.2
- [70] DeepSeek-AI. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *ArXiv*, abs/2501.12948, 2025. URL <https://api.semanticscholar.org/CorpusID:275789950>. 5.1, 6.1, 6.2, 6.5
- [71] Mostafa Dehghani, Stephan Gouws, Oriol Vinyals, Jakob Uszkoreit, and Lukasz Kaiser. Universal transformers. In *International Conference on Learning Representations*, 2019. 7.1, 7.2
- [72] Max Dehn. Über unendliche diskontinuierliche gruppen. *Mathematische Annalen*, 71(1): 116–144, 1911. 5.2.1
- [73] Max Dehn. On infinite discontinuous groups. In *Papers on Group Theory and Topology*, pages 133–178. Springer, 1987. 5.2.1
- [74] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, pages 4171–4186, 2019. 2.1, 2.1.2, 3.3.2, 3.6, 4.1
- [75] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020. 4.1, 4.2.3, 4.2.3, 4.7, 4.7
- [76] Simon S. Du, Wei Hu, and Jason D. Lee. Algorithmic regularization in learning deep homogeneous models: Layers are automatically balanced. In *Advances in Neural Information Processing Systems*, 2018. 7.2
- [77] Simon S Du, Jason D Lee, and Yuandong Tian. When is a convolutional filter easy to learn? In *International Conference on Learning Representations*, 2018. 3.6

- [78] Simon S. Du, Jason D. Lee, Haochuan Li, Liwei Wang, and Xiyu Zhai. Gradient descent finds global minima of deep neural networks. In *36th International Conference on Machine Learning, ICML 2019*, pages 1675–1685, 2019. 2.1.2, 3.6
- [79] Nouha Dziri, Ximing Lu, Melanie Sclar, Xiang Lorraine Li, Liwei Jiang, Bill Yuchen Lin, Sean Welleck, Peter West, Chandra Bhagavatula, and Ronan Le Bras. Faith and fate: Limits of transformers on compositionality. *Advances in Neural Information Processing Systems*, 36: 70293–70332, 2023. 5.1
- [80] Benjamin L Edelman, Surbhi Goel, Sham Kakade, and Cyril Zhang. Inductive biases and variable creation in self-attention mechanisms. In *International Conference on Machine Learning*, pages 5793–5831. PMLR, 2022. 4.6
- [81] Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021. <https://transformer-circuits.pub/2021/framework/index.html>. 5.4
- [82] Aleksandr Ermolov, Aliaksandr Siarohin, Enver Sangineto, and Nicu Sebe. Whitening for self-supervised representation learning. In *International Conference on Machine Learning*, pages 3015–3024. PMLR, 2021. 3.1, 3.1, 3.1.1, 3.6, 3.7
- [83] Carlos Esteves. Theoretical aspects of group equivariant neural networks. *arXiv preprint arXiv:2004.05154*, 2020. 6.A.3
- [84] Ying Fan, Yilun Du, Kannan Ramchandran, and Kangwook Lee. Looped transformers for length generalization. In *International Conference on Learning Representations*, 2025. 7.1, 7.2
- [85] Guhao Feng, Bohang Zhang, Yuntian Gu, Haotian Ye, Di He, and Liwei Wang. Towards revealing the mystery behind chain of thought: a theoretical perspective. *Advances in Neural Information Processing Systems*, 36:70757–70798, 2023. 5.1, 3, 5.2.1
- [86] Peter Földiák and Malcolm P. Young. *Sparse coding in the primate cortex*. 1998. 2.2.1
- [87] Thomas Foster, Anya Sims, Johannes Forkel, Mattie Fellows, and Jakob Foerster. Learning to reason at the frontier of learnability. *arXiv preprint arXiv:2502.12272*, 2025. 6.2
- [88] John B Fraleigh. *A first course in abstract algebra (8th Edition)*. Pearson India, 2023. 5.1.1
- [89] Haoyu Fu, Zixuan Wang, Eshaan Nichani, and Jason D. Lee. Learning hierarchical polynomials of multiple nonlinear features with three-layer networks. In *International Conference on Learning Representations*, 2025. 7.2
- [90] Jason Fulman and Robert Guralnick. Bounds on the number and sizes of conjugacy classes in finite chevalley groups with applications to derangements. *arXiv preprint arXiv:0902.2238*, 2009. 6.B.1

- [91] Jiaxuan Gao, Shusheng Xu, Wenjie Ye, Weilin Liu, Chuyi He, Wei Fu, Zhiyu Mei, Guangju Wang, and Yi Wu. On designing effective RL reward at training time for LLM reasoning. *arXiv preprint arXiv:2410.15115*, 2024. [6.1](#), [6.3.4](#)
- [92] Tianchen Gao, Jiashun Jin, Zheng Tracy Ke, and Gabriel Moryoussef. A comparison of DeepSeek and other LLMs. *arXiv preprint arXiv:2502.03688*, 2025. [5.1](#)
- [93] Tianyu Gao, Xingcheng Yao, and Danqi Chen. Simcse: Simple contrastive learning of sentence embeddings. *arXiv preprint arXiv:2104.08821*, 2021. [3.1](#), [3.6](#)
- [94] Elizabeth Gardner. The space of interactions in neural network models. *Journal of Physics A: Mathematical and General*, 21(1):257–270, 1988. doi: 10.1088/0305-4470/21/1/030. [8.1](#)
- [95] Khashayar Gatmiry, Nikunj Saunshi, Sashank J. Reddi, Stefanie Jegelka, and Sanjiv Kumar. Can looped transformers learn to implement multi-step gradient descent for in-context learning? In *Proceedings of the 41st International Conference on Machine Learning*, 2024. [7.1](#), [7.2](#)
- [96] Antoine Gautier, Quynh N Nguyen, and Matthias Hein. Globally optimal training of generalized polynomial neural networks with nonlinear spectral methods. *Advances in Neural Information Processing Systems*, 29, 2016. [3.3.1](#)
- [97] Rong Ge, Jason D. Lee, and Tengyu Ma. Learning one-hidden-layer neural networks with landscape design. In *6th International Conference on Learning Representations, ICLR 2018*, 2018. [2.1.2](#)
- [98] Jonas Geiping, Sean McLeish, Neel Jain, John Kirchenbauer, Siddharth Singh, Brian R. Bartoldson, Bhavya Kailkhura, Abhinav Bhatle, and Tom Goldstein. Scaling up test-time compute with latent reasoning: A recurrent depth approach. In *Advances in Neural Information Processing Systems*, 2025. [7.1](#), [7.2](#)
- [99] Amin Ghiasi, Hamid Kazemi, Eitan Borgnia, Steven Reich, Manli Shu, Micah Goldblum, Andrew Gordon Wilson, and Tom Goldstein. What do vision transformers learn? a visual exploration. *arXiv preprint arXiv:2212.06727*, 2022. [4.6](#)
- [100] Angeliki Giannou, Shashank Rajput, Jy-yong Sohn, Kangwook Lee, Jason D Lee, and Dimitris Papailiopoulos. Looped transformers as programmable computers. *arXiv preprint arXiv:2301.13196*, 2023. [4.6](#), [7.1](#), [7.2](#)
- [101] Gail H Godbeer, John Lipscomb, and Michael George Luby. *On the computational complexity of finding stable state vectors in connectionist models (Hopfield nets)*. Department of Computer Science, University of Toronto, 1987. [5.2](#)
- [102] Noah Golowich, Samy Jelassi, David Brandfonbrener, Sham M Kakade, and Eran Malach. The role of sparsity for length generalization in transformers. *arXiv preprint arXiv:2502.16792*, 2025. [5.1](#), [5.1.1](#)
- [103] Sven Gowal, Po-Sen Huang, Aaron van den Oord, Timothy Mann, and Pushmeet Kohli. Self-supervised adversarial robustness for the low-label, high-data regime. In *International Conference on Learning Representations*, 2020. [\(document\)](#), [3.5](#)

- [104] Alex Graves. Adaptive computation time for recurrent neural networks. *arXiv preprint arXiv:1603.08983*, 2016. [7.2](#)
- [105] Evan Greene and Jon A Wellner. Exponential bounds for the hypergeometric distribution. *Bernoulli: official journal of the Bernoulli Society for Mathematical Statistics and Probability*, 23(3): 1911, 2017. [4.A.1](#)
- [106] Jean-Bastien Grill, Florian Strub, Florent Altché, Corentin Tallec, Pierre Richemond, Elena Buchatskaya, Carl Doersch, Bernardo Avila Pires, Zhaohan Guo, Mohammad Gheshlaghi Azar, et al. Bootstrap your own latent-a new approach to self-supervised learning. *Advances in Neural Information Processing Systems*, 33:21271–21284, 2020. [2.1](#), [2.1.2](#), [2.2.2](#), [3.1](#), [3.3.2](#), [4.1](#)
- [107] Caglar Gulcehre, Tom Le Paine, Srivatsan Srinivasan, Ksenia Konyushkova, Lotte Weerts, Abhishek Sharma, Aditya Siddhant, Alex Ahern, Miaosen Wang, and Chenjie Gu. Reinforced self-training (rest) for language modeling. *arXiv preprint arXiv:2308.08998*, 2023. [5.1.1](#), [5.5](#)
- [108] Suriya Gunasekar, Jason D Lee, Daniel Soudry, and Nati Srebro. Implicit bias of gradient descent on linear convolutional networks. *Advances in Neural Information Processing Systems*, 31, 2018. [3.6](#)
- [109] Zhaohan Daniel Guo, Bernardo Avila Pires, Bilal Piot, Jean-Bastien Grill, Florent Altché, Rémi Munos, and Mohammad Gheshlaghi Azar. Bootstrap latent-predictive representations for multitask reinforcement learning. In *International Conference on Machine Learning*, pages 3875–3886. PMLR, 2020. ([document](#)), [3.1](#), [3.2](#), [3.2](#), [3.2](#)
- [110] Michael U Gutmann and Aapo Hyvärinen. Noise-contrastive estimation of unnormalized statistical models, with applications to natural image statistics. *Journal of Machine Learning Research*, 13(Feb):307–361, 2012. [2.1.2](#)
- [111] Michael Hahn. Theoretical limitations of self-attention in neural sequence models. *Transactions of the Association for Computational Linguistics*, 8:156–171, 2020. [4.6](#)
- [112] Jeff Z HaoChen, Colin Wei, Adrien Gaidon, and Tengyu Ma. Provable guarantees for self-supervised deep learning with spectral contrastive loss. In *Advances in Neural Information Processing Systems*, volume 34, pages 5000–5011, 2021. [2.1.2](#), [3.1](#), [4.1](#)
- [113] Kaiming He, Haoqi Fan, Yuxin Wu, Saining Xie, and Ross Girshick. Momentum contrast for unsupervised visual representation learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 9729–9738, 2020. [2.1](#), [2.1.2](#), [7](#), [3.1](#), [3.6](#), [4.1](#)
- [114] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 16000–16009, 2022. [3.3.2](#), [3.6](#), [4.1](#), [4.2.1](#), [4.2.1](#), [4.2.3](#), [4.2.2](#), [4.2.3](#)

- [115] Zhiyuan He, Xufang Luo, Yike Zhang, Yuqing Yang, and Lili Qiu. V1 norm: Rethink loss aggregation in rlvr. *arXiv preprint arXiv:2509.07558*, 2025. 6.3.4
- [116] R Devon Hjelm, Alex Fedorov, Samuel Lavoie-Marchildon, Karan Grewal, Phil Bachman, Adam Trischler, and Yoshua Bengio. Learning deep representations by mutual information estimation and maximization. In *International Conference on Learning Representations*, 2019. 3.1
- [117] Namgyu Ho, Laura Schmid, and Se-Young Yun. Large language models are reasoning teachers. *arXiv preprint arXiv:2212.10071*, 2022. 5.3.3
- [118] Kelly Hong, Anton Troynikov, and Jeff Huber. Context rot: How increasing input tokens impacts LLM performance. Technical report, Chroma, July 2025. URL <https://research.trychroma.com/context-rot>. 5.1
- [119] Kaiying Hou, David Brandfonbrener, Sham Kakade, Samy Jelassi, and Eran Malach. Universal length generalization with turing programs. *arXiv preprint arXiv:2407.03310*, 2024. 5.1
- [120] Cheng-Ping Hsieh, Simeng Sun, Samuel Krizan, Shantanu Acharya, Dima Rekesh, Fei Jia, Yang Zhang, and Boris Ginsburg. Ruler: What’s the real context size of your long-context language models? *arXiv preprint arXiv:2404.06654*, 2024. 5.1
- [121] Xinyang Hu, Fengzhuo Zhang, Siyu Chen, and Zhuoran Yang. Unveiling the statistical foundations of chain-of-thought prompting methods. *arXiv preprint arXiv:2408.14511*, 2024. 5.1
- [122] Tianyu Hua, Wenxiao Wang, Zihui Xue, Sucheng Ren, Yue Wang, and Hang Zhao. On feature decorrelation in self-supervised learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 9598–9608, 2021. 3.1, 3.1, 3.1, 3.2
- [123] Audrey Huang, Adam Block, Dylan J Foster, Dhruv Rohatgi, Cyril Zhang, Max Simchowitz, Jordan T Ash, and Akshay Krishnamurthy. Self-improvement in language models: The sharpening mechanism. *arXiv preprint arXiv:2412.01951*, 2024. 5.5
- [124] Ruiquan Huang, Yingbin Liang, and Jing Yang. How transformers learn regular language recognition: A theoretical study on training dynamics and implicit bias. *arXiv preprint arXiv:2505.00926*, 2025. 5.1, 5.1.1, 5.A.3
- [125] Weiran Huang, Mingyang Yi, and Xuyang Zhao. Towards the generalization of contrastive self-supervised learning. *arXiv preprint arXiv:2111.00743*, 2021. 3.1
- [126] Xinting Huang, Andy Yang, Satwik Bhattamishra, Yash Sarrof, Andreas Krebs, Hattie Zhou, Preetum Nakkiran, and Michael Hahn. A formal framework for understanding length generalization in transformers. *arXiv preprint arXiv:2410.02140*, 2024. 5.1, 5.1.1
- [127] Yu Huang, Junyang Lin, Chang Zhou, Hongxia Yang, and Longbo Huang. Modality competition: What makes joint training of multi-modal network fail in deep learning?(provably). In *International Conference on Machine Learning*, pages 9226–9259. PMLR, 2022. 5.3.4, 5.A.3

- [128] Yu Huang, Yuan Cheng, and Yingbin Liang. In-context convergence of transformers. *International Conference on Machine Learning*, pages 19660–19722, 2024. 4.2.3, 4.6, 5.3.2, 5.3.3, 5.A.3, 5.A.3, 6.3.2
- [129] Yu Huang, Zixin Wen, Yuejie Chi, and Yingbin Liang. A theoretical analysis of self-supervised learning for vision transformers. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=Antib6Uovh>. 5.3.2, 5.3.3, 5.A.3, 5.A.3, 6.3.2
- [130] Yu Huang, Zixin Wen, Aarti Singh, Yuejie Chi, and Yuxin Chen. Transformers provably learn chain-of-thought reasoning with length generalization. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. 5.1, 5.1.1, 5.A.3, 6.3.1, 6.3.3, 6.3.4
- [131] Yu Huang, Zixin Wen, Yuejie Chi, Yuting Wei, Aarti Singh, Yingbin Liang, and Yuxin Chen. On the emergence of implicit curriculum in RLVR learning dynamics. In *Proceedings of the 43rd International Conference on Machine Learning*, 2026. 7.2, 7.B.1
- [132] Arthur Jacot, Franck Gabriel, and Clément Hongler. Neural tangent kernel: Convergence and generalization in neural networks. In *Advances in neural information processing systems*, pages 8571–8580, 2018. 2.1.2
- [133] Samy Jelassi and Yuanzhi Li. Towards understanding how momentum improves generalization in deep learning. In *International Conference on Machine Learning*, pages 9965–10040. PMLR, 2022. 3.2, 3.6
- [134] Samy Jelassi, Arthur Mensch, Gauthier Gidel, and Yuanzhi Li. Adam is no better than normalized SGD: Dissecting how adaptivity improves GAN performance, 2022. URL <https://openreview.net/forum?id=D9SuLzhgK9>. 3.2
- [135] Samy Jelassi, Michael Sander, and Yuanzhi Li. Vision transformers provably learn spatial structure. In *Advances in Neural Information Processing Systems*, volume 35, pages 37822–37836, 2022. 4.1, 4.2.3, 4.3, 5.A.3, 6.3.2
- [136] Samy Jelassi, Stéphane d’Ascoli, Carles Domingo-Enrich, Yuhuai Wu, Yuanzhi Li, and François Charton. Length generalization in arithmetic transformers. *arXiv preprint arXiv:2306.15400*, 2023. 5.1
- [137] Wenlong Ji, Zhun Deng, Ryumei Nakada, James Zou, and Linjun Zhang. The power of contrast for feature learning: A theoretical analysis. *Journal of Machine Learning Research*, 24 (330):1–78, 2023. 3.1, 3.3.2, 4.4.2
- [138] Ziwei Ji and Matus Telgarsky. The implicit bias of gradient descent on nonseparable data. In *Conference on Learning Theory*, pages 1772–1798. PMLR, 2019. 3.6
- [139] Li Jing, Pascal Vincent, Yann LeCun, and Yuandong Tian. Understanding dimensional collapse in contrastive self-supervised learning. *arXiv preprint arXiv:2110.09348*, 2021. 3.1.1
- [140] Longlong Jing and Yingli Tian. Self-supervised visual feature learning with deep neural networks: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, pages

1–1, 2020. 2.1

- [141] Cameron R Jones and Benjamin K Bergen. Large language models pass the turing test. *arXiv preprint arXiv:2503.23674*, 2025. 5.1
- [142] Nirmitt Joshi, Gal Vardi, Adam Block, Surbhi Goel, Zhiyuan Li, Theodor Misiakiewicz, and Nathan Srebro. A theory of learning with autoregressive chain of thought. *arXiv preprint arXiv:2503.07932*, 2025. 5.1, 5.1.1
- [143] Stefani Karp, Ezra Winston, Yuanzhi Li, and Aarti Singh. Local signal adaptivity: Provable feature learning in neural networks beyond kernels. *Advances in Neural Information Processing Systems*, 34, 2021. 3.6
- [144] Amirhossein Kazemnejad, Inkit Padhi, Karthikeyan Natesan Ramamurthy, Payel Das, and Siva Reddy. The impact of positional encoding on length generalization in transformers. *Advances in Neural Information Processing Systems*, 36:24892–24928, 2023. 5.1
- [145] Joe Kileel, Matthew Trager, and Joan Bruna. On the expressive power of deep polynomial neural networks. *Advances in neural information processing systems*, 32, 2019. 3.3.1
- [146] Juno Kim and Taiji Suzuki. Transformers provably solve parity efficiently with chain of thought. *arXiv preprint arXiv:2410.08633*, 2024. 5.1, 5.1.1, 5.3.3, 5.A.3, 6.3.2, 6.3.4
- [147] Juno Kim, Denny Wu, Jason Lee, and Taiji Suzuki. Metastable dynamics of chain-of-thought reasoning: Provable benefits of search, rl and distillation. *arXiv preprint arXiv:2502.01694*, 2025. 5.1
- [148] Minseon Kim, Jihoon Tack, and Sung Ju Hwang. Adversarial self-supervised contrastive learning. In *Advances in Neural Information Processing Systems*, volume 33, 2020. (document), 2.1
- [149] Najoung Kim and Sebastian Schuster. Entity tracking in language models. *arXiv preprint arXiv:2305.02363*, 2023. 5.1, 5.2.1
- [150] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *CoRR*, abs/1412.6980, 2014. URL <https://api.semanticscholar.org/CorpusID:6628106>. 5.7, 6.7.2
- [151] Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. Large language models are zero-shot reasoners. *Advances in neural information processing systems*, 35:22199–22213, 2022. 5.1
- [152] Imre Risi Kondor. *Group theoretical methods in machine learning*. Columbia University, 2008. 6.A.3, 6.A.3, 6.B.2
- [153] Yeganeh Kordi, Nihal V. Nayak, Max Zuo, Ilana Nguyen, and Stephen Bach. Revisiting generalization across difficulty levels: It’s not so easy. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7014–7042, 2026. 6.2

- [154] Aviral Kumar, Vincent Zhuang, Rishabh Agarwal, Yi Su, John D Co-Reyes, Avi Singh, Kate Baumli, Shariq Iqbal, Colton Bishop, and Rebecca Roelofs. Training language models to self-correct via reinforcement learning. *arXiv preprint arXiv:2409.12917*, 2024. 5.1
- [155] Tanishq Kumar, Blake Bordelon, Samuel J. Gershman, and Cengiz Pehlevan. Grokking as the transition from lazy to rich training dynamics. In *The Twelfth International Conference on Learning Representations*, 2024. 6.2
- [156] Yury Kuratov, Aydar Bulatov, Petr Anokhin, Ivan Rodkin, Dmitry Sorokin, Artyom Sorokin, and Mikhail Burtsev. Babilong: Testing the limits of llms with long context reasoning-in-a-haystack. *Advances in Neural Information Processing Systems*, 37:106519–106554, 2024. 5.1
- [157] Nathan Lambert, Jacob Daniel Morrison, Valentina Pyatkin, Shengyi Huang, Hamish Ivison, Faeze Brahman, Lester James Validad Miranda, Alisa Liu, Nouha Dziri, Xinxu Lyu, Yuling Gu, Saumya Malik, Victoria Graf, Jena D. Hwang, Jiangjiang Yang, Ronan Le Bras, Oyvind Tafjord, Christopher Wilhelm, Luca Soldaini, Noah A. Smith, Yizhong Wang, Pradeep Dasigi, and Hanna Hajishirzi. Tülu 3: Pushing frontiers in open language model post-training. *arXiv preprint arXiv:2411.15124*, 2024. 6.1
- [158] Jason D Lee, Qi Lei, Nikunj Saunshi, and Jiacheng Zhuo. Predicting what you already know helps: Provable self-supervised learning. In *Advances in Neural Information Processing Systems*, volume 34, pages 309–323, 2021. 2.1.2, 2.2.2, 3.6, 4.6
- [159] Nayoung Lee, Ziyang Cai, Avi Schwarzschild, Kangwook Lee, and Dimitris Papailiopoulos. Self-improving transformers overcome easy-to-hard and length generalization challenges. *arXiv preprint arXiv:2502.01612*, 2025. 5.1, 2, 5.1.1, 5.5, 5.7
- [160] Belinda Z. Li, Zifan Carl Guo, and Jacob Andreas. (how) do language models track state? *ArXiv*, abs/2503.02854, 2025. URL <https://api.semanticscholar.org/CorpusID:276776093>. 5.2.1
- [161] Hongkang Li, Meng Wang, Sijia Liu, and Pin-Yu Chen. A theoretical understanding of shallow vision transformers: Learning, generalization, and sample complexity. *arXiv preprint arXiv:2302.06015*, 2023. 4.6
- [162] Hongkang Li, Meng Wang, Tengfei Ma, Sijia Liu, Zaixi Zhang, and Pin-Yu Chen. What improves the generalization of graph transformer? a theoretical dive into self-attention and positional encoding. In *NeurIPS 2023 Workshop: New Frontiers in Graph Learning*, 2023. 4.6
- [163] Hongkang Li, Meng Wang, Songtao Lu, Xiaodong Cui, and Pin-Yu Chen. Training nonlinear transformers for efficient in-context learning: A theoretical learning and generalization analysis, 2024. 4.6
- [164] Yingcong Li, Kartik Sreenivasan, Angeliki Giannou, Dimitris Papailiopoulos, and Samet Oymak. Dissecting chain-of-thought: Compositionality through in-context filtering and learning. *Advances in Neural Information Processing Systems*, 36:22021–22046, 2023. 5.1

- [165] Yuanzhi Li and Zehao Dou. Making method of moments great again?—how can gans learn distributions. *arXiv preprint arXiv:2003.04033*, 2020. 3.3.1
- [166] Yuanzhi Li and Yingyu Liang. Learning overparameterized neural networks via stochastic gradient descent on structured data. In *Advances in Neural Information Processing Systems*, pages 8157–8166, 2018. 2.1.2
- [167] Yuanzhi Li and Yang Yuan. Convergence analysis of two-layer neural networks with relu activation. In *Advances in neural information processing systems*, pages 597–607, 2017. 2.1.2, 3.6
- [168] Yuanzhi Li, Tengyu Ma, and Hongyang Zhang. Algorithmic regularization in over-parameterized matrix sensing and neural networks with quadratic activations. In *COLT 2018: 31st Annual Conference on Learning Theory*, pages 2–47, 2018. 2.1.2, 3.6
- [169] Yuanzhi Li, Colin Wei, and Tengyu Ma. Towards explaining the regularization effect of initial large learning rate in training neural networks. In *NeurIPS 2019 : Thirty-third Conference on Neural Information Processing Systems*, pages 11674–11685, 2019. 3.6
- [170] Yuanzhi Li, Tengyu Ma, and Hongyang R. Zhang. Learning over-parametrized two-layer relu neural networks beyond ntk. In *COLT*, pages 2613–2682, 2020. 2.1.2, 3.6
- [171] Zhiyuan Li, Hong Liu, Denny Zhou, and Tengyu Ma. Chain of thought empowers transformers to solve inherently serial problems. *arXiv preprint arXiv:2402.12875*, 1, 2024. 5.1, 3, 5.1.1, 5.2.1, 5.5, 5.7
- [172] Zhan Ling, Kang Liu, Kai Yan, Yifan Yang, Weijian Lin, Ting-Han Fan, Lingfeng Shen, Zhengyin Du, and Jiecao Chen. Longreason: A synthetic long-context reasoning benchmark via context expansion. *arXiv preprint arXiv:2501.15089*, 2025. 5.1
- [173] Bingbin Liu, Elan Rosenfeld, Pradeep Ravikumar, and Andrej Risteski. Analyzing and improving the optimization landscape of noise-contrastive estimation. *arXiv preprint arXiv:2110.11271*, 2021.
- [174] Bingbin Liu, Jordan T Ash, Surbhi Goel, Akshay Krishnamurthy, and Cyril Zhang. Transformers learn shortcuts to automata. *arXiv preprint arXiv:2210.10749*, 2022. 5.1, 5.2.1, 5.2.1, 6.3.1
- [175] Bingbin Liu, Daniel J Hsu, Pradeep Ravikumar, and Andrej Risteski. Masked prediction: A parameter identifiability view. In *Advances in Neural Information Processing Systems*, pages 21241–21254, 2022. 3.3.2, 3.6, 4.6
- [176] Mingjie Liu, Shizhe Diao, Ximing Lu, Jian Hu, Xin Dong, Yejin Choi, Jan Kautz, and Yi Dong. Prorl: Prolonged reinforcement learning expands reasoning boundaries in large language models. *arXiv preprint arXiv:2505.24864*, 2025. 6.2
- [177] Ziming Liu, Ouail Kitouni, Niklas S Nolte, Eric Michaud, Max Tegmark, and Mike Williams. Towards understanding grokking: An effective theory of representation learning. *Advances in Neural Information Processing Systems*, 35:34651–34663, 2022. 6.2

- [178] Zhou Lu, Hongming Pu, Feicheng Wang, Zhiqiang Hu, and Liwei Wang. The expressive power of neural networks: A view from the width. *Advances in neural information processing systems*, 30, 2017. [5.4](#)
- [179] Zeping Luo, Cindy Weng, Shiyu Wu, Mo Zhou, and Rong Ge. One objective for all models—self-supervised learning for topic models. *arXiv preprint arXiv:2203.03539*, 2022. [3.1](#)
- [180] Bochen Lyu, Yiyang Jia, Xiaohao Cai, and Zhanxing Zhu. Transformers with RL or SFT provably learn sparse boolean functions, but differently. *arXiv preprint arXiv:2511.17852*, 2025. [6.2](#)
- [181] Kaifeng Lyu and Jian Li. Gradient descent maximizes the margin of homogeneous neural networks. *arXiv preprint arXiv:1906.05890*, 2019. [3.6](#)
- [182] Wolfgang Maass. Lower bounds for the computational power of networks of spiking neurons. *Neural computation*, 8(1):1–40, 1996. [5.2](#)
- [183] Aman Madaan and Amir Yazdanbakhsh. Text and patterns: For effective chain of thought, it takes two to tango. *arXiv preprint arXiv:2209.07686*, 2022. [5.1](#)
- [184] Lucie Charlotte Magister, Jonathan Mallinson, Jakub Adamek, Eric Malmi, and Aliaksei Severyn. Teaching small language models to reason. *arXiv preprint arXiv:2212.08410*, 2022. [5.1](#)
- [185] Julien Mairal, Francis Bach, and Jean Ponce. *Sparse Modeling for Image and Vision Processing*. 2014. [2.2.1](#)
- [186] Giovanni Luca Marchetti, Christopher J Hillar, Danica Kragic, and Sophia Sanborn. Harmonics of learning: Universal fourier features emerge in invariant networks. In *The Thirty Seventh Annual Conference on Learning Theory*, pages 3775–3797. PMLR, 2024. [6.A.3](#)
- [187] Giovanni Luca Marchetti, Daniel Kunin, Alex Myers, Fernando Acosta, and Nina Miolane. Sequential group composition: A window into the mechanics of deep learning. *arXiv preprint arXiv:2602.03655*, 2026. [7.2](#)
- [188] Annie Marsden, Evan Dogariu, Naman Agarwal, Xinyi Chen, Daniel Suo, and Elad Hazan. Provable length generalization in sequence prediction via spectral filtering. *arXiv preprint arXiv:2411.01035*, 2024. [5.1](#), [5.1.1](#)
- [189] Luke Melas-Kyriazi. Do you even need attention? a stack of feed-forward layers does surprisingly well on imagenet. *arXiv preprint arXiv:2105.02723*, 2021. [4.6](#)
- [190] William Merrill and Ashish Sabharwal. The parallelism tradeoff: Limitations of log-precision transformers. *Transactions of the Association for Computational Linguistics*, 11: 531–545, 2023. [5.1](#), [5.2](#), [5.2.1](#)
- [191] William Merrill and Ashish Sabharwal. The expressive power of transformers with chain of thought. *arXiv preprint arXiv:2310.07923*, 2023. [5.1](#), [3](#), [5.1.1](#), [5.2.1](#)
- [192] William Merrill and Ashish Sabharwal. Exact expressive power of transformers with padding. *arXiv preprint arXiv:2505.18948*, 2025. [5.2](#)

- [193] William Merrill and Ashish Sabharwal. A little depth goes a long way: The expressive power of log-depth transformers. *arXiv preprint arXiv:2503.03961*, 2025. [5.2.1](#)
- [194] William Merrill, Ashish Sabharwal, and Noah A Smith. Saturated transformers are constant-depth threshold circuits. *Transactions of the Association for Computational Linguistics*, 10: 843–856, 2022. [5.2.1](#)
- [195] William Merrill, Jackson Petty, and Ashish Sabharwal. The illusion of state in state-space models. *arXiv preprint arXiv:2404.08819*, 2024. [5.1](#), [5.2.1](#), [6.3.1](#)
- [196] Tomas Mikolov, Kai Chen, Greg Corrado, and Jeffrey Dean. Efficient estimation of word representations in vector space. *arXiv preprint arXiv:1301.3781*, 2013. [2.1](#), [2.1.2](#)
- [197] Yingqian Min, Zhipeng Chen, Jinhao Jiang, Jie Chen, Jia Deng, Yiwen Hu, Yiru Tang, Jiapeng Wang, Xiaoxue Cheng, and Huatong Song. Imitate, explore, and self-improve: A reproduction report on slow-thinking reasoning systems. *arXiv preprint arXiv:2412.09413*, 2024. [5.1](#)
- [198] Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. *arXiv preprint arXiv:2301.05217*, 2023. [6.2](#)
- [199] Sanmit Narvekar, Bei Peng, Matteo Leonetti, Jivko Sinapov, Matthew E Taylor, and Peter Stone. Curriculum learning for reinforcement learning domains: A framework and survey. *Journal of Machine Learning Research*, 21(181):1–50, 2020. [6.2](#)
- [200] Thanh V. Nguyen, Raymond K. W. Wong, and Chinmay Hegde. Benefits of jointly training autoencoders: An improved neural tangent kernel analysis. *arXiv preprint arXiv:1911.11983*, 2019. [2.1.2](#)
- [201] Eshaan Nichani, Alexandru Damian, and Jason D. Lee. Provable guarantees for nonlinear feature learning in three-layer neural networks. In *Advances in Neural Information Processing Systems*, 2023. [7.2](#)
- [202] Eshaan Nichani, Alex Damian, and Jason D Lee. How transformers learn causal structure with gradient descent. *arXiv preprint arXiv:2402.14735*, 2024. [4.6](#), [5.A.3](#)
- [203] Daisuke Niizumi, Daiki Takeuchi, Yasunori Ohishi, Noboru Harada, and Kunio Kashino. Byol for audio: Self-supervised learning for general-purpose audio representation. In *2021 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8. IEEE, 2021. [3.1](#)
- [204] Bruno A Olshausen and David J Field. Sparse coding with an overcomplete basis set: A strategy employed by v1? *Vision research*, 37(23):3311–3325, 1997. [2.2.1](#)
- [205] Bruno A Olshausen and David J Field. Sparse coding of sensory inputs. *Current Opinion in Neurobiology*, 14(4):481–487, 2004. [2.2.1](#)
- [206] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018. [2.1.2](#), [3.1](#)

- [207] OpenAI. OpenAI o1 system card. *ArXiv*, abs/2412.16720, 2024. URL <https://api.semanticscholar.org/CorpusID:274611667>. 5.1, 6.1
- [208] Jiachun Pan, Pan Zhou, and YAN Shuicheng. Towards understanding why mask reconstruction pretraining helps in downstream tasks. In *The Eleventh International Conference on Learning Representations*, 2022. 4.1, 4.4.2
- [209] Shubham Parashar, Shurui Gui, Xiner Li, Hongyi Ling, Sushil Vemuri, Blake Olson, Eric Li, Yu Zhang, James Caverlee, Dileep Kalathil, et al. Curriculum reinforcement learning from easy to hard tasks improves llm reasoning. *arXiv preprint arXiv:2506.06632*, 2025. 6.2, 6.5
- [210] Ian Parberry. *Circuit complexity and neural networks*. MIT press, 1994. 5.2
- [211] Namuk Park and Songkuk Kim. How do vision transformers work? *arXiv preprint arXiv:2202.06709*, 2022. 4.6
- [212] Namuk Park, Wonjae Kim, Byeongho Heo, Taekyung Kim, and Sangdoo Yun. What do self-supervised vision transformers learn? In *The Eleventh International Conference on Learning Representations*, 2023. (document), 4.1, 4.1, 4.1, 4.2.2, 4.4.2, 4.6, 4.7
- [213] Simon Park, Simran Kaur, and Sanjeev Arora. How does RL post-training induce skill composition? a case study on countdown. *arXiv preprint arXiv:2512.01775*, 2025. 6.3.2
- [214] Sayak Paul and Pin-Yu Chen. Vision transformers are robust learners. In *Proceedings of the AAAI conference on Artificial Intelligence*, volume 36, pages 2071–2081, 2022. 4.6
- [215] Ashwini Pople, Jinjin Tian, Yuchen Li, and Andrej Risteski. Contrasting the landscape of contrastive and non-contrastive learning. *arXiv preprint arXiv:2203.15702*, 2022. 3.1, 3.1.1, 4.6
- [216] Alethea Power, Yuri Burda, Harrison Edwards, Igor Babuschkin, and Vedant Misra. Grokking: Generalization beyond overfitting on small algorithmic datasets. *arXiv preprint arXiv:2201.02177*, 2022. 6.2, 6.5.2
- [217] M. Protter and M. Elad. Image sequence denoising via sparse and redundant representations. *IEEE Transactions on Image Processing*, 18(1):27–35, 2009. 2.2.1
- [218] Ben Prystawski, Michael Li, and Noah Goodman. Why think step by step? reasoning emerges from the locality of experience. *Advances in Neural Information Processing Systems*, 36:70926–70947, 2023. 5.1
- [219] Yiwei Qin, Xuefeng Li, Haoyang Zou, Yixiu Liu, Shijie Xia, Zhen Huang, Yixin Ye, Weizhe Yuan, Hector Liu, and Yuanzhi Li. O1 replication journey: A strategic progress report—part 1. *arXiv preprint arXiv:2410.18982*, 2024. 5.1
- [220] Yuxiao Qu, Amrith Setlur, Virginia Smith, Ruslan Salakhutdinov, and Aviral Kumar. POPE: Learning to reason on hard problems via privileged on-policy exploration. *arXiv preprint arXiv:2601.18779*, 2026. 6.2
- [221] Martin Raab and Angelika Steger. “balls into bins”—a simple and tight analysis. In *International Workshop on Randomization and Approximation Techniques in Computer Science*,

- pages 159–170. Springer, 1998. 5.B.6
- [222] Ali Rad, Khashayar Filom, Darioush Keivan, Peyman Mohajerin Esfahani, and Ehsan Kamalinejad. Rate or fate? RLV^eR: Reinforcement learning with verifiable noisy rewards. 2026. 6.2
 - [223] Alec Radford, Karthik Narasimhan, Tim Salimans, Ilya Sutskever, et al. Improving language understanding by generative pre-training. 2018. 4.1
 - [224] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PMLR, 2021. 3.6, 4.1
 - [225] Adityanarayanan Radhakrishnan, Karren Yang, Mikhail Belkin, and Caroline Uhler. Memorization in overparameterized autoencoders. *arXiv preprint arXiv:1810.10333*, 2018. 2.1.2
 - [226] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of machine learning research*, 21(140):1–67, 2020. 5.1
 - [227] Maithra Raghu, Thomas Unterthiner, Simon Kornblith, Chiyuan Zhang, and Alexey Dosovitskiy. Do vision transformers see like convolutional neural networks? *Advances in Neural Information Processing Systems*, 34:12116–12128, 2021. 4.6
 - [228] Aditya Ramesh, Mikhail Pavlov, Gabriel Goh, Scott Gray, Chelsea Voss, Alec Radford, Mark Chen, and Ilya Sutskever. Zero-shot text-to-image generation. In *International Conference on Machine Learning*, pages 8821–8831. PMLR, 2021. 3.6
 - [229] Aditya Ramesh, Prafulla Dhariwal, Alex Nichol, Casey Chu, and Mark Chen. Hierarchical text-conditional image generation with clip latents. *OpenAI blog*, 2022. URL <https://cdn.openai.com/papers/dall-e-2.pdf>. 3.6
 - [230] Rahul Ramesh, Ekdeep Singh Lubana, Mikail Khona, Robert P. Dick, and Hidenori Tanaka. Compositional capabilities of autoregressive transformers: A study on synthetic, interpretable tasks. In *Proceedings of the 41st International Conference on Machine Learning*, 2024. 7.2
 - [231] Yuval Ran-milo, Yotam Alexander, Shahar Mendel, and Nadav Cohen. Outcome-based rl provably leads transformers to reason, but only with the right data. 2026. 6.2
 - [232] Noam Razin and Nadav Cohen. Implicit regularization in deep learning may not be explainable by norms. *Advances in neural information processing systems*, 33:21174–21187, 2020. 3.6
 - [233] Yi Ren, Yatin Dandi, Florent Krzakala, and Jason D. Lee. Provable learning of random hierarchy models and hierarchical shallow-to-deep chaining. In *Proceedings of the 39th Conference on Learning Theory*, 2026. 7.2

- [234] Yi Ren, Zixuan Wang, and Jason D. Lee. Algorithmic separation between constant-depth and logarithmic-depth neural networks. *arXiv preprint arXiv:2607.25200*, 2026. 7.2
- [235] Laria Reynolds and Kyle McDonell. Prompt programming for large language models: Beyond the few-shot paradigm. In *Extended abstracts of the 2021 CHI conference on human factors in computing systems*, pages 1–7, 2021. 5.1
- [236] Pierre H. Richemond, Jean-Bastien Grill, Florent Altché, Corentin Tallec, Florian Strub, Andrew Brock, Samuel Smith, Soham De, Razvan Pascanu, Bilal Piot, and Michal Valko. BYOL works even without batch statistics. *arXiv:2010.10241 [cs, stat]*, October 2020. 3.1
- [237] Alessandro Rinaldo. 36-755: Advanced statistical theory, lecture 27. Lecture notes, December 5 2016. Scribed by Xiao Hui Tai. 5.B.2
- [238] Joshua Robinson, Li Sun, Ke Yu, Kayhan Batmanghelich, Stefanie Jegelka, and Suvrit Sra. Can contrastive learning avoid shortcut solutions? *Advances in neural information processing systems*, 34:4974–4986, 2021. 3.6, 4.1, 4.6
- [239] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, et al. Imagenet large scale visual recognition challenge. *International journal of computer vision*, 115:211–252, 2015. 4.7
- [240] Mahdi Sabbaghi, George Pappas, Hamed Hassani, and Surbhi Goel. Explicitly encoding structural symmetry is key to length generalization in arithmetic tasks. *arXiv preprint arXiv:2406.01895*, 2024. 5.1
- [241] Clayton Sanford, Bahare Fatemi, Ethan Hall, Anton Tsitsulin, Mehran Kazemi, Jonathan Halcrow, Bryan Perozzi, and Vahab Mirrokni. Understanding transformer reasoning capabilities via graph algorithms. *Advances in Neural Information Processing Systems*, 37: 78320–78370, 2024. 5.1
- [242] Clayton Sanford, Daniel Hsu, and Matus Telgarsky. Transformers, parallel computation, and logarithmic depth. *arXiv preprint arXiv:2402.09268*, 2024. 4.6
- [243] Clayton Sanford, Daniel J Hsu, and Matus Telgarsky. Representational strengths and limitations of transformers. *Advances in Neural Information Processing Systems*, 36, 2024. 4.6
- [244] Nikunj Saunshi, Sadhika Malladi, and Sanjeev Arora. A mathematical exploration of why language models help solve downstream tasks. *arXiv preprint arXiv:2010.03648*, 2020. 3.6
- [245] Nikunj Saunshi, Jordan Ash, Surbhi Goel, Dipendra Misra, Cyril Zhang, Sanjeev Arora, Sham Kakade, and Akshay Krishnamurthy. Understanding contrastive learning requires incorporating inductive biases. In *International Conference on Machine Learning*, pages 19250–19286. PMLR, 2022. 3.6
- [246] Nikunj Saunshi, Nishanth Dikkala, Zhiyuan Li, Sanjiv Kumar, and Sashank J. Reddi. Reasoning with latent thoughts: On the power of looped transformers. In *International Conference on Learning Representations*, 2025. 7.1, 7.2

- [247] Andrew M. Saxe, James L. McClelland, and Surya Ganguli. Exact solutions to the nonlinear dynamics of learning in deep linear neural networks. In *International Conference on Learning Representations*, 2014. [7.2](#)
- [248] Imanol Schlag, Kazuki Irie, and Jürgen Schmidhuber. Linear transformers are secretly fast weight programmers. In *International conference on machine learning*, pages 9355–9366. PMLR, 2021. [8.1](#)
- [249] Jean-Pierre Serre. Linear representations of finite groups. In *Graduate texts in mathematics*, 1977. [6.B.2](#)
- [250] Amrith Setlur, Zijian Wang, Andrew Cohen, Paria Rashidinejad, and Sang Michael Xie. Reuse your FLOPs: Scaling RL on hard problems by conditioning on very off-policy prefixes. *arXiv preprint arXiv:2601.18795*, 2026. [6.2](#)
- [251] Rulin Shao, Shuyue Stella Li, Rui Xin, Scott Geng, Yiping Wang, Sewoong Oh, Simon Shaolei Du, Nathan Lambert, Sewon Min, Ranjay Krishna, Yulia Tsvetkov, Hanna Hajishirzi, Pang Wei Koh, and Luke S. Zettlemoyer. Spurious rewards: Rethinking training signals in rlvr. *arXiv preprint arXiv:2506.10947*, 2025. [6.2](#)
- [252] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, YK Li, Yang Wu, et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024. [6.1](#), [6.7.2](#)
- [253] Peter Shaw, Jakob Uszkoreit, and Ashish Vaswani. Self-attention with relative position representations. *arXiv preprint arXiv:1803.02155*, 2018. [5.1](#)
- [254] John S Shawe-Taylor, Martin HG Anthony, and Walter Kern. Classes of feedforward neural networks and their circuit complexity. *Neural networks*, 5(6):971–977, 1992. [5.2](#)
- [255] Hava T Siegelmann and Eduardo D Sontag. On the computational power of neural nets. In *Proceedings of the fifth annual workshop on Computational learning theory*, pages 440–449, 1992. [5.2](#)
- [256] Jiří Šíma and Pekka Orponen. General-purpose computation with neural networks: A survey of complexity theoretic results. *Neural Computation*, 15(12):2727–2778, 2003. [5.2](#)
- [257] Avi Singh, John D Co-Reyes, Rishabh Agarwal, Ankesh Anand, Piyush Patil, Xavier Garcia, Peter J Liu, James Harrison, Jaehoon Lee, and Kelvin Xu. Beyond human data: Scaling self-training for problem-solving with language models. *arXiv preprint arXiv:2312.06585*, 2023. [5.1.1](#), [5.5](#)
- [258] Noah A. Smith and Jason Eisner. Contrastive estimation: Training log-linear models on unlabeled data. In *Proceedings of the 43rd Annual Meeting on Association for Computational Linguistics*, ACL ’05, pages 354–362, USA, 2005. Association for Computational Linguistics. doi: 10.3115/1219840.1219884. [2.1.2](#)

- [259] Mahdi Soltanolkotabi. Learning relus via gradient descent. In *Advances in Neural Information Processing Systems*, volume 30, pages 2007–2017, 2017. 2.1.2, 3.6
- [260] Yuda Song, Hanlin Zhang, Carson Eisenach, Sham Kakade, Dean Foster, and Udaya Ghai. Mind the gap: Examining the self-improvement capabilities of large language models. *arXiv preprint arXiv:2412.02674*, 2024. 5.5
- [261] Daniel Soudry, Elad Hoffer, Mor Shpigel Nacson, Suriya Gunasekar, and Nathan Srebro. The implicit bias of gradient descent on separable data. *The Journal of Machine Learning Research*, 19(1):2822–2878, 2018. 3.6
- [262] Yifan Sun, Yushan Liang, Zhen Zhang, and Jiaye Teng. Theoretical modeling of llm self-improvement training dynamics through solver-verifier gap. *arXiv preprint arXiv:2507.00075*, 2025. 5.5
- [263] Yiyu Sun, Yuhan Cao, Pohao Huang, Haoyue Bai, Hanna Hajishirzi, Nouha Dziri, and Dawn Xiaodong Song. RL grokking recipe: How does RL unlock and transfer new algorithms in LLMs? 2025. 6.1, 2, 6.2, 6.2, 6.5.2
- [264] Ilya Sutskever, Oriol Vinyals, and Quoc V. Le. Sequence to Sequence Learning with Neural Networks. In *Advances in Neural Information Processing Systems 27*, volume 27, pages 3104–3112, 2014. 2.1
- [265] Hugo Tabanelli, Yatin Dandi, Luca Pesce, and Florent Krzakala. Deep learning of compositional targets with hierarchical spectral methods. *arXiv preprint arXiv:2602.10867*, 2026. 7.2
- [266] Davoud Ataee Tarzanagh, Yingcong Li, Christos Thrampoulidis, and Samet Oymak. Transformers as support vector machines. *arXiv preprint arXiv:2308.16898*, 2023. 4.6
- [267] Davoud Ataee Tarzanagh, Yingcong Li, Xuechen Zhang, and Samet Oymak. Max-margin token selection in attention mechanism. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023. URL <https://openreview.net/forum?id=WXc808ghLH>. 4.6, 7.2
- [268] Anthropic’s Applied AI team. Effective context engineering for ai agents. Technical report, Anthropic, 2025. URL <https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents>. 5.1
- [269] Jiaye Teng, Weiran Huang, and Haowei He. Can pretext-based self-supervised learning be boosted by downstream data? a theoretical analysis. *arXiv preprint arXiv:2103.03568*, 2021. 3.6
- [270] Audrey Terras. *Fourier analysis on finite groups and applications*. Number 43. Cambridge University Press, 1999. 3, 6.A.3
- [271] Yonglong Tian, Dilip Krishnan, and Phillip Isola. Contrastive multiview coding. In *European conference on computer vision*, pages 776–794. Springer, 2020. 2.1, 2.1.2, 3.6
- [272] Yonglong Tian, Chen Sun, Ben Poole, Dilip Krishnan, Cordelia Schmid, and Phillip Isola. What makes for good views for contrastive learning. In *Advances in Neural Information*

Processing Systems, volume 33, 2020. 2.1, 2.1.2, 3.3.2

- [273] Yuandong Tian. An analytical formula of population gradient for two-layered relu network and its applications in convergence and critical point analysis. In *Proceedings of the 34th International Conference on Machine Learning-Volume 70*, pages 3404–3413. JMLR. org, 2017. 3.6
- [274] Yuandong Tian. Provable scaling laws of feature emergence from learning dynamics of grokking. *arXiv preprint arXiv:2509.21519*, 2025. 6.2
- [275] Yuandong Tian, Lantao Yu, Xinlei Chen, and Surya Ganguli. Understanding self-supervised learning with dual deep networks. *arXiv preprint arXiv:2010.00578*, 2020. 2.1.2, 2.2.2
- [276] Yuandong Tian, Xinlei Chen, and Surya Ganguli. Understanding self-supervised learning dynamics without contrastive pairs. In *International Conference on Machine Learning*, pages 10268–10278. PMLR, 2021. 3.1, 3.1, 3.1.1, 3.1.1, 3.1.1, 3.1.1, 4.1
- [277] Yuandong Tian, Yiping Wang, Beidi Chen, and Simon Du. Scan and snap: Understanding training dynamics and token composition in 1-layer transformer. *arXiv preprint arXiv:2305.16380*, 2023. 4.6
- [278] Yuandong Tian, Yiping Wang, Zhenyu Zhang, Beidi Chen, and Simon Shaolei Du. JoMA: Demystifying multilayer transformers via joint dynamics of MLP and attention. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=LbJqRGNYCf>. 4.6
- [279] Shengbang Tong, Ellis Brown, Penghao Wu, Sanghyun Woo, Manoj Middepogu, Sai Charitha Akula, Jihan Yang, Shusheng Yang, Adithya Iyer, Xichen Pan, et al. Cambrian-1: A fully open, vision-centric exploration of multimodal llms. *arXiv preprint arXiv:2406.16860*, 2024. 4.1
- [280] Christopher Tosh, Akshay Krishnamurthy, and Daniel Hsu. Contrastive estimation reveals topic posterior information to linear models. *arXiv preprint arXiv:2003.02234*, 2020. 2.1.2, 3.1
- [281] Christopher Tosh, Akshay Krishnamurthy, and Daniel Hsu. Contrastive learning, multi-view redundancy, and linear models. In *Algorithmic Learning Theory*, pages 1179–1206. PMLR, 2021. 2.1.2, 3.1
- [282] Hugo Touvron, Matthieu Cord, Matthijs Douze, Francisco Massa, Alexandre Sablayrolles, and Hervé Jégou. Training data-efficient image transformers & distillation through attention. In *International conference on machine learning*, pages 10347–10357. PMLR, 2021. 4.7
- [283] Asher Trockman and J Zico Kolter. Patches are all you need? *arXiv preprint arXiv:2201.09792*, 2022. 4.6
- [284] Yao-Hung Hubert Tsai, Yue Wu, Ruslan Salakhutdinov, and Louis-Philippe Morency. Demystifying self-supervised learning: An information-theoretical framework. 2020. 2.1, 2.1.2, 2.2.2

- [285] Yao-Hung Hubert Tsai, Yue Wu, Ruslan Salakhutdinov, and Louis-Philippe Morency. Self-supervised learning from a multi-view perspective. *arXiv preprint arXiv:2006.05576*, 2020. [3.1](#)
- [286] Nikolaos Tsilivis, Eran Malach, Karen Ullrich, and Julia Kempe. How reinforcement learning after next-token prediction facilitates learning. *arXiv preprint arXiv:2510.11495*, 2025. [6.2](#)
- [287] Bhavya Vasudeva, Puneesh Deora, and Christos Thrampoulidis. Implicit bias and fast convergence rates for self-attention. *arXiv preprint arXiv:2402.05738*, 2024. [4.6](#)
- [288] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017. [4.2.3](#), [5.1](#), [5.3.2](#), [6.1](#), [6.3.4](#)
- [289] William E. Vinje and Jack L. Gallant. Sparse coding and decorrelation in primary visual cortex during natural vision. *Science*, 287(5456):1273–1276, 2000. [2.2.1](#)
- [290] Heribert Vollmer. *Introduction to circuit complexity: a uniform approach*. Springer Science & Business Media, 1999. [1](#), [5.2.1](#)
- [291] Julius Von Kügelgen, Yash Sharma, Luigi Gresele, Wieland Brendel, Bernhard Schölkopf, Michel Besserve, and Francesco Locatello. Self-supervised learning with data augmentations provably isolates content from style. *Advances in Neural Information Processing Systems*, 34, 2021. [3.1](#)
- [292] Johannes Von Oswald, Eyvind Niklasson, Ettore Randazzo, João Sacramento, Alexander Mordvintsev, Andrey Zhmoginov, and Max Vladymyrov. Transformers learn in-context by gradient descent. In *International Conference on Machine Learning*, pages 35151–35174. PMLR, 2023. [4.6](#)
- [293] James Vuckovic, Aristide Baratin, and Remi Tachet des Combes. A mathematical theory of attention. *arXiv preprint arXiv:2007.02876*, 2020. [4.6](#)
- [294] Shenzhi Wang, Le Yu, Chang Gao, Chujie Zheng, Shixuan Liu, Rui Lu, Kai Dang, Xionghui Chen, Jianxin Yang, Zhenru Zhang, Yuqiong Liu, An Yang, Andrew Zhao, Yang Yue, Shiji Song, Bowen Yu, Gao Huang, and Junyang Lin. Beyond the 80/20 rule: High-entropy minority tokens drive effective reinforcement learning for llm reasoning. *arXiv preprint arXiv:2506.01939*, 2025. [6.2](#)
- [295] Tongzhou Wang and Phillip Isola. Understanding contrastive representation learning through alignment and uniformity on the hypersphere. In *ICML*, pages 9929–9939, 2020. [2.1](#), [2.1.2](#), [2.4](#)
- [296] Xiang Wang, Xinlei Chen, Simon S Du, and Yuandong Tian. Towards demystifying representation learning with non-contrastive self-supervision. *arXiv preprint arXiv:2110.04947*, 2021. [3.1](#), [3.1](#), [3.1.1](#), [3.1.1](#), [3.1.1](#), [3.1.1](#), [4.1](#), [4.6](#)

- [297] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. *arXiv preprint arXiv:2203.11171*, 2022. [5.1](#)
- [298] Yiping Wang, Qing Yang, Zhiyuan Zeng, Liliang Ren, Lucas Liu, Baolin Peng, Hao Cheng, Xuehai He, Kuan Wang, Jianfeng Gao, Weizhu Chen, Shuohang Wang, Simon Shaolei Du, and Yelong Shen. Reinforcement learning for reasoning in large language models with one training example. *arXiv preprint arXiv:2504.20571*, 2025. [6.2](#)
- [299] Zixuan Wang, Eshaan Nichani, and Jason D. Lee. Learning hierarchical polynomials with three-layer neural networks. In *International Conference on Learning Representations*, 2024. [7.2](#)
- [300] Zixuan Wang, Shiyu Wei, Daniel Hsu, and Jason D. Lee. Transformers provably learn sparse token selection while fully-connected nets cannot. In *Proceedings of the 41st International Conference on Machine Learning*, 2024. [7.2](#)
- [301] Zixuan Wang, Eshaan Nichani, Alberto Bietti, Alexandru Damian, Daniel Hsu, Jason D. Lee, and Difan Wu. Learning compositional functions with transformers from easy-to-hard data. In *Proceedings of the 38th Conference on Learning Theory*, 2025. [5.2](#), [7.2](#)
- [302] Colin Wei, Kendrick Shen, Yining Chen, and Tengyu Ma. Theoretical Analysis of Self-Training with Deep Networks on Unlabeled Data. *arXiv:2010.03622 [cs, stat]*, October 2020. [2.2.2](#)
- [303] Colin Wei, Sang Michael Xie, and Tengyu Ma. Why do pretrained language models help in downstream tasks? an analysis of head and prompt tuning. *Advances in Neural Information Processing Systems*, 34:16158–16170, 2021. [3.6](#), [4.6](#)
- [304] Colin Wei, Yining Chen, and Tengyu Ma. Statistically meaningful approximation: a case study on approximating turing machines with transformers. *Advances in Neural Information Processing Systems*, 35:12071–12083, 2022. [4.6](#)
- [305] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837, 2022. [5.1](#)
- [306] Yixuan Wei, Han Hu, Zhenda Xie, Zheng Zhang, Yue Cao, Jianmin Bao, Dong Chen, and Baining Guo. Contrastive learning rivals masked image modeling in fine-tuning via feature distillation. *arXiv preprint arXiv:2205.14141*, 2022. [4.1](#), [4.2.2](#), [4.6](#)
- [307] Daphna Weinshall, Gad Cohen, and Dan Amir. Curriculum learning by transfer learning: Theory and experiments with deep networks. In *Proceedings of the 35th International Conference on Machine Learning*, 2018. [7.2](#)
- [308] Gail Weiss, Yoav Goldberg, and Eran Yahav. Thinking like transformers. In *International Conference on Machine Learning*, pages 11080–11090. PMLR, 2021. [4.6](#)
- [309] Kaiyue Wen, Huaqing Zhang, Hongzhou Lin, and Jingzhao Zhang. From sparse dependence to sparse attention: unveiling how chain-of-thought enhances transformer sample efficiency.

arXiv preprint arXiv:2410.05459, 2024. 5.1, 5.1.1, 5.A.3, 6.3.2, 6.3.4

- [310] Xumeng Wen, Zihan Liu, Shun Zheng, Shengyu Ye, Zhirong Wu, Yang Wang, Zhijian Xu, Xiao Liang, Junjie Li, Ziming Miao, et al. Reinforcement learning with verifiable rewards implicitly incentivizes correct reasoning in base LLMs. *arXiv preprint arXiv:2506.14245*, 2025. 6.1, 6.2
- [311] Zixin Wen and Yuanzhi Li. Toward understanding the feature learning process of self-supervised contrastive learning. In *International Conference on Machine Learning*, pages 11112–11122. PMLR, 2021. (document), 3.1, 3.2, 3.2, 3.2, 3.6, 3.3.2, 3.6, 4.1, 4.2.4, 4.4.2, 4.6, 5.A.3
- [312] Zixin Wen and Yuanzhi Li. The mechanism of prediction head in non-contrastive self-supervised learning. In *Advances in Neural Information Processing Systems*, 2022. 4.1, 4.2.4, 4.6, 5.A.3
- [313] Noam Wies, Yoav Levine, and Amnon Shashua. Sub-task decomposition enables learning in sequence to sequence tasks. *arXiv preprint arXiv:2204.02892*, 2022. 5.1, 7.2
- [314] Ronald J Williams. Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine learning*, 8(3):229–256, 1992. 6.1, 6.3.4
- [315] Fang Wu, Weihao Xuan, Ximing Lu, Mingjie Liu, Yi Dong, Zaid Harchaoui, and Yejin Choi. The invisible leash: Why RLVR may or may not escape its origin. *arXiv preprint arXiv:2507.14843*, 2025. 6.1, 6.2
- [316] Lei Wu, Chao Zhang, and Yuan Cao. Looped transformers with layer normalization provably learn the power method. *arXiv preprint arXiv:2606.00605*, 2026. 7.1, 7.2
- [317] Changnan Xiao and Bing Liu. Generalizing reasoning problems to longer lengths. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=zpENPcQSj1>. 5.1
- [318] Yuxi Xie, Anirudh Goyal, Wenyue Zheng, Min-Yen Kan, Timothy P Lillicrap, Kenji Kawaguchi, and Michael Shieh. Monte carlo tree search boosts reasoning via iterative preference learning. *arXiv preprint arXiv:2405.00451*, 2024. 5.1
- [319] Zhenda Xie, Zheng Zhang, Yue Cao, Yutong Lin, Jianmin Bao, Zhuliang Yao, Qi Dai, and Han Hu. Simmim: A simple framework for masked image modeling. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9653–9663, 2022. 4.1, 4.2.1, 4.2.1
- [320] Zhenda Xie, Zigang Geng, Jingcheng Hu, Zheng Zhang, Han Hu, and Yue Cao. Revealing the dark secrets of masked image modeling. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14475–14485, 2023. 4.1, 4.1, 4.6
- [321] Yuchen Yan, Yongliang Shen, Yang Liu, Jin Jiang, Mengdi Zhang, Jian Shao, and Yueting Zhuang. Infthythink: Breaking the length limits of long-context reasoning in large language models. *arXiv preprint arXiv:2503.06692*, 2025. 5.1

- [322] An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024. 6.7.2
- [323] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025. 6.7.2
- [324] Greg Yang and Edward J. Hu. Tensor programs IV: Feature learning in infinite-width neural networks. In *Proceedings of the 38th International Conference on Machine Learning*, 2021. 7.2
- [325] Jianchao Yang, Kai Yu, Yihong Gong, and Thomas Huang. Linear spatial pyramid matching using sparse coding for image classification. In *2009 IEEE Conference on Computer Vision and Pattern Recognition*, pages 1794–1801, 2009. 2.2.1
- [326] Liu Yang, Kangwook Lee, Robert Nowak, and Dimitris Papailiopoulos. Looped transformers are better at learning learning algorithms. In *International Conference on Learning Representations*, 2024. 7.1, 7.2
- [327] Songlin Yang, Bailin Wang, Yu Zhang, Yikang Shen, and Yoon Kim. Parallelizing linear transformers with the delta rule over sequence length. *Advances in neural information processing systems*, 37:115491–115522, 2024. 8.1
- [328] Tong Yang, Yu Huang, Yingbin Liang, and Yuejie Chi. In-context learning with representations: Contextual generalization of trained transformers. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024. 4.6, 5.3.3, 6.3.2
- [329] Tong Yang, Yu Huang, Yingbin Liang, and Yuejie Chi. Multi-head transformers provably learn symbolic multi-step reasoning via gradient descent. *arXiv preprint arXiv:2508.08222*, 2025. 5.3.2, 5.3.3, 6.3.4
- [330] Wang Yang, Zirui Liu, Hongye Jin, Qingyu Yin, Vipin Chaudhary, and Xiaotian Han. Longer context, deeper thinking: Uncovering the role of long-context ability in reasoning. *arXiv preprint arXiv:2505.17315*, 2025. 5.1
- [331] Zhilin Yang, Zihang Dai, Yiming Yang, Jaime Carbonell, Russ R Salakhutdinov, and Quoc V Le. Xlnet: Generalized autoregressive pretraining for language understanding. In *Advances in neural information processing systems*, pages 5754–5764, 2019. 3.6
- [332] Edward Yeo, Yuxuan Tong, Morry Niu, Graham Neubig, and Xiang Yue. Demystifying long chain-of-thought reasoning in llms. *arXiv preprint arXiv:2502.03373*, 2025. 6.1, 6.2
- [333] Lifan Yuan, Weize Chen, Yuchen Zhang, Ganqu Cui, Hanbin Wang, Ziming You, Ning Ding, Zhiyuan Liu, Maosong Sun, and Hao Peng. From $f(x)$ and $g(x)$ to $f(g(x))$: LLMs learn new skills in rl by composing old ones. *arXiv preprint arXiv:2509.25123*, 2025. 6.1, 6.2, 6.3.2

- [334] Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, Shiji Song, and Gao Huang. Does reinforcement learning really incentivize reasoning capacity in LLMs beyond the base model? *arXiv preprint arXiv:2504.13837*, 2025. 6.1, 6.2
- [335] Chulhee Yun, Srinadh Bhojanapalli, Ankit Singh Rawat, Sashank J Reddi, and Sanjiv Kumar. Are transformers universal approximators of sequence-to-sequence functions? *arXiv preprint arXiv:1912.10077*, 2019. 4.6
- [336] Jure Zbontar, Li Jing, Ishan Misra, Yann LeCun, and Stéphane Deny. Barlow twins: Self-supervised learning via redundancy reduction. In *International Conference on Machine Learning*, pages 12310–12320. PMLR, 2021. 3.1, 3.1, 4.1
- [337] Weihao Zeng, Yuzhen Huang, Qian Liu, Wei Liu, Keqing He, Zejun Ma, and Junxian He. Simplerl-zoo: Investigating and taming zero reinforcement learning for open base models in the wild. *arXiv preprint arXiv:2503.18892*, 2025. 6.5
- [338] Chaoning Zhang, Kang Zhang, Chenshuang Zhang, Trung X Pham, Chang D Yoo, and In So Kweon. How does simsiam avoid collapse without negative samples? a unified understanding with self-supervised contrastive learning. *arXiv preprint arXiv:2203.16262*, 2022. 3.1, 3.2
- [339] Charlie Zhang, Graham Neubig, and Xiang Yue. On the interplay of pre-training, mid-training, and RL on reasoning language models. *arXiv preprint arXiv:2512.07783*, 2025. 6.1, 6.2, 6.5.3
- [340] Qi Zhang, Yifei Wang, and Yisen Wang. How mask matters: Towards theoretical understandings of masked autoencoders. *Advances in Neural Information Processing Systems*, 35: 27127–27139, 2022. 4.1
- [341] Ruiqi Zhang, Spencer Frei, and Peter L Bartlett. Trained transformers learn linear models in-context. *Journal of Machine Learning Research*, 25(49):1–55, 2024. 4.2.3, 4.6, 6.3.2
- [342] Yi Zhang, Arturs Backurs, Sébastien Bubeck, Ronen Eldan, Suriya Gunasekar, and Tal Wagner. Unveiling transformers with lego: a synthetic reasoning task. *arXiv preprint arXiv:2206.04301*, 2022. 5.1, 5.1, 5.2.1, 5.2.2, 5.2.2
- [343] Yufeng Zhang, Fengzhuo Zhang, Zhuoran Yang, and Zhaoran Wang. What and how does in-context learning learn? bayesian model averaging, parameterization, and generalization. *arXiv preprint arXiv:2305.19420*, 2023. 4.6
- [344] Rosie Zhao, Alexandru Meterez, Sham M. Kakade, Cengiz Pehlevan, Samy Jelassi, and Eran Malach. Echo chamber: RL post-training amplifies behaviors learned in pretraining. *arXiv preprint arXiv:2504.07912*, 2025. 6.2
- [345] Kai Zhong, Zhao Song, Prateek Jain, Peter L Bartlett, and Inderjit S Dhillon. Recovery guarantees for one-hidden-layer neural networks. In *Proceedings of the 34th International Conference on Machine Learning-Volume 70*, pages 4140–4149. JMLR. org, 2017. 3.6

- [346] Tianyang Zhong, Zhengliang Liu, Yi Pan, Yutong Zhang, Yifan Zhou, Shizhe Liang, Zihao Wu, Yanjun Lyu, Peng Shu, and Xiaowei Yu. Evaluation of OpenAI o1: Opportunities and challenges of AGI. *arXiv preprint arXiv:2409.18486*, 2024. 5.1
- [347] Denny Zhou, Nathanael Schärli, Le Hou, Jason Wei, Nathan Scales, Xuezhi Wang, Dale Schuurmans, Claire Cui, Olivier Bousquet, and Quoc Le. Least-to-most prompting enables complex reasoning in large language models. *arXiv preprint arXiv:2205.10625*, 2022. 5.1
- [348] Hattie Zhou, Arwen Bradley, Etai Littwin, Noam Razin, Omid Saremi, Josh Susskind, Samy Bengio, and Preetum Nakkiran. What algorithms can transformers learn? a study in length generalization. *arXiv preprint arXiv:2310.16028*, 2023. 5.1
- [349] Yongchao Zhou, Uri Alon, Xinyun Chen, Xuezhi Wang, Rishabh Agarwal, and Denny Zhou. Transformers can achieve length generalization but not robustly. *arXiv preprint arXiv:2402.09371*, 2024. 5.1
- [350] Hanqing Zhu, Zhenyu Zhang, Hanxian Huang, DiJia Su, Zechun Liu, Jiawei Zhao, Igor Fedorov, Hamed Pirsiavash, Zhizhou Sha, Jinwon Lee, et al. The path not taken: RLVR provably learns off the principals. *arXiv preprint arXiv:2511.08567*, 2025. 6.2
- [351] Rui-Jie Zhu, Zixuan Wang, Kai Hua, Tianyu Zhang, Ziniu Li, Haoran Que, Boyi Wei, Zixin Wen, Fan Yin, He Xing, et al. Scaling latent reasoning via looped language models. *arXiv preprint arXiv:2510.25741*, 2025. 7.1, 7.2
- [352] Difan Zou, Yuan Cao, Yuanzhi Li, and Quanquan Gu. Understanding the generalization of adam in learning neural networks with proper regularization. *arXiv preprint arXiv:2108.11371*, 2021. 3.6